Informal Notes on Mathematics

reorganized English version

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Read it, absorb it, and forget it.

— P. R. Halmos

Preface

One of my early notes contains a table of dualities that, at the time, I could barely understand. It matches Stone spaces with Boolean algebras, affine schemes with rings, and, further down, topoi with logoi. Keyao Peng shared it at Geek College. The first row was already strange enough: for a Stone space, knowing only the algebra of its subsets that are both open and closed is enough to reconstruct the points and topology. I wanted to know how that could be true, and what on earth the remaining rows meant.

A geometry problem in these pages may need only a diagram and a good observation. A question about sets or spaces may take several definitions before reaching its answer. There is algebra and topology, analysis and probability, and later notes on numerical methods and machine learning. Some pages grew out of talks and conversations; others answered problems from school. The chapters therefore differ in length and in what they expect a reader to know.

The earliest notes gave me an excuse to practice LaTeX. I was learning set theory at the time, and spent an unreasonable amount of time making the equations look right. A friend's talk on category theory helped draw me further into foundations. I also translated Tom Leinster's *Rethinking Set Theory*; corresponding with him encouraged me to keep reading. As school became busier, elementary mathematics and competition problems found their way into the collection as well.

The collection gradually spread beyond TeX files. Handwriting was faster, and much of the later material lived in Flexcil, in book margins, or on pages assembled from earlier notes. Some pieces began in blog posts or correspondence. Written in Chinese, English, and occasionally Russian, they could finally be brought together with the help of tools for reading handwriting and mixed-format documents. I have gathered them into an English TeX edition, revisiting explanations and diagrams along the way. The layout is inspired by Evan Chen's *An Infinitely Large Napkin*, which taught me a great deal and influenced the appearance of my first typed notes.

I have kept the dates on the chapters and grouped the notes by subject. There is no need to read them in order. If a question or a picture catches your eye somewhere in the middle, that is a perfectly good place to begin.

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Foundations of Mathematics

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Sets, Classes, and the First Warnings About Size

N-20210204

§1.1 Why classes enter before functions

The basic language of sets, classes, indexed families, and functions begins with a distinction that is easy to underestimate: sets may be elements of other collections, whereas proper classes cannot be treated in the same way without recreating the classical paradoxes.

Definition 1.1.1. A **set** is a class which can be an element of another class. A class which is not a set is called a **proper class**.

This is not meant to be the whole foundation of set theory, but it explains the working convention of the note: some collections are small enough to be members of other collections, while others are too large and must remain proper classes.

> **Caution**
>
> The distinction between sets and proper classes is formally essential: it is one of the mechanisms that prevents the usual paradoxes. In a class theory such as NBG, sets are precisely the classes that may occur as elements of classes; proper classes may not be elements.

§1.2 Russell's class as the first warning

The first serious example is Russell's construction. Consider

$$M = \{X \mid X \text{ is a set and } X \notin X\}.$$

At first sight this is a natural class: many sets are not members of themselves. For instance, the set of all books is not itself a book.

> **Proposition 1.2.1**
>
> The class M is not a set.

Proof. Assume, for contradiction, that M is a set. Then it makes sense to ask whether $M \in M$. By the defining condition of M,

$$M \in M \quad \Longleftrightarrow \quad M \notin M,$$

which is impossible. Hence M must be a proper class. □

This is the first place where the note is not merely listing definitions. The example shows why unrestricted comprehension cannot simply produce a set from every condition.

§1.3 Power sets and indexed families

Definition 1.3.1. The **power set axiom** asserts that for every set A, the class

$$\mathcal{P}(A) = \{X \mid X \subseteq A\}$$

of all subsets of A is itself a set. It is also common to write 2^A for this power set.

The next construction is an indexed family. Given a family of classes or sets

$$\{A_i \mid i \in I\},$$

its union and intersection are written

$$\bigcup_{i\in I} A_i = \{x \mid x \in A_i \text{ for some } i \in I\},$$

and

$$\bigcap_{i\in I} A_i = \{x \mid x \in A_i \text{ for all } i \in I\}.$$

Remark 1.3.2 — The indexing collection I need not be finite or countable. This is a useful mental shift: the notation $\{A_i\}_{i\in I}$ is not just a long list. It may be a genuinely infinite family.

Caution

If I is allowed to be a proper class, then extra care is needed. In ordinary ZFC, one usually speaks of set-indexed families. In a class-theoretic setting, class-indexed families can be discussed, but one must check separately when their unions or intersections are sets.

If A and B are classes, the relative complement of A in B is

$$B \setminus A = \{x \mid x \in B \text{ and } x \notin A\}.$$

If all classes under discussion are subsets of a fixed universe U, then $U \setminus A$ is often denoted A^c.

§1.4 Set identities

The standard set identities take the following form:

Proposition 1.4.1

Let A, B, and B_i be subsets of a fixed universe U. Then

$$A \cap \left(\bigcup_{i\in I} B_i\right) = \bigcup_{i\in I}(A \cap B_i),$$

$$A \cup \left(\bigcap_{i\in I} B_i\right) = \bigcap_{i\in I}(A \cup B_i),$$

$$\left(\bigcup_{i\in I} A_i\right)^c = \bigcap_{i\in I} A_i^c, \qquad \left(\bigcap_{i\in I} A_i\right)^c = \bigcup_{i\in I} A_i^c.$$

Moreover, $A \cup B = B \cup A$, $A \subseteq B$ means every element of A lies in B, and $A \cap B = A$ when $A \subseteq B$.

Proof. All of these follow by chasing membership. For instance, an element x lies in $A \cap (\bigcup_i B_i)$ exactly when $x \in A$ and $x \in B_i$ for at least one i, which is exactly the condition that $x \in \bigcup_i (A \cap B_i)$. The De Morgan laws are proved similarly after replacing "there exists" by "for all" under negation. $\square$

§1.5 Images, inverse images, and functions

Let $f : A \to B$ be a function. If $S \subseteq A$, the image of S under f is

$$f(S) = \{b \in B \mid b = f(a) \text{ for some } a \in S\}.$$

If $T \subseteq B$, the inverse image of T is

$$f^{-1}(T) = \{a \in A \mid f(a) \in T\}.$$

Example 1.5.1

For $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$, we have

$$f(\mathbb{R}) = [0, \infty), \qquad f^{-1}(\{9\}) = \{-3, 3\}.$$

A map $f : A \to B$ is **injective** if $f(a) = f(a')$ implies $a = a'$. It is **surjective** if $f(A) = B$. It is **bijective** if it is both injective and surjective.

Proposition 1.5.2

For composable maps $A \xrightarrow{f} B \xrightarrow{g} C$:

(i) if f and g are injective, then $g \circ f$ is injective;

(ii) if f and g are surjective, then $g \circ f$ is surjective;

(iii) if $g \circ f$ is injective, then f is injective;

(iv) if $g \circ f$ is surjective, then g is surjective.

§1.6 One-sided inverses

One-sided inverses recognize injective and surjective maps.

Theorem 1.6.1

Let $f : A \to B$ be a function and assume $A \neq \varnothing$.

(i) f is injective if and only if there is a map $g : B \to A$ such that $g \circ f = 1_A$.

(ii) f is surjective if and only if there is a map $h : B \to A$ such that $f \circ h = 1_B$, provided the required choices of elements from the fibres of f can be made.

Proof. For (i), suppose f is injective. For $b \in f(A)$, there is a unique $a \in A$ with $f(a) = b$. Choose once and for all an element $a_0 \in A$. Define

$$g(b) = \begin{cases} a, & b \in f(A) \text{ and } f(a) = b, \\ a_0, & b \notin f(A). \end{cases}$$

Then $g(f(a)) = a$ for all $a \in A$. Conversely, if $g \circ f = 1_A$, then $f(a) = f(a')$ implies

$$a = g(f(a)) = g(f(a')) = a',$$

so f is injective.

For (ii), if h exists and $f \circ h = 1_B$, then every $b \in B$ equals $f(h(b))$, so f is surjective. Conversely, if f is surjective, each fibre $f^{-1}(b)$ is nonempty. Choosing $h(b) \in f^{-1}(b)$ gives a right inverse. □

Caution

The last choice of one element from each fibre is exactly where the Axiom of Choice may enter. This point will reappear in the chapter on choice and cardinal numbers.

§1.7 A right inverse hides a choice function

Consider a surjection

$$f : \{a, b, c\} \longrightarrow \{r, s\}, \qquad f(a) = f(b) = r, \quad f(c) = s.$$

There are two possible right inverses: the image of r may be either a or b, while the image of s must be c. A right inverse does exactly what its formula says: it chooses one representative from every nonempty fibre.

For a general surjection $f : A \to B$, those fibres form a family

$$(f^{-1}(\{b\}))_{b \in B}.$$

A section $s : B \to A$ satisfying $f \circ s = 1_B$ is precisely a simultaneous choice of one element from each member of that family. For a finite B, finitely many individual choices suffice. For an arbitrary family, this is where a set-theoretic axiom enters.

Theorem 1.7.1 (A useful equivalent form of Choice)

Over ZF set theory, the Axiom of Choice is equivalent to the assertion that every surjection of sets admits a right inverse.

Proof. If Choice holds and $f : A \to B$ is onto, choose $s(b) \in f^{-1}(\{b\})$ for every b. Then $f(s(b)) = b$.

Conversely, take any indexed family $(X_i)_{i \in I}$ of nonempty sets. Its disjoint union

$$E = \coprod_{i \in I} X_i = \{(i, x) : i \in I,\ x \in X_i\}$$

has a natural surjection $\pi : E \to I$, $(i, x) \mapsto i$. By hypothesis it has a section $s : I \to E$. Writing $s(i) = (i, c(i))$ defines a choice function $c(i) \in X_i$. The assertion for all surjections therefore supplies a choice function for every such family. □

It is the same elementary fibre picture in both directions. The subtlety is that constructing a right inverse for one explicitly given finite function is easy, whereas asserting their existence for *all* surjections is already the full Axiom of Choice.

Relations, Products, and the First Appearance of Universal Properties

N-20210206

§2.1 Relations and equivalence classes

An equivalence relation identifies elements that represent the same mathematical object. Cartesian products, meanwhile, record independent choices. Both constructions become useful when characterized by the maps they admit.

Definition 2.1.1. For sets or classes A and B, their Cartesian product is

$$A \times B = \{(a, b) \mid a \in A,\ b \in B\}.$$

A **relation from A to B** is a subclass of $A \times B$. A **relation on A** is a subclass of $A \times A$.

> **Caution**
>
> When A, B are sets, a relation is a subset of $A \times B$. For proper classes, a relation may itself be a proper class; it must not be treated as an ordinary set.

Definition 2.1.2. A relation $R \subseteq A \times A$ is an **equivalence relation** if it is:

(i) reflexive: $(a, a) \in R$ for all $a \in A$;

(ii) symmetric: $(a, b) \in R$ implies $(b, a) \in R$;

(iii) transitive: $(a, b), (b, c) \in R$ implies $(a, c) \in R$.

When R is an equivalence relation, we write $a \sim b$ for $(a, b) \in R$, and define the equivalence class of a by

$$[a] = \{x \in A \mid x \sim a\}.$$

The quotient class is

$$A/R = \{[a] \mid a \in A\}.$$

Proposition 2.1.3

If R is an equivalence relation on A, then:

(i) $[a] \neq \varnothing$ for every $a \in A$;

(ii) $A = \bigcup_{a \in A} [a] = \bigcup_{C \in A/R} C$;

(iii) $[a] = [b]$ if and only if $a \sim b$.

Lemma 2.1.4

For any $a, b \in A$, either $[a] \cap [b] = \varnothing$, or $[a] = [b]$.

Proof. If $x \in [a] \cap [b]$, then $x \sim a$ and $x \sim b$. By symmetry and transitivity, $a \sim b$, and hence any element equivalent to a is equivalent to b, and conversely. Thus $[a] = [b]$. □

§2.2 Partitions

Definition 2.2.1. Let A be a nonempty class. A family $\{A_i \mid i \in I\}$ of subclasses of A is a **partition** of A if

(i) $A_i \neq \varnothing$ for every $i \in I$;

(ii) $\bigcup_{i \in I} A_i = A$;

(iii) $A_i \cap A_j = \varnothing$ whenever $i \neq j$.

Remark 2.2.2 — This is one of the most useful ways to remember equivalence relations: an equivalence relation cuts A into disjoint blocks, and those blocks form a partition. Conversely, a partition defines an equivalence relation by declaring two elements equivalent exactly when they lie in the same block.

§2.3 Indexed products and projections

Finite Cartesian products generalize to a set-indexed family $\{A_i \mid i \in I\}$, define

$$\prod_{i \in I} A_i$$

to be the set of functions

$$f : I \to \bigcup_{i \in I} A_i$$

such that $f(i) \in A_i$ for every $i \in I$.

When $I = \{1, 2, \ldots, n\}$, this agrees with the usual product

$$A_1 \times A_2 \times \cdots \times A_n,$$

because such a function is the same information as the ordered n-tuple

$$(f(1), f(2), \ldots, f(n)).$$

Remark 2.3.1 — When $I = \{1, 2\}$, a point of $A_1 \times A_2$ may be written as a pair (a_1, a_2), but it may also be regarded as a function f with $f(1) = a_1$ and $f(2) = a_2$. This is a small but important shift: products are not just lists; they are choice-like functions over an index set.

For each $k \in I$, the **canonical projection**

$$\pi_k : \prod_{i \in I} A_i \to A_k$$

is defined by

$$\pi_k(f) = f(k).$$

Example 2.3.2

If $I = \mathbb{R}$ and $A_i = \mathbb{C}$ for all i, then an element of $\prod_{i \in I} A_i$ is a function $f : \mathbb{R} \to \mathbb{C}$. For example, $f(x) = ix$. The projection at $k \in \mathbb{R}$ returns

$$\pi_k(f) = f(k) = ik.$$

§2.4 The first universal property

The first universal property now appears. Products are not only sets of tuples; they are characterized by how maps into them behave.

Theorem 2.4.1 (Universal property of products)

Let $\{A_i \mid i \in I\}$ be a family of sets. There is a set D, equipped with maps

$$\pi_i : D \to A_i \qquad (i \in I),$$

such that for every set C and every family of maps $f_i : C \to A_i$, there is a unique map

$$f : C \to D$$

with

$$\pi_i \circ f = f_i$$

for every $i \in I$. Moreover, D is unique up to a unique bijection compatible with the projections.

Proof. Take $D = \prod_{i \in I} A_i$. Given maps $f_i : C \to A_i$, define

$$f(c)(i) = f_i(c).$$

Then $f(c) \in \prod_i A_i$, and $\pi_i(f(c)) = f_i(c)$. Uniqueness follows because a point of the product is determined by all of its coordinates. □

Remark 2.4.2 — The proof is short because the concrete product is already known. The deeper point is that the same pattern defines products in arbitrary categories, where elements and coordinates may no longer be available.

§2.5 Integers and induction

The note then turns to the natural numbers and the integers through Peano's axioms. In modern notation the Peano axioms say, roughly:

(i) 0 is a natural number;

(ii) every natural number has a successor;

(iii) 0 is not the successor of any natural number;

(iv) distinct natural numbers have distinct successors;

(v) any property true of 0 and inherited by successors is true of all natural numbers.

The fifth axiom is the principle of mathematical induction.

Theorem 2.5.1 (Induction and strong induction)
Let $S \subseteq \mathbb{N}$ with $0 \in S$. If either

(i) $n \in S$ implies $n+1 \in S$ for all $n \in \mathbb{N}$, or

(ii) $\{0, 1, \ldots, n-1\} \subseteq S$ implies $n \in S$ for all $n \in \mathbb{N}$,

then $S = \mathbb{N}$.

Caution

A proof by taking the least element of $\mathbb{N} \setminus S$ is clean if the well-ordering principle for $\mathbb{N}$ is already available. If one is deriving the theory strictly from Peano's axioms, one should first justify the well-ordering principle or prove the statement directly by induction.

§2.6 Division, gcd, and congruence

Theorem 2.6.1 (Division algorithm)
If $a, b \in \mathbb{Z}$ and $a \neq 0$, then there exist unique integers q, r such that

$$b = aq + r, \qquad 0 \leq r < |a|.$$

Definition 2.6.2. For $a \neq 0$, we write $a \mid b$ if there exists $k \in \mathbb{Z}$ such that $ak = b$. Otherwise we write $a \nmid b$.

Definition 2.6.3. The positive integer c is the greatest common divisor of $a_1, a_2, \ldots, a_n$ if

(i) $c \mid a_i$ for every $1 \leq i \leq n$;

(ii) whenever $d \in \mathbb{Z}$ and $d \mid a_i$ for every i, then $d \mid c$.

It is denoted $(a_1, a_2, \ldots, a_n)$.

Definition 2.6.4. Let $m > 0$. If $a, b \in \mathbb{Z}$ and $m \mid (a-b)$, then a is congruent to b modulo m, written

$$a \equiv b \pmod{m}.$$

Theorem 2.6.5

Let $m > 0$ and $a, b, c, d \in \mathbb{Z}$.

(i) Congruence modulo m is an equivalence relation on $\mathbb{Z}$, with precisely m equivalence classes.

(ii) If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then

$$a + c \equiv b + d \pmod{m}, \qquad ac \equiv bd \pmod{m}.$$

(iii) If $ab \equiv ac \pmod{m}$ and $(a, m) = 1$, then $b \equiv c \pmod{m}$.

Proof. The key point is divisibility. For example,

$$m \mid (a - b), \qquad m \mid (c - d)$$

imply

$$m \mid (a + c) - (b + d).$$

For multiplication, write

$$ac - bd = c(a - b) + b(c - d).$$

For cancellation, $ab \equiv ac \pmod{m}$ means $m \mid a(b - c)$. If $(a, m) = 1$, Euclid's lemma gives $m \mid (b - c)$, hence $b \equiv c \pmod{m}$. □

Choice, Well-Ordering, and the Arithmetic of Cardinal Size

N-20210207

§3.1 Choice enters the story

Choice enters when a construction requires selecting one element from each set in a family. The product formulation states this directly.

Axiom 3.1.1 (Axiom of Choice)
The product of a family of nonempty sets indexed by a nonempty set is nonempty.

Remark 3.1.2 — Equivalently, if S is a set, then there is a choice function defined on the collection of all nonempty subsets of S:

$$f(A) \in A$$

for every nonempty $A \subseteq S$. This is an equivalent choice-function formulation.

§3.2 Partially ordered sets

Definition 3.2.1. A **partially ordered set** is a nonempty set A together with a relation $\leq$ which is reflexive, transitive, and antisymmetric. Antisymmetry means that

$$a \leq b \text{ and } b \leq a \quad \Longrightarrow \quad a = b.$$

Two elements $a, b \in A$ are **comparable** if either $a \leq b$ or $b \leq a$. If every pair of elements is comparable, then the order is a **total order** or **linear order**.

Example 3.2.2
Let $A = \mathcal{P}(\{1, 2, 3, 4, 5\})$, ordered by inclusion. Then $C \leq D$ means $C \subseteq D$. This is a partial order, but not a total order: for example, $\{1, 2\}$ and $\{2, 3\}$ are not comparable.

Let $(A, \leq)$ be a partially ordered set. An element $a \in A$ is **maximal** if whenever $c \in A$ is comparable with a and $a \leq c$, then $c = a$. An **upper bound** of a nonempty subset $B \subseteq A$ is an element $d \in A$ such that $b \leq d$ for every $b \in B$. A nonempty subset $B \subseteq A$ is a **chain** if it is totally ordered by the inherited order.

Theorem 3.2.3 (Zorn's lemma)
Let A be a nonempty partially ordered set. If every chain in A has an upper bound in A, then A has a maximal element.

Caution

The standard term is *maximal element*, not "local maximum". A maximal element need not be greatest: there may be no element above all elements of A.

§3.3 Well-ordering

Definition 3.3.1. A partially ordered set A is **well ordered** if every nonempty subset of A has a least element.

Proposition 3.3.2 (Well-ordering principle)

Every nonempty set A admits a total ordering under which it is well ordered.

Remark 3.3.3 — This extends mathematical induction from $\mathbb{N}$ to arbitrary well-ordered sets. Conceptually this is the beginning of transfinite induction, even before ordinals are developed systematically.

§3.4 Equipollence and cardinal numbers

Definition 3.4.1. Two sets A and B are **equipollent**, written $A \sim B$, if there exists a bijection $A \to B$.

Definition 3.4.2. If a set A is equipollent to

$$I_n = \{1, 2, \ldots, n\}$$

for some $n \in \mathbb{N}$, or to $I_0 = \varnothing$, then A is finite. Otherwise A is infinite.

Definition 3.4.3. The **cardinal number** of a set A, denoted $|A|$, is its equivalence class under equipollence. It is a finite or infinite cardinal according as A is finite or infinite.

Caution

Strictly speaking, the "equivalence class of all sets bijective with A" is usually too large to be a set. A formal treatment either uses representatives such as initial ordinals, or works in a class theory where such collections are proper classes. The equivalence-class language is therefore an informal convention.

If A and B are disjoint sets with $|A| = \alpha$ and $|B| = \beta$, define

$$\alpha + \beta = |A \cup B|.$$

Similarly, cardinal multiplication is defined by

$$\alpha\beta = |A \times B|.$$

§3.5 Cantor's theorem

Theorem 3.5.1 (Cantor)

For every set A,

$$|A| < |\mathcal{P}(A)|.$$

Proof. The map $a \mapsto \{a\}$ is injective from A into $\mathcal{P}(A)$, so $|A| \leq |\mathcal{P}(A)|$.

Suppose for contradiction that $f : A \to \mathcal{P}(A)$ is surjective. Define

$$B = \{a \in A \mid a \notin f(a)\}.$$

Since $B \subseteq A$, surjectivity gives $a_0 \in A$ with $f(a_0) = B$. Then

$$a_0 \in B \quad \Longleftrightarrow \quad a_0 \notin f(a_0) = B,$$

which is impossible. Hence no surjection $A \to \mathcal{P}(A)$ exists, and so $|A| < |\mathcal{P}(A)|$. □

This proof deliberately echoes Russell's paradox from the first chapter. In one case the self-reference shows that a class is too large to be a set; in the other, diagonalization shows that the power set is strictly larger than the original set.

§3.6 Comparing cardinals

Theorem 3.6.1 (Cantor–Bernstein)

If A and B are sets such that $|A| \leq |B|$ and $|B| \leq |A|$, then

$$|A| = |B|.$$

Theorem 3.6.2 (Trichotomy for cardinals)

The class of cardinal numbers is linearly ordered by $\leq$. If α and β are cardinal numbers, then exactly one of the following holds:

$$\alpha < \beta, \qquad \alpha = \beta, \qquad \beta < \alpha.$$

Remark 3.6.3 — This is often called the Law of Trichotomy. A standard proof orders a family of partial functions by extension, making this one of the classical places where Zorn's lemma enters the comparison of cardinalities.

§3.7 Infinite cardinals and finite disturbance

The following cardinality statements are best read through explicit encodings: pairing functions and binary strings make the size comparisons concrete.

Theorem 3.7.1

Every infinite set has a denumerable subset. In particular,

$$\aleph_0 \leq \alpha$$

for every infinite cardinal α, assuming the usual choice principles being used in this chapter.

Lemma 3.7.2

If A is an infinite set and F is a finite set, then

$$|A \cup F| = |A|.$$

In particular,

$$\alpha + n = \alpha$$

for every infinite cardinal α and finite cardinal n.

Theorem 3.7.3

If $\alpha \leq \beta$ and β is infinite, then

$$\alpha + \beta = \beta.$$

Theorem 3.7.4

If $0 < \alpha \leq \beta$ and β is infinite, then

$$\alpha\beta = \beta.$$

In particular,

$$\aleph_0 \cdot \aleph_0 = \aleph_0,$$

and if B is finite then

$$\aleph_0 \cdot |B| = \aleph_0$$

provided $B \neq \varnothing$.

Theorem 3.7.5

Let A be a set and, for each integer $n \geq 1$, let

$$A^n = \underbrace{A \times A \times \cdots \times A}_{n \text{ factors}}.$$

If A is finite, then

$$|A^n| = |A|^n,$$

and if A is infinite, then

$$|A^n| = |A|.$$

Moreover,

$$\left| \bigcup_{n \in \mathbb{N}} A^n \right| = \aleph_0 |A|.$$

Corollary 3.7.6

If A is an infinite set and $\mathcal{F}(A)$ denotes the set of all finite subsets of A, then

$$|\mathcal{F}(A)| = |A|.$$

§3.8 Concrete encodings behind the arithmetic

The equality

$$\aleph_0 \cdot \aleph_0 = \aleph_0$$

is not just a theorem name. A direct bijection is the Cantor pairing function

$$\pi : \mathbb{N} \times \mathbb{N} \to \mathbb{N}, \qquad \pi(m,n) = \frac{(m+n)(m+n+1)}{2} + m.$$

It enumerates lattice points by diagonals. For instance,

$$\pi(0,0) = 0, \qquad \pi(1,0) = 2, \qquad \pi(0,1) = 1.$$

To invert it, start with $k \in \mathbb{N}$. Find the unique integer $s \geq 0$ such that

$$\frac{s(s+1)}{2} \leq k < \frac{(s+1)(s+2)}{2}.$$

Then

$$m = k - \frac{s(s+1)}{2}, \qquad n = s - m.$$

This recovers a unique pair (m,n), so π is a bijection.

There is an equally concrete encoding of finite subsets of $\mathbb{N}$. If

$$S = \{s_1 < s_2 < \cdots < s_k\} \subseteq \mathbb{N}$$

is finite, define

$$f(S) = \sum_{i=1}^{k} 2^{s_i},$$

with $f(\varnothing) = 0$. Binary expansion is unique, so two different finite subsets have different codes. Hence

$$|\mathcal{F}(\mathbb{N})| \leq \aleph_0.$$

The reverse injection is the singleton map

$$\mathbb{N} \hookrightarrow \mathcal{F}(\mathbb{N}), \qquad n \mapsto \{n\}.$$

By Cantor–Bernstein,

$$|\mathcal{F}(\mathbb{N})| = \aleph_0.$$

This is the finite-subset theorem in a form one can actually compute: a finite subset is just the support of the 1's in a binary expansion.

Remark 3.8.1 — The general statement $|\mathcal{F}(A)| = |A|$ for every infinite set A still uses the comparison machinery above, and with arbitrary A one must pay attention to the choice principles being used. But for the countable model case, the pairing function and the binary encoding give a complete countable construction.

A Gateway to Dualities: Space, Algebra, and the Dualizing Object

N-20210721

§4.1 What could a space know about itself?

Imagine a small world with three points a, b, c. We may forget their coordinates and even their names, yet still distinguish them by asking yes-or-no questions. Is a point in $\{a, b\}$? Answers may be combined by *and*, *or*, and *not*, so all the questions form a Boolean algebra.

There is an attractive reversal of perspective here. Ordinarily we start with the points, construct the regions, and ask which regions contain a point. What if we start with the algebra of questions and try to recover the points themselves? A point ought to be a consistent way of answering *every* question. It must say yes to the whole space, no to the empty set, and respect all the logical operations. The remarkable fact is that this idea reconstructs a space.

The finite experiment leads to Stone duality. Its more surprising descendants arise when questions are continuous functions, regular functions, or open sets, and eventually when truth itself can vary from one part of a space to another. The type of observation determines what sort of geometry can be recovered.

The exposition of the first duality follows Marcus Tressl's *Stone Duality for Boolean Algebras*.

Geometry	Algebra	Typical gauge object
Stone spaces	Boolean algebras	$\mathbf{2} = \{0, 1\}$
Compact Hausdorff spaces	Unital commutative C^*-algebras	$\mathbb{C}$
Affine schemes	Commutative rings	$\mathbb{A}^1$
Locales	Frames	Sierpiński space $\mathbb{S}$
Topoi	Logoi	**Set**
∞-topoi	∞-logoi	Spaces $\mathcal{S}$ (∞-groupoids)

Table 4.1: Six geometry–algebra correspondences. In the last two rows the precise duality reverses morphisms between categories of topoi and logoi.

The first four rows can be approached through actual functions taking values in a test object. In the last two, the gauge object becomes a whole category—sets, and then spaces—and the reversal occurs between two ways of reading a geometric morphism. We will meet both versions after the concrete Stone construction.

§4.2 Three Points and Eight Regions

For $X = \{a, b, c\}$ the algebra of all yes-or-no tests is

$$B = \mathcal{P}(X) = \{\varnothing, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, X\}.$$

A point $x \in X$ defines an evaluation

$$e_x : B \longrightarrow \mathbf{2} = \{0, 1\}, \qquad e_x(U) = \begin{cases} 1, & x \in U, \\ 0, & x \notin U. \end{cases}$$

For instance $e_a(\{a, b\}) = 1$ whereas $e_c(\{a, b\}) = 0$. These three evaluations are precisely the Boolean homomorphisms $B \to \mathbf{2}$: each must select exactly one of the three mutually exclusive singleton atoms, and then all eight answers are forced.

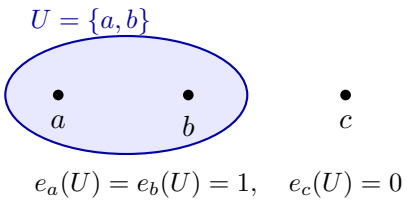


Figure 4.1: A subset is a question; a point is an evaluation of that question. Boolean combinations of the eight subsets are compatible with exactly three evaluations.

This small calculation contains the ambition of duality: reconstruct points as mathematical objects *defined by their observations*. It also suggests when reconstruction could fail: if the allowed questions cannot distinguish two points, the algebra cannot distinguish them either.

§4.3 A map reverses the direction of questions

Let $f : X \to Y$ be a function. A test $V \subseteq Y$ can be performed on a point $x \in X$ by testing whether $f(x) \in V$. Thus a test on Y becomes a test on X:

$$f^{-1} : \mathcal{P}(Y) \longrightarrow \mathcal{P}(X), \qquad V \longmapsto f^{-1}(V).$$

For example, take $X = \{a, b, c\}$, $Y = \{u, v\}$ and

$$f(a) = f(b) = u, \qquad f(c) = v.$$

The question about output u pulls back to the question about inputs a, b:

$$f^{-1}(\{u\}) = \{a, b\}.$$

The reversed arrow is forced by the direction in which *questions* travel.

More generally, for a target object Ω, composition defines

$$\mathrm{Hom}(Y, \Omega) \xrightarrow{\varphi \mapsto \varphi \circ f} \mathrm{Hom}(X, \Omega).$$

This is *contravariance*: $(g \circ f)^* = f^* \circ g^*$. It is a simple identity, but the right shape for a geometry-to-algebra correspondence.

For Stone duality, $\Omega = \mathbf{2}$ wears two compatible costumes. As a *discrete topological space*, maps $X \to \mathbf{2}$ describe clopen regions; as a *Boolean algebra*, homomorphisms $B \to \mathbf{2}$ describe consistent truth assignments. These Hom sets belong to different categories. Stone's theorem states that their constructions recover one another.

Reversing arrows by itself never proves a duality: reconstruction requires sufficiently many observations, an appropriate category of spaces, and the correct algebraic operations on those observations.

§4.4 Stone duality as the first complete example

The first row is the best place to begin because it is the most logical and the most concrete:

$$\text{Stone spaces} \quad \longleftrightarrow \quad \text{Boolean algebras}, \qquad \Omega = \mathbf{2} = \{0, 1\}.$$

Here $\mathbf{2}$ is the space of truth values. A map into $\mathbf{2}$ is a yes-or-no observation.

§4.4.i Boolean algebras as propositional logic

A Boolean algebra is an algebraic model of classical propositional logic.

Definition 4.4.1 (Boolean algebra)**.** A Boolean algebra is a structure

$$B = (B, \vee, \wedge, ', 0, 1)$$

where $\vee$ and $\wedge$ are join and meet, 0 and 1 are bottom and top, and each element $a \in B$ has a complement a'. It satisfies the usual associativity, commutativity, absorption, distributivity, and complementation laws.

The example to keep in mind is the power set $\mathcal{P}(X)$. Then

$$A \vee B = A \cup B, \qquad A \wedge B = A \cap B, \qquad A' = X \setminus A.$$

So Boolean algebra is the algebra of regions that can be combined by union, intersection, and complement.

§4.4.ii Stone spaces as spaces with Boolean topology

A Stone space is a topological space whose topology is controlled by clopen sets.

Definition 4.4.2 (Stone space)**.** A Stone space is a compact Hausdorff totally disconnected space. Equivalently, it is a compact Hausdorff space with a basis of clopen subsets.

A subset is clopen if it is both closed and open. In ordinary connected spaces, clopen sets are rare. In a Stone space, clopen sets are abundant enough to form the logical basis of the space.

For any Stone space X, the clopen subsets form a Boolean algebra:

$$\mathrm{Clop}(X).$$

The Boolean operations are exactly the set operations:

$$U \vee V = U \cup V, \qquad U \wedge V = U \cap V, \qquad U' = X \setminus U.$$

This gives the direction

$$X \longmapsto \mathrm{Clop}(X).$$

§4.4.iii Why 2 is the dualizing object

The set $\mathbf{2} = \{0, 1\}$ is given the discrete topology. A continuous map

$$\chi : X \to \mathbf{2}$$

is the characteristic function of a clopen subset of X. Indeed,

$$\chi^{-1}(\{1\})$$

is clopen because $\{1\} \subset \mathbf{2}$ is clopen.

Thus

$$\mathrm{Hom}_{\mathbf{Top}}(X, \mathbf{2}) \cong \mathrm{Clop}(X).$$

This is the first half of the row: the algebra of logic on a Stone space is the algebra of truth-valued continuous observations.

§4.4.iv Ultrafilters as points

Now start from a Boolean algebra B. A point of the Stone space of B is not given in advance. It must be reconstructed from the algebra.

Definition 4.4.3 (Filter and ultrafilter)**.** A filter $F \subseteq B$ is a subset such that $1 \in F$, $0 \notin F$, it is closed under finite meets, and it is upward closed. An ultrafilter is a maximal proper filter.

An ultrafilter is a complete truth assignment. For every proposition $a \in B$, exactly one of a and a' belongs to the ultrafilter. So an ultrafilter decides every proposition in a way compatible with Boolean logic.

Let

$$S(B) = \{\mathfrak{u} : \mathfrak{u} \text{ is an ultrafilter on } B\}.$$

For each $a \in B$, define

$$U_a = \{\mathfrak{u} \in S(B) : a \in \mathfrak{u}\}.$$

The sets U_a form a basis of clopen subsets. This turns $S(B)$ into a Stone space.

The map

$$a \longmapsto U_a$$

preserves Boolean operations:

$$U_{a \wedge b} = U_a \cap U_b, \qquad U_{a \vee b} = U_a \cup U_b, \qquad U_{a'} = S(B) \setminus U_a.$$

Theorem 4.4.4 (Stone representation theorem)

For every Boolean algebra B, the map

$$B \longrightarrow \mathrm{Clop}(S(B)), \qquad a \longmapsto U_a$$

is an isomorphism of Boolean algebras.

This is the missing geometric half of Boolean algebra. A Boolean algebra is not only a symbolic calculus of propositions. It is the algebra of clopen regions of a canonically reconstructed space.

§4.4.v Why the reconstruction actually works

The displayed Boolean operations are straightforward. The stronger assertion is that $a \mapsto U_a$ loses no information and produces *all* clopen sets. Here is the mechanism.

If $a \neq b$, either $a \wedge b' \neq 0$ or $b \wedge a' \neq 0$. A nonzero Boolean element belongs to some ultrafilter; this is the ultrafilter extension principle, a form of the Boolean prime ideal theorem. Consequently, some ultrafilter contains exactly one of a, b. Their clopen regions U_a, U_b differ. The map is injective.

For compactness, suppose the basic clopen sets U_{a_i} cover $S(B)$ but no finite subfamily covers it. Then every finite meet of complements

$$a'_{i_1} \wedge \cdots \wedge a'_{i_r} \neq 0.$$

The complements have the finite-intersection property, so an ultrafilter contains all of them. It belongs to none of the U_{a_i}, contradicting the cover. Every basic cover thus has a finite subcover. The sets U_a are clopen because $S(B) \setminus U_a = U_{a'}$, and distinct ultrafilters are separated by complementary clopens. Hence $S(B)$ is compact, Hausdorff, and zero-dimensional.

Finally, any clopen $C \subseteq S(B)$ is compact. The basic sets U_a contained in C cover it, and finitely many suffice:

$$C = U_{a_1} \cup \cdots \cup U_{a_r} = U_{a_1 \vee \cdots \vee a_r}.$$

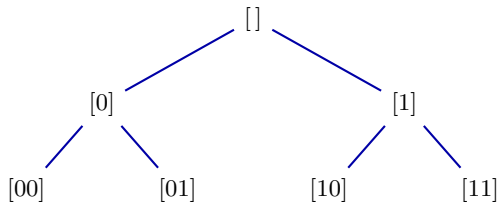
This proves surjectivity. The ultrafilter extension principle is a genuine set-theoretic input for general Boolean algebras; the finite three-point example needed no such principle.

§4.4.vi An infinite example you can draw: the Cantor space

Let $X = \{0,1\}^{\mathbb{N}}$ be the space of infinite binary sequences. To test a point, ask about only finitely many digits. For a finite word w, its *cylinder*

$$[w] = \{x \in X : x_1 \cdots x_{|w|} = w\}$$

is clopen. Cylinders form a basis; every clopen subset is a *finite* union of them, by compactness.



An infinite path is a consistent sequence of answers.

Figure 4.2: Binary cylinder questions form a nested family. Their consistent infinite answers are precisely the points of the Cantor space.

A sequence $x = (x_1, x_2, \ldots)$ chooses, at every level, the unique cylinder recording its first digits. Conversely, any compatible chain $[x_1] \supseteq [x_1 x_2] \supseteq \cdots$ singles out one infinite sequence. Coherent answers reconstruct a point even when there are infinitely many questions. The topology arises because two sequences are close when many early answers agree.

§4.5 Recovering maps as well as points

A duality must reconstruct *arrows*, not only objects. In the finite example, a function $f : X \to Y$ determines a Boolean homomorphism

$$f^{-1} : \mathcal{P}(Y) \to \mathcal{P}(X).$$

Conversely, suppose $h : \mathcal{P}(Y) \to \mathcal{P}(X)$ is a Boolean homomorphism between the power-set algebras of finite sets. The singleton subsets $\{y\}$ of Y are pairwise disjoint and their union is Y. Therefore the sets $h(\{y\})$ partition X (some may be empty). Each $x \in X$ belongs to exactly one of these sets. Define $f(x)$ to be that unique y. The homomorphism is necessarily $h = f^{-1}$.

In the general case, a Boolean homomorphism $h : B \to C$ sends each ultrafilter of C to its inverse-image ultrafilter in B. Hence it induces a continuous map

$$S(C) \longrightarrow S(B), \qquad \mathfrak{u} \longmapsto h^{-1}(\mathfrak{u}).$$

The direction reverses again. Stone duality is the contravariant equivalence

$$\boxed{\mathbf{Stone} \simeq \mathbf{Bool}^{\mathrm{op}}.}$$

Points become consistent truth assignments, predicates become clopen regions, and maps become substitution of predicates. This is a precise sense in which Boolean logic has a topology.

§4.6 What happens when yes-or-no questions are too poor?

Take the connected interval $X = [0, 1]$. Its only clopen subsets are $\varnothing$ and X. If we observe a point only through continuous maps to the *discrete* space $\mathbf{2}$, then every observation is constant. We cannot tell 0 from $1/2$.

A simple repair is to use *complex-valued continuous functions*. The coordinate function $t \mapsto t$ distinguishes every pair of points on the interval. For a compact Hausdorff space X, let $C(X)$ denote the continuous functions $X \to \mathbb{C}$, with pointwise operations, complex conjugation, and

$$\|h\|_\infty = \sup_{x \in X} |h(x)|.$$

This is a commutative unital C^*-algebra: the norm and involution remember precisely the analytic structure we want.

Every point x determines an evaluation character

$$\mathrm{ev}_x : C(X) \to \mathbb{C}, \qquad h \mapsto h(x).$$

A character here is a nonzero unital multiplicative complex-linear functional, and in the C^* setting it respects the $*$-structure. Gelfand's theorem says that for compact Hausdorff X, every character of $C(X)$ is evaluation at a unique point. In this setting functions separate points and remember the topology.

Theorem 4.6.1 (Commutative Gelfand duality)

The category of compact Hausdorff spaces is contravariantly equivalent to the category of commutative unital complex C^*-algebras and unital $*$-homomorphisms. In particular, for such an algebra A,

$$\boxed{A \cong C(\operatorname{Spec} A),}$$

where $\operatorname{Spec} A$ is its compact Hausdorff space of characters.

The finite case can be seen without functional analysis. If $X = \{p, q\}$, then $C(X) \cong \mathbb{C}^2$. A unital multiplicative functional to $\mathbb{C}$ must evaluate one of the two factors, because the idempotents $(1, 0)$ and $(0, 1)$ have images in $\{0, 1\}$ and sum to 1. The characters are exactly $\mathrm{ev}_p, \mathrm{ev}_q$.

A continuous map $f : X \to Y$ again turns observations on Y into observations on X:

$$f^* : C(Y) \to C(X), \qquad h \longmapsto h \circ f.$$

For $f(t) = t^2$ on $[0, 1]$, the observable $h(t) = t$ is pulled back to $h \circ f(t) = t^2$. The same contravariant pattern has acquired analytic information. The difference from Stone duality is substantial: a complex-valued observable carries far more than a binary truth value.

§4.7 Functions beyond Ordinary Points

Continuous functions suggest another sort of observation: *regular functions* in algebraic geometry. An affine scheme is of the form $X = \operatorname{Spec} R$ for a commutative ring R. Its points are prime ideals of R, and the basic open sets are

$$D(f) = \{\mathfrak{p} \in \operatorname{Spec} R : f \notin \mathfrak{p}\}.$$

Ring homomorphisms run opposite to geometric maps:

$$\varphi : R \to S \quad \rightsquigarrow \quad \operatorname{Spec} S \to \operatorname{Spec} R, \qquad \mathfrak{q} \mapsto \varphi^{-1}(\mathfrak{q}).$$

Why prime ideals rather than just numerical solutions? Consider $R = k[t]$, with k algebraically closed. The maximal ideals $(t - a)$ are the familiar points $a \in k$, but the zero ideal (0) is also prime. This *generic point* lies in every nonempty basic open set. It encodes the whole irreducible affine line rather than a particular value of t.

The ring of functions is recovered using the structure sheaf:

$$\Gamma(\operatorname{Spec} R, \mathcal{O}) = R, \qquad \Gamma(D(f), \mathcal{O}) = R_f.$$

For example, over $k[t]$ the open region where $t \neq 0$ admits regular functions such as t^{-1}, even though t^{-1} is not regular on the whole line.

The *affine line scheme* $\mathbb{A}^1_{\mathbb{Z}} = \operatorname{Spec} \mathbb{Z}[t]$ gives a particularly attractive version of the observation principle:

$$\boxed{\operatorname{Hom}_{\mathbf{Sch}}(X, \mathbb{A}^1_{\mathbb{Z}}) \cong \Gamma(X, \mathcal{O}_X).}$$

For an affine $X = \operatorname{Spec} R$, this is R itself: a scheme map to the affine line is determined by the image of its coordinate t, hence by a global function. The computation uses the scheme structure, not merely the underlying point set.

Indeed two affine schemes can have the same underlying topological space and different algebras. The ring of dual numbers

$$R = k[\varepsilon]/(\varepsilon^2)$$

has one prime ideal (ε), so $\operatorname{Spec} R$ has just one topological point. Its global sections still contain a nonzero element ε with $\varepsilon^2 = 0$. The scheme $\operatorname{Spec} k$ has the same one-point topology but lacks that infinitesimal direction. Forgetting the sheaf would lose exactly the information that algebraic geometry needs.

The precise duality is between *affine schemes* and commutative rings, with arrows reversed. The equivalence does not assert that arbitrary schemes are determined by a single ring of global functions. See the Stacks Project, *Affine Schemes*, for the local-section calculation.

§4.8 Why an open set is a different kind of truth

A clopen predicate has a continuous characteristic function into discrete **2**. To represent an *arbitrary open* predicate, the target must change its topology.

Let $\mathbb{S} = \{0, 1\}$ have opens

$$\mathcal{O}(\mathbb{S}) = \{\varnothing, \{1\}, \mathbb{S}\}.$$

This is the Sierpiński space. For any topological space X, the function $\chi_U : X \to \mathbb{S}$ that equals 1 on U and 0 outside is continuous exactly when U is open:

$$\boxed{\operatorname{Hom}_{\mathbf{Top}}(X, \mathbb{S}) \cong \mathcal{O}(X).}$$

One of the two truth values is observable as an open condition; the other need not be. For $U = (0, \infty) \subset \mathbb{R}$, the characteristic function is continuous into $\mathbb{S}$, but not into discrete **2**. This single example explains the asymmetric topology of $\mathbb{S}$.

The algebra of all opens $\mathcal{O}(X)$ is a *frame*: arbitrary unions exist, finite intersections distribute over arbitrary unions, and

$$U \cap \bigcup_i V_i = \bigcup_i (U \cap V_i).$$

A continuous map $f : X \to Y$ gives a frame homomorphism

$$f^{-1} : \mathcal{O}(Y) \longrightarrow \mathcal{O}(X).$$

A locale is a space described directly by such a frame. Thus locales and frames are opposites by their defining conventions. Recovering an actual point-set space from the frame of opens requires a further hypothesis, such as sobriety; point-free locales need not be spatial.

This is a close cousin of the Stone construction, with an important change of expressive power. Clopens encode Boolean logic, where complement always stays open. General opens encode a Heyting algebra, where the logical negation of U is the interior of its complement, $\operatorname{int}(X \setminus U)$.

§4.9 A truth value can vary across a space

A sheaf makes the local character of topology tangible. For each open region $U \subseteq X$, a sheaf $\mathcal{F}$ assigns a set $\mathcal{F}(U)$ of data available over U, together with restriction maps

when $V \subseteq U$. Compatible local data may be glued. The category $\mathbf{Sh}(X)$ of sheaves is an example of a *topos*.

The truth-value object of a topos is called its *subobject classifier* Ω. In **Set** it is simply $\{0,1\}$. In the sheaf topos $\mathbf{Sh}(X)$, it has a beautiful local description:

$$\boxed{\Omega(U) = \{V : V \subseteq U,\ V \text{ open}\}.}$$

Restriction from U to W sends V to $V \cap W$. An assertion can be true over one open patch without being true everywhere, and its truth value is the patch on which it holds.

Here is a concrete logical phenomenon. Let $X = \mathbb{R}$ and take the proposition that the real coordinate is positive. Its truth value is the open set

$$P = (0, \infty).$$

The internal negation corresponds to the largest open set disjoint from P:

$$\neg P = \operatorname{int}(\mathbb{R} \setminus P) = (-\infty, 0).$$

Therefore

$$P \vee \neg P = \mathbb{R} \setminus \{0\} \neq \mathbb{R}.$$

At the boundary point 0, neither side is locally forced. This is the first glimpse of *intuitionistic* internal logic: truth can be local, and Boolean complementation is too strong for general open information.

There is another precise reversal still to come. A geometric map of topoi has an inverse-image functor that preserves finite limits and colimits; reading this functor in the opposite direction is the algebraic map of logoi. This is the topos–logos duality, and it persists for ∞-topoi. Its arrow reversal is genuine, even though it takes a different form from the earlier character-space constructions.

A standard reference for the connection between sheaves and internal logic is Saunders Mac Lane and Ieke Moerdijk, *Sheaves in Geometry and Logic*.

§4.10 The two directions of a sheaf map

Let $X = \{p, q\}$ be a discrete two-point space and $Y = \{*\}$ a single point. A sheaf of sets on X is simply a pair of sets:

$$\mathbf{Sh}(X) \simeq \mathbf{Set} \times \mathbf{Set}, \qquad \mathbf{Sh}(Y) \simeq \mathbf{Set}.$$

The unique continuous map $f : X \to Y$ has an inverse-image functor and a direct-image functor. In this example we can write out both:

$$\begin{aligned} f^* : \mathbf{Set} &\longrightarrow \mathbf{Set} \times \mathbf{Set}, & S &\longmapsto (S, S), \\ f_* : \mathbf{Set} \times \mathbf{Set} &\longrightarrow \mathbf{Set}, & (A, B) &\longmapsto A \times B. \end{aligned}$$

A set on the one-point base pulls back to a copy at each of the two points. A sheaf on X pushes forward to its global sections, which consist of one choice at each point.

The two operations are adjoint:

$$\begin{aligned} \operatorname{Hom}_{\mathbf{Set}\times\mathbf{Set}}((S,S),(A,B)) &\cong \operatorname{Hom}_{\mathbf{Set}}(S,A) \times \operatorname{Hom}_{\mathbf{Set}}(S,B) \\ &\cong \operatorname{Hom}_{\mathbf{Set}}(S, A \times B). \end{aligned}$$

In symbols, $\boxed{f^* \dashv f_*}$. This little calculation is the adjunction behind a geometric morphism.

Why is there a dual description? The geometric arrow goes from the two-point space to the one-point space, and hence from $\mathbf{Sh}(X)$ to $\mathbf{Sh}(Y)$. Its algebraic inverse-image functor goes the other way:

$$\underbrace{\mathbf{Sh}(X) \longrightarrow \mathbf{Sh}(Y)}_{\text{geometric } f} \qquad \longleftrightarrow \qquad \underbrace{\mathbf{Sh}(Y) \xrightarrow{f^*} \mathbf{Sh}(X)}_{\text{algebraic } f^*}.$$

The functor f^* preserves finite limits and all small colimits. For the diagonal functor above this is immediate: limits and colimits in a product category are computed coordinatewise. Its right adjoint f_* is determined by the adjunction and need not preserve colimits.

The same phenomenon holds for arbitrary Grothendieck topoi. The algebraic morphisms of logoi are their left-exact, colimit-preserving functors. The geometric morphisms of topoi read the corresponding adjunctions in the opposite direction:

$$\boxed{\mathbf{Topos} \simeq \mathbf{Logos}^{\mathrm{op}}.}$$

With transformations between functors retained, one uses the corresponding bicategorical or higher-categorical conventions. The reversal here is a genuine equivalence of the categories of structured objects and their chosen morphisms, rather than a supposed one-size-fits-all formula using $\mathrm{Hom}(-, \Omega)$.

Even the role of **Set** in the table is concrete. A point of a topos $\mathcal{E}$ is a geometric morphism $\mathbf{Set} \to \mathcal{E}$. For $\mathcal{E} = \mathbf{Set} \times \mathbf{Set}$, the two familiar points have inverse-image functors

$$(A, B) \longmapsto A, \qquad (A, B) \longmapsto B.$$

They probe a sheaf by asking for its value at one point or the other. An arbitrary topos need not have enough such points to recover its whole sheaf category.

§4.11 Why the gauge changes from sets to spaces

Now replace **Set** by the ∞-category $\mathcal{S}$ of spaces, understood as ∞-groupoids or homotopy types. A space remembers not only its points but paths between them, homotopies between paths, and higher coherence data. A sheaf of spaces can glue that richer information locally.

There is a tiny example which justifies the upgrade. Let $G = \mathbb{Z}/2\mathbb{Z}$ and form the one-object groupoid BG whose automorphisms are the two elements of G. Its classifying space has

$$\boxed{\pi_0(BG) = \{*\}, \qquad \pi_1(BG, *) \cong \mathbb{Z}/2\mathbb{Z}.}$$

If we replace BG by its set of connected components, we obtain only one point. The nontrivial loop has disappeared. This is information that ordinary sets cannot record.

The two-point sheaf experiment still works. Ordinary Hom sets become mapping spaces:

$$\mathbf{Sh}_\infty(\{p, q\}) \simeq \mathcal{S} \times \mathcal{S}, \qquad \Delta : \mathcal{S} \longrightarrow \mathcal{S} \times \mathcal{S}, \qquad \Pi : \mathcal{S} \times \mathcal{S} \longrightarrow \mathcal{S},$$

where $\Delta(T) = (T, T)$ and $\Pi(A, B) = A \times B$. The higher adjunction reads

$$\boxed{\mathrm{Map}_{\mathcal{S} \times \mathcal{S}}(\Delta T, (A, B)) \simeq \mathrm{Map}_{\mathcal{S}}(T, A \times B).}$$

Here the mapping objects and products are themselves spaces, retaining possible homotopies and higher identifications.

The corresponding arrow reversal is

$$\boxed{\mathbf{Topos}_\infty \simeq \mathbf{Logos}_\infty^{\mathrm{op}}.}$$

The logos direction records left-exact colimit-preserving functors, while the topos direction records their geometric morphisms. The mathematical point of view is developed in Mathieu Anel and André Joyal's *Topo-logie*, especially its treatments of the topos–logos and higher topos–logos dualities.

The finite Stone construction recovered points from Boolean characters. In these last two rows, the object of interest has become an entire sheaf category and an arrow carries a pair of adjoint functors. The one-object groupoid BG demonstrates why the higher version matters: even a generalized point can carry a loop that ordinary set-valued observations would forget.

Boolean Algebra as the Geometry of Clopen Sets

N-20210804

The subsets of a finite set can be combined by union, intersection and complement. Nothing in that calculation suggests a new topology until we ask what a Boolean algebra looks like when the underlying set is no longer given. Stone's idea is to regard homomorphisms into the two-element algebra as points. The algebra then manufactures its own compact topological space, and every Boolean operation becomes an operation on clopen sets.[1]

§5.1 A Boolean ring is a calculus of exclusive choices

For a set X, use symmetric difference and intersection as operations on $\mathcal{P}(X)$:

$$A + B = (A \setminus B) \cup (B \setminus A), \qquad AB = A \cap B.$$

Every element is idempotent under multiplication:

$$A^2 = A.$$

Indeed, $\mathcal{P}(X)$ becomes a commutative ring with additive identity $\varnothing$, multiplicative identity X, and $A + A = \varnothing$. Characteristic functions identify it with the ring of functions $X \to \mathbb{F}_2$, with pointwise addition and multiplication.

Conversely, suppose R is a unital ring satisfying $a^2 = a$ for every $a \in R$. Expanding $(a+a)^2 = a + a$ and using $a^2 = a$ shows $2a = 0$. Expanding $(a+b)^2 = a + b$ then shows $ab + ba = 0$, so $ab = ba$ in characteristic two. Thus every Boolean ring is commutative and has characteristic two (unless it is the zero ring).

The Boolean-algebra operations can be recovered from the ring:

$$a \wedge b = ab, \qquad \neg a = 1 + a, \qquad a \vee b = a + b + ab.$$

For subsets these become intersection, complement and union. The ring formalism is useful because algebraic homomorphisms already know how to preserve all three logical operations.

§5.2 Points appear as consistent truth assignments

Let B be a Boolean algebra with $0 \neq 1$. A map

$$\chi : B \longrightarrow \{0, 1\}$$

preserving $0, 1, \wedge, \vee, \neg$ is a complete Boolean truth assignment. Its true elements form

$$U_\chi = \{b \in B : \chi(b) = 1\}.$$

This is an *ultrafilter*: it contains 1 but not 0, is closed under finite meets, is upward closed, and for each $b \in B$ contains exactly one of b and $\neg b$.

[1] The algebraic and topological constructions follow Davey and Priestley, *Introduction to Lattices and Order*, and P. T. Johnstone, *Stone Spaces*.

Conversely, given such an ultrafilter U, put $\chi_U(b) = 1$ if $b \in U$, and 0 otherwise. The ultrafilter axioms verify the Boolean homomorphism laws. Hence

$$\boxed{\{\text{ultrafilters on } B\} \cong \mathrm{Hom}_{\mathrm{Bool}}(B, 2).}$$

The two descriptions are interchangeable: an ultrafilter is a consistent way to declare each Boolean proposition true or false.

For a finite set X, every ultrafilter on $\mathcal{P}(X)$ comes from exactly one point $x \in X$:

$$U_x = \{A \subseteq X : x \in A\}.$$

Indeed, since X is the union of its finitely many singleton subsets, the ultrafilter property forces one singleton to be true; two different singletons cannot both be true because their intersection is empty. A finite Boolean algebra can therefore recover its underlying points without knowing them in advance.

§5.3 An infinite example produces an extra point

Let B be the Boolean algebra of all *finite or cofinite* subsets of $\mathbb{N}$. For each $n \in \mathbb{N}$ we have the principal ultrafilter

$$U_n = \{A \in B : n \in A\}.$$

There is also a different ultrafilter:

$$U_\infty = \{A \in B : \mathbb{N} \setminus A \text{ is finite}\}.$$

It contains every cofinite set, no finite set, and decides every element of B, so it is an ultrafilter.

These are all the ultrafilters. If an ultrafilter contains a finite set F, the finite-union property forces it to contain a singleton $\{n\} \subseteq F$, hence it equals U_n. Otherwise it contains no finite set and therefore must contain the complement of every finite set, making it U_∞.

The ultrafilters have thus produced the space

$$S(B) = \mathbb{N} \cup \{\infty\}.$$

The extra point was not inserted artificially: it is the truth assignment that accepts all cofinite sets and rejects every finite set.

§5.4 The topology is forced by the algebra

For each $b \in B$, define

$$\widehat{b} = \{U \in S(B) : b \in U\}.$$

The ultrafilter laws immediately give

$$\widehat{a \wedge b} = \widehat{a} \cap \widehat{b}, \qquad \widehat{a \vee b} = \widehat{a} \cup \widehat{b}, \qquad \widehat{\neg a} = S(B) \setminus \widehat{a}.$$

These sets are closed under finite unions and complements; declaring them open makes them automatically *clopen*. They form a basis of a topology on $S(B)$.

For our finite–cofinite algebra,

$$\widehat{\{n\}} = \{U_n\},$$

so each ordinary point n is isolated. For a cofinite subset $\mathbb{N}\setminus F$, the corresponding clopen set is

$$\widehat{\mathbb{N}\setminus F} = \{U_\infty\} \cup \{U_n : n \notin F\}.$$

Every neighborhood of U_∞ therefore contains all but finitely many natural-number points. This is exactly the *one-point compactification* of the discrete space $\mathbb{N}$.

Compactness can be checked directly. An open cover has some member containing U_∞, which already covers all but finitely many points. Finitely many additional members cover those remaining points. The topology was constructed from Boolean operations alone, yet it has recovered a classical compact space.

§5.5 Stone's theorem and the role of ultrafilter extension

Stone's representation theorem states that every Boolean algebra B is isomorphic to the Boolean algebra $\mathrm{Clop}(S(B))$ of clopen subsets of its ultrafilter space:

$$\boxed{B \xrightarrow{\sim} \mathrm{Clop}(S(B)), \qquad b \longmapsto \widehat{b}.}$$

We have already verified that the map preserves Boolean operations. The remaining issues are injectivity and surjectivity, which reveal where compactness enters the theory.

If $a \neq b$, their symmetric difference is nonzero. By the Boolean prime ideal theorem, a nonzero element belongs to some ultrafilter. Such an ultrafilter distinguishes a and b, so $\widehat{a} \neq \widehat{b}$. This proves injectivity.

For surjectivity, every clopen $C \subseteq S(B)$ is open, hence covered by basic sets $\widehat{b_i}$ contained in C. The ultrafilter space is compact, so finitely many of them suffice:

$$C = \widehat{b_1} \cup \cdots \cup \widehat{b_m} = \widehat{b_1 \vee \cdots \vee b_m}.$$

Thus every clopen subset comes from an element of B.

The general compactness of $S(B)$ and the ultrafilter-extension step can be obtained from the Boolean prime ideal theorem, a choice principle weaker than the full axiom of choice. In the finite–cofinite example we needed neither an abstract extension lemma nor an appeal to an existence theorem: every point and every basic open set could be listed explicitly.

§5.6 Why the correspondence reverses arrows

Suppose $h : B \to C$ is a Boolean-algebra homomorphism. A point of $S(C)$ is an ultrafilter $U \subseteq C$. Pull it back:

$$h^{-1}(U) = \{b \in B : h(b) \in U\}.$$

It is an ultrafilter of B, so h induces a continuous map

$$S(h) : S(C) \longrightarrow S(B), \qquad U \longmapsto h^{-1}(U).$$

Continuity follows from the actual calculation

$$S(h)^{-1}(\widehat{b}) = \widehat{h(b)}.$$

Notice the direction reversal: an algebra homomorphism $B \to C$ gives a map of spaces $S(C) \to S(B)$, not $S(B) \to S(C)$.

Conversely, if $f : X \to Y$ is a continuous map between compact Hausdorff totally disconnected spaces, inverse image sends clopen sets of Y to clopen sets of X:

$$f^{-1} : \mathrm{Clop}(Y) \longrightarrow \mathrm{Clop}(X).$$

That operation also reverses arrows. Stone duality states that these two contravariant constructions give an equivalence between Boolean algebras and compact Hausdorff zero-dimensional spaces.

This is the deeper reason Boolean algebra belongs simultaneously to logic and topology. The algebra describes which yes/no statements can be combined; the points are consistent evaluations of those statements; and continuity means that truth regions pull back to truth regions. If we replace clopen sets by all open sets, complements need not remain open, and the resulting Heyting algebra may fail excluded middle. Classical Boolean logic corresponds to the geometry of clopen predicates.

Set Theory Through Maps: From Characteristic Functions to Cantor's Theorem

N-20220131

A set can be understood by the ways other sets map into it and by the ways it maps into others. This perspective is especially useful when a proof seems to depend on a mysterious encoding of mathematical objects as sets. Products, subsets, elements and functions can all be recovered from a small collection of mapping principles. Cantor's theorem then becomes a direct consequence of how evaluation and negation interact.[1]

§6.1 An element is a map from a singleton

In familiar set theory, an element $x \in X$ is commonly regarded as a primitive relation. Instead fix a one-element set $1 = \{*\}$. Every element $x \in X$ determines a unique map

$$\widehat{x} : 1 \longrightarrow X, \qquad \widehat{x}(*) = x.$$

Conversely every such map determines one element. Thus

$$\boxed{\text{elements of } X \;\longleftrightarrow\; \mathrm{Hom}(1, X).}$$

This is a small example of a useful change in language: instead of specifying what an object is made of, one specifies maps that probe it.

For example, a loop in a space X is represented by a map $S^1 \to X$, while a sequence of elements of X is a map $\mathbb{N} \to X$. An element is simply a probe with the smallest nonempty domain.

§6.2 The Universal Property of Products

The cartesian product $X \times Y$ comes with projection maps

$$\pi_X : X \times Y \to X, \qquad \pi_Y : X \times Y \to Y.$$

Its decisive property is that, for every set T and maps $f : T \to X$, $g : T \to Y$, there is a unique map $\langle f, g \rangle : T \to X \times Y$ satisfying

$$\pi_X \circ \langle f, g \rangle = f, \qquad \pi_Y \circ \langle f, g \rangle = g.$$

Taking $T = 1$ recovers the usual description of a pair as two elements. But the universal property applies equally well to pairs of functions, pairs of families or pairs of continuously varying parameters in a different category.

It also proves uniqueness of products up to a unique compatible isomorphism. If P and Q both satisfy the same property, the property constructs maps $P \to Q$ and $Q \to P$; their composites must be the respective identities by uniqueness. Thus a product is characterized by what it does, independent of an arbitrary encoding of ordered pairs.

[1] Tom Leinster, *Rethinking Set Theory* (2014), with his supplementary ETCS axioms.

§6.3 Function spaces and currying

An element of Y^X is itself a function $X \to Y$. That observation can be made universal. Given a map

$$q : T \times X \longrightarrow Y,$$

hold the T-coordinate fixed and regard the result as a function of X. We obtain a map

$$\widehat{q} : T \longrightarrow Y^X, \qquad \widehat{q}(t)(x) = q(t, x).$$

Conversely $h : T \to Y^X$ determines $(t, x) \mapsto h(t)(x)$. The constructions are inverse, and they commute with maps between parameter sets:

$$\boxed{\mathrm{Hom}(T \times X, Y) \cong \mathrm{Hom}(T, Y^X).}$$

This is the universal property of an *exponential object*. Just as projections characterize products, evaluation characterizes function spaces. The familiar act of treating a two-variable function as a family of one-variable functions is the map called *currying*.

For a very small example, take $T = \{a, b\}$ and $X = Y = \{0, 1\}$. Define q by the table

$q(t,x)$	$x=0$	$x=1$
$t=a$	0	1
$t=b$	1	0

.

Then $\widehat{q}(a)$ is the identity function on $\{0, 1\}$ and $\widehat{q}(b)$ is Boolean negation. The very same four values can be read row-by-row as two functions, or all at once as one function of two inputs.

The importance of this property is its independence from the chosen description of a function. A category can have exponential objects even when its arrows are continuous maps, group actions, or something less familiar. In each case one must check whether the appropriate function object exists. The evaluation map gives a convenient way to formulate that question.

§6.4 Equations and fibers as universal constructions

There is a similarly economical description of the solution set of an equation. Given maps $f, g : X \to Y$, their *equalizer* is a map $e : E \to X$ such that $fe = ge$, universal among maps on which f and g agree. In ordinary sets,

$$E = \{x \in X : f(x) = g(x)\}.$$

If a map $h : T \to X$ satisfies $fh = gh$, there is exactly one $\bar{h} : T \to E$ with $e\bar{h} = h$. This last sentence still makes sense when the elements of E have never been specified.

A fiber is a special case. Fix $y : 1 \to Y$ and consider $f : X \to Y$. Then $f^{-1}(y)$ sits in a pullback square

$$\begin{array}{ccc} f^{-1}(y) & \xhookrightarrow{\ i\ } & X \\ \downarrow & & \downarrow{\scriptstyle f} \\ 1 & \xrightarrow[\ y\ ]{} & Y. \end{array}$$

One can recognize this square by a factorization test: $h : T \to X$ factors through the fiber exactly when fh is the constant map with value y.

As a calculation, let $f : \mathbb{Z} \to \mathbb{Z}/3\mathbb{Z}$ send each integer to its residue. The fiber over $\overline{1}$ consists of $1, 4, 7, \ldots$ and their negative counterparts, namely $1 + 3\mathbb{Z}$. A map $h : T \to \mathbb{Z}$ factors through this fiber precisely when every value of h is congruent to 1 modulo 3.

Finite products and equalizers give finite limits, including all finite pullbacks. The everyday constructions of pairs, solutions to equations, and inverse images therefore grow from a rather short list of demands on arrows. Later, these same diagrams can be interpreted in categories whose objects have topology, geometry, or varying local data.

§6.5 Subsets as Binary Predicates

Let $2 = \{0, 1\}$. A subset $A \subseteq X$ determines its characteristic function

$$\chi_A : X \to 2, \qquad \chi_A(x) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A. \end{cases}$$

Conversely, a function $\chi : X \to 2$ specifies the subset $\chi^{-1}(\{1\})$. Therefore

$$\boxed{\mathcal{P}(X) \cong 2^X,}$$

where 2^X denotes the set of all maps from X to 2.

Ordinary set operations become pointwise Boolean operations:

$$\chi_{A \cap B} = \chi_A \chi_B, \qquad \chi_{A^c} = 1 - \chi_A,$$

and

$$\chi_{A \cup B} = \chi_A + \chi_B - \chi_A \chi_B.$$

These formulas use arithmetic in the integers, but evaluate only to 0 or 1 on Boolean inputs.

Given $f : X \to Y$, inverse image has a particularly natural description:

$$\boxed{\chi_{f^{-1}(B)} = \chi_B \circ f.}$$

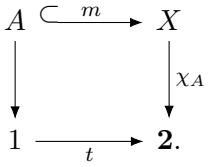
Pulling a subset back along a map is just composition of its predicate with that map. The preimage operation therefore comes from the most basic operation on functions: composition.

§6.6 Membership through a subobject classifier

A subset $A \subseteq X$ is represented by its inclusion $i : A \hookrightarrow X$. Abstractly an inclusion is a *monomorphism*: a map $m : A \to X$ with the property that $mu = mv$ forces $u = v$ for all maps u, v with a common domain. In sets this says exactly that m is injective.

Two injections can represent the same subset without their domains being literally equal. For example, a pair of distinct labels $\{\alpha, \beta\}$ could be mapped injectively to the even residues $\{\overline{0}, \overline{2}\}$ inside $\mathbb{Z}/4\mathbb{Z}$. What matters is the image and how it sits in the ambient set. Categorically we regard monomorphisms into X as equivalent when they differ by an isomorphism over X. Their equivalence classes form the collection $\mathrm{Sub}(X)$ of *subobjects*.

For sets, $\mathbf{2} = \{0, 1\}$ and the distinguished point $t : 1 \to \mathbf{2}$ with value 1 have a universal property. Each monomorphism $m : A \hookrightarrow X$ fits, up to the unique appropriate isomorphism, into a pullback square



Conversely every arrow $\chi : X \to \mathbf{2}$ produces a subobject by pulling back the point t. The object $\mathbf{2}$, together with t, is the *subobject classifier* of **Set**.

The two faces of a subset now meet:

$$\boxed{\mathrm{Sub}(X) \cong \mathrm{Hom}(X, \mathbf{2}) \cong \mathrm{Hom}(1, \mathbf{2}^X).}$$

The first is the classifier property; the second is currying with $T = 1$. The ordinary power set $\mathcal{P}(X)$ and the exponential $\mathbf{2}^X$ are two descriptions of the same object in **Set**.

Membership becomes a question about factorization. An element $x : 1 \to X$ belongs to the subobject represented by $m : A \hookrightarrow X$ when there is a map $a : 1 \to A$ with $ma = x$. Equivalently,

$$x \in A \quad \Longleftrightarrow \quad \chi_A x = t.$$

No membership test is required between two arbitrary mathematical objects. One first names the ambient set X, then a subobject A of X, and finally a point x of X. This is why the categorical viewpoint suits the way one actually uses sets when doing algebra or analysis.

A general elementary topos also has a subobject classifier Ω, although it need not look like $\mathbf{2}$. For sheaves on a space, truth may depend on an open region; a subobject can be locally true without having a single global Boolean value. ETCS singles out a setting in which ordinary set-like points, subsets, and classical truth values behave as expected.

§6.7 Evaluation turns Cantor's paradox into a proof

For sets X, Y, the set Y^X comes with an evaluation map

$$\mathrm{ev} : Y^X \times X \to Y, \qquad (h, x) \longmapsto h(x).$$

Suppose, toward a contradiction, that there is a surjective map

$$s : X \longrightarrow 2^X.$$

Every point $x \in X$ has been assigned a Boolean-valued function $s(x) : X \to 2$. Define the diagonal predicate

$$d(x) = 1 - \mathrm{ev}(s(x), x) = 1 - s(x)(x).$$

Because s is surjective, some $a \in X$ must satisfy $s(a) = d$. Evaluating at a gives

$$d(a) = 1 - s(a)(a) = 1 - d(a),$$

which is impossible for a value in $\{0, 1\}$. Consequently

$$\boxed{\text{there is no surjection } X \to 2^X.}$$

This is Cantor's theorem $|\mathcal{P}(X)| > |X|$. In the map-based presentation, the contradiction comes from evaluation and the operation of changing a truth value, with no need to ask what the elements of X are made of.

The same proof is familiar when written as the diagonal subset

$$D = \{x \in X : x \notin s(x)\}.$$

The categorical-looking version makes clear that the engine of the argument is a self-reference created by evaluation, not any special internal material from which sets are constructed.

For example, on $X = \{a, b\}$ suppose that a proposed list has

$$s(a) = \{b\}, \qquad s(b) = \{a, b\}.$$

The diagonal test asks whether $a \in s(a)$ and $b \in s(b)$, and reverses each answer:

	$a \in s(x)$	$b \in s(x)$	$x \in D$?
$x = a$	no	yes	yes
$x = b$	yes	yes	no

$D = \{a\}$.

Neither $s(a)$ nor $s(b)$ equals D. The calculation does not rely on the particular two assigned subsets: negating the answer on the diagonal guarantees that the newly constructed subset differs from every subset in the list, at its own index.

§6.8 Choice means splitting a surjection

For a surjection $p : E \to B$, a right inverse is a map $s : B \to E$ with

$$p \circ s = \mathrm{id}_B .$$

It chooses one element in each fiber $p^{-1}(b)$, compatibly with the ordinary definition of a function. The axiom of choice is equivalent, in the category of sets, to the assertion that every surjection has such a section.

This is a strong structural statement about maps. It fails automatically in many categories of spaces with extra structure, even when the underlying function is surjective.

Consider the continuous covering map

$$p : S^1 \to S^1, \qquad p(z) = z^2.$$

It is surjective, but there is no continuous right inverse. If s existed, the induced maps on fundamental groups would satisfy

$$p_* \circ s_* = \mathrm{id}_{\pi_1(S^1)} .$$

After identifying $\pi_1(S^1) \cong \mathbb{Z}$, the map p_* is multiplication by 2. Hence we would need an integer m with $2m = 1$, a contradiction.

Set-theoretically one can choose a square root of each z, but no such choice can be continuous around the circle. The example distinguishes the abstract existence of choices from their compatibility with added geometric structure.

§6.9 Natural numbers as a recursion principle

A natural numbers object N comes with a distinguished zero map $z : 1 \to N$ and successor map $s : N \to N$. Its defining property is recursion: given an object X, a point $x_0 : 1 \to X$, and a map $g : X \to X$, there is a unique $h : N \to X$ with

$$h(0) = x_0, \qquad h(n+1) = g(h(n)).$$

In ordinary sets, take $X = \mathbb{N}$, $x_0 = 0$ and $g(k) = k + 2$. The unique recursively defined map is

$$h(n) = 2n.$$

This simple property gives a structural characterization of iteration. It is a basic reason why natural numbers, finite sequences and inductive definitions fit naturally into a map-based foundation.

The same recursion principle can carry more than one number at a time. In **Set** let

$$X = \mathbb{N} \times \mathbb{N}, \quad x_0 = (0, 1), \quad g(a, b) = (b, a + b).$$

The uniquely defined arrow $h : N \to X$ begins

$$h(0) = (0, 1), \quad h(1) = (1, 1), \quad h(2) = (1, 2), \quad h(3) = (2, 3), \ldots.$$

Its two coordinates are consecutive Fibonacci numbers. One recursion on a product object generates an infinite system of linked calculations.

§6.10 The ETCS axioms in action

Lawvere's *Elementary Theory of the Category of Sets* asks for a category of sets and maps with a particularly useful package of properties. One standard modern formulation is

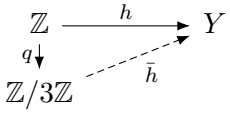
ETCS: a well-pointed elementary topos with a natural numbers object and choice.

Here an *elementary topos* has finite limits, exponential objects, and a subobject classifier. The various constructions already encountered explain what those conditions provide. Finite limits give products, equations, and fibers. Exponentials supply sets of functions. The classifier lets us form and manipulate subsets. The natural numbers object supports recursion and induction, and the choice axiom provides sections of epimorphisms.

A topos also has finite colimits, including quotients of equivalence relations. To see the connection with the elementary theory of congruences, take the relation

$$m \sim n \quad \Longleftrightarrow \quad m - n \in 3\mathbb{Z}$$

on $\mathbb{Z}$. The quotient map $q : \mathbb{Z} \to \mathbb{Z}/3\mathbb{Z}$ identifies the pairs (m, n) with $m \sim n$. Its characteristic property is this: if $h : \mathbb{Z} \to Y$ satisfies $h(m) = h(n)$ whenever $m \sim n$, then h factors uniquely through q:



The relation is the kernel pair of q, and q is the coequalizer of its two projections. Quotients in a topos obey the same universal pattern. Hence the structures named in ETCS give a remarkably broad supply of familiar constructions without selecting particular encodings for them.

§6.11 What well-pointedness buys

ETCS has another hypothesis which deserves more attention. Its terminal object 1 is a *generator*: whenever $f, g : X \to Y$ are different arrows, some point $x : 1 \to X$ witnesses the difference,

$$f \neq g \quad \Longrightarrow \quad fx \neq gx \quad \text{for some } x : 1 \to X.$$

This is the categorical form of the ordinary principle that two functions are equal when they agree at every input. Once this property is assumed, many arrow equations may be verified by element chasing, exactly as in undergraduate mathematics.

In a general category, a map $T \to X$ is called a *generalized element* of X of shape T. If we allow all shapes T, maps can always be distinguished: choose $T = X$ and compare $f \circ \mathrm{id}_X$ with $g \circ \mathrm{id}_X$. Well-pointedness says that the single shape 1 already suffices. It is a substantive requirement, not an automatic consequence of learning the language of arrows.

The axiom of choice also has an arrow formulation:

$$\boxed{\text{every epimorphism } p : E \to B \text{ has a section } s : B \to E, \quad ps = 1_B.}$$

In **Set**, epimorphisms are surjections, so this recovers the ordinary choice axiom. ETCS asserts that the same splitting property holds in its proposed category of sets.

A useful consequence is that the internal logic of an ETCS topos is classical. In fact, Diaconescu's theorem says that the splitting of all epimorphisms in an elementary topos forces its subobject logic to be Boolean. This helps explain why the classifier of ordinary sets has precisely two truth values, while sheaf topoi can use open regions as their local truth values.

§6.12 How much set theory is hidden here?

There is a practical distinction between manipulating an already given set and forming an entirely new collection of sets. ETCS supplies subsets of each set X, quotients, products, and function sets Y^X for given objects X, Y. It can carry out a great deal of the set-based mathematics appearing in algebra, analysis, and geometry.

It does not provide an unrestricted set of all sets, and its standard axioms do not contain the full Replacement schema of ZFC. This is a real difference in proof-theoretic strength. A precise comparison is known: ETCS is *equiconsistent* with bounded Zermelo set theory with Choice, where Separation is restricted to bounded formulas and Replacement is absent. Michael Shulman's *Comparing Material and Structural Set Theories* gives a modern account. Stronger structural set theories add carefully formulated replacement or collection principles when they are needed.

A concrete distinction is useful here. Given a map $f : X \to Y$ and a subset $B \subseteq Y$, the inverse image $f^{-1}(B)$ is part of the ordinary bounded collection of subobjects of X. Replacing an entire class-indexed family of objects by a single new set is a different task: the objects to be collected may range beyond any previously named set. The extra strength required for such arguments cannot be inferred solely from the existence of Y^X.

The categorical language also makes it easier to locate which hypotheses a proof really uses. The existence of a diagonal function and the exponential 2^X is enough to formulate Cantor's argument. Choosing elements simultaneously from many fibers uses the choice axiom. Recursion uses the natural numbers object. When two arrows are compared by evaluating them on points, well-pointedness is doing the work.

§6.13 A square root with no global element

The distinctive force of well-pointedness is clearest in an example where it fails. Cover the unit circle by the map

$$p : S^1 \longrightarrow S^1, \qquad p(z) = z^2.$$

For each open arc $U \subseteq S^1$, let $\mathcal{F}(U)$ consist of continuous choices of square root:

$$\mathcal{F}(U) = \{s : U \to S^1 \text{ continuous} : s(z)^2 = z \text{ for every } z \in U\}.$$

These sets, together with restriction of functions, form a sheaf $\mathcal{F}$ on the circle. On a sufficiently short arc there are two possible continuous branches. Across overlapping arcs, a choice must agree with its restriction.

Over the entire circle there is no continuous square root. Indeed, a global $s : S^1 \to S^1$ with $s(z)^2 = z$ would give, on fundamental groups,

$$(z \mapsto z^2)_* \circ s_* = \mathrm{id}_{\mathbb{Z}},$$

forcing $2\deg(s) = 1$. Therefore

$$\boxed{\mathcal{F}(S^1) = \varnothing.}$$

A global element $1 \to \mathcal{F}$ of the sheaf topos would be exactly such a global section, so $\mathcal{F}$ has no global elements. Yet $\mathcal{F}$ is a genuine, nonempty local object: $\mathcal{F}(U)$ has elements on short arcs.

This difference prevents 1 from being a generator in the sheaf topos. Let $\underline{\mathbf{2}}$ denote the constant sheaf of two values. There are two maps

$$c_0, c_1 : \mathcal{F} \longrightarrow \underline{\mathbf{2}}$$

which assign the constant value 0 or 1 to every local section. They differ on the short arcs where sections exist. But there is no global point $x : 1 \to \mathcal{F}$ available to detect the difference, since both composites $c_0 x, c_1 x$ have no instances at all.

A single loop around the circle exchanges the two local branches. Their obstruction to gluing has a particularly concise categorical form:

$$\boxed{\mathrm{Hom}_{\mathbf{Sh}(S^1)}(1, \mathcal{F}) = \varnothing, \qquad c_0 \neq c_1 \in \mathrm{Hom}_{\mathbf{Sh}(S^1)}(\mathcal{F}, \underline{\mathbf{2}}).}$$

Four Foundations Facing the Same Mathematical Problems

N-20220205

A theory of sets is tested by the constructions it permits. ZFC, von Neumann–Bernays–Goedel class theory (NBG), Kelley–Morse class theory (KM), and Quine's New Foundations (NF) differ most clearly when we ask whether the same collection, function or truth predicate actually exists. Four tests will reveal their distinctive mathematics: Russell's collection, the category of all sets, Cantor's diagonal argument, and a universal satisfaction predicate.[1]

§7.1 Russell's predicate: exactly what may be collected?

The tempting comprehension formula

$$R = \{x : x \notin x\}$$

would imply $R \in R \iff R \notin R$. Each foundational system must reject some step in the construction.

In ZFC, one may *separate* elements from a set already given. If A is a set, the restricted collection

$$R_A = \{x \in A : x \notin x\}$$

is legitimate, and it satisfies the useful theorem

$$\boxed{R_A \notin A}.$$

For if $R_A \in A$, its own defining condition makes $R_A \in R_A$ equivalent to $R_A \notin R_A$. There is no paradox in constructing R_A; it simply lies outside the proposed container A. Consequently no universal *set* exists in ZFC.

NBG permits a Russell *class*

$$\mathcal{R} = \{x : x \text{ is a set and } x \notin x\}.$$

The diagonal argument shows that $\mathcal{R}$ must be a proper class, unable to occur as an element of another class. The universe V of all sets is similarly a proper class. KM gives the same answer to this problem; its stronger comprehension has not been invoked.

NF instead restricts the *syntax* of comprehension. A formula is stratified if variables can be assigned type indices so that $x \in y$ raises the type by one and equality leaves it unchanged. The formula $x = x$ is stratified, and so

$$\boxed{V = \{x : x = x\}}$$

is an actual universal set, with $V \in V$. But $x \notin x$ cannot be stratified: the type of x would have to exceed itself by one. Russell's collection is unavailable for a wholly different reason.

Thus ZFC restricts the domain of comprehension, class theories distinguish what can be a member, and NF restricts which predicates define sets.

[1] For class comprehension and truth predicates, see Victoria Gitman's expositions of second-order set theory. For stratified comprehension and Cantor's argument, see T. E. Forster, *Quine's New Foundations*, in the *Stanford Encyclopedia of Philosophy*, and E. P. Specker, *The Axiom of Choice in Quine's New Foundations* (1953).

§7.2 Large Categories and Expressive Power

For two fixed sets A, B, ZFC constructs the function set B^A without difficulty. It also proves the category laws by evaluating functions at elements:

$$((h \circ g) \circ f)(a) = h(g(f(a))) = (h \circ (g \circ f))(a).$$

But the objects of the category **Set** do not form a set. ZFC states the category laws uniformly about arbitrary sets and functions, using definable collections rather than pretending the entire category is small.

NBG can make the object collection and the collection of set-coded arrows into classes. Each $\mathrm{Hom}(A, B) = B^A$ remains a set, so **Set** is *locally small*. The familiar power-set assignment illustrates what becomes easier to package:

$$\mathcal{P}(f) : \mathcal{P}(A) \longrightarrow \mathcal{P}(B), \qquad S \longmapsto f[S].$$

Because $(g \circ f)[S] = g[f[S]]$, we obtain

$$\mathcal{P}(g \circ f) = \mathcal{P}(g) \circ \mathcal{P}(f).$$

ZFC proves this uniformly, NBG records the functor as class-sized data, and KM permits the same construction. No extra KM comprehension is needed here. Class notation can improve the language in which a theorem is stated without changing the set-level theorem.

The universal set of NF does not automatically settle such a category problem. Standard encodings of functions and ordered pairs must obey stratification; existence of a set of all objects alone does not prove that every intended collection of arrows has the desired set-coded operations.

§7.3 Cantor's Diagonal and the Universal Set

In ZFC, suppose $f : A \to \mathcal{P}(A)$ is surjective. Separation constructs

$$D = \{a \in A : a \notin f(a)\}.$$

Some $d \in A$ must satisfy $f(d) = D$, whence $d \in D \iff d \notin D$, impossible. NBG and KM prove the identical result for each set A.

In NF, however, every set is a subset of the universal set V, and every subset of V is a set. Therefore

$$\boxed{\mathcal{P}(V) = V},$$

so the identity map is a surjection from V onto its power set. Where has Cantor's proof gone? The predicate $a \notin f(a)$ is not stratified for an arbitrary set-coded map f: evaluating a function forces the input and output to have matching syntactic type levels, but the membership test requires the output one level higher. The diagonal set need not exist.

NF does retain a type-shifted Cantor comparison. The set of singletons

$$\{\{a\} : a \in A\}$$

must be distinguished from A itself: the usual singleton map $a \mapsto \{a\}$ need not exist as a set-coded function on an arbitrary NF set. Sets for which this map does exist are called *strongly Cantorian*. This is a concrete mathematical price of admitting a universal set.

§7.4 Uniform Truth in KM

Can one form a single class that says which first-order formulas are true in $(V, \in)$? A satisfaction class T would contain a formula code and an assignment to its free variables, and satisfy the Tarski clauses uniformly:

$$\begin{aligned} T(\ulcorner v_i \in v_j \urcorner, a) &\iff a_i \in a_j, \\ T(\ulcorner \neg\psi \urcorner, a) &\iff \neg T(\ulcorner \psi \urcorner, a), \\ T(\ulcorner \exists v_i\, \psi \urcorner, a) &\iff \exists x\, T(\ulcorner \psi \urcorner, a[i \mapsto x]). \end{aligned}$$

For each *fixed* formula, ZFC can define its truth relation by following its finite syntactic tree. But Tarski's undefinability theorem forbids a single first-order definable predicate handling *all* formulas of the universe at once.

NBG can quantify over classes, but its class comprehension permits only formulas without bound class quantifiers. It cannot prove that there is a full satisfaction class for V: a model containing only its definable classes would violate Tarski's theorem if it had one.

KM permits formulas defining classes to quantify over classes. This impredicative strength proves the existence of a uniform first-order satisfaction class. Its consequences are visible even in arithmetic. Such a truth class supports a formal soundness argument for ZFC: the axioms hold in V, the inference rules preserve truth, and a contradiction cannot be a theorem. Therefore

$$\boxed{\mathrm{KM} \vdash \mathrm{Con(ZFC)}.}$$

Assuming ZFC is consistent, Gödel's second incompleteness theorem means ZFC cannot prove this sentence. NBG is conservative over ZFC for set sentences, so it cannot prove it either. KM's additional comprehension is not simply a more convenient way to talk about the same classes.

§7.5 Choice: global convenience and the NF obstruction

ZFC's axiom of choice says that for each *set-indexed* family of nonempty sets $(A_i)_{i \in I}$, there exists a function c with $c(i) \in A_i$. It does not automatically provide a single class function choosing an element from *every* nonempty set in the universe.

A class theory can express a global choice function C:

$$\boxed{C(A) \in A \quad \text{for every nonempty set } A.}$$

One often writes GBC for Gödel–Bernays class theory augmented by Global Choice. This is a precise additional principle that can streamline uniform class-sized constructions, and it should not be silently identified with the ordinary axiom of choice on sets.

NF has a fundamentally different answer. Specker proved in 1953 that the ordinary axiom of choice fails in NF; indeed its universal set cannot be well-ordered. This is a theorem rather than an unsettled consistency issue. The related theory NFU allows urelements by weakening extensionality. Jensen's relative-consistency theorem shows that NFU can be consistent with Choice, so the contrast is not simply caused by permitting large universal sets.

The comparison now has real mathematical content. Russell's predicate tests the scope of comprehension; the category of all sets tests class-sized organization; Cantor's theorem tests which diagonal subsets may be formed; satisfaction classes test impredicative

comprehension; and Choice tests whether a family of selections can be assembled into a function. Each system answers these questions with different concrete constructions and theorems.

Linear Logic: Resources, Proof Reduction, and Feedback

N-20221005

Linear logic restricts the invisible copying and discarding of hypotheses that ordinary propositional reasoning permits. This becomes useful when a premise represents a consumable resource. A proof can then be read as a resource-transforming program, and cut elimination as the composition of such programs. The Geometry of Interaction develops this dynamical idea through feedback operators and paths in networks.[1]

§8.1 The tensor rule cannot spend one resource twice

In the intuitionistic linear sequent calculus, the rule for tensor splits the available context:

$$\frac{\Gamma \vdash A \qquad \Delta \vdash B}{\Gamma, \Delta \vdash A \otimes B}.$$

A hypothesis allocated to the left premise cannot also be allocated to the right one. Hence there is no general derivation $A \vdash A \otimes A$. The missing step is contraction, the ordinary rule that duplicates an assumption. Weakening, which discards an unused assumption, is also unavailable for a general linear hypothesis.

The exponential modality $!A$ permits controlled reuse. A linear implication $A \multimap B$ represents a transformation consuming an A and producing a B; a reusable transformation is supplied as $!(A \multimap B)$. In a symmetric monoidal closed model, the formula is not just suggestive notation:

$$\mathrm{Hom}(A \otimes B, C) \cong \mathrm{Hom}(A, B \multimap C).$$

Linear implication is the internal function object adjoint to tensoring with an input resource.

§8.2 A small story is an exact transition system

Let P, S, T, B, W represent a pig, straw, sticks, bricks and wolf, and H_S, H_T, H_B the completed houses. Start with

$$M_0 = 3P + S + T + B + W,$$

where addition records multiplicity in a multiset. Consider the six rewrite rules

$$\begin{aligned} P \otimes S &\multimap H_S, & P \otimes T &\multimap H_T, & P \otimes B &\multimap H_B,\\ W \otimes H_S &\multimap W, & W \otimes H_T &\multimap W, & W \otimes H_B &\multimap H_B. \end{aligned}$$

After using each construction rule once, the state is $W + H_S + H_T + H_B$. Applying the three wolf rules gives the completely explicit trace

$$\begin{aligned} W + H_S + H_T + H_B &\longmapsto W + H_T + H_B\\ &\longmapsto W + H_B\\ &\longmapsto H_B. \end{aligned}$$

[1] See J.-Y. Girard, *Linear Logic* (1987), and T. Seiller, *Interaction Graphs: Additives* for a mathematical feedback interpretation.

No pig or material remains: resources have been consumed, while the rules themselves may be treated as reusable assumptions.

The same computation is a Petri-net transition. Record multiplicities by a vector $m \in \mathbb{N}^n$. A rule with input vector a and output vector b can fire exactly when $m \geq a$ coordinatewise, and then

$$\boxed{m \longmapsto m - a + b.}$$

This tests the resource accounting in a way that a narrative analogy cannot. It also distinguishes the persistent availability of a rule from the transient availability of the tokens to which that rule applies.

§8.3 Additive choice is not simultaneous consumption

Tensor $A \otimes B$ supplies both resources, while the additive conjunction $A \,\&\, B$ means that the environment can select either branch. The two right introduction rules have different resource behavior:

$$\frac{\Gamma \vdash A \qquad \Delta \vdash B}{\Gamma, \Delta \vdash A \otimes B}, \qquad \frac{\Gamma \vdash A \qquad \Gamma \vdash B}{\Gamma \vdash A \,\&\, B}.$$

The shared context in the second rule does not license simultaneous consumption twice. The premises describe alternative interactions; the opponent selects which one occurs. Additive disjunction $A \oplus B$ lets the prover instead select a branch. A protocol offering either of two services on demand is therefore different from one physically providing both outputs.

§8.4 Cut elimination composes resource transformers

The cut rule composes a proof producing A with another proof consuming A:

$$\frac{\Gamma \vdash A \qquad \Delta, A \vdash B}{\Gamma, \Delta \vdash B}.$$

Under proofs-as-programs, a principal cut can reduce like a linear beta step

$$(\lambda x.t)\, u \longrightarrow t[u/x],$$

where the bound variable x occurs exactly once in the linear term t. The argument u is therefore neither silently duplicated nor erased.

The first construction and demolition in our resource trace compose into a direct rule $P \otimes S \otimes W \multimap W$. The intermediate object H_S disappears because its producer has been connected to its consumer. A cut-free composite describes the overall transformation without keeping that internal communication visible.

§8.5 The feedback formula counts internal paths

Let a linear process accept an external input x and an internal input v, and return an external output y and an internal output v':

$$\begin{pmatrix} y \\ v' \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} x \\ v \end{pmatrix}.$$

Feedback identifies $v' = v$, so $v = Cx + Dv$. If $I - D$ is invertible, eliminating this hidden state gives

$$\boxed{y = (A + B(I - D)^{-1}C)x.}$$

For nilpotent $D^r = 0$,

$$A + B(I - D)^{-1}C = A + BC + BDC + \cdots + BD^{r-1}C.$$

Each summand records paths passing through the internal connection a specified number of times.

Choose

$$A = 0, \quad B = (1 \;\; 0), \quad C = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Here $D^2 = 0$, $BC = 0$, but $BDC = 1$. Thus the external map is the identity,

$$A + B(I - D)^{-1}C = 1.$$

The input reaches the output only after circulating through the internal connection. The block A alone would entirely miss this effect.

Girard's Geometry of Interaction uses operator-level execution formulas to model cut elimination. This finite-dimensional feedback calculation illustrates the algebraic mechanism; a full logical semantics also requires appropriate typing and proof-correctness conditions. The connection is substantive because eliminating an internal wire is literally an inverse or sum of path operators.

$La \longrightarrow L(^*m) \otimes L((Rm) \parr a)$

may be depicted in the following way:

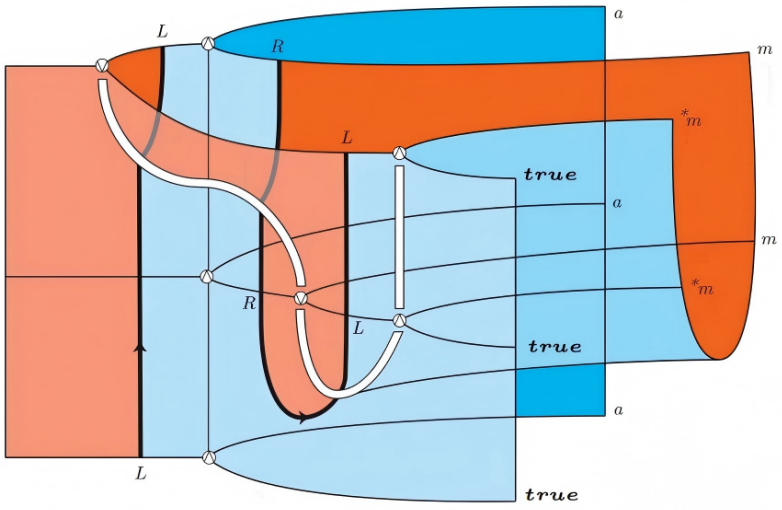


Figure 8.1: Wired inputs and outputs visualize proof composition as controlled interaction.

§8.6 Negation as testing against possible opponents

In Ludics a design can be tested against another design. Write $D \perp E$ if their specified interaction terminates successfully, and for a family $\mathcal{A}$ define

$$\mathcal{A}^{\perp} = \{E : D \perp E \text{ for every } D \in \mathcal{A}\}.$$

The double orthogonal satisfies the closure laws

$$\mathcal{A} \subseteq \mathcal{A}^{\perp\perp}, \qquad \mathcal{A}^{\perp\perp\perp} = \mathcal{A}^{\perp}.$$

These follow from the elementary reversal of inclusion: $\mathcal{A} \subseteq \mathcal{B} \Rightarrow \mathcal{B}^{\perp} \subseteq \mathcal{A}^{\perp}$. Families fixed by double orthogonality are called behaviors.

A linear proof consumes precisely specified resources, cut elimination composes its transformations, feedback semantics describes the internal paths of interaction, and orthogonality characterizes a behavior by the opponents against which it succeeds. The common structure is an explicit operation on processes, rather than a string of analogies about stories or games.

Category Theory Continued and the Question Topology

N-20221227

§9.1 Posets, Categories, and Functors

This note continues the category-theory examples. Abelian groups with group homomorphisms form **Ab**. Modules over a ring A form $\mathbf{Mod}_A$. Rings with unital ring homomorphisms form **Rings**. Topological spaces with continuous maps form **Top**, whose isomorphisms are homeomorphisms.

A partially ordered set $(S, \geq)$ can be regarded as a category: the objects are elements of S, and there is exactly one morphism $x \to y$ iff $x \geq y$. Reflexivity gives identities, transitivity gives composition, and antisymmetry ensures isomorphisms are trivial.

A partially ordered set becomes a category in a very economical way. There is a unique morphism $x \to y$ exactly when $x \leq y$. Identity morphisms come from reflexivity $x \leq x$, and composition comes from transitivity: if $x \leq y$ and $y \leq z$, then $x \leq z$.

This example is important because it shows that a category need not look like a collection of algebraic objects and homomorphisms. A category can encode logic or order. Meets and joins in a poset become limits and colimits in the corresponding category.

A subcategory $\mathcal{A} \subseteq \mathcal{C}$ has some objects and some morphisms of $\mathcal{C}$, closed under identities and composition. A functor

$$F : \mathcal{C} \to \mathcal{D}$$

sends objects to objects and morphisms to morphisms, preserving identities and composition:

$$F(1_A) = 1_{F(A)}, \qquad F(g \circ f) = F(g) \circ F(f).$$

A standard functoriality square says that the image of a composable path under F is still composable and that the composite maps to the composite.

A poset viewed as a category has at most one morphism between any two objects. In this setting, categorical statements become order-theoretic statements. A product is a meet, a coproduct is a join, an initial object is a least element, and a terminal object is a greatest element.

For example, in the poset of open subsets of a space ordered by inclusion, intersections behave like meets and unions behave like joins. This is why topology and order theory are so close: open sets form a lattice, and categorical language packages its universal properties.

§9.2 Forgetful and Free Functors

A forgetful functor forgets structure. For example,

$$\mathbf{Grp} \to \mathbf{Set}$$

sends a group to its underlying set and a homomorphism to its underlying function. Similarly,

$$\mathbf{Top} \to \mathbf{Set}$$

forgets open sets, and $\mathbf{Vect}_k \to \mathbf{Set}$ forgets vector-space operations.

A forgetful functor is not innocent: once structure is forgotten, many questions can no longer be answered. A bijection of underlying sets need not be an isomorphism of groups, vector spaces, or topological spaces.

A forgetful functor loses structure, but it often has a left adjoint that freely adds structure. For instance,

$$U : \mathbf{Grp} \to \mathbf{Set}$$

forgets the group operation. Its left adjoint sends a set S to the free group $F(S)$. The adjunction says that a set map

$$S \to U(G)$$

corresponds uniquely to a group homomorphism

$$F(S) \to G.$$

This is the categorical form of "define a homomorphism out of a free object by choosing images of generators."

§9.3 Initial Topologies from Observations

The second half moves into topology. The page considers a collection of subsets and asks what topology it generates. Given a basis-like family $\mathcal{B}$, the topology generated by it consists of arbitrary unions of finite intersections of elements of $\mathcal{B}$.

The displayed rational-looking function and its graph on the page are used as a reminder: topology is not only about familiar Euclidean open intervals. One can impose a topology by declaring certain sets to be basic neighborhoods.

The topology part of the note concerns how open sets can be generated from a chosen collection. If $\mathcal{B}$ is intended as a basis, two conditions are expected: every point lies in some basis element, and intersections of basis elements are locally refined by basis elements. Then every open set is a union of basis elements.

If one starts with a subbasis $\mathcal{S}$, the generated topology consists of arbitrary unions of finite intersections of elements of $\mathcal{S}$. This explains the phrase "generated by": we first force the chosen sets to be open, then close under the axioms of topology.

The topology generated by a family of maps can be understood as an initial topology. Suppose we have functions

$$f_i : X \to Y_i$$

where each Y_i is already topological. The initial topology on X is the coarsest topology making all f_i continuous. Its subbasis is

$$f_i^{-1}(U), \qquad U \subseteq Y_i \text{ open}.$$

This realizes the idea of a topology generated by observations: each map f_i observes X, and the generated topology is exactly the open-set structure forced by those observations.

The product topology is the initial topology for the projection maps

$$\pi_X : X \times Y \to X, \qquad \pi_Y : X \times Y \to Y.$$

A map $h : Z \to X \times Y$ is continuous iff the coordinate maps $\pi_X \circ h$ and $\pi_Y \circ h$ are continuous. This is a categorical universal property, not a visual fact about rectangles.

The rectangle basis $U \times V$ appears because

$$U \times V = \pi_X^{-1}(U) \cap \pi_Y^{-1}(V).$$

Thus the usual construction of product open sets is forced by the initial-topology viewpoint.

§9.4 Final Topology and Quotients

Topological constructions are full of universal properties. Product topology is the coarsest topology making the projection maps continuous. Quotient topology is the finest topology making the quotient map continuous. These statements are categorical: they characterize objects by how maps into or out of them behave.

The passage from categories to topology is natural. Topology studies spaces through continuous maps, not only through open sets, and this is precisely categorical thinking.

Dually, quotient topology is a final topology. For a quotient map

$$q : X \to Y,$$

the topology on Y is the finest topology making q continuous, and a map $f : Y \to Z$ is continuous iff $f \circ q : X \to Z$ is continuous.

Thus product and quotient are opposite-looking constructions:

initial topology	maps into known spaces	coarsest topology
final topology	maps from known spaces	finest topology

This is the precise categorical version of the topology examples in the note.

§9.5 Functorial invariants turn geometry into algebra

A functor is not merely a map between categories. It is a structure-preserving translation. If $F : \mathcal{C} \to \mathcal{D}$, then for every composable pair

$$A \xrightarrow{f} B \xrightarrow{g} C$$

we require

$$F(g \circ f) = F(g) \circ F(f).$$

This condition says that doing a construction after composing maps gives the same result as first applying the construction to each map and then composing.

Examples clarify the point. The fundamental group construction is a functor from pointed topological spaces to groups. Homology is a functor from spaces to graded abelian groups. The forgetful functor from groups to sets forgets multiplication but still sends homomorphisms to functions.

The fundamental group is functorial. More generally, algebraic topology studies functors such as

$$\pi_1 : \mathbf{Top}_* \to \mathbf{Grp}, \qquad H_n : \mathbf{Top} \to \mathbf{Ab}.$$

A continuous map of spaces induces a homomorphism of groups or abelian groups. This turns geometric questions into algebraic ones.

The power of this move is that algebraic invariants can obstruct homeomorphisms. If two spaces have non-isomorphic fundamental groups or homology groups, they cannot be the same topological space.

Orders, Symmetries, Products, and Categorical Cancellation

N-20230313

An arrow in a category need not be a function. In a partially ordered set it records a comparison; in a group it records a reversible operation. The real test is whether concepts such as product, injection, and surjection can be recovered entirely from arrows. The examples below follow Tom Leinster's *Basic Category Theory*, Chapters 1–2.[1]

§10.1 From reflexivity and transitivity to composition

Let P be a preorder: its relation $\preceq$ is reflexive and transitive. Define

$$\mathrm{Hom}(p,q) = \begin{cases} \{*\}, & p \preceq q, \\ \varnothing, & p \not\preceq q. \end{cases}$$

Reflexivity supplies the identity $p \to p$; transitivity supplies composition. Associativity follows automatically because there is at most one arrow with each source and target.

For example, take the positive divisors of 12, ordered by divisibility. The arrows $2 \to 6$ and $6 \to 12$ compose to $2 \to 12$. A Hasse diagram shows only the generating cover relations; the corresponding category contains all composites as well.

An isomorphism $p \cong q$ requires both $p \preceq q$ and $q \preceq p$. If the preorder is a poset, antisymmetry forces $p = q$. In a general preorder distinct isomorphic objects are possible. Thus a poset category is a special, skeletal form of a preorder category.

§10.2 A group as a one-object category

Given a monoid M, construct $\mathcal{B}M$ with just one object $*$ and

$$\mathrm{Hom}_{\mathcal{B}M}(*,*) = M.$$

Multiplication is composition, and the unit is the identity arrow. These are exactly the associative and unit axioms of a category. If M is a group, each arrow is invertible, making $\mathcal{B}M$ a one-object groupoid.

This also translates group actions into the language of functors. A functor

$$F : \mathcal{B}G \longrightarrow \mathbf{Set}$$

chooses a set $X = F(*)$ and a map $F(g) : X \to X$ for each $g \in G$, satisfying

$$F(gh) = F(g) \circ F(h), \qquad F(e) = \mathrm{id}_X\,.$$

Since g has an inverse, $F(g)$ is a bijection. These conditions are precisely a group action. The functor axioms are doing the mathematical work.

[1] https://arxiv.org/abs/1612.09375

§10.3 Products of categories versus products of objects

In $\mathcal{C} \times \mathcal{D}$, objects are pairs (C, D), arrows are pairs (f, g), and composition is componentwise. There are projection functors

$$\pi_{\mathcal{C}} : \mathcal{C} \times \mathcal{D} \to \mathcal{C}, \qquad \pi_{\mathcal{D}} : \mathcal{C} \times \mathcal{D} \to \mathcal{D}.$$

For any $\mathcal{E}$, a functor $H : \mathcal{E} \to \mathcal{C} \times \mathcal{D}$ is uniquely determined by its projections. Conversely two functors $F : \mathcal{E} \to \mathcal{C}$ and $G : \mathcal{E} \to \mathcal{D}$ define $H(e) = (F(e), G(e))$, with the analogous rule on arrows. This is the product's universal property in the category of small categories.

One must distinguish this from a product *inside* a category. In **Set**, $A \times B$ is a single set with two projection functions, and every pair $X \to A$, $X \to B$ factors through it uniquely. Both constructions satisfy similar diagrams, but their objects live at different levels.

§10.4 Opposite categories and the direction of a test

The opposite category $\mathcal{C}^{\mathrm{op}}$ reverses all arrows, with composition reversed accordingly. The divisor-poset category becomes reverse divisibility; a terminal object becomes initial. Statements phrased entirely by morphisms admit dual statements upon reversing their arrows.

This motivates a definition without reference to individual elements.

Definition 10.4.1. A morphism $f : A \to B$ is *monic* if

$$fg = fh \Longrightarrow g = h \qquad (g, h : X \to A),$$

and *epic* if

$$gf = hf \Longrightarrow g = h \qquad (g, h : B \to X).$$

§10.5 Recovering injection and surjection in sets

In **Set**, monic is equivalent to injective. An injective map cancels pointwise. Conversely, if $f(a) = f(b)$, consider the singleton $\{*\}$ with maps $g(*) = a$, $h(*) = b$. Monicity of f implies $g = h$, hence $a = b$.

Similarly, surjections are epic: agreement of two maps after a surjection forces them to agree everywhere. If $f : A \to B$ omits $b \in B$, define two functions $g, h : B \to \{0, 1\}$ that agree on $f(A)$ and disagree only at b. They satisfy $gf = hf$ but $g \neq h$, so f cannot be epic.

The singleton and the two-point set act as *test objects*. This calculation proves the agreement with familiar set theory, rather than assuming that a category can discuss elements.

§10.6 Two ways the analogy can fail

In a poset category all arrows are simultaneously monic and epic: any two parallel arrows are equal automatically, so both cancellation implications hold. Yet $2 \to 6$ in the divisor-poset of 12 is not invertible, because $6 \nmid 2$.

A less trivial example occurs among commutative unital rings: the inclusion

$$\iota : \mathbb{Z} \longrightarrow \mathbb{Q}$$

is epic without being surjective as a function. If ring homomorphisms $g, h : \mathbb{Q} \to R$ agree on $\mathbb{Z}$, they agree on every $a/b \in \mathbb{Q}$. Indeed $g(b) = h(b)$ is a unit, whose inverse is unique, so

$$g(a/b) = g(a)g(b)^{-1} = h(a)h(b)^{-1} = h(a/b).$$

Thus $g = h$. If no such homomorphisms exist, the cancellation condition is vacuous.

The lesson is a mathematical distinction: categorical cancellation is intrinsic to composition; injectivity and surjectivity refer to an underlying set and can change meaning across categories. The reason to study monomorphisms and epimorphisms is precisely that the arrow test still makes sense when the elements do not.

Free Constructions, Adjunctions, Concrete Categories, and Group Actions

N-20230314

§11.1 A free object solves an extension problem

Let S be a set. The free group $F(S)$ consists of reduced words

$$s_1^{\varepsilon_1} s_2^{\varepsilon_2} \cdots s_n^{\varepsilon_n}, \qquad s_i \in S, \quad \varepsilon_i \in \{1, -1\},$$

with multiplication given by concatenation followed by cancellation of adjacent pairs ss^{-1} and $s^{-1}s$. The empty word is the identity.

There is a canonical function

$$\eta_S : S \to U(F(S)), \qquad s \longmapsto [s],$$

where $U : \mathbf{Grp} \to \mathbf{Set}$ forgets the group structure.

The point of $F(S)$ is not merely its word construction. It satisfies a universal property.

Theorem 11.1.1 (Universal property of the free group)
For every group G and every function $f : S \to U(G)$, there is a unique group homomorphism

$$\widetilde{f} : F(S) \to G$$

such that

$$U(\widetilde{f}) \circ \eta_S = f.$$

Proof. Define

$$\widetilde{f}\,(s_1^{\varepsilon_1} \cdots s_n^{\varepsilon_n}) = f(s_1)^{\varepsilon_1} \cdots f(s_n)^{\varepsilon_n}.$$

Cancelling ss^{-1} in a word does not change this value, so the formula is well defined. Concatenation becomes multiplication, hence $\widetilde{f}$ is a homomorphism. Any homomorphism agreeing with f on the generators must take every word to the displayed product, proving uniqueness. □

The free vector space has the same pattern. For a set S, let

$$k^{(S)} = \{a : S \to k : \operatorname{supp}(a) \text{ finite}\}.$$

A function $S \to U(V)$ extends uniquely to a linear map $k^{(S)} \to V$. The construction changes, but the universal problem is identical.

§11.2 The universal property is a natural Hom-set bijection

The extension theorem gives a bijection

$$\Phi_{S,G} : \operatorname{Hom}_{\mathbf{Grp}}(F(S), G) \longrightarrow \operatorname{Hom}_{\mathbf{Set}}(S, U(G)),$$

with

$$\Phi_{S,G}(\varphi) = U(\varphi) \circ \eta_S.$$

Its inverse sends a function $f : S \to U(G)$ to its unique extension $\widetilde{f}$.

This bijection is natural in both variables. If $a : S' \to S$, then

$$\Phi_{S',G}(\varphi \circ F(a)) = \Phi_{S,G}(\varphi) \circ a.$$

Indeed, both functions send s' to $\varphi([a(s')])$.

If $b : G \to H$ is a group homomorphism, then

$$\Phi_{S,H}(b \circ \varphi) = U(b) \circ \Phi_{S,G}(\varphi).$$

Thus the family $\Phi_{S,G}$ is not a collection of accidental bijections. It is a natural isomorphism

$$\boxed{\mathrm{Hom}_{\mathbf{Grp}}(F(-), -) \cong \mathrm{Hom}_{\mathbf{Set}}(-, U(-)).}$$

This is the adjunction

$$F \dashv U.$$

§11.3 Unit and counit encode the same adjunction

The unit of the adjunction is the natural transformation

$$\eta : \mathrm{Id}_{\mathbf{Set}} \Rightarrow UF,$$

whose component η_S inserts each element as a one-letter word.

The counit is

$$\varepsilon : FU \Rightarrow \mathrm{Id}_{\mathbf{Grp}}.$$

For a group G,

$$\varepsilon_G : F(U(G)) \to G$$

evaluates a word in the underlying elements of G:

$$\varepsilon_G(g_1^{\varepsilon_1} \cdots g_n^{\varepsilon_n}) = g_1^{\varepsilon_1} \cdots g_n^{\varepsilon_n} \quad \text{in } G.$$

The triangle identities are

$$\varepsilon_{F(S)} \circ F(\eta_S) = \mathrm{id}_{F(S)},$$

and

$$U(\varepsilon_G) \circ \eta_{U(G)} = \mathrm{id}_{U(G)}.$$

The second identity sends $g \in G$ to the one-letter word $[g]$, then evaluates it back to g. For the first, a generator $[s] \in F(S)$ is sent by $F(\eta_S)$ to the one-letter word whose letter is itself the word $[s]$; evaluation removes the outer layer. Since both sides are homomorphisms and agree on generators, they agree everywhere.

The Hom-set bijection and the unit–counit description determine one another. For example, given $f : S \to U(G)$, its adjunct is

$$F(S) \xrightarrow{F(f)} F(U(G)) \xrightarrow{\varepsilon_G} G.$$

§11.4 Full and faithful describe maps on arrows, not objects

For a functor $T : \mathcal{C} \to \mathcal{D}$, each pair X, Y gives

$$T_{X,Y} : \mathrm{Hom}_{\mathcal{C}}(X, Y) \to \mathrm{Hom}_{\mathcal{D}}(T(X), T(Y)).$$

The functor is faithful if every $T_{X,Y}$ is injective, full if every $T_{X,Y}$ is surjective, and fully faithful if both hold.

The forgetful functor

$$U : \mathbf{Grp} \to \mathbf{Set}$$

is faithful: two group homomorphisms with the same underlying function are equal. It is not full: an arbitrary function between underlying sets need not preserve multiplication.

The inclusion

$$I : \mathbf{Ab} \hookrightarrow \mathbf{Grp}$$

is fully faithful. A homomorphism between abelian groups is already a morphism in **Ab**, so no arrows are lost or added.

Fullness and faithfulness do not assert that every object of $\mathcal{D}$ is in the image. That additional condition, up to isomorphism, is essential surjectivity. A fully faithful and essentially surjective functor is an equivalence of categories.

§11.5 Concrete Categories and Faithful Functors

A concrete category is a pair

$$(\mathcal{C}, U), \qquad U : \mathcal{C} \to \mathbf{Set} \text{ faithful.}$$

Groups, rings, vector spaces, and topological spaces carry familiar choices of U. Faithfulness ensures that morphisms can be distinguished by their underlying functions.

The functor U is part of the structure. Saying that an object "has underlying elements" is not enough to specify how arrows are represented as functions. A category may admit several inequivalent concrete realizations, and a categorical construction should not silently depend on one unless it is named.

For **Top**, the usual forgetful functor is faithful but not full: many set maps are not continuous. The same pattern appears in **Grp**. Concreteness records access to elements; it does not say that all set-theoretic functions are allowed morphisms.

§11.6 A group is a one-object category

Given a group G, form the category BG with one object $*$ and

$$\mathrm{Hom}_{BG}(*, *) = G.$$

Composition is group multiplication, and every arrow is invertible.

A functor

$$X : BG \to \mathbf{Set}$$

chooses a set $X(*)$, written simply X, and for each $g \in G$ a function

$$X(g) : X \to X.$$

Functoriality says

$$X(e) = \mathrm{id}_X, \qquad X(gh) = X(g) \circ X(h).$$

With the convention

$$g \cdot x = X(g)(x),$$

these are exactly the axioms of a left G-action:

$$e \cdot x = x, \qquad (gh) \cdot x = g \cdot (h \cdot x).$$

Therefore

$$\boxed{\mathrm{Fun}(BG, \mathbf{Set}) \cong G\text{-}\mathbf{Set}.}$$

A functor $BG \to BH$ is exactly a group homomorphism $G \to H$, because there is only one possible object map and functoriality is preservation of multiplication and identity.

§11.7 Natural transformations are equivariant maps

Let $X, Y : BG \to \mathbf{Set}$ be two actions. A natural transformation

$$\alpha : X \Rightarrow Y$$

has one component

$$\alpha_* : X \to Y.$$

For each $g \in G$, naturality is the commutative square

$$\begin{array}{ccc} X & \xrightarrow{X(g)} & X \\ {\scriptstyle \alpha_*}\downarrow & & \downarrow{\scriptstyle \alpha_*} \\ Y & \xrightarrow[Y(g)]{} & Y. \end{array}$$

Elementwise,

$$\alpha_*(g \cdot x) = g \cdot \alpha_*(x).$$

Thus natural transformations between action functors are exactly equivariant maps. The category G-$\mathbf{Set}$ is therefore not merely analogous to a functor category; it is one.

The same idea gives representations. Functors

$$BG \to \mathbf{Vect}_k$$

are linear representations of G, and natural transformations are intertwining linear maps.

§11.8 From Adjunctions to Monads

From $F \dashv U$ one obtains the endofunctor

$$T = UF : \mathbf{Set} \to \mathbf{Set}.$$

The unit is $\eta : \mathrm{Id} \Rightarrow T$. The multiplication

$$\mu = U\varepsilon_F : T^2 \Rightarrow T$$

flattens a word whose letters are themselves words into one word.

The free-monoid version is especially transparent. Let

$$T(S) = \coprod_{n \geq 0} S^n$$

be the set of finite lists in S. Then

$$\eta_S(s) = [s],$$

and

$$\mu_S([[s_{11}, \ldots, s_{1k_1}], \ldots, [s_{m1}, \ldots, s_{mk_m}]])$$

is the single concatenated list

$$[s_{11}, \ldots, s_{1k_1}, \ldots, s_{m1}, \ldots, s_{mk_m}].$$

The monad laws say that inserting singleton brackets and then flattening does nothing, and that flattening three nested levels is independent of the order of flattening. These are the unit and associativity laws of substitution.

Free constructions, group actions, and monads are therefore connected by one recurring mechanism:

$$\text{universal extension} \longleftrightarrow \text{natural Hom-bijection} \longleftrightarrow \text{unit/counit and substitution.}$$

The abstraction is useful because it records not only what objects are built, but why every map out of them is forced.

Foundations of Mathematics: Set Theory, Category Theory, and Type Theory

N-20230710

§12.1 The question of foundations

Three major foundational languages—set theory, category theory, and type theory—answer the same basic questions in different ways. The aim is not to declare a single winner, but to compare their answers:

(i) What are mathematical objects?

(ii) What counts as equality?

(iii) What kind of reasoning is allowed?

(iv) How does computation enter mathematics?

(v) How does one organize structures and transformations between them?

Set theory emphasizes membership; category theory emphasizes morphisms and universal properties; type theory emphasizes construction, judgment, and computation. The most interesting modern developments, especially univalent foundations and homotopy type theory, no longer keep these worlds separate.

§12.2 Set theory

§12.2.i From naive comprehension to axioms

Modern set theory begins with the intuition that mathematical objects can be collected into sets. In naive set theory, one might try to form

$$\{x \mid P(x)\}$$

for any property P. Russell's paradox shows that unrestricted comprehension is impossible: the class

$$R = \{x \mid x \notin x\}$$

leads to contradiction if it is treated as a set.

The response is axiomatization. Zermelo–Fraenkel set theory with Choice, ZFC, restricts set formation and supplies a controlled universe in which ordinary mathematics can be developed.

§12.2.ii ZFC as the standard background

The familiar axioms include extensionality, pairing, union, power set, infinity, replacement, foundation, separation, and choice. Their roles may be summarized as follows:

Extensionality. A set is determined by its elements.

Pairing, union, power set. These build new sets from old ones.

Infinity. This guarantees an infinite set, usually giving the natural numbers.

Replacement. Definable images of sets are sets.

Foundation. Membership is well-founded and circular membership is excluded.

Choice. Products of nonempty families are nonempty; equivalently, every surjection has a right inverse.

Remark 12.2.1 — The strength of ZFC is not that it matches everyday mathematical language perfectly. Its strength is that it gives a stable formal universe large enough for most mathematics and disciplined enough to avoid naive paradoxes.

§12.2.iii Applications and limitations

Set theory supports analysis, algebra, topology, measure theory, model theory, and essentially all ordinary mathematical constructions. It also enables transfinite methods: ordinals, cardinals, large cardinals, forcing, independence, and descriptive set theory.

But ZFC has philosophical and practical limitations. It represents everything as sets, even when this representation feels artificial. A group is not experienced as a nested set; it is experienced as a structure with operations. A topological space is not merely a set of ordered pairs defining open sets; it is a space with continuous maps. This motivates category theory.

§12.3 Category theory

§12.3.i From objects to morphisms

Category theory changes the focus from internal membership to external relations. A category consists of objects and morphisms, with associative composition and identities. In this language, mathematical structures are understood by the maps they admit.

Definition 12.3.1. A category $\mathcal{C}$ consists of objects, morphism sets $\mathcal{C}(X, Y)$, identity morphisms id_X, and composition

$$\mathcal{C}(Y, Z) \times \mathcal{C}(X, Y) \to \mathcal{C}(X, Z),$$

satisfying associativity and identity laws.

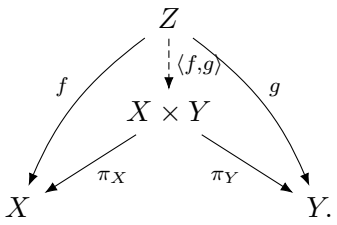
Example 12.3.2

Set, **Grp**, **Top**, **Ring**, and $\mathbf{Vect}_k$ are standard categories. Each has a different notion of structure-preserving map.

§12.3.ii Universal properties

The central categorical idea is the universal property. Products, coproducts, pullbacks, pushouts, quotients, tensor products, free objects, and completions can all be characterized by mapping properties.

For example, a product $X \times Y$ is characterized by



This diagram says that giving a map into a product is the same as giving compatible maps into its two factors.

> **Remark 12.3.3 —** Universal properties explain why the same construction appears in many different subjects. They reveal that the construction is not about the elements of a particular model, but about a structural role.

§12.3.iii Functors and natural transformations

Definition 12.3.4. A functor $F : \mathcal{C} \to \mathcal{D}$ sends objects to objects and morphisms to morphisms, preserving identities and composition.

Definition 12.3.5. A natural transformation $\eta : F \Rightarrow G$ between functors $F, G : \mathcal{C} \to \mathcal{D}$ assigns to each object X a morphism $\eta_X : F(X) \to G(X)$ such that for every $f : X \to Y$,

$$\begin{array}{ccc} F(X) & \xrightarrow{F(f)} & F(Y) \\ {\scriptstyle \eta_X}\downarrow & & \downarrow{\scriptstyle \eta_Y} \\ G(X) & \xrightarrow[G(f)]{} & G(Y) \end{array}$$

commutes.

This square is one of the simplest and most important categorical diagrams. It encodes the idea that a construction is canonical, not dependent on arbitrary choices.

§12.3.iv Categorical foundations

Category theory can function as a language for organizing mathematics, but it can also become foundational. ETCS axiomatizes sets categorically. Topos theory generalizes set-like universes. Higher category theory and infinity-categories extend the categorical language to homotopical and derived contexts.

> **Remark 12.3.6 —** Category theory does not merely replace set theory. It clarifies invariance. It asks not only what an object is made of, but how it transforms and how it participates in universal constructions.

§12.4 Type theory

§12.4.i Judgments and types

Type theory begins from judgments such as

$$a : A,$$

meaning that a is a term of type A. Instead of putting every object into one universe of sets, type theory stratifies objects by the types they inhabit.

Remark 12.4.1 — The phrase "$a \in A$" in set theory and the judgment "$a : A$" in type theory look similar but behave differently. Membership is a proposition; typing is a judgment of well-formedness.

§12.4.ii Dependent types

Dependent type theory allows types to depend on terms. If A is a type and $B(x)$ is a type depending on $x : A$, then one can form:

$$\prod_{x:A} B(x), \qquad \sum_{x:A} B(x).$$

The product type corresponds to universal quantification, while the sum type corresponds to existential quantification.

§12.4.iii Curry–Howard

The Curry–Howard correspondence identifies propositions with types and proofs with terms:

$$\text{proposition} \leftrightarrow \text{type}, \qquad \text{proof} \leftrightarrow \text{term}.$$

For example,

$$A \times B$$

corresponds to conjunction, while

$$A \to B$$

corresponds to implication.

This makes type theory naturally computational. A proof is not merely a static certificate; it may carry an algorithm.

§12.4.iv Constructivity and computation

Classical set theory freely uses excluded middle and choice. Type theory is often constructive: to prove existence, one should construct a witness. This is not only a philosophical choice. It is what makes type theory suitable for proof assistants such as Coq, Lean, Agda, and Isabelle/HOL.

§12.5 Homotopy type theory and univalence

Homotopy type theory interprets types as spaces, terms as points, and identity proofs as paths. Higher identity proofs become higher homotopies. In this interpretation, equality is no longer merely a binary relation; it has structure.

The univalence axiom says, roughly, that equivalent structures may be identified:

$$(A = B) \simeq (A \simeq B).$$

This expresses a structuralist principle: isomorphic or equivalent objects should be interchangeable in mathematics.

Remark 12.5.1 — Univalence is one of the most striking attempts to synthesize set-theoretic rigor, categorical invariance, and type-theoretic computation. It turns the informal mathematical habit of identifying isomorphic objects into a formal principle.

§12.6 Comparative analysis

Framework	Primitive emphasis	Equality	Strength
Set theory	membership	extensional equality	universal coding power
Category theory	morphisms	isomorphism/ equivalence	structural invariance
Type theory	judgments/terms	identity types	computation and construction

Set theory is excellent as a common foundational background. Category theory is excellent for organizing structures and transformations. Type theory is excellent when proofs, programs, and constructions are meant to interact.

§12.7 Modern applications

§12.7.i Proof assistants

Proof assistants make foundations operational. Lean's mathlib, Coq's constructive type theory, Agda's dependent types, and Isabelle's higher-order logic show that foundational choices affect how mathematics is formalized.

§12.7.ii Computer science

Type theory underlies programming language theory, dependent types, formal verification, compiler correctness, and proof-carrying code. Category theory appears in semantics, monads, functional programming, and compositional systems. Set theory remains important in logic, model theory, semantics, and databases.

§12.7.iii Geometry and higher structures

Modern geometry often uses categorical and type-theoretic ideas: stacks, sheaves, derived categories, infinity-categories, and higher topos theory. Foundations are no longer merely a safety net beneath mathematics; they become active tools inside research.

§12.8 Two points expose three notions of sameness

Consider two sets

$$A = \{\text{red}, \text{blue}\}, \qquad B = \{0, 1\}.$$

Set-theoretically they are not equal: extensionality says that sets are equal precisely when their members coincide. Yet there are exactly two bijections $A \to B$:

	red	blue
f	0	1
g	1	0

The fact that the sets have the same cardinality does not tell us which labeling to use. That freedom is information, not an annoying technicality.

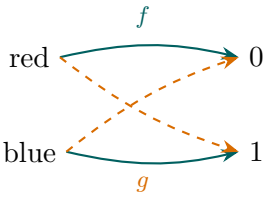


Figure 12.1: Two equally valid bijections. Literal equality, existence of an isomorphism, and choice of a particular isomorphism are different assertions.

Category theory keeps both bijections as distinct morphisms. If one set is viewed as a pointed set, with its first element designated, then morphisms must preserve that point; only one of the two bijections remains. What counts as an isomorphism depends on the structure one has asked the morphisms to respect.

§12.9 Why a product has no favorite encoding

In ZFC an ordered pair can be encoded by Kuratowski's formula

$$(a, b) = \{\{a\}, \{a, b\}\}.$$

Other encodings are possible. A categorical product instead has a role: for every pair of arrows $f : Z \to X$ and $g : Z \to Y$, there is exactly one arrow $\langle f, g\rangle : Z \to X \times Y$ whose projections are f and g.

The uniqueness clause has a concrete consequence. Suppose (P, p_X, p_Y) and (Q, q_X, q_Y) both satisfy the product property for X, Y. It supplies unique maps

$$u : P \to Q, \qquad v : Q \to P$$

such that

$$q_X u = p_X, \quad q_Y u = p_Y, \qquad p_X v = q_X, \quad p_Y v = q_Y.$$

Then vu and 1_P have the same composites with both projections of P. Since the defining property gives a *unique* map with these composites, $vu = 1_P$. Likewise $uv = 1_Q$. Thus the two products are isomorphic by a uniquely determined *projection-preserving* isomorphism.[1]

Without the qualifier about projections, the underlying product object may have other automorphisms. A universal property is powerful precisely because it tells us which comparisons are canonical once the required structure is fixed. This proof never had to inspect the set-theoretic implementation of an ordered pair.

§12.10 A dependent type is an executable promise

Now use the two-element type

$$\mathsf{Bool} = \{\mathsf{false}, \mathsf{true}\}.$$

[1] For this exact proof, see *Five Lectures on Category Theory*, Proposition 1.5.

An inhabitant of the dependent product

$$\prod_{b:\mathsf{Bool}} \mathbb{N}$$

must supply a natural number for *every* Boolean input. Evaluating such a function at the two inputs yields the map

$$h \longmapsto (h(\mathsf{false}), h(\mathsf{true})).$$

Conversely, a pair (m, n) defines

$$h_{m,n}(b) = \begin{cases} m, & b = \mathsf{false}, \\ n, & b = \mathsf{true}. \end{cases}$$

The two constructions are inverse up to function extensionality, so

$$\boxed{\left(\prod_{b:\mathsf{Bool}} \mathbb{N}\right) \simeq \mathbb{N} \times \mathbb{N}.}$$

This is the computational content of a universal statement: a proof that supplies an answer for each input acts as a program.

By contrast,

$$\sum_{b:\mathsf{Bool}} \mathbb{N} \simeq \mathbb{N} \sqcup \mathbb{N}$$

records *one chosen branch* and its number. Both fibers happen to be $\mathbb{N}$, yet choosing Π or Σ makes a decisive difference. The propositions-as-types interpretation similarly distinguishes a dependent sum $\sum_{x:A} P(x)$, containing a witness and evidence for $P(x)$, from its *propositional truncation*, which asserts mere existence without exposing a particular witness. This distinction is essential if one hopes to extract an algorithm from a proof.

§12.11 Univalence preserves the symmetries of an identity

The Boolean type has precisely two self-equivalences:

$$\mathrm{id} : \mathsf{Bool} \simeq \mathsf{Bool}, \qquad \mathrm{swap} : \mathsf{Bool} \simeq \mathsf{Bool}.$$

Intensional type theory represents an identification of two types A, B in a universe $\mathcal{U}$ by the identity type $(A =_{\mathcal{U}} B)$. Every such identification induces an equivalence of types:

$$(A =_{\mathcal{U}} B) \longrightarrow \mathsf{Equiv}(A, B).$$

The univalence axiom says that this canonical map is an equivalence. In a univalent universe,

$$\boxed{(A =_{\mathcal{U}} B) \simeq \mathsf{Equiv}(A, B).}$$

Setting $A = B = \mathsf{Bool}$ makes the meaning surprisingly tangible:

$$(\mathsf{Bool} =_{\mathcal{U}} \mathsf{Bool}) \simeq \mathsf{Equiv}(\mathsf{Bool}, \mathsf{Bool}).$$

There are two distinguishable elements on the right, identity and swap; hence the *universe-level identity type* of Bool with itself has two distinct paths under univalence.

The nontrivial automorphism becomes a way of identifying the type with itself; it does not disappear into one anonymous equality.[2]

The qualification matters twice. This is an identity *of types in a universe*, not a claim that $\mathsf{true} = \mathsf{false}$ inside Bool. And an arbitrary bijection between underlying sets need not be an isomorphism of groups, topological spaces, or ordered sets: preserving the extra structure is part of the type of equivalence one is discussing.

The same little two-point calculation has now exhibited three ways to ask a precise question. Set theory distinguishes membership and equality of sets; category theory distinguishes maps and universal characterizations; dependent type theory turns the existence of data into an explicit construction. Univalence then makes equivalences visible as possible identifications, including their nontrivial symmetries.

[2]See the Univalent Foundations Program, *Homotopy Type Theory: Univalent Foundations of Mathematics*, Section 2.10.

What Is a Topos? From Local Data to a Universe of Variable Sets

N-20240211

Imagine measuring a physical field over a region. One instrument knows what happens on the left, another on the right, and neither has the whole picture. If they agree wherever their coverage overlaps, can we regard their results as one measurement? That is natural for continuous functions, but the same question becomes more subtle for solutions of equations, sections of coverings, or the truth of statements about a changing system.

Sheaf theory organizes local observations into global objects. Topos theory goes further: it finds categories of local data in which we can perform many of the constructions ordinarily performed with sets, functions and logical predicates. The surprise is that truth can depend on an open region, and some classical logical principles need not hold internally. On an open region, a claim may hold locally even when it does not hold globally; restriction maps determine precisely how such local truths fit together.[1]

§13.1 The Locality of Sheaf Elements

Let X be a topological space and $U \subseteq X$ open. The continuous real-valued functions on U form a set

$$\mathcal{C}(U) = \{f : U \to \mathbb{R} : f \text{ is continuous}\}.$$

If $V \subseteq U$, we can restrict a function:

$$\rho_V^U : \mathcal{C}(U) \to \mathcal{C}(V), \qquad f \mapsto f|_V.$$

Restriction obeys two laws:

$$\rho_U^U = \mathrm{id}, \qquad \rho_W^V \circ \rho_V^U = \rho_W^U \quad (W \subseteq V \subseteq U).$$

Keeping only $\mathcal{C}(X)$ would discard local functions that cannot be extended to all of X. We therefore regard the family of sets $\mathcal{C}(U)$, together with its restriction maps, as a single object.

A *presheaf* is an assignment $U \mapsto F(U)$ with exactly these restriction laws. Formally, open sets form a category with an arrow $V \to U$ whenever $V \subseteq U$; the direction of the restriction maps is reversed:

$$\boxed{F : \mathrm{Open}(X)^{\mathrm{op}} \longrightarrow \mathbf{Set}.}$$

An element of $F(U)$ is a *section over* U. It need not be an ordinary function. It can represent a solution to an equation, a choice of coordinates, or a possible state of a system on that region.

[1] For these constructions see Mac Lane and Moerdijk, *Sheaves in Geometry and Logic*, and Fong and Spivak, *Seven Sketches in Compositionality*, Chapter 7.

§13.2 Compatible local descriptions should glue

Suppose $U = U_1 \cup U_2$. Two continuous functions $f_1 \in \mathcal{C}(U_1)$ and $f_2 \in \mathcal{C}(U_2)$ satisfying $f_1 = f_2$ on $U_1 \cap U_2$ define

$$f(x) = \begin{cases} f_1(x), & x \in U_1, \\ f_2(x), & x \in U_2. \end{cases}$$

The overlap condition makes the definition independent of which local formula we choose. It is continuous, and it is the unique function whose restrictions are f_1, f_2.

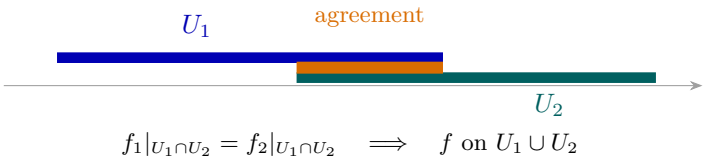


Figure 13.1: The equality on the overlap is precisely what makes one global function from two local functions.

A *sheaf* is a presheaf for which this construction works on every open cover $U = \bigcup_i U_i$. Whenever $s_i \in F(U_i)$ agree on all intersections $U_i \cap U_j$, there exists one and only one $s \in F(U)$ restricting to every s_i. The existence clause and the uniqueness clause are both essential.

Consider a presheaf with

$$F(U) = \{\text{red}, \text{blue}\} \quad (U \neq \varnothing), \qquad F(\varnothing) = \{*\},$$

with the obvious constant restrictions on nonempty opens. Choose two disjoint nonempty intervals U_1, U_2. Red on U_1 and blue on U_2 agree vacuously on the empty overlap, but neither of the two globally constant options in $F(U_1 \cup U_2)$ restricts to both. This presheaf is not a sheaf.

The corresponding sheaf consists of *locally constant functions* $U \to \{\text{red}, \text{blue}\}$. It permits red on one connected component and blue on another. The point of sheafification in this elementary example is to recognize combinations of local choices that the topology really allows.

§13.3 Square Roots around the Circle

Let $X = S^1 \subset \mathbb{C}$. On every open arc U, consider the set

$$\mathcal{R}(U) = \{w : U \to S^1 \text{ continuous} : w(z)^2 = z \text{ for all } z \in U\}.$$

These local roots restrict to smaller arcs and glue uniquely whenever they agree. Thus $\mathcal{R}$ is a sheaf.

Locally there are two roots, but there is no global continuous root $w : S^1 \to S^1$ satisfying $w(z)^2 = z$. If it existed, the degree of circle maps would obey

$$1 = \deg(\mathrm{id}_{S^1}) = \deg(w^2) = 2\deg(w),$$

an impossibility since degrees are integers.

The obstruction is easy to visualize. Write $z = e^{it}$. A local choice is $w = e^{it/2}$. As t makes one turn from 0 to 2π, the root moves from 1 to -1. Its sign is exchanged, so the starting choice fails to close up.

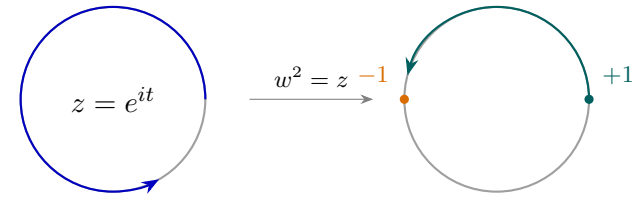


Figure 13.2: One circuit of the base exchanges the two local square roots. Local solvability does not imply the existence of a global choice.

There is an actual geometric object behind this example:

$$E = \{(z, w) \in S^1 \times S^1 : w^2 = z\}, \qquad E \longrightarrow S^1, \quad (z, w) \longmapsto z.$$

This is a two-sheeted covering. The sheaf $\mathcal{R}$ is its sheaf of local sections. Calling a sheaf a set varying over a space now has a concrete meaning: every small region has a set of available choices, but the choices can twist when transported around loops.

§13.4 From subsets to characteristic maps

In **Set**, a subset $A \subseteq S$ can be recorded as a characteristic function

$$\chi_A : S \to \{0, 1\}.$$

Let $1 = \{*\}$ and let $\text{true} : 1 \to \{0, 1\}$ select the value 1. The defining property of χ_A is expressed by the pullback square

$$\begin{array}{ccc} A & \longrightarrow & 1 \\ \downarrow & & \downarrow{\scriptstyle \text{true}} \\ S & \xrightarrow[\chi_A]{} & \{0, 1\}. \end{array}$$

A pullback is the categorical version of imposing two compatible constraints: here it selects the elements of S for which the characteristic function is true.

An injective map into S, considered up to isomorphism over S, is a *subobject*. The pair consisting of $\Omega = \{0, 1\}$ and the truth map is a *subobject classifier*: every subobject arises as the pullback of truth along a unique characteristic map.

This description uses arrows and universal properties rather than elements. So it can be repeated in a sheaf category, even though elements there are locally defined sections rather than points.

§13.5 Open Sets as Truth Values

In $\mathbf{Sh}(X)$, the subobject classifier Ω has a remarkably simple form:

$$\boxed{\Omega(U) = \{V \subseteq U : V \text{ is open in } X\}.}$$

Restrict a truth value $V \subseteq U$ to $W \subseteq U$ by $V \mapsto V \cap W$. Complete truth on U is the whole open set U; complete falsehood is $\varnothing$.

Here is why this classifier works. Given a subsheaf $G \hookrightarrow F$ and $s \in F(U)$, form the largest region on which s belongs to G:

$$\boxed{\chi_{G,U}(s) = \bigcup \{V \subseteq U \text{ open} : s|_V \in G(V)\}.}$$

The sheaf condition for G guarantees that s restricted to this union is genuinely a section of G. Restriction to a smaller open commutes with taking this truth region, so the maps define a morphism $\chi_G : F \to \Omega$. And

$$s \in G(U) \quad \Longleftrightarrow \quad \chi_{G,U}(s) = U.$$

That is exactly the pullback-of-truth condition familiar from characteristic functions of subsets.

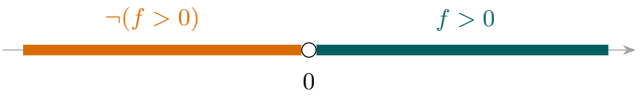
Take continuous real functions $\mathcal{C}(U)$, and let

$$\mathcal{P}(U) = \{f \in \mathcal{C}(U) : f(x) > 0 \text{ for all } x \in U\}.$$

The characteristic morphism of this subsheaf has a very concrete value:

$$\boxed{\chi_{\mathcal{P},U}(f) = \{x \in U : f(x) > 0\}.}$$

The positivity region is open because f is continuous. For the function $f(x) = x$, the predicate $f > 0$ is true on $(0, \infty)$, while its negation is true on $(-\infty, 0)$. Neither one is true throughout an open neighborhood of 0.



no open neighborhood of 0 lies wholly in either region

Figure 13.3: Truth over a sheaf is local. The boundary of a positivity region is not covered by either the predicate or its open negation.

§13.6 Why the internal logic is intuitionistic

Global truth values are elements of $\Omega(X)$, namely open subsets of X. They form a Heyting algebra:

$$\begin{aligned} U \wedge V &= U \cap V, \\ U \vee V &= U \cup V, \\ U \Rightarrow V &= \operatorname{int}((X \setminus U) \cup V), \\ \neg U &= \operatorname{int}(X \setminus U). \end{aligned}$$

The implication is the largest open W for which $W \cap U \subseteq V$. Indeed, that condition is equivalent to $W \subseteq (X \setminus U) \cup V$, so the largest open solution is its interior.

For $U = (0, \infty) \subseteq \mathbb{R}$,

$$\neg U = (-\infty, 0), \qquad U \vee \neg U = \mathbb{R} \setminus \{0\} \neq \mathbb{R}.$$

The law of excluded middle fails internally because 0 has no open neighborhood contained in either truth region.

There is an even more striking phenomenon:

$$\neg\neg U = \operatorname{int}(\overline{U}).$$

For the dense open set $U = \mathbb{R} \setminus \{0\}$,

$$\neg U = \varnothing, \qquad \neg\neg U = \mathbb{R}.$$

A proposition can be true on a dense open set without being true on the whole space, while its double negation is true everywhere. These calculations explain the intuitionistic logic of sheaves through the elementary topology of open sets. We can still reason classically about the category from outside; internal and external truth are different mathematical levels.

§13.7 Existence inside a sheaf topos is a local assertion

Return to the square-root example on S^1. Let $\mathcal{S}(U)$ be the continuous S^1-valued functions on U, and let z denote the identity function $S^1 \to S^1$. Consider the statement

$$\exists w \in \mathcal{S} : \quad w^2 = z.$$

Externally it is false if interpreted as the existence of one globally defined continuous root. Internally, however, the existential statement is true over the whole circle, because the circle has an open cover on each member of which a continuous root exists.

In the Kripke–Joyal semantics of sheaves, a statement is evaluated at an open stage U, written $U \Vdash \varphi$. The rule

$$U \Vdash \exists w\, \varphi(w)$$

means that there exists an open cover $U = \bigcup_i U_i$ and, on each U_i, a local witness w_i satisfying φ. The witnesses need not match on overlaps; if they did, the sheaf law would glue them globally.

Universal quantification is different. The assertion $U \Vdash \forall w\, \varphi(w)$ requires φ to hold for every section w over *every* smaller open $V \subseteq U$. The logical rules are constructed to respect restriction.

Here a real distinction becomes visible. Local solvability, global solvability and the existence of a globally consistent selection are three questions that should not be conflated. Topos logic lets the first be true while the third remains false.

§13.8 What Defines an Elementary Topos

An *elementary topos* is a category equipped with three types of structure.

First, it has *finite limits*: terminal objects, products, and pullbacks. These encode compatible pairs, intersections of constraints and solutions of equations.

Second, it is *cartesian closed*: for objects A, B, there exists an object B^A representing functions out of A. Precisely,

$$\boxed{\mathrm{Hom}(T \times A, B) \cong \mathrm{Hom}(T, B^A)}$$

naturally in T. This makes function spaces into objects of the same universe.

Third, it has a *subobject classifier* $1 \xrightarrow{\text{true}} \Omega$. This is the categorical mechanism for subsets and predicates.

The category **Set** has all three structures. So does the category of sheaves $\mathbf{Sh}(X)$ for any topological space X. Products of sheaves are calculated sectionwise. Their exponentials are sheaves of locally defined morphisms:

$$(G^F)(U) \cong \mathrm{Hom}_{\mathbf{Sh}(U)}(F|_U, G|_U).$$

Thus a function space in a sheaf topos is itself an object whose available functions vary from region to region.

A one-point topological space has a sheaf category equivalent to **Set**, so ordinary sets are recovered as the simplest example. An elementary topos need not satisfy excluded middle or the axiom of choice; these are additional properties. Its basic axioms also do not require a natural-numbers object, which is an extra assumption when needed.

Grothendieck topoi enlarge the geometric viewpoint: instead of open subsets of a topological space, one uses an abstract *site*, a category supplied with designated covering families. Every Grothendieck topos is an elementary topos, but not every elementary topos must arise from a sheaf construction on an ordinary topological space. The distinction prevents the attractive varying-sets picture from being overinterpreted.

§13.9 A two-stage world already has three truth values

There is a finite example requiring no topology or continuous functions. Consider the category with two objects u, v and exactly one nonidentity arrow

$$a : u \longrightarrow v.$$

A presheaf on this category consists of sets $F(v), F(u)$ and a restriction map $F(v) \to F(u)$. The stage v can be read as a broad observation and u as a more restricted one.

A *sieve* on v is a collection of arrows into v closed under precomposition. Here there are exactly three:

$$\boxed{\varnothing, \qquad \{a\}, \qquad \{\mathrm{id}_v, a\}.}$$

The middle sieve says that a property is verified after restriction along a, but not at the broad stage v. The subobject classifier of the presheaf topos $\mathbf{Set}^{\mathcal{C}^{\mathrm{op}}}$ consists of sieves stage by stage; its value $\Omega(v)$ is precisely this three-element ordered set.

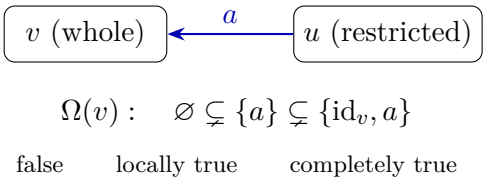


Figure 13.4: Even a category with just two stages supports a non-Boolean local truth value.

The negation of the middle sieve $\{a\}$ is empty: there is no arrow into v along which the sieve becomes empty after pullback. Thus

$$\{a\} \vee \neg\{a\} = \{a\} \neq \{\mathrm{id}_v, a\}.$$

Excluded middle fails in this finite example just as it did for opens of the real line.

For an arbitrary category $\mathcal{C}$, the subobject classifier of its presheaf category assigns to each object the collection of all sieves on that object. The open-set truth values of sheaves and the sieve truth values of presheaves express the same principle: valid information must remain valid when we pass to a more restricted viewpoint.

§13.10 Why the abstraction earns its place

Sheaves allow local models to be combined exactly when their agreement permits it. The double cover of the circle shows that local solutions can carry a global topological obstruction. The subobject classifier turns properties of those local solutions into arrows, just as characteristic functions turn subsets into arrows in ordinary set theory. And the internal logic distinguishes a global witness from a collection of local witnesses.

The same machinery is useful in several mathematical settings. A geometer may consider functions, sections and differential equations over open neighborhoods. A logician can evaluate propositions relative to contexts rather than assume every truth value is decided globally. A computer scientist can represent states or possible behaviors varying across observations or time intervals. A topos provides a common setting in which these objects, their functions and their predicates can be studied coherently.

It is thus helpful to regard a topos as a category with enough of the structural features of sets for ordinary mathematical constructions to make sense internally. The elements, however, may be locally defined, and the truth values may reflect the region or context on which they are asserted. The gluing law, the square-root obstruction and the three-element sieve show those distinctions in computations rather than slogans.

Kernels, Cokernels, Images, and Coimages in Abelian Categories

N-20251109

§14.1 The setting: maps with zero maps

Consider a morphism

$$f : A \to B$$

from the categorical viewpoint. The words domain and codomain mean exactly what they mean for an ordinary function:

$$A = \text{domain}, \qquad B = \text{codomain}.$$

For linear maps one may think of A as the input space and B as the declared output space. The actual values reached by f may form a smaller object inside B; that smaller object is the image.

The categorical definitions below need a zero object and zero morphisms. A zero object is simultaneously initial and terminal. Once a category has a zero object 0, every pair of objects X, Y has a distinguished zero morphism

$$0_{X,Y} : X \to 0 \to Y.$$

This lets equations such as $f \circ k = 0$ make sense without mentioning elements.

The most familiar examples are:

$$\mathbf{Vect}_k, \qquad \mathbf{Ab}, \qquad R\text{-}\mathbf{Mod}.$$

These are abelian categories. In them, kernels, cokernels, images, and coimages behave like the objects from linear algebra and module theory.

§14.2 Kernel: the universal object killed by a map

For a morphism $f : A \to B$, a kernel of f is a morphism

$$k : K \to A$$

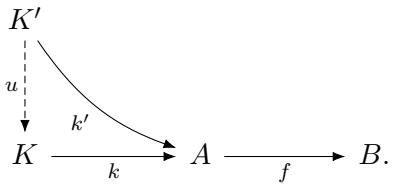
such that

$$f \circ k = 0,$$

and universal with this property: whenever $k' : K' \to A$ also satisfies $f \circ k' = 0$, there is a unique morphism $u : K' \to K$ with

$$k' = k \circ u.$$

In diagram form:



The condition is that both routes from K' to B are zero after applying f.

In vector spaces this recovers

$$\mathrm{Ker}(f) = \{a \in A : f(a) = 0\}.$$

The categorical definition says something stronger than just listing elements: the kernel is the best, largest, and universal way to factor all inputs that become zero.

In an abelian category, the kernel morphism $k : K \to A$ is a monomorphism. It is therefore legitimate to think of K as a subobject of the domain A, though the invariant datum is the arrow k, not merely the object K.

§14.3 Cokernel: the universal quotient killing a map

The cokernel is the dual construction. A cokernel of $f : A \to B$ is a morphism

$$c : B \to C$$

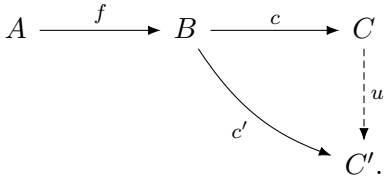
such that

$$c \circ f = 0,$$

and universal with this property: whenever $c' : B \to C'$ satisfies $c' \circ f = 0$, there is a unique morphism $u : C \to C'$ with

$$c' = u \circ c.$$

The diagram is:



In vector spaces or modules,

$$\mathrm{Coker}(f) = B/\mathrm{Im}(f).$$

Thus the cokernel measures what remains in the codomain after the actual output of f has been collapsed to zero.

In an abelian category, the cokernel morphism $c : B \to C$ is an epimorphism. It is therefore the categorical version of a quotient object of the codomain.

§14.4 Image: the part of the codomain actually reached

In linear algebra the image is

$$\mathrm{Im}(f) = \{f(a) : a \in A\} \subseteq B.$$

In an abelian category, one defines the image without element language by first taking the cokernel

$$c : B \to \mathrm{Coker}(f)$$

and then taking the kernel of that cokernel:

$$\mathrm{Im}(f) = \mathrm{Ker}(B \xrightarrow{c} \mathrm{Coker}(f)).$$

So the image is represented by a monomorphism

$$i : \mathrm{Im}(f) \to B.$$

The diagram is:

$$\mathrm{Im}(f) \overset{i}{\rightarrowtail} B \xrightarrow{c} \mathrm{Coker}(f).$$

The equation $c \circ i = 0$ says that everything in the image is killed by the quotient map c. The universal property says that it is the largest subobject of B killed by c.

This gives a useful slogan:

$$\mathrm{Im}(f) = \text{the part of the codomain not visible to the cokernel.}$$

§14.5 Coimage: the effective part of the domain

The coimage is dual to the image. Begin with the kernel

$$k : \mathrm{Ker}(f) \to A.$$

Then define the coimage as the cokernel of the kernel:

$$\mathrm{Coim}(f) = \mathrm{Coker}(\mathrm{Ker}(f) \xrightarrow{k} A).$$

So it is represented by an epimorphism

$$q : A \to \mathrm{Coim}(f).$$

The diagram is:

$$\mathrm{Ker}(f) \overset{k}{\rightarrowtail} A \xrightarrow{q} \mathrm{Coim}(f).$$

In vector spaces this is the quotient

$$\mathrm{Coim}(f) = A/\mathrm{Ker}(f).$$

It is the domain after removing precisely the part that f sends to zero. In that sense, it is the effective input object of f.

§14.6 The standard factorization

In an abelian category, every morphism $f : A \to B$ factors through its coimage and image:

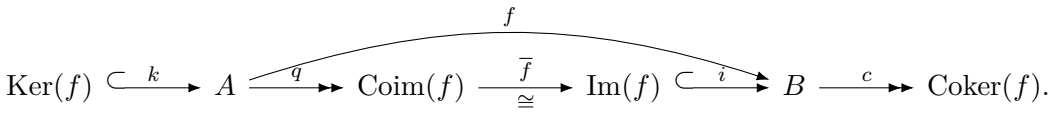
$$A \xrightarrow{q} \mathrm{Coim}(f) \xrightarrow{\overline{f}} \mathrm{Im}(f) \xrightarrow{i} B.$$

Here:

- $q : A \to \mathrm{Coim}(f)$ is the cokernel of $\mathrm{Ker}(f) \to A$, hence an epimorphism;
- $i : \mathrm{Im}(f) \to B$ is the kernel of $B \to \mathrm{Coker}(f)$, hence a monomorphism;
- the middle map $\overline{f} : \mathrm{Coim}(f) \to \mathrm{Im}(f)$ is an isomorphism in an abelian category;
- the original morphism is recovered by

$$f = i \circ \overline{f} \circ q.$$

A compact diagram is:



This is the categorical form of the first isomorphism theorem:

$$A/\operatorname{Ker}(f) \cong \operatorname{Im}(f).$$

The factorization separates the information in a map: q identifies inputs with the same effect, i embeds the actual image into the declared codomain, and $\overline{f}$ is the canonical isomorphism between coimage and image.

§14.7 A vector-space sanity check

Let $T : V \to W$ be a linear map. Then

$$\operatorname{Ker}(T) = \{v \in V : T(v) = 0\}, \qquad \operatorname{Coker}(T) = W/\operatorname{Im}(T).$$

The coimage is

$$\operatorname{Coim}(T) = V/\operatorname{Ker}(T),$$

and the induced map

$$\overline{T} : V/\operatorname{Ker}(T) \to \operatorname{Im}(T), \qquad [v] \mapsto T(v),$$

is an isomorphism. It is well-defined because if $v - v' \in \operatorname{Ker}(T)$, then $T(v) = T(v')$. It is injective because $T(v) = 0$ means $[v] = 0$, and it is surjective by definition of image.

For a matrix $A : k^n \to k^m$, this says that the rank is the dimension of both sides of the middle isomorphism:

$$\dim(k^n/\operatorname{Ker} A) = \dim \operatorname{Im} A = \operatorname{rank} A.$$

This is rank-nullity in categorical clothing.

§14.8 A module example: multiplication by an integer

Consider the group homomorphism

$$f : \mathbb{Z} \to \mathbb{Z}, \qquad f(x) = nx.$$

If $n \neq 0$, then

$$\operatorname{Ker}(f) = 0, \qquad \operatorname{Im}(f) = n\mathbb{Z}, \qquad \operatorname{Coker}(f) = \mathbb{Z}/n\mathbb{Z}.$$

The coimage is

$$\operatorname{Coim}(f) = \mathbb{Z}/0 \cong \mathbb{Z}.$$

The middle isomorphism is

$$\mathbb{Z} \xrightarrow{\sim} n\mathbb{Z}, \qquad x \mapsto nx.$$

This example is useful because the image is literally a subgroup of the codomain, while the coimage is a quotient of the domain. They look different as subobject and quotient, but in an abelian category they are canonically isomorphic through f.

§14.9 Exactness language

Kernel and image are also the grammar of exact sequences. A sequence

$$X \xrightarrow{g} Y \xrightarrow{h} Z$$

is exact at Y when

$$\mathrm{Im}(g) = \mathrm{Ker}(h)$$

as subobjects of Y. The equality means that the elements, vectors, or generalized inputs that h kills are exactly those that came from g.

In categorical language, exactness replaces element chasing by universal properties. For example, saying that

$$\mathrm{Ker}(f) \to A \to B$$

is exact at A is precisely saying that the first arrow is the kernel of f. Saying that

$$A \to B \to \mathrm{Coker}(f)$$

is exact at B is precisely saying that the second arrow is the cokernel of f, after passing through the image.

§14.10 Warnings: what depends on being abelian

The statements above should not be read as valid in every category. Several layers are being used.

- A category must have a zero object and zero morphisms before kernels and cokernels can even be formulated in this form.
- A category may have kernels and cokernels but still fail to make every monomorphism a kernel or every epimorphism a cokernel.
- The canonical map $\mathrm{Coim}(f) \to \mathrm{Im}(f)$ need not be an isomorphism outside abelian categories.

Thus the beautiful factorization

$$\mathrm{Coim}(f) \cong \mathrm{Im}(f)$$

is not a formal tautology. It is one of the defining strengths of abelian categories.

For intuition, **Set** is not the right environment for this theory: it has no zero object in the needed sense. Pointed sets have a zero object and kernels in a pointed sense, but they are not abelian. Vector spaces, abelian groups, and modules are the safe models.

§14.11 From Elements to Universal Properties

The definitions fit into the following table:

object	categorical construction	linear algebra model	where it lives
$\mathrm{Ker}(f)$	universal map into A killed by f	$\{a : f(a) = 0\}$	subobject of A
$\mathrm{Coker}(f)$	universal quotient of B killing f	$B/\mathrm{Im}(f)$	quotient of B
$\mathrm{Im}(f)$	$\mathrm{Ker}(B \to \mathrm{Coker}(f))$	$\{f(a) : a \in A\}$	subobject of B
$\mathrm{Coim}(f)$	$\mathrm{Coker}(\mathrm{Ker}(f) \to A)$	$A/\mathrm{Ker}(f)$	quotient of A

The standard factorization

$$A \twoheadrightarrow \mathrm{Coim}(f) \xrightarrow{\sim} \mathrm{Im}(f) \hookrightarrow B$$

turns a morphism into three conceptual pieces: quotient out the part of the input killed by f, identify the remaining effective input with the subobject actually reached in B, and then include that subobject in the codomain.

For vector spaces this says exactly the familiar rank–nullity picture: $A/\ker f$ is isomorphic to $\mathrm{im}\, f$. For modules and abelian groups the same statement is the first isomorphism theorem. The categorical formulation is useful because it records which part of the argument is universal and which part depends on being in an abelian category: the comparison map $\mathrm{Coim}(f) \to \mathrm{Im}(f)$ is an isomorphism in the safe models listed above, but it should not be assumed in an arbitrary category with kernels and cokernels.

II

Number Theory

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Polygon Motions and Roots of Unity: Modular Hops, Chords, and Vanishing Sums

N-20220723

§15.1 The ant on a decagon is a modular-arithmetic problem

Label the vertices of a regular decagon by

$$0, 1, \ldots, 9$$

in counterclockwise order. Moving forward by k vertices sends a position j to

$$j + k \pmod{10}.$$

Only the residue class of the total displacement matters.

Suppose an ant makes ten jumps of length 1, then ten jumps of length 2, and continues in this way through ten jumps of length 10. One block of ten jumps of length k changes the vertex label by

$$10k \equiv 0 \pmod{10}.$$

Therefore every block returns the ant to its starting vertex. If it begins at vertex 1, it finishes at vertex 1.

The point is not specific to ten sides.

Proposition 15.1.1 (A full block on a regular polygon)
On a regular N-gon, any block of N equal jumps returns to the starting vertex.

Proof. If each jump advances by k vertices, the total displacement is Nk, and

$$Nk \equiv 0 \pmod{N}.$$

□

For an arbitrary list of jump lengths $k_1, \ldots, k_m$, the final vertex is

$$j_0 + k_1 + \cdots + k_m \pmod{N}.$$

A geometric hopping problem has become addition in the cyclic group $\mathbb{Z}/N\mathbb{Z}$.

§15.2 The greatest common divisor determines the orbit

Repeated jumps of a fixed length k produce the orbit

$$j, \quad j + k, \quad j + 2k, \quad \ldots \pmod{N}.$$

The ant returns after the least positive integer t satisfying

$$tk \equiv 0 \pmod{N}.$$

Write

$$d = \gcd(N, k), \qquad N = dN_0, \qquad k = dk_0,$$

with $\gcd(N_0, k_0) = 1$. The condition $N \mid tk$ becomes

$$N_0 \mid tk_0.$$

Since k_0 is invertible modulo N_0, the smallest possible t is N_0. Hence the orbit length is

$$\boxed{\frac{N}{\gcd(N, k)}.}$$

On a decagon, jumps of length 4 visit

$$0, 4, 8, 2, 6, 0.$$

There are five distinct vertices because

$$\frac{10}{\gcd(10, 4)} = 5.$$

Jumps of length 3 visit all ten vertices because 3 is coprime to 10.

Thus a fixed jump length visits every vertex exactly when

$$\gcd(N, k) = 1.$$

This is the first appearance of a general principle: multiplication by k permutes residue classes modulo N exactly when k is invertible modulo N.

§15.3 Roots of Unity as Polygon Vertices

Let

$$\zeta_N = e^{2\pi i/N}.$$

The N-th roots of unity are

$$\mu_N = \{1, \zeta_N, \zeta_N^2, \ldots, \zeta_N^{N-1}\}.$$

They satisfy

$$z^N = 1$$

and form the vertices of a regular N-gon on the unit circle.

Vertex j is represented by ζ_N^j. A jump of k vertices is multiplication by

$$\zeta_N^k : \qquad \zeta_N^j \longmapsto \zeta_N^{j+k}.$$

After m such jumps, the point is

$$\zeta_N^{j+mk}.$$

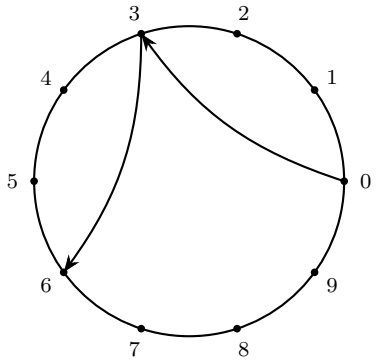
Returning to the starting point means

$$\zeta_N^{mk} = 1,$$

which is the same congruence

$$mk \equiv 0 \pmod{N}.$$

The complex and modular descriptions are two coordinate systems for the same cyclic motion.



Multiplication by ζ_{10}^{3} advances three vertices.

Figure 15.1: A regular decagon encoded by the tenth roots of unity.

§15.4 Power maps have a visible kernel and image

For an integer m, consider the power map

$$p_m : \mu_N \to \mu_N, \qquad p_m(z) = z^m.$$

In exponent coordinates it is multiplication by m:

$$\zeta_N^j \longmapsto \zeta_N^{mj}.$$

The kernel consists of exponents satisfying

$$mj \equiv 0 \pmod{N}.$$

There are exactly $\gcd(m, N)$ such residue classes. Therefore

$$|\ker p_m| = \gcd(m, N).$$

Since $|\mu_N| = N$, the image has size

$$|\operatorname{im} p_m| = \frac{N}{\gcd(m, N)}.$$

Consequently, the following conditions are equivalent:

$$\begin{aligned} &\gcd(m, N) = 1, \\ &p_m \text{ is injective}, \\ &p_m \text{ is surjective}, \\ &p_m \text{ permutes the vertices of the regular } N\text{-gon}. \end{aligned}$$

For example, squaring permutes the vertices of a regular pentagon, but not those of a regular hexagon. On μ_6, both 1 and -1 square to 1, so the map has a nontrivial kernel.

This power-map viewpoint simultaneously explains orbit lengths, vertex permutations, and many elementary root-of-unity sums.

§15.5 The sum of the vertices is zero

Let $\zeta = \zeta_N$ with $N > 1$. The sum of all vertices is

$$1 + \zeta + \zeta^2 + \cdots + \zeta^{N-1}.$$

Multiplying by $\zeta - 1$ gives

$$(\zeta - 1)(1 + \zeta + \cdots + \zeta^{N-1}) = \zeta^N - 1 = 0.$$

Since $\zeta \neq 1$,

$$\boxed{1 + \zeta + \cdots + \zeta^{N-1} = 0.}$$

Geometrically, the centroid of a regular polygon centered at the origin is the origin.

More generally, for any integer m,

$$\sum_{j=0}^{N-1} \zeta^{mj} = \begin{cases} N, & N \mid m, \\ 0, & N \nmid m. \end{cases}$$

If $N \nmid m$, the ratio ζ^m is not 1, so the sum is a finite geometric series:

$$\sum_{j=0}^{N-1} \zeta^{mj} = \frac{(\zeta^m)^N - 1}{\zeta^m - 1} = 0.$$

If $N \mid m$, every summand equals 1.

The case $d = \gcd(m, N) > 1$ has a useful geometric interpretation. The list

$$1, \zeta^m, \zeta^{2m}, \ldots$$

traces a smaller regular polygon with N/d vertices, each vertex repeated d times when j runs through a complete set of residues modulo N. Its vector sum is still zero unless the polygon has only one vertex.

§15.6 Orthogonality is a vanishing polygon sum

For integers r, s,

$$\begin{aligned} \sum_{j=0}^{N-1} \zeta_N^{rj} \overline{\zeta_N^{sj}} &= \sum_{j=0}^{N-1} \zeta_N^{(r-s)j} \\ &= \begin{cases} N, & r \equiv s \pmod N, \\ 0, & r \not\equiv s \pmod N. \end{cases} \end{aligned}$$

The vectors

$$(1, \zeta_N^r, \zeta_N^{2r}, \ldots, \zeta_N^{(N-1)r})$$

are therefore mutually orthogonal for distinct residues $r \pmod N$.

This is the elementary core of the discrete Fourier transform. Fourier orthogonality is not an unrelated analytic miracle: it is the statement that a nontrivial regular polygon has vector sum zero.

As a quick application, let $c_0, \ldots, c_{N-1}$ be complex numbers and define

$$\widehat{c}_r = \sum_{j=0}^{N-1} c_j \zeta_N^{-rj}.$$

Then orthogonality gives the inversion formula

$$c_j = \frac{1}{N}\sum_{r=0}^{N-1} \widehat{c}_r \zeta_N^{rj}.$$

The proof is only a double sum followed by the vanishing identity above.

§15.7 A Fourier Filter for Subset Counts

The vanishing polygon sum is useful far beyond geometry. Suppose we want to count subsets of an n-element set whose sizes are divisible by three:

$$C_3(n) = \sum_{\substack{0\le k\le n\\ 3\mid k}} \binom{n}{k}.$$

The binomial theorem gives every size at once:

$$(1+t)^n = \sum_{k=0}^{n} \binom{n}{k} t^k.$$

If $\omega = e^{2\pi i/3}$, orthogonality says

$$\frac{1+\omega^k+\omega^{2k}}{3} = \begin{cases} 1, & 3 \mid k,\\ 0, & 3 \nmid k. \end{cases}$$

Multiply the k-th binomial coefficient by this expression and sum:

$$\begin{aligned} C_3(n) &= \frac{1}{3}\sum_{k=0}^{n} \binom{n}{k}(1+\omega^k+\omega^{2k}) \\ &= \boxed{\frac{2^n + (1+\omega)^n + (1+\omega^2)^n}{3}}. \end{aligned}$$

The complex terms combine to a real integer. In fact

$$1+\omega = e^{i\pi/3}, \qquad 1+\omega^2 = e^{-i\pi/3},$$

and hence

$$\boxed{C_3(n) = \frac{2^n + 2\cos(n\pi/3)}{3}}.$$

At $n = 6$ this gives $(64+2)/3 = 22$, exactly the direct count

$$\binom{6}{0} + \binom{6}{3} + \binom{6}{6} = 1 + 20 + 1 = 22.$$

The apparently mysterious cube roots are acting as a *filter*: they cancel the binomial coefficients with the wrong residues and keep the desired ones. More generally, for any modulus m and residue r,

$$\boxed{\sum_{\substack{0\le k\le n\\ k\equiv r \pmod m}} \binom{n}{k} = \frac{1}{m}\sum_{j=0}^{m-1} \zeta_m^{-rj}(1+\zeta_m^j)^n.}$$

This follows by inserting the orthogonality identity for the congruence $k \equiv r$. Thus the same roots of unity that describe hops around a polygon also solve counting problems that may have no polygon in their statement.

§15.8 Chord lengths depend only on modular distance

The distance between vertices a and b on the unit circle is

$$|\zeta_N^a - \zeta_N^b|.$$

Factor out ζ_N^b, whose absolute value is one:

$$|\zeta_N^a - \zeta_N^b| = |\zeta_N^{a-b} - 1|.$$

Using

$$e^{i\theta} - 1 = 2ie^{i\theta/2} \sin \frac{\theta}{2},$$

we obtain

$$\boxed{|\zeta_N^a - \zeta_N^b| = 2\left|\sin \frac{\pi(a-b)}{N}\right|}.$$

Thus chord length depends only on the difference $a - b \pmod N$, exactly as rotational symmetry predicts.

For a unit circumradius decagon, the side length is

$$2 \sin \frac{\pi}{10},$$

the chord spanning two edges has length

$$2 \sin \frac{2\pi}{10},$$

and opposite vertices are distance

$$2 \sin \frac{5\pi}{10} = 2.$$

The same modular difference that controls hopping also controls Euclidean distance on the polygon.

§15.9 The Product of Chords

Chord lengths offer another unexpected exact identity. Fix the vertex 1 of the regular N-gon inscribed in the unit circle. The distances to the other vertices are

$$|1 - \zeta_N^j| = 2 \sin \frac{\pi j}{N}, \qquad j = 1, \ldots, N-1.$$

(The sine is positive because $0 < \pi j/N < \pi$.) How can the product of all these trigonometric lengths be something simple?

Factor the polynomial whose roots are the *other* vertices:

$$\frac{z^N - 1}{z - 1} = 1 + z + \cdots + z^{N-1} = \prod_{j=1}^{N-1} (z - \zeta_N^j).$$

Substituting $z = 1$ yields an exact complex identity,

$$N = \prod_{j=1}^{N-1} (1 - \zeta_N^j).$$

Taking absolute values gives

$$\boxed{\prod_{j=1}^{N-1} 2\sin\frac{\pi j}{N} = N,} \qquad \boxed{\prod_{j=1}^{N-1} \sin\frac{\pi j}{N} = \frac{N}{2^{N-1}}.}$$

For the equilateral triangle the two chord lengths from one vertex are each $\sqrt{3}$, so their product is 3. For a square, they are $\sqrt{2}, 2, \sqrt{2}$, and their product is 4. The general polygon identity explains both observations without evaluating any trigonometric product by hand.

One subtle point is that the formula for the original chord lengths involves *Euclidean geometry*, but the product identity came from the value of a polynomial at 1. Geometry supplies the absolute values, while algebra controls their product. It is a good example of how keeping complex phases until the last step can make a real-valued calculation easier.

§15.10 The roots of $z^N - 1$ form the polygon

The polynomial

$$z^N - 1$$

has the N roots

$$1, \zeta_N, \ldots, \zeta_N^{N-1}.$$

They are distinct because the derivative

$$Nz^{N-1}$$

does not vanish at a root of unity. Hence

$$z^N - 1 = \prod_{j=0}^{N-1} (z - \zeta_N^j).$$

Complex conjugation reflects the polygon across the real axis:

$$\overline{\zeta_N^j} = \zeta_N^{-j} = \zeta_N^{N-j}.$$

Therefore nonreal roots occur in conjugate pairs. If N is odd, 1 is the only real root. If N is even, the real roots are 1 and -1.

For instance,

$$\omega = \frac{-1 + i\sqrt{3}}{2} = e^{2\pi i/3}$$

is a nontrivial cubic root of unity, so

$$\omega^3 = 1$$

and

$$1 + \omega + \omega^2 = 0.$$

This single equilateral-triangle identity is enough to solve a basic polynomial-divisibility problem.

§15.11 One elementary divisibility test

Let

$$P_n(x) = x^{2n} + x^n + 1$$

and

$$Q(x) = x^2 + x + 1.$$

The roots of Q are ω and ω^2, where $\omega^3 = 1$ and $\omega \neq 1$. Since the roots are distinct,

$$Q(x) \mid P_n(x)$$

exactly when both roots of Q are roots of P_n.

At $x = \omega$,

$$P_n(\omega) = \omega^{2n} + \omega^n + 1.$$

If $n \equiv 0 \pmod 3$, then $\omega^n = 1$ and

$$P_n(\omega) = 3 \neq 0.$$

If $n \equiv 1$ or $2 \pmod 3$, then ω^n is one of the two nontrivial cubic roots, and

$$1 + \omega^n + \omega^{2n} = 0.$$

The same argument applies to ω^2. Therefore

$$\boxed{x^2 + x + 1 \mid x^{2n} + x^n + 1 \quad \Longleftrightarrow \quad 3 \nmid n.}$$

The proof is elementary: polynomial divisibility is checked on the two vertices of an equilateral triangle, and exponent reduction is arithmetic modulo three.

Arithmetic and Geometric Sequence Exercises

N-20220803

§16.1 The trigonometric example and logarithmic progressions

The first exercise concerns a triangle ABC such that

$$\lg(\sin A), \quad \lg(\sin B), \quad \lg(\sin C)$$

form an arithmetic progression, and A, B, C themselves also form an arithmetic progression. Since

$$A + B + C = \pi,$$

if A, B, C are in arithmetic progression, then

$$B = \frac{\pi}{3}.$$

The logarithmic condition means

$$2\lg(\sin B) = \lg(\sin A) + \lg(\sin C),$$

so

$$\sin^2 B = \sin A \sin C.$$

Because $\sin B = \sqrt{3}/2$, we get

$$\sin A \sin C = \frac{3}{4}.$$

Using $A + C = 2\pi/3$, one can analyze

$$\cos(A - C) = \cos A \cos C + \sin A \sin C.$$

The conclusion can be proved without guessing symmetry. Using the product-to-sum identity,

$$2\sin A \sin C = \cos(A - C) - \cos(A + C) = \cos(A - C) + \frac{1}{2}.$$

Since $\sin A \sin C = \frac{3}{4}$, it follows that $\cos(A - C) = 1$. As $-\pi < A - C < \pi$, we have $A = C$, and the triangle is equilateral: $A = B = C = \pi/3$. The decisive step is the exact identity, together with the allowed interval for angle differences.

If

$$a_n = 10 + \lg 2^n,$$

then

$$a_n = 10 + n \lg 2.$$

Hence

$$a_{n+1} - a_n = \lg 2,$$

so $\{a_n\}$ is an arithmetic progression.

This is a typical use of logarithms: they turn powers into multiples. A geometric pattern in 2^n becomes an arithmetic pattern after applying lg.

§16.2 Term recovery by the finite difference of sums

If S_n is the sum of the first n terms of a sequence, then

$$a_n = S_n - S_{n-1}.$$

This applies to problems involving two arithmetic progressions $\{a_n\}$, $\{b_n\}$ whose sums S_n, T_n satisfy

$$\frac{S_n}{T_n} = \frac{2n}{3n+1}.$$

For arithmetic progressions,

$$S_n = \frac{n}{2}(a_1 + a_n),$$

but the cleaner trick is to use

$$a_n = S_n - S_{n-1}, \qquad b_n = T_n - T_{n-1}.$$

One can derive the answer. Since arithmetic progressions have quadratic partial sums, write $S_n = n(A + Bn)$ and $T_n = n(C + Dn)$. Then $(A + Bn)(3n + 1) = 2n(C + Dn)$ for every positive integer n. Comparing coefficients yields $A = 0$, $B = 2C$, $D = 3C$. Taking consecutive differences gives $a_n = 2C(2n - 1)$ and $b_n = 2C(3n - 1)$, so

$$\boxed{\frac{a_n}{b_n} = \frac{2n-1}{3n-1}}.$$

This calculation also requires $T_n \neq 0$, as assumed in the quotient.

A common type of problem gives a relation between S_n and a_n. For example,

$$S_n = 2a_n - 2.$$

Then

$$a_{n+1} = S_{n+1} - S_n = 2a_{n+1} - 2 - (2a_n - 2),$$

so

$$a_{n+1} = 2a_n.$$

Thus the sequence is geometric. The first term is fixed as well: $S_1 = a_1 = 2a_1 - 2$ implies $a_1 = 2$. Therefore the unique sequence is $a_n = 2^n$ for $n \geq 1$. Whenever a problem relates a term to a partial sum, subtract consecutive equations and then check the initial condition.

If $S_n = a_1 + \cdots + a_n$, then

$$a_n = S_n - S_{n-1} = \Delta S_{n-1}.$$

So recovering terms from partial sums is finite differentiation. Conversely,

$$S_n = S_0 + \sum_{k=1}^{n} a_k$$

is finite integration.

For an arithmetic progression $a_n = a_1 + (n-1)d$, the first difference is constant:

$$\Delta a_n = d.$$

For a quadratic sequence, the second difference is constant. This is why finite-difference tables detect polynomial sequences, just as derivatives detect polynomial degree in calculus.

§16.3 Geometric sequences, sums, and telescoping

If $\{a_n\}$ is a geometric progression, then

$$a_m^2 = a_{m-r}a_{m+r}.$$

This appears in several exercises. It follows from

$$a_n = a_1 q^{n-1}.$$

For example,

$$a_6^2 = a_5 a_7.$$

Such identities compute missing terms and ratios from partial information.

Two sums with visually similar denominators behave quite differently. For a product denominator, partial fractions give

$$\sum_{k=1}^{n} \frac{1}{k(k+1)} = \sum_{k=1}^{n} \left(\frac{1}{k} - \frac{1}{k+1} \right) = 1 - \frac{1}{n+1}.$$

For an additive denominator, the corresponding sum is

$$\sum_{k=1}^{n} \frac{1}{k+(k+1)} = \sum_{k=1}^{n} \frac{1}{2k+1} = H_{2n+1} - 1 - \frac{1}{2}H_n, \qquad H_m = \sum_{j=1}^{m} \frac{1}{j}.$$

Here the odd-denominator terms are isolated by subtracting the even terms from the harmonic sum. No analogous termwise telescoping occurs. The two formulas illustrate why the denominator's algebraic factorization, rather than its visual appearance, determines which summation technique applies.

A telescoping sum has the form

$$a_k = F(k+1) - F(k).$$

Then

$$\sum_{k=m}^{n} a_k = F(n+1) - F(m).$$

Only the boundary terms survive. In this language, a telescoping term is a discrete derivative or coboundary.

The identity

$$\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$$

is exactly this pattern with $F(k) = 1/k$. The cancellation is not accidental; it is the discrete fundamental theorem of calculus.

§16.4 Generating functions and second-order recurrences

A sequence (a_n) can be packaged into a generating function

$$A(x) = \sum_{n \geq 0} a_n x^n.$$

Arithmetic and geometric progressions have especially simple forms. For example,

$$\sum_{n \geq 0} q^n x^n = \frac{1}{1 - qx}.$$

A linear sequence produces a rational generating function with denominator $(1-x)^2$. Recurrences translate into algebraic equations for generating functions.

The standard generating-function and characteristic-root treatments may be found in MIT 6.042J, *Mathematics for Computer Science*, Chapters 10–12.[1] For an arithmetic sequence indexed from zero, differentiating $\sum_{n\geq 0} z^n = 1/(1-z)$ shows $\sum_{n\geq 0}(n+1)z^n = (1-z)^{-2}$. Thus the extra factor n corresponds precisely to a repeated denominator factor in the generating function.

Some sequence problems go beyond arithmetic and geometric progressions. A linear recurrence such as

$$a_{n+2} = pa_{n+1} + qa_n$$

is solved by trying $a_n = \lambda^n$. This gives the characteristic equation

$$\lambda^2 = p\lambda + q.$$

When the roots are distinct, the solution has the form

$$a_n = A\lambda_1^n + B\lambda_2^n.$$

Thus geometric progressions are the eigenvectors of the shift recurrence. This is the bridge from elementary sequences to linear algebra and dynamical systems.

[1] MIT 6.042J, *Mathematics for Computer Science*, readings.

Divisibility as Arithmetic in Quotient Rings

N-20220913

The most productive step in many divisibility problems is to stop treating the divisor as a large integer to be compared with an even larger expression. Once the divisor m is fixed, calculations take place among residue classes in $\mathbb{Z}/m\mathbb{Z}$. The familiar operations of greatest common divisors, Bezout coefficients, and the Chinese remainder theorem describe the structure of that ring.[1]

§17.1 A necessary condition can collapse the search

Suppose a prime has the form

$$p = x(x+1) - 4, \qquad x \in \mathbb{Z}_{>0}.$$

The product of consecutive integers is even, so p is even. The only even positive prime is 2. Therefore $x(x+1) = 6$ and $x = 2$, which indeed gives $p = 2$. The short argument has two parts: parity narrows the candidates, and substitution verifies the surviving candidate.

A similar reduction works even when the divisor depends on the unknown integer. Consider

$$x^2 + 1 \mid x^3 - x, \qquad x \in \mathbb{Z}_{>0}.$$

Since $x^2 \equiv -1 \pmod{x^2+1}$, the condition is equivalent to

$$x^2 + 1 \mid 2x.$$

For $x > 1$, $x^2 + 1 > 2x > 0$, making divisibility impossible. At $x = 1$, the dividend vanishes, so $x = 1$ is the unique positive solution.

§17.2 Ideals explain why remainders preserve information

For nonzero $m \in \mathbb{Z}$, the congruence $a \equiv b \pmod{m}$ means $a - b \in m\mathbb{Z}$. In the quotient ring $\mathbb{Z}/m\mathbb{Z}$, that is simply equality of the classes $[a] = [b]$. Polynomial or exponential expressions respect this equality, so we can simplify them before asking whether the original expression is divisible by m.

The ideal $(a) = a\mathbb{Z}$ contains the multiples of a, and

$$a \mid b \quad \Longleftrightarrow \quad (b) \subseteq (a).$$

The reversal of inclusion is useful. Bezout's identity implies

$$(a) + (b) = (\gcd(a,b)), \qquad (a) \cap (b) = (\operatorname{lcm}(a,b))$$

for nonzero integers, using positive gcd and lcm. Thus gcd and lcm describe two universal ways to combine divisibility conditions.

A unit in $\mathbb{Z}/m\mathbb{Z}$ is a class $[a]$ possessing a multiplicative inverse. Such an inverse exists exactly when $\gcd(a,m) = 1$: the Bezout equation $au + mv = 1$ becomes $[a][u] = [1]$. By contrast, modulo 6, the nonzero classes [2] and [3] multiply to zero. This zero-divisor phenomenon explains why cancellation by an arbitrary nonzero class is invalid.

[1] See the introductory number theory and congruence lectures of MIT 18.781: https://ocw.mit.edu/courses/18-781-theory-of-numbers-spring-2012/pages/lecture-notes/.

§17.3 A polynomial remainder hidden in an integer problem

For integer x, consider

$$x^2 + x + 1 \mid x^4 - 2.$$

The identity

$$x^3 - 1 = (x - 1)(x^2 + x + 1)$$

shows that $x^3 \equiv 1 \pmod{x^2 + x + 1}$, whence

$$x^4 - 2 \equiv x - 2 \pmod{x^2 + x + 1}.$$

For positive x, the divisor exceeds $|x - 2|$ except when $x = 2$, and the small remaining value $x = 1$ also fails. At $x = 2$, $x^4 - 2 = 14$ and $x^2 + x + 1 = 7$, so $x = 2$ is the unique positive solution.

There is a parallel computation in the polynomial quotient $k[t]/(t^2 + t + 1)$, where the symbol t satisfies $t^3 = 1$. That statement is about a class of a polynomial, independent of any integer substituted for t. The integer problem evaluates the same polynomial identity at a particular x, then interprets it as a congruence. Keeping these two uses of a symbol separate prevents a common logical slip.

§17.4 The Chinese remainder theorem builds solutions

Suppose m, n are coprime. Choose $u, v \in \mathbb{Z}$ satisfying $um + vn = 1$. Then the classes

$$e_m = vn, \qquad e_n = um$$

satisfy

$$e_m \equiv 1 \pmod{m}, \quad e_m \equiv 0 \pmod{n}, \qquad e_n \equiv 0 \pmod{m}, \quad e_n \equiv 1 \pmod{n}.$$

Therefore the simultaneous system

$$x \equiv a \pmod{m}, \qquad x \equiv b \pmod{n}$$

has solution

$$\boxed{x \equiv ae_m + be_n \pmod{mn}.}$$

Any two solutions differ by a multiple of both m and n, hence by a multiple of mn; the solution is unique modulo mn.

For example, solve $x \equiv 2 \pmod 3$ and $x \equiv 4 \pmod 5$. The selectors $10 \equiv 1 \pmod 3$, $0 \pmod 5$ and $6 \equiv 0 \pmod 3$, $1 \pmod 5$ give

$$x \equiv 2 \cdot 10 + 4 \cdot 6 = 44 \equiv 14 \pmod{15}.$$

These selectors are idempotents: modulo 15, $10^2 \equiv 10$, $6^2 \equiv 6$, $10 \cdot 6 \equiv 0$, and $10 + 6 \equiv 1$. Their algebraic behavior is exactly that of coordinate projections in the product ring

$$\mathbb{Z}/15\mathbb{Z} \cong \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z}.$$

The general theorem for pairwise coprime moduli follows by the same Bezout construction. It is an explicit decomposition, rather than a mysterious claim that local solutions can be assembled globally.

§17.5 Odd powers and the arithmetic of a reflection

A large example is

$$S = 1^{1983} + 2^{1983} + \cdots + 1983^{1983}.$$

The claim is that S is divisible by the triangular number

$$T = \frac{1983 \cdot 1984}{2} = 1983 \cdot 992.$$

The parity of the number of terms matters. Since 1983 is odd, pairing j with $1983 - j$ for $1 \leq j \leq 1982$ gives $1983 \mid S$. Pairing instead j with $1984 - j$ gives

$$j^{1983} + (1984 - j)^{1983} \equiv 0 \pmod{992},$$

since $1984 - j \equiv -j \pmod{992}$. The middle term $j = 992$, which is its own partner, also vanishes modulo 992. Hence $992 \mid S$. Because 1983 and 992 are coprime, $T \mid S$.

This method is algebra in a quotient ring together with a reflection symmetry of residue classes. The same strategy generalizes: instead of expanding a sum with thousands of terms, search for a transformation of the index that makes contributions cancel after passing to a suitable modulus.

Divisibility by Pairing, Finite Differences, and Harmonic Reflection

N-20220918

Three problems in this note appear unrelated: sums of odd powers, an exponential congruence modulo 512, and an alternating rational sum whose numerator is divisible by 1979. Their solutions become shorter once the expressions are organized by symmetry, telescoping and modular inverses. The number-theoretic conventions follow MIT 18.781 *Theory of Numbers*; for finite differences, see also Graham, Knuth and Patashnik, *Concrete Mathematics*.[1]

§18.1 Odd Powers and Triangular Numbers

Theorem 18.1.1

For every integer $k \geq 1$ and odd positive integer r,

$$\boxed{\frac{k(k+1)}{2} \mid \sum_{j=1}^{k} j^r.}$$

Proof. Write $S = \sum_{j=1}^{k} j^r$. Since r is odd,

$$(m-j)^r \equiv -j^r \pmod{m}.$$

If k is even, pairing j with $k+1-j$ has no fixed point, giving $k+1 \mid S$. Modulo k, the terms $1, \ldots, k-1$ are paired by $j \mapsto k-j$; the only possible fixed point is $k/2$. Hence $k/2 \mid S$. As $\gcd(k/2, k+1) = 1$,

$$(k/2)(k+1) \mid S.$$

If k is odd, instead pair $1, \ldots, k-1$ modulo k, with no fixed point, and note $k^r \equiv 0 \pmod{k}$. Thus $k \mid S$. Pair the terms $1, \ldots, k$ under $j \mapsto k+1-j$; the unique fixed point $(k+1)/2$ contributes a multiple of $(k+1)/2$. So $(k+1)/2 \mid S$. Since $\gcd(k, (k+1)/2) = 1$, their product divides S. □

The parity cases matter. For $k = 3, r = 1$, the sum is 6, divisible by $k(k+1)/2 = 6$, but not by $k+1 = 4$. A proof that claims all terms can simply be paired modulo $k+1$ overlooks the central fixed point.

§18.2 The discrete fundamental theorem of calculus

For an integer-valued sequence f, define the finite difference $\Delta f(n) = f(n+1) - f(n)$. Telescoping yields

$$f(n) = f(1) + \sum_{j=1}^{n-1} \Delta f(j).$$

[1] https://ocw.mit.edu/courses/18-781-theory-of-numbers-spring-2012/pages/lecture-notes/

Thus if $m \mid f(1)$ and $m \mid \Delta f(j)$ for every $j \geq 1$, then $m \mid f(n)$ for every $n \geq 1$. This is induction expressed in a form that separates the initial value from the repeated increment.

The operator also reduces polynomial degree:

$$\Delta n^2 = 2n + 1, \qquad \Delta \binom{n}{k} = \binom{n}{k-1}.$$

The binomial basis is especially convenient because taking a difference simply lowers the index. For a polynomial P of degree d, $\Delta^{d+1} P = 0$.

§18.3 A congruence modulo 512 in two steps

We wish to prove

$$\boxed{512 \mid 3^{2n} - 32n^2 + 24n - 1 \qquad (n \geq 1).}$$

Let $F(n) = 9^n - 32n^2 + 24n - 1$. The base case is $F(1) = 9 - 32 + 24 - 1 = 0$. Its finite difference simplifies to

$$\begin{aligned} \Delta F(n) &= 9^{n+1} - 9^n - 32((n+1)^2 - n^2) + 24 \\ &= 8(9^n - 8n - 1). \end{aligned}$$

Set $H(n) = 9^n - 8n - 1$, so $H(1) = 9 - 8 - 1 = 0$. Another difference gives

$$\Delta H(n) = 8(9^n - 1).$$

Since $9 \equiv 1 \pmod 8$, the bracket is divisible by 8 for every n; hence $\Delta H(n)$ is divisible by 64. Telescoping and $H(1) = 0$ show $64 \mid H(n)$, and consequently $512 \mid \Delta F(n)$. Another telescoping step and $F(1) = 0$ prove the claim.

The two differences were chosen to peel away the polynomial terms and expose increasing powers of 2; no long binomial expansion of 9^n was required.

§18.4 Harmonic reflection modulo a prime

The rational-sum exercise asks us to write

$$S = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots - \frac{1}{1318} + \frac{1}{1319} = \frac{p}{q}$$

in lowest terms and prove $1979 \mid p$. The central observation is that

$$1979 - 1 = 1319 + 659.$$

Let $H_n = \sum_{j=1}^{n} 1/j$ with $H_0 = 0$. Separating odd and even denominators gives

$$S = H_{1319} - 2\sum_{j=1}^{659} \frac{1}{2j} = H_{1319} - H_{659}.$$

The integer 1979 is prime (trial division by primes at most $\sqrt{1979} < 45$ confirms this). Therefore every denominator from 1 to 1978 has an inverse modulo 1979; rational sums with these denominators make sense in the finite field $\mathbb{F}_{1979}$.

For any odd prime ℓ, pair j with $\ell - j$:

$$\frac{1}{\ell - j} \equiv -\frac{1}{j} \pmod{\ell}.$$

Summing over $1 \le j \le \ell - 1$ yields

$$H_{\ell-1} \equiv 0 \pmod{\ell}.$$

More precisely,

$$\begin{aligned} H_{\ell-1-m} &= H_{\ell-1} - \sum_{j=\ell-m}^{\ell-1} \frac{1}{j} \\ &\equiv -\sum_{j=1}^{m} \frac{1}{\ell - j} \equiv H_m \pmod{\ell}. \end{aligned}$$

Choose $\ell = 1979$, $m = 659$. Then $\ell - 1 - m = 1319$, so $H_{1319} \equiv H_{659} \pmod{1979}$ and hence

$$\boxed{S \equiv 0 \pmod{1979}.}$$

No denominator of S is divisible by 1979, and this property survives when the fraction is reduced to lowest terms. Consequently the reduced numerator p is divisible by 1979, as required.

The proof does not require constructing the astronomically large common denominator. The complementary harmonic intervals were deliberately chosen so that modular reflection pairs them.

§18.5 Prime-power valuations clarify the method

For a prime p, write $v_p(N)$ for the exponent of p in a nonzero integer N. Then

$$p^k \mid N \quad \Longleftrightarrow \quad v_p(N) \ge k.$$

The modulus $512 = 2^9$ is a valuation condition: after two finite-difference reductions, we extract three factors of 8, providing nine factors of 2. The harmonic problem instead uses a prime ℓ that is invertible in all denominators, so it is naturally solved by arithmetic in $\mathbb{F}_\ell$.

These two cases explain why modular arithmetic changes shape with the modulus. At a prime, inverses and field symmetries can be used; at a prime power, tracking the number of factors of the prime is often the right invariant.

Finite Fields and Polynomial Factorization

N-20221002

§19.1 Constructing Finite Fields

The construction and factorization results follow the standard organization in Lidl and Niederreiter, *Introduction to Finite Fields and their Applications*, Chapters 2–4, with small examples calculated explicitly below.[1] The central question is: how do polynomials factor when the coefficients live in a finite field rather than in $\mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$?

The question is not only theoretical: it appears in coding theory, cryptography, computational algebra, and the construction of algebraic extensions. The definition of a field comes first because factorization depends heavily on the coefficient domain.

A field F is a set with addition and multiplication such that:

(i) $(F, +)$ is an abelian group with identity 0;

(ii) $F \setminus \{0\}$ is an abelian group under multiplication with identity 1;

(iii) multiplication distributes over addition.

The exclusion of 0 from the multiplicative group is essential. If 0 had a multiplicative inverse, then from $0 \cdot a = 0$ one would get contradictions such as $1 = 0$. So the phrase "every nonzero element has an inverse" is not a technical nuisance; it is what makes division possible.

Examples are

$$\mathbb{Q}, \qquad \mathbb{R}, \qquad \mathbb{C},$$

and, for a prime p,

$$\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}.$$

The requirement that p be prime is crucial. If p is composite, zero divisors appear. For example in $\mathbb{Z}/6\mathbb{Z}$,

$$2 \cdot 3 \equiv 0 \pmod 6,$$

even though neither factor is zero.

A field extension K/F means F is embedded in a larger field K. The point of passing to K is often to make more polynomial equations solvable. For example,

$$x^2 + 1$$

has no root in $\mathbb{R}$, but it splits in $\mathbb{C}$:

$$x^2 + 1 = (x - i)(x + i).$$

If K is finite-dimensional as a vector space over F, its dimension is the degree of the extension:

$$[K : F] = \dim_F K.$$

[1] R. Lidl and H. Niederreiter, *Introduction to Finite Fields and their Applications*, Cambridge University Press, Chapters 2–4.

Finite extensions are tractable because linear algebra applies directly, whereas infinite extensions require substantially heavier tools.

For a prime p, $\mathbb{F}_p$ is the simplest finite field. To construct fields with p^n elements, use irreducible polynomials. If

$$f(x) \in \mathbb{F}_p[x]$$

is irreducible of degree n, then

$$\mathbb{F}_p[x]/(f(x))$$

is a field with p^n elements.

For example, over $\mathbb{F}_2$, the polynomial

$$f(x) = x^3 + x + 1$$

has no root in $\mathbb{F}_2$, hence is irreducible of degree 3. Therefore

$$\mathbb{F}_2[x]/(x^3 + x + 1)$$

is a field with $2^3 = 8$ elements.

In this quotient field, if α denotes the class of x, then

$$\alpha^3 + \alpha + 1 = 0,$$

so

$$\alpha^3 = \alpha + 1$$

in characteristic 2. Every element can be uniquely written as

$$a + b\alpha + c\alpha^2, \qquad a, b, c \in \mathbb{F}_2.$$

§19.2 The first irreducibility test: degrees one and two

In $\mathbb{F}_q[x]$, every nonzero polynomial factors uniquely into irreducible polynomials, up to order and multiplication by nonzero constants. This is analogous to unique prime factorization of integers.

However, a polynomial need not split into linear factors over its base finite field. For example, a polynomial may be irreducible over $\mathbb{F}_p$ but split over an extension field. This is exactly like $x^2 + 1$ over $\mathbb{R}$, except that the extension fields are finite and highly structured.

Consider

$$f(x) = x^4 + x^2 + 2 \in \mathbb{F}_3[x].$$

First test roots:

$$f(0) = 2 \neq 0,$$

$$f(1) = 1 + 1 + 2 = 4 \equiv 1 \pmod 3,$$

$$f(2) = 2^4 + 2^2 + 2 = 16 + 4 + 2 = 22 \equiv 1 \pmod 3.$$

So f has no linear factor. It might still factor as a product of two quadratics. For a degree-four polynomial, root testing is insufficient: a reducible polynomial without roots must factor into two irreducible quadratics. Over $\mathbb{F}_3$, the monic irreducible quadratics are exactly $x^2 + 1$, $x^2 + x + 2$, and $x^2 + 2x + 2$. Dividing f by these yields respective remainders

$$2, \qquad 2x + 2, \qquad x + 2.$$

None vanishes. Since f also has no linear factor, it is irreducible over $\mathbb{F}_3$. This completes the omitted calculation rather than merely asserting that a check exists.

This illustrates the basic workflow: root testing only detects linear factors. For degree 4, one must also check quadratic factors.

A polynomial can factor differently depending on the coefficient field. For example, $x^2 + 1$ is irreducible over $\mathbb{R}$ only if one insists on real linear factors, but over $\mathbb{C}$ it splits. Over finite fields the same phenomenon becomes arithmetic.

For odd p, a quadratic polynomial

$$ax^2 + bx + c, \qquad a \neq 0,$$

has a root (and hence factors) iff its discriminant

$$\Delta = b^2 - 4ac$$

is a square in $\mathbb{F}_p$, including zero. The restriction $p \neq 2$ matters: the quadratic formula divides by $2a$, which is impossible in characteristic 2. In characteristic 2, the polynomial must instead be analyzed using its actual values or an appropriate Artin–Schreier form. Thus irreducibility is controlled by quadratic residues. This links polynomial factorization directly to modular arithmetic.

§19.3 Frobenius is the field's intrinsic symmetry

The standard facts are:

(i) finite fields have order p^n;

(ii) for every prime power p^n, there is a finite field of that order, unique up to isomorphism;

(iii) the nonzero elements $\mathbb{F}_{p^n}^{\times}$ form a cyclic group of order $p^n - 1$;

(iv) the Frobenius map

$$\mathrm{Frob}(a) = a^p$$

is an automorphism of $\mathbb{F}_{p^n}$.

The cyclicity of the multiplicative group is surprisingly powerful. It means every nonzero element is a power of one primitive element. This is why finite field computations often become exponent arithmetic modulo $p^n - 1$.

In characteristic p, the Frobenius map is

$$\varphi(a) = a^p.$$

It is a field homomorphism because

$$(a + b)^p = a^p + b^p$$

in characteristic p. In a finite field, Frobenius is an automorphism. Its fixed field inside $\mathbb{F}_{p^n}$ is $\mathbb{F}_p$.

The polynomial

$$x^{p^n} - x$$

has all elements of $\mathbb{F}_{p^n}$ as roots, and it factors over $\mathbb{F}_p$ as the product of all monic irreducible polynomials whose degrees divide n. This is a central organizing fact for finite-field factorization.

In characteristic p, the Frobenius map

$$F : x \mapsto x^p$$

is a field homomorphism because

$$(x+y)^p = x^p + y^p.$$

In $\mathbb{F}_{p^n}$, it is an automorphism, and

$$F^n = 1.$$

The Galois group is cyclic:

$$\mathrm{Gal}(\mathbb{F}_{p^n}/\mathbb{F}_p) = \langle F \rangle \cong \mathbb{Z}/n\mathbb{Z}.$$

This is the cleanest first example of a Galois group: roots are moved by raising to powers of p, and the fixed field of Frobenius is $\mathbb{F}_p$.

An element $\alpha \in \mathbb{F}_{q^n}$ has conjugates

$$\alpha, \quad \alpha^q, \quad \alpha^{q^2}, \quad \dots$$

under Frobenius. Its minimal polynomial over $\mathbb{F}_q$ is

$$\prod_{i=0}^{d-1} (x - \alpha^{q^i}),$$

where d is the size of the Frobenius orbit. Thus irreducible polynomials over finite fields are Frobenius orbits in disguise.

This explains why the degrees of irreducible factors of a polynomial are controlled by Frobenius cycles.

§19.4 Factor Degrees and Möbius Inversion

Every element of $\mathbb{F}_{q^n}$ is a root of

$$x^{q^n} - x.$$

Over $\mathbb{F}_q$, this polynomial is the product of all monic irreducible polynomials whose degrees divide n. If N_d is the number of monic irreducible polynomials of degree d, then

$$q^n = \sum_{d \mid n} d N_d.$$

By Möbius inversion,

$$N_n = \frac{1}{n} \sum_{d \mid n} \mu(d) q^{n/d}.$$

This formula turns factorization into enumeration. The factor d appears because each degree-d irreducible polynomial contributes d roots.

Finite-field factorization algorithms exploit Frobenius. The polynomial

$$x^{q^m} - x$$

has as roots exactly the elements of $\mathbb{F}_{q^m}$. Therefore

$$\gcd(f(x), x^{q^m} - x)$$

extracts from f the product of irreducible factors whose degrees divide m.

This is the idea behind distinct-degree factorization. First analyze repeated factors by checking $\gcd(f, f')$. If $f' = 0$ over a perfect finite field, this signals that f is a polynomial in x^p; the square-free algorithm must take an appropriate p-th root before continuing, rather than dividing by the zero derivative. Then use Frobenius powers to separate factors by degree. Finally use algorithms such as Cantor–Zassenhaus to split equal-degree parts.

Berlekamp's algorithm turns factorization into a nullspace computation. Over $\mathbb{F}_q$, consider the map on the quotient algebra $A = \mathbb{F}_q[x]/(f)$

$$B : g \mapsto g^q - g.$$

Elements in the kernel satisfy $g^q = g$, and this kernel contains information about decomposing f into factors.

Thus the algorithm uses the finite-dimensional $\mathbb{F}_q$-vector space structure of the quotient ring. This is a concrete instance of a general computational theme: algebraic structure becomes linear algebra after passing to a finite quotient.

§19.5 Evaluation Codes from Polynomials

Finite fields are essential in coding theory because code symbols can be added, multiplied, and evaluated polynomially. Reed–Solomon codes evaluate polynomials over $\mathbb{F}_q$ at distinct field points:

$$f \longmapsto (f(a_1), \ldots, f(a_m)).$$

A nonzero polynomial of degree $< k$ has at most $k - 1$ roots. Thus two distinct polynomials of degree $< k$ differ in at least $m - k + 1$ of the m evaluation coordinates, assuming $m \leq q$ distinct evaluation points. This proves the Reed–Solomon minimum-distance bound $d_{\min} \geq m - k + 1$, with equality for suitable polynomial differences; hence a codeword has enough separation to correct up to $\lfloor (m - k)/2 \rfloor$ symbol errors.

Here the same polynomial-root principle used in factorization becomes a robustness principle for information transmission.

The identity

$$\#\mathbb{A}^1(\mathbb{F}_q) = q$$

and the polynomial $x^q - x$ already hint at a larger story: equations over finite fields define finite sets of points, and Frobenius acts on their solutions. In higher-dimensional algebraic geometry, counting points over $\mathbb{F}_{q^n}$ reveals cohomological information.

Thus elementary finite-field polynomial factorization is a first step toward modern arithmetic geometry: finite fields provide a world where algebraic equations, symmetries, and point counts interact exactly.

Oriented Boundary Sums, Polynomial Remainders, and Congruence Classes

N-20221013

§20.1 Three Computations, Three Structures

A coordinate-area formula, a polynomial remainder problem, and a modular-exponent problem can appear on the same exercise sheet without sharing one master theorem. What they do share is a style of simplification: each calculation throws away information that the question does not need.

For a polygon, the relevant data are the oriented contributions of its edges. For a polynomial modulo $x - a$, all multiples of $x - a$ become invisible and only the value at a remains. For an integer modulo m, numbers differing by a multiple of m are treated as the same residue class. The useful lesson is therefore not that every "remainder" is secretly identical, but that one should identify the map that preserves exactly the requested invariant.

§20.2 Determinants as Signed Area

For vectors

$$p = (x_1, y_1), \qquad q = (x_2, y_2),$$

define

$$\det(p, q) = x_1 y_2 - x_2 y_1.$$

Its absolute value is the area of the parallelogram spanned by p, q, while its sign records orientation. Thus the oriented area of the triangle with vertices $0, p, q$ is

$$\frac{1}{2}\det(p, q).$$

Let a polygon have cyclically ordered vertices

$$p_i = (x_i, y_i), \qquad i = 1, \dots, n, \qquad p_{n+1} = p_1.$$

Triangulating from the origin suggests the signed sum

$$\frac{1}{2}\sum_{i=1}^{n} \det(p_i, p_{i+1}).$$

If the origin lies outside the polygon, some triangles point with the opposite orientation. This is not a defect: the negative pieces cancel the extra positive pieces, leaving the polygon's oriented area.

§20.3 Shoelace as a Boundary Integral

Green's theorem gives an origin-free expression for the oriented area of a positively oriented region D:

$$\operatorname{Area}(D) = \frac{1}{2}\int_{\partial D}(x\,dy - y\,dx).$$

For a polygon, parametrize the edge from p_i to p_{i+1} by

$$r_i(t) = p_i + t(p_{i+1} - p_i), \qquad 0 \le t \le 1.$$

Writing $r_i(t) = (x(t), y(t))$, a direct calculation gives

$$\begin{aligned}\frac{1}{2}\int_{p_i}^{p_{i+1}}(x\,dy - y\,dx) &= \frac{1}{2}\int_0^1 \left(x(t)y'(t) - y(t)x'(t)\right)dt \\ &= \frac{1}{2}(x_i y_{i+1} - x_{i+1}y_i).\end{aligned}$$

Summing over the edges yields

$$\boxed{\operatorname{Area}(D) = \frac{1}{2}\left|\sum_{i=1}^{n}(x_i y_{i+1} - x_{i+1}y_i)\right|.}$$

The familiar crossing columns are therefore a mnemonic for a discrete line integral, not the reason the formula is true.

§20.4 Translation and Orientation Checks

The shoelace sum should not depend on where the coordinate origin is placed. Translate every vertex by a fixed vector t. Then

$$\det(p_i + t, p_{i+1} + t) = \det(p_i, p_{i+1}) + \det(t, p_{i+1} - p_i).$$

After summing around the closed polygon, the extra term vanishes because

$$\sum_{i=1}^{n}(p_{i+1} - p_i) = 0.$$

Thus the signed area is translation invariant even though each individual determinant uses position vectors from the origin.

Reversing the cyclic order changes every edge orientation and therefore changes the sign of the sum. The absolute value in the ordinary shoelace formula forgets this orientation. For a self-intersecting closed polygonal path, the same sum computes signed area with winding multiplicity; overlapping regions may cancel rather than simply add.

§20.5 A complete polygon calculation

Take the vertices

$$(3,4),\ (5,11),\ (12,8),\ (9,5),\ (5,6)$$

in the displayed cyclic order. The signed double area is

$$\begin{aligned}2A = {} & 3\cdot 11 - 5\cdot 4 + 5\cdot 8 - 12\cdot 11 \\ & + 12\cdot 5 - 9\cdot 8 + 9\cdot 6 - 5\cdot 5 \\ & + 5\cdot 4 - 3\cdot 6 \\ = {} & -60.\end{aligned}$$

Hence

$$A = 30.$$

The negative sign only says that the listed orientation is clockwise. Reversing the order would give $+60$ without changing the ordinary area.

§20.6 Evaluation removes every multiple of $x - a$

Let k be a field and $f \in k[x]$. Polynomial division gives unique $q \in k[x]$ and $r \in k$ such that

$$f(x) = (x - a)q(x) + r.$$

Substituting $x = a$ kills the divisible part:

$$f(a) = r.$$

Thus the remainder theorem is the statement

$$f(x) \equiv f(a) \pmod{x - a}.$$

This is naturally encoded by the evaluation homomorphism

$$\mathrm{ev}_a : k[x] \longrightarrow k, \qquad f \longmapsto f(a).$$

Its kernel is the ideal $(x - a)$, so the first isomorphism theorem gives

$$k[x]/(x - a) \cong k.$$

The quotient notation is useful because it says exactly what has been discarded: two polynomials become equal when their difference is divisible by $x - a$.

§20.7 Remainders as Interpolation Data

Suppose $f(x)$ leaves remainder 99 modulo $x - 19$ and remainder 19 modulo $x - 99$. The remainder modulo

$$(x - 19)(x - 99)$$

has degree at most one, so write

$$R(x) = ux + v.$$

It must satisfy

$$R(19) = 99, \qquad R(99) = 19.$$

Hence

$$19u + v = 99, \qquad 99u + v = 19,$$

and therefore

$$u = -1, \qquad v = 118.$$

Thus

$$\boxed{R(x) = 118 - x.}$$

This is not a separate trick from interpolation. A polynomial of degree at most one is uniquely determined by its values at two distinct points. In Lagrange form,

$$R(x) = 99\frac{x - 99}{19 - 99} + 19\frac{x - 19}{99 - 19},$$

which simplifies to the same answer.

§20.8 Polynomial CRT and Uniqueness

Because $19 \neq 99$, the ideals $(x-19)$ and $(x-99)$ are comaximal. The Chinese remainder theorem gives

$$k[x]/((x-19)(x-99)) \cong k[x]/(x-19) \times k[x]/(x-99).$$

Using evaluation on each factor,

$$k[x]/((x-19)(x-99)) \cong k \times k.$$

A residue class modulo the quadratic is therefore exactly a pair of values

$$(f(19), f(99)).$$

The linear remainder found above is the unique degree-<2 representative of the pair $(99, 19)$.

More generally, for distinct $a_1, \ldots, a_n$,

$$k[x]/\prod_{i=1}^{n}(x-a_i) \cong k^n,$$

and the inverse map is Lagrange interpolation. The elementary remainder theorem has grown into a structural statement without changing the computation it was designed to perform.

§20.9 Repeated Factors and Hermite Data

Evaluation modulo $x-a$ forgets everything except $f(a)$. A natural question is what remains visible modulo a higher power $(x-a)^2$. Write the exact polynomial Taylor expansion

$$f(x) = f(a) + f'(a)(x-a) + (x-a)^2 q(x),$$

where $q \in \mathbb{Q}[x]$. Then

$$\boxed{f(x) \equiv f(a) + f'(a)(x-a) \pmod{(x-a)^2}.}$$

Two polynomials agree modulo $(x-a)^2$ precisely when their values *and first derivatives* agree at a. Repeated linear factors remember a little more local information than an ordinary point evaluation.

This observation extends the preceding Chinese-remainder example. Suppose we wish to find a polynomial R of degree less than three such that

$$R(1) = 2, \qquad R'(1) = 3, \qquad R(-1) = 4.$$

Start with the first-order Taylor data at 1:

$$R(x) = 2 + 3(x-1) + c(x-1)^2.$$

The remaining value condition gives

$$4 = R(-1) = 2 - 6 + 4c, \qquad c = 2.$$

Thus

$$\boxed{R(x) = 2x^2 - x + 1.}$$

Check directly that $R(1) = 2$, $R'(1) = 4 - 1 = 3$, and $R(-1) = 4$. The calculation has not secretly used three unrelated equations: it is the polynomial Chinese remainder theorem for the coprime factors $(x-1)^2$ and $(x+1)$,

$$\mathbb{Q}[x]/((x-1)^2(x+1)) \cong \mathbb{Q}[x]/(x-1)^2 \times \mathbb{Q}[x]/(x+1).$$

The first component is the pair of numbers $(f(1), f'(1))$, and the second component is $f(-1)$. The total degree bound is $2 + 1 = 3$, matching the three scalar data.

Why is this solution unique? If R and S both have the three prescribed data, their difference is divisible by $(x-1)^2$ and by $(x+1)$. Since these factors are coprime, their product divides $R - S$. But $\deg(R - S) < 3$; therefore $R = S$. This proves uniqueness without a determinant calculation.

§20.10 Infinitesimals in Quotient Rings

There is a particularly memorable way to interpret the first component. Introduce a symbol ε with $\varepsilon^2 = 0$, but *do not* assume $\varepsilon = 0$. The ring

$$\mathbb{Q}[\varepsilon]/(\varepsilon^2)$$

is called the ring of dual numbers. The substitution $x = 1 + \varepsilon$ makes every quadratic and higher term disappear:

$$f(1 + \varepsilon) = f(1) + f'(1)\varepsilon.$$

For instance,

$$(1 + \varepsilon)^5 = 1 + 5\varepsilon + 10\varepsilon^2 + \cdots = 1 + 5\varepsilon.$$

It is easy to verify that

$$\mathbb{Q}[x]/(x-1)^2 \cong \mathbb{Q}[\varepsilon]/(\varepsilon^2)$$

by sending the class of $x-1$ to ε. More generally, the quotient modulo $(x-a)^k$ remembers the first k coefficients of the Taylor expansion of f at a (over a characteristic-zero field).

This is the algebraic germ of a derivative, or a *jet*. We have not introduced a real number smaller than every positive real number: ε lives in a different ring and satisfies $\varepsilon^2 = 0$. Nevertheless, first-order calculus becomes exact arithmetic there. The same distinction between a point value and its first-order neighborhood later appears in tangent vectors, differential geometry, and automatic differentiation.[1]

§20.11 Integer congruence is another quotient arithmetic

For integers, write

$$a \equiv b \pmod{m}$$

when $m \mid (a - b)$. Arithmetic then takes place in the quotient ring

$$\mathbb{Z}/m\mathbb{Z}.$$

Addition and multiplication descend because replacing an integer by another representative of the same class does not change the resulting class.

[1] For Hermite interpolation viewed through Taylor data and polynomial congruences, see Bossoney and Troyanov, *A Direct Approach to Hermite Interpolation*, Sections 1–3.

Division is more delicate. From

$$ca \equiv cb \pmod{m}$$

one may cancel c only when its residue class is a unit, equivalently when

$$\gcd(c, m) = 1.$$

For example,

$$2x \equiv 2 \pmod{6}$$

has the two solutions

$$x \equiv 1, 4 \pmod{6}.$$

Cancelling 2 would incorrectly discard one of them. Quotient arithmetic preserves addition and multiplication automatically, but inverse operations require invertibility.

Small residue sets give quick impossibility tests. Squares satisfy

$$n^2 \equiv 0, 1 \pmod{3}$$

and

$$n^2 \equiv 0, 1, 4 \pmod{8}.$$

A proposed square lying outside these lists is impossible before any large-number calculation begins.

§20.12 Finite unit groups turn huge powers into short cycles

If p is prime and $p \nmid a$, Fermat's theorem says

$$a^{p-1} \equiv 1 \pmod{p}.$$

The reason is that the nonzero residue classes form the finite group

$$(\mathbb{Z}/p\mathbb{Z})^\times$$

of order $p-1$. Lagrange's theorem makes the exponent of every element divide the group order.

To compute

$$2^{1000} \pmod{13},$$

use

$$2^{12} \equiv 1 \pmod{13}$$

and reduce the exponent:

$$1000 = 83 \cdot 12 + 4.$$

Therefore

$$2^{1000} \equiv 2^4 = 16 \equiv 3 \pmod{13}.$$

The quotient ring has retained precisely the residue information while discarding the enormous integer itself.

The three parts of the chapter now end differently. The polygon formula converts a closed boundary into oriented area. Polynomial evaluation converts divisibility by linear factors into interpolation data. Modular arithmetic converts integers into residue classes and exposes finite cycles. Their common practical habit is to choose the correct invariant map before expanding the calculation.

Integer Arithmetic: Euclidean Descent, Bézout Identities, and the Euler Function

N-20221030

§21.1 Descent is the algorithmic form of well-ordering

Many arguments about integers begin with a simple impossibility: a strictly decreasing sequence of nonnegative integers cannot continue forever. This is the computational face of the well-ordering principle.

Theorem 21.1.1 (Well-ordering principle)
Every nonempty subset of $\mathbb{Z}_{>0}$ has a least element.

Induction, minimal-counterexample arguments, and the Euclidean algorithm all use the same structure. One assumes that some positive integer with a desired property exists, chooses the least such integer, and then produces a smaller one unless the process has already reached its terminal case.

The division algorithm is the first important example.

Theorem 21.1.2 (Division algorithm)
Let $a, b \in \mathbb{Z}$ with $b \neq 0$. There exist unique integers q, r such that

$$a = qb + r, \qquad 0 \leq r < |b|.$$

To see existence, consider all nonnegative integers of the form

$$a - k|b|, \qquad k \in \mathbb{Z}.$$

This set is nonempty: for sufficiently negative k, the number is positive. Let r be its least element, so

$$r = a - q|b|$$

for some integer q. If $r \geq |b|$, then

$$r - |b| = a - (q+1)|b|$$

would be a smaller nonnegative element of the same set, a contradiction. Hence $0 \leq r < |b|$. A sign adjustment replaces $|b|$ by b.

For uniqueness, suppose

$$a = qb + r = q'b + r'$$

with both remainders in $[0, |b|)$. Then

$$(q - q')b = r' - r.$$

The right side has absolute value less than $|b|$, while a nonzero multiple of b has absolute value at least $|b|$. Therefore both sides vanish.

The theorem is more than a formal version of long division. It supplies a remainder whose size is strictly smaller than the divisor, and that strict decrease is what makes Euclidean descent possible.

§21.2 Euclid's GCD Invariant

For integers a, b, not both zero, the greatest common divisor $d = \gcd(a, b)$ is the unique positive integer satisfying

$$d \mid a, \qquad d \mid b,$$

and every common divisor of a, b divides d.

If

$$a = qb + r,$$

then

$$\gcd(a, b) = \gcd(b, r).$$

Indeed, a number divides both a and b exactly when it divides both b and

$$r = a - qb.$$

The set of common divisors has not changed.

This invariant turns division into an algorithm. First take absolute values and arrange the inputs so that $a \geq b > 0$; a zero input has an immediate gcd. Now repeatedly replace a pair by its divisor and remainder:

$$\begin{aligned} a &= q_0 b + r_0, \\ b &= q_1 r_0 + r_1, \\ r_0 &= q_2 r_1 + r_2, \\ &\vdots \end{aligned}$$

with

$$b > r_0 > r_1 > r_2 > \cdots \geq 0.$$

Stop when the next remainder is zero. Every nonzero remainder is smaller than the previous positive divisor, so well-ordering guarantees termination. The last nonzero remainder is the gcd because the common-divisor invariant survives every step.

The important object is not the displayed chain of divisions but the transformation

$$(a, b) \longmapsto (b, a - qb),$$

which preserves the ideal of integer linear combinations as well as the gcd.

§21.3 A complete Euclidean calculation

Take

$$a = 57970, \qquad b = 10353.$$

Repeated division gives

$$\begin{aligned} 57970 &= 5 \cdot 10353 + 6205, \\ 10353 &= 1 \cdot 6205 + 4148, \\ 6205 &= 1 \cdot 4148 + 2057, \\ 4148 &= 2 \cdot 2057 + 34, \\ 2057 &= 60 \cdot 34 + 17, \\ 34 &= 2 \cdot 17. \end{aligned}$$

Hence

$$\gcd(57970, 10353) = 17.$$

The same computation contains more information than the gcd. Reading the equations backward expresses 17 as an integer linear combination of the original numbers:

$$17 = 2057 - 60 \cdot 34.$$

Since

$$34 = 4148 - 2 \cdot 2057,$$

we obtain

$$17 = 121 \cdot 2057 - 60 \cdot 4148.$$

Now use

$$2057 = 6205 - 4148$$

to get

$$17 = 121 \cdot 6205 - 181 \cdot 4148.$$

Then

$$4148 = 10353 - 6205$$

gives

$$17 = 302 \cdot 6205 - 181 \cdot 10353.$$

Finally,

$$6205 = 57970 - 5 \cdot 10353,$$

so

$$\boxed{17 = 302 \cdot 57970 - 1691 \cdot 10353.}$$

This backward pass is the extended Euclidean algorithm. It computes coefficients, not merely a divisor.

§21.4 Bézout's identity is an ideal statement

For integers a, b, consider the set

$$(a, b) = \{ax + by : x, y \in \mathbb{Z}\}.$$

This is an ideal of $\mathbb{Z}$: it is closed under addition and under multiplication by arbitrary integers.

Let d be the least positive element of this ideal. Dividing a by d, write

$$a = qd + r, \qquad 0 \le r < d.$$

Because both a and d lie in the ideal, so does

$$r = a - qd.$$

Minimality of d forces $r = 0$, so $d \mid a$. Similarly, $d \mid b$. Conversely, every common divisor of a, b divides every linear combination $ax + by$, hence divides d. Therefore

$$d = \gcd(a, b).$$

We have proved Bézout's identity:

$$\boxed{\gcd(a,b) = ax + by}$$

for suitable integers x, y.

In ideal notation,

$$(a) + (b) = (\gcd(a,b)).$$

The gcd is the positive generator of the sum of the two principal ideals.

This viewpoint explains why the extended Euclidean algorithm is so useful: it is a constructive proof that a particular ideal is principal. More generally, a Euclidean domain is an integral domain equipped with a size function that permits division with smaller remainder. Euclidean descent then produces gcds and Bézout identities. The rings

$$\mathbb{Z} \qquad \text{and} \qquad k[x]$$

for a field k are the standard examples.

Every Euclidean domain is a principal ideal domain, and every principal ideal domain is a unique factorization domain. The converses do not all hold. The elementary integer algorithm is therefore the visible beginning of a hierarchy of algebraic properties.

§21.5 Coprimality is exactly invertibility modulo n

Suppose

$$\gcd(a,n) = 1.$$

Bézout gives integers x, y such that

$$ax + ny = 1.$$

Reducing modulo n,

$$ax \equiv 1 \pmod{n}.$$

Thus x is a multiplicative inverse of a modulo n.

Conversely, if a has an inverse modulo n, then

$$ax \equiv 1 \pmod{n}$$

for some x, so

$$ax + ny = 1$$

for an integer y. Any common divisor of a, n must divide 1, and therefore

$$\gcd(a,n) = 1.$$

Hence

$$[a] \in (\mathbb{Z}/n\mathbb{Z})^{\times} \quad \Longleftrightarrow \quad \gcd(a,n) = 1.$$

The extended Euclidean algorithm is the computational heart of inversion in the quotient ring.

For example, the identity

$$17 = 302 \cdot 57970 - 1691 \cdot 10353$$

can be divided by 17 after first reducing the numbers:

$$1 = 302 \cdot 3410 - 1691 \cdot 609.$$

Thus

$$302 \cdot 3410 \equiv 1 \pmod{609},$$

so 302 is the inverse of 3410 modulo 609, or equivalently the inverse of its residue class 365.

Modular inverses reveal the connection between Bézout identities and unit groups. Linear congruences and simultaneous congruence systems require a separate treatment through residue-class maps and integer normal forms.

§21.6 Prime divisibility makes factorization rigid

A prime p is an integer greater than 1 whose only positive divisors are 1 and p. The key lemma is not merely the definition but Euclid's divisibility property.

Proposition 21.6.1 (Euclid's lemma)

If p is prime and

$$p \mid ab,$$

then

$$p \mid a \quad \text{or} \quad p \mid b.$$

If $p \nmid a$, then $\gcd(p, a) = 1$. Bézout gives

$$px + ay = 1.$$

Multiplying by b,

$$pbx + aby = b.$$

Both terms on the left are divisible by p, so $p \mid b$.

This short proof shows how uniqueness of prime factorization grows from Bézout's identity.

Theorem 21.6.2 (Fundamental theorem of arithmetic)

Every integer $n > 1$ can be written as a product of primes, and the product is unique up to reordering.

Existence follows by minimal counterexample. If the least integer without a prime factorization were composite, it would split into two smaller positive integers, both of which factor by minimality.

For uniqueness, suppose

$$p_1 p_2 \cdots p_r = q_1 q_2 \cdots q_s.$$

Then p_1 divides the product on the right. Euclid's lemma says it divides one q_j, and since both are prime they are equal. Cancel this factor and continue.

Writing

$$n = \prod_p p^{v_p(n)},$$

the exponent $v_p(n)$ is the p-adic valuation of n. Gcd and lcm become coordinatewise minimum and maximum:

$$v_p(\gcd(a, b)) = \min\{v_p(a), v_p(b)\},$$

$$v_p(\operatorname{lcm}(a,b)) = \max\{v_p(a), v_p(b)\}.$$

Therefore, for positive a, b,

$$\gcd(a,b)\operatorname{lcm}(a,b) = ab.$$

Prime factorization turns divisibility into arithmetic on exponent vectors.

§21.7 Euler's function counts the unit group

Euler's totient function is

$$\varphi(n) = \#\{1 \le a \le n : \gcd(a,n) = 1\}.$$

The preceding section gives its structural meaning:

$$\boxed{\varphi(n) = \left|(\mathbb{Z}/n\mathbb{Z})^\times\right|.}$$

For a prime power p^k, the nonunits are exactly the multiples of p. Among the p^k residue classes, there are p^{k-1} such multiples. Hence

$$\varphi(p^k) = p^k - p^{k-1} = p^k\left(1 - \frac{1}{p}\right).$$

If m, n are coprime, the Chinese remainder theorem gives a ring isomorphism

$$\mathbb{Z}/mn\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}.$$

An element of a product ring is a unit exactly when each component is a unit. Therefore

$$(\mathbb{Z}/mn\mathbb{Z})^\times \cong (\mathbb{Z}/m\mathbb{Z})^\times \times (\mathbb{Z}/n\mathbb{Z})^\times,$$

and

$$\varphi(mn) = \varphi(m)\varphi(n) \qquad \text{when } \gcd(m,n) = 1.$$

Thus φ is a multiplicative arithmetic function. If

$$n = \prod_{p \mid n} p^{v_p(n)},$$

then

$$\boxed{\varphi(n) = n\prod_{p \mid n}\left(1 - \frac{1}{p}\right).}$$

For example,

$$360 = 2^3 \cdot 3^2 \cdot 5,$$

so

$$\varphi(360) = 360\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{5}\right) = 96.$$

This is simultaneously an inclusion–exclusion count and the order of a finite group.

§21.8 Euler's theorem is Lagrange's theorem in disguise

If $\gcd(a, n) = 1$, then $[a]$ belongs to the finite group

$$G = (\mathbb{Z}/n\mathbb{Z})^{\times},$$

whose order is $\varphi(n)$. Lagrange's theorem implies that the order of $[a]$ divides $|G|$, so

$$\boxed{a^{\varphi(n)} \equiv 1 \pmod{n}.}$$

There is also a proof that looks elementary but contains the same group action. Let

$$r_1, \dots, r_{\varphi(n)}$$

be a reduced residue system modulo n. Multiplication by a permutes these residues because $[a]$ is a unit. Hence

$$ar_1, \dots, ar_{\varphi(n)}$$

represent the same residue classes in a different order. Taking products,

$$a^{\varphi(n)} r_1 \cdots r_{\varphi(n)} \equiv r_1 \cdots r_{\varphi(n)} \pmod{n}.$$

The product of the r_i is a unit and can be cancelled, giving Euler's theorem.

The group-theoretic proof explains both the exponent and the permutation. The elementary proof displays the action concretely.

The exponent $\varphi(n)$ is always valid but is not usually minimal. The least positive exponent m satisfying

$$a^m \equiv 1 \pmod{n}$$

for every unit a is the exponent of the group, called the Carmichael function $\lambda(n)$. Group structure refines the coarse bound supplied by group order.

§21.9 Counting Elements by Order

A useful identity is

$$\boxed{\sum_{d \mid n} \varphi(d) = n.}$$

One proof counts the integers

$$1, 2, \dots, n$$

according to their gcd with n.

Fix a divisor $d \mid n$. The integers k satisfying

$$\gcd(k, n) = \frac{n}{d}$$

are exactly those of the form

$$k = \frac{n}{d}\, a$$

with

$$1 \le a \le d, \qquad \gcd(a, d) = 1.$$

There are $\varphi(d)$ of them. Summing over all divisors partitions the n integers and gives the identity.

There is also a group-theoretic interpretation. In a cyclic group of order n, the number of elements of exact order d is $\varphi(d)$ for every $d \mid n$. Counting all group elements by their order gives the same sum.

The identity is an early example of arithmetic convolution. For arithmetic functions f, g, define the Dirichlet convolution

$$(f * g)(n) = \sum_{d|n} f(d)g(n/d).$$

Let $\mathbf{1}(n) = 1$ and $\mathrm{id}(n) = n$. Then

$$\mathbf{1} * \varphi = \mathrm{id}\,.$$

The convolution identity can be inverted using the Möbius function μ, which satisfies

$$\mu * \mathbf{1} = \varepsilon,$$

where $\varepsilon(1) = 1$ and $\varepsilon(n) = 0$ for $n > 1$. Therefore

$$\varphi = \mu * \mathrm{id},$$

or explicitly,

$$\varphi(n) = \sum_{d|n} \mu(d)\frac{n}{d}.$$

For squarefree d, $\mu(d) = (-1)^k$ when d has k prime factors, so this formula is inclusion–exclusion written as an algebraic inverse.

This is the natural point where elementary counting begins to resemble analytic number theory: multiplicative functions form an algebra under Dirichlet convolution, and Euler products arise because prime-power data determine multiplicative functions.

The route from Euclidean descent to arithmetic convolution is continuous rather than a stack of unrelated viewpoints. The step

$$(a, b) \longmapsto (b, a - qb)$$

preserves common divisors and the generated ideal; back-substitution turns it into a Bézout identity and computes modular inverses. Euclid's lemma then converts coprimality into uniqueness of factorization, prime powers determine the unit count $\varphi(n)$, the Chinese remainder theorem explains multiplicativity, and counting by exact divisors leads to Dirichlet convolution and Möbius inversion. The later language does not replace the algorithm: it records the structures the same descent has been preserving from the beginning.

Congruences and Integer Linear Algebra: From Residue Classes to Smith Normal Form

N-20221031

§22.1 A remainder is a coordinate on a quotient

The division algorithm assigns to every integer a a unique remainder

$$r \in \{0, 1, \dots, n-1\}$$

when divided by a positive integer n. Two integers have the same remainder exactly when their difference is divisible by n:

$$a \equiv b \pmod{n} \quad \Longleftrightarrow \quad n \mid (a-b).$$

Congruence is therefore not approximate equality. It is equality in the quotient ring

$$\mathbb{Z}/n\mathbb{Z}.$$

The remainder is a chosen representative, or normal form, for the residue class.

Addition and multiplication descend to the quotient because replacing an integer by a congruent one does not change the resulting class:

$$a \equiv a', \qquad b \equiv b' \pmod{n}$$

implies

$$a+b \equiv a'+b', \qquad ab \equiv a'b' \pmod{n}.$$

The quotient remembers arithmetic while forgetting multiples of n.

Bézout identities and modular inverses are the starting tools, but the main questions are now different: which residue classes can a linear map reach, how do several moduli fit together, and what replaces row reduction when equations must be solved over the integers rather than over a field?

§22.2 A linear congruence is an image problem

Consider

$$ax \equiv b \pmod{n}.$$

Multiplication by a defines a group homomorphism

$$m_a : \mathbb{Z}/n\mathbb{Z} \longrightarrow \mathbb{Z}/n\mathbb{Z}, \qquad [x] \longmapsto [ax].$$

The congruence asks whether $[b]$ lies in the image of this map.

Let

$$d = \gcd(a, n).$$

Every value ax is divisible by d modulo n, so solvability requires

$$d \mid b.$$

Conversely, suppose $b = db'$, and write

$$a = da', \qquad n = dn'.$$

Then $\gcd(a', n') = 1$, so a' has an inverse modulo n'. The reduced congruence

$$a'x \equiv b' \pmod{n'}$$

has a solution, and hence so does the original one.

Thus

$$\boxed{ax \equiv b \pmod{n} \text{ is solvable } \iff \gcd(a, n) \mid b.}$$

The kernel explains the number of solutions. We have

$$ax \equiv 0 \pmod{n}$$

exactly when

$$n' \mid x.$$

Modulo $n = dn'$, this gives the d classes

$$0, n', 2n', \ldots, (d-1)n'.$$

Therefore

$$|\ker m_a| = d.$$

By the finite-group version of rank–nullity,

$$|\operatorname{im} m_a| = \frac{n}{d}.$$

Whenever $[b]$ lies in the image, its fibre is a coset of the kernel and contains exactly d solutions modulo n.

This is the structural form of the elementary theorem: divisibility decides existence, and the kernel decides multiplicity.

§22.3 A complete congruence calculation

Solve

$$14x \equiv 8 \pmod{30}.$$

Since

$$\gcd(14, 30) = 2$$

and $2 \mid 8$, solutions exist. Divide the entire congruence and modulus by 2:

$$7x \equiv 4 \pmod{15}.$$

The inverse of 7 modulo 15 is 13, because

$$7 \cdot 13 = 91 \equiv 1 \pmod{15}.$$

Therefore

$$x \equiv 13 \cdot 4 \equiv 52 \equiv 7 \pmod{15}.$$

This describes one class modulo 15, but the original modulus was 30. Lifting back gives

$$x \equiv 7 \qquad \text{or} \qquad x \equiv 22 \pmod{30}.$$

There are exactly two solutions, as predicted by

$$d = \gcd(14, 30) = 2.$$

A direct check gives

$$14 \cdot 7 = 98 \equiv 8 \pmod{30},$$
$$14 \cdot 22 = 308 \equiv 8 \pmod{30}.$$

The calculation illustrates the general procedure:

(i) compute $d = \gcd(a, n)$;

(ii) test whether $d \mid b$;

(iii) divide by d, producing a coefficient invertible modulo n/d;

(iv) solve the reduced congruence;

(v) lift its single class modulo n/d to d classes modulo n.

§22.4 The Chinese remainder theorem glues local coordinates

Suppose m, n are coprime. The map

$$\Phi : \mathbb{Z}/mn\mathbb{Z} \longrightarrow \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z},$$
$$[x]_{mn} \longmapsto ([x]_m, [x]_n)$$

is a ring homomorphism.

Its kernel consists of classes divisible by both m and n. Coprimality implies that such an integer is divisible by mn, so the kernel is zero. Both rings contain mn elements, hence the injective map is also surjective.

Therefore

$$\boxed{\mathbb{Z}/mn\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}.}$$

This is the Chinese remainder theorem. It does not merely promise a solution to simultaneous congruences; it says that arithmetic modulo a product of coprime moduli is equivalent to independent arithmetic in each factor.

The isomorphism can be made constructive. Choose Bézout coefficients

$$um + vn = 1.$$

Then

$$e_m = vn$$

satisfies

$$e_m \equiv 1 \pmod{m}, \qquad e_m \equiv 0 \pmod{n},$$

while

$$e_n = um$$

satisfies the opposite conditions. The solution to

$$x \equiv a \pmod{m}, \qquad x \equiv b \pmod{n}$$

is

$$x \equiv ae_m + be_n \pmod{mn}.$$

The elements e_m, e_n are orthogonal idempotents:

$$e_m^2 = e_m, \qquad e_n^2 = e_n, \qquad e_m e_n = 0, \qquad e_m + e_n = 1$$

in $\mathbb{Z}/mn\mathbb{Z}$. They project onto the two ring factors.

§22.5 A three-modulus reconstruction

Solve

$$x \equiv 2 \pmod 3, \qquad x \equiv 3 \pmod 5, \qquad x \equiv 2 \pmod 7.$$

The moduli are pairwise coprime, so there is a unique solution modulo

$$3 \cdot 5 \cdot 7 = 105.$$

Set

$$N_1 = 35, \qquad N_2 = 21, \qquad N_3 = 15.$$

We need inverses of these partial products in the corresponding moduli:

$$35 \equiv 2 \pmod 3, \qquad 2^{-1} \equiv 2 \pmod 3,$$

$$21 \equiv 1 \pmod 5, \qquad 1^{-1} \equiv 1 \pmod 5,$$

$$15 \equiv 1 \pmod 7, \qquad 1^{-1} \equiv 1 \pmod 7.$$

Hence

$$x \equiv 2 \cdot 35 \cdot 2 + 3 \cdot 21 \cdot 1 + 2 \cdot 15 \cdot 1 \pmod{105}.$$

The right side is

$$140 + 63 + 30 = 233,$$

so

$$\boxed{x \equiv 23 \pmod{105}.}$$

Indeed,

$$23 \equiv 2 \pmod 3, \qquad 23 \equiv 3 \pmod 5, \qquad 23 \equiv 2 \pmod 7.$$

The reconstruction is a weighted sum of local answers. Each weight equals 1 in one component and 0 in all the others.

§22.6 When moduli are not coprime

Consider

$$x \equiv a \pmod m, \qquad x \equiv b \pmod n.$$

If a solution exists, then

$$a - b = (a - x) + (x - b)$$

is divisible by

$$d = \gcd(m, n).$$

Thus a necessary condition is

$$a \equiv b \pmod d.$$

It is also sufficient. Write

$$a - b = dc, \qquad m = dm', \qquad n = dn'$$

with $\gcd(m', n') = 1$. Searching for $x = a + mt$, the second congruence becomes

$$m't \equiv -c \pmod{n'},$$

which is solvable because m' is invertible modulo n'.

Therefore

$$\boxed{\begin{array}{l} x \equiv a \pmod{m}, \\ x \equiv b \pmod{n} \end{array} \text{ is solvable } \iff a \equiv b \pmod{\gcd(m,n)}.}$$

When solvable, the solution is unique modulo

$$\operatorname{lcm}(m,n).$$

The coprime CRT is the clean product decomposition. The general version is a compatibility-and-gluing theorem: local data must agree on the overlap measured by the gcd.

§22.7 Invertible Integer Row Operations

Over $\mathbb{R}$, Gaussian elimination allows multiplication of a row by any nonzero real number. Over $\mathbb{Z}$, this can destroy the integer lattice. Dividing a row by 2, for example, is invertible over $\mathbb{Q}$ but not as a map

$$\mathbb{Z}^m \longrightarrow \mathbb{Z}^m.$$

The correct coordinate changes are unimodular matrices: integer matrices U with

$$\det U = \pm 1.$$

Their inverses also have integer entries. The elementary unimodular operations are:

(i) exchange two rows or two columns;

(ii) add an integer multiple of one row or column to another;

(iii) multiply a row or column by -1.

These operations change bases of the domain or codomain lattice without changing the underlying homomorphism up to integer coordinates.

This is the essential difference between linear algebra over a field and linear algebra over $\mathbb{Z}$. Rank records the dimension after rational or real scalars are allowed, but it forgets divisibility information inside the lattice.

For example, the maps

$$[2] : \mathbb{Z} \to \mathbb{Z} \qquad \text{and} \qquad [3] : \mathbb{Z} \to \mathbb{Z}$$

both have rank 1, yet their images have indices 2 and 3, and their cokernels are

$$\mathbb{Z}/2\mathbb{Z} \qquad \text{and} \qquad \mathbb{Z}/3\mathbb{Z}.$$

An integer normal form must remember these indices.

§22.8 Smith normal form diagonalizes divisibility

Theorem 22.8.1 (Smith normal form)

For every integer matrix $A \in \mathbb{Z}^{m\times n}$, there exist unimodular matrices

$$P \in GL_m(\mathbb{Z}), \qquad Q \in GL_n(\mathbb{Z})$$

such that

$$PAQ = \operatorname{diag}(d_1, \dots, d_r, 0, \dots, 0),$$

where

$$d_i > 0, \qquad d_1 \mid d_2 \mid \cdots \mid d_r.$$

The nonzero numbers d_i are uniquely determined by A.

The algorithm repeatedly uses gcd and Bézout operations on rows and columns. A nonzero entry is moved to the upper-left corner. Euclidean combinations reduce it until it divides every entry in its row and column. One then clears that row and column and continues on the remaining submatrix. If the divisibility chain fails, adjacent diagonal entries are adjusted by another Bézout step.

Thus Smith reduction is the matrix-sized descendant of the Euclidean algorithm. A single gcd becomes a sequence of invariant factors.

The determinantal divisors provide an intrinsic description. Let Δ_k be the positive gcd of all $k \times k$ minors, with $\Delta_0 = 1$. Then

$$d_k = \frac{\Delta_k}{\Delta_{k-1}}.$$

In particular,

$$d_1 = \gcd\{a_{ij}\},$$

and for a full-rank square matrix,

$$d_1 d_2 \cdots d_n = |\det A|.$$

The invariant factors are therefore encoded in the gcds of minors.

§22.9 A complete Smith reduction

Take

$$A = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}.$$

Apply the integer row operation

$$R_2 \leftarrow R_2 - 3R_1 :$$

$$\begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix} \longrightarrow \begin{bmatrix} 2 & 4 \\ 0 & -4 \end{bmatrix}.$$

Then

$$C_2 \leftarrow C_2 - 2C_1$$

gives

$$\begin{bmatrix} 2 & 0 \\ 0 & -4 \end{bmatrix}.$$

Finally change the sign of the second row:

$$\boxed{PAQ = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}.}$$

One choice is

$$P = \begin{bmatrix} 1 & 0 \\ 3 & -1 \end{bmatrix}, \qquad Q = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}.$$

Both determinants are -1 or 1, so both matrices are unimodular.

The invariant factors are $2, 4$. They also follow from minors:

$$\Delta_1 = \gcd(2, 4, 6, 8) = 2,$$

$$\Delta_2 = |\det A| = 8,$$

so

$$d_1 = 2, \qquad d_2 = \frac{8}{2} = 4.$$

Over $\mathbb{Q}$, the matrix is simply invertible. Over $\mathbb{Z}$, the diagonal entries reveal exactly how its image sits inside $\mathbb{Z}^2$.

§22.10 Integer systems reduce to divisibility tests

Consider

$$Ax = b, \qquad A \in \mathbb{Z}^{m \times n}, \quad x \in \mathbb{Z}^n.$$

If

$$PAQ = D$$

is Smith normal form, set

$$y = Q^{-1}x, \qquad c = Pb.$$

Since P, Q are unimodular, x is integral exactly when y is integral. The system becomes

$$Dy = c.$$

For

$$D = \operatorname{diag}(d_1, \ldots, d_r, 0, \ldots, 0),$$

integer solvability is equivalent to

$$d_i \mid c_i \qquad (1 \le i \le r)$$

and

$$c_i = 0 \qquad (i > r).$$

The normal form separates the problem into independent one-dimensional divisibility conditions. This is the matrix analogue of the criterion

$$ax \equiv b \pmod{n} \iff \gcd(a, n) \mid b.$$

For the example matrix, solving

$$Ax = b$$

reduces after multiplication by P to

$$2y_1 = c_1, \qquad 4y_2 = c_2.$$

The right-hand side must therefore land in the sublattice

$$2\mathbb{Z} \oplus 4\mathbb{Z}.$$

Smith form does not merely decide solvability; it displays the obstruction.

§22.11 The cokernel records what the map cannot reach

For an integer matrix

$$A : \mathbb{Z}^n \longrightarrow \mathbb{Z}^m,$$

define the cokernel

$$\operatorname{coker} A = \mathbb{Z}^m / A\mathbb{Z}^n.$$

Two target vectors represent the same cokernel class when their difference lies in the image of A. The group measures the failure of surjectivity over the integer lattice.

Smith normal form gives

$$\operatorname{coker} A \cong \mathbb{Z}/d_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/d_r\mathbb{Z} \oplus \mathbb{Z}^{m-r}.$$

For

$$A = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix},$$

we obtain

$$\operatorname{coker} A \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}.$$

Its order is

$$2 \cdot 4 = 8 = |\det A|.$$

Geometrically, the image lattice has index 8 in $\mathbb{Z}^2$. A fundamental parallelogram for the image has area 8, matching the determinant.

This is the finite-index lattice meaning of the determinant. Over $\mathbb{R}$, $|\det A|$ is a volume-scaling factor. Over $\mathbb{Z}$, for a nonsingular integer matrix, it is also the number of residue classes left after quotienting by the image.

§22.12 Finite abelian groups inherit the same normal form

Every finitely generated abelian group has a presentation

$$G \cong \mathbb{Z}^m / A\mathbb{Z}^n$$

for some integer matrix A. Smith normal form yields

$$G \cong \mathbb{Z}^{m-r} \oplus \mathbb{Z}/d_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/d_r\mathbb{Z},$$

with

$$d_1 \mid d_2 \mid \cdots \mid d_r.$$

This is the invariant-factor form of the classification theorem for finitely generated abelian groups.

The CRT reappears inside this classification. If a, b are coprime, then

$$\mathbb{Z}/ab\mathbb{Z} \cong \mathbb{Z}/a\mathbb{Z} \oplus \mathbb{Z}/b\mathbb{Z}.$$

Factoring each invariant factor into prime powers gives the elementary-divisor form of the same group.

Thus CRT and Smith normal form are not separate tricks. Both decompose arithmetic objects into independent pieces, one by splitting coprime moduli and the other by diagonalizing an integer homomorphism.

§22.13 Congruences and Smith Form

A system

$$Ax \equiv b \pmod{n}$$

can be rewritten as an integer equation

$$Ax - ny = b$$

with an auxiliary integer vector y. Equivalently,

$$\begin{bmatrix} A & -nI \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = b.$$

Smith normal form of this enlarged integer matrix gives complete solvability conditions and parametrizes all solutions.

Alternatively, reduce A by unimodular row and column operations and interpret those operations modulo n. The system becomes a collection of scalar congruences

$$d_i z_i \equiv c_i \pmod{n}.$$

Each one is decided by

$$\gcd(d_i, n) \mid c_i.$$

The one-variable congruence theorem is therefore not an isolated result. It is the first diagonal component of a general normal-form theory for homomorphisms between finitely generated modules.

Over a field, rank and reduced row echelon form answer most elementary questions about a linear map. Over $\mathbb{Z}$, division is restricted, so the lattice retains divisibility information that rank cannot see. The scalar congruence theorem, CRT, unimodular reduction, Smith normal form, cokernels, and the classification of finitely generated abelian groups are therefore successive stages of one problem: deciding what an integer homomorphism can reach and what discrete obstruction remains. The passage from

$$ax \equiv b \pmod{n}$$

to

$$d_i z_i \equiv c_i \pmod{n}$$

is the matrix form of the same image-and-kernel calculation, while the invariant factors record the obstruction permanently in the quotient group.

Prime Factorization, Divisor Sums, and Arithmetic Functions

N-20221119

The elementary statement that every positive integer factors uniquely into primes has surprisingly different consequences: it gives algorithms, product formulas for divisor counts, inversion formulas, and eventually Euler products. A useful thread is that prime exponents turn multiplication into addition, divisor sums into algebraic convolutions, and finally arithmetic functions into generating series.[1]

§23.1 A divisor below a square root

If an integer $n > 1$ is composite, write $n = ab$ with $1 < a \le b$. Then $a^2 \le n$, hence $a \le \sqrt{n}$. Factoring a supplies a prime divisor $p \le \sqrt{n}$. Thus it suffices to test primes no larger than $\sqrt{n}$ to decide whether n is prime.

For $n = 221$, it is enough to test the primes $2, 3, 5, 7, 11, 13$, because $\sqrt{221} < 15$. The factor 13 succeeds:

$$221 = 13 \cdot 17.$$

Trial division and the sieve of Eratosthenes exploit the same bound in different ways: the sieve marks composites in a whole interval, whereas trial division tests one integer.

§23.2 Euclid's theorem and the prime-divisor lemma

Theorem 23.2.1 (Euclid)
There are infinitely many prime numbers.

Proof. If $p_1, \dots, p_k$ were all the primes, the integer $N = p_1 \cdots p_k + 1 > 1$ would have a prime divisor q. But $N \equiv 1 \pmod{p_i}$ for every i, so q is absent from the supposedly complete list. □

To obtain *uniqueness* of prime factorization, we need another fact.

Lemma 23.2.2 (Euclid's lemma)
If p is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.

Proof. If $p \nmid a$, then $\gcd(a, p) = 1$. Bezout's identity supplies integers u, v with $ua + vp = 1$. Multiplying by b gives $uab + vpb = b$, whose left side is divisible by p because $p \mid ab$. Hence $p \mid b$. □

Repeated division proves the existence of a prime factorization. To prove uniqueness, compare two factorizations of the same integer. Euclid's lemma forces the first prime in one factorization to divide a prime in the other and hence to equal it. Cancel it, and continue by induction.

[1] Compare MIT 18.781, *Theory of Numbers*, lecture notes.

§23.3 Prime exponents as arithmetic coordinates

For $n > 0$, let $v_p(n)$ denote the exponent of a prime p in n. Unique factorization is the assertion that

$$n = \prod_p p^{v_p(n)}, \qquad v_p(n) \geq 0,$$

with finitely many nonzero exponents. It turns multiplication and divisibility into coordinatewise statements:

$$v_p(mn) = v_p(m) + v_p(n), \qquad a \mid b \iff v_p(a) \leq v_p(b) \quad \forall p.$$

Consequently the greatest common divisor and least common multiple satisfy

$$v_p(\gcd(a,b)) = \min(v_p(a), v_p(b)), \quad v_p(\operatorname{lcm}(a,b)) = \max(v_p(a), v_p(b)).$$

For example, $72 = 2^3 3^2$, $90 = 2 \cdot 3^2 \cdot 5$, so

$$\gcd(72,90) = 2 \cdot 3^2 = 18, \qquad \operatorname{lcm}(72,90) = 2^3 3^2 5 = 360.$$

§23.4 Counting divisors and their sum

If $n = \prod_{i=1}^r p_i^{a_i}$, every divisor has the unique form $d = \prod_i p_i^{b_i}$ with $0 \leq b_i \leq a_i$. Choosing each b_i independently gives

$$\tau(n) = \sum_{d \mid n} 1 = \prod_{i=1}^{r} (a_i + 1).$$

Summing the values of all choices instead gives

$$\sigma(n) = \sum_{d \mid n} d = \prod_{i=1}^{r} (1 + p_i + \cdots + p_i^{a_i}) = \prod_i \frac{p_i^{a_i+1} - 1}{p_i - 1}.$$

For $n = 12 = 2^2 \cdot 3$, the divisors are $1, 2, 3, 4, 6, 12$, so $\tau(12) = 6$ and $\sigma(12) = 28$. The formulas agree:

$$(2+1)(1+1) = 6, \qquad (1+2+4)(1+3) = 28.$$

A positive integer n is a perfect square exactly when all its $v_p(n)$ are even; by the formula for τ, this is equivalent to $\tau(n)$ being odd. Likewise, n is called perfect when $\sigma(n) = 2n$, as for $n = 6$.

§23.5 Legendre's formula is a double count

The exponent of p in $n!$ is

$$\boxed{v_p(n!) = \sum_{j \geq 1} \left\lfloor \frac{n}{p^j} \right\rfloor.}$$

Each multiple of p contributes at least one factor; each multiple of p^2 contributes one additional factor, and so on. For example,

$$v_2(10!) = \lfloor 10/2 \rfloor + \lfloor 10/4 \rfloor + \lfloor 10/8 \rfloor = 5 + 2 + 1 = 8.$$

This also yields a practical test for binomial coefficients:

$$v_p\binom{n}{k} = v_p(n!) - v_p(k!) - v_p((n-k)!).$$

The integrality of $\binom{n}{k}$ is reflected in the nonnegativity of every one of these prime exponents.

§23.6 Dirichlet convolution encodes divisor sums

An arithmetic function assigns a complex number to each positive integer. For such functions define

$$(f * g)(n) = \sum_{d \mid n} f(d) g(n/d).$$

The function $\varepsilon(1) = 1$, $\varepsilon(n) = 0$ for $n > 1$ is the identity under convolution. If $\mathbf{1}(n) = 1$ and $\mathrm{id}(n) = n$, then

$$\tau = \mathbf{1} * \mathbf{1}, \qquad \sigma = \mathbf{1} * \mathrm{id}.$$

The first identity counts one ordered pair of divisors d, e with $de = n$ for each divisor d. The second weighs each pair by e; it computes precisely the divisor sum.

Associativity is worth checking rather than assuming. Both $(f * g) * h$ and $f * (g * h)$ sum $f(a)g(b)h(c)$ over the same ordered triples $abc = n$.

§23.7 Möbius inversion removes previous divisor sums

Define the Möbius function by

$$\mu(n) = \begin{cases} 1, & n = 1, \\ (-1)^r, & n \text{ is the product of } r \text{ distinct primes}, \\ 0, & p^2 \mid n \text{ for some prime } p. \end{cases}$$

For $n > 1$ with r distinct prime factors,

$$\sum_{d \mid n} \mu(d) = \sum_{j=0}^{r} \binom{r}{j} (-1)^j = (1-1)^r = 0.$$

For $n = 1$ the sum is 1. Thus

$$\mu * \mathbf{1} = \varepsilon.$$

If $F(n) = \sum_{d \mid n} f(d) = (\mathbf{1} * f)(n)$, convolution by μ gives the *inversion formula*

$$f(n) = \sum_{d \mid n} \mu(d) F(n/d).$$

Example 23.7.1

The integers $1 \le k \le n$ with $\gcd(k, n) = 1$ are counted by Euler's totient $\varphi(n)$. Partitioning $1, \dots, n$ by their gcd with n gives

$$\sum_{d \mid n} \varphi(d) = n.$$

Möbius inversion yields

$$\varphi(n) = \sum_{d \mid n} \mu(d) \frac{n}{d} = n \prod_{p \mid n} \left(1 - \frac{1}{p}\right).$$

For $n = 12$, this predicts $\varphi(12) = 12(1 - \tfrac{1}{2})(1 - \tfrac{1}{3}) = 4$, matching the units $1, 5, 7, 11$ modulo 12.

§23.8 Euler Products and Factorization

For $\mathrm{Re}(s) > 1$, the series

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$$

converges absolutely. Expanding geometric series for every prime p and using unique factorization gives

$$\boxed{\zeta(s) = \prod_{p} (1 - p^{-s})^{-1} \qquad (\mathrm{Re}(s) > 1).}$$

The region of convergence matters: no unrestricted interchange of infinitely many sums and products has been justified outside it.

The simple congruence exercise $2, 4, 8, 6 \pmod{10}$ belongs to another consequence of the same arithmetic structure: units modulo m form a finite multiplicative group. A periodicity claim about powers must therefore be checked against coprimality. For example, 2 is not a unit modulo 10, so its observed eventual cycle is not an instance of Euler's theorem on $(\mathbb{Z}/10\mathbb{Z})^{\times}$.

The progression is from a local divisibility test to unique prime coordinates, then to product formulas and inversion. Each new operation is motivated by a concrete earlier calculation; analytic continuation or deeper zeta-function claims can be studied separately.

Congruence Classes, Units, and the Arithmetic of Exponents

N-20221126

§24.1 Congruence is equality in a quotient ring

For $n \geq 2$, the relation

$$a \equiv b \pmod{n} \quad \Longleftrightarrow \quad n \mid (a-b)$$

partitions $\mathbb{Z}$ into n classes. It is an equivalence relation and respects the two ring operations:

$$a \equiv b, \quad c \equiv d \pmod{n} \implies a+c \equiv b+d, \quad ac \equiv bd \pmod{n}.$$

The residue class $[a]$ is the coset $a + n\mathbb{Z}$, and the quotient

$$\mathbb{Z}/n\mathbb{Z}$$

is a ring. A complete residue system consists of one representative from each of its classes. For example, $0, 1, \ldots, n-1$ is one such system, but replacing any representative by a congruent integer leaves it complete.

The purpose of reducing a complicated expression is therefore algebraic: evaluating a polynomial of integers and then passing to the quotient gives the same answer as reducing every integer first. Modular arithmetic is the image of integer arithmetic under a ring homomorphism, which explains why these reductions compose reliably. For the number-theoretic organization we follow Abhinav Kumar's MIT 18.781 course notes on congruences, Euler's theorem, the Chinese remainder theorem, and primitive roots.[1]

Example 24.1.1 (Three ways to avoid multiplying large numbers)
The last digit of 15^{200} is 5, because $15^r \equiv 5 \pmod{10}$ for every positive integer r. Modulo 17,

$$15^{100} \equiv (-2)^{100} = 2^{100}, \qquad 2^4 \equiv -1, \quad 2^8 \equiv 1,$$

so $2^{100} \equiv (2^8)^{12} 2^4 \equiv 16 \pmod{17}$. Finally,

$$\begin{aligned} 2001 \cdot 2002 + 2003 \cdot 2004 + 2005 \cdot 2006 \\ &\equiv 21 \cdot 22 + 23 \cdot 24 + 25 \cdot 26 \\ &\equiv 12 + 12 + 20 \equiv 44 \pmod{45}. \end{aligned}$$

The examples illustrate reduction of a constant base, reduction by a multiplicative period, and reduction before distributing products.

Quadratic residues provide another useful finite check. Since every integer is congruent to $0, \ldots, 7 \pmod{8}$, squaring gives

$$\{x^2 \bmod 8 : x \in \mathbb{Z}\} = \{0, 1, 4\}.$$

[1] https://ocw.mit.edu/courses/18-781-theory-of-numbers-spring-2012/pages/lecture-notes/

The eight possibilities can be reduced further: an odd integer has square $1 \pmod 8$; an even integer is either divisible by 4, giving square $0 \pmod 8$, or is $2 \pmod 4$, giving square $4 \pmod 8$.

§24.2 Reduced residue systems form the group of units

A class $[a] \in \mathbb{Z}/n\mathbb{Z}$ is invertible if and only if $\gcd(a, n) = 1$. Indeed, Bézout's identity $ua + vn = 1$ supplies an inverse of $[a]$, whereas an inverse modulo n implies $\gcd(a, n) = 1$.

Consequently the reduced residue classes form the multiplicative group

$$(\mathbb{Z}/n\mathbb{Z})^\times.$$

Its cardinality is Euler's totient

$$\varphi(n) = \#(\mathbb{Z}/n\mathbb{Z})^\times.$$

For $n = p^r$, precisely the multiples of p are excluded, so

$$\varphi(p^r) = p^r - p^{r-1}.$$

For coprime m, n, the Chinese remainder theorem gives a ring isomorphism

$$\mathbb{Z}/mn\mathbb{Z} \simeq \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z},$$

which restricts to the unit groups. Their sizes multiply:

$$\varphi(mn) = \varphi(m)\varphi(n).$$

Factoring $n = \prod_{p|n} p^{v_p(n)}$ now derives

$$\boxed{\varphi(n) = n \prod_{p|n} (1 - p^{-1}).}$$

This is a deduction from counting units and the CRT, not an unrelated formula to memorize.

§24.3 Exponent reduction is a statement about element order

For a unit $a \pmod n$, define its multiplicative order by

$$\operatorname{ord}_n(a) = \min\{d \geq 1 : a^d \equiv 1 \pmod n\}.$$

The finite-group theorem of Lagrange implies

$$\operatorname{ord}_n(a) \mid \varphi(n), \qquad a^{\varphi(n)} \equiv 1 \pmod n.$$

This is Euler's theorem; it requires $\gcd(a, n) = 1$. A much smaller order may exist. For example,

$$2^4 \equiv -1 \pmod{17}, \quad 2^8 \equiv 1 \pmod{17},$$

so $\operatorname{ord}_{17}(2) = 8$, whereas $\varphi(17) = 16$. The exponent 100 can be reduced modulo 8, the actual group order of the cyclic subgroup generated by 2. By contrast, $15 \pmod{10}$ is not a unit, so applying Euler's theorem to the last-digit problem would be invalid.

Consider the sequence

$$F_k = 12^{2k+1} + 36^k + 5^k + 3, \qquad k \geq 1.$$

Every term has $F_k \equiv 0 \pmod 4$. A finer residue computation gives

$$F_1 \equiv 12 \pmod{16}, \qquad F_k \equiv 5^k + 3 \pmod{16} \quad (k \geq 2),$$

because $12 \cdot 144^k$ and 36^k are divisible by 16 for $k \geq 2$. The powers $5^k \pmod{16}$ cycle with period four:

$$5, \quad 9, \quad 13, \quad 1.$$

It follows that

$$\boxed{16 \mid F_k \quad \Longleftrightarrow \quad k \equiv 3 \pmod 4} \qquad (k \geq 1).$$

Reducing to a small multiplicative cycle exposes more structure than checking isolated numerical values; it also shows why the precise modulus is central to a divisibility question.

§24.4 Primitive roots give discrete-logarithm coordinates

A primitive root modulo n is a generator of the unit group. If $(\mathbb{Z}/n\mathbb{Z})^\times = \langle g \rangle$, then every unit has the form g^j; multiplication becomes addition of exponents modulo $\varphi(n)$. For odd primes p, the unit group modulo p^r is cyclic, and a generator exists. For $n = 8$, however,

$$(\mathbb{Z}/8\mathbb{Z})^\times = \{1, 3, 5, 7\}$$

and all nonidentity elements have square 1, so the group is not cyclic.

Apart from the trivial modulus 1, the classical classification is that primitive roots exist precisely for

$$n = 2, \ 4, \ p^r, \ 2p^r$$

with p an odd prime. A convenient primitive root is thus not available for every modulus. In other cases the Chinese remainder theorem and the structure theorem for finite abelian groups are the better language.

§24.5 Fourier Theory of the Unit Group

A Dirichlet character modulo n is a homomorphism

$$\chi : (\mathbb{Z}/n\mathbb{Z})^\times \longrightarrow \mathbb{C}^\times,$$

extended to all integers by 0 on nonunits. Since the unit group is finite, the nonzero values of χ are roots of unity. If G is a finite abelian group, its characters satisfy

$$\sum_{\chi \in \widehat{G}} \chi(g) = \begin{cases} |G|, & g = e, \\ 0, & g \neq e. \end{cases}$$

To see the cancellation in the second line, choose a character ψ with $\psi(g) \neq 1$. Multiplying each summand by $\psi(g)$ merely permutes the characters, so the sum equals $\psi(g)$ times itself and must vanish.

For $G = (\mathbb{Z}/n\mathbb{Z})^\times$, this yields an indicator for a specified invertible congruence class:

$$\mathbf{1}_{a\equiv b \pmod n} = \frac{1}{\varphi(n)} \sum_{\chi \pmod n} \chi(a)\overline{\chi(b)}$$

when both a, b are units modulo n. This is the exact connection between the original residue-class exercises and Fourier analysis on a finite group. It is also why characters enter the study of primes in arithmetic progressions.

Cyclotomic Pullbacks: Exact Orders, Galois Conjugacy, and Polynomial Divisibility

N-20230127

§25.1 A plausible generalization fails for a precise reason

Write

$$H_m(x) = 1 + x + x^2 + \cdots + x^{m-1} = \frac{x^m - 1}{x - 1}.$$

The elementary problem

$$x^2 + x + 1 \mid x^{2n} + x^n + 1$$

is the case

$$H_3(x) \mid H_3(x^n).$$

For $m = 3$, the answer is simply $3 \nmid n$. It is tempting to guess that, for general m,

$$H_m(x) \mid H_m(x^n) \quad \text{whenever} \quad m \nmid n.$$

The guess is false.

Take $m = 6$ and $n = 2$. Although $6 \nmid 2$, the number -1 is a root of

$$H_6(x) = 1 + x + \cdots + x^5,$$

because

$$1 - 1 + 1 - 1 + 1 - 1 = 0.$$

But

$$H_6((-1)^2) = H_6(1) = 6,$$

so -1 is not a root of $H_6(x^2)$. Therefore

$$H_6(x) \nmid H_6(x^2).$$

The failed guess looked only at the order m itself. The polynomial H_m, however, contains every nontrivial m-th root of unity, including roots whose exact orders are proper divisors of m. In the counterexample, the troublesome root has order 2, and $2 \mid n$. The correct question is therefore not merely whether m divides n, but how the map

$$z \longmapsto z^n$$

changes the exact order of every root of unity involved.

§25.2 Orders of Powers

Let ζ be a root of unity of exact order d. Thus

$$\zeta^d = 1$$

and no smaller positive power of ζ equals 1.

Theorem 25.2.1 (Order after taking a power)
For every positive integer n,

$$\operatorname{ord}(\zeta^n) = \frac{d}{\gcd(d,n)}.$$

Proof. Set

$$g = \gcd(d,n), \qquad d = gd_0, \qquad n = gn_0,$$

where $\gcd(d_0, n_0) = 1$. Then

$$(\zeta^n)^{d_0} = \zeta^{gn_0d_0} = \zeta^{n_0d} = 1,$$

so the order of ζ^n divides d_0.

Conversely, if $(\zeta^n)^r = 1$, then $d \mid nr$. After dividing by g,

$$d_0 \mid n_0 r.$$

Since d_0 and n_0 are coprime, Euclid's lemma gives $d_0 \mid r$. Hence no smaller positive exponent than d_0 can kill ζ^n. □

This formula explains several phenomena at once. If $\gcd(d,n) = 1$, taking the n-th power preserves exact order. If $d \mid n$, it collapses the root all the way to 1. Between these extremes, the order decreases by exactly the common factor.

For example, let ζ have order 12. Then

$$\operatorname{ord}(\zeta^8) = \frac{12}{\gcd(12,8)} = 3.$$

Thus the eighth-power map sends primitive twelfth roots to primitive cubic roots. It neither preserves all twelve vertices nor collapses them to one point; it folds the twelve-cycle onto a three-cycle.

§25.3 The correct divisibility criterion

The roots of $H_m(x)$ are precisely the nontrivial m-th roots of unity. Indeed,

$$(x-1)H_m(x) = x^m - 1,$$

and $x = 1$ is the only m-th root removed by the quotient.

Now let $\zeta \neq 1$ satisfy $\zeta^m = 1$. Then

$$H_m(\zeta^n) = 0$$

exactly when ζ^n is still a nontrivial m-th root of unity. It is automatically an m-th root because

$$(\zeta^n)^m = (\zeta^m)^n = 1.$$

The only possible failure is $\zeta^n = 1$.

Theorem 25.3.1 (General geometric-sum divisibility)

For positive integers m, n,

$$H_m(x) \mid H_m(x^n) \quad \Longleftrightarrow \quad \gcd(m, n) = 1.$$

Proof. Assume first that $\gcd(m, n) = 1$. Every root ζ of H_m has exact order $d > 1$ dividing m. Therefore $\gcd(d, n) = 1$, and the order formula gives

$$\operatorname{ord}(\zeta^n) = d > 1.$$

Hence $H_m(\zeta^n) = 0$. Every root of H_m is a root of $H_m(x^n)$, and the roots of H_m are simple, so divisibility follows.

Conversely, suppose

$$g = \gcd(m, n) > 1.$$

Choose a primitive g-th root of unity η. Since $g \mid m$, the number $\eta \neq 1$ is a root of H_m. Since $g \mid n$,

$$\eta^n = 1,$$

and therefore

$$H_m(\eta^n) = H_m(1) = m \neq 0.$$

Thus divisibility fails. □

The condition is stronger than $m \nmid n$ because every nontrivial divisor of m matters. Coprimality guarantees that the power map is an automorphism of the entire cyclic group of m-th roots, not merely that a primitive m-th root avoids collapsing to 1.

§25.4 The exact common factor is stronger information

When full divisibility fails, the two polynomials may still share a substantial factor. Exact orders identify it completely.

Let ζ have exact order d, where $d \mid m$ and $d > 1$. Then ζ is a root of $H_m(x^n)$ precisely when

$$\zeta^n \neq 1.$$

By the order formula, this is equivalent to

$$d \nmid n.$$

Thus the roots shared by $H_m(x)$ and $H_m(x^n)$ are exactly the roots whose exact orders divide m but do not divide n.

To turn this root description into a polynomial formula, group roots by exact order. Denote by $\Phi_d(x)$ the polynomial whose roots are the primitive d-th roots of unity. Then

$$H_m(x) = \prod_{\substack{d \mid m \\ d > 1}} \Phi_d(x).$$

Consequently,

$$\boxed{\gcd(H_m(x), H_m(x^n)) = \prod_{\substack{d \mid m,\ d > 1 \\ d \nmid n}} \Phi_d(x).}$$

The gcd is taken monic in $\mathbb{Q}[x]$.

For $m = 6$ and $n = 4$,

$$H_6(x) = \Phi_2(x)\Phi_3(x)\Phi_6(x).$$

The relevant orders are $2, 3, 6$. Since $2 \mid 4$ but $3 \nmid 4$ and $6 \nmid 4$,

$$\gcd\big(H_6(x), H_6(x^4)\big) = \Phi_3(x)\Phi_6(x).$$

Using

$$\Phi_3(x) = x^2 + x + 1, \qquad \Phi_6(x) = x^2 - x + 1,$$

we obtain

$$\gcd\big(H_6(x), H_6(x^4)\big) = x^4 + x^2 + 1.$$

The factor is neither trivial nor all of H_6. It records exactly which order strata survive the fourth-power map.

§25.5 Rational Factors from Primitive Roots

A root test over $\mathbb{C}$ becomes a divisibility statement over $\mathbb{Q}$ only because primitive roots come in algebraically inseparable families.

Let

$$\zeta_d = e^{2\pi i/d}$$

be a primitive d-th root of unity. Its powers

$$\zeta_d^a, \qquad a \in (\mathbb{Z}/d\mathbb{Z})^\times,$$

are exactly the other primitive d-th roots. The polynomial

$$\Phi_d(x) = \prod_{a \in (\mathbb{Z}/d\mathbb{Z})^\times} (x - \zeta_d^a)$$

is the d-th cyclotomic polynomial.

A fundamental theorem says that $\Phi_d(x)$ lies in $\mathbb{Z}[x]$ and is irreducible over $\mathbb{Q}$. Therefore it is the minimal polynomial of ζ_d over $\mathbb{Q}$. If a rational polynomial $P(x)$ satisfies

$$P(\zeta_d) = 0,$$

then polynomial division by the minimal polynomial forces

$$\Phi_d(x) \mid P(x).$$

Indeed, write

$$P(x) = Q(x)\Phi_d(x) + R(x), \qquad \deg R < \deg \Phi_d.$$

Evaluating at ζ_d gives $R(\zeta_d) = 0$. Minimality implies $R = 0$.

The field

$$\mathbb{Q}(\zeta_d)$$

is the cyclotomic field generated by ζ_d. Every automorphism fixing $\mathbb{Q}$ is determined by

$$\zeta_d \longmapsto \zeta_d^a$$

for a unit a mod d, and every such choice occurs. Thus

$$\mathrm{Gal}(\mathbb{Q}(\zeta_d)/\mathbb{Q}) \cong (\mathbb{Z}/d\mathbb{Z})^\times.$$

If $P \in \mathbb{Q}[x]$ vanishes at one primitive root, applying these automorphisms shows that it vanishes at every Galois conjugate primitive root. The minimal-polynomial proof and the Galois proof are two descriptions of the same rigidity.

For $d = 5$, the primitive roots are

$$\zeta_5, \zeta_5^2, \zeta_5^3, \zeta_5^4,$$

and

$$\Phi_5(x) = x^4 + x^3 + x^2 + x + 1.$$

A rational polynomial cannot vanish at ζ_5 alone; it must contain the whole four-root orbit as a factor.

§25.6 Cyclotomic polynomials separate roots by exact order

Every m-th root of unity has a unique exact order d dividing m. Partitioning the roots of $x^m - 1$ by exact order gives

$$\boxed{x^m - 1 = \prod_{d \mid m} \Phi_d(x).}$$

The factors are pairwise coprime because their roots have different exact orders.

Taking degrees yields

$$m = \sum_{d \mid m} \deg \Phi_d.$$

Since the primitive d-th roots are indexed by the units modulo d,

$$\deg \Phi_d = \varphi(d),$$

so the degree identity becomes the familiar divisor sum

$$\sum_{d \mid m} \varphi(d) = m.$$

The arithmetic identity counts elements of a cyclic group by exact order, while the polynomial identity groups roots by the same invariant.

For $m = 12$, the divisors are

$$1, 2, 3, 4, 6, 12.$$

The factorization is

$$\begin{aligned} x^{12} - 1 &= \Phi_1(x)\Phi_2(x)\Phi_3(x)\Phi_4(x)\Phi_6(x)\Phi_{12}(x) \\ &= (x-1)(x+1)(x^2+x+1)(x^2+1) \\ &\quad \cdot (x^2-x+1)(x^4-x^2+1). \end{aligned}$$

The degrees

$$1 + 1 + 2 + 2 + 2 + 4 = 12$$

match the twelve roots. The factor Φ_{12} contains exactly the roots of order twelve, not all roots whose orders merely divide twelve.

This exact-order separation is what the original geometric-sum problem needs. The polynomial H_m contains every Φ_d with $d \mid m$, $d > 1$, and the power map acts differently on each such stratum.

§25.7 Cyclotomic Factors under Powers

To understand $\Phi_r(x^n)$, ask which roots x have the property that x^n has exact order r. Suppose x has exact order s. The order formula says

$$\frac{s}{\gcd(s,n)} = r.$$

Write $s = rd$. The condition becomes

$$\gcd(rd, n) = d.$$

This forces $d \mid n$, and after writing $n = dq$, it becomes

$$\gcd(r, q) = 1.$$

Therefore the possible exact orders of roots of $\Phi_r(x^n)$ are

$$rd \quad \text{with} \quad d \mid n \quad \text{and} \quad \gcd\left(r, \frac{n}{d}\right) = 1.$$

Theorem 25.7.1 (Cyclotomic pullback formula)

For positive integers r, n,

$$\boxed{\Phi_r(x^n) = \prod_{\substack{d \mid n \\ \gcd(r, n/d)=1}} \Phi_{rd}(x).}$$

Proof. The root analysis above shows that both sides have exactly the same roots, grouped by the exact order rd. All these roots are simple because $x^{rn} - 1$ has no repeated roots in characteristic zero. Both sides are monic, so equality follows. □

Take $r = 3$ and $n = 4$. Every divisor d of 4 satisfies

$$\gcd\left(3, \frac{4}{d}\right) = 1,$$

so

$$\Phi_3(x^4) = \Phi_3(x)\Phi_6(x)\Phi_{12}(x).$$

Indeed,

$$x^8 + x^4 + 1 = (x^2 + x + 1)(x^2 - x + 1)(x^4 - x^2 + 1).$$

The three factors correspond to roots of exact orders $3, 6, 12$, all of which become primitive cubic roots after taking fourth powers.

By contrast,

$$\Phi_2(x^4) = x^4 + 1 = \Phi_8(x).$$

Only $d = 4$ contributes because

$$\gcd(2, 4/d) = 1$$

holds only when $d = 4$. An eighth root of exact order eight becomes -1 after the fourth power, while roots of orders two or four collapse to 1 instead.

§25.8 Möbius inversion reconstructs the exact-order factor

The identity

$$x^n - 1 = \prod_{d|n} \Phi_d(x)$$

expresses an "order divides n" object as a product of "exact order d" objects. Möbius inversion reverses this divisor aggregation.

Recall the Möbius function:

$$\mu(n) = \begin{cases} 1, & n = 1, \\ 0, & n \text{ is divisible by a square}, \\ (-1)^k, & n \text{ is a product of } k \text{ distinct primes}. \end{cases}$$

Its key property is

$$\sum_{d|n} \mu(d) = \begin{cases} 1, & n = 1, \\ 0, & n > 1. \end{cases}$$

Multiplicative Möbius inversion gives

$$\boxed{\Phi_n(x) = \prod_{d|n} (x^d - 1)^{\mu(n/d)}.}$$

Although negative exponents appear, the total expression is a polynomial because it is obtained by inverting the already established cyclotomic factorization.

The cancellation can be checked factor by factor. Substitute

$$x^d - 1 = \prod_{e|d} \Phi_e(x)$$

into the right-hand side. The exponent of a fixed $\Phi_e(x)$ becomes

$$\sum_{\substack{d|n \\ e|d}} \mu(n/d).$$

Writing $d = eq$, this is

$$\sum_{q|n/e} \mu\left(\frac{n/e}{q}\right),$$

which equals 1 when $e = n$ and 0 otherwise. Every unwanted exact-order factor cancels.

For $n = 12$, the nonzero values of $\mu(12/d)$ occur for

$$d = 2, 4, 6, 12.$$

Thus

$$\Phi_{12}(x) = \frac{(x^{12} - 1)(x^2 - 1)}{(x^6 - 1)(x^4 - 1)}.$$

Using

$$x^{12} - 1 = (x^6 - 1)(x^6 + 1)$$

and

$$x^4 - 1 = (x^2 - 1)(x^2 + 1),$$

we obtain

$$\Phi_{12}(x) = \frac{x^6 + 1}{x^2 + 1} = x^4 - x^2 + 1.$$

The formula extracts the primitive twelfth roots from all lower-order roots by inclusion–exclusion on the divisor lattice.

§25.9 Exact Formulas for Root Plots

Consider

$$H_m(x^n) = 1 + x^n + x^{2n} + \cdots + x^{(m-1)n}.$$

Using the geometric-series identity,

$$H_m(x^n) = \frac{x^{mn} - 1}{x^n - 1}.$$

Hence its roots are the mn-th roots of unity that are not n-th roots of unity.

Let

$$\xi = e^{2\pi i/(mn)}.$$

Every root has the form

$$\xi^q, \qquad 0 \le q < mn,$$

subject to

$$m \nmid q.$$

Indeed,

$$(\xi^q)^n = e^{2\pi i q/m} = 1$$

exactly when $m \mid q$. Therefore

$$\boxed{\mathrm{Roots}(H_m(x^n)) = \left\{ e^{2\pi i q/(mn)} : 0 \le q < mn,\ m \nmid q \right\}.}$$

There are $mn - n = n(m-1)$ such roots, matching the degree.

For $m = 6, n = 4$, the roots are the twenty fourth roots of unity except

$$1, i, -1, -i,$$

which are precisely the fourth roots of unity. A plot therefore shows a regular twenty-four-gon with every sixth vertex removed. The missing pattern is not numerical noise; it is the denominator $x^4 - 1$ in

$$\frac{x^{24} - 1}{x^4 - 1}.$$

The exact orders present in the plot can be read from the cyclotomic pullback formula. Since

$$H_m(x^n) = \prod_{\substack{r \mid m \\ r > 1}} \Phi_r(x^n),$$

each factor $\Phi_r(x^n)$ splits into the order strata rd permitted by

$$d \mid n, \qquad \gcd(r, n/d) = 1.$$

Thus the visual rings of angles, the algebraic factorization, and the arithmetic of exact orders are the same data in three forms.

Over $\mathbb{R}$, conjugate roots pair into quadratic factors. If $\theta \notin \{0, \pi\}$, then

$$(x - e^{i\theta})(x - e^{-i\theta}) = x^2 - 2\cos\theta\, x + 1.$$

For example,

$$\Phi_5(x) = \left(x^2 - 2\cos(2\pi/5)x + 1\right)\left(x^2 - 2\cos(4\pi/5)x + 1\right).$$

The individual quadratic coefficients lie in $\mathbb{Q}(\sqrt{5})$, while their product lies in $\mathbb{Z}[x]$. This is the real shadow of the Galois orbit of primitive fifth roots.

§25.10 What Resultants Detect

The divisibility question asks whether every root of H_m remains a root after the substitution $x \mapsto x^n$. A weaker question asks whether the two polynomials share at least one root. The resultant detects exactly this weaker condition.

If $P(x)$ is monic with roots $\alpha_1, \ldots, \alpha_r$, define

$$\operatorname{Res}(P, Q) = \prod_{j=1}^{r} Q(\alpha_j).$$

Equivalently, the resultant is the determinant of the Sylvester matrix of P and Q. The product formula shows

$$\operatorname{Res}(P, Q) = 0 \quad \Longleftrightarrow \quad P, Q \text{ have a common root over } \mathbb{C}.$$

Apply this to

$$P(x) = H_m(x), \qquad Q(x) = H_m(x^n).$$

If $m \nmid n$, choose a divisor-order stratum that survives. In fact, a primitive m-th root ζ satisfies

$$\zeta^n \neq 1$$

whenever $m \nmid n$, so

$$H_m(\zeta^n) = 0.$$

Thus the resultant is zero.

If $m \mid n$, then every nontrivial m-th root ζ satisfies

$$\zeta^n = 1,$$

so

$$H_m(\zeta^n) = H_m(1) = m.$$

Since H_m has $m - 1$ roots,

$$\operatorname{Res}(H_m(x), H_m(x^n)) = m^{m-1}.$$

Therefore

$$\boxed{\operatorname{Res}(H_m(x), H_m(x^n)) = \begin{cases} 0, & m \nmid n, \\ m^{m-1}, & m \mid n. \end{cases}}$$

This result sharply separates three levels of interaction:

condition	polynomial consequence
$m \mid n$	$\gcd(H_m, H_m(x^n)) = 1$,
$1 < \gcd(m, n) < m$	a proper nontrivial common factor may remain,
$\gcd(m, n) = 1$	$H_m(x) \mid H_m(x^n)$.

The resultant detects whether the gcd is nontrivial; the exact-order factorization determines the gcd itself; the power-map automorphism criterion determines when the entire divisor survives.

The original factorization problem has therefore grown without changing its central question. A false congruence guess forced us to track exact orders. Exact orders forced a decomposition into cyclotomic factors. Rational divisibility forced us to understand minimal polynomials and Galois conjugacy. Pulling those factors through $x \mapsto x^n$ led to a precise factorization theorem, while Möbius inversion reconstructed each exact-order piece. Finally, the resultant clarified the difference between one surviving orbit and all surviving orbits. Each new object entered because the previous language could no longer state the next question precisely.

III

Combinatorics and Discrete Structures

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Counting Principles, Binomial Identities, and Mixed Problem Solving

N-20220720

§26.1 Adding cases and multiplying independent choices

The standard counting principles follow the same decomposition principles developed in MIT's *Mathematics for Computer Science*, especially the counting chapters on bijections and inclusion–exclusion.[1] We will treat each method as a rule for a different kind of combinatorial construction:

Addition principle. If a task can be done in disjoint cases with $m_1, m_2, \dots, m_n$ possibilities, then the total number is $m_1 + \cdots + m_n$.

Multiplication principle. If a task is done in stages with $m_1, m_2, \dots, m_n$ choices at each stage, then the total number is $m_1 m_2 \cdots m_n$.

Inclusion–exclusion. For sets $A_1, \dots, A_n$, one adds single sizes, subtracts pairwise intersections, adds triple intersections, and so on.

A Venn diagram in the note reviews the three-set formula:

$$\begin{aligned}|A \cup B \cup C| = {} & |A| + |B| + |C| \\ & - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.\end{aligned}$$

The addition principle applies to disjoint alternatives. The multiplication principle applies to successive independent choices. Many counting mistakes come from using multiplication when cases overlap, or using addition when a task has stages. A good habit is to write a sentence: "first choose ..., then choose ..." for multiplication, and "either ..., or ..." for addition.

The addition and multiplication principles are not merely classroom rules. They are the first operations on combinatorial classes. If $\mathcal{A}$ and $\mathcal{B}$ are disjoint classes, then

$$|\mathcal{A} \sqcup \mathcal{B}| = |\mathcal{A}| + |\mathcal{B}|.$$

If an object is built by choosing one object from $\mathcal{A}$ and one from $\mathcal{B}$, then

$$|\mathcal{A} \times \mathcal{B}| = |\mathcal{A}||\mathcal{B}|.$$

Thus addition corresponds to disjoint union, and multiplication corresponds to Cartesian product.

This viewpoint is useful because it explains why formulas should follow from constructions. Before computing a number, identify whether the object is a disjoint alternative, a staged product, a quotient by symmetry, or a coefficient in a generating function.

[1] MIT 6.042J, *Mathematics for Computer Science*, readings.

§26.2 Inclusion-exclusion in arithmetic counting

One problem asks how many positive integers not exceeding 2021 are neither multiples of 6 nor multiples of 8. The excluded set is

$$A \cup B,$$

where A is the set of multiples of 6, and B is the set of multiples of 8. Since

$$\operatorname{lcm}(6, 8) = 24,$$

we have

$$|A| = 336, \qquad |B| = 252, \qquad |A \cap B| = 84.$$

Thus the number excluded is

$$336 + 252 - 84 = 504,$$

and the answer is

$$2021 - 504 = 1517.$$

The calculation also has a short direct verification:

$$\#\{1 \le k \le 2021 : 6 \nmid k,\ 8 \nmid k\} = 2021 - \left\lfloor \frac{2021}{6} \right\rfloor - \left\lfloor \frac{2021}{8} \right\rfloor + \left\lfloor \frac{2021}{24} \right\rfloor = 1517.$$

Inclusion–exclusion corrects the double subtraction of common multiples.

The formula

$$|A \cup B| = |A| + |B| - |A \cap B|$$

is a small instance of Möbius inversion on the Boolean lattice. For a finite family $A_1, \dots, A_n$, the number of elements lying in at least one A_i is

$$\left| \bigcup_{i=1}^{n} A_i \right| = \sum_{\varnothing \neq S \subseteq [n]} (-1)^{|S|+1} \left| \bigcap_{i \in S} A_i \right|.$$

The alternating signs are the Möbius function of the subset poset:

$$\mu(S, T) = (-1)^{|T|-|S|}.$$

So inclusion–exclusion is not a trick for Venn diagrams; it is inversion of the operation that replaces exact membership data by cumulative membership data.

For a finite universe U, a set $A \subseteq U$ has indicator function

$$\mathbf{1}_A(x) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A. \end{cases}$$

Then

$$|A| = \sum_{x \in U} \mathbf{1}_A(x).$$

The two-set inclusion–exclusion identity is the pointwise identity

$$\mathbf{1}_{A \cup B} = \mathbf{1}_A + \mathbf{1}_B - \mathbf{1}_A \mathbf{1}_B.$$

Summing over U gives the cardinality formula.

This is also the beginning of probability. If X is uniformly chosen from U, then

$$\mathbb{P}(X \in A) = \frac{|A|}{|U|} = \mathbb{E}[\mathbf{1}_A(X)].$$

Counting and expectation become the same operation after normalization.

§26.3 Permutations, positions, and binomial coefficients

The page distinguishes arrangements and selections:

$$A_n^m = \frac{n!}{(n-m)!}, \qquad C_n^m = \frac{n!}{m!(n-m)!}.$$

A sequence problem asks for the number of 10-term sequences with $x + y = 10$ and $x - y = 4$, giving $x = 7, y = 3$. Therefore the number of sequences is

$$\binom{10}{3} = 120.$$

The thought process is: once the number of one kind of symbol is fixed, the sequence is determined by choosing its positions.

Counting problems involving subject choices, sign-chart inequalities, and similar-triangle configurations share one skill: case management. One must decide whether the structure is a sum of cases, a product of stages, or a coefficient extraction.

The binomial theorem is

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k.$$

Several standard identities follow immediately:

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n,$$

$$\sum_{k=0}^{n} (-1)^k \binom{n}{k} = 0,$$

and by differentiating

$$(1+x)^n$$

and setting $x = 1$,

$$\sum_{k=0}^{n} k \binom{n}{k} = n2^{n-1}.$$

The page writes these identities as examples of how a coefficient formula becomes a counting tool.

A worked example asks for the constant term or a particular coefficient in an expansion such as

$$(2x - 2y)^5.$$

The term involving $x^{5-k}y^k$ is

$$\binom{5}{k} (2x)^{5-k} (-2y)^k.$$

Thus the coefficient of x^2y^3 comes from $k = 3$:

$$\binom{5}{3} 2^2 (-2)^3 = -320.$$

The note's point is that coefficient problems become simple once the target exponent determines k.

The identity

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$$

comes from choosing a committee of size k from n people. Either a distinguished person is not chosen, or that person is chosen. This kind of proof is often clearer than algebraic manipulation because it explains what the two terms mean.

The binomial theorem says

$$(1+x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k.$$

The coefficient of x^k counts subsets of size k. Differentiating and setting $x = 1$ gives

$$\sum_{k=0}^{n} k\binom{n}{k} = n2^{n-1},$$

which counts pairs (S, s) where $S \subseteq [n]$ and $s \in S$: choose s first, then choose the rest of S.

For another concrete coefficient calculation, the coefficient of x^4 in $(1+x)^3(1+x^2)^2$ counts choices from three objects of weight one and two objects of weight two totaling weight four. It equals $\binom{3}{0}\binom{2}{2} + \binom{3}{2}\binom{2}{1} = 1 + 6 = 7$. Each factor records an independent choice and multiplication records the summed weight. This illustrates the right use of generating functions: algebraic operations on series encode operations on counted objects.

§26.4 Counting orbits under symmetry

If a finite group G acts on a finite set X, the number of orbits is

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |\operatorname{Fix}(g)|.$$

This is Burnside's lemma. It explains why one cannot always divide by the number of symmetries: objects with nontrivial stabilizers have smaller orbits.

For a square necklace of four binary beads, count colorings up to rotations. The identity fixes all $2^4 = 16$ colorings, the half-turn fixes $2^2 = 4$, and each quarter-turn fixes 2. Burnside's formula therefore gives

$$\#\text{binary length-four necklaces} = \frac{16 + 4 + 2 + 2}{4} = 6.$$

Simply dividing 16 by 4 would give the wrong answer because symmetric colorings have nontrivial stabilizers. This small calculation shows exactly when symmetry changes the unit of counting from individual objects to orbits.

Set Counting Through Residues, Recurrences, and Finite Fourier Analysis

N-20220808

Residue classes turn arithmetic constraints on large sets of integers into small finite combinatorial problems. Sometimes this gives an extremal construction; sometimes it gives an exact count. A particularly effective method is to encode every subset by a monomial, then use roots of unity to select those subsets whose sums satisfy a prescribed congruence. The resulting formula can also be read as a diagonalization of a recurrence on residue classes.[1]

§27.1 An extremal construction becomes a graph on residues

Consider integers $1, \dots, 11m$. Choose a set A so that no two elements of A have residues differing by 1 or -1 modulo 11. Elements with the same residue are allowed.

The eleven residue classes can be arranged around a cycle:

$$0-1-2-\cdots-10-0.$$

If two adjacent vertices were occupied by residues of elements of A, the condition would fail. Thus the occupied residue classes form an independent set in the cycle graph C_{11}.

An independent set of k vertices in an odd cycle of length 11 leaves at least k unoccupied neighbors, so $2k \le 11$ and $k \le 5$. Equality is achieved, for instance, by the residues $\{0, 2, 4, 6, 8\}$. Each residue appears exactly m times among $1, \dots, 11m$, hence

$$\boxed{\max |A| = 5m.}$$

The extremal argument has two distinct pieces: a quotient of the integers identifies the finite constraint graph, and a sharp construction occupies every element in the five selected residue classes. The graph exposes the combinatorial obstruction without requiring a search over $11m$ individual integers.

§27.2 Encode all subsets at once

For $n \ge 1$, let $N_r(n)$ denote the number of subsets $S \subseteq \{1, \dots, n\}$ with

$$\sum_{k \in S} k \equiv r \pmod 3, \qquad r \in \{0, 1, 2\}.$$

The empty subset is included, with sum zero.

Introduce the polynomial

$$P_n(z) = \prod_{k=1}^{n} (1 + z^k).$$

[1] For the root-of-unity filter and its discrete Fourier interpretation see the University of Illinois Chicago notes on discrete Fourier transforms: https://homepages.math.uic.edu/~jan/mcs472/mcs472notes/lec09.html; for generating-function methods see Stanley, *Enumerative Combinatorics*, Volume 1.

In each factor, choosing 1 means omitting k, while choosing z^k means including k. Therefore

$$P_n(z) = \sum_{S \subseteq \{1,\dots,n\}} z^{\sum_{k \in S} k}.$$

The coefficient of z^j counts the subsets whose sum is exactly j. Evaluating at $z = 1$ gives the familiar count $P_n(1) = 2^n$.

Our problem requires grouping exponents modulo 3. It would be wasteful to calculate all coefficients and add one third of them. We need an operation that keeps precisely one residue class of exponents.

§27.3 The roots of unity are perfect residue detectors

Let $\omega = e^{2\pi i/3}$, so

$$1 + \omega + \omega^2 = 0, \qquad \omega^3 = 1.$$

The geometric-sum identity gives

$$\frac{1}{3}(1 + \omega^j + \omega^{2j}) = \begin{cases} 1, & j \equiv 0 \pmod 3, \\ 0, & j \not\equiv 0 \pmod 3. \end{cases}$$

This formula is the indicator function of a congruence condition, expressed using complex numbers.

Apply it term by term to P_n:

$$\begin{aligned} N_0(n) &= \sum_{S \subseteq [n]} \frac{1}{3}(1 + \omega^{\sum_{k \in S} k} + \omega^{2\sum_{k \in S} k}) \\ &= \frac{1}{3}(P_n(1) + P_n(\omega) + P_n(\omega^2)). \end{aligned}$$

Thus

$$\boxed{N_0(n) = \frac{2^n + P_n(\omega) + P_n(\omega^2)}{3}.}$$

The large combinatorial count has become a small number of special polynomial evaluations.

§27.4 An exact formula for all n

Group consecutive integers into blocks of three. Since $\omega^3 = 1$,

$$\begin{aligned} (1 + \omega^{3j+1})(1 + \omega^{3j+2})(1 + \omega^{3j+3}) &= (1 + \omega)(1 + \omega^2) \cdot 2 \\ &= 2. \end{aligned}$$

The same is true at ω^2. Consequently, for $n = 3m$,

$$P_{3m}(\omega) = P_{3m}(\omega^2) = 2^m, \qquad \boxed{N_0(3m) = \frac{2^{3m} + 2^{m+1}}{3}.}$$

For $n = 3m + 1$, the extra factor has residue 1, and

$$\begin{aligned} P_n(\omega) + P_n(\omega^2) &= 2^m((1 + \omega) + (1 + \omega^2)) \\ &= 2^m. \end{aligned}$$

Therefore

$$\boxed{N_0(3m+1) = \frac{2^{3m+1} + 2^m}{3}.}$$

For $n = 3m + 2$, the two extra factors $(1 + \omega)(1 + \omega^2)$ have product 1, so

$$\boxed{N_0(3m+2) = \frac{2^{3m+2} + 2^{m+1}}{3}.}$$

These three formulas cover every nonnegative n, taking $N_0(0) = 1$.

For a quick check, the subsets of $\{1, 2, 3\}$ whose sums are multiples of three are

$$\varnothing, \quad \{3\}, \quad \{1, 2\}, \quad \{1, 2, 3\}.$$

There are four, in agreement with $(2^3 + 2^2)/3 = 4$.

§27.5 Induction Meets Fourier Analysis

Every subset of $\{1, \ldots, n\}$ either excludes n or includes it. Thus

$$\boxed{N_r(n) = N_r(n-1) + N_{r-n \bmod 3}(n-1),}$$

where residue indices are read modulo three. The initial vector is

$$(N_0(0), N_1(0), N_2(0)) = (1, 0, 0).$$

This recurrence could be used for a direct induction proof of the formulas above.

The deeper structure becomes visible in vector notation. Write

$$v_n = \begin{pmatrix} N_0(n) \\ N_1(n) \\ N_2(n) \end{pmatrix}, \qquad (Sv)_r = v_{r-1 \bmod 3}.$$

Then

$$\boxed{v_n = (I + S^n)v_{n-1}.}$$

The matrix S cyclically permutes residue classes. Its eigenvalues are $1, \omega, \omega^2$. Passing to its eigenbasis, the discrete Fourier transform, diagonalizes every factor $I + S^n$. The eigenvalue products are exactly

$$P_n(1), \quad P_n(\omega), \quad P_n(\omega^2).$$

The roots-of-unity formula is therefore the closed-form solution of the induction recurrence in a better coordinate system.

§27.6 Why subset sums become nearly uniform

There are 2^n subsets in total, so a perfectly uniform distribution of their sums modulo three would place $2^n/3$ subsets in each class. The formulas show that the discrepancy for residue zero is only

$$N_0(n) - \frac{2^n}{3} = \begin{cases} 2^{m+1}/3, & n = 3m, \\ 2^m/3, & n = 3m+1, \\ 2^{m+1}/3, & n = 3m+2. \end{cases}$$

It grows like $2^{n/3}$, while the total number of subsets grows like 2^n. The relative error therefore decays exponentially.

In Fourier language, the constant mode $P_n(1) = 2^n$ determines the uniform component. The nontrivial modes $P_n(\omega)$ and $P_n(\omega^2)$ measure imbalance between residue classes, and their relative size shrinks rapidly.

This example explains the value of the finite Fourier transform beyond a clever counting trick. A modular condition makes the problem finite, a generating function packages all subsets simultaneously, and the Fourier basis separates the uniform count from the oscillatory corrections. Induction then reappears as the same operation written in the original residue coordinates.

Counting by Logic, Projection, Inversion, and Generating Series

N-20221205

§28.1 Set Operations as Logic

For a finite universe U, the indicator of a set $A \subseteq U$ is

$$\mathbf{1}_A(x) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A. \end{cases}$$

Then

$$|A| = \sum_{x \in U} \mathbf{1}_A(x).$$

Union, intersection, and complement become pointwise arithmetic:

$$\mathbf{1}_{A \cap B} = \mathbf{1}_A \mathbf{1}_B,$$

$$\mathbf{1}_{A^c} = 1 - \mathbf{1}_A,$$

$$\mathbf{1}_{A \cup B} = \mathbf{1}_A + \mathbf{1}_B - \mathbf{1}_A \mathbf{1}_B.$$

De Morgan's law follows immediately:

$$1 - \mathbf{1}_{A \cup B} = (1 - \mathbf{1}_A)(1 - \mathbf{1}_B).$$

The set identity

$$(A \cup B)^c = A^c \cap B^c$$

is therefore the logical identity

$$\neg(P \vee Q) = (\neg P) \wedge (\neg Q).$$

This translation is more than notation. It allows a set identity to be verified pointwise and then summed to obtain a counting identity. The same habit will later turn inclusion–exclusion into an algebraic inversion formula.

§28.2 Addition and Multiplication Principles

If a finite collection is partitioned into disjoint cases C_i, then

$$\left| \bigsqcup_i C_i \right| = \sum_i |C_i|.$$

If an object is built through successive independent choices, with m_i options at stage i, then the number of constructions is

$$\prod_i m_i.$$

The first principle counts alternatives; the second counts stages.

Arranging r distinct objects chosen from n gives

$$n(n-1)\cdots(n-r+1)=\frac{n!}{(n-r)!}.$$

If order is forgotten, every r-element subset has $r!$ internal orders, so

$$\binom{n}{r}=\frac{n!}{r!(n-r)!}.$$

The symmetry

$$\binom{n}{r}=\binom{n}{n-r}$$

comes from the complement bijection: choosing the selected elements is equivalent to choosing the omitted ones.

The pigeonhole principle is an averaging consequence of the same counting. If N objects occupy r boxes, some box contains at least

$$\left\lceil \frac{N}{r} \right\rceil$$

objects; otherwise the total would be smaller than N.

§28.3 The binomial theorem counts choices inside a product

Expanding

$$(a+b)^n=(a+b)\cdots(a+b)$$

requires choosing either a or b from each factor. A term $a^{n-r}b^r$ occurs once for every choice of the r factors that contribute b. Hence

$$\boxed{(a+b)^n=\sum_{r=0}^{n}\binom{n}{r}a^{n-r}b^r.}$$

Setting $a=b=1$ gives

$$\sum_{r=0}^{n}\binom{n}{r}=2^n.$$

This is also the count of all subsets of an n-element set: each element independently receives the decision “in” or “out.”

Pascal’s identity

$$\binom{n}{r}=\binom{n-1}{r}+\binom{n-1}{r-1}$$

comes from partitioning r-subsets according to whether they contain one distinguished element. The identity is a decomposition of the counted set, not merely a factorial manipulation.

§28.4 Double Counting via Incidences

Let X be a finite set of pairs. The two coordinate projections partition X in different ways, and equality of the resulting totals produces an identity.

For example, count pairs

$$(S, s), \qquad S \subseteq [n], \quad s \in S.$$

Counting first by S gives

$$|X| = \sum_{k=0}^{n} k \binom{n}{k}.$$

Counting first by s, choose the distinguished element in n ways and then choose any subset of the remaining $n-1$ elements:

$$|X| = n2^{n-1}.$$

Thus

$$\boxed{\sum_{k=0}^{n} k \binom{n}{k} = n2^{n-1}.}$$

Vandermonde's identity is another projection count. Split a population into disjoint groups of sizes r and s. Choosing n people altogether can be done directly in

$$\binom{r+s}{n}$$

ways, or by choosing k from the first group and $n-k$ from the second:

$$\boxed{\sum_{k} \binom{r}{k}\binom{s}{n-k} = \binom{r+s}{n}.}$$

Naming the incidence object explains the summation index and the range automatically.

§28.5 Inclusion–exclusion is a pointwise product expansion

For sets $A_1, \dots, A_m$, the indicator of the complement of their union is

$$\mathbf{1}_{(\cup_i A_i)^c} = \prod_{i=1}^{m} (1 - \mathbf{1}_{A_i}).$$

Expanding and subtracting from one gives

$$\begin{aligned} \mathbf{1}_{\cup_i A_i} = &\sum_i \mathbf{1}_{A_i} - \sum_{i<j} \mathbf{1}_{A_i \cap A_j} \\ &+ \sum_{i<j<k} \mathbf{1}_{A_i \cap A_j \cap A_k} - \cdots . \end{aligned}$$

Summing over U yields the full inclusion–exclusion formula

$$\left| \bigcup_{i=1}^{m} A_i \right| = \sum_{\varnothing \neq S \subseteq [m]} (-1)^{|S|+1} \left| \bigcap_{i \in S} A_i \right|.$$

The alternating signs arise from expanding a product, and every element is checked locally according to how many sets contain it.

As a quick application, the number of surjections from an n-element set onto an m-element set is

$$\sum_{j=0}^{m}(-1)^j\binom{m}{j}(m-j)^n.$$

Start with all m^n functions and exclude those missing at least one target value. Intersections correspond to missing a specified set of j values, leaving $(m-j)^n$ functions.

§28.6 Möbius Inversion on Subsets

Let functions f, g be defined on subsets of $[m]$, with

$$g(S)=\sum_{T\subseteq S} f(T).$$

This cumulative transform is the zeta transform of the Boolean lattice. Möbius inversion recovers the exact data:

$$\boxed{f(S)=\sum_{T\subseteq S}(-1)^{|S|-|T|}g(T).}$$

The Boolean Möbius function is therefore

$$\mu(T,S)=(-1)^{|S|-|T|}\qquad (T\subseteq S).$$

For $m=3$, the top value is

$$\begin{aligned}f(123)=\ & g(123)-g(12)-g(13)-g(23)\\ & +g(1)+g(2)+g(3)-g(\varnothing).\end{aligned}$$

The familiar inclusion–exclusion signs are exactly the coefficients of the inverse transform.

To verify the formula, substitute the definition of g. The coefficient of a fixed $f(R)$ is

$$\sum_{R\subseteq T\subseteq S}(-1)^{|S|-|T|}=(1-1)^{|S|-|R|},$$

which is 1 when $R=S$ and 0 otherwise. Inversion works because all unwanted lower-level contributions cancel.

§28.7 Products of Generating Functions

For a sequence $(a_n)_{n\ge 0}$, its ordinary generating function is

$$A(x)=\sum_{n\ge 0}a_n x^n.$$

The notation

$$[x^n]A(x)=a_n$$

extracts a coefficient. If an object of size n splits into an object of size k and one of size $n-k$, the number of pairs is the convolution

$$c_n=\sum_{k=0}^{n}a_k b_{n-k},$$

and

$$C(x) = A(x)B(x).$$

The multiplication principle has become multiplication of series.

Vandermonde's identity follows algebraically from

$$(1+x)^r(1+x)^s = (1+x)^{r+s}.$$

Extracting $[x^n]$ from the left gives

$$\sum_k \binom{r}{k}\binom{s}{n-k},$$

while the right gives $\binom{r+s}{n}$. The same identity now has both an incidence-set proof and a coefficient proof.

§28.8 Recurrences and Coin Change

Let $F_0 = 0$, $F_1 = 1$, and

$$F_n = F_{n-1} + F_{n-2}.$$

Set

$$F(x) = \sum_{n \ge 0} F_n x^n.$$

Multiplying the recurrence by x^n and summing for $n \ge 2$ gives

$$F(x) - x = xF(x) + x^2F(x).$$

Therefore

$$\boxed{F(x) = \frac{x}{1 - x - x^2}.}$$

Factoring the denominator recovers Binet's formula; alternatively, the rational function generates the sequence without solving each recurrence anew.

For unlimited coins of values $1, 2, 5$, choosing $a, b, c \ge 0$ coins contributes value

$$a + 2b + 5c.$$

The generating function is

$$\frac{1}{(1-x)(1-x^2)(1-x^5)}.$$

The coefficient of x^n counts solutions of

$$a + 2b + 5c = n.$$

For $n = 5$, the four representations are

$$1+1+1+1+1, \quad 2+1+1+1, \quad 2+2+1, \quad 5,$$

so

$$[x^5]\frac{1}{(1-x)(1-x^2)(1-x^5)} = 4.$$

A product of geometric series encodes independent multiplicities of allowed parts.

§28.9 Averaging and random choice can prove existence

The pigeonhole principle says that some box is at least as full as the average. The probabilistic method applies the same idea to a random object: if the expected value of a statistic is M, some object has value at least M.

Let a graph have m edges. Place each vertex independently into one of two parts with probability $1/2$. An edge crosses the cut exactly when its endpoints land in different parts, which occurs with probability $1/2$. If X is the number of crossing edges, linearity of expectation gives

$$\mathbb{E}[X] = \sum_{e} \mathbb{P}(e \text{ crosses}) = \frac{m}{2}.$$

Therefore at least one cut satisfies

$$X \geq \frac{m}{2}.$$

No independence between edge indicators is required for linearity of expectation.

The argument is constructive in logic even when it does not exhibit the cut: if every cut had fewer than $m/2$ crossing edges, the average could not equal $m/2$.

§28.10 Stirling and Central Binomial Growth

The largest coefficient in the n-th row of Pascal's triangle lies near $n/2$. For even $n = 2m$, Stirling's approximation

$$m! \sim \sqrt{2\pi m}\left(\frac{m}{e}\right)^m$$

gives

$$\begin{aligned}\binom{2m}{m} &= \frac{(2m)!}{(m!)^2} \\ &\sim \frac{\sqrt{4\pi m}(2m/e)^{2m}}{2\pi m(m/e)^{2m}} \\ &= \frac{4^m}{\sqrt{\pi m}}.\end{aligned}$$

Equivalently,

$$\boxed{\binom{n}{\lfloor n/2 \rfloor} \sim 2^n\sqrt{\frac{2}{\pi n}}.}$$

Thus the largest layer of the Boolean lattice contains about $2^n/\sqrt{n}$ subsets. The exact identity

$$\sum_{k} \binom{n}{k} = 2^n$$

counts the whole lattice; the asymptotic formula describes how that mass concentrates around the middle ranks.

§28.11 Species and Relabelling

A combinatorial species F assigns to each finite label set U a set $F[U]$ of structures, and every bijection

$$\sigma : U \to V$$

transports structures through a bijection

$$F[\sigma] : F[U] \to F[V]$$

compatible with composition. The definition remembers not only how many structures exist, but how they behave under relabelling.

The exponential generating function is

$$F(x) = \sum_{n \geq 0} |F[n]| \frac{x^n}{n!}.$$

For the species SET, there is one set structure on every label set, so

$$\mathrm{SET}(x) = \sum_{n \geq 0} \frac{x^n}{n!} = e^x.$$

For the species SEQ, a label set of size n has $n!$ linear orders, so

$$\mathrm{SEQ}(x) = \sum_{n \geq 0} n! \frac{x^n}{n!} = \frac{1}{1-x}.$$

For $n \geq 1$, a labelled cycle has $(n-1)!$ cyclic orders, hence

$$\mathrm{CYC}(x) = \sum_{n \geq 1} (n-1)! \frac{x^n}{n!} = \sum_{n \geq 1} \frac{x^n}{n} = -\log(1-x).$$

Every permutation decomposes uniquely into an unordered set of disjoint cycles. Species composition therefore gives

$$\mathrm{PERM} = \mathrm{SET}(\mathrm{CYC}).$$

At the level of exponential generating functions,

$$\begin{aligned} \mathrm{PERM}(x) &= \exp(\mathrm{CYC}(x)) \\ &= \exp\bigl(-\log(1-x)\bigr) \\ &= \frac{1}{1-x}. \end{aligned}$$

The coefficient of $x^n/n!$ is $n!$, recovering the number of permutations. Addition, product, and composition of labelled structures have become algebraic operations on generating series because relabelling symmetry is built into the objects from the beginning.

Sumset Growth, Boolean Coordinates, and Polynomial Counting

N-20221207

A finite set can carry two very different kinds of algebra. If its elements are integers, adding two of them produces a sumset whose size reflects additive structure. If its elements are subsets of a fixed universe, taking symmetric difference produces a vector space over $\mathbb{F}_2$. Both settings transform set manipulation into algebra, but they make distinct predictions about how quickly a set grows under repeated operations.[1]

§29.1 A sharp extremal bound from sums and differences

Let $A = \{a_1 < \cdots < a_k\} \subseteq \{1, \ldots, 2021\}$, and define

$$S = A + A = \{a + b : a, b \in A\}, \qquad T = A - A = \{a - b : a, b \in A,\ a \geq b\}.$$

Suppose $S \cap T = \varnothing$. The $2k - 1$ sums

$$2a_1 < a_1 + a_2 < 2a_2 < \cdots < a_{k-1} + a_k < 2a_k$$

are distinct. Thus $|S| \geq 2k - 1$. The k nonnegative differences

$$0 < a_2 - a_1 < \cdots < a_k - a_1$$

are also distinct, giving $|T| \geq k$. As the two sets are disjoint and lie in $\{0, \ldots, 2a_k\}$,

$$3k - 1 \leq |S \cup T| \leq 2a_k + 1 \leq 4043.$$

Therefore $k \leq 1348$.

The bound is attained:

$$A = \{674, 675, \ldots, 2021\}.$$

Its nonnegative difference set is $\{0, \ldots, 1347\}$, while its sumset is $\{1348, \ldots, 4042\}$. They are disjoint, and $|A| = 1348$.

The calculation is instructive because every inequality is sharp at the same configuration. The extremal set is a long arithmetic progression. Its sumset is as small as an integer sumset can be, and its large starting point separates the sums from the differences.

§29.2 Why arithmetic progressions have minimal sumset growth

For any two finite nonempty subsets $A, B \subset \mathbb{Z}$, write $A = \{a_1 < \cdots < a_r\}$ and $B = \{b_1 < \cdots < b_s\}$. The chain

$$a_1 + b_1 < a_2 + b_1 < \cdots < a_r + b_1 < a_r + b_2 < \cdots < a_r + b_s$$

[1] For additive-combinatorial bounds and their polynomial proofs, see Tao and Vu, *Additive Combinatorics*, and Alon, *Combinatorial Nullstellensatz*, https://www.math.tau.ac.il/~nogaa/PDFS/comb_null.pdf.

contains $r + s - 1$ distinct elements of $A + B$, so

$$\boxed{|A + B| \geq |A| + |B| - 1.}$$

For arithmetic progressions with the same positive common difference, equality holds.

The inequality has a geometric interpretation: adding a second independent choice usually expands a set. When equality holds, the choices are aligned so that many distinct pairs yield the same sum. More elaborate inverse theorems in additive combinatorics ask whether small $|A + A|$ forces a set to resemble a generalized arithmetic progression. The extremal example above is the simplest visible case of that question.

§29.3 The finite-field version is Cauchy–Davenport

Over $\mathbb{F}_p$, where p is prime, addition wraps around. The correct bound for nonempty $A, B \subseteq \mathbb{F}_p$ is

$$\boxed{|A + B| \geq \min(p, |A| + |B| - 1).}$$

This is the Cauchy–Davenport theorem. The appearance of p reflects the impossibility of a sumset containing more than the entire field.

There is a striking polynomial proof of the essential case $|A|+|B|-2 < p$. Put $r = |A|$, $s = |B|$, and suppose for contradiction that $|A+B| \leq r+s-2$. Choose a set $C \supseteq A+B$ of exactly $r + s - 2$ field elements, and form

$$P(X, Y) = \prod_{c \in C} (X + Y - c).$$

The polynomial vanishes on every point of $A \times B$. Yet the coefficient of $X^{r-1}Y^{s-1}$ is

$$[X^{r-1}Y^{s-1}]P = \binom{r + s - 2}{r - 1},$$

which is nonzero modulo p, since $r + s - 2 < p$.

The contradiction comes from polynomial interpolation. For any polynomial P of total degree at most $r + s - 2$, the coefficient above equals

$$\sum_{\substack{a \in A \\ b \in B}} \frac{P(a, b)}{\prod\limits_{a' \in A \setminus \{a\}} (a - a') \prod\limits_{b' \in B \setminus \{b\}} (b - b')}.$$

To see this, apply the one-variable Lagrange interpolation formula first in X and then in Y; terms with total degree below $r + s - 2$ cannot contribute to the requested bidegree. Every summand on the right is zero because P vanishes on the grid, contradicting the nonzero binomial coefficient.

If $r+s-1 \geq p$, choose smaller nonempty subsets $A' \subseteq A$, $B' \subseteq B$ with $|A'|+|B'|-1 = p$, and apply the preceding argument to show $A' + B' = \mathbb{F}_p$. This establishes the general bound.

An additive-combinatorial inequality has become a statement about the coefficient of a polynomial. The fields $\mathbb{Z}$ and $\mathbb{F}_p$ behave differently, but the same cardinality growth has an algebraic explanation in both.

§29.4 Subsets of a universe are vectors over $\mathbb{F}_2$

Now let $U = \{1, \dots, n\}$, and associate to each $A \subseteq U$ its indicator vector

$$\mathbf{1}_A = (\mathbf{1}_{1\in A}, \dots, \mathbf{1}_{n\in A}) \in \mathbb{F}_2^n.$$

Under this encoding, symmetric difference becomes vector addition:

$$\mathbf{1}_{A\triangle B} = \mathbf{1}_A + \mathbf{1}_B \pmod 2.$$

Indeed an element belongs to $A\triangle B$ precisely when it belongs to exactly one of A, B.

The collection $\mathcal{P}(U)$ is therefore an n-dimensional vector space over $\mathbb{F}_2$. It has 2^n elements, and an independent family of r subsets generates exactly 2^r distinct symmetric differences.

For example, let $U = \{1, 2, 3\}$ and take $A = \{1, 2\}$, $B = \{2, 3\}$. Then

$$A\triangle B = \{1, 3\}, \qquad \langle A, B\rangle_\triangle = \{\varnothing, A, B, \{1, 3\}\}.$$

Only four distinct subsets appear because the two indicator vectors generate a two-dimensional subspace.

Unlike ordinary sumsets in the integers, a vector subspace can be perfectly stable under addition: $V + V = V$. A small doubling set may therefore signal algebraic closure rather than the geometry of a progression, depending on the ambient group.

§29.5 Venn diagrams are coordinate decompositions

Given m subsets $A_1, \dots, A_m$ of U, each element $u \in U$ has a membership signature

$$\sigma(u) = (\mathbf{1}_{u\in A_1}, \dots, \mathbf{1}_{u\in A_m}) \in \{0, 1\}^m.$$

The 2^m possible signatures partition U into disjoint Boolean atoms. Every expression built from unions, intersections and complements is a union of some of these atoms.

This gives a rigorous algebraic alternative to visually reading a complicated Venn diagram. For example,

$$A \setminus (B \cup C) = A \cap B^c \cap C^c$$

selects exactly the signature $(1, 0, 0)$. The proof is a one-line statement about membership coordinates.

There is also a useful distinction between two operations. Intersection acts like coordinatewise multiplication of indicator vectors, while symmetric difference acts like addition modulo two. These are precisely the ring operations of a Boolean ring. Its algebra is different from integer addition, but both arise from asking which combinations of finite objects preserve information and which combinations force growth.

Discrete Counting by Floors, Binomials, Incidences, and Symmetries

N-20221209

A set-builder expression is an instruction to select objects. A binomial coefficient counts choices of positions; an incidence table can be counted by rows or columns; a group action identifies objects that a symmetry makes equivalent. The natural order for this exercise sheet is to turn every symbolic condition into a concrete count. The standard techniques follow Richard Stanley's *Enumerative Combinatorics, Volume 1* and the MIT 18.212 combinatorics notes.[1]

§30.1 Set-builder notation forces us to fix the universe

The expression

$$S = \{x \in \mathbb{R} : 3\lfloor x \rfloor - 5 < 0\}$$

is read by solving the condition $3\lfloor x \rfloor < 5$. Since $\lfloor x \rfloor$ is integral, this is equivalent to $\lfloor x \rfloor \le 1$, hence

$$S = (-\infty, 2).$$

The endpoint 2 is excluded: $\lfloor 2 \rfloor = 2$ violates the inequality. If the universe had been $\mathbb{Z}$, the answer would instead be the integers at most 1. The universe is part of the problem, not an irrelevant heading.

Set notation also distinguishes $a \in A$ from $\{a\} \subseteq A$. The first is a membership statement about an element; the second is a containment statement about a set. When $a \in A$, both can be true, but they have different types and behave differently under operations on sets.

§30.2 A floor sum counts lattice points

For integers $a, m > 0$, the sum

$$S(a, m) = \sum_{k=0}^{m-1} \left\lfloor \frac{ak}{m} \right\rfloor$$

counts the integer lattice points

$$\{(k, j) : 0 \le k < m, \quad 1 \le j \le ak/m\}.$$

A useful exact formula arises when $\gcd(a, m) = 1$:

$$\boxed{\sum_{k=1}^{m-1} \left\lfloor \frac{ak}{m} \right\rfloor = \frac{(a-1)(m-1)}{2}.}$$

[1] MIT 18.212, *Algebraic Combinatorics*, lecture notes.

For $1 \le k \le m-1$, the rational number ak/m is not integral. Pair the terms indexed by k and $m-k$:

$$\left\lfloor \frac{ak}{m} \right\rfloor + \left\lfloor \frac{a(m-k)}{m} \right\rfloor = a - 1.$$

Summing this identity over $k = 1, \dots, m-1$, each floor term is counted twice. The displayed formula follows. The coprimality hypothesis matters; it ensures there are no lattice points on the diagonal at interior integer coordinates.

For $a = 3, m = 5$, the floors are $0, 0, 1, 1, 2$, summing to 4, and the formula gives $(3-1)(5-1)/2 = 4$. Here a floor function became a lattice-point problem with a symmetry proof.

§30.3 Binomial identities count nested choices

The identity

$$\sum_{j=0}^{n} \binom{n}{j} = 2^n$$

counts all subsets of an n-element set by partitioning them according to cardinality. Its signed companion

$$\sum_{j=0}^{n} (-1)^j \binom{n}{j} = 0, \qquad n \ge 1,$$

follows by pairing each subset not containing a fixed element with its version containing that element. The two subsets have opposite parity.

Another identity,

$$\binom{n}{r}\binom{n-r}{k-r} = \binom{n}{k}\binom{k}{r} \qquad (0 \le r \le k \le n),$$

counts ordered pairs $R \subseteq K \subseteq [n]$ with $|R| = r$ and $|K| = k$. The left side chooses R first, then extends it to K; the right side chooses K first, then selects R inside it. The equality is a double count with the same combinatorial objects on each side.

§30.4 Finite Differences and Binomials

Define $\Delta f(n) = f(n+1) - f(n)$. Pascal's identity gives

$$\Delta \binom{n}{k} = \binom{n}{k-1}.$$

For a polynomial f of degree at most d, its forward differences satisfy $\Delta^{d+1} f = 0$. Successive application at $n = 0$ yields the Newton expansion

$$\boxed{f(n) = \sum_{k=0}^{d} (\Delta^k f)(0) \binom{n}{k}}$$

for nonnegative integers n, with the polynomial identity extending it to all n.

For $f(n) = n^2$, the initial differences are $f(0) = 0$, $\Delta f(0) = 1$, $\Delta^2 f(0) = 2$, so

$$n^2 = \binom{n}{1} + 2\binom{n}{2} = n + n(n-1).$$

This is why the binomial basis naturally belongs to discrete counting: its differences behave like the derivatives of ordinary powers but with no denominators.

§30.5 Recursive sets and finite stabilization

Suppose a rule $F : \mathcal{P}(U) \to \mathcal{P}(U)$ is extensive, so $S \subseteq F(S)$, and U is a finite set of size N. Starting from S_0 and setting $S_{k+1} = F(S_k)$ gives an increasing chain

$$S_0 \subseteq S_1 \subseteq S_2 \subseteq \cdots .$$

Every strict step adds at least one new element. After at most $N - |S_0|$ strict steps the process stabilizes at $S_* = F(S_*)$. The bound does not require that F be monotone on all subsets; extensivity is enough for this particular increasing chain.

If a recursive rule is not extensive, the sets may cycle. For $U = \{1\}$, the operation $F(S) = U \setminus S$ alternates between $\varnothing$ and U forever. Hence finiteness of the universe alone does not imply stabilization of an arbitrary recurrence.

§30.6 Incidence counting is a matrix identity

Let X be a finite set of elements and $\mathcal{A}$ a finite family of subsets of X. Define the incidence set

$$I = \{(x, A) \in X \times \mathcal{A} : x \in A\}.$$

Counting first by x, then by A, gives

$$\boxed{\sum_{x \in X} \#\{A \in \mathcal{A} : x \in A\} = \sum_{A \in \mathcal{A}} |A|.}$$

For example, if $X = \{1, 2, 3, 4\}$ and $\mathcal{A}$ consists of all two-element subsets, there are $\binom{4}{2} = 6$ members, so the right side is $6 \cdot 2 = 12$. On the left, each point belongs to $\binom{3}{1} = 3$ two-element subsets, giving $4 \cdot 3 = 12$. In matrix language, both sides sum all entries of the same 0-1 incidence matrix, in different orders.

§30.7 Symmetry counting needs stabilizers

Suppose a finite group G acts on a finite set X. Simply dividing $|X|$ by $|G|$ assumes every orbit has the same size $|G|$. In reality,

$$|Gx| = \frac{|G|}{|\operatorname{Stab}(x)|}.$$

A correct general formula is Burnside's lemma:

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |\operatorname{Fix}(g)|.$$

For binary necklaces of length three, $G = C_3$ acts on the $2^3 = 8$ binary strings by cyclic rotation. The identity fixes all 8. Each nontrivial rotation fixes only 000 and 111, hence exactly 2 strings. Therefore the number of rotation classes is

$$\frac{8 + 2 + 2}{3} = 4.$$

The naive quotient $8/3$ was not even an integer. Fixed-point counting repairs it because not every string has an orbit of size three.

§30.8 The main diagnostic for a counting problem

The correct move depends on what is being counted. Floors encode lattice-point thresholds; binomial coefficients encode nested finite choices; finite differences encode discrete polynomial changes; recursive sets demand explicit stability conditions; incidence arguments count a single relation in two ways; symmetry counts require orbit and stabilizer information. A useful explanation always identifies the actual objects and proves a formula for them. Additional connections to Ehrhart theory, polynomial interpolation or graph spectra are worthwhile only when they lead to a new mathematical computation.

Binomial Coefficients as a Calculus of Finite Choices

N-20221210

The binomial coefficient $\binom{n}{k}$ counts the k-element subsets of an n-element set. That familiar count becomes much more powerful when it is treated as a change of coordinates: a function of an integer can be expanded in the basis $\binom{x}{k}$, just as a smooth function is expanded in powers of x. The inverse operation is a finite difference, and the same subset counting leads to inclusion–exclusion, probability, and labelled combinatorial structures.[1]

§31.1 The binomial expansion is a decomposition into choices

Expanding $(a+b)^n$ means choosing either a or b from each of n factors. If exactly k factors contribute a, the resulting monomial is $a^k b^{n-k}$, and there are $\binom{n}{k}$ choices. Therefore

$$\boxed{(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}.}$$

Putting $a = b = 1$ gives $\sum_k \binom{n}{k} = 2^n$: every subset is counted once. Putting $a = 1, b = -1$ gives, for $n \geq 1$,

$$\sum_{k=0}^{n} (-1)^k \binom{n}{k} = 0.$$

Thus the even-sized and odd-sized subsets are equally numerous. There is also a direct bijection: choose a fixed element and toggle whether it belongs to the subset, thereby reversing its cardinality's parity.

The triangle inequality yields the useful bound

$$|a+b|^n \leq (|a| + |b|)^n = \sum_{k=0}^{n} \binom{n}{k} |a|^k |b|^{n-k}.$$

The validity of the inequality does not depend on formally moving absolute values through an equality; it follows from the ordinary triangle inequality applied to $a + b$.

§31.2 Inclusion–exclusion is a binomial cancellation

For finitely many sets $A_1, \ldots, A_n$ inside a finite universe U,

$$\left| \bigcup_{i=1}^{n} A_i \right| = \sum_{\varnothing \neq I \subseteq [n]} (-1)^{|I|+1} \left| \bigcap_{i \in I} A_i \right|.$$

To prove it, fix $u \in U$ belonging to exactly r of these sets. It is counted on the right with total weight

$$\sum_{k=1}^{r} (-1)^{k+1} \binom{r}{k} = 1 - (1-1)^r.$$

[1] The binomial-basis and combinatorial generating-function perspective follows Stanley, *Enumerative Combinatorics*, Vol. 1, Chapters 1 and 3, and Flajolet–Sedgewick, *Analytic Combinatorics*, Chapter II: https://algo.inria.fr/flajolet/Publications/book.pdf.

The weight is 1 if $r \geq 1$ and 0 if $r = 0$, exactly the indicator of membership in the union.

This computation is the elementary core of Möbius inversion on the Boolean lattice of subsets: alternate signs cancel all the overcounted choices.

§31.3 Binomial inversion recovers hidden coefficients

Suppose two sequences are related by

$$b_n = \sum_{k=0}^{n} \binom{n}{k} a_k.$$

Then the inverse transform is

$$\boxed{a_n = \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} b_k.}$$

To verify it, substitute the first relation into the second and reverse the finite sums:

$$\begin{aligned}\sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} b_k &= \sum_{j=0}^{n} a_j \sum_{k=j}^{n} (-1)^{n-k} \binom{n}{k} \binom{k}{j} \\ &= \sum_{j=0}^{n} a_j \binom{n}{j} \sum_{r=0}^{n-j} (-1)^{n-j-r} \binom{n-j}{r}.\end{aligned}$$

The last inner sum is $(1-1)^{n-j}$. Only the term $j = n$ survives, giving a_n.

For instance, if $b_n = 2^n$, the inverse formula yields

$$a_n = \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} 2^k = (2-1)^n = 1.$$

The direct transform of the constant sequence $a_k = 1$ is indeed 2^n.

§31.4 Finite differences take the place of derivatives

For a function $f : \mathbb{Z}_{\geq 0} \to \mathbb{R}$, define its forward difference

$$\Delta f(n) = f(n+1) - f(n).$$

Iterating produces

$$\boxed{\Delta^k f(0) = \sum_{j=0}^{k} (-1)^{k-j} \binom{k}{j} f(j).}$$

The coefficients are the same as in binomial inversion. The analogy with differentiation becomes exact on the binomial polynomials

$$\binom{x}{k} = \frac{x(x-1)\cdots(x-k+1)}{k!}.$$

Pascal's identity says

$$\Delta \binom{x}{k} = \binom{x}{k-1},$$

where $\binom{x}{0} = 1$. Thus the binomial basis is adapted to discrete differentiation, much as $x^k/k!$ is adapted to ordinary differentiation.

If $f(x)$ is a polynomial of degree at most d, its Newton expansion is

$$\boxed{f(x) = \sum_{k=0}^{d} \Delta^k f(0) \binom{x}{k}.}$$

Indeed, successive finite differences of this expression at 0 agree with those of f; two polynomials of degree at most d with the same first $d+1$ Newton coefficients are identical.

Take $f(x) = x^3$. Directly,

$$\Delta f(0) = 1, \quad \Delta^2 f(0) = 6, \quad \Delta^3 f(0) = 6.$$

Hence

$$\boxed{x^3 = \binom{x}{1} + 6\binom{x}{2} + 6\binom{x}{3}.}$$

This identity is a discrete analogue of Taylor's theorem. It expresses the cubic in basis functions that respond to Δ by lowering their index.

§31.5 Summation is the inverse of taking differences

Pascal's identity also gives

$$\binom{k+1}{r+1} - \binom{k}{r+1} = \binom{k}{r}.$$

Summing from $k = 0$ to $n-1$ telescopes:

$$\boxed{\sum_{k=0}^{n-1} \binom{k}{r} = \binom{n}{r+1}.}$$

This is the hockey-stick identity. In discrete calculus, the binomial polynomial $\binom{x}{r+1}$ plays the role of an antiderivative of $\binom{x}{r}$.

Combining this with the Newton expansion immediately evaluates sums of polynomial sequences. Using the cubic identity above,

$$\sum_{k=0}^{n-1} k^3 = \binom{n}{2} + 6\binom{n}{3} + 6\binom{n}{4} = \left(\frac{n(n-1)}{2}\right)^2.$$

Thus familiar power-sum formulas are instances of a systematic discrete integration method.

§31.6 The same coefficients form a probability distribution

Let X be the number of selected elements when each of n labels is independently included with probability p. Then

$$\Pr(X = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

The generating polynomial is

$$\mathbb{E}[z^X] = (1 - p + pz)^n.$$

Differentiating at $z = 1$ gives the factorial moments:

$$\mathbb{E}[X] = np, \qquad \mathbb{E}[X(X-1)] = n(n-1)p^2.$$

Since $X^2 = X(X-1) + X$, we obtain

$$\boxed{\mathrm{Var}(X) = np(1-p).}$$

For $p = 1/2$, the central binomial coefficients control the most likely subset sizes. Chebyshev's inequality already gives a quantitative concentration estimate:

$$\Pr\left(\left|\frac{X}{n} - \frac{1}{2}\right| \geq \varepsilon\right) \leq \frac{1}{4n\varepsilon^2}.$$

The binomial theorem therefore simultaneously normalizes a probability distribution and gives access to its fluctuations.

§31.7 From choosing subsets to assembling structures

Subset counting asks us to label each element in one of two ways: selected or not selected. Other labelled structures can be assembled from components, with a generating function recording the number of ways to make each choice.

A permutation decomposes uniquely into disjoint cycles. The number of cycles of length k on a fixed k-element label set is $(k-1)!$, so the exponential generating function of one cycle is

$$C(z) = \sum_{k\geq 1} \frac{(k-1)!}{k!} z^k = \sum_{k\geq 1} \frac{z^k}{k} = -\log(1-z).$$

A permutation is an *unordered set of cycles*. The exponential formula for labelled structures therefore gives

$$P(z) = \exp(C(z)) = \frac{1}{1-z}.$$

If p_n denotes the number of permutations of n labels, the coefficient convention is $P(z) = \sum_{n\geq 0} p_n z^n/n!$, so $p_n = n!$, as expected.

The exponential formula can be checked directly in this case: expanding $\exp(C(z))$ counts unordered collections of cycles, and the division by $r!$ in the exponential removes the ordering of r components. The division by $k!$ in the generating function records the relabelling of an individual size-k component. The algebra captures precisely the combinatorial symmetries involved.

Binomial coefficients begin as counts of subsets, but their usefulness comes from several compatible structures. Alternating signs implement inversion; finite differences provide discrete derivatives; generating functions record independent choices; and exponential generating functions account for the relabelling of assembled components. The same small counting identity has grown into a calculus for finite structures.

Local Information, Global Structure: Pigeonhole, Ramsey, Radon, and Helly

N-20230118

§32.1 Pigeonhole as Averaging

If N objects are placed into r boxes, some box contains at least

$$\left\lceil \frac{N}{r} \right\rceil$$

objects. The proof is the contrapositive: if every box held at most q objects, the total would be at most rq.

The theorem is easy; choosing the boxes is the real work. Suppose nine elements are each recorded by membership in three sets A, B, C. Every element receives a bit pattern

$$(\mathbf{1}_A(x), \mathbf{1}_B(x), \mathbf{1}_C(x)) \in \{0,1\}^3.$$

There are only

$$2^3 = 8$$

possible patterns. Among nine elements, two have the same pattern and therefore belong to exactly the same subcollection of A, B, C.

The conclusion is not merely "two objects coincide." A carefully chosen finite state space forces two complicated objects to become indistinguishable with respect to all recorded tests.

§32.2 Partial Sums and Divisibility

Let $a_1, \dots, a_n$ be integers. There is always a nonempty consecutive block whose sum is divisible by n.

Consider the partial sums

$$s_k = a_1 + \cdots + a_k, \qquad k = 1, \dots, n,$$

and reduce them modulo n. If some

$$s_k \equiv 0 \pmod{n},$$

then $a_1 + \cdots + a_k$ is the required block. Otherwise the n partial sums occupy only the $n-1$ nonzero residue classes, so two satisfy

$$s_i \equiv s_j \pmod{n}, \qquad i < j.$$

Subtracting gives

$$a_{i+1} + \cdots + a_j = s_j - s_i \equiv 0 \pmod{n}.$$

The objects are partial histories, the boxes are residue states, and a collision cancels the common prefix. This pattern appears throughout additive combinatorics and automata: encode a running process in finitely many states, then use repetition to extract a structured segment.

§32.3 Pigeonhole and Sequential Compactness

Let (x_n) be an infinite sequence in $[0,1]$. Divide the interval into two closed halves. At least one half contains infinitely many terms; call it I_1. Divide I_1 into two halves and choose one, I_2, that again contains infinitely many terms. Continuing gives nested intervals

$$I_1 \supset I_2 \supset I_3 \supset \cdots$$

with

$$|I_k| = 2^{-k}.$$

Choose indices

$$n_1 < n_2 < \cdots$$

so that

$$x_{n_k} \in I_k.$$

The nested interval theorem gives a unique point

$$x \in \bigcap_{k=1}^{\infty} I_k.$$

For $j \geq k$, both x_{n_j} and x lie in I_k, so

$$|x_{n_j} - x| \leq 2^{-k}.$$

Hence

$$x_{n_j} \to x.$$

This is Bolzano–Weierstrass on $[0,1]$. The infinite conclusion is built by iterating a finite pigeonhole step at finer and finer scales. In a compact metric space, finite covers by small balls play the role of interval subdivisions.

§32.4 Ramsey's Forced Patterns

Color every edge of the complete graph K_6 red or blue. Choose a vertex v. Among the five edges incident to v, at least three have the same color; suppose

$$va, vb, vc$$

are red.

If any of

$$ab, bc, ca$$

is red, that edge together with v forms a red triangle. If none is red, then all three are blue and abc is a blue triangle. Thus every red–blue coloring of K_6 contains a monochromatic triangle:

$$R(3,3) \leq 6.$$

The bound is sharp. Color the edges of a five-cycle red and its five diagonals blue. The red graph is a cycle and has no triangle; the blue graph is another five-cycle and also has no triangle. Therefore

$$R(3,3) > 5,$$

so

$$\boxed{R(3,3) = 6.}$$

Pigeonhole forced three edges of one color around a vertex. The graph structure then amplified that local repetition into a monochromatic triangle. Ramsey theory studies how large an ambient configuration must be before a specified organized substructure becomes unavoidable.

§32.5 Intervals give the one-dimensional Helly theorem

Let

$$I_i = [a_i, b_i] \qquad (i = 1, \dots, m)$$

be closed intervals, and assume every pair intersects. Set

$$a = \max_i a_i, \qquad b = \min_i b_i.$$

Choose indices p, q with

$$a = a_p, \qquad b = b_q.$$

Since $I_p \cap I_q \neq \varnothing$,

$$a_p \leq b_q,$$

so $a \leq b$. Every interval contains the segment $[a, b]$, and therefore

$$\bigcap_{i=1}^{m} I_i \neq \varnothing.$$

Pairwise intersection is enough in one dimension because the order lets one reduce all left endpoints to their maximum and all right endpoints to their minimum. In higher dimensions there is no total order of boundaries; convexity and affine dependence replace it.

§32.6 Radon's Theorem from Affine Dependence

Theorem 32.6.1 (Radon)

Any $d+2$ points in $\mathbb{R}^d$ can be partitioned into two disjoint sets whose convex hulls intersect.

Let the points be $p_1, \dots, p_{d+2}$. The lifted vectors

$$(p_i, 1) \in \mathbb{R}^{d+1}$$

are linearly dependent, so there are scalars λ_i, not all zero, with

$$\sum_i \lambda_i p_i = 0, \qquad \sum_i \lambda_i = 0.$$

Let

$$I = \{i : \lambda_i > 0\}, \qquad J = \{i : \lambda_i < 0\}.$$

Both are nonempty. Put

$$s = \sum_{i \in I} \lambda_i = -\sum_{j \in J} \lambda_j > 0.$$

Then

$$\sum_{i \in I} \frac{\lambda_i}{s} p_i = \sum_{j \in J} \frac{-\lambda_j}{s} p_j.$$

Both sides are convex combinations. Their common value lies in

$$\operatorname{conv}\{p_i : i \in I\} \cap \operatorname{conv}\{p_j : j \in J\}.$$

The Radon partition is the geometric form of a minimal affine dependence. In oriented-matroid language, such a signed dependence is the prototype of a circuit.

§32.7 Helly's Theorem via Radon

Theorem 32.7.1 (Finite Helly)

Let $C_1, \dots, C_m$ be convex subsets of $\mathbb{R}^d$, with $m \geq d+1$. If every $d+1$ of them have nonempty intersection, then

$$\bigcap_{i=1}^{m} C_i \neq \varnothing.$$

We prove the theorem by induction on m. The case $m = d+1$ is the hypothesis. Assume $m > d+1$ and the result holds for $m-1$ sets.

For each $i = 1, \dots, d+2$, the family obtained by omitting C_i has $m-1$ members and still satisfies the local intersection hypothesis. By induction, choose

$$x_i \in \bigcap_{j \neq i} C_j.$$

Apply Radon's theorem to $x_1, \dots, x_{d+2}$. There is a partition

$$I \sqcup J = [d+2]$$

and a point

$$x \in \operatorname{conv}\{x_i : i \in I\} \cap \operatorname{conv}\{x_j : j \in J\}.$$

Fix a set C_k. If $k \in I$, then every x_j with $j \in J$ belongs to C_k, because $j \neq k$. Convexity gives

$$x \in C_k.$$

If $k \in J$, use the convex hull indexed by I. If $k > d+2$, every x_i lies in C_k. Thus x belongs to every C_k, completing the induction.

The proof shows exactly where the number $d+1$ comes from: Radon's affine dependence begins at $d+2$ points.

§32.8 Carathéodory's Reduction

Theorem 32.8.1 (Carathéodory)

If $x \in \operatorname{conv} S \subset \mathbb{R}^d$, then x lies in the convex hull of at most $d+1$ points of S.

Write

$$x = \sum_{i=1}^{m} \alpha_i p_i, \qquad \alpha_i > 0, \qquad \sum_i \alpha_i = 1,$$

with $m > d+1$. The points are affinely dependent, so there are λ_i, not all zero, satisfying

$$\sum_i \lambda_i p_i = 0, \qquad \sum_i \lambda_i = 0.$$

After changing all signs if necessary, some $\lambda_i > 0$. Let

$$t = \min_{\lambda_i > 0} \frac{\alpha_i}{\lambda_i}.$$

Then the new coefficients

$$\alpha_i' = \alpha_i - t\lambda_i$$

remain nonnegative, still sum to one, and give the same point x. At least one coefficient becomes zero. Repeating removes points until at most $d+1$ remain.

Helly compresses an intersection certificate to small subfamilies. The preceding convex-hull theorem compresses a representation to at most $d+1$ points. Both reductions are driven by affine dependence.

§32.9 Fractional Helly tolerates failed local tests

The ordinary Helly theorem assumes that every $(d+1)$-tuple intersects. Fractional Helly allows failures.

For each d and $\alpha > 0$, there is $\beta = \beta(d, \alpha) > 0$ such that the following holds. If at least an α-fraction of the $(d+1)$-tuples in a family of n convex sets in $\mathbb{R}^d$ have nonempty intersection, then some point belongs to at least βn members of the family. One may take

$$\beta = 1 - (1-\alpha)^{1/(d+1)}.$$

In one dimension, the sets are intervals. Intersecting pairs are edges of an interval graph, while intervals containing a common point form a clique. If a positive fraction of all pairs intersect, the theorem guarantees a clique containing a positive fraction of all intervals. Thus numerous local successes cannot be dispersed arbitrarily; they concentrate around a common witness.

The conclusion is weaker than total intersection but robust under noise. It is an extremal version of Helly rather than a different slogan about local-to-global behavior.

§32.10 Three disks and one missing triangle

For a finite family of sets $\mathcal{F} = \{U_i\}_{i \in I}$, its *nerve* is the simplicial complex

$$N(\mathcal{F}) = \left\{ J \subseteq I : \bigcap_{j \in J} U_j \neq \varnothing \right\}.$$

Vertices record sets, edges record pairwise intersections, and a filled triangle appears precisely when three sets share a point. The nerve theorem says that if the sets form a suitable open *good cover*—every nonempty finite intersection is contractible—then

$$\boxed{\bigcup_{i \in I} U_i \simeq N(\mathcal{F})}$$

up to homotopy equivalence. For convex sets the intersection condition is automatic. This is the topological mechanism behind Helly-type theorems: information about intersections of small subfamilies constrains which simplices, and hence which holes, the nerve may have.

Here is also a short homological explanation of the number $d+1$ in a topological version of Helly's theorem. Suppose a finite good cover by open subsets of $\mathbb{R}^d$ has at least $d+1$ members, every $d+1$ members meet, yet the whole family has empty intersection. Choose a *smallest* subfamily with empty intersection, of size $m \geq d+2$. Every proper subfamily meets, so its nerve is the boundary of an $(m-1)$-simplex, a sphere S^{m-2}. The nerve theorem would then give

$$\widetilde{H}_{m-2}\left(\bigcup_{i=1}^{m} U_i\right) \cong \widetilde{H}_{m-2}(S^{m-2}) \neq 0.$$

But an open subset of $\mathbb{R}^d$ has no nonzero homology in degrees $j \geq d$. Therefore $m-2 < d$, or $m \leq d+1$. In particular, if every $d+1$ members meet, no empty intersection can exist. Convex families are good covers, so classical Helly is included.

A small geometric experiment makes the nerve theorem almost visible. Put three centers at the vertices of an equilateral triangle of side 1,

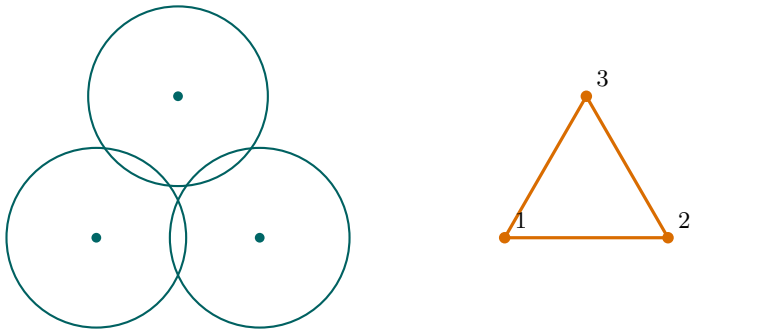
$$p_1 = (0,0), \qquad p_2 = (1,0), \qquad p_3 = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right),$$

and draw open disks $U_i = B(p_i, r)$ of the same radius.

Two disks meet as soon as $2r > 1$, so $r > \frac{1}{2}$ ensures *every pair* intersects. But a point can belong to all three disks only if its maximum distance from the three centers is less than r. Symmetry, or a short smallest-enclosing-circle argument, shows that the least possible maximum distance is the circumradius $1/\sqrt{3}$. Thus if

$$\boxed{\frac{1}{2} < r < \frac{1}{\sqrt{3}},}$$

every pair intersects but the triple intersection is empty. Take $r = 11/20$.



three overlapping disks — their nerve: an unfilled triangle

Figure 32.1: All three pairwise intersections are nonempty, so every nerve edge is present. The missing triple intersection leaves out the two-dimensional face.

The nerve is exactly the boundary of a triangle:

$$N(\{U_1, U_2, U_3\}) = \{\varnothing, \{1\}, \{2\}, \{3\}, \{1,2\}, \{2,3\}, \{1,3\}\}.$$

Since intersections of disks in the plane are convex whenever nonempty, this is a good cover. The nerve theorem now says that the union $U_1 \cup U_2 \cup U_3$ has the homotopy type of a circle. Its central hole is recorded by the one-dimensional homology of the *missing face*.

At $r > 1/\sqrt{3}$, the triple intersection becomes nonempty. The nerve gains the triangle $\{1, 2, 3\}$, and that cycle becomes the boundary of a 2-simplex. An intersection condition that changes at one explicit radius has removed a topological hole. For open disks the endpoint $r = 1/\sqrt{3}$ still lacks the triple intersection; the distinction between open and closed balls matters exactly at this threshold.

Combinatorics: Identities, Generating Functions, Catalan Numbers, and More

N-20240323

These exercises move from binomial identities and recurrences to convexity and graph walks. What first looks like a mixed problem set gradually becomes a catalogue of ways to encode counting: by subsets, generating functions, recursive decompositions, convex combinations, and matrix powers. The Catalan-number derivations and their combinatorial models follow Alexander Postnikov's MIT 18.212 algebraic-combinatorics lectures.[1]

§33.1 Binomial identities

Several binomial identities become clearer once their combinatorial meanings are made explicit. The basic identity

$$\sum_{i=0}^{n} \binom{n}{i} = 2^n$$

counts all subsets of an n-element set: choosing a subset means choosing its size i and then choosing i elements.

Pascal's identity

$$\binom{n}{i} + \binom{n}{i+1} = \binom{n+1}{i+1}$$

can be proved by fixing a distinguished element in an $(n+1)$-element set. An $(i+1)$-subset either contains the distinguished element or it does not. The two cases give the two terms on the left.

Vandermonde's identity

$$\sum_{i} \binom{m}{i}\binom{n}{r-i} = \binom{m+n}{r}$$

counts choosing r balls from two boxes with m and n balls. If i balls are chosen from the first box, then $r-i$ are chosen from the second.

Another useful identity is

$$\binom{n}{r}\binom{n-r}{k-r} = \binom{n}{k}\binom{k}{r}.$$

Both sides count a pair (R, K) with $R \subseteq K \subseteq [n]$, $|R| = r$, and $|K| = k$.

§33.2 Good subsets

A nonempty subset $A \subseteq \{1, 2, \ldots, n\}$ is called good if

$$|A| \leq \min_{x \in A} x.$$

[1] MIT 18.212, *Algebraic Combinatorics*, lecture notes.

Let a_n be the number of good subsets. The problem asks one to prove

$$a_{n+2} = a_{n+1} + a_n + 1.$$

A clean proof splits good subsets of $[n+2]$ into cases.

First, if $n+2 \notin A$, then A is just a good subset of $[n+1]$, giving a_{n+1} possibilities.

Second, if $A = \{n+2\}$, this gives one extra subset.

Third, suppose $n+2 \in A$ and $|A| \geq 2$. Remove $n+2$ and subtract 1 from every remaining element. If

$$A' = A \setminus \{n+2\},$$

then the condition becomes

$$|A'| + 1 \leq \min A'.$$

After shifting $B = \{x - 1 : x \in A'\}$, this is exactly

$$|B| \leq \min B,$$

so B is a good subset of $[n]$. This gives a_n possibilities. Hence

$$a_{n+2} = a_{n+1} + a_n + 1.$$

A summation proof fixes $|A| = k$. Every element must then be at least k, so the remaining data are chosen from a tail interval, and repeated Pascal identities yield the recurrence.

§33.3 Generating functions

For a sequence $(a_n)_{n\geq 0}$, its ordinary generating function is

$$f(x) = \sum_{n\geq 0} a_n x^n.$$

The note emphasizes a key principle: coefficients can encode counting problems.

For example, the number of nonnegative integer solutions of

$$x_1 + x_2 + \cdots + x_k = n$$

is the coefficient of x^n in

$$(1 + x + x^2 + \cdots)^k = \frac{1}{(1-x)^k}.$$

Using the binomial expansion,

$$\frac{1}{(1-x)^k} = \sum_{n\geq 0} \binom{n+k-1}{k-1} x^n,$$

so the number of solutions is

$$\binom{n+k-1}{k-1}.$$

The notes also use generating functions to separate parity or color constraints. A typical trick is to multiply several series, each recording the allowed choices for one box or color, and then read off the coefficient of the target power.

§33.4 Independent Trials and Generating Functions

For a nonnegative integer-valued random variable X, its probability generating function is

$$G_X(t) = \mathbb{E}[t^X] = \sum_{k \ge 0} \Pr(X = k) t^k.$$

If X, Y are independent, then

$$G_{X+Y}(t) = \mathbb{E}[t^{X+Y}] = \mathbb{E}[t^X]\mathbb{E}[t^Y] = G_X(t) G_Y(t).$$

For example, one Bernoulli(p) trial has generating function $1 - p + pt$. Two independent batches of 2 and 3 such trials have respective generating functions $(1 - p + pt)^2$ and $(1 - p + pt)^3$. Their sum therefore has generating function

$$(1 - p + pt)^5, \qquad \Pr(X + Y = 2) = \binom{5}{2} p^2 (1 - p)^3.$$

This is the probabilistic analogue of ordinary generating-function convolution: independence turns counting or probability of a sum into multiplication of series.

§33.5 Catalan numbers

The note then turns to Catalan numbers. One model is the number of ways to triangulate a convex $(n + 2)$-gon:

$$C_n = \frac{1}{n+1} \binom{2n}{n}.$$

For example, a hexagon has 14 triangulations.

Let h_n denote the number of triangulations of an $(n + 2)$-gon. Choose the triangle incident with a fixed side. If its third vertex cuts the polygon into two smaller polygons, the numbers of triangulations multiply. Therefore

$$h_n = \sum_{i=0}^{n-1} h_i h_{n-1-i}, \qquad h_0 = 1.$$

If

$$H(x) = \sum_{n \ge 0} h_n x^n,$$

then the recurrence gives

$$H(x) = 1 + x H(x)^2.$$

Solving the quadratic and choosing the branch with $H(0) = 1$ gives

$$H(x) = \frac{1 - \sqrt{1 - 4x}}{2x}, \qquad h_n = C_n.$$

This derivation explains why Catalan numbers appear in triangulations, parentheses, Dyck paths, and many noncrossing structures: all of them are recursively split into a left and a right part.

§33.6 Convex sets and convex hulls

The geometry part defines a convex set S by the condition that for any $x, y \in S$ and any $t \in [0,1]$,

$$tx + (1-t)y \in S.$$

Equivalently, S contains the line segment between any two of its points. The note contrasts convex and non-convex examples.

For finitely many points $x_1, \dots, x_k$, their convex hull is

$$\left\{ \sum_{i=1}^{k} t_i x_i : \ t_i \geq 0, \ \sum_{i=1}^{k} t_i = 1 \right\}.$$

Induction on k proves the claim: the case $k = 2$ is the definition of a segment, and the induction step writes the convex combination as a combination of one point with a convex combination of the remaining points.

§33.7 Separating a point from a closed convex set

Let $C \subset \mathbb{R}^n$ be nonempty, closed and convex, and let $x \notin C$. Because squared distance to x is continuous and grows without bound at infinity, it attains a minimum at some $y \in C$. For any $z \in C$, convexity gives $y + t(z-y) \in C$ for $0 \leq t \leq 1$. The function

$$\varphi(t) = \|x - y - t(z-y)\|^2$$

has its minimum at $t = 0$. Its right derivative therefore satisfies

$$0 \leq \varphi'(0) = -2\langle x - y, z - y \rangle.$$

Let $\ell(v) = \langle x - y, v \rangle$. Then every $z \in C$ obeys

$$\ell(z) \leq \ell(y) < \ell(x),$$

since $\ell(x) - \ell(y) = \|x - y\|^2 > 0$. Thus an affine hyperplane strictly separates the point x from C. The proof also explains the geometric role of projection onto a convex set.

For two arbitrary disjoint convex sets, a closest pair need not exist; additional closedness, compactness or other hypotheses must be checked before applying this proof to both sets.

§33.8 Planar graphs and Euler-type counting

Graph theory and planar configurations provide another counting mechanism: a planar drawing constrains vertices, edges, and faces through Euler-type formulas.

This connects naturally with the previous convex-geometry material: triangulations, planar graphs, polygon decompositions, and convex hulls are different languages for controlling incidence data.

§33.9 Graphs and adjacency matrices

A graph G consists of a vertex set $V = \{v_1, \dots, v_n\}$ and an edge relation. For a simple graph, the adjacency matrix $A(G) = (a_{ij})$ is defined by

$$a_{ij} = \begin{cases} 1, & \text{if there is an edge from } v_i \text{ to } v_j, \\ 0, & \text{otherwise.} \end{cases}$$

For a multigraph, a_{ij} records the number of edges from v_i to v_j.

The source quotes the standard theorem: the (i,j)-entry of $A(G)^\ell$ equals the number of walks of length ℓ from v_i to v_j. The proof is matrix multiplication. For $\ell = 2$,

$$(A^2)_{ij} = \sum_k a_{ik} a_{kj},$$

which counts all intermediate vertices v_k. The general case follows by induction.

§33.10 Catalan objects share a splitting rule

The parenthesizations of $n+1$ ordered factors and the triangulations of a convex $(n+2)$-gon are both counted by C_n. Choosing the final binary operation or the triangle incident with a fixed edge splits the remaining object into independent left and right pieces. Both satisfy

$$C_0 = 1, \qquad C_n = \sum_{i=0}^{n-1} C_i C_{n-1-i}.$$

A separate derivation uses Dyck paths. There are $\binom{2n}{n}$ paths with n upward and n downward steps. The reflection principle bijects those that ever cross below the horizontal axis with paths containing $n+1$ downward and $n-1$ upward steps, giving

$$C_n = \binom{2n}{n} - \binom{2n}{n+1} = \frac{1}{n+1}\binom{2n}{n}.$$

Recursion, generating functions and reflection now calculate the same number by distinct mechanisms; there is no need to restart the Catalan definition in several chapters of the same note.

§33.11 Adjacency powers count walks

For a finite undirected graph with adjacency matrix A, the entry

$$(A^k)_{ij}$$

counts length-k walks from vertex i to vertex j. For $k = 2$, this is

$$(A^2)_{ij} = \sum_h A_{ih} A_{hj},$$

which counts intermediate vertices. Inductively, a walk of length $k+1$ consists of a length-k walk followed by one edge; matrix multiplication counts exactly these decompositions.

If the graph is d-regular with $d > 0$, then $P = A/d$ is the transition matrix for simple random walk. Because P is symmetric and each row sums to one, the uniform distribution is stationary. If the graph is also connected and non-bipartite, its remaining eigenvalues have modulus strictly less than one:

$$1 = \lambda_1(P), \qquad \rho = \max_{2 \le j \le n} |\lambda_j(P)| < 1.$$

Expanding a probability vector in an orthonormal eigenbasis shows that its deviation from the uniform distribution decreases in Euclidean norm by at least the factor ρ^k after k steps. In a bipartite connected graph, the eigenvalue -1 causes period-two oscillation, so convergence without averaging fails.

This distinction matters: the graph's second-largest signed eigenvalue is not enough; mixing is controlled by the largest *absolute* nontrivial eigenvalue.

§33.12 Which encoding should a problem use?

Subsets and chains naturally invite binomial coefficients; independent decomposition produces products of generating functions; recursively split polygon triangulations produce Catalan numbers; convex hulls encode finite weighted averages; graph walks are encoded by powers of an adjacency matrix. The point of identifying the encoding is that it yields a calculable identity or a proof, with its conditions made explicit, rather than an analogy based on shared vocabulary.

Self-Avoiding Walks I: Connective Constants, Bridges, Polygons, and Critical Exponents

N-20251126

§34.1 What the model is trying to measure

A self-avoiding walk is the simplest discrete model of a polymer chain with excluded volume. The walk is allowed to move from one lattice point to a neighbouring lattice point, but it is not allowed to visit the same vertex twice. Thus the model keeps the most basic feature of a real chain: two monomers cannot occupy the same place.

Let $\mathbb{Z}^d$ be the hypercubic lattice. An n-step walk is a sequence

$$\omega = (\omega_0, \omega_1, \dots, \omega_n),$$

where $\omega_i \in \mathbb{Z}^d$, $\omega_0 = 0$, and

$$|\omega_{i+1} - \omega_i| = 1.$$

It is self-avoiding if

$$\omega_i \neq \omega_j \qquad (i \neq j).$$

Write c_n for the number of n-step self-avoiding walks starting at the origin.

The ordinary simple random walk has exactly $(2d)^n$ possible n-step paths. For self-avoiding walks, there is no such closed formula, because each new step depends on the entire past of the path. This is the first reason the subject is difficult: the model is local in its rule but non-local in its memory.

Remark 34.1.1 — The phrase "self-avoiding" sounds like a small constraint, but it changes the problem qualitatively. The number of walks is no longer obtained by multiplying independent choices, and the endpoint distribution is no longer governed by the usual central limit theorem in low dimensions.

The very first counts already show where the dependence on the whole past begins. On the square lattice $\mathbb{Z}^2$ there are four choices for the first step. As long as the path has fewer than four steps, forbidding *immediate reversal* is enough to prevent every repeated vertex: the square is the shortest lattice cycle. Thus

$$c_1 = 4, \qquad c_2 = 4 \cdot 3 = 12, \qquad c_3 = 4 \cdot 3^2 = 36.$$

At step four, blindly allowing three directions gives $4 \cdot 3^3 = 108$ non-backtracking paths. But there are four unit squares incident to the origin, and two ways to traverse each square back to the starting vertex. Those eight paths revisit the origin, so

$$\boxed{c_4 = 108 - 8 = 100.}$$

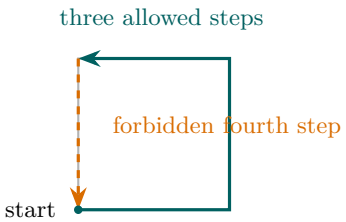


Figure 34.1: A three-step square-lattice walk is self-avoiding, but the dashed fourth edge returns to its first vertex. Local avoidance starts to depend on history at the first available cycle.

This tiny discrepancy, $108 - 100$, is the start of the large-scale difficulty: every later step has to remember *all* visited vertices, not just the previous one. The eventual connective constant measures exactly the accumulated effect of those excluded returns.

§34.2 Submultiplicativity and the connective constant

The first invariant is the exponential growth rate of c_n. Concatenation gives the basic inequality

$$c_{m+n} \le c_m c_n.$$

Indeed, if an $(m+n)$-step self-avoiding walk is cut after m steps, then the first part is an m-step self-avoiding walk, and the translated second part is an n-step self-avoiding walk. Not every pair can be concatenated without intersection, so this gives an inequality rather than an equality.

By Fekete's lemma, the limit

$$\mu = \lim_{n \to \infty} c_n^{1/n}$$

exists. This number is called the connective constant. Equivalently,

$$\log \mu = \lim_{n \to \infty} \frac{1}{n} \log c_n.$$

The rough meaning is

$$c_n \approx \mu^n \times \text{a subexponential correction.}$$

Definition 34.2.1. The connective constant μ is the exponential growth rate of self-avoiding walks on the lattice. It depends on the lattice, not only on the dimension.

The reciprocal

$$z_c = \mu^{-1}$$

is the critical point of the generating function

$$\chi(z) = \sum_{n \geq 0} c_n z^n.$$

For $z < z_c$, the series converges. For $z > z_c$, it diverges. Thus the connective constant is both a combinatorial growth constant and the location of a statistical-mechanical phase transition.

§34.3 Bridges and why they help

Counting all self-avoiding walks directly is hard because a walk can wander in all directions. A bridge is a self-avoiding walk with a preferred direction. For example, in the first coordinate direction, a bridge is a walk whose first coordinate starts at its minimum and ends at its maximum:

$$\omega_0^{(1)} < \omega_i^{(1)} \leq \omega_n^{(1)} \qquad (1 \leq i \leq n).$$

The precise convention varies slightly, but the guiding idea is stable: a bridge progresses from left to right without going below its starting level in that coordinate.

The following picture is the right mental picture for the Hammersley–Welsh argument. A half-space walk is not yet a bridge, but it can be chopped into alternating bridge-like pieces. The dotted reflected part indicates the trick: reflect suitable tails of the walk so that the pieces line up into one bridge.

Let b_n denote the number of n-step bridges. Bridges are useful because they concatenate better than arbitrary walks. If one bridge is followed by another translated bridge, the result is still a bridge. Hence

$$b_{m+n} \geq b_m b_n.$$

This supermultiplicativity implies that the bridge growth rate exists:

$$\lim_{n \to \infty} b_n^{1/n} = \sup_n b_n^{1/n}.$$

A central fact is that bridges have the same exponential growth rate as all self-avoiding walks:

$$\lim_{n \to \infty} b_n^{1/n} = \mu.$$

The Hammersley–Welsh method turns this bridge idea into a quantitative estimate. A general walk is first cut at a suitable extremal point into two half-space walks. Each of those is then *unfolded*: reflecting certain pieces makes its successive widths strictly decreasing and produces bridge-like pieces. The original walk can be recovered when the unfolding choices are recorded.

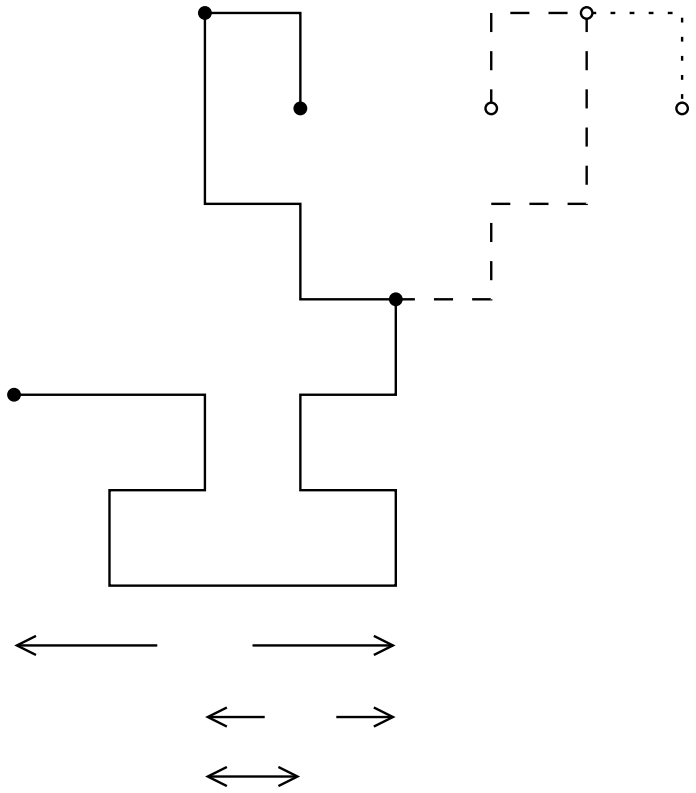


Figure 34.2: a half-space walk decomposed into bridges and reflected into a single bridge.

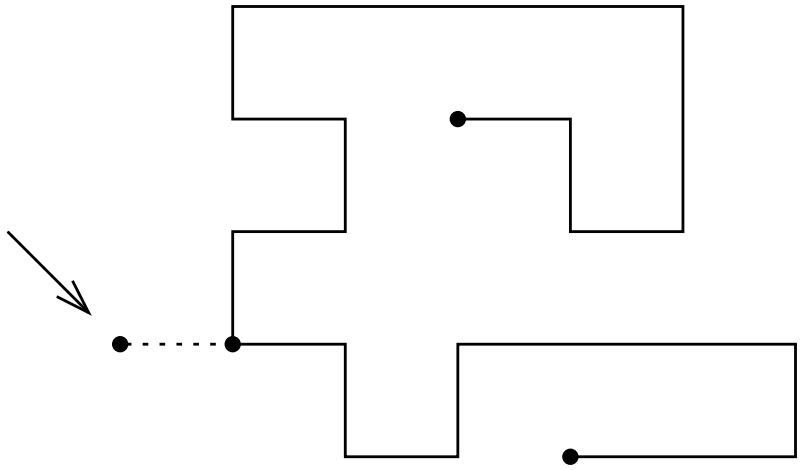


Figure 34.3: Cutting a general walk into two half-space walks before unfolding their successive excursions into bridges.

Why should the number of possible unfoldings grow only subexponentially? The sequence of distinct positive widths is constrained by the total length n: it is essentially an integer partition of a number no larger than n. The number $p(n)$ of integer partitions satisfies

$$p(n) \le \exp(C\sqrt{n})$$

for some constant C. Each set of bridge pieces can therefore be encoded with a multiplicity cost bounded at this scale. The resulting Hammersley–Welsh upper bound has the form

$$\boxed{c_n \le \exp(C'\sqrt{n})\,\mu^n} \qquad (n \ge 1,\ d \ge 2),$$

with C' depending on the lattice and dimension.[1]

Taking n-th roots shows immediately why this is useful:

$$\bigl(\exp(C'\sqrt{n})\mu^n\bigr)^{1/n} = \exp(C'/\sqrt{n})\mu \longrightarrow \mu.$$

The organizing information (a sequence of excursion widths) grows far more slowly than the walks themselves. This is a recurring technique in combinatorics: replace an enormous family by a structured family with a controlled multiplicity of encodings.

§34.4 Polygons

A self-avoiding polygon is a closed self-avoiding path, considered up to translation and often up to choice of starting point and orientation. If p_n denotes the number of n-edge self-avoiding polygons, then $p_n = 0$ when n has the wrong parity on bipartite lattices such as $\mathbb{Z}^d$.

Polygons have the same connective constant:

$$\lim_{n\to\infty} p_n^{1/n} = \mu,$$

[1] See Tom Hutchcroft, *The Hammersley–Welsh Bound for Self-Avoiding Walk Revisited*, for a modern proof and context.

with the usual convention of restricting to the allowed lengths. Intuitively, a long polygon is like a long self-avoiding walk whose two ends happen to meet. The closing condition changes the polynomial correction, not the exponential growth rate.

The polygon viewpoint is important because it connects self-avoiding walks with loop models and planar statistical mechanics. In two dimensions, loops are not just a counting device; they interact with conformal geometry.

The bridge picture has an analogue for polygons. The following construction turns two bridges into a longer self-avoiding walk closely related to a polygon-counting lower bound. The useful point is again not exact enumeration. It is a controlled way to build large objects out of smaller ones while keeping self-avoidance visible.

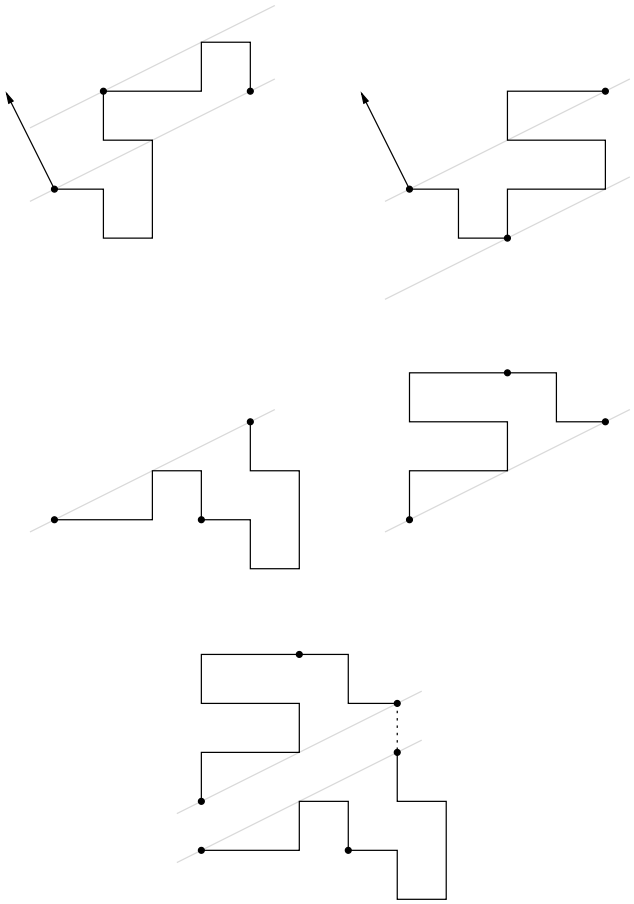


Figure 34.4: bridge data used in a polygon lower-bound construction.

§34.5 Critical exponents

The connective constant describes only the exponential part of c_n. The more delicate question is the subexponential correction. The expected form is

$$c_n \sim A\mu^n n^{\gamma-1},$$

where γ is a critical exponent. Another exponent ν describes the typical size of an n-step walk:

$$\mathbb{E}(|\omega_n|^2) \approx Dn^{2\nu}.$$

The exponent ν measures how swollen the polymer is. For simple random walk one has $\nu = 1/2$. Self-avoidance pushes the walk away from itself, so in low dimensions ν is expected to be larger.

There is also a two-point function

$$G_z(x) = \sum_{\omega:0\to x} z^{|\omega|},$$

where the sum is over self-avoiding walks from 0 to x. Summing over endpoints gives the susceptibility

$$\chi(z) = \sum_{x\in\mathbb{Z}^d} G_z(x) = \sum_{n\geq 0} c_n z^n.$$

Near z_c, one expects

$$\chi(z) \approx (z_c - z)^{-\gamma}.$$

So the same exponent γ can be read from either the coefficient asymptotics of c_n or the singularity of the susceptibility.

§34.6 How dimension changes the story

The dimension d controls how strongly a walk intersects itself. This is why the self-avoiding walk behaves differently in different dimensions.

In dimension 1, the model is trivial: after the first step, the walk cannot turn around without revisiting a vertex. Thus there are only two n-step self-avoiding walks for $n \geq 1$.

In dimension 2, self-avoidance is strong. The expected scaling limit is related to Schramm–Loewner evolution with parameter 8/3. The predicted exponents are

$$\nu = \frac{3}{4}, \qquad \gamma = \frac{43}{32}.$$

These values are part of the conformal-invariance picture, but the full square-lattice scaling limit remains a major open problem.

In dimension 3, the model is physically central but mathematically difficult. The exponents are not expected to be mean-field, and no exact solution is known.

Dimension 4 is critical. The leading power laws are mean-field-like, but logarithmic corrections appear. This is the dimension where renormalisation group methods become essential.

In dimensions $d \geq 5$, the model has mean-field behaviour. The lace expansion proves, in suitable settings, that self-avoiding walks behave diffusively:

$$\nu = \frac{1}{2}, \qquad \gamma = 1.$$

The walk is still self-avoiding, but high-dimensional space gives it enough room that self-intersections become perturbative rather than dominant.

§34.7 The broader lesson

The first layer of the theory is the following dictionary:

combinatorial object	analytic quantity
c_n	$\chi(z) = \sum c_n z^n$
μ	$z_c = 1/\mu$
bridges	controlled concatenation
polygons	closed-loop version of the model
$G_z(x)$	endpoint-resolved two-point function

The connective constant is the exponential growth rate. Critical exponents describe what remains after this exponential growth has been factored out. Bridges and polygons are not side topics: they are the combinatorial tools that make the basic growth theory work.

Self-Avoiding Walks II: The Hexagonal Lattice, Holomorphic Observables, and the Exact Connective Constant

N-20251127

§35.1 Why the hexagonal lattice is special

For most lattices, the connective constant of self-avoiding walks is not known exactly. The hexagonal lattice is exceptional. Duminil-Copin and Smirnov proved that its connective constant is

$$\mu_{\mathrm{hex}} = \sqrt{2+\sqrt{2}}.$$

Equivalently, the critical step weight is

$$x_c = \frac{1}{\mu_{\mathrm{hex}}} = \frac{1}{\sqrt{2+\sqrt{2}}}.$$

This result is not an enumeration trick in the elementary sense. It comes from a discrete complex-analytic observable. The proof is one of the cleanest examples of the modern two-dimensional philosophy: at the critical point, a lattice model may possess a discrete holomorphic structure that forces exact identities.

Remark 35.1.1 — The number $\sqrt{2+\sqrt{2}}$ should not be memorised as a miracle constant. It is the value for which the parafermionic observable satisfies a local cancellation identity on the hexagonal lattice.

§35.2 When does self-avoidance first change the count?

At very short lengths, the hexagonal lattice makes counting look almost trivial. From a fixed vertex there are three choices for the first step. After that, a walk that merely avoids immediately reversing direction has two possible next steps. A regular hexagon is the shortest cycle in this lattice. Consequently, for $1 \le n \le 5$, a non-backtracking n-step walk cannot revisit any earlier vertex, so

$$\boxed{c_1 = 3, \quad c_2 = 6, \quad c_3 = 12, \quad c_4 = 24, \quad c_5 = 48.}$$

At step six something genuinely new happens. There would be $3 \cdot 2^5 = 96$ non-backtracking paths if we kept ignoring the older part of the walk. But three hexagonal faces meet at the starting vertex, and each can be traversed in two directions to return to that vertex in six steps. Those six paths are forbidden. Hence

$$\boxed{c_6 = 96 - 3 \cdot 2 = 90.}$$

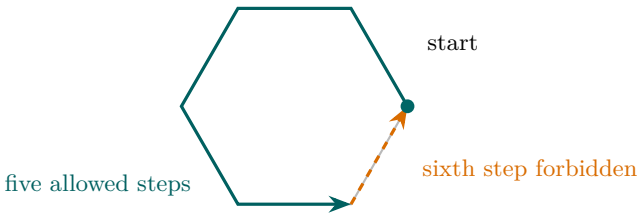


Figure 35.1: A length-five walk around five sides of a hexagonal face visits six distinct vertices; the dashed sixth edge would return to its starting vertex. This is the earliest time that the honeycomb lattice remembers more than the previous step.

This is the difficulty in miniature: a local rule with two immediate choices becomes a history-dependent counting problem as soon as cycles become possible. Finding the exponential growth rate μ requires something stronger than any fixed-length enumeration. The parafermionic observable achieves that by using winding to make entire families of walks cancel.

§35.3 A domain version of the counting problem

Instead of counting walks in the whole plane immediately, one works in a finite domain cut from the hexagonal lattice. Fix a boundary point a. For another boundary point or mid-edge z, consider self-avoiding walks γ starting at a and ending at z, staying inside the domain. Their weighted sum has the form

$$\sum_{\gamma : a \to z} x^{|\gamma|} \times \text{a complex phase depending on the winding of } \gamma.$$

The length weight $x^{|\gamma|}$ remembers the usual self-avoiding-walk generating function. The complex phase is the new ingredient.

The winding of a planar path records how much the direction of the path turns. On the hexagonal lattice the allowed turns are highly constrained, and this makes it possible to tune the phase so that local contributions cancel.

The first picture for this part is especially helpful. On the left, the domain is made of hexagonal-lattice cells and the boundary is described by mid-edges. On the right, the same picture explains winding: the path is not only counted by its length, but also by the total amount it turns.

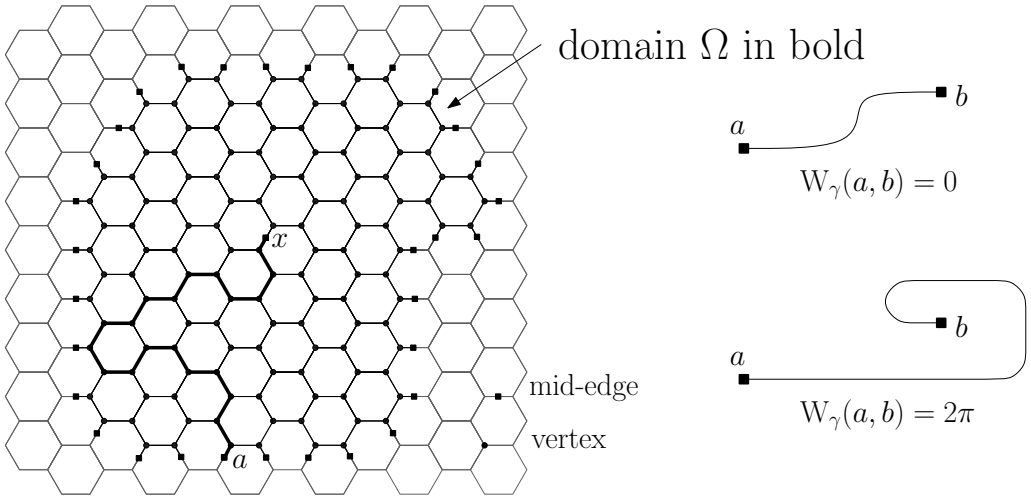


Figure 35.2: a hexagonal-lattice domain and the winding of a self-avoiding walk.

§35.4 The parafermionic observable

The observable can be remembered schematically as

$$F(z) = \sum_{\gamma : a \to z} x^{|\gamma|} e^{-i\sigma W(\gamma)},$$

where $W(\gamma)$ is the total winding and σ is a spin parameter. The exact theorem uses the special choice

$$\sigma = \frac{5}{8}, \qquad x = x_c = \frac{1}{\sqrt{2+\sqrt{2}}}.$$

At this value, the observable satisfies a discrete Cauchy–Riemann type relation around each vertex.

To see where the exact constant comes from, let p, q, r be the three mid-edges touching an interior vertex v, represented as complex points in the plane. The observable satisfies the local relation

$$\boxed{(p-v)F(p) + (q-v)F(q) + (r-v)F(r) = 0.}$$

Duminil-Copin and Smirnov prove it by grouping walks into pairs (that traverse a local loop in opposite directions) and triplets (one walk and its two possible one-step extensions). In a triplet, write

$$j = e^{2\pi i/3}, \qquad \lambda = e^{-5\pi i/24}.$$

After the common length and winding weight is factored out, its contribution to the left-hand side is proportional to

$$1 + x j \overline{\lambda} + x \bar{j} \lambda = 1 + 2x \cos\frac{7\pi}{8} = 1 - 2x \cos\frac{\pi}{8}.$$

This vanishes exactly when

$$x = x_c = \frac{1}{2\cos(\pi/8)} = \frac{1}{\sqrt{2+\sqrt{2}}}.$$

The other paired contributions cancel because of the chosen winding exponent $\sigma = 5/8$. Thus the otherwise unexpected square root is forced by the angles of a hexagon and the cancellation of three possible local paths.[1]

The term *holomorphic* needs a qualification. The displayed identity is a *partial* discrete Cauchy–Riemann relation. It has the useful cancellation property, but by itself does not imply classical holomorphicity of a scaling limit, nor even the existence of that limit. This is exactly why the connective-constant theorem can be proved while convergence of self-avoiding walks to $\mathrm{SLE}_{8/3}$ remains a separate problem.

Remark 35.4.1 — The observable is not merely a generating function. The phase remembers geometry. Without the winding phase, the local cancellation identity would not hold.

§35.5 From local cancellation to a global identity

The local identity is summed over all vertices in a finite domain. Interior terms cancel, leaving only boundary contributions. This is the discrete analogue of using Green's theorem or Cauchy's theorem: local holomorphicity turns an interior statement into a boundary statement.

The cancellation is proved by grouping local contributions. The picture below shows the two types of groupings: pairs of walks that visit all three adjacent mid-edges, and triplets obtained by extending or erasing the last step. At the special values of x and σ, each such group has total contribution zero.

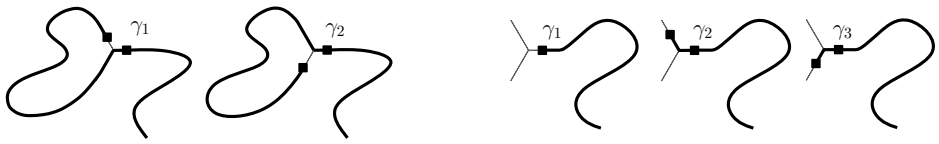


Figure 35.3: pair and triplet cancellations in the proof of the local observable identity.

In a finite trapezoidal strip $S_{T,L}$, partition the endpoints among the left boundary (excluding the start), the right boundary, and the upper and lower boundaries. Call their ordinary, *positive* weighted walk sums $A_{T,L}(x)$, $B_{T,L}(x)$ and $E_{T,L}(x)$, respectively. The domain picture records which walks contribute to which sum.

[1] The derivation follows Duminil-Copin and Smirnov, *The connective constant of the honeycomb lattice equals* $\sqrt{2+\sqrt{2}}$, Sections 2–3.

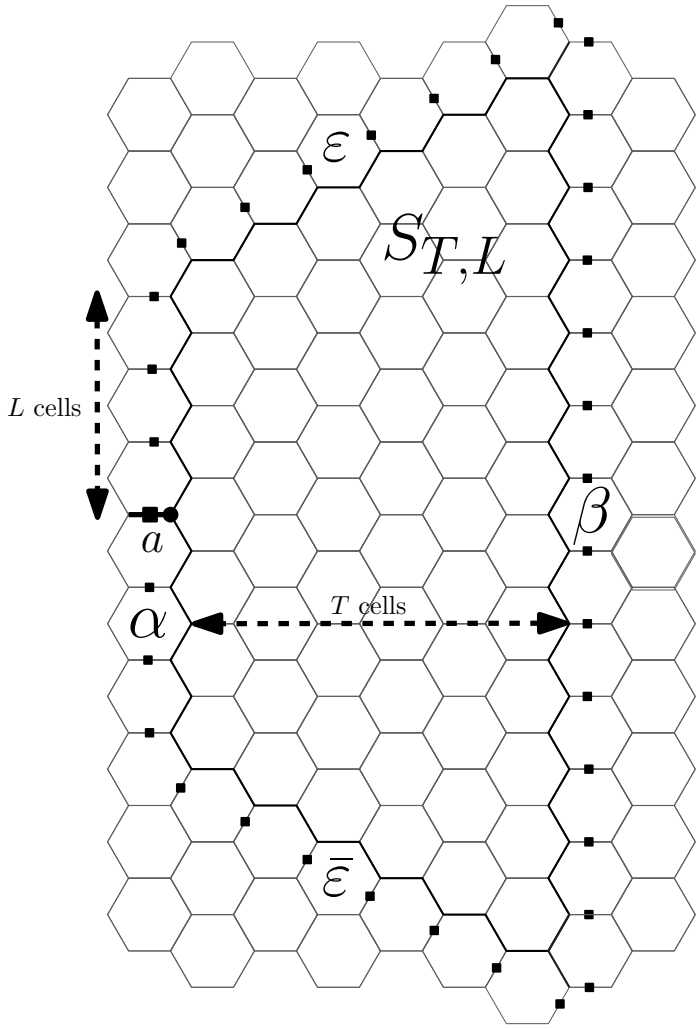


Figure 35.4: the trapezoidal domain $S_{T,L}$ and its boundary pieces.

Summing the local relation over the strip cancels every interior mid-edge. The winding phase on each remaining boundary arc is known, so the complex identity becomes an exact positive real identity:

$$\boxed{1 = \underbrace{\cos\frac{3\pi}{8}}_{c_\alpha} A_{T,L}(x_c) + B_{T,L}(x_c) + \underbrace{\cos\frac{\pi}{4}}_{c_\varepsilon} E_{T,L}(x_c).}$$

All three coefficients are positive. In particular,

$$B_{T,L}(x_c) \le 1, \qquad A_{T,L}(x_c) \le \left(\cos\frac{3\pi}{8}\right)^{-1}.$$

The initial value 1 on the left comes from the empty walk at the starting mid-edge; the trigonometric coefficients are the real parts of its boundary winding weights. The exact coefficients matter, because the same identity will provide both a bound for bridges and a mechanism forcing divergence at the critical point.

As the strip height L increases, the ordinary generating functions for walks reaching a fixed boundary arc have limits, allowing us to work with A_T, B_T and E_T. Their positivity survives the limit.

§35.6 Why divergence gives the lower bound

There is an easy logical trap here. The radius of convergence of the full walk-generating function

$$Z(x) = \sum_{n \geq 0} c_n x^n$$

is $R = 1/\mu_{\text{hex}}$ by the root test. If $Z(x_c)$ diverges, the radius satisfies $R \leq x_c$, and therefore

$$\boxed{\mu_{\text{hex}} \geq x_c^{-1} = \sqrt{2+\sqrt{2}}.}$$

The inequality points this way because taking positive reciprocals reverses order.

The local identity provides a surprisingly concrete route to that divergence. Let A_T, B_T, E_T be the limits of the positive boundary generating functions as the strip height tends to infinity. If $E_T > 0$ for some width T, walks reaching arbitrarily high in that strip already force $Z(x_c) = \infty$: every sufficiently tall boundary contributes at least E_T to disjoint classes of walks.

What if $E_T = 0$ for every width? Then the remaining identity is

$$1 = c_\alpha A_T + B_T.$$

In the original proof, cutting a walk counted by $A_{T+1} - A_T$ at the first visit to the farther edge bounds the difference by two crossing bridges:

$$A_{T+1} - A_T \leq x_c B_{T+1}^2.$$

Subtracting the two boundary identities gives

$$\boxed{c_\alpha x_c B_{T+1}^2 + B_{T+1} \geq B_T.}$$

An elementary induction then implies $B_T \geq c/T$ for some $c > 0$ (one may take $c = \min(B_1, (c_\alpha x_c)^{-1})$). Walks crossing different strip widths form disjoint classes, so

$$Z(x_c) \;\geq\; \sum_{T \geq 1} B_T \;\geq\; c \sum_{T \geq 1} \frac{1}{T} \;=\; \infty.$$

Either way, the generating function diverges at x_c. The harmonic-series bound is a rather beautiful place for a discrete holomorphic identity to end up: all those complex phases have led to a statement about *positive counts*.

§35.7 Why convergence gives the upper bound

For the reverse inequality, we must prove that $Z(x)$ converges at every $x < x_c$. This means $R \geq x_c$, or

$$\boxed{\mu_{\text{hex}} \leq x_c^{-1}.}$$

The critical boundary identity bounds the bridge sum by $B_T(x_c) \leq 1$. A bridge crossing a strip of width T takes at least T steps, so for $0 < x < x_c$,

$$B_T(x) \leq \left(\frac{x}{x_c}\right)^T B_T(x_c) \leq \left(\frac{x}{x_c}\right)^T.$$

Consequently $\sum_{T\geq 1} B_T(x) < \infty$. The bridge decomposition explained in the preceding note represents every walk by two ordered collections of bridges with strictly decreasing widths. This controls the complete partition function:

$$\boxed{Z(x) \leq 2 \prod_{T\geq 1} (1 + B_T(x))^2 < \infty.}$$

The product converges because $\log(1+u) \leq u$ for $u \geq 0$ and the bridge weights have a convergent sum. Thus $R \geq x_c$, as desired.

The two halves of the proof use the boundary identity in different ways: it creates *enough* walks to force divergence at the boundary, while simultaneously placing *strong enough* bounds on crossing bridges to guarantee convergence below it. Together they give

$$\boxed{\mu_{\text{hex}} = \sqrt{2+\sqrt{2}}.}$$

Theorem 35.7.1 (Connective constant of the hexagonal lattice)
For self-avoiding walks on the hexagonal lattice,

$$\lim_{n\to\infty} c_n^{1/n} = \sqrt{2+\sqrt{2}}.$$

§35.8 Relation with SLE

The same observable also points toward the expected scaling limit. The conjectural statement is that two-dimensional self-avoiding walk, after suitable rescaling, converges to $\text{SLE}_{8/3}$. This would imply many predicted critical exponents, such as

$$\nu = \frac{3}{4}, \qquad \gamma = \frac{43}{32}.$$

The exact connective constant result is much more modest than a full scaling-limit theorem, but it belongs to the same conceptual world. The local observable is designed so that at criticality the lattice model begins to look conformally natural.

Remark 35.8.1 — The theorem gives the exact exponential growth constant on the hexagonal lattice. It does not by itself prove the $\text{SLE}_{8/3}$ scaling limit. The former is a rigorous theorem; the latter is the guiding two-dimensional conjectural picture.

§35.9 Loop models and the same mechanism

Self-avoiding walks can be viewed as the $n = 0$ member of the $O(n)$ loop model. In the loop model, configurations consist of nonintersecting loops, and each loop receives a weight n. Formally setting $n = 0$ kills configurations with closed loops and leaves a single open path, which is the self-avoiding walk.

This viewpoint explains why holomorphic observables appear. For special values of the loop weight and the step weight, the model has a parafermionic observable satisfying local relations. The hexagonal self-avoiding walk is the most famous case, but the mechanism belongs to a broader family of two-dimensional lattice models.

The phase diagram situates this in the $O(n)$ phase diagram. For this note, the picture should be read only as orientation: self-avoiding walk corresponds to the $n = 0$ edge of the loop-model story, while the exactly solvable point belongs to a broader two-dimensional critical curve.

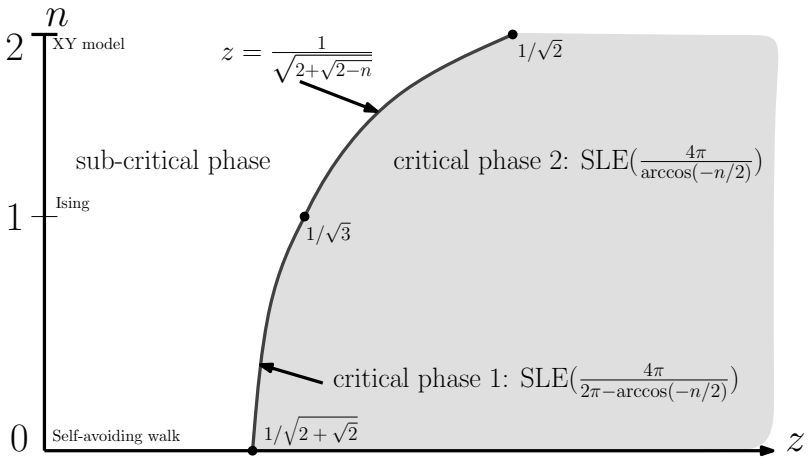


Figure 35.5: phase diagram for $O(n)$ loop models.

§35.10 From example to method

The proof of the hexagonal connective constant has a clear conceptual shape:

choose a winding-weighted observable
$\Downarrow$
tune x and σ so local cancellations hold
$\Downarrow$
sum over a finite domain
$\Downarrow$
obtain positive boundary identities
$\Downarrow$
pass to the infinite lattice and identify μ.

The striking part is that an exact enumerative constant is obtained through a complex-analytic idea rather than through exact enumeration of c_n.

Self-Avoiding Walks III: Lace Expansion, Functional Integrals, and the Critical Dimension Four

N-20251128

§36.1 The high-dimensional philosophy

In low dimensions, self-avoidance dominates the geometry of the walk. In high dimensions, the walk has more room, and self-intersections become less frequent. The expected result is that the self-avoiding walk behaves like simple random walk at large scales:

$$|\omega_n| \text{ is typically of order } n^{1/2}.$$

This is called mean-field behaviour.

The rigorous tool that proves this in dimensions $d \geq 5$, for nearest-neighbour self-avoiding walk, is the lace expansion. It is a systematic inclusion–exclusion method that turns the global self-avoidance constraint into a perturbative correction to the simple random walk equation.

§36.2 The two-point function

The central analytic object is the two-point function

$$G_z(x) = \sum_{\omega : 0 \to x} z^{|\omega|},$$

where the sum is over self-avoiding walks from 0 to x. The susceptibility is

$$\chi(z) = \sum_{x \in \mathbb{Z}^d} G_z(x).$$

The critical point is $z_c = 1/\mu$.

For simple random walk, the two-point function satisfies a renewal equation. If D is the one-step distribution, then the generating function C_z satisfies

$$C_z = \delta_0 + zD * C_z.$$

Equivalently, in Fourier space,

$$\widehat{C}_z(k) = \frac{1}{1 - z\widehat{D}(k)}.$$

For self-avoiding walk, this exact renewal equation fails because the future path is constrained by the past. The lace expansion repairs the equation by adding correction terms.

§36.3 The lace expansion as corrected random walk

The output of the lace expansion can be remembered schematically as

$$G_z = \delta_0 + zD * G_z + \Pi_z * G_z,$$

where Π_z is the lace-expansion correction. This formula is not meant as a definition of Π_z; it is the structural form to remember. The term $zD * G_z$ is what simple random walk would have. The term $\Pi_z * G_z$ records the accumulated effect of forbidding self-intersections.

The correction terms are represented by small diagrams. The following two examples should be read as diagrammatic shorthand rather than literal pictures of the walk. A line represents a two-point function, and extra chords represent constraints coming from self-intersection corrections.

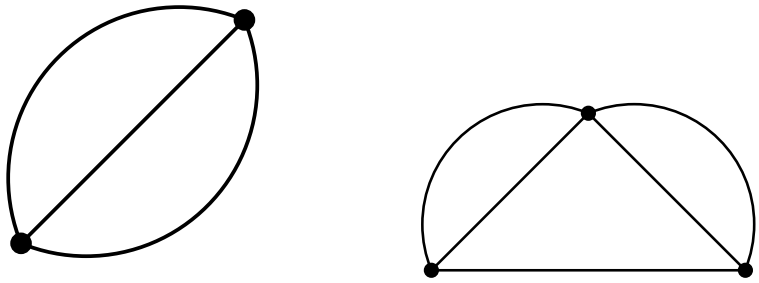


Figure 36.1: typical lace-expansion diagrams.

In Fourier space the relation becomes morally

$$\widehat{G}_z(k) = \frac{1}{1 - z\widehat{D}(k) - \widehat{\Pi}_z(k)}.$$

Thus the proof of mean-field behaviour becomes a problem of showing that Π_z is small and regular enough near criticality.

Remark 36.3.1 — The lace expansion is often intimidating because the combinatorics of laces is elaborate. Conceptually, however, its role is simple: it isolates the first random-walk term and packages all self-intersection corrections into a controlled perturbation.

§36.4 Bubble diagrams and diagrammatic estimates

The perturbative parameter is measured by diagrammatic sums. A basic example is the bubble diagram

The word “bubble” is literal in the diagrammatic notation. One sums over two independent-looking connections between the same endpoints. Finiteness of the corresponding sum is what makes the high-dimensional expansion behave like a small perturbation.

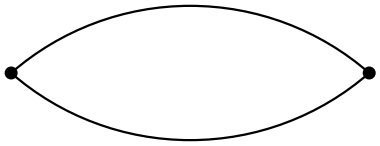


Figure 36.2: the bubble diagram.

$$B(z) = \sum_{x \in \mathbb{Z}^d} G_z(x)^2.$$

Here is a calculation that makes the number *four* emerge rather than appear as a fact to memorize. Use the nearest-neighbor simple random walk on $\mathbb{Z}^d$, with

$$D(e_j) = D(-e_j) = \frac{1}{2d} \quad (1 \le j \le d), \qquad \widehat{D}(k) = \frac{1}{d} \sum_{j=1}^{d} \cos k_j.$$

For this *probability-normalized* random walk, the critical step parameter is $z = 1$. Its critical two-point function $C_1(x)$ has Fourier transform

$$\widehat{C}_1(k) = \frac{1}{1 - \widehat{D}(k)}.$$

When the bubble sum is finite, Parseval's identity identifies it with

$$\sum_{x \in \mathbb{Z}^d} C_1(x)^2 = \frac{1}{(2\pi)^d} \int_{[-\pi,\pi]^d} \frac{dk}{(1 - \widehat{D}(k))^2}.$$

The only possible singularity of this integral is at $k = 0$. The ordinary Taylor expansion of cosine tells us how strong that singularity is:

$$\begin{aligned} 1 - \widehat{D}(k) &= \frac{1}{d} \sum_{j=1}^{d} (1 - \cos k_j) \\ &= \frac{|k|^2}{2d} + O(|k|^4). \end{aligned}$$

Thus the integrand behaves like $|k|^{-4}$ near the origin. In d dimensions a thin radial shell of radius r has volume proportional to $r^{d-1} dr$. The dangerous part of the integral is therefore comparable to

$$\boxed{\int_0^1 r^{d-1}\, r^{-4}\, dr = \int_0^1 r^{d-5}\, dr.}$$

We can read the threshold directly:

$d < 4$	power-law divergence
$d = 4$	$\int_0^1 dr/r$ diverges logarithmically
$d > 4$	the integral converges.

This is a calculation for *simple random walk*; it identifies the dimension in which a self-avoidance perturbation can plausibly become small. Establishing a corresponding

bubble bound for self-avoiding walk itself is a much deeper conclusion of lace-expansion analysis.[1]

In dimensions $d > 4$, one can prove estimates of the following kind:

$$\sum_x G_{z_c}(x)^2 < \infty.$$

Such bounds imply that the lace-expansion correction behaves perturbatively. They lead to the mean-field critical exponents

$$\gamma = 1, \qquad \nu = \frac{1}{2}.$$

More concretely, the susceptibility diverges like

$$\chi(z) \asymp \frac{1}{1 - z/z_c},$$

and the mean-square displacement grows linearly in n.

§36.5 Differential inequalities

One route to controlling the susceptibility is through differential inequalities. The generating function $\chi(z)$ is increasing in z, and differentiating it gives information about the average length of a walk under the z-weighted measure. The lace expansion supplies inequalities that compare $\chi'(z)$ to powers of $\chi(z)$.

The mean-field picture corresponds to the ordinary differential equation

$$\chi'(z) \approx C\chi(z)^2.$$

Solving this heuristic equation gives a simple pole at the critical point, hence $\gamma = 1$. The rigorous proof replaces the heuristic approximation by upper and lower differential inequalities with controlled error terms.

§36.6 Functional integral representations

The lecture notes also explain a different representation of the model. A weakly self-avoiding walk can be written using a self-intersection local time. In continuous time, one can weight a walk by

$$\exp\left(-g\sum_x L_x^2\right),$$

where L_x is the local time spent at x and $g > 0$ penalises self-intersections. The strict self-avoiding walk corresponds morally to the limit where self-intersections are forbidden completely.

There is a supersymmetric functional integral representation in which walk quantities are encoded by integrals involving bosonic and fermionic variables. The point is not that one wants to compute these integrals explicitly. The point is that the representation turns the probabilistic walk model into a field-theoretic object suitable for renormalisation group analysis.

The renormalisation-group part organises space into blocks and then into polymers made of blocks. This terminology is unfortunately overloaded: these "polymers" are not the original self-avoiding walks. They are bookkeeping regions used to localise the effective interaction at each scale.

[1] For the bubble condition and its Fourier interpretation, see Gordon Slade, *Self-avoiding walk, spin systems and renormalization*; compare the original lecture notes *Lectures on Self-Avoiding Walks* by Bauerschmidt, Duminil-Copin, Goodman and Slade.

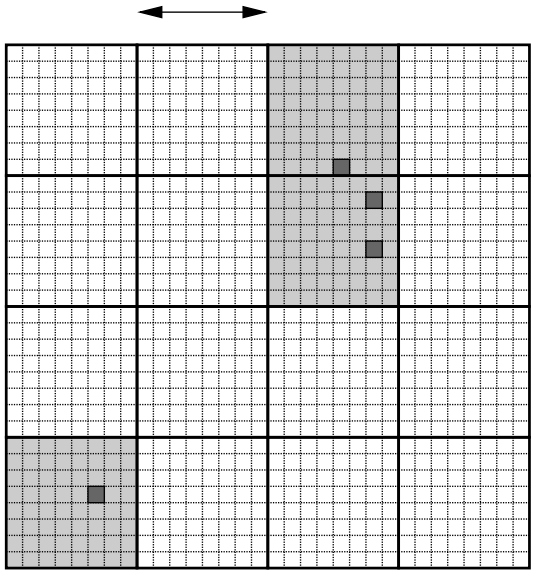


Figure 36.3: a polymer at one scale and its closure at the next scale.

Remark 36.6.1 — The fermionic variables are not an extra physical particle in the polymer. They are an algebraic device that makes cancellations exact. In this representation, supersymmetry is a bookkeeping symmetry for the walk model.

§36.7 Dimension four and logarithmic corrections

Dimension 4 is the upper critical dimension for self-avoiding walk. Above four dimensions, the bubble diagram is finite and the lace expansion gives mean-field behaviour. At four dimensions, the relevant diagram is marginal: it fails only logarithmically. This is why the leading powers look mean-field but acquire logarithmic corrections.

For the continuous-time weakly self-avoiding walk in $d = 4$, the renormalisation group tracks how coupling constants change across length scales. One decomposes the covariance into scale pieces and integrates out fluctuations scale by scale. The effective interaction evolves under a map of the form

$$Z_0 \mapsto Z_1 \mapsto Z_2 \mapsto \cdots .$$

The essential phenomenon is that the self-interaction coupling is marginally irrelevant: it flows slowly to zero. The walk becomes asymptotically Gaussian, but only after logarithmic corrections are taken into account.

A typical conclusion has the form

$$\chi(g, \nu) \sim A\, \varepsilon^{-1} (\log \varepsilon^{-1})^{1/4}$$

as the killing parameter $\varepsilon \downarrow 0$, with model-dependent constants. The exact formulation depends on the version of the weakly self-avoiding walk, but the memory aid is stable: four dimensions means mean-field powers multiplied by logarithms.

§36.8 How the three methods fit together

The three major techniques in these lectures have different roles:

method	dimension or setting	main output
bridges and polygons	all dimensions	growth constants and subexponential bounds
holomorphic observable	$d = 2$, hexagonal lattice	exact connective constant
lace expansion	$d > 4$	mean-field critical behaviour
renormalisation group	$d = 4$	logarithmic corrections

This is a useful mental map because the methods are not interchangeable. The two-dimensional proof uses special planar structure. The high-dimensional proof uses small intersection probabilities. The four-dimensional proof needs a scale-by-scale analysis because the perturbation is exactly marginal at first order.

§36.9 The conceptual turn

The lace expansion should be remembered as the high-dimensional counterpart of the two-dimensional holomorphic observable. Both are ways of making the self-avoidance constraint tractable, but they exploit opposite regimes. In two dimensions, geometry is rigid and special identities appear. In high dimensions, geometry is loose and self-intersections become a small perturbation.

The critical dimension 4 is the boundary between these worlds. Above it, self-avoiding walk is close to simple random walk. At it, the same statement remains morally true, but logarithms record the slow decay of the self-interaction.

Twelvefold Counting: What Labels Remember

N-20261015

Suppose four people must be placed into three nonempty groups. With group names, there are 36 assignments; without group names, there are 6 partitions of the four people; if the people are also indistinguishable, the only possible pattern is $2 + 1 + 1$. All three answers are correct. What changed was the amount of information we agreed to remember.

The *twelvefold way* arranges such questions into a single scheme. More unexpectedly, its entries lead to group actions, Stirling numbers, finite differences and two rather different sorts of generating functions. We will try to recover the formulas from the objects they count, so that the table can eventually be remembered without memorizing its entries.[1]

§37.1 A distribution is a function

Take n balls and k boxes, initially with both sorts of labels. Every distribution is a function

$$f : [n] \longrightarrow [k], \qquad i \longmapsto \text{the box containing ball } i.$$

It is worth asking which extra properties of f can be imposed. With no restriction, there are k^n functions. If no two balls may share a box, f must be injective; if every box must be used, f must be surjective.

For $n = 4$, $k = 3$, the distribution

$$A, B \longmapsto 1, \qquad C \longmapsto 2, \qquad D \longmapsto 3$$

has a particularly useful fourfold description. If we remove the names from the balls while retaining the names of the boxes, we remember only the *occupancy vector* $(2, 1, 1)$. If we keep the ball names but remove the box names, we remember which balls were together:

$$\{\{A, B\}, \{C\}, \{D\}\}.$$

Finally, with neither sort of name, we remember just the multiset of positive occupancies, customarily written as an integer partition $4 = 2 + 1 + 1$.

One subtlety will matter throughout: an unused box still exists. In the occupancy vector it appears as a zero; in a set partition or an integer partition it disappears. Thus k boxes without names permit *at most* k nonempty blocks, unless we insist that all boxes are occupied.

§37.2 The twelve entries

Let $S(n, j)$ denote the number of ways to partition an n-element labelled set into exactly j nonempty, unordered blocks. Let $p_{\le k}(n)$ and $p_{=k}(n)$ count integer partitions of n into

[1] The classical table and its precise mapping formulation appear in NIST Digital Library of Mathematical Functions, §26.17, following Richard Stanley's *Enumerative Combinatorics*, Vol. 1.

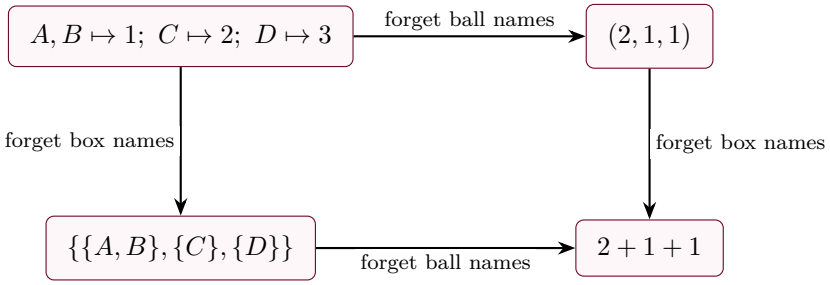


Figure 37.1: Four descriptions of one assignment. The arrows discard information; reversing them requires choices and usually does not give a unique answer.

at most k, respectively exactly k, positive parts. Put

$$(k)_{\underline{n}} = k(k-1)\cdots(k-n+1), \qquad [P] = \begin{cases} 1, & P \text{ is true,} \\ 0, & P \text{ is false.} \end{cases}$$

Here $n, k \geq 1$, and an impossible configuration has count zero.

Labels retained	Unrestricted	Injective	Surjective
Balls and boxes	k^n	$(k)_{\underline{n}}$	$k!S(n,k)$
Boxes only	$\binom{n+k-1}{k-1}$	$\binom{k}{n}$	$\binom{n-1}{k-1}$
Balls only	$\sum_{j=1}^{k} S(n,j)$	$[n \leq k]$	$S(n,k)$
Neither	$p_{\leq k}(n)$	$[n \leq k]$	$p_{=k}(n)$

Table 37.1: The twelvefold way for $n, k \geq 1$. Empty boxes may occur in the unrestricted and injective columns.

It is enough to know four kinds of objects—functions, occupancy vectors, set partitions, integer partitions—and three restrictions on the fibers of a function. We now derive the rows, partly to check the table and partly to understand where their formulas come from.

§37.3 Reconstructing the table

With both sorts of labels, unrestricted functions give k^n; injections give $(k)_{\underline{n}}$. For surjections, first partition the n balls into k nonempty blocks, then match those blocks bijectively with the k named boxes. This gives $k!S(n,k)$. The factor $k!$ is justified because every box is used: the blocks really can be assigned $k!$ different lists of names.

If only boxes are labelled, write a_i for the number of balls in box i. An unrestricted arrangement is a solution

$$a_1 + \cdots + a_k = n, \qquad a_i \geq 0.$$

Arrange n stars in a row and insert $k-1$ separators among the $n+k-1$ available positions. Consecutive separators indicate empty boxes, and we obtain

$$\#\{(a_i) \in \mathbb{Z}_{\geq 0}^k : \sum a_i = n\} = \binom{n+k-1}{k-1}.$$

For injections, the occupancies belong to $\{0,1\}$, so we choose n of the k boxes. For surjections, $a_i \geq 1$; setting $b_i = a_i - 1$ gives $\binom{n-1}{k-1}$. The difference between weak and positive compositions is just a shift of one in every coordinate.

If only balls are labelled, the nonempty fibers $f^{-1}(i)$ form an unordered set partition. In the unrestricted case it may have $1, 2, \ldots, k$ blocks, giving $\sum_{j=1}^{k} S(n,j)$. Surjectivity fixes the number of blocks at k. Injectivity makes every block a singleton, leaving exactly one partition when $k \geq n$.

Finally, if neither sort of label remains, the only information is a nonincreasing sequence of positive occupancies

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_\ell \geq 1, \qquad \lambda_1 + \cdots + \lambda_\ell = n.$$

An unrestricted arrangement has $\ell \leq k$; a surjection has $\ell = k$; an injection has only parts equal to 1. These are precisely the last three entries.

Remark 37.3.1 — A useful memory test is to redraw the table from a blank sheet using the words

function, weak composition, set partition, integer partition.

The middle column asks that every fiber have size at most one; the last column requires every fiber to be nonempty. No separate list of twelve formulas is needed.

§37.4 Forgetting names

Why is removing a label an operation that can be studied mathematically? Let $\mathcal{F} = \{f : [n] \to [k]\}$. Permutations $\sigma \in S_n$ and $\tau \in S_k$ act on a function by

$$(\sigma, \tau) \cdot f = \tau \circ f \circ \sigma^{-1}.$$

Relabelling the balls changes which input carries a given identity; relabelling the boxes changes the names of the outputs. The injective and surjective subsets are invariant under these actions.

Two functions lie in the same S_n-orbit exactly when they have the same occupancy vector $(a_1, \ldots, a_k)$. Their fibers have the same labelled sizes, and a bijection can be chosen between corresponding fibers. Two functions lie in the same S_k-orbit exactly when they have the same partition of the labelled domain into nonempty fibers. Allowing both actions leaves only the integer partition of the nonzero occupancies.

Thus the first three lines of our diagram can be written as quotients:

$$\mathcal{F}, \qquad \mathcal{F}/S_n, \qquad \mathcal{F}/S_k, \qquad \mathcal{F}/(S_n \times S_k).$$

It is tempting to divide k^n by $k!$ after forgetting the names of the boxes. Take one ball and three boxes: there are three labelled assignments, but just one unlabelled assignment. Dividing by $3! = 6$ would give $1/2$. The two empty boxes may be swapped without changing the assignment, so this assignment has fewer than six distinct relabellings.

For a group G acting on a set X, write Gx for the orbit of x and G_x for its stabilizer. Counting the group elements mapping x to the same point gives

$$|Gx| = \frac{|G|}{|G_x|}.$$

Orbits need not have equal sizes. A particularly useful way to count them is the following.

Theorem 37.4.1 (Burnside's lemma)

For a finite group G acting on a finite set X,

$$\boxed{|X/G| = \frac{1}{|G|}\sum_{g\in G} |\operatorname{Fix}(g)|.}$$

Proof. Count pairs (g, x) satisfying $gx = x$, once by the group element and once by the point:

$$\sum_{g\in G} |\operatorname{Fix}(g)| = \sum_{x\in X} |G_x|.$$

Within each orbit, every stabilizer has cardinality $|G|/|Gx|$; summing $|G_x|$ over that orbit gives $|G|$. Divide by $|G|$. □

For example, color six beads black or white on a circular necklace; rotations count as the same pattern. A rotation by r positions breaks the bead positions into $\gcd(6, r)$ cycles. A coloring survives the rotation precisely when every such cycle is monochromatic. Its fixed-point count is $2^{\gcd(6,r)}$. Hence

$$\#\text{necklaces} = \frac{2^6 + 2^1 + 2^2 + 2^3 + 2^2 + 2^1}{6} = 14.$$

The acting group here is the cyclic group C_6, rather than the full symmetric group S_6. This is the beginning of Pólya counting: specify which symmetries are allowed, then count colorings constant on their cycles. It is also why a geometrically arranged collection of boxes often requires a different counting theory from the twelvefold table.

§37.5 Stirling numbers appear when a new ball arrives

Return to $S(n, k)$. Insert the element n into a partition of $[n]$ with k blocks. Either n forms a singleton block, leaving a partition of $n-1$ elements into $k-1$ blocks, or it joins one of the k existing blocks of a partition of $n-1$ elements into k blocks. Therefore

$$\boxed{S(n, k) = S(n-1, k-1) + kS(n-1, k).}$$

With $S(0, 0) = 1$, $S(n, 0) = 0$ for $n > 0$, this gives

$n\backslash k$	0	1	2	3	4
0	1				
1	0	1			
2	0	1	1		
3	0	1	3	1	
4	0	1	7	6	1

In particular $S(4, 3) = 6$, explaining the initial group problem: labelling the three nonempty groups gives $3! \cdot 6 = 36$ assignments.

There is another formula, useful when n is large and k is fixed. A function $[n] \to [k]$ fails to be surjective when at least one box is missed. Inclusion–exclusion over the missing boxes gives

$$\boxed{k!S(n, k) = \sum_{j=0}^{k} (-1)^j \binom{k}{j} (k-j)^n.}$$

For $n = 4, k = 3$, the right side is $3^4 - 3 \cdot 2^4 + 3 = 36$. The recurrence and the alternating sum compute the same quantity through different decompositions: one inserts a new element, the other excludes forbidden images.

§37.6 From counting to polynomials

Fix an integer $m \geq 0$. There are m^n functions from $[n]$ to $[m]$. Count these functions by the number j of distinct values they actually use. Their fibers form a partition into j blocks, and their j distinct images can be assigned to these blocks in $(m)_{\underline{j}}$ ways. Therefore

$$m^n = \sum_{j=0}^{n} S(n, j)(m)_{\underline{j}}.$$

Both sides are polynomials in m. Since they agree at every nonnegative integer, they agree for an indeterminate x:

$$\boxed{x^n = \sum_{j=0}^{n} S(n, j)(x)_{\underline{j}}.}$$

For example,

$$x^3 = (x)_{\underline{3}} + 3(x)_{\underline{2}} + (x)_{\underline{1}}$$

and

$$x^4 = (x)_{\underline{4}} + 6(x)_{\underline{3}} + 7(x)_{\underline{2}} + (x)_{\underline{1}}.$$

This has a second interpretation. The usual monomials $1, x, x^2, \ldots$ and the falling factorials

$$1, \quad x, \quad x(x-1), \quad x(x-1)(x-2), \ldots$$

are two bases of the polynomial ring. The Stirling numbers of the second kind are the coefficients for changing from the first basis to the second. Reversing the change of basis defines the signed Stirling numbers of the first kind $s(n, j)$:

$$(x)_{\underline{n}} = \sum_{j=0}^{n} s(n, j)x^j.$$

For instance, $(x)_{\underline{3}} = x^3 - 3x^2 + 2x$. The two triangular arrays are inverse matrices. The reason is algebraic and concrete: each is a change of basis, and changing back recovers the original polynomial.

§37.7 The discrete derivative

Define the forward difference

$$\Delta f(x) = f(x+1) - f(x).$$

Ordinary differentiation is particularly simple on powers: $Dx^j = jx^{j-1}$. For finite differences, falling factorials play the same role:

$$\boxed{\Delta(x)_{\underline{j}} = j(x)_{\underline{j-1}}.}$$

Indeed, using $(x)_{\underline{j}} = j!\binom{x}{j}$, this follows from Pascal's identity

$$\binom{x+1}{j} - \binom{x}{j} = \binom{x}{j-1},$$

which is a polynomial identity in x.

Repeated differences now provide a discrete version of Taylor's formula. If f is a polynomial of degree at most d, then

$$\boxed{f(x) = \sum_{j=0}^{d} (\Delta^j f)(0) \binom{x}{j}.}$$

To see this, expand $f = \sum_j c_j \binom{x}{j}$ in the binomial basis. Successive differences satisfy $\Delta\binom{x}{j} = \binom{x}{j-1}$; evaluating $\Delta^r f$ at $x = 0$ leaves precisely c_r.

Apply the formula to $f(x) = x^n$, and compare with the Stirling expansion. We obtain the striking identity

$$\boxed{S(n,j) = \frac{(\Delta^j x^n)|_{x=0}}{j!}.}$$

As $\Delta^j f(0) = \sum_{i=0}^{j} (-1)^{j-i}\binom{j}{i} f(i)$, this is exactly

$$S(n,j) = \frac{1}{j!} \sum_{i=0}^{j} (-1)^{j-i} \binom{j}{i} i^n.$$

The inclusion–exclusion formula for surjections has reappeared as the coefficient formula for a discrete derivative.

Example 37.7.1

Let $f(x) = x^3$. Its successive values at $0, 1, 2, 3$ are $0, 1, 8, 27$. The difference rows begin with $1, 6, 6$, so

$$x^3 = \binom{x}{1} + 6\binom{x}{2} + 6\binom{x}{3}.$$

Dividing the coefficients by $1!, 2!, 3!$ recovers $S(3,1) = 1$, $S(3,2) = 3$, $S(3,3) = 1$. The recurrence triangle, inclusion–exclusion and Newton interpolation are three views of the same coefficients.

There is even an operator identity. Write $D = d/dx$. Since xD sends x^m to mx^m, the polynomial identity already proved implies, on polynomials,

$$\boxed{(xD)^n = \sum_{j=0}^{n} S(n,j) x^j D^j.}$$

For $n = 2$, this reads $(xD)^2 = x^2D^2 + xD$. The two sides agree on each monomial x^m, hence on every polynomial. The same Stirling numbers that count ways of grouping balls therefore appear when we rearrange compositions of differentiation and multiplication by x.

§37.8 A partition's geometry

The final row of the twelvefold table deserves its own geometry. Consider

$$7 = 4 + 2 + 1.$$

Its Ferrers diagram is a collection of left-justified unit squares with row lengths $4, 2, 1$. Reflect the diagram across its main diagonal. Its column lengths are $3, 2, 1, 1$, so the transposed picture represents

$$7 = 3 + 2 + 1 + 1.$$

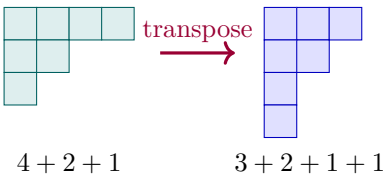


Figure 37.2: Conjugating an integer partition exchanges the number of rows and the largest row length.

The diagram proves a useful equinumerosity:

$$\boxed{p_{\le k}(n) = \#\{\lambda \vdash n : \text{every part of } \lambda \text{ is at most } k\}.}$$

The left side limits the number of rows, while the right side limits their width. Diagram transposition is a bijection, even though there is no obvious term-by-term algebraic correspondence between the two descriptions.

For $n = 7, k = 3$, the partition $4 + 2 + 1$ has three parts and is admissible on the left, while its transpose $3 + 2 + 1 + 1$ has largest part three and is admissible on the right. These are different partitions related by a canonical geometric operation.

§37.9 Products for parts

Why do integer partitions naturally suggest products? For every permitted part size r, we may use zero, one, two or more copies. Recording the total size by the exponent of x, this independent choice contributes

$$1 + x^r + x^{2r} + \cdots = \frac{1}{1 - x^r}.$$

Consequently, as an identity of formal power series,

$$\boxed{\sum_{n \ge 0} p(n) x^n = \prod_{r \ge 1} \frac{1}{1 - x^r}.}$$

If we permit only part sizes $1, \ldots, k$, the product truncates:

$$\sum_{n \ge 0} p_{\le k}(n) x^n = \prod_{r=1}^{k} \frac{1}{1 - x^r},$$

where the equality uses the conjugation bijection from the previous section.

To record *exactly* how many parts we use, introduce a second variable:

$$\boxed{\sum_{n,\ell\geq 0} p_{=\ell}(n)u^{\ell}x^{n} = \prod_{r\geq 1}\frac{1}{1-ux^{r}}.}$$

Here u marks the number of parts. For a fixed positive k, removing one square from every row of a partition into k positive parts produces a partition of $n-k$ into at most k parts. Hence

$$\sum_{n\geq k} p_{=k}(n)x^{n} = \frac{x^{k}}{(1-x)(1-x^{2})\cdots(1-x^{k})}.$$

These formulas encode both restrictions appearing in the unlabelled–unlabelled row of the twelvefold table.

§37.10 Exponentials for blocks

A set partition is built from *labelled* elements. Let $B_n = \sum_{k=0}^{n} S(n,k)$, the n-th Bell number. For a fixed k, the exponential generating function for a nonempty labelled block is

$$e^{x}-1 = \sum_{r\geq 1}\frac{x^{r}}{r!}.$$

If we arrange k nonempty blocks in a definite order, the labelled product construction gives $(e^{x}-1)^{k}$. Forgetting the order of the blocks divides by $k!$, since every block is nonempty and therefore distinct as a set. We get

$$\boxed{\sum_{n\geq 0} S(n,k)\frac{x^{n}}{n!} = \frac{(e^{x}-1)^{k}}{k!}.}$$

Summing over all k gives the exponential formula

$$\boxed{\sum_{n\geq 0} B_{n}\frac{x^{n}}{n!} = \exp(e^{x}-1).}$$

For a quick consistency check, fix the block containing element $n+1$ in a partition of $[n+1]$. Choose r of the other n elements to join it, and partition the remaining $n-r$ elements in any way:

$$B_{n+1} = \sum_{r=0}^{n}\binom{n}{r}B_{n-r}.$$

Differentiating $B(x)=\exp(e^{x}-1)$ produces $B'(x)=e^{x}B(x)$, and comparing coefficients reproduces the same recurrence.

The contrast with Euler's product is informative. For an integer partition, each numerical part size can be repeated independently, so we multiply geometric series. For a set partition, every block consumes a subset of a common pool of labelled elements; choosing how those labels are shared introduces binomial coefficients and factorial denominators. This is why exponential generating functions are adapted to labelled constructions and ordinary generating functions to the unrestricted multiplicities of integer parts.

The twelvefold table began with four ways of remembering the same distribution. It has now supplied two polynomial bases and two kinds of generating functions. There is a remaining question that the table alone cannot answer: if we select a labelled distribution uniformly at random and then forget its labels, are all the resulting unlabelled objects equally likely? The answer depends on the sizes of the orbits, and it leads to an entirely different side of combinatorics.

After Forgetting Labels: Random Occupancy and Hidden Weights

N-20261016

Three balls are independently thrown into two boxes, each ball choosing either box with probability $1/2$. After the experiment, we forget the identities of the balls. There are just four possible occupancy vectors:

$$(3,0), \qquad (2,1), \qquad (1,2), \qquad (0,3).$$

It would be natural, at first glance, to assign probability $1/4$ to each. In the experiment we actually described, their probabilities are

$$\frac{1}{8}, \qquad \frac{3}{8}, \qquad \frac{3}{8}, \qquad \frac{1}{8}.$$

The information we forget is precisely where the unequal weights come from. There are three labelled assignments realizing $(2,1)$, and only one realizing $(3,0)$.

In the twelvefold way, labels determined which objects were considered equal. Here we will ask a more delicate question: *what probability measure survives after the labels have been forgotten?* It leads from exact enumerations to probability thresholds, Poisson approximation and the cycle structure of a random permutation.[1]

§38.1 Three probability models

Keep n and k positive, and consider three ways of choosing a distribution of n balls into k boxes. Each way is perfectly legitimate, but they define different experiments.

First, the balls can be labelled and can independently choose one of the k labelled boxes. We choose one of the k^n functions $f : [n] \to [k]$, all equally likely. If A_i is the occupancy of box i, then

$$\Pr(A_1 = a_1, \dots, A_k = a_k) = \frac{n!}{k^n a_1! \cdots a_k!}, \qquad \sum_{i=1}^{k} a_i = n.$$

The factorials count which ball identities give the specified occupancies. This is the multinomial distribution.

Second, we can ignore ball identities *before sampling*, and choose each weak composition

$$(a_1, \dots, a_k), \qquad a_i \geq 0, \quad \sum_i a_i = n,$$

with equal probability. There are $\binom{n+k-1}{k-1}$ such vectors, so each has probability

$$\Pr_{\text{comp}}(a_1, \dots, a_k) = \binom{n+k-1}{k-1}^{-1}.$$

[1] For the occupancy and coupon-collector calculations, see MIT's 6.856 lecture notes on deviations and Djalil Chafaï, *Back to basics: Coupon collector*. The combinatorics of permutation cycles also appears in Sedgewick and Flajolet's *Analysis of Algorithms*, Chapter 7.

This sampling rule is different from throwing labelled balls independently.

Third, we might disregard the labels of the boxes as well, and choose uniformly among the integer partitions $\lambda \vdash n$ having at most k positive parts. Each such partition then receives probability $1/p_{\le k}(n)$. This is a further distinct probability model.

For the first two models, take $n = 3, k = 2$. The numbers make the distinction difficult to miss:

(a_1, a_2)	$(3,0)$	$(2,1)$	$(1,2)$	$(0,3)$
independent throws	$1/8$	$3/8$	$3/8$	$1/8$
uniform compositions	$1/4$	$1/4$	$1/4$	$1/4$

If box names are subsequently removed, the two possible integer partitions are 3 and $2+1$. Under independent throws their probabilities are $1/4$ and $3/4$, whereas a uniform choice of integer partition assigns $1/2$ to each.

§38.2 Orbit weights

The discrepancy can be calculated for every possible integer partition, not just for small examples.

Let $\lambda = (\lambda_1, \dots, \lambda_\ell)$ be a partition of n into $\ell \le k$ positive parts. Let m_r denote the number of parts of size r in λ. How many functions $f : [n] \to [k]$ produce the unlabelled occupancy pattern λ?

Choose ℓ distinct box names, assign the ℓ part sizes to them, and then assign the n labelled balls to boxes with these occupancies. A convenient combined formula is

$$\boxed{w_k(\lambda) = \frac{n!(k)_{\underline{\ell}}}{\left(\prod_{i=1}^{\ell} \lambda_i!\right)\left(\prod_{r \ge 1} m_r!\right)}.}$$

The $m_r!$ denominators account for the indistinguishable equal parts. The formula agrees with the orbit–stabilizer principle: symmetries that exchange equal-sized groups, and permutations of balls within a group, affect how many labelled functions represent the same pattern.

Example 38.2.1

For $n = 4, k = 3$, there are four possible unlabelled occupancy patterns. The table records the numbers of labelled functions producing them.

λ	(4)	$(3,1)$	$(2,2)$	$(2,1,1)$	total
$w_3(\lambda)$	3	24	18	36	81

Their probabilities under four independent throws are respectively

$$\frac{3}{81}, \quad \frac{24}{81}, \quad \frac{18}{81}, \quad \frac{36}{81}.$$

Under a uniform choice of integer partition, each would have probability $1/4$. The difference is large even at this tiny scale.

There is a general principle here. If a finite group G acts on a finite set X, and we choose an element $x \in X$ uniformly before replacing it by its orbit, then

$$\Pr(\text{orbit } O) = \frac{|O|}{|X|}.$$

Uniformity on X induces an *orbit-size-biased* distribution on X/G. To make the orbits themselves uniformly likely, one must change the sampling procedure. Dividing counts by symmetry and pushing forward a probability measure are related operations, but their normalization rules are different.

§38.3 Counting occupied boxes

Now return to the independent-throw model, with n labelled balls and k labelled boxes. Let K_n be the number of occupied boxes and $Z_n = k - K_n$ the number of empty boxes.

There are $\binom{k}{j}$ choices of which j boxes are occupied. For any fixed choice, the balls must map surjectively onto those boxes, and there are $j!S(n,j)$ such maps. The twelvefold table therefore gives the full distribution:

$$\boxed{\Pr(K_n = j) = \frac{\binom{k}{j} j! S(n,j)}{k^n} = \frac{(k)_j S(n,j)}{k^n},} \qquad 1 \le j \le \min(n,k).$$

In particular,

$$\Pr(Z_n = 0) = \frac{k! S(n,k)}{k^n}.$$

The old counting formula has become an exact probability of covering every box.

To find the expected number of empty boxes, introduce indicators

$$I_i = \begin{cases} 1, & \text{box } i \text{ is empty,} \\ 0, & \text{otherwise.} \end{cases} \qquad Z_n = \sum_{i=1}^{k} I_i.$$

A particular box remains empty precisely when all n balls avoid it. Thus

$$q := \Pr(I_i = 1) = \left(1 - \frac{1}{k}\right)^n, \qquad \boxed{\mathbb{E}Z_n = k\left(1 - \frac{1}{k}\right)^n.}$$

Notice how this expectation was easier than the complete probability distribution. Linearity of expectation avoids the need to describe all possible joint occupancy patterns.

The indicators are dependent: if one box is known to be empty, the remaining balls must occupy the other boxes. For distinct i, j,

$$\mathbb{E}[I_i I_j] = \left(1 - \frac{2}{k}\right)^n.$$

Consequently,

$$\boxed{\operatorname{Var}(Z_n) = kq(1-q) + k(k-1)\left[\left(1 - \frac{2}{k}\right)^n - q^2\right].}$$

The covariance term is nonpositive for $k \ge 2$. Empty-box events are negatively correlated in this model.

What happens in the *uniform-composition* model? A particular box is empty in $\binom{n+k-2}{k-2}$ of the $\binom{n+k-1}{k-1}$ weak compositions. Therefore, for $k \ge 2$,

$$\boxed{\mathbb{E}_{\text{comp}} Z_n = k\frac{k-1}{n+k-1}.}$$

When $n = k \to \infty$, independent throws give approximately k/e empty boxes, while uniform weak compositions give approximately $k/2$. One can meaningfully ask for a uniformly random allocation only after fixing the underlying space of equally likely outcomes.

§38.4 The first collision

Throw balls independently until some box receives a second ball. Before the first collision, the assignment is injective. After n throws, the probability of no collision is exactly

$$\Pr(\text{no collision after } n) = \frac{(k)_{\underline{n}}}{k^n} = \prod_{j=0}^{n-1}\left(1-\frac{j}{k}\right) \qquad (n \le k).$$

Taking logarithms and using $\log(1-t) = -t + O(t^2)$ for small t gives

$$\begin{aligned}\log \Pr(\text{no collision}) &= -\frac{1}{k}\sum_{j=0}^{n-1} j + O\left(\frac{1}{k^2}\sum_{j=0}^{n-1} j^2\right) \\ &= -\frac{n(n-1)}{2k} + O\left(\frac{n^3}{k^2}\right)\end{aligned}$$

when $n = o(k)$. In the interesting regime $n \sim t\sqrt{k}$ with fixed $t > 0$, the error tends to zero. Hence

$$\boxed{\Pr(\text{a collision by } n) \longrightarrow 1 - e^{-t^2/2} \quad \text{if } n/\sqrt{k} \longrightarrow t.}$$

The first collision occurs on the scale $\sqrt{k}$, long before most boxes have been occupied. This is the mathematical core of the birthday paradox.

Why the square root? There are $\binom{n}{2}$ pairs of balls, and each pair has probability $1/k$ of landing in the same box. The expected number of colliding pairs is

$$\mathbb{E}C = \frac{\binom{n}{2}}{k}.$$

It becomes order one when n is of order $\sqrt{k}$. The exact injection formula turns this heuristic into the exponential limit above.

§38.5 The last empty box

Let T_k be the first throw number at which all k boxes are occupied. Once j different boxes have appeared, the next throw reaches a new box with probability

$$\frac{k-j}{k}.$$

Write $T_k = G_0 + \cdots + G_{k-1}$, where G_j is the number of further throws needed to move from j to $j+1$ occupied boxes. The G_j are independent geometric waiting times with means $k/(k-j)$. Thus

$$\boxed{\mathbb{E}T_k = \sum_{j=0}^{k-1}\frac{k}{k-j} = kH_k, \qquad H_k = \sum_{r=1}^{k}\frac{1}{r}.}$$

Since $H_k = \log k + \gamma + o(1)$, the expected cover time is $k\log k + O(k)$. One needs roughly $k\log k$ throws to reach every box, compared with only $\sqrt{k}$ throws to encounter the first repeated box.

The geometric decomposition also computes the variance. A geometric waiting time on $\{1,2,\dots\}$ with success probability p has variance $(1-p)/p^2$. Therefore

$$\boxed{\operatorname{Var}(T_k) = k^2\sum_{r=1}^{k}\frac{1}{r^2} - kH_k = \left(\frac{\pi^2}{6} + o(1)\right)k^2.}$$

The natural fluctuations around the main term $k \log k$ have scale k.

There is also an exact formula for the distribution function. The event $T_k \le n$ says precisely that the n throws form a surjection onto the k boxes:

$$\boxed{\Pr(T_k \le n) = \frac{k!S(n,k)}{k^n} = \sum_{j=0}^{k} (-1)^j \binom{k}{j} \left(1 - \frac{j}{k}\right)^n.}$$

We have returned to the inclusion–exclusion formula for Stirling numbers, now with a time parameter. This will let us refine the estimate $k \log k$ into a limiting distribution.

§38.6 A Gumbel limit

Place the throw count at

$$n_k = \lfloor k \log k + ck \rfloor$$

for a fixed real number c. The expected number of boxes still empty is

$$\mathbb{E} Z_{n_k} = k \left(1 - \frac{1}{k}\right)^{n_k} \longrightarrow e^{-c}.$$

It is already suggestive that a finite expected number of missing boxes remains exactly on the scale where the cover time changes from unlikely to likely.

To see the full limiting law, consider the falling factorial moment

$$(Z_n)_{\underline{r}} = Z_n(Z_n - 1) \cdots (Z_n - r + 1).$$

It counts ordered r-tuples of *distinct empty boxes*. A specified set of r boxes is simultaneously empty when every throw avoids all of them. Hence

$$\boxed{\mathbb{E}(Z_n)_{\underline{r}} = (k)_{\underline{r}} \left(1 - \frac{r}{k}\right)^n.}$$

With $n = n_k$ and r fixed,

$$(k)_{\underline{r}} \sim k^r, \qquad \left(1 - \frac{r}{k}\right)^{n_k} \sim k^{-r} e^{-cr},$$

and therefore

$$\mathbb{E}(Z_{n_k})_{\underline{r}} \longrightarrow e^{-cr}.$$

These are exactly the factorial moments of a Poisson random variable with mean e^{-c}. The factorial-moment criterion gives

$$Z_{n_k} \xrightarrow{\mathrm{d}} \mathrm{Poisson}(e^{-c}).$$

In particular, the probability that *no* empty boxes remain tends to $\exp(-e^{-c})$. Therefore

$$\boxed{\Pr\left(\frac{T_k - k \log k}{k} \le c\right) \longrightarrow \exp(-e^{-c}).}$$

This is the Gumbel distribution. Its appearance has a simple reading: the final stage of collecting all boxes is governed by the extreme waiting time for the last missing possibilities, rather than by an average waiting time.

The scales from the previous two sections are markedly different. The first collision is already likely after $O(\sqrt{k})$ throws. The last empty box disappears around $k \log k$ throws. They are both properties of the same random function, revealed by applying different events to its fibers.

§38.7 Poissonization

In fixed-n occupancy, knowing that one box has received many balls affects the allocations of the others. There is an elegant way to remove this dependence: randomize the *number* of balls.

Let N be Poisson with mean $k\lambda$, and conditionally on N, throw all N balls independently into k labelled boxes. Let A_i be the resulting occupancies. Their joint probability generating function is

$$\begin{aligned}\mathbb{E}\left[\prod_{i=1}^{k} t_i^{A_i}\right] &= \mathbb{E}\left[\left(\frac{t_1+\cdots+t_k}{k}\right)^N\right] \\ &= \exp\left(k\lambda\left(\frac{t_1+\cdots+t_k}{k}-1\right)\right) \\ &= \prod_{i=1}^{k}\exp(\lambda(t_i-1)).\end{aligned}$$

The last expression factors completely. Each A_i is a Poisson random variable with mean λ, and the k occupancies are independent.

An individual box is now empty with probability $e^{-\lambda}$, so

$$\boxed{Z \sim \text{Binomial}(k, e^{-\lambda}), \qquad \Pr(Z=0) = (1-e^{-\lambda})^k.}$$

Set $\lambda = \log k + c$. Then

$$(1-e^{-\lambda})^k = \left(1-\frac{e^{-c}}{k}\right)^k \longrightarrow \exp(-e^{-c}).$$

The Gumbel expression emerges immediately from independent Poisson occupancies. To recover the fixed-number-of-balls result rigorously, one can compare Poisson total counts with $\lfloor k\log k + ck\rfloor \pm \varepsilon k$. The Poisson total has standard deviation $O(\sqrt{k\log k}) = o(k)$, and the event that every box is occupied is monotone in the number of throws. Taking the limit and then letting $\varepsilon \downarrow 0$ transfers the Poissonized result back to the fixed-n model.

Poissonization is useful far beyond boxes and balls. It replaces a fixed-total constraint by independent local counts; analytic calculations become simpler, and concentration of the randomized total can return us to the original problem.

§38.8 Random set partitions

So far we have forgotten the identities of the balls and occasionally the boxes. There is another quotient worth considering: keep the balls labelled and forget only the names of the boxes. The outcome is a set partition of $[n]$, with a number J of blocks between 1 and k.

Suppose we first sample a uniform random function $[n] \to [k]$, and only then forget box names. A fixed set partition with j blocks has

$$(k)_{\underline{j}}$$

preimages: we can assign its j distinct nonempty blocks injectively to k labelled boxes. Therefore that particular partition has probability

$$\boxed{\Pr(\Pi = \pi) = \frac{(k)_{\underline{|\pi|}}}{k^n},}$$

where $|\pi|$ is the number of blocks of π.

In contrast, a uniformly random set partition of $[n]$ chooses among the B_n partitions with equal weight, and its block-count distribution is

$$\Pr_{\text{uniform}}(J = j) = \frac{S(n,j)}{B_n}.$$

For the partition induced by a random function, the corresponding law is

$$\Pr_{\text{function}}(J = j) = \frac{S(n,j)(k)_{\underline{j}}}{k^n}.$$

Even though both random objects are called a random partition, their block counts need not behave similarly. The construction leading to the partition is part of its probability law.

This is a useful distinction whenever a combinatorial structure is obtained by forgetting hidden labels in a random process. Counting the unlabelled structures and describing their natural sampling distribution are separate tasks.

§38.9 Permutation cycles

Integer partitions arise in another setting that brings a new probability measure with them. Every permutation $\sigma \in S_n$ can be written as a product of disjoint cycles. For example,

$$\sigma = (1\,4\,2)(3\,5)(6)$$

has cycle-length partition $3 + 2 + 1$ of 6.

The lengths alone do not specify the permutation: a cycle of length r has $(r-1)!$ possible cyclic orderings on a given set of r labelled elements. The cost of forgetting those cyclic orderings is another factorial weight.

Let $\lambda \vdash n$ have m_r parts of size r. The number of permutations of cycle type λ is

$$\boxed{\frac{n!}{z_\lambda}, \qquad z_\lambda = \prod_{r \geq 1} r^{m_r} m_r!.}$$

To derive it, arrange the n labels in a linear order ($n!$ choices), cut the list into successive groups of lengths prescribed by λ, and close each group into a cycle. Rotating an r-cycle does not change it, producing a factor r for every such cycle. Permuting the m_r cycles of equal length also changes nothing, producing the factor $m_r!$.

If σ is uniformly sampled from S_n, its cycle type has the distribution

$$\boxed{\Pr(\text{cycle type} = \lambda) = \frac{1}{z_\lambda}.}$$

In particular, $\sum_{\lambda \vdash n} 1/z_\lambda = 1$. Integer partitions have appeared again, but with weights that differ from both uniform integer partitions and the random occupancy patterns.

For $n = 4$, this produces a surprisingly uneven distribution:

λ	(4)	$(3,1)$	$(2,2)$	$(2,1,1)$	$(1,1,1,1)$
#permutations	6	8	3	6	1
$\Pr(\lambda)$	1/4	1/3	1/8	1/4	1/24

If we select one of the five integer partitions uniformly, every type would instead have probability 1/5.

§38.10 Counting random cycles

Let $C_r(\sigma)$ be the number of cycles of length r in a uniformly random permutation of n elements. Choose an r-element subset to be the support of a specified cycle. There are $(r-1)!$ cyclic orders on this subset and $(n-r)!$ permutations on the remaining labels. Dividing by $n!$ gives

$$\boxed{\mathbb{E}C_r = \frac{\binom{n}{r}(r-1)!(n-r)!}{n!} = \frac{1}{r}, \qquad 1 \le r \le n.}$$

Summing over all lengths,

$$\boxed{\mathbb{E}[\#\text{cycles of } \sigma] = \sum_{r=1}^{n} \frac{1}{r} = H_n.}$$

A random permutation of a very large set typically has far fewer cycles than elements: its expected number of cycles grows only logarithmically.

There is a generating-function version that explains the exact weights $1/r$. A cycle of length r on labelled elements contributes z^r/r to an exponential generating function: there are $(r-1)!$ cyclic orders, and we divide by $r!$. A permutation is an unordered set of such cycles. If the variable u_r records the number of r-cycles, the exponential construction gives

$$\boxed{\sum_{n\ge 0} \frac{z^n}{n!} \sum_{\sigma\in S_n} \prod_{r\ge 1} u_r^{C_r(\sigma)} = \exp\left(\sum_{r\ge 1} \frac{u_r z^r}{r}\right).}$$

Set every $u_r = 1$:

$$\exp\left(\sum_{r\ge 1} \frac{z^r}{r}\right) = \exp(-\log(1-z)) = \frac{1}{1-z}.$$

The coefficient of z^n is 1, as expected from $n!/n! = 1$.

More interestingly, setting every $u_r = u$ records the total number $C(\sigma)$ of cycles:

$$\frac{1}{n!} \sum_{\sigma\in S_n} u^{C(\sigma)} = [z^n](1-z)^{-u} = \frac{u(u+1)\cdots(u+n-1)}{n!}.$$

This is the probability generating function of $C(\sigma)$. Differentiating at $u = 1$ yields

$$\mathbb{E}C(\sigma) = \frac{d}{du} \frac{u(u+1)\cdots(u+n-1)}{n!}\bigg|_{u=1} = H_n,$$

agreeing with our direct cycle-count argument.

The harmonic number has appeared twice in this chapter, for two unrelated random quantities:

$$\mathbb{E}T_k = kH_k \qquad \text{and} \qquad \mathbb{E}C(\sigma) = H_n.$$

In the coupon-collector problem it comes from successive geometric waiting times; in random permutations it comes from the chance of seeing a cycle of each length. The shared expression is revealing precisely because the counting mechanisms are so different.

§38.11 Long cycles and the hundred prisoners

There is a wonderful application of the cycle-counting formula that looks, at first, like a problem about search strategies.

A hundred numbered prisoners face a hundred numbered boxes. The boxes contain the numbers $1, \dots, 100$, arranged in a uniformly random permutation. Each prisoner, alone, may open at most fifty boxes and must find the number matching their own identity. The group succeeds only if *every* prisoner succeeds. They may agree on a strategy before the experiment but cannot communicate afterward.

If each prisoner independently opens fifty randomly chosen boxes, their individual success probabilities are $1/2$, and the probability of everyone succeeding is 2^{-100}. There is a far more effective strategy using the permutation hidden in the boxes.

Prisoner i first opens box i. If it contains the number j, they next open box j, and repeat. Their search follows the unique cycle of the permutation containing i. They find their own number upon completing that cycle, so prisoner i succeeds within fifty openings exactly when their cycle has length at most 50. Every prisoner succeeds exactly when the permutation has no cycle longer than 50.

Why can we compute the probability of this global event so easily? For $r > 50$, a permutation of 100 elements cannot contain two cycles of length r, or even two cycles whose lengths both exceed 50. Thus the events

$$\{\text{there is a cycle of length } r\}, \qquad r = 51, \dots, 100,$$

are pairwise disjoint. Their probabilities are $\mathbb{E}C_r = 1/r$, by the calculation in the previous section. Therefore

$$\boxed{\Pr(\text{everyone succeeds}) = 1 - \sum_{r=51}^{100} \frac{1}{r} = 1 - H_{100} + H_{50} \approx 0.311828.}$$

A probability of more than 31% is astonishing compared with 2^{-100}. The prisoners do not need to know the permutation in advance; they exploit its cycle structure as they reveal it.

The same calculation works for $2m$ prisoners who may open m boxes each. Under the cycle-following strategy,

$$\Pr(\text{all succeed}) = 1 - \sum_{r=m+1}^{2m} \frac{1}{r} \longrightarrow 1 - \log 2 \approx 0.306853.$$

We have avoided any direct enumeration of the huge family of successful permutations. The global search constraint has been translated into an invariant of a permutation, and the expected number $1/r$ of r-cycles has done all the counting. This is a good example of why the cycle type of a random permutation is a mathematically useful description rather than merely a convenient notation.

§38.12 What is random?

We have encountered several ways to produce a partition. Throw labelled balls independently and forget the box names; choose a set partition uniformly; forget ball identities too and obtain an integer partition; or take cycle lengths from a uniformly random permutation. Their possible outputs overlap, but their probability measures retain traces of the procedures that generated them.

For uniform independent throws, the weight of an occupancy vector (a_i) is proportional to $1/\prod_i a_i!$. For unlabelled occupancy patterns λ, the orbit-size weight is $w_k(\lambda)$. For cycle types of a uniform random permutation, the weight is $1/z_\lambda$. Uniform integer partitions instead give every allowed λ equal weight.

All these formulas are instances of a simple idea: when several labelled realizations collapse into the same visible object, the number of hidden realizations determines how likely that visible object is to appear. Once that multiplicity is understood, counting identities become probability distributions, and the study of typical behavior begins.

Two particularly instructive consequences have emerged. A birthday-type collision is seen after order $\sqrt{k}$ samples, while the disappearance of the last empty box takes order $k \log k$ samples, with Gumbel fluctuations. And among the very different distributions carried by integer partitions, the cycle types of uniform random permutations lead naturally to a logarithmic number of cycles. The word random needs a construction just as much as the word count needs a definition of equality.

Waiting for a Pattern

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Toss a fair coin until three consecutive heads appear. The expected number of tosses is 14. If we wait for HTH instead, the answer is 10; for HHT, it is 8. All three words have length three, and each has probability $1/8$ of appearing in any specified block of three tosses. Evidently, knowing the probability of a single block is not enough to predict how long we must wait.

The discrepancy is already present in ordinary string matching. A partial match need not be forgotten after a mismatch, and a completed word may overlap another copy of itself. We will use that observation to turn a small probability puzzle into a finite automaton, then into a Markov chain and a generating function. The same calculation will eventually reveal the entire waiting-time distribution through its tail probabilities.[1]

§39.1 Three words, three waiting times

Let $X_1, X_2, \dots$ be independent and uniformly distributed on $\{\mathrm{H}, \mathrm{T}\}$. For a fixed word $w = w_1 \cdots w_m$, define

$$T_w = \min\{n \ge m : X_{n-m+1} \cdots X_n = w_1 \cdots w_m\}.$$

Thus T_w counts tosses up to and including the first completed occurrence of w. We count *consecutive* patterns: HHT is absent from HTHT.

A tempting calculation is

$$\Pr(X_{n-2}X_{n-1}X_n = \mathrm{HHH}) = \frac{1}{8}, \qquad \text{so perhaps } \mathbb{E}T_{\mathrm{HHH}} = 8.$$

The guess overlooks two facts. Windows ending at successive positions use many of the same tosses, and the first two tosses do not yet form a candidate block. Dividing a sequence into disjoint blocks of length three gives independent trials, but discards patterns that straddle a block boundary.

We can find the correct expectation by remembering just how much of the target has matched at the *end* of the sequence seen so far.

For HHH, there are three unfinished states:

$$0 = \text{no useful suffix}, \qquad 1 = \mathrm{H}, \qquad 2 = \mathrm{HH}.$$

Let e_0, e_1, e_2 be the expected additional tosses before success, starting in these states. Conditioning on the next toss gives

$$\begin{aligned} e_0 &= 1 + \tfrac{1}{2}e_1 + \tfrac{1}{2}e_0, \\ e_1 &= 1 + \tfrac{1}{2}e_2 + \tfrac{1}{2}e_0, \\ e_2 &= 1 + \tfrac{1}{2}\cdot 0 + \tfrac{1}{2}e_0. \end{aligned}$$

[1] The underlying theory of overlapping words is developed in Leonidas Guibas and Andrew Odlyzko, *String Overlaps, Pattern Matching, and Nontransitive Games* (1981). For the prefix machinery see Knuth, Morris and Pratt, *Fast Pattern Matching in Strings* (1977). Our calculations below are self-contained.

The 1 in each equation accounts for the toss we are about to make; the 0 in the last line reflects success if the next toss is H. The first equation says $e_0 = 2 + e_1$, and the last says $e_2 = 1 + e_0/2$. Substituting into the middle equation gives

$$e_1 = \frac{3}{2} + \frac{3}{4}e_0, \qquad e_0 = \frac{7}{2} + \frac{3}{4}e_0, \qquad \boxed{\mathbb{E}T_{\mathrm{HHH}} = 14}.$$

For HTH, the unfinished states are $0, \mathrm{H}, \mathrm{HT}$. After reading H, another head still leaves the useful suffix H; a tail advances the match to HT. From HT, a head completes the pattern and a tail leaves no useful suffix. The equations become

$$\begin{aligned} u_0 &= 1 + \tfrac{1}{2}u_1 + \tfrac{1}{2}u_0, \\ u_1 &= 1 + \tfrac{1}{2}u_1 + \tfrac{1}{2}u_2, \\ u_2 &= 1 + \tfrac{1}{2}\cdot 0 + \tfrac{1}{2}u_0. \end{aligned}$$

Now $u_0 = 2 + u_1$, $u_1 = 2 + u_2$ and $u_2 = 1 + u_0/2$, so

$$u_0 = 5 + \frac{u_0}{2}, \qquad \boxed{\mathbb{E}T_{\mathrm{HTH}} = 10}.$$

For HHT the same method gives 8. It is worth understanding why these states work for *every* word, rather than solving a new system from scratch each time.

§39.2 The useful part of a failed match

Take a word w of length m. After observing a finite string s, let

$$q_w(s) = \max\{0 \le j < m : \text{the length-}j \text{ suffix of } s \text{ equals } w_1 \cdots w_j\}, \tag{39.1}$$

provided that w has not already appeared. The integer $q_w(s)$ records the longest suffix of what we have read that could be the beginning of the desired pattern.

The next state is obtained by appending the new letter a and retaining the longest suffix which is a prefix of w. Write this transition as $\delta(j, a)$. If the appended string completes w, we enter a separate absorbing success state †. Otherwise,

$$q_w(sa) = \delta(q_w(s), a).$$

Why does the entire past reduce to a single integer? Every shorter candidate suffix that could become a prefix match after the next letter is itself a suffix of the current longest matching prefix. We can compute its fallback from that prefix alone.

This is the idea behind the failure links in the Knuth–Morris–Pratt algorithm. With a zero-based prefix-function convention,

$$\pi[i] = \text{length of the longest proper border of } w_1 \cdots w_{i+1},$$

where a *border* is simultaneously a prefix and a suffix. In particular,

$$\pi[0] = 0, \qquad \pi_{\mathrm{HHH}} = (0, 1, 2), \qquad \pi_{\mathrm{HTH}} = (0, 0, 1).$$

There is no nonempty *proper* border of a one-letter word. That explains the initial zero; it says nothing about whether arrays are indexed from zero or one.

An instructive check is the string HHT while looking for HTH. The first two heads leave state 1, and the following tail leaves state 2. The earlier failed attempt to match HT did not destroy the opportunity to start matching at the *second* head. For a general pattern, the prefix function organizes precisely these fallback possibilities.

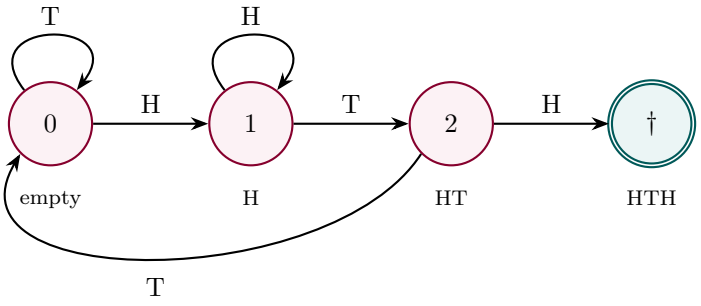


Figure 39.1: The prefix automaton for the first occurrence of HTH. Each arrow consumes one toss. Under a fair coin, each of the two possible outgoing letters has probability $1/2$; † stops the process.

§39.3 A finite automaton becomes a Markov chain

The state transition depends only on the current state and on the next independently sampled letter. Thus a random stream of letters turns our deterministic matching automaton into a finite-state Markov chain, stopped on entering †.

For the three transient states $0, 1, 2$, list the probabilities of transitions *between transient states* in a matrix Q. In the order $0, 1, 2$, the two patterns give

$$Q_{\mathrm{HHH}} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & 0 \end{pmatrix}, \qquad Q_{\mathrm{HTH}} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & 0 \end{pmatrix}. \tag{39.2}$$

Rows corresponding to a state from which success is possible sum to less than 1, because we have removed transitions to †. This is why Q is *substochastic*.

Let $\mathbf{1} = (1, 1, 1)^{\top}$ and let $\mathbf{e}_0 = (1, 0, 0)^{\top}$. If we start in state 0, the probability of *not* seeing the pattern during the first n tosses is exactly

$$\Pr(T_w > n) = \mathbf{e}_0^{\top} Q^n \mathbf{1}. \tag{39.3}$$

This identity is already much more informative than the mean. For example,

$$\Pr(T_{\mathrm{HHH}} = 3) = \Pr(T_{\mathrm{HTH}} = 3) = \frac{1}{8},$$

whereas

$$\Pr(T_{\mathrm{HHH}} = 4) = \frac{1}{16}, \qquad \Pr(T_{\mathrm{HTH}} = 4) = \frac{1}{8}.$$

At toss four, the two waiting-time distributions have visibly diverged.

Why can we invert $I - Q$? For a word w of length m, every disjoint block of m tosses has probability 2^{-m} of equaling w. If any such block equals w, then w must already have appeared by the end of that block. The blocks are independent, so

$$\Pr(T_w > km) \le (1 - 2^{-m})^k. \tag{39.4}$$

In particular, $T_w < \infty$ almost surely and $\mathbb{E}T_w < \infty$. The same bound applies from any transient state, ensuring that the powers Q^n tend to zero. Consequently,

$$N = (I - Q)^{-1} = \sum_{n \ge 0} Q^n$$

exists. Its entry N_{ij} is the expected number of visits to transient state j, starting from state i, before absorption. Summing over the possible states gives

$$\mathbf{t} = N\mathbf{1}, \qquad \mathbf{t} = \mathbf{1} + Q\mathbf{t}, \qquad \mathbb{E}T_w = \mathbf{e}_0^\top (I - Q)^{-1}\mathbf{1}. \tag{39.5}$$

Our earlier recurrences were just these matrix equations written componentwise.

This is an algorithm for any fixed pattern and any finite alphabet with positive letter probabilities. There is, however, a much shorter expression for the answer if the letters are independent and equally likely. To find it, we should look at *how a completed word can overlap itself.*

§39.4 How a word overlaps itself

A *border* of a word $w = w_1 \cdots w_m$ is a prefix which is also a suffix. In this section we allow the whole word itself as a border; a *proper* border has length less than m. Write

$$\mathcal{B}(w) = \{k \in \{1, \ldots, m\} : w_1 \cdots w_k = w_{m-k+1} \cdots w_m\}.$$

An overlap can also be described by the shift between two copies of w. Define

$$\mathcal{D}(w) = \{d \in \{0, \ldots, m-1\} : w_{d+1} \cdots w_m = w_1 \cdots w_{m-d}\}.$$

The two sets are equivalent: $d \in \mathcal{D}(w)$ precisely when $m - d \in \mathcal{B}(w)$. In particular, $0 \in \mathcal{D}(w)$, corresponding to the full-length border.

For our three examples,

w	$\mathcal{B}(w)$	$\mathcal{D}(w)$
HHH	$\{1, 2, 3\}$	$\{0, 1, 2\}$
HTH	$\{1, 3\}$	$\{0, 2\}$
HHT	$\{3\}$	$\{0\}$

The first two heads of HHH can also be the last two heads of a preceding copy. For HTH, only one head can be shared between copies at distinct starting positions. The word HHT has no nonempty proper border at all.

Package the self-overlaps into the polynomial

$$R_w(z) = \sum_{d \in \mathcal{D}(w)} 2^{-d} z^d. \tag{39.6}$$

The exponent records the shift, while 2^{-d} records the probability of obtaining the d fresh letters required to finish a second copy. Thus

$$R_{\mathrm{HHH}}(z) = 1 + \frac{z}{2} + \frac{z^2}{4}, \quad R_{\mathrm{HTH}}(z) = 1 + \frac{z^2}{4}, \quad R_{\mathrm{HHT}}(z) = 1.$$

A polynomial extracted solely from the letters of a word will turn out to determine its entire first-occurrence distribution.

§39.5 One identity for the entire distribution

Write

$$f_n = \Pr(T_w = n), \qquad G_w(z) = \sum_{n \geq m} f_n z^n = \mathbb{E}[z^{T_w}], \qquad 0 \leq z < 1.$$

The event that w *ends* at time n has probability 2^{-m} for every $n \geq m$, irrespective of whether w appeared earlier. We can partition this event according to the first occurrence time $T_w = t \leq n$.

Suppose $T_w = t$, and put $d = n - t$. If $d < m$, an occurrence ending at n overlaps the completed word ending at t. It is possible only when $d \in \mathcal{D}(w)$; in that case the d new tosses must reproduce a specified sequence, giving probability 2^{-d}. If $d \geq m$, the m letters in the later occurrence are entirely new and have probability 2^{-m}. Accordingly define

$$c_d = \begin{cases} 2^{-d}, & 0 \leq d < m, \quad d \in \mathcal{D}(w), \\ 0, & 0 \leq d < m, \quad d \notin \mathcal{D}(w), \\ 2^{-m}, & d \geq m. \end{cases}$$

Independence of future tosses, conditional on $T_w = t$, now gives the *renewal identity*

$$2^{-m} = \sum_{t=m}^{n} f_t \, c_{n-t}, \qquad n \geq m. \tag{39.7}$$

On the left we count every occurrence ending at n. On the right we first locate the earliest occurrence, then check how a later copy can follow it. This is where the word's overlaps enter the calculation.

The generating function of the left side is

$$B(z) = \sum_{n \geq m} 2^{-m} z^n = \frac{2^{-m} z^m}{1 - z}.$$

For the coefficients c_d,

$$C(z) = \sum_{d \geq 0} c_d z^d = R_w(z) + \frac{2^{-m} z^m}{1 - z}.$$

Equation (39.7) is precisely the coefficient identity $B(z) = G_w(z) C(z)$. Solving for G_w yields:

Theorem 39.5.1 (First-occurrence generating function)

For any nonempty binary word w of length m in an independent sequence of fair coin tosses,

$$\boxed{G_w(z) = \frac{2^{-m} z^m}{(1 - z) R_w(z) + 2^{-m} z^m}.} \tag{39.8}$$

The polynomial R_w is determined by the self-overlaps of w as in (39.6).

The formula looks rather generous: from a few overlap shifts, it produces every number $\Pr(T_w = n)$, by expanding a rational function as a power series.

The geometric bound (39.4) gives exponential tails, so we may differentiate G_w at $z = 1$ to compute $\mathbb{E}T_w$. Since

$$G_w(1) = 1, \qquad \frac{d}{dz} \log\bigl((1 - z) R_w(z) + 2^{-m} z^m\bigr)\Big|_{z=1} = m - 2^m R_w(1),$$

we obtain

$$\boxed{\mathbb{E}T_w = G_w'(1) = 2^m R_w(1) = \sum_{k \in \mathcal{B}(w)} 2^k.} \tag{39.9}$$

Now the three original answers are almost visible without calculation:

$$\mathbb{E}T_{\mathrm{HHH}} = 2 + 4 + 8 = 14, \quad \mathbb{E}T_{\mathrm{HTH}} = 2 + 8 = 10, \quad \mathbb{E}T_{\mathrm{HHT}} = 8.$$

For an alphabet of size q, with letters chosen independently and uniformly, the same argument replaces 2 by q:

$$\mathbb{E}T_w = \sum_{k \in \mathcal{B}(w)} q^k.$$

If the letters are independent with unequal probabilities $p(a) > 0$, the corresponding expression is

$$\mathbb{E}T_w = \sum_{k \in \mathcal{B}(w)} \frac{1}{p(w_1)p(w_2)\cdots p(w_k)}. \tag{39.10}$$

Indeed, when a later copy overlaps in $m - d$ letters, the d fresh letters have probability $\prod_{i=m-d+1}^{m} p(w_i)$, while the entire word has probability $\prod_{i=1}^{m} p(w_i)$. The ratio is exactly the reciprocal probability of the border prefix of length $m - d$.

§39.6 A gambler who bets on prefixes

There is a rather charming explanation of the sum in (39.9). Before each toss, a new gambler arrives and stakes one unit on the first letter of w. A winning bet doubles the gambler's capital, which is then wagered on the second letter of w, and so on. An unsuccessful bet ends that gambler's participation; a gambler who correctly predicts all m letters is paid and leaves.

Every individual bet is fair. Each arrival contributes one unit, so after n tosses the casino's net wealth is

$$W_n = n - \text{total capital still held by winning gamblers.}$$

Treating a completed winner's payout as part of this total at the stopping instant, the process W_n, up to the first occurrence of w, is a martingale.

When the first complete w arrives at time T_w, which gamblers can still be holding capital? Exactly those who have correctly predicted a suffix of the observed sequence which is also a prefix of w. Since the last m symbols equal w, their successful run lengths are the elements of $\mathcal{B}(w)$. A gambler who has made k correct predictions has capital 2^k. Thus

$$W_{T_w} = T_w - \sum_{k \in \mathcal{B}(w)} 2^k.$$

There are at most m live gamblers, with capital at most 2^m each. Together with the exponential waiting-time bound, this justifies stopping the fair-game martingale at T_w by truncation and dominated convergence. Taking expectations gives the same formula:

$$\mathbb{E}T_w = \sum_{k \in \mathcal{B}(w)} 2^k.$$

The betting picture makes the self-overlaps tangible. When HHH finally arrives, the casino pays one gambler 8, another 4, and another 2. When HHT arrives, only the gambler who predicted all three letters survives to collect 8.

§39.7 Avoiding a word

We can now recover the probabilities of *not yet* having seen w. Let

$$S_n = \Pr(T_w > n), \qquad n \geq 0, \quad \text{and} \quad H_w(z) = \sum_{n\geq 0} S_n z^n.$$

For any positive integer-valued T,

$$\sum_{n\geq 0} z^n \mathbf{1}_{\{T>n\}} = 1 + z + \cdots + z^{T-1} = \frac{1-z^T}{1-z}.$$

Take expectations with $T = T_w$ to obtain

$$H_w(z) = \frac{1 - G_w(z)}{1-z} = \frac{R_w(z)}{(1-z)R_w(z) + 2^{-m} z^m}. \tag{39.11}$$

Compare this with (39.3): the automaton computes the survival probabilities by matrix multiplication, while the overlap polynomial packages the same sequence into a rational function.

For HHH,

$$H_{\mathrm{HHH}}(z) = \frac{1 + z/2 + z^2/4}{1 - z/2 - z^2/4 - z^3/8}.$$

Let a_n count length-n binary words with no occurrence of HHH. Since every length-n word is equally likely,

$$\frac{a_n}{2^n} = \Pr(T_{\mathrm{HHH}} > n).$$

Consequently,

$$\sum_{n\geq 0} a_n x^n = H_{\mathrm{HHH}}(2x) = \frac{1 + x + x^2}{1 - x - x^2 - x^3}. \tag{39.12}$$

Comparing coefficients gives

$$a_0 = 1, \qquad a_1 = 2, \qquad a_2 = 4, \qquad a_n = a_{n-1} + a_{n-2} + a_{n-3} \quad (n \geq 3).$$

For $n \geq 3$, we can see this recurrence directly. A binary word avoiding HHH must contain a last tail. It therefore ends with exactly one of the blocks T, TH, or THH. Removing that final block leaves a shorter word avoiding HHH, and conversely each shorter avoiding word can be extended by any one of these blocks. Thus the initial counts are

$$1,\ 2,\ 4,\ 7,\ 13,\ 24,\ 44, \ldots.$$

A waiting-time problem has produced a Tribonacci-type counting problem, using exactly the same forbidden word.

§39.8 The tail remembers the distribution

The expectation itself can be recovered from the survival probabilities:

$$\mathbb{E}T_w = \sum_{n=0}^{\infty} \Pr(T_w > n). \tag{39.13}$$

To prove it, write the elementary identity

$$T_w = \sum_{n=0}^{\infty} \mathbf{1}_{\{T_w > n\}}$$

for each sample outcome, then interchange expectation and the nonnegative sum. It is also the limit $H_w(1^-)$ of (39.11).

A single expectation summarizes the tail in one number. An entire family of *truncated expectations* retains much more. For integers $k \geq 0$, define

$$C_w(k) = \mathbb{E}[(T_w - k)_+], \qquad (x)_+ = \max\{x, 0\}.$$

For each integer $t \geq 0$,

$$(t-k)_+ = \sum_{n=k}^{\infty} \mathbf{1}_{\{t>n\}}.$$

Consequently,

$$C_w(k) = \sum_{n=k}^{\infty} \Pr(T_w > n). \tag{39.14}$$

Subtract successive values:

$$\boxed{C_w(k) - C_w(k+1) = \Pr(T_w > k).} \tag{39.15}$$

A second difference recovers individual probabilities:

$$\boxed{C_w(k) - 2C_w(k+1) + C_w(k+2) = \Pr(T_w = k+1).} \tag{39.16}$$

Thus the sequence $C_w(0), C_w(1), C_w(2), \ldots$ determines the entire law of T_w. It begins with $C_w(0) = \mathbb{E}T_w$, but each new entry reveals another piece of the tail. Its successive slopes are nondecreasing, because the right side of (39.16) is nonnegative: C_w is discretely convex.

There is a continuous version of the same picture. For any nonnegative integrable random variable X and real $a \geq 0$,

$$C_X(a) := \mathbb{E}[(X-a)_+] = \int_a^{\infty} \Pr(X > t)\, dt. \tag{39.17}$$

Indeed, pointwise, $(X-a)_+ = \int_a^{\infty} \mathbf{1}_{\{X>t\}}\, dt$; Tonelli's theorem allows us to exchange the integral and the expectation. Where the survival function is continuous,

$$C_X'(a) = -\Pr(X > a).$$

At an atom a, the right and left derivatives differ:

$$C_{X,+}'(a) = -\Pr(X > a), \qquad C_{X,-}'(a) = -\Pr(X \geq a).$$

The jump in slope therefore gives $\Pr(X = a)$. A convex function built from expected excesses above thresholds encodes the distribution itself.

For our integer-valued hitting times, $a \mapsto C_{T_w}(a)$ is likewise piecewise linear between consecutive integers. Its values at the integers already suffice to recover the law by (39.16). The waiting-time puzzle has led us to a general principle about random variables: a tail integral is a cumulative record of probabilities, and differentiation can read them back.

§39.9 When two patterns race

There is one final surprise. Suppose two players choose different length-three words and we toss the same fair coin until either pattern appears. The first completed pattern wins.

Let Alice choose HHH and Bob choose THH. Alice wins if and only if the first three tosses are all heads. Once a tail has appeared, any later run of three heads must first pass through the consecutive pattern THH: the first two heads after the last tail already complete Bob's word. Since some winning word appears almost surely,

$$\Pr(\text{THH occurs before HHH}) = 1 - \Pr(X_1X_2X_3 = \text{HHH}) = \frac{7}{8}. \tag{39.18}$$

Bob wins seven times out of eight, even though both players chose a word of length three.

The larger family of such races, often called *Penney's game*, exhibits nontransitive preferences among patterns. Analyzing several competing words requires cross-overlaps between *different* patterns, just as our single-pattern analysis required self-overlaps. The same prefix-automaton method extends by making the success state record *which* word has appeared first.

For two competing words A, B of the same length, we can make a prefix automaton whose transient states are the proper prefixes of *either* word. There are now two absorbing states, one for each winner. If Q is the transient transition matrix and r_B is the column of one-step transition probabilities into Bob's winning state, the vector of Bob's eventual winning probabilities satisfies

$$\mathbf{p}_B = Q\mathbf{p}_B + r_B, \qquad \mathbf{p}_B = (I - Q)^{-1} r_B.$$

Each state records a possible partial match of either competitor. The calculation is as finite as the one for expected waiting time.

As a concrete example, set $A = \text{HHT}$ and $B = \text{THH}$. The transient states are

$$\varnothing, \qquad \text{H}, \qquad \text{HH}, \qquad \text{T}, \qquad \text{TH}.$$

Let p_s be the probability that B wins first when we are in state s. A direct check of the next letter gives

$$\begin{aligned} p_\varnothing &= \tfrac{1}{2}(p_\text{H} + p_\text{T}), & p_\text{H} &= \tfrac{1}{2}(p_\text{HH} + p_\text{T}), \\ p_\text{HH} &= \tfrac{1}{2}p_\text{HH} + \tfrac{1}{2}\cdot 0, & p_\text{T} &= \tfrac{1}{2}(p_\text{TH} + p_\text{T}), \\ p_\text{TH} &= \tfrac{1}{2}\cdot 1 + \tfrac{1}{2}p_\text{T}. \end{aligned}$$

The two boundary values 0 and 1 correspond to completing A or B. The equations yield $p_\text{HH} = 0$, $p_\text{T} = p_\text{TH} = 1$, $p_\text{H} = 1/2$, and finally

$$\boxed{\Pr(\text{THH appears before HHT}) = p_\varnothing = \frac{3}{4}.}$$

The same calculation for four choices produces a small cycle:

First choice A	Reply B	$\Pr(B \text{ wins})$
HHT	THH	3/4
THH	TTH	2/3
TTH	HTT	3/4
HTT	HHT	2/3

Each reply is more likely to occur first than the word it answers, yet following the replies brings us back to the starting word. The overlapping suffix of one word and prefix of another now play the role that a border played for a single word.

This cycle says something subtler than its table of odds. The words HHT and THH have *equal individual expected waiting times*,

$$\mathbb{E}T_{\mathrm{HHT}} = \mathbb{E}T_{\mathrm{THH}} = 8,$$

yet THH beats HHT with probability $3/4$ when both compete on the *same* stream of coin tosses. Separate expectations do not determine the probability that one stopping time precedes another; this probability depends on their joint construction.

Inverse Ackermann and Path Compression

N-20261108

Some functions grow so slowly that there is no sensible way to watch them grow on a computer. The inverse Ackermann function is one of them. What makes it interesting is that it appears in the complexity of a very ordinary data structure: a collection of rooted trees used to maintain a partition of a set.

The route from a frighteningly fast-growing function to almost constant-time algorithms is not obvious. We will first build a hierarchy of growth rates, then look carefully at path compression in a disjoint-set forest. A complete charging argument will give the bound $O(m \log^* n)$. The sharper inverse-Ackermann theorem requires a more intricate version of the rank analysis; we will identify what needs to be sharpened rather than disguising an incomplete argument as its proof.[1]

§40.1 Beyond exponentials

Repeatedly adding n produces multiplication; repeatedly multiplying n produces powers. Repeating exponentiation gives power towers. The operation of *iteration* makes it possible to keep going.

For a function f, let $f^{\circ k}$ be its k-fold composition, with $f^{\circ 0}(x) = x$. Define

$$A_0(n) = n + 1, \qquad A_{r+1}(n) = A_r^{\circ(n+1)}(1), \qquad r, n \geq 0. \tag{40.1}$$

This is one convention for an Ackermann hierarchy. Other books shift the indices or change the initial argument.

The first few functions can be calculated directly:

$$\begin{aligned} A_0(n) &= n + 1, \\ A_1(n) &= n + 2, \\ A_2(n) &= 2n + 3, \\ A_3(n) &= 2^{n+3} - 3. \end{aligned}$$

For example, $A_2 = A_1^{\circ(n+1)}(1)$, and every application of A_1 adds two, so $A_2(n) = 1 + 2(n + 1) = 2n + 3$. Applying $x \mapsto 2x + 3$ exactly k times to 1 gives $4 \cdot 2^k - 3$. Taking $k = n + 1$ produces the fourth line.

The next level is already unpleasant. At $n = 1$,

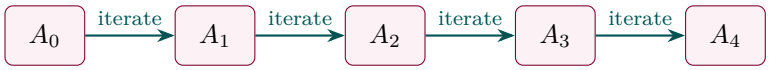
$$\begin{aligned} &A_0(1) = 2, && A_1(1) = 3, \quad A_2(1) = 5, \quad A_3(1) = 13, \\ &A_4(1) = A_3(A_3(1)) = A_3(13) = 2^{16} - 3 = 65533. \end{aligned} \tag{40.2}$$

Going one level higher means

$$A_5(1) = A_4(A_4(1)) = A_4(65533).$$

Even the notation becomes more useful than decimal digits. The function A_4 itself is obtained by iterating a roughly exponential function, $n + 1$ times.

[1] Robert E. Tarjan, *Efficiency of a Good But Not Linear Set Union Algorithm* (1975). The rank-block argument for the iterated-logarithm bound follows the standard exposition in *Introduction to Algorithms*, §21.4, with charging details worked out below.



Each level is built from repeated composition of the preceding one

Figure 40.1: Each new Ackermann level outgrows the previous one. Inverting the hierarchy asks how many such levels are needed to reach a threshold.

For definiteness, put

$$\alpha(n) = \min\{r \geq 1 : A_r(1) \geq n\}, \qquad n \geq 1. \tag{40.3}$$

With our convention,

$$\alpha(n) = \begin{cases} 1, & n \leq 3, \\ 2, & 3 < n \leq 5, \\ 3, & 5 < n \leq 13, \\ 4, & 13 < n \leq 65533, \\ 5, & 65533 < n \leq A_5(1). \end{cases}$$

The last range extends to an extraordinary bound. The function is unbounded in the mathematical sense, yet increasing its value by one may require a number far larger than any quantity we can physically store.

In sharp data-structure theorems, the notation $\alpha(m, n)$ often denotes a *two-parameter* inverse Ackermann variant, measuring m operations on n elements. Different normalizations can produce different small numerical values. A claim about $\alpha(n)$ being less than four, for instance, must be read with its particular definition in mind. We use (40.3) when computing values and use Tarjan's conventional asymptotic notation for the sharp complexity result.

§40.2 A forest that keeps changing shape

A disjoint-set union structure maintains a partition into nonoverlapping sets. It supports MAKESET(x), creating a singleton; FIND(x), returning the representative of the set containing x; and UNION(x, y), merging the sets containing x and y.

Represent every set by a rooted tree. Each node x has a parent pointer $p[x]$, and a root satisfies $p[x] = x$. A find follows pointers to the root.

If every union attaches an arbitrary root below another, a tree can become a chain of length $n - 1$. A find can then traverse $\Theta(n)$ pointers. The first improvement is to choose the direction of attachment carefully.

Union by rank stores a nonnegative integer rank$[x]$. Initially every rank is zero. When combining distinct roots r, s, attach the lower-rank root below the higher-rank root. If the ranks are equal, attach either below the other and increase the new root's rank by one. Rank is maintained as a historical bound, not as the current depth after compression.

```
MAKESET(x):
   parent[x] = x; rank[x] = 0
FIND(x):
   if parent[x] != x:
      parent[x] = FIND(parent[x])
   return parent[x]
UNION(x, y):
   a = FIND(x); b = FIND(y)
   if a == b: return
   if rank[a] < rank[b]: swap(a, b)
   parent[b] = a
   if rank[a] == rank[b]: rank[a] += 1
```

The recursive call implements *full path compression*. Once the root is known, all visited vertices on that path become its direct children.

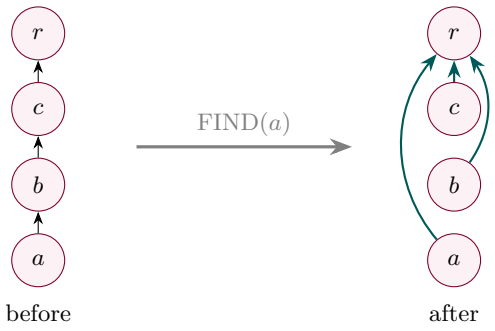


Figure 40.2: One find traverses a chain and redirects all nonroot nodes to the common root. A later find cannot keep paying for the same chain unless subsequent unions change the forest.

§40.3 Rank is an accounting device

Two elementary lemmas explain why rank is so useful.

Lemma 40.3.1 (Strict ranks on every parent chain)

For a nonroot x,

$$\operatorname{rank}[x] < \operatorname{rank}[p[x]].$$

Ranks never decrease. Once a node becomes a nonroot, its rank never changes.

Proof. The claims hold when every node is a singleton root. A link attaches a lower-rank root below a higher-rank root, or attaches equal-rank roots and then increases the new root's rank; the strict inequality holds in either case. Ranks change only for roots involved in ties. Path compression replaces a parent by an ancestor, whose rank is strictly larger than the replaced parent's rank unless the parent was already the root. □

Lemma 40.3.2 (Exponential size)

Whenever x is a root,

$$|\operatorname{tree}(x)| \geq 2^{\operatorname{rank}[x]}.$$

Consequently every rank is at most $\lfloor \log_2 n \rfloor$ if a total of n elements have been created.

Proof. A singleton has size $1 = 2^0$. If two roots have different ranks, the larger-rank root keeps its rank and gains nodes, so the bound remains true. If both have rank r, their disjoint trees each have at least 2^r nodes; the new root has rank $r+1$ and at least 2^{r+1} nodes. Path compression changes no set sizes. Induction proves the claim. □

A stronger counting statement holds over an entire operation sequence.

Lemma 40.3.3 (Ranks are rare)

Among n created elements, at most $n/2^r$ are ever assigned rank r, for every integer $r \geq 0$.

Proof. Whenever a root first receives rank r, mark every element in its then-current tree with that root's name. The tree has at least 2^r elements. Different roots first attaining rank r have disjoint marked groups: once the first such root is attached to another, all its future ancestors have rank greater than r. None of its elements can later belong to a different root that *first* attains rank r. Therefore no element is marked twice at the same rank. Since there are n elements, there can be at most $n/2^r$ such groups. Every node first acquires a rank while it is a root, so this counts all nodes ever assigned that rank. □

Here are two valuable resources for the analysis: rank increases strictly as we climb a parent chain, and very few nodes can carry a high rank.

§40.4 Why one find can still be logarithmic

Even union by rank and full path compression do not make *every individual find* constant time. Before any compressing find is performed, repeatedly combine trees of equal rank. Two singletons form a tree of rank 1; two rank-1 trees form one of rank 2, and so on. After building a set of 2^r elements this way, a leaf can lie at depth r. Its first find takes $\Theta(r) = \Theta(\log n)$ time.

The point of path compression is that the long find modifies the structure so that subsequent finds on the same path become cheap. The interesting upper bound is therefore *amortized*: it controls the total cost of a long sequence of operations, not the cost of every operation separately.

It is useful to recall the weaker logarithmic bound first. By the strict-rank lemma, a parent chain has at most $\lfloor \log_2 n \rfloor + 1$ distinct ranks, so every find costs $O(\log n)$. The analysis we really want must count how often a node can make us pay for the *same portion* of that rank range.

§40.5 Grouping ranks into towers

Define the iterated binary logarithm by

$$\log_2^* x = \min\{k \geq 0 : \underbrace{\log_2 \circ \cdots \circ \log_2}_{k \text{ times}}(x) \leq 1\}.$$

For example,

$$65536 \longmapsto 16 \longmapsto 4 \longmapsto 2 \longmapsto 1, \qquad \log_2^*(65536) = 4.$$

It is much slower growing than $\log_2 n$, but still grows faster than an inverse Ackermann function.

Construct tower boundaries

$$B_{-1} = -1, \quad B_0 = 1, \qquad B_{j+1} = 2^{B_j}. \tag{40.4}$$

Thus

$$B_0 = 1, \quad B_1 = 2, \quad B_2 = 4, \quad B_3 = 16, \quad B_4 = 65536, \ldots.$$

We partition the possible ranks into consecutive groups

$$G_j = \{B_{j-1} + 1, \ldots, B_j\}, \qquad j \geq 0.$$

The first groups are

$$G_0 = \{0, 1\}, \quad G_1 = \{2\}, \quad G_2 = \{3, 4\}, \quad G_3 = \{5, \ldots, 16\}.$$

Only those groups intersecting the range $0, \ldots, \lfloor \log_2 n \rfloor$ can occur. Their number is

$$b = O(1 + \log_2^* n).$$

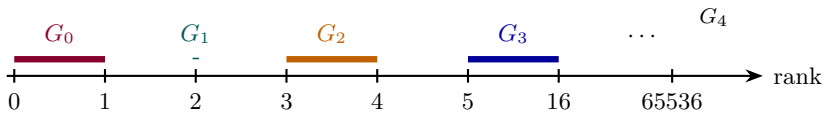


Figure 40.3: Successive rank groups end at tower values $1, 2, 4, 16, 65536, \ldots$. The geometric spacing here is schematic, not numerical: the last group is enormously wider than it looks.

Why choose these apparently absurd group widths? Because the rank-counting lemma makes the high-rank groups correspondingly rare. The rapid growth of B_j will cancel the scarcity of nodes having ranks in G_j.

§40.6 A complete iterated-logarithm bound

Consider a sequence of m operations after at most n MAKESET operations, with $m \geq n$. We count the parent pointers traversed by every FIND. A UNION uses two FINDs and one constant-time link, so replacing unions by those primitive actions changes the operation count by at most a constant factor.

Take one FIND path

$$x_0 \longrightarrow x_1 \longrightarrow \cdots \longrightarrow x_\ell, \qquad x_\ell = \text{root}.$$

The ranks strictly increase. Classify its traversed edges in three ways:

1. An edge $x_i \to x_{i+1}$ that passes from one rank group G_j to a higher group. There can be at most $b-1$ of these per find.

2. The last edge $x_{\ell-1} \to x_\ell$, which we pay for directly if it has not already been counted. This contributes at most one extra edge per find.

3. Every other edge stays inside a single rank group. We charge this edge to its lower endpoint x_i.

Only the third type needs a global argument. Suppose $x = x_i$ has rank in group G_j, and its edge to its parent x_{i+1} is of the third type. Since the parent is *not* the final root, full path compression redirects x all the way to x_ℓ. By strict increase along the original chain,

$$\operatorname{rank}[x_\ell] > \operatorname{rank}[x_{i+1}].$$

The rank of x is now fixed forever, because x is a nonroot; the rank of its parent can only increase over time. On every future occasion when the same type of edge is charged to x, its parent's rank has increased by at least one. As long as that parent's rank stays in G_j, it cannot exceed B_j.

Therefore each nonroot x whose rank lies in G_j receives at most

$$B_j - B_{j-1}$$

charges of the third type during the entire sequence. This may be a huge number for a large j, but it applies to very few nodes.

Let N_j be the number of nodes whose ranks ever belong to G_j. The rank-count lemma gives, for $j \geq 1$,

$$\begin{aligned} N_j &\leq \sum_{r=B_{j-1}+1}^{B_j} \frac{n}{2^r} \\ &< \frac{n}{2^{B_{j-1}}} = \frac{n}{B_j}. \end{aligned}$$

For $j = 0$, the two possible ranks $0, 1$ give $N_0 \leq 3n/2$. Thus the total number of third-type charges from a nonzero group is at most

$$N_j(B_j - B_{j-1}) \leq \frac{n}{B_j} B_j = n,$$

and the group G_0 contributes at most a constant times n. Summing over all at most b groups, there are only $O(nb)$ charged edges of the third type in the *entire sequence*.

The other two edge types contribute $O(mb)$, because they occur at most $O(b)$ times per find. Every traversed edge has been paid for, giving the promised theorem.

Theorem 40.6.1 (A fully proved bound)

With union by rank and full path compression, a sequence of $m \geq n$ MAKESET, UNION and FIND operations on n elements takes

$$\boxed{O(m(1 + \log_2^* n))} \tag{40.5}$$

total time.

The important step is the cancellation

$$\underbrace{\frac{n}{B_j}}_{\text{few nodes}} \cdot \underbrace{B_j}_{\text{many charges per node}} = n.$$

Even though the groups rapidly become huge, each one can account for only $O(n)$ expensive same-group traversals. The number of groups controls the answer.

§40.7 Where the inverse Ackermann function enters

Tarjan's sharper theorem replaces the iterated logarithm by an inverse Ackermann function. In standard forms it states that a sequence of $m \geq n$ operations can be carried out in

$$O(m\,\alpha(m,n)) \quad \text{or, with a one-variable upper bound,} \quad O(m\,\alpha(n)). \tag{40.6}$$

Here α denotes one of the conventional functional inverses of an Ackermann hierarchy. The two-parameter function adjusts its threshold to the ratio m/n; the precise definition affects small values, though not the nearly constant character of the complexity.

Our rank-block proof already explains why this kind of hierarchy appears. A traversed edge is inexpensive if it crosses to a higher rank class; if it stays in one class, compression makes the parent's rank grow, so the edge can be charged only a limited number of times. To improve $\log^* n$ further, a single tier of tower-sized classes is insufficient. The refined analysis assigns several *levels* inside rank intervals, with advancement thresholds defined recursively by iterating the preceding level's threshold function. A hierarchy of repeated exponentials and their iterations naturally appears. The number of levels required to dominate all ranks is then controlled by the inverse Ackermann function. Establishing the sharp bound needs a carefully defined multilevel potential or charging scheme and a proof that the level transitions account for every traversed edge; we do not silently substitute the tower-block estimate for that argument.

There is an asymptotic point worth keeping. The bound (40.6) grows without limit as $n \to \infty$. It is mathematically different from $O(m)$, and in relevant computational models there are matching inverse-Ackermann lower bounds for sufficiently general sequences of unions and finds. Calling the running time constant per operation is an extremely good engineering approximation, while the theorem says something more delicate.

If you construct a rank-r tree by repeatedly merging equal-rank trees, an unlucky first FIND can still cost $\Theta(r) = \Theta(\log n)$. The near-constant guarantee comes from adding costs over the complete sequence. In this sense the inverse Ackermann function measures the remaining cost of a long history of *self-repairing* pointer traversals, not the height of a particular tree.

IV

Functions and Inequalities

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Designing Inequality Proofs: Equality Cases, Homogeneity, and Cyclic Structure

N-20220417

§41.1 Finding the likely answer is only half of a proof

An inequality problem often has an exploratory phase and a certification phase. During exploration one tests special cases, guesses the equality configuration, normalizes variables, and searches for a familiar pattern. A complete proof must then establish the bound for every admissible input and verify that equality can actually occur.

For an optimization statement, the two obligations are

$$F(x) \geq L$$

for all allowed x, and

$$F(x_0) = L$$

for at least one allowed x_0. The first proves that L is a lower bound; the second proves it is sharp.

A simple example is

$$a^2 + b^2 + c^2 \geq ab + bc + ca.$$

The symmetric point $a = b = c$ suggests the equality case. The certification is the identity

$$a^2 + b^2 + c^2 - ab - bc - ca = \frac{1}{2}((a-b)^2 + (b-c)^2 + (c-a)^2).$$

The right-hand side is nonnegative, and it vanishes exactly when

$$a = b = c.$$

The equality case is not a decorative sentence after the proof. It confirms that the proposed bound is the best possible and checks that no step has introduced unnecessary slack.

The same habit matters in existence and classification problems. “Find all” requires both producing candidates and excluding everything else. “Find the minimum” requires both a universal lower bound and an example attaining it. The proof strategy should be designed around both obligations from the beginning.

§41.2 Homogeneity tells us what may be normalized

An expression $F(a, b, c)$ is homogeneous of degree d if

$$F(ta, tb, tc) = t^d F(a, b, c)$$

for every positive t. If both sides of an inequality have the same degree, scaling all variables does not change its truth. One degree of freedom may then be removed by a normalization such as

$$a + b + c = 1, \qquad abc = 1, \qquad a^2 + b^2 + c^2 = 1.$$

The best choice is the one that simplifies the structure already present.

For example, the inequality

$$(a+b+c)^2 \geq 3(ab+bc+ca)$$

is homogeneous of degree two. Normalizing $a+b+c=1$ turns it into

$$ab+bc+ca \leq \frac{1}{3}.$$

Using

$$1=(a+b+c)^2=a^2+b^2+c^2+2(ab+bc+ca)$$

and

$$a^2+b^2+c^2 \geq ab+bc+ca$$

gives the result immediately.

If a condition says

$$abc=1,$$

the substitution

$$a=\frac{x}{y}, \qquad b=\frac{y}{z}, \qquad c=\frac{z}{x}$$

builds the constraint automatically. This substitution is useful only when the resulting cyclic ratios simplify the expression; otherwise the normalization $abc=1$ may be better used directly.

Homogeneity is therefore not a command to normalize mechanically. It is permission to choose a convenient cross-section of the scaling orbits.

§41.3 Cyclic versus Symmetric Expressions

For three variables, cyclic summation means

$$\sum_{\text{cyc}} f(a,b,c)=f(a,b,c)+f(b,c,a)+f(c,a,b).$$

A symmetric sum includes every distinct permutation. For instance,

$$\sum_{\text{cyc}} a^2b=a^2b+b^2c+c^2a,$$

whereas

$$\sum_{\text{sym}} a^2b=a^2b+a^2c+b^2a+b^2c+c^2a+c^2b.$$

A symmetric expression is unchanged by every permutation of the variables. A cyclic expression remembers an orientation. The two orientations differ by

$$\begin{aligned} &a^2b+b^2c+c^2a-(ab^2+bc^2+ca^2) \\ &\quad =-(a-b)(b-c)(c-a). \end{aligned}$$

Thus a cyclic sum becomes equal to its reverse when two variables coincide, but not in general.

This identity is a useful diagnostic. If a proposed proof replaces a cyclic expression by a symmetric one, it may be discarding the orientation term

$$(a-b)(b-c)(c-a).$$

Sometimes that loss is harmless because the symmetric estimate is strong enough. Sometimes the sign of the orientation term is exactly what an ordering assumption such as

$$a \geq b \geq c$$

was meant to control.

The notation should therefore follow the invariance of the problem. A cyclic problem often rewards cyclic substitutions and ordered cases; a symmetric homogeneous problem may invite smoothing, majorization, or a reduction to elementary symmetric polynomials.

§41.4 Equality cases guide the choice of inequality

Suppose a positive symmetric expression appears to attain equality at

$$a = b = c.$$

That suggests methods whose equality conditions also force equal variables: AM–GM, Jensen with strict convexity or concavity, Cauchy in a proportional configuration, or smoothing two variables toward their average.

Consider

$$\frac{a^3}{bc} + \frac{b^3}{ca} + \frac{c^3}{ab} \geq a + b + c, \qquad a, b, c > 0.$$

Set

$$A = \frac{a^3}{bc}, \qquad B = \frac{b^3}{ca}, \qquad C = \frac{c^3}{ab}.$$

To produce the term a, apply AM–GM to

$$A, A, B, C.$$

Their product is

$$A^2BC = a^4,$$

so

$$\frac{2A + B + C}{4} \geq a.$$

Cyclically,

$$\frac{A + 2B + C}{4} \geq b,$$

$$\frac{A + B + 2C}{4} \geq c.$$

Adding gives

$$A + B + C \geq a + b + c.$$

Equality in the first AM–GM step requires

$$A = B = C,$$

and the same condition appears cyclically. For positive variables this gives

$$a = b = c.$$

The proof was designed backward from the desired right-hand terms and the expected equality configuration. The repeated copy of A was chosen so that the geometric mean became exactly a.

§41.5 Denominators often reveal an Engel-form Cauchy step

Fractions with positive denominators frequently suggest the Engel form

$$\sum_{i=1}^{n} \frac{x_i^2}{y_i} \geq \frac{(x_1 + \cdots + x_n)^2}{y_1 + \cdots + y_n}.$$

The numerators need not initially be squares; one may rewrite

$$\frac{a}{b+c} = \frac{a^2}{ab+ac}.$$

This gives a short proof of Nesbitt's inequality:

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}, \qquad a, b, c > 0.$$

Indeed,

$$\begin{aligned} \sum_{\text{cyc}} \frac{a}{b+c} &= \sum_{\text{cyc}} \frac{a^2}{ab+ac} \\ &\geq \frac{(a+b+c)^2}{2(ab+bc+ca)}. \end{aligned}$$

Since

$$(a+b+c)^2 \geq 3(ab+bc+ca),$$

we obtain

$$\sum_{\text{cyc}} \frac{a}{b+c} \geq \frac{3}{2}.$$

The equality conditions also match. Engel equality requires

$$\frac{a}{ab+ac} = \frac{b}{bc+ab} = \frac{c}{ca+bc},$$

which reduces to

$$a = b = c.$$

The second inequality has the same equality case, so no slack is lost between the two steps.

A denominator pattern is therefore not merely an obstacle. It often announces which quantities should become the y_i in a Cauchy quotient.

§41.6 Smoothing explains why equal variables often extremize

Fix the sum

$$x + y = s.$$

Replacing x, y by their average

$$\frac{s}{2}, \frac{s}{2}$$

is called a smoothing step. For a convex function f,

$$f(x) + f(y) \geq 2f\left(\frac{x+y}{2}\right).$$

Thus smoothing decreases a convex symmetric sum. For a concave function, the direction reverses.

The elementary identity

$$x^2 + y^2 - \frac{s^2}{2} = \frac{(x-y)^2}{2}$$

shows this directly for $f(t) = t^2$. With the sum fixed, the square sum is minimized when the variables are equal.

The product behaves oppositely:

$$xy = \frac{s^2 - (x-y)^2}{4} \le \frac{s^2}{4}.$$

Hence smoothing increases the product. Iterating the argument yields AM–GM for three variables with fixed sum:

$$abc \le \left(\frac{a+b+c}{3}\right)^3.$$

A useful calculus diagnostic reaches the same candidate. To maximize

$$\log(abc) = \log a + \log b + \log c$$

subject to

$$a + b + c = s,$$

Lagrange multipliers give

$$\frac{1}{a} = \frac{1}{b} = \frac{1}{c},$$

so the only interior critical point is

$$a = b = c = \frac{s}{3}.$$

Because log is strictly concave and the product tends to zero at the boundary, this point is the unique global maximum. The calculus calculation discovers the candidate; AM–GM or concavity supplies the global certificate.

§41.7 Triangle substitutions encode the boundary conditions

If a, b, c are side lengths of a nondegenerate triangle, then

$$a < b + c, \qquad b < c + a, \qquad c < a + b.$$

The substitution

$$a = y + z, \qquad b = z + x, \qquad c = x + y, \qquad x, y, z > 0$$

builds these inequalities automatically. Conversely,

$$x = \frac{b+c-a}{2}, \quad y = \frac{c+a-b}{2}, \quad z = \frac{a+b-c}{2}.$$

As a simple application, Heron's semiperimeter differences become

$$s - a = x, \qquad s - b = y, \qquad s - c = z,$$

where

$$s = x + y + z.$$

The area formula

$$\Delta^2 = s(s-a)(s-b)(s-c)$$

becomes

$$\Delta^2 = (x+y+z)xyz.$$

A geometric constraint has turned into positivity of three independent variables.

The substitution is useful when the expression repeatedly contains

$$b+c-a, \quad c+a-b, \quad a+b-c.$$

It is not automatically beneficial for every triangle inequality. As with normalization, the substitution should simplify the actual algebra rather than merely certify that the solver recognizes a standard trick.

§41.8 Schur's Inequality and Ordering

Schur's inequality of degree three states

$$a^3+b^3+c^3+3abc \geq \sum_{\text{sym}} a^2b$$

for nonnegative a, b, c. Equivalently,

$$\sum_{\text{cyc}} a(a-b)(a-c) \geq 0.$$

Because the expression is symmetric, assume without loss of generality that

$$a \geq b \geq c.$$

Then group the first two terms:

$$\begin{aligned} &a(a-b)(a-c)+b(b-c)(b-a) \\ &\quad = (a-b)\big(a(a-c)-b(b-c)\big) \\ &\quad = (a-b)^2(a+b-c). \end{aligned}$$

The remaining term is

$$c(c-a)(c-b) = c(a-c)(b-c).$$

Therefore

$$\sum_{\text{cyc}} a(a-b)(a-c) = (a-b)^2(a+b-c)+c(a-c)(b-c) \geq 0.$$

Every factor is nonnegative under the chosen ordering. Equality occurs when

$$a=b=c$$

or when one variable is zero and the other two are equal.

This proof illustrates a general pattern. A symmetric inequality permits an ordering assumption. The ordered variables control signs that would otherwise be ambiguous, and a cyclic expression can then be regrouped into visibly nonnegative pieces.

§41.9 A complete proof keeps the equality case alive

Consider the homogeneous inequality

$$\frac{a^2}{b+c}+\frac{b^2}{c+a}+\frac{c^2}{a+b}\geq\frac{a+b+c}{2},\qquad a,b,c>0.$$

The denominators suggest Engel form, and the symmetric right-hand side suggests equality at $a=b=c$. One line gives

$$\begin{aligned}\sum_{\text{cyc}}\frac{a^2}{b+c}&\geq\frac{(a+b+c)^2}{(b+c)+(c+a)+(a+b)}\\&=\frac{(a+b+c)^2}{2(a+b+c)}\\&=\frac{a+b+c}{2}.\end{aligned}$$

The Cauchy equality condition is

$$\frac{a}{b+c}=\frac{b}{c+a}=\frac{c}{a+b},$$

which for positive variables forces

$$a=b=c.$$

The candidate survives the only inequality step, so the bound is sharp.

This small example contains the full design process. Homogeneity permits normalization if desired; symmetry predicts the equality configuration; the denominator pattern identifies Engel form; and the equality condition verifies sharpness. A good inequality proof is not a list of named weapons. It is a sequence of transformations chosen so that the structure of the problem and the structure of the theorem line up without losing the extremal case.

Conditioning as Projection: Probability, Jensen's Inequality, and Lost Information

N-20220721

Conditioning changes the weights assigned to outcomes. When the condition is a single event this is a new probability measure; when the available information is a partition of the sample space it is a new function, the conditional expectation. That function turns out to be an orthogonal projection in L^2. The same operation also satisfies Jensen's inequality and, in information theory, can only decrease the relative entropy between distributions after outcomes have been merged.[1]

§42.1 How Conditioning Reweights Outcomes

On a finite probability space Ω, let $p(\omega) > 0$ denote the weight of each outcome. For an event $B \subseteq \Omega$ with $\Pr(B) > 0$, the conditional law is

$$p_B(\omega) = \frac{p(\omega)\mathbf{1}_B(\omega)}{\Pr(B)}.$$

Thus for a random variable X,

$$\mathbb{E}[X \mid B] = \sum_{\omega \in \Omega} X(\omega) p_B(\omega) = \frac{\mathbb{E}[X\mathbf{1}_B]}{\Pr(B)}.$$

Conditioning is an explicit reweighting of the same outcomes.

Bayes' theorem follows by comparing two ways of multiplying conditional and marginal probabilities:

$$\Pr(H \mid E) = \frac{\Pr(E \mid H)\Pr(H)}{\Pr(E)}.$$

The odds form is especially informative:

$$\boxed{\frac{\Pr(H \mid E)}{\Pr(H^c \mid E)} = \frac{\Pr(E \mid H)}{\Pr(E \mid H^c)} \frac{\Pr(H)}{\Pr(H^c)}.}$$

It says that evidence multiplies prior odds by a likelihood ratio. For a rare event with prior probability $1/100$, a test of sensitivity $9/10$ and false-positive rate $1/10$ gives

$$\Pr(H \mid +) = \frac{(9/10)(1/100)}{(9/10)(1/100) + (1/10)(99/100)} = \frac{1}{12}.$$

High sensitivity alone cannot overcome very small prior odds and a substantial false-positive rate.

[1] Stanford Stats 310A, §4.3 (conditional expectation as projection); MIT 6.441, Chapters 2–3 (information-theoretic data processing).

§42.2 Observing a partition gives a random estimate

Suppose the observation only tells us which cell of a partition $\Omega = C_1 \sqcup \cdots \sqcup C_m$ contains the outcome. If $\Pr(C_j) > 0$, define

$$M(\omega) = \frac{\sum_{\eta \in C_j} p(\eta) X(\eta)}{\Pr(C_j)} \qquad (\omega \in C_j).$$

The function M is the conditional expectation of X given this observation. It is constant on each information cell and averages X using the conditional weights within that cell.

For example, take four outcomes of probabilities

$$(p_1, p_2, p_3, p_4) = \frac{1}{10}(1, 2, 3, 4),$$

with $X = (0, 2, 3, 7)$. If the observation distinguishes only $\{1, 2\}$ from $\{3, 4\}$, the best cellwise predictions are

$$M|_{\{1,2\}} = \frac{1 \cdot 0 + 2 \cdot 2}{1 + 2} = \frac{4}{3}, \qquad M|_{\{3,4\}} = \frac{3 \cdot 3 + 4 \cdot 7}{3 + 4} = \frac{37}{7}.$$

The mean is preserved:

$$\mathbb{E}[M] = \frac{3}{10}\frac{4}{3} + \frac{7}{10}\frac{37}{7} = \frac{41}{10} = \mathbb{E}[X].$$

§42.3 Conditional Expectation as Projection

Equip real-valued functions on Ω with the inner product

$$\langle X, Z \rangle = \mathbb{E}[XZ].$$

The functions constant on each information cell form a vector subspace V of this finite-dimensional Hilbert space. For any $Z \in V$, say $Z = z_j$ on C_j,

$$\begin{aligned} \mathbb{E}[(X - M)Z] &= \sum_j z_j \sum_{\omega \in C_j} p(\omega)(X(\omega) - M(\omega)) \\ &= 0. \end{aligned}$$

Thus the residual $X - M$ is orthogonal to every function measurable from the observation:

$$\boxed{M = \mathrm{Proj}_V X.}$$

For any other predictor $Z \in V$, Pythagoras gives

$$\mathbb{E}[(X - Z)^2] = \mathbb{E}[(X - M)^2] + \mathbb{E}[(M - Z)^2].$$

It follows that M uniquely minimizes the mean squared prediction error among all predictors using the available observation. No Gaussian distribution or linear regression assumption was needed.

The same geometry yields the law of total variance. Writing $X - \mathbb{E}[X] = (X - M) + (M - \mathbb{E}[X])$ and using orthogonality,

$$\boxed{\mathrm{Var}(X) = \mathbb{E}[\mathrm{Var}(X \mid Y)] + \mathrm{Var}(\mathbb{E}[X \mid Y]),}$$

where Y denotes the partition observation. Total variability splits into the uncertainty left inside each cell and the variability explained by knowing the cell.

§42.4 Jensen turns changes of viewpoint into inequalities

A convex function φ satisfies, for nonnegative weights summing to one,

$$\varphi\left(\sum_i \lambda_i x_i\right) \le \sum_i \lambda_i \varphi(x_i).$$

Applying this separately within each information cell gives the conditional Jensen inequality:

$$\boxed{\varphi(\mathbb{E}[X \mid Y]) \le \mathbb{E}[\varphi(X) \mid Y].}$$

This follows directly from convexity; the inequality requires the relevant expectations to exist.

The familiar Cauchy inequality

$$(a+b+c)^2 \le 3(a^2+b^2+c^2)$$

is Jensen for $\varphi(x) = x^2$ and a uniform random variable whose three possible values are a, b, c. Similarly, concavity of log gives weighted AM–GM:

$$\sum_i \lambda_i \log x_i \le \log\left(\sum_i \lambda_i x_i\right) \quad \Longleftrightarrow \quad \boxed{\prod_i x_i^{\lambda_i} \le \sum_i \lambda_i x_i}$$

for $x_i > 0$. Choosing the weights and the random variable is therefore a systematic way of finding the right normalization in many inequality problems.

§42.5 Merging outcomes can only erase relative entropy

Let $p = (p_i)$ and $q = (q_i)$ be strictly positive probability vectors on a finite set. Their relative entropy is

$$D(p\|q) = \sum_i p_i \log \frac{p_i}{q_i}.$$

Define the likelihood ratio $R(i) = p_i/q_i$ and take expectation under q. Then

$$D(p\|q) = \mathbb{E}_q[R \log R].$$

The function $\varphi(r) = r \log r$ is strictly convex on $r > 0$, because $\varphi''(r) = 1/r$. By ordinary Jensen,

$$D(p\|q) \ge \varphi(\mathbb{E}_q R) = \varphi(1) = 0,$$

with equality precisely when $p = q$.

Now suppose an observation merges individual outcomes into cells C_j. Let

$$\overline{p}_j = \sum_{i \in C_j} p_i, \qquad \overline{q}_j = \sum_{i \in C_j} q_i.$$

The conditional mean of R within the j-th cell, under weights $q_i/\overline{q}_j$, is

$$\mathbb{E}_q[R \mid C_j] = \frac{\overline{p}_j}{\overline{q}_j}.$$

Apply conditional Jensen and average over all cells:

$$\begin{aligned} D(p\|q) &= \mathbb{E}_q[\varphi(R)] \\ &\geq \sum_j \overline{q}_j \varphi\left(\frac{\overline{p}_j}{\overline{q}_j}\right) \\ &= \sum_j \overline{p}_j \log \frac{\overline{p}_j}{\overline{q}_j}. \end{aligned}$$

Therefore

$$\boxed{D(p\|q) \geq D(\overline{p}\|\overline{q}).}$$

This is the data-processing inequality for a deterministic coarse-graining. Distinguishing fewer outcomes cannot make two distributions more distinguishable in relative entropy. Equality holds when the ratio p_i/q_i is constant on each cell.

The projection interpretation and data processing describe the same loss of information from two angles. For squared prediction error, conditioning keeps the component visible to the observation and discards an orthogonal residual. For relative entropy, conditioning averages the likelihood ratio, and convexity shows that its distinguishing power cannot increase. Both calculations begin with reweighting on the information cells and end with a precise inequality.

Set Exercises, Functions, and Elementary Problem-Solving

N-20220801

§43.1 Sets as predicates, Boolean operations, and counting

If a set A has n elements, then it has

$$2^n$$

subsets. One way to see this is to use the characteristic function

$$\chi_S(x) = \begin{cases} 1, & x \in S, \\ 0, & x \notin S. \end{cases}$$

Choosing a subset $S \subseteq A$ is the same as choosing a function

$$\chi_S : A \to \{0, 1\}.$$

There are 2 choices for each element of A, hence 2^n choices in total.

This is a useful conceptual improvement over simply memorizing 2^n. A subset is a yes/no decision for every element. The subsequent algebraic and measurable-set material follows standard Boolean-algebra and probability presentations, in particular Kruckman's *Model Theory* notes on Stone duality and Scott Sheffield's MIT 18.175 lecture on sigma-algebras.[1]

De Morgan's laws are:

$$C_U(A \cap B) = C_U A \cup C_U B,$$
$$C_U(A \cup B) = C_U A \cap C_U B.$$

It also recalls inclusion–exclusion:

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

The exercises ask for set descriptions such as:

$$\{2n : 1 \le n \le 5\},$$

$$\{x : x \le 5\},$$

and graphs such as

$$\{(x, y) : y = x^2 - x - 1\}.$$

This is a good reminder that sets can be finite lists, solution sets of inequalities, or geometric loci.

A subset $A \subseteq U$ is equivalently a predicate or a characteristic function

$$\mathbf{1}_A : U \to \{0, 1\}.$$

Set operations become operations on 0-1 functions:

$$\mathbf{1}_{A \cap B} = \mathbf{1}_A \mathbf{1}_B, \qquad \mathbf{1}_{A^c} = 1 - \mathbf{1}_A,$$

[1] A. Kruckman's logic lecture notes; S. Sheffield's MIT probability lectures.

and

$$\mathbf{1}_{A\cup B} = \mathbf{1}_A + \mathbf{1}_B - \mathbf{1}_A\mathbf{1}_B.$$

For finite U, cardinality is summation:

$$|A| = \sum_{x\in U} \mathbf{1}_A(x).$$

This converts set identities into ordinary algebraic identities.

§43.2 Two-set exercises and the atoms of information

One exercise gives

$$U = A \cup B = \{1,2,3,4\}, \qquad A \cap C_U B = \{1,2\},$$

and asks for B. Since $A \setminus B = \{1,2\}$, the remaining elements $3,4$ must lie in B, so

$$B = \{3,4\}.$$

Another exercise has

$$A = \{x : 1 < x < 7\}, \qquad B = \{x : x^2 - 4x - 5 \le 0\}.$$

Since

$$x^2 - 4x - 5 = (x-5)(x+1),$$

we get

$$B = [-1,5].$$

Therefore

$$A \cap C_{\mathbb{R}} B = (5,7).$$

Venn diagrams force the essential distinction between "inside A, inside B, inside both, inside neither."

One problem describes a survey of 100 students reading classical novels. If 80 read either X or Y, 60 read X, and 40 read both, then the number reading Y is found by

$$|X \cup Y| = |X| + |Y| - |X \cap Y|.$$

Thus

$$80 = 60 + |Y| - 40,$$

so

$$|Y| = 60.$$

Again, the important point is not the story, but the relation among union, intersection, and individual counts.

A Venn diagram is a visual partition of a universe into atoms determined by a finite family of sets. For two sets, the atoms are

$$A \cap B, \quad A \cap B^c, \quad A^c \cap B, \quad A^c \cap B^c.$$

For n sets, there are at most 2^n atoms. Every Boolean expression in the sets is a union of some of these atoms.

This gives a systematic method for elementary set identities: reduce both sides to the same atoms. In logic, the same method is truth tables. In probability, the atoms are elementary events. In database theory, they are the records satisfying combinations of predicates.

Thus the elementary operation of shading regions is already the algebra of information partitions. A set is a yes/no question about elements; a finite family of sets is a finite collection of yes/no questions; the atoms are the complete answer patterns.

§43.3 What the ambient number system allows

The note includes exercises with sets such as

$$A = \{a : a = x^2 - y^2,\ x, y \in \mathbb{Z}\},$$

and

$$B = \{a : a = x^2 - y^2,\ x, y \in \mathbb{Q}\}.$$

Since

$$x^2 - y^2 = (x + y)(x - y),$$

every difference of squares is a product of two numbers of the same parity in the integer case. This can be made into a complete characterization. An integer a is a difference of two integer squares exactly when $a \not\equiv 2 \pmod 4$. Indeed, $x+y$ and $x-y$ have the same parity. If both are odd, their product is odd; if both are even, the product is divisible by 4. Conversely, every odd integer $2k+1$ is $(k+1)^2 - k^2$, and every multiple $4k$ is $(k+1)^2 - (k-1)^2$. Over $\mathbb{Q}$ every rational q is a difference of squares, since

$$q = \left(\frac{q+1}{2}\right)^2 - \left(\frac{q-1}{2}\right)^2.$$

The representation problem is therefore controlled exactly by the ambient ring.

For rationals, the factorization is more flexible, and quotients of such expressions can often remain in B. The page is training the habit of factoring before trying random examples.

A floor-function example studies values of

$$y = \lfloor x \rfloor + \lfloor 2x \rfloor + \lfloor 3x \rfloor, \qquad 1 \le y \le 100.$$

The natural move is to write

$$x = k + t, \qquad k \in \mathbb{Z}, \quad 0 \le t < 1.$$

Then the three floor terms change only when t crosses multiples of $1/2$ or $1/3$. Hence the possible values in one period come from the intervals determined by

$$0, \quad \frac{1}{3}, \quad \frac{1}{2}, \quad \frac{2}{3}, \quad 1.$$

Indeed, $\lfloor x \rfloor + \lfloor 2x \rfloor + \lfloor 3x \rfloor = 6k + r(t)$, where $r(t) = 0, 1, 2, 3$ on the consecutive intervals $[0, \frac{1}{3})$, $[\frac{1}{3}, \frac{1}{2})$, $[\frac{1}{2}, \frac{2}{3})$, and $[\frac{2}{3}, 1)$. Hence the attainable integers are exactly those congruent to $0, 1, 2, 3 \pmod 6$; among the integers 1 through 100, exactly 67 are attained. This finishes the piecewise calculation rather than leaving it as a method hint.

§43.4 Functions as maps and the contravariance of preimages

A function is not merely a formula. It consists of a domain, codomain, and assignment rule. The same formula can define different functions if the domain or codomain changes. This is why image, preimage, injectivity, and surjectivity must always be stated relative to a specified domain and codomain.

For domains, ranges, monotonicity, and extrema of functions, typical tools include:

(i) rewriting a square root or logarithm condition as a domain restriction;

(ii) using monotonicity to locate maxima and minima;

(iii) drawing a rough graph to understand intervals;

(iv) using substitution when a composite expression repeats.

For an iterated piecewise map, tracing successive images across intervals reveals possible periodic orbits. The transition pattern is a small instance of symbolic dynamics: a finite graph of interval-to-interval transitions encodes which itineraries the iterates can follow.

For a function $f : X \to Y$, the preimage operation

$$f^{-1} : \mathcal{P}(Y) \to \mathcal{P}(X)$$

preserves Boolean operations:

$$f^{-1}(B \cup C) = f^{-1}(B) \cup f^{-1}(C),$$
$$f^{-1}(B \cap C) = f^{-1}(B) \cap f^{-1}(C), \qquad f^{-1}(Y \setminus B) = X \setminus f^{-1}(B).$$

The distinction can be checked on a two-point counterexample. Let $f : \{a, b\} \to \{0\}$ be constant and take $A = \{a\}$, $B = \{b\}$. Then $f(A \cap B) = \varnothing$, whereas $f(A) \cap f(B) = \{0\}$. In contrast, inverse images commute with arbitrary unions and intersections by logical substitution: $x \in f^{-1}(B \cap C)$ iff $f(x) \in B$ and $f(x) \in C$. The stability of inverse images is the reason topology defines continuity through preimages of open sets.

§43.5 Sigma-algebras make countable questions measurable

In probability and measure theory, one does not always allow every subset of a universe. A sigma-algebra $\mathcal{A}$ on U is a collection of subsets satisfying

$$U \in \mathcal{A}, \qquad A \in \mathcal{A} \Rightarrow A^c \in \mathcal{A}, \qquad A_1, A_2, \ldots \in \mathcal{A} \Rightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{A}.$$

This is a Boolean algebra closed under countable unions. The elements of $\mathcal{A}$ are the measurable sets or events.

The elementary set laws still hold, but the important new question is: which subsets are allowed to be measured? Probability begins by assigning a number $P(A)$ to each $A \in \mathcal{A}$ such that

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} P(A_n)$$

for disjoint events. Thus basic set operations become the grammar of probability.

§43.6 Topology from Boolean Algebra

Stone duality says that Boolean algebras can be represented as algebras of clopen subsets of topological spaces. Given a Boolean algebra B, its Stone space consists of ultrafilters on B. Each element $b \in B$ determines a clopen set

$$\widehat{b} = \{\mathfrak{u} : b \in \mathfrak{u}\}.$$

For a finite Boolean algebra $\mathcal{P}(U)$, the ultrafilters are precisely the families of subsets containing a fixed element $u \in U$; the resulting Stone space is just the finite discrete set U. In general the Stone representation theorem identifies a Boolean algebra with the clopen subsets of its compact, Hausdorff, totally disconnected Stone space. This concrete finite case gives a precise bridge between truth-table atoms and the more abstract ultrafilter construction.

Sets, Divisibility Chains, and Closure Operators

N-20220806

Several elementary set exercises in this note are really questions about order: what does a condition select, when is a family closed under an operation, and how do we construct the smallest closed object? We begin by checking the concrete exercises, then develop their common structure. The order-theoretic language follows Davey and Priestley, *Introduction to Lattices and Order*, particularly the chapters on closure operations, Galois connections, and fixed points.[1]

§44.1 Membership comes before manipulation

Let

$$A = \{0, 1, 2, 3\}, \qquad B = \{x \in A : 2 - x^2 \notin A\}.$$

There are exactly four candidates, all taken from A:

x	0	1	2	3
$2 - x^2$	2	1	-2	-7
$2 - x^2 \in A$	yes	yes	no	no

Thus $B = \{2, 3\}$. Checking negative inputs would be invalid because the defining condition already restricts $x \in A$.

More generally, for $f : X \to Y$, the image and preimage of a subset are

$$f(S) = \{f(s) : s \in S\}, \qquad f^{-1}(T) = \{x \in X : f(x) \in T\}.$$

They behave differently. For any $T_1, T_2 \subseteq Y$,

$$f^{-1}(T_1 \cap T_2) = f^{-1}(T_1) \cap f^{-1}(T_2),$$

whereas always

$$f(S_1 \cap S_2) \subseteq f(S_1) \cap f(S_2).$$

The latter can be strict: with $f(x) = x^2$, $S_1 = \{-1\}$, $S_2 = \{1\}$, the left side is empty and the right side is $\{1\}$.

§44.2 Closure under the Chosen Operation

A subset $S \subseteq \mathbb{R}$ is closed under addition if $x, y \in S$ implies $x + y \in S$. Examples include $\mathbb{Z}, \mathbb{Q}, [0, \infty)$, and $(-\infty, 0]$. The last two are proper additive-closed subsets whose union is all of $\mathbb{R}$. Therefore the claim that two proper additive-closed sets can never cover $\mathbb{R}$ is false.

There is a related correct statement if the sets are *subgroups*. A group is not the union of two proper subgroups. To prove it, let H, K be proper subgroups. If $H \subseteq K$ or $K \subseteq H$, their union is plainly proper. Otherwise choose $h \in H \setminus K$ and $k \in K \setminus H$. Then

[1] Davey and Priestley, *Introduction to Lattices and Order*, chapter on complete lattices and Galois connections.

$h + k \notin H$, since otherwise $k = (h + k) - h \in H$; similarly $h + k \notin K$. This contradicts $H \cup K = G$.

The difference is structural: an additive submonoid need not contain additive inverses. Removing that hypothesis changes the conclusion.

§44.3 An extremal divisibility-chain exercise

Suppose $M \subseteq \{1, \dots, 2011\}$ is nonempty and for each $a, b \in M$ at least one of $a \mid b$, $b \mid a$, or $|a - b| = \max M$ holds. Write $m = \max M$. Since $1 \le a, b \le m$, one always has $|a - b| \le m - 1$, so the third option is impossible.

Thus all pairs in M must be comparable under divisibility. Enumerate its distinct elements in increasing order:

$$a_1 < a_2 < \cdots < a_r, \qquad a_i \mid a_{i+1}.$$

Every proper positive multiple is at least twice the original number; hence $a_{i+1} \ge 2a_i$ and inductively $a_r \ge 2^{r-1}$. Since $a_r \le 2011 < 2^{11}$, it follows that $r \le 11$. Equality occurs for

$$M = \{1, 2, 4, \dots, 1024\}.$$

Therefore the exact maximum cardinality is 11. The argument is an order-theoretic chain bound derived from the original arithmetic condition, without importing unrelated extremal theorems.

§44.4 Generated objects are intersections of closed objects

A closure operator $c : P \to P$ on a poset is extensive, monotone, and idempotent:

$$x \le c(x), \quad x \le y \Rightarrow c(x) \le c(y), \quad c(c(x)) = c(x).$$

For subsets of a vector space, $c(S) = \operatorname{span}(S)$ is one example; for a group, $c(S) = \langle S \rangle$ is another.

To see why intersections naturally arise, suppose $\mathcal{F}$ is a family of subsets closed under arbitrary intersections and containing the entire universe. Define

$$c(S) = \bigcap \{C \in \mathcal{F} : S \subseteq C\}.$$

The intersection is a member of $\mathcal{F}$ and contains S. Monotonicity follows because a larger input imposes a stricter condition on the indexing family; idempotence follows because $c(S)$ is already in $\mathcal{F}$. This gives an actual construction of the smallest closed object, rather than merely naming it.

§44.5 Image and preimage form an adjunction

The image/preimage relationship is more precise than a slogan about intersections. For $f : X \to Y$, $A \subseteq X$, and $B \subseteq Y$,

$$\boxed{f(A) \subseteq B \quad \Longleftrightarrow \quad A \subseteq f^{-1}(B).}$$

Both directions follow immediately from the definition of image and preimage. In the language of ordered sets, direct image is left adjoint to inverse image.

Consequently, the composite

$$c(A) = f^{-1}(f(A))$$

is a closure operator on the powerset of X: it includes A, is monotone, and is idempotent because $f(f^{-1}(f(A))) = f(A)$. What does it add? Exactly the points indistinguishable from points of A under f:

$$c(A) = \{x \in X : \exists a \in A,\ f(x) = f(a)\}.$$

For $f(x) = x^2$ and $A = \{2\}$ in $\mathbb{R}$, we get $c(A) = \{-2, 2\}$. This concrete calculation realizes an abstract Galois connection without invoking sophisticated machinery.

§44.6 Closure, fixed points, and stable iteration

A subset C is closed precisely when $c(C) = C$. The fixed points of a closure operator are closed under arbitrary intersections, because if $C_i = c(C_i)$ then monotonicity gives

$$c\left(\bigcap_i C_i\right) \subseteq \bigcap_i c(C_i) = \bigcap_i C_i,$$

and extensivity gives the reverse containment.

An apparently similar but distinct topic is fixed points of a general monotone map $F : \mathcal{P}(X) \to \mathcal{P}(X)$. There is a useful direct proof of existence of a *least* fixed point. Set

$$L = \bigcap \{A \subseteq X : F(A) \subseteq A\}.$$

The family is nonempty because X belongs to it. By monotonicity, $F(L) \subseteq L$. Applying F gives $F(F(L)) \subseteq F(L)$, so $F(L)$ itself belongs to the family defining L. Its minimality therefore implies $L \subseteq F(L)$. Hence $F(L) = L$. This is the least-fixed-point construction underlying the Knaster–Tarski theorem on complete lattices.

For a finite X, if the monotone process is also extensive and begins at S, the chain $S \subseteq F(S) \subseteq F^2(S) \subseteq \cdots$ must stabilize after finitely many strict increases. On an infinite universe, monotonicity alone does not warrant a claim of finite stabilization; a more careful limit construction may be required.

§44.7 A small algebraic side problem

The pair of elementary symmetric expressions

$$u = x + y, \qquad v = xy$$

determines the unordered pair $\{x, y\}$ as the roots of

$$t^2 - ut + v = 0.$$

Over $\mathbb{R}$, such a pair exists exactly when $u^2 - 4v \geq 0$. The discriminant is therefore a precise consistency condition on the proposed sums and products, rather than an unexplained substitution trick.

This is an example of the same general practice: specify which quantities are preserved under a transformation (here swapping x, y), then write the conditions in those invariants. For the set problems, the corresponding invariant information is closure under an operation or compatibility with image and preimage.

Three Elementary Inequalities and Their Geometric Meaning

N-20220810

Cauchy–Schwarz, AM–GM and the triangle inequality appear in almost every early inequality exercise. Their proofs have different mechanisms: positivity of a square, concavity of logarithm, and the geometry of a norm. The presentation follows the standard inequality toolkit of Hardy, Littlewood and Pólya's *Inequalities*, with the vector interpretation aligned with MIT 18.06.[1]

§45.1 Squares give sharp inequalities and equality cases

For real a, b, nonnegativity of $(a-b)^2$ gives

$$a^2 + b^2 \geq 2ab,$$

with equality if and only if $a = b$. Adding this identity over pairs yields

$$2(a^2 + b^2 + c^2 - ab - bc - ca) = (a-b)^2 + (b-c)^2 + (c-a)^2 \geq 0.$$

Equality holds precisely at $a = b = c$. This pattern is called a sum-of-squares certificate: it proves the inequality and simultaneously explains when the deficit vanishes.

§45.2 Cauchy–Schwarz from the discriminant

For vectors $u, v \in \mathbb{R}^n$, the quadratic

$$q(t) = \|u - tv\|^2 = \|v\|^2 t^2 - 2\langle u, v\rangle t + \|u\|^2$$

is nonnegative for every real t. When $v \neq 0$, its discriminant must be nonpositive, so

$$\langle u, v\rangle^2 \leq \|u\|^2 \|v\|^2.$$

When $v = 0$, both sides vanish. Equality for nonzero v occurs exactly when $u = tv$ for some scalar t. The equality condition is a dependence relation, not merely a coincidence of numbers.

For $a_i > 0$, substitute

$$u_i = \frac{x_i}{\sqrt{a_i}}, \qquad v_i = \sqrt{a_i}.$$

Cauchy then becomes its Engel form:

$$\boxed{\sum_i \frac{x_i^2}{a_i} \geq \frac{(\sum_i x_i)^2}{\sum_i a_i}.}$$

Equality holds when x_i/a_i is independent of i.

[1] https://web.mit.edu/18.06/www/index.shtml

Example 45.2.1

For $a, b, c > 0$, applying Engel with denominators $b + c, c + a, a + b$ gives

$$\frac{a^2}{b+c} + \frac{b^2}{c+a} + \frac{c^2}{a+b} \geq \frac{(a+b+c)^2}{2(a+b+c)} = \frac{a+b+c}{2}.$$

Equality requires $a/(b + c) = b/(c + a) = c/(a + b)$, equivalent in this positive setting to $a = b = c$.

§45.3 The triangle inequality is the norm form of Cauchy

For Euclidean vectors,

$$\|u + v\|^2 = \|u\|^2 + 2\langle u, v\rangle + \|v\|^2 \leq (\|u\| + \|v\|)^2,$$

so $\|u + v\| \leq \|u\| + \|v\|$. In $\mathbb{R}$ this is $|a + b| \leq |a| + |b|$; substituting $u = a - b$, $v = b - c$ gives

$$|a - c| \leq |a - b| + |b - c|.$$

Rearranging the triangle inequality in both directions gives the reverse form

$$\big|\|u\| - \|v\|\big| \leq \|u - v\|.$$

In Euclidean space, equality in the forward triangle inequality occurs when the nonzero vectors point in the same direction. In a general normed space equality can also occur for other configurations; for example $\|(1, 0) + (0, 1)\|_1 = 2 = \|(1, 0)\|_1 + \|(0, 1)\|_1$. This shows why the equality criterion needs to mention the norm.

§45.4 AM–GM follows from concavity

For positive $x_1, \ldots, x_n$, the second derivative of the logarithm is

$$(\log x)'' = -x^{-2} < 0,$$

so log is strictly concave. Jensen's inequality gives

$$\log\left(\frac{x_1 + \cdots + x_n}{n}\right) \geq \frac{1}{n}\sum_i \log x_i = \log\left(\prod_i x_i\right)^{1/n}.$$

Exponentiation yields

$$\boxed{\frac{x_1 + \cdots + x_n}{n} \geq (x_1 \cdots x_n)^{1/n}.}$$

Strict concavity forces equality exactly when all x_i are equal. The formula extends to nonnegative variables by continuity; when any variable is zero, the right-hand side is zero, and equality requires all variables to be zero.

For $x > 0$, apply this to x and $1/x$:

$$x + \frac{1}{x} \geq 2,$$

with equality at $x = 1$. For $x, y, z > 0$, the product of x/y, y/z, z/x is 1, so

$$\frac{x}{y} + \frac{y}{z} + \frac{z}{x} \geq 3,$$

with equality at $x = y = z$.

§45.5 Coordinate geometry is an inner-product calculation

Let $A = (x_1, y_1)$, $B = (x_2, y_2)$. The distance formula is

$$|AB|^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2 = \langle B - A, B - A \rangle.$$

For direction vectors u, v, perpendicularity is the invariant relation $\langle u, v \rangle = 0$. When both lines have finite slopes m_1, m_2, direction vectors may be chosen as $(1, m_1)$, $(1, m_2)$, so perpendicularity becomes $1 + m_1 m_2 = 0$.

The slope rule $m_1 m_2 = -1$ does not cover vertical lines, but the dot-product rule does. Likewise, checking parallelism via a determinant $\det[u\ v] = 0$ works whether the slopes exist or not.

§45.6 A matrix version of the same square argument

For a real symmetric matrix A, the condition

$$x^T A x \geq 0 \qquad \text{for all } x$$

defines positive semidefiniteness, denoted $A \succeq 0$. For

$$A = \begin{pmatrix} a & b \\ b & c \end{pmatrix},$$

the exact criterion is

$$a \geq 0, \quad c \geq 0, \quad ac - b^2 \geq 0.$$

If $a > 0$, completing a square gives

$$ax^2 + 2bxy + cy^2 = a(x + (b/a)y)^2 + (c - b^2/a)y^2;$$

the case $a = 0$ requires $b = 0$ and $c \geq 0$. This is the two-dimensional analogue of proving an inequality by writing a sum of squares.

The three elementary methods now have distinct mechanisms: Cauchy is controlled by a nonnegative quadratic and linear dependence; AM–GM is controlled by concavity and equality of inputs; the triangle inequality describes metric geometry. Choosing the method means recognizing which mathematical structure the problem actually supplies.

Inequality Exercises: Constraints, Equality, and Vector Geometry

N-20220811

The central skill in an inequality problem is finding the correct representation of the expression. Fractions can be rewritten as a Cauchy–Schwarz inner product; a radical can encode a Euclidean distance; a quadratic constraint defines an ellipsoid or sphere. The standard methods and equality conditions follow Hardy, Littlewood and Pólya's *Inequalities* and Strang's MIT 18.06 linear algebra material.[1]

§46.1 Fraction inequalities and their domains

For positive denominators u, v, Cauchy–Schwarz gives

$$\frac{a^2}{u} + \frac{b^2}{v} \geq \frac{(a+b)^2}{u+v}.$$

To see equality, write

$$\frac{a^2}{u} + \frac{b^2}{v} - \frac{(a+b)^2}{u+v} = \frac{(av-bu)^2}{uv(u+v)}.$$

Thus equality occurs exactly when $a/u = b/v$.

The original expression

$$F(x,y) = \frac{4}{4-x^2} + \frac{9}{9-y^2}$$

can be bounded by the same mechanism *only* when $|x| < 2$ and $|y| < 3$. Under these conditions,

$$F(x,y) \geq \frac{25}{13-x^2-y^2}.$$

If in addition $xy = -1$, then $x^2 + y^2 \geq 2|xy| = 2$, so

$$\boxed{F(x,y) \geq \frac{25}{11}.}$$

However this last bound need not be sharp: equality in Cauchy would require $2/(4-x^2) = 3/(9-y^2)$, and equality in $x^2 + y^2 \geq 2$ would require $x^2 = y^2 = 1$, conditions which cannot both hold. It is an instructive valid *estimate*, not necessarily the optimum.

§46.2 A square-root comparison is Cauchy in disguise

Let $a, b, c, d, m, n > 0$. Define

$$P = \sqrt{ab} + \sqrt{cd}, \qquad Q = \sqrt{ma+nc}\sqrt{b/m + d/n}.$$

Apply Cauchy to

$$u = (\sqrt{ma}, \sqrt{nc}), \qquad v = (\sqrt{b/m}, \sqrt{d/n}).$$

[1] https://web.mit.edu/18.06/www/index.shtml

Their inner product is P, and $\|u\|\|v\| = Q$. Hence $P \le Q$. Equality occurs precisely when u and v are positive scalar multiples, equivalently

$$\frac{\sqrt{ma}}{\sqrt{b/m}} = \frac{\sqrt{nc}}{\sqrt{d/n}} \quad \Longleftrightarrow \quad \boxed{m^2ad = n^2bc}.$$

The equality condition is a useful check on the expansion of Q^2.

§46.3 Maximizing two radicals on a fixed interval

For $3 \le x \le 5$, consider

$$f(x) = 2\sqrt{x-3} + \sqrt{5-x}.$$

Cauchy–Schwarz gives

$$f(x)^2 \le (2^2 + 1^2)((x-3) + (5-x)) = 10.$$

Equality holds when

$$(\sqrt{x-3}, \sqrt{5-x}) = \lambda(2,1) \quad \text{with } \lambda \ge 0.$$

Thus $(x-3)/(5-x) = 4$, yielding $x = 23/5$. Direct substitution confirms

$$\boxed{\max_{3\le x\le 5} f(x) = \sqrt{10}.}$$

This proof establishes both a universal upper bound and a point where it is attained; there is no need to differentiate and then check endpoints separately.

§46.4 Equality in Cauchy recovers unknown vectors

Suppose $u = (a,b,c)$ and $v = (x,y,z)$ satisfy

$$\|u\|^2 = 25, \quad \|v\|^2 = 36, \quad \langle u, v\rangle = 30.$$

Cauchy gives $\langle u, v\rangle \le \|u\|\|v\| = 5 \cdot 6 = 30$. Equality forces $u = \lambda v$ for some $\lambda \ge 0$. Comparing norms gives

$$u = \frac{5}{6}v.$$

Consequently $a+b+c = \frac{5}{6}(x+y+z)$. The ratio $(a+b+c)/(x+y+z)$ equals $5/6$ *only if* the denominator is nonzero. When $x+y+z = 0$, both sums vanish, so their ratio is $0/0$ and is undefined. The proportionality of the vectors remains valid.

§46.5 Triangle angles and sharp symmetric bounds

If $A, B, C > 0$ are the angles of a Euclidean triangle, $A + B + C = \pi$. Engel's inequality gives

$$\frac{1}{A} + \frac{1}{B} + \frac{1}{C} \ge \frac{(1+1+1)^2}{A+B+C} = \frac{9}{\pi}.$$

Cauchy also gives

$$A^2 + B^2 + C^2 \ge \frac{(A+B+C)^2}{3} = \frac{\pi^2}{3}.$$

The equality cases agree: both are attained by the equilateral triangle $A = B = C = \pi/3$.

§46.6 Two spheres: subtraction gives the radical plane

Let $c_1, c_2 \in \mathbb{R}^3$ and consider spheres

$$\|x - c_1\|^2 = r_1^2, \qquad \|x - c_2\|^2 = r_2^2.$$

Subtracting their equations cancels $\|x\|^2$ and leaves

$$2(c_2 - c_1) \cdot x = r_1^2 - r_2^2 + \|c_2\|^2 - \|c_1\|^2.$$

If $c_1 \neq c_2$, this is a plane, called the radical plane. Depending on how that plane meets either sphere, the intersection is a circle, a single point, or empty.

For a separate constrained optimization, maximize $w \cdot x$ on $\|x - c\| = r$. Writing $x = c + ru$, $\|u\| = 1$, Cauchy gives

$$w \cdot x = w \cdot c + r(w \cdot u) \leq w \cdot c + r\|w\|,$$

with equality at $u = w/\|w\|$ when $w \neq 0$. For $w = (-8, 6, -24)$, $\|w\| = 26$. Thus the maximum is $w \cdot c + 26r$, with its location determined explicitly by the coefficient vector.

§46.7 A product constraint and AM–GM

Suppose $a, b, c > 0$ and $(1 + a)(1 + b)(1 + c) = 8$. Since

$$(1 + a)(1 + b)(1 + c) \geq \left(1 + (abc)^{1/3}\right)^3,$$

we have $2 \geq 1 + (abc)^{1/3}$ and therefore $abc \leq 1$.

To justify the middle inequality, let $t = (abc)^{1/3}$. AM–GM gives $a + b + c \geq 3t$ and $ab + bc + ca \geq 3t^2$, so expansion of the left side is at least $1 + 3t + 3t^2 + t^3$. Equality requires $a = b = c = 1$, where $abc = 1$ and the constraint holds. The bound is therefore sharp.

§46.8 Three rays turn a radical into a triangle

Consider the inequality, valid for arbitrary real x, y, z,

$$\sqrt{x^2 + xy + y^2} + \sqrt{x^2 + xz + z^2} \geq \sqrt{y^2 + yz + z^2}.$$

Choose three unit vectors e_1, e_2, e_3 in the plane separated by $120°$, so $e_i \cdot e_j = -1/2$ for $i \neq j$. Put

$$A = xe_1, \qquad B = ye_2, \qquad C = ze_3.$$

Then

$$\|A - B\|^2 = x^2 + xy + y^2, \quad \|A - C\|^2 = x^2 + xz + z^2, \quad \|B - C\|^2 = y^2 + yz + z^2.$$

The proposed inequality is exactly $\|A - B\| + \|A - C\| \geq \|B - C\|$. It is the triangle inequality for three explicit points. The three-ray construction is the missing geometric detail: simply identifying each radical separately as a vector norm would not prove the relationship among them.

§46.9 Quadratic constraints and eigenvalues

If H is a real symmetric matrix, the extremal values of $x^T Hx$ on the unit sphere equal its smallest and largest eigenvalues. Indeed, choose an orthonormal eigenbasis, write $x = \sum_i c_i e_i$, and calculate

$$\frac{x^T Hx}{x^T x} = \frac{\sum_i \lambda_i c_i^2}{\sum_i c_i^2}.$$

The quotient is a weighted average of the eigenvalues and lies between their minimum and maximum. Equality occurs on the corresponding eigenspaces.

This links inequality practice to constrained optimization without replacing concrete proofs by a generic appeal to Lagrange multipliers. When a question actually contains a quadratic form and a sphere constraint, the eigenvalue calculation is its natural solution.

Classical Inequalities from Convexity, Duality, and Norm Geometry

N-20221012

§47.1 A convex graph lies above each supporting tangent

A differentiable function f on an interval is convex when its slopes increase. Equivalently, for every point x_0, the graph lies above the tangent line:

$$\boxed{f(x) \geq f(x_0) + f'(x_0)(x - x_0).}$$

This one-dimensional picture already contains Bernoulli's inequality. Let

$$f(x) = (1 + x)^r, \qquad x > -1, \qquad r \geq 1.$$

Since

$$f''(x) = r(r - 1)(1 + x)^{r-2} \geq 0,$$

the function is convex. Its tangent at $x = 0$ is

$$f(0) + f'(0)x = 1 + rx.$$

Therefore

$$\boxed{(1 + x)^r \geq 1 + rx.}$$

For an integer $r = n$, this is the classical Bernoulli inequality. If $r > 1$, equality holds only at $x = 0$.

The proof matters more than the formula. Bernoulli is not an isolated binomial estimate; it is the first example of a general principle: convexity converts local first-order information into a global lower bound.

§47.2 Jensen averages the supporting-line inequality

Let f be differentiable and convex, and let

$$\lambda_i \geq 0, \qquad \sum_{i=1}^{n} \lambda_i = 1.$$

Set

$$m = \sum_{i=1}^{n} \lambda_i x_i.$$

The supporting line at m gives

$$f(x_i) \geq f(m) + f'(m)(x_i - m).$$

Multiply by λ_i and sum. The linear terms cancel because

$$\sum_i \lambda_i (x_i - m) = 0.$$

Hence

$$\boxed{f\left(\sum_i \lambda_i x_i\right) \le \sum_i \lambda_i f(x_i).}$$

This is Jensen's inequality.

If f is strictly convex and all weights are positive, equality requires

$$x_1 = \cdots = x_n.$$

For a concave function the inequality reverses.

Take the concave function

$$f(x) = \log x$$

on $(0, \infty)$, with equal weights $1/n$. Jensen gives

$$\frac{1}{n}\sum_{i=1}^{n} \log a_i \le \log\left(\frac{a_1 + \cdots + a_n}{n}\right).$$

Exponentiating yields AM–GM:

$$\boxed{\sqrt[n]{a_1 \cdots a_n} \le \frac{a_1 + \cdots + a_n}{n}.}$$

Equality holds exactly when all a_i are equal.

The same tangent-line mechanism therefore proves both an additive estimate such as Bernoulli and a multiplicative estimate such as AM–GM; the difference lies in the chosen convex or concave function.

§47.3 Smoothing leads from Jensen to majorization

Jensen compares a list with its weighted average. Majorization compares two lists with the same total sum and asks which one is more spread out.

For decreasing rearrangements

$$x_1^{\downarrow} \ge \cdots \ge x_n^{\downarrow}, \qquad y_1^{\downarrow} \ge \cdots \ge y_n^{\downarrow},$$

we say that x majorizes y, written $x \succ y$, when

$$\sum_{i=1}^{k} x_i^{\downarrow} \ge \sum_{i=1}^{k} y_i^{\downarrow} \qquad (1 \le k < n),$$

and the total sums are equal.

The elementary move behind this order is smoothing. Keep the mean m fixed and compare

$$m + t, \qquad m - t.$$

For differentiable convex f, define

$$g(t) = f(m + t) + f(m - t), \qquad t \ge 0.$$

Then

$$g'(t) = f'(m + t) - f'(m - t) \ge 0$$

because f' is increasing. Moving two entries closer together decreases the sum of their convex images.

A finite sequence of such balancing moves transforms a majorizing vector into a more even one. This gives the content of Karamata's inequality:

$$x \succ y \quad \Longrightarrow \quad \sum_i f(x_i) \geq \sum_i f(y_i)$$

for convex f.

For example,

$$(4,1,1) \succ (2,2,2).$$

With $f(t) = t^2$,

$$4^2 + 1^2 + 1^2 = 18 \geq 2^2 + 2^2 + 2^2 = 12.$$

With $f(t) = e^t$, the same ordering gives

$$e^4 + 2e \geq 3e^2.$$

Majorization formalizes the intuition that convex functions penalize unevenness.

§47.4 Young's Inequality and Tangents

Let

$$p, q > 1, \qquad \frac{1}{p} + \frac{1}{q} = 1.$$

Young's inequality states that for $a, b \geq 0$,

$$\boxed{ab \leq \frac{a^p}{p} + \frac{b^q}{q}.}$$

A direct derivation shows where the conjugate exponents come from. Fix a and consider

$$\Phi(b) = ab - \frac{b^q}{q}.$$

Its derivative is

$$\Phi'(b) = a - b^{q-1}.$$

The maximum occurs at

$$b = a^{1/(q-1)} = a^{p-1},$$

because

$$q - 1 = \frac{1}{p-1}.$$

At this point,

$$\Phi(b) = a^p - \frac{a^p}{q} = \frac{a^p}{p}.$$

Thus

$$ab - \frac{b^q}{q} \leq \frac{a^p}{p},$$

which is Young's inequality.

Equality holds when

$$b = a^{p-1},$$

or equivalently

$$a^p = b^q.$$

The two powers are paired so that each is the optimal supporting term for the other. This duality is what later makes Hölder's inequality exact.

§47.5 Hölder follows by normalizing Young's inequality

For vectors $x = (x_i)$ and $y = (y_i)$, define

$$\|x\|_p = \left(\sum_i |x_i|^p\right)^{1/p}, \qquad \|y\|_q = \left(\sum_i |y_i|^q\right)^{1/q}.$$

If either norm is zero, Hölder is trivial. Otherwise set

$$X_i = \frac{|x_i|}{\|x\|_p}, \qquad Y_i = \frac{|y_i|}{\|y\|_q}.$$

Young's inequality gives

$$X_i Y_i \le \frac{X_i^p}{p} + \frac{Y_i^q}{q}.$$

Summing and using

$$\sum_i X_i^p = 1, \qquad \sum_i Y_i^q = 1$$

yields

$$\sum_i X_i Y_i \le \frac{1}{p} + \frac{1}{q} = 1.$$

Therefore

$$\boxed{\sum_i |x_i y_i| \le \|x\|_p \|y\|_q.}$$

This is Hölder's inequality.

Equality requires equality in Young for every nonzero component:

$$X_i^p = Y_i^q.$$

Thus the normalized mass distributions $(|x_i|^p)$ and $(|y_i|^q)$ must agree.

For $p = q = 2$, Hölder becomes Cauchy–Schwarz:

$$\left|\sum_i x_i y_i\right| \le \left(\sum_i x_i^2\right)^{1/2} \left(\sum_i y_i^2\right)^{1/2}.$$

The equality condition reduces to linear dependence of the two vectors.

§47.6 Hölder identifies the dual norm exactly

Hölder shows that every y with $\|y\|_q = 1$ satisfies

$$\left|\sum_i x_i y_i\right| \le \|x\|_p.$$

The bound is attained. For $x \neq 0$, define

$$y_i = \frac{\operatorname{sgn}(x_i)|x_i|^{p-1}}{\|x\|_p^{p-1}}.$$

Since

$$(p-1)q = p,$$

we have

$$\sum_i |y_i|^q = \frac{\sum_i |x_i|^p}{\|x\|_p^p} = 1.$$

Moreover,

$$\sum_i x_i y_i = \frac{\sum_i |x_i|^p}{\|x\|_p^{p-1}} = \|x\|_p.$$

Hence

$$\boxed{\|x\|_p = \sup_{\|y\|_q=1} \sum_i x_i y_i.}$$

This is the dual-norm formula. An ℓ^p norm is recovered by testing the vector against all unit vectors in the conjugate ℓ^q norm. Hölder is therefore not only an inequality; it identifies the exact linear functionals that measure ℓ^p geometry.

The case $p = 2$ is self-dual. The maximizing vector is parallel to x, which is the familiar equality condition in Cauchy–Schwarz.

§47.7 Minkowski's Inequality via Duality

Using the dual-norm formula,

$$\begin{aligned}\|x+z\|_p &= \sup_{\|y\|_q=1} \sum_i (x_i + z_i) y_i \\ &\le \sup_{\|y\|_q=1} \sum_i x_i y_i + \sup_{\|y\|_q=1} \sum_i z_i y_i \\ &= \|x\|_p + \|z\|_p.\end{aligned}$$

Thus

$$\boxed{\left(\sum_i |x_i + z_i|^p\right)^{1/p} \le \left(\sum_i |x_i|^p\right)^{1/p} + \left(\sum_i |z_i|^p\right)^{1/p}.}$$

This is Minkowski's inequality.

For $p = 2$, it is the Euclidean triangle inequality. For general $p > 1$, it proves that ℓ^p is a normed vector space.

Equality in the strictly convex ℓ^p norm occurs when the two nonzero vectors point in the same nonnegative direction:

$$x = \lambda z, \qquad \lambda \ge 0.$$

The algebraic equality condition is the norm-geometric statement that travelling along two segments is as short as travelling along their sum only when the segments are aligned.

§47.8 Cauchy's Engel form is projection geometry in disguise

Cauchy–Schwarz can be rearranged into the Engel form

$$\boxed{\sum_{i=1}^n \frac{a_i^2}{b_i} \ge \frac{(a_1 + \cdots + a_n)^2}{b_1 + \cdots + b_n}} \qquad (b_i > 0).$$

Apply Cauchy to the vectors

$$\left(\frac{a_1}{\sqrt{b_1}}, \ldots, \frac{a_n}{\sqrt{b_n}}\right)$$

and

$$(\sqrt{b_1}, \dots, \sqrt{b_n}).$$

Their inner product is $a_1 + \cdots + a_n$, giving the formula.

Equality holds when the vectors are proportional:

$$\frac{a_i}{\sqrt{b_i}} = \lambda \sqrt{b_i},$$

so

$$\frac{a_i}{b_i} = \lambda$$

for every i.

The same result can be read as an optimization problem. Among all vectors with prescribed weighted sum, the squared weighted norm is minimized by the vector parallel to the constraint direction. The frequently used contest lemma is therefore a projection theorem in a weighted Euclidean space.

§47.9 Power means form one monotone scale

For positive $a_1, \dots, a_n$ and $r \neq 0$, define

$$M_r(a) = \left(\frac{1}{n} \sum_{i=1}^{n} a_i^r \right)^{1/r}.$$

For $0 < r < s$, the function

$$t \longmapsto t^{s/r}$$

is convex. Jensen gives

$$\left(\frac{1}{n} \sum_i a_i^r \right)^{s/r} \leq \frac{1}{n} \sum_i a_i^s.$$

Taking the s-th root yields

$$M_r(a) \leq M_s(a).$$

As $r \to 0$, write

$$a_i^r = e^{r \log a_i} = 1 + r \log a_i + o(r).$$

Then

$$\log M_r = \frac{1}{r} \log \left(\frac{1}{n} \sum_i a_i^r \right) \longrightarrow \frac{1}{n} \sum_i \log a_i.$$

Hence

$$M_0(a) = \sqrt[n]{a_1 \cdots a_n}.$$

For negative orders,

$$M_{-r}(a_1, \dots, a_n) = \frac{1}{M_r(a_1^{-1}, \dots, a_n^{-1})}.$$

Combining these observations gives the full monotonicity:

$$r < s \quad \Longrightarrow \quad M_r \leq M_s.$$

The classical chain is therefore one theorem:

$$M_{-1} \leq M_0 \leq M_1 \leq M_2,$$

or

$$\text{harmonic} \leq \text{geometric} \leq \text{arithmetic} \leq \text{quadratic}.$$

Strict inequality holds unless all variables are equal.

§47.10 Chebyshev is a covariance identity

Suppose

$$a_1 \le \cdots \le a_n, \qquad b_1 \le \cdots \le b_n.$$

Chebyshev's sum inequality states

$$\frac{1}{n}\sum_i a_i b_i \ge \left(\frac{1}{n}\sum_i a_i\right)\left(\frac{1}{n}\sum_i b_i\right).$$

The difference has the exact form

$$\begin{aligned}\frac{1}{n}\sum_i a_i b_i - \left(\frac{1}{n}\sum_i a_i\right)\left(\frac{1}{n}\sum_i b_i\right) \\ = \frac{1}{2n^2}\sum_{i,j}(a_i - a_j)(b_i - b_j).\end{aligned}$$

Every summand is nonnegative because the two sequences have the same ordering. Hence the difference is nonnegative.

If one sequence is increasing and the other decreasing, every product changes sign and the inequality reverses. Chebyshev is therefore a finite covariance statement: similarly ordered variables have nonnegative covariance.

Equality occurs when one sequence is constant, or more generally when every pair with a nonzero difference in one sequence has zero difference in the other.

§47.11 Log-Sum Inequality and Relative Entropy

Let $a_i, b_i > 0$, and set

$$A = \sum_i a_i, \qquad B = \sum_i b_i.$$

The function

$$\phi(t) = t \log t$$

is convex on $(0, \infty)$. Apply Jensen with weights

$$\lambda_i = \frac{b_i}{B}$$

and points

$$x_i = \frac{a_i}{b_i}.$$

Since

$$\sum_i \lambda_i x_i = \frac{A}{B},$$

Jensen gives

$$\sum_i \frac{b_i}{B}\frac{a_i}{b_i}\log\frac{a_i}{b_i} \ge \frac{A}{B}\log\frac{A}{B}.$$

Multiplying by B yields the log-sum inequality:

$$\boxed{\sum_i a_i \log\frac{a_i}{b_i} \ge A \log\frac{A}{B}.}$$

Equality holds exactly when the ratios

$$\frac{a_i}{b_i}$$

are all equal.

For probability distributions p, q, we have $A = B = 1$, so

$$D(p\|q) = \sum_i p_i \log \frac{p_i}{q_i} \geq 0.$$

The quantity $D(p\|q)$ is relative entropy or Kullback–Leibler divergence. It vanishes exactly when $p = q$.

The route from a tangent line to relative entropy is continuous. Convexity gives Jensen; the same averaging principle yields AM–GM and majorization; convex conjugacy gives Young; normalization produces Hölder; dual norms produce Minkowski; and the convex function $t \log t$ measures the cost of replacing one probability distribution by another. The named inequalities are not a list of unrelated tricks but consequences of how convex functions and dual norms control averaging, pairing, and distance.

Functional Symmetry, Tangency, and Sharp Inequalities

N-20221105

A functional equation can be inconsistent before one attempts to solve it. A polynomial can change the number of roots as a parameter moves. And an inequality may become sharp only when its constraint is used. These simple problems reward structural calculations that will later reappear as group actions, fold singularities, and convex geometry.[1]

§48.1 A translation and a reflection need to be compatible

Consider the drift equation

$$f(x+4) = f(x) + 2.$$

Subtracting the linear function $x/2$ makes the behavior periodic:

$$h(x) = f(x) - \frac{x}{2}, \qquad h(x+4) = h(x).$$

Thus every solution consists of a linear drift plus an arbitrary 4-periodic function.

What happens if we also demand the reflection symmetry

$$f(2-x) = f(x)?$$

There are no solutions. The reflection condition at $x-4$ gives

$$f(6-x) = f(x-4) = f(x) - 2,$$

while translation applied to $2-x$ gives

$$f(6-x) = f(2-x) + 2 = f(x) + 2.$$

The same value cannot equal both $f(x)-2$ and $f(x)+2$.

The structural reason is a relation in the infinite dihedral group. The translations $T(x) = x+4$ and reflection $R(x) = 2-x$ satisfy

$$RTR = T^{-1}.$$

A reflection reverses the translation, so their corresponding transformations of function values must also be compatible. Shifting the output by $+2$ is not compatible with leaving the output unchanged under reflection.

A point-reflection condition *is* compatible:

$$f(x+4) = f(x) + 2, \qquad f(2-x) = 2 - f(x).$$

Its complete solution family is

$$\boxed{f(x) = \frac{x+1}{2} + h(x),} \qquad h(x+4) = h(x), \quad h(2-x) = -h(x).$$

For instance $h(x) = A\sin(\pi(x-1)/2)$ works for every real A. The computation has converted a functional equation into an invariance and anti-invariance problem under two generators.

[1] For the classical inequality methods see Hardy, Littlewood and Pólya, *Inequalities*; the parameter analysis below uses only standard calculus and the cubic discriminant.

§48.2 A logarithm comparison is also an analytic expansion

For $a > 0$, $a \neq 1$, and $-1 < x < 1$, consider

$$F_a(x) = \log_a(1+x) - \log_a(1-x) = \frac{\log((1+x)/(1-x))}{\log a}.$$

The numerator has the same sign as x, so the sign of $F_a(x)$ changes when a crosses 1. In particular, inequalities reverse when the base is between zero and one.

The numerator has a useful expansion. Since

$$\frac{d}{dx}\log\frac{1+x}{1-x} = \frac{2}{1-x^2},$$

integrate the geometric series to obtain, for $|x| < 1$,

$$\boxed{\log\frac{1+x}{1-x} = 2\sum_{n=0}^{\infty}\frac{x^{2n+1}}{2n+1}.}$$

For $0 < x < 1$, the truncation through $n = m$ is a strict lower bound, and the remainder satisfies

$$0 < R_m(x) \leq \frac{2x^{2m+3}}{(2m+3)(1-x^2)}.$$

Thus a simple sign problem naturally yields the Taylor series of $2\operatorname{artanh}(x)$ and a rigorous approximation error.

§48.3 Two roots merge at a tangent level

Consider the parameter-dependent cubic

$$F_p(x) = x^3 - 3px + 2.$$

A tangent zero satisfies both $F_p(x) = 0$ and $F_p'(x) = 0$. The second equation gives $p = x^2$, and substitution into the first gives

$$-2x^3 + 2 = 0.$$

Hence

$$\boxed{x = 1, \qquad p = 1, \qquad F_1(x) = (x-1)^2(x+2).}$$

The root at 1 is double; the root at -2 is simple.

The discriminant of $x^3 + bx + c$ is $-4b^3 - 27c^2$. For this cubic,

$$\Delta(p) = 108(p^3 - 1).$$

The equation has one distinct real root if $p < 1$, two distinct roots when $p = 1$, and three distinct real roots if $p > 1$.

Write $x = 1 + y$ and $p = 1 + \delta$. A direct expansion gives

$$F_{1+\delta}(1+y) = 3y^2 + y^3 - 3\delta(1+y).$$

When $\delta > 0$ is small, the two roots that meet at 1 obey

$$y = \pm\sqrt{\delta} + O(\delta).$$

A polynomial exercise has exposed the local square-root geometry of a fold: solutions appear in pairs across a critical parameter.

§48.4 The product constraint produces a sharp inequality

For $a, b, c > 0$ with $abc = 1$, prove

$$\boxed{\frac{a^2}{b+c} + \frac{b^2}{c+a} + \frac{c^2}{a+b} \geq \frac{3}{2}.}$$

Cauchy–Schwarz in Engel form yields

$$\sum_{\text{cyc}} \frac{a^2}{b+c} \geq \frac{(a+b+c)^2}{2(a+b+c)} = \frac{a+b+c}{2}.$$

AM–GM and $abc = 1$ give $a+b+c \geq 3$, so the desired bound follows. Equality in both steps requires $a = b = c = 1$; the constant is sharp.

There is also a useful local geometric interpretation of the constraint. Put

$$a = e^u, \qquad b = e^v, \qquad c = e^{-u-v}.$$

The condition $abc = 1$ is now automatic. By convexity of the exponential,

$$e^u + e^v + e^{-u-v} \geq 3.$$

A second-order expansion around the equality point gives

$$a+b+c = 3 + \frac{1}{2}\bigl(u^2 + v^2 + (u+v)^2\bigr) + O\bigl((|u|+|v|)^3\bigr).$$

The positive quadratic correction quantifies how the sum grows away from the equal-variable configuration. In logarithmic coordinates, the product constraint has become a linear plane and AM–GM is simply a Jensen inequality on that plane.

§48.5 Quadratics give the exact local model

For comparison, completing the square in a one-variable problem yields

$$x^2 + x + \frac{5}{3} = \left(x + \frac{1}{2}\right)^2 + \frac{17}{12}.$$

The excess above the minimum $17/12$ is exactly a square. At the constrained exponential minimum the excess is only approximately a positive quadratic form, but it plays the same local role.

The method across these problems is to identify what cannot change. A functional equation respects the relations of its symmetry group, a critical root is marked by the simultaneous vanishing of a function and its derivative, and the equality case of an inequality identifies the point about which a convexity argument can be sharpened. Each is a useful route from elementary algebra to structural mathematics.

Function Spaces: Domains, Orbit Symmetry, and Changing Numbers of Solutions

N-20221217

The same function may be studied as a formula, as a map between topological spaces, or as an object carrying symmetry. A denominator determines where its formula makes sense; the image records attainable values; a functional equation can transport information across infinitely many inputs; a parameter can change the topology of a solution set. The following examples make these perspectives meet in actual calculations.

§49.1 Domains from Constraints

For the real-valued expression

$$F(x) = \frac{\sqrt{x(3-x)}}{\log_{10}((x-3)^2)},$$

the square root requires $0 \le x \le 3$, the logarithm requires $x \neq 3$, and its value cannot be zero, excluding $(x-3)^2 = 1$. Hence

$$\operatorname{dom} F = [0,2) \cup (2,3).$$

The open and closed endpoints encode different facts: $x = 0$ is allowed, while $x = 2$ gives a zero denominator and $x = 3$ gives an undefined logarithm.

A floor expression changes its form at integer thresholds:

$$\lfloor x \rfloor = k \quad \Longleftrightarrow \quad k \le x < k+1.$$

It is therefore often best handled on intervals with fixed k. The discontinuities of the floor function are a geometric record of that case split.

§49.2 Ranges and Compactness

Consider

$$G(x) = \sqrt{x-4} + \sqrt{15-3x}, \qquad 4 \le x \le 5.$$

With $u = x - 4$, this becomes

$$G = \sqrt{u} + \sqrt{3(1-u)}, \qquad 0 \le u \le 1.$$

Cauchy–Schwarz gives $G^2 \le (1+3)(u+1-u) = 4$, and equality holds at $u = 1/4$, where $G = 2$. The only interior critical point is this maximum; the endpoint values are $G(0) = \sqrt{3}$, $G(1) = 1$. Thus

$$\boxed{G([4,5]) = [1,2].}$$

There is a topological step hidden in the final interval. A continuous image of a compact interval is compact and connected, so once the extremal values are known the entire range between them is attained. On a noncompact or punctured domain, endpoint values might be approached without being attained.

§49.3 Absolute values are shadows of switching branches

Knowing a range requires knowing where a function changes its formula. An especially useful example is the upper envelope

$$M(x) = \max\{(x-1)^2, (x+1)^2\}.$$

Subtract the candidates: $(x+1)^2 - (x-1)^2 = 4x$. Thus the second branch is active for $x \geq 0$, the first for $x \leq 0$, and

$$\boxed{M(x) = x^2 + 1 + 2|x|.}$$

The identity $\max\{u, v\} = (u + v + |u - v|)/2$ is a reliable way to derive such a formula, with no guesswork about the active branch.

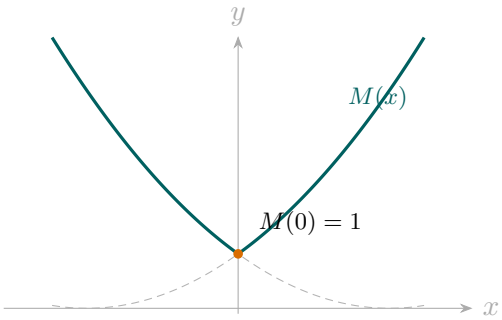


Figure 49.1: The solid upper envelope switches quadratic branches at $x = 0$. The minimum is a corner, so it cannot be found by solving the ordinary equation $M'(0) = 0$.

For $x < 0$ the derivative is $2x - 2$, and for $x > 0$ it is $2x + 2$. At the minimum $x = 0$, the one-sided derivatives are -2 and 2: the ordinary derivative does not exist. Yet convexity provides a better replacement. A number s is a *subgradient* of M at zero if

$$M(y) \geq M(0) + sy \quad \text{for every } y.$$

Since $M(y) - M(0) = y^2 + 2|y|$, every $s \in [-2, 2]$ satisfies the condition. If $|s| > 2$, a sufficiently small y with the same sign as s violates it. Hence

$$\boxed{\partial M(0) = [-2, 2].}$$

The familiar smooth necessary condition $M'(x_*) = 0$ becomes $0 \in \partial M(x_*)$ for convex nonsmooth functions. Here the subgradient interval describes *all* supporting slopes at the corner.[1]

Switching can also change the number of solutions at one parameter value. The simpler function

$$H(x) = |x-1| + |x+1| = \begin{cases} -2x, & x < -1, \\ 2, & -1 \leq x \leq 1, \\ 2x, & x > 1 \end{cases}$$

has no solutions to $H(x) = c$ for $c < 2$, a whole *interval* of solutions for $c = 2$, and precisely two solutions $x = \pm c/2$ for $c > 2$. A critical value can create an entire plateau, in contrast to the pair of roots created at the smooth tangency in the logarithmic example below.

[1] For maxima of differentiable convex functions and subgradients of active branches, see Boyd and Vandenberghe, *Subgradients*, EE364b notes.

§49.4 Dense Sets and Integrability

Let $D(x) = \mathbf{1}_{\mathbb{Q}}(x)$. Rationals and irrationals are dense in every interval, so D is discontinuous at every point. On any nontrivial closed interval, each subinterval has supremum 1 and infimum 0; its upper and lower Riemann sums differ by the whole interval length. Hence D is nowhere Riemann integrable on a nontrivial interval.

Yet $\mathbb{Q}$ has Lebesgue measure zero, so its Lebesgue integral is zero:

$$\int_a^b D(x)\,dx = 0 \quad \text{in the Lebesgue sense.}$$

This is a structural distinction between two integrals. Riemann sums examine the oscillation on every subinterval, while Lebesgue integration can disregard a measure-zero set.

Thomae's function makes the boundary sharper:

$$T(x) = \begin{cases} 1/q, & x = p/q \text{ in lowest terms, } q > 0, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

It is discontinuous exactly at the rational points and continuous at every irrational point. To see continuity at an irrational x_0, fix $\varepsilon > 0$ and choose N with $1/N < \varepsilon$. A bounded neighborhood contains only finitely many rationals with denominator $q \le N$. Shrink the neighborhood to avoid them. All remaining values of T are less than ε. Since the discontinuity set is countable, the Lebesgue criterion for Riemann integrability shows T is Riemann integrable with integral 0.

§49.5 A functional equation is transport along an orbit

Suppose $f : \mathbb{R} \to \mathbb{R}$ satisfies

$$f(x+1) = 1 - f(x).$$

Applying the shift twice gives $f(x+2) = f(x)$. Every x has a unique decomposition $x = t + n$ with $0 \le t < 1$ and $n \in \mathbb{Z}$. If $h = f|_{[0,1)}$, then

$$f(t+n) = \begin{cases} h(t), & n \text{ even}, \\ 1 - h(t), & n \text{ odd}. \end{cases}$$

Thus all solutions are determined by arbitrary data on a fundamental interval, transported by the integer translation action.

Continuity is an additional constraint. If h is continuous on $[0,1)$ and has a limit as $t \uparrow 1$, the extension is continuous on $\mathbb{R}$ precisely when

$$\lim_{t \uparrow 1} h(t) = 1 - h(0).$$

For $h(t) = 2t - t^2$, both sides are 1, so the resulting period-two function is continuous.

Write $g = f - \frac{1}{2}$. Then $g(x+1) = -g(x)$, an *antiperiodicity* relation. If g is square-integrable on one period and expanded in the 2-periodic Fourier basis $e^{\pi i k x}$, the shift multiplies the k-th mode by $(-1)^k$. Antiperiodicity therefore forces the even Fourier coefficients to vanish. The alternating-value functional equation has become a spectral selection rule.

§49.6 A quadratic recurrence hides a circle-doubling map

An iteration problem looks quite different from an ordinary range problem, but it becomes manageable when the coordinates expose the repeated operation. Consider

$$x_{n+1} = x_n^2 - 2, \qquad x_0 \in [-2, 2].$$

Expanding $x_2 = (x_0^2 - 2)^2 - 2$ is easy, but expanding x_{10} is a hopeless way to understand its behavior. Instead write $x_0 = 2\cos\theta$, which is possible for every $x_0 \in [-2, 2]$. The double-angle identity gives

$$(2\cos\theta)^2 - 2 = 4\cos^2\theta - 2 = 2\cos(2\theta).$$

Applying the same calculation repeatedly proves by induction that

$$\boxed{x_n = 2\cos(2^n\theta).}$$

The quadratic recurrence has become the doubling of an angle on the circle. This is a *semiconjugacy*

$$h(2\theta) = F(h(\theta)), \qquad h(\theta) = 2\cos\theta, \quad F(x) = x^2 - 2.$$

It is not a conjugacy: θ and $-\theta$ generally represent distinct angles but have the same image under h.

The trigonometric coordinate gives exact periodic orbits. Take

$$x_0 = 2\cos\frac{2\pi}{5} = \frac{\sqrt{5}-1}{2}.$$

Then

$$x_1 = 2\cos\frac{4\pi}{5} = -\frac{\sqrt{5}+1}{2}, \qquad x_2 = 2\cos\frac{8\pi}{5} = x_0.$$

This is an exact two-cycle. The angle is multiplied by two each time, but taking the cosine forgets reflection of the angle; that is precisely why $8\pi/5$ and $2\pi/5$ give the same x.

The interval $[-2, 2]$ is invariant because the cosine formula always stays inside it. Outside it, the behavior is completely different. If $x_0 > 2$, then

$$x_{n+1} - x_n = (x_n - 2)(x_n + 1) > 0;$$

a bounded increasing orbit would converge to a fixed point of F, but its only fixed points are 2 and -1. Since every iterate is greater than two, the orbit must diverge to $+\infty$. If $x_0 < -2$, the first iterate exceeds 2, and the same conclusion follows.

This is a good example of a natural substitution doing more work than brute force. The algebraic recurrence, the geometry of the circle, and the distinction between bounded and escaping orbits all become explicit from a single identity. The relation with the doubling map is a standard example in one-dimensional dynamics; compare the discussion of the Chebyshev quadratic in *The Doubling Map and Chaos*.

§49.7 Solving a reciprocal functional equation completely

Suppose on $\mathbb{R} \setminus \{0\}$ a function satisfies

$$f(x) = f(1)x + \frac{f(2)}{x} - 1.$$

Set $A = f(1)$, $B = f(2)$. Substituting $x = 1$ gives $B = 1$; substituting $x = 2$ then gives

$$1 = 2A + \frac{1}{2} - 1, \qquad A = \frac{3}{4}.$$

Consequently the function is uniquely determined:

$$\boxed{f(x) = \frac{3x}{4} + \frac{1}{x} - 1.}$$

On $x > 0$, AM–GM gives

$$f(x) \geq 2\sqrt{\frac{3}{4}} - 1 = \sqrt{3} - 1,$$

with equality when $3x/4 = 1/x$, at $x = 2/\sqrt{3}$. A relation that initially seems to involve two unknown function values has reduced to two algebraic consistency equations.

§49.8 Tangency Changes Solution Counts

For $a > 0$, consider

$$\log(ax) = 2\log(x + 1).$$

The real domain is $x > 0$. Exponentiating yields

$$ax = (x + 1)^2 \quad \Longleftrightarrow \quad a = x + 2 + \frac{1}{x}.$$

The last expression is at least 4, with equality only at $x = 1$. Therefore there are no solutions when $0 < a < 4$, exactly one when $a = 4$, and two when $a > 4$.

The same conclusion arises from the discriminant of $x^2 + (2 - a)x + 1$:

$$\Delta = a(a - 4).$$

But the function $a(x) = x + 2 + 1/x$ adds geometric information: the two branches coalesce at the nondegenerate minimum $x = 1$. Near the critical point,

$$a(x) = 4 + (x - 1)^2 + O((x - 1)^3),$$

so if $a = 4 + \varepsilon$ with $\varepsilon > 0$ small, the two solutions satisfy $x = 1 \pm \sqrt{\varepsilon} + O(\varepsilon)$. A discriminant-zero event is a simple fold bifurcation in the geometry of the zero set.

Domains and ranges determine where a function exists and which values it attains; the density of exceptional sets determines regularity and integrability; group actions transport functional equations; and critical values explain how solutions appear and disappear. Each viewpoint becomes useful exactly when the corresponding calculation demands it.

Symmetry, Folding, and the Birth of New Solutions

N-20230107

A graph transformation becomes much easier to understand once the formula is treated as an operation on the domain or the plane. Reflection acts as an involution, absolute value folds one half-space onto another, and intersections of a moving horizontal line with a graph can change only at specific critical values or boundaries. These ideas connect elementary function problems to group actions and the local geometry of bifurcations.[1]

§50.1 A graph center is an involutive symmetry

A graph $y = f(x)$ is invariant under a half-turn about (a, b) precisely when

$$(x, f(x)) \longmapsto (2a - x, 2b - f(x))$$

takes every point of the graph to another point of the graph. Equivalently, the domain is invariant under $x \mapsto 2a - x$, and

$$\boxed{f(a + t) + f(a - t) = 2b.}$$

After the translation $g(t) = f(a + t) - b$, the condition becomes

$$g(-t) = -g(t).$$

Thus a center of symmetry is exactly an odd-function structure after translating the coordinate origin. In particular, the transformation is an involution: applying it twice returns every point to its original position.

Consider the rational function

$$f(x) = \frac{1}{x + 1} + \frac{1}{x + 2}.$$

Its poles are paired around $a = -3/2$. Let $t = x + 3/2$. Then

$$f\left(t - \frac{3}{2}\right) = \frac{1}{t - \frac{1}{2}} + \frac{1}{t + \frac{1}{2}} = \frac{2t}{t^2 - \frac{1}{4}}.$$

The last expression is odd, so the graph has center $(-3/2, 0)$. Notice that a center can be outside the domain: the function is undefined at its two poles, and neither pole needs to be a point of the graph.

The analogous criterion for symmetry about a vertical line $x = a$ is $f(a+t) = f(a-t)$, or evenness after translation. These equations are more reliable than guessing symmetry from a partial sketch.

[1] For the derivative and critical-value viewpoint, see V. I. Arnold, *Catastrophe Theory*, Chapter 1, and the inverse/implicit-function material in MIT 18.02SC: https://ocw.mit.edu/courses/18-02sc-multivariable-calculus-fall-2010/.

§50.2 Absolute values are quotient maps of reflections

The map $q : \mathbb{R} \to [0, \infty)$, $q(x) = |x|$, identifies the reflection pair $\{x, -x\}$. In this sense $[0, \infty)$ is the orbit space of the $\mathbb{Z}/2\mathbb{Z}$ action generated by $x \mapsto -x$.

The graphs of $f(|x|)$ and $|f(x)|$ encode two different foldings:

$$f(|x|) = \begin{cases} f(x), & x \geq 0, \\ f(-x), & x < 0, \end{cases} \qquad |f(x)| = \begin{cases} f(x), & f(x) \geq 0, \\ -f(x), & f(x) < 0. \end{cases}$$

The first folds the *input* across $x = 0$, while the second folds the *output* across $y = 0$. These need not commute.

For $f(x) = x - 1$,

$$|f(|x|)| = ||x| - 1|, \qquad (|f|)(|x|) = ||x| - 1|,$$

which happen to agree. But replacing the input transformation with $|f(x)|$ before folding gives

$$f(|f(x)|) = |x - 1| - 1,$$

a genuinely different function. The order of operations matters because the fold is many-to-one.

A nested absolute value becomes transparent after locating its breakpoints. Set

$$h(x) = \big||x| - 2\big| - 1.$$

Its zeros occur where $\big||x| - 2\big| = 1$, namely at $x \in \{-3, -1, 1, 3\}$. The further folding $H(x) = |h(x)|$ is even, nonnegative, piecewise linear, and has exactly these four zeros. Instead of expanding sixteen cases, one may first analyze $x \geq 0$ on the intervals cut out by $1, 2, 3$, then reflect the result.

§50.3 Intersections change count when transversality fails

Let $F(x, t) = 0$ describe a family of scalar equations depending on a parameter t. If

$$F(x_0, t_0) = 0, \qquad F_x(x_0, t_0) \neq 0,$$

the implicit function theorem gives a unique nearby solution branch $x = x(t)$. Thus a simple root cannot suddenly disappear under a small parameter perturbation.

For roots to appear or merge *inside a smooth region*, one must encounter

$$F(x, t) = 0, \qquad F_x(x, t) = 0.$$

If the graph has corners, jumps, or excluded endpoints, root counts can also change there. These separate mechanisms are visible in a familiar example:

$$|x^2 - 1| = t.$$

Because the left side is nonnegative, there are no roots for $t < 0$. For $t \geq 0$, the equation splits into

$$x^2 = 1 + t, \qquad x^2 = 1 - t,$$

with the second equation admissible only when $t \le 1$. Counting distinct real roots gives

$$\#\{x : |x^2 - 1| = t\} = \begin{cases} 0, & t < 0, \\ 2, & t = 0, \\ 4, & 0 < t < 1, \\ 3, & t = 1, \\ 2, & t > 1. \end{cases}$$

At $t = 0$, the two absolute-value branches meet at the corners $x = \pm 1$. At $t = 1$, the two inner roots coalesce at the smooth maximum $x = 0$ of $1 - x^2$. The root count changes for different geometric reasons at these two levels.

§50.4 The local model is a fold

Near the smooth critical point of the preceding example, write $F(x, t) = 1 - x^2 - t$. At $(0, 1)$,

$$F = 0, \quad F_x = 0, \quad F_t = -1, \quad F_{xx} = -2.$$

The zero set is $t = 1 - x^2$, so for $t < 1$ its two branches are

$$x_\pm(t) = \pm\sqrt{1 - t}.$$

The square-root separation is characteristic of a nondegenerate fold. More generally, suppose $F(x_0, t_0) = F_x(x_0, t_0) = 0$, while F_t and F_{xx} are nonzero. Solving locally for $t = t(x)$ and Taylor expanding gives

$$t - t_0 = -\frac{F_{xx}(x_0, t_0)}{2F_t(x_0, t_0)}(x - x_0)^2 + o((x - x_0)^2).$$

The coefficient determines on which side of t_0 the two nearby solutions exist.

One must still check admissibility: a fold may lie outside the original domain or on a branch discarded by an absolute-value case split. An algebraic multiple root of an auxiliary equation does not automatically describe a valid solution of the original problem.

§50.5 An intersection diagram can be replaced by a proof

For a continuous piecewise monotone function on a compact interval, count the intersections with the level $y = t$ by splitting at all endpoints, corners, and stationary points. On each open monotonicity interval the function takes each intermediate value at most once; the intermediate value theorem determines whether it is taken at all. The only possible changes in the count occur at the finitely many corresponding critical heights.

This procedure explains why symmetry and absolute values are useful *before* computation. Symmetry identifies equal branches, absolute-value foldings create predictable breakpoints, and monotonicity turns a sketch into an exact counting theorem. The higher-level language of involutions and folds merely records the mechanisms already visible in the elementary examples.

Functions through Fibers, Iteration, and Functional Equations

N-20230116

§51.1 A function includes its source and target

A function is data

$$f : X \to Y$$

consisting of a domain X, a codomain Y, and a rule assigning to every $x \in X$ exactly one value $f(x) \in Y$. A formula by itself is not enough. The same expression

$$x \longmapsto x^2$$

defines different functions when its domain or codomain changes. For example,

$$\mathbb{R} \to \mathbb{R}, \qquad \mathbb{R} \to [0, \infty), \qquad \mathbb{C} \to \mathbb{C}$$

have different surjectivity and inverse-image behavior even though the displayed formula is identical.

Two functions are equal when they have the same domain, the same codomain, and the same value at every input. This convention prevents hidden changes of target from being treated as harmless notation.

If $X = \{1, \dots, n\}$ and $Y = \{1, \dots, m\}$, then a function $f : X \to Y$ is completely recorded by the list

$$(f(1), \dots, f(n)).$$

Each of the n entries has m choices, so the number of such functions is

$$m^n.$$

This simple count is the starting point for more refined counts of injections, surjections, and functions satisfying iteration equations.

§51.2 Images and preimages obey different algebraic laws

For $A \subseteq X$, the direct image is

$$f(A) = \{f(a) : a \in A\}.$$

For $B \subseteq Y$, the inverse image is

$$f^{-1}(B) = \{x \in X : f(x) \in B\}.$$

The notation $f^{-1}(B)$ does not require f to be invertible; it denotes a subset of the domain.

Inverse image preserves all ordinary set operations. For any family $(B_i)_{i\in I}$ of subsets of Y,

$$f^{-1}\left(\bigcup_{i\in I} B_i\right) = \bigcup_{i\in I} f^{-1}(B_i),$$

$$f^{-1}\left(\bigcap_{i\in I} B_i\right) = \bigcap_{i\in I} f^{-1}(B_i),$$

and

$$f^{-1}(Y \setminus B) = X \setminus f^{-1}(B).$$

Each identity follows by checking membership. For example,

$$\begin{aligned} x \in f^{-1}(Y \setminus B) &\iff f(x) \notin B \\ &\iff x \notin f^{-1}(B). \end{aligned}$$

Thus inverse image is a Boolean-algebra homomorphism between power sets.

Direct image preserves arbitrary unions,

$$f\left(\bigcup_i A_i\right) = \bigcup_i f(A_i),$$

but for intersections one only has

$$f(A \cap C) \subseteq f(A) \cap f(C).$$

The inclusion can be strict. Let $f(x) = x^2$, $A = \{-1\}$, and $C = \{1\}$. Then

$$A \cap C = \varnothing,$$

while

$$f(A) \cap f(C) = \{1\}.$$

Equality for all subsets $A, C \subseteq X$ holds precisely when f is injective.

§51.3 Direct and inverse image form a Galois connection

Order the power sets $\mathcal{P}(X)$ and $\mathcal{P}(Y)$ by inclusion. Both direct image and inverse image are monotone:

$$A_1 \subseteq A_2 \Longrightarrow f(A_1) \subseteq f(A_2),$$

$$B_1 \subseteq B_2 \Longrightarrow f^{-1}(B_1) \subseteq f^{-1}(B_2).$$

Their deeper relation is

$$\boxed{f(A) \subseteq B \quad \Longleftrightarrow \quad A \subseteq f^{-1}(B).}$$

The two statements say the same thing element by element: every point of A is sent into B. In the language of ordered sets, this equivalence is a Galois connection, or adjunction,

$$f(-) \dashv f^{-1}(-).$$

The formula has a useful extremal interpretation. Among all subsets $B \subseteq Y$ whose inverse image contains A, the direct image $f(A)$ is the smallest. Dually, among all subsets $A \subseteq X$ whose direct image lies in B, the inverse image $f^{-1}(B)$ is the largest.

Thus the two operations are not merely convenient notation: each solves an optimization problem in the inclusion order.

Taking $B = f(A)$ gives

$$A \subseteq f^{-1}(f(A)).$$

The right-hand side adds every point lying in the same fiber as some point of A. Define

$$c(A) = f^{-1}(f(A)).$$

Then c is a closure operator on $\mathcal{P}(X)$: it is extensive,

$$A \subseteq c(A),$$

monotone, and idempotent,

$$c(c(A)) = c(A).$$

Its fixed points are exactly the subsets that are unions of whole fibers of f. Such sets are often called *saturated*.

In the other direction,

$$f(f^{-1}(B)) = B \cap f(X) \subseteq B.$$

Define

$$i(B) = f(f^{-1}(B)).$$

Then i is contractive, monotone, and idempotent. Its fixed points are precisely the subsets of the actual image $f(X)$. The two composites therefore remove different kinds of irrelevant information: c forgets distinctions inside fibers, while i discards target points that the function never reaches.

For a concrete example, take

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = x^2.$$

For every $A \subseteq \mathbb{R}$,

$$f^{-1}(f(A)) = A \cup (-A).$$

A set becomes saturated by adding the reflected point $-a$ whenever it contains a. For every $B \subseteq \mathbb{R}$,

$$f(f^{-1}(B)) = B \cap [0, \infty).$$

Negative target values disappear because no real number squares to them. This example makes the two failures visible: noninjectivity enlarges A along fibers, while nonsurjectivity shrinks B to the image.

The inclusion

$$A \subseteq f^{-1}(f(A))$$

is an equality for every A exactly when f is injective. The equality

$$f(f^{-1}(B)) = B$$

holds for every B exactly when f is surjective. The same adjunction also explains why continuity and measurability are defined using inverse images: inverse images preserve all unions, intersections, and complements without requiring either injectivity or surjectivity. The closure operator c now leads naturally to the next question: what are the fibers whose unions it is forcing us to take?

§51.4 Fibers measure information loss

For $y \in Y$, the fiber over y is

$$f^{-1}(y) = \{x \in X : f(x) = y\}.$$

A function is injective exactly when every fiber has at most one element, and surjective exactly when every fiber is nonempty.

The fibers partition the domain after empty fibers are discarded. Define

$$x \sim_f x' \quad \Longleftrightarrow \quad f(x) = f(x').$$

Then $\sim_f$ is an equivalence relation, and f factors canonically as

$$X \twoheadrightarrow X/\!\sim_f \xrightarrow{\cong} f(X) \hookrightarrow Y.$$

The first map collapses each fiber to one point. The middle map sends a fiber class to its common value. The last map includes the image into the declared codomain.

This factorization separates two kinds of failure to be bijective. Noninjectivity occurs in the quotient step, where distinct inputs are identified. Nonsurjectivity occurs in the final inclusion, where some target points are missed.

For finite X, counting by fibers gives

$$|X| = \sum_{y \in f(X)} |f^{-1}(y)|.$$

This elementary identity is the discrete prototype of decomposing a space or measure along fibers.

§51.5 Counting injections, surjections, and bijections

Let $|X| = n$ and $|Y| = m$. An injection $X \to Y$ exists only when $n \le m$. Its values can be chosen successively without repetition, so the number of injections is

$$m(m-1)\cdots(m-n+1) = \frac{m!}{(m-n)!}.$$

When $n = m$, every injection is automatically surjective, and the number of bijections is

$$n!.$$

Surjections can be counted by inclusion–exclusion. There are m^n total functions. For each target point y, let E_y be the set of functions that miss y. If a specified set of j target points is missed, the function has only $m - j$ possible values at each input. Therefore the number of surjections is

$$\boxed{\sum_{j=0}^{m} (-1)^j \binom{m}{j} (m-j)^n.}$$

For example, the number of surjections from a four-element set onto a three-element set is

$$3^4 - \binom{3}{1}2^4 + \binom{3}{2}1^4 = 81 - 48 + 3 = 36.$$

The same number is $3!S(4,3)$, where $S(n,m)$ is a Stirling number of the second kind: first partition the domain into m nonempty fibers, then label those fibers by the target elements.

§51.6 A finite function is a directed dynamical system

Let $f : X \to X$ with X finite. Draw one directed edge

$$x \longrightarrow f(x)$$

from every vertex. Each vertex has out-degree one, though its in-degree may be arbitrary.

Theorem 51.6.1 (Structure of a finite functional graph)
Every connected component contains exactly one directed cycle, with directed rooted trees feeding into the vertices of that cycle.

Proof. Starting at any $x \in X$, the sequence

$$x, f(x), f^2(x), \dots$$

must repeat because X is finite. Once a value repeats, the subsequent orbit is periodic and therefore enters a directed cycle.

A connected component cannot contain two disjoint directed cycles. Since every vertex has exactly one outgoing edge, a forward orbit can enter only one cycle. If two cycles lay in one weakly connected component, any undirected path between them would force some vertex to have two different forward destinations, contradicting the uniqueness of its outgoing edge. □

The nonperiodic vertices form trees oriented toward the cycle. Thus every finite orbit has a transient part followed by a periodic part.

A finite self-map is injective exactly when every vertex has in-degree one. In that case there are no attached trees, and every component is a pure cycle. This proves again that, for finite sets of equal size,

$$\text{injective} \iff \text{surjective} \iff \text{permutation}.$$

§51.7 Iterates and Cycle Lengths

Suppose

$$f^r = \mathrm{id}_X .$$

Then f has the inverse f^{r-1}, so it is a permutation. On a cycle of length d, the map f^r is the identity exactly when

$$d \mid r.$$

Hence the equation $f^r = \mathrm{id}$ means that every cycle length divides r.

For an involution,

$$f^2 = \mathrm{id},$$

all cycles have length one or two. To count involutions on an n-element set, choose $2k$ elements to belong to transpositions and pair them. This gives

$$\boxed{\sum_{k=0}^{\lfloor n/2 \rfloor} \frac{n!}{2^k k!(n-2k)!}.}$$

For $n = 4$, the count is

$$1 + 6 + 3 = 10 :$$

one identity, six permutations with one transposition, and three products of two disjoint transpositions.

Now consider the idempotent equation

$$f^2 = f.$$

Every point in the image is fixed, because if $y = f(x)$, then

$$f(y) = f(f(x)) = f(x) = y.$$

Conversely, any function that fixes every point of its image is idempotent. Its functional graph consists of fixed points with depth-one trees feeding into them.

If the image has size r, first choose its r fixed points and then send each of the remaining $n - r$ elements to any image point. Therefore the number of idempotent self-maps on a nonempty n-element set is

$$\boxed{\sum_{r=1}^{n} \binom{n}{r} r^{n-r}.}$$

For $n = 3$, this gives

$$\binom{3}{1}1^2 + \binom{3}{2}2^1 + \binom{3}{3}3^0 = 3 + 6 + 1 = 10.$$

The equation is thus not merely an algebraic string; it specifies an exact graph shape and leads to an exact enumeration.

§51.8 Left inverses, right inverses, and what they prove

Let $f : X \to Y$. If there is a map $g : Y \to X$ with

$$g \circ f = \mathrm{id}_X,$$

then f is injective. Indeed,

$$f(x) = f(x') \Longrightarrow g(f(x)) = g(f(x')) \Longrightarrow x = x'.$$

Such a g is a left inverse of f.

If there is a map $h : Y \to X$ with

$$f \circ h = \mathrm{id}_Y,$$

then f is surjective, because every $y \in Y$ equals $f(h(y))$. Such an h is a right inverse.

Conversely, an injective function with nonempty domain admits a left inverse after assigning arbitrary values away from its image. A surjective function admits a right inverse only after choosing one element from every fiber; for arbitrary families this assertion is equivalent to a form of the axiom of choice. In finite sets the choices can always be made directly.

A map with both a left and a right inverse is bijective, and the two inverses coincide. If

$$g \circ f = \mathrm{id}_X, \qquad f \circ h = \mathrm{id}_Y,$$

then

$$g = g \circ (f \circ h) = (g \circ f) \circ h = h.$$

§51.9 Functional equations transport values along orbits

Many functional equations involve a transformation $T : X \to X$ of the input. An equation such as

$$F(Tx) = \Phi(F(x))$$

propagates values along the orbit

$$x, Tx, T^2x, \dots.$$

If the orbit closes after r steps, consistency requires

$$F(x) = F(T^r x) = \Phi^r(F(x)).$$

Thus cycles in the input action impose algebraic constraints on possible values.

For example, solve

$$f(x+1) = f(x) + 2$$

on $\mathbb{R}$. Subtract the particular solution $2x$ and define

$$p(x) = f(x) - 2x.$$

Then

$$p(x+1) = f(x+1) - 2x - 2 = f(x) - 2x = p(x).$$

Hence every solution has the form

$$\boxed{f(x) = 2x + p(x),}$$

where p is an arbitrary one-periodic function. The equation determines the change along each translation orbit but leaves one period of initial data free.

By contrast, suppose a transformation satisfies $T^r = \mathrm{id}$ and one asks for

$$f(Tx) = f(x) + c$$

with real-valued f. Iterating around the cycle gives

$$f(x) = f(T^r x) = f(x) + rc,$$

so necessarily $c = 0$. The obstruction comes from consistency around a closed orbit.

§51.10 Regularity in Cauchy's Equation

Consider

$$f(x+y) = f(x) + f(y) \qquad (x, y \in \mathbb{R}).$$

Without any regularity assumption, this equation does not force the familiar form $f(x) = cx$. It first forces only $\mathbb{Q}$-linearity.

Set $y = 0$ to get $f(0) = 0$. Then

$$0 = f(x + (-x)) = f(x) + f(-x),$$

so $f(-x) = -f(x)$. Repeated addition gives

$$f(nx) = nf(x) \qquad (n \in \mathbb{Z}).$$

For a positive integer q,

$$qf(x/q) = f(x),$$

so

$$f(rx) = rf(x) \qquad (r \in \mathbb{Q}).$$

In particular,

$$f(q) = qf(1) \qquad (q \in \mathbb{Q}).$$

Now assume f is continuous at one point a. Since

$$f(h) = f(a+h) - f(a),$$

continuity at a implies continuity at 0, and additivity then implies continuity everywhere. Choose rationals $q_n \to x$. We obtain

$$f(x) = \lim_{n\to\infty} f(q_n) = \lim_{n\to\infty} q_n f(1) = xf(1).$$

Thus

$$\boxed{f(x) = cx}$$

with $c = f(1)$.

The same conclusion follows from several weaker regularity hypotheses, including measurability or boundedness on a sufficiently large set. Without regularity, choose a Hamel basis of $\mathbb{R}$ over $\mathbb{Q}$, prescribe values arbitrarily on the basis, and extend $\mathbb{Q}$-linearly. The resulting additive functions are generally discontinuous everywhere. The lesson is that an algebraic functional equation and an analytic regularity condition play distinct roles.

§51.11 Monomorphisms and Epimorphisms

In the category **Set**, a function $f : X \to Y$ is injective exactly when it is left-cancellable:

$$f \circ g_1 = f \circ g_2 \quad \Longrightarrow \quad g_1 = g_2$$

for all maps $g_1, g_2 : Z \to X$. This is the definition of a monomorphism.

To see the converse, suppose f is not injective and choose distinct $x, x' \in X$ with $f(x) = f(x')$. Taking Z to be a one-point set and sending its point to x or x' gives two different maps whose composites with f agree. Hence f is not monic.

Similarly, a function in **Set** is surjective exactly when it is right-cancellable:

$$h_1 \circ f = h_2 \circ f \quad \Longrightarrow \quad h_1 = h_2.$$

This is the definition of an epimorphism. If f misses a point $y \in Y$, two maps out of Y can agree on $f(X)$ but differ at y, so f is not epic.

These identifications depend on the category. In the category of commutative rings with identity, the inclusion

$$\mathbb{Z} \hookrightarrow \mathbb{Q}$$

is an epimorphism although it is not surjective. Any two ring homomorphisms out of $\mathbb{Q}$ that agree on $\mathbb{Z}$ must agree on every fraction. The set-theoretic notions are therefore prototypes of categorical cancellation, not universal descriptions of it.

In **Set**, however, the preceding viewpoints now fit together exactly. The fiber relation records which inputs a map identifies; the quotient–image factorization separates that information loss from the target points that are missed; monomorphism and epimorphism express the same two phenomena through cancellation. Finite iteration and real functional equations then study what happens when such maps are composed repeatedly, but the basic structural data remain the fibers, the image, and the way composition preserves or destroys distinguishability.

More Function Exercises: Periodicity, Piecewise Definitions, Images, and Functional Equations

N-20230117

§52.1 Piecewise Maps and Their Domains

Function exercises of this kind are solved by translating each problem into one of several languages:

formula, graph, image/preimage, iteration, functional equation.

Many errors come from using a formula while ignoring its domain or treating an identity valid for all inputs as though it were a formula at a few special values. We follow the standard real-analysis approach to continuous images and the Cauchy equation, as in Gressman's *Advanced Analysis* notes and the lecture notes on Cauchy's functional equation.[1]

A piecewise-defined function may use different formulas on intervals such as

$$x < 0, \qquad x = 0, \qquad x > 0.$$

When studying continuity or solving equations involving such a function, one must first decide which case the input belongs to. If the input itself is an expression such as $f(x)$ or $x + 1$, the relevant case may change during the calculation.

The safe method is:

(i) identify the possible regions of the input;

(ii) solve the equation in each region;

(iii) check that the solution actually belongs to the assumed region.

The pages include small sketches of graphs. The standard transformations are:

$$y = f(x - a) \quad \text{shift right by } a,$$

$$y = f(x) + b \quad \text{shift up by } b,$$

$$y = -f(x) \quad \text{reflect across the } x\text{-axis},$$

$$y = f(-x) \quad \text{reflect across the } y\text{-axis}.$$

Absolute values fold graphs:

$$y = |f(x)|$$

reflects the negative part upward, while

$$y = f(|x|)$$

copies the right half to the left.

[1] P. Gressman's intermediate-value theorem notes, with complementary calculus exercises from Carleton University.

A piecewise function is a function defined after partitioning the domain into regions. The mathematical issue is not only which formula applies, but how the pieces meet. Discontinuities occur at boundaries of the partition; differentiability requires matching first-order behavior across boundaries.

For example, if

$$f(x) = \begin{cases} f_1(x), & x < a, \\ f_2(x), & x \geq a, \end{cases}$$

then continuity at a requires

$$\lim_{x \to a^-} f_1(x) = f_2(a).$$

Differentiability additionally requires matching slopes. This is a local gluing condition. For example, let $f(x) = ax^2$ for $x < 1$ and $f(x) = bx + c$ for $x \geq 1$. Continuity at 1 requires $a = b + c$, while differentiability requires in addition $2a = b$. Hence differentiability holds exactly when $b = 2a$ and $c = -a$. The boundary test is an actual linear system for the unknown coefficients, rather than an informal statement that the formulas should match.

§52.2 Images of intervals and finite maps

For a continuous function on an interval, the image is also an interval. Indeed, intervals are the connected subsets of $\mathbb{R}$, and continuous images of connected sets remain connected. This is the intermediate value theorem in set language. However, the image of an open interval need not be open: $f(x) = x^2$ sends $(-1, 1)$ to $[0, 1)$. If the function is monotone on the interval, then the endpoints determine the image directly.

To find the range of an expression, rewrite it as a one-variable function, determine monotonicity or extrema, and translate the result back into an interval.

The printed exercise page also touches finite functions. If $|A| = m$ and $|B| = n$, then the number of functions $A \to B$ is

$$n^m,$$

because each element of A independently chooses one image in B.

If one requires injectivity and $m \leq n$, the number is

$$n(n-1)\cdots(n-m+1).$$

If one requires bijectivity and $m = n$, the number is

$$n!.$$

If surjectivity is required for $m \geq n$, inclusion–exclusion gives

$$\#\operatorname{Surj}(A, B) = \sum_{j=0}^{n} (-1)^j \binom{n}{j} (n-j)^m.$$

Indeed, count all n^m functions and subtract those omitting at least one of the n target elements. This explains why injectivity and surjectivity lead to different counting problems.

§52.3 Functional equations as transformations

A relation such as

$$f(x+1) = 1 - f(x)$$

implies

$$f(x+2) = 1 - f(x+1) = f(x).$$

Thus the function is periodic with period 2. The general method is to apply the same equation repeatedly until the input returns to a familiar form.

If instead

$$f(x+T) = f(x)$$

for all x, then T is a period. The smallest positive period, if it exists, is called the fundamental period.

Substituting special values such as $x = 0$, $y = 0$, or $y = x$ is a standard strategy. For a functional equation in two variables,

$$F(x, y, f(x), f(y), f(x+y)) = 0,$$

choosing simple values isolates pieces of the function.

For example, if

$$f(x+y) = f(x) + f(y) + c,$$

then setting $x = y = 0$ determines $f(0)$, and subtracting a constant can reduce the equation to the additive Cauchy equation.

A functional equation often specifies a transformation rule rather than a direct formula. For example, equations involving $f(x+1)$, $f(2x)$, or $f(f(x))$ are naturally studied by iteration. The orbit of x under a transformation T is

$$x,\ T(x),\ T^2(x),\ \ldots.$$

A functional equation may prescribe how f behaves along these orbits.

For instance, if

$$f(x+1) = f(x),$$

then f is constant along orbits of the translation $T(x) = x + 1$. Periodicity is invariance under a group action of $\mathbb{Z}$.

A functional equation often says that f is invariant or equivariant under transformations. The equation

$$f(x+T) = f(x)$$

says f is invariant under the translation action of $T\mathbb{Z}$. The equation

$$f(ax) = bf(x)$$

says f is homogeneous with respect to scaling.

Thus solving a functional equation often means finding the invariants of a group or semigroup action. This perspective makes the original examples more natural: periodicity, parity, and scaling are all symmetry constraints.

§52.4 Regularity makes Cauchy's equation rigid

Cauchy's equation is

$$f(x+y) = f(x) + f(y).$$

For an additive map $f : \mathbb{R} \to \mathbb{R}$, continuity at one point, Lebesgue measurability, or boundedness on a nondegenerate interval each implies

$$f(x) = cx.$$

Here is the continuous case. Additivity gives $f(0) = 0$, $f(nx) = nf(x)$, and $f(p/q) = (p/q)f(1)$ for rationals p/q. If $q_k \in \mathbb{Q}$ approaches real x, continuity yields $f(x) = \lim_k f(q_k) = xf(1)$. The discontinuous Hamel-basis solutions show exactly what continuity rules out. But without regularity assumptions, there are wild solutions constructed using a Hamel basis of $\mathbb{R}$ over $\mathbb{Q}$.

This shows why elementary functional equations often include hidden conditions such as continuity or monotonicity. These conditions are not technical decorations; they exclude pathological solutions arising from the axiom of choice.

§52.5 From iterated intervals to symbolic dynamics

Piecewise-defined functions and iteration can lead to symbolic dynamics. If an interval map sends subintervals across one another, one can encode an orbit by a sequence of symbols recording which subinterval contains each iterate.

The shift map

$$\sigma((a_0, a_1, a_2, \ldots)) = (a_1, a_2, a_3, \ldots)$$

is a central model. Complicated interval dynamics can often be compared to shifts on symbol spaces.

For a specific example, partition $[0,1)$ into $I_0 = [0, 1/2)$ and $I_1 = [1/2, 1)$, and set $T(x) = 2x \pmod 1$. The itinerary of x is the sequence $a_k \in \{0,1\}$ recording which interval contains $T^k(x)$. Under the binary expansion convention that avoids trailing strings of ones, this itinerary is precisely the binary digits of x; the map T shifts them left. Thus symbolic dynamics is the exact encoding of a concrete interval map, with the usual care about dyadic endpoints.

From Tangent Lines to Convex Duality and the Geometry of L^p

N-20230310

A derivative is a local linear approximation. For a differentiable function f, the tangent approximation at a is

$$f(a+h) = f(a) + f'(a)h + o(|h|).$$

When f is convex, that local line provides a *global* lower bound. This observation is the route from elementary calculus to Young's inequality, then to Holder's and Minkowski's inequalities, and finally to the duality between L^p norms. Our convex-analytic conventions follow Boyd and Vandenberghe, *Convex Optimization*, Section 3.3, together with the L^p material in MIT 18.102.[1]

§53.1 What Tangent Lines Remember

The derivative begins as a limit of secant slopes,

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h},$$

and differentiability means

$$f(a+h) = f(a) + f'(a)h + r(h), \qquad \lim_{h \to 0} \frac{r(h)}{|h|} = 0.$$

The linear term is the best first-order local description of the function. For a twice differentiable function, the next term is

$$f(a+h) = f(a) + f'(a)h + \frac{1}{2} f''(a)h^2 + o(h^2).$$

This expansion reveals why the usual first-derivative test looks for changes in the sign of f' and the second-derivative test looks for the curvature f'': both are ways of detecting what first survives when the leading term disappears.

One should not silently identify a stationary point with an extremum. The function $f(x) = x^3$ has $f'(0) = 0$, but takes both signs on either side of zero. Meanwhile $f(x) = |x|$ has a strict minimum at zero *without* having a derivative there. Fermat's theorem says that differentiability plus an *interior extremum* implies $f' = 0$; the converse need not hold, and nondifferentiable extrema are not excluded.

Convexity adds something much stronger. For a differentiable convex function and $0 < t < 1$,

$$f(x + t(y-x)) \le (1-t)f(x) + tf(y).$$

Rearranging, dividing by $t > 0$, and letting $t \downarrow 0$ gives

$$f'(x)(y-x) \le f(y) - f(x).$$

So the tangent line at x is a *global* supporting line. This is exactly the inequality used below to derive Young's inequality. Convexity converts local slope data into an everywhere-valid comparison.

[1] Boyd and Vandenberghe, *Convex Optimization*, Chapter 3; MIT 18.102, functional-analysis lecture notes.

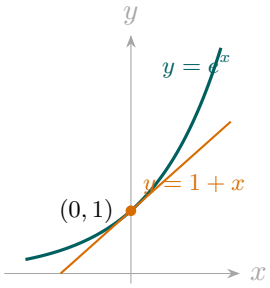


Figure 53.1: The exponential function lies globally above its tangent at zero: $e^x \geq 1 + x$. This one geometric fact is a prototype for the convex-analytic inequalities that follow.

§53.2 Convexity turns a tangent into an inequality

A differentiable function f on an interval is convex if and only if its graph lies above every tangent:

$$\boxed{f(y) \geq f(x) + f'(x)(y - x).}$$

For a twice differentiable function, $f'' \geq 0$ guarantees this property. Indeed, by the mean value theorem applied to f', its derivative is nondecreasing, so the secant slopes to the right of x are no smaller than $f'(x)$; the same estimate follows on the left.

The simple convex function $f(t) = e^t$ satisfies $e^t \geq 1 + t$ by taking the tangent at 0. For $f(t) = -\log t$ on $t > 0$, the tangent at 1 gives

$$-\log t \geq 1 - t, \qquad \text{or} \qquad \log t \leq t - 1.$$

These are global inequalities obtained from a first-order local calculation. Equality holds at the tangency point.

§53.3 Young's inequality is a conjugate-function calculation

Let $p > 1$ and $q = p/(p-1)$. Consider the convex function

$$\phi(u) = \frac{|u|^p}{p}.$$

For each real v, its convex conjugate is

$$\phi^*(v) = \sup_{u \in \mathbb{R}} \{uv - \phi(u)\}.$$

The optimizing u has the sign of v, and the stationarity equation is

$$|u|^{p-1} = |v|.$$

Thus $u = \operatorname{sgn}(v)|v|^{q-1}$, and substitution gives

$$\boxed{\phi^*(v) = \frac{|v|^q}{q}.}$$

By the definition of a supremum, $uv - \phi(u) \leq \phi^*(v)$. We obtain Young's inequality

$$\boxed{uv \leq \frac{|u|^p}{p} + \frac{|v|^q}{q}.}$$

Equality holds exactly when $v = \operatorname{sgn}(u)|u|^{p-1}$.

This explains the conjugate exponents more sharply than memorizing $1/p + 1/q = 1$. The power q appears because the inverse of $u \mapsto |u|^{p-1}\operatorname{sgn}(u)$ is $v \mapsto |v|^{q-1}\operatorname{sgn}(v)$.

§53.4 Holder's inequality is Young after normalization

For nonnegative numbers a_i, b_i, put

$$A = \left(\sum_i a_i^p\right)^{1/p}, \qquad B = \left(\sum_i b_i^q\right)^{1/q}.$$

If $A = 0$ or $B = 0$, the inequality below is immediate. Otherwise Young's inequality applied to a_i/A and b_i/B gives

$$\frac{a_i b_i}{AB} \le \frac{a_i^p}{pA^p} + \frac{b_i^q}{qB^q}.$$

Sum over i. The right-hand side becomes $1/p + 1/q = 1$, yielding

$$\boxed{\sum_i a_i b_i \le \left(\sum_i a_i^p\right)^{1/p} \left(\sum_i b_i^q\right)^{1/q}.}$$

For arbitrary signed or complex sequences, replace each product by its absolute value. Exactly the same pointwise argument, with integration in place of summation, proves

$$\int |fg| \le \|f\|_{L^p}\|g\|_{L^q}$$

whenever the respective norms are finite.

There is a revealing dual formulation in finite-dimensional real spaces (for complex vectors, use the absolute value of the Hermitian pairing):

$$\boxed{\|x\|_p = \sup_{\|y\|_q \le 1} \sum_i x_i y_i.}$$

Holder gives the upper bound. If $x \neq 0$, choose

$$y_i = \frac{\operatorname{sgn}(x_i)|x_i|^{p-1}}{\|x\|_p^{p-1}}.$$

Then $\|y\|_q = 1$, and $\sum_i x_i y_i = \|x\|_p$. The dual vector is exactly the equality case of Young's inequality, now interpreted as the linear measurement achieving the norm.

§53.5 Minkowski makes ℓ^p into a normed space

For $p > 1$, let $z = x + y$. By Holder with exponents p, q,

$$\begin{aligned}
\|z\|_p^p &= \sum_i |z_i|^p \\
&\le \sum_i |x_i||z_i|^{p-1} + \sum_i |y_i||z_i|^{p-1} \\
&\le (\|x\|_p + \|y\|_p)\left(\sum_i |z_i|^{(p-1)q}\right)^{1/q} \\
&= (\|x\|_p + \|y\|_p)\|z\|_p^{p-1}.
\end{aligned}$$

If $z = 0$, the conclusion is trivial; otherwise divide by $\|z\|_p^{p-1}$ to obtain

$$\boxed{\|x + y\|_p \le \|x\|_p + \|y\|_p.}$$

The same reasoning applies to L^p integrals, while $p = 1$ follows directly from the pointwise triangle inequality.

There is a conceptual difference between proving an inequality and proving a norm axiom. Holder controls the pairing of vectors from conjugate exponent spaces; Minkowski uses that control to show that ℓ^p is closed under addition with a subadditive notion of size. The norm's geometry is therefore built from its duality inequality.

§53.6 Why the exponent threshold really matters

Young's inequality and its convex-conjugate proof require $p > 1$, or $p = 1$ with a limiting dual exponent $q = \infty$. What happens if we blindly extend the expression

$$\|x\|_p = \left(\sum_i |x_i|^p \right)^{1/p}$$

to $0 < p < 1$? The expression is homogeneous and still vanishes only at zero, but it can fail the triangle inequality.

For example, take $p = 1/2$ and

$$x = (1, 0), \qquad y = (0, 1).$$

Then

$$\|x\|_{1/2} = \|y\|_{1/2} = 1, \qquad \boxed{\|x + y\|_{1/2} = (1 + 1)^2 = 4 > 2.}$$

The unit ball for this *quasi-norm* is not convex. The proof of Minkowski breaks at precisely the same place: one would need the conjugate exponent $q = p/(p-1) < 0$, and the convexity argument for the power function no longer applies.

The equality case of Holder is also geometric. When $p = q = 2$, it reduces to Cauchy–Schwarz; equality for nonzero real vectors means that the vectors are proportional. The resulting Minkowski triangle inequality is strict for two nonzero vectors pointing in genuinely different directions. For $x = (1, 0)$ and $y = (0, 1)$,

$$\|x + y\|_2 = \sqrt{2} < 2 = \|x\|_2 + \|y\|_2,$$

whereas if $y = tx$ with $t \ge 0$ the triangle inequality becomes equality. This is why p-norms are not merely formulas for adding powers: their convexity properties determine the geometry of the unit ball and which inequalities may be used.

§53.7 Jensen's inequality and entropy

For a convex function f, nonnegative weights α_i with $\sum_i \alpha_i = 1$ satisfy

$$f\left(\sum_i \alpha_i x_i \right) \le \sum_i \alpha_i f(x_i).$$

This is Jensen's inequality, obtained by repeatedly applying the two-point definition of convexity. The direction reverses for concave functions such as log.

A particularly useful application involves probability vectors $p = (p_i)$, $q = (q_i)$ with strictly positive coordinates and total mass 1. Concavity of log gives

$$\sum_i p_i \log \frac{q_i}{p_i} \le \log\left(\sum_i p_i \frac{q_i}{p_i}\right) = \log \sum_i q_i = 0.$$

Therefore the relative entropy, or Kullback–Leibler divergence, satisfies

$$\boxed{D_{\mathrm{KL}}(p\|q) = \sum_i p_i \log \frac{p_i}{q_i} \ge 0.}$$

Equality, by strict concavity, holds if and only if $p = q$. The result extends to zeros by the standard conventions and appropriate support conditions.

Taking q uniform on N points gives

$$H(p) = -\sum_i p_i \log p_i \le \log N,$$

with equality at the uniform distribution. If there are 2^n states and base-two logarithms are used, this reads $H(p) \le n$.

The connection to entropy emerges from precisely the same tangent and convexity technology that produced Young's inequality. Convexity controls a function by linear data; conjugacy controls a bilinear pairing; Jensen controls averages. These three operations provide a common analytic language for norms, optimization, and information.

Integral Inequalities, Symmetry, and the Beta Function

N-20260106

Integral inequalities become especially transparent when one interprets integration as averaging. Convexity compares the average of a function with its values at special points; symmetry replaces an integral with another integral of exactly the same value; and repeated integration by parts produces recurrences whose limiting behavior is encoded by special functions. We will use these three mechanisms to connect quadrature errors, a sharp reciprocal inequality, and Wallis' formula.[1]

§54.1 Hermite–Hadamard is a theorem about averages

Suppose f is convex on $[a,b]$, $a<b$, and put $m=(a+b)/2$. Convexity gives

$$f(m)\le \frac{f(x)+f(a+b-x)}{2}.$$

Integrate in x to obtain the lower bound in

$$\boxed{f(m)\le \frac{1}{b-a}\int_a^b f(x)\,dx\le \frac{f(a)+f(b)}{2}.}$$

For the upper bound, use the fact that a convex function lies below its endpoint chord:

$$f(x)\le \frac{b-x}{b-a}f(a)+\frac{x-a}{b-a}f(b).$$

Integrating the chord gives the endpoint average.

The two inequalities measure different approximations to the same integral. In numerical analysis they are the midpoint and trapezoidal rules, each exact for affine functions. The sign of the quadrature error is already determined by convexity.

§54.2 Quadrature errors are weighted second derivatives

Let $f\in C^2([a,b])$. Twice integrating by parts gives the exact identity

$$\frac{b-a}{2}\bigl(f(a)+f(b)\bigr)-\int_a^b f(x)\,dx=\frac{1}{2}\int_a^b (x-a)(b-x)f''(x)\,dx.$$

The kernel $(x-a)(b-x)$ is nonnegative. By the integral mean value theorem there is $\xi\in(a,b)$ such that

$$\boxed{\frac{f(a)+f(b)}{2}-\frac{1}{b-a}\int_a^b f=\frac{(b-a)^2}{12}f''(\xi).}$$

[1] For the beta–gamma identity, see NIST Digital Library of Mathematical Functions, §5.12: https://dlmf.nist.gov/5.12.

For the midpoint approximation, expand around m with the integral remainder. The odd first-order term integrates to zero, and the remaining second-order remainder has a nonnegative kernel. Another application of the integral mean value theorem yields

$$\boxed{\frac{1}{b-a}\int_a^b f - f(m) = \frac{(b-a)^2}{24} f''(\eta)}$$

for some $\eta \in (a, b)$. These formulas refine Hermite–Hadamard: convexity determines the signs, while the curvature f'' determines the magnitudes.

For $f(x) = x^2$ on $[-1, 1]$, the true average is $1/3$, the midpoint value is 0, and the endpoint average is 1. Since $f'' = 2$, the displayed formulas predict gaps $1/3$ and $2/3$, exactly as observed.

§54.3 A sharp reciprocal inequality from one pointwise bound

Assume $0 < m \le f(x) \le M$ almost everywhere on $[a, b]$, and let

$$\mathbb{E}[g] = \frac{1}{b-a}\int_a^b g(x)\,dx.$$

The product $(M - f)(f - m)$ is nonnegative, so

$$f + \frac{mM}{f} \le m + M.$$

Averaging and applying AM–GM to the two nonnegative terms gives

$$2\sqrt{mM\,\mathbb{E}[f]\mathbb{E}[1/f]} \le \mathbb{E}[f] + mM\,\mathbb{E}[1/f] \le m + M.$$

Therefore

$$\boxed{\mathbb{E}[f]\,\mathbb{E}[1/f] \le \frac{(m+M)^2}{4mM}.}$$

This is the integral version of the Kantorovich inequality. Cauchy–Schwarz supplies the reverse inequality $\mathbb{E}[f]\mathbb{E}[1/f] \ge 1$.

The upper bound is sharp: choose $f = m$ on half the interval and $f = M$ on the other half. Then

$$\mathbb{E}[f] = \frac{m+M}{2}, \qquad \mathbb{E}[1/f] = \frac{1/m + 1/M}{2}.$$

The product equals the claimed constant. When continuity is imposed and $m < M$, the sharp constant is approached by continuous approximations of this two-level function.

§54.4 An involution can evaluate a logarithmic integral

Consider

$$I = \int_0^1 \frac{\log(1+x)}{1+x^2}\,dx.$$

Set $x = \tan t$, so $0 \le t \le \pi/4$, and obtain

$$I = \int_0^{\pi/4} \log(1 + \tan t)\,dt.$$

The reflection $t \mapsto \pi/4 - t$ preserves the integration interval. Moreover,

$$1 + \tan\left(\frac{\pi}{4} - t\right) = \frac{2}{1 + \tan t}.$$

Consequently

$$I = \int_0^{\pi/4} \bigl(\log 2 - \log(1 + \tan t)\bigr)\, dt = \frac{\pi}{4}\log 2 - I.$$

The exact value follows without expanding any logarithm:

$$\boxed{I = \frac{\pi}{8}\log 2.}$$

The essential move was to find a measure-preserving reflection under which the integrand transforms into a constant minus itself.

The same pairing philosophy proves positivity of an oscillatory example:

$$\begin{aligned}\int_0^{\sqrt{2\pi}} \sin(x^2)\, dx &= \frac{1}{2}\int_0^{2\pi} \frac{\sin t}{\sqrt{t}}\, dt \\ &= \frac{1}{2}\int_0^{\pi} \sin t \left(\frac{1}{\sqrt{t}} - \frac{1}{\sqrt{t+\pi}}\right) dt > 0.\end{aligned}$$

The two half-periods have opposite signs, but the positive one carries the larger weight. This is an elementary instance of oscillatory cancellation with monotone amplitude.

§54.5 Opposite-Point Averages

The reflection $t \mapsto -t$ can turn a symmetric pair of integrals into a derivative test. For f continuous near 0 and $x > 0$, write

$$H(x) = \int_0^x f(t)\, dt + \int_0^{-x} f(t)\, dt = \int_0^x \bigl(f(t) - f(-t)\bigr)\, dt.$$

The integral mean-value theorem supplies some $\vartheta_x \in (0,1)$ with

$$H(x) = x\bigl(f(\vartheta_x x) - f(-\vartheta_x x)\bigr).$$

What can we say about ϑ_x as the interval shrinks? If f is differentiable at 0 and $f'(0) \neq 0$, then

$$f(t) - f(-t) = 2tf'(0) + o(t), \qquad H(x) = x^2 f'(0) + o(x^2).$$

The remainder estimate is uniform for $0 \le t \le x$, so the mean-value identity gives

$$x^2 f'(0) + o(x^2) = 2\vartheta_x x^2 f'(0) + o(x^2); \qquad \boxed{\vartheta_x \longrightarrow \tfrac{1}{2}}.$$

The nonzero-derivative hypothesis matters. For $f(t) = t^3$, the identity is exactly

$$\frac{x^4}{2} = 2\vartheta_x^3 x^4, \qquad \vartheta_x = 4^{-1/3},$$

a fixed value different from $1/2$. The representative point of an integral is governed by the first nonvanishing odd-order term, not automatically by the midpoint.

§54.6 Two antiderivatives hiding in an arcsine integral

Consider

$$A = \int_0^1 \frac{\arcsin\sqrt{x}}{\sqrt{x(1-x)}}\,dx.$$

The weight looks singular at the endpoints, but it is integrable. Put $u = \arcsin\sqrt{x}$, so $x = \sin^2 u$ and $dx = 2\sin u\cos u\,du$. Thus the apparent singularity disappears:

$$A = 2\int_0^{\pi/2} u\,du = \boxed{\frac{\pi^2}{4}}.$$

There is a second proof worth keeping, because it anticipates beta integrals. The involution $x \mapsto 1-x$ preserves $dx/\sqrt{x(1-x)}$, while

$$\arcsin\sqrt{x} + \arcsin\sqrt{1-x} = \frac{\pi}{2}.$$

Averaging the transformed and untransformed integrals yields

$$2A = \frac{\pi}{2}\int_0^1 \frac{dx}{\sqrt{x(1-x)}} = \frac{\pi}{2}B\left(\tfrac12,\tfrac12\right) = \frac{\pi^2}{2}.$$

This is also the arcsine probability density in disguise: $\left(\pi\sqrt{x(1-x)}\right)^{-1}$ is a probability density on $(0,1)$, and symmetry makes its mean transformed angle $\mathbb{E}[\arcsin\sqrt{X}] = \pi/4$. The same integral has an elementary substitution, a symmetry proof, and a probabilistic interpretation.

§54.7 Why a cosine integral leads to the Poisson kernel

For $a > |b|$ with $a > 0$, the family

$$J(a,b) = \int_0^{\pi} \frac{dx}{a + b\cos x}$$

has a surprisingly clean value. The substitution $t = \tan(x/2)$ gives

$$\cos x = \frac{1-t^2}{1+t^2}, \qquad dx = \frac{2\,dt}{1+t^2}, \qquad 0 < t < \infty.$$

Consequently

$$\begin{aligned} J(a,b) &= 2\int_0^{\infty} \frac{dt}{(a+b) + (a-b)t^2} \\ &= \boxed{\frac{\pi}{\sqrt{a^2-b^2}}}. \end{aligned}$$

Changing the interval changes the answer: the integral over $[-\pi/2, \pi/2]$ is instead

$$\int_{-\pi/2}^{\pi/2} \frac{dx}{a+b\cos x} = \frac{4}{\sqrt{a^2-b^2}}\arctan\sqrt{\frac{a-b}{a+b}}.$$

One can check both formulas immediately at $b = 0$.

A more structural derivation explains the radical. Set $\Delta = \sqrt{a^2-b^2}$, $C = (a+\Delta)/2$, and $r = -b/(a+\Delta)$. Then $|r| < 1$ and

$$a + b\cos x = C(1 - 2r\cos x + r^2), \qquad C(1-r^2) = \Delta.$$

Therefore

$$\frac{1}{a+b\cos x} = \frac{1}{\Delta}\,\underbrace{\frac{1-r^2}{1-2r\cos x+r^2}}_{P_r(x)}.$$

The factor P_r is the Poisson kernel for the disk. Its Fourier expansion

$$P_r(x) = 1 + 2\sum_{n=1}^{\infty} r^n \cos(nx)$$

converges absolutely for $|r| < 1$; integrating from 0 to π kills every nonconstant cosine. We recover $J(a,b) = \pi/\Delta$ and see why a rational trigonometric integral is naturally related to harmonic functions on the disk.

§54.8 Variation Bounds from Derivatives

Convexity is not the only source of integral inequalities. Let $f \in C^2([a,b])$ satisfy $f(a) = f(b) = 0$ and $f > 0$ on (a,b). Write $M = f(x_0) = \max f$ for an interior maximizer. The mean-value theorem gives points $\xi_1 \in (a, x_0)$ and $\xi_2 \in (x_0, b)$ with

$$f'(\xi_1) = \frac{M}{x_0 - a}, \qquad f'(\xi_2) = -\frac{M}{b - x_0}.$$

Since $f \le M$ where it is positive, we have (allowing the left side to be infinite)

$$\begin{aligned}\int_a^b \left|\frac{f''(x)}{f(x)}\right| dx &\ge \frac{1}{M}\int_a^b |f''(x)|\,dx \\ &\ge \frac{f'(\xi_1) - f'(\xi_2)}{M} \\ &= \frac{1}{x_0 - a} + \frac{1}{b - x_0} \ge \boxed{\frac{4}{b-a}}.\end{aligned}$$

The middle inequality is the total-variation estimate $\int |f''| \ge |f'(\xi_2) - f'(\xi_1)|$. This is a different kind of argument from Hermite–Hadamard: a function forced to rise from zero and return to zero must pay a definite amount of curvature. For the smooth profile $f(x) = \sin(\pi(x-a)/(b-a))$, the integral is $\pi^2/(b-a)$, which confirms the scale $1/(b-a)$ and shows why the dimensions of the estimate are natural.

§54.9 Wallis integrals become beta integrals

Define

$$I_n = \int_0^{\pi/2} \sin^n x\,dx, \qquad n \ge 0.$$

Integration by parts shows

$$I_n = \frac{n-1}{n} I_{n-2} \quad (n \ge 2), \qquad I_0 = \frac{\pi}{2}, \quad I_1 = 1.$$

Hence the even and odd subsequences are controlled by alternating products of consecutive integers.

There is a useful closed form that does not require treating the two parities separately. With $t = \sin^2 x$,

$$\begin{aligned} I_n &= \frac{1}{2}\int_0^1 t^{(n-1)/2}(1-t)^{-1/2}\,dt \\ &= \frac{1}{2}B\left(\frac{n+1}{2},\frac{1}{2}\right) = \frac{\sqrt{\pi}\,\Gamma((n+1)/2)}{2\Gamma((n+2)/2)}. \end{aligned}$$

The beta function is Euler's integral $B(u,v) = \int_0^1 t^{u-1}(1-t)^{v-1}dt$, and the last equality uses $B(u,v) = \Gamma(u)\Gamma(v)/\Gamma(u+v)$.

The asymptotic behavior has a geometric explanation. Writing $x = \pi/2 - t$, we have

$$I_n = \int_0^{\pi/2} \cos^n t\,dt.$$

For large n, the mass concentrates near $t = 0$, where $\log\cos t = -t^2/2 + O(t^4)$. The resulting Gaussian scaling, justified by Laplace's method, gives

$$\boxed{I_n \sim \sqrt{\frac{\pi}{2n}}.}$$

The recurrence produces Wallis' product for π, while the beta–gamma formula and Gaussian approximation describe the same sequence from special-function and asymptotic viewpoints.

There is one more payoff. Regard $I_s = \int_0^{\pi/2} \sin^s x\,dx$ as a function of a real parameter $s > -1$. Differentiation under the integral sign gives

$$I_s' = \int_0^{\pi/2} \sin^s x \log(\sin x)\,dx.$$

At $s = 1$ this is the logarithmic sine integral. It has an elementary evaluation that checks the parameter method without invoking derivatives of the gamma function:

$$\begin{aligned} I_1' &= \int_0^{\pi/2} \sin x \log(\sin x)\,dx \\ &= \frac{1}{2}\int_0^1 \log(1-u^2)\,du \qquad (u = \cos x) \\ &= \frac{1}{2}\left(\int_0^1 \log(1-u)\,du + \int_0^1 \log(1+u)\,du\right) \\ &= \boxed{\log 2 - 1}. \end{aligned}$$

The parameter s has converted an integral with a logarithm into the derivative of an ordinary beta integral. This is a useful general strategy for moments and normalizing constants in probability.

Averaging, reflection and recurrence all reduce an integral to simpler data: a convex bound, an invariant pair of points, or a relation between neighboring exponents. Here the special-function language grows directly from calculations one can still check by elementary substitutions.

V

Linear and Multilinear Algebra

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Graphical Linear Algebra: From Matrices to Lagrangian Relations

N-20210621

A matrix is an economical way to describe a linear map, but matrix multiplication hides a remarkably simple fact. In a network, a signal can reach an output along several paths. Each path contributes the product of its weights, while the paths ending at the same output have their contributions added. Thus

$$(AB)_{ij} = \sum_r A_{ir} B_{rj}$$

is already a wiring instruction: the repeated index r marks a collection of intermediate wires that are connected and then hidden.

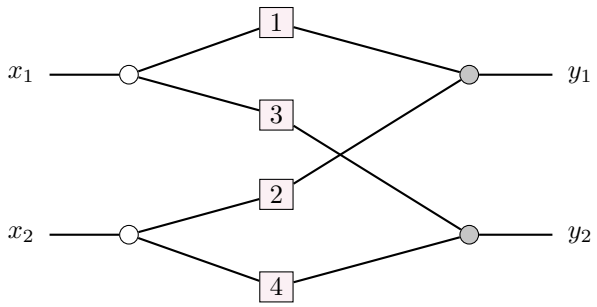
Could one perform the algebra with the wires themselves, transforming one network into another by small local moves? The question becomes even more interesting when a network specifies an equation which has several solutions, or no solution at all. We shall begin with the matrix operations, then allow equations to be read in either direction, and finally ask which such relations preserve a symplectic form. The resulting path leads from the PROP $\mathbf{cb}_k$ to $\mathbf{ih}_k$, and then to Lagrangian relations.[1]

§55.1 A linear map on four paths

Start with

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \qquad \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

The two outputs can be assembled with three kinds of operation: copy each input, multiply each copy by a weight, and add all contributions to an output. We draw white circles for copying and grey circles for addition. In every diagram of this note, *inputs are on the left and outputs on the right.*



A crossing without a node leaves the two wires separate.

Figure 55.1: The matrix A expressed as a weighted network. Reading from left to right gives $y_1 = x_1 + 2x_2$ and $y_2 = 3x_1 + 4x_2$.

[1] The first part follows Paweł Sobociński, *Graphical Linear Algebra*, especially Episodes 7–30. Our white-copy/grey-add convention follows the ZX-style colouring used by Comfort and Kissinger. The complete matrix and relation presentations are developed in Bonchi, Sobociński and Zanasi, *Interacting Hopf Algebras* (2017).

A diagram of this kind is a mathematical expression: connecting two diagrams ought to mean composing the transformations. If we want to *prove* equalities by moving its nodes, however, we must specify both the types of the diagrams and the equations that justify moving them.

§55.2 Why a number of wires is an object

Fix a commutative ring k for the elementary calculations; a field will be required later for some geometric results. The category Mat_k uses natural numbers $0, 1, 2, \ldots$ as objects. Its arrows are

$$\mathsf{Mat}_k(n, m) = \{m \times n \text{ matrices over } k\}.$$

The number n stands for a free module k^n. An arrow $A : n \to m$ is therefore a linear map $k^n \to k^m$.

There are two ways to combine arrows. For $A : n \to m$ and $B : m \to p$, connecting the output of A to the input of B means $B \circ A = BA : n \to p$. For $C : r \to s$, putting A and C next to each other means the block diagonal matrix

$$A \oplus C = \begin{pmatrix} A & 0 \\ 0 & C \end{pmatrix} : n + r \to m + s.$$

The switch of two adjacent wires is a permutation matrix $\sigma : 2 \to 2$, and permutations of larger collections of wires are built from such switches.

A *PROP* is a strict symmetric monoidal category with object set $\mathbb{N}$ and monoidal product on objects given by addition. Here its monoidal product of arrows is the *direct sum*. There is an important distinction:

$$\text{two parallel wires} \quad \longleftrightarrow \quad k \oplus k,$$

which has rank 2. It does not mean $k \otimes_k k$, which has rank 1. Confusing these operations would completely alter the meaning of graphical copying.

§55.3 Five operations and their exact types

The graphical theory $\mathbf{cb}_k$ has four basic structural operations, plus a scalar operation for every $a \in k$. These are maps of finite free modules:

operation	type	value on inputs
Δ (copy)	$1 \to 2$	$x \mapsto (x, x)$
ϵ (discard)	$1 \to 0$	$x \mapsto 0 \in k^0$
μ (add)	$2 \to 1$	$(x, y) \mapsto x + y$
η (zero)	$0 \to 1$	$0 \in k^0 \mapsto 0$
$[a]$ (scalar)	$1 \to 1$	$x \mapsto ax$

The zero-dimensional module k^0 has only one element. In ordinary matrices these generators are

$$\Delta = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \epsilon = 0_{0\times 1}, \quad \mu = \begin{pmatrix} 1 & 1 \end{pmatrix}, \quad \eta = 0_{1\times 0}, \quad [a] = \begin{pmatrix} a \end{pmatrix}.$$

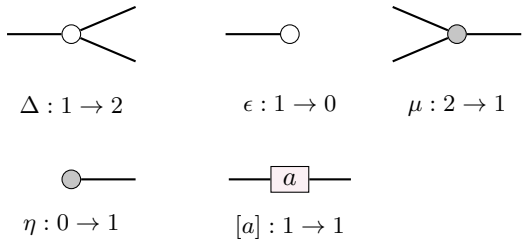


Figure 55.2: Five elementary diagrams. Their boundary types are part of the notation, and the two colours carry different operations.

The copying operation is genuinely linear:

$$\Delta(x+y) = (x+y, x+y) = \Delta(x) + \Delta(y)$$

where addition on the right is componentwise. By comparison, the map $v \mapsto v \otimes v$ is generally nonlinear, since $(v+w)\otimes(v+w)$ contains cross terms. We can copy a classical value into a direct sum without assuming that arbitrary vectors may be copied in a tensor product.

The types already catch elementary mistakes. The composite $\eta\epsilon : 1 \to 1$ is the zero map $x \mapsto 0$; reversing the order gives $\epsilon\eta : 0 \to 0$, which has an entirely different domain and codomain.

§55.4 The equations behind moving the nodes

The grey addition operation is associative, commutative and has zero as its unit. Diagrammatically this says

$$\mu(\mu \oplus \mathrm{id}) = \mu(\mathrm{id} \oplus \mu), \qquad \mu\sigma = \mu, \qquad \mu(\eta \oplus \mathrm{id}) = \mathrm{id}.$$

Copy and discard obey the equations with all arrows reversed:

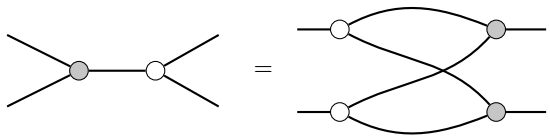
$$(\Delta \oplus \mathrm{id})\Delta = (\mathrm{id} \oplus \Delta)\Delta, \quad \sigma\Delta = \Delta, \quad (\epsilon \oplus \mathrm{id})\Delta = \mathrm{id}.$$

We call the former a commutative monoid and the latter a cocommutative comonoid. The equations authorize local substitution *inside* larger diagrams: adding three values in two bracketings really is the same map.

The first interesting law concerns the interaction of the two colours. If we add x and y, then copy the result, we obtain $(x+y, x+y)$. Alternatively we may copy both inputs, exchange the middle two resulting wires and add corresponding pairs:

$$\boxed{\Delta\mu = (\mu \oplus \mu)(\mathrm{id} \oplus \sigma \oplus \mathrm{id})(\Delta \oplus \Delta).}$$

Both sides have type $2 \to 2$; the switch is essential for the types and the intended pairing.



add, then copy

copy, exchange, then add

Figure 55.3: The bialgebra law. On either side the output is $(x+y, x+y)$; crossing wires without a node remain distinct.

In ordinary matrices the left side is

$$\begin{pmatrix}1\\1\end{pmatrix}\begin{pmatrix}1 & 1\end{pmatrix}=\begin{pmatrix}1 & 1\\1 & 1\end{pmatrix}.$$

The right side gives the same matrix by multiplying the appropriate block and permutation matrices. There are also laws saying that the zero can be copied, that sums can be discarded, and that scalar nodes cooperate with the two structures.

Scalars are particularly instructive. Two beads in series multiply, while a copy followed by two weighted branches and an addition produces their sum:

$$[b]\,[a]=[ba],\qquad [1]=\mathrm{id}_1,\qquad \boxed{[a+b]=\mu([a]\oplus[b])\Delta},\qquad [0]=\eta\epsilon.$$

These identities explain how several paths can collapse to a single scalar. They are examples of the full equations for a bicommutative bialgebra with the scalar family; our displayed selection is meant to explain the mechanisms, not silently replace the complete presentation.

§55.5 A normal form for every matrix

For an arbitrary $m\times n$ matrix $A=(a_{ij})$, there is a systematic recipe. Copy each input x_j into m branches, multiply its i-th branch by a_{ij}, permute branches so that the n contributions to output i sit together, and add them. The result is the map

$$y_i=\sum_{j=1}^{n}a_{ij}x_j.$$

Write $\Delta_m:1\to m$ for repeated copying, $\mu_n:n\to 1$ for repeated addition, and $P_{m,n}$ for the wire permutation changing input-major to output-major order. Then the diagram can be written as a typed composite:

$$\mathcal{D}(A)=\left(\bigoplus_{i=1}^{m}\mu_n\right)P_{m,n}\left(\bigoplus_{j=1}^{n}\bigoplus_{i=1}^{m}[a_{ij}]\right)\left(\bigoplus_{j=1}^{n}\Delta_m\right).$$

Its intermediate types are

$$n\longrightarrow mn\longrightarrow mn\longrightarrow mn\longrightarrow m.$$

The conventions $\Delta_0=\epsilon$ and $\mu_0=\eta$ accommodate empty boundaries.

This construction shows that *all matrices have diagrams*. The converse is a subtler theorem: the equations suffice to prove equality of any two diagrams which denote the same matrix. A normalization argument repeatedly moves copying past addition using the bialgebra law, groups wires using associativity and permutation laws, and combines repeated weighted paths using the scalar laws. Eventually the diagram takes the copy–scalar–add form corresponding to its matrix entries. A complete proof must show that each normalization step is licensed by the full list of equations.

Theorem 55.5.1 (The matrix presentation)

For a commutative ring k, the PROP $\mathbf{cb}_k$ of bicommutative bialgebra diagrams with its complete scalar equations presents the PROP Mat_k:

$$\boxed{\mathbf{cb}_k\cong\mathsf{Mat}_k.}$$

In particular, graphical provability and equality of denoted matrices coincide.

The essential distinction is between *expressiveness*, which the construction $\mathcal{D}(A)$ establishes directly, and *completeness*, which requires the full equational normal-form argument. An attractive circuit drawing alone gives neither a proof of faithfulness nor an algorithm for rewriting every equivalent diagram.

§55.6 Matrix multiplication counts weighted paths

Let

$$B = \begin{pmatrix} 3 & 0 \\ 4 & 1 \end{pmatrix}, \qquad A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}.$$

Passing signals first through B and then through A means multiplying in the order

$$AB = \begin{pmatrix} 11 & 2 \\ 4 & 1 \end{pmatrix}.$$

The coefficient 11 is a sum of contributions from two paths: $1 \cdot 3$ through the first intermediate wire and $2 \cdot 4$ through the second.

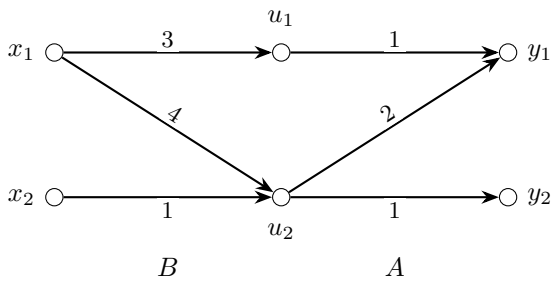


Figure 55.4: The path-sum rule for two connected linear networks. Zero-weight edges have been omitted; each path contributes a product and parallel paths contribute a sum.

The path-sum rule follows by induction for finite acyclic networks built from copy, scalars and addition. Composition concatenates paths, while addition merges their contributions:

$$(AB)_{ij} = \sum_{r} A_{ir} B_{rj}.$$

It is important not to extrapolate this formula to an arbitrary directed feedback loop by declaring an infinite sum of paths. Feedback introduces simultaneous equations; without invertibility or convergence it may fail to determine a function. A language of *relations* will handle those cases without pretending that every feedback network has a unique solution.

§55.7 Negative coefficients and the Hopf antipode

The scalar $[-1]$ deserves attention because it permits cancellation. Write $S = [-1] : 1 \to 1$. Then

$$\boxed{\mu(\mathrm{id} \oplus S)\Delta = \eta\epsilon : 1 \to 1.}$$

Both sides send x to zero: copy it, negate one copy, and add. In Hopf-algebra terminology S is the *antipode*.

There is a genuine hierarchy here. The copy/add calculus with natural-number coefficients describes nonnegative integer matrices; allowing negative coefficients reaches integer matrices; admitting scalar beads indexed by a ring k yields matrices over that ring. Cancellation cannot be used in an equational system whose available scalars do not admit additive inverses. The antipode is the structural form of subtraction, and it will also explain sign conventions when arrows are reversed.

§55.8 What happens when an arrow is reversed?

Consider a perfectly ordinary equation:

$$z = x + y.$$

When x, y are the inputs and z the output, this is the linear map $\mu : k^2 \to k$. But suppose we are given z and asked to recover (x, y). Over a field, there are many pairs satisfying the equation:

$$(x, y) = (t, z - t), \qquad t \in k.$$

The same constraint makes sense in both directions, although it is a function in only one of them.

A *linear relation* $R : n \to m$ is a linear subspace

$$R \subseteq k^n \oplus k^m.$$

We write $x\, R\, y$ when $(x, y) \in R$. An input may be related to no outputs, one output, or many outputs. A linear map $A : k^n \to k^m$ determines the relation

$$\Gamma_A = \{(x, Ax) : x \in k^n\}.$$

The *converse* of a relation reverses its two boundaries:

$$R^\dagger = \{(y, x) : (x, y) \in R\}.$$

This operation always exists. When $R = \Gamma_A$, however, its converse need not be the graph of a function.

For our two basic generators, their converses behave quite differently:

$$\mu^\dagger = \{(z, (x, y)) : x + y = z\},$$
$$\Delta^\dagger = \{((x, y), z) : x = y = z\}.$$

Thus the converse of addition splits a given value into *any* two summands, while the converse of copying tests that two values agree. Merely observing that both $\mu^\dagger$ and Δ have type $1 \to 2$ does not make them equal.

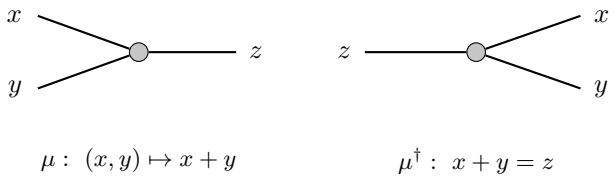


Figure 55.5: The addition constraint oriented in opposite directions. A grey node with two outputs means the *converse relation* of addition, and has no deterministic output.

This is a useful warning about diagrams: reflection is an operation on *relations*, and its semantics must be stated. It is generally neither matrix inversion nor matrix transposition.

§55.9 Connecting constraints means eliminating variables

Let $R : n \to m$ and $S : m \to p$ be linear relations. Connecting the output of R to the input of S means

$$\boxed{S \circ R = \{(x, z) : \exists y \in k^m,\ (x, y) \in R,\ (y, z) \in S\}.}$$

The intermediate wire is an existentially quantified variable. Relational composition therefore performs a familiar operation: *eliminating hidden variables in a system of equations*.

For example, let

$$R = \{(x, (u, v)) : u = x,\ v = x\}, \quad S = \{((u, v), z) : z = u + 2v\}.$$

The composite eliminates u, v and leaves the graph $z = 3x$. But if R says $u + v = x$, then for each x there are many allowed pairs (u, v); this is nevertheless a perfectly good linear relation.

Here is a powerful normal form, derived from ordinary linear algebra.

Proposition 55.9.1 (Every linear relation is a span of two maps)

For each linear relation $R \subseteq k^n \oplus k^m$ over a field, there are linear maps $X : k^r \to k^n$, $Y : k^r \to k^m$ with

$$\boxed{R = \Gamma_Y \circ \Gamma_X^\dagger.}$$

Proof. Take a basis of R, let $r = \dim R$, and collect the first and second components of the basis vectors as the columns of X and Y. The set of all pairs generated by this basis is

$$R = \{(Xt, Yt) : t \in k^r\}.$$

But $(x, y) \in \Gamma_Y \circ \Gamma_X^\dagger$ means precisely that there exists $t \in k^r$ with $x = Xt$ and $y = Yt$. □

The same relation may be represented by *equations* rather than by parameters. Choose a matrix C with kernel R, and write

$$R = \{(x, y) : C \begin{pmatrix} x \\ y \end{pmatrix} = 0\}.$$

The span representation describes solutions; the kernel representation describes constraints. Gaussian elimination translates one into the other.

These elementary facts already explain why one wants to add reflected copies of the matrix generators to a diagrammatic language. They generate every relation, but generation alone does not settle whether two diagrams expressing the same subspace can be transformed into each other by the rules.

Theorem 55.9.2 (Interacting Hopf algebras)

For a field k, the PROP $\mathbf{ih}_k$ formed from forward and reversed matrix generators with the complete interacting-Hopf equations presents the PROP of linear relations:

$$\boxed{\mathbf{ih}_k \cong \mathsf{LinRel}_k.}$$

The completeness theorem must account for redundant parameters in a span, different matrices cutting out the same kernel, and all the ways these choices interact during composition. The equations implement those changes of presentation graphically. A proof that all relations have *some* diagram, such as the proposition above, is only half the story.

§55.10 The zero scalar is not invertible, but it has a converse

This distinction between a function and a relation explains the provocative expression *division by zero* in the graphical linear algebra series.

Let $[a]$ denote multiplication by a. Reversing its graph gives

$$[a]^\dagger = \{(y, x) : y = ax\}.$$

When $a \neq 0$ in a field, this is the ordinary function $y \mapsto a^{-1}y$. When $a = 0$, instead,

$$\boxed{[0]^\dagger = \{(0, x) : x \in k\}.}$$

It accepts the input 0, and there it allows *every* output. At a nonzero input it has no solution. There is no numerical object $1/0$ hiding in the picture; the reversed relation remains meaningful because equations do not need to have unique solutions.

This becomes especially useful in feedback. Suppose a value satisfies the loop equation

$$x = u + rx,$$

where u is an external input. The network denotes

$$F_r = \{(u, x) : (1 - r)x = u\}.$$

For $r \neq 1$, it is the ordinary graph $x = (1 - r)^{-1}u$. For $r = 1$, it becomes the relation $\{(0, x) : x \in k\}$. The loop is then inconsistent for $u \neq 0$ and underdetermined for $u = 0$.

A vector-valued version makes the link to classical linear algebra even clearer:

$$F_M = \{(u, x) : (I - M)x = u\}.$$

If $I - M$ is invertible, the relation is a function. Otherwise there can be no solutions when $u \notin \mathrm{im}(I - M)$, and an affine family of solutions indexed by $\ker(I - M)$ when u lies in the image. A relational circuit retains both possibilities without imposing invertibility by fiat.

§55.11 Two Frobenius structures and the spider rule

Some very short calculations explain why interacting graphical calculi admit dramatic simplifications. Relational converse gives

$$\boxed{\Delta^\dagger\Delta = \mathrm{id}_1, \qquad \mu\mu^\dagger = \mathrm{id}_1.}$$

To verify the first, copy x into (x, x), then apply the equality test; the only possible result is x. For the second, split z into any two summands and add them back; every possible route returns z.

Reversing the order does *not* give identities:

$$\Delta\Delta^\dagger = \{((a,b),(c,d)) : a = b = c = d\},$$
$$\mu^\dagger\mu = \{((a,b),(c,d)) : a + b = c + d\}.$$

The second relation preserves only the total sum, while the first demands equality of all four entries. Their difference is a concrete reason why arbitrary nodes cannot be cancelled as if they were isomorphisms.

The copy operation with its converse, and the addition operation with its converse, each satisfy the equations of a *special commutative Frobenius algebra* in LinRel_k. The corresponding spider theorem says that a connected network consisting exclusively of one such family of nodes depends only on the number of its exposed inputs and outputs. For the equality family, a connected spider is simply the relation that all incident variables have the same value. A tree of equality tests and copies can therefore be contracted into one node.

This theorem is a powerful *local* rewriting principle. Mixing the two colours introduces additional interaction equations; one cannot fuse a copy spider with an add spider just because their pictures look alike.

§55.12 The missing structure: phase space

The graphical language is now expressive enough to handle general linear constraints, including reversed arrows and feedback. A further problem arises in mechanics and in certain electrical and quantum systems: two kinds of variables travel together. A configuration variable q is paired with a momentum-like or covector variable p.

Let $V = k^n$ over a field k, and write its cotangent phase space in coordinates as

$$T^*V \cong k^n_q \oplus k^n_p.$$

The canonical symplectic form is the alternating, nondegenerate bilinear form

$$\boxed{\omega((q,p),(q',p')) = q^\mathsf{T}p' - p^\mathsf{T}q'.}$$

Its matrix is

$$J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}.$$

The word *alternating*, meaning $\omega(v,v) = 0$, is important in characteristic 2, where an equation involving only skew-symmetry can be misleading.

For a subspace $L \subseteq T^*V$, its symplectic orthogonal is

$$L^\omega = \{w : \omega(w,\ell) = 0 \text{ for all } \ell \in L\}.$$

We call L *Lagrangian* when $L = L^\omega$. Because ω is nondegenerate, a subspace is Lagrangian precisely when ω vanishes on it and its dimension equals n.

What does this condition mean for a familiar matrix? Given $A \in k^{n\times n}$, consider the graph

$$L_A = \{(q,p) : p = Aq\}.$$

It has dimension n, and for its two generic points we calculate

$$\begin{aligned}\omega((q,Aq),(q',Aq')) &= q^\mathsf{T}Aq' - (Aq)^\mathsf{T}q' \\ &= q^\mathsf{T}(A - A^\mathsf{T})q'.\end{aligned}$$

Thus

$$\boxed{L_A \text{ is Lagrangian} \iff A = A^{\mathsf{T}}.}$$

We have found matrix symmetry again, now as the necessary and sufficient condition for a graph in phase space to be compatible with the symplectic pairing.

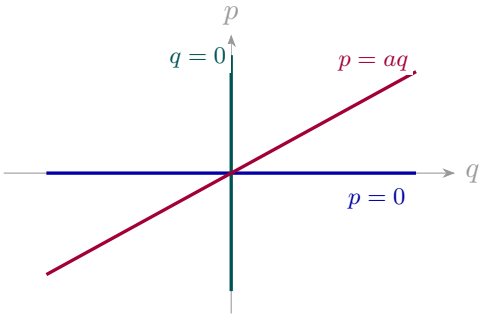


Figure 55.6: Lagrangian lines in the real phase plane. The two axes will arise from doubling zero and full subspaces; a nonhorizontal, nonvertical line exposes the need for an additional generator.

In one degree of freedom, every line through the origin is isotropic and therefore Lagrangian. In two degrees of freedom, the symmetric matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ gives the Lagrangian plane $p_1 = q_2,\ p_2 = q_1$, while the nonsymmetric matrix $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ does not.

§55.13 A matrix on positions determines a relation on momenta

A linear map $A : k^n \to k^m$ sends a position q to $Q = Aq$. But what should happen to a momentum-like covector? Its natural transformation runs in the other direction:

$$p = A^{\mathsf{T}} P.$$

Together the two equations define the *cotangent-lift relation*

$$\boxed{\mathcal{L}_A = \{((q,p),(Q,P)) : Q = Aq,\ p = A^{\mathsf{T}}P\}.}$$

The construction is a relation even when A is singular or rectangular. It has as many solutions as its freely chosen parameters $q \in k^n$ and $P \in k^m$ permit.

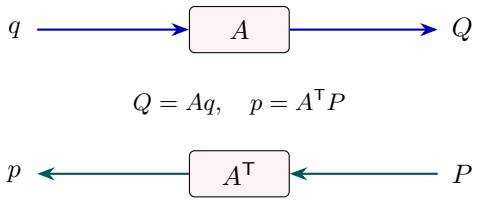


Figure 55.7: A position map A and its transpose on covectors form a Lagrangian relation. The opposing arrows indicate how to solve the defining constraints, without asserting that the relation is a function.

The ambient space for a relation between phase spaces is

$$\overline{T^*k^n} \oplus T^*k^m,$$

where the bar reverses the symplectic form on the input. We can verify explicitly that $\mathcal{L}_A$ is Lagrangian. It has dimension $n+m$, half the dimension $2(n+m)$ of the ambient space. Its symplectic pairing is zero:

$$\begin{aligned}&\omega_m((Aq,P),(Aq',P')) - \omega_n((q,A^\mathsf{T}P),(q',A^\mathsf{T}P'))\\ &= q^\mathsf{T}A^\mathsf{T}P' - P^\mathsf{T}Aq' - q^\mathsf{T}A^\mathsf{T}P' + P^\mathsf{T}Aq' = 0.\end{aligned}$$

Furthermore, for $B : k^m \to k^r$, interconnecting these lifted relations gives

$$Q = Aq, \quad Z = BQ, \quad p = A^\mathsf{T}P, \quad P = B^\mathsf{T}R.$$

Eliminating Q, P yields $Z = (BA)q,\ p = (BA)^\mathsf{T}R$, hence

$$\boxed{\mathcal{L}_B \circ \mathcal{L}_A = \mathcal{L}_{BA}.}$$

The transpose arises from the symplectic pairing, and the reversed momentum flow makes the composition work without requiring either A or B to be invertible.

§55.14 Doubling a linear relation with its annihilator

We can remove the assumption that our starting object was the graph of a matrix. Let $U = k^n,\ V = k^m$ and let $R \subseteq U \oplus V$ be any linear relation. Define its annihilator

$$R^\perp = \{(\alpha,\beta) \in U^* \oplus V^* : \alpha(u) + \beta(v) = 0 \text{ for all } (u,v) \in R\}.$$

Now construct

$$\boxed{\mathcal{L}_R = \left\{((u,p_U),(v,p_V)) : (u,v) \in R, \quad (-p_U,p_V) \in R^\perp\right\}.}$$

The minus sign is essential: the symplectic form on the *input* phase space has been reversed. Without it, even the matrix case would not give the cotangent-lift equation $p_U = A^\mathsf{T}p_V$.

Here is a direct proof that the construction is Lagrangian. Since

$$\dim R + \dim R^\perp = n + m,$$

the doubled relation has dimension $n+m$, half that of its ambient symplectic space. Let (r,s) and (r',s') represent two of its elements with $r, r' \in R$ and $s, s' \in R^\perp$, using the sign convention above. The ambient symplectic pairing is

$$s'(r) - s(r') = 0$$

because the covectors annihilate R. The relation is isotropic and half-dimensional, hence Lagrangian.

For $R = \Gamma_A$, a covector (α,β) annihilates its graph precisely when $\alpha = -A^\mathsf{T}\beta$. Thus

$$(-p_U,p_V) \in \Gamma_A^\perp \quad \Longleftrightarrow \quad p_U = A^\mathsf{T}p_V.$$

The earlier cotangent lift was the graph case of this more general *doubling* construction.

One also needs to know that doubling respects composition and tensoring. For finite-dimensional vector spaces over a field, orthogonal-complement identities under linear projection and intersection give

$$\mathcal{L}_{S\circ R} = \mathcal{L}_S \circ \mathcal{L}_R.$$

Together with the natural reordering of position and momentum summands, this makes doubling a faithful strong symmetric monoidal functor

$$\mathsf{LinRel}_k \longrightarrow \mathsf{LagRel}_k.$$

Faithfulness is immediate from the position component: one can recover R by projecting $\mathcal{L}_R$ onto the original position variables. The compositional statement is a verified theorem, and rests on the annihilator calculus for finite-dimensional relations.[2]

§55.15 A Lagrangian relation that doubling cannot produce

The doubling functor is faithful, but it is not *full*. The simplest obstruction appears in just one degree of freedom.

A linear relation $R : 0 \to 1$ is a linear subspace of the one-dimensional space k. There are only two choices: $R = 0$ or $R = k$. Their doubles in T^*k are

$$\mathcal{L}_0 = \{(q,p) : q = 0\}, \qquad \mathcal{L}_k = \{(q,p) : p = 0\}.$$

Yet for any nonzero a, the line

$$L_a = \{(q,p) : p = aq\}$$

is Lagrangian, as we established earlier, and is neither axis. Therefore it is *not* in the image of doubling.

How can we create it? Consider the shear

$$S_a : (q,p) \longmapsto (q, p + aq).$$

For two vectors,

$$\begin{aligned}\omega(S_a(q,p), S_a(q',p')) &= q(p' + aq') - (p + aq)q' \\ &= qp' - pq'.\end{aligned}$$

Thus S_a is symplectic, and it sends the horizontal axis $p = 0$ to $p = aq$. Another symplectic operation is the Fourier-type rotation

$$F : (q,p) \longmapsto (-p, q).$$

It exchanges the two coordinate directions, with the sign necessary for preservation of the symplectic form.

These are concrete new operations which the purely doubled subcategory lacks. Comfort and Kissinger prove that, over any field, doubled linear relations together with the indicated Fourier and shear generators generate *all* linear Lagrangian relations. Their constructive proof uses symplectic Gaussian-elimination operations on a matrix representation of a Lagrangian subspace.

The distinction between generation and a stand-alone equation system matters here, just as it did for matrices. The paper gives a complete description via the already complete graphical calculus of linear relations, supplemented by appropriate generators. It does not provide a small, independent complete list of diagrammatic equations written solely in its chosen Lagrangian generators; this is explicitly left as a further direction.

[2] Cole Comfort and Aleks Kissinger, *A Graphical Calculus for Lagrangian Relations*, 2022, especially Lemma 2.7 and Theorems 3.1–3.2. Their paper also states the matrix and linear-relation presentations in Section 1.

§55.16 Affine shifts and a carefully scoped quantum connection

Every linear relation is a subspace, so it contains the zero vector. To describe $p = aq + b$ with $b \neq 0$, we must allow an *affine* Lagrangian subspace, or a translate of a linear Lagrangian subspace. The translations

$$(q, p) \longmapsto (q + b, p + c)$$

preserve the symplectic form on displacement vectors. One extra affine-shift generator supplies the additional expressive power needed for affine Lagrangian relations.

A major application now becomes intelligible. Over an odd prime field $\mathbb{F}_p$, the paper proves a symmetric monoidal equivalence between affine Lagrangian relations and the corresponding p-dimensional qudit stabilizer process theory, *modulo invertible scalars*. The restriction to odd primes is substantive; the statement does not claim an equivalence with all quantum computations or an identical theorem for qubits.

Another application concerns electrical networks, where potential/current or voltage/current variables form conjugate pairs. Interconnecting ports gives linear constraints and suitable ideal components admit a Lagrangian-relational semantics. To describe resistors, sources or dynamical components, the chosen coefficient field and the individual constitutive laws still matter. The diagrams do not magically make every physical component symplectic or lossless.

This also clarifies the family resemblance to ZX-calculus. Both use coloured generators, spiders and equations that permit local replacements. The intended semantic objects are nevertheless different:

$$\begin{aligned} \mathbf{cb}_k &\longleftrightarrow \text{matrices}, \\ \mathbf{ih}_k &\longleftrightarrow \text{linear relations}, \\ \mathsf{LagRel}_k &\longleftrightarrow \text{Lagrangian phase-space relations}. \end{aligned}$$

A node is given its mathematical meaning by its interpretation, not by its colour or visual resemblance to some other diagram.

§55.17 Returning to the first network

The original weighted network represented

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

In $\mathbf{cb}_k$, it can be compared with any other weighted copy–scalar–add circuit using a sound and complete equational calculus. Viewed in LinRel_k, its graph can be reversed, connected with a zero test, or placed inside feedback; the resulting system may no longer be a function. Finally, as a doubled relation in phase space it carries the pair of constraints

$$Q = Aq, \qquad p = A^{\mathsf{T}} P.$$

The transpose and the reversed momentum wire now come from a mathematical requirement: compatibility with the symplectic pairing.

At each stage the pictures require a precise rule for what their boundaries mean and for what counts as equality. The diagrammatic method earns its value when it permits us to manipulate a network while preserving its linear or symplectic semantics; a visually pleasing picture without those rules would be little more than an illustration.

Fields, Affine Fibers, and the Linear Algebra of Forgetting Directions

N-20220407

A vector space owes much of its simplicity to division by nonzero scalars. Over $\mathbb{C}$ this is already visible in the identity

$$z^{-1} = \frac{\overline{z}}{|z|^2} \qquad (z \neq 0).$$

Yet the most useful distinction in early linear algebra concerns equations: a homogeneous equation defines a subspace, while an inhomogeneous equation defines either an empty set or a translate of a subspace. From that observation, quotient spaces and duality arise with surprisingly little extra machinery.[1]

§56.1 Complex multiplication is a real linear transformation

Write $z = a + bi$. Multiplication by z is an $\mathbb{R}$-linear map $\mathbb{C} \to \mathbb{C}$, represented in the basis $1, i$ by

$$M_z = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}.$$

The identity $M_z M_w = M_{zw}$ follows from complex multiplication itself, while

$$M_z^T M_z = (a^2 + b^2) I, \qquad \det M_z = a^2 + b^2.$$

For $z = re^{i\theta}$, the map is a rotation through θ followed by dilation by r. The algebraic inverse $1/z = \overline{z}/|z|^2$ and the matrix inverse $M_z^{-1} = M_{\overline{z}}/|z|^2$ are therefore two descriptions of the same operation.

For example, $\omega = e^{2\pi i/3}$ satisfies

$$1 + \omega + \omega^2 = 0, \qquad M_\omega^3 = I.$$

The nontrivial cube roots of unity are rotations of order three. They provide an early example of a representation of the cyclic group C_3 on a real vector space.

§56.2 What a linear equation actually determines

Let $T : V \to W$ be linear and consider $T(v) = b$. If v_0 is one solution, then

$$T(v) = b \quad \Longleftrightarrow \quad T(v - v_0) = 0,$$

so the full solution set is

$$\boxed{T^{-1}(b) = v_0 + \ker T.}$$

The fiber is a translate of the kernel. It is a vector subspace precisely when $b = 0$, provided that the equation is solvable.

[1] For the linear-algebraic setting, see Axler, *Linear Algebra Done Right*, and MIT 18.700 lecture readings: https://ocw.mit.edu/courses/18-700-linear-algebra-fall-2013/pages/readings/.

This viewpoint resolves a family of elementary tests at once. In F^4, the condition

$$x_3 - 5x_4 = b$$

defines a nonempty affine hyperplane; it is a linear subspace only when $b = 0$. In the vector space $C^1((0,3),\mathbb{R})$, the condition $f'(2) = b$ has the same structure: evaluation of the derivative at 2 is a linear functional, and its fibers are translates of its kernel.

The equations need not involve finite coordinates. If $T : C^1([0,1]) \to \mathbb{R}$ sends f to $\int_0^1 f(t)\,dt$, the functions with integral 1 form an affine fiber of the same kind.

§56.3 What Survives the Quotient

For a subspace $U \subseteq V$, two vectors are equivalent when $v - w \in U$. Their equivalence classes form the quotient V/U, with projection $q(v) = v + U$.

Proposition 56.3.1 (Factorization through a quotient)
If $T : V \to W$ is linear and $U \subseteq \ker T$, there is a unique linear map $\overline{T} : V/U \to W$ such that $T = \overline{T} \circ q$.

Proof. There is only one possible definition:

$$\overline{T}(v + U) = T(v).$$

If $v+U = w+U$, then $v-w \in U \subseteq \ker T$, hence $T(v) = T(w)$. The definition is therefore independent of the representative. Linearity and uniqueness follow from surjectivity of q. □

Taking $U = \ker T$ makes $\overline{T}$ injective, with image equal to $\operatorname{im} T$. Thus

$$\boxed{V/\ker T \cong \operatorname{im} T.}$$

The first isomorphism theorem says that a linear map records exactly the information left after the directions invisible to it have been identified. For finite-dimensional spaces, taking dimensions recovers rank–nullity.

§56.4 Annihilators and Quotients

The dual $V^* = \operatorname{Hom}_F(V, F)$ is the space of linear measurements on V. A functional $\lambda \in V^*$ descends to V/U precisely when it is constant on each coset, or equivalently when $\lambda(u) = 0$ for all $u \in U$.

Define the annihilator

$$U^\circ = \{\lambda \in V^* : \lambda|_U = 0\}.$$

Pullback along q gives the map

$$q^* : (V/U)^* \longrightarrow V^*, \qquad \varphi \longmapsto \varphi \circ q.$$

It is injective because q is surjective, and its image is exactly U°, by the quotient factorization just proved. Hence

$$\boxed{(V/U)^* \cong U^\circ.}$$

For $V = \mathbb{R}^3$ and $U = \operatorname{span}\{(1,1,0)\}$, a linear functional $\lambda(x,y,z) = ax + by + cz$ lies in U° exactly when $a + b = 0$. The two functionals $x - y$ and z are independent measurements on the quotient; they distinguish its cosets and form a basis of its dual.

This is a concrete reason to learn covectors. When one solves linear constraints, it is often easier to describe the surviving information as linear measurements than to choose arbitrary coordinates on the quotient.

§56.5 Why changing the scalars changes the theory

A vector space over a field has bases, and every linearly independent list extends to a basis. The mechanism is that every nonzero pivot coefficient can be inverted. For a module over a ring this may fail.

For example, $\mathbb{Z}/2\mathbb{Z}$ is a $\mathbb{Z}$-module but has no basis over $\mathbb{Z}$: if a putative free generator e existed, the relation $2e = 0$ would contradict uniqueness of integer coordinates. This is torsion. More subtly, even a torsion-free module need not be free over an arbitrary ring.

The distinction points directly toward integral matrix reduction. Over $\mathbb{Q}$, a nonzero integer pivot may be normalized to 1; over $\mathbb{Z}$, only ± 1 are units, and row and column operations preserve arithmetic divisibility information. Smith normal form is the corresponding normal form over $\mathbb{Z}$.

Quotients, annihilators, and affine fibers already make sense before any matrices are written down. Fields then guarantee that dimension and coordinates behave especially well, explaining both the efficiency and the limitations of elementary Gaussian elimination.

Indices, Dual Maps, Adjoints, Trace, and Singular Values

N-20220412

§57.1 Indices as Type Information

Let V have basis $e_1, \dots, e_n$, with dual basis $e^1, \dots, e^n$. The Kronecker delta is

$$\delta_i{}^j = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

It is the coordinate representation of the identity map:

$$I(e_j) = e_i \delta_i{}^j = e_j.$$

The position of an index records which kind of slot it occupies. A vector has components x^j, a covector has components α_i, and contraction pairs one of each type:

$$\alpha(x) = \alpha_i x^i.$$

Einstein convention suppresses the summation sign when an index occurs once upstairs and once downstairs:

$$\alpha_i x^i = \sum_i \alpha_i x^i.$$

An index that survives once in every term is free and determines the output type. A repeated index is dummy and may be renamed. Expressions such as

$$A_i{}^j x^i$$

are ill typed if i is intended to be the output index, because it occurs twice while j remains unmatched. Index notation is valuable precisely because many algebraic mistakes become visible as type errors.

§57.2 Matrix Products as Contractions

Let

$$B : U \to V, \qquad A : V \to W.$$

Choose bases and write

$$B(e_k) = B_j{}^k v_j, \qquad A(v_j) = A_i{}^j w_i.$$

Then

$$\begin{aligned} (A \circ B)(e_k) &= B_j{}^k A(v_j) \\ &= A_i{}^j B_j{}^k w_i. \end{aligned}$$

Thus

$$(AB)_i{}^k = A_i{}^j B_j{}^k.$$

The repeated index j is the intermediate vector-space direction that is produced by B and consumed by A.

The identity tensor acts neutrally because

$$A_i{}^j \delta_j{}^k = A_i{}^k, \qquad \delta_i{}^j A_j{}^k = A_i{}^k.$$

Trace is another contraction:

$$\operatorname{tr} A = A_i{}^i.$$

Matrix multiplication and trace are therefore not unrelated recipes; both pair an output index with a compatible input index.

§57.3 Basis Changes and Contractions

Let P be the change-of-basis matrix, so coordinate columns satisfy

$$[x]_{\text{old}} = P[x]_{\text{new}}.$$

An endomorphism matrix transforms by similarity:

$$A' = P^{-1}AP.$$

In indices,

$$A'^j_i = (P^{-1})_i{}^r A_r{}^s P_s{}^j.$$

Contracting the free indices gives

$$\begin{aligned} A'^i_i &= (P^{-1})_i{}^r A_r{}^s P_s{}^i \\ &= A_r{}^s P_s{}^i (P^{-1})_i{}^r \\ &= A_r{}^s \delta_s{}^r \\ &= A_r{}^r. \end{aligned}$$

Hence

$$\operatorname{tr}(P^{-1}AP) = \operatorname{tr} A.$$

The diagonal sum is invariant because it is a complete contraction: the basis-change matrix and its inverse cancel along the closed index loop.

By contrast, an individual diagonal entry $A_i{}^i$ with i fixed is not invariant. Only the sum over the repeated index has intrinsic meaning.

§57.4 Transpose is the matrix of the dual map

Let

$$A : V \to W$$

and define the dual map

$$A^\vee : W^* \to V^*, \qquad A^\vee(\varphi) = \varphi \circ A.$$

Choose bases e_j of V and f_i of W, with

$$A(e_j) = \sum_i A_{ij} f_i.$$

For the dual basis vector f^i,

$$\begin{aligned} A^\vee(f^i)(e_j) &= f^i(Ae_j) \\ &= A_{ij}. \end{aligned}$$

Therefore

$$A^\vee(f^i) = \sum_j A_{ij} e^j.$$

The matrix of $A^\vee$ has entry A_{ij} in row j, column i, so it is

$$[A^\vee] = A^T.$$

The reversal law now follows without entry manipulation. For

$$U \xrightarrow{B} V \xrightarrow{A} W,$$

one has

$$(A \circ B)^\vee = B^\vee \circ A^\vee.$$

In matrices,

$$(AB)^T = B^T A^T.$$

Transposition reverses order because a functional is pulled backward through a composite map.

§57.5 Adjoints from the Dual Space

The transpose needs only vector spaces and duality. An adjoint additionally needs inner products. Let

$$A : V \to W$$

be a linear map between finite-dimensional complex inner-product spaces. The adjoint

$$A^\dagger : W \to V$$

is characterized by

$$\langle Ax, y\rangle_W = \langle x, A^\dagger y\rangle_V.$$

There is a subtlety over $\mathbb{C}$. In this note the inner product is *conjugate-linear in the first slot* and linear in the second:

$$\langle x, z\rangle = x^* G z.$$

The ordinary algebraic dual V^* consists of *complex-linear* functionals. Riesz representation therefore gives the map

$$R_V : V \longrightarrow V^*, \qquad R_V(y)(x) = \langle y, x\rangle_V.$$

It identifies covectors with unique vectors, but the map R_V is *conjugate-linear*:

$$R_V(cy)(x) = \langle cy, x\rangle = \overline{c}\,\langle y, x\rangle = \overline{c}\, R_V(y)(x).$$

We may now express the adjoint precisely as

$$\boxed{A^\dagger = R_V^{-1} \circ A^\vee \circ R_W : W \longrightarrow V.}$$

Indeed, evaluating the right side at $y \in W$ gives the unique vector $z \in V$ with

$$\langle z, x\rangle_V = (A^\vee R_W(y))(x) = R_W(y)(Ax) = \langle y, Ax\rangle_W$$

for every $x \in V$. Taking conjugates yields $\langle x, z\rangle_V = \langle Ax, y\rangle_W$, so $z = A^\dagger y$.

The two conjugate-linear Riesz maps cancel one another's conjugation, which is why $A^\dagger : W \to V$ is an ordinary *complex-linear* map. This is easy to miss if one calls the dual map and adjoint the same operation. For the one-dimensional operator

$$A : \mathbb{C} \to \mathbb{C}, \qquad A(z) = iz,$$

the dual map multiplies the coefficient of a linear functional by i, whereas the Hilbert adjoint is

$$\boxed{A^\dagger(z) = -iz.}$$

For instance, $\langle i, 1\rangle = -i = \langle 1, -i\rangle$. Over $\mathbb{R}$, conjugation disappears and both Riesz maps are linear.[1]

In orthonormal bases,

$$[A^\dagger] = \overline{A}^{\,T} = A^*.$$

The operations should be kept distinct:

operation	matrix
transpose	A^T
entrywise conjugate	$\overline{A}$
conjugate transpose	$\overline{A}^{\,T}$
adjoint in orthonormal bases	A^*.

Over $\mathbb{R}$, conjugation disappears and the adjoint matrix in orthonormal bases is the transpose.

§57.6 Adjoints in Nonorthogonal Frames

Suppose the chosen bases have Gram matrices

$$G_V = (\langle e_i, e_j\rangle_V), \qquad G_W = (\langle f_i, f_j\rangle_W).$$

For coordinate columns,

$$\langle x, z\rangle_V = x^* G_V z, \qquad \langle y, w\rangle_W = y^* G_W w.$$

The defining identity for the adjoint becomes

$$x^* A^* G_W y = x^* G_V [A^\dagger] y$$

for all x, y. Hence

$$\boxed{[A^\dagger] = G_V^{-1} A^* G_W.}$$

The formula $A^\dagger = A^*$ is therefore a feature of orthonormal coordinates, not the definition of adjointness.

For example, on $\mathbb{R}^2$ use the inner product with

$$G = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}$$

[1] Compare Sergei Treil, *Linear Algebra Done Wrong*, Chapter 8, on complex duality and the Riesz map.

and the linear map

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Although $A^T = A$, its adjoint in this metric is

$$\begin{aligned} A^\dagger &= G^{-1} A^T G \\ &= \begin{pmatrix} 0 & 4 \\ 1/4 & 0 \end{pmatrix}. \end{aligned}$$

The coordinate swap is not self-adjoint when the second coordinate has four times the metric weight.

§57.7 Rank and Minimal Factorizations

For

$$A : V \to W,$$

define

$$\operatorname{rank} A = \dim \operatorname{im} A.$$

The first isomorphism theorem gives

$$V/\ker A \cong \operatorname{im} A.$$

Thus rank is the dimension of the effective quotient after directions killed by A have been removed.

If $\operatorname{rank} A = r$, choose a basis of the column space and place it in a matrix

$$B \in k^{m \times r}.$$

Express every column of A in that basis and collect the coordinates in

$$C \in k^{r \times n}.$$

Then

$$A = BC, \qquad \operatorname{rank} B = \operatorname{rank} C = r.$$

Conversely, any factorization through an r-dimensional space has rank at most r. Hence rank is the smallest intermediate dimension through which the map can factor.

This factorization also proves equality of row and column rank. Since $A = BC$, every row of A is a linear combination of rows of C, so

$$\operatorname{row}(A) \subseteq \operatorname{row}(C).$$

Because B has full column rank, it has a left inverse L with

$$LB = I_r.$$

Hence

$$C = LA,$$

so every row of C is a linear combination of rows of A. Thus

$$\operatorname{row}(A) = \operatorname{row}(C),$$

and its dimension is r, the same as the column rank.

§57.8 Trace and Cyclic Contraction

For endomorphisms A, B,

$$\begin{aligned}\operatorname{tr}(AB) &= (AB)_i{}^i \\ &= A_i{}^j B_j{}^i \\ &= B_j{}^i A_i{}^j \\ &= \operatorname{tr}(BA).\end{aligned}$$

The middle equality only renames dummy indices and commutes scalar entries. Consequently,

$$\operatorname{tr}(P^{-1}AP) = \operatorname{tr}(APP^{-1}) = \operatorname{tr} A.$$

For a rank-one operator

$$A = uv^*,$$

one has

$$\boxed{\operatorname{tr}(uv^*) = v^*u.}$$

Indeed,

$$(uv^*)_{ij} = u_i\overline{v_j},$$

so

$$\operatorname{tr}(uv^*) = \sum_i u_i\overline{v_i} = v^*u.$$

If u is a unit vector, the orthogonal projector onto its span is $P = uu^*$, and

$$\operatorname{tr} P = 1.$$

More generally, an orthogonal projector has eigenvalues $0, 1$, hence

$$\operatorname{tr} P = \operatorname{rank} P.$$

Trace counts the dimension of a projected subspace because it sums the projector's surviving modes.

§57.9 Hilbert–Schmidt and Schatten Norms

Matrices themselves form an inner-product space with

$$\langle A, B\rangle_{HS} = \operatorname{tr}(A^*B).$$

Its norm is the Frobenius norm:

$$\begin{aligned}\|A\|_F^2 &= \operatorname{tr}(A^*A) \\ &= \sum_{i,j} |a_{ij}|^2.\end{aligned}$$

If the singular values of A are

$$\sigma_1(A), \dots, \sigma_r(A),$$

then the Schatten norms are

$$\|A\|_{S_p} = \left(\sum_i \sigma_i(A)^p\right)^{1/p}, \qquad 1 \le p < \infty,$$

and

$$\|A\|_{S_\infty} = \max_i \sigma_i(A).$$

The three common cases are

$$\|A\|_{S_1} = \text{nuclear norm}, \qquad \|A\|_{S_2} = \|A\|_F, \qquad \|A\|_{S_\infty} = \|A\|_{op}.$$

For

$$A = \begin{pmatrix} 0 & 3 \\ 4 & 0 \end{pmatrix},$$

one has

$$A^*A = \begin{pmatrix} 16 & 0 \\ 0 & 9 \end{pmatrix},$$

so the singular values are $4, 3$. Therefore

$$\|A\|_{S_1} = 7, \qquad \|A\|_{S_2} = 5, \qquad \|A\|_{S_\infty} = 4.$$

Rank counts how many singular values are nonzero; Schatten norms measure their size with different emphases.

The Hilbert–Schmidt product also gives the estimate

$$|\operatorname{tr}(A^*B)| \le \|A\|_F \|B\|_F,$$

which is ordinary Cauchy–Schwarz applied in the matrix space.

§57.10 Categorical Trace as Wire Closure

Let V be finite dimensional with basis e_i and dual basis e^i. The coevaluation map is

$$\operatorname{coev} : k \to V \otimes V^*, \qquad 1 \longmapsto \sum_i e_i \otimes e^i.$$

The evaluation map is

$$\operatorname{ev} : V^* \otimes V \to k, \qquad \varphi \otimes v \longmapsto \varphi(v).$$

For an endomorphism $A : V \to V$, compose

$$k \xrightarrow{\text{coev}} V \otimes V^* \xrightarrow{A \otimes I} V \otimes V^* \xrightarrow{\tau} V^* \otimes V \xrightarrow{\text{ev}} k,$$

where $\tau(v \otimes \varphi) = \varphi \otimes v$. Acting on 1, the composite gives

$$\begin{aligned} 1 &\longmapsto \sum_i e_i \otimes e^i \\ &\longmapsto \sum_i Ae_i \otimes e^i \\ &\longmapsto \sum_i e^i \otimes Ae_i \\ &\longmapsto \sum_i e^i(Ae_i). \end{aligned}$$

If

$$Ae_i = A_j{}^i e_j,$$

then

$$\sum_i e^i(Ae_i) = \sum_i A_i{}^i = \operatorname{tr} A.$$

The abstract operation of closing an output back into an input reproduces the ordinary matrix trace exactly. Basis invariance is built into the evaluation–coevaluation pairing, while the coordinate calculation reveals the same contraction $A_i{}^i$ encountered at the beginning of the chapter.

Permutation Signs, Determinants, and Exterior Powers

N-20220725

An $n \times n$ determinant has $n!$ terms in its usual expansion. Yet Gaussian elimination computes it with far fewer operations. The reason is that all those terms implement one simple rule: alternating multilinear dependence on the columns. This treatment follows the determinant and elimination chapters of MIT 18.06 (Strang), with the exterior-power interpretation as a structural continuation.[1]

§58.1 Inversions determine permutation signs

A permutation $\sigma = (\sigma(1), \dots, \sigma(n))$ has an inversion at $i < j$ when $\sigma(i) > \sigma(j)$. Put

$$\operatorname{inv}(\sigma) = \#\{(i,j) : i < j,\ \sigma(i) > \sigma(j)\}, \qquad \operatorname{sgn}(\sigma) = (-1)^{\operatorname{inv}(\sigma)}.$$

For the original example $\sigma = 426315$, the numbers of smaller entries to the right of each position are $3, 1, 3, 1, 0, 0$. Hence

$$\operatorname{inv}(426315) = 3 + 1 + 3 + 1 = 8, \qquad \operatorname{sgn}(\sigma) = +1.$$

Swapping two adjacent positions changes the inversion count by exactly one. Every permutation can be reduced to the identity by adjacent swaps (for example, bubble sort), and consequently its sign is the product of -1 over the swaps. In particular, composing permutations multiplies signs:

$$\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}(\sigma)\operatorname{sgn}(\tau).$$

Although the number of swaps in a reduction is not unique, its parity is fixed by the inversion count.

§58.2 Why the Leibniz formula must alternate

For $A = (a_{ij})$, define

$$\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^{n} a_{i,\sigma(i)}.$$

The terms select one matrix entry in each row and each column. If two columns are interchanged, reindexing permutations by the corresponding transposition pairs the same products with opposite signs; hence the determinant changes sign. It is also linear in each column separately.

A useful characterization is that there is exactly one alternating multilinear function $D(v_1, \dots, v_n)$ with $D(e_1, \dots, e_n) = 1$. Indeed, expand every $v_j = \sum_i a_{ij} e_i$ by multilinearity. Terms with repeated basis vectors vanish; the remaining terms are indexed by permutations and have the signs above.

This argument derives the $n!$-term formula from a short list of geometric axioms, rather than memorizing the formula and reading its consequences backward.

[1] https://web.mit.edu/18.06/www/index.shtml

§58.3 Elementary operations and triangular determinants

Alternation gives $D(\dots, v_i, \dots, v_i, \dots) = 0$. Therefore

$$\det(\dots, v_j + cv_i, \dots) = \det(\dots, v_j, \dots) \quad (i \neq j).$$

Swapping columns changes sign and multiplying one column by c multiplies the determinant by c. Exactly analogous rules hold for rows, since $\det A = \det A^T$.

For a triangular matrix U, the only nonzero term in the Leibniz formula is the identity permutation, so

$$\det U = \prod_{i=1}^{n} U_{ii}.$$

Row reduction is thus a determinant algorithm: record the effects of row exchanges and scalings, and finish by multiplying diagonal entries.

§58.4 A rank-one matrix calculation

Consider the $n \times n$ matrix

$$A = \begin{pmatrix} x & a & \cdots & a \\ a & x & \cdots & a \\ \vdots & \vdots & \ddots & \vdots \\ a & a & \cdots & x \end{pmatrix} = (x-a)I + aJ,$$

where $J = \mathbf{1}\mathbf{1}^T$ has rank one. On the line spanned by $\mathbf{1} = (1, \dots, 1)^T$, J acts by multiplication by n. On the $(n-1)$-dimensional subspace

$$W = \{v : v_1 + \cdots + v_n = 0\},$$

it acts by zero. Since the direct sum of these spaces is $\mathbb{R}^n$,

$$\boxed{\det A = (x + (n-1)a)(x-a)^{n-1}.}$$

For $n = 3$, $x = 2$, $a = 1$, this predicts $\det A = 4$. Direct row elimination on $\left(\begin{smallmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{smallmatrix}\right)$ gives the same result. The symmetry decomposition explains the factorization for every n.

§58.5 Exterior powers remove the choice of coordinates

Let V be n-dimensional. The top exterior power $\bigwedge^n V$ is a one-dimensional vector space. A linear map $A : V \to V$ induces

$$\bigwedge^n A : \bigwedge^n V \longrightarrow \bigwedge^n V, \qquad v_1 \wedge \cdots \wedge v_n \mapsto Av_1 \wedge \cdots \wedge Av_n.$$

An endomorphism of a one-dimensional space is multiplication by a scalar; this scalar is $\det A$. In a basis $e_1, \dots, e_n$,

$$Ae_1 \wedge \cdots \wedge Ae_n = (\det A)\, e_1 \wedge \cdots \wedge e_n.$$

Now multiplicativity has a direct proof: functoriality of the exterior power gives

$$\bigwedge^n (AB) = (\bigwedge^n A)(\bigwedge^n B), \qquad \det(AB) = \det A \det B.$$

We do not need to manipulate two nested sums over permutations.

§58.6 Minors are the matrix of an exterior-power map

For $A : V \to W$ with bases e_i and f_j, the induced map $\bigwedge^k A$ has matrix entries

$$(\bigwedge^k A)_{J,I} = \det A_{J,I},$$

the $k \times k$ minors. This follows by expanding

$$(Ae_{i_1}) \wedge \cdots \wedge (Ae_{i_k})$$

and collecting coefficients of $f_{j_1} \wedge \cdots \wedge f_{j_k}$. Therefore

$$\operatorname{rank} A < k \quad \Longleftrightarrow \quad \bigwedge^k A = 0 \quad \Longleftrightarrow \quad \text{all } k \times k \text{ minors vanish.}$$

It is possible to recognize a rank condition directly from polynomial equations in the matrix entries. This is why minors reappear in algebraic geometry.

§58.7 Cramer's rule is alternating multilinearity again

If A is invertible with columns $a_1, \ldots, a_n$, write

$$b = \sum_{j=1}^{n} x_j a_j.$$

Let $A_i(b)$ be obtained by replacing column i of A by b. Multilinearity yields

$$\det A_i(b) = \sum_j x_j \det(a_1, \ldots, a_{i-1}, a_j, a_{i+1}, \ldots, a_n).$$

All terms with $j \neq i$ have repeated columns and vanish, so $\det A_i(b) = x_i \det A$. Thus

$$x_i = \frac{\det A_i(b)}{\det A}.$$

This proof explains the volume-ratio interpretation precisely, with $\det A \neq 0$ essential. The Leibniz sum, row reduction, minors, exterior powers and Cramer's rule are not five independent pieces of determinant folklore; each is forced by the same alternating multilinear operation.

What Does "Product" Mean? Spatial Vectors, Cross Products, and Universal Products

N-20220728

§59.1 One word can hide different mathematical types

In elementary mathematics, the word "product" often means the output of a multiplication. With vectors, however, several constructions carry that name:

$$a \cdot b \in \mathbb{R}, \qquad a \times b \in \mathbb{R}^3, \qquad (a, b) \in V \times W.$$

These expressions do not merely compute different values. They have different *types*. The dot product returns a scalar, the cross product returns a vector, and a Cartesian product is a new space whose elements are ordered pairs.

The distinction becomes even sharper in category theory. A categorical product is not primarily a binary operation on elements. It is an object characterized by projection maps and a universal mapping property. The phrase "product of X and Y" then means an object that packages maps into X and Y in the most economical possible way.

This chapter follows the word through two settings. Spatial vector geometry shows how metric and orientation create the dot and cross products. Categorical products show how a construction can instead be characterized by the maps it receives. The comparison is useful only if the two meanings are developed separately before they are placed side by side.

§59.2 Vectors, Coordinates, and Bases

Let V be a three-dimensional real vector space. A basis

$$e_1, e_2, e_3$$

means that every vector $v \in V$ has a unique expression

$$v = v^1 e_1 + v^2 e_2 + v^3 e_3.$$

The triple (v^1, v^2, v^3) is not the vector itself; it is the coordinate list of the vector in this basis. Changing the basis changes the list while leaving the geometric vector unchanged.

In an orthonormal basis of Euclidean space, vector addition and scalar multiplication act componentwise:

$$(a^1, a^2, a^3) + (b^1, b^2, b^3) = (a^1 + b^1, a^2 + b^2, a^3 + b^3),$$

$$\lambda(a^1, a^2, a^3) = (\lambda a^1, \lambda a^2, \lambda a^3).$$

The zero vector has coordinates $(0, 0, 0)$. Nonzero vectors a, b are parallel exactly when

$$a = \lambda b$$

for some nonzero scalar λ. Three vectors form a basis exactly when no one lies in the plane spanned by the other two.

A coordinate list therefore answers a decomposition problem: which scalar multiples of the basis vectors assemble the given vector? Later, an ordered pair in a categorical product will answer a different packaging problem: which two component maps assemble a map into the product?

§59.3 The dot product adds metric information

A bare vector space has addition and scalar multiplication, but no intrinsic lengths or angles. An inner product adds that geometry. In an orthonormal basis,

$$a \cdot b = a^1 b^1 + a^2 b^2 + a^3 b^3.$$

The norm is

$$\|a\| = \sqrt{a \cdot a},$$

and for nonzero vectors the angle $\theta \in [0, \pi]$ is determined by

$$\cos\theta = \frac{a \cdot b}{\|a\|\|b\|}.$$

Perpendicularity becomes the algebraic condition

$$a \cdot b = 0.$$

The dot product also computes projection. To approximate v by a multiple of a nonzero vector u, minimize

$$\|v - tu\|^2.$$

Expanding gives

$$\|v - tu\|^2 = \|v\|^2 - 2t(u \cdot v) + t^2\|u\|^2.$$

The quadratic is minimized at

$$t = \frac{u \cdot v}{u \cdot u}.$$

Thus

$$\mathrm{proj}_u(v) = \frac{u \cdot v}{u \cdot u} u.$$

The formula is not merely a coordinate trick: the inner product measures the component of v along u.

For example, with

$$u = (1, 1, 0), \qquad v = (2, 0, 1),$$

we have

$$u \cdot v = 2, \qquad u \cdot u = 2,$$

so

$$\mathrm{proj}_u(v) = u = (1, 1, 0).$$

The residual

$$v - u = (1, -1, 1)$$

is orthogonal to u, as the minimizing condition predicts.

§59.4 The cross product packages area and orientation

In an oriented Euclidean three-space, the cross product of a and b is the unique vector $a \times b$ satisfying three conditions:

(i) it is perpendicular to both a and b;

(ii) its length is
$$\|a \times b\| = \|a\|\|b\| \sin\theta;$$

(iii) its direction follows the chosen orientation.

The length is the area of the parallelogram spanned by the two vectors. The sign convention distinguishes a right-handed frame from a left-handed one. Reversing the inputs reverses the orientation:

$$a \times b = -(b \times a).$$

In a positively oriented orthonormal basis,

$$a \times b = \begin{pmatrix} a^2 b^3 - a^3 b^2 \\ a^3 b^1 - a^1 b^3 \\ a^1 b^2 - a^2 b^1 \end{pmatrix}.$$

Take
$$a = (1, 2, 0), \qquad b = (0, 1, 3).$$

Then
$$a \times b = (6, -3, 1).$$

A direct check gives
$$a \cdot (a \times b) = 6 - 6 = 0,$$
$$b \cdot (a \times b) = -3 + 3 = 0.$$

Moreover,
$$\|a \times b\|^2 = 36 + 9 + 1 = 46.$$

The Gram determinant of a, b is
$$\det \begin{pmatrix} a \cdot a & a \cdot b \\ b \cdot a & b \cdot b \end{pmatrix} = \det \begin{pmatrix} 5 & 2 \\ 2 & 10 \end{pmatrix} = 46,$$
confirming that the squared area equals $\|a \times b\|^2$.

The cross product therefore depends on more structure than vector addition: it uses the metric to define perpendicularity and length, the orientation to choose a sign, and the fact that the dimension is three to turn an oriented area into a perpendicular vector.

§59.5 The scalar triple product measures oriented volume

Pairing the cross product with a third vector gives

$$[a, b, c] = a \cdot (b \times c).$$

This scalar is the oriented volume of the parallelepiped spanned by a, b, c. In coordinates,

$$a \cdot (b \times c) = \det \begin{pmatrix} a^1 & a^2 & a^3 \\ b^1 & b^2 & b^3 \\ c^1 & c^2 & c^3 \end{pmatrix}.$$

Swapping two vectors reverses the sign. The value vanishes exactly when the three vectors are coplanar.

For

$$a = (1, 0, 1), \quad b = (0, 2, 0), \quad c = (1, 1, 3),$$

we obtain

$$b \times c = (6, 0, -2),$$

so

$$a \cdot (b \times c) = 4.$$

The parallelepiped has volume 4, and the positive sign says that the ordered triple has the chosen orientation.

The determinant, cross product, and scalar triple product are thus not three unrelated formulas. They are three ways of reading the same oriented-volume structure in three dimensions.

§59.6 A cross product can generate a rotation

Fix $\omega \in \mathbb{R}^3$. The map

$$v \longmapsto \omega \times v$$

is linear, so in coordinates it is represented by the skew-symmetric matrix

$$[\omega]_\times = \begin{pmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{pmatrix}.$$

Skew-symmetry follows from

$$u \cdot (\omega \times v) = -v \cdot (\omega \times u).$$

If a rigid body rotates with angular velocity ω, the instantaneous velocity of a point at position r is

$$r'(t) = \omega \times r(t) = [\omega]_\times r(t).$$

The solution is

$$r(t) = e^{t[\omega]_\times} r(0).$$

Suppose ω is a unit vector. Let

$$K = [\omega]_\times.$$

The vector triple-product identity gives

$$K^2 v = \omega \times (\omega \times v) = \omega(\omega \cdot v) - v.$$

If

$$P = \omega\omega^T$$

is projection onto the rotation axis, then

$$K^2 = P - I, \qquad K^3 = -K.$$

The exponential series therefore collapses to

$$\boxed{e^{tK} = I + \sin t\, K + (1 - \cos t)K^2.}$$

This is Rodrigues' rotation formula. It fixes the component parallel to ω and rotates the perpendicular plane by angle t.

For the axis $\omega = e_3$,

$$K = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

and Rodrigues' formula gives

$$e^{tK} = \begin{pmatrix} \cos t & -\sin t & 0 \\ \sin t & \cos t & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The same operation that first measured oriented area now appears as the infinitesimal generator of spatial rotation.

§59.7 Why three dimensions are exceptional

The elementary coordinate formula can make the cross product look universal, but its target being another vector is special. Two vectors naturally span an oriented parallelogram. The algebraic object recording that area is a bivector, written

$$a \wedge b \in \Lambda^2 V.$$

In dimension three,

$$\dim \Lambda^2 V = \binom{3}{2} = 3 = \dim V.$$

Once a metric and orientation are chosen, oriented area elements can be identified with perpendicular vectors. The cross product is the result of that identification.

In dimension four,

$$\dim \Lambda^2 V = \binom{4}{2} = 6,$$

so a general oriented two-dimensional area has six independent components and cannot be represented by one ordinary four-vector. In dimension two, $\Lambda^2 V$ is one-dimensional, so the oriented area is naturally a scalar multiple of the plane's area form rather than another vector in the plane.

The following note develops this exterior-algebra explanation and derives vector identities from wedge product, contraction, and the Hodge star. Here the dimension count already tells us why the familiar binary cross product should not be copied blindly into arbitrary dimensions.

§59.8 Cartesian Products of Data

Now change settings. For sets X and Y, the Cartesian product is

$$X \times Y = \{(x, y) : x \in X,\ y \in Y\}.$$

Its elements are ordered pairs. The projections

$$\pi_X : X \times Y \to X, \qquad \pi_Y : X \times Y \to Y$$

recover the two components:

$$\pi_X(x, y) = x, \qquad \pi_Y(x, y) = y.$$

A function

$$h : Z \to X \times Y$$

is therefore the same information as a pair of functions

$$f : Z \to X, \qquad g : Z \to Y.$$

From h, take

$$f = \pi_X \circ h, \qquad g = \pi_Y \circ h.$$

Conversely, from f, g, define

$$\langle f, g \rangle(z) = (f(z), g(z)).$$

These constructions are inverse to one another.

For example, a parametrized plane curve

$$h(t) = (\cos t, \sin t)$$

is exactly the pairing of its coordinate functions:

$$h = \langle \cos, \sin \rangle.$$

The Cartesian product does not multiply the coordinates. It stores them side by side while allowing each to be recovered independently.

§59.9 Products without Ordered Pairs

A category has objects, morphisms between objects, identity morphisms, and associative composition. In the category **Set**, objects are sets and morphisms are functions. In $\mathbf{Vect}_{\mathbb{R}}$, objects are real vector spaces and morphisms are linear maps. In **Grp**, objects are groups and morphisms are group homomorphisms.

The elementwise definition of $X \times Y$ makes sense in **Set**, but category theory asks for a characterization using only morphisms.

Definition 59.9.1 (Categorical product)**.** A product of objects X, Y is an object P with morphisms

$$\pi_X : P \to X, \qquad \pi_Y : P \to Y$$

such that for every object Z and every pair of morphisms

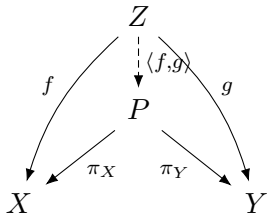
$$f : Z \to X, \qquad g : Z \to Y,$$

there exists a unique morphism

$$\langle f, g \rangle : Z \to P$$

with

$$\pi_X \circ \langle f, g \rangle = f, \qquad \pi_Y \circ \langle f, g \rangle = g.$$



Existence says that the two component maps can be packaged into one map. Uniqueness says that no additional hidden data are present. Any map into P is completely determined by its two projections.

In **Set**, the ordinary ordered-pair construction satisfies this property. If

$$h : Z \to X \times Y$$

has projections f, g, then for every z,

$$h(z) = (\pi_X h(z), \pi_Y h(z)) = (f(z), g(z)).$$

Thus

$$h = \langle f, g \rangle$$

and the mediating map is unique.

§59.10 The universal property makes products unique

Suppose

$$(P, \pi_X, \pi_Y) \quad \text{and} \quad (Q, \rho_X, \rho_Y)$$

are both products of X, Y. The product property of P, applied to the maps ρ_X, ρ_Y, gives a unique map

$$u : Q \to P$$

with

$$\pi_X u = \rho_X, \qquad \pi_Y u = \rho_Y.$$

Similarly there is a unique map

$$v : P \to Q$$

with

$$\rho_X v = \pi_X, \qquad \rho_Y v = \pi_Y.$$

Now both $u \circ v$ and id_P have the same projections:

$$\pi_X uv = \pi_X, \qquad \pi_Y uv = \pi_Y.$$

Uniqueness in the product property forces

$$u \circ v = \mathrm{id}_P\,.$$

Likewise,

$$v \circ u = \mathrm{id}_Q\,.$$

Therefore P and Q are uniquely isomorphic in a way that preserves their projections.

This is why category theory says "the product" even though many concrete models may exist. The universal property determines the object up to a unique structure-preserving relabelling.

§59.11 Products in vector spaces and groups

In $\mathbf{Vect}_{\mathbb{R}}$, the product of V and W is the vector space

$$V \times W$$

with componentwise operations:

$$(v, w) + (v', w') = (v + v', w + w'),$$

$$\lambda(v, w) = (\lambda v, \lambda w).$$

The projections are linear. Given linear maps

$$f : U \to V, \qquad g : U \to W,$$

the pairing

$$\langle f, g\rangle(u) = (f(u), g(u))$$

is linear and is the unique map with the required projections.

For finite-dimensional spaces,

$$\dim(V \times W) = \dim V + \dim W.$$

Choosing bases of V and W concatenates them into a basis of the product. A vector in $V \times W$ is therefore decomposed into a V-component and a W-component, just as a spatial vector is decomposed into scalar coordinates after a basis is chosen. The similarity lies in packaging components; the universal property, not a coordinate formula, defines the product.

In **Grp**, the product of groups G, H has elements (g, h) and multiplication

$$(g, h)(g', h') = (gg', hh').$$

The projections are homomorphisms, and a pair of homomorphisms into G, H combines uniquely into a homomorphism into $G \times H$.

These examples show why the categorical definition is useful: the same diagram characterizes products in settings whose elements carry different algebraic structures.

The three main constructions in this chapter can now be compared without relying on the shared English word:

construction	input	output
dot product	$V \times V$	$\mathbb{R}$
cross product	$\mathbb{R}^3 \times \mathbb{R}^3$	$\mathbb{R}^3$
categorical product	X, Y	an object P with projections

The categorical product $V \times V$ may serve as the *domain* of the dot or cross product, but neither bilinear operation is itself a categorical product. The type signatures already prevent the confusion:

$$\cdot : V \times V \to \mathbb{R}, \qquad \times : \mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R}^3.$$

A categorical product is characterized by maps *into* it, while a dot or cross product is a particular map *out of* a product object.

The difference can also be seen as information preservation. An element of $V \times V$ retains both inputs because the projections recover them exactly:

$$\pi_1(u, v) = u, \qquad \pi_2(u, v) = v.$$

The cross product deliberately discards information. For example,

$$e_1 \times e_2 = e_3$$

and

$$(2e_1) \times \left(\frac{1}{2}e_2\right) = e_3.$$

The same output comes from different ordered pairs, so no pair of projections from the output vector can reconstruct the original inputs. The dot product loses still more: every orthogonal pair has output zero. These operations are valuable precisely because they extract a selected geometric invariant rather than preserving all component data.

In categorical language, the cross product is a morphism

$$V \times V \longrightarrow V$$

whose domain happens to be a product object. A product cone consists instead of an object together with projection morphisms *from* that object. Reversing these arrows changes the mathematical question completely.

The dependencies are equally different. The dot product depends on a metric. The cross product depends on metric, orientation, and three-dimensionality. The categorical product depends only on the morphisms of the category and its universal property. Understanding a construction therefore means more than knowing a formula: one must know its type, the structure it requires, and the property that makes it canonical.

Vector Identities from Exterior Algebra: Wedge, Contraction, and the Hodge Star

N-20230205

§60.1 Why the vector triple product is easy to misremember

The identity

$$u \times (v \times w) = (u \cdot w)v - (u \cdot v)w$$

is often compressed into the mnemonic "BAC minus CAB." The mnemonic remembers the order of the letters, but it does not explain why the result lies in the v, w-plane or why the two coefficients have those signs.

The geometry already determines most of the answer. Since

$$v \times w$$

is perpendicular to the plane spanned by v, w, the vector

$$u \times (v \times w)$$

is perpendicular to $v \times w$ and therefore lies in that plane. Hence

$$u \times (v \times w) = Av + Bw$$

for some scalars A, B.

A quick test fixes the coefficients. If $u = v$ and $v \perp w$, then

$$v \times (v \times w) = -\|v\|^2 w.$$

Thus the coefficient of w must involve $-u \cdot v$. If $u = w$ with $v \perp w$, then

$$w \times (v \times w) = \|w\|^2 v,$$

so the coefficient of v must involve $u \cdot w$.

This reasoning reveals the projection structure, but it still relies on familiar cross-product behavior. The rest of the chapter builds a framework in which the identity follows from two systematic operations: contraction and Hodge duality.

§60.2 The scalar triple product is a volume form

The scalar triple product is

$$[u, v, w] = u \cdot (v \times w).$$

It equals the oriented volume of the parallelepiped spanned by the ordered triple. In a positively oriented orthonormal basis,

$$[u, v, w] = \det \begin{pmatrix} u^1 & u^2 & u^3 \\ v^1 & v^2 & v^3 \\ w^1 & w^2 & w^3 \end{pmatrix}.$$

The expression is multilinear and alternating. If two inputs are exchanged, the sign changes; if the vectors are linearly dependent, the value is zero.

These properties are better packaged by the Euclidean volume form

$$\mathrm{vol} = e^1 \wedge e^2 \wedge e^3,$$

where e^1, e^2, e^3 is the dual coframe. Then

$$\mathrm{vol}(u, v, w) = u \cdot (v \times w).$$

The cross product can therefore be characterized without coordinates: $v \times w$ is the unique vector whose inner product with every u equals the oriented volume of u, v, w.

For example, let

$$u = (1, 0, 1), \quad v = (0, 2, 0), \quad w = (1, 1, 3).$$

Then

$$v \times w = (6, 0, -2),$$

and

$$u \cdot (v \times w) = 4.$$

The sign is positive because the ordered triple agrees with the chosen orientation, and the absolute value is the geometric volume.

§60.3 Wedge Products and Oriented Planes

Given vectors u, v in a vector space V, their wedge product

$$u \wedge v \in \Lambda^2 V$$

is a bivector. It represents the oriented parallelogram spanned by the pair. The defining properties are bilinearity and alternation:

$$(au + bv) \wedge w = a(u \wedge w) + b(v \wedge w),$$

$$u \wedge v = -v \wedge u, \qquad u \wedge u = 0.$$

The vanishing condition

$$u \wedge v = 0$$

holds exactly when u, v are linearly dependent.

In a three-dimensional basis e_1, e_2, e_3, the bivectors

$$e_1 \wedge e_2, \qquad e_2 \wedge e_3, \qquad e_3 \wedge e_1$$

form a basis of $\Lambda^2 V$. If

$$u = u^i e_i, \qquad v = v^j e_j,$$

then

$$\begin{aligned} u \wedge v = {} & (u^1 v^2 - u^2 v^1) e_1 \wedge e_2 \\ & + (u^2 v^3 - u^3 v^2) e_2 \wedge e_3 \\ & + (u^3 v^1 - u^1 v^3) e_3 \wedge e_1. \end{aligned}$$

The three coefficients are the familiar 2×2 minors appearing in the cross product. At this stage, however, the result is an oriented area element, not yet a perpendicular vector.

That distinction matters in other dimensions. In four dimensions,

$$\dim \Lambda^2 V = \binom{4}{2} = 6,$$

so an oriented plane carries six independent components and cannot be encoded by an ordinary four-vector. The wedge product is the dimension-independent object; the cross product is a special identification available in three-dimensional metric geometry.

§60.4 The metric turns vectors into covectors

The Hodge star is most naturally defined on differential forms, so first use the Euclidean metric to convert vectors into covectors. For $u \in V$, define

$$u^\flat(v) = u \cdot v.$$

This is a one-form. The inverse operation is denoted by $\sharp$:

$$(u^\flat)^\sharp = u.$$

In an orthonormal basis the coordinate lists happen to agree, but the types do not. A vector is an input to a one-form; a one-form is a linear measurement of vectors.

The metric also induces an inner product on k-forms. On decomposable two-forms,

$$\left\langle u^\flat \wedge v^\flat, w^\flat \wedge z^\flat \right\rangle = \det \begin{pmatrix} u \cdot w & u \cdot z \\ v \cdot w & v \cdot z \end{pmatrix}.$$

In particular,

$$\|u^\flat \wedge v^\flat\|^2 = \|u\|^2 \|v\|^2 - (u \cdot v)^2.$$

The right-hand side is the Gram determinant of u, v, so the norm of the wedge product is exactly the area of their parallelogram.

Thus the metric does two jobs: it lets us compare forms by an inner product, and it identifies vectors with one-forms. Orientation will supply the remaining step that turns a two-form into a one-form.

§60.5 The Hodge star converts complementary dimensions

Let V be an oriented Euclidean n-space with volume form vol. The Hodge star is the linear map

$$* : \Lambda^k V^* \to \Lambda^{n-k} V^*$$

defined by

$$\alpha \wedge *\beta = \langle \alpha, \beta \rangle \, \mathrm{vol}$$

for all k-forms α, β.

In an oriented orthonormal coframe of three-space,

$$*(e^1 \wedge e^2) = e^3, \qquad *(e^2 \wedge e^3) = e^1, \qquad *(e^3 \wedge e^1) = e^2.$$

The cross product is precisely

$$\boxed{(u \times v)^\flat = *(u^\flat \wedge v^\flat).}$$

The wedge product forms the oriented area, the Hodge star selects the complementary one-form using metric and orientation, and $\sharp$ converts that one-form back into a vector.

This formula immediately gives perpendicularity. Pairing with u,

$$\begin{aligned} u \cdot (u \times v)\,\mathrm{vol} &= u^\flat \wedge *(u \times v)^\flat \\ &= u^\flat \wedge u^\flat \wedge v^\flat = 0. \end{aligned}$$

The same argument works with v. Since the Hodge star preserves norms,

$$\|u \times v\| = \|u^\flat \wedge v^\flat\|,$$

so the cross-product norm equals the parallelogram area.

The ordinary right-hand rule is therefore not an extra mnemonic attached to a coordinate determinant. It is the visible consequence of the chosen orientation in the definition of the Hodge star.

§60.6 Lagrange's identity is an inner product of bivectors

Because the Hodge star is an isometry,

$$\begin{aligned} (u \times v) \cdot (w \times z) &= \left\langle *(u^\flat \wedge v^\flat), *(w^\flat \wedge z^\flat) \right\rangle \\ &= \left\langle u^\flat \wedge v^\flat, w^\flat \wedge z^\flat \right\rangle. \end{aligned}$$

Using the induced inner product on two-forms yields

$$\boxed{(u \times v) \cdot (w \times z) = (u \cdot w)(v \cdot z) - (u \cdot z)(v \cdot w).}$$

This is Lagrange's identity.

Setting $w = u$, $z = v$ gives

$$\|u \times v\|^2 = \|u\|^2 \|v\|^2 - (u \cdot v)^2.$$

The nonnegativity of the left-hand side gives the Cauchy–Schwarz inequality

$$|u \cdot v| \le \|u\| \|v\|.$$

Equality holds exactly when the wedge product vanishes, that is, when u, v are linearly dependent.

What looks like a special identity about a three-dimensional operation is therefore the determinant formula for the inner product of two oriented area elements.

§60.7 Contraction of Alternating Forms

For a vector u, the interior product or contraction

$$\iota_u : \Lambda^k V^* \to \Lambda^{k-1} V^*$$

is defined by inserting u into the first slot:

$$(\iota_u \alpha)(v_2, \dots, v_k) = \alpha(u, v_2, \dots, v_k).$$

For one-forms α, β,

$$\iota_u(\alpha \wedge \beta) = \alpha(u)\beta - \beta(u)\alpha.$$

Taking $\alpha = v^\flat$, $\beta = w^\flat$ gives

$$\iota_u(v^\flat \wedge w^\flat) = (u \cdot v)w^\flat - (u \cdot w)v^\flat.$$

In an oriented Euclidean three-space, the identity

$$*(u^\flat \wedge *\eta) = -\iota_u \eta$$

holds for every two-form η. Apply it to

$$\eta = v^\flat \wedge w^\flat.$$

Then

$$\begin{aligned}(u \times (v \times w))^\flat &= *\left(u^\flat \wedge (v \times w)^\flat\right) \\ &= *\left(u^\flat \wedge *(v^\flat \wedge w^\flat)\right) \\ &= -\iota_u(v^\flat \wedge w^\flat) \\ &= (u \cdot w)v^\flat - (u \cdot v)w^\flat.\end{aligned}$$

Applying $\sharp$ gives

$$\boxed{u \times (v \times w) = (u \cdot w)v - (u \cdot v)w.}$$

The BAC–CAB identity is no longer a mnemonic. It is the contraction rule for an oriented area form, translated back into vectors by the Hodge star.

§60.8 Volume and the Levi–Civita Symbol

In a positively oriented orthonormal basis, define

$$\varepsilon_{ijk} = \begin{cases} 1, & (i,j,k) \text{ is an even permutation of } (1,2,3), \\ -1, & (i,j,k) \text{ is an odd permutation}, \\ 0, & \text{two indices coincide.} \end{cases}$$

These are the components of the volume form in that basis. The cross product becomes

$$(u \times v)_i = \varepsilon_{ijk} u_j v_k.$$

The repeated indices are summed, while i remains free and labels the output component.

The key contraction is

$$\varepsilon_{ijk}\varepsilon_{klm} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}.$$

Therefore

$$\begin{aligned}(u \times (v \times w))_i &= \varepsilon_{ijk} u_j \varepsilon_{klm} v_l w_m \\ &= (\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}) u_j v_l w_m \\ &= v_i(u_j w_j) - w_i(u_j v_j).\end{aligned}$$

This is the component proof of the same contraction identity.

The notation must be interpreted carefully. The array with entries $0, \pm 1$ is tied to an oriented orthonormal basis. In arbitrary curvilinear coordinates, the components of the geometric volume form include the metric factor

$$\sqrt{\det g}.$$

Treating the permutation symbol as though it were an unchanged tensor array in every coordinate system leads to missing Jacobian factors.

§60.9 The same operators produce curl and divergence

The Hodge star is not limited to static vector identities. On oriented Euclidean three-space, a vector field X becomes a one-form $X^\flat$. Its exterior derivative

$$d(X^\flat)$$

is a two-form measuring infinitesimal circulation. Applying the Hodge star and converting back to a vector gives curl:

$$\boxed{(\operatorname{curl} X)^\flat = *d(X^\flat).}$$

Consider the rotation field

$$X(x, y, z) = (-y, x, 0).$$

Then

$$X^\flat = -y\,dx + x\,dy.$$

Taking the exterior derivative,

$$\begin{aligned} d(X^\flat) &= -dy \wedge dx + dx \wedge dy \\ &= 2\,dx \wedge dy. \end{aligned}$$

Since

$$*(dx \wedge dy) = dz,$$

we obtain

$$(\operatorname{curl} X)^\flat = 2\,dz,$$

so

$$\operatorname{curl} X = (0, 0, 2).$$

The field rotates around the z-axis with constant vorticity.

Divergence uses the complementary construction:

$$d(*X^\flat) = (\operatorname{div} X)\,\mathrm{vol}\,.$$

For the radial field

$$Y = (x, y, z),$$

we have

$$Y^\flat = x\,dx + y\,dy + z\,dz$$

and

$$*Y^\flat = x\,dy \wedge dz - y\,dx \wedge dz + z\,dx \wedge dy.$$

Differentiating gives

$$d(*Y^\flat) = 3\,dx \wedge dy \wedge dz,$$

so

$$\operatorname{div} Y = 3.$$

§60.10 A small square measures the same curl

The formula for curl has a directly measurable interpretation. Take the same field

$$X = (-y, x, 0)$$

and orient the rectangle $S = [0, a] \times [0, b]$ in the xy-plane counterclockwise. On its four boundary edges, the line integrals are

edge	$\int X \cdot d\ell$
$y = 0,\ x : 0 \to a$	0
$x = a,\ y : 0 \to b$	ab
$y = b,\ x : a \to 0$	ab
$x = 0,\ y : b \to 0$	0

so $\oint_{\partial S} X \cdot d\ell = 2ab$. We already computed $\operatorname{curl} X = (0, 0, 2)$; its flux through S is also $2ab$.

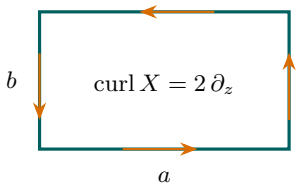


Figure 60.1: The boundary integral for the rotating field is $2ab$, exactly the curl flux through the rectangle.

Stokes' theorem packages this equality as

$$\oint_{\partial S} X^\flat = \int_S d(X^\flat).$$

One more exterior derivative produces two classical identities with almost no calculation:

$$\operatorname{curl}(\nabla f) = 0, \qquad \operatorname{div}(\operatorname{curl} X) = 0.$$

Both come from $d^2 = 0$. The first applies d^2 to the 0-form f; the second applies it to the 1-form $X^\flat$ and uses the Hodge star to translate a 2-form into a vector field. The apparent miracles of vector-calculus notation become ordinary algebraic consequences of taking the boundary of a boundary.

Linear Maps, Matrix Representations, and Tensor Invariance

N-20230209

§61.1 From vector identities to linear representations

Cross products, scalar triple products, exterior algebra, and the Hodge star motivate a broader question: how do coordinate arrays represent invariant linear or multilinear objects?

A linear map $T : V \to W$ is an object independent of coordinates. A matrix is what appears only after choosing a basis of V and a basis of W. Similarly, a tensor is not the displayed array of numbers; the array is a coordinate representation whose transformation law tells us what object is being represented.

Remark 61.1.1 (The type of a matrix matters) — The same rectangular array may represent a bilinear form, an endomorphism, or a tensor with several kinds of slots. The distinction is visible in the transformation law: similarity acts on an operator, while congruence acts on a bilinear form. The next calculations make this difference explicit.

§61.2 Coordinates as isomorphisms

A basis is not just a list of vectors; it is an isomorphism from an abstract vector space to a coordinate space. If $B = (b_1, \dots, b_n)$ is a basis of V, then

$$[\,]_B : V \longrightarrow k^n$$

sends a vector to its coordinate column in that basis. The vector v is intrinsic; the column $[v]_B$ depends on B.

This distinction is the source of all basis-change formulas. A matrix is not the linear map itself. A matrix is the expression of a linear map after choosing coordinate isomorphisms.

Definition 61.2.1 (Matrix of a linear map)**.** Let $T : V \to W$ be a linear map, let B be a basis of V, and let C be a basis of W. The matrix of T in these bases is the matrix $[T]_{C,B}$ satisfying

$$[T(v)]_C = [T]_{C,B}[v]_B$$

for all $v \in V$. Equivalently,

$$[T]_{C,B} = [\,]_C \circ T \circ [\,]_B^{-1}.$$

The last formula is often the cleanest one. It says that the bottom arrow in coordinate space is obtained by transporting the actual map T through coordinate isomorphisms.

§61.3 The basis-change formula

Suppose B, B' are two bases of V, and C, C' are two bases of W. Let

$$[v]_B = P[v]_{B'}, \qquad [w]_C = Q[w]_{C'}.$$

Thus P converts new V-coordinates into old V-coordinates, and Q converts new W-coordinates into old W-coordinates.

Proposition 61.3.1 (Matrix transformation rule)

If

$$A = [T]_{C,B}, \qquad A' = [T]_{C',B'},$$

then

$$A' = Q^{-1}AP.$$

Proof. Starting with new coordinates $[v]_{B'}$, first convert to old coordinates:

$$[v]_B = P[v]_{B'}.$$

Apply the old matrix A:

$$[T(v)]_C = A[v]_B = AP[v]_{B'}.$$

Finally convert the output from old C-coordinates to new C'-coordinates. Since $[w]_C = Q[w]_{C'}$, we have $[w]_{C'} = Q^{-1}[w]_C$. Hence

$$[T(v)]_{C'} = Q^{-1}AP[v]_{B'}.$$

Therefore $A' = Q^{-1}AP$. □

For an endomorphism $T : V \to V$, if the same basis change is used on domain and codomain, then $P = Q$ and

$$A' = P^{-1}AP.$$

This is similarity. Similar matrices are not two different operators; they are the same operator written in two coordinate systems.

§61.4 A worked basis-change example

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ have standard-basis matrix

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}.$$

Use the new basis

$$b_1 = (1, 0), \qquad b_2 = (1, 1).$$

The change-of-basis matrix whose columns are the new basis vectors in the old basis is

$$P = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Then

$$A' = P^{-1}AP = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}.$$

The operator did not change. The new basis reveals that b_1 and b_2 are eigenvectors, so the matrix becomes diagonal.

This example also explains the meaning of diagonalization: it is not a magical simplification of numbers; it is the search for a coordinate system adapted to the invariant directions of the operator.

§61.5 Which quantities are invariant?

For an endomorphism, changing basis replaces A by $P^{-1}AP$. Any expression unchanged under this replacement belongs to the operator rather than to the coordinate matrix.

Proposition 61.5.1 (Similarity invariants)
If $A' = P^{-1}AP$, then

$$\operatorname{tr}(A') = \operatorname{tr}(A), \qquad \det(A') = \det(A),$$

and A and A' have the same characteristic polynomial.

Proof. For trace,

$$\operatorname{tr}(P^{-1}AP) = \operatorname{tr}(APP^{-1}) = \operatorname{tr}(A),$$

using $\operatorname{tr}(XY) = \operatorname{tr}(YX)$. For determinant,

$$\det(P^{-1}AP) = \det(P^{-1})\det(A)\det(P) = \det(A).$$

For the characteristic polynomial,

$$\det(\lambda I - P^{-1}AP) = \det(P^{-1}(\lambda I - A)P) = \det(\lambda I - A).$$

□

Rank is invariant for maps between different spaces as well, because left and right multiplication by invertible matrices do not change the dimension of the image. Trace and determinant, however, are naturally invariants of endomorphisms, not arbitrary maps $V \to W$.

§61.6 Covectors and the inverse-transpose rule

A covector is a linear functional $\alpha : V \to k$. It pairs with a vector to give a scalar:

$$\alpha(v) \in k.$$

The scalar must not depend on coordinates.

Assume again that

$$[v]_B = P[v]_{B'}.$$

Write a_B and $a_{B'}$ for the coordinate columns of α in the dual bases, so that

$$\alpha(v) = a_B^T[v]_B = a_{B'}^T[v]_{B'}.$$

Substituting $[v]_B = P[v]_{B'}$, we get

$$a_B^T P[v]_{B'} = a_{B'}^T[v]_{B'}$$

for all v, hence

$$a_{B'} = P^T a_B.$$

Equivalently, if one writes coordinate changes in the opposite direction, the familiar inverse-transpose appears. The important point is not the slogan "covectors transform oppositely", but the invariant pairing

$$\alpha(v)$$

that forces the formula.

§61.7 Bilinear forms are not linear operators

A common source of confusion is that both a linear operator and a bilinear form may be represented by square matrices. Their basis-change laws are different.

Let $g : V \times V \to k$ be a bilinear form. In basis B, write

$$g(v, w) = [v]_B^T G [w]_B.$$

If $[v]_B = P[v]_{B'}$, then

$$g(v, w) = [v]_{B'}^T P^T G P [w]_{B'}.$$

Thus the matrix of the bilinear form in the new basis is

$$G' = P^T G P.$$

Compare this with the operator rule $A' = P^{-1} A P$. A matrix representing a bilinear form changes by congruence; a matrix representing an endomorphism changes by similarity.

Example 61.7.1 (The Euclidean metric in a non-orthonormal basis)
In the standard basis of $\mathbb{R}^2$, the Euclidean inner product has matrix I. For the basis

$$b_1 = (1, 0), \qquad b_2 = (1, 1),$$

with

$$P = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

the metric matrix becomes

$$G' = P^T I P = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

The inner product is still the same geometric object, but because the new basis is not orthonormal, its coordinate matrix is no longer the identity.

This is why statements such as "$A^T = A$" must be interpreted carefully. A symmetric matrix may represent a symmetric bilinear form in a chosen basis, or it may represent a self-adjoint operator relative to a chosen inner product. These are related but not identical ideas.

§61.8 Symmetric and skew parts

For an ordinary square matrix over a field of characteristic not 2, one can write

$$A = \frac{A + A^T}{2} + \frac{A - A^T}{2}.$$

The first part is symmetric, and the second part is skew-symmetric. But the coordinate-free meaning depends on what A represents.

If A represents a bilinear form, the symmetric part corresponds to

$$g_s(v, w) = \frac{g(v, w) + g(w, v)}{2},$$

and the skew part corresponds to

$$g_a(v, w) = \frac{g(v, w) - g(w, v)}{2}.$$

The skew part is an alternating 2-form.

If A represents a linear operator on an inner-product space, then self-adjointness is the condition

$$\langle Av, w\rangle = \langle v, Aw\rangle.$$

In an orthonormal basis this becomes $A^T = A$. In a non-orthonormal basis with metric matrix G, it becomes

$$A^T G = GA.$$

So the naive transpose test is valid only after the metric and basis have been fixed.

§61.9 Tensor slots and transformation laws

A tensor is best understood by its slots. The basic examples are:

object	tensor type
vector	V
covector	V^*
bilinear form	$V^* \otimes V^*$
linear operator	$V \otimes V^*$

The components transform according to which slots are vector slots and which slots are covector slots.

If $[v]_B = P[v]_{B'}$, then schematically:

object	components	basis-change law
vector	x	$x' = P^{-1}x$
covector	a	$a' = P^T a$
operator	A	$A' = P^{-1}AP$
bilinear form	G	$G' = P^T GP$

The formulas differ because the objects have different tensor types. Treating every square array as the same kind of object is the conceptual mistake tensor notation is designed to prevent.

§61.10 Tensor products and universal property

A bilinear map

$$B : V \times W \to U$$

is equivalently a linear map

$$\widetilde{B} : V \otimes W \to U$$

satisfying

$$\widetilde{B}(v \otimes w) = B(v, w).$$

This is not just notation. It says that the tensor product is the linear object that represents bilinear dependence.

For example, matrix multiplication can be viewed as contraction of repeated indices:

$$(AB)^i{}_j = A^i{}_k B^k{}_j.$$

The repeated index k is summed. The output has one upper and one lower index, so it is again an operator-type tensor.

Trace is another contraction:

$$\operatorname{tr}(A) = A^i{}_i.$$

This explains why trace is coordinate-invariant for an endomorphism: it contracts one vector slot with one covector slot, producing a scalar.

§61.11 Coordinate-free reading of the matrix identities

The scattered formulas involving A^{-1}, A^T, and A^{-T} should now be read as transformation laws:

(i) $P^{-1}AP$ belongs to linear operators under basis change;

(ii) P^TGP belongs to bilinear forms and quadratic forms;

(iii) inverse-transpose behavior belongs to covectors and dual bases;

(iv) trace is contraction of an operator-type tensor;

(v) determinant is the induced scalar on the determinant line.

The mature lesson is not merely that "coordinates may change". The real lesson is that each mathematical object has a type, and its type determines the correct transformation law. Tensor notation is useful because it keeps this type information visible.

Tensors in Nonorthogonal Coordinates: Dual Bases, Metrics, and Contraction

N-20230217

§62.1 Nonorthogonal Coordinates

In an orthonormal basis, vector calculus can hide a great deal of structure. The same list of numbers may be used for a vector and for the linear functional obtained by taking its dot product. Upper and lower indices look interchangeable. The identity matrix also serves as the metric matrix, and the dual basis is visually indistinguishable from the original basis.

All of these coincidences disappear as soon as the coordinate directions are tilted or stretched. That is not an inconvenience to be avoided. It is the setting in which tensor notation begins to explain what is genuinely geometric and what is merely a consequence of a particularly friendly basis.

Tilt the coordinate axes, and a simple geometric question becomes surprisingly delicate: what does the dot product of two vectors look like when their coordinate basis is no longer orthonormal? Answering it brings in the dual basis, the Gram matrix, raised and lowered indices, and tensor contraction, each for a concrete reason.

The basic lesson is that a geometric object does not change when coordinates change, but its components must change in a precisely compensating way.

§62.2 Tensor products package bilinear dependence

Suppose

$$B : V \times W \longrightarrow U$$

is bilinear. It is linear in each argument separately, but it is not a linear map on the Cartesian product in the usual sense. The tensor product is constructed so that bilinear data can be handled by ordinary linear algebra.

There is a vector space $V \otimes W$ and a bilinear map

$$\otimes : V \times W \longrightarrow V \otimes W$$

with the following universal property: for every bilinear map B, there is a unique linear map

$$\widetilde{B} : V \otimes W \longrightarrow U$$

such that

$$\widetilde{B}(v \otimes w) = B(v, w).$$

Equivalently,

$$\mathrm{Bil}(V \times W, U) \cong \mathrm{Lin}(V \otimes W, U).$$

The symbol $v \otimes w$ is therefore not an ordered pair. It is a generator subject to the relations required by bilinearity:

$$(v_1 + v_2) \otimes w = v_1 \otimes w + v_2 \otimes w,$$

$$v \otimes (w_1 + w_2) = v \otimes w_1 + v \otimes w_2,$$
$$(cv) \otimes w = v \otimes (cw) = c(v \otimes w).$$

If $e_1, \dots, e_m$ is a basis of V and $f_1, \dots, f_n$ is a basis of W, then

$$e_i \otimes f_j, \qquad 1 \le i \le m, \quad 1 \le j \le n,$$

form a basis of $V \otimes W$. Hence

$$\dim(V \otimes W) = mn.$$

A general element has the form

$$T = T^{ij} e_i \otimes f_j.$$

The repeated indices are summed. The array (T^{ij}) is a coordinate description; the tensor T is the basis-independent object being described.

Not every tensor is a single pure tensor $v \otimes w$. For example,

$$e_1 \otimes f_1 + e_2 \otimes f_2$$

generally cannot be factored into one product. This is analogous to the difference between a rank-one matrix and a general matrix.

§62.3 Vectors and covectors are different kinds of objects

A vector belongs to V. A covector belongs to the dual space

$$V^* = \mathrm{Lin}(V, \mathbb{R}).$$

A covector consumes a vector and returns a scalar.

Given a basis

$$e_1, \dots, e_n$$

of V, the dual basis

$$\varepsilon^1, \dots, \varepsilon^n$$

of V^* is defined by

$$\varepsilon^i(e_j) = \delta^i_j.$$

If

$$v = v^i e_i,$$

then

$$v^i = \varepsilon^i(v).$$

Thus the dual basis is the device that extracts coordinates.

A covector α is written

$$\alpha = \alpha_i \varepsilon^i,$$

and its action on v is

$$\alpha(v) = \alpha_i v^i.$$

This pairing is a contraction: one lower index meets one upper index and disappears.

The placement of indices is not decorative. Under a change of basis, vector components and covector components transform in opposite ways so that the scalar $\alpha_i v^i$ remains unchanged. The general transformation laws were developed in the preceding basis-change note; here we will see their concrete geometric meaning through a nonorthogonal frame.

§62.4 A nonorthogonal basis exposes the dual basis

Consider $\mathbb{R}^2$ with the basis

$$g_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad g_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

The coordinate directions meet at $45°$, not $90°$.

Let

$$v = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

Writing

$$v = v^1 g_1 + v^2 g_2$$

gives

$$\begin{bmatrix} 3 \\ 2 \end{bmatrix} = v^1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + v^2 \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Therefore

$$v^1 = 1, \qquad v^2 = 2.$$

These are the contravariant components of v in the basis (g_i).

Now seek vectors g^1, g^2 satisfying

$$g^i \cdot g_j = \delta^i_j.$$

A direct calculation gives

$$g^1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \qquad g^2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Indeed,

$$g^1 \cdot g_1 = 1, \quad g^1 \cdot g_2 = 0, \quad g^2 \cdot g_1 = 0, \quad g^2 \cdot g_2 = 1.$$

The coordinate coefficients can now be recovered by dot products with the reciprocal vectors:

$$v^1 = g^1 \cdot v = 1, \qquad v^2 = g^2 \cdot v = 2.$$

The reciprocal vectors are the Euclidean representatives of the abstract dual covectors. The dual basis lives naturally in V^*; the inner product identifies each covector with a vector, allowing us to draw it in the same plane.

This is the first important distinction:

dual basis is intrinsic to a chosen basis,

whereas

reciprocal vectors also use the metric.

Without an inner product, the dual basis still exists, but there is no canonical way to turn its covectors into vectors.

§62.5 The metric is the Gram matrix of the frame

In the basis (g_i), define

$$g_{ij} = g_i \cdot g_j.$$

For the example above,

$$(g_{ij}) = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

This is the Gram matrix of the basis. It records lengths and angles of the coordinate directions.

If

$$u = u^i g_i, \qquad v = v^j g_j,$$

then

$$u \cdot v = g_{ij} u^i v^j.$$

The familiar formula

$$u \cdot v = \sum_i u^i v^i$$

is valid only when $g_{ij} = \delta_{ij}$.

The inverse matrix is

$$(g^{ij}) = (g_{ij})^{-1} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}.$$

It reconstructs the reciprocal basis:

$$g^i = g^{ij} g_j.$$

For example,

$$g^1 = 2g_1 - g_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix},$$

$$g^2 = -g_1 + g_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

The inverse relations are

$$g^{ik} g_{kj} = \delta^i_j.$$

They express that the metric and inverse metric undo one another when used to change index type.

§62.6 Raising and Lowering Indices

The metric is not merely a table of dot products. It defines a linear map

$$\flat : V \longrightarrow V^*, \qquad v^\flat(w) = g(v, w).$$

In differential geometry this is called the flat musical isomorphism. Its inverse is

$$\sharp : V^* \longrightarrow V.$$

In components,

$$v_i = g_{ij} v^j,$$

which is called lowering an index. Conversely,

$$v^i = g^{ij} v_j,$$

raises the index.

For the vector in the running example,

$$(v^1, v^2) = (1, 2).$$

Lowering gives

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}.$$

These numbers can also be read directly:

$$v_1 = v \cdot g_1 = 3, \qquad v_2 = v \cdot g_2 = 5.$$

They are not the coefficients used to assemble v from the basis vectors. They are the values of the covector $v^\flat$ on those basis vectors.

Raising them again returns the original coordinates:

$$\begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

The same example can now verify the transformation laws rather than merely state them. Replace the frame (g_1, g_2) by

$$h_1 = g_1 + g_2, \qquad h_2 = g_2.$$

If

$$P = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix},$$

then the column relation is

$$\begin{bmatrix} h_1 & h_2 \end{bmatrix} = \begin{bmatrix} g_1 & g_2 \end{bmatrix} P.$$

Vector components therefore transform by the inverse matrix:

$$[v]_h = P^{-1}[v]_g = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Covector components transform in the opposite direction:

$$[v^\flat]_h = P^T[v^\flat]_g = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 8 \\ 5 \end{bmatrix}.$$

The metric matrix becomes

$$G_h = P^T G_g P = \begin{bmatrix} 5 & 3 \\ 3 & 2 \end{bmatrix}.$$

A direct check closes the diagram:

$$G_h[v]_h = \begin{bmatrix} 5 & 3 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 8 \\ 5 \end{bmatrix} = [v^\flat]_h.$$

The object v and the covector $v^\flat$ did not change. Three component arrays changed together so that lowering an index still commutes with change of frame.

In an orthonormal basis, both component lists happen to coincide because $g_{ij} = \delta_{ij}$. Tensor notation refuses to build that coincidence into the definitions, which is why it remains valid in arbitrary coordinates and on curved spaces.

§62.7 Inputs and Outputs of Tensors

A tensor of type (r, s) can be viewed as a multilinear map

$$T : \underbrace{V^* \times \cdots \times V^*}_{r \text{ copies}} \times \underbrace{V \times \cdots \times V}_{s \text{ copies}} \longrightarrow \mathbb{R}.$$

Equivalently, it belongs to

$$V^{\otimes r} \otimes (V^*)^{\otimes s}.$$

A vector is a tensor of type $(1, 0)$. A covector has type $(0, 1)$. A bilinear form such as the metric has type $(0, 2)$:

$$g = g_{ij}\, \varepsilon^i \otimes \varepsilon^j.$$

A linear map $A : V \to V$ can be represented as a tensor of type $(1, 1)$:

$$A = A^i{}_j\, e_i \otimes \varepsilon^j.$$

It consumes a vector through the covector slot and produces a vector through the vector slot:

$$Av = A^i{}_j v^j e_i.$$

This notation explains why a matrix can represent different kinds of objects. The same rectangular array could be the components of a bilinear form, a linear map, or a tensor with another slot arrangement. Its transformation law and index type tell us which object it represents.

The metric can change tensor type. For example, from a $(1, 1)$-tensor $A^i{}_j$, lowering the upper index gives

$$A_{ij} = g_{ik} A^k{}_j.$$

Whether A_{ij} is symmetric is therefore a metric-dependent statement about the corresponding operator. In Euclidean orthonormal coordinates this subtlety is hidden because lowering an index leaves the numerical array unchanged.

§62.8 Contractions and Invariants

Contraction pairs one upper and one lower index. For a $(1, 1)$-tensor,

$$A^i{}_j,$$

contracting the indices gives

$$A^i{}_i.$$

This is the trace:

$$\operatorname{tr} A = A^i{}_i.$$

Why is the result basis-independent? Under a basis change, one index transforms with the change-of-basis matrix and the other with its inverse. When the indices are contracted, these factors cancel. The invariant scalar is what remains after the coordinate dependence closes on itself.

The inner product is also a contraction:

$$u \cdot v = g_{ij} u^i v^j.$$

First the metric lowers one vector index, then the resulting covector is paired with the other vector.

For a rank-two contravariant tensor T^{ij}, contraction with the metric gives

$$g_{ij}T^{ij}.$$

For a differential operator in Euclidean coordinates, this pattern produces the Laplacian from the Hessian:

$$\Delta f = \delta^{ij}\partial_i\partial_j f.$$

On a curved Riemannian manifold, the same idea persists, but partial derivatives must be replaced by covariant derivatives and the metric varies from point to point.

Contraction is therefore more than compact notation. It is one of the principal mechanisms by which tensors produce coordinate-independent scalars and lower-rank tensors.

§62.9 Orientation Beyond the Metric

In three-dimensional Euclidean space, the scalar triple product

$$u \cdot (v \times w)$$

measures oriented volume. In a basis g_1, g_2, g_3, the basis volume is

$$J = g_1 \cdot (g_2 \times g_3).$$

The reciprocal vectors can then be written

$$g^1 = \frac{g_2 \times g_3}{J}, \qquad g^2 = \frac{g_3 \times g_1}{J}, \qquad g^3 = \frac{g_1 \times g_2}{J}.$$

The denominator appears because the cross product creates a normal vector with the wrong scale; division by the oriented volume normalizes it to evaluate the corresponding basis vector as 1.

The Levi–Civita symbol is

$$\varepsilon_{ijk} = \begin{cases} 1, & (i,j,k) \text{ is an even permutation of } (1,2,3), \\ -1, & (i,j,k) \text{ is an odd permutation of } (1,2,3), \\ 0, & \text{two indices coincide.} \end{cases}$$

In a right-handed orthonormal basis,

$$(u \times v)_i = \varepsilon_{ijk}u^j v^k.$$

In general coordinates, however, the bare alternating symbol

$$[i_1 \cdots i_n] \in \{-1, 0, 1\}$$

is not by itself the component array of an ordinary tensor. If $\varepsilon^1, \dots, \varepsilon^n$ is an oriented coordinate coframe, the metric volume form is

$$\mathrm{vol}_g = \sqrt{\det(g_{ij})}\, \varepsilon^1 \wedge \cdots \wedge \varepsilon^n.$$

Equivalently, the covariant Levi–Civita tensor has components

$$\epsilon_{i_1 \cdots i_n} = \sqrt{\det(g_{ij})}\, [i_1 \cdots i_n].$$

The square root supplies the coordinate-volume scale, while orientation chooses the sign. Keeping the alternating symbol, the tensor, and the differential form distinct prevents the common mistake of assigning tensorial transformation laws to a fixed array of zeros and signs.

This is also why the cross product is special. It requires a metric to identify vectors and covectors, an orientation to choose a sign, and dimension three so that two-dimensional area elements can be identified with vectors. The exterior product $u \wedge v$ is more fundamental and exists in every dimension; the cross product is its three-dimensional metric-dependent avatar.

A reciprocal lattice is a concrete use of the same dual geometry. Given primitive crystal vectors a_1, a_2, a_3, define reciprocal vectors b^i by

$$b^i \cdot a_j = 2\pi \delta^i_j.$$

In three dimensions,

$$b^1 = 2\pi \frac{a_2 \times a_3}{a_1 \cdot (a_2 \times a_3)},$$

with cyclic formulas for b^2, b^3. A reciprocal vector

$$k = h_i b^i, \qquad h_i \in \mathbb{Z},$$

defines planes of constant phase

$$k \cdot x = 2\pi m.$$

Because $k \cdot a_j = 2\pi h_j$, translating by a lattice vector changes the phase by an integer multiple of 2π. Diffraction indices are therefore covector-like data: they measure lattice directions rather than assemble a physical-space displacement.

§62.10 Moving frames turn the same algebra into geometry

On a manifold, the basis vectors may vary from point to point. A local frame is written

$$e_1(x), \ldots, e_n(x),$$

and its dual coframe satisfies

$$\theta^i(e_j) = \delta^i_j.$$

A vector field and a one-form take the forms

$$X = X^i e_i, \qquad \alpha = \alpha_i \theta^i.$$

The metric is

$$g = g_{ij}\, \theta^i \otimes \theta^j.$$

At each point, raising and lowering indices is exactly the finite-dimensional computation already performed above.

The new difficulty is differentiation. Differentiating the components alone is not enough because the frame itself changes:

$$\partial_k (X^i e_i) = (\partial_k X^i) e_i + X^i \partial_k e_i.$$

A connection packages the derivatives of the basis into coefficients $\Gamma^i{}_{jk}$, producing the covariant derivative

$$\nabla_j X^i = \partial_j X^i + \Gamma^i{}_{jk} X^k.$$

For a covector, the correction has the opposite sign:

$$\nabla_j \alpha_i = \partial_j \alpha_i - \Gamma^k{}_{ji} \alpha_k.$$

The signs differ for the same reason vector and covector components transform oppositely.

In moving-frame notation one writes

$$\nabla e_j = \omega^i{}_j \otimes e_i,$$

where the matrix of one-forms $\omega^i{}_j$ records how the frame turns. For the Levi–Civita connection in an orthonormal frame,

$$\omega_{ij} + \omega_{ji} = 0,$$

so infinitesimal frame change is skew-symmetric, just like an infinitesimal rotation. The torsion-free first structure equation is

$$d\theta^i + \omega^i{}_j \wedge \theta^j = 0.$$

This is not a new layer of index decoration: it says that the variation of the coframe and the connection correction cancel in the geometric derivative.

This is the natural continuation of reciprocal bases. Once coordinates vary with position, tensor notation does not merely keep formulas tidy; it ensures that differentiation produces another tensor rather than a coordinate artifact.

§62.11 A curvilinear example: polar coordinates

In the Euclidean plane, polar coordinates are

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

The coordinate basis vectors are

$$e_r = \frac{\partial(x,y)}{\partial r} = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix},$$

$$e_\theta = \frac{\partial(x,y)}{\partial \theta} = \begin{bmatrix} -r\sin\theta \\ r\cos\theta \end{bmatrix}.$$

They are orthogonal but not both unit length:

$$e_r \cdot e_r = 1, \qquad e_r \cdot e_\theta = 0, \qquad e_\theta \cdot e_\theta = r^2.$$

Thus

$$(g_{ij}) = \begin{bmatrix} 1 & 0 \\ 0 & r^2 \end{bmatrix}, \qquad (g^{ij}) = \begin{bmatrix} 1 & 0 \\ 0 & r^{-2} \end{bmatrix}.$$

The reciprocal basis vector corresponding to e_θ is

$$e^\theta = r^{-2} e_\theta.$$

The area form is

$$\sqrt{\det g}\, dr \wedge d\theta = r\, dr \wedge d\theta.$$

The familiar Jacobian factor r is therefore not an isolated integration trick. It is the square root of the metric determinant.

For a scalar field f, the differential is

$$df = (\partial_r f)\, dr + (\partial_\theta f)\, d\theta.$$

Raising its index with the inverse metric gives the gradient:

$$\nabla f = (\partial_r f) e_r + r^{-2} (\partial_\theta f) e_\theta.$$

Since the unit angular vector is $\widehat{e}_\theta = r^{-1} e_\theta$, this is the familiar formula

$$\nabla f = (\partial_r f) \widehat{e}_r + \frac{1}{r} (\partial_\theta f) \widehat{e}_\theta.$$

The factor $1/r$ is a metric effect, not a memorized correction.

§62.12 A deformation makes the distinction physical

In continuum mechanics, one compares a reference body $\mathcal{B}$ with its deformed configuration $\mathcal{S}$. A deformation is a map

$$\varphi : \mathcal{B} \to \mathcal{S},$$

and its derivative at a material point X is the deformation gradient

$$F = D\varphi(X) : T_X \mathcal{B} \to T_{\varphi(X)} \mathcal{S}.$$

A small material vector dX is pushed forward to

$$dx = F\, dX.$$

A covector transforms in the opposite direction. If α measures current tangent vectors, its pullback to the reference configuration is

$$F^T \alpha,$$

because

$$(F^T \alpha)(dX) = \alpha(F dX).$$

This identity is the physical version of the vector–covector distinction developed earlier: vectors are transported forward, while measurements of vectors are pulled back.

The current Euclidean metric can also be pulled back to the reference body. In coordinates the result is the right Cauchy–Green tensor

$$C = F^T F.$$

For a reference vector a, the squared length after deformation is

$$\|Fa\|^2 = (Fa)^T (Fa) = a^T C a.$$

Thus C stores all changes of lengths and angles while acting entirely on the reference tangent space.

Consider the simple shear

$$\varphi(X, Y) = (X + \gamma Y, Y).$$

Its deformation gradient is

$$F = \begin{bmatrix} 1 & \gamma \\ 0 & 1 \end{bmatrix},$$

so

$$C = F^T F = \begin{bmatrix} 1 & \gamma \\ \gamma & 1+\gamma^2 \end{bmatrix}.$$

The reference basis vectors behave differently:

$$Fe_1 = e_1, \qquad Fe_2 = \gamma e_1 + e_2.$$

The first direction keeps unit length, while

$$\|Fe_2\|^2 = 1 + \gamma^2.$$

Their current inner product is

$$(Fe_1) \cdot (Fe_2) = \gamma.$$

These three quantities are exactly the entries of C. The determinant satisfies

$$\det F = 1,$$

so the shear preserves area even though it changes lengths and angles. A single tensor therefore separates shape change from volume change more clearly than a list of coordinate formulas.

Stress makes the index placement even more consequential. The Cauchy stress σ acts in the current configuration: after using the current metric to identify a unit normal covector with a normal vector n, the traction is

$$t = \sigma n.$$

The first Piola–Kirchhoff stress P, by contrast, accepts reference-area data and produces a current force. Its two slots belong to different configurations. The relation

$$P = J\sigma F^{-T}, \qquad J = \det F,$$

contains exactly the required pullback of the area covector.

For the simple shear, $J = 1$. If, only for illustration, the current Cauchy stress is the identity matrix, then

$$P = F^{-T} = \begin{bmatrix} 1 & 0 \\ -\gamma & 1 \end{bmatrix}.$$

Although $\sigma = I$, the reference-to-current stress array is not the identity because its input lives in a different tangent space. Writing the two tensors with the same undecorated pair of indices would hide this distinction.

The same lesson was already visible in the nonorthogonal coordinate example. There,

$$(v^1, v^2) = (1, 2)$$

and

$$(v_1, v_2) = (3, 5)$$

represented the same geometric vector in contravariant and covariant form. Mechanics adds another layer: the upper or lower index must also remember whether it belongs to the reference or current configuration. Tensor notation is therefore not decorative bookkeeping. It prevents illegal pairings, records which metric performs an identification, and makes clear which quantities are pushed forward, pulled back, or compared only after a deformation map has connected their spaces.

Linear Maps: Free Variables, Obstructions, and Canonical Splittings

N-20251108

A matrix equation $Ax = b$ raises two logically independent questions. Does a solution exist? If it does, is it unique? The image answers the first question, and the kernel answers the second. This elementary observation leads to exact sequences and quotient spaces; with an inner product, it leads further to the pseudoinverse and an optimal choice among the many possible solutions.[1]

§63.1 Free variables describe the kernel as a graph

Consider the row-reduced matrix

$$R = \begin{pmatrix} 1 & 3 & 0 & 0 & 3 \\ 0 & 0 & 1 & 0 & 9 \\ 0 & 0 & 0 & 1 & -4 \end{pmatrix}.$$

Its pivot columns are $1, 3, 4$; columns $2, 5$ are free. The system $Rx = 0$ reads

$$x_1 = -3x_2 - 3x_5, \qquad x_3 = -9x_5, \qquad x_4 = 4x_5.$$

Thus

$$\boxed{\ker R = \operatorname{span}\left\{ \begin{pmatrix} 3 \\ -1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 0 \\ 9 \\ -4 \\ -1 \end{pmatrix} \right\}.}$$

The negative signs in the free coordinates are immaterial; they merely give the negatives of the usual kernel basis vectors. This is the practical minus-one trick.

The structural calculation is more useful than the trick. After permuting pivot coordinates before free coordinates, every RREF system has the form

$$\begin{pmatrix} I_r & C \end{pmatrix} \begin{pmatrix} x_p \\ x_f \end{pmatrix} = 0.$$

Therefore

$$\boxed{\ker \begin{pmatrix} I_r & C \end{pmatrix} = \left\{ \begin{pmatrix} -Cu \\ u \end{pmatrix} : u \in \mathbb{F}^{n-r} \right\}.}$$

The kernel is the graph of the map $u \mapsto -Cu$. Its dimension is $n - r$ because the free coordinate u can be chosen arbitrarily.

A matrix changes with the bases in its domain and codomain. If the new coordinate matrix is $T^{-1}RS$, with S, T invertible, its kernel is $S^{-1} \ker R$. Consequently kernel *dimension* and rank are basis-independent, even though the displayed basis vectors change.

[1] For the linear-map, quotient, orthogonal-complement and pseudoinverse framework see Sheldon Axler, *Linear Algebra Done Right*, Chapters 3 and 6.

§63.2 Rank–nullity is the dimension of an exact sequence

For any linear map $T : V \to W$, factor it as

$$V \xrightarrow{q} V/\ker T \xrightarrow{\overline{T}} \operatorname{im} T \hookrightarrow W,$$

where q sends v to its equivalence class and

$$\overline{T}([v]) = T(v).$$

This is well defined: if $v - v' \in \ker T$, then $T(v) = T(v')$. It is also injective and surjective onto the image, so

$$\boxed{V/\ker T \cong \operatorname{im} T.}$$

In finite dimensions, counting dimensions gives

$$\boxed{\dim V = \dim \ker T + \dim \operatorname{im} T.}$$

The underlying exact sequence is

$$0 \longrightarrow \ker T \longrightarrow V \xrightarrow{T} \operatorname{im} T \longrightarrow 0.$$

Exactness means that the image of each arrow is precisely the kernel of the next. It is not a decorative way of displaying rank–nullity: it remains meaningful in settings where numerical dimensions are unavailable.

§63.3 The cokernel measures equations that cannot be solved

The cokernel of $T : V \to W$ is

$$\operatorname{coker} T = W/\operatorname{im} T.$$

A right-hand side $b \in W$ is in the image of T precisely when its class in the cokernel is zero.

To see this computationally, consider

$$B = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}.$$

Every output of B lies on the line

$$(u, v) = (t, 2t), \qquad t \in \mathbb{R}.$$

The vector $y = (-2, 1)^T$ spans the left null space because $y^T B = 0$. Hence the equation $Bx = b$ is solvable exactly when

$$\boxed{-2b_1 + b_2 = 0.}$$

This condition is both necessary and sufficient: if $b_2 = 2b_1$, simply take $x = (b_1, 0, 0)^T$.

Here

$$\dim \ker B = 2, \qquad \dim \operatorname{im} B = 1, \qquad \dim \operatorname{coker} B = 1.$$

The two-dimensional kernel records the nonuniqueness of solutions, while the one-dimensional cokernel records the single compatibility condition on b.

In finite-dimensional inner-product spaces,

$$\boxed{(\operatorname{im} T)^{\perp} = \ker T^{*}.}$$

Indeed, $w \perp \operatorname{im} T$ exactly when $\langle Tv, w\rangle = 0$ for all v, or $T^{*}w = 0$. This gives a concrete realization of cokernel constraints by left null vectors.

§63.4 An arbitrary solution is not an optimal solution

A surjective linear map admits many right inverses, because one may choose many complementary subspaces to its kernel. An inner product singles out one canonical choice: the solution orthogonal to the kernel.

Take

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

Its kernel is spanned by $k = (1, -1, 1)^T$, since $Ak = 0$. The matrix has full row rank, so AA^T is invertible:

$$AA^T = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad (AA^T)^{-1} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

The Moore–Penrose right inverse is

$$\boxed{A^+ = A^T(AA^T)^{-1} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ 1 & 1 \\ -1 & 2 \end{pmatrix}.}$$

Direct multiplication gives $AA^+ = I_2$.

For $b = (1, 2)^T$, the preferred solution is

$$x_* = A^+ b = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}.$$

All solutions are

$$x = x_* + tk, \qquad t \in \mathbb{R}.$$

Since $x_* \cdot k = 0$, Pythagoras gives

$$\|x\|^2 = \|x_*\|^2 + t^2\|k\|^2 = 2 + 3t^2.$$

Thus x_* is the unique solution of minimum Euclidean norm. Choosing the free coordinate $x_2 = 0$ would instead produce $(1, 0, 2)^T$, whose squared norm is 5, strictly larger than 2.

The orthogonal splitting can also be read from two projectors:

$$P = A^+A, \qquad Q = I_3 - P.$$

They satisfy

$$P^2 = P = P^T, \qquad Q^2 = Q = Q^T, \qquad PQ = 0.$$

Here P projects onto the row space $\operatorname{im} A^T$, and Q projects onto $\ker A$. Row reduction selects one arbitrary complement of the kernel; the Euclidean inner product singles out its orthogonal complement.

§63.5 A dimension identity survives the choice of rank

Combine the kernel and cokernel in the longer exact sequence

$$0 \longrightarrow \ker T \longrightarrow V \xrightarrow{T} W \longrightarrow \operatorname{coker} T \longrightarrow 0.$$

Taking the alternating sum of dimensions gives

$$\boxed{\dim \ker T - \dim \operatorname{coker} T = \dim V - \dim W.}$$

For either of the 2×3 matrices above the right side equals 1. The surjective matrix A has kernel dimension 1 and cokernel dimension 0, whereas the rank-one matrix B has kernel dimension 2 and cokernel dimension 1. Both yield the same difference 1.

This finite-dimensional invariant motivates the *Fredholm index* of suitable linear operators between infinite-dimensional spaces. There one requires finite-dimensional kernel and cokernel and closed range, so their difference remains meaningful even though domain and codomain may both have infinite dimension. The definition records the same balance between nonuniqueness and obstruction that began in a row-reduction exercise.

The computation has moved through several compatible descriptions. Free variables parametrize the kernel; the first isomorphism theorem identifies the information that survives the map; left null vectors detect unreachable right-hand sides; and the pseudoinverse uses an inner product to choose the unique minimum-norm solution. Exact sequences express what all these calculations have in common without requiring a preferred coordinate system.

Determinants and Rank: From Alternating Volume to Elimination and Rank Strata

N-20251110

§64.1 Why one scalar can encode both volume and independence

A determinant problem often looks like a request to simplify a large array. The useful starting point is more geometric. If the columns of

$$A = (a_1, \dots, a_n)$$

are vectors in k^n, then $\det A$ is the oriented volume scale of the parallelotope they span. Three properties determine this quantity uniquely:

(i) it is linear in each column while the others are fixed;

(ii) it changes sign when two columns are exchanged;

(iii) $\det I = 1$.

Alternation immediately gives

$$a_i = a_j \text{ for some } i \neq j \quad \Longrightarrow \quad \det A = 0.$$

More generally,

$$\det A \neq 0 \quad \Longleftrightarrow \quad a_1, \dots, a_n \text{ are linearly independent.}$$

Thus determinant computation and rank detection are not separate topics. A nonzero determinant says that no direction has collapsed; a zero determinant says that the image volume has fallen into a lower-dimensional subspace.

Elementary column operations now have conceptual explanations. Adding a multiple of one column to another leaves the wedge volume unchanged. Scaling one column scales the volume by the same factor. Exchanging two columns reverses orientation. Every safe determinant trick is an application of these three rules.

§64.2 Cofactors as Exterior Coordinates

A linear map $A : V \to V$ induces maps

$$\Lambda^r A : \Lambda^r V \to \Lambda^r V.$$

In a basis, the entries of $\Lambda^r A$ are the $r \times r$ minors of A. At the top degree, $\Lambda^n V$ is one-dimensional and

$$\Lambda^n A = (\det A) I.$$

This is the coordinate-free source of multiplicativity:

$$\det(AB) = \det A \det B,$$

because

$$\Lambda^n(AB) = \Lambda^n A \circ \Lambda^n B.$$

For $r = n-1$, the minors assemble into the adjugate. If C_{ij} is the cofactor of a_{ij}, then

$$\operatorname{adj}(A) = (C_{ji})$$

and

$$A \operatorname{adj}(A) = \operatorname{adj}(A) A = (\det A) I.$$

The diagonal entries of this product are ordinary Laplace expansions. An off-diagonal entry is the determinant obtained by replacing one row by another already present, hence it vanishes.

The same exterior-power bookkeeping gives the Cauchy–Binet formula. For

$$X \in k^{m\times n}, \qquad Y \in k^{n\times m}, \qquad m \le n,$$

one has

$$\boxed{\det(XY) = \sum_{|S|=m} \det(X_{[:,S]}) \det(Y_{[S,:]}).}$$

General Laplace expansion is a special case: selecting a group of rows and summing products of complementary minors is exactly composition on an exterior power.

§64.3 Vandermonde and Uniqueness

Let $\mathcal{P}_{n-1}$ be the vector space of polynomials of degree at most $n-1$. Evaluation at points $x_1, \dots, x_n$ defines a linear map

$$\operatorname{ev} : \mathcal{P}_{n-1} \longrightarrow k^n, \qquad p \longmapsto (p(x_1), \dots, p(x_n)).$$

In the monomial basis $1, t, \dots, t^{n-1}$, its matrix is the Vandermonde matrix

$$V = \begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{pmatrix}.$$

Its determinant is

$$\boxed{\det V = \prod_{1\le i<j\le n} (x_j - x_i).}$$

The factorization can be proved by observing that the determinant is alternating in the x_i, hence divisible by every $x_j - x_i$, and that both sides have the same total degree and leading coefficient.

This formula has a direct meaning:

$$\det V \neq 0 \quad \Longleftrightarrow \quad x_1, \dots, x_n \text{ are distinct}$$

$$\Longleftrightarrow \quad \text{values at the } x_i \text{ determine } p \in \mathcal{P}_{n-1} \text{ uniquely.}$$

Lagrange interpolation is simply the inverse of this evaluation map.

The square of the Vandermonde product,

$$\prod_{i<j}(x_i - x_j)^2,$$

is the discriminant of a monic polynomial with roots x_i, up to the conventional sign. It vanishes exactly when two roots collide. The same determinant therefore detects both failure of interpolation and repeated roots.

§64.4 Finite differences expose hidden polynomial dependence

Suppose columns are formed by evaluating a polynomial sequence at consecutive integers. Repeated column subtraction is the matrix version of the finite-difference operator

$$(\Delta f)(x) = f(x+1) - f(x).$$

If f has degree d, then Δf has degree $d-1$, and

$$\Delta^{d+1} f = 0.$$

This explains why many determinants built from low-degree polynomial data vanish after a few adjacent differences.

For example, consider

$$D = \det \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0^2 & 1^2 & 2^2 & 3^2 \\ 0^2+0 & 1^2+1 & 2^2+2 & 3^2+3 \end{pmatrix}.$$

The last row is the sum of the second and third rows, so $D = 0$. Repeated differences reveal the same fact without spotting the relation immediately: every row is generated by polynomials of degree at most two, so four evaluation rows cannot be independent.

A nonzero finite-difference determinant is equally informative. For a polynomial p of exact degree $n-1$ with leading coefficient c, the determinant

$$\det\bigl(p_i(j)\bigr)_{0\le i,j<n}$$

can often be reduced to a triangular matrix whose diagonal contains successive leading finite differences. The determinant records the product of the new independent polynomial degrees. The row operations are not a bag of tricks; they are a change from monomials to the Newton finite-difference basis.

§64.5 Block elimination produces the Schur complement

Let

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

with A square and invertible. Multiplying on the left by

$$L = \begin{pmatrix} I & 0 \\ -CA^{-1} & I \end{pmatrix}$$

performs block Gaussian elimination:

$$LM = \begin{pmatrix} A & B \\ 0 & D - CA^{-1}B \end{pmatrix}.$$

Since $\det L = 1$,

$$\boxed{\det M = \det A \, \det(D - CA^{-1}B).}$$

The matrix

$$S = D - CA^{-1}B$$

is the Schur complement of A.

Its role is clearer in a linear system. From

$$Ax + By = u, \qquad Cx + Dy = v,$$

the first equation gives

$$x = A^{-1}(u - By).$$

Substitution leaves the effective equation

$$Sy = v - CA^{-1}u.$$

The same object controls determinant, invertibility, and elimination of the x-variables.

The hypothesis that A is invertible cannot be silently discarded. If A is singular, the displayed formula is not defined; one must pivot to another block, use a limiting argument only when justified, or return to rank and minors.

§64.6 Low-Rank Determinant Updates

Suppose A is invertible and the perturbation has rank one:

$$A + uv^T.$$

Factor out A:

$$A + uv^T = A(I + A^{-1}uv^T).$$

Sylvester's determinant identity

$$\det(I_m + UV) = \det(I_r + VU)$$

for $U \in k^{m\times r}$, $V \in k^{r\times m}$ gives, with $r = 1$,

$$\boxed{\det(A + uv^T) = \det A \, (1 + v^T A^{-1}u).}$$

This is the matrix determinant lemma.

Consider the arrowhead matrix

$$M = \begin{pmatrix} d_1 & 0 & \cdots & 0 & u_1 \\ 0 & d_2 & \cdots & 0 & u_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & d_n & u_n \\ v_1 & v_2 & \cdots & v_n & \alpha \end{pmatrix},$$

where every $d_i \neq 0$. Taking

$$A = \operatorname{diag}(d_1, \dots, d_n)$$

in the Schur complement formula yields

$$\det M = \left(\prod_{i=1}^{n} d_i\right)\left(\alpha - \sum_{i=1}^{n} \frac{u_i v_i}{d_i}\right).$$

A determinant of order $n+1$ has become one scalar correction to the diagonal product.

For a rank-r update,

$$\det(A + UV^T) = \det A \, \det(I_r + V^T A^{-1} U).$$

The computational gain is structural: only the coupled r-dimensional directions need a new determinant.

§64.7 Recurrences of Determinants

Let

$$D_n = \det \begin{pmatrix} a & b & 0 & \cdots & 0 \\ c & a & b & \ddots & \vdots \\ 0 & c & a & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & b \\ 0 & \cdots & 0 & c & a \end{pmatrix}.$$

Expansion along the first row gives

$$D_n = aD_{n-1} - bcD_{n-2}, \qquad D_0 = 1, \quad D_1 = a.$$

Rather than treating each order as a new determinant, package the recurrence as

$$\begin{pmatrix} D_n \\ D_{n-1} \end{pmatrix} = \begin{pmatrix} a & -bc \\ 1 & 0 \end{pmatrix} \begin{pmatrix} D_{n-1} \\ D_{n-2} \end{pmatrix}.$$

The 2×2 transfer matrix carries the entire family.

If the characteristic roots of

$$r^2 - ar + bc = 0$$

are distinct, say r_+, r_-, then

$$D_n = \frac{r_+^{n+1} - r_-^{n+1}}{r_+ - r_-}.$$

When the roots coincide, the solution contains a polynomial factor in n. Repeated-root behavior of the transfer matrix is the same phenomenon as a Jordan block in a recurrence. A patterned determinant has become a small linear dynamical system.

§64.8 Cramer's rule lives only on the invertible open set

For a square system

$$Ax = b$$

with $\det A \neq 0$, the unique solution is

$$x = A^{-1} b.$$

Using

$$A^{-1} = \frac{\operatorname{adj}(A)}{\det A}$$

gives Cramer's formula

$$x_j = \frac{\det A_j}{\det A},$$

where A_j is obtained by replacing column j by b.

The formula is elegant but its domain is important. It describes the open set

$$\{A : \det A \neq 0\}$$

of invertible matrices. On the hypersurface $\det A = 0$, ratios of determinants cease to be the right language. Solvability is instead the image condition

$$b \in \operatorname{im} A,$$

or equivalently

$$\operatorname{rank} A = \operatorname{rank}[A \mid b].$$

If the common rank is $r < n$, the solution set is an affine space

$$x_0 + \ker A$$

of dimension $n - r$. Rank is therefore not a fallback after determinant tricks fail; it is the correct invariant on the singular locus.

§64.9 Minors cut out the geometry of rank failure

A matrix satisfies

$$\operatorname{rank} A \leq r$$

exactly when all $(r+1) \times (r+1)$ minors vanish. Thus matrices of rank at most r form an algebraic set, the determinantal variety

$$\mathcal{D}_r = \{A \in k^{m \times n} : \operatorname{rank} A \leq r\}.$$

For 2×3 matrices of rank at most one, write

$$A = \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}.$$

The defining equations are its three 2×2 minors:

$$ae - bd = 0, \qquad af - cd = 0, \qquad bf - ce = 0.$$

Away from the zero matrix, every such matrix factors as

$$A = uv^T, \qquad u \in k^2 \setminus \{0\}, \quad v \in k^3 \setminus \{0\}.$$

The factorization has one scaling redundancy

$$(u, v) \sim (\lambda u, \lambda^{-1} v),$$

so the rank-one stratum has dimension

$$2 + 3 - 1 = 4.$$

This agrees with the general formula

$$\dim \mathcal{D}_r = r(m + n - r).$$

The same example lets us see the singular locus rather than merely name it. On the open chart where $a \neq 0$, the equations

$$ae - bd = 0, \qquad af - cd = 0$$

solve uniquely for

$$e = \frac{bd}{a}, \qquad f = \frac{cd}{a}.$$

The third minor then vanishes automatically:

$$bf - ce = b\frac{cd}{a} - c\frac{bd}{a} = 0.$$

Thus (a, b, c, d) are four local coordinates near every rank-one point with $a \neq 0$. If a different entry is nonzero, the same argument uses the corresponding chart. Every nonzero rank-one matrix is therefore a smooth point of this four-dimensional variety.

At the zero matrix, all first derivatives of the three defining minors vanish. Their Jacobian has rank zero there, so the equations have no nontrivial linear part and the tangent space is the entire six-dimensional ambient space. The origin is singular. In this example,

$$\operatorname{Sing}(\mathcal{D}_1) = \mathcal{D}_0 = \{0\}.$$

More generally, the singular locus of the rank-at-most-r determinantal variety is the lower stratum $\mathcal{D}_{r-1}$.

For square matrices, $\det A = 0$ is the boundary where one direction collapses; deeper rank strata record how many independent directions have collapsed. Cofactors, Schur complements, determinant lemmas, and rank conditions are therefore different local descriptions of the same geometry of elimination and loss of dimension.

Inner-Product Geometry and Spectral Decomposition: From QR to Quadratic Forms

N-20251202

§65.1 Diagonalization and Invariant Lines

A matrix $A \in k^{n\times n}$ is diagonalizable when there is an invertible matrix P such that

$$P^{-1}AP = D$$

is diagonal. If the columns of P are $v_1, \dots, v_n$, then

$$Av_i = \lambda_i v_i.$$

Thus diagonalization is the decomposition

$$k^n = E_{\lambda_1} \oplus \cdots \oplus E_{\lambda_s}$$

into eigenspaces. The question is not merely whether eigenvalues exist, but whether their eigenspaces supply enough independent directions.

Several properties that are often confused are logically independent. The zero matrix is diagonalizable but not invertible. The Jordan block

$$J = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

is invertible but not diagonalizable. Repeated eigenvalues cause no difficulty for the identity matrix, whereas the same repeated eigenvalue leaves J with only one eigendirection. Rank, invertibility, eigenvalue multiplicity, and diagonalizability answer different questions.

§65.2 Minimal Polynomials and Eigenspaces

The minimal polynomial $m_A(t)$ is the monic polynomial of least degree satisfying

$$m_A(A) = 0.$$

If $A = PDP^{-1}$, then m_A must vanish on every diagonal entry of D, so it is a product of distinct linear factors:

$$m_A(t) = \prod_{i=1}^{s} (t - \lambda_i).$$

Conversely, suppose

$$m_A(t) = \prod_{i=1}^{s} (t - \lambda_i)$$

with distinct roots. Define the Lagrange polynomials

$$\ell_i(t) = \prod_{j \neq i} \frac{t - \lambda_j}{\lambda_i - \lambda_j}.$$

They satisfy

$$\sum_i \ell_i(t) = 1 \pmod{m_A(t)}.$$

Hence

$$I = \sum_i \ell_i(A).$$

Moreover,

$$(A - \lambda_i I)\ell_i(A) = 0,$$

so $\ell_i(A)V \subseteq E_{\lambda_i}$. Every vector is therefore a sum of eigenvectors, and A is diagonalizable.

Thus

$$\boxed{A \text{ is diagonalizable } \iff m_A \text{ splits with no repeated root.}}$$

For the Jordan block above,

$$m_J(t) = (t-1)^2.$$

The repeated factor records the nilpotent motion along the generalized eigendirection.

§65.3 Projection as Best Approximation

An inner product adds lengths and angles to a vector space. For a subspace $W \subset \mathbb{R}^n$, the orthogonal projection of x onto W is the unique vector $p \in W$ for which

$$x - p \perp W.$$

If $Q \in \mathbb{R}^{n\times r}$ has orthonormal columns spanning W, then

$$p = QQ^T x.$$

Indeed, $QQ^T x \in W$, while

$$Q^T(x - QQ^T x) = Q^T x - Q^T x = 0.$$

The matrix

$$P_W = QQ^T$$

is symmetric and idempotent:

$$P_W^T = P_W, \qquad P_W^2 = P_W.$$

Orthogonality also proves optimality. For any $w \in W$,

$$x - w = (x - p) + (p - w)$$

is an orthogonal decomposition, so

$$\|x - w\|^2 = \|x - p\|^2 + \|p - w\|^2 \geq \|x - p\|^2.$$

Projection is therefore the closest point in the subspace. Least squares, Fourier approximation, and Gram–Schmidt all begin with this Pythagorean identity.

§65.4 Gram–Schmidt and QR

Given independent vectors $a_1, \dots, a_r$, define

$$u_1 = a_1,$$

$$u_j = a_j - \sum_{i<j} \frac{\langle a_j, u_i \rangle}{\langle u_i, u_i \rangle} u_i.$$

Each u_j is orthogonal to the previous vectors, and independence guarantees $u_j \neq 0$. Normalizing gives orthonormal vectors

$$q_j = \frac{u_j}{\|u_j\|}.$$

For the columns

$$a_1 = (1,0,0,1)^T, \quad a_2 = (1,1,0,1)^T, \quad a_3 = (1,1,1,0)^T,$$

one obtains

$$q_1 = \frac{1}{\sqrt{2}}(1,0,0,1)^T, \qquad q_2 = (0,1,0,0)^T,$$

$$q_3 = \frac{1}{\sqrt{6}}(1,0,2,-1)^T.$$

Writing $A = (a_1, a_2, a_3)$ and $Q = (q_1, q_2, q_3)$, the coefficients

$$r_{ij} = \langle q_i, a_j \rangle$$

form the upper-triangular matrix

$$R = \begin{pmatrix} \sqrt{2} & \sqrt{2} & 1/\sqrt{2} \\ 0 & 1 & 1 \\ 0 & 0 & \sqrt{6}/2 \end{pmatrix},$$

and

$$\boxed{A = QR, \qquad Q^T Q = I.}$$

QR factorization is Gram–Schmidt written as a matrix identity: Q stores the orthonormal geometry, while R stores how the original columns were assembled from it.

§65.5 Householder QR

Classical Gram–Schmidt can lose accuracy when columns are nearly dependent, because it subtracts nearly equal vectors. Modified Gram–Schmidt reduces the damage by updating residuals after each projection. Householder QR takes a different route: reflect an entire column onto a coordinate axis, with a crucial choice of sign that prevents a small subtraction.

For a nonzero real vector x, define

$$s = \begin{cases} 1, & x_1 \geq 0, \\ -1, & x_1 < 0, \end{cases} \qquad \alpha = -s\|x\|_2, \qquad v = x - \alpha e_1 = x + s\|x\|_2 e_1.$$

The sign convention at $x_1 = 0$ is intentional: the usual sign function with $\operatorname{sign}(0) = 0$ would make the proposed target $\alpha e_1 = 0$, which no orthogonal map can produce from

nonzero x. Moreover $|v_1| = |x_1| + \|x\|_2$, so the first component is formed by adding magnitudes rather than subtracting nearly equal numbers.

Now set

$$\boxed{H = I - 2\frac{vv^T}{v^Tv}.}$$

Because vv^T is symmetric and $(vv^T)^2 = (v^Tv)vv^T$,

$$H^T = H, \qquad H^2 = I.$$

It is therefore an orthogonal reflection. The choice $\alpha^2 = \|x\|_2^2$ also gives

$$v^Tv = 2(\|x\|_2^2 - \alpha x_1), \qquad 2v^Tx = v^Tv.$$

Consequently

$$Hx = x - \frac{2v(v^Tx)}{v^Tv} = x - v = \boxed{\alpha e_1}.$$

So the first column is sent exactly to the coordinate axis without repeatedly subtracting projections.

For a complete example, take

$$A = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix}, \qquad x = \begin{pmatrix} 3 \\ 4 \end{pmatrix}.$$

Our stable sign choice is $\alpha = -5$ and $v = (8, 4)^T$. Thus

$$H = \begin{pmatrix} -3/5 & -4/5 \\ -4/5 & 3/5 \end{pmatrix}, \qquad HA = \begin{pmatrix} -5 & -4 \\ 0 & 3 \end{pmatrix} = R.$$

Since $H^T = H = H^{-1}$, putting $Q = H$ gives the exact QR factorization

$$\boxed{A = QR, \qquad Q^TQ = I.}$$

The upper-triangular factor need not have positive diagonal entries; if positive diagonals are preferred, multiplying a column of Q and the corresponding row of R by -1 does not change their product.

In exact arithmetic either target sign $\alpha = \pm\|x\|_2$ can work when $v \neq 0$; the *opposite* sign used here avoids cancellation in floating-point computation. This small choice helps explain why a numerically stable Householder implementation is more than the geometric formula for a reflection. The same mechanism underlies practical QR algorithms.[1]

§65.6 Orthogonal Maps and Reflections

A real matrix U is orthogonal when

$$U^TU = I.$$

It preserves inner products:

$$\langle Ux, Uy\rangle = x^TU^TUy = \langle x, y\rangle.$$

[1] For Householder QR and the noncancelling sign convention, see Trefethen and Bau, *Numerical Linear Algebra*, Lecture 10.

Hence it preserves lengths, angles, and orthogonality. Its determinant satisfies

$$(\det U)^2 = 1,$$

so

$$\det U = \pm 1.$$

The sign records orientation: rotations have determinant $+1$, while a single reflection has determinant -1.

Householder matrices show that reflections are not merely examples. Every orthogonal matrix is a product of reflections. QR factorization repeatedly uses this fact to rotate or reflect successive column spaces into coordinate position without changing their metric geometry.

Over $\mathbb{C}$, the corresponding maps are unitary:

$$U^*U = I.$$

Complex conjugation in the Hermitian inner product is essential; without it, $\langle x, x\rangle$ need not be positive.

§65.7 Spectral Theorem by Induction

Let $A = A^T$ be real symmetric. On the unit sphere, the Rayleigh quotient

$$R_A(x) = x^T Ax$$

attains a maximum because the sphere is compact. At a maximizing unit vector v, Lagrange multipliers give

$$2Av = 2\lambda v,$$

so v is an eigenvector.

Its orthogonal complement is invariant. If $w \perp v$, then

$$\langle Aw, v\rangle = \langle w, Av\rangle = \lambda\langle w, v\rangle = 0.$$

Thus A restricts to a symmetric operator on $v^\perp$. Repeating the argument inductively produces an orthonormal eigenbasis.

Therefore

$$\boxed{A = A^T \quad \Longrightarrow \quad A = Q\,\mathrm{diag}(\lambda_1, \ldots, \lambda_n)Q^T}$$

for an orthogonal matrix Q. The proof explains both parts of the theorem: symmetry makes the orthogonal complement invariant, while compactness supplies the first eigenvector.

The extreme Rayleigh values are the extreme eigenvalues:

$$\lambda_{\min}(A) = \min_{\|x\|=1} x^T Ax, \qquad \lambda_{\max}(A) = \max_{\|x\|=1} x^T Ax.$$

The spectral theorem is therefore also an optimization theorem.

§65.8 Spectral Projectors and Functions

Suppose a diagonalizable matrix has distinct eigenvalues $\lambda_1, \dots, \lambda_s$. The Lagrange polynomials from the minimal-polynomial argument give projectors

$$P_i = \prod_{j \neq i} \frac{A - \lambda_j I}{\lambda_i - \lambda_j}.$$

They satisfy

$$P_i^2 = P_i, \qquad P_i P_j = 0 \ (i \neq j), \qquad \sum_i P_i = I,$$

and $P_i V = E_{\lambda_i}$. Hence

$$A = \sum_i \lambda_i P_i.$$

For any function f defined on the spectrum, set

$$f(A) = \sum_i f(\lambda_i) P_i.$$

This is finite-dimensional functional calculus.

If $A^2 = I$, the eigenvalues are ± 1, and

$$P_+ = \frac{I + A}{2}, \qquad P_- = \frac{I - A}{2}.$$

Therefore

$$\begin{aligned} e^{tA} &= e^t P_+ + e^{-t} P_- \\ &= \cosh t \, I + \sinh t \, A. \end{aligned}$$

A matrix exponential has been computed without multiplying matrices repeatedly. The projectors separate the invariant modes first, and the scalar function is then applied to each mode.

§65.9 Similarity versus Congruence

Similarity

$$A \longmapsto P^{-1} A P$$

changes the basis of a linear operator and preserves eigenvalues. Congruence

$$A \longmapsto P^T A P$$

changes variables in a quadratic form

$$q(x) = x^T A x, \qquad x = Py.$$

It need not preserve eigenvalues.

Consider

$$q(x, y) = 2x^2 + 4xy - y^2.$$

Completing the square gives

$$q(x, y) = 2(x + y)^2 - 3y^2.$$

With

$$u = x + y, \qquad v = y,$$

the form becomes

$$q = 2u^2 - 3v^2.$$

Thus it has one positive and one negative square. Sylvester's law of inertia says that the triple

$$(n_+, n_-, n_0)$$

of positive, negative, and zero squares is invariant under every invertible congruence. Here the inertia is

$$(1, 1, 0).$$

Orthogonal diagonalization is a special congruence that also happens to be a similarity for symmetric matrices. It preserves Euclidean geometry as well as the quadratic form's spectral data; a general completing-square substitution preserves only the inertia classification.

§65.10 Tests for Positive Definiteness

For a symmetric matrix A, the following are equivalent:

$$x^T A x > 0 \quad (x \neq 0),$$

$$\lambda_i(A) > 0 \quad \text{for every } i,$$

$$\Delta_1 > 0, \dots, \Delta_n > 0,$$

where Δ_k are the leading principal minors, and

$$A = LL^T$$

for an invertible lower-triangular matrix L with positive diagonal. These are the quadratic-form, spectral, Sylvester, and Cholesky criteria.

For

$$A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix},$$

the leading principal minors are

$$\Delta_1 = 2, \qquad \Delta_2 = 3, \qquad \Delta_3 = 4.$$

Hence $A \succ 0$. Cholesky elimination gives

$$L = \begin{pmatrix} \sqrt{2} & 0 & 0 \\ -1/\sqrt{2} & \sqrt{3/2} & 0 \\ 0 & -\sqrt{2/3} & 2/\sqrt{3} \end{pmatrix},$$

and a direct multiplication verifies

$$A = LL^T.$$

The factorization makes positivity immediate:

$$x^T A x = \|L^T x\|^2 > 0$$

for $x \neq 0$.

§65.11 Principal axes organize conics, Hessians, and PCA

A quadratic polynomial in two variables can be written

$$F(x) = x^T Ax + b^T x + c, \qquad A = A^T.$$

An orthogonal change aligns the coordinates with eigenvectors of A, removing cross terms. A translation then completes squares and moves the center when one exists. Rotation and translation perform different jobs: the first changes directions, the second changes the origin.

The same matrix appears as a Hessian in optimization. Near a critical point p,

$$f(p+h) = f(p) + \frac{1}{2} h^T \nabla^2 f(p) h + o(\|h\|^2).$$

The inertia of the Hessian distinguishes a local minimum, maximum, or saddle when the Hessian is nonsingular.

PCA applies the same spectral geometry to data. Consider the centered points

$$(2,1),\ (1,1),\ (-1,-1),\ (-2,-1).$$

Their covariance matrix is

$$C = \frac{1}{4} \sum_{i=1}^{4} x_i x_i^T = \begin{pmatrix} 5/2 & 3/2 \\ 3/2 & 1 \end{pmatrix}.$$

Its eigenvalues are

$$\lambda_\pm = \frac{7 \pm 3\sqrt{5}}{4}.$$

For λ_+, an eigenvector can be chosen proportional to

$$\begin{pmatrix} 1 \\ (\sqrt{5}-1)/2 \end{pmatrix}.$$

Projection onto this direction preserves the largest possible average squared variation. PCA is therefore the Rayleigh-quotient extremum of the covariance quadratic form, not a separate algorithmic miracle.

§65.12 Projection in Hilbert Spaces

In a Hilbert space H, every closed subspace M still has an orthogonal projection. To see why closedness matters, take a minimizing sequence $m_n \in M$ for the distance from x to M. The parallelogram identity shows that (m_n) is Cauchy; completeness of M gives a limit $m \in M$, and the usual first-variation argument gives

$$x - m \perp M.$$

Thus

$$H = M \oplus M^\perp.$$

Finite-dimensional least squares is the first instance of this projection theorem.

The spectral theorem also survives, but not always as an eigenbasis. On

$$H = L^2([0,1]),$$

consider the self-adjoint multiplication operator

$$(Mf)(t) = tf(t).$$

If $Mf = \lambda f$, then

$$(t - \lambda)f(t) = 0$$

almost everywhere. The set $\{t = \lambda\}$ has measure zero, so $f = 0$ in L^2. The operator has no nonzero eigenvectors, even though its spectrum is the whole interval $[0, 1]$.

The correct infinite-dimensional statement uses a projection-valued measure:

$$M = \int_{[0,1]} \lambda \, dE(\lambda).$$

Finite diagonal matrices are the discrete case of this integral. Orthogonal projection, self-adjointness, and functional calculus remain; what can disappear is a basis made from individual eigenvectors.

VI

Matrix Analysis and Computation

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Invertibility, Orthogonal Motion, and Polar Decomposition

N-20220411

A matrix may be invertible while compressing one direction so severely that recovering it from noisy data becomes impossible in practice. Gaussian elimination detects whether a direction is destroyed; singular values quantify how close that destruction is. Between these two questions lies the geometry of orthogonal transformations and polar decomposition.[1]

§66.1 An inverse requires a trivial kernel

For $A \in M_n(F)$, invertibility is equivalent to $\ker A = \{0\}$. Indeed, rank–nullity shows that an injective endomorphism of F^n must also be surjective. Gaussian elimination detects this by seeking a pivot in every column. For the two-dimensional case,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}, \quad ad - bc \neq 0.$$

The determinant is the signed volume factor, so a zero determinant means that some nonzero direction has been collapsed.

The inverse can also be found without a closed formula. To solve $AX = I$, use the same row operations on both blocks of the augmented matrix:

$$\left[\begin{array}{cc|cc} 1 & 4 & 1 & 0 \\ 2 & 9 & 0 & 1 \end{array}\right] \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{cc|cc} 1 & 4 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{array}\right] \xrightarrow{R_1 \leftarrow R_1 - 4R_2} \left[\begin{array}{cc|cc} 1 & 0 & 9 & -4 \\ 0 & 1 & -2 & 1 \end{array}\right].$$

Thus $A = \left(\begin{smallmatrix} 1 & 4 \\ 2 & 9 \end{smallmatrix}\right)$ has $A^{-1} = \left(\begin{smallmatrix} 9 & -4 \\ -2 & 1 \end{smallmatrix}\right)$. Every row operation is multiplication by an invertible elementary matrix on the left. Writing the entire computation as $E_k \cdots E_1 A = I$ explains both why the right block becomes $A^{-1} = E_k \cdots E_1$ and why elimination is an algorithm for constructing the inverse.

The order of composition is worth checking once: $(AB)^{-1} = B^{-1}A^{-1}$, because reversing a sequence of operations requires undoing the last one first. In contrast, a diagonal matrix $\operatorname{diag}(d_1, \dots, d_n)$ is inverted entry by entry if every d_i is nonzero.

§66.2 The adjugate records the missing direction

The adjugate $\operatorname{adj}(A)$ is the transpose of the cofactor matrix. Expanding the determinant along a row gives the identity

$$A \operatorname{adj}(A) = \operatorname{adj}(A) A = (\det A) I.$$

When $\det A \neq 0$, this produces A^{-1}. If $\operatorname{rank} A = n - 1$, some $(n-1)$-minor is nonzero, so $\operatorname{adj}(A) \neq 0$; the identity then shows every column of the adjugate is a kernel vector. In fact, its rank is one. If $\operatorname{rank} A \leq n - 2$, all these minors vanish and $\operatorname{adj}(A) = 0$.

The adjugate is algebraic. The *adjoint* A^* instead comes from an inner product; it is A^T over $\mathbb{R}$ and $\overline{A}^T$ over $\mathbb{C}$. Distinguishing the two matters when interpreting geometry.

[1] See MIT 18.700, supplementary notes on Gaussian elimination and the spectral theorem: https://ocw.mit.edu/courses/18-700-linear-algebra-fall-2013/pages/study-materials/.

§66.3 Rotations as one-parameter subgroups

An orthogonal matrix Q satisfies $Q^TQ = I$ and preserves all Euclidean lengths and angles. The two-dimensional rotations are

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad R_\theta R_\varphi = R_{\theta+\varphi}.$$

Differentiating at $\theta = 0$ gives

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad J^T = -J.$$

This is a Lie-algebraic phenomenon that can be checked directly. If $K^T = -K$, then $Q(t) = e^{tK}$ obeys

$$\frac{d}{dt}(Q(t)^TQ(t)) = Q(t)^T(K^T + K)Q(t) = 0.$$

Since $Q(0) = I$, every $Q(t)$ is orthogonal; continuity of determinant shows $\det Q(t) = 1$. Thus skew-symmetric matrices generate smooth paths of rotations.

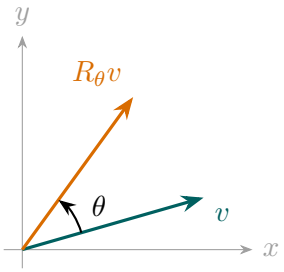


Figure 66.1: A rotation changes the direction of a vector while preserving its length; the angle between the two rays is θ.

§66.4 The polar factorization of an invertible map

Let $A \in GL(n, \mathbb{R})$. The matrix A^TA is positive definite, so by the spectral theorem it has a unique positive-definite square root

$$P = (A^TA)^{1/2}.$$

Define $Q = AP^{-1}$. Then

$$Q^TQ = P^{-1}A^TAP^{-1} = I, \qquad \boxed{A = QP}.$$

The factor P is a stretching along orthogonal eigendirections; Q is an isometry, possibly reversing orientation. Because $P^2 = A^TA$, the two factors are uniquely determined.

For the shear $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, the decomposition is explicit:

$$A^TA = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix},$$

$$P = \frac{1}{\sqrt{5}}\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, \quad Q = \frac{1}{\sqrt{5}}\begin{pmatrix} 2 & 1 \\ -1 & 2 \end{pmatrix}.$$

Indeed, $P^2 = A^TA$, P has positive eigenvalues, $Q^TQ = I$, and multiplying QP recovers the original shear. Here Q is a clockwise rotation by $\arctan(1/2)$: even a shear has a genuine rotational factor when described using the Euclidean metric. Polar decomposition lets us quantify that geometry instead of attaching a vague arrow to a matrix.

§66.5 Conditioning is more precise than the determinant

The singular values of A are the square roots of the eigenvalues of A^TA, ordered $\sigma_1 \geq \cdots \geq \sigma_n > 0$. They give the sharp estimates

$$\sigma_n\|x\|_2 \leq \|Ax\|_2 \leq \sigma_1\|x\|_2, \qquad \|A^{-1}\|_2 = \sigma_n^{-1}.$$

A perturbation in the data of $Ax = b$ produces

$$\|\delta x\|_2 = \|A^{-1}\delta b\|_2 \leq \frac{\|\delta b\|_2}{\sigma_n}.$$

For relative errors the spectral condition number is

$$\kappa_2(A) = \frac{\sigma_1}{\sigma_n}.$$

The determinant can remain harmless even when the condition number diverges. For

$$A_\varepsilon = \begin{pmatrix} \varepsilon & 0 \\ 0 & \varepsilon^{-1} \end{pmatrix}, \qquad 0 < \varepsilon < 1,$$

we have $\det A_\varepsilon = 1$ but $\kappa_2(A_\varepsilon) = \varepsilon^{-2}$. The map preserves area while nearly destroying information along one axis.

Elimination determines whether an inverse exists, polar decomposition isolates stretching from orthogonal motion, and singular values show how reliable that inverse is. These are progressively finer questions about a single linear transformation.

Rank, Projections, and Long-Term Matrix Dynamics

N-20260103

§67.1 Polynomial Relations of Matrices

A large matrix power or rank identity rarely becomes easier by multiplying entries. The decisive information is often an algebraic relation:

$$P^2 = P, \qquad N^k = 0, \qquad A^2 + c_1 A + c_0 I = 0, \qquad A = UV^T.$$

Each relation compresses the matrix into a small algebra. An idempotent splits the space into image and kernel. A nilpotent has a finite geometric series. A polynomial relation reduces every high power to lower powers. A low-rank factorization says that only a few independent directions pass through the map.

This chapter therefore organizes matrix problems by structure rather than by the surface form of an exercise. Rank describes which directions survive, projections describe how the space splits, polynomial identities describe repeated iteration, and spectral projectors describe what remains after transient modes decay.

§67.2 Rank Factorization and Quotients

Let $A : k^n \to k^m$ have rank r. Choose a basis $c_1, \dots, c_r$ of its column space and place these columns in

$$B \in k^{m \times r}.$$

Every column of A has unique coordinates in this basis; collect those coordinates in

$$C \in k^{r \times n}.$$

Then

$$\boxed{A = BC,}$$

with

$$\operatorname{rank} B = \operatorname{rank} C = r.$$

This is a rank factorization. It is the matrix form of

$$k^n \longrightarrow k^n / \ker A \cong \operatorname{im} A \hookrightarrow k^m.$$

For example,

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{pmatrix}$$

has rank two. Its first two columns are independent, and the third satisfies

$$c_3 = -c_1 + 2c_2.$$

Thus

$$A = \underbrace{\begin{pmatrix} 1 & 2 \\ 2 & 4 \\ 1 & 1 \end{pmatrix}}_{B} \underbrace{\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \end{pmatrix}}_{C}.$$

Equivalently,

$$A = c_1(1, 0, -1) + c_2(0, 1, 2),$$

a sum of two rank-one matrices.

The factorization is not unique. For any $S \in GL_r(k)$,

$$A = (BS)(S^{-1}C).$$

The ambiguity is exactly a change of coordinates in the intermediate r-dimensional image space.

§67.3 Four Fundamental Subspaces

For a real matrix $A \in \mathbb{R}^{m\times n}$, the row space and null space are orthogonal complements:

$$\mathrm{row}(A) = (\ker A)^{\perp}.$$

Indeed, $Ax = 0$ says that x is orthogonal to every row. Similarly,

$$\mathrm{col}(A) = (\ker A^T)^{\perp}.$$

Hence

$$\mathbb{R}^n = \mathrm{row}(A) \oplus \ker A,$$
$$\mathbb{R}^m = \mathrm{col}(A) \oplus \ker A^T.$$

These are the four fundamental subspaces.

Positivity gives the useful identities

$$\ker(A^TA) = \ker A, \qquad \ker(AA^T) = \ker A^T.$$

For example,

$$A^TAx = 0 \quad \Longrightarrow \quad x^TA^TAx = \|Ax\|^2 = 0,$$

so $Ax = 0$. Therefore

$$\mathrm{rank}(A^TA) = \mathrm{rank}(AA^T) = \mathrm{rank}\, A.$$

A rank identity has been reduced to equality of null spaces, and that equality is certified by a norm square.

If two homogeneous systems $Ax = 0$ and $Bx = 0$ have the same solution space, then

$$\mathrm{row}(A) = (\ker A)^{\perp} = (\ker B)^{\perp} = \mathrm{row}(B).$$

Thus equality of solution sets and equality of row spaces are the same statement viewed from opposite sides.

§67.4 Adjugates at Corank One

For every square matrix,

$$A\,\mathrm{adj}(A) = \mathrm{adj}(A)A = (\det A)I.$$

If $\operatorname{rank} A = n-1$, then $\det A = 0$ but at least one $(n-1)$-minor is nonzero, so

$$\mathrm{adj}(A) \neq 0.$$

The identity

$$A\,\mathrm{adj}(A) = 0$$

shows that every column of $\mathrm{adj}(A)$ lies in $\ker A$, which is one-dimensional. Hence all columns are proportional and

$$\operatorname{rank}\mathrm{adj}(A) = 1.$$

Likewise, every row lies in $\ker A^T$. If

$$Au = 0, \qquad v^T A = 0,$$

with nonzero u, v, then

$$\mathrm{adj}(A) = cuv^T$$

for some nonzero scalar c.

For

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix},$$

one has

$$\mathrm{adj}(A) = \begin{pmatrix} 4 & -2 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \end{pmatrix}\begin{pmatrix} -2 & 1 \end{pmatrix}.$$

The same null vector appears on the right and left because A is symmetric. Any nonzero column of the adjugate solves $Ax = 0$.

If $\operatorname{rank} A \leq n-2$, every $(n-1)$-minor vanishes and

$$\mathrm{adj}(A) = 0.$$

The adjugate therefore distinguishes corank one from deeper singularity.

§67.5 Idempotents and Oblique Projections

Suppose

$$P^2 = P.$$

Every vector splits as

$$x = Px + (x - Px).$$

The first term belongs to $\operatorname{im} P$, while

$$P(x - Px) = Px - P^2x = 0,$$

so the second belongs to $\ker P$. Their intersection is zero, hence

$$V = \operatorname{im} P \oplus \ker P.$$

The matrix acts as identity on its image and zero on its kernel. Its minimal polynomial divides

$$t(t-1),$$

so it is diagonalizable with eigenvalues $0, 1$.

The projection is orthogonal exactly when

$$P^T = P.$$

Without symmetry it is oblique. For example,

$$P = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}$$

satisfies $P^2 = P$. It projects onto the x-axis along the line

$$\ker P = \operatorname{span}(1, 1).$$

The image and kernel are complementary but not orthogonal.

If P, Q are idempotent over a field of characteristic not two, then

$$P + Q \text{ is idempotent} \quad \Longleftrightarrow \quad PQ = QP = 0.$$

Indeed, expanding $(P+Q)^2 = P + Q$ gives

$$PQ + QP = 0.$$

Multiplying on the left and right by P shows $PQ = QP$; the displayed sum then gives $2PQ = 0$. Geometrically, each image lies in the other's kernel, so the two projections occupy noninteracting components.

§67.6 Finite Formulas from Nilpotence

If $N^k = 0$, then

$$(I - N)(I + N + \cdots + N^{k-1}) = I - N^k = I.$$

Therefore

$$\boxed{(I - N)^{-1} = I + N + \cdots + N^{k-1}.}$$

The inverse is a polynomial because the geometric series terminates.

For the nilpotent Jordan block

$$N = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \qquad N^3 = 0,$$

one has

$$(I - N)^{-1} = I + N + N^2 = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

More generally,

$$(\lambda I + N)^m = \sum_{j=0}^{k-1} \binom{m}{j} \lambda^{m-j} N^j.$$

A Jordan block therefore contributes an exponential factor λ^m multiplied by polynomials in m. The nilpotent part measures the failure of pure eigendirection scaling.

§67.7 The minimal polynomial reduces every high power

Cayley–Hamilton says that a matrix satisfies its characteristic polynomial. The sharper computational object is the minimal polynomial. If

$$m_A(t) = t^d + c_{d-1}t^{d-1} + \cdots + c_0,$$

then

$$A^d = -c_{d-1}A^{d-1} - \cdots - c_0 I,$$

and every higher power reduces to a linear combination of

$$I, A, \ldots, A^{d-1}.$$

Equivalently, compute the remainder of t^n modulo $m_A(t)$ and substitute A.

Take

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}.$$

Its minimal polynomial is

$$m_A(t) = (t-1)(t-2).$$

The degree-one remainder $r_n(t) = \alpha t + \beta$ of t^n satisfies

$$r_n(1) = 1, \qquad r_n(2) = 2^n.$$

Thus

$$r_n(t) = (2^n - 1)t + (2 - 2^n),$$

and

$$\boxed{A^n = (2^n - 1)A + (2 - 2^n)I.}$$

Even A^{100} is no harder than evaluating two scalar powers.

§67.8 Primary Projectors from CRT

Suppose

$$m_A(t) = p_1(t)^{e_1} \cdots p_s(t)^{e_s}$$

with pairwise coprime factors. The polynomial Chinese remainder theorem provides idempotent classes $e_i(t)$ satisfying

$$e_i \equiv 1 \pmod{p_i^{e_i}}, \qquad e_i \equiv 0 \pmod{p_j^{e_j}} \quad (j \neq i).$$

Substituting A gives commuting projectors

$$E_i = e_i(A), \qquad \sum_i E_i = I, \qquad E_i E_j = 0.$$

Their images are the primary components

$$\ker p_i(A)^{e_i}.$$

For a concrete example, suppose

$$m_A(t) = (t-1)^2(t+2).$$

Polynomials with the required congruences are

$$e_1(t) = \frac{(t+2)(4-t)}{9}, \qquad e_2(t) = \frac{(t-1)^2}{9}.$$

Indeed,

$$e_1 \equiv 1 \pmod{(t-1)^2}, \qquad e_1 \equiv 0 \pmod{t+2},$$

and $e_2 = 1 - e_1$. Therefore

$$E_1 = \frac{(A+2I)(4I-A)}{9}, \qquad E_2 = \frac{(A-I)^2}{9}$$

project onto the generalized 1- and -2-primary subspaces.

This is the structural source of generalized eigenspace decomposition. The vector space becomes a module over $k[t]$, with t acting as A; coprime polynomial factors split that module into independent primary pieces.

§67.9 Convergence of Matrix Powers

On a Jordan block

$$J = \lambda I + N, \qquad N^k = 0,$$

one has

$$J^n = \sum_{j=0}^{k-1} \binom{n}{j} \lambda^{n-j} N^j.$$

This formula gives precise long-term behavior.

- If $|\lambda| < 1$, every term tends to zero despite the polynomial factor.
- If $|\lambda| > 1$, some vectors grow exponentially.
- If $|\lambda| = 1$ and $N \neq 0$, polynomial growth prevents boundedness.
- A semisimple eigenvalue 1 contributes a persistent projection.
- Any other eigenvalue on the unit circle produces oscillation rather than convergence.

Consequently, A^n converges exactly when every eigenvalue lies inside the unit disk except possibly the eigenvalue 1, and the eigenvalue 1 is semisimple. In that case,

$$\lim_{n\to\infty} A^n = P_1,$$

where P_1 is the spectral projector onto $\ker(A - I)$ along the direct sum of the decaying primary components.

The phrase “only the eigenvalue 1 remains” is therefore incomplete. What remains is a specific projection determined by the decomposition of the whole space.

§67.10 A Two-State Markov Projector

Let

$$T = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}, \qquad 0 < a, b < 1.$$

Each row sums to one, so T evolves row probability vectors. Its eigenvalues are

$$1, \qquad \rho = 1 - a - b,$$

with $|\rho| < 1$. The stationary distribution is

$$\pi = \left(\frac{b}{a+b}, \frac{a}{a+b} \right),$$

because $\pi T = \pi$.

Let

$$P = \mathbf{1}\,\pi = \begin{pmatrix} b/(a+b) & a/(a+b) \\ b/(a+b) & a/(a+b) \end{pmatrix}.$$

Then

$$P^2 = P, \qquad TP = PT = P,$$

and $I - P$ is the projector onto the decaying mode. Since T acts by ρ there,

$$\boxed{T^n = P + \rho^n (I - P).}$$

Hence

$$T^n \longrightarrow P.$$

Every initial probability row converges to π. The Markov limit is a spectral projector, while the convergence rate is the magnitude of the subdominant eigenvalue.

§67.11 Diagonal Dominance and Invertibility

Suppose

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}| \qquad (i = 1, \ldots, n).$$

If $Ax = 0$ had a nonzero solution, choose k with

$$|x_k| = \max_i |x_i| > 0.$$

The k-th equation gives

$$a_{kk} x_k = - \sum_{j \neq k} a_{kj} x_j.$$

Therefore

$$|a_{kk}||x_k| \leq \sum_{j \neq k} |a_{kj}||x_j| \leq \left(\sum_{j \neq k} |a_{kj}| \right) |x_k|,$$

contradicting strict diagonal dominance. Thus A is invertible.

The same argument is the zero-eigenvalue case of Gershgorin's theorem. Every eigenvalue lies in at least one disk

$$|z - a_{ii}| \leq \sum_{j \neq i} |a_{ij}|.$$

Strict diagonal dominance keeps all disks away from zero. Before computing a determinant or inverse, the row geometry can already certify that no kernel direction exists.

§67.12 Drazin Inverses and Nilpotent Modes

A singular matrix has no ordinary inverse, but its primary decomposition separates the nonzero spectrum from the generalized zero eigenspace. On the nonzero primary components, A is invertible; on the zero component, it is nilpotent. The Drazin inverse A^D acts as the ordinary inverse on the first part and as zero on the second.

If the nilpotent zero component has index k, the Drazin inverse is characterized by

$$AA^D = A^DA,$$

$$A^DAA^D = A^D,$$

$$A^{k+1}A^D = A^k.$$

For

$$A = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 2 \end{pmatrix} \oplus \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

the nilpotent block has index two and

$$A^D = \begin{pmatrix} 1/2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The Moore–Penrose pseudoinverse is different:

$$A^+ = \begin{pmatrix} 1/2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

The pseudoinverse is determined by orthogonal least-squares geometry, whereas the Drazin inverse is determined by the algebraic decomposition into invertible and nilpotent dynamics.

This distinction completes the path from rank to iteration. Rank factorization identifies the active image, idempotents split complementary components, primary projectors isolate spectral modes, powers reveal persistent and transient behavior, and the Drazin inverse inverts exactly the modes that are dynamically invertible while discarding the nilpotent transient part.

Spectral Calculus and Positive Matrix Geometry: From Basis Changes to Lyapunov Equations

N-20260105

§68.1 Coordinates change, but the operator does not

Let V be a finite-dimensional vector space. A basis turns a linear map $T : V \to V$ into a matrix, but a different basis gives a different matrix for the same operator. Suppose the ordered bases α and β are related by

$$(\beta_1, \dots, \beta_n) = (\alpha_1, \dots, \alpha_n)P.$$

For every vector v,

$$[v]_\alpha = P[v]_\beta.$$

If $A = [T]_\alpha$ and $B = [T]_\beta$, then

$$B = P^{-1}AP.$$

Thus similarity is not an arbitrary matrix manipulation. It is the coordinate shadow of keeping the operator fixed while changing the basis.

Quantities preserved by similarity belong to the operator rather than to its coordinates. The characteristic polynomial, eigenvalues, trace, determinant, rank, and minimal polynomial are all similarity invariants. This is why spectral methods can replace long coordinate computations: they work with data that survive every basis change.

§68.2 A complete transition-matrix calculation

Let $\varepsilon = (\varepsilon_1, \varepsilon_2, \varepsilon_3)$ be a reference basis and define

$$\alpha_1 = \varepsilon_1, \qquad \alpha_2 = \varepsilon_1 + \varepsilon_2, \qquad \alpha_3 = \varepsilon_1 + \varepsilon_2 + \varepsilon_3,$$

while

$$\beta_1 = \varepsilon_1 + \varepsilon_2, \qquad \beta_2 = 2\varepsilon_2 + \varepsilon_3, \qquad \beta_3 = \varepsilon_1 + 3\varepsilon_3.$$

The basis matrices relative to ε are

$$P_\alpha = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \qquad P_\beta = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & 0 \\ 0 & 1 & 3 \end{pmatrix}.$$

Since

$$(\beta_1, \beta_2, \beta_3) = (\alpha_1, \alpha_2, \alpha_3)P,$$

one has

$$P = P_\alpha^{-1}P_\beta = \begin{pmatrix} 0 & -2 & 1 \\ 1 & 1 & -3 \\ 0 & 1 & 3 \end{pmatrix}.$$

A coordinate column transforms by

$$[v]_\alpha = P[v]_\beta.$$

The question "which vectors have the same coordinate column in the ε- and α-bases?" is also structural. Such a column t must satisfy

$$t = P_\alpha t.$$

Thus it lies in the fixed subspace of the transition matrix. Here

$$(P_\alpha - I)t = 0$$

forces

$$t_2 = t_3 = 0,$$

so the common-coordinate vectors are exactly the multiples of ε_1. A basis exercise has become an eigenspace calculation for eigenvalue 1.

§68.3 Matrix Functions from Polynomials

Suppose

$$Av = \lambda v.$$

For every polynomial p,

$$p(A)v = p(\lambda)v.$$

Consequently, if a matrix satisfies

$$p(A) = 0,$$

then every eigenvalue of A is a root of p.

This simple observation explains several common shortcuts. If

$$A^k = 0,$$

then every eigenvalue is zero, so for an $n \times n$ matrix

$$\det(I + 3A) = \prod_{i=1}^{n}(1 + 3\lambda_i) = 1.$$

The matrix need not be diagonalizable; the determinant conclusion uses only the eigenvalues counted with algebraic multiplicity.

If a real matrix satisfies

$$A^2 + aA + bI = 0$$

and

$$a^2 - 4b < 0,$$

then it has no real eigenvector. Indeed, a real eigenvalue λ would satisfy

$$\lambda^2 + a\lambda + b = 0,$$

contradicting the negative discriminant. The matrix may still have complex eigenvalues and become diagonalizable over $\mathbb{C}$; the obstruction is specifically to real eigendirections.

§68.4 The minimal polynomial decides diagonalizability

The characteristic polynomial lists eigenvalues with algebraic multiplicity, but the minimal polynomial records the largest Jordan blocks. A matrix is diagonalizable over a field precisely when its minimal polynomial splits into distinct linear factors.

For example, if

$$A^3 = A,$$

then

$$A(A - I)(A + I) = 0.$$

Over a field of characteristic different from 2, the polynomial

$$x(x - 1)(x + 1)$$

has three distinct roots. The minimal polynomial divides this square-free polynomial, so A is diagonalizable and

$$V = \ker A \oplus \ker(A - I) \oplus \ker(A + I).$$

This conclusion is stronger than merely saying that the possible eigenvalues are $-1, 0, 1$.

Repeated eigenvalues require more care. If an $n \times n$ matrix is known to be diagonalizable and λ has algebraic multiplicity m, then

$$\dim \ker(A - \lambda I) = m.$$

Thus a parameter problem with a repeated eigenvalue can often be solved by imposing a rank condition on $A - \lambda I$. The rank is not an unrelated trick; it measures the missing eigenspace dimension.

§68.5 Spectral mapping also controls inverses and adjugates

If A is invertible and $Av = \lambda v$, then

$$A^{-1}v = \lambda^{-1}v.$$

Since

$$\operatorname{adj}(A) = \det(A)A^{-1},$$

the same eigenvector satisfies

$$\operatorname{adj}(A)v = \frac{\det A}{\lambda}v.$$

Thus the eigenvalues of the adjugate are the products of all eigenvalues except one:

$$\frac{\lambda_1 \cdots \lambda_n}{\lambda_i} = \prod_{j \neq i} \lambda_j.$$

Trace and determinant are the first two symmetric spectral statistics:

$$\operatorname{tr} A = \sum_i \lambda_i, \qquad \det A = \prod_i \lambda_i.$$

They can be read from diagonal entries, from a triangular form, or from the eigenvalues because these are different coordinate descriptions of the same invariants.

A row-sum shortcut fits the same pattern. If every row sum of A equals c, then

$$A\mathbf{1} = c\mathbf{1}, \qquad \mathbf{1} = (1, \ldots, 1)^T.$$

The aggregate direction $\mathbf{1}$ is an invariant one-dimensional subspace. The shortcut works because the matrix preserves a visible symmetry direction.

§68.6 The Symmetric Spectral Theorem

For a real symmetric matrix A, there is an orthogonal matrix Q such that

$$A = Q\,\mathrm{diag}(\lambda_1, \dots, \lambda_n)Q^T.$$

The eigenspaces are mutually orthogonal, and every geometric statement can be checked along independent principal directions.

Suppose a symmetric 3×3 matrix has eigenvalues

$$-1, 1, 1$$

and a -1-eigenvector

$$u = \frac{1}{\sqrt{2}}(0, 1, 1)^T.$$

The orthogonal complement $u^\perp$ is the $+1$-eigenspace. Therefore A is reflection across $u^\perp$:

$$A = I - 2uu^T.$$

Computing the rank-one term gives

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}.$$

This reconstructs the matrix without guessing two additional eigenvectors. The spectral decomposition is the geometry of a reflection.

§68.7 Functions on Eigenspaces

For a symmetric matrix

$$A = Q\,\mathrm{diag}(\lambda_i)Q^T$$

and a function f defined on its spectrum, set

$$f(A) = Q\,\mathrm{diag}(f(\lambda_i))Q^T.$$

This defines powers, exponentials, logarithms, and roots without choosing coordinates beyond the orthonormal eigenbasis.

Every real number has a unique real odd root. Hence, for any positive integer k, define

$$B = Q\,\mathrm{diag}(\sqrt[2k+1]{\lambda_1}, \dots, \sqrt[2k+1]{\lambda_n})Q^T.$$

Then B is real symmetric and

$$B^{2k+1} = A.$$

Even roots behave differently. A real symmetric matrix has a real symmetric square root exactly when

$$A \succeq 0.$$

In that case it has a unique positive semidefinite square root, although other symmetric square roots may choose signs independently on eigenspaces. For a positive definite matrix, one may similarly define

$$A^t, \qquad \log A, \qquad e^A$$

by applying the scalar functions to positive eigenvalues.

§68.8 Rayleigh quotients define the Loewner order

For symmetric A, the Rayleigh quotient is

$$R_A(x) = \frac{x^T A x}{x^T x}.$$

Writing $x = Qy$ gives

$$R_A(x) = \frac{\sum_i \lambda_i y_i^2}{\sum_i y_i^2}.$$

It is a weighted average of the eigenvalues, so

$$\lambda_{\min}(A) \le R_A(x) \le \lambda_{\max}(A).$$

The extreme values occur exactly on the corresponding eigenspaces.

For symmetric matrices, define the Loewner order by

$$A \succeq B \quad \Longleftrightarrow \quad x^T(A-B)x \ge 0 \quad \text{for every } x.$$

Thus

$$A \succeq B \quad \Longleftrightarrow \quad \lambda_{\min}(A-B) \ge 0.$$

If

$$\lambda_{\min}(A) > \lambda_{\max}(B),$$

then for every unit vector x,

$$x^T(A-B)x \ge \lambda_{\min}(A) - \lambda_{\max}(B) > 0,$$

so

$$A - B \succ 0.$$

The eigenvalue comparison is exactly an order statement about quadratic forms.

§68.9 Positive Matrices and Ellipsoids

A symmetric matrix A is positive definite when

$$x^T A x > 0 \qquad (x \neq 0).$$

Spectrally, this means

$$\lambda_i(A) > 0$$

for all i. Geometrically,

$$E_A = \{x : x^T A x \le 1\}$$

is an ellipsoid whose semi-axis lengths are

$$\lambda_i(A)^{-1/2}.$$

If $A \succeq B \succ 0$, then

$$E_A \subseteq E_B.$$

Larger quadratic forms produce smaller ellipsoids.

The positive semidefinite matrices form the convex cone

$$S_+^n = \{A = A^T : A \succeq 0\}.$$

Indeed, if $A, B \succeq 0$ and $s, t \ge 0$, then

$$x^T(sA + tB)x = s\,x^T A x + t\,x^T B x \ge 0.$$

Sylvester's criterion, which tests positivity through leading principal minors, is a coordinate certificate for membership in the interior of this cone.

§68.10 A Block-Matrix Spectral Trick

Let $B = B^T$ and consider

$$M = \begin{pmatrix} I & B \\ B & I \end{pmatrix}.$$

Use the orthogonal matrix

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} I & I \\ I & -I \end{pmatrix}.$$

A direct multiplication gives

$$U^T M U = \begin{pmatrix} I+B & 0 \\ 0 & I-B \end{pmatrix}.$$

Therefore

$$M \succ 0 \quad \iff \quad I+B \succ 0 \text{ and } I-B \succ 0.$$

Since the eigenvalues of $I \pm B$ are $1 \pm \lambda_i(B)$,

$$\boxed{M \succ 0 \quad \iff \quad |\lambda_i(B)| < 1 \text{ for every } i.}$$

The block congruence trick is really a decomposition into symmetric and antisymmetric subspaces

$$(x, x), \qquad (x, -x).$$

§68.11 Noncommutativity is where scalar order intuition fails

For positive numbers,

$$0 < a \le b \quad \Longrightarrow \quad a^2 \le b^2.$$

The analogous matrix statement is false without commutativity. Take

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}.$$

Both are positive definite, and

$$B - A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \succeq 0.$$

However,

$$B^2 - A^2 = \begin{pmatrix} 6 & 5 \\ 5 & 4 \end{pmatrix}$$

has determinant

$$24 - 25 = -1,$$

so it is indefinite. Thus $A \preceq B$ does not imply $A^2 \preceq B^2$.

Functions that preserve Loewner order on every matrix size are called operator monotone. The inverse function is order reversing:

$$0 \prec A \preceq B \quad \Longrightarrow \quad B^{-1} \preceq A^{-1}.$$

To see this, conjugate by $B^{-1/2}$:

$$0 \prec B^{-1/2} A B^{-1/2} \preceq I.$$

Taking reciprocal eigenvalues reverses the inequality, and conjugating back gives the result. The proof works because inversion can be reduced to an ordinary spectral inequality after a congruence.

§68.12 A Positive-Matrix Inequality

For positive definite A, B, consider

$$(A+B)(A^{-1}+B^{-1}).$$

It need not be symmetric. Nevertheless, it is similar to a symmetric positive definite matrix. Set

$$X = A^{-1/2}BA^{-1/2} \succ 0.$$

Conjugation by $A^{1/2}$ gives

$$A^{-1/2}(A+B)(A^{-1}+B^{-1})A^{1/2} = 2I + X + X^{-1}.$$

If $\lambda > 0$ is an eigenvalue of X, the corresponding eigenvalue of the right-hand side is

$$2 + \lambda + \lambda^{-1} \geq 4.$$

Hence every eigenvalue of the original product is real and at least 4.

The successful reduction does not require A and B to commute. It uses a carefully chosen similarity that packages their relative position into the single positive matrix X.

§68.13 Solving Sylvester Equations

Consider

$$AX + XB = C.$$

If A and B are diagonalizable with eigenvalues α_i and β_j, the equation becomes entrywise in their eigenbases:

$$(\alpha_i + \beta_j)Y_{ij} = K_{ij}.$$

Thus the solution is unique when

$$\alpha_i + \beta_j \neq 0$$

for every pair, equivalently when

$$\sigma(A) \cap \sigma(-B) = \varnothing.$$

This is the spectral separation condition for the Sylvester operator

$$\mathcal{S}(X) = AX + XB.$$

For the symmetric equation

$$AX + XA = C$$

with $A \succ 0$, uniqueness follows because all sums $\lambda_i(A) + \lambda_j(A)$ are positive. If also $C \succ 0$, the solution has the integral representation

$$\boxed{X = \int_0^\infty e^{-tA} C e^{-tA}\, dt.}$$

Indeed,

$$\frac{d}{dt}(e^{-tA}Ce^{-tA}) = -Ae^{-tA}Ce^{-tA} - e^{-tA}Ce^{-tA}A.$$

Integrating from 0 to ∞ gives

$$AX + XA = C,$$

because $e^{-tA} \to 0$. Moreover, for $v \neq 0$,

$$v^T X v = \int_0^\infty (e^{-tA}v)^T C (e^{-tA}v)\, dt > 0.$$

Thus the unique solution is positive definite. The integral proves positivity without requiring A and C to commute.

§68.14 Lyapunov Equations and Decay

Let a linear system satisfy

$$x'(t) = Mx(t).$$

Assume every eigenvalue of M has negative real part. For any $Q \succ 0$, define

$$P = \int_0^\infty e^{tM^T} Q e^{tM}\, dt.$$

The exponential decay makes the integral converge, and the same differentiation argument yields

$$M^T P + PM = -Q.$$

The matrix P is positive definite because

$$x^T P x = \int_0^\infty (e^{tM}x)^T Q (e^{tM}x)\, dt > 0$$

for $x \neq 0$.

Now set

$$V(x) = x^T P x.$$

Along a trajectory,

$$\begin{aligned} \frac{d}{dt} V(x(t)) &= x(t)^T (M^T P + PM) x(t) \\ &= -x(t)^T Q x(t) < 0 \end{aligned}$$

away from the origin. The solution of a matrix equation has become a decreasing energy that proves asymptotic stability.

This is where the chapter's themes meet. Similarity isolates coordinate-independent spectral data; symmetric functional calculus turns that data into matrix functions and quadratic geometry; Loewner order records positivity; and the Lyapunov integral converts spectral decay into a concrete positive quadratic form.

Two-Dimensional Linear Maps, Spectral Geometry, and Singular Value Decomposition

N-20260303

§69.1 Concept map of the chapter

The main linear-algebra concepts form a dependency map. Determinants test invertibility and prepare the spectral problem; eigenvalues determine eigenspaces; orthonormal eigenvectors produce orthogonal diagonalization in favorable cases; diagonalization and singular value decomposition then become computational languages for dimensionality reduction.

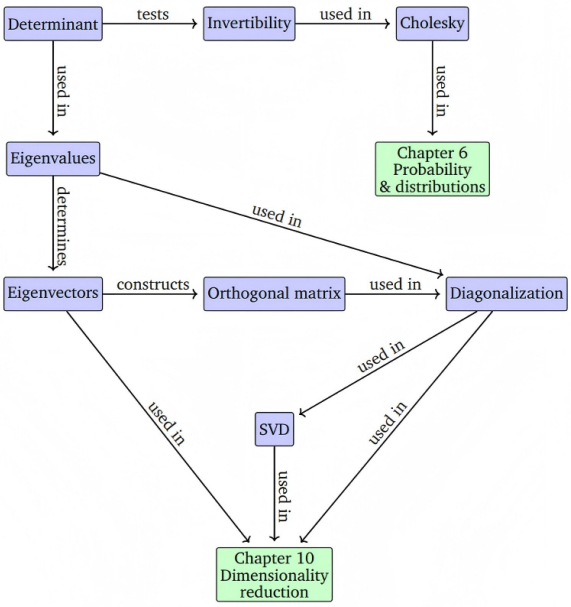


Figure 4.1 A mind map of the concepts introduced in this chapter, along with where they are used in other parts of the book.

Figure 69.1: A concept map linking determinant, invertibility, eigenvalues, eigenvectors, orthogonal matrices, diagonalization, Cholesky factorization, SVD, and dimensionality reduction.

The diagram should not be read as a strict dependency graph. It is a guide to how these objects are reused: determinant and eigenvalues are algebraic invariants; eigenvectors and orthogonal matrices give coordinate systems; SVD is the robust replacement when diagonalization is unavailable or when the matrix is rectangular.

§69.2 Five model transformations

A 2×2 matrix is not only an array of four numbers; it is a rule that moves every point of the plane. The determinant records signed area scaling, while eigenvectors record directions that survive the transformation without turning:

$$|A(e_1), A(e_2)| = \det A, \qquad Av = \lambda v.$$

The source figure compares five concrete linear maps by applying each to a grid of colored points. The left column is the input, the right column is the transformed output, and the middle column marks eigenvector directions and eigenvalue data where they are real.

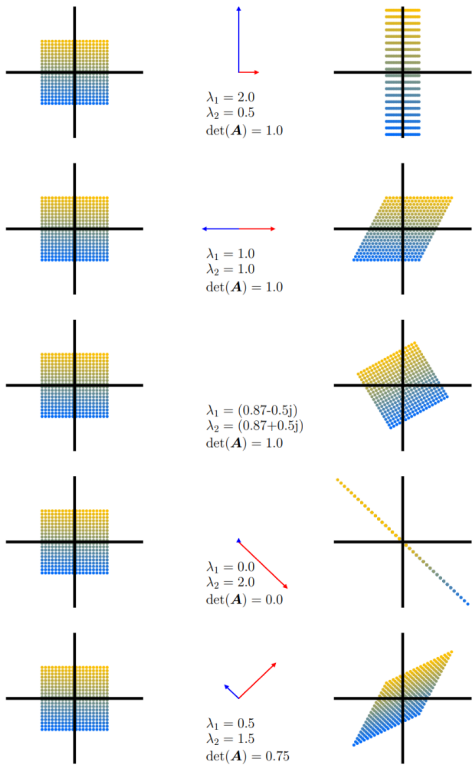


Figure 4.4 Determinants and eigenspaces. Overview of five linear mappings and their associated transformation matrices $\boldsymbol{A}_i \in \mathbb{R}^{2\times 2}$ projecting 400 color-coded points $\boldsymbol{x} \in \mathbb{R}^2$ (left column) onto target points $\boldsymbol{A}_i\boldsymbol{x}$ (right column). The central column depicts the first eigenvector, stretched by its associated eigenvalue λ_1, and the second eigenvector stretched by its eigenvalue λ_2. Each row depicts the effect of one of five transformation matrices $\boldsymbol{A}_i$ with respect to the standard basis.

Figure 69.2: Five planar linear mappings and their eigenspace behavior. Each row shows colored input points, eigenvalue/eigenvector information, and the transformed output points.

The five rows illustrate the standard possibilities: axis-aligned stretching, shear, rotation with complex eigenvalues, collapse onto a line, and symmetric stretching. The determinant and the eigenvalues answer different questions. The determinant says how area changes; eigenvectors say which directions are preserved.

§69.3 A spectral example: a biological neural network

The note then moves from geometry to data. For the *C. elegans* neural network, one may form a connectivity matrix. Since the directed connection matrix is generally not symmetric, one often studies a symmetrized matrix, for example

$$A_{\text{sym}} = A + A^T.$$

The eigenvalues of this matrix describe dominant modes of connectivity.

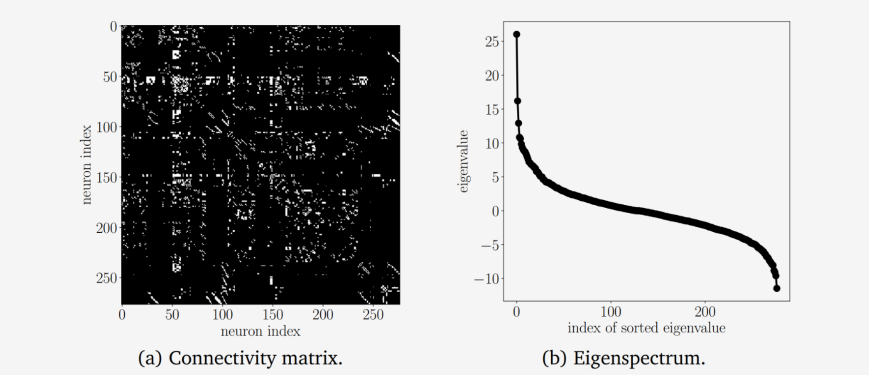


Figure 4.5 Caenorhabditis elegans neural network (Kaiser and Hilgetag, 2006).(a) Symmetrized connectivity matrix; (b) Eigenspectrum.

Figure 69.3: The symmetrized connectivity matrix of the C. elegans neural network and its sorted eigenspectrum.

This example is a useful warning: eigenvectors are not only arrows in the plane. In a network or a dataset, an eigenvector can indicate a coherent global mode, and the corresponding eigenvalue measures its strength.

§69.4 SVD as a sequence of geometric operations

Eigenvalue decomposition is powerful but fragile: it is best behaved for square matrices with enough eigenvectors, and especially for symmetric matrices. Singular value decomposition is more stable. Every real matrix $A \in \mathbb{R}^{m\times n}$ admits

$$A = U\Sigma V^T,$$

where U and V are orthogonal and Σ has nonnegative diagonal entries $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$. Geometrically, V^T changes the input coordinates, Σ stretches by the singular values, and U changes the output coordinates.

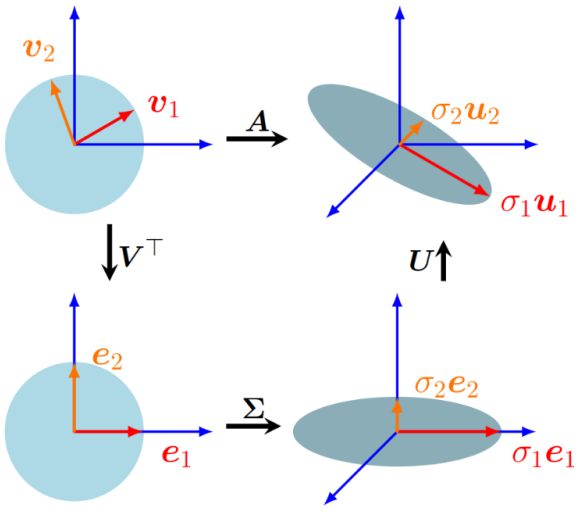


Figure 69.4: Geometric interpretation of SVD: the unit disk is first expressed in the right-singular-vector coordinates, then stretched by the singular values, and finally rotated into the output coordinates.

Thus SVD does not ask for directions that remain invariant under A. Instead, it asks for orthogonal input directions that are sent to orthogonal output axes. The unit circle becomes an ellipse, and the semi-axis lengths are the singular values.

§69.5 SVD and latent structure in data

A rating table can be viewed as a matrix. Rows may represent movies and columns may represent users. The raw entries are ratings, but the singular vectors can reveal latent directions such as genre preference or taste profile.

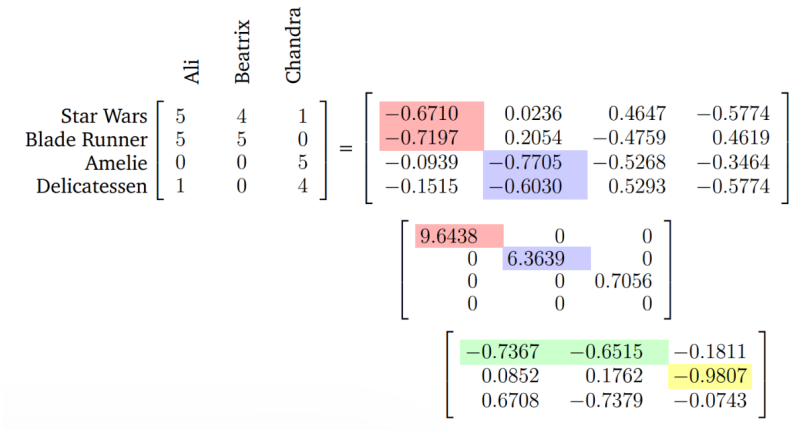


Figure 4.10 Movie ratings of three people for four movies and its SVD decomposition.

Figure 69.5: A movie-rating matrix and its SVD decomposition. Dominant singular directions highlight latent structure in user and movie preferences.

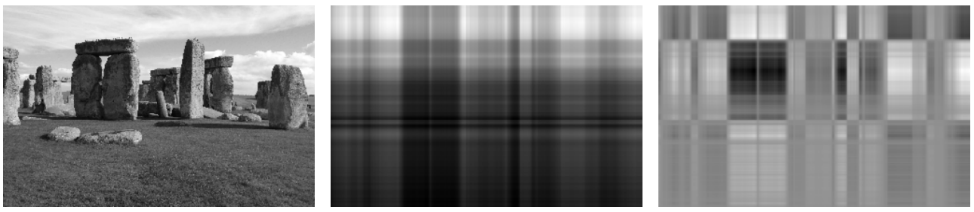
The labels placed on singular directions are interpretations, not part of the theorem. The theorem says that the first singular directions give the most efficient orthogonal explanation of the matrix energy.

§69.6 Rank-one blocks and image compression

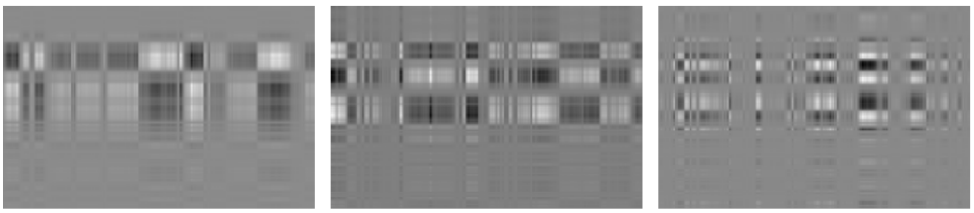
SVD decomposes a matrix into rank-one pieces:

$$A = \sum_{i=1}^{r} \sigma_i u_i v_i^T.$$

For an image matrix, each term $\sigma_i u_i v_i^T$ is a weighted rank-one visual pattern.



(a) Original image $\boldsymbol{A}$. (b) $\boldsymbol{A}_1,\ \sigma_1 \approx 228,052$. (c) $\boldsymbol{A}_2,\ \sigma_2 \approx 40,647$.



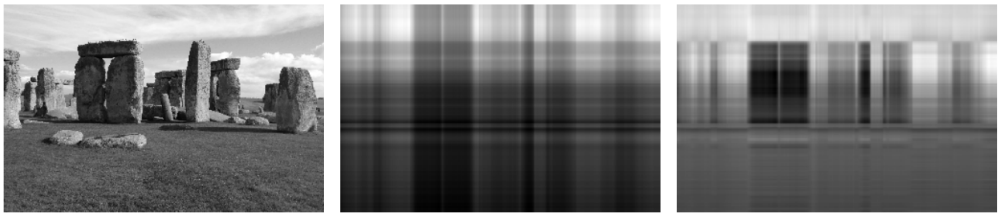
(d) $\boldsymbol{A}_3,\ \sigma_3 \approx 26,125$. (e) $\boldsymbol{A}_4,\ \sigma_4 \approx 20,232$. (f) $\boldsymbol{A}_5,\ \sigma_5 \approx 15,436$.

Figure 69.6: The first rank-one SVD components of an image. Each component captures one weighted visual pattern rather than a recognizable full image.

Truncated SVD keeps only the first k terms:

$$\widehat{A}(k) = \sum_{i=1}^{k} \sigma_i u_i v_i^T.$$

The Eckart–Young theorem says that this is the best rank-k approximation in the spectral norm and the Frobenius norm.



(a) Original image $\boldsymbol{A}$. (b) Rank-1 approximation $\widehat{\boldsymbol{A}}(1)$. (c) Rank-2 approximation $\widehat{\boldsymbol{A}}(2)$.



(d) Rank-3 approximation $\widehat{\boldsymbol{A}}(3)$. (e) Rank-4 approximation $\widehat{\boldsymbol{A}}(4)$. (f) Rank-5 approximation $\widehat{\boldsymbol{A}}(5)$.

Figure 69.7: Low-rank image reconstruction from the first few singular components. As the rank increases, the approximation recovers more recognizable structure.

This example explains why SVD is useful beyond pure linear algebra: it gives a principled compression method by separating large-scale structure from smaller-scale details.

§69.7 A taxonomy of matrix properties

The final source figure is a taxonomy of real matrices. It distinguishes square from nonsquare matrices, singular from regular matrices, defective from nondefective matrices, normal from non-normal matrices, and the special positions of symmetric, positive definite, orthogonal, diagonal, and identity matrices.

Figure 4.13 A functional phylogeny of matrices encountered in machine learning.

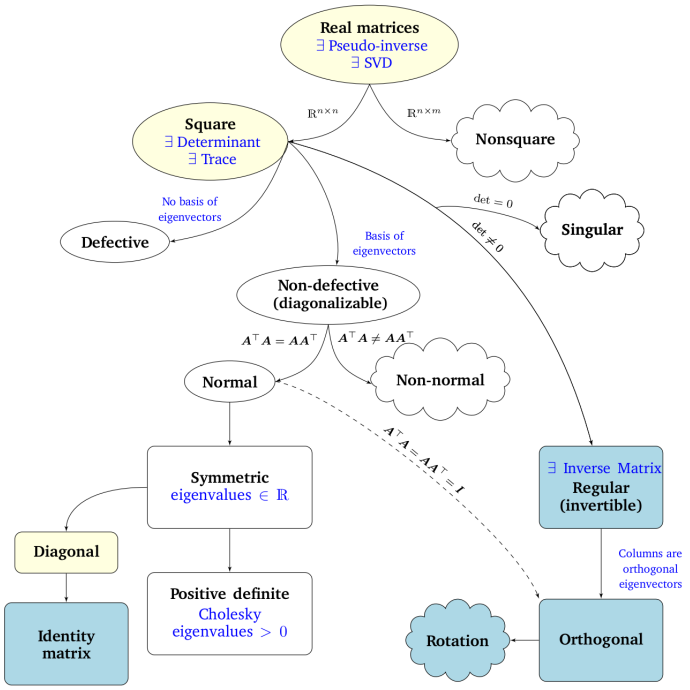


Figure 69.8: A functional phylogeny of real matrices encountered in machine learning, indicating which tools apply to which matrix classes.

The diagram prevents a common error: applying the wrong theorem to the wrong class. Determinants and eigenvalues belong naturally to square matrices; orthogonal diagonalization belongs to symmetric real matrices; pseudoinverses and SVD work for arbitrary real matrices.

§69.8 Choosing the Right Matrix Invariant

The unifying theme is that a matrix should be read through the invariant appropriate to the question. The determinant measures signed volume scaling; the trace and eigenvalues measure intrinsic square-matrix behavior; rank measures the dimension of the image; singular values measure metric distortion; and the pseudoinverse measures least-squares inverse behavior when an exact inverse is unavailable.

Similarity

$$A' = P^{-1}AP$$

preserves the linear operator but changes coordinates on the same space. A general matrix representation

$$\widetilde{A} = T^{-1}AS$$

may change both input and output coordinates. SVD chooses these input and output coordinates orthogonally, which is why it is numerically stable.

For the examples in this note, the practical reading rule is:

question	tool
area/orientation	$\det A$
invariant directions	eigenvectors
collapse or information loss	$\operatorname{rank} A$
principal stretching	singular values
best low-rank compression	truncated SVD

The same linear map has an algebraic signature (rank and spectrum), a geometric action (stretches and invariant directions), and a numerical role (conditioning and low-rank approximation). The choice of invariant determines which of these aspects can be read from a matrix.

Orthogonal Projection, Least Squares, and the Geometry of Approximation

N-20260305

§70.1 The nearest point is found by a right angle

Suppose a vector y does not lie in a subspace W. There are infinitely many vectors in W, so what could make one of them the best approximation to y?

The answer is visible before it is algebraic. If $p \in W$ is the closest point, then the error

$$r = y - p$$

must be perpendicular to every direction in W. Otherwise one could move a little inside W in a direction having positive inner product with r, shortening the distance.

This gives the projection principle:

$$p = \operatorname{proj}_W y \qquad \Longleftrightarrow \qquad p \in W, \quad y - p \perp W.$$

The right angle is not a decorative consequence. It is the condition that singles out the nearest point.

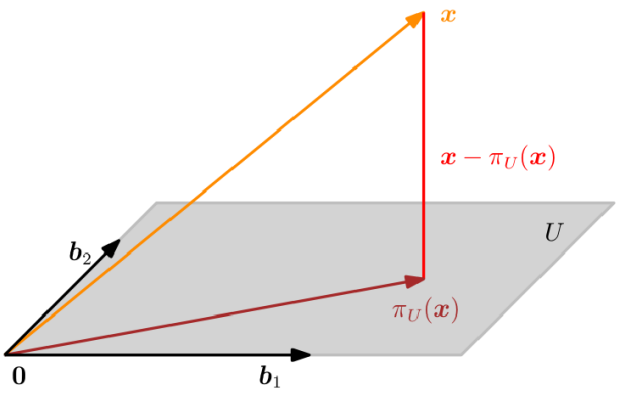


Figure 3.11 Projection onto a two-dimensional subspace U with basis b_1, b_2. The projection $\pi_U(x)$ of $x \in \mathbb{R}^3$ onto U can be expressed as a linear combination of b_1, b_2 and the displacement vector $x - \pi_U(x)$ is orthogonal to both b_1 and b_2.

Figure 70.1: The best approximation is characterized by an orthogonal residual. Moving within the subspace cannot reduce the remaining error.

§70.2 Projection onto one direction

Let $u \neq 0$. A candidate point on the line $\operatorname{span}\{u\}$ has the form $p = \alpha u$. Orthogonality of the error says

$$\langle y - \alpha u, u \rangle = 0.$$

Hence

$$\alpha = \frac{\langle y, u \rangle}{\langle u, u \rangle}, \qquad \operatorname{proj}_u y = \frac{\langle y, u \rangle}{\langle u, u \rangle} u.$$

If u is a unit vector, this simplifies to

$$\operatorname{proj}_u y = \langle y, u\rangle u.$$

Take

$$y = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \qquad u = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Then

$$\langle y, u\rangle = 5, \qquad \langle u, u\rangle = 5,$$

so

$$p = u = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad r = y - p = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

Indeed,

$$r^T u = 2 - 2 = 0.$$

The decomposition

$$y = p + r$$

separates the part visible inside the line from the part the line cannot represent.

§70.3 A basis turns the geometric condition into equations

Let the columns of

$$B = \begin{bmatrix} b_1 & \cdots & b_m \end{bmatrix} \in \mathbb{R}^{n\times m}$$

form a basis of W. Every candidate approximation is Bc for some coefficient vector c. The residual condition

$$y - Bc \perp W$$

means that it is orthogonal to each basis vector. In matrix form,

$$B^T(y - Bc) = 0.$$

Therefore

$$B^T Bc = B^T y.$$

If the columns of B are independent, the Gram matrix $B^T B$ is positive definite and invertible, giving

$$c = (B^T B)^{-1} B^T y$$

and

$$\boxed{\operatorname{proj}_W y = B(B^T B)^{-1} B^T y.}$$

This formula looks more complicated than the one-dimensional version only because the basis vectors need not be orthogonal. The Gram matrix records their mutual angles and corrects for the fact that the coordinate directions interfere with one another.

§70.4 Projection Matrices and Subspaces

Define

$$P = B(B^TB)^{-1}B^T.$$

Then

$$Py = \text{proj}_W\, y.$$

Two identities reveal its geometry:

$$P^T = P, \qquad P^2 = P.$$

Symmetry says the projection is orthogonal. Idempotence says that once a vector has been projected, projecting again does nothing.

Although the formula uses a basis, the matrix depends only on W. If $C = BR$ is another basis matrix with R invertible, then

$$C(C^TC)^{-1}C^T = BR(R^TB^TBR)^{-1}R^TB^T = B(B^TB)^{-1}B^T.$$

Coordinates changed, but the geometric operation did not.

There is also a complementary projection:

$$I - P.$$

It sends y to the residual $r = y - Py$, which lies in $W^\perp$. Thus

$$\mathbb{R}^n = W \oplus W^\perp, \qquad y = Py + (I - P)y.$$

§70.5 Least Squares as Projection

Now consider a system

$$Ax \approx b$$

with more equations than unknowns. Exact equality may be impossible because $b \notin R(A)$, the column space of A. The least-squares problem asks for

$$\min_x \|Ax - b\|_2.$$

But every vector Ax lies in $R(A)$. Therefore we are simply looking for the point in the column space closest to b.

At the minimizer $\widehat{x}$, the residual

$$r = b - A\widehat{x}$$

must be orthogonal to every column of A. Hence

$$A^Tr = 0,$$

which gives the normal equations

$$\boxed{A^TA\widehat{x} = A^Tb.}$$

The phrase “normal equations” comes from this normality: the residual is normal, or perpendicular, to the allowable model space.

§70.6 A complete line-fitting example

Suppose three measured points are

$$(0,1), \qquad (1,2), \qquad (2,2).$$

We fit a line

$$y \approx \alpha + \beta x.$$

The design matrix and observation vector are

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

The normal equations are

$$\begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}.$$

Solving gives

$$\widehat{\alpha} = \frac{7}{6}, \qquad \widehat{\beta} = \frac{1}{2}.$$

Thus the fitted line is

$$y = \frac{7}{6} + \frac{1}{2}x.$$

The predicted data are

$$A\widehat{x} = \begin{bmatrix} 7/6 \\ 5/3 \\ 13/6 \end{bmatrix},$$

and the residual is

$$r = b - A\widehat{x} = \begin{bmatrix} -1/6 \\ 1/3 \\ -1/6 \end{bmatrix}.$$

A direct check gives

$$A^T r = \begin{bmatrix} -1/6 + 1/3 - 1/6 \\ 0 + 1/3 - 2/6 \end{bmatrix} = 0.$$

The fit is not exact, but no infinitesimal change in either the intercept or slope can improve the squared error to first order.

§70.7 The Pythagorean proof of optimality

Let $p = Py$ be the orthogonal projection of y onto W. For any other $w \in W$, write

$$y - w = (y - p) + (p - w).$$

The first term lies in $W^{\perp}$, while the second lies in W. Therefore they are orthogonal, and Pythagoras gives

$$\|y - w\|^2 = \|y - p\|^2 + \|p - w\|^2.$$

Hence

$$\|y - w\| \geq \|y - p\|,$$

with equality only when $w = p$.

This proof explains both existence and uniqueness in finite dimensions. It also shows why the residual condition is stronger than a derivative calculation: the whole excess error is exactly the squared distance from w to the projection.

For least squares, the same identity becomes

$$\|b - Ax\|^2 = \|b - A\widehat{x}\|^2 + \|A(x - \widehat{x})\|^2.$$

Every competing parameter vector pays the irreducible residual plus an additional nonnegative model-space error.

§70.8 Orthonormal coordinates remove the Gram matrix

If the columns of Q are orthonormal, then

$$Q^TQ = I$$

and projection simplifies to

$$P = QQ^T.$$

The coefficients are simply inner products:

$$c = Q^Ty.$$

This is one reason orthonormal bases are so useful: geometry and coordinates agree without a correction matrix.

For a full-column-rank matrix A, a QR factorization writes

$$A = QR,$$

where $Q^TQ = I$ and R is upper triangular. Then

$$\|Ax - b\|_2 = \|QRx - b\|_2.$$

Multiplying by an orthogonal completion of Q separates the reachable and unreachable parts of b, and the least-squares solution is obtained from

$$Rx = Q^Tb.$$

This is computationally preferable to explicitly forming A^TA. The condition number is squared by the normal equations:

$$\kappa_2(A^TA) = \kappa_2(A)^2.$$

Thus the normal equations are conceptually illuminating but can be numerically fragile. QR preserves the projection geometry without amplifying the conditioning in this way.

§70.9 Rank deficiency changes uniqueness, not the geometry

If the columns of A are dependent, the fitted vector $A\widehat{x}$ can still be unique even though the coefficient vector is not. Indeed, all least-squares minimizers produce the same projection

$$A\widehat{x} = Pb,$$

but two parameter vectors may differ by an element of the null space:

$$x_1 - x_2 \in N(A).$$

This distinction is easy to miss in applications. The model prediction can be completely determined while the internal coefficients remain ambiguous. Collinear features in regression are a familiar example.

The Moore–Penrose inverse, studied later in more detail, chooses the minimizer orthogonal to $N(A)$. It is therefore the least-squares solution with minimum Euclidean norm:

$$\widehat{x} = A^{+}b.$$

Here the pseudoinverse is a canonical coordinate choice, while the geometric heart of the problem remains projection onto $R(A)$.

§70.10 Weighted Least Squares

Sometimes residual coordinates should not be treated equally. If W is symmetric positive definite, define

$$\langle u, v\rangle_W = u^T W v, \qquad \|u\|_W^2 = u^T W u.$$

Weighted least squares minimizes

$$\|Ax - b\|_W^2 = (Ax - b)^T W (Ax - b).$$

Differentiation or weighted orthogonality gives

$$A^T W(A\widehat{x} - b) = 0,$$

so

$$A^T W A\widehat{x} = A^T W b.$$

The algebra resembles ordinary least squares, but the geometry has changed. The residual is no longer Euclidean-orthogonal to the column space; it is orthogonal in the W-inner product. Equivalently, after writing $W = C^T C$, one solves the ordinary least-squares problem

$$\min_x \|CAx - Cb\|_2.$$

This is how measurement confidence, covariance information, and anisotropic geometry enter the same projection framework.

§70.11 Projection beyond finite-dimensional matrices

The finite-dimensional picture survives in Hilbert spaces. If H is a Hilbert space and $M \subseteq H$ is a closed subspace, then every $y \in H$ has a unique decomposition

$$y = p + r, \qquad p \in M, \quad r \in M^{\perp}.$$

The point p is the unique best approximation to y from M.

The word “closed” matters. Without it, an infimum distance may exist without being attained inside the subspace. Completeness and closedness are the infinite-dimensional replacements for the compact geometric intuition that works automatically in finite dimensions.

Fourier series are a central example. Truncating a Fourier expansion is orthogonal projection onto a finite-dimensional space of trigonometric modes. Polynomial approximation, finite elements, and Galerkin methods use the same principle with different trial spaces. The matrix normal equations are therefore one coordinate shadow of a much broader approximation theorem.

§70.12 Projecting a curved function onto a straight line

The same perpendicularity equation works when the vectors are *functions*. In the Hilbert space $L^2([-1,1])$, use the inner product

$$\langle f, g\rangle = \int_{-1}^{1} f(t)g(t)\,dt.$$

What is the best linear polynomial $p(t) = a + bt$ approximating $f(t) = t^2$ in mean-square error? We are projecting onto the two-dimensional subspace $\operatorname{span}\{1, t\}$.

The residual $r(t) = t^2 - a - bt$ must be orthogonal to both basis functions. Thus

$$\begin{pmatrix} \langle 1,1\rangle & \langle 1,t\rangle \\ \langle t,1\rangle & \langle t,t\rangle \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \langle t^2,1\rangle \\ \langle t^2,t\rangle \end{pmatrix} \quad\Longleftrightarrow\quad \begin{pmatrix} 2 & 0 \\ 0 & 2/3 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 2/3 \\ 0 \end{pmatrix}.$$

Therefore

$$\boxed{p(t) = \frac{1}{3}, \qquad r(t) = t^2 - \frac{1}{3}.}$$

A line is allowed to tilt, but the optimal one is horizontal. The reason is symmetry: t^2 is even and t is odd, so their inner product vanishes.

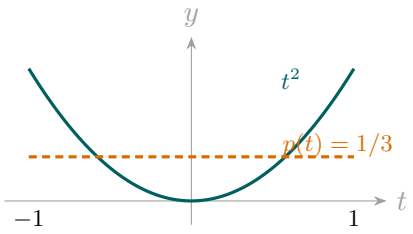


Figure 70.2: The best linear least-squares approximation to the parabola on $[-1,1]$ is the horizontal line $1/3$. The signed residual integrates to zero, and so does its first moment.

Both normal equations can be checked directly:

$$\int_{-1}^{1} r(t)\,dt = 0, \qquad \int_{-1}^{1} t\,r(t)\,dt = 0.$$

The remaining squared error is

$$\boxed{\|r\|_{L^2}^2 = \int_{-1}^{1} \left(t^2 - \frac{1}{3}\right)^2 dt = \frac{8}{45}.}$$

The familiar Legendre polynomial $P_2(t) = (3t^2 - 1)/2$ is exactly a rescaled version of this residual: it is the part of a quadratic function that cannot be represented by degree-one polynomials.

Matrix Computation I: Norms, Amplification, and Numerical Size

N-20260311

§71.1 Entrywise versus Operator Norms

A matrix is not merely a rectangular table of numbers. It is a map

$$x \longmapsto Ax,$$

and numerical analysis cares about what that map can do to perturbations.

Suppose the input is changed from x to $x + \delta x$. The output changes by

$$A(x + \delta x) - Ax = A\delta x.$$

To say anything useful, we need a notion of size for both δx and $A\delta x$. A vector norm measures the size of data; an operator norm measures the largest factor by which a matrix can enlarge that size.

This distinction matters. The matrix

$$A = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{bmatrix}$$

has entries no larger than 1, yet in the all-ones direction it multiplies Euclidean length by n. Looking at the largest entry alone misses the collective action of the columns.

§71.2 Different norms ask different questions

For $x = (x_1, \dots, x_n)^T$, the standard norms are

$$\|x\|_1 = \sum_{i=1}^{n} |x_i|, \qquad \|x\|_2 = \left(\sum_{i=1}^{n} |x_i|^2 \right)^{1/2},$$

$$\|x\|_\infty = \max_i |x_i|.$$

More generally,

$$\|x\|_p = \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p}, \qquad 1 \le p < \infty.$$

These norms emphasize different features. The 1-norm measures total mass. The 2-norm is rotationally invariant and fits inner-product geometry. The ∞-norm records the worst coordinate. For

$$x = (1, 1, \dots, 1)^T,$$

we have

$$\|x\|_1 = n, \qquad \|x\|_2 = \sqrt{n}, \qquad \|x\|_\infty = 1.$$

None of these answers is more correct in isolation. They answer different questions about what counts as a large perturbation.

In finite dimensions all norms are equivalent: for any two norms $\|\cdot\|_a$ and $\|\cdot\|_b$, there exist constants $c, C > 0$ such that

$$c\|x\|_a \le \|x\|_b \le C\|x\|_a.$$

Thus they define the same convergent sequences, but the constants may depend strongly on dimension. Numerical estimates therefore do depend on the chosen norm even when qualitative convergence does not.

§71.3 Hölder duality explains the familiar inequalities

Let $1 \le p, q \le \infty$ satisfy

$$\frac{1}{p} + \frac{1}{q} = 1.$$

Hölder's inequality states

$$|y^T x| \le \|y\|_q \|x\|_p.$$

The exponent q is called the conjugate exponent of p.

This inequality says that a linear functional $x \mapsto y^T x$ has size $\|y\|_q$ when inputs are measured in p-norm. More precisely,

$$\max_{\|x\|_p = 1} |y^T x| = \|y\|_q.$$

For $1 < p < \infty$, equality is achieved by choosing the phases of x_i to match y_i and setting

$$|x_i| \propto |y_i|^{q-1}.$$

For $p = 1$, the maximizing vector concentrates on a coordinate where $|y_i|$ is largest; hence the dual of ℓ^1 is ℓ^∞. For $p = \infty$, choose each coordinate with maximal allowed magnitude and matching sign; the dual is ℓ^1.

The Cauchy–Schwarz inequality is simply the case $p = q = 2$:

$$|y^T x| \le \|y\|_2 \|x\|_2.$$

This dual viewpoint is useful because matrix norms can be read as collections of linear functionals acting on columns or rows.

§71.4 An induced norm measures worst-case amplification

Given norms on the input and output spaces, define

$$\|A\| = \max_{x \ne 0} \frac{\|Ax\|}{\|x\|} = \max_{\|x\| = 1} \|Ax\|.$$

This is the induced, or subordinate, operator norm. By definition,

$$\|Ax\| \le \|A\| \, \|x\|.$$

It is the smallest constant for which this inequality holds for every x.

Induced norms are automatically submultiplicative:

$$\|AB\| = \max_{x\neq 0} \frac{\|ABx\|}{\|x\|} \leq \max_{x\neq 0} \frac{\|A\|\,\|Bx\|}{\|x\|} \leq \|A\|\|B\|.$$

Thus a product of linear maps cannot amplify more than the product of their individual worst-case factors.

The standard formulas are

$$\|A\|_1 = \max_j \sum_i |a_{ij}|, \qquad \|A\|_\infty = \max_i \sum_j |a_{ij}|.$$

The first is the largest absolute column sum; the second is the largest absolute row sum.

To see the 1-norm formula, write

$$\|Ax\|_1 \leq \sum_i \sum_j |a_{ij}|\,|x_j| = \sum_j \left(\sum_i |a_{ij}|\right) |x_j|.$$

This is bounded by

$$\left(\max_j \sum_i |a_{ij}|\right) \|x\|_1.$$

Equality is obtained by putting all mass on a column with maximal sum and choosing the sign or phase appropriately.

§71.5 A worked comparison of three operator norms

Let

$$A = \begin{bmatrix} 1 & -2 \\ 2 & 2 \end{bmatrix}.$$

The absolute column sums are

$$3, \qquad 4,$$

so

$$\|A\|_1 = 4.$$

The absolute row sums are

$$3, \qquad 4,$$

so

$$\|A\|_\infty = 4.$$

For the Euclidean norm, compute

$$A^T A = \begin{bmatrix} 5 & 2 \\ 2 & 8 \end{bmatrix}.$$

Its eigenvalues are

$$\lambda_\pm = \frac{13 \pm 5}{2} = 9, 4.$$

Therefore

$$\|A\|_2 = \sqrt{\lambda_{\max}(A^T A)} = 3.$$

The three values differ because they optimize over different unit balls. The 1- and ∞-balls have corners aligned with coordinate directions; the Euclidean ball is rotationally symmetric.

The right singular vector corresponding to singular value 3 is a direction in which A stretches Euclidean length exactly by a factor of 3. Other input directions stretch less.

§71.6 The spectral norm comes from singular values

The Euclidean operator norm satisfies

$$\|A\|_2 = \max_{\|x\|_2=1} \|Ax\|_2.$$

Squaring gives

$$\|Ax\|_2^2 = x^T A^T A x.$$

The maximum of this Rayleigh quotient over the unit sphere is the largest eigenvalue of $A^T A$. Hence

$$\boxed{\|A\|_2 = \sqrt{\lambda_{\max}(A^T A)} = \sigma_{\max}(A).}$$

This formula explains why singular values, rather than eigenvalues, govern Euclidean amplification. Eigenvalues describe invariant directions satisfying $Ax = \lambda x$. Singular vectors solve a different optimization problem: which input direction produces the longest output?

For a nonnormal matrix these questions can diverge dramatically. Consider

$$N_M = \begin{bmatrix} 0 & M \\ 0 & 0 \end{bmatrix}.$$

Both eigenvalues are zero, so the spectral radius is zero, yet

$$\|N_M\|_2 = M.$$

The matrix is nilpotent, but it can cause a large one-step transient amplification. This is why eigenvalues alone are not a complete measure of numerical size.

§71.7 Unitary transformations preserve Euclidean geometry

If $U^T U = I$, then

$$\|Ux\|_2^2 = x^T U^T U x = \|x\|_2^2.$$

Thus orthogonal or unitary matrices are changes of orthonormal coordinates. They rotate or reflect without changing length.

Consequently,

$$\|UAV\|_2 = \|A\|_2$$

for unitary U, V. Singular value decomposition writes

$$A = U\Sigma V^T,$$

so the Euclidean operator norm is simply the largest diagonal entry of Σ.

The Frobenius norm

$$\|A\|_F = \left(\sum_{i,j} |a_{ij}|^2 \right)^{1/2}$$

is also unitarily invariant. Since

$$\|A\|_F^2 = \operatorname{tr}(A^T A),$$

we obtain

$$\|UAV\|_F^2 = \operatorname{tr}(V^T A^T U^T U A V) = \operatorname{tr}(V^T A^T A V) = \operatorname{tr}(A^T A).$$

Equivalently,

$$\|A\|_F^2 = \sum_i \sigma_i(A)^2.$$

The spectral norm asks for the single strongest direction. The Frobenius norm combines energy across all singular directions. They satisfy

$$\|A\|_2 \le \|A\|_F \le \sqrt{r}\,\|A\|_2,$$

where $r = \operatorname{rank} A$.

§71.8 Schatten Norms

For $1 \le p < \infty$, define the Schatten p-norm by applying a vector p-norm to the singular values:

$$\|A\|_{S_p} = \left(\sum_i \sigma_i(A)^p\right)^{1/p}.$$

Then

$$\|A\|_{S_2} = \|A\|_F, \qquad \|A\|_{S_\infty} = \|A\|_2.$$

The case $p = 1$ is the nuclear norm,

$$\|A\|_* = \sum_i \sigma_i(A).$$

It is used as a convex surrogate for rank because it penalizes the total singular-value mass.

These norms separate two notions that are often conflated. Entrywise sparsity concerns how many coordinates are nonzero. Low rank concerns how many singular directions are active. The 1-norm of entries encourages the former; the nuclear norm encourages the latter.

§71.9 Norms turn perturbation statements into inequalities

If x is perturbed by δx, then

$$\|A(x + \delta x) - Ax\| \le \|A\|\,\|\delta x\|.$$

This is the basic Lipschitz estimate for a linear map.

For a product,

$$F(x) = W_L W_{L-1} \cdots W_1 x,$$

we have

$$\|F(x + \delta x) - F(x)\|_2 \le \left(\prod_{\ell=1}^{L} \|W_\ell\|_2\right) \|\delta x\|_2.$$

The bound can be pessimistic because the maximally amplified direction of one layer may not align with that of the next. Still, it gives a rigorous worst-case control and explains why spectral normalization constrains the Lipschitz constant of a network layer.

Norms also bound truncation and rounding errors. If a computed matrix is $A + E$, then

$$\|(A + E)x - Ax\| \le \|E\|\,\|x\|.$$

The norm of the perturbation is meaningful only relative to the chosen geometry. A small Frobenius perturbation controls average singular energy, while a small spectral perturbation controls every Euclidean input direction.

§71.10 Distance to Singularity

For an invertible square matrix,

$$\sigma_{\min}(A) = \min_{\|x\|_2=1} \|Ax\|_2.$$

This is the least amount by which A stretches any unit direction. If it is small, there is a direction that is almost collapsed.

The Euclidean distance from A to the set of singular matrices is exactly

$$\sigma_{\min}(A)$$

when distance is measured by the spectral norm. Indeed, if u_n, v_n are singular vectors for $\sigma_{\min}$, then

$$E = -\sigma_{\min} u_n v_n^T$$

has norm $\sigma_{\min}$ and makes

$$(A + E)v_n = 0.$$

No smaller perturbation can make the matrix singular, because

$$\|(A + E)x\|_2 \geq \|Ax\|_2 - \|E\|_2\|x\|_2 \geq \sigma_{\min}(A) - \|E\|_2$$

for unit x.

Thus the smallest singular value is not merely one more number in an SVD table. It measures how close invertibility is to failure.

§71.11 Two norm-one operators whose product can be zero

Submultiplicativity says

$$\|AB\|_2 \leq \|A\|_2\|B\|_2,$$

but it gives no lower bound. To see precisely what is missing, take two unit vectors $u, v \in \mathbb{R}^2$ meeting at angle θ, and let

$$P_u = uu^T, \qquad P_v = vv^T$$

be the orthogonal projections onto their lines. Each has operator norm 1. Their composition, however, is

$$P_u P_v = u(u^T v)v^T, \qquad \boxed{\|P_u P_v\|_2 = |u^T v| = |\cos\theta|.}$$

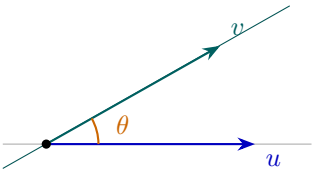


Figure 71.1: Project first onto the v-line and then onto the u-line. Only the part aligned with u survives the second projection.

If the lines coincide, $P_u P_v = P_u$ and equality holds in the norm bound. If they are perpendicular, $P_u P_v = 0$ even though both factors have norm 1. At intermediate angles the product norm is exactly the cosine loss.

This calculation also shows why multiplying layerwise Lipschitz constants in a matrix product can be extremely pessimistic. Each separate layer may stretch some direction maximally, while the next layer may discard precisely that direction.

Matrix Computation II: Spectral Radius, Stability, and Conditioning

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§72.1 One step and many steps are different questions

A matrix norm asks how much a vector can be amplified in one application:

$$\|Ax\| \le \|A\|\|x\|.$$

The spectral radius

$$\rho(A) = \max_{\lambda \in \sigma(A)} |\lambda|$$

asks about the exponential rate that remains after the matrix is applied many times.

These quantities agree for normal matrices in the Euclidean norm, but not in general. A nonnormal matrix can produce enormous short-time growth even when all its eigenvalues lie well inside the unit disk. Conversely, a large one-step norm does not by itself imply that powers grow forever.

This chapter separates four issues that are often compressed into the word "stability":

$\|A\|$	:	one-step amplification,
$\rho(A)$	:	asymptotic exponential growth,
$\kappa(A)$	:	sensitivity of solving or inversion,
backward stability	:	whether an algorithm solves a nearby problem.

A stable algorithm cannot make an ill-conditioned problem well-conditioned, and a well-conditioned problem can still be mishandled by an unstable algorithm.

§72.2 Spectral radius is not a norm

If $Ax = \lambda x$ with $x \neq 0$, then every induced norm satisfies

$$|\lambda|\|x\| = \|Ax\| \le \|A\|\|x\|.$$

Hence

$$\rho(A) \le \|A\|.$$

The inequality is fundamental, but ρ itself is not a matrix norm.

The nonzero nilpotent matrix

$$N = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

has

$$\rho(N) = 0.$$

Thus positive definiteness fails. Moreover, with

$$N_1 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \qquad N_2 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix},$$

we have

$$\rho(N_1) = \rho(N_2) = 0, \qquad \rho(N_1 + N_2) = 1.$$

So the triangle inequality fails as well.

The spectral radius is nevertheless the right asymptotic quantity. If A is diagonalizable,

$$A = S\Lambda S^{-1},$$

then

$$A^k = S\Lambda^k S^{-1}.$$

The eigenvalues determine the exponential factors λ_i^k, while the geometry of S determines how strongly the corresponding modes can interfere at finite times.

§72.3 Gelfand's formula extracts the exponential rate

For every matrix norm,

$$\boxed{\rho(A) = \lim_{k\to\infty} \|A^k\|^{1/k}.}$$

This is Gelfand's formula.

The lower bound is immediate from

$$\rho(A^k) = \rho(A)^k \le \|A^k\|,$$

which gives

$$\rho(A) \le \|A^k\|^{1/k}.$$

For the opposite direction, put A into Jordan form. On a Jordan block

$$J = \lambda I + N, \qquad N^m = 0,$$

we have

$$J^k = \sum_{j=0}^{m-1} \binom{k}{j} \lambda^{k-j} N^j.$$

The binomial factors grow only polynomially in k, while $|\lambda|^k$ carries the exponential rate. Taking the k-th root makes every fixed polynomial factor tend to 1. The largest $|\lambda|$ therefore survives, yielding the formula.

Gelfand's formula should not be confused with the power method. The power method follows a particular vector

$$x, Ax, A^2x, \dots$$

and tries to isolate a dominant eigenvector. It can fail if the starting vector misses the dominant invariant direction, if dominant eigenvalues tie in modulus, or if nonnormal transient behavior obscures the asymptotic regime. Gelfand's formula is a statement about the norm of the whole operator power, not a promise about one trajectory.

§72.4 Jordan blocks explain the boundary

For a 2×2 Jordan block

$$J = \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix},$$

direct calculation gives

$$J^k = \begin{bmatrix} \lambda^k & k\lambda^{k-1} \\ 0 & \lambda^k \end{bmatrix}.$$

Three regimes are visible.

If $|\lambda| < 1$, both λ^k and $k\lambda^{k-1}$ tend to zero. Exponential decay eventually defeats the polynomial factor. If $|\lambda| > 1$, powers grow exponentially. If $|\lambda| = 1$, the off-diagonal term grows like k, even though every eigenvalue has modulus one.

Consequently,

$$A^k \to 0 \quad \Longleftrightarrow \quad \rho(A) < 1.$$

The strict inequality matters. Eigenvalues on the unit circle are harmless only when the associated Jordan blocks have size one and the remaining geometry is controlled.

The same calculation explains why a matrix can look stable by eigenvalues and still exhibit large transient effects. Let

$$B = \begin{bmatrix} 0.9 & 10 \\ 0 & 0.9 \end{bmatrix}.$$

Then

$$B^k = \begin{bmatrix} 0.9^k & 10k\,0.9^{k-1} \\ 0 & 0.9^k \end{bmatrix}.$$

Compare it with the normal matrix $C = 0.9I$, which has the same eigenvalues.

k	$\|C^k\|_2$	lower bound for $\|B^k\|_2$
1	0.9000	10.0000
5	0.5905	32.8050
10	0.3487	38.7420
20	0.1216	27.0170
50	0.00515	2.8632

The lower bound comes from applying B^k to e_2. Both matrices converge to zero, but only one does so monotonically in operator norm. Eigenvalues describe the destination; nonnormality can make the route violently indirect.

§72.5 Neumann series turns repeated feedback into an inverse

If $\|A\| < 1$, then

$$(I - A)^{-1} = \sum_{k=0}^{\infty} A^k.$$

Indeed,

$$(I - A)\sum_{k=0}^{m} A^k = I - A^{m+1},$$

and $A^{m+1} \to 0$.

The truncation error satisfies

$$(I - A)^{-1} - \sum_{k=0}^{m} A^k = A^{m+1}(I - A)^{-1}.$$

Therefore

$$\left\| (I-A)^{-1} - \sum_{k=0}^{m} A^k \right\| \le \frac{\|A\|^{m+1}}{1-\|A\|}.$$

Take

$$A = \begin{bmatrix} 0.2 & 0.1 \\ 0 & 0.2 \end{bmatrix}.$$

Its induced ∞-norm is 0.3. The exact inverse is

$$(I-A)^{-1} = \begin{bmatrix} 1.25 & 0.15625 \\ 0 & 1.25 \end{bmatrix}.$$

The cubic truncation is

$$I + A + A^2 + A^3 = \begin{bmatrix} 1.248 & 0.152 \\ 0 & 1.248 \end{bmatrix}.$$

Its actual ∞-norm error is

$$0.00625,$$

while the general bound gives

$$\frac{0.3^4}{1-0.3} \approx 0.01157.$$

The bound is not exact, but it is guaranteed without diagonalizing the matrix.

The same idea defines the resolvent

$$R(z,A) = (zI-A)^{-1}.$$

Whenever $|z| > \|A\|$,

$$(zI-A)^{-1} = \frac{1}{z} \sum_{k=0}^{\infty} \left(\frac{A}{z}\right)^k.$$

The resolvent packages the behavior of all powers into one analytic matrix-valued function.

§72.6 Condition number is the geometry of inversion

For an invertible matrix,

$$\kappa(A) = \|A\|\|A^{-1}\|.$$

In the Euclidean norm,

$$\kappa_2(A) = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)}.$$

The image of the unit sphere under A is an ellipsoid. The longest semiaxis has length $\sigma_{\max}$, and the shortest has length $\sigma_{\min}$. Inversion must expand the shortest output direction by $1/\sigma_{\min}$. A highly elongated ellipsoid therefore corresponds to a sensitive inverse problem.

Suppose only the right-hand side is perturbed:

$$Ax = b, \qquad A(x+\delta x) = b + \delta b.$$

Then

$$A\delta x = \delta b, \qquad \delta x = A^{-1}\delta b.$$

Using

$$\|b\| = \|Ax\| \le \|A\|\|x\|,$$

we obtain the relative error bound

$$\boxed{\frac{\|\delta x\|}{\|x\|} \le \kappa(A)\frac{\|\delta b\|}{\|b\|}.}$$

The condition number is a worst-case amplification factor. It does not say that every perturbation is amplified maximally; the dangerous perturbations align with the smallest singular directions.

§72.7 A tiny data change can reverse the solution

Let

$$A_\varepsilon = \begin{bmatrix} 1 & 1 \\ 1 & 1+\varepsilon \end{bmatrix}, \qquad b = \begin{bmatrix} 2 \\ 2+\varepsilon \end{bmatrix}.$$

The exact solution is

$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Now perturb only the second observation by ε:

$$\delta b = \begin{bmatrix} 0 \\ \varepsilon \end{bmatrix}.$$

Since

$$A_\varepsilon^{-1} = \frac{1}{\varepsilon}\begin{bmatrix} 1+\varepsilon & -1 \\ -1 & 1 \end{bmatrix},$$

we get

$$\delta x = A_\varepsilon^{-1}\delta b = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

For $\varepsilon = 10^{-4}$, the relative data perturbation is about

$$3.54 \times 10^{-5},$$

while the relative solution perturbation is 1. The amplification is roughly 2.8×10^4.

The matrix is not singular; it is nearly singular. Its two columns are almost parallel, so the data barely distinguish the two parameters. No algorithm can recover information that the model geometry has nearly erased.

In this example,

$$\kappa_2(A_\varepsilon) \sim \frac{4}{\varepsilon}.$$

For $\varepsilon = 10^{-4}$, the condition number is about 4×10^4, consistent with the observed amplification.

§72.8 Forward error, residual, and backward error

Given an approximate solution $\widehat{x}$, the residual is

$$r = b - A\widehat{x}.$$

The forward error is

$$x - \widehat{x} = A^{-1}r.$$

Thus a small residual implies a small forward error only when A^{-1} is not too large.

Backward error asks a different question: what is the smallest perturbation of the data for which $\widehat{x}$ is exact? If only the right-hand side is perturbed, then

$$A\widehat{x} = b - r,$$

so $-r$ is exactly the required perturbation. A backward-stable algorithm produces the exact solution of a nearby problem.

The distinction can be summarized as follows:

$$\begin{array}{rcl} \text{small backward error} & \Rightarrow & \text{the algorithm behaved stably,} \\ \text{small condition number} & \Rightarrow & \text{nearby problems have nearby solutions,} \\ \text{both together} & \Rightarrow & \text{small forward error.} \end{array}$$

If the condition number is enormous, a backward-stable computation may still have a large forward error. That is not a contradiction; it is the correct response to an ill-conditioned problem.

When both A and b are perturbed, a standard first-order estimate is

$$\frac{\|\delta x\|}{\|x\|} \lesssim \kappa(A)\left(\frac{\|\delta A\|}{\|A\|} + \frac{\|\delta b\|}{\|b\|}\right).$$

A rigorous finite-perturbation bound contains the denominator

$$1 - \kappa(A)\frac{\|\delta A\|}{\|A\|},$$

which must remain positive. The denominator records the possibility that the perturbed matrix crosses into singularity.

§72.9 Eigenvalues can be sensitive too

If A is normal, each eigenvalue of $A + E$ lies within $\|E\|_2$ of the spectrum of A. For a diagonalizable nonnormal matrix

$$A = S\Lambda S^{-1},$$

the Bauer–Fike theorem gives the weaker bound

$$\min_{\lambda \in \sigma(A)} |\mu - \lambda| \le \kappa_2(S)\|E\|_2$$

for every eigenvalue μ of $A + E$.

The factor $\kappa_2(S)$ measures how far the eigenvector basis is from orthonormal. Nearly parallel eigenvectors make eigenvalues highly sensitive.

Take

$$S = \begin{bmatrix} 1 & 1 \\ 0 & 0.01 \end{bmatrix}, \qquad \Lambda = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}.$$

Then

$$A = S\Lambda S^{-1} = \begin{bmatrix} 1 & 100 \\ 0 & 2 \end{bmatrix},$$

and

$$\kappa_2(S) \approx 200.$$

Perturb by

$$E = \begin{bmatrix} 0 & 0 \\ 10^{-4} & 0 \end{bmatrix}.$$

The characteristic polynomial of $A + E$ is

$$(1 - \mu)(2 - \mu) - 0.01,$$

so the eigenvalues are approximately

$$0.9901, \qquad 2.0099.$$

A perturbation of norm 10^{-4} has moved the eigenvalues by about 10^{-2}. Bauer–Fike permits movement up to roughly

$$200 \times 10^{-4} = 0.02.$$

The sensitivity comes from the eigenvectors, not from the separation of the displayed diagonal entries alone.

§72.10 Pseudospectrum records nearby spectra

For $\varepsilon > 0$, the ε-pseudospectrum can be defined by

$$\sigma_\varepsilon(A) = \{z : \|(zI - A)^{-1}\|_2 > \varepsilon^{-1}\} \cup \sigma(A).$$

Equivalently, it is the set of eigenvalues of matrices $A + E$ with $\|E\|_2 < \varepsilon$.

For a normal matrix, the resolvent norm is exactly

$$\|(zI - A)^{-1}\|_2 = \frac{1}{\operatorname{dist}(z, \sigma(A))}.$$

Its pseudospectrum is simply a union of small disks around the eigenvalues. For a nonnormal matrix, the resolvent can be much larger.

Consider

$$B = \begin{bmatrix} 0.8 & 10 \\ 0 & 0.8 \end{bmatrix}.$$

For $z \neq 0.8$, write $d = z - 0.8$. Then

$$(zI - B)^{-1} = \begin{bmatrix} d^{-1} & 10d^{-2} \\ 0 & d^{-1} \end{bmatrix}.$$

Therefore

$$\|(zI - B)^{-1}\|_2 \geq \frac{10}{|d|^2}.$$

z	$\lvert z-0.8\rvert$	resolvent lower bound
1.0	0.2	250
1.2	0.4	62.5
1.6	0.8	15.625

With $\varepsilon = 0.02$, the threshold is $1/\varepsilon = 50$. The point $z = 1.2$ therefore lies in the 0.02-pseudospectrum, even though the only eigenvalue is 0.8. The pseudospectrum has expanded far beyond a radius-0.02 disk.

This expansion is the frequency-domain counterpart of transient growth. A large resolvent means that a small forcing or perturbation at a particular complex frequency can produce a large response. It also warns that computed eigenvalues of a nonnormal matrix may be dominated by tiny perturbations.

§72.11 A practical diagnostic order

When a computation involving powers, inverses, or eigenvalues looks unstable, several questions should be asked in order.

First, identify the quantity of interest. Powers A^k, solutions $A^{-1}b$, and eigenvalues have different condition theories. Next, inspect structure: normality, symmetry, positive definiteness, sparsity, and rank can radically change the relevant bounds. Then compare spectral radius with an operator norm. A large gap suggests nonnormal transient behavior. For solving, estimate $\kappa(A)$ and report a residual or backward error rather than relying on residual size alone. For eigenvalue sensitivity, inspect eigenvector conditioning or the resolvent; a pseudospectrum may be more informative than a list of eigenvalues.

The conceptual map is

$$\text{powers} \longleftrightarrow \rho(A) \text{ and Jordan structure},$$

$$\text{inverse} \longleftrightarrow \sigma_{\min}(A) \text{ and } \kappa(A),$$

$$\text{nearby spectra} \longleftrightarrow \|(zI - A)^{-1}\| \text{ and pseudospectrum}.$$

The same matrix may be harmless for one question and difficult for another. Stability begins by deciding which question is actually being asked.

Matrix Computation III: Canonical Forms, Spectral Localization, and Eigenvalue Iteration

N-20260318

§73.1 Before computing an eigenvalue, decide what can change

A matrix changes when the basis changes. If

$$B = S^{-1}AS,$$

then the entries of B may look nothing like those of A, yet both matrices represent the same linear operator in different coordinates. A useful theory of eigenvalues therefore begins by separating coordinate accidents from invariant structure.

Three questions guide this chapter.

First, what algebraic data survive every change of basis? Canonical forms answer this. Second, before carrying out an expensive computation, where can the eigenvalues possibly lie? Gershgorin's theorem gives a cheap geometric answer. Third, once a spectral region or dominant mode has been identified, how can one actually approximate the corresponding eigenvector? Power and inverse iterations turn localization into computation.

These are different levels of information:

$$\text{canonical structure} \longrightarrow \text{spectral localization} \longrightarrow \text{numerical approximation}.$$

Trying to use one level as a substitute for another causes confusion. Jordan form explains exact structure but is usually a poor numerical algorithm. Gershgorin discs are robust and cheap but rarely exact. Iteration computes selected eigenpairs but only under separation and stability assumptions.

§73.2 Polynomial Matrices and Invariant Factors

The characteristic polynomial

$$\chi_A(\lambda) = \det(\lambda I - A)$$

records the eigenvalues with algebraic multiplicity. It does not, by itself, reveal how many Jordan blocks occur or how large they are. That missing information is already present in the whole polynomial matrix

$$\lambda I - A,$$

not merely in its determinant.

Work over the principal ideal domain $\mathbb{C}[\lambda]$. Row and column operations are allowed when they are invertible over this ring: exchange two rows or columns, add a polynomial

multiple of one to another, or multiply by a nonzero constant. These operations produce polynomial matrices

$$U(\lambda)(\lambda I - A)V(\lambda)$$

with unimodular U, V, meaning their determinants are nonzero constants.

Every polynomial matrix of rank r can be reduced to Smith normal form

$$\operatorname{diag}\big(d_1(\lambda), \dots, d_r(\lambda), 0, \dots, 0\big),$$

where the nonzero polynomials are monic and satisfy

$$d_1 \mid d_2 \mid \cdots \mid d_r.$$

The polynomials d_i are the invariant factors.

If $D_k(\lambda)$ denotes the monic greatest common divisor of all nonzero $k \times k$ minors, then

$$d_1 = D_1, \qquad d_k = \frac{D_k}{D_{k-1}}.$$

The determinant is only the last determinantal divisor:

$$D_n = d_1 d_2 \cdots d_n = \chi_A(\lambda).$$

The smaller minors are what remember how the characteristic polynomial is distributed among cyclic pieces.

§73.3 Invariant factors are a module decomposition

Let $V = \mathbb{C}^n$, and let the indeterminate λ act on vectors by

$$\lambda \cdot v = Av.$$

This turns V into a $\mathbb{C}[\lambda]$-module. The structure theorem for finitely generated modules over a principal ideal domain gives

$$V \cong \bigoplus_i \mathbb{C}[\lambda]/(d_i(\lambda)),$$

where the nonconstant invariant factors are those of $\lambda I - A$.

This description explains why Smith form belongs naturally beside canonical matrix forms. A cyclic summand

$$\mathbb{C}[\lambda]/(d(\lambda))$$

is generated by one vector

$$v, Av, A^2v, \dots,$$

subject to the relation $d(A)v = 0$. In a suitable cyclic basis, the matrix of A on this summand is the companion matrix of d. Taking the direct sum over invariant factors gives the rational canonical form.

The largest invariant factor is the minimal polynomial:

$$m_A(\lambda) = d_n(\lambda),$$

while their product is the characteristic polynomial. Thus

$$m_A \mid \chi_A.$$

The minimal polynomial records the strongest polynomial relation satisfied by the operator and controls every polynomial expression in A, including high powers and matrix functions.

§73.4 From invariant factors to Jordan blocks

Over $\mathbb{C}$, factor each invariant factor into powers of linear terms. The resulting powers

$$(\lambda - \mu)^s$$

are called elementary divisors. Each elementary divisor corresponds to one Jordan block

$$J_s(\mu).$$

For example, suppose the elementary divisors are

$$(\lambda - 2)^2, \qquad (\lambda - 2), \qquad (\lambda - 3).$$

Then the Jordan form is

$$J_2(2) \oplus J_1(2) \oplus J_1(3).$$

The characteristic polynomial is

$$(\lambda - 2)^3(\lambda - 3),$$

but the minimal polynomial is only

$$(\lambda - 2)^2(\lambda - 3).$$

The exponent 2 of $\lambda - 2$ in the minimal polynomial records the largest Jordan block for eigenvalue 2, not the total multiplicity of that eigenvalue.

A concrete polynomial-matrix example makes the mechanism visible. Let

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

Then

$$\lambda I - A = \begin{bmatrix} \lambda - 2 & -1 & 0 \\ 0 & \lambda - 2 & 0 \\ 0 & 0 & \lambda - 3 \end{bmatrix}.$$

Because one entry is the unit -1, the gcd of the entries is $D_1 = 1$. Among the 2×2 minors appear both

$$(\lambda - 2)^2 \qquad \text{and} \qquad -(\lambda - 3),$$

whose gcd is 1. Hence $D_2 = 1$. Finally,

$$D_3 = (\lambda - 2)^2(\lambda - 3).$$

The invariant factors are therefore

$$1, \quad 1, \quad (\lambda - 2)^2(\lambda - 3),$$

and the elementary divisors are

$$(\lambda - 2)^2, \qquad (\lambda - 3).$$

They recover the block structure $J_2(2) \oplus J_1(3)$.

§73.5 Jordan Chains and Growth

A Jordan chain for eigenvalue μ is a sequence

$$v_1, v_2, \dots, v_s$$

satisfying

$$(A - \mu I)v_1 = 0,$$
$$(A - \mu I)v_{j+1} = v_j.$$

The first vector is an eigenvector; the others are generalized eigenvectors. In this basis the operator acts by

$$Av_1 = \mu v_1, \qquad Av_{j+1} = \mu v_{j+1} + v_j.$$

Write a Jordan block as

$$J = \mu I + N, \qquad N^s = 0.$$

Then

$$J^k = \sum_{j=0}^{s-1} \binom{k}{j} \mu^{k-j} N^j.$$

The exponential term μ^k is decorated by polynomial factors in k. This is the precise origin of terms such as

$$k\mu^{k-1}, \qquad \binom{k}{2}\mu^{k-2}.$$

For

$$J = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix},$$

we obtain

$$J^k = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}.$$

Every eigenvalue equals 1, yet powers grow linearly. The Jordan chain carries dynamical information that the list of eigenvalues alone cannot express.

This also explains why Jordan form is mathematically revealing but numerically delicate. A tiny perturbation can split a multiple eigenvalue and destroy a large Jordan block. Computing exact block sizes from floating-point data is therefore ill-conditioned. Stable numerical algorithms usually prefer Schur form,

$$A = QTQ^*,$$

where Q is unitary and T is upper triangular. The diagonal of T gives the eigenvalues while unitary similarity avoids an ill-conditioned eigenvector basis.

§73.6 Gershgorin from Row Sums

Canonical forms describe exact similarity invariants, but they are not a cheap first diagnostic. Gershgorin's theorem locates every eigenvalue using only row sums.

For $A = (a_{ij})$, define

$$R_i = \sum_{j \neq i} |a_{ij}|.$$

The i-th Gershgorin disc is

$$G_i = \{z \in \mathbb{C} : |z - a_{ii}| \le R_i\}.$$

Every eigenvalue lies in

$$\bigcup_i G_i.$$

The proof is a one-line maximum-coordinate argument. If

$$Ax = \lambda x$$

and $|x_i| = \max_j |x_j|$, then

$$(\lambda - a_{ii})x_i = \sum_{j \ne i} a_{ij} x_j.$$

Therefore

$$|\lambda - a_{ii}| \le \sum_{j \ne i} |a_{ij}| \frac{|x_j|}{|x_i|} \le R_i.$$

The chosen coordinate may depend on the eigenvector, so the theorem gives a union of discs rather than one fixed disc.

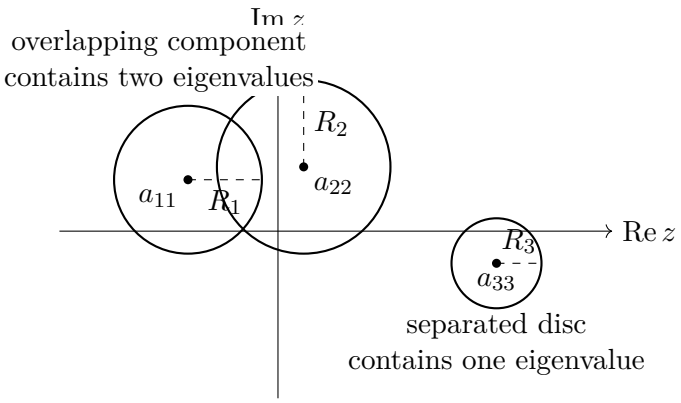


Figure 73.1: A separated component containing exactly m Gershgorin discs contains exactly m eigenvalues, counted with algebraic multiplicity.

§73.7 Counting Eigenvalues with Gershgorin

A stronger form of the theorem says that if a connected component of

$$\bigcup_i G_i$$

contains exactly m discs and is disjoint from the remaining discs, then it contains exactly m eigenvalues, counted with algebraic multiplicity.

The idea is deformation. Consider matrices

$$A(t) = D + t(A - D), \qquad 0 \le t \le 1,$$

where D is the diagonal part of A. At $t = 0$, every eigenvalue is a diagonal entry, and each disc has radius zero. As t increases, the discs expand continuously. If one cluster stays separated from the rest, eigenvalues cannot cross the gap between components. The number inside that component is therefore conserved.

For example, let

$$A = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 4 & 0 \\ 0 & 0 & 9 \end{bmatrix}.$$

The first two discs are both the interval-like discs centered at 4 with radius 1, while the third is the point 9. Hence two eigenvalues lie in the component around 4, and one lies at 9. Direct calculation gives

$$\sigma(A) = \{3, 5, 9\},$$

exactly matching the count.

This separated-component principle is particularly useful for proving that a matrix is nonsingular. If every Gershgorin disc avoids the origin, then $0 \notin \sigma(A)$. Strict diagonal dominance is the familiar special case

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}|.$$

§73.8 Gershgorin after Diagonal Scaling

Gershgorin discs depend on the chosen basis even though eigenvalues do not. This is not a defect; it is an opportunity. A diagonal similarity

$$B = D^{-1}AD$$

preserves the spectrum while changing off-diagonal magnitudes:

$$b_{ij} = d_i^{-1} a_{ij} d_j.$$

Consider

$$A = \begin{bmatrix} 1 & 100 \\ 0.01 & 2 \end{bmatrix}.$$

The row discs have radii 100 and 0.01, giving a very poor picture. Choose

$$D = \begin{bmatrix} 100 & 0 \\ 0 & 1 \end{bmatrix}.$$

Then

$$D^{-1}AD = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

Now both radii are 1. The eigenvalues have not moved, but the coordinates have been balanced so that the row equations display their coupling more honestly.

Choosing an optimal diagonal scaling is itself a nontrivial problem. The example nonetheless illustrates a broad numerical principle: poor coordinates can hide useful structure, and similarity transformations can expose it without changing the underlying operator.

§73.9 Power iteration extracts a dominant direction

Suppose A is diagonalizable with eigenpairs (λ_i, v_i), ordered so that

$$|\lambda_1| > |\lambda_2| \geq \cdots.$$

Write the initial vector as

$$x_0 = c_1 v_1 + c_2 v_2 + \cdots + c_n v_n, \qquad c_1 \neq 0.$$

Then

$$A^k x_0 = \lambda_1^k \left(c_1 v_1 + \sum_{j \geq 2} c_j \left(\frac{\lambda_j}{\lambda_1} \right)^k v_j \right).$$

After normalization, the smaller ratios decay, so the direction approaches v_1. The asymptotic rate is governed by

$$\left| \frac{\lambda_2}{\lambda_1} \right|.$$

Take

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Its eigenvectors are

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix},$$

with eigenvalues 3 and 1. Starting from

$$x_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{2} v_1 + \frac{1}{2} v_2,$$

we obtain

$$A^k x_0 = \frac{1}{2} \begin{bmatrix} 3^k + 1 \\ 3^k - 1 \end{bmatrix}.$$

The ratio of the two coordinates tends to 1, and the normalized vector approaches the dominant direction $(1, 1)^T$.

A convenient eigenvalue estimate is the Rayleigh quotient

$$\rho(x) = \frac{x^* A x}{x^* x}.$$

For Hermitian A, if x_k approaches a normalized eigenvector, then $\rho(x_k)$ approaches its eigenvalue. The residual

$$r_k = A x_k - \rho(x_k) x_k$$

provides an a posteriori check: an exact eigenvector has zero residual.

§73.10 Power iteration has precise failure modes

The method is not guaranteed merely because a matrix has eigenvalues.

If the initial vector has no component in the dominant eigenspace, the dominant mode never appears. If two eigenvalues tie in modulus, the direction may fail to converge. For example,

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

has two eigenvalues of equal modulus. Starting from a generic vector, the second coordinate changes sign at every step, so the normalized sequence oscillates.

Nonnormality creates another complication. Even when one eigenvalue is asymptotically dominant, an ill-conditioned eigenvector basis can produce large transient mixtures

before the asymptotic regime becomes visible. A small residual is therefore more informative than a visually stable sequence of normalized coordinates.

Finally, normalization prevents overflow but does not repair loss of information. If roundoff repeatedly suppresses a weak component or if the dominant eigenvalue is poorly separated, convergence may stagnate.

§73.11 Inverse iteration turns a target into a dominant mode

Power iteration naturally finds the eigenvalue of largest modulus. To find an eigenvalue near a chosen shift μ, apply power iteration to

$$(A-\mu I)^{-1}.$$

Its eigenvalues are

$$\frac{1}{\lambda_j-\mu}.$$

The eigenvalue of A closest to μ becomes the dominant eigenvalue of the inverse operator.

The practical iteration is not to form the inverse. Instead, solve

$$(A-\mu I)y_{k+1}=x_k$$

and normalize

$$x_{k+1}=\frac{y_{k+1}}{\|y_{k+1}\|}.$$

If the shift stays fixed, factor $A-\mu I$ once and reuse the triangular solves. This makes shifted inverse iteration efficient when several steps are required.

The apparent paradox is that the method becomes stronger as μ approaches an eigenvalue, while the linear system becomes more ill-conditioned. The near singularity is exactly what amplifies the desired eigendirection. Stable pivoting, residual monitoring, and an appropriate stopping criterion are therefore essential.

§73.12 Rayleigh quotient iteration lets the shift move

A natural refinement updates the shift using the current Rayleigh quotient:

$$\mu_k=\frac{x_k^*Ax_k}{x_k^*x_k},$$

then solves

$$(A-\mu_k I)y_{k+1}=x_k$$

and normalizes.

For Hermitian matrices and a sufficiently good initial vector, Rayleigh quotient iteration converges cubically. This remarkable speed comes from two facts: the Rayleigh quotient already approximates the eigenvalue to second order in the angular error, and inverse iteration strongly amplifies the nearest eigendirection.

Take again

$$A=\begin{bmatrix}2&1\\1&2\end{bmatrix}.$$

If

$$x_0=\frac{1}{\sqrt{5}}\begin{bmatrix}2\\1\end{bmatrix},$$

then

$$\mu_0 = x_0^T A x_0 = \frac{14}{5} = 2.8,$$

already close to the dominant eigenvalue 3. Solving with $A - 2.8I$ strongly favors the $(1,1)^T$ direction, and the next Rayleigh quotient is dramatically more accurate.

The method should not be described as universally cubic. Symmetry or normality, a nearby simple eigenvalue, and a sufficiently accurate starting vector are important. For a general nonnormal matrix, convergence can be less regular and the moving linear systems can become difficult.

§73.13 Large matrices lead from vectors to Krylov subspaces

Power iteration uses the one-dimensional space spanned by the newest vector. A more informative approach retains the whole Krylov space

$$\mathcal{K}_k(A, x_0) = \operatorname{span}\{x_0, Ax_0, \ldots, A^{k-1}x_0\}.$$

Instead of asking one vector to represent all spectral information, project A onto this growing subspace.

For Hermitian matrices, the Lanczos process constructs an orthonormal basis of the Krylov space and produces a small tridiagonal matrix. Its eigenvalues, called Ritz values, approximate selected eigenvalues of A. For general matrices, Arnoldi iteration produces an upper Hessenberg projection.

These methods are the large-scale descendants of power iteration. They retain the same polynomial idea—vectors in $\mathcal{K}_k$ have the form $p_{k-1}(A)x_0$—but optimize over many polynomials rather than repeatedly using only $p(t) = t^k$. This is why Krylov eigensolvers can resolve several eigenvalues and converge much faster than plain power iteration.

§73.14 A cubic improvement you can calculate by hand

For the symmetric matrix already used above,

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \sigma(A) = \{3, 1\},$$

Rayleigh quotient iteration has an especially transparent calculation. Write a vector in the two eigenvector directions as

$$x \;\propto\; \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \delta \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

The number δ measures the relative unwanted eigendirection. Since the two directions are orthogonal, the Rayleigh quotient is

$$\mu = \frac{3 + \delta^2}{1 + \delta^2}.$$

One shifted inverse solve changes the ratio of the two eigencomponents to

$$\delta_{\text{new}} = \delta \frac{3 - \mu}{1 - \mu} = \boxed{-\delta^3}.$$

There is the cubic improvement: in this two-dimensional Hermitian example it is an *exact identity*, provided the shifted linear system is nonsingular.

Take $x_0 = (2,1)^T/\sqrt{5}$, for which $\delta = (2-1)/(2+1) = 1/3$. The original quotient is $14/5$. One step gives

$$\delta_1 = -\frac{1}{27}, \qquad x_1 = \frac{1}{\sqrt{365}}\begin{pmatrix}13\\14\end{pmatrix}, \qquad \mu_1 = \frac{1094}{365} \approx 2.99726.$$

The unwanted component has fallen from $1/3$ to $1/27$ in a single iteration. A second step cubes it again.

There is also a simple certificate for any approximate Hermitian eigenpair. If $\|x\|_2 = 1$, $\mu = x^{*}Ax$, and $r = Ax - \mu x$, then expansion in an orthonormal eigenbasis gives

$$\|r\|_2^2 = \sum_{j} |c_j|^2 |\lambda_j - \mu|^2 \;\geq\; \operatorname{dist}(\mu, \sigma(A))^2.$$

Thus some eigenvalue lies within $\|r\|_2$ of the computed quotient. For our x_1, the residual norm is $27/365$, so the interval centered at μ_1 with that radius contains the eigenvalue 3. A rapid iteration and a reliable error check come from the same two-by-two calculation.

Matrix Decompositions: LU, Cholesky, Rank Factorization, and SVD

N-20260602

§74.1 One matrix, several kinds of hidden structure

A matrix is often presented as a rectangular array of numbers, but an algorithm rarely wants to work with that array in its original form. It would rather expose a structure that the matrix is hiding. A triangular matrix can be solved from one end to the other. A positive definite matrix carries an inner-product geometry. A low-rank matrix really passes through a smaller space. An arbitrary linear map stretches some orthogonal directions more than others.

These observations lead to four decompositions:

$$PA = LU, \qquad A = LL^*, \qquad A = FG, \qquad A = U\Sigma V^*.$$

They are not four versions of the same trick. Each answers a different question.

Factorization	Question it answers	Structure it exposes
$PA = LU$	How can elimination be reused?	successive triangular solves
$A = LL^*$	What extra benefit comes from positivity?	a Gram-type square root
$A = FG$	Through how many coordinates does the map really pass?	exact rank
$A = U\Sigma V^*$	Which orthogonal directions are stretched most?	singular directions and scales

The point of this chapter is therefore not to memorize four formulas. It is to watch four different pieces of linear algebra become algorithms.

§74.2 Elimination remembers its work

Consider a square system

$$Ax = b.$$

Gaussian elimination replaces rows of A until the matrix becomes upper triangular. Every elimination step is multiplication on the left by a lower triangular elementary matrix. If

$$E_{n-1} \cdots E_2 E_1 A = U,$$

then

$$A = E_1^{-1} E_2^{-1} \cdots E_{n-1}^{-1} U.$$

The product of the inverse elimination matrices is lower triangular. This is the origin of

$$A = LU.$$

The factor L is therefore a record of what elimination did. Its entries below the diagonal are the multipliers used to cancel entries of A; U is the echelon-shaped matrix left behind.

Proposition 74.2.1 (Existence without row exchanges)
Let $A \in \mathbb{C}^{n\times n}$. If every leading principal minor

$$\Delta_k = \det A_{1:k,1:k}, \qquad k = 1, \dots, n,$$

is nonzero, then there is a factorization

$$A = LU,$$

where L is unit lower triangular and U is upper triangular.

The condition has a direct computational meaning: elimination reaches a nonzero pivot at every stage without exchanging rows. It is not a mysterious determinant test added after the algorithm; it is the determinant form of saying that the algorithm never gets stuck.

The factors are unique under this normalization. Indeed, if

$$A = LU = \widetilde{L}\widetilde{U},$$

then

$$L^{-1}\widetilde{L} = U\widetilde{U}^{-1}.$$

The left side is unit lower triangular and the right side is upper triangular. A matrix of both kinds must be the identity, so $L = \widetilde{L}$ and $U = \widetilde{U}$.

§74.3 A complete LU calculation

Take

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix}.$$

The first pivot is 2. To clear the first column, use the multipliers

$$m_{21} = 2, \qquad m_{31} = -1.$$

After

$$R_2 \leftarrow R_2 - 2R_1, \qquad R_3 \leftarrow R_3 + R_1,$$

we obtain

$$\begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 8 & 3 \end{bmatrix}.$$

The second multiplier is

$$m_{32} = \frac{8}{-8} = -1.$$

Thus

$$R_3 \leftarrow R_3 + R_2$$

produces

$$U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & 1 \end{bmatrix}.$$

The multipliers, placed in the corresponding positions below a diagonal of ones, give

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix}.$$

A direct multiplication confirms that $A = LU$.

Now solve

$$Ax = \begin{bmatrix} 5 \\ -2 \\ 9 \end{bmatrix}.$$

Instead of eliminating again, first solve

$$Ly = b.$$

Forward substitution gives

$$y_1 = 5, \qquad 2y_1 + y_2 = -2, \qquad -y_1 - y_2 + y_3 = 9,$$

so

$$y = \begin{bmatrix} 5 \\ -12 \\ 2 \end{bmatrix}.$$

Then solve

$$Ux = y.$$

Backward substitution gives

$$x_3 = 2, \qquad -8x_2 - 2x_3 = -12, \qquad 2x_1 + x_2 + x_3 = 5,$$

and hence

$$x = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}.$$

This is why factorization matters when several right-hand sides are present. The expensive elimination is performed once. Each new vector b requires only one forward and one backward triangular solve.

§74.4 When the first pivot betrays us

The matrix

$$A_0 = \begin{bmatrix} 0 & 2 & 1 \\ 1 & -2 & -3 \\ 2 & 3 & 1 \end{bmatrix}$$

is invertible, but elimination cannot begin with its upper-left entry. Swapping the first and third rows gives

$$PA_0 = \begin{bmatrix} 2 & 3 & 1 \\ 1 & -2 & -3 \\ 0 & 2 & 1 \end{bmatrix},$$

where

$$P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

Elimination now yields

$$PA_0 = LU$$

with

$$L = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ 0 & -\frac{4}{7} & 1 \end{bmatrix}, \qquad U = \begin{bmatrix} 2 & 3 & 1 \\ 0 & -\frac{7}{2} & -\frac{7}{2} \\ 0 & 0 & -1 \end{bmatrix}.$$

Row exchanges are also useful when a pivot is nonzero but dangerously small. For example,

$$B_\varepsilon = \begin{bmatrix} \varepsilon & 1 \\ 1 & 1 \end{bmatrix}, \qquad 0 < \varepsilon \ll 1,$$

would use the multiplier $1/\varepsilon$ if elimination proceeded in the displayed order. That large multiplier creates subtraction between large nearby numbers and can amplify rounding error. Partial pivoting exchanges the rows and uses the entry 1 instead.

Partial pivoting chooses the largest available entry in magnitude from the current column. It keeps all elimination multipliers at most 1. This is one reason Gaussian elimination with partial pivoting is reliable in ordinary numerical work. The statement needs one caveat: specially constructed matrices can still exhibit a large growth factor. Pivoting greatly improves stability, but it does not turn every matrix into a harmless one.

There is also a distinction that should not be blurred:

$$\text{algorithmic stability} \qquad \text{versus} \qquad \text{conditioning of the problem.}$$

Pivoting controls the computation. It cannot repair a linear system whose solution is intrinsically extremely sensitive to perturbations.

§74.5 Positive definiteness buys a square root

Suppose now that A is Hermitian positive definite:

$$A = A^*, \qquad x^*Ax > 0 \quad (x \neq 0).$$

Then no row exchange is needed, and symmetry allows us to use essentially one triangular factor instead of two.

Theorem 74.5.1 (Cholesky decomposition)

Every Hermitian positive definite matrix has a unique factorization

$$A = LL^*,$$

where L is lower triangular with positive diagonal entries.

The proof reveals more than the formula. Write

$$A = \begin{bmatrix} a & c^* \\ c & B \end{bmatrix}.$$

Positive definiteness gives $a > 0$, so set

$$\ell_{11} = \sqrt{a}, \qquad w = \frac{c}{\sqrt{a}}.$$

The trailing matrix that remains is the Schur complement

$$S = B - ww^*.$$

It is again positive definite. To see this, for any nonzero y, choose

$$t = -\frac{c^* y}{a}.$$

Then

$$\begin{bmatrix} t \\ y \end{bmatrix}^* A \begin{bmatrix} t \\ y \end{bmatrix} = y^* S y > 0.$$

The problem has therefore reduced to a smaller positive definite matrix. Repeating the argument constructs L recursively.

This induction explains why every square root appearing in Cholesky is real and positive. Positive definiteness is exactly the hypothesis preventing the algorithm from taking the square root of a negative pivot.

§74.6 A Cholesky solve from beginning to end

Consider

$$A = \begin{bmatrix} 4 & 2 & 2 \\ 2 & 5 & 1 \\ 2 & 1 & 5 \end{bmatrix}.$$

The first Cholesky entry is

$$\ell_{11} = \sqrt{4} = 2.$$

The entries below it are

$$\ell_{21} = \frac{2}{2} = 1, \qquad \ell_{31} = \frac{2}{2} = 1.$$

The next diagonal entry is

$$\ell_{22} = \sqrt{5 - 1^2} = 2,$$

and

$$\ell_{32} = \frac{1 - 1 \cdot 1}{2} = 0.$$

Finally,

$$\ell_{33} = \sqrt{5 - 1^2 - 0^2} = 2.$$

Thus

$$L = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}, \qquad A = LL^T.$$

For

$$Ax = \begin{bmatrix} 8 \\ 8 \\ 8 \end{bmatrix},$$

solve first

$$Ly = b.$$

This gives

$$y = \begin{bmatrix} 4 \\ 2 \\ 2 \end{bmatrix}.$$

Then solve

$$L^T x = y,$$

which gives

$$x = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Cholesky uses about half the leading-order arithmetic and half the off-diagonal storage of a general LU factorization. The gain is not a clever coding accident; it comes from using symmetry instead of computing the same information twice.

The hypothesis cannot be discarded. For

$$C = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix},$$

the first step would give $\ell_{11} = 1$ and $\ell_{21} = 2$, but then

$$\ell_{22}^2 = 1 - 2^2 = -3.$$

The failure reflects the negative eigenvalue of C. Cholesky is not merely an LU decomposition with matching factors; it is a certificate of positive definiteness.

There is a geometric interpretation as well. If $A = LL^*$, then

$$x^* A x = \|L^* x\|_2^2.$$

Thus a positive definite quadratic form is Euclidean length after an invertible change of coordinates. Equivalently, A is a Gram matrix of independent vectors.

§74.7 Rank factorization removes redundant coordinates

A rectangular matrix may contain exact redundancy rather than a difficult pivot. If

$$A \in \mathbb{C}^{m \times n}, \qquad \operatorname{rank} A = r,$$

then there are matrices

$$F \in \mathbb{C}^{m \times r}, \qquad G \in \mathbb{C}^{r \times n}$$

with

$$\operatorname{rank} F = \operatorname{rank} G = r$$

such that

$$A = FG.$$

This is a rank factorization.

The map represented by A now visibly passes through an r-dimensional space:

$$\mathbb{C}^n \xrightarrow{G} \mathbb{C}^r \xrightarrow{F} \mathbb{C}^m.$$

Because G has full row rank, it is onto. Because F has full column rank, it is one-to-one. Consequently,

$$\ker A = \ker G, \qquad \operatorname{im} A = \operatorname{im} F.$$

The factorization separates two jobs: G discards the directions that carry no information, and F places the remaining coordinates into the output space.

For example, let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}.$$

The first two columns are independent, while the third satisfies

$$A_{:3} = -A_{:1} + 2A_{:2}.$$

Choose

$$F = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 1 & 1 \end{bmatrix}, \qquad G = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \end{bmatrix}.$$

Then $A = FG$, and both factors have rank 2.

The factors are not unique. For every invertible $S \in \mathbb{C}^{r\times r}$,

$$A = (FS)(S^{-1}G).$$

This freedom is a change of coordinates in the intermediate space. What is intrinsic is not a particular pair (F, G), but the fact that the map factors through an r-dimensional object. In coordinate-free language, this is the familiar passage

$$\mathbb{C}^n / \ker A \cong \operatorname{im} A.$$

§74.8 SVD chooses orthogonal coordinates on both sides

Rank factorization exposes the correct dimension, but it does not say which coordinates in that dimension are geometrically best. The singular value decomposition makes a special choice: it uses orthonormal coordinates in both the domain and codomain.

Theorem 74.8.1 (Singular value decomposition)

For every $A \in \mathbb{C}^{m\times n}$, there are unitary matrices U and V such that

$$A = U\Sigma V^*,$$

where Σ is rectangular diagonal with entries

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$$

and all remaining entries zero. The number r is $\operatorname{rank} A$.

The construction begins with the positive semidefinite matrix A^*A. Choose an orthonormal eigenbasis

$$A^*Av_i = \sigma_i^2 v_i.$$

For every nonzero singular value, define

$$u_i = \frac{Av_i}{\sigma_i}.$$

Then

$$Av_i = \sigma_i u_i,$$

and the vectors u_i are orthonormal because

$$u_i^* u_j = \frac{v_i^* A^* A v_j}{\sigma_i \sigma_j} = \delta_{ij}.$$

Complete the two orthonormal families to bases. This yields $A = U\Sigma V^*$.

The equation

$$Av_i = \sigma_i u_i$$

is the most useful way to read the SVD. The input direction v_i is sent to the output direction u_i and stretched by σ_i. Directions corresponding to zero singular values lie in the kernel.

§74.9 A tilted ellipse and its best rank-one shadow

Let

$$A = \frac{1}{\sqrt{2}} \begin{bmatrix} 3 & -1 \\ 3 & 1 \end{bmatrix}.$$

Its two columns are orthogonal. The first has length 3, and the second has length 1. Define

$$u_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad u_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

Then

$$Ae_1 = 3u_1, \qquad Ae_2 = u_2.$$

Therefore an SVD is

$$A = \underbrace{\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}}_{U} \underbrace{\begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}}_{\Sigma} \underbrace{I}_{V^T}.$$

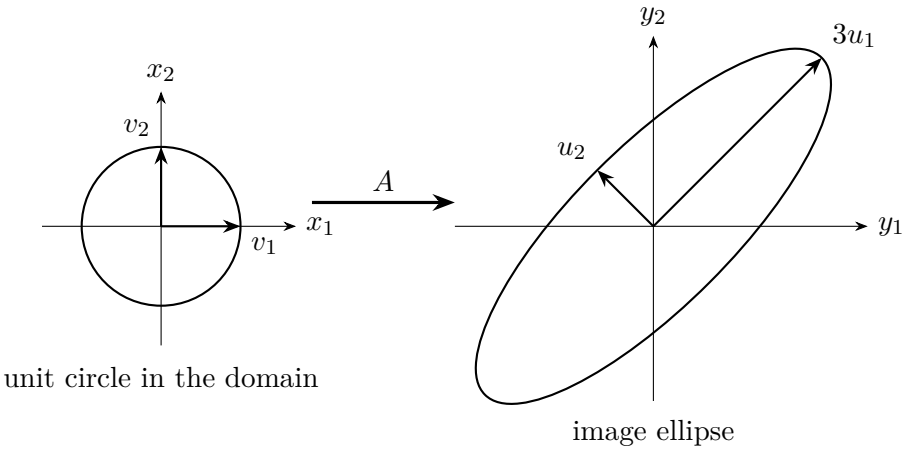


Figure 74.1: The right singular vectors give orthogonal input directions; the left singular vectors give the principal axes of the image ellipse.

The compact SVD is also a rank factorization:

$$A = U_r \Sigma_r V_r^* = (U_r \Sigma_r) V_r^*.$$

Unlike an arbitrary rank factorization, it orders the intermediate coordinates by geometric importance.

This ordering leads to a remarkable optimality statement.

Theorem 74.9.1 (Eckart–Young, spectral-norm form)

If

$$A = \sum_{i=1}^{r} \sigma_i u_i v_i^*, \qquad \sigma_1 \geq \cdots \geq \sigma_r > 0,$$

then among all matrices B of rank at most k, the truncated SVD

$$A_k = \sum_{i=1}^{k} \sigma_i u_i v_i^*$$

minimizes $\|A - B\|_2$, and

$$\|A - A_k\|_2 = \sigma_{k+1}.$$

For the matrix above, the best rank-one approximation is

$$A_1 = 3u_1 e_1^T = \frac{3}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}.$$

The discarded part is

$$A - A_1 = u_2 e_2^T,$$

so

$$\|A - A_1\|_2 = 1.$$

No other rank-one matrix can do better. This is why singular values appear in compression, denoising, latent-factor models, and reduced-order simulation: truncation is not merely plausible; for these norms it is optimal.

Small singular values carry a second message. If A is invertible, then

$$\|A^{-1}\|_2 = \frac{1}{\sigma_{\min}(A)}.$$

A direction that is almost collapsed by A must be enormously expanded by A^{-1}. Thus the same SVD that provides a low-rank approximation also diagnoses instability in solving linear systems.

§74.10 Choosing the factorization by the question

The four decompositions now fit together without becoming interchangeable.

Use $PA = LU$ when a general square system must be solved, especially for several right-hand sides. The essential advantage is reusable elimination. Use Cholesky when the matrix is Hermitian positive definite. It is faster, stores less, and certifies a geometric property of the quadratic form. Use a rank factorization when exact redundancy is the main issue and one wants an explicit passage through the image dimension. Use the SVD when orthogonal geometry, conditioning, or optimal low-rank approximation matters enough to justify its greater cost.

A decomposition should therefore be read as a mathematical answer to a computational question:

$$\begin{array}{rcl} \text{elimination} & \rightsquigarrow & \text{triangular factors,} \\ \text{positive quadratic geometry} & \rightsquigarrow & \text{Cholesky,} \\ \text{quotient by the kernel} & \rightsquigarrow & \text{rank factorization,} \\ \text{orthogonal principal directions} & \rightsquigarrow & \text{SVD.} \end{array}$$

The formulas are useful because each one exposes exactly the structure needed by the next calculation. That is the sense in which a matrix decomposition is an algorithm rather than a decorative identity.

Generalized Inverses, Projections, and Least-Squares Geometry

N-20260603

§75.1 What an inverse was supposed to do

For a square nonsingular matrix, the inverse performs two jobs at once. It solves every equation

$$Ax = b$$

and it undoes the action of A:

$$A^{-1}A = I, \qquad AA^{-1} = I.$$

Once A is rectangular or rank deficient, these two jobs separate. Some vectors b may not lie in the column space, so an exact solution does not exist. If the null space is nontrivial, several vectors x may produce the same output, so an exact solution need not be unique.

The Moore–Penrose inverse does not pretend that these failures have disappeared. It makes two geometric choices:

(i) replace an unreachable target by its orthogonal projection onto the column space;

(ii) among all inputs producing that projected target, choose the one orthogonal to the null space.

The resulting vector

$$x^+ = A^+ b$$

is therefore canonical even when ordinary inversion is impossible.

The familiar cases are recovered as special formulas:

matrix type	formula for A^+	interpretation
$m = n = \operatorname{rank} A$	A^{-1}	ordinary inverse
$m > n,\ \operatorname{rank} A = n$	$(A^*A)^{-1}A^*$	full-column-rank least squares
$m < n,\ \operatorname{rank} A = m$	$A^*(AA^*)^{-1}$	full-row-rank minimum norm
$\operatorname{rank} A < \min(m, n)$	$V\Sigma^+U^*$	rank-deficient case

The last line is the real definition in computational practice; the first three are consequences of it.

§75.2 The four fundamental subspaces reveal the answer

Let

$$A : \mathbb{C}^n \longrightarrow \mathbb{C}^m.$$

The domain and codomain split orthogonally as

$$\mathbb{C}^n = R(A^*) \oplus N(A),$$

and

$$\mathbb{C}^m = R(A) \oplus N(A^*).$$

These are the four fundamental subspaces of the matrix.

The restriction

$$A\big|_{R(A^*)} : R(A^*) \longrightarrow R(A)$$

is a bijection. It is injective because

$$R(A^*) \cap N(A) = \{0\},$$

and the two spaces have the same dimension, namely $\operatorname{rank} A$. Thus A has a genuine inverse after we discard precisely the directions on which it vanishes and restrict the target to the vectors it can actually reach.

This gives a coordinate-free description of the pseudoinverse. Decompose

$$b = b_R + b_\perp, \qquad b_R \in R(A), \quad b_\perp \in N(A^*).$$

Then $A^+ b$ is the unique vector in $R(A^*)$ satisfying

$$A(A^+ b) = b_R.$$

The orthogonal component $b_\perp$ is ignored because no input can produce it.

The two central products are now transparent:

$$AA^+ = P_{R(A)}, \qquad A^+ A = P_{R(A^*)}.$$

The first projects targets onto the reachable output space. The second projects inputs away from the null-space ambiguity.

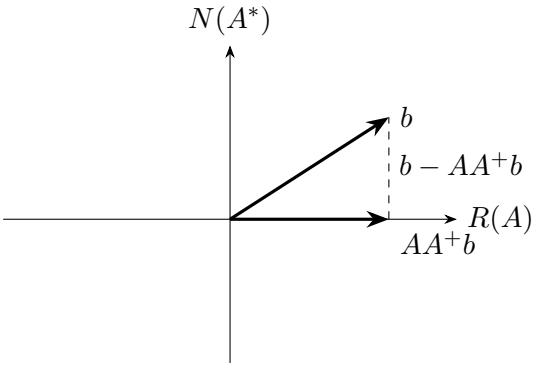


Figure 75.1: The pseudoinverse first replaces b by the closest reachable vector AA^+b.

§75.3 Why the four Penrose equations appear

The Moore–Penrose inverse is usually introduced as the unique matrix X satisfying

$$AXA = A, \qquad XAX = X,$$

$$(AX)^* = AX, \qquad (XA)^* = XA.$$

Read without context, these equations look ceremonial. The projection picture explains every one of them.

If $X = A^{+}$, then AX should be the orthogonal projection onto $R(A)$. An orthogonal projection is idempotent and self-adjoint. The relation

$$AXA = A$$

says that AX fixes every vector already in $R(A)$, while

$$(AX)^{*} = AX$$

forces the projection to be orthogonal rather than oblique.

Similarly, XA should be the orthogonal projection onto $R(A^{*})$. The equations

$$XAX = X, \qquad (XA)^{*} = XA$$

encode that requirement on the input side.

Dropping some of the four conditions produces broader families of generalized inverses. There are 15 nonempty subsets of the four equations, which explains the traditional count of generalized-inverse classes. The count itself is less important than the distinction: algebraic retraction conditions alone allow many choices, while the two adjoint conditions select the orthogonal geometry and make the answer unique.

Proposition 75.3.1

The Moore–Penrose inverse, if it exists, is unique.

One way to see uniqueness is to use the two projection identities. Any matrix satisfying the four equations must act as the inverse of A from $R(A)$ to $R(A^{*})$, must vanish on $N(A^{*})$, and must return values orthogonal to $N(A)$. Those requirements determine its value on the orthogonal decomposition

$$\mathbb{C}^{m} = R(A) \oplus N(A^{*}).$$

§75.4 SVD turns the definition into a calculation

Suppose

$$A = U\Sigma V^{*}$$

is a singular value decomposition, with positive singular values

$$\sigma_1 \geq \cdots \geq \sigma_r > 0.$$

Write

$$\Sigma = \begin{bmatrix} \operatorname{diag}(\sigma_1, \ldots, \sigma_r) & 0 \\ 0 & 0 \end{bmatrix}.$$

Then

$$A^{+} = V\Sigma^{+}U^{*},$$

where

$$\Sigma^{+} = \begin{bmatrix} \operatorname{diag}(\sigma_1^{-1}, \ldots, \sigma_r^{-1}) & 0 \\ 0 & 0 \end{bmatrix}^{T}.$$

Only the nonzero singular values are inverted.

The formula is easier to remember mode by mode. Since

$$Av_i = \sigma_i u_i,$$

the pseudoinverse must satisfy

$$A^+ u_i = \frac{1}{\sigma_i} v_i \qquad (i \le r),$$

and

$$A^+ u = 0 \qquad (u \in N(A^*)).$$

Thus A^+ literally reverses the active singular directions and does nothing to unreachable directions.

The projection formulas follow at once:

$$AA^+ = U_r U_r^*, \qquad A^+ A = V_r V_r^*.$$

Here the columns of U_r form an orthonormal basis of $R(A)$, and the columns of V_r form an orthonormal basis of $R(A^*)$.

A rank factorization also gives a useful formula. If

$$A = FG,$$

where $F \in \mathbb{C}^{m\times r}$ has full column rank and $G \in \mathbb{C}^{r\times n}$ has full row rank, then

$$A^+ = G^*(GG^*)^{-1}(F^*F)^{-1}F^*.$$

The middle two inverses are only $r \times r$. This expression makes the factor-through-the-image viewpoint explicit, although SVD is usually the cleaner numerical language when rank is uncertain.

§75.5 One rank-one matrix contains the whole geometry

Let

$$u = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \qquad v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad A = uv^T = \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 0 & 0 \end{bmatrix}.$$

For a nonzero rank-one matrix uv^*,

$$(uv^*)^+ = \frac{vu^*}{\|u\|_2^2 \|v\|_2^2}.$$

Since

$$\|u\|_2^2 = 5, \qquad \|v\|_2^2 = 2,$$

we obtain

$$A^+ = \frac{1}{10}\begin{bmatrix} 1 & 2 & 0 \\ 1 & 2 & 0 \end{bmatrix}.$$

The two products are

$$AA^+ = \frac{1}{5}\begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \frac{uu^T}{u^T u},$$

and

$$A^+A = \frac{1}{2}\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix} = \frac{vv^T}{v^Tv}.$$

They are the orthogonal projections onto $\operatorname{span}(u)$ and $\operatorname{span}(v)$, respectively.

Now take

$$b = \begin{bmatrix}1\\ 0\\ 1\end{bmatrix}.$$

The closest reachable vector is

$$\widehat{b} = AA^+b = \frac{1}{5}\begin{bmatrix}1\\ 2\\ 0\end{bmatrix}.$$

The residual is

$$r = b - \widehat{b} = \begin{bmatrix}\frac{4}{5}\\ -\frac{2}{5}\\ 1\end{bmatrix},$$

and indeed

$$u^Tr = 0.$$

The pseudoinverse solution is

$$x^+ = A^+b = \begin{bmatrix}\frac{1}{10}\\ \frac{1}{10}\end{bmatrix}.$$

Every vector satisfying

$$x_1 + x_2 = \frac{1}{5}$$

produces $\widehat{b}$, but x^+ is the one with smallest Euclidean norm. The equality of its two coordinates is not accidental: it is the orthogonal projection of the origin onto that affine line.

This single example displays all four subspaces. The column space is $\operatorname{span}(u)$; the left null space is its orthogonal complement. The row space is $\operatorname{span}(v)$; the null space consists of vectors proportional to $(1,-1)^T$. The pseudoinverse moves only between the two one-dimensional active spaces.

§75.6 Exact solutions, least squares, and minimum norm

For an arbitrary b, the vector

$$x^+ = A^+b$$

minimizes

$$\|Ax - b\|_2.$$

Indeed,

$$Ax^+ = AA^+b = P_{R(A)}b,$$

which is the closest point in the column space to b.

All least-squares minimizers have the form

$$x = A^+b + (I - A^+A)z, \qquad z \in \mathbb{C}^n.$$

The second term lies in $N(A)$, so it does not change Ax. Since

$$A^+b \in R(A^*) = N(A)^\perp,$$

the two terms are orthogonal. Therefore

$$\|x\|_2^2 = \|A^+b\|_2^2 + \|(I - A^+A)z\|_2^2.$$

This proves that A^+b is the unique minimum-norm least-squares solution.

Exact solvability is detected by

$$AA^+b = b.$$

If this holds, the same formula describes every exact solution:

$$x = A^+b + (I - A^+A)z.$$

Thus one expression covers the overdetermined, underdetermined, and rank-deficient cases. What changes is only whether the projection AA^+b equals b and whether the null-space term is present.

The common identities

$$(A^+)^+ = A, \qquad (A^*)^+ = (A^+)^*, \qquad \operatorname{rank} A^+ = \operatorname{rank} A$$

are immediate from the SVD. They are consequences of the singular-direction picture, not a separate collection of tricks.

§75.7 Why inverse-like algebra can fail

Ordinary inverses reverse products:

$$(AB)^{-1} = B^{-1}A^{-1}.$$

The corresponding pseudoinverse identity is false in general. Let

$$A = \begin{bmatrix} 1 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

Then

$$AB = \begin{bmatrix} 1 & 0 \end{bmatrix}, \qquad (AB)^+ = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

On the other hand,

$$A^+ = \frac{1}{2}\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad B^+ = B,$$

so

$$B^+A^+ = \begin{bmatrix} \frac{1}{2} \\ 0 \end{bmatrix} \neq (AB)^+.$$

The failure is geometric. Each pseudoinverse inserts orthogonal projections onto row and column spaces. In a product, the active output space of B need not align with the active input space of A. Reversing the two pseudoinverses introduces projections in the wrong places. Reverse-order laws do hold under additional range and commutation hypotheses, but they should never be assumed from notation alone.

This is a useful general warning: the pseudoinverse behaves like an inverse only on the active singular subspaces. Algebraic identities that move through kernels or orthogonal complements require extra compatibility.

§75.8 Updating a pseudoinverse one column at a time

Recomputing an SVD after every new data column may be wasteful. Greville's recursion updates the pseudoinverse when

$$A_{k+1} = \begin{bmatrix} A_k & a \end{bmatrix}.$$

Set

$$d = A_k^{+} a, \qquad c = a - A_k d = (I - A_k A_k^{+})a.$$

The vector c is the part of the new column orthogonal to the old column space. If $c \neq 0$, define

$$b^{*} = \frac{c^{*}}{c^{*}c}.$$

Then

$$A_{k+1}^{+} = \begin{bmatrix} A_k^{+} - d b^{*} \\ b^{*} \end{bmatrix}.$$

For a small example, begin with

$$A_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad A_1^{+} = \begin{bmatrix} 1 & 0 \end{bmatrix},$$

and append

$$a = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Then

$$d = A_1^{+} a = 1, \qquad c = a - A_1 d = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad b^{*} = \begin{bmatrix} 0 & 1 \end{bmatrix}.$$

Therefore

$$A_2^{+} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}.$$

But

$$A_2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

is invertible, and the update has recovered its ordinary inverse exactly.

If $c = 0$, the new column already lies in the old column space and a different scalar correction is used. That branch is not a technical nuisance; it records whether the incoming data adds a genuinely new direction.

§75.9 Iteration sees the pseudoinverse as a limit

The pseudoinverse need not be formed explicitly in a large problem. Consider Landweber iteration

$$x_{k+1} = x_k + \alpha A^{*}(b - A x_k),$$

with

$$0 < \alpha < \frac{2}{\sigma_1^2}.$$

This is gradient descent on

$$\frac{1}{2}\|Ax - b\|_2^2.$$

In singular-vector coordinates, each active component evolves independently. If

$$x_k = \sum_i \xi_{i,k} v_i,$$

then

$$\xi_{i,k+1} = (1 - \alpha\sigma_i^2)\xi_{i,k} + \alpha\sigma_i u_i^* b.$$

The step-size condition makes

$$|1 - \alpha\sigma_i^2| < 1$$

for every nonzero singular value. Hence

$$\xi_{i,k} \longrightarrow \frac{u_i^* b}{\sigma_i}.$$

If the iteration starts from $x_0 = 0$, the null-space components remain zero, so

$$x_k \longrightarrow A^+ b.$$

The minimum-norm choice is therefore encoded by the initialization: gradient descent never invents a component in a direction that the data cannot see.

This observation is useful beyond numerical linear algebra. In an underdetermined linear model, the optimization dynamics themselves can impose an implicit bias toward the minimum-norm solution.

§75.10 A tiny noisy direction: three spectral filters

Take a matrix which almost erases its second coordinate:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 10^{-2} \end{pmatrix}.$$

Suppose the exact signal is $x_* = (1, 0)^T$, so $Ax_* = (1, 0)^T$, but we observe

$$b = \begin{pmatrix} 1 \\ 0.02 \end{pmatrix}.$$

The perturbation of the data is only 0.02 in norm. Exact pseudoinversion turns it into

$$\boxed{A^+ b = (1, 2)^T,}$$

an error of size 2 in the recovered vector. The small singular direction has amplified the noise a hundredfold.

A Tikhonov penalty with $\lambda = 10^{-3}$ changes the answer to

$$x_\lambda = \begin{pmatrix} \dfrac{1}{1 + 10^{-3}} \\ \dfrac{10^{-2}}{10^{-4} + 10^{-3}}\, 0.02 \end{pmatrix} = \begin{pmatrix} \dfrac{1000}{1001} \\ \dfrac{2}{11} \end{pmatrix}.$$

It suppresses the nearly invisible coordinate, but at a cost: if the true signal *had* a large component in that direction, regularization would erase part of it. There is no universally correct cutoff without an assumption about signal and noise.

The preceding Landweber iteration hides a related kind of regularization. Starting at $x_0 = 0$, solving its scalar recurrence in each singular direction gives an exact finite-step formula:

$$\boxed{x_k = \sum_{\sigma_i > 0} \frac{1 - (1 - \alpha\sigma_i^2)^k}{\sigma_i} (u_i^* b) v_i.}$$

For $\alpha = 1$ and $k = 1000$ in our example, the first coordinate is 1, while the second is

$$2\left(1 - \left(1 - 10^{-4}\right)^{1000}\right) \approx 0.1903.$$

The iterative solver has not yet amplified that direction all the way to 2. Near $\sigma = 0$, the finite-step filter behaves as

$$\frac{1 - (1 - \alpha\sigma^2)^k}{\sigma} \sim k\alpha\sigma;$$

Tikhonov's filter behaves as σ/λ. They are different functions with the same useful feature: both attenuate the tiniest singular directions.

In this example, running the iteration *to convergence* would recover the noisy pseudoinverse $(1, 2)^T$. Stopping early is therefore a meaningful part of the inverse problem, not simply an unfinished numerical computation.

Solving Linear Systems: Elimination, Iteration, and Energy

N-20260604

§76.1 One equation, three ways of thinking

The equation

$$Ax = b$$

looks like a single problem, but numerically it can mean three rather different things.

If the matrix is moderate in size and used several times, it is natural to *factor* it. If the matrix is huge and sparse, it may be better to improve a guess one step at a time. If A is symmetric positive definite, the equation is also the stationarity condition of a convex quadratic energy. These are not three unrelated collections of formulas. They are three views of the same linear map:

eliminate variables, propagate corrections, minimize energy.

A useful solver therefore starts by asking what structure is available. Dense or sparse? Symmetric or nonsymmetric? Positive definite or indefinite? One right-hand side or many? The answer often matters more than the nominal matrix size.

§76.2 Triangular systems are the basic currency

Nearly every direct method tries to reduce a general system to triangular ones. If L is lower triangular, then the first equation contains only x_1, the second contains x_1, x_2, and so on. Forward substitution gives

$$x_i = \frac{1}{l_{ii}} \left(b_i - \sum_{j<i} l_{ij} x_j \right).$$

For an upper triangular matrix U, backward substitution runs in the opposite direction:

$$x_i = \frac{1}{u_{ii}} \left(b_i - \sum_{j>i} u_{ij} x_j \right).$$

The arithmetic cost is only $O(n^2)$. The expensive part of a direct solve is not substitution but producing the triangular factors. This explains why an LU or Cholesky factorization is especially valuable when many vectors b must be solved against the same matrix: the cubic work is paid once, while each new right-hand side needs only triangular solves.

§76.3 Gaussian elimination is a factorization in disguise

Consider

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix}, \qquad b = \begin{bmatrix} 5 \\ -2 \\ 9 \end{bmatrix}.$$

The first pivot is 2. Replacing

$$R_2 \leftarrow R_2 - 2R_1, \qquad R_3 \leftarrow R_3 + R_1$$

gives

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 0 & -8 & -2 & -12 \\ 0 & 8 & 3 & 14 \end{array}\right].$$

One more elimination step,

$$R_3 \leftarrow R_3 + R_2,$$

produces

$$U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & 1 \end{bmatrix}.$$

Back substitution yields

$$x_3 = 2, \qquad x_2 = 1, \qquad x_1 = 1.$$

The multipliers used during elimination were $2, -1, -1$. Placed in a unit lower triangular matrix, they form

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix}, \qquad A = LU.$$

Thus elimination is not merely a sequence of row tricks. It constructs a reusable representation of the linear map.

§76.4 Why pivoting is part of the algorithm

Exact arithmetic only objects to a zero pivot. Floating-point arithmetic also objects to a very small one. For example,

$$A = \begin{bmatrix} 10^{-12} & 1 \\ 1 & 1 \end{bmatrix}$$

is perfectly invertible, yet elimination without row exchange uses the multiplier 10^{12}. A tiny rounding error in the first row can then be magnified enormously.

Partial pivoting swaps the largest available entry in the current column into the pivot position. Algebraically the result is

$$PA = LU,$$

where P is a permutation matrix. Pivoting does not change the problem; it changes the order in which information is eliminated so that intermediate numbers remain under control.

This distinction is worth keeping: conditioning belongs to the original problem, whereas numerical stability belongs to the algorithm. A well-conditioned system can still be damaged by a poor elimination order, and a stable algorithm cannot make an intrinsically ill-conditioned system insensitive.

§76.5 Structure can make elimination cheaper

A general dense factorization costs $O(n^3)$, but special matrices often require much less.

If A is symmetric positive definite, Cholesky factorization writes

$$A = LL^T.$$

Only one triangular factor is needed, symmetry halves the storage, and the work is about half that of a general LU factorization. An LDL^T factorization separates scaling from elimination:

$$A = LDL^T,$$

with L unit lower triangular and D diagonal or block diagonal.

For a tridiagonal matrix,

$$A = \begin{bmatrix} d_1 & c_1 & & & \\ a_1 & d_2 & c_2 & & \\ & a_2 & d_3 & \ddots & \\ & & \ddots & \ddots & c_{n-1} \\ & & & a_{n-1} & d_n \end{bmatrix},$$

elimination preserves the band. The resulting Thomas algorithm needs only $O(n)$ work and storage. This is a first example of a general numerical principle: preserve sparsity whenever possible, because unnecessary fill-in can cost more than the original equations.

§76.6 Iteration versus Factorization

For a very large sparse matrix, even storing the factors may be too expensive. Iterative methods instead generate

$$x^{(0)}, x^{(1)}, x^{(2)}, \dots$$

and hope that these vectors approach the solution.

Write

$$A = M - N.$$

Then

$$Mx = Nx + b,$$

so a natural iteration is

$$Mx^{(k+1)} = Nx^{(k)} + b.$$

Equivalently,

$$x^{(k+1)} = Bx^{(k)} + c, \qquad B = M^{-1}N, \quad c = M^{-1}b.$$

If x_* is the exact solution and $e^{(k)} = x^{(k)} - x_*$, then

$$e^{(k+1)} = Be^{(k)}, \qquad e^{(k)} = B^k e^{(0)}.$$

Therefore convergence for every initial guess is equivalent to

$$\rho(B) < 1.$$

This one equation explains why stationary iterations are spectral problems rather than merely recursive formulas.

§76.7 Jacobi and Gauss–Seidel seen on the same example

Let

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

The exact solution is

$$x_* = \begin{bmatrix} 1/10 \\ 3/5 \end{bmatrix}.$$

Starting from $x^{(0)} = (0, 0)^T$, Jacobi updates both coordinates using only the old iterate:

$$x_1^{(k+1)} = \frac{1 - x_2^{(k)}}{4}, \qquad x_2^{(k+1)} = \frac{2 - 2x_1^{(k)}}{3}.$$

The first two iterates are

$$x^{(1)} = \begin{bmatrix} 1/4 \\ 2/3 \end{bmatrix}, \qquad x^{(2)} = \begin{bmatrix} 1/12 \\ 1/2 \end{bmatrix}.$$

Gauss–Seidel immediately reuses the newest value of x_1:

$$x_1^{(k+1)} = \frac{1 - x_2^{(k)}}{4}, \qquad x_2^{(k+1)} = \frac{2 - 2x_1^{(k+1)}}{3}.$$

Thus

$$x^{(1)} = \begin{bmatrix} 1/4 \\ 1/2 \end{bmatrix}, \qquad x^{(2)} = \begin{bmatrix} 1/8 \\ 7/12 \end{bmatrix}.$$

Both converge here, but Gauss–Seidel moves more quickly because each sweep contains a small amount of implicit coupling.

Strict diagonal dominance,

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}|,$$

is a convenient sufficient condition for convergence of both methods. It is not the only condition, and failure of diagonal dominance does not automatically imply divergence. The exact question remains $\rho(B) < 1$.

§76.8 Residuals are visible; errors are not

During an iteration the true error

$$e^{(k)} = x^{(k)} - x_*$$

is usually unknown, because knowing x_* would make the computation unnecessary. What can be computed is the residual

$$r^{(k)} = b - Ax^{(k)}.$$

Since

$$Ae^{(k)} = -r^{(k)},$$

we have

$$e^{(k)} = -A^{-1}r^{(k)}.$$

Hence a small residual implies a small error only after the inverse is controlled. In norm form,

$$\|e^{(k)}\| \le \|A^{-1}\|\,\|r^{(k)}\|.$$

For an ill-conditioned matrix, a visually tiny residual can still hide a noticeable solution error. Stopping criteria should therefore be interpreted together with scaling and condition estimates, not as universal certificates.

§76.9 Positive Systems and Convexity

Suppose $A = A^T$ is positive definite. Define

$$\phi(x) = \frac{1}{2}x^T Ax - b^T x.$$

Then

$$\nabla\phi(x) = Ax - b, \qquad \nabla^2\phi(x) = A \succ 0.$$

Thus ϕ is strictly convex, and its unique minimizer satisfies

$$Ax = b.$$

Completing the square gives an even clearer identity:

$$\phi(x) - \phi(x_*) = \frac{1}{2}(x - x_*)^T A(x - x_*) = \frac{1}{2}\|x - x_*\|_A^2,$$

where

$$\|v\|_A = \sqrt{v^T Av}.$$

Solving the linear system is therefore exactly the same as driving the A-energy error to zero.

Steepest descent follows the negative residual direction. With

$$r_k = b - Ax_k,$$

choose

$$x_{k+1} = x_k + \alpha_k r_k.$$

Minimizing $\phi(x_k + \alpha r_k)$ along this line gives

$$\alpha_k = \frac{r_k^T r_k}{r_k^T Ar_k}.$$

The method always decreases the energy, but for elongated level sets it tends to zigzag. The elongation is governed by the condition number $\kappa_2(A)$.

§76.10 Conjugate gradients remembers previous progress

Steepest descent repeatedly chooses the locally best direction, yet may undo part of earlier progress. Conjugate gradients avoids this by constructing search directions p_k that are A-orthogonal:

$$p_i^T Ap_j = 0 \qquad (i \ne j).$$

Starting from x_0, set

$$r_0 = b - Ax_0, \qquad p_0 = r_0.$$

Then iterate

$$\alpha_k = \frac{r_k^T r_k}{p_k^T A p_k}, \qquad x_{k+1} = x_k + \alpha_k p_k,$$

$$r_{k+1} = r_k - \alpha_k A p_k, \qquad \beta_k = \frac{r_{k+1}^T r_{k+1}}{r_k^T r_k},$$

$$p_{k+1} = r_{k+1} + \beta_k p_k.$$

After k steps, x_k lies in the affine Krylov space

$$x_0 + \mathcal{K}_k(A, r_0), \qquad \mathcal{K}_k(A, r_0) = \operatorname{span}\{r_0, Ar_0, \ldots, A^{k-1} r_0\},$$

and minimizes the energy over that space. In exact arithmetic, conjugate gradients reaches the exact solution in at most n steps. Its practical value is that useful accuracy often arrives much sooner, especially when the eigenvalues are clustered.

A standard bound is

$$\frac{\|x_k - x_*\|_A}{\|x_0 - x_*\|_A} \le 2 \left(\frac{\sqrt{\kappa_2(A)} - 1}{\sqrt{\kappa_2(A)} + 1} \right)^k.$$

This makes the role of conditioning explicit: poor geometry slows the polynomial approximation carried out by the Krylov method.

§76.11 Preconditioning changes the geometry, not the solution

A preconditioner M is a matrix that is easy to solve with and approximates A. Instead of attacking

$$Ax = b$$

directly, one solves an equivalent system such as

$$M^{-1} A x = M^{-1} b.$$

The solution is unchanged, but the spectrum may become much more favorable.

For symmetric positive definite problems, a symmetric preconditioner can be understood through

$$M^{-1/2} A M^{-1/2} y = M^{-1/2} b, \qquad x = M^{-1/2} y.$$

Good preconditioning clusters the eigenvalues of the transformed matrix and reduces its condition number. Diagonal scaling, incomplete Cholesky factors, and multigrid operators are increasingly sophisticated realizations of the same idea: solve an easier nearby problem to correct the hard one.

§76.12 Choosing a Numerical Method

The methods above are best remembered through the obstacle each one removes.

A dense system with several right-hand sides rewards a stable factorization such as pivoted LU. Positive definiteness makes Cholesky cheaper and supplies an energy interpretation. Tridiagonal or banded structure should be preserved rather than destroyed. Very large sparse systems favor iteration because matrix–vector products are cheap while fill-in is expensive. Stationary methods are simple smoothers, but their speed is tied to one fixed iteration matrix. Krylov methods adapt a polynomial to the observed residuals

and are usually far more effective. Preconditioning is the decisive step when the original coordinates make convergence slow.

The deeper unity is that every solver tries to expose directions in which the equation is easy. Elimination removes them one by one, stationary iteration repeatedly damps them, and Krylov methods build a subspace tailored to them. The matrix is the same; the algorithm decides how its geometry is revealed.

VII

Real and Complex Analysis

Part VII: Contents

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Sequences, Limits, Derivatives, and Riemann Integration

N-20220319

A limit identifies which changes eventually become negligible; a derivative selects the best linear approximation; an integral accumulates such changes. This development follows Rudin's *Principles of Mathematical Analysis* and MIT 18.100B real-analysis notes, with each conclusion tied to its hypotheses.[1]

§77.1 A limit is an eventually quantified statement

For a sequence $a : \mathbb{N} \to \mathbb{R}$, write $a_n \to \ell$ when

$$\forall \varepsilon > 0 \; \exists N \in \mathbb{N} \; \forall n \geq N : \quad |a_n - \ell| < \varepsilon.$$

The cutoff N depends on the requested error but must work for every later index. For $a_n = n/(n+1)$, we have $|a_n - 1| = 1/(n+1)$; choosing $N + 1 > 1/\varepsilon$ establishes convergence.

By contrast, $(-1)^n$ cannot converge. Both its even subsequence, constantly 1, and its odd subsequence, constantly -1, would have to approach the same limit. Uniqueness of limits contradicts this.

§77.2 Limits ignore the value at the point

Let a be an accumulation point of the domain A. The statement $\lim_{x \to a} f(x) = L$ means

$$\forall \varepsilon > 0 \; \exists \delta > 0 : \quad x \in A, \quad 0 < |x - a| < \delta \Longrightarrow |f(x) - L| < \varepsilon.$$

The puncture is essential. If $f(x) = x^2$ for $x \neq 2$, but $f(2) = 17$, then $\lim_{x \to 2} f(x) = 4$, even though $f(2) = 17$.

Theorem 77.2.1 (Sequential criterion)
The limit $f(x) \to L$ as $x \to a$ exists if and only if every sequence $x_n \in A \setminus \{a\}$ with $x_n \to a$ satisfies $f(x_n) \to L$.

Proof. The forward direction follows by applying the δ condition once x_n is close to a. Conversely, if no suitable δ exists for some $\varepsilon_0 > 0$, choose for each n a point $x_n \in A$ such that

$$0 < |x_n - a| < 1/n, \qquad |f(x_n) - L| \geq \varepsilon_0.$$

Then $x_n \to a$, but its image does not converge to L, a contradiction. □

This proves discontinuity of the rational-indicator function at every real point: rational and irrational sequences converge to any chosen point but have constantly different function values.

[1] https://ocw.mit.edu/courses/18-100b-real-analysis-spring-2025/pages/lecture-notes/

§77.3 The derivative is a linear approximation

A function f is differentiable at a if

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

exists. Equivalently,

$$f(a+h) = f(a) + f'(a)h + o(|h|),$$

where $r(h) = o(|h|)$ means $r(h)/|h| \to 0$. The linear part $f'(a)h$ is unique, because two proposed coefficients would differ by $o(|h|)/h \to 0$.

For $f(x) = x^2$,

$$f(a+h) = a^2 + 2ah + h^2 = a^2 + 2ah + o(|h|).$$

The little-o term measures the actual approximation error. By contrast $r(h) = O(h^2)$ asserts only that $|r(h)| \le Ch^2$ near zero, a stronger scale-specific bound in this example.

The product rule follows from the exact identity

$$u(a+h)v(a+h) - u(a)v(a) = u(a+h)(v(a+h) - v(a)) + v(a)(u(a+h) - u(a)).$$

Divide by h and use differentiability of both factors to obtain

$$(uv)' = u'v + uv'.$$

Applying this to $v(1/v) = 1$ gives the quotient rule when $v \neq 0$. Composing two first-order expansions gives the chain rule $(F \circ g)' = F'(g)g'$. Repeated product differentiation leads by induction to

$$(uv)^{(n)} = \sum_{k=0}^{n} \binom{n}{k} u^{(k)} v^{(n-k)}.$$

§77.4 An integral is a partition-independent sum

For $P : a = x_0 < \cdots < x_m = b$, select $\xi_i \in [x_{i-1}, x_i]$ and define

$$S(f, P, \xi) = \sum_{i=1}^{m} f(\xi_i)(x_i - x_{i-1}).$$

Riemann integrability requires these sums to approach one limit as the maximum interval length tends to zero, independently of the sample points. Every continuous function on a compact interval is integrable.

For $f(x) = x$ on $[0, 1]$, equal partitions and right-endpoint samples give

$$S_n = \frac{1}{n^2} \sum_{i=1}^{n} i = \frac{n(n+1)}{2n^2} \to \frac{1}{2}.$$

This computes the value once integrability has been established by continuity.

§77.5 The fundamental theorem connects both operations

Theorem 77.5.1 (Fundamental theorem of calculus)
For continuous $f : [a, b] \to \mathbb{R}$, the function

$$F(x) = \int_a^x f(t)\,dt$$

has derivative $F'(x) = f(x)$ at interior points. If $G' = f$, then $\int_a^b f(t)\,dt = G(b) - G(a)$.

For $h > 0$,

$$\frac{F(x+h) - F(x)}{h} = \frac{1}{h}\int_x^{x+h} f(t)\,dt.$$

The difference from $f(x)$ has absolute value at most $\sup_{|t-x|\le h} |f(t) - f(x)|$, which tends to zero by continuity. The case $h < 0$ is analogous. The second identity follows from the first because $F - G$ has derivative zero on the connected interval, and the mean value theorem makes it constant.

§77.6 What changes when one changes the norm

Metric convergence replaces $|a_n - a|$ by $d(a_n, a)$. A useful example is $f_n(x) = x^n$ on $[0, 1]$. Although

$$\|f_n\|_{L^1} = \int_0^1 x^n\,dx = \frac{1}{n+1} \to 0,$$

its supremum norm is $\|f_n\|_\infty = 1$ for every n. Thus it converges to zero in L^1 but not uniformly. Different notions of distance ask for different notions of small error, even though every convergence proof follows the same epsilon-quantifier pattern.

Mean Value Theorems: How Local Slopes Control an Interval

N-20220320

A derivative is information at one point, whereas an estimate such as

$$|f(b) - f(a)| \le M|b - a|$$

compares two distinct points. The mean value theorem explains how to pass between these scales. We follow the classical route through extrema, Rolle's theorem, and its two-function extension, as developed in Rudin's *Principles of Mathematical Analysis*, Chapter 5, and the MIT 18.100B real-analysis lectures.[1]

§78.1 Why a compact interval enters the proof

Suppose $h : [a, b] \to \mathbb{R}$ is continuous and $h(a) = h(b)$. The extreme value theorem guarantees that h attains a maximum and a minimum. If both extrema occur only at endpoints, their common endpoint value makes h constant. Otherwise one extremum occurs at an interior point c. Provided h is differentiable there, the two one-sided difference quotients have opposite allowable signs, so $h'(c) = 0$.

Theorem 78.1.1 (Rolle)
If h is continuous on $[a, b]$, differentiable on (a, b), and $h(a) = h(b)$, then $h'(c) = 0$ for some $c \in (a, b)$.

The hypotheses cannot be omitted casually. For $h(x) = |x|$ on $[-1, 1]$, the endpoints agree but the sole interior minimum occurs at a point where the derivative does not exist. If continuity fails at an endpoint, the extreme value argument also breaks.

§78.2 Removing the secant line

Theorem 78.2.1 (Mean value theorem)
If f is continuous on $[a, b]$ and differentiable on (a, b), then there is $c \in (a, b)$ with

$$f(b) - f(a) = f'(c)(b - a).$$

Proof. The affine interpolant between the endpoint values is

$$\ell(x) = f(a) + \frac{f(b) - f(a)}{b - a}(x - a).$$

The difference $h = f - \ell$ satisfies $h(a) = h(b) = 0$. Rolle's theorem supplies c with

$$0 = h'(c) = f'(c) - \frac{f(b) - f(a)}{b - a}.$$

□

[1] https://ocw.mit.edu/courses/18-100b-real-analysis-spring-2025/pages/lecture-notes/

This proof suggests the correct geometric operation: subtract the simplest function matching the boundary data. For example, $f(x) = x^2$ on $[1, 3]$ has secant slope 4; the equality $2c = 4$ identifies $c = 2$.

§78.3 The two-function theorem and a nonzero denominator

Theorem 78.3.1 (Cauchy's mean value theorem)

Suppose f, g are continuous on $[a, b]$ and differentiable on (a, b). Then some $c \in (a, b)$ satisfies

$$(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c).$$

Proof. Put $A = f(b) - f(a)$, $B = g(b) - g(a)$, and set

$$h(x) = Bf(x) - Ag(x).$$

Its endpoint difference is $BA - AB = 0$. Applying Rolle to h yields $Bf'(c) - Ag'(c) = 0$. □

The cross-multiplied identity is the theorem's basic form. To write a ratio one must justify the divisions. If $g' \neq 0$ throughout (a, b), Rolle applied to g forces $g(b) \neq g(a)$, and Cauchy's identity becomes

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}.$$

Without this extra condition, a quotient formulation may be meaningless although the cross-multiplied theorem remains true.

§78.4 Where l'Hopital's rule comes from

Consider the right-hand $0/0$ situation. Assume f, g are continuous on $[a, a + \delta)$, differentiable on $(a, a + \delta)$, with

$$f(a) = g(a) = 0, \qquad g'(x) \neq 0 \quad (a < x < a + \delta).$$

For every $x > a$, Cauchy's theorem on $[a, x]$ supplies $c_x \in (a, x)$ such that

$$\frac{f(x)}{g(x)} = \frac{f'(c_x)}{g'(c_x)}.$$

Consequently, if $f'(t)/g'(t) \to L$ as $t \downarrow a$, then $f(x)/g(x) \to L$, because $c_x \downarrow a$ in the sense that $a < c_x < x \to a$.

Example 78.4.1

Let $f(x) = \sin x - x$ and $g(x) = x^3$, with $a = 0$. Direct substitution gives $0/0$, but

$$\frac{f'(x)}{g'(x)} = \frac{\cos x - 1}{3x^2} \longrightarrow -\frac{1}{6}.$$

The derivative ratio here still uses the familiar limit $(1 - \cos x)/x^2 \to \frac{1}{2}$; it does not license a circular application of l'Hopital to itself.

The theorem is a statement about limits under explicit hypotheses. Taking derivatives of numerator and denominator is not a general algebraic operation on fractions.

§78.5 Derivative bounds give global estimates

Proposition 78.5.1 (Lipschitz estimate)

If f is continuous on an interval and differentiable in its interior, with $|f'(x)| \le M$, then

$$|f(x) - f(y)| \le M|x - y|$$

for all x, y in the interval.

Proof. Apply the mean value theorem to the subinterval with endpoints x, y, then take absolute values. □

If $f' = 0$ throughout an interval, this forces f to be constant. If $f' \ge 0$, the same reasoning shows f is nondecreasing. These are stronger conclusions than a picture of tangent lines: a pointwise differential inequality implies a global comparison.

The Lipschitz condition also enters uniqueness arguments for ordinary differential equations. When $|F(t, u) - F(t, v)| \le L|u - v|$, two solutions with the same initial value satisfy

$$e(t) := |u(t) - v(t)| \le L \int_{t_0}^{t} e(s)\, ds \qquad (t \ge t_0).$$

For $H(t) = \int_{t_0}^{t} e(s)\, ds$, one has $H(t_0) = 0$ and $H'(t) \le LH(t)$. Therefore $(e^{-Lt}H(t))' \le 0$. Since $H \ge 0$, it follows that $H = 0$ and the solutions coincide. This is the elementary core of the Gronwall argument.

§78.6 Continuity versus uniform continuity

Continuity at a point x allows the δ in the ε–δ definition to depend on x. Uniform continuity asks for the same $\delta(\varepsilon)$ across the domain.

Theorem 78.6.1 (Heine–Cantor)

A continuous function from a compact metric space to a metric space is uniformly continuous.

Proof. Suppose otherwise. There is $\varepsilon_0 > 0$ and sequences x_n, y_n with

$$d(x_n, y_n) < 1/n, \qquad d(f(x_n), f(y_n)) \ge \varepsilon_0.$$

By compactness, a subsequence x_{n_j} converges to some x. The triangle inequality gives $y_{n_j} \to x$ as well. Continuity at x forces both image subsequences to converge to $f(x)$, a contradiction. □

Compactness matters. The function $x \mapsto x^2$ on $\mathbb{R}$ is continuous but not uniformly continuous: take $x_n = n$, $y_n = n + 1/n$. The inputs approach one another, while the output difference tends to 2.

§78.7 Taylor's formula as an error statement

For $f \in C^2$ between a and x, Taylor's theorem gives some ξ between them such that

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(\xi)}{2}(x-a)^2.$$

In particular, if $|f''| \le K$, the error of the tangent-line approximation satisfies

$$|f(x) - f(a) - f'(a)(x-a)| \le \frac{K}{2}|x-a|^2.$$

The logical progression is now visible: extrema force a zero derivative; removing a secant produces a mean slope; bounding derivatives controls finite differences; controlling a higher derivative quantifies the error of a local polynomial approximation.

Motion on the Unit Circle, Roots of Unity, and Finite Fourier Analysis

N-20220321

§79.1 A Differential Equation for Rotation

Consider the complex-valued initial-value problem

$$Z'(t) = iZ(t), \qquad Z(0) = 1.$$

Its solution is

$$Z(t) = e^{it}.$$

Multiplication by i sends

$$x + iy \longmapsto -y + ix,$$

which is a counterclockwise quarter-turn. Thus the velocity $Z'(t) = iZ(t)$ is always perpendicular to the radius vector.

The equation also proves that the radius is constant. Since

$$|Z|^2 = Z\overline{Z},$$

one has

$$\begin{aligned}\frac{d}{dt}|Z(t)|^2 &= Z'(t)\overline{Z(t)} + Z(t)\overline{Z'(t)} \\ &= iZ(t)\overline{Z(t)} - iZ(t)\overline{Z(t)} = 0.\end{aligned}$$

Because $|Z(0)| = 1$,

$$|Z(t)| = 1$$

for every t. Moreover,

$$|Z'(t)| = |iZ(t)| = 1,$$

so the particle moves around the unit circle with unit speed.

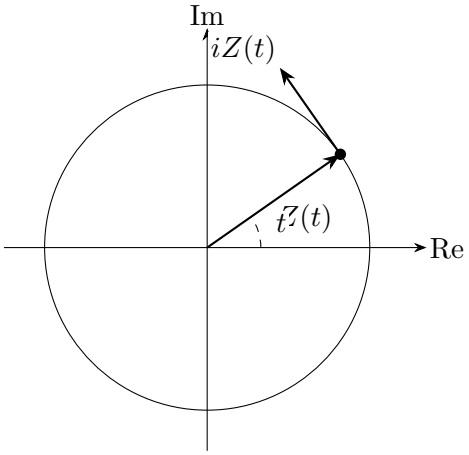


Figure 79.1: The equation $Z' = iZ$ rotates the radius vector by 90°, producing tangent velocity of constant length.

§79.2 The exponential map wraps a line onto a circle

The map

$$\exp_i : \mathbb{R} \longrightarrow S^1, \qquad t \longmapsto e^{it}$$

is a homomorphism from addition to multiplication:

$$e^{i(s+t)} = e^{is}e^{it}.$$

It is surjective because every unit complex number has the form

$$\cos t + i \sin t.$$

Its kernel is

$$\ker \exp_i = 2\pi\mathbb{Z}.$$

Therefore the first isomorphism theorem gives an isomorphism of groups

$$\boxed{\mathbb{R}/2\pi\mathbb{Z} \cong S^1.}$$

Angles differing by a full number of turns are not merely numerically similar; they represent the same coset in the quotient.

Locally, $t \mapsto e^{it}$ is one-to-one, but globally it wraps the real line around the circle infinitely many times. For a continuous loop

$$\gamma : [0,1] \to S^1, \qquad \gamma(0) = \gamma(1),$$

one can choose a continuous lift $\theta : [0,1] \to \mathbb{R}$ with

$$\gamma(t) = e^{i\theta(t)}.$$

The endpoint difference satisfies

$$\theta(1) - \theta(0) = 2\pi k$$

for an integer k, the winding number of the loop. The kernel of the exponential map is what makes this integer possible.

§79.3 Roots and Branch Choices

Let

$$w = \rho e^{i\theta}, \qquad \rho > 0.$$

The equation

$$z^n = w$$

has the n solutions

$$z_k = \rho^{1/n} e^{i(\theta + 2\pi k)/n}, \qquad k = 0, 1, \ldots, n-1.$$

The radii are fixed by

$$|z|^n = \rho,$$

while the arguments solve

$$n \arg z \equiv \theta \pmod{2\pi}.$$

The solutions form a regular n-gon on the circle of radius $\rho^{1/n}$.

For example, the fourth roots of -4 are

$$\sqrt{2}\, e^{i(\pi+2\pi k)/4}, \qquad k = 0, 1, 2, 3.$$

They occur at angles

$$\frac{\pi}{4}, \quad \frac{3\pi}{4}, \quad \frac{5\pi}{4}, \quad \frac{7\pi}{4}.$$

The multivalued appearance of complex roots is therefore the geometry of lifting one point through the covering map $z \mapsto z^n$.

§79.4 The roots of unity form a finite cyclic group

Let

$$\zeta_n = e^{2\pi i/n}.$$

The n-th roots of unity are

$$\mu_n = \{1, \zeta_n, \ldots, \zeta_n^{n-1}\}.$$

Multiplication adds exponents modulo n:

$$\zeta_n^a \zeta_n^b = \zeta_n^{a+b}.$$

Hence

$$\boxed{\mathbb{Z}/n\mathbb{Z} \longrightarrow \mu_n, \qquad [k] \longmapsto \zeta_n^k}$$

is a group isomorphism.

The order of ζ_n^k is

$$\frac{n}{\gcd(n, k)}.$$

Indeed, $(\zeta_n^k)^m = 1$ exactly when

$$n \mid km.$$

The smallest positive such m is $n/\gcd(n, k)$.

For $n = 12$, the powers ζ_{12}^k with

$$\gcd(k, 12) = 1$$

are the primitive roots:

$$\zeta_{12}, \quad \zeta_{12}^5, \quad \zeta_{12}^7, \quad \zeta_{12}^{11}.$$

Exact rotational period has become an elementary gcd calculation.

§79.5 Divisibility on Finite Subgroups

Let

$$\omega = e^{2\pi i/3}.$$

The roots of

$$x^2 + x + 1$$

are ω and ω^2. Therefore

$$x^2 + x + 1 \mid x^{2n} + x^n + 1$$

exactly when the latter polynomial vanishes at ω.

If $n \equiv 0 \pmod 3$, then $\omega^n = 1$ and

$$\omega^{2n} + \omega^n + 1 = 3.$$

If $n \equiv 1 \pmod 3$, then

$$\omega^{2n} + \omega^n + 1 = \omega^2 + \omega + 1 = 0.$$

If $n \equiv 2 \pmod 3$, then

$$\omega^{2n} + \omega^n + 1 = \omega + \omega^2 + 1 = 0.$$

Thus

$$\boxed{x^2 + x + 1 \mid x^{2n} + x^n + 1 \iff 3 \nmid n.}$$

This is the first prototype of a character test. Powers of ω remember only the exponent modulo three, so evaluation converts a polynomial identity into arithmetic on a finite cyclic group.

§79.6 Character orthogonality filters one congruence class

For $m \in \mathbb{Z}$, the geometric series gives

$$\sum_{j=0}^{n-1} \zeta_n^{jm} = \begin{cases} n, & n \mid m, \\ 0, & n \nmid m. \end{cases}$$

Equivalently,

$$\frac{1}{n} \sum_{j=0}^{n-1} \zeta_n^{j(k-r)} = \begin{cases} 1, & k \equiv r \pmod n, \\ 0, & k \not\equiv r \pmod n. \end{cases}$$

This is the roots-of-unity filter.

Let

$$P(x) = \sum_{k \geq 0} a_k x^k.$$

The sum of the coefficients with indices congruent to $r \pmod n$ is

$$\sum_{k \equiv r\,(n)} a_k = \frac{1}{n} \sum_{j=0}^{n-1} \zeta_n^{-jr} P(\zeta_n^j).$$

For example, let

$$P(x) = (1 + x)^8$$

and ask for the sum of coefficients whose exponent is divisible by four. With $\zeta_4 = i$,

$$\begin{aligned} a_0 + a_4 + a_8 &= \frac{1}{4}\big(P(1) + P(i) + P(-1) + P(-i)\big) \\ &= \frac{1}{4}\big(256 + (1 + i)^8 + 0 + (1 - i)^8\big). \end{aligned}$$

Since

$$1 \pm i = \sqrt{2} e^{\pm i\pi/4},$$

one has

$$(1 + i)^8 = (1 - i)^8 = 16.$$

Therefore

$$a_0 + a_4 + a_8 = 72,$$

matching

$$\binom{8}{0} + \binom{8}{4} + \binom{8}{8} = 1 + 70 + 1.$$

The filter replaces a restricted coefficient sum by a short list of evaluations.

§79.7 The DFT as Character Coordinates

For a function

$$f : \mathbb{Z}/n\mathbb{Z} \to \mathbb{C},$$

define

$$\widehat{f}(m) = \sum_{k=0}^{n-1} f(k)\zeta_n^{-mk}.$$

The inverse formula is

$$\boxed{f(k) = \frac{1}{n}\sum_{m=0}^{n-1} \widehat{f}(m)\zeta_n^{mk}.}$$

To verify it, substitute the definition:

$$\begin{aligned}\frac{1}{n}\sum_m \widehat{f}(m)\zeta_n^{mk} &= \frac{1}{n}\sum_m \sum_{j=0}^{n-1} f(j)\zeta_n^{m(k-j)} \\ &= \sum_{j=0}^{n-1} f(j)\left(\frac{1}{n}\sum_m \zeta_n^{m(k-j)}\right) \\ &= f(k).\end{aligned}$$

The roots-of-unity filter is exactly the orthogonality step in this proof.

For $n = 4$, take

$$f = (1, 2, 0, -1).$$

Using $\zeta_4 = i$,

$$\begin{aligned}\widehat{f}(0) &= 1 + 2 + 0 - 1 = 2, \\ \widehat{f}(1) &= 1 + 2(-i) + (-1)i = 1 - 3i, \\ \widehat{f}(2) &= 1 - 2 + 0 + 1 = 0, \\ \widehat{f}(3) &= 1 + 2i - i = 1 + 3i.\end{aligned}$$

Thus

$$\widehat{f} = (2, 1 - 3i, 0, 1 + 3i).$$

Applying the inverse formula recovers all four original entries. The transform is not merely evaluation at roots of unity; orthogonality makes those evaluations a complete coordinate system.

§79.8 Fourier Characters and Circulants

Let S be the cyclic shift on $\mathbb{C}^n$:

$$S(x_0, x_1, \ldots, x_{n-1}) = (x_{n-1}, x_0, \ldots, x_{n-2}).$$

For each m, define the Fourier vector

$$v_m = (1, \zeta_n^{-m}, \zeta_n^{-2m}, \ldots, \zeta_n^{-(n-1)m})^T.$$

Then

$$Sv_m = \zeta_n^m v_m.$$

Hence the Fourier basis diagonalizes the shift.

Every circulant matrix is a polynomial in S:

$$C = c_0 I + c_1 S + \cdots + c_{n-1} S^{n-1}.$$

Therefore

$$Cv_m = \lambda_m v_m,$$

where

$$\lambda_m = c_0 + c_1 \zeta_n^m + \cdots + c_{n-1} \zeta_n^{m(n-1)}.$$

The eigenvalues are the DFT of the first row, up to the chosen sign convention.

For the four-cycle adjacency matrix

$$C = S + S^{-1},$$

the eigenvalues are

$$\lambda_m = \zeta_4^m + \zeta_4^{-m} = 2\cos\frac{2\pi m}{4},$$

so

$$\sigma(C) = \{2, 0, -2, 0\}.$$

A graph operator with coupled coordinates becomes four independent scalar multipliers in the Fourier basis.

Circular convolution is diagonalized for the same reason. If

$$(f * g)(r) = \sum_{k \in \mathbb{Z}/n\mathbb{Z}} f(k) g(r-k),$$

then

$$\widehat{f * g}(m) = \widehat{f}(m)\widehat{g}(m).$$

Convolution in physical coordinates is multiplication in character coordinates.

§79.9 Heat on a Four-Node Cycle

The Fourier basis earns its keep when we let an operator evolve rather than merely diagonalize it. Imagine four sites arranged on a circle, with heat (or a probability distribution) allowed to move between nearest neighbors. If $u_k(t)$ is the quantity at vertex $k \in \mathbb{Z}/4\mathbb{Z}$, the discrete heat equation is

$$\frac{d}{dt} u_k(t) = u_{k-1}(t) - 2u_k(t) + u_{k+1}(t), \qquad u_k(0) = \delta_{k0}.$$

Indices are taken modulo four. In matrix form, $\dot{u} = -Lu$, where

$$L = 2I - S - S^{-1}.$$

The Fourier characters turn the *coupled* system of four differential equations into independent scalar ones. Indeed, for $w_m(k) = \zeta_4^{mk} = i^{mk}$,

$$Lw_m = (2 - \zeta_4^m - \zeta_4^{-m}) w_m = \left(2 - 2\cos\frac{2\pi m}{4}\right) w_m.$$

Thus the four eigenvalues are

$$\lambda_0 = 0, \qquad \lambda_1 = 2, \qquad \lambda_2 = 4, \qquad \lambda_3 = 2.$$

The point source δ_{k0} has Fourier transform $\widehat{u}_0(m) = 1$ for every m. Each coefficient solves $\dot{\widehat{u}}_t(m) = -\lambda_m \widehat{u}_t(m)$, giving

$$\boxed{u_k(t) = \frac{1}{4}\sum_{m=0}^{3} e^{-\lambda_m t}\, i^{mk}.}$$

Returning to real coordinates and writing $a = e^{-2t}$, we get the exact four-site solution

$$u_0(t) = \frac{1 + 2a + a^2}{4}, \quad u_1(t) = u_3(t) = \frac{1 - a^2}{4},$$
$$u_2(t) = \frac{1 - 2a + a^2}{4}.$$

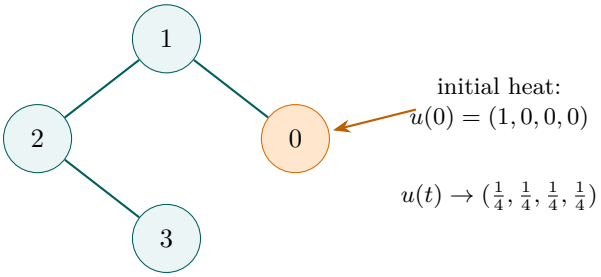


Figure 79.2: The four-cycle has one constant Fourier mode and three decaying modes. Mass spreads from the highlighted vertex until the distribution becomes uniform.

These expressions are automatically nonnegative for $t \geq 0$ because $0 \leq a \leq 1$, and they sum to one:

$$\sum_{k=0}^{3} u_k(t) = 1.$$

At $t = 0$, $a = 1$ and all heat starts at vertex zero. As $t \to \infty$, the nonconstant Fourier modes decay and each site approaches $1/4$. The surviving zero-eigenvalue mode represents conservation of total mass, while the first nonzero eigenvalue 2 controls the slowest exponential relaxation. This is the spectral gap of the graph Laplacian.

The computation has unified three pictures with no extra black box: roots of unity are characters of a cyclic group, characters diagonalize circulant matrices, and those same eigenvectors solve heat flow and continuous-time random walks on a periodic graph.[1]

§79.10 Cyclotomic factors separate roots by exact period

The factorization

$$x^n - 1 = \prod_{d \mid n} \Phi_d(x)$$

groups roots of unity according to exact order. For $n = 12$,

$$\begin{aligned} x^{12} - 1 &= \Phi_1(x)\Phi_2(x)\Phi_3(x)\Phi_4(x)\Phi_6(x)\Phi_{12}(x) \\ &= (x-1)(x+1)(x^2+x+1)(x^2+1) \\ &\quad \cdot (x^2 - x + 1)(x^4 - x^2 + 1). \end{aligned}$$

[1] For the diagonalization of circulant matrices by the Fourier matrix, see Gilbert Strang, MIT 18.065, Lecture 31.

The last factor contains exactly the primitive twelfth roots. This is enough for the present chapter: finite Fourier analysis uses all characters of the cyclic group, while cyclotomic factorization separates those characters by exact order. The arithmetic of the factors themselves belongs to the dedicated cyclotomic chapter.

§79.11 Fifth Roots and the Pentagon

Let

$$\zeta = e^{2\pi i/5}$$

and set

$$u = \zeta + \zeta^{-1} = 2\cos\frac{2\pi}{5}.$$

Because

$$1 + \zeta + \zeta^2 + \zeta^3 + \zeta^4 = 0,$$

dividing by ζ^2 gives

$$\zeta^2 + \zeta + 1 + \zeta^{-1} + \zeta^{-2} = 0.$$

Now

$$\zeta^2 + \zeta^{-2} = u^2 - 2,$$

so

$$u^2 + u - 1 = 0.$$

The positive root is

$$u = \frac{\sqrt{5} - 1}{2}.$$

Therefore

$$\cos\frac{2\pi}{5} = \frac{\sqrt{5} - 1}{4}.$$

The central angle of a regular pentagon is obtained using only a square root, which is why the pentagon is constructible by straightedge and compass.

This small calculation is the appropriate arithmetic endpoint here. The finite cyclic group supplies the roots, Fourier characters use all their powers as coordinates, and the real combination $\zeta + \zeta^{-1}$ collapses a complex symmetry orbit to the quadratic number controlling the pentagon.

Complex Differentiability, Cauchy–Riemann Equations, and Holomorphic Rigidity

N-20220323

§80.1 Complex Differentiability in Every Direction

Let $U \subseteq \mathbb{C}$ be open and let $f : U \to \mathbb{C}$. The complex derivative at $z_0 \in U$ is

$$f'(z_0) = \lim_{h \to 0} \frac{f(z_0 + h) - f(z_0)}{h},$$

provided the limit exists. The increment h approaches zero through the whole plane. This is much stronger than asking for two one-sided real limits.

Write

$$z = x + iy, \qquad f(z) = u(x, y) + iv(x, y).$$

If f is complex differentiable at $z_0 = x_0 + iy_0$, then the quotient has the same limit along the real and imaginary axes.

Along $h = t \in \mathbb{R}$,

$$f'(z_0) = u_x(x_0, y_0) + iv_x(x_0, y_0).$$

Along $h = it$,

$$\begin{aligned} f'(z_0) &= \lim_{t \to 0} \frac{u(x_0, y_0 + t) - u(x_0, y_0) + i\big(v(x_0, y_0 + t) - v(x_0, y_0)\big)}{it} \\ &= v_y(x_0, y_0) - iu_y(x_0, y_0). \end{aligned}$$

Equality gives the Cauchy–Riemann equations

$$\boxed{u_x = v_y, \qquad u_y = -v_x.}$$

The derivative can then be written in either form:

$$f' = u_x + iv_x = v_y - iu_y.$$

§80.2 When Cauchy–Riemann Is Sufficient

The equations alone at one point do not automatically imply complex differentiability. A clean sufficient hypothesis is real differentiability.

Proposition 80.2.1

Suppose $u, v : \mathbb{R}^2 \to \mathbb{R}$ are differentiable at (x_0, y_0) and satisfy the Cauchy–Riemann equations there. Then $f = u + iv$ is complex differentiable at $z_0 = x_0 + iy_0$.

Proof. Let $h = \Delta x + i\Delta y$. Real differentiability gives

$$\Delta u = u_x \Delta x + u_y \Delta y + r_1(h),$$
$$\Delta v = v_x \Delta x + v_y \Delta y + r_2(h),$$

where

$$\frac{|r_1(h)| + |r_2(h)|}{|h|} \to 0.$$

Using $u_x = v_y$ and $u_y = -v_x$,

$$\begin{aligned}\Delta f &= \Delta u + i\Delta v \\ &= (u_x + iv_x)(\Delta x + i\Delta y) + r_1(h) + ir_2(h).\end{aligned}$$

Hence

$$\frac{\Delta f}{h} = u_x + iv_x + \frac{r_1(h) + ir_2(h)}{h} \longrightarrow u_x + iv_x.$$

□

If the first partial derivatives are continuous in a neighborhood, real differentiability is automatic. Thus the familiar C^1 version is:

$$f \text{ holomorphic} \quad \Longleftrightarrow \quad u_x = v_y, \quad u_y = -v_x.$$

§80.3 Wirtinger derivatives isolate the holomorphic part

Define

$$\partial_z = \frac{1}{2}\left(\partial_x - i\partial_y\right), \qquad \partial_{\bar{z}} = \frac{1}{2}\left(\partial_x + i\partial_y\right).$$

Since

$$x = \frac{z + \bar{z}}{2}, \qquad y = \frac{z - \bar{z}}{2i},$$

these operators treat z and $\bar{z}$ as formally independent coordinates:

$$\partial_z z = 1, \quad \partial_z \bar{z} = 0, \quad \partial_{\bar{z}} z = 0, \quad \partial_{\bar{z}} \bar{z} = 1.$$

For $f = u + iv$,

$$2\partial_{\bar{z}} f = (u_x - v_y) + i(v_x + u_y).$$

Therefore the Cauchy–Riemann equations are exactly

$$\boxed{\partial_{\bar{z}} f = 0.}$$

Under real differentiability, this is equivalent to holomorphicity.

The operator $\partial_{\bar{z}}$ measures failure of holomorphicity. For example,

$$\partial_{\bar{z}} \bar{z} = 1,$$

so $\bar{z}$ is nowhere holomorphic. Also

$$|z|^2 = z\bar{z}, \qquad \partial_{\bar{z}} |z|^2 = z.$$

Hence $|z|^2$ is complex differentiable only at $z = 0$, and it is not holomorphic on any neighborhood of that point.

§80.4 Complex Jacobians as Rotations and Scalings

The real derivative of $f = u + iv$ is the Jacobian matrix

$$Df = \begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix}.$$

Under the Cauchy–Riemann equations,

$$Df = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}, \qquad a = u_x, \quad b = v_x.$$

This is precisely the real matrix of multiplication by

$$f'(z_0) = a + ib.$$

If $f'(z_0) \neq 0$, write

$$f'(z_0) = \rho e^{i\theta}.$$

Then

$$Df(z_0) = \rho \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

Infinitesimally, f scales every direction by ρ and rotates every direction by θ. Thus a holomorphic map with nonzero derivative preserves oriented angles. This is the local meaning of conformality.

At a critical point $f'(z_0) = 0$, the first-order map collapses. For instance, $f(z) = z^m$ near 0 multiplies angles by m but has zero linear part when $m \geq 2$.

§80.5 Derivatives from Cauchy's Formula

The rigidity of holomorphic functions does not follow from the Cauchy–Riemann equations alone by a short algebraic argument. The decisive analytic input is Cauchy's integral formula.

If f is holomorphic on a neighborhood of the closed disk

$$\overline{D(a, R)},$$

then for $|z - a| < R$,

$$\boxed{f(z) = \frac{1}{2\pi i} \int_{|\zeta - a| = R} \frac{f(\zeta)}{\zeta - z}\, d\zeta.}$$

Differentiating under the integral sign gives

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_{|\zeta - a| = R} \frac{f(\zeta)}{(\zeta - z)^{n+1}}\, d\zeta.$$

One complex derivative therefore forces derivatives of every order.

The same formula yields the Cauchy estimate. If

$$M_R = \max_{|\zeta - a| = R} |f(\zeta)|,$$

then

$$|f^{(n)}(a)| \leq \frac{n! M_R}{R^n}.$$

Boundary control becomes quantitative control of all derivatives in the interior.

§80.6 Holomorphic functions are analytic

For $|z-a|<R$, expand the kernel as a geometric series:

$$\begin{aligned}\frac{1}{\zeta-z} &= \frac{1}{\zeta-a}\frac{1}{1-\frac{z-a}{\zeta-a}}\\ &= \sum_{n=0}^{\infty}\frac{(z-a)^n}{(\zeta-a)^{n+1}}.\end{aligned}$$

The series converges uniformly on the integration circle for $|z-a|<R$. Substitution into Cauchy's formula gives

$$f(z)=\sum_{n=0}^{\infty}c_n(z-a)^n,$$

where

$$c_n=\frac{1}{2\pi i}\int_{|\zeta-a|=R}\frac{f(\zeta)}{(\zeta-a)^{n+1}}\,d\zeta=\frac{f^{(n)}(a)}{n!}.$$

Thus

$$\boxed{f(z)=\sum_{n=0}^{\infty}\frac{f^{(n)}(a)}{n!}(z-a)^n.}$$

Holomorphicity implies analyticity. This is the sharp contrast with smooth real functions such as

$$g(x)=\begin{cases}e^{-1/x^2}, & x\neq 0,\\ 0, & x=0,\end{cases}$$

whose Taylor series at 0 vanishes identically although the function does not.

§80.7 Zeros are isolated unless the whole function vanishes

Suppose f is holomorphic near a and not identically zero. Its Taylor series has a first nonzero coefficient:

$$f(z)=c_m(z-a)^m+c_{m+1}(z-a)^{m+1}+\cdots, \qquad c_m\neq 0.$$

Factor

$$f(z)=(z-a)^m g(z), \qquad g(a)=c_m\neq 0.$$

By continuity, g is nonzero near a, so a is an isolated zero of multiplicity m.

This proves the identity theorem. If two holomorphic functions agree on a set with a limit point inside a connected domain, their difference has non-isolated zeros and must vanish identically.

The chain of rigidity is now explicit:

$$\begin{aligned}\partial_{\bar z}f=0 &\longrightarrow \text{Cauchy formula} \longrightarrow \text{power series},\\ &\longrightarrow \text{isolated zeros and unique continuation}.\end{aligned}$$

The operator $\partial_{\bar z}$ is the local differential equation; Cauchy's formula supplies the global analytic mechanism that makes its solutions rigid.

Techniques of Integration: Partial Fractions, Parts, and Substitution

N-20220628

§81.1 Integration methods are reversed differentiation rules

Before selecting a method, it helps to ask what derivative could have produced the integrand. Three common sources are the chain rule, product rule, and differentiation of logarithms or inverse trigonometric functions. Their inverses are substitution, integration by parts, and integration of partial fractions. The elementary organization here follows *OpenStax, Calculus Volume 2*, Chapter 3 (Integration Techniques).[1]

Linearity is the permission to deal with the resulting pieces independently. On any interval on which the functions have antiderivatives,

$$\int (f+g-h)\,dx = \int f\,dx + \int g\,dx - \int h\,dx.$$

All equalities between *indefinite* integrals are understood up to an additive constant on each connected interval of definition. This qualification matters when a denominator vanishes: crossing a pole need not preserve the same choice of constant.

§81.2 Partial fractions begin with polynomial algebra

The first example is

$$I = \int \frac{dx}{x^2 - a^2}, \qquad a \neq 0.$$

The factorization, rather than any rule of integration, determines the answer:

$$\frac{1}{(x-a)(x+a)} = \frac{1}{2a}\left(\frac{1}{x-a} - \frac{1}{x+a}\right).$$

Consequently, on an interval avoiding the two poles,

$$\boxed{I = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + C.}$$

The signs in this formula can be checked by differentiating the logarithm, which returns $1/(x^2-a^2)$. The factor $1/(2a)$ is not a memorized integration constant: it comes from solving a polynomial identity.

In general, for a rational function $P(x)/Q(x)$, first perform polynomial division if $\deg P \geq \deg Q$. Factor Q over the field in which the decomposition is sought. A real quadratic irreducible factor requires a linear numerator; a repeated linear factor $(x-a)^m$ requires *every* term

$$\frac{c_1}{x-a} + \frac{c_2}{(x-a)^2} + \cdots + \frac{c_m}{(x-a)^m}.$$

This is forced by the algebra of rational functions, independently of the later integration.

[1] https://openstax.org/books/calculus-volume-2/pages/3-introduction

Example 81.2.1 (A repeated pole really needs two coefficients)
Consider the original problem

$$I = \int \frac{3x+5}{x^3 - x^2 - x + 1}\, dx.$$

The denominator factors as

$$x^3 - x^2 - x + 1 = (x+1)(x-1)^2.$$

The correct ansatz is

$$\frac{3x+5}{(x+1)(x-1)^2} = \frac{A}{x+1} + \frac{B}{x-1} + \frac{C}{(x-1)^2}.$$

Multiplying through gives

$$3x + 5 = A(x-1)^2 + B(x+1)(x-1) + C(x+1).$$

Putting $x = -1$ and $x = 1$ yields $A = \frac{1}{2}$ and $C = 4$; comparing the remaining coefficient yields $B = -\frac{1}{2}$. Thus

$$\boxed{I = \tfrac{1}{2}\ln|x+1| - \tfrac{1}{2}\ln|x-1| - \frac{4}{x-1} + C.}$$

Differentiating the three summands is a direct independent check. Omitting the square-pole term would make the polynomial identity unsolvable.

For an irreducible quadratic, completing the square is the other algebraic step. Write

$$t = x + \frac{p}{2}, \qquad x^2 + px + q = t^2 + \Delta, \qquad \Delta = q - \frac{p^2}{4},$$

and $mx + n = mt + (n - \frac{mp}{2})$. The term with numerator t is a logarithm; the constant numerator produces an arctangent if $\Delta > 0$, while $\Delta < 0$ reduces to real linear factors. For $\Delta > 0$,

$$\int \frac{mx+n}{x^2+px+q}\, dx = \frac{m}{2}\ln(t^2+\Delta) + \frac{n - mp/2}{\sqrt{\Delta}} \arctan\frac{t}{\sqrt{\Delta}} + C.$$

This also identifies precisely which denominator sign produces which type of antiderivative.

§81.3 Substitution is the chain rule in reverse

Suppose $x = \rho(t)$ is differentiable. The chain rule says that if $F' = f$, then

$$\frac{d}{dt} F(\rho(t)) = f(\rho(t))\rho'(t).$$

Hence

$$\int f(\rho(t))\rho'(t)\, dt = F(\rho(t)) + C.$$

For a definite integral, the same argument gives

$$\int_{\rho(a)}^{\rho(b)} f(x)\,dx = \int_a^b f(\rho(t))\rho'(t)\,dt$$

under the usual regularity hypotheses; the signed derivative is intentional.

For example, the inner derivative in

$$\int \sin^3 x \cos x\,dx$$

is visible. Taking $t = \sin x$ gives

$$\int t^3\,dt = \frac{t^4}{4} + C = \boxed{\frac{\sin^4 x}{4} + C}.$$

By contrast, $\int \sin^2 x\,dx$ is easier after a trigonometric identity:

$$\sin^2 x = \frac{1-\cos 2x}{2}, \qquad \int \sin^2 x\,dx = \frac{x}{2} - \frac{\sin 2x}{4} + C.$$

The preliminary algebraic transformation is as important as the substitution.

Example 81.3.1 (A square root becomes a trigonometric identity)
For $a > 0$ and $|x| < a$, put $x = a\sin t$ with $t \in (-\pi/2, \pi/2)$. Then

$$\sqrt{a^2 - x^2} = a\cos t, \qquad dx = a\cos t\,dt,$$

so

$$\int \sqrt{a^2 - x^2}\,dx = a^2 \int \cos^2 t\,dt = \frac{a^2}{2}\,(t + \sin t\cos t) + C.$$

Returning to x gives

$$\boxed{\int \sqrt{a^2 - x^2}\,dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\arcsin\frac{x}{a} + C.}$$

One obtains related substitutions

$$\sqrt{a^2 + x^2} : x = a\tan t, \qquad \sqrt{x^2 - a^2} : x = a\sec t$$

on intervals where the signs and inverse branches are chosen consistently.

§81.4 Integration by parts is the product rule in reverse

For differentiable u, v, integrate $(uv)' = u'v + uv'$ to obtain

$$\boxed{\int u\,dv = uv - \int v\,du.}$$

For $\int \ln x\,dx$ on $x > 0$, choosing $u = \ln x$ and $dv = dx$ turns the logarithm into $1/x$:

$$\int \ln x\,dx = x\ln x - \int 1\,dx = x\ln x - x + C.$$

The useful choice is the one that makes the *new* integral simpler, not necessarily the one that makes one factor shorter.

Repeated integration by parts yields a finite identity. If u has $n+1$ derivatives and $v^{(n+1)}$ is the other factor, then

$$\int u\, v^{(n+1)}\, dx = uv^{(n)} - u'v^{(n-1)} + \cdots + (-1)^n u^{(n)} v + (-1)^{n+1} \int u^{(n+1)} v\, dx.$$

For a polynomial $u = P$ of degree n, the last integral vanishes. Thus

$$\int P(x) e^x\, dx = e^x (P - P' + P'' - \cdots + (-1)^n P^{(n)}) + C.$$

In particular, for $P = 2x^3 + 3x^2 + 4x + 5$,

$$P' = 6x^2 + 6x + 4, \quad P'' = 12x + 6, \quad P''' = 12,$$

and

$$\boxed{\int (2x^3 + 3x^2 + 4x + 5) e^x\, dx = e^x (2x^3 - 3x^2 + 10x - 5) + C.}$$

Differentiate this answer once to check every coefficient. Repeated parts with $P(x)\sin x$ or $P(x)\cos x$ works similarly, but the successive antiderivatives cycle through sines and cosines.

§81.5 The same calculus rules survive in differential forms

Substitution is the one-dimensional instance of the pullback of a 1-form:

$$\rho^*(f(x)\, dx) = f(\rho(t))\rho'(t)\, dt.$$

This language records the derivative factor automatically. The higher-dimensional change-of-variables formula has an absolute Jacobian determinant for ordinary area,

$$\iint_{\rho(U)} f(x,y)\, dx\, dy = \iint_U f(\rho(u,v))\, |\det D\rho(u,v)|\, du\, dv$$

when ρ is a suitable one-to-one regular change of coordinates. Oriented differential forms use the *signed* determinant instead. The distinction avoids confusing substitution with an arbitrary change of labels.

The definite form of integration by parts also keeps track of boundary information:

$$\int_a^b (Du) v\, dx = [uv]_a^b - \int_a^b u(Dv)\, dx, \qquad D = \frac{d}{dx}.$$

If the boundary term vanishes on a chosen domain of functions, D is formally skew-adjoint with respect to the real L^2 pairing. Without boundary conditions it would be misleading to write simply $D^* = -D$. This is the first elementary example of a boundary term controlling an adjoint differential operator.

Finally, partial fractions have their complex-analytic interpretation: the coefficient of $(z-a)^{-1}$ in the Laurent principal part is the residue. When f is meromorphic inside a positively oriented closed contour γ and has no poles on γ,

$$\oint_\gamma f(z)\, dz = 2\pi i \sum_{a \text{ inside } \gamma} \operatorname{Res}(f, a).$$

This gives the precise link from the algebra of rational denominators to complex integration. The three elementary methods have different mechanisms—factorization, the product rule, and the chain rule—and these mechanisms remain identifiable in more advanced mathematics.

Vectors, Ordered Fields, and the First Axioms of Real Numbers

N-20220710

§82.1 Vectors express geometry using a field

A vector in the plane may be written

$$\vec{a} = (x, y) = x\mathbf{i} + y\mathbf{j}.$$

Addition and scalar multiplication are coordinatewise:

$$(x, y) + (z, w) = (x + z, y + w), \qquad \lambda(x, y) = (\lambda x, \lambda y).$$

The length and dot product are

$$|(x, y)| = \sqrt{x^2 + y^2}, \qquad (x, y) \cdot (z, w) = xz + yw.$$

Thus the angle between two nonzero vectors satisfies

$$\cos\theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|\,|\vec{b}|} = \frac{xz + yw}{\sqrt{x^2 + y^2}\sqrt{z^2 + w^2}}.$$

If

$$\vec{x} = (\cos\alpha, \sin\alpha), \qquad \vec{y} = (\cos\beta, \sin\beta),$$

then

$$\cos\theta = \cos\alpha\cos\beta + \sin\alpha\sin\beta = \cos(\alpha - \beta).$$

This gives a vector proof of the cosine subtraction formula. The coordinate expressions depend on the field of scalars: the dot product and positive length require the ordered real-number setting, whereas the vector-space axioms themselves make sense over any field. This distinction motivates separating linear algebra from the extra order and completeness assumptions of Euclidean geometry.

A plane vector (x, y) packages a displacement into two real numbers. Vector addition corresponds to composing displacements; scalar multiplication corresponds to stretching. The algebraic laws of vectors are therefore not arbitrary axioms. They express basic geometric operations in coordinate form.

§82.2 Field laws make coordinate calculations consistent

Axiom 82.2.1 (Field axioms)

A field is a set F with operations $+$ and $\cdot$ such that $(F, +)$ is an abelian group, $(F \setminus \{0\}, \cdot)$ is an abelian group, and multiplication distributes over addition.

The axioms below specify the algebraic rules shared by $\mathbb{Q}$, $\mathbb{R}$ and other fields; they do not yet imply that square roots exist or that Cauchy sequences converge. Their role here follows the structure of MIT 18.100A *Real Analysis*, lectures on the characterization of the reals and the Archimedean property.[1] The real numbers form a field. Its axioms are:

[1] MIT 18.100A, *Real Analysis*, lecture notes.

(F1) $x + (y + z) = (x + y) + z$;

(F2) $x + y = y + x$;

(F3) there exists 0 such that $0 + x = x$;

(F4) for each x, there is $-x$ such that $x + (-x) = 0$;

(F5) $x(yz) = (xy)z$;

(F6) $xy = yx$;

(F7) there exists $1 \neq 0$ such that $1x = x$;

(F8) for $x \neq 0$, there is x^{-1} such that $xx^{-1} = 1$;

(F9) $x(y + z) = xy + xz$.

The note emphasizes that a "space" in mathematics is usually a set equipped with extra structure. A field is not merely a set; it is a set equipped with addition, multiplication, identities, inverses, and compatibility laws.

§82.3 Ordered fields: what algebra and order can establish

The order axioms include:

(O1) Transitivity: if $x \leq y$ and $y \leq z$, then $x \leq z$.

(O2) Antisymmetry: if $x \leq y$ and $y \leq x$, then $x = y$.

(O3) Totality: for all x, y, either $x \leq y$ or $y \leq x$.

(O4) Compatibility with addition: $x \leq y \Rightarrow x + z \leq y + z$.

(O5) Compatibility with multiplication: $0 \leq x$ and $0 \leq y \Rightarrow 0 \leq xy$.

The Archimedean axiom says that for every $x > 0$ and every y, there exists a positive integer n such that

$$nx \geq y.$$

A typical consequence is that every interval $(0, a)$ with $a > 0$ is nonempty: choose n so large that $n > a^{-1}$, then $1/n < a$.

Proposition 82.3.1 (Consequences of an ordered field)
In an ordered field, addition preserves order and multiplication by a positive element preserves order. In particular, squares are nonnegative:

$$x^2 \geq 0.$$

Representative exercises include:

(i) $x \geq 0$ iff $-x \leq 0$, and $y > 1$ implies $0 < 1/y < 1$.

(ii) $1 > 0$ and $-1 \neq 1$.

(iii) If $x \leq y$ and $a \leq 0$, then $ax \geq ay$.

(iv) If $a \le b$ and $x \le y$, then $a + x \le b + y$, with equality iff $a = b$ and $x = y$.

(v) If $a < x$ for every $a < x$, then $x \le y$; this is a typical contradiction argument using an interval between two distinct real numbers.

(vi) $x^2 \ge 0$ for every real x.

(vii) If $a^2 < a$, then $0 < a < 1$.

(viii) If nonzero x, y have the same sign, then

$$(x + y)^2 > (x - y)^2.$$

The rationals satisfy the ordered-field axioms but lack the least-upper-bound property. A concrete witness is the set $S = \{q \in \mathbb{Q} : q \ge 0,\ q^2 < 2\}$. If it had a rational supremum r, the order density of $\mathbb{Q}$ and continuity of $x \mapsto x^2$ over the reals would force $r^2 = 2$, which is impossible for rational r. Hence an ordered field supports algebra and order without necessarily supporting the analytic limits needed for completeness.

§82.4 Embedding rational numbers into an Archimedean field

The note then proves that $\mathbb{R}$ contains a copy of $\mathbb{Q}$. Define for integers n

$$\iota(n) = \underbrace{1 + \cdots + 1}_{n \text{ times}}$$

for $n > 0$, $\iota(0) = 0$, and $\iota(n) = -\iota(-n)$ for $n < 0$. Then define

$$\iota\left(\frac{p}{q}\right) = \frac{\iota(p)}{\iota(q)}.$$

One must check that this is well-defined: if $p/q = s/t$, then $pt = qs$, so

$$\iota(p)\iota(t) = \iota(q)\iota(s),$$

and therefore

$$\frac{\iota(p)}{\iota(q)} = \frac{\iota(s)}{\iota(t)}.$$

This embedding preserves addition, multiplication, and order.

In the real numbers, the Archimedean property says no positive infinitesimal exists inside $\mathbb{R}$: for every x, some integer n satisfies $n > x$. This leads to density of rationals and helps connect arithmetic approximations with the geometry of the line.

§82.5 Completeness begins where rational arithmetic stops

The final part introduces the nested interval axiom. If

$$I_n = [a_n, b_n]$$

is a descending sequence of closed intervals with real endpoints, then

$$I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$$

implies

$$\bigcap_{n\geq 1} I_n \neq \varnothing.$$

The real closed intervals are compact, so nested nonempty closed intervals have nonempty intersection. When the lengths shrink to zero, the intersection is a single point. In $\mathbb{Q}$ with its inherited order and metric, there is a counterexample: take nested rational-endpoint brackets $[a_n, b_n] \cap \mathbb{Q}$ of $\sqrt{2}$ whose lengths tend to zero. Every bracket contains rational points, yet their common intersection inside $\mathbb{Q}$ is empty. This distinguishes the completeness property from the ordinary order axioms, rather than merely listing it as another axiom. The ordered field of reals can be characterized by this additional completeness principle.

The supremum property, nested intervals, Cauchy completeness, and Dedekind completeness are equivalent ways of saying that the real line has no gaps. This is the feature that turns ordered algebra into analysis.

Completeness, Supremum, Dedekind Cuts, and Metric Spaces

N-20220713

§83.1 Order, bounds, and the least-upper-bound property

A simple but important fact is that for every positive real A, there is a positive real $M > A$, and for every positive real a, there is a positive real $\varepsilon < a$. For instance,

$$M = A + 1, \qquad \varepsilon = \frac{a}{2}.$$

This is the first appearance of the ε-language of analysis. Our treatment follows the ordered-field and metric-completion development in the MIT 18.100A notes and Jiří Lebl's *Basic Analysis I*; see especially the discussion of supremum and Cauchy completeness.[1]

We think of $\mathbb{R}$ as a line, where order is represented by left and right. This picture is helpful, but the note emphasizes that all conclusions must ultimately come from axioms and rigorous reasoning, not from the drawing itself.

A set $X \subseteq \mathbb{R}$ is bounded above if there exists $A \in \mathbb{R}$ such that

$$x \le A \quad \text{for all } x \in X.$$

Such an A is an upper bound. Similarly, X is bounded below if there exists a such that

$$a \le x \quad \text{for all } x \in X.$$

The least upper bound property says: if $X \subseteq \mathbb{R}$ is nonempty and bounded above, then the set of upper bounds has a least element. This least upper bound is denoted

$$\sup X.$$

Equivalently, $M = \sup X$ iff:

(i) $x \le M$ for all $x \in X$;

(ii) for every $\varepsilon > 0$, there exists $x \in X$ such that

$$x > M - \varepsilon.$$

The note sketches a proof of the supremum property from the nested interval principle by bisecting intervals and selecting the half that still contains upper bounds or points of the set.

The dual notion is the infimum:

$$\inf X.$$

One always has

$$\inf X \le \sup X$$

[1] MIT 18.100A, *Real Analysis*, lecture notes.

for a nonempty bounded set.

The supremum of a set $A \subseteq \mathbb{R}$ is not merely a maximum. It is the least number s such that every element of A is at most s. A maximum must belong to A; a supremum need not.

For example,

$$\sup(0,1) = 1,$$

but $(0,1)$ has no maximum. Completeness has a concrete obstruction over $\mathbb{Q}$. The set $S = \{q \in \mathbb{Q} : q > 0,\ q^2 < 2\}$ is nonempty and bounded above but has no supremum in $\mathbb{Q}$; any putative rational boundary would satisfy $r^2 = 2$, contradicting the elementary parity argument that $\sqrt{2}$ is irrational. Therefore the ordered-field axioms alone do not imply the supremum property.

§83.2 Completeness and nested-interval bisection

Let $X \subseteq \mathbb{R}$ be nonempty and bounded above. One can construct nested intervals

$$I_n = [a_n, b_n]$$

so that a_n is not an upper bound of X, while b_n is an upper bound, and

$$b_n - a_n \leq 2^{-n}.$$

The nested interval axiom gives a point

$$M \in \bigcap_{n=1}^{\infty} I_n.$$

At each bisection, retain the half whose lower endpoint is not an upper bound and whose upper endpoint is an upper bound; the interval length halves. The nested-interval property then supplies M. The shrinking length forces M to be the least upper bound of X. If M were not an upper bound, some $x \in X$ would lie above it; for large n, the right endpoint b_n would be too close to M, contradiction. If M were not least, a smaller upper bound would contradict the construction of the a_n's.

The supremum property says every nonempty bounded-above set has a least upper bound. This is the order-theoretic heart of the real line. It is what makes monotone convergence and intermediate-value reasoning possible.

§83.3 Dedekind cuts and the order completion

A Dedekind cut is a subset $X \subset \mathbb{Q}$ such that:

(i) $X \neq \varnothing$ and $\mathbb{Q} \setminus X \neq \varnothing$;

(ii) if $x \in X$ and $y < x$, then $y \in X$;

(iii) X has no greatest element.

A rational number p/q defines the cut

$$X_{p/q} = \{x \in \mathbb{Q} : x < p/q\}.$$

An irrational such as $\sqrt{2}$ is represented by

$$X_{\sqrt{2}} = \{x \in \mathbb{Q}_{>0} : x^2 < 2\} \cup \mathbb{Q}_{\leq 0}.$$

This is an actual lower cut: it is nonempty, proper, downward closed and has no greatest rational element. The last property uses density of the rationals. The cut does not come from any rational endpoint, because 2 is not a square in $\mathbb{Q}$. Thus real numbers can be constructed from rational cuts.

A Dedekind cut separates rationals into a lower part and an upper part with no gap. Irrational numbers appear as cuts not produced by a rational boundary. Completeness says that every such cut corresponds to a real number.

This is the philosophical point of constructing $\mathbb{R}$: the real line is obtained by filling all rational gaps.

Dedekind cuts construct reals by filling order gaps; Cauchy completion constructs reals by adding missing limits of Cauchy sequences. The two constructions produce isomorphic complete ordered fields, but they emphasize different structures: order versus metric.

§83.4 Metric spaces, dense subspaces, and Cauchy completion

A metric space is a set X equipped with a function

$$d : X \times X \to \mathbb{R}_{\geq 0}$$

satisfying:

(i) $d(x, y) = 0$ iff $x = y$;

(ii) $d(x, y) = d(y, x)$;

(iii) $d(x, z) \leq d(x, y) + d(y, z)$.

The usual metric on $\mathbb{R}$ is

$$d(x, y) = |x - y|.$$

On $\mathbb{R}^q$, common metrics are:

$$d_2(x, y) = \left(\sum_{n=1}^{q} |x_n - y_n|^2 \right)^{1/2},$$

$$d_1(x, y) = \sum_{n=1}^{q} |x_n - y_n|,$$

and

$$d_\infty(x, y) = \max_{1 \leq n \leq q} |x_n - y_n|.$$

The note remarks that d_2 requires Cauchy–Schwarz for the triangle inequality, d_1 is the Manhattan metric, and d_∞ is often called the Chebyshev metric.

A final example is the discrete metric:

$$d(x, y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$$

In the discrete metric, open balls are especially simple:

$$B_r(x) = \begin{cases} \{x\}, & 0 < r \leq 1, \\ X, & r > 1. \end{cases}$$

Boundary points are not included in an open ball, which explains why the case $r = 1$ gives $\{x\}$, not all of X.

If $Y \subseteq X$ and X has metric d, then Y inherits the subspace metric

$$d_Y(y_1, y_2) = d(y_1, y_2).$$

A subset $Y \subseteq X$ is dense if for every $x \in X$ and every $\varepsilon > 0$, there exists $y \in Y$ such that

$$d(x, y) < \varepsilon.$$

This formalizes the idea that Y comes arbitrarily close to every point of X.

A metric space can be completed by adding equivalence classes of Cauchy sequences. Two Cauchy sequences are equivalent if their distance tends to zero. The real numbers can be constructed as the completion of $\mathbb{Q}$ under the usual metric. For example, set $q_n = \lfloor 10^n \sqrt{2} \rfloor / 10^n$. Each q_n is rational and $|q_n - \sqrt{2}| < 10^{-n}$; hence (q_n) is Cauchy in $\mathbb{Q}$, but has no rational limit. The completion adjoins its equivalence class. Two Cauchy sequences represent the same real number precisely when their termwise difference tends to zero. This turns the philosophical claim about "filling gaps" into an explicit construction.

Contour Integrals: Parametrization, Exactness, Winding, and Residues

N-20220813

§84.1 Contour Integrals and Tangents

A contour is a piecewise C^1 path

$$\gamma : [a, b] \to \mathbb{C}.$$

For a continuous function f on the image of γ, define

$$\int_\gamma f(z)\,dz = \int_a^b f(\gamma(t))\gamma'(t)\,dt.$$

The right-hand side is an ordinary complex-valued Riemann integral.

Let

$$a = t_0 < t_1 < \cdots < t_n = b$$

and choose $\tau_j \in [t_{j-1}, t_j]$. The Riemann sum is

$$\sum_{j=1}^{n} f(\gamma(\tau_j))(\gamma(t_j) - \gamma(t_{j-1})).$$

Each directed increment is multiplied by f, hence scaled and rotated before the increments are added. A contour integral is not an area under a graph; it is a limit of transformed tangent vectors.

The basic estimate follows immediately:

$$\left|\int_\gamma f(z)\,dz\right| \le \int_a^b |f(\gamma(t))|\,|\gamma'(t)|\,dt \le L(\gamma)\max_{z\in\gamma}|f(z)|,$$

where

$$L(\gamma) = \int_a^b |\gamma'(t)|\,dt.$$

This ML-estimate is the main error-control tool for contour deformation and limiting arguments.

§84.2 Orientation, concatenation, and reparametrization

The reverse contour is

$$\overline{\gamma}(t) = \gamma(a + b - t).$$

Substitution gives

$$\boxed{\int_{\overline{\gamma}} f(z)\,dz = -\int_\gamma f(z)\,dz.}$$

If $\gamma_1(b_1) = \gamma_2(a_2)$, the concatenated contour $\gamma_1 * \gamma_2$ traverses the first and then the second. Splitting the parameter interval yields

$$\boxed{\int_{\gamma_1 * \gamma_2} f(z)\,dz = \int_{\gamma_1} f(z)\,dz + \int_{\gamma_2} f(z)\,dz.}$$

Let $\phi : [c, d] \to [a, b]$ be a piecewise C^1, increasing surjection with $\phi(c) = a$, $\phi(d) = b$. Then

$$\begin{aligned}\int_{\gamma\circ\phi} f(z)\,dz &= \int_c^d f(\gamma(\phi(s)))\gamma'(\phi(s))\phi'(s)\,ds \\ &= \int_a^b f(\gamma(t))\gamma'(t)\,dt.\end{aligned}$$

Thus the integral is unchanged by orientation-preserving reparametrization. If ϕ reverses orientation, the sign changes.

For the unit circle, all three parametrizations

$$e^{it}, \quad 0 \le t \le 2\pi; \qquad e^{2\pi it}, \quad 0 \le t \le 1; \qquad e^{2\pi it^5}, \quad 0 \le t \le 1$$

give the same oriented contour integral.

§84.3 The complex integral is the pullback of a one-form

Write

$$z = x + iy, \qquad f(z) = u(x, y) + iv(x, y), \qquad dz = dx + i\,dy.$$

Then

$$\begin{aligned}f(z)\,dz &= (u + iv)(dx + i\,dy) \\ &= (u\,dx - v\,dy) + i(v\,dx + u\,dy).\end{aligned}$$

Therefore

$$\int_\gamma f(z)\,dz = \int_\gamma (u\,dx - v\,dy) + i\int_\gamma (v\,dx + u\,dy).$$

A complex contour integral is two real line integrals coupled by the complex multiplication rule.

In differential-form notation, $f(z)\,dz$ is a complex-valued one-form. The parametrized formula is exactly pullback:

$$\int_\gamma f(z)\,dz = \int_{[a,b]} \gamma^*(f(z)\,dz).$$

This explains reparametrization invariance: integration is applied to the pulled-back one-form on the parameter interval.

§84.4 Primitives are equivalent to path independence

Let $D \subseteq \mathbb{C}$ be a domain and let $f : D \to \mathbb{C}$ be continuous.

Theorem 84.4.1

The following are equivalent.

(i) There is a holomorphic function $F : D \to \mathbb{C}$ with $F' = f$.

(ii) The integral of f depends only on the endpoints of the contour.

(iii) Every closed contour γ in D satisfies

$$\int_\gamma f(z)\,dz = 0.$$

Proof. If $F' = f$, the chain rule gives

$$\frac{d}{dt}F(\gamma(t)) = f(\gamma(t))\gamma'(t).$$

Hence

$$\int_\gamma f(z)\,dz = F(\gamma(b)) - F(\gamma(a)),$$

which proves (i)⇒(ii), and (ii)⇒(iii) is immediate.

Assume (iii). Fix $z_0 \in D$, and for $z \in D$ define

$$F(z) = \int_{\gamma_{z_0,z}} f(\zeta)\,d\zeta,$$

where $\gamma_{z_0,z}$ is any contour from z_0 to z. If two contours are chosen, following one and reversing the other gives a closed contour, so (iii) makes F well defined.

For small h, use the same contour to z and then the line segment from z to $z+h$:

$$\frac{F(z+h)-F(z)}{h} = \int_0^1 f(z+th)\,dt.$$

By continuity, the right-hand side tends to $f(z)$. Thus $F'(z) = f(z)$. □

The topology of the domain enters in condition (iii): local primitives may exist while a global primitive fails because closed loops can carry nonzero periods.

§84.5 The Cauchy–Goursat Theorem

Suppose $f = u + iv$ has continuous first partial derivatives on a region containing a positively oriented simple closed contour $\gamma = \partial\Omega$. Using the decomposition of $f\,dz$ and Green's theorem,

$$\begin{aligned}\operatorname{Re}\int_\gamma f(z)\,dz &= \int_\gamma u\,dx - v\,dy \\ &= \iint_\Omega (-v_x - u_y)\,dx\,dy,\end{aligned}$$

and

$$\begin{aligned}\operatorname{Im}\int_\gamma f(z)\,dz &= \int_\gamma v\,dx + u\,dy \\ &= \iint_\Omega (u_x - v_y)\,dx\,dy.\end{aligned}$$

The Cauchy–Riemann equations

$$u_x = v_y, \qquad u_y = -v_x$$

make both integrands zero. Therefore

$$\int_\gamma f(z)\,dz = 0.$$

Goursat's theorem removes the assumption of continuous partial derivatives: complex differentiability alone is enough. On a simply connected domain, every closed contour is deformable through the domain to a point, and Cauchy–Goursat implies

$$\int_\gamma f(z)\,dz = 0.$$

By the preceding theorem, every holomorphic function on a simply connected domain has a primitive.

The conclusion fails when the domain contains a hole around a singularity. The model obstruction is $1/z$ on $\mathbb{C}^*$.

§84.6 Powers on a circle isolate one Fourier mode

For the circle traversed k times,

$$\gamma_k(t) = e^{ikt}, \qquad 0 \le t \le 2\pi,$$

one has

$$\begin{aligned}\int_{\gamma_k} z^m\,dz &= \int_0^{2\pi} e^{imkt} ike^{ikt}\,dt \\ &= ik\int_0^{2\pi} e^{ik(m+1)t}\,dt.\end{aligned}$$

If $m \neq -1$, the oscillation completes an integer number of periods and cancels. If $m = -1$, the integrand is the constant ik. Hence

$$\boxed{\int_{\gamma_k} z^m\,dz = \begin{cases} 2\pi ik, & m = -1, \\ 0, & m \neq -1. \end{cases}}$$

The exceptional exponent -1 is the coefficient that survives contour averaging. This is the local algebra behind residues.

§84.7 Winding number is the period of $dz/(z-a)$

Let γ be a closed contour avoiding a. Define

$$\operatorname{Ind}(\gamma, a) = \frac{1}{2\pi i}\int_\gamma \frac{dz}{z-a}.$$

For a circle centered at a traversed k times, the preceding computation gives

$$\operatorname{Ind}(\gamma, a) = k.$$

In general the index is an integer, equal to the endpoint change of a lifted argument of $\gamma(t) - a$.

It is invariant under homotopies that avoid a. For a smooth homotopy $\Gamma(t,s)$, differentiate under the integral sign:

$$\begin{aligned}\frac{d}{ds}\int_0^1 \frac{\partial_t\Gamma(t,s)}{\Gamma(t,s)-a}\,dt &= \int_0^1 \partial_t\left(\frac{\partial_s\Gamma(t,s)}{\Gamma(t,s)-a}\right)dt\\ &= 0,\end{aligned}$$

because the loop has the same initial and terminal point. Approximation or covering-space lifting gives the general continuous-homotopy statement.

Thus the integral depends on the homotopy class of the loop in $\mathbb{C}\setminus\{a\}$, not on its detailed shape.

§84.8 Closed need not mean exact on a punctured domain

On $\mathbb{R}^2\setminus\{0\}$, consider the real one-form

$$\alpha = \frac{1}{2\pi}\frac{-y\,dx + x\,dy}{x^2+y^2}.$$

A direct calculation gives

$$d\alpha = 0.$$

So α is closed. On the unit circle $x=\cos t$, $y=\sin t$,

$$\alpha = \frac{dt}{2\pi},$$

and therefore

$$\int_{S^1}\alpha = 1.$$

If $\alpha = dF$ were exact, every closed-loop integral would vanish. Hence α is closed but not exact.

This period represents the generator of

$$H^1_{\mathrm{dR}}(\mathbb{C}^*)\cong\mathbb{R}.$$

The complex form dz/z contains the same obstruction:

$$\frac{1}{2\pi i}\int_\gamma \frac{dz}{z} = \operatorname{Ind}(\gamma,0).$$

The failure of a global logarithm on $\mathbb{C}^*$ is the same phenomenon. Locally,

$$d(\log z) = \frac{dz}{z},$$

but no single-valued logarithm exists around a loop of nonzero winding.

§84.9 Recovering Values from the Boundary

Let f be holomorphic on a neighborhood of a positively oriented simple closed contour γ and its interior, and let z lie inside. Then

$$\boxed{f(z) = \frac{1}{2\pi i}\int_\gamma \frac{f(\zeta)}{\zeta - z}\,d\zeta.}$$

To see the mechanism, write

$$\frac{f(\zeta)}{\zeta - z} = \frac{f(\zeta) - f(z)}{\zeta - z} + \frac{f(z)}{\zeta - z}.$$

The first term has a removable singularity at $\zeta = z$, with value $f'(z)$, so its closed-contour integral vanishes. The second term contributes

$$f(z) \int_\gamma \frac{d\zeta}{\zeta - z} = 2\pi i \, f(z) \operatorname{Ind}(\gamma, z).$$

For a simple positively oriented contour containing z, the index is 1.

Differentiating under the integral sign gives

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_\gamma \frac{f(\zeta)}{(\zeta - z)^{n+1}} \, d\zeta.$$

For example,

$$\int_{|\zeta|=2} \frac{e^\zeta}{(\zeta - 1)^3} \, d\zeta = \frac{2\pi i}{2!} e = \pi i e.$$

§84.10 Residues and Global Integrals

If f has an isolated singularity at a, its Laurent expansion is

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z - a)^n.$$

The residue is

$$\operatorname{Res}(f, a) = c_{-1}.$$

All other powers integrate to zero around a small circle, while $(z - a)^{-1}$ contributes $2\pi i$. Therefore

$$\int_{|z-a|=r} f(z) \, dz = 2\pi i \operatorname{Res}(f, a).$$

For a contour γ avoiding finitely many poles $a_1, \ldots, a_N$, the residue theorem is

$$\boxed{\int_\gamma f(z) \, dz = 2\pi i \sum_{j=1}^{N} \operatorname{Ind}(\gamma, a_j) \operatorname{Res}(f, a_j).}$$

Local Laurent coefficients are paired with global winding numbers.

Consider

$$f(z) = \frac{z + 1}{z(z - 1)}.$$

The residues are

$$\operatorname{Res}(f, 0) = -1, \qquad \operatorname{Res}(f, 1) = 2.$$

Hence

$$\int_{|z|=2} f(z) \, dz = 2\pi i(-1 + 2) = 2\pi i.$$

For the smaller contour $|z| = 1/2$, only the pole at 0 is enclosed, so

$$\int_{|z|=1/2} f(z) \, dz = -2\pi i.$$

The integrand has not changed; the answer changes because the winding data relative to the poles has changed.

§84.11 Counting zeros without solving a polynomial

Residues can count zeros, not only measure poles. Suppose p is a polynomial with no zero on a counterclockwise simple closed contour γ. Near a zero a of multiplicity m,

$$p(z) = (z-a)^m h(z), \qquad h(a) \neq 0, \qquad \frac{p'(z)}{p(z)} = \frac{m}{z-a} + \frac{h'(z)}{h(z)}.$$

The logarithmic derivative therefore has residue m at this zero. Applying the residue theorem gives the *argument principle*:

$$\boxed{\frac{1}{2\pi i}\int_\gamma \frac{p'(z)}{p(z)}\,dz = \#\{\text{zeros inside } \gamma\},}$$

counted with multiplicity. Equivalently, this number is the winding number of the *image curve* $p\circ\gamma$ about the origin. The integral is well defined only if no zero lies on the contour.

The striking step is that one can often count zeros without evaluating that integral. If g and h are holomorphic inside and on γ, and

$$\boxed{|h(z)| < |g(z)| \quad (z \in \gamma),}$$

then the curves $g(\gamma(t))$ and $(g+h)(\gamma(t))$ have the same winding number around zero. To see why, consider the continuous deformation

$$H_s(z) = g(z) + s h(z), \qquad 0 \le s \le 1.$$

On the boundary,

$$|H_s(z)| \ge |g(z)| - s|h(z)| > 0.$$

Throughout the deformation the image contour avoids zero, and winding number is integer-valued and cannot change continuously. The argument principle implies that g and $g+h$ have the same number of zeros inside. This is Rouché's theorem, derived directly from the homotopy invariance already used for contour integrals.

For a worked problem, consider

$$p(z) = z^4 + 3z + 1.$$

On the smaller circle $|z| = 1/2$, choose $g(z) = 3z$ and $h(z) = z^4 + 1$. Then

$$|g(z)| = \frac{3}{2}, \qquad |h(z)| \le 1 + \frac{1}{16} = \frac{17}{16} < \frac{3}{2}.$$

The function g has exactly one zero there, so p has exactly *one zero in* $|z| < 1/2$. On $|z| = 2$, choose $g(z) = z^4$ and $h(z) = 3z + 1$. This time

$$|g(z)| = 16, \qquad |h(z)| \le 7 < 16.$$

Therefore p has exactly *four zeros in* $|z| < 2$, namely all its zeros. Subtracting the counts proves that the intervening annulus

$$\boxed{\tfrac{1}{2} < |z| < 2}$$

contains exactly three zeros. No quartic formula was needed.

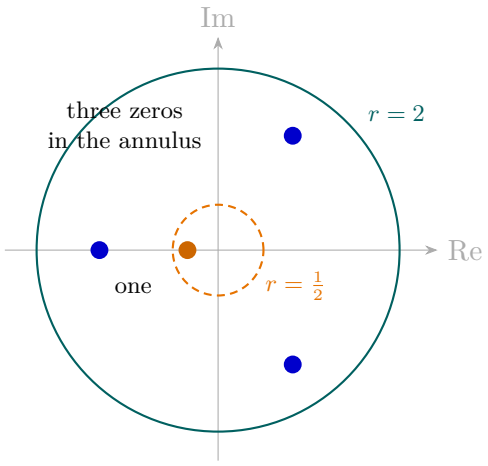


Figure 84.1: Approximate positions of the four zeros of $z^4 + 3z + 1$. The two root counts follow rigorously from boundary inequalities; the plotted positions are only a visual check.

We have followed a useful logical chain:

$$\begin{aligned} \text{contour integrals} &\longrightarrow \text{winding numbers} \\ &\longrightarrow \text{residues of } p'/p \\ &\longrightarrow \text{root counts stable under perturbation.} \end{aligned}$$

This is one reason analytic methods can tell us global facts about the solutions of algebraic equations without producing their individual roots.[1]

[1] For the argument principle and Rouché's theorem with contour examples, see Jeremy Orloff, MIT 18.04, Topic 11.

Trigonometric Limits, Derivatives, and Inverse Trigonometric Examples

N-20221107

§85.1 Radians are built into the derivative

The familiar formula

$$\frac{d}{dx}\sin x = \cos x$$

is true only when the variable is measured in radians. This is not a typographical convention. A derivative compares a small change in function value with a small change in the input, and radian measure is the angle coordinate whose small increment agrees with arc length on the unit circle.

To see what changes in degrees, write

$$\sin_{\text{deg}}(x) = \sin\left(\frac{\pi x}{180}\right),$$

where the sine on the right uses radians. The chain rule gives

$$\frac{d}{dx}\sin_{\text{deg}}(x) = \frac{\pi}{180}\cos\left(\frac{\pi x}{180}\right).$$

The extra factor is the derivative of the unit conversion. Radians are the unique constant multiple of degrees for which that factor becomes one.

The analytical reason is the limit

$$\lim_{h\to 0}\frac{\sin h}{h} = 1.$$

The rest of the trigonometric derivative table can be reconstructed from this limit, the addition formulas, and ordinary differentiation rules.

§85.2 The circle squeeze proves the first basic limit

Let $0 < h < \pi/2$. On the unit circle, compare the area of the inscribed triangle, the circular sector, and the tangent triangle. Their areas are respectively

$$\frac{1}{2}\sin h, \qquad \frac{1}{2}h, \qquad \frac{1}{2}\tan h.$$

Therefore

$$\sin h < h < \tan h.$$

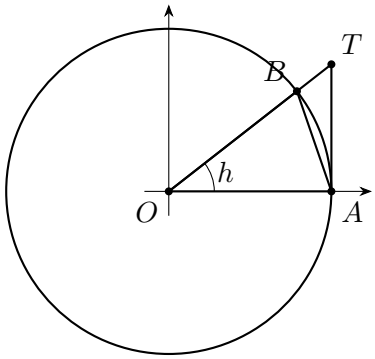


Figure 85.1: The area comparison gives $\sin h < h < \tan h$ for a positive acute angle measured in radians.

Since $\sin h > 0$, division gives

$$1 < \frac{h}{\sin h} < \frac{1}{\cos h},$$

or equivalently

$$\cos h < \frac{\sin h}{h} < 1.$$

As $h \to 0^+$, both outer terms tend to 1. Hence

$$\lim_{h \to 0^+} \frac{\sin h}{h} = 1.$$

The quotient is even because

$$\frac{\sin(-h)}{-h} = \frac{\sin h}{h},$$

so the same limit holds from the left:

$$\boxed{\lim_{h \to 0} \frac{\sin h}{h} = 1.}$$

The conclusion depends on radian measure at the area-of-sector step. In degrees, the sector area would be proportional to $\pi h/180$, and the limit would be $\pi/180$.

§85.3 The Second Trigonometric Limit

The identity

$$1 - \cos h = 2 \sin^2 \frac{h}{2}$$

gives

$$\frac{1 - \cos h}{h} = \frac{h}{2} \left(\frac{\sin(h/2)}{h/2} \right)^2.$$

The factor in parentheses tends to 1, while $h/2 \to 0$. Therefore

$$\boxed{\lim_{h \to 0} \frac{1 - \cos h}{h} = 0.}$$

Dividing by one more factor of h gives

$$\frac{1 - \cos h}{h^2} = \frac{1}{2} \left(\frac{\sin(h/2)}{h/2} \right)^2,$$

so

$$\boxed{\lim_{h\to 0}\frac{1-\cos h}{h^2}=\frac{1}{2}.}$$

These limits are often summarized as

$$\sin h \sim h, \qquad 1-\cos h \sim \frac{h^2}{2} \qquad (h\to 0).$$

The symbol $f \sim g$ means $f/g \to 1$. It can safely replace factors inside products and quotients when the surrounding expression respects the limit. It cannot be substituted blindly inside a cancellation.

For example,

$$\frac{\sin h - h}{h}$$

cannot be evaluated by replacing $\sin h$ with h, because the leading terms cancel and the next-order behavior matters. Small-angle equivalence records the first nonzero scale, not a full Taylor expansion.

§85.4 Sine and cosine emerge from their addition laws

Using

$$\sin(x+h)=\sin x\cos h+\cos x\sin h,$$

we obtain

$$\frac{\sin(x+h)-\sin x}{h}=\sin x\frac{\cos h-1}{h}+\cos x\frac{\sin h}{h}.$$

The two basic limits give

$$\lim_{h\to 0}\frac{\cos h-1}{h}=0, \qquad \lim_{h\to 0}\frac{\sin h}{h}=1.$$

Therefore

$$\boxed{(\sin x)'=\cos x.}$$

Similarly,

$$\cos(x+h)=\cos x\cos h-\sin x\sin h,$$

so

$$\frac{\cos(x+h)-\cos x}{h}=\cos x\frac{\cos h-1}{h}-\sin x\frac{\sin h}{h}.$$

Hence

$$\boxed{(\cos x)'=-\sin x.}$$

The derivation exhibits a tight dependency chain:

$$\text{radian geometry} \Longrightarrow \frac{\sin h}{h}\to 1 \Longrightarrow \frac{1-\cos h}{h}\to 0 \Longrightarrow \begin{cases}(\sin x)'=\cos x,\\ (\cos x)'=-\sin x.\end{cases}$$

The addition formulas are also required. A separate note compares their geometric, matrix, and analytic proofs; here they are part of the input to the differentiation argument.

Differentiating once more gives the coupled differential equations

$$(\sin x)''=-\sin x, \qquad (\cos x)''=-\cos x.$$

Thus sine and cosine are not merely periodic graphs. They are the two normalized solutions of

$$y'' + y = 0$$

with initial data

$$\cos 0 = 1, \quad \cos' 0 = 0, \qquad \sin 0 = 0, \quad \sin' 0 = 1.$$

§85.5 Derivatives without Memorization

Once sine and cosine are known, quotient and reciprocal rules produce the remaining functions. Where $\cos x \neq 0$,

$$\tan x = \frac{\sin x}{\cos x},$$

so

$$(\tan x)' = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \sec^2 x.$$

Where $\sin x \neq 0$,

$$(\cot x)' = \left(\frac{\cos x}{\sin x}\right)' = -\csc^2 x.$$

The reciprocal rules give

$$(\sec x)' = \left(\frac{1}{\cos x}\right)' = \sec x \tan x,$$

$$(\csc x)' = \left(\frac{1}{\sin x}\right)' = -\csc x \cot x.$$

The domains belong to the formulas:

Function	Domain	Derivative	Excluded points
$\sin x$	$\mathbb{R}$	$\cos x$	none
$\cos x$	$\mathbb{R}$	$-\sin x$	none
$\tan x$	$\cos x \neq 0$	$\sec^2 x$	$\pi/2 + k\pi$
$\cot x$	$\sin x \neq 0$	$-\csc^2 x$	$k\pi$
$\sec x$	$\cos x \neq 0$	$\sec x \tan x$	$\pi/2 + k\pi$
$\csc x$	$\sin x \neq 0$	$-\csc x \cot x$	$k\pi$

A derivative formula does not repair a missing function value. At a pole of tangent or secant, the function itself is undefined, so there is no ordinary real derivative to discuss.

§85.6 Inverse functions require branches

A periodic function cannot have a global inverse on all of $\mathbb{R}$. Inverse trigonometric functions are obtained by restricting to intervals on which the original function is one-to-one:

Inverse	Domain	Principal-value range	Derivative domain
$\arcsin x$	$[-1, 1]$	$[-\pi/2, \pi/2]$	$(-1, 1)$
$\arccos x$	$[-1, 1]$	$[0, \pi]$	$(-1, 1)$
$\arctan x$	$\mathbb{R}$	$(-\pi/2, \pi/2)$	$\mathbb{R}$

Let

$$y = \arcsin x.$$

Then

$$x = \sin y, \qquad y \in [-\pi/2, \pi/2].$$

For $|x| < 1$, the inverse function theorem applies because

$$\cos y \neq 0.$$

Differentiating gives

$$1 = \cos y \, y'.$$

On the chosen branch, $\cos y \geq 0$, and

$$\cos y = \sqrt{1 - \sin^2 y} = \sqrt{1 - x^2}.$$

Therefore

$$\boxed{(\arcsin x)' = \frac{1}{\sqrt{1 - x^2}}.}$$

The positive square root is forced by the principal range. Without the branch information, the sign would be ambiguous.

For $y = \arccos x$, the principal branch lies in $[0, \pi]$, where $\sin y \geq 0$. Differentiating $x = \cos y$ gives

$$\boxed{(\arccos x)' = -\frac{1}{\sqrt{1 - x^2}}.}$$

For $y = \arctan x$,

$$x = \tan y, \qquad y \in (-\pi/2, \pi/2),$$

so

$$1 = \sec^2 y \, y'$$

and

$$\boxed{(\arctan x)' = \frac{1}{1 + x^2}.}$$

§85.7 Endpoint blow-up is a failure of local invertibility

At $x = 1$, the derivative formula for $\arcsin x$ diverges. This is not an algebraic accident. The original sine curve has a horizontal tangent at $y = \pi/2$:

$$\cos \frac{\pi}{2} = 0.$$

The inverse function theorem fails precisely when the derivative of the original map vanishes.

The local shape can be quantified. Put

$$y = \frac{\pi}{2} - t, \qquad t \to 0^+.$$

Then

$$\sin y = \cos t = 1 - \frac{t^2}{2} + o(t^2).$$

If $x = \sin y$, this gives

$$1 - x \sim \frac{t^2}{2}.$$

Therefore

$$\boxed{\frac{\pi}{2} - \arcsin x \sim \sqrt{2(1-x)} \qquad (x \to 1^-).}$$

Differentiating the square-root model predicts the vertical tangent:

$$(\arcsin x)' \sim \frac{1}{\sqrt{2(1-x)}}.$$

The same phenomenon occurs near -1.

The square root in this asymptotic is also what appears when the inverse is continued to complex arguments. Starting from

$$z = \sin w = \frac{e^{iw} - e^{-iw}}{2i},$$

set $u = e^{iw}$. Multiplying by $2iu$ gives the quadratic equation

$$u^2 - 2izu - 1 = 0.$$

Hence

$$u = iz \pm \sqrt{1 - z^2},$$

and solving for w gives a local inverse of the form

$$\boxed{\arcsin z = -i \operatorname{Log}\left(iz + \sqrt{1 - z^2}\right).}$$

Neither the logarithm nor the square root has a single-valued inverse on all of $\mathbb{C} \setminus \{0\}$, so this formula requires compatible branch choices. The points $z = \pm 1$ are special because the two roots of the quadratic merge there:

$$1 - z^2 = 0.$$

They are square-root branch points, and the real vertical tangents are their visible restriction to the interval $[-1, 1]$. Differentiating on any consistent local branch recovers

$$\frac{d}{dz} \arcsin z = \frac{1}{\sqrt{1 - z^2}},$$

with the sign determined by the chosen branch. Thus the real endpoint blow-up, the inverse-function theorem, and the complex branch structure are three descriptions of the same local degeneracy.

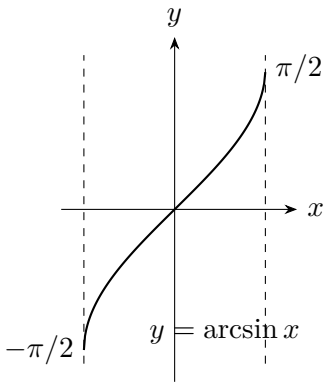


Figure 85.2: The principal inverse has vertical tangents at $x = \pm 1$, where the derivative of sine on the corresponding branch vanishes.

§85.8 Three examples that use different ideas

First consider the composite expression

$$f(x) = \arcsin(2x - x^2).$$

Its real domain is determined before differentiation:

$$-1 \le 2x - x^2 \le 1.$$

Since

$$2x - x^2 = 1 - (x-1)^2 \le 1,$$

the lower inequality gives

$$(x-1)^2 \le 2.$$

Thus

$$x \in [1 - \sqrt{2}, 1 + \sqrt{2}].$$

At interior points where $|2x - x^2| < 1$, the chain rule gives

$$f'(x) = \frac{2 - 2x}{\sqrt{1 - (2x - x^2)^2}}.$$

The apparent zero-over-zero at $x = 1$ should be simplified from the original function or treated by a local limit; blindly substituting into the unsimplified derivative formula hides the cusp-like branch behavior.

For an implicit example, let

$$\sin(x + y) = x.$$

Differentiating along a differentiable solution curve gives

$$\cos(x + y)(1 + y') = 1.$$

Where $\cos(x + y) \neq 0$,

$$y' = \sec(x + y) - 1.$$

The excluded points are exactly where the implicit function theorem cannot solve locally for y as a function of x.

For a parametric example, take

$$x(t) = \cos^3 t, \qquad y(t) = \sin^3 t.$$

Where $dx/dt \neq 0$,

$$\frac{dy}{dx} = \frac{3\sin^2 t \cos t}{-3\cos^2 t \sin t} = -\tan t.$$

At $t = 0$, both numerator and denominator of the ratio vanish, so the simplified expression must be interpreted by a limit or by local expansion. Parametric differentiation is not permission to divide by a zero velocity component.

A final warning concerns units. If

$$y = \sin(30x^\circ),$$

then in radian notation

$$y = \sin\left(\frac{\pi x}{6}\right),$$

so

$$y' = \frac{\pi}{6}\cos\left(\frac{\pi x}{6}\right).$$

Writing $30\cos(30x)$ mixes degree notation with the radian derivative rule and is wrong.

§85.9 Derivatives from the Circle Generator

The elementary argument above began with circle geometry and the limit

$$\frac{\sin h}{h} \longrightarrow 1.$$

There is a second route once matrix exponentials and linear differential equations have been developed independently. Let

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

The matrix J performs a positive quarter-turn and satisfies

$$J^2 = -I.$$

Separating even and odd powers in the exponential series gives

$$\begin{aligned} e^{xJ} &= I + xJ + \frac{x^2J^2}{2!} + \frac{x^3J^3}{3!} + \cdots \\ &= I\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots\right) + J\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots\right) \\ &= I\cos x + J\sin x. \end{aligned}$$

Therefore

$$R(x) = e^{xJ} = \begin{pmatrix} \cos x & -\sin x \\ \sin x & \cos x \end{pmatrix}.$$

Because all scalar multiples of J commute,

$$R(x+y) = e^{(x+y)J} = e^{xJ}e^{yJ} = R(x)R(y).$$

Thus $R : \mathbb{R} \to SO(2)$ is a one-parameter group. Differentiating the exponential series term by term gives

$$R'(x) = JR(x) = R(x)J.$$

If the first column is written

$$u(x) = \begin{pmatrix} c(x) \\ s(x) \end{pmatrix},$$

then

$$u'(x) = Ju(x)$$

means

$$c'(x) = -s(x), \qquad s'(x) = c(x),$$

with initial condition

$$c(0) = 1, \qquad s(0) = 0.$$

Uniqueness for the linear system identifies c and s with cosine and sine. The derivative formulas are therefore the infinitesimal form of the rotation-group law.

This route also explains the word *generator*. The tangent vector to the curve $R(x)$ at the identity is

$$R'(0) = J.$$

Every later rotation is obtained by transporting the same infinitesimal turning rule along the group:

$$R'(x) = JR(x).$$

The matrix J is the Lie-algebra element that generates the entire one-parameter subgroup.

Logically, this is an independent foundation only when the matrix exponential, its differentiation rule, and uniqueness of the linear ODE have been proved without using the trigonometric derivative formulas. Otherwise it remains a valuable consistency check rather than a new proof. The two routes meet at the same system,

$$y'' + y = 0,$$

but arrive there from different starting data: one from the local metric geometry of the circle, the other from exponentiating an infinitesimal rotation.

Derivatives: Local Linearization, Computation Graphs, and Higher-Order Data

N-20221211

§86.1 A tangent line is a local model, not only a slope

The difference quotient

$$\frac{f(a+h)-f(a)}{h}$$

compares the actual change in f with the input change h. If the quotient approaches a finite number L, then

$$f(a+h)-f(a)\approx Lh$$

for small h. The derivative is therefore the coefficient of the best first-order linear model near a.

This viewpoint is stronger than the phrase "slope of the tangent line." A slope describes a graph in the plane. Linearization describes how the function transports small perturbations:

$$h \longmapsto f'(a)h.$$

It survives in several variables, on manifolds, and inside automatic differentiation systems where no graph is ever drawn.

The tangent-line approximation is

$$f(a+h)\approx f(a)+f'(a)h.$$

To make the word "approximately" precise, define a remainder

$$r(h)=f(a+h)-f(a)-Lh.$$

The linear model is first-order accurate when

$$\frac{r(h)}{h}\longrightarrow 0.$$

This is written

$$r(h)=o(h).$$

Thus differentiability at a is equivalently the existence of a number L such that

$$\boxed{f(a+h)=f(a)+Lh+o(h).}$$

The number L is necessarily $f'(a)$.

Indeed, subtract $f(a)$, divide by h, and let $h\to 0$:

$$\frac{f(a+h)-f(a)}{h}=L+\frac{o(h)}{h}\longrightarrow L.$$

Conversely, if the difference quotient tends to L, then

$$r(h)=h\left(\frac{f(a+h)-f(a)}{h}-L\right)=o(h).$$

The limit definition and the linearization definition contain exactly the same information.

§86.2 The algebra of small remainders generates the rules

Suppose f and g are differentiable at a. Write

$$f(a+h) = f(a) + f'(a)h + r_f(h), \qquad r_f(h) = o(h),$$

$$g(a+h) = g(a) + g'(a)h + r_g(h), \qquad r_g(h) = o(h).$$

The sum rule is immediate:

$$(f+g)(a+h) = (f+g)(a) + (f'(a)+g'(a))h + o(h).$$

Hence

$$(f+g)'(a) = f'(a) + g'(a).$$

For the product, multiply the two expansions:

$$\begin{aligned} f(a+h)g(a+h) = {} & f(a)g(a) \\ & + (f'(a)g(a) + f(a)g'(a))h \\ & + f'(a)g'(a)h^2 + f(a)r_g(h) + g(a)r_f(h) \\ & + f'(a)hr_g(h) + g'(a)hr_f(h) + r_f(h)r_g(h). \end{aligned}$$

Every term after the first-order line is $o(h)$. Therefore

$$\boxed{(fg)' = f'g + fg'.}$$

The product rule is not an arbitrary mnemonic. It is what remains after second-order-small terms are discarded.

If $g(a) \neq 0$, continuity gives $g(a+h) \neq 0$ for sufficiently small h. Applying the product rule to

$$g \cdot \frac{1}{g} = 1$$

gives

$$\left(\frac{1}{g}\right)' = -\frac{g'}{g^2}.$$

Combining this with the product rule yields the quotient rule.

The chain rule has the same interpretation. Suppose $b = f(a)$. Then

$$f(a+h) = b + f'(a)h + o(h).$$

Let

$$k = f'(a)h + o(h).$$

Since $k = O(h)$, differentiability of g at b gives

$$g(b+k) = g(b) + g'(b)k + o(k).$$

Because $o(k) = o(h)$,

$$g(f(a+h)) = g(f(a)) + g'(f(a))f'(a)h + o(h).$$

Thus

$$\boxed{(g \circ f)'(a) = g'(f(a))f'(a).}$$

The chain rule is composition of local linear maps.

§86.3 Powers, exponentials, and logarithms

For a positive integer n, the binomial theorem gives

$$(x+h)^n - x^n = nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \cdots + h^n.$$

After division by h, every term except the first vanishes as $h \to 0$. Hence

$$(x^n)' = nx^{n-1}.$$

Negative integer powers follow from the reciprocal rule.

For a rational exponent $r = m/n$, one may differentiate

$$y^n = x^m$$

on a real domain where the chosen power is defined and locally single-valued. This gives

$$ny^{n-1}y' = mx^{m-1},$$

so

$$(x^r)' = rx^{r-1}.$$

The domain matters. A denominator with even n restricts real inputs, while an odd denominator permits negative inputs but may still create a singular derivative at zero.

For arbitrary real α, define on $x > 0$

$$x^\alpha = e^{\alpha \ln x}.$$

Then the chain rule gives

$$\frac{d}{dx}x^\alpha = e^{\alpha \ln x}\frac{\alpha}{x} = \alpha x^{\alpha-1}.$$

This proof explains why the unrestricted real-power formula naturally lives on the positive half-line.

The exponential can be characterized by the differential equation

$$y' = y, \qquad y(0) = 1.$$

Its flow property is

$$e^{x+t} = e^x e^t.$$

Differentiating with respect to t at $t = 0$ gives

$$(e^x)' = e^x.$$

More generally,

$$(a^x)' = a^x \ln a$$

for $a > 0$.

The logarithm is the inverse of the exponential. If

$$y = \ln x, \qquad x = e^y,$$

then

$$1 = e^y y' = xy',$$

so

$$\boxed{(\ln x)' = \frac{1}{x}.}$$

The identity

$$\ln(xy) = \ln x + \ln y$$

shows that logarithm converts multiplication into addition. Its differential

$$d(\ln x) = \frac{dx}{x}$$

is the infinitesimal quantity invariant under multiplicative rescaling.

§86.4 The inverse rule and where it fails

Let f be differentiable near a, one-to-one there, and suppose

$$f'(a) \neq 0.$$

Put

$$b = f(a).$$

If the local inverse is differentiable at b, the identity

$$f^{-1}(f(x)) = x$$

and the chain rule give

$$(f^{-1})'(b) f'(a) = 1.$$

Therefore

$$\boxed{(f^{-1})'(b) = \frac{1}{f'(a)}.}$$

The one-dimensional inverse function theorem supplies the missing existence statement: under suitable continuity hypotheses, nonvanishing derivative makes the map locally invertible with a differentiable inverse.

The condition $f'(a) \neq 0$ is sufficient, not logically necessary for every kind of inverse. The function

$$f(x) = x^3$$

is globally one-to-one and has an inverse, even though $f'(0) = 0$. Its inverse

$$f^{-1}(y) = y^{1/3}$$

has no finite derivative at zero. The vertical tangent of the inverse records the collapse of first-order information in the original map.

Inverse trigonometric derivatives are applications rather than a separate theory. Once sine is restricted to

$$[-\pi/2, \pi/2],$$

the inverse rule gives, for $|x| < 1$,

$$(\arcsin x)' = \frac{1}{\sqrt{1 - x^2}}.$$

Likewise,

$$(\arccos x)' = -\frac{1}{\sqrt{1 - x^2}}, \qquad (\arctan x)' = \frac{1}{1 + x^2}.$$

The branch choices and endpoint singularities are developed in the preceding trigonometric note. Here they illustrate the general inverse rule and its failure when the original derivative vanishes.

§86.5 A small derivative library and its dependency map

A derivative table is useful when it remembers how each entry was produced.

Function	Derivative	Structural source
x^α, $x > 0$	$\alpha x^{\alpha-1}$	exponential and logarithm
e^x	e^x	one-parameter flow
$\ln x$	$1/x$	inverse function rule
$\sin x$, $\cos x$	$\cos x$, $-\sin x$	basic limits and addition laws
$\tan x$	$\sec^2 x$	quotient rule
$\arcsin x$	$(1-x^2)^{-1/2}$	branch-restricted inverse rule

The general rules form a compact dependency graph:

$$\text{linearization} \longrightarrow \begin{cases} \text{sum and product rules}, \\ \text{chain rule}, \\ \text{inverse rule}, \end{cases} \longrightarrow \text{elementary derivative library}.$$

This ordering is more reusable than memorizing a long page of formulas. New functions enter the library by exposing their algebraic composition or defining differential equation.

§86.6 Forward mode: propagate a perturbation

Automatic differentiation applies the chain rule to a computation graph without symbolic expansion and without finite-difference approximation. Consider

$$F(x) = \ln\left(1 + \sin^2(x^2)\right).$$

Break the expression into nodes:

$$u = x^2, \qquad v = \sin u, \qquad w = 1 + v^2, \qquad F = \ln w.$$

Forward mode attaches to every value its derivative with respect to the chosen input. Starting from

$$\dot{x} = 1,$$

we propagate

$$\begin{aligned} \dot{u} &= 2x\dot{x} = 2x, \\ \dot{v} &= \cos u\, \dot{u} = 2x\cos(x^2), \\ \dot{w} &= 2v\dot{v} = 4x\sin(x^2)\cos(x^2), \\ \dot{F} &= \frac{\dot{w}}{w} = \frac{4x\sin(x^2)\cos(x^2)}{1+\sin^2(x^2)}. \end{aligned}$$

The dot notation represents a tangent perturbation. If the input changes by ε, then to first order each node changes by its dotted value times ε.

This mechanism can be encoded by dual numbers. Introduce a symbol ϵ with

$$\epsilon^2 = 0.$$

Then

$$f(x+\epsilon) = f(x) + f'(x)\epsilon.$$

Ordinary arithmetic on pairs

$$x + \dot{x}\epsilon$$

automatically propagates values and first derivatives. The nilpotent relation discards all terms beyond first order, exactly as the $o(h)$ calculus did analytically.

§86.7 Reverse mode: pull sensitivity backward

Reverse mode evaluates the same graph forward to store intermediate values, then propagates output sensitivity backward. Write

$$\bar{z} = \frac{\partial F}{\partial z}$$

for the adjoint associated with a node z. Begin at the output with

$$\bar{F} = 1.$$

The local derivatives give

$$\bar{w} = \bar{F}\frac{\partial F}{\partial w} = \frac{1}{w},$$
$$\bar{v} = \bar{w}\frac{\partial w}{\partial v} = \frac{2v}{w},$$
$$\bar{u} = \bar{v}\frac{\partial v}{\partial u} = \frac{2v\cos u}{w},$$
$$\bar{x} = \bar{u}\frac{\partial u}{\partial x} = \frac{4x\sin(x^2)\cos(x^2)}{1+\sin^2(x^2)}.$$

The final adjoint $\bar{x}$ is $F'(x)$, matching forward mode.

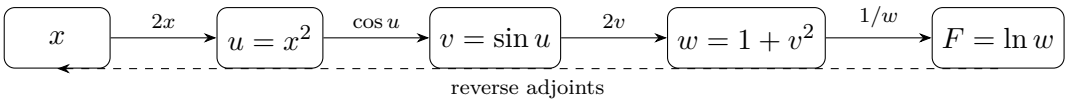


Figure 86.1: Forward mode pushes a tangent from left to right; reverse mode pulls an output covector from right to left through the same local derivatives.

Forward mode is efficient when there are few input directions and many outputs. Reverse mode is efficient for a scalar output with many inputs, which is why it underlies backpropagation. The two modes are not different chain rules. They choose opposite orders for composing the same local linear maps and their duals.

§86.8 Second-Order Automatic Differentiation

Forward mode used dual numbers

$$\mathbb{R}[\epsilon]/(\epsilon^2)$$

to retain a value and one first-order perturbation. Keeping one more power gives

$$\mathbb{R}[\epsilon]/(\epsilon^3),$$

which stores second-order local data. Represent an input curve through x by

$$X(\epsilon) = x + v\epsilon + \frac{1}{2}a\epsilon^2.$$

The coefficients v and a are its velocity and acceleration at $\epsilon = 0$.

Taylor expansion followed by the rule $\epsilon^3 = 0$ gives

$$f(X(\epsilon)) = f(x) + f'(x)\left(v\epsilon + \frac{1}{2}a\epsilon^2\right) + \frac{1}{2}f''(x)(v\epsilon)^2 \pmod{\epsilon^3}.$$

Therefore

$$\boxed{f(X(\epsilon)) = f(x) + f'(x)v\epsilon + \frac{1}{2}(f''(x)v^2 + f'(x)a)\epsilon^2.}$$

The first-order coefficient is the transported velocity. The second-order coefficient contains two effects: curvature of the function, $f''(x)v^2$, and acceleration already present in the input curve, $f'(x)a$. This is the same distinction that appears in the second derivative of a composition.

For a concrete computation, take

$$F(x) = e^{\sin x}$$

and seed a unit-speed straight perturbation at $x = 0$:

$$X(\epsilon) = \epsilon, \qquad v = 1, \qquad a = 0.$$

First propagate through sine:

$$\sin \epsilon = \epsilon \pmod{\epsilon^3},$$

because the first discarded term is $-\epsilon^3/6$. Then propagate through the exponential:

$$e^{\epsilon} = 1 + \epsilon + \frac{1}{2}\epsilon^2 \pmod{\epsilon^3}.$$

The three stored coefficients give

$$F(0) = 1, \qquad F'(0) = 1, \qquad F''(0) = 1.$$

No symbolic second derivative of the full composite was formed; each node merely propagated a truncated local polynomial.

This procedure is precisely composition of second jets. The second jet of f at x can be represented by

$$j_x^2 f(h) = f(x) + f'(x)h + \frac{1}{2}f''(x)h^2 \pmod{h^3}.$$

Composing functions means substituting one truncated polynomial into another and discarding terms of degree three or higher. At higher orders the coefficient bookkeeping is governed by set partitions and the Faà di Bruno formula; the dedicated higher-derivative chapter develops that combinatorics in full.

The same idea works in several variables. For a smooth scalar function

$$f : \mathbb{R}^n \to \mathbb{R}$$

and a direction v, substitute

$$x + \epsilon v$$

into its second-order expansion:

$$f(x + \epsilon v) = f(x) + \epsilon \nabla f(x)^T v + \frac{1}{2}\epsilon^2 v^T \nabla^2 f(x) v \pmod{\epsilon^3}.$$

Thus one second-order forward sweep computes a Hessian quadratic form $v^T H v$ without constructing every entry of H. Mixed terms can be recovered by polarization:

$$u^T H v = \frac{1}{2}\big((u+v)^T H (u+v) - u^T H u - v^T H v\big).$$

Higher-order automatic differentiation is therefore not an unrelated extension of the chain rule. It is ordinary computation performed in a richer truncated algebra that retains more of the local Taylor model.

§86.9 Differentials are covectors

For a function $f : \mathbb{R} \to \mathbb{R}$, the differential at a is the linear functional

$$df_a(h) = f'(a)h.$$

In the coordinate x, this is written

$$df = f'(x)\,dx.$$

The symbol dx is the basic covector that sends an increment h to h. It is not necessary to treat dy/dx as an ordinary fraction in order to justify the notation.

If g is another function, then

$$d(g \circ f)_a = dg_{f(a)} \circ df_a.$$

In pullback notation,

$$d(g \circ f) = f^*(dg).$$

This is the chain rule for one-forms. Reverse-mode automatic differentiation follows the same direction: an output covector is pulled backward through the transpose or adjoint of each derivative map.

For a function of several variables, the differential is no longer multiplication by one number. If

$$f : \mathbb{R}^n \to \mathbb{R},$$

then

$$df_x(h) = \nabla f(x)^T h.$$

The gradient vector depends on the chosen inner product, while the differential is intrinsically a covector. The one-dimensional notation hides this distinction because every linear map $\mathbb{R} \to \mathbb{R}$ is represented by a scalar.

§86.10 Differentiable does not mean smoothly differentiable

Continuity is necessary for differentiability, but it is not sufficient. The function

$$f(x) = |x|$$

is continuous at zero, yet

$$\lim_{h \to 0^+} \frac{|h|}{h} = 1, \qquad \lim_{h \to 0^-} \frac{|h|}{h} = -1.$$

No single linear map approximates the function from both sides.

A derivative may exist without varying continuously. Define

$$g(x) = \begin{cases} x^2 \sin(1/x), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

At zero,

$$\frac{g(h) - g(0)}{h} = h \sin(1/h) \longrightarrow 0,$$

so $g'(0) = 0$. For $x \neq 0$,

$$g'(x) = 2x \sin(1/x) - \cos(1/x),$$

which has no limit as $x \to 0$. The function is differentiable everywhere but not continuously differentiable at zero.

These examples separate three notions:

$$\text{continuity}, \qquad \text{differentiability}, \qquad C^1 \text{ regularity}.$$

Each additional level controls more than the existence of a tangent at one point.

§86.11 Fréchet Differentiability

The formula

$$f(a + h) = f(a) + f'(a)h + o(h)$$

contains the multidimensional definition almost unchanged. For a map between normed spaces,

$$F : X \to Y,$$

we seek a bounded linear map

$$DF(a) : X \to Y$$

such that

$$\boxed{\|F(a + h) - F(a) - DF(a)h\|_Y = o(\|h\|_X).}$$

This is the Fréchet derivative. The estimate must hold uniformly as h approaches zero from every direction, not along one line at a time.

Consider

$$F(x, y) = \begin{pmatrix} e^x \cos y \\ e^x \sin y \end{pmatrix}.$$

At $(0, 0)$, use

$$e^h = 1 + h + O(h^2), \qquad \cos k = 1 + O(k^2), \qquad \sin k = k + O(k^3).$$

Then

$$e^h \cos k = 1 + h + O(h^2 + k^2),$$
$$e^h \sin k = k + O(|hk| + |k|^3).$$

Hence

$$F(h, k) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} h \\ k \end{pmatrix} + O(h^2 + k^2),$$

so

$$DF(0, 0) = I.$$

The Jacobian is not merely a table of partial derivatives; it is the linear map that simultaneously approximates the function for every small vector (h, k).

Directional derivatives alone do not guarantee such a map. Define

$$G(x,y) = \begin{cases} \dfrac{x^3}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

Along a direction $v = (a,b)$,

$$G(ta,tb) = t\frac{a^3}{a^2+b^2}.$$

Therefore every directional derivative at the origin exists and equals

$$D_v G(0,0) = \frac{a^3}{a^2+b^2}.$$

But this expression is not linear in (a,b). For example,

$$D_{(1,0)}G = 1, \qquad D_{(0,1)}G = 0, \qquad D_{(1,1)}G = \frac{1}{2},$$

whereas linearity would require the last value to be 1. No single linear map can represent all directional changes, so G is not Fréchet differentiable at the origin.

Once the Fréchet derivative exists, the chain rule becomes literal composition:

$$D(H \circ F)(a) = DH(F(a)) \circ DF(a).$$

After bases are chosen, this is Jacobian multiplication. Forward mode applies the composite map to tangent vectors; reverse mode applies its dual maps to covectors. The differential df_a is the scalar-valued covector version, while jets retain the higher-order terms discarded by the first linear model.

The chapter therefore ends with the same object with which it began: the best local linear map. Elementary differentiation rules explain how it behaves under algebra and composition; inverse-function formulas ask when it can be inverted; automatic differentiation transports it through a graph; and Fréchet differentiability specifies what it means for one approximation to control all nearby directions at once.

Completed Decimals, Hyperfinite Truncations, and Nonstandard Analysis

N-20230225

§87.1 One notation can hide three different objects

The expression

$$0.999\ldots$$

is often discussed as though there were only one possible mathematical object behind it. In fact, three nearby constructions must be separated.

For an ordinary natural number n, the finite decimal with n nines is

$$d_n = \sum_{k=1}^{n} 9 \cdot 10^{-k} = 1 - 10^{-n}.$$

It is strictly less than 1. The real infinite decimal is the limit

$$d = \lim_{n\to\infty} d_n,$$

so it equals 1. A third object becomes available in nonstandard analysis: if H is an infinite hyperinteger, one may form the hyperfinite sum

$$d_H = \sum_{k=1}^{H} 9 \cdot 10^{-k}.$$

It is still a finite sum inside the hyperreal universe, but it has more terms than every standard finite truncation.

The familiar disagreement is therefore usually not about algebra. It is about which of these three objects the notation is intended to denote.

§87.2 The real decimal is a completed limit

The real-number definition is direct. Since

$$d_n = 1 - 10^{-n},$$

we have

$$|1 - d_n| = 10^{-n}.$$

Given $\varepsilon > 0$, choose N so large that $10^{-N} < \varepsilon$. Then for $n \geq N$,

$$|1 - d_n| < \varepsilon.$$

Hence

$$\boxed{0.999\ldots = \lim_{n\to\infty} d_n = 1.}$$

The standard subtraction argument says the same thing more compactly. If $x = 0.999\ldots$, then

$$10x = 9.999\ldots,$$

and subtraction gives $9x = 9$. This computation is valid because both decimals denote completed real limits. It is not a calculation with an unfinished string.

The uniqueness of decimal expansion fails only at terminating endpoints. More generally,

$$0.a_1a_2\cdots a_{m-1}(a_m - 1)999\ldots = 0.a_1a_2\cdots a_m000\ldots$$

whenever $a_m > 0$. This is not a defect of the real line; it is the usual nonuniqueness of positional notation at boundary points of adjacent decimal intervals.

§87.3 An ultrapower turns sequences into numbers

To make "larger than every ordinary integer" precise, begin with all real sequences

$$(x_1, x_2, x_3, \ldots) \in \mathbb{R}^{\mathbb{N}}.$$

Fix a nonprincipal ultrafilter $\mathcal{U}$ on $\mathbb{N}$. Two sequences are declared equivalent when they agree on a set belonging to $\mathcal{U}$:

$$(x_n) \sim (y_n) \quad \Longleftrightarrow \quad \{n : x_n = y_n\} \in \mathcal{U}.$$

The hyperreal field is the quotient

$${}^*\mathbb{R} = \mathbb{R}^{\mathbb{N}}/\mathcal{U}.$$

Addition and multiplication are performed coordinatewise before passing to equivalence classes.

Ordinary reals embed as constant sequences:

$$r \longmapsto [r, r, r, \ldots].$$

The class

$$H = [1, 2, 3, \ldots]$$

is larger than every standard integer m, because

$$\{n : n > m\}$$

is cofinite and therefore belongs to every nonprincipal ultrafilter. Thus H is an infinite hyperinteger. Its reciprocal

$$\eta = \frac{1}{H} = [1, 1/2, 1/3, \ldots]$$

is positive but smaller than every positive standard real number.

This construction does not adjoin a mysterious symbol by decree. It packages asymptotic behavior of sequences into algebraic elements of a larger ordered field.

§87.4 Infinitesimal closeness and standard part

A hyperreal x is infinitesimal when

$$|x| < r$$

for every positive standard real r. We write

$$x \approx y$$

when $x - y$ is infinitesimal. A hyperreal is finite when its absolute value is bounded by some standard integer.

Every finite hyperreal is infinitely close to a unique real number. This real number is its standard part:

$$\operatorname{st}(x) \in \mathbb{R}, \qquad x \approx \operatorname{st}(x).$$

Uniqueness is easy. If $x \approx r$ and $x \approx s$ for real numbers r, s, then $r - s$ is a real infinitesimal. The only real infinitesimal is 0, so $r = s$.

Existence uses completeness of $\mathbb{R}$. For finite x, consider the standard set

$$A = \{r \in \mathbb{R} : r < x\}.$$

It is nonempty and bounded above. Let $a = \sup A$. If $x - a$ were appreciably positive or negative, the least-upper-bound property would be contradicted. Therefore $x \approx a$.

The standard-part map is not defined on infinite hyperreals, and it is not an order-field homomorphism on all of ${}^*\mathbb{R}$. On finite hyperreals, however, it respects addition and multiplication:

$$\operatorname{st}(x + y) = \operatorname{st}(x) + \operatorname{st}(y),$$

$$\operatorname{st}(xy) = \operatorname{st}(x)\operatorname{st}(y).$$

It is the operation that turns an infinitesimally accurate hyperreal computation back into an ordinary real answer.

§87.5 Transfer explains why finite algebra still works

The transfer principle says, roughly, that every first-order statement true in the ordered field $\mathbb{R}$ remains true in its hyperreal extension. In particular, polynomial identities, order laws, and statements quantified over finitely many elements transfer.

For every natural number n,

$$\sum_{k=1}^{n} 9 \cdot 10^{-k} = 1 - 10^{-n}.$$

This is a first-order statement about finite sums. Transfer therefore gives, for every hypernatural H,

$$\sum_{k=1}^{H} 9 \cdot 10^{-k} = 1 - 10^{-H}.$$

If H is infinite, then 10^{-H} is a positive infinitesimal. Consequently,

$$d_H < 1, \qquad d_H \approx 1, \qquad \operatorname{st}(d_H) = 1.$$

There is no contradiction with the real identity. The real decimal is the completed limit 1; the hyperfinite decimal is a different hyperreal number infinitesimally below 1. Standard part sends the latter to the former.

§87.6 Internal sets are the domain of transfer

Transfer cannot be applied indiscriminately to every subset of the hyperreals. The relevant distinction is between internal and external objects.

An internal set is represented by a sequence of standard sets. For example, if

$$A_n = \{1, 2, \dots, n\},$$

then the class $[A_n]$ is the hyperfinite interval

$$\{1, 2, \dots, H\}$$

for $H = [n]$. Although it contains more elements than any standard finite set, it behaves internally like a finite set: transferred finite-sum and finite-counting statements apply to it.

By contrast, the set of all infinitesimals

$$\mu(0) = \{x \in {}^*\mathbb{R} : x \approx 0\}$$

is external. So is the set of all finite hyperreals. These sets are mathematically legitimate subsets, but they are not themselves produced by the ultrapower in the internal way required by transfer.

A useful test comes from overspill. Suppose an internal set $A \subseteq {}^*\mathbb{N}$ contains every standard natural number. Then A must contain some infinite hypernatural. To see the idea, write

$$A = [A_n].$$

For each standard m, membership of m means that $m \in A_n$ for $\mathcal{U}$-many n. The internal statement "$1, \dots, m \in A$" holds for every standard m, and transfer forces it to hold up to some nonstandard bound.

Now consider the set of standard natural numbers inside ${}^*\mathbb{N}$. It contains every standard natural but no infinite hypernatural, so overspill shows that it cannot be internal. The same phenomenon explains why one cannot transfer a statement that quantifies over "all infinitesimals" as though that collection were an internal finite-scale object.

§87.7 Derivatives from Standard Parts

Let $f : \mathbb{R} \to \mathbb{R}$ and let ${}^*f$ denote its hyperreal extension. A standard function is differentiable at a precisely when there is a real number L such that for every nonzero infinitesimal η,

$$\frac{{}^*f(a+\eta) - {}^*f(a)}{\eta} \approx L.$$

Then

$$f'(a) = \operatorname{st}\left(\frac{{}^*f(a+\eta) - {}^*f(a)}{\eta}\right).$$

For $f(x) = x^2$,

$$\frac{(a+\eta)^2 - a^2}{\eta} = 2a + \eta,$$

so

$$f'(a) = \operatorname{st}(2a + \eta) = 2a.$$

The condition must hold for every nonzero infinitesimal, not merely one convenient choice. This is the nonstandard form of the ordinary limit requiring all sufficiently small increments to give the same first-order behavior.

For $f(x) = |x|$ at 0, a positive infinitesimal gives quotient 1, while a negative infinitesimal gives -1. No single real standard part works for all infinitesimal directions, so the function is not differentiable at 0. The usual left-right failure appears without introducing two separate limit symbols.

§87.8 Hyperfinite sums recover the Riemann integral

Let f be continuous on $[a, b]$, choose an infinite hyperinteger H, and set

$$\Delta x = \frac{b-a}{H}, \qquad x_i = a + i\Delta x.$$

The hyperfinite Riemann sum is

$$S_H = \sum_{i=0}^{H-1} {}^*f(x_i)\Delta x.$$

Continuity on a compact interval implies uniform continuity. Hence replacing each infinitesimal cell by any other point in the same cell changes the sum only infinitesimally. The standard part is therefore independent of the hyperfinite sampling choice, and

$$\boxed{\int_a^b f(x)\,dx = \operatorname{st}(S_H).}$$

For $f(x) = x$ on $[0, 1]$, take $x_i = i/H$. Then

$$S_H = \sum_{i=0}^{H-1} \frac{i}{H}\frac{1}{H} = \frac{H(H-1)}{2H^2} = \frac{1}{2} - \frac{1}{2H}.$$

Since $1/H$ is infinitesimal,

$$\operatorname{st}(S_H) = \frac{1}{2}.$$

The phrase “add infinitely many infinitesimal rectangles” has become an ordinary finite-sum identity transferred to a hyperfinite index set.

§87.9 Loeb measure turns counting into a standard measure

The hyperfinite grid also explains how a genuinely standard measure can emerge from internal counting. Let

$$\Omega_H = \left\{0, \frac{1}{H}, \ldots, \frac{H-1}{H}\right\}.$$

For an internal subset $A \subseteq \Omega_H$, define the internal counting probability

$$\nu_0(A) = \frac{|A|}{H}.$$

This value is a finite hyperreal. Taking standard part gives

$$\nu(A) = \operatorname{st}\nu_0(A).$$

On the algebra of internal subsets, ν is finitely additive. The Loeb construction applies the usual measure-extension machinery to obtain a countably additive standard measure on a larger sigma-algebra.

Consider the interval $[0, t) \subseteq [0, 1]$ with standard $0 \leq t \leq 1$. Its grid approximation contains

$$\lfloor tH \rfloor$$

points, so

$$\nu_0([0, t) \cap \Omega_H) = \frac{\lfloor tH \rfloor}{H} \approx t.$$

Therefore its Loeb measure is t, exactly the ordinary length of the interval.

Thus the same hyperfinite idea that distinguishes d_H from the completed decimal also builds a bridge from finite counting to continuum measure. Hyperfinite objects are not completed real limits, but after standard part and measure completion they can encode the standard analytic structures those limits describe.

From Oscillation to Stable Limits: Riemann and Lebesgue Integration

N-20230311

§88.1 Integrating Pointwise Limits

For a single continuous function on a compact interval, integration is rarely mysterious. The difficulty appears when analysis constructs a function as a limit:

$$f_n(x) \longrightarrow f(x).$$

One then wants to exchange two limiting operations:

$$\int \lim_{n\to\infty} f_n \, d\mu \stackrel{?}{=} \lim_{n\to\infty} \int f_n \, d\mu.$$

Pointwise convergence alone does not justify this exchange. A sequence may carry a fixed amount of area farther and farther away, or compress it into narrower and taller spikes.

Riemann integration begins by controlling oscillation on small intervals. Lebesgue integration changes the basic pieces from intervals to measurable sets and builds the integral so that monotone approximation is automatic. The two theories therefore differ less in the picture of rectangles than in the limiting processes they are designed to support.

§88.2 Riemann integration controls oscillation on intervals

Let $f : [a, b] \to \mathbb{R}$ be bounded and let

$$P : a = x_0 < x_1 < \cdots < x_m = b$$

be a partition. On $I_i = [x_{i-1}, x_i]$, define

$$M_i = \sup_{x\in I_i} f(x), \qquad m_i = \inf_{x\in I_i} f(x).$$

The upper and lower Darboux sums are

$$U(P, f) = \sum_{i=1}^{m} M_i |I_i|, \qquad L(P, f) = \sum_{i=1}^{m} m_i |I_i|.$$

Their difference is

$$U(P, f) - L(P, f) = \sum_{i=1}^{m} \omega(f; I_i)|I_i|,$$

where

$$\omega(f; I_i) = M_i - m_i$$

is the oscillation of f on the interval.

The Darboux criterion says

$$f \text{ is Riemann integrable} \quad \Longleftrightarrow \quad \forall \varepsilon > 0 \ \exists P : U(P, f) - L(P, f) < \varepsilon.$$

Thus the theory succeeds when intervals on which the function oscillates substantially can be made to have small total length.

A function may have discontinuities and still satisfy this condition. If

$$f(x) = \begin{cases} 1, & x = c, \\ 0, & x \neq c, \end{cases}$$

then every lower sum is 0. Only one interval of a partition contains c, and its length can be made arbitrarily small, so the upper sums also approach 0.

§88.3 Dense oscillation defeats every interval partition

Consider the Dirichlet function on $[a, b]$:

$$D(x) = \mathbf{1}_{\mathbb{Q}}(x).$$

Every nonempty interval contains rational and irrational points. Hence on every partition interval,

$$\sup D = 1, \qquad \inf D = 0.$$

Therefore

$$U(P, D) = b - a, \qquad L(P, D) = 0$$

for every partition. No refinement reduces the oscillation gap.

The Thomae function gives a subtler comparison. Define on $[0, 1]$

$$T(x) = \begin{cases} 1/q, & x = p/q \text{ in lowest terms}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

It is discontinuous at every rational and continuous at every irrational. Nevertheless it is Riemann integrable with integral 0.

Fix $\varepsilon > 0$. There are only finitely many reduced fractions in $[0, 1]$ with denominator at most $2/\varepsilon$. Cover these finitely many points by intervals of total length less than $\varepsilon/2$. Outside those intervals,

$$T(x) < \frac{\varepsilon}{2}.$$

A partition adapted to the cover has upper sum bounded by

$$1 \cdot \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \cdot 1 = \varepsilon.$$

Thus dense discontinuities alone are not the obstruction; what matters is the size and strength of the oscillation.

Lebesgue's criterion makes this exact: a bounded function on $[a, b]$ is Riemann integrable if and only if its set of discontinuities has Lebesgue measure zero. The theorem is stated using measure because the Riemann criterion was already measuring how much interval length supports persistent oscillation.

§88.4 Measure spaces decide which sets may be counted

Lebesgue integration begins with a measure space

$$(X, \mathcal{M}, \mu).$$

The sigma-algebra $\mathcal{M}$ specifies the measurable subsets of X, and the measure assigns sizes to them. Countable additivity means that for pairwise disjoint measurable sets E_k,

$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = \sum_{k=1}^{\infty} \mu(E_k).$$

A function $f : X \to \overline{\mathbb{R}}$ is measurable when sets of the form

$$\{x : f(x) > t\}$$

are measurable for every real t. This condition ensures that the regions on which the function lies in a prescribed range have a defined size.

Indicator functions connect sets and functions:

$$\int_X \mathbf{1}_E \, d\mu = \mu(E).$$

For the Dirichlet function on an interval,

$$D = \mathbf{1}_{\mathbb{Q}\cap[a,b]}.$$

Since the rationals are countable and every singleton has Lebesgue measure zero,

$$\mu(\mathbb{Q} \cap [a, b]) = 0.$$

Therefore

$$\int_a^b D(x)\, dx = 0.$$

The function that defeats every interval partition becomes simple once its exceptional set can be measured directly.

§88.5 Integration from Simple Functions

A nonnegative simple function has the form

$$s = \sum_{j=1}^{r} a_j \mathbf{1}_{E_j}, \qquad a_j \geq 0,$$

where the measurable sets E_j are pairwise disjoint. Define

$$\int_X s \, d\mu = \sum_{j=1}^{r} a_j \mu(E_j).$$

For a nonnegative measurable function f, define

$$\boxed{\int_X f \, d\mu = \sup\left\{\int_X s \, d\mu : 0 \leq s \leq f,\ s \text{ simple}\right\}.}$$

The integral is therefore designed around monotone approximation from below.

For example, on $[0,1]$, let

$$s_n(x) = 2^{-n}\lfloor 2^n x \rfloor.$$

Then s_n is simple,

$$0 \le s_n(x) \le x, \qquad s_n(x) \uparrow x.$$

Its integral is

$$\begin{aligned}\int_0^1 s_n(x)\,dx &= \sum_{k=0}^{2^n-1} \frac{k}{2^n}\cdot\frac{1}{2^n} \\ &= \frac{2^n(2^n-1)}{2\cdot 2^{2n}} \\ &= \frac{1}{2} - \frac{1}{2^{n+1}}.\end{aligned}$$

Taking the supremum gives

$$\int_0^1 x\,dx = \frac{1}{2}.$$

Here the value is not imported from Riemann integration; it follows directly from measurable step approximations.

For a real-valued measurable function, write

$$f = f^+ - f^-, \qquad f^+ = \max(f,0), \qquad f^- = \max(-f,0).$$

The function is Lebesgue integrable when

$$\int f^+ < \infty, \qquad \int f^- < \infty,$$

or equivalently when $\int |f| < \infty$.

§88.6 Monotone convergence is built into the definition

Suppose

$$0 \le f_1 \le f_2 \le \cdots, \qquad f_n(x) \to f(x).$$

Since $f_n \le f$, monotonicity gives

$$\lim_{n\to\infty} \int f_n\,d\mu \le \int f\,d\mu.$$

For the reverse inequality, take a simple function $0 \le s \le f$ and a number $0 < c < 1$. Define

$$E_n = \{x : f_n(x) \ge cs(x)\}.$$

The sets E_n increase and cover the region on which $s > 0$. Hence

$$\int f_n\,d\mu \ge c\int_{E_n} s\,d\mu \longrightarrow c\int s\,d\mu.$$

Letting $c \uparrow 1$ and then taking the supremum over all simple $s \le f$ yields

$$\boxed{\int f\,d\mu = \lim_{n\to\infty}\int f_n\,d\mu.}$$

This is the Monotone Convergence Theorem.

The proof works because nonnegative mass only accumulates. No positive and negative parts can cancel, and no contribution already present in f_n is later removed.

§88.7 Fatou and dominated convergence control general limits

For nonnegative measurable functions f_n, define

$$g_k(x) = \inf_{n \geq k} f_n(x).$$

Then

$$g_k \uparrow \liminf_{n\to\infty} f_n.$$

Monotone convergence gives

$$\int \liminf f_n \, d\mu = \lim_{k\to\infty} \int g_k \, d\mu.$$

Since $g_k \leq f_n$ for every $n \geq k$,

$$\int g_k \, d\mu \leq \inf_{n \geq k} \int f_n \, d\mu.$$

Therefore

$$\boxed{\int \liminf f_n \, d\mu \leq \liminf_{n\to\infty} \int f_n \, d\mu.}$$

This is Fatou's lemma.

Now suppose $f_n \to f$ almost everywhere and

$$|f_n| \leq g$$

for some integrable g. Apply Fatou to the nonnegative sequences

$$g + f_n \quad \text{and} \quad g - f_n.$$

The first gives

$$\int g + \int f \leq \int g + \liminf \int f_n,$$

so

$$\int f \leq \liminf \int f_n.$$

The second gives

$$\int g - \int f \leq \int g - \limsup \int f_n,$$

so

$$\limsup \int f_n \leq \int f.$$

Consequently,

$$\boxed{\lim_{n\to\infty} \int f_n \, d\mu = \int f \, d\mu.}$$

This is the Dominated Convergence Theorem.

A basic example is

$$f_n(x) = x^n \qquad (0 \leq x \leq 1).$$

Then $f_n(x) \to 0$ for every $x < 1$, and the exceptional point $x = 1$ has measure zero. Since $|f_n| \leq 1$, dominated convergence gives

$$\int_0^1 x^n \, dx \longrightarrow 0,$$

consistent with the exact value $1/(n+1)$.

§88.8 Escape and Concentration of Mass

On $\mathbb{R}$, let

$$f_n = \mathbf{1}_{[n,n+1]}.$$

For each fixed x, eventually $x \notin [n, n+1]$, so

$$f_n(x) o 0.$$

Yet

$$\int_{\mathbb{R}} f_n \, dx = 1$$

for every n. The mass has escaped to infinity. No single integrable function can dominate all translated indicators.

On $[0, 1]$, let

$$h_n = n\mathbf{1}_{[0,1/n]}.$$

Then $h_n(x) o 0$ for every $x > 0$, but

$$\int_0^1 h_n \, dx = 1.$$

The mass has concentrated near 0. Any pointwise dominating function would need to be at least of order $1/x$ near zero and therefore could not be integrable.

These examples show what the hypothesis $|f_n| \le g \in L^1$ prevents. It supplies a fixed integrable envelope that forbids both translation of unit mass to infinity and compression of unit mass into arbitrarily narrow spikes.

§88.9 Improper versus Lebesgue Integrability

The improper integral

$$\int_1^\infty \frac{\sin x}{x} \, dx$$

converges by cancellation. Integration by parts on $[1, R]$ gives

$$\int_1^R \frac{\sin x}{x} \, dx = \left[-\frac{\cos x}{x}\right]_1^R - \int_1^R \frac{\cos x}{x^2} \, dx.$$

The boundary term has a limit and the remaining integral converges absolutely.

However,

$$\int_1^\infty \frac{|\sin x|}{x} \, dx = \infty.$$

Indeed, on a fixed positive fraction of each period, $|\sin x|$ is bounded below by a positive constant, and the resulting comparison is with the harmonic series.

Thus $\sin x/x$ is not Lebesgue integrable on $[1, \infty)$, because Lebesgue integrability of a signed function requires

$$\int |f| < \infty.$$

This is not a weakness. Absolute integrability makes the integral insensitive to rearrangement and places the function in a normed space. Conditional improper integration preserves cancellation along a specified exhaustion of the domain, which is a different structure.

§88.10 The spaces L^p record quantitative integrability

For $1 \le p < \infty$, define

$$L^p(X) = \left\{ f : \int_X |f|^p \, d\mu < \infty \right\} / \sim,$$

where functions equal almost everywhere are identified. The norm is

$$\|f\|_p = \left(\int_X |f|^p \, d\mu \right)^{1/p}.$$

The quotient by almost-everywhere equality is forced by integration: changing a function on a null set changes neither its integral nor its L^p norm. The Dirichlet function and the zero function therefore represent the same element of $L^p([a,b])$.

If $1/p + 1/q = 1$, Hölder's inequality gives

$$\int_X |fg| \, d\mu \le \|f\|_p \|g\|_q.$$

Thus the integral defines a continuous pairing between L^p and L^q. In the case $p = q = 2$,

$$\langle f, g \rangle = \int_X f \overline{g} \, d\mu$$

is an inner product, and integration becomes part of Hilbert-space geometry.

Dominated convergence also has a norm interpretation. If $f_n \to f$ almost everywhere and $|f_n| \le g \in L^p$, then

$$|f_n - f|^p \le (2g)^p.$$

Applying dominated convergence to $|f_n - f|^p$ yields

$$\|f_n - f\|_p o 0.$$

A pointwise limit has become convergence in a function-space norm because one integrable envelope controls the entire sequence.

§88.11 Probability reveals the role of uniform integrability

A probability space is a measure space

$$(\Omega, \mathcal{F}, \mathbb{P})$$

with $\mathbb{P}(\Omega) = 1$. A random variable is a measurable function, and expectation is integration:

$$\mathbb{E}[X] = \int_\Omega X \, d\mathbb{P}.$$

For an event A,

$$\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A).$$

Suppose $X_n \to X$ almost surely and $|X_n| \le Y$ with $\mathbb{E}[Y] < \infty$. Dominated convergence gives

$$\mathbb{E}[X_n] \to \mathbb{E}[X].$$

A single dominating random variable is sufficient, but it is stronger than necessary. What the proof really needs is uniform control of the tails.

A family $\{X_n\} \subset L^1$ is *uniformly integrable* if

$$\lim_{K\to\infty} \sup_n \mathbb{E}\left[|X_n|\mathbf{1}_{\{|X_n|>K\}}\right] = 0.$$

This condition says that no member of the family can hide a fixed amount of expected mass at arbitrarily large values.

Domination implies uniform integrability. If $|X_n| \le Y \in L^1$, then

$$\{|X_n| > K\} \subseteq \{Y > K\},$$

and therefore

$$\mathbb{E}\left[|X_n|\mathbf{1}_{\{|X_n|>K\}}\right] \le \mathbb{E}\left[Y\mathbf{1}_{\{Y>K\}}\right].$$

The right-hand side tends to zero by dominated convergence and is independent of n.

The concentrating sequence

$$X_n = n\mathbf{1}_{(0,1/n]}$$

on $(0, 1]$ converges almost surely to zero, but

$$\mathbb{E}[X_n] = 1.$$

It is not uniformly integrable. For any fixed K, choose $n > K$. Then $|X_n| > K$ on its whole support, so

$$\mathbb{E}\left[|X_n|\mathbf{1}_{\{|X_n|>K\}}\right] = 1.$$

The supremum over n never decreases. Uniform integrability detects exactly the mass concentration that pointwise convergence misses.

Vitali's convergence theorem gives the corresponding exchange principle: if $X_n \to X$ in probability and $\{X_n\}$ is uniformly integrable, then

$$\mathbb{E}|X_n - X| \to 0.$$

In particular,

$$\mathbb{E}[X_n] \to \mathbb{E}[X].$$

The mechanism is a bounded-tail decomposition. Choose K so that the large-value tails contribute little uniformly. On the region where $|X_n|$ and $|X|$ are at most K, convergence in probability controls the integral after separating the set on which $|X_n - X|$ exceeds a small threshold. The tails are then controlled by uniform integrability.

The development has therefore returned to the question that motivated the two integration theories. Riemann integration controls spatial oscillation well enough to integrate a single bounded function on an interval. Lebesgue integration reorganizes the construction around measurable sets and monotone approximation, making precise when limits may pass through integrals, norms, and expectations. Dominated convergence supplies one fixed envelope; uniform integrability isolates the more general tail condition that the exchange actually requires.

Calculus Review I: Limits, Completeness, Asymptotic Scales, and Local Approximation

N-20251112

§89.1 Cauchy–Schwarz and Minkowski

For vectors $x, y \in \mathbb{R}^n$, the Cauchy–Schwarz inequality is

$$|\langle x, y\rangle| \le \|x\|_2 \|y\|_2.$$

If $y \neq 0$, consider the nonnegative quadratic

$$0 \le \|x - ty\|_2^2 = \|x\|_2^2 - 2t\langle x, y\rangle + t^2\|y\|_2^2.$$

Its discriminant is nonpositive, so

$$\langle x, y\rangle^2 \le \|x\|_2^2 \|y\|_2^2.$$

Equality holds exactly when x and y are linearly dependent.

Minkowski's inequality follows:

$$\begin{aligned} \|x + y\|_2^2 &= \|x\|_2^2 + 2\langle x, y\rangle + \|y\|_2^2 \\ &\le \|x\|_2^2 + 2\|x\|_2\|y\|_2 + \|y\|_2^2 \\ &= (\|x\|_2 + \|y\|_2)^2. \end{aligned}$$

Hence

$$\boxed{\|x + y\|_2 \le \|x\|_2 + \|y\|_2.}$$

For nonzero vectors, equality holds exactly when x and y point in the same direction.

These inequalities control accumulated errors. If $e_1, \dots, e_m$ are perturbations, then

$$\left\| \sum_{j=1}^{m} e_j \right\| \le \sum_{j=1}^{m} \|e_j\|.$$

If an error is represented by an inner product, Cauchy–Schwarz converts it into a product of sizes. The same two patterns survive in Hilbert and normed spaces.

§89.2 Limits through Neighborhoods

For $a \in \mathbb{R}$, let

$$N_\varepsilon(a) = \{x : |x - a| < \varepsilon\}.$$

A sequence (x_n) converges to a when

$$\forall \varepsilon > 0, \quad x_n \in N_\varepsilon(a) \text{ eventually.}$$

Here "eventually" means that there exists N such that the statement holds for every $n \ge N$.

For a function f,

$$\lim_{x \to x_0} f(x) = L$$

means that for every neighborhood V of L, there is a punctured neighborhood U of x_0 such that

$$x \in U \quad \Longrightarrow \quad f(x) \in V.$$

In epsilon–delta notation,

$$0 < |x - x_0| < \delta \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

The neighborhood families form filters: intersections of finitely many neighborhoods are neighborhoods, and larger sets remain neighborhoods. But a *punctured* limit has a subtlety that an ordinary continuity statement does not. The tempting rule

$$f(x) \longrightarrow L, \qquad \lim_{y \to L} g(y) = M \quad \Longrightarrow \quad g(f(x)) \longrightarrow M$$

is false without an extra assumption. For instance, set $f(x) = 0$ for every x, and

$$g(y) = \begin{cases} 1, & y = 0, \\ 0, & y \neq 0. \end{cases}$$

Then $\lim_{x \to 0} f(x) = 0$ and $\lim_{y \to 0} g(y) = 0$, but $g(f(x)) = 1$ everywhere. The inner function lands exactly on the point that the outer limit *excludes*.

There are two common ways to repair the composition rule. If g is continuous at L, then $g(L) = M$, and the ordinary neighborhood definition proves $g(f(x)) \to M$. Alternatively, even if g is discontinuous at L, it suffices that $f(x) \neq L$ on some punctured neighborhood of x_0: the image then stays in the punctured neighborhood where the outer limit estimate is valid. This is why one must pay attention to which point a limit quantifier allows us to visit.

For example, the continuous outer function $g(y) = y^2$ always satisfies the familiar rule $\lim_{x \to x_0} g(f(x)) = (\lim_{x \to x_0} f(x))^2$, with no restriction on whether $f(x) = L$ along the way. In contrast, an outer function with an isolated removable discontinuity cannot be composed blindly. The filter language makes this distinction explicit: a neighborhood of L and a *deleted* neighborhood of L are different kinds of test sets.

§89.3 Uniqueness and Algebra of Limits

Suppose $x_n \to a$ and $x_n \to b$. If $a \neq b$, choose

$$\varepsilon = \frac{|a - b|}{3}.$$

For large n,

$$|a - b| \leq |a - x_n| + |x_n - b| < 2\varepsilon < |a - b|,$$

a contradiction. Thus limits are unique.

If $x_n \to a$ and $y_n \to b$, then

$$x_n + y_n \to a + b, \qquad x_n y_n \to ab.$$

For the product,

$$|x_n y_n - ab| \leq |x_n||y_n - b| + |b||x_n - a|.$$

A convergent sequence is bounded, so $|x_n| \le M$ eventually, and both terms tend to zero. If $b \neq 0$, then $y_n \neq 0$ eventually and

$$\frac{x_n}{y_n} \to \frac{a}{b}.$$

The definition of $x_n \to +\infty$ is also eventual:

$$\forall M > 0, \quad x_n > M \text{ eventually.}$$

It is not convergence to a real number, but it becomes ordinary convergence after adjoining points at infinity or applying a compactifying coordinate such as $x \mapsto \arctan x$.

§89.4 Heine's Sequential Criterion

Theorem 89.4.1 (Heine criterion)

Let f be defined on a punctured neighborhood of x_0. Then

$$\lim_{x \to x_0} f(x) = L$$

iff for every sequence $x_n \to x_0$ with $x_n \neq x_0$, one has

$$f(x_n) \to L.$$

Proof. Assume the epsilon–delta limit. Given $\varepsilon > 0$, choose $\delta > 0$ so that

$$0 < |x - x_0| < \delta \Longrightarrow |f(x) - L| < \varepsilon.$$

Since $x_n \to x_0$, the hypothesis on x_n holds eventually, so $f(x_n) \to L$.

Conversely, suppose the epsilon–delta condition fails. Then there exists $\varepsilon_0 > 0$ such that for every n one can choose x_n with

$$0 < |x_n - x_0| < \frac{1}{n}, \qquad |f(x_n) - L| \ge \varepsilon_0.$$

Then $x_n \to x_0$, but $f(x_n) \not\to L$, contradicting the sequential assumption. □

To disprove a limit, construct incompatible approaches. For

$$f(x) = \cos \frac{1}{x},$$

take

$$x_n = \frac{1}{2\pi n}, \qquad y_n = \frac{1}{(2n+1)\pi}.$$

Both tend to zero, but

$$f(x_n) = 1, \qquad f(y_n) = -1.$$

Therefore $\lim_{x \to 0} f(x)$ does not exist.

§89.5 Asymptotic Scales

Let $g(x) > 0$ near a limiting point. The notation

$$f = O(g)$$

means that there are constants $C > 0$ and a neighborhood on which

$$|f(x)| \le Cg(x).$$

The notation

$$f = o(g)$$

means

$$\frac{f(x)}{g(x)} \to 0.$$

We write

$$f = \Theta(g)$$

when both $f = O(g)$ and $g = O(f)$, and

$$f \sim g$$

when

$$\frac{f(x)}{g(x)} \to 1.$$

The implications are

$$f \sim g \Longrightarrow f = \Theta(g), \qquad f = o(g) \Longrightarrow f = O(g),$$

but the converses fail.

Equivalent infinitesimals may be substituted safely in products and quotients when the remaining factors have controlled limits. For example,

$$\sin x \sim x, \qquad 1 - \cos x \sim \frac{x^2}{2} \quad (x \to 0).$$

Hence

$$\lim_{x\to 0} \frac{1-\cos x}{x \sin x} = \lim_{x\to 0} \frac{x^2/2}{x^2} = \frac{1}{2}.$$

Substitution is not valid inside arbitrary sums where cancellation may change the leading order. Although

$$\sqrt{1+x} \sim 1,$$

one cannot replace it by 1 in

$$\sqrt{1+x} - 1.$$

The difference is of smaller scale:

$$\sqrt{1+x} - 1 = \frac{x}{\sqrt{1+x}+1} \sim \frac{x}{2}.$$

§89.6 Squeeze and monotone convergence use order structure

The squeeze theorem says that if

$$a_n \le b_n \le c_n$$

eventually and

$$a_n \to L, \qquad c_n \to L,$$

then $b_n \to L$. Indeed,

$$|b_n - L| \le \max\{|a_n - L|, |c_n - L|\}$$

when L lies between the bounding terms.

A monotone bounded sequence converges because $\mathbb{R}$ has suprema.

Theorem 89.6.1 (Monotone convergence)
If (x_n) is increasing and bounded above, then

$$x_n \to \sup_n x_n.$$

Proof. Let

$$L = \sup\{x_n : n \in \mathbb{N}\}.$$

Given $\varepsilon > 0$, the number $L - \varepsilon$ is not an upper bound. Hence some N satisfies

$$x_N > L - \varepsilon.$$

For $n \ge N$, monotonicity gives

$$L - \varepsilon < x_N \le x_n \le L.$$

Thus $|x_n - L| < \varepsilon$. □

The proof exposes the completeness input: without the least-upper-bound property, the candidate L may be missing from the number system.

§89.7 The Cauchy Criterion

A sequence is Cauchy if

$$\forall \varepsilon > 0, \quad |x_m - x_n| < \varepsilon \text{ for all sufficiently large } m, n.$$

Every convergent sequence is Cauchy by the triangle inequality.

For the converse in $\mathbb{R}$, first note that a Cauchy sequence is bounded. Choose N so that

$$|x_n - x_N| < 1 \quad (n \ge N).$$

The tail lies in a bounded interval, and the finitely many earlier terms are also bounded.

By Bolzano–Weierstrass, a bounded sequence has a convergent subsequence

$$x_{n_k} \to L.$$

Given $\varepsilon > 0$, choose K so that

$$|x_m - x_n| < \frac{\varepsilon}{2} \quad (m, n \ge K)$$

and choose k with $n_k \geq K$ and

$$|x_{n_k} - L| < \frac{\varepsilon}{2}.$$

Then for every $n \geq K$,

$$|x_n - L| \leq |x_n - x_{n_k}| + |x_{n_k} - L| < \varepsilon.$$

Thus $x_n \to L$.

The same statement fails in $\mathbb{Q}$. Let

$$q_n = \frac{\lfloor 10^n \sqrt{2} \rfloor}{10^n}.$$

Then (q_n) is Cauchy in the rational metric, but it has no limit in $\mathbb{Q}$. Completion adds the missing limit $\sqrt{2}$.

A Banach space is exactly a normed vector space in which every Cauchy sequence converges. Completeness is therefore not a technical appendix; it is what turns internal error control into the existence of a limit.

§89.8 Pointwise versus Uniform Continuity

A function $f : X \to Y$ between metric spaces is uniformly continuous if

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad d_X(x, y) < \delta \Longrightarrow d_Y(f(x), f(y)) < \varepsilon$$

for all $x, y \in X$. The same δ must work everywhere.

The function $f(x) = x^2$ is continuous on $\mathbb{R}$ but not uniformly continuous. Set

$$x_n = n, \qquad y_n = n + \frac{1}{n}.$$

Then

$$|x_n - y_n| = \frac{1}{n} \to 0,$$

but

$$|x_n^2 - y_n^2| = 2 + \frac{1}{n^2} \not\to 0.$$

On compact domains, continuity becomes uniform.

Theorem 89.8.1 (Heine–Cantor)

A continuous function from a compact metric space to a metric space is uniformly continuous.

Proof. If uniform continuity failed, there would be $\varepsilon_0 > 0$ and pairs x_n, y_n with

$$d(x_n, y_n) < \frac{1}{n}, \qquad d(f(x_n), f(y_n)) \geq \varepsilon_0.$$

Compactness gives a convergent subsequence $x_{n_k} \to x$. Since

$$d(y_{n_k}, x) \leq d(y_{n_k}, x_{n_k}) + d(x_{n_k}, x) \to 0,$$

one also has $y_{n_k} \to x$. Continuity then forces both image subsequences to converge to $f(x)$, contradicting their separation by ε_0. □

Uniform continuity preserves Cauchy sequences and is therefore the natural hypothesis for extending a map to a completion.

§89.9 Derivatives as Linear Approximations

The derivative at a can be written as

$$f(a+h) = f(a) + f'(a)h + r(h), \qquad \frac{r(h)}{h} \to 0.$$

Equivalently,

$$f(a+h) = f(a) + f'(a)h + o(h).$$

This is stronger than the existence of a slope formula: it says the nonlinear error is negligible compared with the increment.

The differential

$$df_a(h) = f'(a)h$$

is the linear map giving the best first-order approximation. In several variables, the correct extension is the Fréchet derivative:

$$f(a+h) = f(a) + Df(a)h + o(\|h\|).$$

The derivative is therefore a local linear model, not merely a quotient of small numbers.

The chain rule follows by composing the two local models. If

$$g(a+h) = g(a) + g'(a)h + o(h)$$

and

$$f(g(a)+k) = f(g(a)) + f'(g(a))k + o(k),$$

then $k = g'(a)h + o(h) = O(h)$, so

$$f(g(a+h)) = f(g(a)) + f'(g(a))g'(a)h + o(h).$$

Thus

$$(f \circ g)'(a) = f'(g(a))g'(a).$$

§89.10 Taylor's Theorem with Remainder

If $f \in C^{n+1}$ near a, Taylor's theorem gives

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + R_n(x).$$

The Lagrange remainder has the form

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-a)^{n+1}$$

for some ξ between a and x.

The integral remainder is

$$\boxed{R_n(x) = \frac{1}{n!}\int_a^x f^{(n+1)}(t)(x-t)^n\,dt.}$$

It follows by repeated use of the fundamental theorem of calculus. For example,

$$\begin{aligned} f(x) - f(a) &= \int_a^x f'(t_1)\,dt_1 \\ &= f'(a)(x-a) + \int_a^x \int_a^{t_1} f''(t_2)\,dt_2\,dt_1, \end{aligned}$$

and exchanging the order of integration produces the kernel $(x-t)$. Iterating gives the general formula.

If

$$|f^{(n+1)}(t)| \le M$$

between a and x, then

$$|R_n(x)| \le \frac{M|x-a|^{n+1}}{(n+1)!}.$$

For e^x on $[0,1]$, the degree-n Maclaurin approximation satisfies

$$\left| e^x - \sum_{k=0}^{n} \frac{x^k}{k!} \right| \le \frac{e}{(n+1)!}.$$

Taylor expansion is therefore an asymptotic statement with a certified error, not permission to replace every smooth function by an infinite series.

§89.11 Rotation matrices generate trigonometric identities

Let

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad J^2 = -I.$$

The matrix exponential separates into even and odd powers:

$$\begin{aligned} e^{tJ} &= I\cos t + J\sin t \\ &= \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}. \end{aligned}$$

It preserves the Euclidean quadratic form:

$$(e^{tJ})^T e^{tJ} = I.$$

The group law

$$e^{(s+t)J} = e^{sJ}e^{tJ}$$

gives the addition formulas by matrix multiplication:

$$\cos(s+t) = \cos s\cos t - \sin s\sin t,$$

$$\sin(s+t) = \sin s\cos t + \cos s\sin t.$$

Thus the identities are coordinates of the rotation group, not an isolated table to memorize.

§89.12 Lorentz boosts generate hyperbolic identities

Let

$$K = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad K^2 = I.$$

Then

$$\begin{aligned} e^{tK} &= I\cosh t + K\sinh t \\ &= \begin{pmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{pmatrix}. \end{aligned}$$

Let

$$\eta = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

A direct multiplication gives

$$(e^{tK})^T \eta e^{tK} = \eta.$$

Therefore the boost preserves

$$x^2 - y^2.$$

Applying it to $(1,0)^T$ yields

$$\begin{pmatrix} \cosh t \\ \sinh t \end{pmatrix},$$

so invariance gives

$$\boxed{\cosh^2 t - \sinh^2 t = 1.}$$

Again the group law

$$e^{(s+t)K} = e^{sK} e^{tK}$$

produces

$$\cosh(s+t) = \cosh s \cosh t + \sinh s \sinh t,$$

$$\sinh(s+t) = \sinh s \cosh t + \cosh s \sinh t.$$

Circular and hyperbolic functions are parallel because both arise from one-parameter matrix groups. The sign difference comes from

$$J^2 = -I, \qquad K^2 = I,$$

and from the quadratic forms they preserve: Euclidean $x^2 + y^2$ versus Lorentzian $x^2 - y^2$.

Calculus Review II: Monotonicity, Integration Methods, Applications, and More

N-20251210

§90.1 Derivative signs, extrema, and the mean value theorem

If f is differentiable on an interval I and

$$f'(x) > 0 \quad (x \in I),$$

then f is strictly increasing on I. If

$$f'(x) < 0 \quad (x \in I),$$

then f is strictly decreasing.

We follow the standard presentation of differentiation and integration from OpenStax, *Calculus*, Volumes 1 and 2, with the assumptions and error estimates made explicit below.[1] The proof uses the Lagrange mean value theorem. For $x_2 > x_1$,

$$f(x_2) - f(x_1) = f'(\xi)(x_2 - x_1)$$

for some $\xi \in (x_1, x_2)$. If $f'(\xi) > 0$, then the difference is positive.

Suppose f is continuous near x_0 and differentiable in the punctured neighborhood. If the sign of f' changes from negative to positive at x_0, then $f(x_0)$ is a local minimum. If the sign changes from positive to negative, then $f(x_0)$ is a local maximum.

The compact criterion is:

$$(x - x_0)f'(x) > 0 \implies f(x_0) \text{ is a local minimum},$$
$$(x - x_0)f'(x) < 0 \implies f(x_0) \text{ is a local maximum}.$$

If

$$f'(x_0) = 0, \qquad f''(x_0) \neq 0,$$

then

$$f''(x_0) > 0 \implies x_0 \text{ is a local minimum},$$
$$f''(x_0) < 0 \implies x_0 \text{ is a local maximum}.$$

A useful mnemonic is that $f'' > 0$ looks like a smile and $f'' < 0$ looks like a frown.

If $f' \geq 0$ on an interval, the mean value theorem gives

$$f(y) - f(x) = f'(c)(y - x) \geq 0 \qquad (x < y).$$

So monotonicity is not a separate rule; it is a global consequence of local derivative sign.

Strict monotonicity needs care: $f' > 0$ everywhere is sufficient, not necessary. For $f(x) = x^3$, we have $f'(x) = 3x^2 \geq 0$ and $f'(0) = 0$, yet f is strictly increasing on all of $\mathbb{R}$. Similarly $f(x) = x^4$ has a strict local minimum at zero even though $f''(0) = 0$, so the second-derivative test is inconclusive in the zero case.

[1] OpenStax, *Calculus*, Vol. 1, Chapter 4, together with Vol. 2, Chapter 3.

§90.2 Concavity, asymptotes, and a global curve sketch

A curve segment is concave up if it lies above its tangent lines, and concave down if it lies below its tangent lines. In the usual calculus language,

$$f''(x) > 0$$

indicates concave up, while

$$f''(x) < 0$$

indicates concave down.

An inflection point is a point where the concavity changes. A necessary condition, when f'' exists, is

$$f''(x_0) = 0.$$

This condition alone is insufficient: vanishing or nonexistence of $f''(x_0)$ merely identifies a possible inflection point. The actual criterion is a change of concavity across the point.

An inflection point may have a horizontal or non-horizontal tangent, and in some examples the tangent is vertical or absent. What all cases share is a reversal of concavity, independent of the tangent's existence.

The asymptotic geometry of a graph has several distinct forms.

A vertical asymptote occurs when

$$\lim_{x \to x_0} f(x) = \pm\infty.$$

Then $x = x_0$ is a vertical asymptote.

A horizontal asymptote occurs when

$$\lim_{x \to +\infty} f(x) = c \quad \text{or} \quad \lim_{x \to -\infty} f(x) = c.$$

Then $y = c$ is a horizontal asymptote.

For an oblique asymptote

$$y = ax + b,$$

we require

$$\lim_{x \to \infty} [f(x) - (ax + b)] = 0.$$

The page derives

$$a = \lim_{x \to \infty} \frac{f(x)}{x},$$

and then

$$b = \lim_{x \to \infty} (f(x) - ax).$$

Both formulas follow from the defining limit after isolating the slope and intercept.

A systematic curve study proceeds through:

(i) determine the domain;

(ii) check symmetry and periodicity;

(iii) compute $f'(x)$ and $f''(x)$;

(iv) find critical points and possible inflection points;

(v) determine intervals of monotonicity and concavity;

(vi) find asymptotes;

(vii) locate intercepts or special points if useful;

(viii) draw the graph according to this information.

This is not a mechanical ritual. The derivative table tells how the curve moves; the second derivative table tells how it bends; asymptotes determine the large-scale shape.

Convexity can be read geometrically: the graph lies below chords and above supporting tangent lines. If f is differentiable and convex, then

$$f(y) \geq f(x) + f'(x)(y - x).$$

For $f(x) = x^2$, the tangent-line inequality reads $y^2 \geq x^2 + 2x(y - x)$, equivalently $(y - x)^2 \geq 0$. Thus the supporting-line property is a concrete algebraic consequence of convexity rather than a geometric slogan.

§90.3 Integration techniques as reverse differentiation

The integration section records standard formulas:

$$\int 0\,dx = C, \qquad \int x^{\mu}\,dx = \frac{x^{\mu+1}}{\mu+1} + C \quad (\mu \neq -1),$$

$$\int \frac{1}{x}\,dx = \ln|x| + C, \qquad \int a^x\,dx = \frac{a^x}{\ln a} + C.$$

Trigonometric formulas include

$$\int \sin x\,dx = -\cos x + C, \qquad \int \cos x\,dx = \sin x + C,$$

$$\int \sec^2 x\,dx = anx + C, \qquad \int \csc^2 x\,dx = -\cot x + C,$$

$$\int \frac{dx}{\sqrt{1-x^2}} = \arcsin x + C, \qquad \int \frac{dx}{1+x^2} = \arctan x + C.$$

These antiderivative identities are valid only on intervals where their integrands and the chosen logarithm or inverse-trigonometric branches are defined; in particular the exponential formula requires $a > 0$, $a \neq 1$. The page also lists logarithmic forms such as

$$\int \frac{dx}{\sqrt{x^2+a^2}} = \ln|x + \sqrt{x^2+a^2}| + C.$$

For a composite function, the substitution rule is

$$\int f(\varphi(v))\varphi'(v)\,dv = \int f(u)\,du, \qquad u = \varphi(v).$$

Equivalently,

$$\int f(u)\,du = \int f(\varphi(v))\varphi'(v)\,dv.$$

One direction recognizes a derivative already present, while the other introduces a variable that simplifies the integrand. The first recognizes a derivative already present; the second introduces a new variable to simplify the integral.

From

$$d(uv) = u\,dv + v\,du,$$

one obtains

$$\int u\,dv = uv - \int v\,du.$$

Integration by parts complements substitution. Its use depends on choosing u so that differentiating it simplifies the expression and choosing dv so that it can be integrated.

A rational function is

$$\frac{P(x)}{Q(x)},$$

where P, Q are polynomials and $Q \neq 0$. If $\deg P \geq \deg Q$, first divide polynomials to write it as a polynomial plus a proper rational function.

The pages list the elementary partial fraction forms. A real linear factor $(x-a)^k$ contributes

$$\frac{A_1}{x-a} + \frac{A_2}{(x-a)^2} + \cdots + \frac{A_k}{(x-a)^k}.$$

An irreducible quadratic factor $(x^2+px+q)^k$ with $p^2 - 4q < 0$ contributes

$$\frac{M_1 x + N_1}{x^2+px+q} + \cdots + \frac{M_k x + N_k}{(x^2+px+q)^k}.$$

Thus integration of rational functions reduces to two basic integral types:

$$\int \frac{dx}{(x-a)^k}, \qquad \int \frac{Mx+N}{(x^2+px+q)^k}\,dx.$$

For the second type, split the numerator as a derivative part plus a remainder:

$$Mx + N = \frac{M}{2}(2x+p) + \left(N - \frac{M}{2}p\right).$$

Then

$$\int \frac{Mx+N}{(x^2+px+q)^k}\,dx = \frac{M}{2} I_k^1 + \left(N - \frac{M}{2}p\right) I_k^2,$$

where

$$I_k^1 = \int \frac{d(x^2+px+q)}{(x^2+px+q)^k},$$

and

$$I_k^2 = \int \frac{dx}{(x^2+px+q)^k}.$$

Complete the square by setting

$$t = x + \frac{p}{2}, \qquad a^2 = q - \frac{p^2}{4} > 0.$$

Then

$$x^2 + px + q = t^2 + a^2.$$

For $k = 1$,

$$I_1^2 = \frac{1}{a}\arctan\frac{t}{a} + C.$$

For higher k, the recurrence shown on the page is

$$I_{k+1}^2 = \frac{1}{2ka^2}\left(\frac{t}{(t^2+a^2)^k} + (2k-1)I_k^2\right).$$

The notes next handle rational functions of $\sin x$ and $\cos x$. Use the tangent half-angle substitution

$$t = an\frac{x}{2}, \qquad -\pi < x < \pi.$$

Then

$$\sin x = \frac{2t}{1+t^2}, \qquad \cos x = \frac{1-t^2}{1+t^2}, \qquad anx = \frac{2t}{1-t^2},$$

and

$$dx = \frac{2\,dt}{1+t^2}.$$

Thus

$$\int R(\sin x, \cos x)\,dx = \int R\left(\frac{2t}{1+t^2}, \frac{1-t^2}{1+t^2}\right)\frac{2\,dt}{1+t^2},$$

which becomes a rational integral in t.

§90.4 Definite Integrals and Averages

The page defines the definite integral. Divide $[a, b]$ by

$$a = x_0 < x_1 < \cdots < x_n = b,$$

let

$$\Delta x_i = x_i - x_{i-1},$$

and choose sample points $\xi_i \in [x_{i-1}, x_i]$. The Riemann sum is

$$\sum_{i=1}^{n} f(\xi_i)\Delta x_i.$$

If the limit exists as the mesh

$$\lambda = \max_i \Delta x_i$$

tends to zero, then

$$\int_a^b f(x)\,dx$$

is the definite integral.

The pages list sufficient conditions for integrability: continuous functions, monotone bounded functions, and bounded functions with finitely many discontinuities are integrable on closed intervals.

The standard properties are:

$$\int_a^b [\alpha f(x) + \beta g(x)]\,dx = \alpha\int_a^b f(x)\,dx + \beta\int_a^b g(x)\,dx,$$

$$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx, \qquad \int_a^a f(x)\,dx = 0,$$

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx.$$

If $f(x) \geq 0$ on $[a, b]$, then

$$\int_a^b f(x)\,dx \geq 0.$$

If $f \le g$, then

$$\int_a^b f(x)\,dx \le \int_a^b g(x)\,dx.$$

Theorem 90.4.1 (Integral mean value theorem)

If f is continuous on $[a,b]$, then there exists $c \in [a,b]$ such that

$$\int_a^b f(x)\,dx = f(c)(b-a).$$

If f is continuous on $[a,b]$, then there exists $\xi \in [a,b]$ such that

$$\int_a^b f(x)\,dx = f(\xi)(b-a).$$

If g does not change sign, the weighted form is

$$\int_a^b f(x)g(x)\,dx = f(\xi)\int_a^b g(x)\,dx.$$

For a continuous f, let $m = \min_{[a,b]} f$, $M = \max_{[a,b]} f$. Then $m(b-a) \le \int_a^b f \le M(b-a)$ when $a < b$. Dividing by $b-a$ places the average inside $[m,M]$, and the intermediate value theorem supplies $f(c)$ equal to that average. This proves the integral mean value theorem; the weighted variant also needs g integrable with constant sign.

Area between curves is computed by

$$A = \int_a^b [f(x) - g(x)]\,dx$$

when $f \ge g$. If horizontal slicing is better,

$$A = \int_c^d [x_{right}(y) - x_{left}(y)]\,dy.$$

The volume of a solid of revolution around the x-axis is

$$V = \pi \int_a^b [f(x)]^2\,dx.$$

The shell method has the form

$$V = 2\pi \int_a^b x f(x)\,dx$$

when rotating a region around the y-axis.

Arc length of $y = f(x)$ is

$$L = \int_a^b \sqrt{1 + [f'(x)]^2}\,dx.$$

For a parametric curve $x = x(t), y = y(t)$,

$$L = \int_\alpha^\beta \sqrt{(x'(t))^2 + (y'(t))^2}\,dt.$$

§90.5 Improper integration and quadrature error

The last pages include reminders about improper integrals. If the interval is unbounded,

$$\int_a^{+\infty} f(x)\,dx = \lim_{R\to+\infty} \int_a^R f(x)\,dx.$$

If the integrand has a singularity at b, then

$$\int_a^b f(x)\,dx = \lim_{t\to b^-} \int_a^t f(x)\,dx.$$

Convergence depends on the existence of these limits.

A standard comparison is the p-integral:

$$\int_1^{\infty} \frac{dx}{x^p}$$

converges iff $p > 1$, while

$$\int_0^1 \frac{dx}{x^p}$$

converges iff $p < 1$.

For $f \in C^2([a,b])$, the one-panel midpoint and trapezoid approximations satisfy the error bounds

$$\left| \int_a^b f(x)\,dx - (b-a) f\left(\frac{a+b}{2}\right) \right| \le \frac{(b-a)^3}{24} \|f''\|_\infty,$$

$$\left| \int_a^b f(x)\,dx - \frac{b-a}{2}(f(a)+f(b)) \right| \le \frac{(b-a)^3}{12} \|f''\|_\infty.$$

For $f(x) = x^2$ on $[0,1]$, the exact integral is $1/3$; the midpoint estimate $1/4$ errs by $1/12$, and the trapezoidal estimate $1/2$ errs by $1/6$. The sign of each error is consistent with the graph's convexity. These computations expose the actual role of the second derivative in quadrature.

Infinite Sums at the Boundary: Uniform Convergence, Analyticity, and Abel Limits

N-20260616

The same infinite series can converge at one point and diverge at another. Even where it converges pointwise, differentiation or integration may fail to commute with summation. Power series are unusually well behaved inside their convergence disks, while interesting phenomena accumulate on the boundary. Abel's theorem connects these two regimes by approaching the boundary from inside, and an oscillatory integral gives a continuous analogue of the same regularization.[1]

§91.1 Pointwise convergence is too weak for calculus

The convergence of $\sum_{n\geq 1} a_n$ means that its partial sums have a finite limit; $a_n \to 0$ is necessary but not sufficient. The harmonic series diverges because

$$\frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \cdots + \frac{1}{8}\right) + \cdots$$

contains infinitely many blocks of sum at least $1/2$.

For a series of functions $\sum_n f_n(x)$, uniform convergence means

$$\sup_{x\in X} \left| \sum_{n=1}^{N} f_n(x) - f(x) \right| \longrightarrow 0.$$

The Weierstrass M-test supplies a practical criterion: if $|f_n(x)| \leq M_n$ on X and $\sum M_n < \infty$, then $\sum f_n$ converges uniformly and absolutely.

Uniform convergence of continuous functions preserves continuity. On a compact interval, it also allows integration term by term:

$$\int_a^b \sum_{n=1}^{\infty} f_n(x)\,dx = \sum_{n=1}^{\infty} \int_a^b f_n(x)\,dx.$$

Differentiation requires stronger information, typically uniform convergence of the derivative series together with convergence at one reference point.

An elementary contrast is $f_n(x) = x^n$ on $[0,1]$. It converges pointwise to zero for $x < 1$ but stays equal to 1 at $x = 1$. The limit is discontinuous, so the convergence cannot be uniform. The phenomenon occurs at the boundary where all the smallness estimates break down simultaneously.

§91.2 Power series are analytic inside a disk

Consider

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n.$$

[1] For complex power series and analytic continuation, see MIT 18.04, *Complex Variables with Applications*; for Abelian summation methods, see G. H. Hardy, *Divergent Series*. https://ocw.mit.edu/courses/18-04-complex-variables-with-applications-spring-2018/.

Its radius of convergence is determined by

$$R^{-1} = \limsup_{n\to\infty} |a_n|^{1/n},$$

with the usual conventions for $R = 0$ and $R = \infty$. For every $r < R$, the root test provides a summable majorant for $|a_n|r^n$; hence the series converges uniformly on the closed disk $|z - z_0| \le r$.

The differentiated series

$$f'(z) = \sum_{n=1}^{\infty} n a_n (z - z_0)^{n-1}$$

has the same convergence radius, because $n^{1/n} \to 1$. Thus a complex power series can be differentiated termwise any number of times inside its disk. In particular, its sum is holomorphic there.

The real function

$$f(x) = \frac{1}{1 + x^2}$$

is smooth on the entire real line, but its Taylor series at zero is

$$f(x) = 1 - x^2 + x^4 - x^6 + \cdots, \qquad |x| < 1.$$

The obstruction at radius 1 is invisible on the real axis: as a complex function, $1/(1+z^2)$ has poles at $z = \pm i$. This explains why real smoothness is insufficient to guarantee a large Taylor radius. Complex singularities govern the analytic power series.

§91.3 Abel's theorem recovers an ordinary sum from inside

Suppose $\sum_{n=0}^{\infty} a_n = S$, and write

$$s_n = \sum_{k=0}^{n} a_k.$$

For $0 < r < 1$, summation by parts gives

$$\sum_{n=0}^{\infty} a_n r^n = (1 - r) \sum_{n=0}^{\infty} s_n r^n.$$

The right-hand series converges because the convergent partial sums s_n are bounded. Since $s_n \to S$,

$$\begin{aligned}(1 - r) \sum_{n=0}^{\infty} s_n r^n - S &= (1 - r) \sum_{n=0}^{\infty} (s_n - S) r^n \\ &\longrightarrow 0 \qquad (r \uparrow 1).\end{aligned}$$

To justify the last step, split the sum at a fixed N. The finitely many early terms are multiplied by $1 - r$ and vanish; the remaining terms are bounded by $\sup_{n\ge N} |s_n - S|$, which can be made arbitrarily small.

We have proved Abel's theorem:

$$\boxed{\sum_{n=0}^{\infty} a_n = S \implies \lim_{r\uparrow 1} \sum_{n=0}^{\infty} a_n r^n = S.}$$

The factors r^n damp the distant terms, turning a boundary sum into a family of absolutely convergent interior sums.

§91.4 The converse needs more information

The series $1 - 1 + 1 - 1 + \cdots$ does not converge in the ordinary sense; its partial sums alternate between 1 and 0. Yet for $0 < r < 1$,

$$1 - r + r^2 - r^3 + \cdots = \frac{1}{1+r} \longrightarrow \frac{1}{2} \qquad (r \uparrow 1).$$

The value $1/2$ is its *Abel sum*, not its usual sum. Abel summability is strictly more permissive than ordinary convergence.

For nonnegative terms $a_n \geq 0$, however, a finite Abel limit does force convergence. Indeed

$$\sum_{n=0}^{\infty} a_n r^n \uparrow \sum_{n=0}^{\infty} a_n \quad (r \uparrow 1),$$

by monotone convergence applied to the counting measure on $\mathbb{N}$. Therefore, if the left side tends to $L < \infty$, then the ordinary series converges and equals L.

This is a simple Tauberian principle: the positivity of the coefficients prevents oscillatory cancellation from concealing divergence. Stronger Tauberian theorems replace positivity by quantitative assumptions on the decay or sign changes of coefficients.

§91.5 Oscillatory improper integrals behave similarly

The improper integral

$$\int_1^{\infty} \frac{\sin x}{x}\, dx$$

converges by integration by parts:

$$\int_A^B \frac{\sin x}{x}\, dx = \left[-\frac{\cos x}{x}\right]_A^B - \int_A^B \frac{\cos x}{x^2}\, dx.$$

The boundary term converges as $B \to \infty$, and the remaining integral is absolutely convergent. But the original integral is not absolutely convergent: over successive half-periods the integral of $|\sin x|/x$ is bounded below by a constant multiple of $1/k$, giving a divergent harmonic comparison.

For $t > 0$, introduce exponential damping:

$$F_t(a) = \int_0^{\infty} e^{-tx} \frac{\sin(ax)}{x}\, dx.$$

Differentiating with respect to a, justified by the integrable majorant e^{-tx}, gives

$$F_t'(a) = \int_0^{\infty} e^{-tx} \cos(ax)\, dx = \frac{t}{t^2 + a^2}.$$

Since $F_t(0) = 0$, integration in a yields

$$\boxed{F_t(a) = \arctan \frac{a}{t}.}$$

At $a = 1$, the limit $t \downarrow 0$ is $\pi/2$. To pass to the undamped integral, observe that for every $A > 0$ and $t \geq 0$, integration by parts together with the positivity and monotonicity of e^{-tx}/x gives

$$\left| \int_A^{\infty} e^{-tx} \frac{\sin x}{x}\, dx \right| \leq \frac{2}{A}.$$

The tail is therefore uniformly small for all $t \geq 0$. On the bounded interval $[0, A]$, dominated convergence permits $t \downarrow 0$ under the integral. Combining the two bounds gives

$$\lim_{t \downarrow 0} F_t(1) = \int_0^\infty \frac{\sin x}{x}\, dx = \boxed{\frac{\pi}{2}}.$$

For comparison, the simpler oscillatory integral $\int_0^\infty \sin x\, dx$ does not converge as an ordinary improper integral, even though

$$\int_0^\infty e^{-tx} \sin x\, dx = \frac{1}{1+t^2} \longrightarrow 1.$$

The regularized value 1 is not an ordinary integral. As with $1 - 1 + 1 - \cdots$, damping can create a boundary value without making the undamped object convergent.

Power series, Abel limits and oscillatory integrals are governed by the same issue: operations performed safely in an interior convergence regime may require additional information at the boundary. Uniform estimates justify passage to a limit, positivity can prevent hidden cancellation, and complex analyticity reveals obstructions that a purely real calculation may overlook.

Fourier Series: Projection, Summation, and Regularity

N-20260617

The Fourier series of a periodic function is simultaneously a set of coordinates, a way of reconstructing a signal, and a spectral decomposition of differentiation. These interpretations agree, but only if we distinguish three notions of convergence: convergence in mean square, pointwise convergence, and uniform convergence. A single function with a jump already forces the distinction.[1]

§92.1 Fourier coefficients are orthogonal coordinates

Work on the circle $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$ with the normalized inner product

$$\langle f, g \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x)\overline{g(x)}\, dx.$$

For $n \in \mathbb{Z}$, put $e_n(x) = e^{inx}$. Then

$$\langle e_m, e_n \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{i(m-n)x}\, dx = \delta_{mn}.$$

Thus the characters e_n form an orthonormal system; for $f \in L^2(\mathbb{T})$ define

$$\boxed{\widehat{f}(n) = \langle f, e_n \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx}\, dx.}$$

The truncated Fourier series is

$$S_N f(x) = \sum_{|n| \le N} \widehat{f}(n) e^{inx}.$$

It is the orthogonal projection of f onto the $(2N+1)$-dimensional space of trigonometric polynomials of degree at most N. Indeed, for every such polynomial P,

$$\langle f - S_N f, P \rangle = 0,$$

and Pythagoras gives the best-approximation identity

$$\boxed{\|f - P\|_2^2 = \|f - S_N f\|_2^2 + \|S_N f - P\|_2^2.}$$

With this normalized inner product, the norm and Fourier coefficients will obey Parseval's formula without any extra 2π factors.

[1] The normalization, Fejér argument and applications follow Stein and Shakarchi, *Fourier Analysis: An Introduction*, Chapters 1–3.

§92.2 Fourier Coefficients versus Point Values

Take $f(x) = x$ on $(-\pi, \pi)$, extended 2π-periodically. It is odd, so its cosine coefficients vanish. Integration by parts gives, for $n \geq 1$,

$$\begin{aligned} b_n &= \frac{2}{\pi} \int_0^\pi x \sin(nx)\, dx \\ &= \frac{2}{\pi} \left[-\frac{x\cos(nx)}{n} + \frac{\sin(nx)}{n^2} \right]_0^\pi = \frac{2(-1)^{n+1}}{n}. \end{aligned}$$

Consequently,

$$\boxed{x \sim 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin(nx).}$$

For $-\pi < x < \pi$, the series converges to x. At $x = \pi$, the periodic extension jumps from π to $-\pi$; the series takes the midpoint 0, since every term is zero there.

The coefficients decay only like $1/n$. This slow decay is a trace of the jump. Compare the continuous even function $g(x) = |x|$. Again by integration by parts,

$$a_0 = \pi, \qquad a_n = \frac{2}{\pi} \int_0^\pi x \cos(nx)\, dx = \frac{2((-1)^n - 1)}{\pi n^2}.$$

Thus

$$\boxed{|x| = \frac{\pi}{2} - \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{\cos((2k+1)x)}{(2k+1)^2}.}$$

The corner has produced coefficients of order $1/n^2$, and the series now converges uniformly by the Weierstrass test. Evaluating at $x = 0$ yields

$$\sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} = \frac{\pi^2}{8}.$$

Since the odd terms form three quarters of $\sum_{n\geq 1} n^{-2}$, this also gives the Basel sum $\sum_{n\geq 1} n^{-2} = \pi^2/6$.

§92.3 The Dirichlet kernel explains both power and danger

The partial sum has a convolution representation:

$$S_N f(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x-t) D_N(t)\, dt,$$

where

$$\boxed{D_N(t) = \sum_{n=-N}^{N} e^{int} = \frac{\sin((N+\frac{1}{2})t)}{\sin(t/2)}.}$$

The integral of D_N is 2π, but it is not nonnegative, and its L^1 norm grows on the order of $\log N$. Ordinary partial sums are therefore not averages with positive weights. Oscillation can help reconstruct a function at a point while simultaneously allowing overshoots near a jump.

For a piecewise C^1 periodic function, Dirichlet's convergence theorem gives

$$\lim_{N\to\infty} S_N f(x) = \frac{f(x+0) + f(x-0)}{2}.$$

This theorem is consistent with the sawtooth calculation; it does not assert uniform convergence in the presence of jumps.

In fact, the overshoot has an exact limiting profile. For the sawtooth, put $x_N = \pi - t/N$ with fixed $t > 0$. The sine identity $\sin(n(\pi - y)) = (-1)^{n+1}\sin(ny)$ yields

$$S_N f(x_N) = 2\sum_{n=1}^{N} \frac{\sin(nt/N)}{n} = \frac{2}{N}\sum_{n=1}^{N} \frac{\sin(tn/N)}{n/N}.$$

This is a Riemann sum, so

$$\boxed{\lim_{N\to\infty} S_N f(\pi - t/N) = 2\int_0^t \frac{\sin u}{u}\, du.}$$

At $t = \pi$, the limit is approximately 3.7039, exceeding the left-hand value π by about 0.5623. Relative to the jump height 2π, the overshoot is about 8.95%. Its spatial width shrinks like $1/N$, but its height does not vanish: this is the Gibbs phenomenon.

§92.4 Fejér averaging repairs the kernel

Instead of taking the last partial sum, average all the partial sums through N:

$$\sigma_N f = \frac{1}{N+1}\sum_{k=0}^{N} S_k f.$$

Its convolution kernel is

$$\boxed{F_N(t) = \frac{1}{N+1}\left(\frac{\sin((N+1)t/2)}{\sin(t/2)}\right)^2.}$$

Unlike D_N, the Fejér kernel satisfies

$$F_N(t) \geq 0, \qquad \frac{1}{2\pi}\int_{-\pi}^{\pi} F_N(t)\, dt = 1.$$

Moreover, for any fixed $\delta > 0$,

$$\sup_{\delta \leq |t| \leq \pi} F_N(t) \leq \frac{1}{(N+1)\sin^2(\delta/2)}.$$

The mass of the kernel concentrates near $t = 0$.

For continuous periodic f, let $\omega_f(\delta) = \sup_{|x-y|\leq\delta} |f(x) - f(y)|$, with distance understood on the circle. Positivity allows us to estimate

$$\begin{aligned} |\sigma_N f(x) - f(x)| &\leq \frac{1}{2\pi}\int_{-\pi}^{\pi} F_N(t)|f(x-t) - f(x)|\, dt \\ &\leq \omega_f(\delta) + \frac{2\|f\|_\infty}{(N+1)\sin^2(\delta/2)}. \end{aligned}$$

First choose δ so the modulus of continuity is small, then choose N large. The bound is uniform in x, proving

$$\boxed{\|\sigma_N f - f\|_\infty \longrightarrow 0.}$$

Fejér's theorem is therefore more than a different summation convention: it changes a sign-changing convolution kernel into a positive approximate identity.

§92.5 Parseval and the precise meaning of completeness

Trigonometric polynomials are uniformly dense in the continuous periodic functions, by Fejér's theorem. Continuous functions are dense in $L^2(\mathbb{T})$, so trigonometric polynomials are dense in $L^2(\mathbb{T})$. Together with orthogonality, this proves that e_n, $n \in \mathbb{Z}$, form an orthonormal basis.

Consequently the orthogonal projections satisfy $S_N f \to f$ in L^2 for every $f \in L^2(\mathbb{T})$, and Parseval's identity is

$$\boxed{\frac{1}{2\pi}\int_{-\pi}^{\pi} |f(x)|^2\,dx = \sum_{n\in\mathbb{Z}} |\widehat{f}(n)|^2.}$$

For the sawtooth, its complex coefficients satisfy

$$|\widehat{f}(n)| = \frac{1}{|n|} \quad (n \neq 0).$$

Therefore

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} x^2\,dx = \frac{\pi^2}{3} = 2\sum_{n=1}^{\infty}\frac{1}{n^2}.$$

This is a second, independent route to the Basel identity, now through orthogonal projection rather than evaluating a uniformly convergent cosine series.

Notice what Parseval does *not* claim: the Fourier series need not reproduce the prescribed point value of f at a discontinuity. An L^2 function is an equivalence class modulo changes on null sets; its Hilbert-space coordinates are insensitive to isolated point values.

§92.6 Regularity and the Sobolev threshold

For a sufficiently regular periodic function, integration by parts gives

$$\widehat{f'}(n) = in\widehat{f}(n).$$

Each derivative multiplies the n-th Fourier coefficient by in. This motivates the Sobolev norm

$$\boxed{\|f\|_{H^s(\mathbb{T})}^2 = \sum_{n\in\mathbb{Z}} (1+n^2)^s |\widehat{f}(n)|^2.}$$

It is initially defined for trigonometric polynomials and extends to distributions for which the sum is finite.

For $s > 1/2$, Cauchy–Schwarz gives

$$\begin{aligned}\sum_{n\in\mathbb{Z}} |\widehat{f}(n)| &\le \left(\sum_{n}(1+n^2)^{-s}\right)^{1/2}\left(\sum_{n}(1+n^2)^{s}|\widehat{f}(n)|^2\right)^{1/2}\\ &\le C_s\|f\|_{H^s}.\end{aligned}$$

The first sum is finite exactly when $s > 1/2$. Hence every $H^s(\mathbb{T})$ function with $s > 1/2$ has an absolutely and uniformly convergent Fourier series, determining a continuous representative. This proves the one-dimensional Sobolev embedding

$$H^s(\mathbb{T}) \hookrightarrow C^0(\mathbb{T}), \qquad s > \frac{1}{2}.$$

The threshold is not an arbitrary label: it comes directly from the summability of inverse frequency weights.

§92.7 Heat and waves act differently on the same coordinates

Fourier characters diagonalize the Laplacian:

$$-\partial_x^2 e^{inx} = n^2 e^{inx}.$$

For initial data $f \in L^2(\mathbb{T})$, the periodic heat equation $u_t = \kappa u_{xx}$ has solution

$$\boxed{u(x,t) = \sum_{n\in\mathbb{Z}} e^{-\kappa n^2 t}\,\widehat{f}(n)e^{inx}}, \qquad t > 0.$$

For every positive time, the Gaussian decay in n beats every polynomial growth factor n^m. The heat solution is therefore smooth in x, even if its initial data have jumps.

For the wave equation $u_{tt} = c^2 u_{xx}$ with initial displacement f and velocity g, each nonzero Fourier mode solves a harmonic oscillator equation. The resulting formula is

$$\begin{aligned} u(x,t) = {} & (\widehat{f}(0) + t\widehat{g}(0)) \\ & + \sum_{n\neq 0} \left(\widehat{f}(n)\cos(c|n|t) + \frac{\widehat{g}(n)}{c|n|}\sin(c|n|t) \right) e^{inx}. \end{aligned}$$

The heat semigroup damps high frequencies; wave propagation rotates their phases without smoothing their amplitudes. The same orthogonal basis detects a fundamental difference between irreversible diffusion and reversible oscillation.

§92.8 Fourier series can represent distributions

Point masses are not L^2 functions, but their Fourier coefficients are meaningful by duality. For the periodic Dirac distribution at zero, define

$$\langle \delta_0, \varphi \rangle = \varphi(0)$$

on smooth periodic test functions. Its Fourier coefficients in the integral normalization above are all $1/(2\pi)$, so formally

$$\boxed{\delta_0 = \frac{1}{2\pi}\sum_{n\in\mathbb{Z}} e^{inx}.}$$

This equality is true as an equality of periodic distributions: pairing either side with a smooth test function gives $\varphi(0)$. It is not a pointwise sum of ordinary functions.

The distributional viewpoint explains why frequency methods continue to work for singular initial data and fundamental solutions. Orthogonal projection controls mean-square approximation, positive kernels control uniform summation, Sobolev weights measure regularity, and weak Fourier series extend these constructions to objects more singular than functions.

Random Chords, Couplings, and Copulas

N-20261105

A chord is drawn at random in a circle. What is the probability that it is longer than a side of an inscribed equilateral triangle? There is a familiar answer of $1/3$, another of $1/2$, and a third of $1/4$. Each follows from a perfectly elementary calculation; the disagreement begins before any calculation, when someone decides what *at random* means.

This is Bertrand's chord paradox. I want to follow its mathematical question a little further. A random experiment requires a probability measure on its possible outcomes. Even when we specify the distribution of each coordinate of an experiment, we can still leave its joint law undetermined. These two forms of ambiguity lead naturally to measure-theoretic pushforwards, couplings, copulas and a small corner of optimal transport.[1]

§93.1 How close is the chord to the center?

Use a circle of radius 1. If a chord has midpoint at distance $r \in [0, 1]$ from the center, a right triangle gives

$$\ell(r) = 2\sqrt{1 - r^2}, \tag{93.1}$$

where $\ell(r)$ is the length of the chord. The side of an inscribed equilateral triangle has length $\sqrt{3}$, hence

$$\ell(r) > \sqrt{3} \quad \Longleftrightarrow \quad r < \frac{1}{2}. \tag{93.2}$$

Every proposed random-chord construction therefore reduces to one question:

$$\Pr(\text{long chord}) = \Pr\left(R < \frac{1}{2}\right),$$

where R is the distance of the chord's midpoint from the center. The choice of how to draw the chord has disappeared from the geometry, but it survives in the *distribution of* R.

§93.2 Three laws on the same set of chords

Two random endpoints. Choose two points independently and uniformly on the circumference, and join them. By rotational symmetry we may fix the first endpoint at angle 0, and let the angle of the other endpoint be $\Theta \sim \mathrm{Unif}[0, 2\pi)$. The chord length is

$$L = 2\left|\sin\frac{\Theta}{2}\right|.$$

[1] For the historical discussion of Bertrand's chord problem and an influential proposal based on invariance, see E. T. Jaynes, *The Well-Posed Problem* (1973). Jaynes proposes additional invariance assumptions under which a distinguished answer emerges. Our interest here begins with the different experiments and the laws they actually generate.

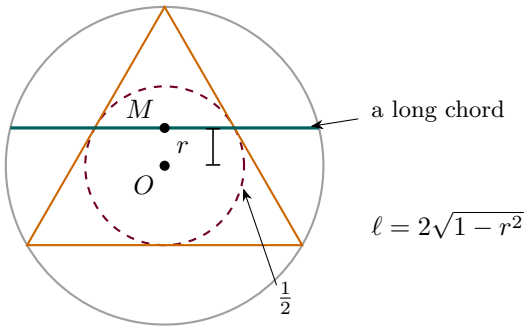


Figure 93.1: For a unit circle, a chord is longer than the inscribed equilateral triangle's side precisely when its midpoint lies inside the dashed disk of radius 1/2. The three sampling rules below agree on this geometry and differ on the distribution of the midpoint.

It exceeds $\sqrt{3}$ precisely when $\Theta \in (2\pi/3, 4\pi/3)$. Thus

$$\boxed{\Pr(L > \sqrt{3}) = \frac{2\pi/3}{2\pi} = \frac{1}{3}.}$$

The distance of its midpoint is

$$R = \left|\cos\frac{\Theta}{2}\right|.$$

For $0 \le r \le 1$,

$$\Pr(R \le r) = \frac{2}{\pi}\arcsin r, \qquad f_R(r) = \frac{2}{\pi\sqrt{1-r^2}}.$$

Here chord midpoints are more likely to be found near the boundary than near the center.

A random line through the disk. Choose a normal direction Φ uniformly on $[0, \pi)$, and independently choose a signed distance S uniformly on $[-1, 1]$. The equation

$$x\cos\Phi + y\sin\Phi = S$$

describes a line crossing the disk, and its intersection with the disk is a chord. The midpoint distance is $R = |S|$. Since $|S| \sim \mathrm{Unif}[0, 1]$,

$$\boxed{\Pr(L > \sqrt{3}) = \Pr(|S| < 1/2) = \frac{1}{2}.}$$

The adjective *uniform* now applies to a one-dimensional offset, not to area in the disk.

A random midpoint. Choose a point uniformly with respect to area in the entire unit disk, and declare it to be the midpoint of a chord perpendicular to its radius. The disk of radius r occupies the fraction r^2 of the unit disk's area, so

$$\Pr(R \le r) = r^2, \qquad f_R(r) = 2r.$$

Consequently,

$$\boxed{\Pr(L > \sqrt{3}) = \Pr(R < 1/2) = \frac{1}{4}.}$$

All three experiments are invariant under rotating the circle. One of them concentrates midpoint probability near the circumference, one makes the radius uniform, and one is uniform with respect to area. Rotational symmetry leaves room for every one of them.

Sampling rule	Density of midpoint radius r	$\Pr(r < 1/2)$
Independent endpoints	$\dfrac{2}{\pi\sqrt{1-r^2}}$	$1/3$
Uniform line offset	1	$1/2$
Uniform midpoint area	$2r$	$1/4$

Table 93.1: Three probability laws on the same geometric space of unoriented chords. Each density is on $0 < r < 1$.

§93.3 Uniform in which coordinates?

It is worth making the hidden choice of measure explicit. Let $\mathcal{C}$ be the set of unoriented chords of the unit disk, let Z be a parameter space, and let $F : Z \to \mathcal{C}$ be a procedure for turning parameters into chords. If P_Z is the law used to generate the parameters, the actual chord law is its *pushforward*,

$$P_{\mathcal{C}} = F_{\#}P_Z, \qquad (F_{\#}P_Z)(A) = P_Z(F^{-1}(A)). \tag{93.3}$$

The geometric map alone does not specify $P_{\mathcal{C}}$. We must also specify P_Z.

For instance, choosing R uniformly on $[0,1]$ yields density $f_R(r) = 1$. Choosing R^2 uniformly on $[0,1]$ yields $f_R(r) = 2r$, which is precisely the uniform-area model. In polar coordinates, this is the Jacobian:

$$dx\,dy = r\,dr\,d\theta.$$

A distribution that is uniform in one coordinate does not generally remain uniform after a nonlinear change of coordinates. The density transforms along with the coordinates.

We can even exhibit a continuum of rotationally symmetric chord laws. Choose a uniform normal direction and an independent radius with density

$$f_a(r) = (a+1)r^a, \qquad 0 < r < 1, \quad a > -1.$$

Every such law has the same geometric outcomes and the same rotational symmetry, but

$$\Pr_a(L > \sqrt{3}) = \int_0^{1/2} (a+1)r^a\,dr = 2^{-(a+1)}.$$

As a varies, the answer varies continuously between 0 and 1. Symmetry under rotations alone does not determine a probability measure.

Jaynes suggested looking for a law compatible with additional transformations, in particular translations and changes of scale. One can see why the random-line model is relevant. The natural Euclidean-motion-invariant measure on the *space of lines in the plane* has, up to scale, density

$$d\Phi\,dS.$$

It is an infinite measure on all lines, as a uniform offset over the entire real line cannot define a probability distribution. Once we restrict this measure to the lines intersecting the unit disk and normalize it, the permissible offsets satisfy $|S| \le 1$, and we obtain exactly the random-line construction yielding $1/2$.

This supplies a principled way to specify a particular experiment. Whether its assumptions describe the intended physical procedure is a further question. An experiment that samples two boundary points is another precise procedure with a different induced measure.

§93.4 Same margins, different joint laws

Here is another kind of underdetermination. Suppose someone tells us that two random variables U, V are both uniformly distributed on $[0,1]$. We know all the probabilities

$$\Pr(U \le u) = u, \qquad \Pr(V \le v) = v \qquad (0 \le u, v \le 1).$$

Do these facts determine the probability that *both* inequalities hold?

Take a single uniform random variable Z on $[0,1]$, and consider three constructions:

$$(U,V) = (Z,Z), \qquad (U,V) = (Z, 1-Z), \qquad (U,V) = (Z_1, Z_2), \tag{93.4}$$

where Z_1, Z_2 are independent uniforms in the third construction. Both coordinates are uniform in all three cases. Their joint probabilities are different:

$$\begin{aligned} \Pr(Z \le u, Z \le v) &= \min(u,v), \\ \Pr(Z \le u, 1-Z \le v) &= \max(u+v-1, 0), \\ \Pr(Z_1 \le u, Z_2 \le v) &= uv. \end{aligned}$$

For example,

$$\Pr(U \le \tfrac{1}{4}, V \le \tfrac{1}{4}) = \begin{cases} \frac{1}{4}, & V = U, \\ 0, & V = 1-U, \\ \frac{1}{16}, & U, V \text{ independent.} \end{cases}$$

Knowledge of each coordinate's distribution has not determined their dependence.

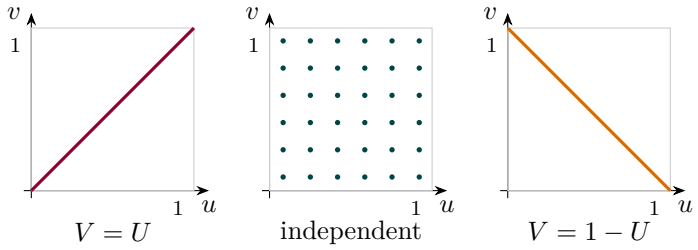


Figure 93.2: Three couplings of the same two uniform marginals. The left and right laws are concentrated on lines, while the middle law spreads mass uniformly over the square. The dots are a schematic depiction of uniform area, not an empirical sample.

A *coupling* of two probability distributions μ and ν is a probability measure π on the product space whose coordinate marginals are μ and ν. For example, the joint law of any one of the pairs in (93.4) is a coupling of two uniform measures. For any sets A, B,

$$\pi(A \times [0,1]) = \mu(A), \qquad \pi([0,1] \times B) = \nu(B).$$

The set of couplings is typically very large. It is even convex: if π_0, π_1 share the same marginals, then so does

$$\pi_\lambda = \lambda \pi_0 + (1-\lambda)\pi_1, \qquad 0 \le \lambda \le 1.$$

A random choice between two joint experiments preserves each coordinate's distribution, while varying their dependence.

§93.5 Copulas and the sharp bounds on dependence

The function

$$C(u,v) = \Pr(U \le u, V \le v)$$

for a pair of uniform random variables is called its *copula*. This name is apt: a copula joins two one-dimensional distributions into a joint distribution. It also makes precise what information is missing when the marginals are known.[2]

What conditions must a function $C : [0,1]^2 \to [0,1]$ satisfy in order to be a copula? Its restrictions to the edges are forced by the uniform marginals:

$$C(u,0) = C(0,v) = 0, \qquad C(u,1) = u, \qquad C(1,v) = v. \tag{93.5}$$

The remaining condition says that every rectangle has nonnegative probability. If $u_1 \le u_2$ and $v_1 \le v_2$, then

$$\begin{aligned} \Delta C &:= C(u_2,v_2) - C(u_1,v_2) - C(u_2,v_1) + C(u_1,v_1) \\ &= \Pr(u_1 < U \le u_2,\ v_1 < V \le v_2) \ge 0. \end{aligned} \tag{93.6}$$

A function satisfying the boundary conditions and this rectangle inequality is a copula. The requirement $\Delta C \ge 0$ is sometimes called *2-increasingness*; it is the two-dimensional analogue of the monotonicity of an ordinary distribution function.

Every copula obeys remarkably simple bounds. For the events

$$A = \{U \le u\}, \qquad B = \{V \le v\}$$

we have $\Pr(A) = u$ and $\Pr(B) = v$. Since an intersection is contained in either event,

$$C(u,v) = \Pr(A \cap B) \le \min(u,v).$$

On the other hand, inclusion–exclusion and $\Pr(A \cup B) \le 1$ give

$$C(u,v) = u + v - \Pr(A \cup B) \ge u + v - 1.$$

Nonnegative probabilities must also be at least zero. Combining these inequalities,

$$\boxed{\max(u+v-1,0) \le C(u,v) \le \min(u,v).} \tag{93.7}$$

These are the *Fréchet–Hoeffding bounds*. Neither can be improved, because the constructions $V = 1 - U$ and $V = U$ attain them for all u, v. The independent coupling sits between the extremes:

$$W(u,v) = \max(u+v-1,0), \qquad \Pi(u,v) = uv, \qquad M(u,v) = \min(u,v).$$

Notice how little we needed to prove the inequalities. We did not assume that a joint density existed. This matters because the extreme copulas W and M are supported on diagonal lines, sets having two-dimensional area zero.

[2] The proofs and standard terminology here follow Roger B. Nelsen, *An Introduction to Copulas*, 2nd ed. (2006), especially Chapter 2.

§93.6 Moving between marginals and dependence

The example with uniform random variables is much more general than it first appears. Suppose X, Y have marginal distribution functions

$$F(x) = \Pr(X \le x), \qquad G(y) = \Pr(Y \le y)$$

and joint distribution function $H(x, y) = \Pr(X \le x, Y \le y)$.

For the moment suppose F, G are continuous. Then the probability integral transforms

$$U = F(X), \qquad V = G(Y)$$

are both uniform on $[0, 1]$. To see this, take any $u \in (0, 1)$, choose a quantile x_u with $F(x_u) = u$, and use the monotonicity and continuity of F:

$$\Pr(F(X) \le u) = F(x_u) = u.$$

If F has a constant interval, equality between the two threshold events may fail there pointwise, but only on a set to which X assigns probability zero. The same argument applies to G.

Let C be the joint distribution function of (U, V). By the same almost-sure equivalence of threshold events,

$$H(x, y) = C(F(x), G(y)). \tag{93.8}$$

Furthermore, because continuous distribution functions attain every value strictly between 0 and 1, the formula determines C uniquely.

Theorem 93.6.1 (Sklar's theorem)

Given a bivariate distribution function H with marginal distribution functions F, G, there exists a copula C such that

$$H(x, y) = C(F(x), G(y)).$$

If F, G are continuous, C is unique. For general marginals the values of C are uniquely determined only on $\mathrm{Ran}(F) \times \mathrm{Ran}(G)$. Conversely, every copula C and pair of distribution functions F, G define a bivariate distribution function by this formula.

We have proved the continuous-marginal case. For discontinuous marginals, one first defines a *subcopula* on $\mathrm{Ran}(F) \times \mathrm{Ran}(G)$ and then extends it to $[0, 1]^2$, as in the standard proof of Sklar's theorem. The extension is generally nonunique.

There is also a constructive converse. Given a copula C, sample a uniform pair (U, V) with joint law C, and set

$$X = F^{-1}(U), \qquad Y = G^{-1}(V),$$

using generalized quantile functions $F^{-1}(u) = \inf\{x : F(x) \ge u\}$ and similarly for G. The resulting variables have exactly the prescribed marginals, and their joint distribution is $C(F(x), G(y))$.

An explicit discrete example shows why the uniqueness qualification matters. Let U be uniform on $[0, 1]$, and define a measure-preserving map

$$\tau(u) = \begin{cases} \frac{1}{2} - u, & 0 \le u \le \frac{1}{2}, \\ \frac{3}{2} - u, & \frac{1}{2} < u \le 1. \end{cases}$$

Both pairs (U, U) and $(U, \tau(U))$ have uniform marginals, but their copulas differ: for the former,

$$\Pr(U \le \tfrac{1}{4}, V \le \tfrac{1}{4}) = \frac{1}{4},$$

whereas for the latter the same probability is zero. Now threshold both coordinates at $1/2$, defining $X = \mathbf{1}_{\{U>1/2\}}$ and $Y = \mathbf{1}_{\{V>1/2\}}$. Because τ preserves each half of the unit interval, both constructions give $X = Y$, each a fair Bernoulli variable. Their *discrete* joint distributions are identical even though the underlying copulas are not.

Sklar's theorem separates two choices that were tangled together in the original random-chord puzzle: which distribution to assign to each coordinate, and how to couple those coordinates.

§93.7 A dependence with zero correlation

There is a particularly memorable way to see how much a correlation coefficient can miss. Let

$$X \sim \mathrm{Unif}(-1, 1), \qquad Y = X^2.$$

The relation between X and Y is deterministic. Yet symmetry gives

$$\mathrm{Cov}(X, Y) = \mathbb{E}[X^3] - \mathbb{E}X\,\mathbb{E}[X^2] = 0.$$

Since both variances are positive, their Pearson correlation is exactly zero. Independence would imply that every event in X is unrelated to every event in Y, which is plainly false here. For example,

$$\Pr(Y \le \tfrac{1}{4} \mid |X| \le \tfrac{1}{2}) = 1, \qquad \Pr(Y \le \tfrac{1}{4}) = \frac{1}{2}.$$

The marginal distribution functions are explicit:

$$F_X(x) = \frac{x+1}{2} \quad (-1 \le x \le 1), \qquad F_Y(y) = \sqrt{y} \quad (0 \le y \le 1).$$

Applying their probability integral transforms yields

$$U = F_X(X) = \frac{X+1}{2}, \qquad V = F_Y(Y) = |X| = |2U - 1|. \tag{93.9}$$

Both U, V are uniform on $[0, 1]$, but their joint law is concentrated on a V-shaped curve in the unit square.

We can compute its copula directly. For $0 \le u, v \le 1$,

$$\begin{aligned} C(u, v) &= \Pr(X \le 2u - 1,\ -v \le X \le v) \\ &= \frac{1}{2} \max\{0, \min(2u - 1, v) + v\} \\ &= \boxed{\max\left\{0, \min\left(v, u + \frac{v-1}{2}\right)\right\}}. \end{aligned}$$

At $(u, v) = (1/4, 1/2)$ this copula equals zero, whereas independence would give $uv = 1/8$.

Now suppose we tried to recover the joint density by differentiating:

$$f_{X,Y}(x, y) \overset{?}{=} c(F_X(x), F_Y(y))\, f_X(x) f_Y(y), \qquad c(u, v) = \frac{\partial^2 C}{\partial u\, \partial v}.$$

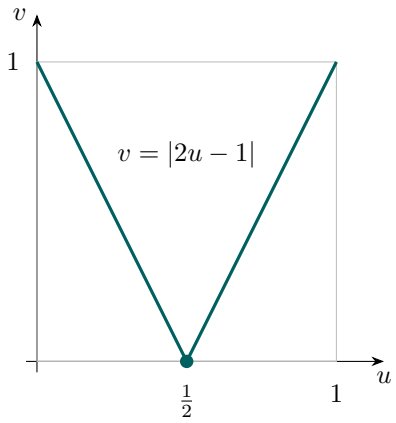


Figure 93.3: A singular copula with uniform marginal distributions and zero correlation in the original variables: probability lies on the two line segments $v = |2u - 1|$.

That formula requires an *absolutely continuous copula* and appropriate marginal densities. It does not apply here. The entire joint distribution of (X, Y) lies on the parabola $y = x^2$, which has two-dimensional Lebesgue measure zero. Thus no ordinary two-dimensional joint density $f_{X,Y}$ can describe it. There are perfectly ordinary one-dimensional marginal densities:

$$f_X(x) = \tfrac{1}{2}\mathbf{1}_{(-1,1)}(x), \qquad f_Y(y) = \frac{1}{2\sqrt{y}}\mathbf{1}_{(0,1)}(y).$$

The second formula follows by differentiating $F_Y(y) = \sqrt{y}$, or by adding the contributions from the two inverse branches $x = \pm\sqrt{y}$.

This is also a useful place to distinguish conditioning on a *value* from dividing by the probability of a point. For $0 < y < 1$, the conditional distribution of X given $Y = y$, in the regular conditional-distribution sense, is

$$\mathcal{L}(X \mid Y = y) = \frac{1}{2}\delta_{-\sqrt{y}} + \frac{1}{2}\delta_{\sqrt{y}}. \tag{93.10}$$

The two preimages have equal Jacobian factors $1/(2\sqrt{y})$ and equal original densities. Hence

$$\mathbb{E}[X \mid Y = y] = 0, \qquad \mathbb{E}[X^2 \mid Y = y] = y.$$

No event $\{Y = y\}$ has positive probability; the displayed conditional law specifies the relevant disintegration rather than an elementary ratio of point probabilities. Conversely, $\mathbb{E}[Y \mid X = x] = x^2$. The direction in which we condition matters enormously.

§93.8 The geometry of choosing a coupling

A coupling can also be selected by asking it to optimize a mathematical quantity. This is the idea behind optimal transport. For example, among all pairs of uniform $U, V \in [0, 1]$, how small or large can

$$\mathbb{E}[(U - V)^2]$$

be?

We have $\mathbb{E}U^2 = \mathbb{E}V^2 = 1/3$, so

$$\mathbb{E}[(U - V)^2] = \frac{2}{3} - 2\mathbb{E}[UV]. \tag{93.11}$$

It is clearly nonnegative. Equality is attained by the diagonal coupling $V = U$, for which the cost is zero.

For the maximum, use another nonnegative square:

$$0 \le \mathbb{E}[(U + V - 1)^2] = 2\mathbb{E}[UV] - \frac{1}{3}.$$

Thus $\mathbb{E}[UV] \ge 1/6$, and (93.11) gives the sharp upper bound $1/3$. Equality is attained by $V = 1 - U$. For independent U, V, $\mathbb{E}[UV] = 1/4$ and the cost is $1/6$. We have proved, with no appeal to a general transport theorem, that

$$0 \le \mathbb{E}[(U - V)^2] \le \frac{1}{3}, \qquad \begin{aligned} V = U &: \ 0, \\ U \perp V &: \ \frac{1}{6}, \\ V = 1 - U &: \ \frac{1}{3}. \end{aligned}$$

This is a small instance of a more general phenomenon: fixing the marginals leaves an entire geometric space of couplings, and a cost function can select one of them. In one dimension, quantile coupling supplies the optimal alignment for many convex costs.

One last construction is worth keeping. Let $Z \sim \mathrm{Unif}[0, 1]$ and independently let J be Bernoulli with parameter λ. Set

$$(U, V) = \begin{cases} (Z, Z), & J = 1, \\ (Z, 1 - Z), & J = 0. \end{cases}$$

Both marginals remain uniform, and

$$\mathrm{Corr}(U, V) = 2\lambda - 1, \qquad \mathbb{E}[(U - V)^2] = \frac{1 - \lambda}{3}.$$

Indeed, the covariance is the weighted average of $1/12$ and $-1/12$, while each marginal variance is $1/12$. At $\lambda = 1/2$, the correlation vanishes, but the pair is confined to the *union of two diagonals*. It is far from independent. Even a numerical summary of dependence that reaches exactly zero can conceal a singular geometric arrangement.

Our two opening questions now have a common mathematical language. A chord construction pushes a chosen measure on parameters forward to geometric outcomes. A joint experiment selects a coupling of its marginal laws. Both require information that the description of the possible outcomes alone does not contain. Sometimes invariance or an optimization principle supplies that information; sometimes the physical sampling mechanism must supply it.

VIII

Calculus in Several Variables

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Tangency, Discriminants, and Critical Values of a Projection

N-20220315

A parameter-dependent equation can have one, two, or several real solutions as its parameter changes. The number of solutions can change when roots collide or escape through the boundary of the admissible domain. In smooth geometry this is a critical-value phenomenon: a curve ceases to project regularly onto its parameter axis. The same rank condition lies behind constrained extrema.

§94.1 From a double root to a critical parameter

Consider

$$F(x,t) = x^2 - t = 0.$$

For $t < 0$ there are no real roots; at $t = 0$ the two branches meet; for $t > 0$ the roots are $x = \pm\sqrt{t}$. At the transition,

$$F(0,0) = 0, \qquad F_x(0,0) = 0,$$

while $F_t(0,0) = -1 \neq 0$. The zero set is a smooth parabola $t = x^2$; only its projection onto the t-axis has become singular at the vertex.

For a smooth family $F(x,t) = 0$, suppose $F_t \neq 0$. The implicit function theorem solves locally for $t = t(x)$, and differentiation gives

$$t'(x) = -\frac{F_x}{F_t}.$$

Therefore a critical point of the parameter projection satisfies $F = F_x = 0$. If in addition $F_{xx} \neq 0$, the local Taylor expansion has the form

$$t - t_0 = -\frac{F_{xx}(x_0,t_0)}{2F_t(x_0,t_0)}(x - x_0)^2 + o((x - x_0)^2).$$

The quadratic leading term explains why pairs of solutions generically appear or disappear with square-root scaling.

§94.2 The discriminant describes a concrete bifurcation

A more instructive example is

$$F(x,t) = x^3 - 3x + t.$$

The derivative $F_x = 3(x^2 - 1)$ vanishes at $x = \pm 1$. The corresponding parameter values follow from $F = 0$:

$$(x,t) = (-1,-2), \qquad (1,2).$$

For $-2 < t < 2$, the cubic has a local maximum of value $t + 2 > 0$ and a local minimum of value $t - 2 < 0$, hence three distinct real roots. For $|t| > 2$ it has exactly one real root. At $t = \pm 2$, two roots coalesce and a third remains simple.

The discriminant of $x^3 - 3x + t$ is

$$\Delta(t) = 108 - 27t^2 = 27(4 - t^2).$$

Its zeros locate precisely the two transition values. Algebraically, the discriminant vanishes when F and F_x share a root. Geometrically, those are the points at which the map from the solution curve to the parameter line develops a critical value.

For higher-degree polynomials, the *resultant* $\mathrm{Res}_x(F, F_x)$ generalizes this test. A zero resultant detects a common root over an algebraic closure, but whether that root is real and lies in the problem's domain must still be checked.

§94.3 Constrained extrema are the same rank failure

Let $g : \mathbb{R}^2 \to \mathbb{R}$ have nonzero gradient on the constraint curve $g(x, y) = 0$, and let h be the objective. The tangent space at a constraint point consists of vectors v satisfying

$$Dg(v) = \nabla g \cdot v = 0.$$

At a constrained extremum, the first variation of h vanishes along every such direction. Therefore ∇h is proportional to ∇g:

$$\nabla h = \lambda \nabla g.$$

Equivalently, the Jacobian of the map $(g, h) : \mathbb{R}^2 \to \mathbb{R}^2$ has rank below two:

$$\det \begin{pmatrix} g_x & g_y \\ h_x & h_y \end{pmatrix} = 0.$$

This is the constrained analogue of $F_x = 0$ for a parameter projection. In both cases, a map to the quantity being measured loses its local first-order freedom.

§94.4 Stationary points do not imply global extrema

Consider the objective $h(x, y) = x + y$ on the curve

$$g(x, y) = x^2 + y^3 - 2 = 0.$$

Since $\nabla g = (2x, 3y^2)$ never vanishes on this curve, every constrained stationary point satisfies the multiplier equations. Eliminating the multiplier gives

$$2x = 3y^2, \qquad x = \frac{3}{2}y^2,$$

and therefore

$$\frac{9}{4}y^4 + y^3 = 2.$$

There are exactly two real solutions for y: one positive and one below $-1/3$. To see this, differentiate the left-hand side:

$$\frac{d}{dy}\left(\frac{9}{4}y^4 + y^3\right) = 3y^2(3y + 1).$$

It decreases on $(-\infty, -1/3)$, increases on $(-1/3, \infty)$, and takes a value below 2 at its minimum. The two solutions thus yield the only constrained stationary points.

To classify them, work on the positive-x branch

$$x = \sqrt{2 - y^3}, \qquad H(y) = y + \sqrt{2 - y^3}.$$

At a stationary point $x = 3y^2/2$, one obtains

$$H''(y) = -\frac{2(3y + 1)}{3y^2}.$$

The negative solution gives $H'' > 0$, a local minimum; the positive one gives $H'' < 0$, a local maximum. The negative-x branch has no stationary points.

Neither is a global extremum on the entire constraint curve. Taking $y = -R$ with $R \to \infty$ gives

$$x = \pm\sqrt{2 + R^3}, \qquad h(x, y) = \pm\sqrt{2 + R^3} - R,$$

which tends to $+\infty$ and $-\infty$ along the two branches. The feasible set is noncompact, so local stationarity says nothing about existence of a global maximum or minimum.

A discriminant tells us where algebraic branches collide; the implicit function theorem explains their local geometry; Lagrange multipliers detect the same loss of rank along a constraint. The distinctions among local, global, real, and admissible solutions determine which conclusions actually follow from the equations.

Fréchet Differentiability, Hessians, and Constrained Tests

N-20220405

§95.1 Fréchet versus Complex Differentiability

Complex differentiability is much stronger than ordinary real differentiability. In one real direction a limit may exist, whereas in the complex plane the difference quotient must have the same limit along every direction.

Consider

$$f(z) = \overline{z}, \qquad z = a + bi.$$

The complex derivative would be

$$f'(z) = \lim_{h \to 0} \frac{\overline{z+h} - \overline{z}}{h} = \lim_{h \to 0} \frac{\overline{h}}{h}.$$

Along the real direction $h = c \to 0$,

$$\frac{\overline{h}}{h} = \frac{c}{c} = 1.$$

Along the imaginary direction $h = di \to 0$,

$$\frac{\overline{h}}{h} = \frac{-di}{di} = -1.$$

Along the diagonal direction $h = c + ci$,

$$\frac{\overline{h}}{h} = \frac{c - ci}{c + ci} = -i.$$

The directional limits disagree, so $f(z) = \overline{z}$ is not complex differentiable. This is the basic reason complex differentiability is rigid.

If $f(z) = u(x, y) + iv(x, y)$, the Cauchy–Riemann equations are

$$u_x = v_y, \qquad u_y = -v_x.$$

When these hold with suitable regularity, the complex derivative is

$$f'(z_0) = u_x(x_0, y_0) + iv_x(x_0, y_0).$$

§95.2 Fréchet derivative

Let V, W be normed vector spaces, and let $U \subseteq V$ be open. A function $f : U \to W$ is Fréchet differentiable at $x \in U$ if there exists a bounded linear operator $A : V \to W$ such that

$$\lim_{\|h\| \to 0} \frac{\|f(x+h) - f(x) - Ah\|_W}{\|h\|_V} = 0.$$

Equivalently,

$$f(x + h) = f(x) + Ah + o(\|h\|).$$

The operator A is unique and is denoted

$$Df(x) = A.$$

The note compares this with the Gâteaux differential. If $k = \varepsilon h$, then

$$f(x + \varepsilon h) = f(x) + \varepsilon (Df)h + h\, o(\varepsilon),$$

and, in the limit, one gets a directional derivative. But the Fréchet derivative is stronger: it requires one linear map to approximate the function uniformly in all directions.

§95.3 Visualizing complex maps

A conformal grid illustrates how a rectangular coordinate net is sent to curved orthogonal families under maps such as

$$f(z) = z^2, \qquad f(z) = \frac{1}{z}, \qquad f(z) = \sin z.$$

The key geometric point is that holomorphic maps with nonzero derivative preserve angles locally. They may bend the grid, but they preserve the orthogonality of the two coordinate families.

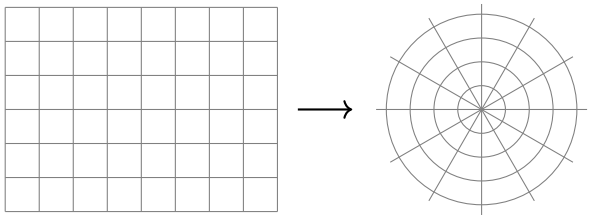


Figure 95.1: A rectangular grid and its image under a locally conformal map. The curves meet orthogonally where the derivative is nonzero.

§95.4 Hessian matrices and Taylor expansion

The second main topic is the relationship between Hessians, Taylor expansion, and extrema. For a smooth function $f : \mathbb{R}^n \to \mathbb{R}$, Taylor expansion at a critical point p gives

$$f(p + h) = f(p) + \frac{1}{2} h^T H_f(p) h + O(\|h\|^3).$$

Thus the quadratic form associated to the Hessian controls the local behavior.

For unconstrained extrema, positive definiteness of $H_f(p)$ gives a local minimum, negative definiteness gives a local maximum, and indefiniteness gives a saddle. For constrained extrema the situation is subtler, because one must restrict the second variation to the tangent space of the constraint.

§95.5 Bordered Hessian

For a constrained problem

$$\text{extremize } f(x) \quad \text{subject to } g_1(x) = \cdots = g_m(x) = 0,$$

one introduces the Lagrangian

$$\mathcal{L}(x,\lambda) = f(x) - \sum_{i=1}^{m} \lambda_i g_i(x).$$

The bordered Hessian packages the second derivatives of $\mathcal{L}$ together with the first derivatives of the constraints:

$$H_B = \begin{pmatrix} 0 & Dg \\ Dg^{T} & D_x^2 \mathcal{L} \end{pmatrix}.$$

The sign pattern of certain principal minors then diagnoses constrained maxima and minima.

§95.6 Example

The note considers the inequality problem

$$a,b,c>0, \qquad ab+bc+ca = a+b+c,$$

and asks to prove

$$\sqrt{ab}+\sqrt{bc}+\sqrt{ca}-\sqrt{abc} \ge 2.$$

Let

$$f(a,b,c) = \sqrt{ab}+\sqrt{bc}+\sqrt{ca}-\sqrt{abc},$$

and

$$g(a,b,c) = a+b+c-ab-bc-ca.$$

The Lagrangian is

$$F(a,b,c,\lambda) = f(a,b,c) + \lambda g(a,b,c).$$

The first-order equations are

$$\begin{cases} \dfrac{\sqrt{abc}-\sqrt{ab}+\sqrt{ac}}{2a} + \lambda(1-b-c) = 0, \\ \dfrac{\sqrt{abc}+\sqrt{ab}+\sqrt{bc}}{2b} + \lambda(1-a-c) = 0, \\ \dfrac{\sqrt{abc}+\sqrt{ac}+\sqrt{bc}}{2c} + \lambda(1-a-b) = 0, \\ a+b+c-ab-bc-ca = 0. \end{cases}$$

The symmetric point

$$a=b=c=1$$

is a critical point. At that point the bordered Hessian has the schematic form

$$H_B = \begin{pmatrix} 0 & -1 & -1 & -1 \\ -1 & -\tfrac{1}{4} & 0 & 0 \\ -1 & 0 & -\tfrac{1}{4} & 0 \\ -1 & 0 & 0 & -\tfrac{1}{4} \end{pmatrix}.$$

The relevant minors have the required sign pattern, so the symmetric point gives the constrained extremum. Since

$$f(1,1,1) = 2,$$

we obtain the desired inequality.

§95.7 Finite fields and Galois theory

A final aside in the note jumps to finite fields and Galois theory. The basic facts listed are:

(i) the finite field $\mathbb{F}_p$ has arithmetic modulo p;

(ii) polynomial factorization over $\mathbb{F}_p$ can differ dramatically from factorization over $\mathbb{Q}$;

(iii) adjoining roots creates finite extensions such as $\mathbb{F}_{p^n}$;

(iv) quotient rings $\mathbb{F}_p[x]/(q(x))$ construct finite fields when q is irreducible.

Thus the same algebraic idea reappears: to solve an equation, enlarge the field, then study the symmetries of the roots.

§95.8 Why complex analysis meets Hessians

Complex differentiability and Hessian tests share a common theme: first-order information determines the allowed linear approximation, while second-order information controls what happens after the first-order obstruction vanishes.

For $f(z) = \overline{z}$, the real derivative exists as a real-linear map, but complex differentiability asks for multiplication by one complex number. The failure is exactly the failure of a real-linear map to commute with multiplication by i. In modern language, complex differentiability means the derivative is $\mathbb{C}$-linear, not merely $\mathbb{R}$-linear.

For constrained extrema, the first-order condition removes the tangent freedom, so the Hessian test must be performed only on tangent directions to the constraint. This is why the bordered Hessian appears: it is not a mysterious determinant recipe, but a bookkeeping device for second variation under first-order constraints.

§95.9 Frechet derivative as linear approximation

For a map $F : V \to W$ between normed spaces, differentiability at a means there is a bounded linear map L such that

$$F(a+h) = F(a) + Lh + o(\|h\|).$$

This L is the Frechet derivative. It is stronger and cleaner than checking directional derivatives separately, because it requires one linear map to approximate the function uniformly in all directions.

§95.10 Hessian as second derivative

For scalar-valued functions, the Hessian is the derivative of the gradient. It records the quadratic part of the Taylor expansion:

$$f(a+h) = f(a) + Df(a)h + \frac{1}{2}h^T H_f(a)h + o(\|h\|^2).$$

At a critical point, the linear term vanishes, so the Hessian controls the first visible local shape.

Composition and Elimination: Higher Derivatives, Determinants, and Schur Complements

N-20220409

§96.1 What remains when a variable is allowed to adjust?

Consider an energy with two real variables:

$$\Phi(x, z) = x^2 + xz + \frac{3}{2}z^2.$$

If we hold z fixed and allow x to find its cheapest position, completing the square gives

$$\Phi(x, z) = \left(x + \frac{z}{2}\right)^2 + \frac{5}{4}z^2.$$

The minimizing hidden coordinate is $x_*(z) = -z/2$, and the remaining energy is

$$g(z) = \min_x \Phi(x, z) = \frac{5}{4}z^2, \qquad g''(z) = \frac{5}{2}.$$

Why is the surviving curvature $5/2$ rather than the original second derivative $\Phi_{zz} = 3$? The hidden coordinate moves when z moves. Its response absorbs part of the energy change.

There are two ways to reach the answer, and they will meet in a useful theorem. Differentiating through the moving solution requires the chain rule and, at higher orders, careful management of derivatives of derivatives. Eliminating the hidden variable requires a little linear algebra; for a quadratic energy it produces a *Schur complement*. The equality between those computations is a precise geometric fact.

At the scalar level we can already see it:

$$\begin{pmatrix} \Phi_{xx} & \Phi_{xz} \\ \Phi_{zx} & \Phi_{zz} \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, \qquad \boxed{g'' = 3 - \frac{1^2}{2} = \frac{5}{2}.}$$

First we need to understand how derivatives compose. Then we can ask why eliminating variables changes the Hessian in exactly this way.

§96.2 Repeated derivations produce the higher Leibniz rule

Let D be a derivation on a commutative algebra, so

$$D(ab) = D(a)b + aD(b).$$

Applying D repeatedly gives

$$D^2(ab) = D^2(a)b + 2D(a)D(b) + aD^2(b),$$

and

$$D^3(ab) = D^3(a)b + 3D^2(a)D(b) + 3D(a)D^2(b) + aD^3(b).$$

The coefficients are not accidental.

Theorem 96.2.1 (Higher Leibniz rule)

For every integer $n \geq 0$,

$$D^n(ab) = \sum_{k=0}^{n} \binom{n}{k} D^k(a) D^{n-k}(b).$$

Proof. The result is clear for $n = 0$. Assume it holds for n. Differentiating once more gives

$$D^{n+1}(ab) = \sum_{k=0}^{n} \binom{n}{k} (D^{k+1}(a)D^{n-k}(b) + D^k(a)D^{n-k+1}(b)).$$

After shifting the index in the first sum, the coefficient of $D^k(a)D^{n+1-k}(b)$ becomes

$$\binom{n}{k-1} + \binom{n}{k} = \binom{n+1}{k}.$$

This proves the induction step. □

There is also a combinatorial reading. During n differentiations, each derivative must land on either the first factor or the second. Choosing the k stages that land on a gives $\binom{n}{k}$. The formula is therefore a count of derivative-placement histories.

For a product of r factors, the same argument gives the multinomial rule

$$D^n(a_1 \cdots a_r) = \sum_{\alpha_1 + \cdots + \alpha_r = n} \frac{n!}{\alpha_1! \cdots \alpha_r!} \prod_{j=1}^{r} D^{\alpha_j}(a_j).$$

This form will reappear when differentiating a determinant row by row.

§96.3 Faà di Bruno records partitions of derivative labels

The ordinary chain rule has one outer derivative and one inner derivative:

$$(f \circ g)' = f'(g)g'.$$

At second and third order,

$$(f \circ g)'' = f''(g)(g')^2 + f'(g)g'',$$

$$(f \circ g)''' = f'''(g)(g')^3 + 3f''(g)g'g'' + f'(g)g'''.$$

The coefficients and monomials are controlled by set partitions. Think of the n derivative operations as carrying labels $1, \ldots, n$. Labels in the same block ultimately combine into one derivative of g; the number of blocks tells us which derivative of f appears.

For nonnegative integers $k_1, \ldots, k_n$, impose

$$k_1 + 2k_2 + \cdots + nk_n = n.$$

Here k_j is the number of blocks of size j. The total number of blocks is

$$m = k_1 + \cdots + k_n.$$

The number of set partitions with these block sizes is

$$\frac{n!}{k_1! \cdots k_n! (1!)^{k_1} \cdots (n!)^{k_n}}.$$

This gives the one-variable Faà di Bruno formula.

Theorem 96.3.1 (Faà di Bruno)

For sufficiently differentiable functions f and g,

$$\frac{d^n}{dx^n} f(g(x)) = \sum_{\substack{k_1,\ldots,k_n \geq 0 \\ k_1 + 2k_2 + \cdots + nk_n = n}} \frac{n!}{k_1! \cdots k_n!} f^{(k_1+\cdots+k_n)}(g(x)) \prod_{j=1}^{n} \left(\frac{g^{(j)}(x)}{j!} \right)^{k_j}.$$

Proof. There is a short way to derive all the coefficients at once. Expand $g(x+h)$ around x and then expand f around $g(x)$, retaining only powers through h^n:

$$f(g(x+h)) \equiv \sum_{m=0}^{n} \frac{f^{(m)}(g(x))}{m!} \left(\sum_{j=1}^{n} \frac{g^{(j)}(x)}{j!} h^j \right)^m \pmod{h^{n+1}}.$$

To obtain h^n in the term with outer derivative $f^{(m)}$, choose k_j copies of the degree-j term. Necessarily

$$\sum_j k_j = m, \qquad \sum_j j k_j = n.$$

The multinomial coefficient for these choices is $m! / \prod_j k_j!$. It cancels the prefactor $1/m!$. Multiplying the coefficient of h^n by $n!$ gives exactly the stated formula. The blocks in the combinatorial proof are the same selections, with derivative labels restored. □

The same information is often compressed using the partial exponential Bell polynomials

$$B_{n,m}(x_1, \ldots, x_{n-m+1}) = \sum \frac{n!}{k_1! \cdots k_{n-m+1}!} \prod_{j=1}^{n-m+1} \left(\frac{x_j}{j!} \right)^{k_j},$$

where the sum satisfies

$$\sum_j k_j = m, \qquad \sum_j j k_j = n.$$

Then

$$(f \circ g)^{(n)} = \sum_{m=1}^{n} f^{(m)}(g) B_{n,m}(g', g'', \ldots, g^{(n-m+1)}).$$

Bell polynomials separate the combinatorics of composition from the particular functions being composed.

§96.4 A Fourth-Derivative Calculation

The integer partitions of 4 are

$$1+1+1+1, \qquad 2+1+1, \qquad 2+2, \qquad 3+1, \qquad 4.$$

They correspond respectively to the block-size data

$$(k_1,k_2,k_3,k_4) = (4,0,0,0),(2,1,0,0),(0,2,0,0),(1,0,1,0),(0,0,0,1).$$

Substitution into Faà di Bruno gives

$$\begin{aligned}(f\circ g)^{(4)} = {} & f^{(4)}(g)(g')^4 + 6f^{(3)}(g)(g')^2g'' + 3f''(g)(g'')^2 \\ & + 4f''(g)g'g''' + f'(g)g^{(4)}.\end{aligned}$$

The coefficients $1,6,3,4,1$ count set partitions of the five displayed block types. For example, the type $2+1+1$ has

$$\frac{4!}{1!2!(2!)^1} = 6$$

partitions, while the type $2+2$ has

$$\frac{4!}{2!(2!)^2} = 3.$$

Take

$$h(x) = e^{\sin x}.$$

Since every derivative of $f(u) = e^u$ equals e^u,

$$\begin{aligned}h^{(4)}(x) = e^{\sin x}\big(& (\cos x)^4 + 6(\cos x)^2(-\sin x) + 3(-\sin x)^2 \\ & + 4(\cos x)(-\cos x) + \sin x\big).\end{aligned}$$

At $x=0$, this reduces to

$$h^{(4)}(0) = 1 - 4 = -3.$$

The Taylor series confirms the calculation:

$$e^{\sin x} = 1 + x + \frac{x^2}{2} - \frac{x^4}{8} + O(x^5),$$

so the fourth derivative at the origin is

$$4!\left(-\frac{1}{8}\right) = -3.$$

This check is useful because a missing block type in Faà di Bruno usually appears as an incorrect Taylor coefficient.

§96.5 Jets as Truncated Taylor Algebra

Leibniz notation becomes subtle beyond first order because the symbols $dx, d^2x, \dots$ do not behave like ordinary powers of one number. The clean way to see what they encode is to work with Taylor polynomials modulo terms that are too small to matter at the chosen order.

Fix a point a. Two smooth functions have the same n-jet at a when their derivatives through order n agree there. The jet can be represented by the truncated Taylor polynomial

$$j_a^n f(h) = f(a) + f'(a)h + \frac{f''(a)}{2!}h^2 + \cdots + \frac{f^{(n)}(a)}{n!}h^n,$$

where calculations are performed modulo h^{n+1}. In other words, one works in the truncated polynomial algebra

$$\mathbb{R}[h]/(h^{n+1}).$$

Terms of degree greater than n are discarded because they cannot affect derivatives of order at most n.

This makes composition completely explicit. Suppose $g(a) = b$, and write the second-order jets as

$$j_a^2 g(h) = b + b_1 h + \frac{b_2}{2}h^2,$$
$$j_b^2 f(u) = c + c_1 u + \frac{c_2}{2}u^2.$$

Here

$$b_1 = g'(a), \qquad b_2 = g''(a), \qquad c_1 = f'(b), \qquad c_2 = f''(b).$$

Set

$$u = b_1 h + \frac{b_2}{2}h^2.$$

Since

$$u^2 = b_1^2 h^2 \pmod{h^3},$$

substitution gives

$$\begin{aligned} j_a^2(f \circ g)(h) &\equiv c + c_1 u + \frac{c_2}{2}u^2 \pmod{h^3} \\ &= c + c_1 b_1 h + \frac{c_2 b_1^2 + c_1 b_2}{2}h^2. \end{aligned}$$

Reading off coefficients recovers

$$(f \circ g)'(a) = f'(b)g'(a)$$

and

$$(f \circ g)''(a) = f''(b)(g'(a))^2 + f'(b)g''(a).$$

The second-order chain rule is therefore ordinary polynomial substitution followed by truncation.

At third order, write

$$j_a^3 g(h) = b + b_1 h + \frac{b_2}{2}h^2 + \frac{b_3}{6}h^3$$

and

$$j_b^3 f(u) = c + c_1 u + \frac{c_2}{2}u^2 + \frac{c_3}{6}u^3.$$

Modulo h^4,

$$u^2 = b_1^2 h^2 + b_1 b_2 h^3, \qquad u^3 = b_1^3 h^3.$$

Hence the cubic coefficient of the composition is

$$\frac{c_3 b_1^3 + 3c_2 b_1 b_2 + c_1 b_3}{6},$$

so

$$(f \circ g)'''(a) = f'''(b)(g'(a))^3 + 3f''(b)g'(a)g''(a) + f'(b)g'''(a).$$

The coefficient 3 is no longer mysterious: it comes from the two cross terms in u^2 together with the factorial normalization in the Taylor coefficients.

For a concrete check, take $f(u) = e^u$, $g(x) = \sin x$, and expand at 0. The relevant jets are

$$j_0^3 \sin(h) = h - \frac{h^3}{6}$$

and

$$j_0^3 e^u = 1 + u + \frac{u^2}{2} + \frac{u^3}{6}.$$

Substitution gives

$$j_0^3 (e^{\sin x})(h) = 1 + h + \frac{h^2}{2},$$

because the cubic contributions $-h^3/6$ and $+h^3/6$ cancel. Thus

$$(e^{\sin x})'''\big|_{x=0} = 0,$$

which agrees with the third-order chain rule.

The differential notation is a compressed version of this bookkeeping. If $y = f(x)$, then

$$dy = f'(x)\,dx$$

and

$$d^2y = f''(x)(dx)^2 + f'(x)d^2x.$$

When x itself depends on a parameter t, the first two pieces of its jet are

$$dx = x'(t)dt, \qquad d^2x = x''(t)(dt)^2.$$

Therefore

$$d^2y = (f''(x)(x')^2 + f'(x)x'')(dt)^2.$$

Only when x is chosen as an affine coordinate in the parameter does d^2x vanish. Under a nonlinear reparametrization it carries genuine second-order information. Jets explain precisely why treating d^2x as though it were merely $(dx)^2$ loses part of the chain rule.

§96.6 Differentiating Implicit Solutions

An equation may specify a function without ever solving for it. Suppose

$$F(x(z), z) = 0,$$

where x can itself be a vector and the square Jacobian F_x is invertible. The implicit function theorem gives a locally defined differentiable branch $x(z)$; differentiating the equation rather than explicitly solving it gives

$$F_x x' + F_z = 0, \qquad \boxed{x' = -F_x^{-1} F_z.}$$

For one real parameter z, differentiating again gives

$$F_x x'' + F_{xx}[x', x'] + 2F_{xz}x' + F_{zz} = 0,$$

hence

$$\boxed{x'' = -F_x^{-1}\big(F_{xx}[x', x'] + 2F_{xz}x' + F_{zz}\big)\,.}$$

The bracket $F_{xx}[x', x']$ means the second derivative in the direction x' twice. The expression is exactly the second-order chain rule, with one extra linear solve. At still higher orders the terms are organized by the same partitions that appear in Faà di Bruno's formula.

For an example that is small enough to check to third order, let

$$F(x, z) = x + x^2 - z, \qquad x(0) = 0.$$

Then $(1 + 2x)x' = 1$, so $x'(0) = 1$. Another derivative gives

$$(1 + 2x)x'' + 2(x')^2 = 0, \qquad x''(0) = -2.$$

A third derivative gives $(1 + 2x)x''' + 6x'x'' = 0$, so $x'''(0) = 12$. The resulting jet is

$$\boxed{x(z) = z - z^2 + 2z^3 + O(z^4),}$$

which also follows by expanding the positive branch

$$x(z) = \frac{-1 + \sqrt{1 + 4z}}{2}.$$

The implicit calculation scales to high-dimensional problems because each derivative order asks us to solve with the same linearized operator F_x. This will be especially helpful when the equation expresses stationarity of an energy.

§96.7 Parametric Differentiation

Let a regular plane curve be given by

$$x = x(t), \qquad y = y(t),$$

with $x'(t) \neq 0$. Differentiation with respect to x means applying

$$\frac{d}{dx} = \frac{1}{x'(t)}\frac{d}{dt}.$$

Hence

$$\frac{dy}{dx} = \frac{y'}{x'},$$

and

$$\frac{d^2y}{dx^2} = \frac{x'y'' - y'x''}{(x')^3}.$$

The numerator is a determinant:

$$x'y'' - y'x'' = \det\begin{pmatrix} x' & y' \\ x'' & y'' \end{pmatrix}.$$

Applying the operator once more gives

$$\frac{d^3y}{dx^3} = \frac{(x')^2y''' - 3x'x''y'' + \big(3(x'')^2 - x'x'''\big)y'}{(x')^5}.$$

The growing expression is another composition problem: one repeatedly composes multiplication by $1/x'$ with d/dt.

For the semicubical parabola

$$x = t^2, \qquad y = t^3, \qquad t > 0,$$

we obtain

$$\frac{dy}{dx} = \frac{3t}{2}, \qquad \frac{d^2y}{dx^2} = \frac{3}{4t}, \qquad \frac{d^3y}{dx^3} = -\frac{3}{8t^3}.$$

Since $y = x^{3/2}$ on this branch, direct differentiation gives exactly the same values. The singular behavior as $t \to 0$ records the failure of x to be a valid local coordinate at the cusp.

§96.8 Determinants inherit a multinomial Leibniz rule

Write an $m \times m$ matrix-valued function by rows:

$$A(t) = \begin{pmatrix} r_1(t) \\ \vdots \\ r_m(t) \end{pmatrix}.$$

Because the determinant is multilinear in its rows,

$$\frac{d}{dt} \det A(t) = \sum_{i=1}^{m} \det(r_1, \dots, r_i', \dots, r_m).$$

Repeated differentiation and the multinomial Leibniz rule give

$$\frac{d^n}{dt^n} \det A(t) = \sum_{\alpha_1 + \cdots + \alpha_m = n} \frac{n!}{\alpha_1! \cdots \alpha_m!} \det(r_1^{(\alpha_1)}, \dots, r_m^{(\alpha_m)}).$$

Thus the higher derivative of a determinant is organized by distributing n derivative operations among m rows.

At first order there is also a basis-independent formula. Let $\operatorname{adj}(A)$ denote the adjugate matrix. Cofactor expansion gives

$$\frac{d}{dt} \det A(t) = \operatorname{tr}\big(\operatorname{adj}(A(t)) A'(t)\big).$$

If $A(t)$ is invertible, then

$$\operatorname{adj}(A) = \det(A) A^{-1},$$

so

$$\boxed{\frac{d}{dt} \det A(t) = \det A(t) \operatorname{tr}\big(A(t)^{-1} A'(t)\big).}$$

This is Jacobi's formula. Equivalently,

$$\frac{d}{dt} \log \det A(t) = \operatorname{tr}\big(A(t)^{-1} A'(t)\big)$$

whenever the logarithm is defined.

For

$$A(t) = \begin{pmatrix} e^t & t \\ 0 & 1 + t \end{pmatrix},$$

we have

$$\det A(t) = e^t(1+t),$$

and therefore

$$\frac{d}{dt}\det A(t) = e^t(2+t).$$

Jacobi's formula gives the same result because

$$\operatorname{tr}(A^{-1}A') = 1 + \frac{1}{1+t}.$$

Multiplication by $\det A = e^t(1+t)$ yields $e^t(2+t)$.

§96.9 Matrix Products as Composition

Let

$$B : U \to V, \qquad A : V \to W$$

be linear maps. After bases are chosen, their composition is represented by the product AB. Its entries are

$$(AB)_{ij} = \sum_k A_{ik}B_{kj}.$$

The index k is internal: it labels the intermediate coordinate in V, and summation removes it from the final answer. This is a contraction between the input index of A and the output index of B.

Associativity follows because finite sums may be regrouped:

$$\begin{aligned}((AB)C)_{ij} &= \sum_k \left(\sum_\ell A_{i\ell}B_{\ell k}\right) C_{kj} \\ &= \sum_\ell A_{i\ell}\left(\sum_k B_{\ell k}C_{kj}\right) \\ &= (A(BC))_{ij}.\end{aligned}$$

Commutativity fails because reversing the order reverses the composition. Even when both products exist, AB and BA generally represent maps on different spaces or different operations on the same space.

Block multiplication is the same rule with each scalar entry replaced by a compatible linear map. For example,

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix}\begin{pmatrix} x \\ z \end{pmatrix} = \begin{pmatrix} Ax + Bz \\ Cx + Dz \end{pmatrix}.$$

This notation becomes powerful when one block of variables is to be eliminated.

§96.10 Schur complements are exact block elimination

Consider

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

where A is invertible. Multiplying on the left by a block elimination matrix gives

$$\begin{pmatrix} I & 0 \\ -CA^{-1} & I \end{pmatrix}\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} A & B \\ 0 & S \end{pmatrix},$$

where

$$\boxed{S = D - CA^{-1}B}$$

is the Schur complement of A in M.

This identity immediately gives

$$\det M = \det A \, \det S.$$

It also gives a block solve. For

$$Ax + Bz = b, \qquad Cx + Dz = c,$$

the first equation gives

$$x = A^{-1}(b - Bz).$$

Substitution into the second equation produces the reduced system

$$Sz = c - CA^{-1}b.$$

After solving for z, one recovers x by back-substitution. The Schur complement is exactly what remains after the variable x has been eliminated.

§96.11 A complete block-elimination example

Take

$$M = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 3 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

with

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad C = \begin{pmatrix} 1 & 0 \end{pmatrix}, \qquad D = (3).$$

Since

$$A^{-1} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix},$$

we obtain

$$CA^{-1}B = \frac{2}{3}, \qquad S = 3 - \frac{2}{3} = \frac{7}{3}.$$

Therefore

$$\det M = \det A \, S = 3 \cdot \frac{7}{3} = 7.$$

Now solve

$$M \begin{pmatrix} x_1 \\ x_2 \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}.$$

The reduced right-hand side is

$$2 - CA^{-1}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = 2 - \frac{2}{3} = \frac{4}{3}.$$

Hence

$$\frac{7}{3}z = \frac{4}{3}, \qquad z = \frac{4}{7}.$$

Back-substitution gives

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = A^{-1}\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \end{pmatrix}\frac{4}{7}\right) = A^{-1}\begin{pmatrix} 3/7 \\ 0 \end{pmatrix} = \begin{pmatrix} 2/7 \\ -1/7 \end{pmatrix}.$$

A direct multiplication verifies

$$M\begin{pmatrix} 2/7 \\ -1/7 \\ 4/7 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}.$$

The calculation shows concretely that the reduced scalar $7/3$ contains the full effect of the eliminated two-dimensional block.

§96.12 Block inverse and positive definiteness

If both A and S are invertible, block factorization gives

$$M^{-1} = \begin{pmatrix} A^{-1} + A^{-1}BS^{-1}CA^{-1} & -A^{-1}BS^{-1} \\ -S^{-1}CA^{-1} & S^{-1} \end{pmatrix}.$$

The formula is best remembered as elimination followed by back-substitution, not as four unrelated blocks.

For a symmetric block matrix

$$M = \begin{pmatrix} A & B \\ B^T & D \end{pmatrix}$$

with A positive definite, completing the square gives

$$\begin{pmatrix} x \\ z \end{pmatrix}^T M \begin{pmatrix} x \\ z \end{pmatrix} = (x + A^{-1}Bz)^T A(x + A^{-1}Bz) + z^T(D - B^TA^{-1}B)z.$$

Consequently,

$$M \succ 0 \quad \Longleftrightarrow \quad A \succ 0 \text{ and } D - B^TA^{-1}B \succ 0.$$

Thus the same Schur complement controls elimination, determinants, inverses, and positivity. In statistics it also appears as a conditional covariance; in constrained optimization it appears after eliminating primal or dual variables.

§96.13 Optimized Curvature and Schur Complements

Let $\Phi(x, z)$ be a twice continuously differentiable real function, with $x \in \mathbb{R}^p$ and $z \in \mathbb{R}^q$. Suppose at (x_0, z_0) we have

$$\nabla_x \Phi(x_0, z_0) = 0, \qquad A = \nabla^2_{xx}\Phi(x_0, z_0) \succ 0.$$

The implicit function theorem gives a local stationary branch $x_*(z)$ through x_0, and positive definiteness makes each nearby point a strict local minimum in the x-directions. Put

$$g(z) = \Phi(x_*(z), z)$$

and write the full Hessian, at a point of this branch, as

$$\nabla^2\Phi = \begin{pmatrix} A & B \\ B^T & D \end{pmatrix}, \qquad B = \nabla^2_{xz}\Phi.$$

Theorem 96.13.1 (Hessian after eliminating stationary variables)
At every point of the branch where A remains positive definite,

$$\boxed{\nabla^2 g(z) = D - B^T A^{-1} B.}$$

Proof. Differentiate the stationarity equation $\nabla_x\Phi(x_*(z), z) = 0$:

$$A\, D_z x_* + B = 0, \qquad D_z x_* = -A^{-1}B.$$

By the ordinary chain rule,

$$D_z g = (\nabla_x\Phi)^T D_z x_* + \nabla_z\Phi = \nabla_z\Phi.$$

This is the envelope theorem in its simplest form: the first-order movement of the minimizer makes no contribution because the x-gradient vanishes there. Differentiate once more:

$$\nabla^2 g = D + B^T D_z x_* = D - B^T A^{-1} B.$$

The Schur complement is therefore exactly the Hessian of the reduced energy. □

In particular $B^T A^{-1} B$ is positive semidefinite, so $\nabla^2 g \preceq D$: when a hidden variable is allowed to relax, the visible energy cannot become *stiffer* at second order than it was with that hidden coordinate held fixed.

Return to the two-variable energy from the beginning. With $A = 2$, $B = 1$, $D = 3$, the formula gives $g'' = 5/2$ without needing to complete a square. For the three-dimensional matrix of the block-elimination example, take

$$\Phi(x, z) = \frac{1}{2}\begin{pmatrix} x \\ z \end{pmatrix}^T \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 3 \end{pmatrix} \begin{pmatrix} x \\ z \end{pmatrix}, \qquad x \in \mathbb{R}^2.$$

The minimizing response is

$$x_*(z) = -A^{-1}Bz = \begin{pmatrix} -2/3 \\ 1/3 \end{pmatrix} z.$$

The effective energy becomes

$$g(z) = \frac{1}{2}\left(3 - \frac{2}{3}\right) z^2 = \frac{7}{6} z^2.$$

Its second derivative is $7/3$, precisely the Schur complement that previously appeared in the determinant and block solve. This is the promised meeting of implicit differentiation, Hessian geometry, and elimination.

§96.14 A Gaussian ellipse makes the elimination visible

The same quadratic calculation explains a familiar probability picture. Let

$$M = \begin{pmatrix} A & B \\ B^T & D \end{pmatrix} \succ 0, \qquad p(x,z) \propto \exp\left[-\frac{1}{2}\begin{pmatrix} x \\ z \end{pmatrix}^T M \begin{pmatrix} x \\ z \end{pmatrix}\right].$$

The matrix M is the *precision*, or inverse covariance, of a centered Gaussian distribution. Completing the square gives

$$\begin{pmatrix} x \\ z \end{pmatrix}^T M \begin{pmatrix} x \\ z \end{pmatrix} = (x + A^{-1}Bz)^T A (x + A^{-1}Bz) + z^T (D - B^T A^{-1} B) z.$$

For a fixed z, the conditional Gaussian $x \mid z$ is centered at $-A^{-1}Bz$ and has covariance A^{-1}. Integrating out x instead gives

$$\int_{\mathbb{R}^p} e^{-\frac{1}{2}(x+A^{-1}Bz)^T A (x+A^{-1}Bz)}\, dx = \frac{(2\pi)^{p/2}}{\sqrt{\det A}},$$

which does not depend on z. Therefore the *marginal* law of z has precision

$$\boxed{S = D - B^T A^{-1} B}$$

and covariance S^{-1}. Conditional behavior and marginal behavior are different, yet both are read off from the same completed square.

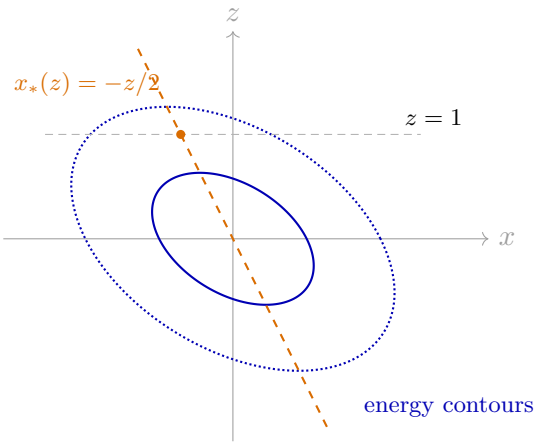


Figure 96.1: Exact contours of the joint Gaussian energy $\Phi(x,z) = x^2 + xz + \frac{3}{2}z^2$, at levels $\Phi = 1/2$ and $\Phi = 2$. Each horizontal slice has its lowest energy at $x = -z/2$ (orange line); eliminating x leaves the quadratic profile $g(z) = 5z^2/4$.

For this particular matrix, $x \mid z$ has mean $-z/2$ and variance $1/2$, whereas the marginal z has variance $2/5$. The ellipse encodes both statements. It also explains why eliminating hidden Gaussian variables is an exact operation, unlike approximating an arbitrary nonlinear energy by a quadratic.

§96.15 Differentiating through Elimination

The derivative rule for matrix inverses follows immediately from differentiating $A(t)A(t)^{-1} = I$:

$$\boxed{\frac{d}{dt}A^{-1} = -A^{-1}A'A^{-1}.}$$

If the blocks of $M(t)$ vary differentiably, then $S(t) = D - CA^{-1}B$ satisfies

$$\begin{aligned} S' = D' - C'A^{-1}B - CA^{-1}B' \\ + CA^{-1}A'A^{-1}B. \end{aligned}$$

Thus one can differentiate a *reduced* system without ever constructing a full matrix inverse. The same inverse derivative gives a compact second-order identity for any positive definite matrix curve:

$$\boxed{\frac{d^2}{dt^2}\log\det A = \operatorname{tr}\left(A^{-1}A'' - A^{-1}A'A^{-1}A'\right).}$$

Indeed, the first derivative is $\operatorname{tr}(A^{-1}A')$ by Jacobi's formula. The product rule and the inverse derivative produce the two displayed terms. When M, A, S are all positive definite,

$$\log\det M = \log\det A + \log\det S,$$

so differentiating the small Schur complement also differentiates the contribution of the eliminated variables to the log-determinant. These are standard block-elimination identities; an accessible general reference is Boyd and Vandenberghe, *Convex Optimization*.

The most satisfying link is now an exact one. Taylor jets and Faà di Bruno describe how a composed response changes; the implicit-function theorem finds the response of variables determined by equations; and block Gaussian elimination removes those variables. For an optimized quadratic problem the two routes agree in the single matrix $D - B^T A^{-1}B$. That matrix is simultaneously an effective Hessian, a reduced linear system, and the precision of a Gaussian marginal. The combinatorics of higher derivatives still has its own interest, but there is now a concrete problem in which it and linear algebra genuinely work together.

Constrained Extrema: Tangent Spaces, Second Variation, and Dual Variables

N-20220501

At a constrained extremum, one is free to move along the feasible set but not away from it. Lagrange multipliers express the vanishing of the objective's first-order change along every allowed direction. The Hessian test must then measure curvature along those same directions, including the curvature of the constraint itself. This leads naturally to bordered Hessians and to the role of multipliers as sensitivities in convex optimization.[1]

§97.1 First variation is normal to the feasible surface

Let $f, g : \mathbb{R}^n \to \mathbb{R}$ be C^2, and suppose $g(x_*) = 0$ with $\nabla g(x_*) \neq 0$. The implicit function theorem says that the feasible set near x_* is a smooth hypersurface, with tangent space

$$T_{x_*}M = \{v \in \mathbb{R}^n : \nabla g(x_*) \cdot v = 0\}.$$

If $f|_M$ has a local extremum at x_*, then $Df(x_*)v = 0$ for every $v \in T_{x_*}M$. The annihilator of this tangent hyperplane is the line spanned by $Dg(x_*)$. Hence

$$\boxed{\nabla f(x_*) = \lambda \nabla g(x_*)}$$

for some scalar λ. The multiplier is the coefficient expressing the objective's differential in the normal direction.

For several constraints $g = (g_1, \dots, g_m)$, the same argument works if the rows of $Dg(x_*)$ are independent:

$$\nabla f(x_*) = Dg(x_*)^T \lambda.$$

Independence is a constraint qualification, ensuring that every vector satisfying the linearized constraints is tangent to an actual feasible motion.

Example 97.1.1 (Finding the nearest point is already a multiplier calculation)
Find the point on $x + y = 1$ nearest the origin. Minimize $f(x, y) = x^2 + y^2$ subject to $g(x, y) = x + y - 1 = 0$. The multiplier equations are

$$(2x, 2y) = \lambda(1, 1), \qquad x + y = 1,$$

giving $\boxed{x = y = \frac{1}{2}}$ and $\lambda = 1$. Substitution $y = 1 - x$ independently verifies that

$$x^2 + (1 - x)^2 = 2\left(x - \frac{1}{2}\right)^2 + \frac{1}{2}$$

has its unique global minimum there. The tangent vector $(1, -1)$ gives a positive restricted quadratic form:

$$(1, -1) H_f (1, -1)^T = 4 > 0.$$

[1] For constrained second-order conditions and multiplier sensitivity, see Boyd and Vandenberghe, *Convex Optimization*, Chapters 5 and 10: https://web.stanford.edu/~boyd/cvxbook/.

For an affine constraint $H_g = 0$, so here the Lagrangian Hessian agrees with H_f.

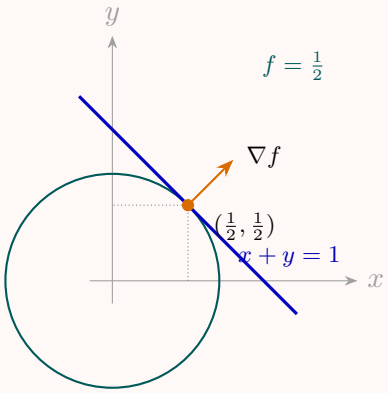


Figure 97.1: The smallest squared-distance level set touching $x + y = 1$ has its normal aligned with the constraint normal.

§97.2 Why the ordinary Hessian is not the right one

Choose the Lagrangian $L(x, \lambda) = f(x) - \lambda g(x)$, and let $x(t)$ be any twice-differentiable feasible curve with $x(0) = x_*$, $x'(0) = v \in T_{x_*}M$. Differentiating $g(x(t)) = 0$ twice gives

$$Dg(x_*)x''(0) + D^2g(x_*)[v, v] = 0.$$

Meanwhile,

$$\frac{d^2}{dt^2} f(x(t))\Big|_{t=0} = D^2 f(x_*)[v, v] + Df(x_*)x''(0).$$

At a stationary point $Df = \lambda Dg$, so the second equation becomes

$$\boxed{\frac{d^2}{dt^2} f(x(t))\Big|_{t=0} = v^T \big(H_f(x_*) - \lambda H_g(x_*)\big)v.}$$

The curvature of the constraint has cancelled the acceleration of the chosen path. This is the *restricted Hessian of the Lagrangian*. For a strict constrained minimum, it suffices that the quadratic form be positive definite on nonzero tangent vectors; negative definiteness similarly gives a strict local maximum. Indefiniteness on the tangent space rules out either kind of extremum.

§97.3 The sphere reveals the correction term

Maximize or minimize $f(x, y) = x$ on the unit circle $g(x, y) = x^2 + y^2 - 1 = 0$. The stationarity equations are

$$(1, 0) = \lambda(2x, 2y), \qquad x^2 + y^2 = 1,$$

giving the two points $p_+ = (1, 0)$, $p_- = (-1, 0)$, with $\lambda = 1/2$ and $-1/2$ respectively.

Since $H_f = 0$ and $H_g = 2I$, the Lagrangian Hessian is

$$H_L = -2\lambda I.$$

At either point the tangent space is the vertical line, spanned by $v = (0, 1)$. At p_+, $v^T H_L v = -1 < 0$, so the point is a strict constrained maximum. At p_-, the value is $+1 > 0$, a strict constrained minimum.

The objective is linear and its ordinary Hessian vanishes identically. All the relevant second-order behavior comes from the curvature of the circle. This is why checking H_f alone would miss the entire problem.

§97.4 Bordered Hessians encode the same restriction

In two variables with one regular constraint, write $J = (g_x, g_y)$ and let $H = H_L$. A nonzero tangent vector is $v = (-g_y, g_x)$. The bordered matrix is

$$B = \begin{pmatrix} 0 & g_x & g_y \\ g_x & H_{11} & H_{12} \\ g_y & H_{12} & H_{22} \end{pmatrix}.$$

Direct expansion shows

$$\boxed{\det B = -v^T H v.}$$

Thus, for this particular dimension and convention, a negative bordered determinant corresponds to positive constrained second variation, while a positive determinant corresponds to negative second variation. In higher dimensions the signs and lists of principal minors depend on the number of constraints; the invariant statement is always definiteness of H_L on $\ker Dg$. This avoids memorizing ambiguous bordered-Hessian sign tables.

§97.5 Singular constraints change the rules

Take the cusp

$$g(x, y) = y^2 - x^3 = 0$$

and the objective $f(x, y) = x$. Every feasible point satisfies $x \geq 0$, so the origin is a strict global minimum. Yet

$$\nabla g(0, 0) = 0, \qquad \nabla f(0, 0) = (1, 0),$$

and no multiplier can satisfy $\nabla f = \lambda \nabla g$ there. The feasible set is singular at the origin, so the regular Lagrange theorem does not apply.

Even with a regular constraint, a vanishing restricted Hessian does not determine the type of stationary point. For $f(x, y) = x^4 - y^4$ on the whole plane, the gradient and Hessian vanish at the origin, but positive and negative values approach along the coordinate axes. Higher-order terms resolve the degeneracy.

§97.6 Multipliers also measure the price of a constraint

Consider a convex differentiable $f : \mathbb{R}^n \to \mathbb{R}$ minimized under an affine constraint

$$Ax = b.$$

Use $L(x, \lambda) = f(x) + \lambda^T (Ax - b)$. If x_* is feasible and $\nabla f(x_*) + A^T \lambda_* = 0$, convexity implies, for any feasible x,

$$f(x) \geq f(x_*) + \nabla f(x_*)^T (x - x_*) = f(x_*) - \lambda_*^T A(x - x_*) = f(x_*).$$

Thus the multiplier stationarity condition is globally sufficient in this convex setting.

Let $v(b)$ denote the optimal value when the constraint is $Ax = b$, and assume locally that a smooth optimum and multiplier exist. Differentiating the optimal-value equation gives

$$dv(b) = -\lambda_*^T \, db.$$

A multiplier is therefore a shadow price: it measures the first-order change of the optimal objective when the permitted constraint value moves. In the circle example the multiplier encodes normality; in convex optimization the same normal coefficient becomes a quantitative sensitivity.

§97.7 When a boundary pushes back: an inequality example

The nearest-point problem changes subtly if we replace $x + y = 1$ by the half-plane $x + y \geq 1$. Minimize $x^2 + y^2$ on this set. Define

$$h(x, y) = 1 - x - y \leq 0, \qquad L(x, y, \mu) = x^2 + y^2 + \mu(1 - x - y).$$

For this convex problem the Karush–Kuhn–Tucker conditions read

$$\begin{cases} 2x - \mu = 0, & 2y - \mu = 0, \\ x + y \geq 1, & \mu \geq 0, \\ \mu(1 - x - y) = 0. \end{cases}$$

The last condition is complementary slackness: an inactive inequality has multiplier zero, while a positive multiplier requires equality at the boundary. If $\mu = 0$, then $x = y = 0$, which is infeasible. Thus the constraint is active and

$$\boxed{(x_*, y_*) = (\tfrac{1}{2}, \tfrac{1}{2}), \qquad \mu_* = 1.}$$

Strict convexity and feasibility certify the global minimum. The multiplier's nonnegative sign has a geometric meaning: the active boundary can only push toward the feasible side.

A warning about numerical methods emerges from this example. Squaring a constraint expression h as ρh^2 punishes even strictly feasible points with $h < 0$. A quadratic penalty for $h \leq 0$ should instead use $\rho[h]_+^2$ with $[h]_+ = \max(h, 0)$. Even then a finite quadratic penalty generally does not enforce exact feasibility. An interior-point method takes a different route, adding a barrier $-\tau \log(-h)$ only where $h < 0$. These constructions approximate or enforce the feasible boundary in genuinely different ways.

§97.8 Second-order silence and global existence

A zero or semidefinite restricted Hessian is a statement that the second-order test is inconclusive. At the origin,

$$f_+(x, y) = x^4 + y^4, \qquad f_-(x, y) = x^4 - y^4$$

have identical zero gradients and Hessians, but f_+ has a strict minimum and f_- a saddle. The same contrast can be realized on a regular two-dimensional constraint manifold by adding an extraneous coordinate z with constraint $z = 0$. Higher-order terms matter when the quadratic approximation cannot distinguish signs.

Existence of global extrema asks a logically separate question. The function x has no extremum on the open interval $(0, 1)$; there is no stationary point to find. On a compact feasible set, however, every continuous objective attains minimum and maximum. The circle in the earlier example is compact, so comparing its two stationary values actually settles the global question.

A deeper connection is Morse theory: at a nondegenerate critical point on a smooth manifold, the number of negative eigenvalues of the restricted Hessian is its Morse index. The very quadratic form used to test constrained extrema also gives local information from which changes in the manifold's topology can be studied. The difficult global theorems lie beyond this example, but the local datum entering them is explicit.

The tangent space tells us which infinitesimal motions are allowed, the restricted Hessian classifies their second-order cost, and the multiplier measures the cost of changing the constraint itself. These three views belong to one differential-geometric calculation.

Newton's Shell Theorem: Solid Angle, Potential, and Gauss Law

N-20230201

§98.1 First orientation

The elementary statement is familiar: the gravitational field produced by a uniform spherical shell is zero at every point inside the shell. But this result is not a trivial symmetry statement. If the field point is not the center, the near side of the shell is closer and should pull more strongly, while the far side has more area visible in the same direction. The theorem says that these two effects cancel exactly.

The problem admits three coherent languages:

solid-angle cancellation,
scalar potential computation,
Gauss law and harmonic potential.

The point is not only to remember that the answer is zero. The point is to understand why the inverse-square law, spherical geometry, and potential theory fit together so perfectly.

§98.2 Statement of Newton's shell theorem

Let a spherical shell of radius R have uniform surface density σ. Its total mass is

$$M = 4\pi R^2 \sigma.$$

Let $\mathbf{x} \in \mathbb{R}^3$, and write $r = |\mathbf{x}|$, with the shell centered at the origin.

Theorem 98.2.1 (Newton's shell theorem)
The gravitational field of the shell is

$$\mathbf{g}(\mathbf{x}) = \begin{cases} \mathbf{0}, & 0 \le r < R, \\ -\dfrac{GM}{r^2}\dfrac{\mathbf{x}}{r}, & r > R. \end{cases}$$

Equivalently, outside the shell the field is the same as if all mass were concentrated at the center, while inside the shell the field is zero.

The interior and exterior cases should be proved together because the same potential calculation gives both at once.

§98.3 Why symmetry alone is not enough

At the center of the shell, cancellation is immediate: every mass element has an opposite mass element at the same distance. Away from the center, this argument fails. A point P inside the shell is closer to one side than the other.

The correct local cancellation uses three facts at once:

the same narrow cone cuts two opposite surface patches,
patch area grows like distance squared,
gravity decays like distance squared.

There is also a small geometric correction: the patch may be tilted relative to the cone. This is the obliquity factor. Omitting it makes the proof feel like a slogan rather than a proof.

§98.4 Solid-angle proof in full detail

Let P be an interior point. Draw a line through P, meeting the shell at two points A and B. Around this line take a very narrow cone with vertex P and solid angle $d\Omega$. It cuts two small patches of the shell near A and B.

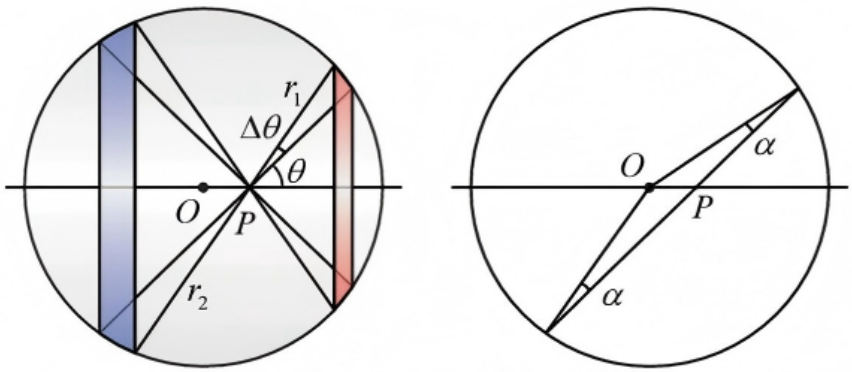


Figure 98.1: The geometric cancellation proof: the same narrow cone from an interior point cuts two opposite shell patches, and the equal angle geometry supplies the common obliquity factor.

Let

$$s_A = PA, \qquad s_B = PB.$$

Let β_A and β_B be the angles between the ray and the outward normal to the shell, taken with absolute cosine. For a small surface patch, solid angle and area satisfy

$$d\Omega = \frac{|\cos\beta|}{s^2}\, dS,$$

so

$$dS = \frac{s^2}{|\cos\beta|}\, d\Omega.$$

Therefore

$$dS_A = \frac{s_A^2}{|\cos\beta_A|}\, d\Omega, \qquad dS_B = \frac{s_B^2}{|\cos\beta_B|}\, d\Omega.$$

The missing geometric point is that the two obliquity factors are equal. Since $OA = OB = R$, the triangle OAB is isosceles. Thus the base angles at A and B are equal. Because P lies on the chord AB, the angles between the chord direction and the radii at A and B have the same absolute cosine:

$$|\cos\beta_A| = |\cos\beta_B|.$$

The mass of a patch is $dm = \sigma dS$. Its gravitational field magnitude at P is

$$|d\mathbf{g}| = G\frac{dm}{s^2} = G\sigma\frac{dS}{s^2}.$$

Substituting the area formula gives

$$|d\mathbf{g}_A| = G\sigma\frac{d\Omega}{|\cos\beta_A|}, \qquad |d\mathbf{g}_B| = G\sigma\frac{d\Omega}{|\cos\beta_B|}.$$

These magnitudes are equal, and the directions are opposite along the same line through P. Hence the two contributions cancel.

Pairing all such cones through P gives

$$\mathbf{g}(P) = \mathbf{0}.$$

The proof is now precise: the inverse-square law cancels the square growth of apparent area, and spherical geometry cancels the obliquity factor.

§98.5 Potential proof: doing the integral

The force integral is a vector integral. The potential is scalar, so it is cleaner. Let the field point have distance r from the center, and put it on the positive z-axis. A point of the shell at polar angle θ has distance

$$\ell(\theta) = \sqrt{R^2 + r^2 - 2Rr\cos\theta}$$

from the field point.

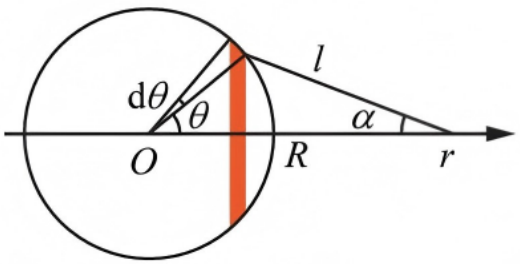


Figure 98.2: Geometry of the shell integral: the distance ℓ from the field point to a surface element and the projection angle used when passing from scalar potential to radial force.

The gravitational potential is

$$V(r) = -G\int\frac{dm}{\ell}.$$

For the shell,

$$dm = \sigma R^2\sin\theta\,d\theta\,d\phi.$$

Hence

$$V(r) = -G\sigma R^2\int_0^{2\pi}\int_0^{\pi}\frac{\sin\theta\,d\theta\,d\phi}{\sqrt{R^2 + r^2 - 2Rr\cos\theta}}.$$

Integrating over ϕ and putting $u = \cos\theta$, we obtain

$$V(r) = -2\pi G\sigma R^2 \int_{-1}^{1} \frac{du}{\sqrt{R^2 + r^2 - 2Rru}}.$$

Now compute the elementary integral:

$$\begin{aligned}\int_{-1}^{1} \frac{du}{\sqrt{R^2 + r^2 - 2Rru}} &= \left[-\frac{1}{Rr}\sqrt{R^2 + r^2 - 2Rru}\right]_{-1}^{1} \\ &= \frac{(R + r) - |R - r|}{Rr}.\end{aligned}$$

Therefore

$$V(r) = -2\pi G\sigma R^2 \cdot \frac{(R + r) - |R - r|}{Rr}.$$

Since $M = 4\pi R^2\sigma$, this becomes

$$V(r) = \begin{cases} -\dfrac{GM}{R}, & 0 \le r < R, \\ -\dfrac{GM}{r}, & r > R. \end{cases}$$

Now recover the field by

$$\mathbf{g} = -\nabla V.$$

For a radial potential, this means

$$\mathbf{g}(r) = -\frac{dV}{dr}\,\hat{\mathbf{r}}.$$

Inside the shell, $V(r)$ is constant, so

$$\mathbf{g} = \mathbf{0}.$$

Outside the shell,

$$V(r) = -\frac{GM}{r},$$

so

$$\mathbf{g}(r) = -\frac{GM}{r^2}\hat{\mathbf{r}}.$$

This proves both halves of Newton's shell theorem.

§98.6 The direct force integral

The direct force calculation is possible, but it is less elegant. By symmetry only the radial component survives. The radial component from a ring at polar angle θ is proportional to the projection of the force onto the radial axis.

One obtains

$$g_r(r) = 2\pi G\sigma R^2 \int_0^{\pi} \frac{(R\cos\theta - r)\sin\theta}{(R^2 + r^2 - 2Rr\cos\theta)^{3/2}}\,d\theta.$$

The sign convention here takes the positive direction away from the center. This integral is exactly

$$-\frac{dV}{dr}.$$

So the potential method does not avoid the physics; it packages the same computation in a scalar form and differentiates only at the end.

This explains why potential theory is the natural language for shell problems. Instead of separately tracking force directions, one integrates a scalar quantity whose gradient gives the field.

§98.7 Gauss-law viewpoint

There is a still shorter proof once the field is known to be spherically symmetric. The gravitational Gauss law is

$$\nabla \cdot \mathbf{g} = -4\pi G\rho,$$

or, in integral form,

$$\oint_S \mathbf{g} \cdot d\mathbf{S} = -4\pi G M_{\text{enc}}.$$

For a spherical shell, symmetry implies that the field at radius r is radial and has constant magnitude on each sphere:

$$\mathbf{g}(r) = g(r)\hat{\mathbf{r}}.$$

Taking S to be the sphere of radius r, the flux is

$$\oint_S \mathbf{g} \cdot d\mathbf{S} = 4\pi r^2 g(r).$$

Thus

$$4\pi r^2 g(r) = -4\pi G M_{\text{enc}}(r),$$

so

$$g(r) = -\frac{GM_{\text{enc}}(r)}{r^2}.$$

Inside the shell, $M_{\text{enc}}(r) = 0$, hence $g(r) = 0$. Outside the shell, $M_{\text{enc}}(r) = M$, hence

$$g(r) = -\frac{GM}{r^2}.$$

The nuance is important: Gauss law alone gives the result quickly only because spherical symmetry has already reduced the field to one radial function. The uniform spherical shell provides that symmetry.

§98.8 Harmonic potential

In an empty region, the gravitational potential satisfies Laplace's equation:

$$\Delta V = 0.$$

Inside the shell there is no mass. Moreover the shell is spherically symmetric, so the interior potential must be radial:

$$V = V(r).$$

For a radial function in $\mathbb{R}^3$, Laplace's equation becomes

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dV}{dr}\right) = 0.$$

Integrating once gives

$$r^2 V'(r) = C.$$

If $C \neq 0$, then $V'(r) = C/r^2$, which is singular at $r = 0$. The potential inside the shell should be regular at the center, so $C = 0$. Therefore

$$V'(r) = 0,$$

and the potential is constant inside.

This is the PDE version of the theorem. The absence of mass gives harmonicity; spherical symmetry reduces the harmonic equation to an ODE; regularity at the center forces the constant solution.

§98.9 Why the inverse-square law is special

The solid-angle proof shows exactly why the exponent 2 matters. For a patch cut out by a fixed solid angle,

$$dS \sim s^2\, d\Omega$$

up to the common obliquity factor. Gravity contributes a factor $1/s^2$. These cancel.

If the force law were $1/s^p$, the patch contribution would scale like

$$\frac{dS}{s^p} \sim s^{2-p} d\Omega.$$

The near and far patches would have equal magnitudes for all distances only when

$$p = 2.$$

Thus the shell theorem is not a generic fact about attractive forces. It is a special feature of inverse-square fields.

The other hypotheses also matter. If the shell is not uniform, the two paired patches do not have masses proportional only to area. If the surface is not spherical, the obliquity factors do not match in the same way. The theorem is the result of three matching structures:

$$\text{uniform density} + \text{spherical geometry} + \text{inverse-square law}.$$

§98.10 Application: a uniform solid sphere

A uniform solid sphere can be regarded as a stack of thin spherical shells. At a point of radius r, all shells outside radius r contribute zero field. Only the mass enclosed inside radius r matters.

If the solid sphere has volume density ρ, then

$$M(r) = \frac{4\pi}{3}\rho r^3.$$

Thus the interior field is

$$g(r) = -\frac{GM(r)}{r^2} = -\frac{4\pi G\rho}{3} r.$$

So the field inside a uniform ball grows linearly with distance from the center.

This has a famous dynamical consequence. If one imagines a straight tunnel through a uniform spherical planet and ignores friction and rotation, a particle moving along the tunnel satisfies

$$\ddot{r} = -\frac{4\pi G\rho}{3} r.$$

This is simple harmonic motion. The shell theorem turns a gravitational problem into the equation of a spring.

§98.11 Multipole viewpoint

There is also a useful advanced memory aid. The Newtonian kernel has the expansion

$$\frac{1}{|x-y|} = \sum_{\ell=0}^{\infty} \frac{r_<^\ell}{r_>^{\ell+1}} P_\ell(\cos\gamma),$$

where

$$r_< = \min(|x|, |y|), \qquad r_> = \max(|x|, |y|).$$

For a uniformly distributed spherical shell, integrating over the angles kills all higher spherical harmonics. Only the monopole term survives outside. That is why the exterior field sees only total mass.

Inside the shell, the same expansion leaves a constant potential. A constant potential has zero gradient, hence zero field. This is the harmonic-analysis version of the same theorem.

§98.12 A broader reading

The shell theorem should not be remembered as "obvious by symmetry." For an off-center interior point, ordinary pairwise symmetry is not enough. The real mechanism is:

spherical geometry controls apparent area and obliquity,
the inverse-square law cancels the distance-squared factor,
potential theory turns cancellation into harmonicity,
and Gauss law turns it into flux conservation.

The geometric proof explains the local cancellation. The potential proof computes the whole field. Gauss law explains the flux structure. Laplace's equation explains why the interior potential must be constant.

Together these four views make Newton's shell theorem more than a trick: it is a meeting point of geometry, analysis, and field theory.

Local Linear Geometry: A Mapmaker's View of Derivatives and Tangent Spaces

N-20240413

§99.1 The mapmaker's problem

Suppose one wants to draw a map of a curved surface. A small enough region of the Earth can be treated as flat, even though the Earth is not flat. This is not a lie; it is a local approximation.

Differential geometry begins from the same idea. Curved objects are understood by replacing them, near one point, with linear objects. The derivative is the machine that performs this replacement.

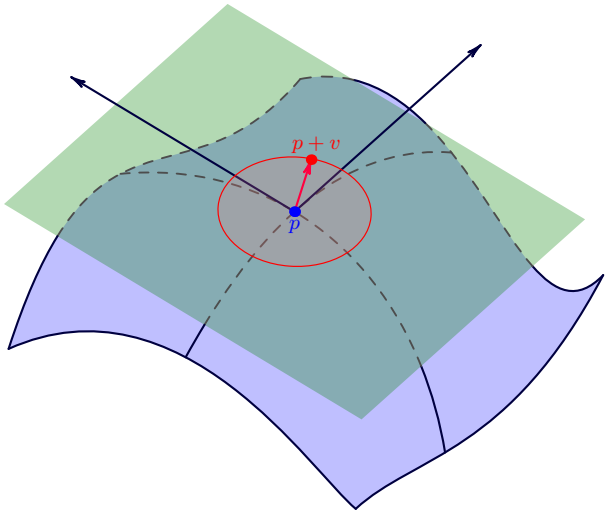


Figure 99.1: The tangent line is the first linear shadow of a curve. Higher-order details are deliberately ignored.

The guiding principle for this chapter is:

> to see a curved object clearly, first learn its best linear shadow.

This principle sounds elementary, but it is the starting point for manifolds, tangent spaces, differential forms, and eventually curvature.

§99.2 Derivative first, partial derivatives second

Let

$$f : \mathbb{R}^n \to \mathbb{R}^m.$$

At a point p, the derivative should be thought of as a linear map

$$Df_p : \mathbb{R}^n \to \mathbb{R}^m$$

satisfying

$$f(p+h) = f(p) + Df_p(h) + o(\|h\|).$$

This formula says that $Df_p(h)$ is the best first-order prediction of how f changes when the input moves by h.

The Jacobian matrix is only a coordinate display of this linear map. If $f = (f_1, \dots, f_m)$, then

$$J_f(p) = \begin{bmatrix} \partial f_1/\partial x_1 & \cdots & \partial f_1/\partial x_n \\ \vdots & \ddots & \vdots \\ \partial f_m/\partial x_1 & \cdots & \partial f_m/\partial x_n \end{bmatrix}_p.$$

The matrix is useful, but the linear map is the concept.

A useful correction. Students often learn multivariable calculus as a list of partial derivatives. Geometry reverses the priority: the total derivative is the object, and partial derivatives are its coordinates.

§99.3 The chain rule is not a formula trick

If

$$\mathbb{R}^n \xrightarrow{f} \mathbb{R}^m \xrightarrow{g} \mathbb{R}^k,$$

then

$$D(g \circ f)_p = Dg_{f(p)} \circ Df_p.$$

In coordinates this becomes matrix multiplication. Geometrically it says something simpler: if f is locally approximated by Df_p, and g is locally approximated by $Dg_{f(p)}$, then their composition is locally approximated by the composition of the two linear maps.

This is why the chain rule is one of the first genuinely structural theorems in calculus. It says that differentiation respects composition.

§99.4 A small laboratory: polar coordinates

Consider

$$F(r, \theta) = (r\cos\theta, r\sin\theta).$$

Then

$$DF_{(r,\theta)} = \begin{bmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{bmatrix}.$$

The two columns are

$$\frac{\partial F}{\partial r} = (\cos\theta, \sin\theta), \qquad \frac{\partial F}{\partial \theta} = (-r\sin\theta, r\cos\theta).$$

The first vector points outward. The second vector points around the circle. Thus a coordinate change is not only a substitution rule; it turns coordinate motion into tangent motion.

Notice the factor r in the angular direction. Moving by $d\theta$ near the origin is physically shorter than moving by the same $d\theta$ far away. The derivative records this distortion.

§99.5 Tangent planes from equations

A graph $z = f(x, y)$ has tangent plane

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

A level surface

$$F(x, y, z) = c$$

has tangent plane

$$\nabla F(p) \cdot (X - p) = 0.$$

Both formulas are versions of the same principle: differentiate the defining data and keep only the first-order part.

For the sphere

$$S^2 = \{x \in \mathbb{R}^3 : \|x\|^2 = 1\},$$

let $p \in S^2$. If $\gamma(t) \subset S^2$ and $\gamma(0) = p$, then

$$\gamma(t) \cdot \gamma(t) = 1.$$

Differentiating gives

$$p \cdot \gamma'(0) = 0.$$

So

$$T_p S^2 = \{v \in \mathbb{R}^3 : p \cdot v = 0\}.$$

The tangent plane is perpendicular to the radius vector. This is the most concrete form of the slogan "differentiate the constraint."

§99.6 A manifold is a space with local Euclidean windows

A smooth manifold is a space that can be covered by coordinate windows

$$\varphi : U \to \mathbb{R}^n$$

whose overlaps are smoothly compatible. A chart is not the manifold itself. It is a local language for talking about it.

The circle, sphere, torus, and projective spaces all share this feature: no single coordinate system describes them perfectly, but many local coordinate systems can be patched together.

Why charts matter. Charts let us import calculus into spaces that are not globally Euclidean. Differentiation, tangent vectors, and smooth maps are all defined by moving into a chart, doing the Euclidean calculation, and checking that the answer does not depend on the chart in a bad way.

§99.7 Two pictures of tangent vectors

There are two useful definitions of a tangent vector at p.

The first is dynamic: a tangent vector is the velocity of a curve through p:

$$v = \gamma'(0), \qquad \gamma(0) = p.$$

The second is operational: a tangent vector differentiates functions:

$$v(f) = \left.\frac{d}{dt}\right|_{t=0} f(\gamma(t)).$$

The second definition is more intrinsic. It does not require the manifold to be sitting inside a larger Euclidean space.

A smooth map $F : M \to N$ sends tangent vectors forward:

$$dF_p : T_pM \to T_{F(p)}N.$$

This is the derivative on manifolds.

§99.8 Regular values: one clean machine for making manifolds

Suppose

$$F : \mathbb{R}^n \to \mathbb{R}^k$$

is smooth. If c is a regular value, meaning DF_p is surjective for every $p \in F^{-1}(c)$, then

$$F^{-1}(c)$$

is a smooth manifold of dimension $n - k$.

This theorem is useful because many geometric objects are naturally described by equations. Spheres, matrix groups, and many surfaces arise this way. The theorem says that if the equations cut cleanly, the solution set is a manifold.

§99.9 One chart in detail: stereographic projection

The sphere is the first place where one really feels why a single coordinate system is not enough. Remove the north pole

$$N = (0, 0, 1)$$

from S^2. Stereographic projection sends a point $(X, Y, Z) \in S^2 \setminus \{N\}$ to the point where the line through N and (X, Y, Z) meets the plane $Z = 0$:

$$(X, Y, Z) \longmapsto \left(\frac{X}{1 - Z}, \frac{Y}{1 - Z}\right).$$

The inverse map is

$$(u, v) \longmapsto \left(\frac{2u}{u^2 + v^2 + 1}, \frac{2v}{u^2 + v^2 + 1}, \frac{u^2 + v^2 - 1}{u^2 + v^2 + 1}\right).$$

This formula is worth seeing once because it makes the word “chart” concrete. The sphere is not being flattened globally; one point is missing, and the remaining part is described by ordinary coordinates (u, v).

A second chart, obtained by projecting from the south pole, covers the missing point. The transition map between the two charts is smooth away from the origin. This is the atlas idea in its most visible form.

§99.10 Angle-Preserving Charts

A coordinate chart gives a map from a flat page into a curved surface. But how much does a small square on that page stretch? The inverse stereographic map introduced above provides an exact calculation. Write

$$\sigma(u,v) = \left(\frac{2u}{1+r^2}, \frac{2v}{1+r^2}, \frac{r^2-1}{1+r^2} \right), \qquad r^2 = u^2 + v^2.$$

A small displacement $h = (a, b)$ in the drawing plane is carried to the tangent vector $D\sigma_{(u,v)}h$ on the sphere. Let $D = 1 + r^2$ and $t = ua + vb$. Differentiating the formula componentwise yields

$$D\sigma(h) = \left(\frac{2a}{D} - \frac{4ut}{D^2}, \frac{2b}{D} - \frac{4vt}{D^2}, \frac{4t}{D^2} \right).$$

Its squared Euclidean length simplifies in a rather satisfying way:

$$\begin{aligned} \|D\sigma(h)\|^2 &= \frac{4(a^2+b^2)}{D^2} - \frac{16t^2}{D^3} + \frac{16t^2(r^2+1)}{D^4} \\ &= \frac{4}{(1+r^2)^2}\,(a^2+b^2). \end{aligned}$$

Polarization of this identity shows that *all* tangent inner products are scaled by the same positive factor:

$$\boxed{(D\sigma)^{T}(D\sigma) = \frac{4}{(1+u^2+v^2)^2}\, I_2.}$$

Angles are therefore preserved, although lengths are multiplied locally by $2/(1+r^2)$. This is the meaning of a conformal coordinate chart. Near the origin, corresponding to the south pole, it magnifies lengths by a factor of two; at $r = 1$, along the equator, the length factor equals one. Far from the origin, arbitrarily large planar distances can correspond to tiny movements close to the missing north pole.

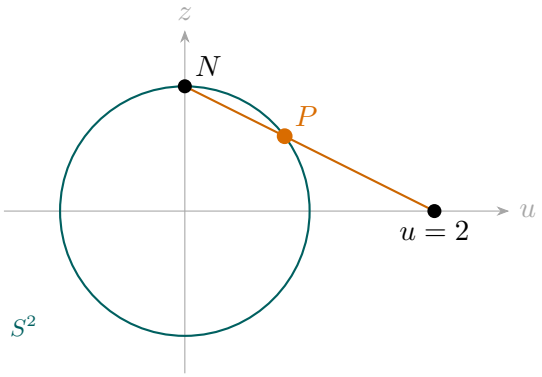


Figure 99.2: Stereographic projection from the north pole. The line through $N = (0, 1)$ and $P = (0.8, 0.6)$ meets the horizontal coordinate plane at $u = 2$, exactly as the projection formula predicts.

The metric factor also gives an area element on the sphere:

$$dS = \sqrt{\det\!\left((D\sigma)^{T} D\sigma\right)}\, du\, dv = \frac{4}{(1+r^2)^2}\, du\, dv.$$

As a useful check, integrating over the whole coordinate plane recovers the area of the unit sphere:

$$\begin{aligned}\mathrm{Area}(S^2) &= \int_0^{2\pi}\int_0^{\infty} \frac{4r}{(1+r^2)^2}\,dr\,d\theta \\ &= 2\pi\left[-\frac{2}{1+r^2}\right]_0^{\infty} = 4\pi.\end{aligned}$$

The missing north pole has area zero. This example joins the derivative, the tangent metric, conformality and integration without needing an abstract tensor formalism: the local linear map tells us exactly how much geometry survives a coordinate change.[1]

§99.11 Vector fields: tangent spaces glued along a space

A tangent vector lives at one point. A vector field chooses a tangent vector at every point:

$$p \longmapsto V(p) \in T_pM.$$

On the plane, one may draw a vector field as little arrows attached to points. On a manifold, this picture is still correct, but each arrow belongs to a different tangent space.

For example, on the unit circle $S^1 \subset \mathbb{R}^2$, the vector field

$$V(x,y) = (-y, x)$$

is tangent to the circle because

$$(x,y)\cdot(-y,x) = 0.$$

It rotates counterclockwise around the circle. This example shows a useful habit: to check whether a vector field is tangent to a constraint, check whether it preserves the constraint to first order.

§99.12 Pushforward in a concrete example

Let

$$F : \mathbb{R}^2 \to \mathbb{R}^2, \qquad F(x,y) = (x^2 - y^2, 2xy).$$

This is the complex squaring map $z \mapsto z^2$ written in real coordinates. Its derivative is

$$DF_{(x,y)} = \begin{bmatrix} 2x & -2y \\ 2y & 2x \end{bmatrix}.$$

At the point $(1,0)$, a tangent vector (a,b) is sent to

$$(2a, 2b).$$

At the point $(0,1)$, it is sent to

$$(-2b, 2a).$$

The same coordinate vector can be stretched and rotated differently depending on the base point. This is why the derivative of a nonlinear map is a family of linear maps, one for each point.

[1] A systematic treatment of stereographic coordinates and their first fundamental form appears in *Differential Geometry*, Section 2.7.

§99.13 What curvature will later add

This chapter only keeps first-order information. A tangent plane cannot distinguish a sphere from a plane at the point of tangency. Curvature begins when one asks how tangent spaces change from point to point.

So the local-linear story is not the end of geometry. It is the first layer. The later layers ask:

How do tangent spaces vary?
How do we compare vectors at different points?
How does a manifold bend?

Those questions require connections, metrics, and curvature. But without the tangent-space layer, none of them can even be stated cleanly.

The local-linear picture becomes useful through the chain

$$\text{derivative} \longrightarrow \text{linear approximation} \longrightarrow \text{tangent space} \longrightarrow \text{manifold calculus}.$$

In a calculation this chain has a definite order. First choose a chart or a defining equation. Then replace the nonlinear object by its first-order part: for a parametrized surface this means differentiating the parametrization; for a level set $F^{-1}(c)$ this means studying dF_p. The tangent space is the kernel of the linear constraint when the manifold is cut out by equations, and it is the span of coordinate velocity vectors when the manifold is parametrized.

The stereographic-projection example and the regular-value criterion are two versions of the same habit. A chart lets one compute on $\mathbb{R}^n$; a regular value tells one that the kernel of a derivative has the expected dimension. Curvature, connections, and metrics will later add second-order and comparison data, but the first check remains the same: before asking how a manifold bends, identify the tangent space in which the infinitesimal question lives.

Differential Forms: Pullback, Boundary Signs, and Stokes' Theorem

N-20240427

The fundamental theorem of calculus relates an integral of a derivative to an expression on the boundary. Green's theorem, the divergence theorem and the classical Stokes theorem have the same structure, but their coordinates obscure it. Following Victor Guillemin's MIT 18.952 *Theory of Differential Forms* notes, we build the formula from pullback and exterior differentiation before stating it on a manifold.[1]

§100.1 A one-form eats a velocity vector

A covector at p is a linear function $T_pM \to \mathbb{R}$; a smooth one-form varies such covectors with p. In the plane write

$$\omega = P(x,y)\,dx + Q(x,y)\,dy.$$

For a curve $\gamma(t) = (x(t), y(t))$, the velocity is $\gamma'(t)$ and

$$\omega_{\gamma(t)}(\gamma'(t)) = P(x(t),y(t))x'(t) + Q(x(t),y(t))y'(t).$$

The pullback by γ packages this into a form on the parameter interval:

$$\gamma^*\omega = (P(\gamma(t))x'(t) + Q(\gamma(t))y'(t))\,dt.$$

Consequently $\int_\gamma \omega = \int_a^b \gamma^*\omega$. This is precisely the usual line-integral formula; it includes the chain rule and the orientation automatically.

For example, on the unit circle traversed counterclockwise,

$$\gamma(t) = (\cos t, \sin t), \qquad \gamma^*(x\,dy - y\,dx) = dt,$$

so $\int_\gamma (x\,dy - y\,dx) = 2\pi$. Reversing the parametrization changes the sign.

§100.2 An oriented area needs an alternating form

The wedge product is alternating:

$$dx \wedge dy = -dy \wedge dx, \qquad dx \wedge dx = 0.$$

For tangent vectors $u = (u_1, u_2)$, $v = (v_1, v_2)$, it evaluates as

$$(dx \wedge dy)(u, v) = u_1 v_2 - u_2 v_1 = \det[u\ v].$$

The determinant detects both oriented area and the order of the two vectors. Hence $F(x,y)\,dx \wedge dy$ is an appropriate integrand for an oriented surface, unlike an arbitrary scalar density stripped of its orientation.

A metric measures length by $ds = \|\gamma'(t)\|\,dt$. This is not the pullback of an ordinary one-form depending linearly on the velocity, because the norm satisfies $\|-v\| = \|v\|$. Reversing a curve changes a one-form integral's sign but leaves its length unchanged.

[1] https://math.mit.edu/classes/18.952/2015SP/materials.html

§100.3 Exterior differentiation contains the planar curl

For a function f, define $df = f_x\,dx + f_y\,dy$. For the one-form $\omega = P\,dx + Q\,dy$,

$$d\omega = dP \wedge dx + dQ \wedge dy = (Q_x - P_y)\,dx \wedge dy.$$

The equality follows by writing $dP = P_x\,dx + P_y\,dy$, and using $dx \wedge dx = 0$.

If $f \in C^2$, then

$$d(df) = (f_{yx} - f_{xy})\,dx \wedge dy = 0.$$

This is the local manifestation of $d^2 = 0$. In higher dimensions the corresponding identities include curl of a gradient being zero and divergence of a curl being zero, with the appropriate form identification.

§100.4 Green's theorem in a rectangle

For $R = [a,b] \times [c,d]$ with counterclockwise boundary and a C^1 form $P\,dx + Q\,dy$, calculate each edge:

$$\begin{aligned}\int_{\partial R} P\,dx + Q\,dy = &\int_a^b P(x,c)\,dx + \int_c^d Q(b,y)\,dy \\ &- \int_a^b P(x,d)\,dx - \int_c^d Q(a,y)\,dy.\end{aligned}$$

Apply the fundamental theorem of calculus first in y to the two P-terms and then in x to the Q-terms:

$$\int_{\partial R} \omega = \int_a^b \int_c^d (Q_x - P_y)\,dy\,dx = \int_R d\omega.$$

This calculation reveals the mechanism. If a region is subdivided into rectangles, each interior oriented edge appears twice with opposite signs and cancels. Only the outer boundary survives. For a general smooth domain, parametrization, subdivision, approximation and compatibility on overlaps make that cancellation rigorous.

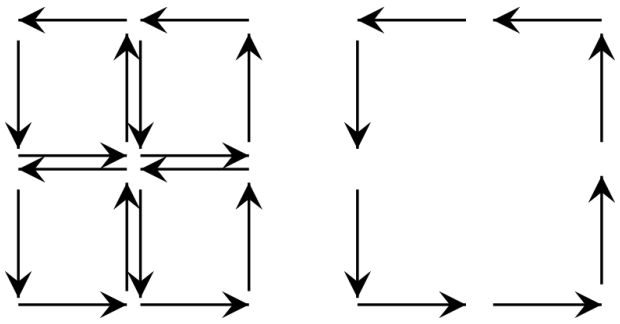


Figure 100.1: Orientation of the patch induces an orientation on the boundary, determining the signs in Stokes' theorem.

§100.5 The exterior derivative commutes with pullback

If $\Phi : U \to M$ is a smooth map and f a smooth function on M, the chain rule says $d(f \circ \Phi) = \Phi^*(df)$. Since both the pullback and d respect the alternating product, the identity extends to arbitrary forms:

$$\boxed{\Phi^*(d\omega) = d(\Phi^*\omega)}.$$

For a two-dimensional parameterization $\Phi(u,v) = (x(u,v), y(u,v))$, one sees the Jacobian directly:

$$\Phi^*(dx \wedge dy) = (x_u y_v - x_v y_u)\, du \wedge dv.$$

The determinant's sign is exactly what records whether a parametrization preserves or reverses orientation. This is the mechanism behind change of variables, rather than an extra correction added to a formula after the fact.

§100.6 Stokes' theorem and induced orientation

Theorem 100.6.1 (Generalized Stokes)

Let M be a compact oriented smooth n-manifold with smooth boundary, and let ω be a smooth $(n-1)$-form defined near M. With the induced boundary orientation,

$$\int_M d\omega = \int_{\partial M} \omega.$$

At a boundary point, the outward normal is placed first: a tangent basis $v_1, \dots, v_{n-1}$ is positively oriented on ∂M when (outward normal,$v_1, \dots, v_{n-1}$) is positively oriented on M. In the oriented plane this makes the boundary run counterclockwise. The proof reduces the manifold locally to parameterized pieces in Euclidean space; interior faces cancel as in the rectangle, while a partition of unity controls the local-to-global assembly. We have demonstrated the rectangular computation above, not a complete proof of that manifold theorem.

§100.7 The divergence theorem as a three-form computation

For a vector field $F = (P, Q, R)$ on $\mathbb{R}^3$, its flux form is

$$\eta = P\, dy \wedge dz + Q\, dz \wedge dx + R\, dx \wedge dy.$$

Exterior differentiation yields

$$d\eta = (P_x + Q_y + R_z)\, dx \wedge dy \wedge dz.$$

Thus general Stokes specializes to the divergence theorem. Take $F(x,y,z) = (x,y,z)$ and M the unit ball. The divergence equals 3, so the volume side is

$$\int_M d\eta = 3\frac{4\pi}{3} = 4\pi.$$

On the unit sphere the outward unit normal equals F; the flux density is $F \cdot n = 1$, and the boundary integral is its area, again 4π. In a single example, differential forms and the familiar vector calculus formula give the same computable answer.

§100.8 A closed form can remember a hole

Not every closed form is globally a derivative. On $\mathbb{R}^2 \setminus \{0\}$, consider

$$\alpha = \frac{-y\, dx + x\, dy}{x^2 + y^2}.$$

Differentiation shows $d\alpha = 0$. On the unit circle, however,

$$\gamma(t) = (\cos t, \sin t), \qquad \gamma^*\alpha = dt, \qquad \int_\gamma \alpha = 2\pi.$$

If $\alpha = df$ globally for a single-valued smooth function, the fundamental theorem of calculus would force its integral around every closed curve to be zero. Therefore α is closed but not exact.

Locally $\alpha = d\theta$, where θ is an angular coordinate. One can choose an angle on a small simply connected patch, but no smooth single-valued angular function goes all the way around the missing origin. This is the elementary obstruction measured by the first de Rham cohomology group: the distinction between closed one-forms and exact one-forms records genuine topology.

The generalized Stokes formula is now more than a slogan. A form can be pulled back along a parametrization, its exterior derivative is compatible with that pullback, and induced boundary orientations ensure that the interior contributions cancel. The failure of closed forms to be exact explains why global topology still matters even when the local derivative vanishes.

Point-Set Language in $\mathbb{R}^n$: Neighborhoods, Boundaries, and Compactness

N-20260304

§101.1 Why point-set language appears

In one-variable calculus, the domain is often an interval, so the topology is mostly hidden. In several variables, a domain may have holes, corners, isolated points, missing boundary pieces, or several connected components. Pictures help, but pictures can lie. The reliable language is the language of balls, neighborhoods, boundary points, and sequences.

Throughout this note, points are in $\mathbb{R}^n$, and distance means Euclidean distance:

$$\rho(P,Q) = \|P-Q\|_2 = \sqrt{\sum_{i=1}^{n}(p_i-q_i)^2}.$$

§101.2 Balls and neighborhoods

Definition 101.2.1. For $P_0 \in \mathbb{R}^n$ and $\delta > 0$, the open ball of radius δ around P_0 is

$$B_\delta(P_0) = \{P \in \mathbb{R}^n : \rho(P,P_0) < \delta\}.$$

A set U is a **neighborhood** of P_0 if it contains some open ball $B_\delta(P_0)$.

A neighborhood is the formal version of "all points sufficiently close to P_0." This phrase is the engine behind limits, continuity, differentiability, and local extrema.

Example 101.2.2

In $\mathbb{R}^2$, the disk

$$D = \{(x,y) : x^2+y^2 < 1\}$$

is a neighborhood of $(0,0)$, because it contains $B_{1/2}(0,0)$. The punctured disk

$$D \setminus \{(0,0)\}$$

is not a neighborhood of $(0,0)$, although every small ball around the origin meets it.

§101.3 Interior, exterior, and boundary

Definition 101.3.1. Let $E \subseteq \mathbb{R}^n$. A point P is

(i) an **interior point** of E if some ball around P is contained in E;

(ii) an **exterior point** of E if some ball around P is disjoint from E;

(iii) a **boundary point** of E if every ball around P meets both E and $\mathbb{R}^n \setminus E$.

The set of all boundary points is denoted ∂E.

These three classes partition $\mathbb{R}^n$. The boundary is where local membership cannot be decided by zooming in.

Example 101.3.2

For

$$E = \{(x, y) : x^2 + y^2 < 1\},$$

the interior is E itself, the exterior is $\{(x, y) : x^2 + y^2 > 1\}$, and the boundary is the circle

$$\partial E = \{(x, y) : x^2 + y^2 = 1\}.$$

For the closed disk $\{x^2 + y^2 \leq 1\}$, the boundary is the same circle. Boundary is not the same as "points excluded from the set."

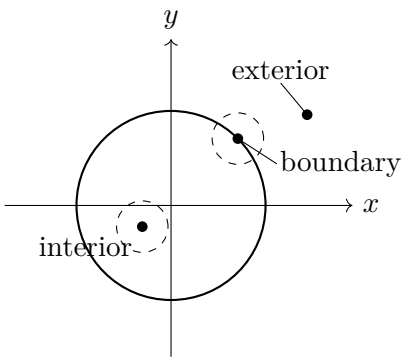


Figure 101.1: Interior points have a small ball inside the set; boundary points never do.

§101.4 Open and closed sets

Definition 101.4.1. A set $E \subseteq \mathbb{R}^n$ is **open** if every point of E is an interior point. It is **closed** if its complement is open.

The two words are not opposites in the everyday sense. A set can be both open and closed, or neither open nor closed.

Example 101.4.2

In $\mathbb{R}$, the interval $(0, 1)$ is open but not closed, $[0, 1]$ is closed but not open, $[0, 1)$ is neither open nor closed, and both $\varnothing$ and $\mathbb{R}$ are both open and closed.

Proposition 101.4.3

A set $E \subseteq \mathbb{R}^n$ is closed if and only if it contains all its boundary points.

Proof. If E is closed, its complement is open, so no boundary point can lie in the complement; hence every boundary point lies in E. Conversely, if E contains all boundary points, then any point outside E is not an interior point of E and not a boundary point, so it is exterior. Thus the complement is open. □

§101.5 Accumulation points and closure

Definition 101.5.1. A point P is an **accumulation point** of E if every punctured ball

$$B_\delta(P) \setminus \{P\}$$

contains a point of E. The **closure** of E, denoted $\overline{E}$, is the union of E with all its accumulation points.

Proposition 101.5.2 (Sequential test)

For $E \subseteq \mathbb{R}^n$, a point P lies in $\overline{E}$ if and only if there is a sequence (P_k) in E such that $P_k \to P$.

Proof. Suppose $P \in \overline{E}$. If $P \in E$, the constant sequence $P_k = P$ converges to P. Otherwise every ball around P meets E, so choose $P_k \in E \cap B_{1/k}(P)$; then $P_k \to P$. Conversely, if $P_k \in E$ and $P_k \to P$, every ball around P contains some P_k. Hence P lies in the closure of E. □

This gives the most useful closed-set criterion:

$$E \text{ is closed} \iff \text{every convergent sequence in } E \text{ has its limit in } E.$$

§101.6 Boundedness and compactness

Definition 101.6.1. A set $E \subseteq \mathbb{R}^n$ is **bounded** if it is contained in some ball $B_R(0)$. It is **compact** if every open cover of E has a finite subcover.

In $\mathbb{R}^n$, compactness has a simpler working form.

Theorem 101.6.2 (Heine–Borel)

A subset $K \subseteq \mathbb{R}^n$ is compact if and only if it is closed and bounded.

For computation, compactness is an existence machine. It makes ”largest,” ”smallest,” and ”convergent subsequence” statements true.

Theorem 101.6.3 (Extreme value theorem)

If $K \subseteq \mathbb{R}^n$ is compact and $f : K \to \mathbb{R}$ is continuous, then f attains a maximum and a minimum on K.

Example 101.6.4

The function $f(x,y) = x^2 + y^2$ has a maximum and a minimum on the closed disk $x^2 + y^2 \leq 1$. It has a minimum but no maximum on the open disk $x^2 + y^2 < 1$. The missing boundary is exactly where the supremum would be reached.

§101.7 Connectedness and path-connectedness

Definition 101.7.1. A set E is **connected** if it cannot be written as the union of two nonempty separated pieces. It is **path-connected** if for any two points $P, Q \in E$, there is a continuous curve $\gamma : [0,1] \to E$ with $\gamma(0) = P$ and $\gamma(1) = Q$.

Path-connectedness implies connectedness. In the elementary domains of multivariable calculus, the two often coincide, but they are not the same in general.

A **region** in many calculus texts usually means an open connected subset of $\mathbb{R}^n$, sometimes together with a piecewise smooth boundary when integration is involved.

§101.8 What this vocabulary controls

The words in this note are not decorative definitions. They decide which theorems are legal:

open set	local perturbations stay inside the domain
closed set	limits of internal sequences remain inside
compact set	continuous functions attain extrema
boundary	where constraints and integration surfaces live
connectedness	where one can move without jumping

When later notes discuss limits, tangent planes, extrema, or multiple integrals, these are the hidden hypotheses that make the calculations legitimate.

Multivariable Limits, Continuity, Differentiability, and the Total Derivative

N-20260306

§102.1 What changes from one variable to many

In one variable, approaching a point means moving from the left or from the right. In several variables, there are infinitely many ways to approach a point: lines, curves, spirals, parabolas, and irregular paths. This is why multivariable limits are stricter than they first appear.

Definition 102.1.1. For $f : D \subseteq \mathbb{R}^n \to \mathbb{R}^m$, the statement

$$\lim_{x \to a} f(x) = L$$

means that for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$0 < \|x - a\| < \delta,\ x \in D \quad \Longrightarrow \quad \|f(x) - L\| < \varepsilon.$$

The definition requires all sufficiently close points, not only points on convenient lines.

§102.2 How to prove and disprove limits

There are two basic strategies.

To prove a limit, dominate the expression by a function of $r = \|x - a\|$ that tends to zero:

$$|f(x) - L| \le g(\|x - a\|), \qquad g(r) \to 0.$$

To disprove a limit, find two paths approaching the same point that give different limiting values.

Example 102.2.1 (Two paths are enough to disprove)

Let

$$f(x, y) = \begin{cases} \dfrac{xy}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

Along the x-axis, $f(x, 0) = 0$. Along the line $y = x$,

$$f(x, x) = \frac{x^2}{2x^2} = \frac{1}{2}.$$

Thus the limit at the origin does not exist.

Example 102.2.2 (Lines are not enough)

Let

$$f(x,y)=\begin{cases}\dfrac{x^2y}{x^4+y^2}, & (x,y)\neq(0,0),\\ 0, & (x,y)=(0,0).\end{cases}$$

Along every line $y=mx$, the value tends to 0. But along $y=x^2$,

$$f(x,x^2)=\frac{x^4}{2x^4}=\frac{1}{2}.$$

So checking all straight lines still does not prove a two-variable limit.

§102.3 Continuity

Definition 102.3.1. A map $f:D\to\mathbb{R}^m$ is **continuous** at $a\in D$ if

$$\lim_{x\to a}f(x)=f(a).$$

It is continuous on D if it is continuous at every point of D.

Continuity is not just a pointwise property; it interacts strongly with compactness and connectedness. A continuous real-valued function on a compact set attains maxima and minima; a continuous image of a connected set is connected.

§102.4 Partial and directional derivatives

For $f:\mathbb{R}^2\to\mathbb{R}$, the partial derivatives at (x_0,y_0) are

$$f_x(x_0,y_0)=\lim_{h\to0}\frac{f(x_0+h,y_0)-f(x_0,y_0)}{h},$$
$$f_y(x_0,y_0)=\lim_{h\to0}\frac{f(x_0,y_0+h)-f(x_0,y_0)}{h}.$$

For a unit vector u, the directional derivative is

$$D_uf(a)=\lim_{t\to0}\frac{f(a+tu)-f(a)}{t}.$$

Partial derivatives measure coordinate-direction changes. Directional derivatives measure changes along chosen lines. Neither, by itself, guarantees differentiability.

Example 102.4.1

For the function

$$f(x,y)=\begin{cases}\dfrac{xy}{x^2+y^2}, & (x,y)\neq(0,0),\\ 0, & (x,y)=(0,0),\end{cases}$$

both partial derivatives at the origin exist and equal 0, because the function is zero on both axes. But the function is not continuous at the origin, hence cannot be differentiable there.

§102.5 Differentiability as linear approximation

The correct higher-dimensional analogue of differentiability is not "all partial derivatives exist." It is "there is a good linear approximation."

Definition 102.5.1. A map $F : \mathbb{R}^n \to \mathbb{R}^m$ is **differentiable** at a if there is a linear map $L : \mathbb{R}^n \to \mathbb{R}^m$ such that

$$\lim_{h \to 0} \frac{\|F(a+h) - F(a) - Lh\|}{\|h\|} = 0.$$

The linear map L is the **total derivative** or **Fréchet derivative**, denoted $DF(a)$.

In coordinates, $DF(a)$ is the Jacobian matrix

$$DF(a) = \left(\frac{\partial F_i}{\partial x_j}(a) \right)_{i,j}.$$

For scalar-valued f, the derivative is represented by the row vector $\nabla f(a)^T$, so

$$f(a+h) = f(a) + \nabla f(a) \cdot h + o(\|h\|).$$

Proposition 102.5.2

If F is differentiable at a, then F is continuous at a. Moreover, for every direction u, the directional derivative exists and is

$$D_u F(a) = DF(a)u.$$

Proof. Differentiability gives

$$F(a+h) - F(a) = DF(a)h + r(h), \qquad \frac{\|r(h)\|}{\|h\|} \to 0.$$

Both $DF(a)h$ and $r(h)$ tend to 0, so continuity follows. For the directional derivative, put $h = tu$ and divide by t. □

Theorem 102.5.3 (A useful sufficient condition)

If all first partial derivatives of F exist in a neighborhood of a and are continuous at a, then F is differentiable at a.

The safe hierarchy is therefore

$$C^1 \implies \text{differentiable} \implies \text{continuous}.$$

Existence of partial derivatives alone sits outside this chain.

§102.6 Tangent planes and normal vectors

For a graph $z = f(x, y)$, differentiability at (x_0, y_0) gives the tangent plane

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0),$$

where $z_0 = f(x_0, y_0)$. This is just the linear approximation written geometrically.

For an implicit surface $F(x, y, z) = 0$, if $\nabla F(P_0) \neq 0$, the tangent plane is

$$\nabla F(P_0) \cdot (P - P_0) = 0.$$

The gradient is normal to the level surface.

Example 102.6.1

For the sphere $F(x, y, z) = x^2 + y^2 + z^2 - 1 = 0$, at $P_0 = (0, 0, 1)$,

$$\nabla F(P_0) = (0, 0, 2).$$

The tangent plane is therefore

$$(0, 0, 2) \cdot (x, y, z - 1) = 0,$$

so $z = 1$.

§102.7 Mixed partial derivatives

For a surface $z = f(x, y)$, the mixed partial derivative

$$f_{xy} = \partial_x(\partial_y f)$$

asks how the y-direction slope changes when one moves in the x-direction. Under the usual smoothness hypotheses, the order does not matter.

Theorem 102.7.1 (Clairaut)

If f_{xy} and f_{yx} are continuous in a neighborhood of a point, then

$$f_{xy} = f_{yx}$$

at that point.

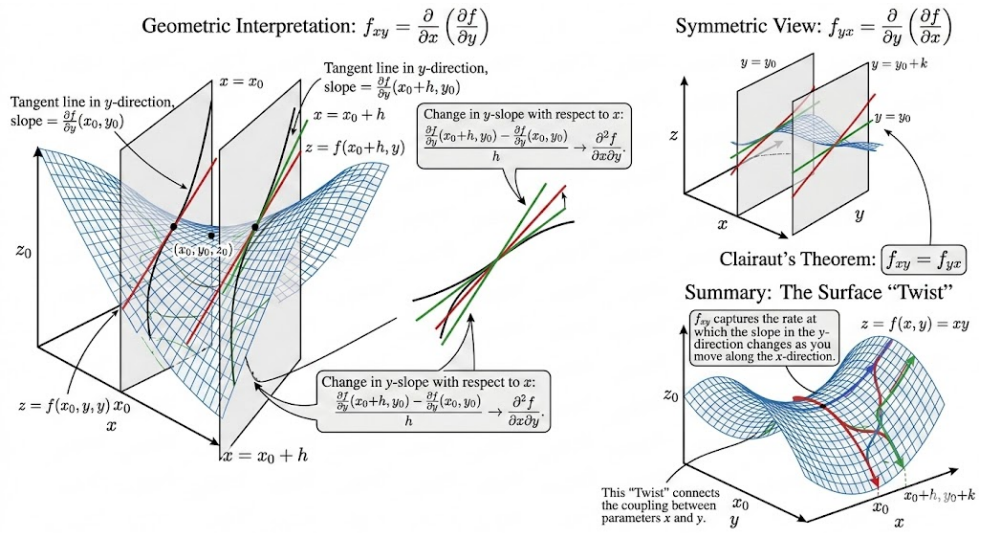


Figure 102.1: A mixed partial derivative measures how one coordinate slope changes as we move in the other coordinate direction.

§102.8 The main lesson

The word ”derivative” in several variables means ”linear map that captures first-order change.” Partial derivatives are entries of that map, directional derivatives are evaluations of that map, tangent planes are pictures of that map, and the Jacobian is its matrix representation.

Multivariable Chain Rule, Total Differentials, and Implicit Differentiation

N-20260313

§103.1 The chain rule is composition of linear maps

The chain rule is not a collection of unrelated formulas. It says that first-order approximations compose.

Theorem 103.1.1 (Chain rule)
Let $F : \mathbb{R}^n \to \mathbb{R}^m$ be differentiable at x, and let $G : \mathbb{R}^m \to \mathbb{R}^k$ be differentiable at $F(x)$. Then $G \circ F$ is differentiable at x, and

$$D(G \circ F)(x) = DG(F(x))\, DF(x).$$

Proof. Write

$$F(x+h) = F(x) + DF(x)h + r(h), \qquad \|r(h)\| = o(\|h\|).$$

Then apply the differentiability expansion for G at $F(x)$:

$$G(F(x)+u) = G(F(x)) + DG(F(x))u + s(u), \qquad \|s(u)\| = o(\|u\|).$$

Substitute $u = DF(x)h + r(h)$. Since $u = O(\|h\|)$, the remainder is still $o(\|h\|)$. The linear part is $DG(F(x))DF(x)h$. □

This theorem is the source of all ordinary partial-derivative chain rules.

§103.2 One input variable, two intermediate variables

Suppose

$$z = f(u,v), \qquad u = u(x), \qquad v = v(x).$$

Then

$$\frac{dz}{dx} = f_u \frac{du}{dx} + f_v \frac{dv}{dx}.$$

In matrix form, this is

$$\frac{dz}{dx} = \begin{pmatrix} f_u & f_v \end{pmatrix} \begin{pmatrix} u_x \\ v_x \end{pmatrix}.$$

The row vector is Df, the column vector is the derivative of (u,v), and their product is the derivative of the composite.

Example 103.2.1

Let $z = u^2 + uv$, $u = x^2$, and $v = \sin x$. Then

$$z_u = 2u + v, \qquad z_v = u, \qquad u' = 2x, \qquad v' = \cos x.$$

Hence

$$\frac{dz}{dx} = (2u + v)2x + u\cos x.$$

Substituting $u = x^2$, $v = \sin x$ gives

$$\frac{dz}{dx} = 2x(2x^2 + \sin x) + x^2 \cos x.$$

§103.3 Two input variables

Suppose

$$z = f(u, v), \qquad u = u(x, y), \qquad v = v(x, y).$$

Then

$$\begin{pmatrix} z_x & z_y \end{pmatrix} = \begin{pmatrix} f_u & f_v \end{pmatrix} \begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix}.$$

Equivalently,

$$z_x = f_u u_x + f_v v_x, \qquad z_y = f_u u_y + f_v v_y.$$

Example 103.3.1

Let $z = e^u \cos v$, where $u = x^2 + y^2$ and $v = x - y$. Then

$$f_u = e^u \cos v, \qquad f_v = -e^u \sin v,$$

and

$$\begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix} = \begin{pmatrix} 2x & 2y \\ 1 & -1 \end{pmatrix}.$$

Thus

$$z_x = e^u \cos v \cdot 2x - e^u \sin v, \qquad z_y = e^u \cos v \cdot 2y + e^u \sin v.$$

§103.4 Total differentials

The notation

$$dz = f_x\, dx + f_y\, dy$$

is best read as a compact way to write a linear approximation. If (x, y) changes by (dx, dy), then the first-order change in $z = f(x, y)$ is

$$dz \approx f_x\, dx + f_y\, dy.$$

For a vector-valued map F, the same statement is

$$dF = J_F\, dx.$$

The differential notation is safe when one remembers that it represents a linear map, not ordinary fraction arithmetic.

§103.5 Implicit differentiation in one equation

Suppose a relation

$$F(x, y) = 0$$

defines y locally as a differentiable function of x. Differentiating the identity $F(x, y(x)) = 0$ gives

$$F_x + F_y \frac{dy}{dx} = 0.$$

If $F_y \neq 0$, then

$$\frac{dy}{dx} = -\frac{F_x}{F_y}.$$

Example 103.5.1

For the circle $F(x, y) = x^2 + y^2 - 1 = 0$, where $y \neq 0$,

$$\frac{dy}{dx} = -\frac{2x}{2y} = -\frac{x}{y}.$$

This is the slope of the tangent line to the circle.

§103.6 Implicit differentiation for systems

The same idea becomes matrix algebra. Suppose

$$F(x, u) = 0, \qquad F : \mathbb{R}^p \times \mathbb{R}^q \to \mathbb{R}^q,$$

and u is locally a function of x. Differentiating gives

$$F_x \, dx + F_u \, du = 0.$$

If F_u is invertible, then

$$du = -F_u^{-1} F_x \, dx,$$

so

$$Du(x) = -F_u^{-1} F_x.$$

Theorem 103.6.1 (Implicit function theorem, working form)

Let $F : \mathbb{R}^p \times \mathbb{R}^q \to \mathbb{R}^q$ be C^1, and suppose

$$F(a, b) = 0, \qquad \det F_u(a, b) \neq 0.$$

Then near (a, b), the equation $F(x, u) = 0$ determines u as a C^1 function of x, and

$$Du(a) = -F_u(a, b)^{-1} F_x(a, b).$$

The theorem says when the formal linearized computation is legitimate. The condition $\det F_u \neq 0$ means the constraint can be solved in the u-directions.

§103.7 A shape table

Keeping track of shapes prevents many mistakes:

map	derivative shape	meaning
$f:\mathbb{R}^n\to\mathbb{R}$	$1\times n$	row gradient
$F:\mathbb{R}^n\to\mathbb{R}^m$	$m\times n$	Jacobian
$G\circ F$	$k\times n$	$(k\times m)(m\times n)$
$F(x,u)=0$	$q\times p,\ q\times q$	F_x, F_u

The formulas work because the linear maps have compatible domains and codomains.

§103.8 When Implicit Linearization Fails

The invertibility condition in the implicit function theorem cannot simply be omitted. Consider

$$F(x,y)=y^2-x.$$

At the origin,

$$F(0,0)=0, \qquad F_y(0,0)=0.$$

The zero set is

$$x=y^2,$$

or, for $x\geq 0$, the two branches

$$y=\sqrt{x}, \qquad y=-\sqrt{x}.$$

No neighborhood of the origin in the zero set is the graph of a single function $y=u(x)$, and the two branch derivatives become unbounded as $x\downarrow 0$. The formal formula

$$u'(x)=-\frac{F_x}{F_y}=\frac{1}{2y}$$

therefore signals its own breakdown at $y=0$.

The same zero set can be solved smoothly in the other direction:

$$x=y^2,$$

because

$$F_x(0,0)=-1\neq 0.$$

Implicit solvability is directional. The Jacobian block that is invertible determines which variables may be expressed as functions of the others.

A Problem-Solving Guide for Multivariable Limits and Local Geometry

N-20260314

§104.1 Purpose of this guide

Multivariable problems become simpler when the first question is methodological: does the task ask for a limit, a linear approximation, a tangent direction, or an extremum? Each requires a different test. The examples below make explicit which conclusions are justified and where a tempting shortcut breaks down.

§104.2 Classifying points relative to a set

For a set $G \subseteq \mathbb{R}^n$, classify a point P by inspecting small balls around it:

classification	ball test
interior	$\exists r > 0,\ B_r(P) \subseteq G$
exterior	$\exists r > 0,\ B_r(P) \cap G = \varnothing$
boundary	$\forall r > 0,\ B_r(P) \cap G \neq \varnothing$ and $B_r(P) \cap G^c \neq \varnothing$

Example 104.2.1

For

$$G = \{(x, y) : 0 < x^2 + y^2 < 1\},$$

the origin is not in G, but it is a boundary point. Every ball around the origin contains points with $0 < x^2 + y^2 < 1$, and also contains the origin itself, which lies outside G.

§104.3 Limit templates

To prove a limit at the origin, try to convert the expression into polar form

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

If the expression becomes

$$r^{\alpha} g(\theta)$$

where $\alpha > 0$ and g is bounded, then the limit is 0.

Example 104.3.1

For

$$f(x,y)=\frac{x^2y}{x^2+y^2},$$

we have

$$f(r\cos\theta,r\sin\theta)=r\cos^2\theta\sin\theta.$$

Since $|\cos^2\theta\sin\theta|\le 1$, the limit is 0.

To disprove a limit, find paths giving different values. The common path choices are:

$$y=0,\quad x=0,\quad y=mx,\quad y=x^2,\quad x=y^2.$$

Caution

Path testing can disprove a limit, but it usually cannot prove one. Even checking every straight line is not enough, because a curved path may still detect different behavior.

§104.4 Degree comparison for rational expressions

For rational expressions near the origin, compare the lowest total degrees of numerator and denominator only after checking whether the denominator is comparable to a power of r. The estimate

$$x^2+y^2=r^2$$

is reliable. The expression x^4+y^2, however, is anisotropic: along $y=x^2$, the two terms have the same size.

Example 104.4.1

The function

$$f(x,y)=\frac{x^2y}{x^4+y^2}$$

looks as if the numerator has degree 3 and the denominator has degree 2, but the denominator changes scale depending on direction. Along $y=x^2$, the limit is $1/2$. Degree-counting alone fails.

§104.5 Differentiability checklist

For a scalar function $f(x,y)$, use the following hierarchy:

$$C^1 \Longrightarrow \text{differentiable} \Longrightarrow \text{continuous}.$$

Partial derivatives alone do not imply continuity. Directional derivatives in every direction alone do not imply differentiability.

To prove differentiability at a, the definition asks for

$$f(a+h)=f(a)+L(h)+o(\|h\|).$$

The candidate linear map is usually

$$L(h)=\nabla f(a)\cdot h.$$

Then one must estimate the remainder.

§104.6 Tangent and normal templates

For a graph $z = f(x, y)$, the tangent plane at (x_0, y_0, z_0) is

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

For an implicit surface $F(x, y, z) = 0$, the normal vector is ∇F, and the tangent plane is

$$\nabla F(P_0) \cdot (P - P_0) = 0.$$

For a space curve given by the intersection

$$F(x, y, z) = 0, \qquad G(x, y, z) = 0,$$

a tangent direction is

$$\nabla F(P_0) \times \nabla G(P_0),$$

provided the two normals are not parallel.

Example 104.6.1

The curve

$$x^2 + y^2 + z^2 = 1, \qquad z = x$$

at a point P_0 has tangent direction

$$\nabla(x^2 + y^2 + z^2 - 1)(P_0) \times \nabla(z - x)(P_0).$$

This works because the curve lies on both surfaces, so its tangent vector must be perpendicular to both surface normals.

§104.7 Local extrema template

For an unconstrained interior extremum of a differentiable function, first solve

$$\nabla f = 0.$$

Then classify candidates using the Hessian when possible. For two variables,

$$H = \begin{pmatrix} A & B \\ B & C \end{pmatrix}, \qquad D = AC - B^2.$$

At a critical point,

condition	classification
$D > 0,\ A > 0$	local minimum
$D > 0,\ A < 0$	local maximum
$D < 0$	saddle
$D = 0$	inconclusive

For a constraint $g(x, y) = 0$, first try

$$\nabla f = \lambda \nabla g.$$

Geometrically, the level curve of f is tangent to the constraint curve at a constrained extremum.

§104.8 Common failure modes

(i) Checking axes proves almost nothing about a two-variable limit.

(ii) Checking all lines still may not prove a limit.

(iii) Existence of partial derivatives is not differentiability.

(iv) The Hessian test classifies only critical points, and it is inconclusive when $D = 0$.

(v) Lagrange multipliers require checking regularity and comparing all candidate points, including boundary or endpoint cases when the constraint has them.

The common idea is local replacement: balls replace vague closeness, radial estimates replace path guessing, and linear maps replace nonlinear functions to first order.

Extrema of Two-Variable Functions and the Hessian Test

N-20260325

§105.1 Local extrema

Let $f : G \subseteq \mathbb{R}^2 \to \mathbb{R}$. A point $P_0 \in G$ is a **local maximum** if there is a neighborhood U of P_0 such that

$$f(P) \le f(P_0) \qquad (P \in U \cap G).$$

It is a **local minimum** if the inequality is reversed. If the inequality is strict for all nearby $P \neq P_0$, the extremum is strict.

The word "local" matters. A point can be best among nearby points without being best on the whole domain.

§105.2 First-order necessary condition

Theorem 105.2.1 (Fermat theorem in several variables)
If f is differentiable at an interior point $P_0 \in G$ and P_0 is a local extremum, then

$$\nabla f(P_0) = 0.$$

Proof. For every direction u, the one-variable function

$$\varphi(t) = f(P_0 + tu)$$

has a local extremum at $t = 0$. Hence $\varphi'(0) = 0$. But

$$\varphi'(0) = \nabla f(P_0) \cdot u.$$

This vanishes for every u, so $\nabla f(P_0) = 0$. □

A point with $\nabla f = 0$ is called a **critical point**. Critical points are candidates, not conclusions.

§105.3 Quadratic model and the Hessian

At a critical point P_0, the Taylor expansion begins with the quadratic term:

$$f(P_0 + h) = f(P_0) + \frac{1}{2} h^T H(P_0) h + o(\|h\|^2),$$

where

$$H(P_0) = \begin{pmatrix} f_{xx}(P_0) & f_{xy}(P_0) \\ f_{yx}(P_0) & f_{yy}(P_0) \end{pmatrix}.$$

When the second partial derivatives are continuous, $f_{xy} = f_{yx}$, so the Hessian is symmetric.

Thus the classification problem becomes a problem about a quadratic form.

§105.4 Second derivative test

Let

$$A = f_{xx}(P_0), \qquad B = f_{xy}(P_0), \qquad C = f_{yy}(P_0), \qquad D = AC - B^2.$$

Theorem 105.4.1 (Second derivative test)

Assume $\nabla f(P_0) = 0$ and the second partial derivatives are continuous near P_0. Then:

condition	conclusion
$D > 0,\ A > 0$	P_0 is a strict local minimum
$D > 0,\ A < 0$	P_0 is a strict local maximum
$D < 0$	P_0 is a saddle point
$D = 0$	the test is inconclusive

Proof. The quadratic part is

$$q(h,k) = Ah^2 + 2Bhk + Ck^2.$$

A symmetric 2×2 matrix is positive definite exactly when $A > 0$ and $D = AC - B^2 > 0$. It is negative definite exactly when $A < 0$ and $D > 0$. If $D < 0$, the quadratic form takes both positive and negative values. The Taylor expansion shows that the sign of the quadratic term controls the sign of $f(P_0 + h) - f(P_0)$ for sufficiently small h, except in the inconclusive case $D = 0$. □

Example 105.4.2

For

$$f(x,y) = x^2 + 3y^2,$$

the only critical point is $(0,0)$, and

$$H = \begin{pmatrix} 2 & 0 \\ 0 & 6 \end{pmatrix}.$$

Here $D = 12 > 0$ and $A = 2 > 0$, so $(0,0)$ is a strict local minimum.

Example 105.4.3

For

$$f(x,y) = x^2 - y^2,$$

the origin is critical, but

$$H = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}, \qquad D = -4 < 0.$$

The origin is a saddle point. Indeed, along the x-axis the function is positive, while along the y-axis it is negative.

Example 105.4.4 (Inconclusive case)
For $f(x,y) = x^4 + y^4$, the Hessian at the origin is the zero matrix, so $D = 0$. The test is inconclusive, but the origin is still a strict local minimum. For $g(x,y) = x^3 - y^3$, the Hessian at the origin is also zero, but the origin is not an extremum. This is why $D = 0$ really means "use another method."

§105.5 A picture of the Hessian test

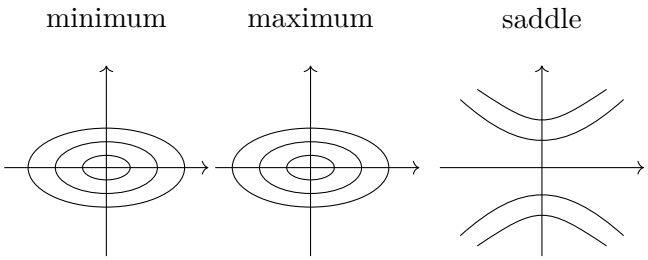


Figure 105.1: Near a nondegenerate critical point, the Hessian determines the local quadratic shape.

§105.6 Constrained extrema

Suppose one wants to optimize $f(x,y)$ subject to a constraint

$$g(x,y) = 0.$$

At a regular point of the constraint curve, $\nabla g \neq 0$, so ∇g is normal to the constraint. At a constrained extremum, moving along the constraint produces no first-order change in f. Thus ∇f must also be normal to the constraint.

Theorem 105.6.1 (Lagrange multiplier condition)
If P_0 is a constrained local extremum of f on $g = 0$, and $\nabla g(P_0) \neq 0$, then there exists λ such that

$$\nabla f(P_0) = \lambda \nabla g(P_0).$$

For several constraints $g_1 = \cdots = g_k = 0$, the gradient of f lies in the span of the constraint gradients:

$$\nabla f = \lambda_1 \nabla g_1 + \cdots + \lambda_k \nabla g_k.$$

Example 105.6.2

Find extrema of $f(x,y)=x+y$ on the circle $x^2+y^2=1$. The equations are

$$(1,1)=\lambda(2x,2y), \qquad x^2+y^2=1.$$

Thus $x=y$, and $2x^2=1$, so

$$(x,y)=\left(\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\right) \quad \text{or} \quad (x,y)=\left(-\frac{1}{\sqrt{2}},-\frac{1}{\sqrt{2}}\right).$$

The corresponding values are $\sqrt{2}$ and $-\sqrt{2}$, the maximum and minimum.

§105.7 Endpoint and boundary warnings

The equation $\nabla f=0$ applies only to interior unconstrained extrema. The equation $\nabla f=\lambda\nabla g$ applies only at regular constrained points. If the domain has corners, endpoints, singular constraint points, or boundary pieces, they must be checked separately.

A reliable workflow is:

1. find interior critical points;
2. find constrained or boundary candidates;
3. check singular points and endpoints;
4. compare values when the problem asks for absolute extrema.

Double Integrals: Riemann Sums, Area, Symmetry, and Average Value

N-20260403

§106.1 From area sums to double integrals

A double integral is the two-dimensional version of adding many small contributions. If a quantity has density $f(x,y)$ over a plane region D, then a small cell of area ΔA_i contributes approximately

$$f(\xi_i,\eta_i)\Delta A_i.$$

Adding all cells and refining the partition gives the integral.

Definition 106.1.1 (Riemann double integral)**.** Let $D \subseteq \mathbb{R}^2$ be a bounded Jordan-measurable region and let $f : D \to \mathbb{R}$ be bounded. Partition D into small subregions ΔD_i, choose sample points $(\xi_i,\eta_i) \in \Delta D_i$, and form

$$S(P,f) = \sum_i f(\xi_i,\eta_i)\Delta A_i.$$

If $S(P,f)$ tends to a limit independent of the partition and sample points as the mesh tends to 0, then the limit is denoted

$$\iint_D f(x,y)\,dA.$$

The notation dA or $d\sigma$ represents an infinitesimal area element. It is not an independent variable; it tells us what geometric measure is being used.

§106.2 Geometric meanings

The meaning depends on the integrand.

integrand	meaning of $\iint_D f\,dA$
$f = 1$	area of D
$f \geq 0$	volume under $z = f(x,y)$
f sign-changing	signed volume
$f = \rho$	mass of a lamina with density ρ

Example 106.2.1 (Area as an integral)
For the disk $D = \{(x, y) : x^2 + y^2 \le R^2\}$,

$$\text{area}(D) = \iint_D 1 \, dA.$$

Using polar coordinates later gives

$$\int_0^{2\pi} \int_0^R r \, dr \, d\theta = \pi R^2.$$

§106.3 Basic properties

The following properties are the main algebra of double integrals.

Proposition 106.3.1
Assume the displayed integrals exist.

(i) Linearity:

$$\iint_D (af + bg) \, dA = a \iint_D f \, dA + b \iint_D g \, dA.$$

(ii) Additivity over regions: if $D = D_1 \cup D_2$ and the overlap has area zero, then

$$\iint_D f \, dA = \iint_{D_1} f \, dA + \iint_{D_2} f \, dA.$$

(iii) Monotonicity: if $f \le g$ on D, then

$$\iint_D f \, dA \le \iint_D g \, dA.$$

(iv) Estimate: if $m \le f \le M$ on D, then

$$m \ \text{area}(D) \le \iint_D f \, dA \le M \ \text{area}(D).$$

Proof idea. Each statement is first true for finite Riemann sums. Passing to the limit preserves addition, scalar multiplication, and inequalities. The estimate comes from multiplying $m \le f(\xi_i, \eta_i) \le M$ by ΔA_i and summing. □

§106.4 Existence of the integral

For most calculus computations, the following sufficient condition is enough.

Theorem 106.4.1
If f is continuous on a bounded closed elementary region D, then f is Riemann integrable on D.

The theorem is often used silently. It is the reason that ordinary continuous examples can be computed by choosing convenient coordinates and orders of integration.

Remark 106.4.2 — The statement is not saying that only continuous functions are integrable. Piecewise continuous functions and many functions with mild discontinuities are also integrable. Continuity is simply the safest elementary hypothesis.

§106.5 Symmetry before computation

Symmetry is one of the fastest ways to evaluate an integral.

Proposition 106.5.1 (Odd cancellation)

Let D be symmetric under a measure-preserving transformation $T : D \to D$. If

$$f(T(x, y)) = -f(x, y),$$

then

$$\iint_D f \, dA = 0.$$

If $f \circ T = f$, then the integral over D can often be reduced to a fundamental half or sector of D.

Example 106.5.2

Let D be the disk centered at the origin. Since D is symmetric under $(x, y) \mapsto (-x, y)$,

$$\iint_D xe^{y^2} \, dA = 0.$$

The integrand is odd in x, while the region is unchanged by reflecting across the y-axis.

§106.6 Mean value theorem for double integrals

Theorem 106.6.1 (Mean value theorem)

Let D be compact, connected, Jordan-measurable, and have strictly positive area, and let $f : D \to \mathbb{R}$ be continuous. Then there exists a point $P_0 \in D$ such that

$$\iint_D f \, dA = f(P_0) \operatorname{area}(D).$$

Equivalently, the average value

$$f_{\text{avg}} = \frac{1}{\operatorname{area}(D)} \iint_D f \, dA$$

is attained somewhere on D.

Proof skeleton. By compactness, f attains a minimum m and maximum M on D. The

estimate property gives

$$m \le f_{\text{avg}} \le M.$$

Since D is connected and f is continuous, the image $f(D)$ is an interval. Therefore the intermediate value theorem gives a point P_0 where $f(P_0) = f_{\text{avg}}$. The positive-area hypothesis is essential for dividing by $\text{area}(D)$. Connectedness cannot simply be omitted: take two disjoint equal-area squares and the continuous function which is zero on one and two on the other. Its average equals one, a value the function never attains. □

§106.7 A worked average-value example

Example 106.7.1

Let $D = [0,1] \times [0,2]$ and $f(x,y) = x + y$. Then

$$\iint_D (x+y)\,dA = \int_0^1 \int_0^2 (x+y)\,dy\,dx = \int_0^1 (2x+2)\,dx = 3.$$

Since $\text{area}(D) = 2$, the average value is $3/2$. The mean-value theorem says there is a point in the rectangle with $x + y = 3/2$, for example $(1/2, 1)$.

§106.8 The region decides the order of integration

Now take a region that is deliberately *not* a rectangle:

$$D = \{(x,y) : 0 \le x \le 1, \quad x^2 \le y \le x\}.$$

It is the strip between the parabola $y = x^2$ and the line $y = x$. Drawing it first is the essential step: for a fixed x, the lower and upper heights are x^2 and x; for a fixed y, the left and right endpoints are y and $\sqrt{y}$. Both descriptions name the same set:

$$\begin{aligned} D &= \{0 \le x \le 1,\ x^2 \le y \le x\} \\ &= \{0 \le y \le 1,\ y \le x \le \sqrt{y}\}. \end{aligned}$$

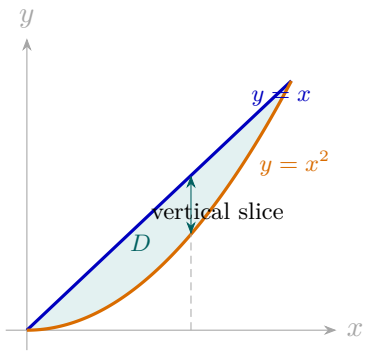


Figure 106.1: One region, two slicing directions. On vertical slices, $x^2 \le y \le x$; on horizontal slices, $y \le x \le \sqrt{y}$. Both boundaries must change together when reversing the integrals.

For the density $f(x,y) = x + 2y$, compute with vertical slices:

$$\begin{aligned} I &= \iint_D (x+2y)\,dA = \int_0^1 \int_{x^2}^{x} (x+2y)\,dy\,dx \\ &= \int_0^1 \Big[xy + y^2\Big]_{x^2}^{x}\,dx \\ &= \int_0^1 (2x^2 - x^3 - x^4)\,dx \\ &= \frac{2}{3} - \frac{1}{4} - \frac{1}{5} = \boxed{\frac{13}{60}}. \end{aligned}$$

Reversing the order is a useful independent check:

$$\begin{aligned} I &= \int_0^1 \int_{y}^{\sqrt{y}} (x+2y)\,dx\,dy \\ &= \int_0^1 \left[\frac{x^2}{2} + 2yx\right]_{y}^{\sqrt{y}}\,dy \\ &= \int_0^1 \left(\frac{y}{2} - \frac{5y^2}{2} + 2y^{3/2}\right)\,dy = \frac{1}{4} - \frac{5}{6} + \frac{4}{5} = \frac{13}{60}. \end{aligned}$$

These are two evaluations of the *same* absolutely integrable function on the *same* region, exactly the setting in which Fubini's theorem licenses the exchange. Nothing analogous would justify an arbitrary exchange in the singular, non-absolutely-integrable example below.[1]

The area and average are just as concrete:

$$\operatorname{area}(D) = \int_0^1 (x - x^2)\,dx = \frac{1}{6}, \qquad f_{\text{avg}} = \frac{I}{\operatorname{area}(D)} = \boxed{\frac{13}{10}}.$$

The point $(\frac{1}{2}, \frac{2}{5})$ really lies in D, since $\frac{1}{4} \le \frac{2}{5} \le \frac{1}{2}$, and its value $\frac{1}{2} + 2 \cdot \frac{2}{5} = 13/10$ realizes the average predicted by the mean-value theorem.

There is a small probabilistic bonus. If (X, Y) is chosen uniformly in D, the density of X is proportional to the vertical slice length:

$$p_X(x) = 6(x - x^2), \qquad 0 \le x \le 1.$$

Conditional on $X = x$ in the interior, Y is uniform between x^2 and x. Therefore

$$\mathbb{E}[X] = \frac{1}{2}, \qquad \mathbb{E}[Y \mid X = x] = \frac{x + x^2}{2}, \qquad \mathbb{E}[Y] = \frac{2}{5}.$$

The tower property now reproduces $\mathbb{E}[X + 2Y] = \frac{1}{2} + 2 \cdot \frac{2}{5} = \frac{13}{10}$ without setting up a new double integral. Describing a region by its slices is also describing the marginal and conditional laws of a uniform random point in that region.

§106.9 Relation with product measure

The elementary Riemann picture divides a region into small rectangles. In measure-theoretic language, this is integration with respect to two-dimensional Lebesgue measure:

$$\iint_D f(x,y)\,dA = \int_{\mathbb{R}^2} f(x,y)\mathbf{1}_D(x,y)\,d\lambda^2.$$

[1] For the geometric mechanics of changing order, see MIT OpenCourseWare, *Multivariable Calculus*, Session 49.

This viewpoint separates two choices:

(i) the function being accumulated;

(ii) the measure that tells how large each piece of the region is.

For nonnegative measurable functions, Tonelli's theorem justifies iterated integration even when the value may be infinite. For sign-changing functions, Fubini's theorem requires integrability of $|f|$. The elementary rule "switch the order of integration" is therefore backed by real hypotheses.

§106.10 Fubini without Absolute Convergence

It is tempting to treat order-switching as a harmless algebraic move. The advanced point is sharper: for sign-changing functions, order-switching is licensed by absolute integrability. Without that hypothesis, different limiting procedures can see different cancellations.

A standard improper example on the square near the origin is

$$f(x,y) = \frac{x^2 - y^2}{(x^2+y^2)^2}, \qquad (x,y) \in (0,1]^2.$$

For fixed $x > 0$,

$$\int_0^1 \frac{x^2 - y^2}{(x^2+y^2)^2}\, dy = \left[\frac{y}{x^2+y^2}\right]_0^1 = \frac{1}{1+x^2}.$$

Hence

$$\int_0^1 \int_0^1 f(x,y)\, dy\, dx = \frac{\pi}{4}.$$

But for fixed $y > 0$,

$$\int_0^1 \frac{x^2 - y^2}{(x^2+y^2)^2}\, dx = \left[-\frac{x}{x^2+y^2}\right]_0^1 = -\frac{1}{1+y^2},$$

so

$$\int_0^1 \int_0^1 f(x,y)\, dx\, dy = -\frac{\pi}{4}.$$

The two answers differ because the singularity at $(0,0)$ is not absolutely integrable. This is the concrete content of the distinction between Tonelli's theorem and Fubini's theorem.

§106.11 Product measure behind rectangular sums

The phrase "divide the region into rectangles" hides a construction. Rectangles $A \times B$ are assigned area by

$$\lambda^2(A \times B) = \lambda(A)\lambda(B).$$

This rule first defines area on finite unions of rectangles. One then extends it to the sigma-algebra generated by rectangles. The double integral is integration with respect to this product measure.

This explains why iterated integrals work first for indicator functions of rectangles:

$$\int \mathbf{1}_{A\times B}(x,y)\, d(\mu \times \nu) = \mu(A)\nu(B) = \int_A \left(\int_B 1\, d\nu \right) d\mu.$$

Linearity handles simple functions, monotone convergence handles nonnegative measurable functions, and absolute integrability handles signed functions. Thus the elementary Riemann-sum picture is not abandoned in measure theory; it is completed.

§106.12 Symmetry as projection onto invariant functions

If a finite group G acts on D by area-preserving transformations, averaging over the group defines the Reynolds operator

$$\mathcal{R}f = \frac{1}{|G|}\sum_{g\in G} f\circ g.$$

Because the action preserves area,

$$\iint_D f\,dA = \iint_D \mathcal{R}f\,dA.$$

So integration only sees the invariant part of the function. Odd functions cancel because their invariant part is zero. This is the same idea that later appears in representation theory as projection onto the trivial representation, but here it is already visible in a disk or rectangle symmetry problem.

For example, on the disk, the rotation group kills all angular Fourier modes except the constant mode:

$$f(r,\theta) = a_0(r) + \sum_{n\geq 1}\left(a_n(r)\cos n\theta + b_n(r)\sin n\theta\right)$$

gives

$$\iint_D f\,dA = \int_0^R \int_0^{2\pi} a_0(r)\,r\,d\theta\,dr.$$

The elementary symmetry shortcut is the first case of harmonic decomposition.

§106.13 Average value as a pushforward distribution

For continuous $f : D \to \mathbb{R}$, the average value can be viewed probabilistically. Put the uniform probability measure on D:

$$\mathbb{P}(E) = \frac{\text{area}(E)}{\text{area}(D)}.$$

Then

$$f_{\text{avg}} = \mathbb{E}[f].$$

The pushforward measure $f_*\mathbb{P}$ on $\mathbb{R}$ records how much area of D lies at each range of values of f. The mean value theorem says that if D is connected and f is continuous, this expectation lies inside the interval $f(D)$, hence equals $f(P_0)$ for some point.

This interpretation is useful later: integration can be studied either by integrating over the original region or by pushing measure forward through a function and integrating over its values.

§106.14 How the double-integral setup is checked

A double integral begins as a limit of weighted area sums, so the first data to identify are the region D, the area element, and the function being averaged over that region. Linearity and additivity come directly from cutting the region into pieces; this is why splitting a domain along a change in boundary formula is legitimate rather than cosmetic.

The examples in this note give three practical tests. If the region has a symmetry and the integrand is odd under that symmetry, the corresponding contribution vanishes because the area measure is preserved. If the question asks for an average value, rewrite

$$\iint_D f\,dA = f_{\text{avg}}\,\text{area}(D)$$

and check connectedness and continuity before invoking the mean value theorem. If the computation is rearranged by Fubini or by a change of variables, the real issue is not the order of symbols but whether the slices describe the same region and whether the Jacobian accounts for area distortion. This is the point at which Tonelli, Fubini, and change of variables turn the geometric setup into a justified calculation.

Changing Variables in Double Integrals: Geometry, Jacobians, and Multiplicity

N-20260405

The bounds of an iterated integral are descriptions of a region, not decorations attached to the integration symbols. A good coordinate change transforms a difficult region into a simple one, but it must also measure how area changes. The determinant of the derivative gives this local area factor. When the coordinate map covers a point several times, an additional global issue appears: multiplicity.[1]

§107.1 A reversed integral should describe the same region

Consider the region between $y = x^2$ and $y = x$ for $0 \le x \le 1$. It is

$$D = \{(x, y) : 0 \le x \le 1,\ x^2 \le y \le x\}.$$

For each fixed x, the vertical slice has length $x - x^2$, so

$$\operatorname{area}(D) = \int_0^1 (x - x^2)\,dx = \frac{1}{6}.$$

If we choose horizontal slices, the conditions $x^2 \le y \le x$ become

$$y \le x \le \sqrt{y}, \qquad 0 \le y \le 1.$$

Therefore the same area is

$$\boxed{\int_0^1 \int_{x^2}^{x} 1\,dy\,dx = \int_0^1 \int_{y}^{\sqrt{y}} 1\,dx\,dy = \frac{1}{6}.}$$

Nothing about the integrand changed. We merely described the same measurable set as a union of different one-dimensional fibers. The equality is an instance of Fubini's theorem.

§107.2 Jacobian as an Area Scale

Let $\Phi(u, v) = (x(u, v), y(u, v))$ be a continuously differentiable map. Near a point (u_0, v_0), a tiny rectangle of sides du, dv is sent, to first order, to a parallelogram spanned by

$$\Phi_u\,du, \qquad \Phi_v\,dv.$$

Its area is

$$\left|\det \begin{pmatrix} x_u & x_v \\ y_u & y_v \end{pmatrix}\right| du\,dv.$$

[1] For Fubini and the multivariable change-of-variables theorem, see MIT 18.02SC, *Multivariable Calculus*, Units 4–5: https://ocw.mit.edu/courses/18-02sc-multivariable-calculus-fall-2010/. The general multiplicity formula is discussed in Evans and Gariepy, *Measure Theory and Fine Properties of Functions*.

Thus the local area scale is the absolute Jacobian determinant

$$J_\Phi = \det D\Phi.$$

For a one-to-one C^1 coordinate change with nonzero Jacobian on a suitable region, the change-of-variables theorem reads

$$\boxed{\iint_{\Phi(U)} f(x,y)\,dx\,dy = \iint_U f(\Phi(u,v))|J_\Phi(u,v)|\,du\,dv.}$$

The absolute value ensures that ordinary area is positive whether the coordinate transformation preserves or reverses orientation. For oriented differential forms, the sign of the determinant is retained instead.

§107.3 Coordinates adapted to two families of straight lines

Suppose a region is bounded by lines of the form $x+y=$ constant and $x-y=$ constant. Set

$$u = x+y, \qquad v = x-y.$$

Then

$$x = \frac{u+v}{2}, \qquad y = \frac{u-v}{2}, \qquad \det\frac{\partial(x,y)}{\partial(u,v)} = -\frac{1}{2}.$$

Consider the parallelogram

$$D = \{(x,y) : 0 \le x+y \le 2, \quad 0 \le x-y \le 1\}.$$

It becomes the rectangle $0 \le u \le 2,\ 0 \le v \le 1$. Its area is immediately

$$\operatorname{area}(D) = \int_0^2 \int_0^1 \frac{1}{2}\,dv\,du = 1.$$

The same substitution simplifies nonconstant integrands. Since

$$x^2 - y^2 = (x+y)(x-y) = uv,$$

we get

$$\boxed{\iint_D (x^2-y^2)\,dx\,dy = \frac{1}{2}\int_0^2 \int_0^1 uv\,dv\,du = \frac{1}{2}.}$$

The region and the integrand become simple at the same time. This is the real criterion for a useful substitution.

§107.4 Polar coordinates derive a Gaussian integral

The polar map is

$$\Phi(r,\theta) = (r\cos\theta, r\sin\theta), \qquad r \ge 0, \quad 0 \le \theta < 2\pi.$$

Its derivative has determinant

$$\det\begin{pmatrix}\cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta\end{pmatrix} = r.$$

Thus, away from the origin and the seam at $\theta = 0$, polar coordinates give

$$dx\,dy = r\,dr\,d\theta.$$

The factor r is not arbitrary: angular displacements of size $d\theta$ sweep arcs whose length is approximately $r\,d\theta$.

A particularly elegant use is to evaluate

$$I = \int_{-\infty}^{\infty} e^{-x^2}\,dx.$$

The integral is positive and finite. Tonelli's theorem gives

$$I^2 = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)}\,dx\,dy.$$

Polar coordinates convert the radial Gaussian into a one-dimensional exponential:

$$I^2 = \int_0^{2\pi}\int_0^{\infty} e^{-r^2} r\,dr\,d\theta = 2\pi\cdot\frac{1}{2} = \pi.$$

Hence

$$\boxed{\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}.}$$

The two-dimensional detour solves a one-dimensional integral because the product of two Gaussian factors becomes rotationally symmetric.

§107.5 Global Overcounting by a Jacobian

The usual formula needs a one-to-one parametrization, at least apart from negligible boundary sets. To see why, identify the plane with the complex numbers and consider

$$\Phi(z) = z^2.$$

On the annulus $A = \{z : 1 < |z| < 2\}$, the map is smooth and its real Jacobian is

$$|J_\Phi(z)| = |\Phi'(z)|^2 = 4|z|^2.$$

Its image is the annulus $B = \{w : 1 < |w| < 4\}$. But each point of B has *two* preimages in A, the two square roots.

If we blindly apply the one-to-one formula to the constant function 1, we obtain

$$\int_A 4|z|^2\,dA = \int_0^{2\pi}\int_1^2 4r^3\,dr\,d\theta = 30\pi.$$

The actual area of B is

$$|B| = \pi(4^2 - 1^2) = 15\pi.$$

The integral with the Jacobian is twice the image area, exactly because the annulus is covered twice:

$$\boxed{\int_A |J_\Phi(z)|\,dA = 2\,|\Phi(A)|.}$$

More generally, the area formula for a smooth finite-to-one map involves the number of preimages of almost every image point. A nonzero local Jacobian guarantees local invertibility, but it does not guarantee global one-to-one behavior.

§107.6 Coarea counts fibers of a nonlinear function

The same distinction appears when the new variable is a scalar function rather than a full coordinate system. For a sufficiently regular $g : \mathbb{R}^2 \to \mathbb{R}$, the coarea formula relates a planar integral to integrals over the level curves $g^{-1}(t)$:

$$\int_{\mathbb{R}^2} h(x)|\nabla g(x)|\,dx = \int_{\mathbb{R}} \left(\int_{g^{-1}(t)} h(x)\,d\mathcal{H}^1(x) \right) dt.$$

For $g(x, y) = x^2 + y^2$, the level $g = t > 0$ is a circle of radius $\sqrt{t}$, and $|\nabla g| = 2\sqrt{t}$. Consequently a radial function $\psi(g)$ has

$$\begin{aligned} \int_{\mathbb{R}^2} \psi(x^2 + y^2)\,dx\,dy &= \int_0^\infty \psi(t) \frac{2\pi\sqrt{t}}{2\sqrt{t}}\,dt \\ &= \boxed{\pi \int_0^\infty \psi(t)\,dt}. \end{aligned}$$

Taking $\psi(t) = e^{-t}$ recovers the Gaussian calculation.

The lesson extends from elementary region sketches to geometric measure theory. Reversing integration order changes the slicing map; the Jacobian measures local area distortion; the area formula accounts for how often a parametrization covers the image; and coarea integrates over the fibers of a nonlinear function. Before making a substitution, the decisive questions are what the new fibers look like, how they scale, and whether they overlap.

Triple Integrals: Solids, Cylindrical Coordinates, and Spherical Coordinates

N-20260406

§108.1 Triple integrals as weighted volume

A triple integral accumulates a scalar field over a spatial region $\Omega \subseteq \mathbb{R}^3$:

$$\iiint_{\Omega} f(x,y,z)\,dV.$$

If $f = 1$, the integral is volume. If $f = \rho$ is a mass density, the integral is mass. If f is source density, the integral is total source.

Definition 108.1.1. An xy-simple solid has the form

$$\Omega = \{(x,y,z) : (x,y) \in D,\ z_1(x,y) \le z \le z_2(x,y)\}.$$

Then

$$\iiint_{\Omega} f\,dV = \iint_{D} \int_{z_1(x,y)}^{z_2(x,y)} f(x,y,z)\,dz\,dA.$$

There are analogous yz-simple and zx-simple descriptions. If no single description works, split the solid.

§108.2 Rectangular-coordinate example

Example 108.2.1

Let Ω be the tetrahedron

$$x \ge 0, \quad y \ge 0, \quad z \ge 0, \quad x + y + z \le 1.$$

One description is

$$0 \le x \le 1, \qquad 0 \le y \le 1 - x, \qquad 0 \le z \le 1 - x - y.$$

Thus

$$\operatorname{vol}(\Omega) = \int_0^1 \int_0^{1-x} \int_0^{1-x-y} 1\,dz\,dy\,dx = \frac{1}{6}.$$

§108.3 Cylindrical coordinates

Cylindrical coordinates keep the vertical coordinate and use polar coordinates in the horizontal plane:

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad z = z.$$

The volume element is

$$dV = r\,dr\,d\theta\,dz.$$

They are natural for cylinders, cones around the z-axis, and solids where x^2+y^2 appears.

Example 108.3.1
For the cylinder $x^2+y^2 \le R^2$, $0 \le z \le h$,

$$V = \int_0^{2\pi}\int_0^R\int_0^h r\,dz\,dr\,d\theta = \pi R^2 h.$$

§108.4 Spherical coordinates

Spherical coordinates are

$$x = \rho\sin\varphi\cos\theta, \qquad y = \rho\sin\varphi\sin\theta, \qquad z = \rho\cos\varphi,$$

where usually $\rho \ge 0$, $0 \le \varphi \le \pi$, and $0 \le \theta < 2\pi$. The volume element is

$$dV = \rho^2\sin\varphi\,d\rho\,d\varphi\,d\theta.$$

The factor has a geometric interpretation: small coordinate lengths are approximately

$$d\rho, \qquad \rho\,d\varphi, \qquad \rho\sin\varphi\,d\theta.$$

Multiplying them gives $\rho^2\sin\varphi$.

Example 108.4.1
The ball $x^2+y^2+z^2 \le R^2$ has volume

$$\int_0^{2\pi}\int_0^{\pi}\int_0^R \rho^2\sin\varphi\,d\rho\,d\varphi\,d\theta = \frac{4}{3}\pi R^3.$$

§108.5 Integrating a Spherical Sector

A hemisphere and a cone are elementary solids, but their intersection is a useful test of spherical coordinates. Fix $R > 0$ and an angle $0 < \alpha \le \pi$. Let Ω_α be the portion of the ball of radius R lying within polar angle $\varphi \le \alpha$ of the positive z-axis. Its spherical bounds are simply

$$0 \le \rho \le R, \qquad 0 \le \varphi \le \alpha, \qquad 0 \le \theta < 2\pi.$$

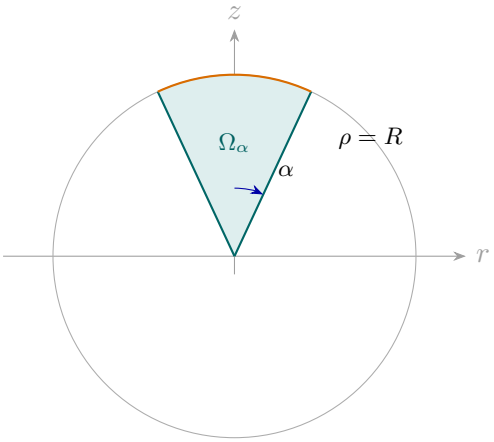


Figure 108.1: A meridional section of a ball cut out by a cone. Rotating the shaded sector about the z-axis produces Ω_α.

For uniform density, its volume is

$$V_\alpha = \int_0^{2\pi}\int_0^{\alpha}\int_0^{R} \rho^2 \sin\varphi \, d\rho \, d\varphi \, d\theta$$
$$= \boxed{\frac{2\pi R^3}{3}(1-\cos\alpha)}.$$

At $\alpha = \pi/2$ this is a hemisphere, whose volume is $2\pi R^3/3$; at $\alpha = \pi$ it is the full ball.

Now find its center of mass. Axial symmetry gives $\bar{x} = \bar{y} = 0$. Since $z = \rho\cos\varphi$, the first moment is

$$\iiint_{\Omega_\alpha} z \, dV = \int_0^{2\pi}\int_0^{\alpha}\int_0^{R} \rho^3 \cos\varphi \sin\varphi \, d\rho \, d\varphi \, d\theta$$
$$= \frac{\pi R^4}{4}\sin^2\alpha.$$

Dividing by the volume and simplifying with $\sin^2\alpha = (1-\cos\alpha)(1+\cos\alpha)$, we obtain

$$\boxed{\bar{z} = \frac{3R}{8}(1+\cos\alpha).}$$

For the solid hemisphere $\alpha = \pi/2$, its center of mass lies $3R/8$ above the equatorial plane. A formula involving a variable cone angle has explained a familiar but non-obvious geometric constant.

§108.6 Where the factors r and $\rho^2\sin\varphi$ really come from

Coordinate transformations act on infinitesimal boxes. If T takes parameter coordinates (u, v, w) to Euclidean coordinates, a tiny parameter box is carried approximately to the parallelepiped spanned by the columns of DT. Its volume is multiplied by $|\det DT|$.

For cylindrical coordinates,

$$T(r,\theta,z) = (r\cos\theta, r\sin\theta, z), \qquad |\det DT| = r.$$

For spherical coordinates,

$$T(\rho,\varphi,\theta) = (\rho\sin\varphi\cos\theta, \rho\sin\varphi\sin\theta, \rho\cos\varphi).$$

The coordinate directions are orthogonal, with lengths per unit coordinate increment

$$1, \qquad \rho, \qquad \rho \sin \varphi.$$

Multiplying gives the familiar Jacobian $\rho^2 \sin \varphi$.

A more invariant derivation takes the Euclidean inner products of the coordinate tangent vectors. The resulting metric is

$$ds^2 = d\rho^2 + \rho^2 d\varphi^2 + \rho^2 \sin^2 \varphi \, d\theta^2,$$

so the metric matrix is diagonal with entries $1, \rho^2, \rho^2 \sin^2 \varphi$. Its volume density is

$$\boxed{\sqrt{\det g} = \rho^2 \sin \varphi.}$$

This construction still works for a Riemannian manifold with a curved metric, where there may be no global Cartesian coordinates. The Jacobian formula is the Euclidean instance of integration against the metric volume form.

§108.7 Probability and Spherical Shells

The same Jacobian tells us something unexpected about random vectors. Let

$$X = (X_1, X_2, X_3), \qquad X_i \sim \mathcal{N}(0, 1)$$

independently. Its density in $\mathbb{R}^3$ is

$$f_X(x) = \frac{1}{(2\pi)^{3/2}} e^{-|x|^2/2}.$$

What is the distribution of its distance $R_X = |X|$ from the origin?

The density depends only on radius, but a shell of radius r and thickness dr has volume approximately $4\pi r^2 \, dr$. Integrating over angular coordinates therefore gives

$$\boxed{f_{R_X}(r) = \frac{4\pi r^2}{(2\pi)^{3/2}} e^{-r^2/2}, \qquad r \geq 0.}$$

The factor r^2 is geometry, and the Gaussian factor is probability. Their competition creates a peak away from the origin:

$$\frac{d}{dr} \log f_{R_X}(r) = \frac{2}{r} - r = 0 \quad \Longrightarrow \quad r = \sqrt{2}.$$

Although the three-dimensional Gaussian density f_X is maximal at $x = 0$, its *radius* is most likely near $\sqrt{2}$. There are simply many more points in a larger shell than in a very small one.

In $\mathbb{R}^d$, the radial density of a standard Gaussian is proportional to $r^{d-1} e^{-r^2/2}$, whose mode for $d > 1$ is $\sqrt{d-1}$. Also

$$\mathbb{E}|X|^2 = \sum_{i=1}^{d} \mathbb{E} X_i^2 = d.$$

This is the elementary geometric origin of the concentration of high-dimensional Gaussian mass near a sphere of radius about $\sqrt{d}$. An ordinary triple-integral Jacobian has become the beginning of high-dimensional probability.

§108.8 Divergence measures volume growth of a moving solid

There is another natural question about volume: how does it change as a region moves? Let a vector field F generate a flow Φ_t. The infinitesimal rate of expansion of volume at a point is $\nabla \cdot F$, and in differential-form notation

$$\mathcal{L}_F(dV) = (\nabla \cdot F)\, dV.$$

For the simple radial field $F(x, y, z) = (x, y, z)$, the flow is

$$\Phi_t(p) = e^t p.$$

The linear expansion factor is e^t in each coordinate, so volume is multiplied by e^{3t}. Its instantaneous rate at $t = 0$ is 3, exactly

$$\nabla \cdot F = 1 + 1 + 1 = 3.$$

On a ball of radius R, the divergence theorem gives a numerical identity that can be checked from both sides:

$$\underbrace{\iiint_{|x|\le R} 3\, dV}_{3(4\pi R^3/3)} = \underbrace{\iint_{|x|=R} F \cdot n\, dS}_{R(4\pi R^2)} = 4\pi R^3.$$

On the boundary, $F = Rn$, so the outward flux density is R. The total expansion inside agrees exactly with the amount passing through the boundary.

Finally, volume and orientation must be distinguished. The ordinary change-of-variables formula uses $|\det DT|$ to count positive volume. The oriented form

$$dx \wedge dy \wedge dz$$

pulls back with the signed determinant:

$$T^*(dx \wedge dy \wedge dz) = \det(DT)\, du \wedge dv \wedge dw.$$

For the reflection $T(u, v, w) = (u, -v, w)$, the absolute Jacobian is one but the oriented determinant is -1. Scalar volume remains unchanged, while orientation is reversed. This sign is exactly the data needed when applying Stokes' theorem to oriented boundaries.

The geometric progression is now continuous: describe a solid, choose coordinates adapted to it, derive the Jacobian, integrate mass or volume, and interpret the same factor as the number of possible points in a spherical shell or as the instantaneous change of an oriented volume. The advanced viewpoints do the same bookkeeping as the elementary integral, but in settings where Cartesian coordinates are no longer available.

Volume by Slicing: Fubini, Cavalieri, and Coarea

N-20260408

Every triple integral starts with a region. Choosing the order of integration means choosing how that region will be decomposed into fibers: vertical columns, horizontal slices, or curved level sets. Fubini's theorem guarantees that legitimate decompositions give the same integral; geometry determines which decomposition is useful. The connection with probability and the coarea formula arises because every slicing operation pushes volume onto a lower-dimensional set.[1]

§109.1 A region is its indicator function

Let $\Omega \subset \mathbb{R}^3$ be a measurable bounded solid. Its volume is

$$|\Omega| = \iiint_{\mathbb{R}^3} \mathbf{1}_\Omega(x, y, z)\, dx\, dy\, dz.$$

For fixed (x, y), the vertical fiber is

$$\Omega_{x,y} = \{z : (x, y, z) \in \Omega\}.$$

If almost every such fiber is the interval $[z_-(x, y), z_+(x, y)]$, Tonelli's theorem gives

$$|\Omega| = \iint_D \big(z_+(x, y) - z_-(x, y)\big)\, dx\, dy,$$

where D is the projection of the solid onto the xy-plane.

The height formula is therefore an integral of a one-dimensional fiber length. In the opposite direction, define the horizontal cross-sectional area

$$A(z) = \iint_{\mathbb{R}^2} \mathbf{1}_\Omega(x, y, z)\, dx\, dy.$$

The very same theorem gives

$$\boxed{|\Omega| = \int_{\mathbb{R}} A(z)\, dz.}$$

These are two different coordinate descriptions of the same nonnegative indicator function; no formal permutation of the integration symbols is needed.

§109.2 One paraboloid, two calculations

Consider

$$\Omega = \{(x, y, z) : x^2 + y^2 \le 4, \quad 0 \le z \le 4 - x^2 - y^2\}.$$

Vertical fibers have height $4 - r^2$. Using polar coordinates in the projection plane,

$$|\Omega| = \int_0^{2\pi} \int_0^2 (4 - r^2) r\, dr\, d\theta = 8\pi.$$

[1] For Fubini–Tonelli and integration along level sets, see Evans and Gariepy, *Measure Theory and Fine Properties of Functions*, and MIT 18.02SC's multiple-integral material: https://ocw.mit.edu/courses/18-02sc-multivariable-calculus-fall-2010/.

Now fix z. The inequality $z \le 4 - x^2 - y^2$ describes a disk of radius $\sqrt{4-z}$, with area $A(z) = \pi(4-z)$ for $0 \le z \le 4$. Thus

$$|\Omega| = \int_0^4 \pi(4-z)\,dz = 8\pi.$$

The first calculation integrates vertical column heights; the second integrates horizontal disk areas. Their equality is Cavalieri's principle made quantitative.

§109.3 Layer cake turns height into level sets

For a nonnegative measurable function $h : \mathbb{R}^n \to [0,\infty]$, the identity

$$h(x) = \int_0^\infty \mathbf{1}_{\{h(x)>t\}}\,dt$$

holds pointwise, with extended values allowed. Integrating and applying Tonelli gives the *layer-cake formula*

$$\boxed{\int_{\mathbb{R}^n} h(x)\,dx = \int_0^\infty |\{x : h(x) > t\}|\,dt.}$$

This is the general principle behind the paraboloid example: the superlevel set of $h(x,y) = \max(4 - x^2 - y^2, 0)$ at height $t \in (0,4)$ is a disk of area $\pi(4-t)$. The integral of those areas again equals 8π.

The formula is also useful when the distribution of heights is known but no explicit parametrization is convenient. In probability, if $X \ge 0$ is a random variable,

$$\mathbb{E}[X] = \int_0^\infty \Pr(X > t)\,dt,$$

the same identity with probability measure replacing area. Cross-sectional area and survival probability are two instances of integrating fiber masses.

§109.4 Choosing Effective Slices

The intersection of two perpendicular unit cylinders is the solid

$$\Omega = \{(x,y,z) : x^2 + z^2 \le 1, \quad y^2 + z^2 \le 1\}.$$

Integrating first in x and y gives an awkward description if written without examining the intersection. But for fixed $z \in [-1,1]$, its cross-section is simply a square:

$$|x| \le \sqrt{1-z^2}, \qquad |y| \le \sqrt{1-z^2}.$$

The side length is $2\sqrt{1-z^2}$, so its area is $4(1-z^2)$. Therefore

$$\boxed{|\Omega| = \int_{-1}^1 4(1-z^2)\,dz = \frac{16}{3}.}$$

This classical Steinmetz solid illustrates why a carefully chosen slicing direction can replace a difficult curved region by elementary cross-sections.

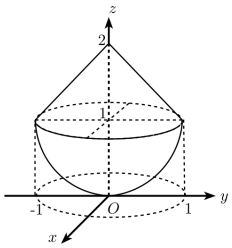


Figure 109.1: A composite solid motivates choosing its cross-sections before committing to an order of integration.

§109.5 Projection changes can force a split of the domain

A region may be simple when projected onto one coordinate plane but require several pieces in another projection. For example, the tetrahedron

$$T = \{(x, y, z) : x, y, z \geq 0, \quad x + y + z \leq 1\}$$

has vertical fibers $0 \leq z \leq 1 - x - y$ above the triangle $x, y \geq 0$, $x + y \leq 1$. Thus

$$|T| = \int_0^1 \int_0^{1-x} (1 - x - y)\, dy\, dx = \frac{1}{6}.$$

Its horizontal slice at height z is instead a right triangle with legs $1-z$, of area $(1-z)^2/2$, so

$$|T| = \int_0^1 \frac{(1 - z)^2}{2}\, dz = \frac{1}{6}.$$

A more complicated region may have a cross-section whose boundary changes from a graph to a circle or whose number of connected components changes. At such transition levels, the formula for the fiber area changes, and the outer integral must be split accordingly. The need to split arises from actual geometry, not from the syntax of an iterated integral.

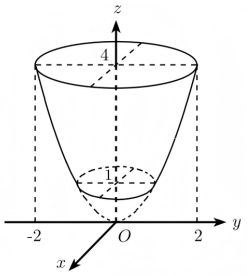


Figure 109.2: A truncated rotational solid may be more naturally described by radial or horizontal slices than by vertical columns.

§109.6 Coarea is slicing along nonlinear level sets

Horizontal slicing uses the function $g(x, y, z) = z$, whose level sets are planes and whose gradient has norm one. The coarea theorem generalizes this idea: for a sufficiently regular $g : \mathbb{R}^n \to \mathbb{R}$ and suitable nonnegative h,

$$\int_{\mathbb{R}^n} h(x) |\nabla g(x)|\, dx = \int_{\mathbb{R}} \left(\int_{g^{-1}(t)} h(x)\, d\mathcal{H}^{n-1}(x) \right) dt.$$

Here $\mathcal{H}^{n-1}$ is the natural surface measure on the level set. The factor $|\nabla g|$ accounts for how densely the levels are packed.

For $g(x, y) = \sqrt{x^2 + y^2} = r$ in the plane, $|\nabla g| = 1$ away from the origin. The level sets are circles, with arc-length element $r\, d\theta$. Coarea therefore gives polar integration:

$$\int_{\mathbb{R}^2} h(x, y)\, dx\, dy = \int_0^{\infty} \int_0^{2\pi} h(r\cos\theta, r\sin\theta)\, r\, d\theta\, dr.$$

The familiar polar Jacobian r is the length scaling of a circular level set, rather than an arbitrary factor to memorize.

Vertical integration, Cavalieri's horizontal slices, layer cake and coarea all express the same operation: a measure is disassembled along fibers of a map. For z-projection the fibers are plane sections, for a height function they are superlevel sets, and for a radius map they are circles. The art of changing integration order is to choose the map whose fibers have the simplest geometry.

Integrating Along Curves and Across Surfaces: Densities and Forms

N-20260416

A wire carrying mass and a force acting along the same wire lead to different integrals. Mass accumulates according to length, while work is sensitive to the direction of motion. The analogous distinction on a surface is between accumulated scalar density and signed flux. The formulas become much easier to remember once we derive them from parametrizations and recognize that only the oriented integrals are differential forms.[1]

§110.1 Length versus Displacement Integrals

Let $\gamma : [a, b] \to \mathbb{R}^3$ be a continuously differentiable regular curve. A small parameter interval of length dt corresponds to arc length

$$ds = \|\gamma'(t)\|\, dt.$$

Thus a scalar density ρ distributed along a wire has total mass

$$\boxed{\int_\gamma \rho\, ds = \int_a^b \rho(\gamma(t))\|\gamma'(t)\|\, dt.}$$

The norm is essential: it makes the contribution nonnegative when the orientation is reversed.

A force field F performs signed work

$$\boxed{\int_\gamma F \cdot d\mathbf{r} = \int_a^b F(\gamma(t)) \cdot \gamma'(t)\, dt.}$$

Its integrand is a signed projection onto the displacement direction. Traversing the path backward changes the sign.

On a circle $\gamma(t) = (R\cos t, R\sin t)$, $0 \le t \le 2\pi$, a linear density $\rho = x^2 + y^2$ gives

$$\int_\gamma \rho\, ds = 2\pi R^3.$$

For the tangential field $F = (-y, x)$, however,

$$F(\gamma(t)) \cdot \gamma'(t) = R^2, \qquad \int_\gamma F \cdot d\mathbf{r} = 2\pi R^2.$$

The distinction can be observed physically: mass does not change when the wire is traversed in the opposite direction; work does.

[1] MIT 18.02, *Surface Integrals*, supplementary note V9.

§110.2 Why Arc-Length Uses an Absolute Value

Suppose $\widetilde{\gamma}(s) = \gamma(\phi(s))$, where ϕ is a smooth monotone bijection of parameter intervals. Then

$$\widetilde{\gamma}'(s) = \gamma'(\phi(s))\phi'(s).$$

For length, the speed transforms with $|\phi'(s)|$. The absolute value compensates for any reversal of parameter direction, leaving the mass integral unchanged.

For work, the derivative $\phi'(s)$ retains its sign. If ϕ reverses the interval, the work integral changes sign. This difference between absolute and signed Jacobian factors is the local reason the two line-integral formulas have different orientation behavior.

The work integral can be written using the differential one-form

$$\alpha = F_1\,dx + F_2\,dy + F_3\,dz.$$

Along γ, its pullback is

$$\gamma^*\alpha = \big(F(\gamma(t)) \cdot \gamma'(t)\big)\,dt.$$

Thus $\int_\gamma \alpha = \int_{[a,b]} \gamma^*\alpha$. Pullback explains the whole calculation without requiring the parameter to have any intrinsic physical meaning.

§110.3 Area is the Gram determinant of the tangent vectors

For a smooth parametrized surface $r : D \subset \mathbb{R}^2 \to \mathbb{R}^3$, the vectors r_u, r_v span a tangent parallelogram. Its area is

$$\|r_u \times r_v\|\,du\,dv = \sqrt{\langle r_u, r_u\rangle\langle r_v, r_v\rangle - \langle r_u, r_v\rangle^2}\,du\,dv.$$

The square root is the Gram determinant of the induced metric on the parameter domain. Therefore the scalar surface integral is

$$\boxed{\iint_S f\,dS = \iint_D f(r(u,v))\|r_u \times r_v\|\,du\,dv.}$$

Like arc length, area is insensitive to a reversal of orientation.

For the lateral surface of a cylinder of radius R and height H, take

$$r(\theta, z) = (R\cos\theta, R\sin\theta, z), \quad 0 \le \theta \le 2\pi, \quad 0 \le z \le H.$$

Since $\|r_\theta \times r_z\| = R$, the integral of $f(x,y,z) = z^2$ is

$$\iint_S z^2\,dS = R\int_0^{2\pi}\int_0^H z^2\,dz\,d\theta = \frac{2\pi R H^3}{3}.$$

The calculation is just length around the circumference times the integral of density over height, but the Gram factor makes it independent of which parameters describe the cylinder.

§110.4 Flux uses the oriented area vector

A flux integral measures the component of a vector field F crossing the chosen side of a surface:

$$\iint_S F \cdot n \, dS = \iint_D F(r(u,v)) \cdot (r_u \times r_v) \, du \, dv,$$

provided $r_u \times r_v$ gives the prescribed orientation.

For the same cylinder, $r_\theta \times r_z = (R\cos\theta, R\sin\theta, 0)$ points outward. Let $F = (x, y, 0)$, so the field points radially away from the axis. Then

$$F(r) \cdot (r_\theta \times r_z) = R^2, \qquad \boxed{\iint_{\text{lateral } S} F \cdot n \, dS = 2\pi R^2 H.}$$

The top and bottom caps contribute zero because F is horizontal. Since $\operatorname{div} F = 2$ and the cylinder's volume is $\pi R^2 H$, the divergence theorem gives the same total flux $2\pi R^2 H$. The direct parametrization and the volume calculation provide independent checks of the orientation and scale.

§110.5 A vector field gives a one-form and a two-form

In oriented Euclidean $\mathbb{R}^3$, the vector field $F = (P, Q, R)$ gives a one-form

$$\alpha_F = P \, dx + Q \, dy + R \, dz$$

and a two-form

$$\beta_F = P \, dy \wedge dz + Q \, dz \wedge dx + R \, dx \wedge dy.$$

The one-form measures work along tangent vectors; the two-form measures flux across oriented tangent parallelograms.

These are related by the Euclidean Hodge star:

$$\boxed{\beta_F = \star \alpha_F.}$$

The star depends on the metric and orientation. It encodes the operation of converting a tangent direction into the complementary oriented area direction.

For a parametrization $r(u, v)$, the two-form pulls back to

$$r^* \beta_F = \big(F(r(u,v)) \cdot (r_u \times r_v)\big) \, du \wedge dv.$$

The vector-calculus flux formula is exactly this pullback formula.

Exterior differentiation then reproduces the two standard differential operators:

$$d\alpha_F = (\operatorname{curl} F) \cdot d\mathbf{S}, \qquad d\beta_F = (\operatorname{div} F) \, dx \wedge dy \wedge dz.$$

Here $(\operatorname{curl} F) \cdot d\mathbf{S}$ means the two-form built from the curl by the same rule used for β_F. Integrating these identities yields the familiar Stokes and divergence theorems as different degree cases of $\int_{\partial M} \omega = \int_M d\omega$.

§110.6 Nonorientability is a genuine obstruction

The factor $\|r_u \times r_v\|$ makes scalar surface integration possible even on a Möbius strip, because local area elements agree after changing charts. To define signed flux, however, the normal must vary continuously and consistently across the entire surface. A Möbius strip has no global continuous choice of unit normal.

One can still integrate a scalar density over it, and a chosen local patch can carry flux relative to a local normal. But a global signed flux through the whole Möbius strip has no canonical meaning for a general field. The failure is topological: the obstruction remains even with arbitrarily accurate parametrization formulas.

This separates the roles of metric, orientation and topology. The metric supplies lengths and areas. Orientation supplies the signs for work and flux. Topology determines whether an orientation can be chosen globally. Once those roles are clear, the first-kind and second-kind integration formulas follow naturally from the geometry of what is being accumulated.

Green's Theorem: From Circulation to Winding and Residues

N-20260417

The local derivative of a vector field determines circulation around a contractible boundary. What happens when the field is defined on a region with a hole? The answer leads from Green's theorem to winding numbers and, after expressing a planar vector field as a complex differential, to the residue theorem. A single punctured-plane calculation will connect all three subjects.[1]

§111.1 Green's theorem is cancellation along internal edges

Let $D \subset \mathbb{R}^2$ be a bounded region with piecewise smooth boundary, oriented so that the region is on the left as one walks along each component. Suppose $P, Q \in C^1$ on a neighborhood of the closure of D. Green's theorem states

$$\boxed{\int_{\partial D} P\,dx + Q\,dy = \iint_D (Q_x - P_y)\,dx\,dy.}$$

On a rectangle, the result follows by the one-variable fundamental theorem of calculus applied to opposite sides. Subdividing the region creates edges traversed once in each direction, so their contributions cancel. Outer boundary components are counterclockwise and boundary components enclosing holes are clockwise.

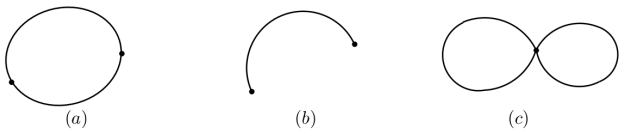


Figure 111.1: Boundary orientation distinguishes an enclosed region from an arbitrary traversal of a curve.

Writing $\omega = P\,dx + Q\,dy$, the differential is

$$d\omega = (Q_x - P_y)\,dx \wedge dy,$$

and Green's theorem becomes $\int_{\partial D} \omega = \int_D d\omega$. The identity $d^2 = 0$ is related to the cancellation principle: applying two consecutive boundary operations also gives zero. The diagrammatic cancellation has an algebraic counterpart.

§111.2 Curl zero is local; a potential is global

A field $F = (P, Q)$ has a potential ϕ when $\omega = d\phi$, or

$$P = \phi_x, \qquad Q = \phi_y.$$

[1] MIT 18.02SC (Green's theorem); MIT 18.04, *Complex Variables with Applications* (holomorphic forms).

Such a form is called exact. It satisfies $d\omega = d^2\phi = 0$, so $Q_x = P_y$. Conversely, on a simply connected open region U, a C^1 form with $d\omega = 0$ is exact.

One way to construct the potential is to choose a base point $x_0 \in U$ and define

$$\phi(x) = \int_{\gamma_{x_0,x}} \omega,$$

where $\gamma_{x_0,x}$ is a piecewise smooth path from x_0 to x. For a simply connected region, any two such paths are homotopic with fixed endpoints. Green's theorem, applied through a homotopy or by subdivision, makes their integrals agree when $d\omega = 0$. The resulting ϕ is well defined and has $d\phi = \omega$.

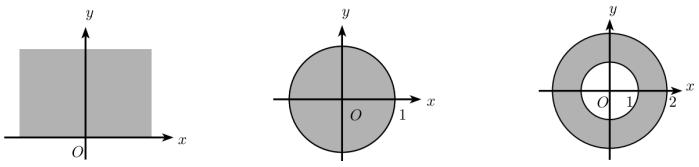


Figure 111.2: A loop around a hole need not bound a region contained in the domain of the field.

§111.3 The angular form detects winding

On $U = \mathbb{R}^2 \setminus \{0\}$, consider

$$\alpha = \frac{-y\,dx + x\,dy}{x^2 + y^2}.$$

A direct derivative calculation gives $d\alpha = 0$, but on the counterclockwise unit circle $\gamma(t) = (\cos t, \sin t)$,

$$\gamma^*\alpha = dt, \qquad \int_\gamma \alpha = 2\pi.$$

Thus no globally defined real potential can have differential α.

The value is topological. Any piecewise smooth closed curve γ avoiding the origin has a continuous polar angle along its parameter interval, say $\gamma(t) = r(t)(\cos\theta(t), \sin\theta(t))$ with $r(t) > 0$. Along the curve,

$$\gamma^*\alpha = d\theta, \qquad \int_\gamma \alpha = \theta(1) - \theta(0).$$

Because the curve is closed, the difference is an integer multiple of 2π. This integer is its winding number:

$$\boxed{\operatorname{wind}(\gamma, 0) = \frac{1}{2\pi}\int_\gamma \alpha \in \mathbb{Z}.}$$

It is invariant under deformations of the curve that avoid the origin.

§111.4 Cohomology of the Punctured Plane

Suppose ω is any smooth closed one-form on U. Let

$$c = \frac{1}{2\pi}\int_{S^1} \omega.$$

The form $\omega - c\alpha$ has zero integral around the unit circle. Every closed loop in U is homotopic, up to orientation and repeated traversal, to an integer number of such circles. Since integrals of closed forms are homotopy invariant, every closed-loop integral of $\omega - c\alpha$ is zero.

The same path-integral construction as above therefore gives a global smooth function f with

$$\boxed{\omega = df + c\alpha.}$$

The coefficient c is unique because α has nonzero period. Hence the first de Rham cohomology group is

$$H^1_{\mathrm{dR}}(\mathbb{R}^2 \setminus \{0\}) \cong \mathbb{R},$$

with $[\alpha]$ as a generator.

This does more than say that the punctured plane has a hole. It computes exactly one independent obstruction to constructing a potential: every closed one-form differs from an exact one by a multiple of the angular form.

§111.5 Holomorphic one-forms lead to Cauchy and residues

Write $z = x + iy$. A complex-valued form $f(z)\,dz$, for $f = u + iv$, splits into real and imaginary parts:

$$f(z)\,dz = (u\,dx - v\,dy) + i(v\,dx + u\,dy).$$

If f is holomorphic, the Cauchy–Riemann equations $u_x = v_y$, $u_y = -v_x$ show that both real one-forms are closed:

$$d(f\,dz) = 0.$$

Consequently, integrals of a holomorphic function around contractible loops vanish: this is Cauchy's theorem in the language of Green's theorem.

Near an isolated simple pole a, suppose

$$f(z) = \frac{c}{z - a} + h(z),$$

with h holomorphic. For a small positively oriented circle C around a,

$$\int_C f(z)\,dz = c \int_C \frac{dz}{z - a} = 2\pi i\, c.$$

Indeed $dz/(z-a)$ has imaginary part equal to the angular form centered at a, and its real part $d\log|z-a|$ has zero period. The coefficient c is the residue at a.

If a meromorphic function has finitely many poles in a bounded region and is holomorphic on the boundary, remove small disks around its poles. Cauchy's theorem on the punctured region, with clockwise inner boundary components, gives

$$\boxed{\int_{\partial D} f(z)\,dz = 2\pi i \sum_{a \in D} \mathrm{Res}(f, a).}$$

The residue theorem is therefore a local-to-global circulation law. Holomorphicity makes the exterior derivative zero away from singularities, while each isolated pole contributes a quantized boundary period.

The same geometry also computes area: choosing $P = -y/2$, $Q = x/2$ gives

$$\mathrm{area}(D) = \frac{1}{2} \int_{\partial D} (x\,dy - y\,dx).$$

In that formula the derivative is 1, so area is accumulated locally. In the winding and residue formulas the derivative vanishes away from singularities, so all the information comes from what a loop encloses. Green's theorem encompasses both phenomena through the orientation and topology of the boundary.

Gauss and Stokes: Surface Geometry, Singular Sources, and Weak Laws

N-20260615

A surface integral can measure mass distributed over a sheet or signed flow through it. In the first case, the metric determines area; in the second, orientation determines the sign. Gauss' theorem and Stokes' theorem then connect these surface measurements to local differential quantities. Their most revealing examples arise when smoothness breaks down: a point charge has zero ordinary divergence everywhere it is defined, yet carries a nonzero flux through every surrounding sphere. The resolution leads directly to weak derivatives and distributions.[1]

§112.1 Area has no preferred orientation

If a surface is parametrized by $r(u, v)$, its scalar area element is

$$dS = \|r_u \times r_v\| \, du \, dv,$$

and the integral of a surface density f is

$$\iint_S f \, dS = \iint_D f(r(u,v)) \|r_u \times r_v\| \, du \, dv.$$

The norm makes this independent of how the parameter coordinates are oriented.

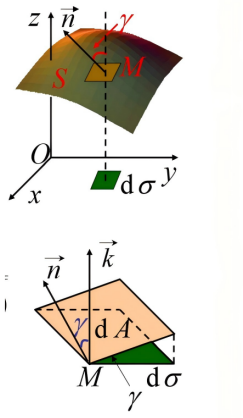


Figure 112.1: A small patch of a surface has area determined by its tangent parallelogram. Its planar projection generally has a different area.

Consider a spherical cap cut from the sphere $x^2 + y^2 + z^2 = R^2$ by $z = h$, with $0 \le h < R$. Parametrize the upper sphere by

$$r(\theta, \varphi) = (R \sin\varphi \cos\theta, \; R \sin\varphi \sin\theta, \; R \cos\varphi).$$

[1] MIT 18.02, supplementary notes V9–V13; Oregon State University, *The Geometry of Static Fields*, §15.2.

The cap corresponds to

$$0 \le \theta \le 2\pi, \qquad 0 \le \varphi \le \arccos(h/R),$$

and $\|r_\theta \times r_\varphi\| = R^2 \sin\varphi$. Therefore

$$\boxed{\text{area(cap)} = 2\pi R^2 \left(1 - \frac{h}{R}\right) = 2\pi R(R-h).}$$

It depends linearly on the cap height $R-h$, although the horizontal cross-sectional radius depends nonlinearly on that height.

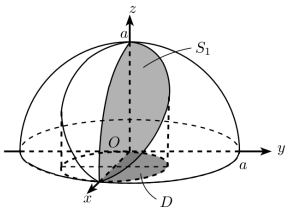


Figure 112.2: Spherical coordinates adapt the parameter domain to the height boundary of a spherical cap.

For a torus with major radius $R > r > 0$, parametrize

$$X(u,v) = ((R + r\cos v)\cos u,\ (R + r\cos v)\sin u,\ r\sin v).$$

Here $u, v \in [0, 2\pi]$, and

$$\|X_u \times X_v\| = r(R + r\cos v).$$

The total area is

$$\boxed{\text{area}(T) = \int_0^{2\pi}\int_0^{2\pi} r(R + r\cos v)\,dv\,du = 4\pi^2 Rr.}$$

The mean of $\cos v$ over a full period is zero, so the long circular direction contributes only its centerline circumference on average.

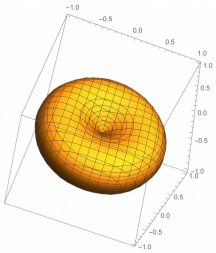


Figure 112.3: The two angular directions of a torus have different local stretch factors, which combine into the area element $r(R + r\cos v)\,du\,dv$.

§112.2 Flux measures outward flow and local sources

For a vector field F, the flux through an oriented surface is

$$\int_S F \cdot n\,dS = \iint_D F(r(u,v)) \cdot (r_u \times r_v)\,du\,dv$$

when $r_u \times r_v$ agrees with the prescribed normal. Unlike scalar area, this changes sign when orientation is reversed.

If $\Omega \subset \mathbb{R}^3$ is bounded by a piecewise smooth closed surface and F is C^1 on a neighborhood of $\overline{\Omega}$, the divergence theorem states

$$\boxed{\int_{\partial\Omega} F \cdot n \, dS = \iiint_{\Omega} \operatorname{div} F \, dV.}$$

The unit normal on the boundary points outward.

For $F(x, y, z) = (x, y, z)$, the sphere of radius R has outward unit normal $n = (x, y, z)/R$, so $F \cdot n = R$. Its total outward flux is

$$R \cdot 4\pi R^2 = 4\pi R^3.$$

The divergence is 3, and integration over the ball gives

$$3 \cdot \frac{4}{3}\pi R^3 = 4\pi R^3,$$

the same answer. The field's local source strength is constant; the boundary records its total over the enclosed volume.

§112.3 Closing Surfaces for Gauss's Theorem

The divergence theorem controls the boundary of a solid. A hemisphere by itself is only part of that boundary. Let S_+ be the upper hemisphere

$$x^2 + y^2 + z^2 = R^2, \qquad z \geq 0,$$

with outward (and therefore upward) normal, and let D be the flat disk in $z = 0$. The boundary of the upper half-ball is $S_+ \cup D$, where the disk carries the *downward* normal $-e_z$. For the field

$$F(x, y, z) = (0, 0, 1 + z), \qquad \operatorname{div} F = 1,$$

the flat cap contributes

$$\iint_{D} F \cdot (-e_z) \, dS = -\pi R^2.$$

The divergence theorem applied to the half-ball therefore gives

$$\iint_{S_+} F \cdot n \, dS - \pi R^2 = \iiint_{\text{half-ball}} 1 \, dV = \frac{2\pi R^3}{3},$$

and hence

$$\boxed{\iint_{S_+} F \cdot n \, dS = \pi R^2 + \frac{2\pi R^3}{3}.}$$

Omitting the cap would lose the whole term πR^2. As a check, spherical coordinates give $z = R\cos\varphi$, $n_z = \cos\varphi$, and

$$\iint_{S_+} F \cdot n \, dS = 2\pi R^2 \int_0^{\pi/2} (1 + R\cos\varphi)\cos\varphi \sin\varphi \, d\varphi$$

$$= \pi R^2 + \frac{2\pi R^3}{3}.$$

For $F = (x, y, z)$ instead, the cap flux is zero and the hemisphere flux is simply 3 times the half-ball volume, or $2\pi R^3$. The geometry is identical; the field decides whether the missing cap matters.

§112.4 Orientation can turn the same cone flux negative

Consider the lateral surface of the cone $z = \sqrt{x^2 + y^2}$, from its apex to height a, together with the horizontal disk $z = a$. It bounds the solid $\{0 \le r \le z \le a\}$. Write a parametrization of the lateral surface as

$$X(r, \theta) = (r\cos\theta, r\sin\theta, r), \qquad 0 \le r \le a, \quad 0 \le \theta \le 2\pi.$$

The cross product in the parameter order (r, θ) is

$$X_r \times X_\theta = (-r\cos\theta, -r\sin\theta, r).$$

Its vertical component is positive, but its radial component points *into* the cone. Thus it gives the inward normal to the solid, not the outward one. For $F = e_z = (0, 0, 1)$,

$$\iint_{\text{side, up}} F \cdot n \, dS = \int_0^{2\pi} \int_0^a r \, dr \, d\theta = \pi a^2.$$

With outward orientation the side flux is $-\pi a^2$. The top disk, outward oriented upward, contributes $+\pi a^2$. The total outward flux is zero, agreeing with $\operatorname{div} e_z = 0$.

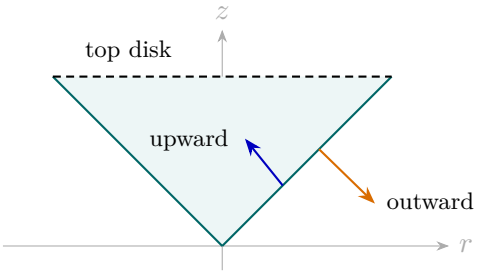


Figure 112.4: In this cross-section of the solid cone, the upward normal on the sloping side points into the solid. The outward normal points downward.

This sign check is also a warning about projecting a curved surface onto a plane. For the constant field e_z, the outward flux through the upper hemisphere is $+\pi R^2$ while through the lower hemisphere it is $-\pi R^2$. Both sheets project to the *same* disk, but their induced orientations are opposite. The two contributions cancel on the complete sphere. The formula $F_z \, dx \, dy$ is an oriented two-form integral; it cannot be used as if $dx \, dy$ were an unsigned surface area element.

§112.5 An inverse-square field exposes a hidden point source

Now let

$$F(x) = \frac{x}{\|x\|^3} \qquad (x \in \mathbb{R}^3 \setminus \{0\}).$$

At every $x \neq 0$, a direct calculation gives

$$\operatorname{div} F = 0.$$

Nevertheless, on the sphere S_R centered at the origin,

$$F \cdot n = \frac{1}{R^2}, \qquad \boxed{\int_{S_R} F \cdot n \, dS = 4\pi.}$$

There is no contradiction with the divergence theorem: the field is undefined at the origin, so it fails the regularity hypothesis on the enclosed ball.

For a region with a small ball around the origin removed, the divergence theorem does apply. Its inner boundary has the normal pointing toward the origin. The outer flux is balanced by an inner flux of -4π. Shrinking the omitted ball retains a nonzero boundary contribution.

This is the mathematical structure of an electrostatic point charge. For charge q and permittivity ε_0, the Coulomb field is $E = \frac{q}{4\pi\varepsilon_0}F$, and every surrounding surface carries flux q/ε_0.

§112.6 Distributional Point Sources

The vector field $F = x/\|x\|^3$ is locally integrable in $\mathbb{R}^3$, so its divergence can be defined as a distribution. For a smooth compactly supported test function φ, put

$$\langle \operatorname{div} F, \varphi \rangle = -\int_{\mathbb{R}^3} F \cdot \nabla\varphi \, dV.$$

Remove a ball of radius ε. Integration by parts on its complement, where $\operatorname{div} F = 0$, gives

$$\begin{aligned} -\int_{\|x\|>\varepsilon} F \cdot \nabla\varphi \, dV &= \int_{\|x\|=\varepsilon} \varphi(x)\frac{1}{\varepsilon^2}\, dS \\ &= \int_{S^2} \varphi(\varepsilon\omega)\, d\omega. \end{aligned}$$

The right-hand side tends to $4\pi\varphi(0)$. Consequently

$$\boxed{\operatorname{div}\frac{x}{\|x\|^3} = 4\pi\delta_0} \qquad \text{in the sense of distributions.}$$

The Dirac delta captures a source concentrated at a point, while the ordinary derivative describes the smooth behavior elsewhere. This identity is the rigorous form of the flux computation.

§112.7 Stokes turns local rotation into boundary circulation

For an oriented surface S with piecewise smooth boundary, Stokes' theorem is

$$\boxed{\oint_{\partial S} F \cdot d\mathbf{r} = \iint_S (\nabla \times F) \cdot n \, dS,}$$

assuming F is C^1 near the surface. The boundary orientation is induced by the surface normal.

A triangle in space is a useful example because its curl flux can be computed without any complicated surface parametrization. Let its ordered vertices be

$$A = (0,0,0), \quad B = (1,0,0), \quad C = (0,1,1)$$

and take $F = (-y/2, x/2, 0)$. Its curl is constant:

$$\nabla \times F = (0,0,1).$$

The oriented area vector of the triangle is

$$\mathbf{A}_S = \frac{1}{2}(B-A)\times(C-A) = \left(0, -\frac{1}{2}, \frac{1}{2}\right).$$

Stokes gives

$$\oint_{A\to B\to C\to A} F\cdot d\mathbf{r} = (\nabla\times F)\cdot \mathbf{A}_S = \frac{1}{2}.$$

One can check this directly. Along AB and CA, the work vanishes because the relevant coordinate is zero. Along BC, write $(x,y,z) = (1-t,t,t)$ for $0\le t\le 1$. Then

$$F\cdot d\mathbf{r} = \left(-\frac{t}{2}, \frac{1-t}{2}, 0\right)\cdot(-1,1,1)\,dt = \frac{1}{2}\,dt.$$

The direct line integral is $1/2$, confirming the sign and area-vector calculation.

The surface spanning the loop need not be flat. Any other smoothly oriented surface sharing the same boundary has the same curl flux, provided the field is smooth on a neighborhood of that surface. Stokes' theorem makes the flux of a curl depend only on the boundary curve.

§112.8 Circulation across Two Surfaces

The line integral in Stokes' theorem may look hard even when the curl is simple. Consider the counterclockwise ellipse

$$C: \quad x = 2\cos t, \quad y = \sin t, \quad z = 0, \qquad 0\le t\le 2\pi,$$

and the field $F = (-y/2, x/2, 0)$. Since

$$\nabla\times F = (0,0,1),$$

the flat elliptical disk $D = \{x^2/4 + y^2 \le 1,\ z = 0\}$, oriented upward, gives

$$\oint_C F\cdot d\mathbf{r} = \iint_D (\nabla\times F)\cdot e_z\,dA = \mathrm{Area}(D) = 2\pi.$$

One can verify the result directly without Stokes. Along C,

$$F = \left(-\tfrac{1}{2}\sin t, \cos t, 0\right), \qquad \mathbf{r}' = (-2\sin t, \cos t, 0),$$

so

$$\boxed{\oint_C F\cdot d\mathbf{r} = \int_0^{2\pi} (\sin^2 t + \cos^2 t)\,dt = 2\pi.}$$

There is even a curved surface with the same boundary: the cap

$$S: \quad z = 1 - \frac{x^2}{4} - y^2, \qquad \frac{x^2}{4} + y^2 \le 1.$$

Its upward vector area element is

$$n\,dS = (-z_x, -z_y, 1)\,dx\,dy = \left(\frac{x}{2}, 2y, 1\right)\,dx\,dy.$$

The vertical curl $(0,0,1)$ therefore sees the same projected area:

$$\iint_S (\nabla\times F)\cdot n\,dS = \iint_D 1\,dx\,dy = 2\pi.$$

These computations make surface independence precise. For this curl field, swapping one spanning surface for another does not change the value; for a *general flux field* it need not be true. Indeed, Gauss says that for two surfaces enclosing a region the flux difference is the integral of $\operatorname{div} F$ over that region. Stokes applies to a curl, whose divergence vanishes identically when the field is smooth. The two theorems thus explain each other's surface-independence hypothesis.

§112.9 The single boundary law behind both theorems

In Euclidean coordinates, associate to $F = (P, Q, R)$ the work one-form

$$\alpha_F = P\,dx + Q\,dy + R\,dz$$

and the flux two-form

$$\beta_F = P\,dy \wedge dz + Q\,dz \wedge dx + R\,dx \wedge dy.$$

Exterior differentiation gives

$$d\alpha_F = \beta_{\nabla \times F}, \qquad d\beta_F = (\nabla \cdot F)\,dx \wedge dy \wedge dz.$$

Thus Stokes' circulation theorem applies the boundary formula to a one-form on a surface, and Gauss' theorem applies it to a two-form on a three-dimensional region:

$$\boxed{\int_{\partial M} \omega = \int_M d\omega.}$$

The difference is the dimension of the manifold M and the degree of the form ω. The orientation of ∂M is induced by that of M.

The algebraic identity $d^2 = 0$ contains the familiar differential identity $\nabla \cdot (\nabla \times F) = 0$. In the smooth setting, it says that an exact flux two-form has zero exterior derivative. Distributional formulations preserve this principle and make it meaningful even when fields are too singular for classical differentiation.

§112.10 Conservation Laws from Divergence

Suppose a time-dependent density $\rho(x, t)$ and flux $J(x, t)$ satisfy

$$\partial_t \rho + \nabla \cdot J = 0.$$

Integrating over a fixed region Ω and applying the divergence theorem gives

$$\boxed{\frac{d}{dt} \int_\Omega \rho(x, t)\,dV = -\int_{\partial\Omega} J(x, t) \cdot n\,dS.}$$

This states that the amount leaving through the boundary accounts for the decrease inside. The same equation describes conservation of mass, charge and probability, with different choices of density and flux.

For a smooth compactly supported test function φ in space, the divergence term can be transferred to the test function:

$$\int_{\mathbb{R}^3} (\nabla \cdot J)\varphi\,dV = -\int_{\mathbb{R}^3} J \cdot \nabla\varphi\,dV.$$

This integration-by-parts identity is the starting point for defining weak solutions when pointwise derivatives do not exist. It is also exactly the definition used above to discover the point source hidden in the inverse-square field.

Smooth surface integration, Stokes' theorem, concentrated charges and weak conservation laws therefore share one mechanism: derivatives describe local changes, while their integrated effect is measured against boundaries. Singularities require a more flexible meaning of derivative, but the boundary accounting remains intact.

IX

Differential Equations and Dynamics

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Recurrences as Orbits: Linear Dynamics, Normalization, and Reversible Transformations

N-20220717

§113.1 A recurrence generates an orbit

A first-order recurrence

$$a_{n+1} = F(a_n)$$

is the iteration of a map $F : X \to X$. Once an initial value a_0 is chosen, the sequence is the forward orbit

$$a_0, \quad F(a_0), \quad F^2(a_0), \quad \dots.$$

This point of view separates two questions. The recurrence rule determines the dynamics, while the initial value selects one orbit. A closed formula is an explicit description of the iterate F^n.

Not every recurrence is reversible. If F is not injective, different states can merge and no inverse iteration is possible. This distinction will later separate semigroup actions, which describe forward iteration, from group actions, which describe reversible transformations.

Higher-order recurrences can also be written as first-order systems by enlarging the state. For example,

$$a_{n+2} = 5a_{n+1} - 6a_n$$

becomes

$$\begin{pmatrix} a_{n+2} \\ a_{n+1} \end{pmatrix} = \begin{pmatrix} 5 & -6 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_{n+1} \\ a_n \end{pmatrix}.$$

Thus recurrence theory is already a form of discrete dynamical systems and linear algebra.

§113.2 Arithmetic progressions are translation orbits

An arithmetic progression satisfies

$$a_{n+1} = a_n + d.$$

The update map is the translation

$$T_d(x) = x + d.$$

Since

$$T_d^n(x) = x + nd,$$

we obtain

$$\boxed{a_n = a_0 + nd.}$$

With indexing beginning at 1, this becomes

$$a_n = a_1 + (n-1)d.$$

The sum of the first n terms follows by pairing terms from opposite ends:

$$S_n = a_1 + a_2 + \cdots + a_n,$$
$$S_n = a_n + a_{n-1} + \cdots + a_1.$$

Every column sums to $a_1 + a_n$, so

$$\boxed{S_n = \frac{n}{2}(a_1 + a_n) = na_1 + \frac{n(n-1)}{2}d.}$$

Arithmetic progressions convert additive relations among indices into additive relations among terms. If

$$i_1 + \cdots + i_k = j_1 + \cdots + j_k,$$

then

$$\begin{aligned} a_{i_1} + \cdots + a_{i_k} &= ka_1 + (i_1 + \cdots + i_k - k)d \\ &= ka_1 + (j_1 + \cdots + j_k - k)d \\ &= a_{j_1} + \cdots + a_{j_k}. \end{aligned}$$

The equal-index-sum rule is therefore a direct consequence of translation dynamics, not an isolated trick.

§113.3 Geometric progressions are scaling orbits

A geometric progression satisfies

$$a_{n+1} = qa_n.$$

The update map is the scaling

$$S_q(x) = qx,$$

whose iterates are

$$S_q^n(x) = q^n x.$$

Hence

$$\boxed{a_n = a_0 q^n}$$

or, with indexing beginning at 1,

$$a_n = a_1 q^{n-1}.$$

For $q \neq 1$, write

$$S_n = a_1 + a_1 q + \cdots + a_1 q^{n-1}.$$

Then

$$qS_n = a_1 q + a_1 q^2 + \cdots + a_1 q^n.$$

Subtracting gives

$$(q-1)S_n = a_1(q^n - 1),$$

so

$$\boxed{S_n = a_1 \frac{q^n - 1}{q - 1}.}$$

The exceptional case $q = 1$ must be handled separately and gives $S_n = na_1$.

If $q \neq 0$, scaling is reversible, with inverse $S_{1/q}$. If $q = 0$, all states collapse to zero after one step. The same-looking recurrence can therefore represent either a group action or a genuinely irreversible map depending on the parameter.

§113.4 Affine Recurrences and Fixed Points

Consider

$$a_{n+1} = ra_n + b.$$

The update map

$$T(x) = rx + b$$

is affine. If $r \neq 1$, it has a unique fixed point L satisfying

$$L = rL + b,$$

namely

$$L = \frac{b}{1-r}.$$

Subtracting the fixed point gives

$$a_{n+1} - L = r(a_n - L).$$

The shifted sequence is geometric, so

$$\boxed{a_n = L + r^n(a_0 - L).}$$

Equivalently,

$$a_n = r^n a_0 + b\frac{1-r^n}{1-r}.$$

If $r = 1$, the recurrence is arithmetic:

$$a_n = a_0 + nb.$$

For example,

$$a_{n+1} = 2a_n + 1, \qquad a_0 = 0.$$

The fixed point is $L = -1$, so

$$a_n + 1 = 2^n(a_0 + 1) = 2^n,$$

and therefore

$$\boxed{a_n = 2^n - 1.}$$

The first values

$$0, 1, 3, 7, 15, \ldots$$

confirm the formula.

The fixed-point form also gives the stability classification. When $|r| < 1$, every orbit converges to L. When $|r| > 1$, every orbit except the fixed point moves away exponentially. When $r = -1$, nonfixed orbits oscillate with period two. Thus a closed formula also reveals the qualitative dynamics.

§113.5 Discrete Variation of Constants

For the nonautonomous recurrence

$$a_{n+1} = ra_n + b_n,$$

repeated substitution gives

$$\begin{aligned} a_1 &= ra_0 + b_0, \\ a_2 &= r^2 a_0 + rb_0 + b_1, \\ a_3 &= r^3 a_0 + r^2 b_0 + rb_1 + b_2. \end{aligned}$$

The pattern is

$$\boxed{a_n = r^n a_0 + \sum_{k=0}^{n-1} r^{n-1-k} b_k.}$$

This is the discrete variation-of-constants formula. A forcing term introduced at time k is propagated through the remaining $n-1-k$ iterations by the factor r^{n-1-k}.

If $b_k = b$ is constant, the sum becomes geometric and recovers the affine fixed-point formula. If the forcing has a special shape, such as $b_k = c\lambda^k$, the sum reveals resonance when $\lambda = r$.

§113.6 Normalization should match the coefficient growth

The recurrence

$$a_n = na_{n-1} + 1, \qquad a_1 = 1,$$

does not become simpler after subtracting a constant. The coefficient n suggests dividing by $n!$. Define

$$b_n = \frac{a_n}{n!}.$$

Then

$$b_n = \frac{na_{n-1} + 1}{n!} = \frac{a_{n-1}}{(n-1)!} + \frac{1}{n!} = b_{n-1} + \frac{1}{n!}.$$

The normalized recurrence telescopes:

$$b_n = 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!}.$$

Hence

$$\boxed{a_n = n!\left(1 + \frac{1}{2!} + \cdots + \frac{1}{n!}\right).}$$

For example,

$$a_1 = 1, \quad a_2 = 3, \quad a_3 = 10, \quad a_4 = 41,$$

and the formula gives

$$4!\left(1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24}\right) = 41.$$

The method is structural: multiplication by n points toward factorial normalization, just as multiplication by a constant points toward geometric normalization.

§113.7 Second-order recurrences are matrix powers

Consider

$$a_{n+2} = 5a_{n+1} - 6a_n.$$

Set

$$v_n = \begin{pmatrix} a_{n+1} \\ a_n \end{pmatrix}.$$

Then

$$v_{n+1} = Cv_n, \qquad C = \begin{pmatrix} 5 & -6 \\ 1 & 0 \end{pmatrix},$$

so

$$v_n = C^n v_0.$$

The characteristic polynomial is

$$\det(\lambda I - C) = \lambda^2 - 5\lambda + 6 = (\lambda - 2)(\lambda - 3).$$

The eigenvalues 2 and 3 explain the two exponential solutions. Thus

$$a_n = A2^n + B3^n.$$

Take

$$a_0 = 0, \qquad a_1 = 1.$$

The initial conditions give

$$A + B = 0, \qquad 2A + 3B = 1,$$

so $A = -1$ and $B = 1$. Therefore

$$\boxed{a_n = 3^n - 2^n.}$$

Indeed,

$$0, 1, 5, 19, 65, \ldots$$

satisfies the recurrence.

The trial solution $a_n = \lambda^n$ is therefore not a guess without justification. It searches for eigenvectors of the shift operator, or equivalently eigenvectors of the companion matrix.

If the characteristic polynomial has a repeated root λ, the matrix may have a Jordan block and the second independent solution becomes

$$n\lambda^n.$$

For example,

$$a_{n+2} = 2a_{n+1} - a_n$$

has characteristic polynomial $(t-1)^2$, so

$$a_n = A + Bn.$$

Arithmetic progressions are exactly this repeated-root case.

§113.8 Generating functions turn shifts into multiplication

The same recurrence can be solved by a generating function. Let

$$A(z) = \sum_{n \geq 0} a_n z^n$$

for the initial data $a_0 = 0$, $a_1 = 1$. Multiply

$$a_{n+2} - 5a_{n+1} + 6a_n = 0$$

by z^{n+2} and sum over $n \geq 0$. The shifted sums are

$$\sum_{n \geq 0} a_{n+2} z^{n+2} = A(z) - z,$$

$$\sum_{n \geq 0} a_{n+1} z^{n+2} = zA(z),$$

and

$$\sum_{n \geq 0} a_n z^{n+2} = z^2 A(z).$$

Hence

$$A(z) - z - 5zA(z) + 6z^2 A(z) = 0,$$

so

$$A(z) = \frac{z}{1 - 5z + 6z^2} = \frac{z}{(1 - 2z)(1 - 3z)}.$$

Partial fractions give

$$A(z) = \frac{1}{1 - 3z} - \frac{1}{1 - 2z}.$$

Since

$$\frac{1}{1 - cz} = \sum_{n \geq 0} c^n z^n,$$

we recover

$$a_n = 3^n - 2^n.$$

The characteristic roots appear as poles of the generating function. Matrix powers, exponential trial solutions, and rational generating functions are three views of the same linear recurrence.

§113.9 Forward iteration is a semigroup action

The nonnegative integers form a monoid under addition: they have an associative operation and an identity element 0, but they do not contain additive inverses. The iterates of any map $F : X \to X$ satisfy

$$F^{m+n} = F^m \circ F^n, \qquad F^0 = \mathrm{id}_X .$$

Therefore every recurrence

$$a_{n+1} = F(a_n)$$

defines an action of $(\mathbb{N}, +)$ on X:

$$n \cdot x = F^n(x).$$

This language says exactly that advancing by n steps and then by m steps is the same as advancing by $m + n$ steps.

No inverse is required. For example,

$$F(x) = x^2$$

on $\mathbb{R}$ identifies x and $-x$ after one step. Once their orbits merge, the previous state cannot be recovered from the current one. The recurrence still has perfectly good forward iterates, but there is no globally defined F^{-1}.

If F is bijective, negative iterates are available and the action extends from $\mathbb{N}$ to $\mathbb{Z}$:

$$(-n) \cdot x = F^{-n}(x).$$

This is the precise boundary at which groups enter. A group action describes transformations that can be composed and undone; a monoid action describes forward evolution whether or not it is reversible.

Definition 113.9.1 (Group)**.** A group is a set G with an associative binary operation, an identity element e, and an inverse g^{-1} for every $g \in G$. If the operation is commutative, the group is abelian.

The phrase "binary operation on G" already includes closure:

$$* : G \times G \to G.$$

The axioms say that compositions are unambiguous, doing nothing is allowed, and every permitted transformation can be reversed.

§113.10 Recurrences in the Affine Group

For $r \neq 0$ and $b \in \mathbb{R}$, define the affine transformation

$$T_{r,b}(x) = rx + b.$$

Composing two such maps gives

$$\begin{aligned} T_{r,b}(T_{s,c}(x)) &= r(sx + c) + b \\ &= rsx + (rc + b), \end{aligned}$$

so

$$T_{r,b} \circ T_{s,c} = T_{rs,rc+b}.$$

The inverse is obtained by solving $y = rx + b$:

$$T_{r,b}^{-1}(y) = \frac{y - b}{r} = T_{1/r,-b/r}(y).$$

Thus all affine transformations with nonzero slope form a group.

Translations and scalings sit inside this group:

$$\tau_b = T_{1,b}, \qquad \sigma_r = T_{r,0}.$$

They satisfy

$$\tau_b \tau_c = \tau_{b+c}, \qquad \sigma_r \sigma_s = \sigma_{rs}.$$

However, they do not commute in general. Conjugating a translation by a scaling gives

$$\begin{aligned}\sigma_r\tau_c\sigma_r^{-1}(x) &= \sigma_r\tau_c(x/r)\\ &= \sigma_r(x/r+c)\\ &= x+rc\\ &= \tau_{rc}(x).\end{aligned}$$

Scaling changes the size of a translation. This interaction is the reason the affine group is a semidirect product rather than a direct product:

$$\mathrm{Aff}(1,\mathbb{R}) \cong \mathbb{R} \rtimes \mathbb{R}^\times.$$

The multiplication law

$$(b,r)(c,s) = (b+rc, rs)$$

records exactly how the scaling component r acts on the translation component c.

Now return to the recurrence

$$a_{n+1} = ra_n + b.$$

Its sequence is the orbit of a_0 under the cyclic subgroup generated by $T_{r,b}$. The fixed-point trick from the first half of the note has a group-theoretic meaning. If $r \neq 1$, let

$$L = \frac{b}{1-r},$$

so $T_{r,b}(L) = L$. Translate the fixed point to the origin using

$$\tau_{-L}(x) = x - L.$$

Then

$$\begin{aligned}\tau_{-L} \circ T_{r,b} \circ \tau_L(x) &= \tau_{-L}(r(x+L)+b)\\ &= rx + rL + b - L\\ &= rx,\end{aligned}$$

because $(1-r)L = b$. Hence

$$\tau_{-L}T_{r,b}\tau_L = \sigma_r.$$

The affine recurrence is not merely similar to a geometric progression after a clever substitution; its update map is conjugate to a pure scaling. Taking powers gives

$$T_{r,b}^n = \tau_L\sigma_r^n\tau_{-L},$$

and therefore

$$T_{r,b}^n(x) = L + r^n(x-L).$$

This is exactly the closed formula derived earlier.

§113.11 Homogeneous Coordinates for Iteration

Affine maps are not linear on the one-dimensional vector space because they do not necessarily preserve the origin. They become linear after adding one extra coordinate. Represent $x \in \mathbb{R}$ by the homogeneous vector

$$\begin{pmatrix} x \\ 1 \end{pmatrix}$$

and define

$$M_{r,b} = \begin{pmatrix} r & b \\ 0 & 1 \end{pmatrix}.$$

Then

$$M_{r,b} \begin{pmatrix} x \\ 1 \end{pmatrix} = \begin{pmatrix} rx + b \\ 1 \end{pmatrix}.$$

Matrix multiplication reproduces affine composition:

$$M_{r,b} M_{s,c} = M_{rs,rc+b}.$$

Thus the affine group is realized as a subgroup of $GL_2(\mathbb{R})$.

The fixed-point conjugacy becomes an ordinary matrix factorization. Let

$$P_L = \begin{pmatrix} 1 & L \\ 0 & 1 \end{pmatrix}, \qquad D_r = \begin{pmatrix} r & 0 \\ 0 & 1 \end{pmatrix}.$$

Since $(1 - r)L = b$,

$$M_{r,b} = P_L D_r P_L^{-1}.$$

Therefore

$$M_{r,b}^n = P_L D_r^n P_L^{-1} = \begin{pmatrix} r^n & (1 - r^n)L \\ 0 & 1 \end{pmatrix}.$$

Substituting $L = b/(1 - r)$ gives

$$\boxed{M_{r,b}^n = \begin{pmatrix} r^n & b\dfrac{1 - r^n}{1 - r} \\ 0 & 1 \end{pmatrix}} \qquad (r \neq 1).$$

Applying this matrix to $(a_0, 1)^T$ recovers the affine recurrence formula.

The case $r = 1$ is different because there is no finite fixed point to translate to the origin. Now

$$M_{1,b} = I + N, \qquad N = \begin{pmatrix} 0 & b \\ 0 & 0 \end{pmatrix}, \qquad N^2 = 0.$$

The binomial theorem truncates:

$$(I + N)^n = I + nN.$$

Hence

$$M_{1,b}^n = \begin{pmatrix} 1 & nb \\ 0 & 1 \end{pmatrix},$$

which gives

$$a_n = a_0 + nb.$$

Arithmetic progressions are therefore governed by a unipotent Jordan block, while affine recurrences with $r \neq 1$ are governed by the two eigenvalues r and 1. The matrix model explains why the formulas split exactly at $r = 1$.

For the concrete recurrence

$$a_{n+1} = 2a_n + 1, \qquad a_0 = 0,$$

we have $L = -1$ and

$$M = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Thus

$$M^n = \begin{pmatrix} 2^n & 2^n - 1 \\ 0 & 1 \end{pmatrix},$$

and the upper-right entry is precisely the orbit value $a_n = 2^n - 1$.

§113.12 Dihedral symmetry is affine geometry modulo n

Label the vertices of a regular n-gon by $\mathbb{Z}/n\mathbb{Z}$. A rotation by b vertices is

$$i \longmapsto i + b,$$

while a reflection followed by a rotation has the form

$$i \longmapsto -i + b.$$

Thus every symmetry can be written uniformly as

$$i \longmapsto \varepsilon i + b, \qquad \varepsilon \in \{1, -1\}, \quad b \in \mathbb{Z}/n\mathbb{Z}.$$

This is an affine transformation over the cyclic ring $\mathbb{Z}/n\mathbb{Z}$.

Compose two such maps:

$$(b, \varepsilon)(c, \delta)(i) = \varepsilon(\delta i + c) + b = (\varepsilon\delta)i + (b + \varepsilon c).$$

Hence

$$(b, \varepsilon)(c, \delta) = (b + \varepsilon c, \varepsilon\delta).$$

The sign acts on translations by either preserving or reversing them. Therefore

$$D_{2n} \cong C_n \rtimes C_2,$$

where the nontrivial element of C_2 acts on C_n by inversion.

Let

$$r(i) = i + 1, \qquad s(i) = -i.$$

Then

$$r^n = 1, \qquad s^2 = 1,$$

and

$$\begin{aligned} srs(i) &= sr(-i) \\ &= s(-i + 1) \\ &= i - 1 \\ &= r^{-1}(i). \end{aligned}$$

Thus

$$D_{2n} = \langle r, s \mid r^n = s^2 = 1,\ srs = r^{-1} \rangle.$$

Every element is uniquely either r^k or sr^k, so the group has $2n$ elements.

For a regular pentagon, the ten maps are

$$i \mapsto i + b \quad \text{and} \quad i \mapsto -i + b, \qquad b \in \mathbb{Z}/5\mathbb{Z}.$$

The formula makes the geometry easy to compute. For example,

$$(2,-1)(3,1) = (2-3,-1) = (4,-1) \pmod 5.$$

A reflection centered differently is obtained because the preceding rotation changes the translation parameter with a sign.

The dihedral group is therefore not an unrelated second example of a group. It is another affine semidirect product, now over a finite cyclic space rather than the real line.

§113.13 Direct and Semidirect Products

If transformations on two coordinates are genuinely independent, the correct construction is a direct product. Suppose

$$x_{n+1} = F(x_n), \qquad y_{n+1} = G(y_n).$$

The combined update is

$$(x,y) \longmapsto (F(x), G(y)),$$

and the two components can be iterated separately. If F and G are invertible, their transformation groups act through

$$(g,h)\cdot(x,y) = (g\cdot x, h\cdot y).$$

The multiplication law

$$(g_1,h_1)(g_2,h_2) = (g_1g_2, h_1h_2)$$

contains no cross-term because neither component modifies the other.

The affine and dihedral groups are different. In

$$(b,r)(c,s) = (b+rc, rs),$$

the scaling r changes the later translation c. In

$$(b,\varepsilon)(c,\delta) = (b+\varepsilon c, \varepsilon\delta),$$

the reflection sign changes the direction of the later rotation. These cross-terms are the signature of a semidirect product: one subgroup acts on the other.

An isomorphism is a bijective homomorphism

$$\varphi : G \to H$$

that preserves multiplication. The maps

$$b \longmapsto \tau_b$$

and

$$(b,\varepsilon) \longmapsto (i \mapsto \varepsilon i + b)$$

are useful because they replace abstract group elements by transformations whose composition can be seen directly.

This completes the bridge from recurrences to groups without identifying the two subjects. Every recurrence gives a forward $\mathbb{N}$-action. Bijective update rules extend to $\mathbb{Z}$-actions. First-order linear recurrences sit inside the affine group, where shifting a fixed point is conjugacy and closed forms are power formulas. Polygon symmetries form a finite affine semidirect product. The common structure is not repetition in the vague sense, but the algebra of composing transformations and deciding whether their effects are independent, interacting, or reversible.

Nonlinear Recurrences, Fixed Points, and Metric-Space Exercises

N-20220718

§114.1 Finding a fixed point is only the beginning

An iteration takes the form

$$a_{n+1} = f(a_n).$$

If the sequence converges to L and f is continuous at L, then

$$L = \lim_{n\to\infty} a_{n+1} = \lim_{n\to\infty} f(a_n) = f(L).$$

This necessary condition does not guarantee convergence. The map $f(x) = -x$ fixes 0, yet almost every orbit alternates between a_1 and $-a_1$. The map $f(x) = 2x$ also fixes 0, but nonzero orbits move away from it.

For a differentiable map near a fixed point L, the derivative is the first local test. By continuity of f', if $|f'(L)| < 1$ then one can find an interval about L in which f is a contraction, so iterations sufficiently close to L converge geometrically. If $|f'(L)| > 1$, the fixed point is locally repelling. The boundary case $|f'(L)| = 1$ requires more analysis and is particularly relevant to the Möbius example below. This local stability viewpoint is standard in one-dimensional discrete dynamics; the linear-recursion method later in the note follows MIT's *Mathematics for Computer Science*, Chapter 10.[1]

An independent elementary proof method is monotone convergence. If one proves by induction that the iterates stay in a closed interval and that $a_{n+1} \geq a_n$, boundedness gives a limit by completeness of the real numbers. Continuity then identifies the limit with a fixed point. Finding the fixed point without first proving these hypotheses would reverse the logical argument.

§114.2 Linearizing Möbius Recurrences

A fractional-linear map is

$$F(x) = \frac{px + q}{rx + t}, \qquad pt - qr \neq 0.$$

It is associated with the invertible matrix

$$M = \begin{pmatrix} p & q \\ r & t \end{pmatrix}$$

acting on the projective line. Iteration corresponds to the matrix powers M^n up to a common nonzero scalar. This gives an exact algebraic representation of a nonlinear-looking recurrence.

Fixed points satisfy

$$rx^2 + (t - p)x - q = 0.$$

[1] For the recurrence calculation see MIT 6.042J, *Mathematics for Computer Science*, Chapter 10.

When this equation has two distinct finite roots α, β, the coordinate

$$h(x) = \frac{x-\alpha}{x-\beta}$$

maps the fixed points to 0 and ∞. The conjugate map $h \circ F \circ h^{-1}$ fixes both 0 and ∞, so it has the form $u \mapsto \lambda u$. In particular,

$$\frac{a_{n+1}-\alpha}{a_{n+1}-\beta} = \lambda \frac{a_n-\alpha}{a_n-\beta}$$

where the expressions are defined. The new sequence is geometric and its behavior can be read from $|\lambda|$.

Example 114.2.1 (A repeated fixed point reveals a finite-domain obstruction)
The original recurrence is

$$a_1 = \frac{1}{2}, \qquad a_{n+1} = \frac{5a_n - 1}{a_n + 3}.$$

Its fixed-point equation reduces to

$$x^2 - 2x + 1 = (x-1)^2 = 0.$$

The repeated fixed point is 1. The usual quotient $(x-\alpha)/(x-\beta)$ degenerates, but the reciprocal coordinate

$$b_n = \frac{1}{a_n - 1}$$

gives a translation:

$$a_{n+1} - 1 = \frac{4(a_n-1)}{a_n+3}, \qquad b_{n+1} = \frac{a_n+3}{4(a_n-1)} = b_n + \frac{1}{4}.$$

Since $b_1 = -2$, we obtain

$$b_n = -2 + \frac{n-1}{4} = \frac{n-9}{4}, \qquad \boxed{a_n = 1 + \frac{4}{n-9}.}$$

There is an important restriction: the displayed real recurrence is defined only while the denominators are nonzero. In fact,

$$a_8 = -3,$$

so the rule for a_9 requires division by 0. The original real-valued sequence stops at $n = 8$, and it would be incorrect to claim that the closed formula describes an infinite real sequence. On the projective line one can instead interpret $a_9 = \infty$ and continue with $a_{10} = 5$.
The derivative

$$F'(x) = \frac{16}{(x+3)^2}$$

satisfies $F'(1) = 1$. The fixed point is therefore neutral at first order, which is precisely why the reciprocal coordinate and its additive drift are more informative than the elementary contraction test.

The single calculation brings together the two cases: distinct fixed points produce a multiplicative coordinate, while a repeated fixed point produces a translation in a suitable reciprocal coordinate. It also shows why the domain of iteration must be specified before asserting a limit.

§114.3 Repeated Roots of Linear Recurrences

A linear recurrence

$$a_{n+2} = pa_{n+1} + qa_n$$

has characteristic polynomial $P(\lambda) = \lambda^2 - p\lambda - q$. Substituting $a_n = \lambda^n$ verifies why each nonzero characteristic root yields a geometric solution. If the two roots λ_1, λ_2 are distinct, every solution has the form

$$a_n = A\lambda_1^n + B\lambda_2^n,$$

with constants determined by two initial conditions.

For a repeated root $\lambda \neq 0$, a second independent solution is $n\lambda^n$, yielding

$$a_n = (A + Bn)\lambda^n.$$

The factor n is the discrete counterpart of a generalized eigenvector: the companion matrix

$$\begin{pmatrix} p & q \\ 1 & 0 \end{pmatrix}$$

has a Jordan block when the characteristic root repeats. Thus repeated roots are not merely an exception in a formula; they are a test of whether diagonalization succeeds.

The matrix picture also explains the Möbius case. Fractional-linear iteration uses a projective 2×2 matrix, while a second-order linear recurrence uses a companion 2×2 matrix. Both problems are governed by eigenvalues and invariant directions, but the transformation from a vector to its ratio changes the representation of the dynamics.

§114.4 Two Constructions of Spaces

The original material also includes problems about Dedekind cuts and metric subspaces. These concern *different* questions from recurrence linearization and deserve their own mathematical explanation rather than being forced into the recurrence story.

For a Dedekind cut $X \subset \mathbb{Q}$, use the usual convention that X is a nonempty proper downward-closed set without a greatest element. For two cuts X, Y, define

$$X + Y = \{x + y : x \in X,\ y \in Y\}.$$

Why is this again a cut? It is nonempty by choosing $x \in X, y \in Y$. It is downward closed: if $q < x + y$, then $q = (q - y) + y$ with $q - y < x$, hence $q - y \in X$. It has no greatest element because X has none: for $x + y$, choose $x' > x$ still in X. Finally, choose rational upper bounds B_X, B_Y of X, Y; then every element of $X + Y$ is less than $B_X + B_Y$, proving it is proper. This checks the four cut axioms explicitly.

For a metric space (X, d) and a subset $Y \subseteq X$, the restriction

$$d_Y(y_1, y_2) = d(y_1, y_2)$$

is again a metric, since each metric axiom holds for all points in X, hence in Y. In Euclidean space,

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2},$$

and the triangle inequality follows from the inner-product Cauchy–Schwarz inequality:

$$\|u + v\|_2^2 = \|u\|_2^2 + 2\langle u, v\rangle + \|v\|_2^2 \leq (\|u\|_2 + \|v\|_2)^2.$$

These exercises establish how an existing order or distance structure can be extended or restricted, without pretending that the resulting objects are instances of the same iteration theorem.

From Work Integrals to Phase-Space Area and Action Variables

N-20230123

An integral measures accumulation, but the geometry of what is accumulated matters. Work is integrated along an oriented path, and the same line-integral mechanism reappears when one measures the area enclosed by a periodic trajectory in phase space. In that setting the integral $\oint p\,dq$ turns out to determine the oscillation period without solving the equation of motion explicitly. This is a concrete route from first-year calculus to Hamiltonian mechanics.[1]

§115.1 Work is the accumulation of force along a path

In one dimension, when a force $F(x)$ acts while a particle moves from a to b,

$$W = \int_a^b F(x)\,dx.$$

For example, the spring force $F(x) = -kx$ performs work

$$W_{a\to b} = -\frac{k}{2}(b^2 - a^2).$$

This calculation naturally suggests the potential energy $U(x) = kx^2/2$, for which $F = -U'$ and $W = U(a) - U(b)$.

For a curve $\gamma(t) = (x(t), y(t), z(t))$ in space, the displacement is $d\mathbf{r} = \gamma'(t)\,dt$. The signed work becomes

$$\boxed{W_\gamma = \int_\gamma F \cdot d\mathbf{r} = \int_a^b F(\gamma(t)) \cdot \gamma'(t)\,dt.}$$

It depends on the path's orientation. If $F = -\nabla U$, the chain rule gives

$$F(\gamma(t)) \cdot \gamma'(t) = -\frac{d}{dt} U(\gamma(t)),$$

so

$$\boxed{W_\gamma = U(\gamma(a)) - U(\gamma(b)).}$$

For a conservative force, the work depends only on the endpoints and vanishes around every closed loop.

§115.2 Circulation converts a line integral to area

Consider the planar field $F = (-y, x)$, which rotates counterclockwise around the origin. On a circle of radius R,

$$\gamma(t) = (R\cos t, R\sin t), \quad 0 \le t \le 2\pi,$$

[1] MIT 18.02 (line integrals and Green's theorem); MIT 8.09, Classical Mechanics III (action variables); V. I. Arnold, *Mathematical Methods of Classical Mechanics*.

one obtains

$$\int_\gamma F \cdot d\mathbf{r} = \int_0^{2\pi} R^2 \, dt = 2\pi R^2.$$

The curl of this field is constant:

$$\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} = 2.$$

Green's theorem therefore explains the result as

$$\oint_{\partial D} (-y \, dx + x \, dy) = \iint_D 2 \, dx \, dy = 2 \text{ area}(D).$$

The calculation is exact for any sufficiently regular positively oriented planar region D, not just circles.

Choosing instead the one-form $x \, dy$, whose exterior derivative is $dx \wedge dy$, gives the particularly simple area formula

$$\boxed{\text{area}(D) = \oint_{\partial D} x \, dy = -\oint_{\partial D} y \, dx.}$$

A planar region can therefore be measured by integrating around its boundary. This is the bridge from accumulation along a path to geometric area.

§115.3 The canonical one-form on phase space

Now consider a particle moving in one dimension, with position q, momentum $p = m\dot{q}$, and mechanical energy

$$H(q, p) = \frac{p^2}{2m} + V(q).$$

The pair (q, p) describes a point in *phase space*. A periodic mechanical motion traces a closed curve there. Along this curve, define the integral

$$\boxed{J(E) = \oint_{H=E} p \, dq.}$$

Unlike the mechanical work $\int F \, dq$, this integral measures momentum against displacement. Its units are momentum times length, or action.

The one-form $\vartheta = p \, dq$ is the canonical one-form in these coordinates. Its differential is

$$d\vartheta = dp \wedge dq = -dq \wedge dp.$$

Green's theorem thus interprets $J(E)$ as an oriented phase-space area. For the usual Hamiltonian evolution of a one-dimensional oscillation, the energy orbit is traversed clockwise in (q, p)-coordinates; the integral $J(E)$ equals the ordinary positive area enclosed by that orbit.

§115.4 Action of the Harmonic Oscillator

Take $V(q) = kq^2/2$ with $\omega = \sqrt{k/m}$. The energy curve is

$$\frac{p^2}{2mE} + \frac{kq^2}{2E} = 1.$$

It is an ellipse with horizontal semiaxis $q_{\max} = \sqrt{2E/k}$ and vertical semiaxis $p_{\max} = \sqrt{2mE}$. Its area is

$$J(E) = \pi q_{\max} p_{\max} = \pi \sqrt{\frac{2E}{k}} \sqrt{2mE} = \boxed{\frac{2\pi E}{\omega}}.$$

The same answer can be checked by following one orbit:

$$q(t) = A\cos\omega t, \qquad p(t) = -m\omega A \sin\omega t, \qquad E = \frac{1}{2} m\omega^2 A^2.$$

Since $dq/dt = -A\omega \sin\omega t$,

$$\oint p\, dq = m\omega^2 A^2 \int_0^{2\pi/\omega} \sin^2 \omega t\, dt = \pi m \omega A^2 = \frac{2\pi E}{\omega}.$$

The boundary-integral interpretation and the direct time integral are equivalent because the particle traces precisely the ellipse.

§115.5 Differentiating enclosed area recovers the period

The oscillator identity is an instance of a general fact. Suppose

$$H(q,p) = \frac{p^2}{2m} + V(q)$$

has a regular family of periodic energy orbits between simple turning points $q_-(E) < q_+(E)$. On the upper half of the orbit,

$$p_+(q,E) = \sqrt{2m(E - V(q))},$$

and the lower half has momentum $-p_+$. The enclosed area is therefore

$$J(E) = 2\int_{q_-(E)}^{q_+(E)} \sqrt{2m(E - V(q))}\, dq.$$

Differentiating with respect to energy is legitimate away from critical energy levels. The integrand vanishes at the moving turning points, so their endpoint contributions vanish. We obtain

$$\begin{aligned} \frac{dJ}{dE} &= 2\int_{q_-}^{q_+} \frac{m}{\sqrt{2m(E - V(q))}}\, dq \\ &= 2\int_{q_-}^{q_+} \frac{dq}{\dot q}. \end{aligned}$$

Here $\dot q = p/m > 0$ on the upper branch. The last integral is twice the travel time from the left turning point to the right turning point, hence the full period:

$$\boxed{\frac{dJ}{dE} = T(E).}$$

For the harmonic oscillator $J(E) = 2\pi E/\omega$, this immediately yields $T = 2\pi/\omega$, without solving $m\ddot q + kq = 0$.

The formula is useful for anharmonic potentials: the period's energy dependence is encoded in the changing area of the energy contours. If an energy approaches a separatrix, the period may diverge because motion slows near an unstable equilibrium; the derivative of the phase-space area captures that divergence.

§115.6 Canonical Coordinate Changes

The phase-space area form is $\omega = dq \wedge dp$. A canonical change of coordinates $(q, p) \mapsto (Q, P)$ preserves this two-form:

$$dq \wedge dp = dQ \wedge dP.$$

Locally, the difference between the canonical one-forms is therefore closed:

$$d(p\,dq - P\,dQ) = 0.$$

On a simply connected coordinate patch it is exact, say $p\,dq - P\,dQ = dS$. Around a closed orbit lying in such a patch,

$$\oint p\,dq - \oint P\,dQ = \oint dS = 0.$$

Consequently the action is a coordinate-invariant measure of the enclosed symplectic area. The requirement about the coordinate patch matters: a closed form on a multiply connected domain need not have a globally defined potential.

The path integral has now done three kinds of work. Integrating $F\,dq$ gives the mechanical work performed by a force, integrating around the boundary of a planar region measures ordinary area, and integrating $p\,dq$ around a periodic trajectory measures phase-space area. In a one-degree-of-freedom Hamiltonian system, differentiating this last area with respect to energy gives a directly observable quantity: the period.

Fixed Points, Convergent Iteration, and Compactness Methods

N-20230316

§116.1 Fixed Points versus Iterations

A fixed point of a map $T : X \to X$ is a point x^* satisfying

$$T(x^*) = x^*.$$

If an iteration

$$x_{n+1} = T(x_n)$$

converges to x and T is continuous, then

$$T(x) = T\left(\lim_{n\to\infty} x_n\right) = \lim_{n\to\infty} T(x_n) = \lim_{n\to\infty} x_{n+1} = x.$$

Thus every convergent orbit ends at a fixed point. The converse is false: the existence of a fixed point says nothing by itself about whether an orbit converges to it.

For example,

$$T(x) = -x$$

has the fixed point 0, but an orbit starting from $x_0 \neq 0$ alternates between x_0 and $-x_0$. The map

$$T(x) = x$$

has every point fixed, so there is no unique target for iteration. A useful fixed-point theorem must therefore specify which mechanism supplies existence, uniqueness, and convergence.

§116.2 Geometry of Contractions

Let (X, d) be a metric space. A map $T : X \to X$ is a contraction when there is a constant $0 < c < 1$ such that

$$d(Tx, Ty) \le c\, d(x, y)$$

for all $x, y \in X$.

Starting from x_0, define

$$x_{n+1} = T(x_n).$$

The contraction estimate gives

$$d(x_{n+1}, x_n) \le c\, d(x_n, x_{n-1}) \le \cdots \le c^n d(x_1, x_0).$$

For $m > n$, the triangle inequality then yields

$$\begin{aligned} d(x_m, x_n) &\le \sum_{k=n}^{m-1} d(x_{k+1}, x_k) \\ &\le d(x_1, x_0) \sum_{k=n}^{m-1} c^k \\ &\le \frac{c^n}{1-c} d(x_1, x_0). \end{aligned}$$

The orbit is therefore Cauchy. This is the key analytic fact; all later conclusions depend on where that Cauchy sequence is allowed to converge.

§116.3 The Role of Completeness

A metric space is complete when every Cauchy sequence converges to a point of the space.

Theorem 116.3.1 (Banach contraction principle)
Let (X, d) be a nonempty complete metric space and let $T : X \to X$ be a contraction. Then T has a unique fixed point x^*. For every $x_0 \in X$, the iterates $x_{n+1} = T(x_n)$ converge to x^*.

Proof. The previous estimate shows that (x_n) is Cauchy, so completeness gives a limit $x^* \in X$. A contraction is Lipschitz and hence continuous, so

$$T(x^*) = T\left(\lim x_n\right) = \lim T(x_n) = \lim x_{n+1} = x^*.$$

If x and y are fixed points, then

$$d(x, y) = d(Tx, Ty) \leq c\, d(x, y).$$

Since $c < 1$, this forces $d(x, y) = 0$, hence $x = y$. □

Completeness cannot be omitted. On the incomplete space $(0, 1)$, the map

$$T(x) = \frac{x}{2}$$

is a contraction with constant $1/2$ and maps $(0, 1)$ into itself. Its iterates satisfy

$$x_n = 2^{-n} x_0 \longrightarrow 0,$$

but $0 \notin (0, 1)$. The contraction has no fixed point in the declared space. The estimates have located a missing boundary point.

§116.4 The proof gives two practical stopping bounds

Let x^* be the fixed point. Passing $m \to \infty$ in the Cauchy estimate gives the a priori bound

$$\boxed{d(x_n, x^*) \leq \frac{c^n}{1 - c} d(x_1, x_0).}$$

It predicts in advance how many iterations are sufficient.

There is also an a posteriori bound that uses the most recent step. Since

$$d(x_{n+k+1}, x_{n+k}) \leq c^k d(x_{n+1}, x_n),$$

we obtain

$$\begin{aligned} d(x_n, x^*) &\leq \sum_{k=0}^{\infty} d(x_{n+k+1}, x_{n+k}) \\ &\leq \frac{1}{1 - c} d(x_{n+1}, x_n). \end{aligned}$$

Thus

$$\boxed{d(x_n, x^*) \le \frac{d(x_{n+1}, x_n)}{1-c}.}$$

This gives a rigorous stopping rule using quantities already computed by the iteration.

§116.5 The equation $x = \cos x$ is a complete example

Consider

$$T(x) = \cos x$$

on the interval $[0, 1]$. Since

$$\cos 1 \le \cos x \le 1,$$

the map sends $[0, 1]$ into itself. By the mean value theorem,

$$|\cos x - \cos y| \le \sin 1 \, |x - y|$$

for $x, y \in [0, 1]$. Because $\sin 1 < 1$, the map is a contraction with

$$c = \sin 1 \approx 0.84147.$$

The interval is complete, so there is a unique solution

$$x^* = \cos x^*$$

in $[0, 1]$, and every initial value in the interval converges to it.

Starting from $x_0 = 1$,

$$x_1 = \cos 1, \qquad x_2 = \cos(\cos 1), \qquad \dots$$

The limit is

$$x^* \approx 0.7390851332.$$

If two successive iterates differ by less than $(1 - c)10^{-6}$, the a posteriori estimate guarantees an absolute error below 10^{-6}. The theorem therefore supplies not only existence and uniqueness but also a certified numerical algorithm.

The choice of fixed-point formulation matters. The equation $f(x) = 0$ can often be rewritten as $x = T(x)$ in many ways. One formulation may be contractive while another diverges. Fixed-point iteration is therefore partly the art of choosing a map with a small Lipschitz constant on an invariant complete set.

§116.6 Picard iteration moves the theorem to a function space

Consider the initial-value problem

$$y'(t) = f(t, y(t)), \qquad y(t_0) = y_0.$$

Integrating gives the equivalent equation

$$y(t) = y_0 + \int_{t_0}^{t} f(s, y(s)) \, ds.$$

On the Banach space $C([t_0 - h, t_0 + h])$ with the supremum norm, define

$$(Ty)(t) = y_0 + \int_{t_0}^{t} f(s, y(s)) \, ds.$$

Assume that f is Lipschitz in its second variable:

$$|f(t,u) - f(t,v)| \le L|u - v|.$$

For t on *either side* of t_0, the integral estimate must account for orientation:

$$\begin{aligned}|(Ty)(t) - (Tz)(t)| &= \left| \int_{t_0}^{t} [f(s, y(s)) - f(s, z(s))]\, ds \right| \\ &\le \int_{\min(t,t_0)}^{\max(t,t_0)} |f(s, y(s)) - f(s, z(s))|\, ds \\ &\le L|t - t_0|\, \|y - z\|_\infty.\end{aligned}$$

Taking the supremum over the symmetric interval gives

$$\boxed{\|Ty - Tz\|_\infty \le Lh\, \|y - z\|_\infty.}$$

It would be incorrect to omit the absolute value on the integral when $t < t_0$: the oriented integral of a nonnegative function from t_0 down to t is nonpositive, and cannot bound a norm from above. If $h < 1/L$, the operator is a contraction. One also chooses a closed ball of functions on which T maps the ball into itself; this follows from a bound on $|f|$ on a suitable rectangle around (t_0, y_0).

Banach's theorem now gives a unique fixed point of T, hence a unique local solution of the differential equation. The iterates

$$y_{n+1}(t) = y_0 + \int_{t_0}^{t} f(s, y_n(s))\, ds$$

are the Picard approximations. The same geometric contraction estimate that solved $x = \cos x$ now proves existence and uniqueness in an infinite-dimensional space.

The operator also provides an explicit sequence of approximations. For the small initial-value problem

$$y'(t) = y(t), \qquad y(0) = 1,$$

the Picard map is

$$(Ty)(t) = 1 + \int_0^t y(s)\, ds.$$

Start with the constant function $y_0(t) = 1$. Repeated integration gives

$$\begin{aligned}y_1(t) &= 1 + t, \\ y_2(t) &= 1 + t + \frac{t^2}{2!}, \\ y_3(t) &= 1 + t + \frac{t^2}{2!} + \frac{t^3}{3!}.\end{aligned}$$

Inductively, if $y_n(t) = \sum_{k=0}^{n} t^k/k!$, then

$$y_{n+1}(t) = 1 + \int_0^t \sum_{k=0}^{n} \frac{s^k}{k!}\, ds = \sum_{k=0}^{n+1} \frac{t^k}{k!}.$$

We have therefore discovered the exponential series by applying the fixed-point operator:

$$\boxed{y_n(t) = \sum_{k=0}^{n} \frac{t^k}{k!} \xrightarrow[n\to\infty]{} e^t.}$$

On a compact interval $[-h, h]$, the error is bounded by a factorial tail, for example

$$|e^t - y_n(t)| \le e^h \frac{h^{n+1}}{(n+1)!}.$$

The convergence is uniform for every fixed h. In the local Banach argument, choosing $h < 1$ makes the Lipschitz constant $Lh = h < 1$; the explicit factorial bound reveals even more about this particular equation. The infinite-dimensional contraction proof and the elementary Taylor series have converged on the same object.

§116.7 Existence versus Banach Iteration

Banach's theorem is not the only route to a fixed point. Brouwer's theorem says that every continuous map from a nonempty compact convex subset of $\mathbb{R}^n$ to itself has a fixed point. No metric contraction is required.

The difference in conclusions is substantial. The identity map on a closed ball satisfies Brouwer's hypotheses and has every point fixed. A rotation of a disk fixes its center but preserves distances rather than shrinking them. Brouwer gives existence, but not uniqueness, a convergence rate, or even a canonical iteration that approaches a fixed point.

This distinction matters in applications. If the goal is a certified numerical method, the Banach hypotheses are valuable precisely because they are stronger. If the map is continuous on a compact convex region but cannot be made contractive, Brouwer may still prove that a solution exists, though a separate method is needed to locate it.

§116.8 Schauder's Fixed-Point Theorem

Closed bounded sets in infinite-dimensional Banach spaces are usually not compact, so Brouwer cannot be applied directly. Schauder's fixed-point theorem uses compactness of the operator instead.

Theorem 116.8.1 (Schauder fixed-point theorem)

Let C be a nonempty closed convex subset of a Banach space. If $T : C \to C$ is continuous and $T(C)$ is relatively compact, then T has a fixed point.

A typical source of compactness is an integral operator. On $C([0, 1])$, let

$$(Tu)(x) = g(x) + \int_0^1 K(x, s)F(s, u(s))\, ds,$$

where K, F, and g are continuous and a closed ball C is chosen so that $T(C) \subseteq C$. Uniform bounds on Tu and uniform continuity of K give equicontinuity of $T(C)$. Arzelà–Ascoli then shows that $T(C)$ is relatively compact.

Schauder therefore yields a solution of the integral equation even when no Lipschitz estimate makes T contractive. What is lost is uniqueness and convergence of the naive iteration. Several fixed points may exist.

§116.9 Krasnoselskii's Decomposition

Many nonlinear equations produce an operator that is neither contractive nor compact as a whole but decomposes as

$$T = A + B.$$

Krasnoselskii's theorem applies on a closed bounded convex set C when A is a contraction, B is continuous and compact, and

$$Ax + By \in C$$

for all $x, y \in C$ under the relevant version of the theorem.

The decomposition reflects two different mechanisms. The map A controls metric instability; the map B supplies compactness. One way to see the proof strategy is to fix $y \in C$ and solve

$$x = Ax + By.$$

For fixed y, the right-hand side is contractive in x, so Banach gives a unique solution $x = R(y)$. Compactness of B is then used to show that the induced map R has a fixed point by a Schauder-type argument.

This theorem is not merely a longer list entry after Banach and Schauder. It is designed for equations whose linear or dissipative part shrinks distances while an integral or smoothing term is compact.

When an equation has been rewritten as $x = T(x)$, the next step is to identify which of these mechanisms is actually available.

If one can prove

$$d(Tx, Ty) \le c\, d(x, y), \qquad c < 1,$$

then Banach gives existence, uniqueness, global convergence of iteration, and explicit error bounds. If only continuity and finite-dimensional compact convex invariance are available, Brouwer gives existence. In an infinite-dimensional problem, relative compactness of the image points toward Schauder. A natural decomposition into a contraction plus a compact map points toward Krasnoselskii.

The hypotheses are therefore not technical decorations around the same conclusion. They encode different reasons a fixed point must exist, and those reasons determine what else one is entitled to claim about uniqueness, approximation, and stability.

First-Order and Higher-Order Differential Equations

N-20260618

§117.1 First-Order ODEs and Substitutions

A differential equation relates an unknown function to its derivatives. The order is the highest derivative appearing. A general first-order equation has the form

$$F(x, y, y') = 0.$$

A solution is a function $y(x)$ satisfying the equation on an interval. When the coefficients and regularity conditions support existence and uniqueness, initial data select a particular solution from a family with integration constants. The development follows the MIT 18.03 *Differential Equations* lectures and Arthur Mattuck's supplementary notes on operators and impulse responses.[1] Initial conditions alone do not guarantee uniqueness: the equation $y' = 2\sqrt{|y|}$, $y(0) = 0$ has both $y \equiv 0$ and a solution that stays zero for $t \le c$ and then equals $(t - c)^2$ for $t > c$, for any $c \ge 0$.

A separable equation has

$$\frac{dy}{dx} = g(x)h(y).$$

If $h(y) \neq 0$, then

$$\frac{dy}{h(y)} = g(x)\,dx.$$

Integrating gives an implicit solution. One must also check equilibrium solutions coming from $h(y) = 0$, because division by $h(y)$ can lose them.

A homogeneous first-order equation has the form

$$y' = F\left(\frac{y}{x}\right).$$

Set $v = y/x$, so $y = vx$ and

$$y' = v + xv'.$$

Then

$$v + x\frac{dv}{dx} = F(v),$$

which is separable:

$$\frac{dv}{F(v) - v} = \frac{dx}{x}.$$

Some equations become homogeneous after translating the origin. If the linear expressions in x, y meet at a point, shift that point to the origin.

[1] https://ocw.mit.edu/courses/18-03-differential-equations-spring-2010/pages/readings/

§117.2 Integrating Factors and Exact Forms

A first-order linear equation is

$$y' + p(x)y = q(x).$$

The integrating factor is

$$\mu(x) = e^{\int p(x)\,dx}.$$

Multiplying the equation by μ gives

$$(\mu y)' = \mu q.$$

Thus

$$y = e^{-\int p}\left(\int q e^{\int p}\,dx + C\right).$$

The integrating factor is not a trick; it conjugates the operator $D+p(x)$ into the ordinary derivative D.

A Bernoulli equation has

$$y' + p(x)y = q(x)y^n.$$

For $n \neq 0, 1$, set

$$u = y^{1-n}.$$

Then

$$u' = (1-n)y^{-n}y',$$

and the equation becomes linear in u. Again, check special solutions lost by division. For example $y' + y = y^2$ becomes, on intervals where $y \neq 0$, $u' - u = -1$ under $u = 1/y$. Thus $u = 1 + Ce^x$ and $y = (1 + Ce^x)^{-1}$, but the equilibrium solution $y \equiv 0$ was lost during division and must be added separately.

An equation

$$M(x,y)\,dx + N(x,y)\,dy = 0$$

is exact if there is a potential Φ with

$$d\Phi = M\,dx + N\,dy.$$

The test is

$$M_y = N_x$$

on a suitable simply connected region. Then the solution is

$$\Phi(x,y) = C.$$

If the equation is not exact, one may look for an integrating factor, often depending only on x or only on y.

§117.3 Higher-Order ODEs and Characteristic Roots

If a second-order equation does not explicitly contain y, set $p = y'$ and reduce the order. If it does not explicitly contain x, set $p = p(y) = y'$, so

$$y'' = \frac{dp}{dx} = \frac{dp}{dy}\frac{dy}{dx} = p\frac{dp}{dy}.$$

These substitutions are not random: they use a missing variable as a symmetry.

A second-order linear equation has

$$y'' + p(x)y' + q(x)y = f(x).$$

The associated homogeneous equation has a solution space. If y_1, y_2 are independent homogeneous solutions, then

$$y_h = C_1 y_1 + C_2 y_2.$$

A general nonhomogeneous solution is

$$y = y_h + y_p.$$

This is the affine-space principle: one particular solution plus the kernel of a linear operator.

The Wronskian

$$W(y_1, y_2) = y_1 y_2' - y_1' y_2$$

measures independence. If it is nonzero at one point for solutions of a linear homogeneous equation, the two solutions form a fundamental system.

For

$$y'' + ay' + by = 0,$$

try $y = e^{rx}$. The characteristic equation is

$$r^2 + ar + b = 0.$$

Distinct real roots r_1, r_2 give

$$y = C_1 e^{r_1 x} + C_2 e^{r_2 x}.$$

A repeated root r gives

$$y = (C_1 + C_2 x)e^{rx}.$$

Complex roots $\alpha \pm i\beta$ give

$$y = e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x).$$

For

$$L[y] = f(x), \qquad L = D^2 + aD + b,$$

the method of undetermined coefficients works when f is built from polynomials, exponentials, sines, and cosines. If the trial function overlaps with a homogeneous solution, multiply by a sufficient power of x.

Variation of parameters works more generally. If y_1, y_2 are fundamental solutions, seek

$$y_p = u_1(x)y_1(x) + u_2(x)y_2(x).$$

The auxiliary condition $u_1' y_1 + u_2' y_2 = 0$ produces a solvable linear system for u_1', u_2'.

§117.4 Initial and boundary conditions select solutions

An initial-value problem specifies

$$y(x_0) = a, \qquad y'(x_0) = b.$$

For regular linear equations, existence and uniqueness say that these data select exactly one solution. This is conceptually important: a second-order equation needs two pieces of initial data because its solution space is two-dimensional.

§117.5 Differential operators and response kernels

Let

$$L[y] = y'' + p(x)y' + q(x)y.$$

Then a differential equation is an operator equation

$$L[y] = f.$$

The homogeneous solutions form $\ker L$. A nonhomogeneous solution set, if nonempty, is an affine translate of $\ker L$. Initial conditions are linear functionals that choose one element from that affine space. Thus a linear differential equation resembles a linear system, but the domain of the differential operator and its boundary conditions must be specified before asking whether an inverse exists. The Wronskian plays the local role of a determinant for fundamental solutions.

For a linear operator L, a Green function $G(x, \xi)$ is a kernel that represents the inverse problem. Formally,

$$L_x G(x, \xi) = \delta(x - \xi),$$

where δ is the Dirac delta distribution. Then a solution of

$$L[y] = f$$

can often be written as

$$y(x) = \int G(x, \xi) f(\xi)\, d\xi$$

with boundary conditions built into G.

The boundary conditions determine the kernel. For the simple Dirichlet problem $-y'' = f$ on $(0, 1)$, with $y(0) = y(1) = 0$, one can calculate

$$G(x, \xi) = \begin{cases} x(1 - \xi), & x \le \xi, \\ \xi(1 - x), & x \ge \xi. \end{cases}$$

It vanishes at the endpoints; its first derivative jumps by -1 at $x = \xi$, so $-\partial_x^2 G = \delta_\xi$ distributionally. Hence $y(x) = \int_0^1 G(x, \xi) f(\xi)\, d\xi$ solves the boundary-value problem for sufficiently regular forcing. This is the precise operator-theoretic form of superposition.

For first-order evolution equations, one often writes

$$\frac{du}{dt} = Au.$$

If A were a number, the solution would be $u(t) = e^{tA}u(0)$. In a Banach or Hilbert space, A may be an unbounded operator. If A is a generator of a strongly continuous semigroup, the solution (in the appropriate mild or strong sense) is described by a semigroup

$$T(t) = e^{tA}, \qquad T(t + s) = T(t)T(s).$$

The heat equation is the model example: the Laplacian generates a smoothing semigroup. The elementary exponential solution of constant-coefficient ODEs is the finite-dimensional shadow of this theory.

A second-order operator of the form

$$L[y] = -(p(x)y')' + q(x)y$$

with suitable boundary conditions is often self-adjoint. Its eigenvalue problem

$$L[y] = \lambda w(x) y$$

produces orthogonal eigenfunctions with respect to the weighted inner product

$$\langle u, v \rangle_w = \int_a^b u(x) v(x) w(x)\, dx,$$

provided $p > 0$, $w > 0$, and self-adjoint separated boundary conditions hold. Integration by parts then gives

$$(\lambda_m - \lambda_n) \int_a^b w u_m u_n\, dx = \left[p(u_m u_n' - u_m' u_n) \right]_a^b.$$

The boundary term vanishes under self-adjoint boundary conditions, forcing orthogonality for distinct eigenvalues. Fourier sine and cosine series are special cases of this principle.

Thus the elementary method of solving $y'' + \lambda y = 0$ under boundary conditions is not an isolated trick. It is the beginning of spectral theory for differential operators.

Characteristics, Caustics, and the Selection of Shocks

N-20260619

A first-order PDE often becomes an ordinary differential equation when followed along the right curves. For linear transport these curves simply carry an initial profile; for nonlinear conservation laws their speeds depend on the transported value itself. When faster characteristics catch slower ones, the classical solution ceases to exist. The weak conservation law still makes sense, but it admits too many solutions until an entropy condition selects the physically relevant one.[1]

§118.1 The chain rule chooses the path

Consider

$$u_t + a(x,t)u_x = b(x,t), \qquad u(x,0) = f(x),$$

where the coefficients are sufficiently smooth to give unique characteristic curves. If $x(t)$ solves

$$x'(t) = a(x(t),t),$$

then the chain rule gives

$$\frac{d}{dt}u(x(t),t) = u_t + a(x(t),t)u_x = b(x(t),t).$$

The PDE is therefore equivalent, along each characteristic, to a pair of ODEs:

$$x'(t) = a(x(t),t), \qquad \frac{d}{dt}u(x(t),t) = b(x(t),t).$$

For constant $a = c$ and no forcing, $x(t) = \xi + ct$ and

$$\boxed{u(x,t) = f(x-ct).}$$

This is a translating profile: neither its height nor its shape changes with time.

As a small example, take $u_t + cu_x = x + t$. Along the characteristic through (x,t), the earlier position is $x - c(t-s)$. Thus

$$\begin{aligned} u(x,t) &= f(x-ct) + \int_0^t (x - c(t-s) + s)\, ds \\ &= \boxed{f(x-ct) + xt + \frac{1-c}{2}t^2}. \end{aligned}$$

The force accumulates along a moving path, rather than at a fixed spatial point.

[1] G. Strang, MIT 18.086 (*Characteristics, Shocks, and Fans*); MIT 18.306, lectures on shock and entropy solutions.

§118.2 Characteristic data require transversality

Suppose initial values are prescribed on a parametrized curve $s \mapsto (x_0(s), t_0(s))$. The directional derivative specified by the PDE is along $(a, 1)$ in the (x, t)-plane. To recover an entire neighborhood from the initial curve, that curve must cross the characteristics:

$$\det \begin{pmatrix} x_0'(s) & a(x_0(s), t_0(s)) \\ t_0'(s) & 1 \end{pmatrix} = x_0'(s) - a(x_0(s), t_0(s)) t_0'(s) \neq 0.$$

This is the noncharacteristic condition. It ensures the characteristic flow gives local coordinates built from the initial curve parameter and time.

For $u_t + u_x = 0$, prescribing data on $x - t = 0$ is not enough to determine the solution nearby: that line is itself a characteristic. Any compatible data must be constant along it, and the values on neighboring lines remain unspecified. The failure comes from geometry, not from an algebraic defect in the formula.

§118.3 Nonlinear characteristics can collide

The inviscid Burgers equation is

$$u_t + u u_x = 0, \qquad u(x, 0) = u_0(x).$$

Along a characteristic, the value u stays constant. If the characteristic begins at ξ, its speed is $u_0(\xi)$, so

$$\boxed{x = \xi + t\, u_0(\xi), \qquad u(x, t) = u_0(\xi).}$$

As long as the map $\xi \mapsto x$ remains invertible, this implicitly defines a classical solution.

Differentiate the characteristic relation with respect to ξ:

$$\frac{\partial x}{\partial \xi} = 1 + t\, u_0'(\xi).$$

The first crossing occurs when this Jacobian vanishes. Differentiating the solution itself gives

$$\boxed{u_x(x, t) = \frac{u_0'(\xi)}{1 + t\, u_0'(\xi)}.}$$

A negative initial slope therefore steepens: the denominator shrinks toward zero and the spatial derivative blows up while the solution values remain bounded.

Take the smooth initial profile $u_0(\xi) = -\tanh \xi$. Since

$$u_0'(\xi) = -\operatorname{sech}^2 \xi,$$

the first characteristic crossing occurs at

$$t_* = -\frac{1}{\min_\xi u_0'(\xi)} = 1.$$

Near $\xi = 0$, the characteristic map is

$$x = \xi - t \tanh \xi = (1 - t)\xi + \frac{t}{3}\xi^3 + O(\xi^5).$$

At $t = 1$ the linear term disappears, making the projection of the characteristic family singular. This is a caustic: the map from initial labels to physical positions loses invertibility. For $t > 1$, the formal characteristic solution becomes multivalued near $x = 0$.

§118.4 Conservation determines the speed of a jump

Burgers' equation is also the conservation law

$$\boxed{u_t + \partial_x\left(\frac{u^2}{2}\right) = 0.}$$

A discontinuity is not differentiable in the classical sense, but the integral conservation law survives.

For a general flux F, suppose u has a jump at $x = s(t)$, with left and right states u_L, u_R. Integrate the conservation law over a narrow interval straddling the moving jump. As the interval shrinks, the balance is

$$s'(t)(u_L - u_R) = F(u_L) - F(u_R).$$

Thus the Rankine–Hugoniot jump speed is

$$\boxed{s'(t) = \frac{F(u_L) - F(u_R)}{u_L - u_R}} \qquad (u_L \neq u_R).$$

For Burgers' flux $F(u) = u^2/2$, this becomes

$$\boxed{s' = \frac{u_L + u_R}{2}.}$$

Conservation fixes a discontinuity's speed, but it does not by itself say whether a proposed discontinuity is physically admissible.

§118.5 Shocks versus Rarefactions

Consider the step initial data

$$u(x,0) = \begin{cases} u_L, & x < 0, \\ u_R, & x > 0. \end{cases}$$

When $u_L > u_R$, left characteristics travel faster than right ones and converge. The entropy solution has a shock moving at $s = (u_L + u_R)/2$:

$$u(x,t) = \begin{cases} u_L, & x < st, \\ u_R, & x > st. \end{cases}$$

For the concrete data $u_L = 1$, $u_R = 0$, the shock speed is $1/2$. The characteristics move into it:

$$1 = F'(u_L) > \frac{1}{2} > F'(u_R) = 0.$$

This inequality is the Lax compressivity condition for a convex flux.

When $u_L < u_R$, characteristics separate, leaving a fan. For $u_L = 0, u_R = 1$, the entropy solution is

$$\boxed{u(x,t) = \begin{cases} 0, & x \le 0, \\ x/t, & 0 < x < t, \\ 1, & x \ge t, \end{cases} \qquad t > 0.}$$

Inside the fan, $u = x/t$ satisfies $u_t + uu_x = 0$ because

$$u_t = -\frac{x}{t^2}, \qquad uu_x = \frac{x}{t} \cdot \frac{1}{t} = \frac{x}{t^2}.$$

The profile is continuous across both fan edges and has the appropriate limiting initial data.

For the same increasing step, a discontinuity of speed $1/2$ also satisfies Rankine–Hugoniot, yet is not the entropy solution. The missing selection criterion is therefore substantive, not merely a preferred way of drawing characteristics.

§118.6 Entropy and Time Direction

For smooth Burgers solutions, every convex twice-differentiable function $\eta(u)$ satisfies another conservation law:

$$\partial_t \eta(u) + \partial_x q(u) = 0, \qquad q'(u) = \eta'(u)F'(u) = u\eta'(u).$$

For discontinuous solutions, the entropy condition replaces equality by

$$\boxed{\partial_t \eta(u) + \partial_x q(u) \le 0}$$

in the distributional sense, for every convex entropy pair.

Take $\eta(u) = u^2/2$, so $q(u) = u^3/3$. Across a Burgers shock of speed $s = (u_L + u_R)/2$, the coefficient of the delta distribution in $\eta_t + q_x$ is

$$[q] - s[\eta] = \frac{u_R^3 - u_L^3}{3} - \frac{u_L + u_R}{2}\frac{u_R^2 - u_L^2}{2} = \frac{(u_R - u_L)^3}{12}.$$

It is nonpositive exactly when $u_L \ge u_R$. The decreasing step creates an entropy-admissible shock; the increasing step would give positive entropy production in this convention and is replaced by a rarefaction fan.

This calculation makes an important distinction. Characteristics explain how smooth information travels, and their crossing predicts failure of a classical solution. Conservation fixes the possible jump speed, while entropy rules out the unphysical weak solutions. The full theory needs all three levels because each answers a different mathematical question.

The Wave Equation: Travelling Signals, Energy, and Reflection

N-20260620

The one-dimensional wave equation can be understood almost entirely through two directions of transport. Its characteristic variables move at speeds c and $-c$, so initial disturbances influence only a finite region of space. The same geometry explains d'Alembert's formula, the domain of dependence, reflection at a fixed boundary and the conversion between standing and travelling waves. Energy conservation makes this propagation picture a uniqueness argument rather than merely a method of guessing solutions.[1]

§119.1 Factoring the Wave Operator

For a vibrating string, $u(x,t)$ denotes its transverse displacement. In the linear small-amplitude model,

$$\boxed{u_{tt} = c^2 u_{xx}, \qquad c > 0.}$$

The spatial curvature determines acceleration rather than the rate of change of displacement. Because the constant-coefficient operators commute,

$$\partial_t^2 - c^2\partial_x^2 = (\partial_t - c\partial_x)(\partial_t + c\partial_x).$$

Thus the wave equation is the composition of the two opposite transport operators.

Introduce characteristic coordinates

$$\xi = x + ct, \qquad \eta = x - ct.$$

Then

$$\partial_t = c(\partial_\xi - \partial_\eta), \qquad \partial_x = \partial_\xi + \partial_\eta,$$

and the wave equation becomes $u_{\xi\eta} = 0$. Integrating twice shows that every sufficiently smooth solution on a characteristic rectangle can be written as

$$\boxed{u(x,t) = F(x-ct) + G(x+ct).}$$

The F-component travels right, while the G-component travels left. The formula is an exact consequence of factorization, not a superposition guess.

§119.2 Displacement and Initial Velocity

On the whole real line, prescribe

$$u(x,0) = f(x), \qquad u_t(x,0) = g(x).$$

At $t = 0$, the travelling-wave representation gives

$$F(x) + G(x) = f(x), \qquad -cF'(x) + cG'(x) = g(x).$$

[1] MIT 18.152 (PDE lecture notes); MIT 2.062J, *Wave Propagation*, Chapter 2.

Differentiate the first equation and solve for

$$F' = \frac{1}{2}f' - \frac{g}{2c}, \qquad G' = \frac{1}{2}f' + \frac{g}{2c}.$$

Substitute these into the travelling-wave formula to obtain d'Alembert's solution:

$$\boxed{u(x,t) = \frac{f(x-ct) + f(x+ct)}{2} + \frac{1}{2c}\int_{x-ct}^{x+ct} g(s)\,ds.}$$

The first term splits the initial displacement into two travelling profiles; the second accumulates the initial velocity over an expanding interval.

If $f(x) = e^{-x^2}$ and $g = 0$, the initial Gaussian pulse splits into two Gaussian pulses:

$$u(x,t) = \tfrac{1}{2}e^{-(x-ct)^2} + \tfrac{1}{2}e^{-(x+ct)^2}.$$

Unlike heat diffusion, the widths of the travelling components do not grow in this ideal nondispersive model.

§119.3 Finite Propagation Speed

The value $u(x,t)$ in d'Alembert's formula depends only on f at $x \pm ct$ and on g inside

$$[x - ct, x + ct].$$

If the initial data vanish on this interval, then $u(x,t) = 0$. In particular, initial data supported in $[-R, R]$ produce a solution supported within

$$[-R - ct, R + ct]$$

at time $t \geq 0$.

The interval $[x - ct, x + ct]$ is the backward domain of dependence of the event (x,t). In the space-time plane its boundary consists of characteristic lines of slopes $\pm c$. The sharp signal front is a consequence of the hyperbolic operator's factorization; the heat kernel, by contrast, is positive at every point for any positive time.

§119.4 Uniqueness by Energy Conservation

On an interval $[a,b]$, define the local wave energy density and flux

$$e = \frac{1}{2}(u_t^2 + c^2u_x^2), \qquad j = -c^2u_tu_x.$$

Differentiate:

$$\begin{aligned} e_t + j_x &= u_tu_{tt} + c^2u_xu_{xt} - c^2(u_{xt}u_x + u_tu_{xx}) \\ &= u_t(u_{tt} - c^2u_{xx}) = 0. \end{aligned}$$

Therefore

$$\boxed{e_t + j_x = 0,}$$

an exact local conservation law. Integrating gives

$$\frac{d}{dt}\int_a^b e(x,t)\,dx = j(a,t) - j(b,t).$$

Energy changes within a fixed segment only through its boundaries.

For a finite string $0 < x < L$ with fixed endpoints $u(0,t) = u(L,t) = 0$, differentiating the boundary condition gives $u_t = 0$ there, so the boundary flux vanishes. Hence

$$\boxed{E(t) = \frac{1}{2}\int_0^L (u_t^2 + c^2 u_x^2)\,dx = E(0).}$$

If two solutions have the same initial and boundary data, their difference has zero initial energy. Since its energy stays zero, both its derivatives vanish, and the zero boundary value forces the difference itself to vanish. Thus the solution is unique.

This energy proof works without explicitly constructing the travelling waves. It also suggests how to handle variable coefficients and higher-dimensional problems where d'Alembert's elementary formula is unavailable.

§119.5 Forcing contributes energy at a calculable rate

Consider the forced equation

$$u_{tt} - c^2 u_{xx} = h(x,t)$$

on the real line with initial data f, g. Duhamel's principle gives

$$\boxed{\begin{aligned} u(x,t) = \frac{f(x-ct) + f(x+ct)}{2} + \frac{1}{2c}\int_{x-ct}^{x+ct} g(s)\,ds \\ + \frac{1}{2c}\int_0^t \int_{x-c(t-\tau)}^{x+c(t-\tau)} h(y,\tau)\,dy\,d\tau. \end{aligned}}$$

The forcing at time τ affects the event (x,t) only if its spatial position y lies within the backward characteristic cone. This makes finite propagation explicit for the forcing as well as for the initial data.

To check the formula, set $f = g = 0$ and $h \equiv 1$. The inner integral is $2c(t-\tau)$, so

$$u(x,t) = \int_0^t (t-\tau)\,d\tau = \frac{t^2}{2}.$$

Indeed $u_{tt} = 1$ and $u_{xx} = 0$. In the energy identity, forcing adds the source term

$$e_t + j_x = h\,u_t,$$

which is the local rate of work done by the external force.

§119.6 Wave Reflection at a Fixed End

On the half-line $x > 0$ with fixed endpoint $u(0,t) = 0$, extend the initial displacement and velocity to the whole line as odd functions:

$$\widetilde{f}(-x) = -\widetilde{f}(x), \qquad \widetilde{g}(-x) = -\widetilde{g}(x).$$

The whole-line d'Alembert solution from these odd data remains odd in x, hence vanishes at $x = 0$. Restricting it to $x > 0$ solves the fixed-end problem.

This is the method of images. A right-moving incoming pulse is reflected into a left-moving pulse with reversed sign so that the endpoint displacement stays zero. For a free endpoint with Neumann condition $u_x(0,t) = 0$, the analogous construction uses even extensions, giving reflection without sign reversal.

The boundary condition therefore selects which symmetry extension makes the whole-line solution compatible with the physical endpoint.

§119.7 Fourier modes are reflected travelling waves

On $0 < x < L$, the fixed endpoint conditions select the sine eigenfunctions

$$X_n(x) = \sin \frac{n\pi x}{L}, \qquad \lambda_n = \left(\frac{n\pi}{L}\right)^2.$$

For sufficiently regular data satisfying the compatibility conditions, the solution is

$$\boxed{u(x,t) = \sum_{n\geq 1} \left(a_n \cos \frac{n\pi ct}{L} + b_n \sin \frac{n\pi ct}{L}\right) \sin \frac{n\pi x}{L},}$$

with coefficients

$$a_n = \frac{2}{L}\int_0^L f(x) \sin \frac{n\pi x}{L}\, dx, \qquad b_n = \frac{2}{n\pi c}\int_0^L g(x) \sin \frac{n\pi x}{L}\, dx.$$

Each mode is an independent harmonic oscillator in time. Its fixed spatial nodes are caused by interference between oppositely travelling components:

$$\sin(kx)\cos(kct) = \tfrac{1}{2}\big(\sin(k(x+ct)) + \sin(k(x-ct))\big).$$

A standing wave is therefore not a third kind of propagation. It is an exact superposition of two travelling waves of the same frequency, constrained by reflections at the endpoints.

§119.8 Damping shows exactly what energy conservation uses

If the string experiences linear damping,

$$u_{tt} + \gamma u_t = c^2 u_{xx}, \qquad \gamma > 0,$$

the same multiplication by u_t gives, under fixed endpoints,

$$\boxed{\frac{dE}{dt} = -\gamma \int_0^L u_t^2\, dx \leq 0.}$$

The boundary still contributes no energy, but damping removes kinetic energy in the interior.

The contrast with the heat equation is now precise. Both equations admit Fourier decompositions, yet a heat mode has multiplier $e^{-\kappa\lambda_n t}$, while a wave mode oscillates at frequency $c\sqrt{\lambda_n}$. The two transport directions of the wave equation combine into reversible oscillatory motion; heat flow defines a dissipative semigroup with no time-reversed smoothing operation. The geometry of characteristics and the energy identity both reveal this distinction.

Heat Flow: Gaussian Scaling, Spectral Decay, and Entropy

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Why does a localized temperature profile become Gaussian, and why does the heat equation smooth irregular data so quickly? Both questions are answered by the same evolution operator. Conservation fixes the mass of the heat kernel, parabolic scaling fixes its characteristic width, and the Fourier transform explains its damping of high frequencies. When the evolving profile is regarded as a probability density, this same process has an exact entropy-dissipation law.[1]

§120.1 Conservation and Fick's law

For a temperature or concentration $u(x,t)$ on a homogeneous one-dimensional medium, write the flux as $J = -\kappa u_x$, with $\kappa > 0$. Conservation on an interval $[a,b]$ says

$$\frac{d}{dt}\int_a^b u(x,t)\,dx = J(a,t) - J(b,t).$$

Since the interval is arbitrary, the local equation is

$$\boxed{u_t + \partial_x J = 0, \qquad u_t = \kappa u_{xx}.}$$

Thus diffusion is conservation driven by a gradient-dependent constitutive law. At a local maximum $u_{xx} \le 0$, so the temperature there cannot increase through the diffusion term.

Integrating over the whole line, when the flux vanishes at infinity, gives

$$\frac{d}{dt}\int_{\mathbb{R}} u(x,t)\,dx = 0.$$

The profile can spread, but its total mass is fixed. A natural fundamental solution is therefore a family of nonnegative functions of mass one that approaches a point mass as $t \downarrow 0$.

§120.2 The Gaussian follows from parabolic scaling

The equation is invariant under $x \mapsto \lambda x$, $t \mapsto \lambda^2 t$. A mass-one similarity solution should have the form

$$G(x,t) = \frac{1}{2\sqrt{\kappa t}}\Phi(\eta), \qquad \eta = \frac{x}{2\sqrt{\kappa t}}.$$

Substitution into $G_t = \kappa G_{xx}$ yields

$$\Phi'' + 2\eta\Phi' + 2\Phi = 0 \quad\Longleftrightarrow\quad (\Phi' + 2\eta\Phi)' = 0.$$

For a nonnegative integrable profile decaying at both ends, the integration constant must be zero. Hence

$$\Phi' = -2\eta\Phi, \qquad \Phi(\eta) = Ce^{-\eta^2}.$$

[1] MIT 18.152, Lectures 2–5; K. Tang, *Topics in Heat Equations*, §9 (entropy and Fisher information).

The mass normalization $\int_{\mathbb{R}} G\,dx = \int_{\mathbb{R}} \Phi(\eta)\,d\eta = 1$ fixes $C = 1/\sqrt{\pi}$:

$$\boxed{G(x,t) = \frac{1}{\sqrt{4\pi\kappa t}} \exp\left(-\frac{x^2}{4\kappa t}\right).}$$

Its variance is $2\kappa t$, so the diffusion length scale is $\sqrt{2\kappa t}$. For sufficiently regular and integrable initial data f, the solution on the line is

$$u(x,t) = \int_{\mathbb{R}} G(x-y,t) f(y)\,dy.$$

§120.3 The same Gaussian is a probability transition kernel

For $s, t > 0$, Gaussian convolution obeys

$$G_t * G_s = G_{t+s}.$$

The convolution of two centered normal distributions with variances $2\kappa t$ and $2\kappa s$ has variance $2\kappa(t+s)$; this proves the identity and also explains it probabilistically.

Let B_t be standard Brownian motion, with $B_t \sim N(0,t)$. Then for bounded continuous f,

$$(S_t f)(x) := (G_t * f)(x) = \mathbb{E}\big[f(x + \sqrt{2\kappa} B_t)\big].$$

Brownian motion has independent increments, so

$$S_t S_s f = S_{t+s} f.$$

The operator family S_t is a Markov semigroup, and its infinitesimal generator is $\kappa\partial_x^2$. Indeed, a Taylor expansion under the Gaussian expectation, for smooth f with bounded derivatives, gives

$$\lim_{t \downarrow 0} \frac{S_t f - f}{t} = \kappa f''.$$

Heat diffusion and Brownian motion thus share one Gaussian transition rule: a deterministic PDE describes how the probability distribution of random motion evolves.

§120.4 Fourier modes make smoothing quantitative

Write $\widehat{u}(\xi,t) = \int_{\mathbb{R}} u(x,t) e^{-i\xi x}\,dx$. Fourier transformation of $u_t = \kappa u_{xx}$ gives an ordinary differential equation at each frequency:

$$\partial_t \widehat{u} = -\kappa\xi^2 \widehat{u}, \qquad \boxed{\widehat{u}(\xi,t) = e^{-\kappa\xi^2 t}\widehat{f}(\xi).}$$

The multiplier rapidly suppresses large $|\xi|$. For example, a Fourier mode of frequency 10 decays a hundred times faster in the exponent than one of frequency 1. This is the spectral explanation for instantaneous smoothness at positive times.

The semigroup relation becomes multiplication of multipliers:

$$e^{-\kappa\xi^2 t} e^{-\kappa\xi^2 s} = e^{-\kappa\xi^2 (t+s)}.$$

The Gaussian kernel, Brownian increments and Fourier damping are three exact realizations of the same operation.

§120.5 Boundary Heating and Error Functions

Consider the half-line $x > 0$, initially at temperature zero, with a boundary maintained at unit temperature:

$$u(x,0) = 0 \ (x > 0), \quad u(0,t) = 1 \ (t > 0), \quad \lim_{x\to\infty} u(x,t) = 0.$$

The parabolic scaling suggests $u(x,t) = \theta(\eta)$, where $\eta = x/(2\sqrt{\kappa t})$. Substitution yields

$$\theta'' + 2\eta\theta' = 0.$$

Therefore $\theta'(\eta) = Ce^{-\eta^2}$. The conditions $\theta(0) = 1$, $\theta(\infty) = 0$ give

$$\boxed{u(x,t) = \operatorname{erfc}\left(\frac{x}{2\sqrt{\kappa t}}\right), \qquad \operatorname{erfc}(z) = \frac{2}{\sqrt{\pi}}\int_z^\infty e^{-r^2}\,dr.}$$

The same Gaussian integral that normalized the whole-line kernel now describes the moving transition layer next to a boundary. The profile is exact for $x, t > 0$; its incompatible values at the corner $(0,0)$ reflect the instantaneous change of boundary temperature.

§120.6 Finite intervals have a spectral gap

For $0 < x < L$, impose the fixed-temperature conditions $u(0,t) = u(L,t) = 0$. Separation of variables and orthogonality of sines give

$$u(x,t) = \sum_{n\ge1} b_n e^{-\kappa(n\pi/L)^2 t}\sin\frac{n\pi x}{L}, \qquad b_n = \frac{2}{L}\int_0^L f(x)\sin\frac{n\pi x}{L}\,dx.$$

The lowest spatial eigenvalue is $(\pi/L)^2$. Hence Parseval's identity implies

$$\boxed{\|u(\cdot,t)\|_{L^2(0,L)} \le e^{-\kappa\pi^2 t/L^2}\|f\|_{L^2(0,L)}.}$$

This decay rate can also be obtained without solving explicitly. Multiplying the PDE by u, integrating by parts, and applying the sharp Dirichlet Poincare inequality yields

$$\frac{d}{dt}\frac{1}{2}\int_0^L u^2\,dx = -\kappa\int_0^L u_x^2\,dx \le -\kappa\frac{\pi^2}{L^2}\int_0^L u^2\,dx.$$

Gronwall's inequality reproduces the same exponential estimate. The PDE and its eigenfunction expansion express the same *spectral gap*.

For fixed nonzero endpoint temperatures A, B, first subtract the steady profile

$$h(x) = A + \frac{B-A}{L}x.$$

Then $v = u - h$ obeys homogeneous endpoint conditions and decays toward zero. For insulated endpoints $u_x(0,t) = u_x(L,t) = 0$, the total mass is conserved and the zero Fourier mode survives. The solution approaches its spatial mean rather than zero.

§120.7 Diffusion increases entropy and dissipates information

Now assume $u(x,t) > 0$ is a smooth probability density on $\mathbb{R}$, with sufficient decay to integrate by parts. Its differential Shannon entropy is

$$H(u_t) = -\int_{\mathbb{R}} u(x,t) \log u(x,t)\, dx.$$

Using mass conservation, the heat equation, and integration by parts,

$$\begin{aligned}
\frac{d}{dt} H(u_t) &= -\int_{\mathbb{R}} (\partial_t u)(1 + \log u)\, dx \\
&= -\kappa \int_{\mathbb{R}} u_{xx} \log u\, dx \\
&= \kappa \int_{\mathbb{R}} \frac{u_x^2}{u}\, dx \\
&= \kappa \int_{\mathbb{R}} u\, |\partial_x \log u|^2\, dx.
\end{aligned}$$

The last integral is the *Fisher information* $I(u_t)$, so

$$\boxed{\frac{d}{dt} H(u_t) = \kappa I(u_t) \geq 0.}$$

Entropy increases because the density becomes more spread out; the rate is precisely its Fisher information. This identity is a concrete bridge between heat flow and information geometry.

For the Gaussian kernel $G_t = N(0, 2\kappa t)$,

$$H(G_t) = \frac{1}{2} \log(4\pi e \kappa t), \qquad I(G_t) = \frac{1}{2\kappa t}.$$

Direct differentiation gives $H'(G_t) = 1/(2t) = \kappa I(G_t)$, confirming the general identity on the fundamental solution.

The wave equation $w_{tt} = c^2 w_{xx}$ behaves differently. Its Fourier modes oscillate instead of acquiring the damping factor $e^{-\kappa \xi^2 t}$; appropriate wave energy is conserved instead of dissipated. This difference between first- and second-order time evolution explains why heat flow smooths and loses detail while wave motion preserves oscillatory information.

X

Classical and Projective Geometry

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Geometry Under Constraints: Projections, Packing, and Symmetric Coordinates

N-20220615

A geometric extremum depends as much on the feasible set as on the formula being optimized. Squared distances invite orthogonal projection; circle packing requires distance constraints between centers; symmetric rational equations become simpler after removing an irrelevant common scale. Each situation suggests a different set of coordinates.

§121.1 A distance problem is a projection problem

Let $A = (0, -2)$, $B = (2, -2)$, and let $M = (x, y)$ lie on $x + y = 1$. We want to minimize

$$|M - A|^2 + |M - B|^2.$$

Put $C = (A + B)/2 = (1, -2)$. Expanding inner products gives the exact identity

$$\boxed{|M - A|^2 + |M - B|^2 = 2|M - C|^2 + \frac{1}{2}|A - B|^2.}$$

It holds in any inner-product space, not just in the plane. The distance $|A - B| = 2$ is fixed, so minimizing the sum is equivalent to projecting C orthogonally onto the constraint line.

The line has normal vector $n = (1, 1)$. Its signed equation is $n \cdot M = 1$, while $n \cdot C = -1$. Therefore the projection is

$$M_* = C + \frac{1 - n \cdot C}{\|n\|^2} n = (1, -2) + (1, 1) = (2, -1).$$

The squared distance from C to the line equals 2. Hence

$$\min_{x+y=1} (|M - A|^2 + |M - B|^2) = 2 \cdot 2 + 2 = 6.$$

There is no maximum because the constraint line is unbounded and the squared distance grows quadratically along it.

This same identity underlies least squares: minimizing sums of squared distances amounts to projecting onto an affine subspace, with the residual orthogonal to all allowed directions.

§121.2 The Parabola's Farthest Point

Take a downward-opening parabola

$$y = ax^2 + bx, \qquad a < 0,\ b > 0,$$

restricted to the interval between its two x-intercepts, $0 \le x \le -b/a$. The squared distance to the origin is

$$F(x) = x^2 + (ax^2 + bx)^2, \qquad F'(x) = 2x + 2(ax^2 + bx)(2ax + b).$$

Although the height of the parabola is maximal at $x = -b/(2a)$, the distance to the origin also depends on x^2. The vertex therefore need not maximize F.

For the concrete parabola $y = 2x - x^2$ on $[0, 2]$,

$$F(x) = x^2 + x^2(2 - x)^2, \qquad F'(x) = 2x(2x^2 - 6x + 5).$$

The quadratic factor has discriminant $36 - 40 = -4$ and positive leading coefficient, so $F'(x) > 0$ for $0 < x \le 2$. Thus the farthest point is $(2, 0)$, at squared distance 4, while the vertex $(1, 1)$ has squared distance only 2.

An extremum of one coordinate is not generally an extremum of a nonlinear function of all coordinates. The correct objective must be differentiated on the actual feasible domain.

§121.3 Packing circles becomes a configuration-space problem

Suppose n closed disks of radius $1/2$ must fit inside a container $C \subseteq \mathbb{R}^2$. Instead of manipulating the disks directly, represent each by its center c_i. The conditions are

$$\|c_i - c_j\| \ge 1 \quad (i \ne j), \qquad c_i \in C \ominus B_{1/2},$$

where $C \ominus B_{1/2}$ is the set of points whose radius-1/2 disk is wholly contained in C. This operation is called morphological erosion or Minkowski subtraction.

For two unit-diameter disks inside a round container of radius R, both centers must lie within radius $R - \frac{1}{2}$ of its center. Their greatest possible separation is $2R - 1$, which must be at least 1. Thus $R \ge 1$, and placing the disk centers at opposite ends of a diameter attains equality. The minimal container diameter is exactly 2.

For two disks inside a square of side s, each center must lie in the smaller square of side $s - 1$. The largest possible distance between two such centers is $\sqrt{2}(s - 1)$. Hence

$$\sqrt{2}(s - 1) \ge 1 \quad \Longleftrightarrow \quad \boxed{s \ge 1 + \frac{1}{\sqrt{2}}.}$$

Opposite corners of the allowed center square achieve the bound.

Already for $n = 2$, the exact results are much stronger than the area comparison $n\pi/4 \le \text{area}(C)$. For larger n, the feasible configurations form a constrained subset of C^n from which all pairwise collision regions have been removed. This configuration-space geometry, rather than area alone, explains the difficulty of optimal packings.

§121.4 Three equal disks force an equilateral configuration

The two-disk examples reveal the constraints, but a third disk makes the geometry more interesting. What is the smallest circular container that can hold three disks of radius $1/2$ without overlap? A naive area estimate would only give

$$\pi R^2 \ge 3\pi(1/2)^2, \qquad R \ge \frac{\sqrt{3}}{2},$$

which is far from sharp because it ignores how centers are separated.

Write c_1, c_2, c_3 for the disk centers, measured from the container center, and let

$$r = R - \frac{1}{2}.$$

Containment implies $\|c_i\| \le r$, while nonoverlap implies $\|c_i - c_j\| \ge 1$ for each pair $i \ne j$. The key is a simple identity:

$$\boxed{\sum_{i<j} \|c_i - c_j\|^2 = 3\sum_{i=1}^{3} \|c_i\|^2 - \left\|\sum_{i=1}^{3} c_i\right\|^2.}$$

Expand the squares to verify it; each $\|c_i\|^2$ occurs twice on the left, and each pairwise dot product occurs with coefficient -2. The right-hand side rearranges exactly those terms.

The three separation constraints give a lower bound of 3 for the left side. For the right side,

$$3 \le \sum_{i<j} \|c_i - c_j\|^2 \le 3\sum_{i} \|c_i\|^2 \le 9r^2.$$

Therefore $r \ge 1/\sqrt{3}$, or

$$\boxed{R \ge \frac{1}{2} + \frac{1}{\sqrt{3}}.}$$

This lower bound is achieved. Place the three centers at the vertices of an equilateral triangle of side 1, centered at the origin. Each center lies at distance $1/\sqrt{3}$ from the origin, so the outer container of radius $1/2+1/\sqrt{3}$ touches all three disks while the disks touch one another.

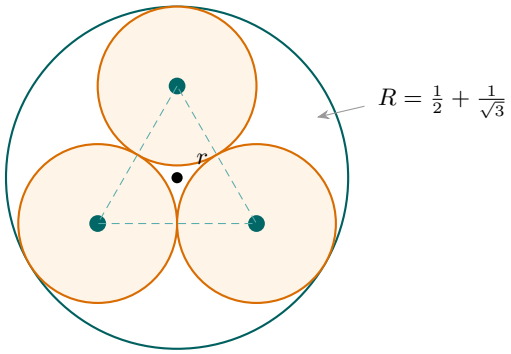


Figure 121.1: The sharp three-disk packing. The dashed equilateral triangle has side length 1 and circumradius $1/\sqrt{3}$; the disks have radius $1/2$.

The inequality is stronger than an area bound because it keeps a *pairwise distance invariant*. Equality in the algebra forces all centers onto the allowed circle, their vector sum to vanish, and all pairwise separations to equal one. Thus the symmetric arrangement is not merely a plausible picture; the calculation explains the rigidity of the optimizer.

§121.5 Symmetric rational equations live on projective ratios

Consider positive a, b, c satisfying

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} = 4.$$

If (a, b, c) solves the equation, so does (ta, tb, tc) for every $t > 0$. Hence the equation constrains only the ratios among a, b, c. It is natural to introduce the elementary symmetric quantities

$$s = a + b + c, \qquad p = ab + bc + ca, \qquad q = abc.$$

Because $\frac{a}{b+c} = \frac{s}{s-a} - 1$, the original equation becomes

$$s\left(\frac{1}{s-a} + \frac{1}{s-b} + \frac{1}{s-c}\right) = 7.$$

The pairwise products of the denominators sum to $s^2 + p$, while their product equals $sp - q$. Clearing the strictly positive denominator yields the homogeneous cubic

$$\boxed{s^3 - 6sp + 7q = 0.}$$

There is an actual family of positive solutions, not just a formal projective equation. Fix the overall scale by taking $b = c = 1$ and write $a = t > 0$. The original condition becomes

$$\frac{t}{2} + \frac{2}{t+1} = 4.$$

Clearing the positive denominator gives

$$t(t+1) + 4 = 8(t+1), \qquad t^2 - 7t - 4 = 0.$$

The unique positive solution on this symmetric ray is

$$\boxed{t = \frac{7 + \sqrt{65}}{2}.}$$

The other quadratic root is negative. Thus all positive triples proportional to

$$\left(\frac{7 + \sqrt{65}}{2},\ 1,\ 1\right)$$

satisfy the rational equation. The cubic derived in s, p, q gives the same result: when $s = t + 2,\ p = 2t + 1,\ q = t$, its left side factors as

$$s^3 - 6sp + 7q = (t+1)(t^2 - 7t - 4).$$

The factor $t+1$ is never zero in the positive region. This explicit section of the projective solution set illustrates why removing scale helps: one homogeneous equation in three variables becomes a scalar quadratic after intersecting with a geometrically meaningful symmetry ray.

This equation defines a projective cubic locus in the ratios $[a : b : c]$, restricted to the positive region. An elementary inequality still gives a useful geometric boundary: by Nesbitt's inequality,

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2},$$

with equality at $a = b = c$. The level 4 therefore lies well away from the balanced point.

The projection calculation, the disk-center constraints, and the homogeneous cubic all remove irrelevant coordinates. For squared distances we eliminate translation using a midpoint; for packing we replace congruent disks by their centers; for a symmetric rational equation we quotient out overall scale. In each case the improved coordinates reveal the geometry of the feasible set before optimization begins.

A Triangle Through a Looking Glass: Reflections, Circles, and Miquel's Point

N-20220708

One of the pleasures of triangle geometry is that a construction which looks accidental from one viewpoint becomes unavoidable from another. Reflecting the orthocenter over a side lands it on the circumcircle; an angle bisector hides an excenter inside a second circle; and three apparently unrelated circles can be forced through one point. The same two moves keep appearing: change the picture without changing an angle, then recognize four points on a circle.[1]

§122.1 A reflection that knows the circumcircle

Let H be the orthocenter of a nondegenerate triangle ABC. Reflect H across the line BC to obtain X. Why should X lie on the circumcircle? Two perpendicular altitudes already determine the required angle.

For any point Q off the three sidelines, write $\angle BQC$ for the directed angle between the lines QB and QC, taken modulo 180°. Reflection in BC reverses the direction of angles, and hence

$$\measuredangle BXC \equiv -\measuredangle BHC \pmod{180^\circ}.$$

But $BH \perp AC$ and $CH \perp AB$, so

$$\measuredangle BHC \equiv -\measuredangle BAC \pmod{180^\circ}.$$

It follows that

$$\boxed{\measuredangle BXC \equiv \measuredangle BAC \pmod{180^\circ}},$$

which is exactly the criterion for A, B, C, X to be concyclic. Directed angles make the argument independent of whether the triangle is acute or obtuse.

There is a second reflection with an even more surprising outcome. Let M be the midpoint of BC, and reflect H through M, obtaining Y. Then

$$\overrightarrow{BY} = \overrightarrow{HC}, \qquad \overrightarrow{CY} = \overrightarrow{HB}.$$

Since $CH \perp AB$ and $BH \perp AC$, we get

$$BY \perp BA, \qquad CY \perp CA.$$

Thus both $\angle ABY$ and $\angle ACY$ are right angles. By Thales' theorem, B and C lie on the circle whose diameter is AY. This circle is the circumcircle of ABC. In particular, Y is the point on the circumcircle opposite A across a diameter.

[1] The standard reflection and Miquel configurations are developed in R. A. Johnson, *Advanced Euclidean Geometry*, and Evan Chen, *Euclidean Geometry in Mathematical Olympiads*.

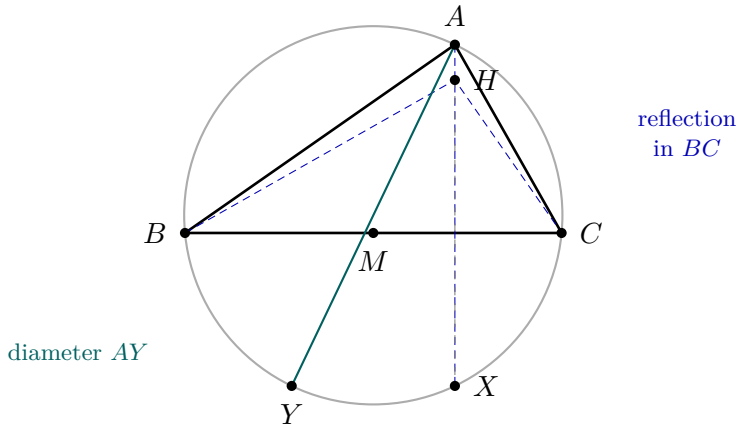


Figure 122.1: Two reflections of the same orthocenter give two distinct points on the circumcircle. The line AY is a diameter.

§122.2 The midpoint of an arc hides an excenter

Let I be the incenter, and let the internal angle bisector AI meet the circumcircle again at L. The equal inscribed angles at A show that L is the midpoint of the arc BC not containing A, so $LB = LC$.

A slightly more useful identity is

$$\boxed{LI = LB = LC.}$$

For example, in triangle BIL,

$$\angle IBL = \frac{A+B}{2} = 90^\circ - \frac{C}{2}.$$

Also, since $\angle BIC = 90^\circ + A/2$ and A, I, L are collinear with I between A and L, the angle $\angle BIL$ equals $90^\circ - C/2$. Hence $LI = LB$; symmetry gives $LI = LC$.

Now reflect I *through the point* L, not across a line, and call the image I_A. Since $L = (I + I_A)/2$, a simple vector identity gives

$$(B - I) \cdot (B - I_A) = |B - L|^2 - |I - L|^2 = 0.$$

The same holds with C in place of B. Thus $BI_A \perp BI$ and $CI_A \perp CI$. The lines BI and CI are internal bisectors, so BI_A and CI_A are their external bisectors. Their intersection I_A is precisely the A-excenter.

The right angles also show that B and C lie on the circle with diameter II_A, whose center is L. A small reflection has simultaneously produced the excenter and a new cyclic quadrilateral. This is often more efficient than introducing excentral distances from scratch.

§122.3 Directed Angles and Invariance

A circle test may be written as

$$\boxed{A, B, C, D \text{ concyclic} \iff \measuredangle ABC \equiv \measuredangle ADC \pmod{180^\circ},}$$

provided the four points are distinct. If one point crosses a side extension, an ordinary acute-angle diagram may reverse an angle or introduce a supplement. Working modulo 180° automatically handles both changes.

The algebra is particularly convenient because adjacent directed angles add:

$$\measuredangle(\ell_1, \ell_2) + \measuredangle(\ell_2, \ell_3) \equiv \measuredangle(\ell_1, \ell_3) \pmod{180^\circ}.$$

But one must not divide arbitrary congruences by two. From $2\alpha \equiv 2\beta \pmod{180^\circ}$ one can infer only $\alpha \equiv \beta \pmod{90^\circ}$. The choice of modulus carries real mathematical information.

The cyclicity criterion is a useful model of a *local test*: a single equality of directions detects a global circle through four points. Miquel's theorem follows by applying this test three times and noticing that the final equality is forced by the first two.

§122.4 Three circles that cannot help meeting

Choose points $D \in BC$, $E \in CA$, $F \in AB$, on the full lines if necessary. Consider the circumcircles

$$\omega_A = (AEF), \qquad \omega_B = (BFD), \qquad \omega_C = (CDE).$$

Assume for the moment that the points and circles are nondegenerate and that ω_A, ω_B have a second intersection P distinct from F. Miquel's theorem asserts that ω_C passes through that same P.

The proof is short enough to see the entire mechanism. Because A, E, F, P are concyclic,

$$\measuredangle EPF \equiv \measuredangle EAF.$$

Because B, F, D, P are concyclic,

$$\measuredangle FPD \equiv \measuredangle FBD.$$

Add the equations and use A, E, C collinear, A, F, B collinear and B, D, C collinear:

$$\begin{aligned} \measuredangle EPD &\equiv \measuredangle EAF + \measuredangle FBD \\ &\equiv \measuredangle(AC, AB) + \measuredangle(AB, BC) \\ &\equiv \measuredangle(AC, BC) \equiv \measuredangle ECD \pmod{180^\circ}. \end{aligned}$$

Therefore C, D, E, P are concyclic, as required. The nondegenerate case proves the theorem; coincident and tangent limiting configurations can be treated by continuity where the limiting circles remain defined.

The interesting point is that the third circle was never independently solved for. Two cyclicity constraints already leave no freedom in the last directed angle. Miquel's point is where this compatibility becomes visible.

§122.5 Inversion of the Three Circles

There is a delightful second way to look at the same configuration *after* finding P. Invert the plane about P. Every circle through P becomes a line, so the three Miquel circles become three straight lines. Their pairwise intersections are exactly the images of F, D, E.

The transformation can be verified without a picture. Put P at the origin and use complex coordinates. A circle through the origin with center $c \neq 0$ satisfies

$$|z - c|^2 = |c|^2 \quad \Longleftrightarrow \quad |z|^2 - 2\,\mathrm{Re}(\bar{c}z) = 0.$$

Under unit inversion $w = z/|z|^2$, the equation becomes

$$\boxed{2\,\mathrm{Re}(\bar{c}w) = 1,}$$

the equation of a straight line. Three circles sharing P therefore become three lines forming an ordinary triangle, with D', E', F' at its vertices.

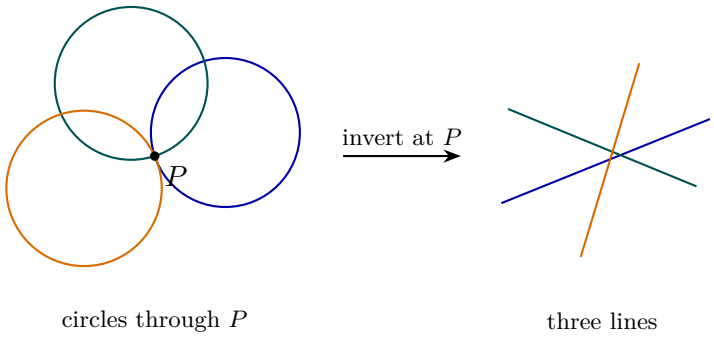


Figure 122.2: A structural view of Miquel's theorem: inversion at the common point straightens all three circles. The diagram illustrates the transformation rather than specific labelled coordinates.

Notice what inversion does *not* do: it is not a projective transformation preserving arbitrary lengths and ratios. It is a conformal transformation away from its center, carrying generalized circles to generalized circles. This is the natural geometry of Miquel configurations.

§122.6 A circle has an algebraic test, too

For distinct complex numbers z_1, z_2, z_3, z_4, define their cross ratio by

$$[z_1, z_2; z_3, z_4] = \frac{(z_3 - z_1)(z_4 - z_2)}{(z_3 - z_2)(z_4 - z_1)}.$$

The four points lie on a generalized circle (a circle or a line) precisely when this quantity is real. To see why, apply the Möbius transformation

$$T(z) = \frac{(z - z_1)(z_2 - z_4)}{(z - z_4)(z_2 - z_1)}.$$

It sends z_1, z_2, z_4 to $0, 1, \infty$. The unique generalized circle through these three images is the real line together with infinity. The fourth point lies on the original generalized circle exactly when its image is real. Up to a real fractional-linear rearrangement, that image is the cross ratio above.

This criterion explains the shared algebraic structure behind directed-angle chasing and inversion. An equality of inscribed angles makes four points concyclic; reality of a cross ratio detects the same property in complex coordinates; a Möbius map can turn the circle into a line where the statement is immediate.

For ordinary triangle problems, a two-line directed-angle proof is usually the most efficient. The higher-level interpretation becomes worthwhile when circles proliferate: instead of chasing every arc, we can choose a conformal coordinate system that turns the relevant circles into straight lines.

Conics and the Dedekind Construction of Addition

N-20220715

§123.1 Conics as metric loci and their tangent geometry

The metric definitions and coordinates of conics below follow OpenStax, *Calculus Volume 3*, Section 1.5; the real-number material later in this chapter follows MIT 18.100A.[1] Conic geometry and the construction of real numbers use coordinates in different ways: conics study metric loci, while Dedekind cuts construct the ordered scalar field in which those coordinates live. A cone cut by a plane produces the classical conics:

$$\text{circle}, \quad \text{ellipse}, \quad \text{parabola}, \quad \text{hyperbola}.$$

An ellipse can be defined as the locus of points P for which the sum of distances to two foci is constant:

$$PF_1 + PF_2 = 2a.$$

With foci $F_1 = (-c, 0)$, $F_2 = (c, 0)$, this gives the standard equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \qquad c^2 = a^2 - b^2.$$

The eccentricity is

$$e = \frac{c}{a}, \qquad 0 < e < 1.$$

A hyperbola is defined by a constant difference of distances to the two foci:

$$|PF_1 - PF_2| = 2a,$$

with standard equations

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

or

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1,$$

depending on the transverse axis.

A parabola is the locus of points equidistant from a focus and a directrix. Its standard equations are

$$y^2 = 2px, \qquad y^2 = -2px, \qquad x^2 = 2py, \qquad x^2 = -2py.$$

For $y^2 = 2px$, the focus is $(p/2, 0)$, the directrix is $x = -p/2$, and the eccentricity is 1.

For an ellipse, a vertical line through a focus intersects the ellipse in endpoints of a latus rectum. The length is

$$\frac{2b^2}{a}.$$

[1] OpenStax, *Calculus*, Vol. 3, §1.5 (conics); MIT 18.100A real-analysis notes.

To derive it, substitute the focal abscissa $x = c$ into $x^2/a^2 + y^2/b^2 = 1$. Since $c^2 = a^2 - b^2$, this gives $y^2 = b^4/a^2$. The two endpoints $(c, \pm b^2/a)$ are separated by the claimed length. The tangent at $P = (x_0, y_0)$ to

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

is

$$\frac{xx_0}{a^2} + \frac{yy_0}{b^2} = 1.$$

The note also records the focus-directrix relation. For an ellipse with eccentricity $e = c/a$, the directrices are

$$x = \pm \frac{a^2}{c}.$$

The ratio

$$\frac{PF}{\operatorname{dist}(P, \text{directrix})} = e$$

characterizes a nondegenerate conic with a chosen focus and its corresponding directrix. For the parabola $y^2 = 2px$ with $p > 0$, choose $F = (p/2, 0)$ and directrix $x = -p/2$. Equality of squared distances gives

$$(x - p/2)^2 + y^2 = (x + p/2)^2,$$

which simplifies to $y^2 = 2px$. This makes the metric-to-algebra transformation explicit.

A conic can be defined by the ratio

$$e = \frac{\text{distance to focus}}{\text{distance to directrix}}.$$

If $e < 1$, the conic is an ellipse. If $e = 1$, it is a parabola. If $e > 1$, it is a hyperbola. This unifies the three classical curves by one parameter: eccentricity.

§123.2 Affine Coordinates for Conics

A conic can be represented by a quadratic equation

$$ax^2 + bxy + cy^2 + dx + ey + f = 0.$$

Affine transformations preserve conic type up to degeneracy, while projective transformations unify ellipses, parabolas, and hyperbolas as sections of a cone.

The elementary focus-directrix and latus-rectum formulas are metric shadows of a more invariant projective object.

§123.3 Rational density and the real coordinate line

The analytic side begins with the density of rational and irrational numbers in $\mathbb{R}$. For rationals, given $x \in \mathbb{R}$ and $\varepsilon > 0$, choose $q \in \mathbb{N}$ with

$$\frac{1}{q} < \varepsilon.$$

Then choose an integer p such that

$$p \le qx < p + 1.$$

Thus

$$\left|x - \frac{p}{q}\right| < \frac{1}{q} < \varepsilon.$$

For irrationals, take a rational approximation r to $x - \sqrt{2}/N$, then

$$r + \frac{\sqrt{2}}{N}$$

is irrational and can be made arbitrarily close to x.

The density of $\mathbb{Q}$ in $\mathbb{R}$ says every real interval contains rationals. Dedekind cuts use order rather than decimal expansions to construct real numbers. Addition of cuts must be defined so that order and completeness are preserved.

This is the structural reason the note can move from conics to real-number construction: analytic geometry needs a complete ordered field.

The note includes the vector space axioms. Let F be a field and V a set equipped with addition and scalar multiplication:

$$+ : V \times V \to V, \qquad \cdot : F \times V \to V.$$

The axioms are associativity and commutativity of addition, existence of 0, additive inverses, compatibility of scalar multiplication, distributivity, and $1v = v$. With the usual coordinate operations,

$$\mathbb{R}^n$$

is an $\mathbb{R}$-linear space.

§123.4 Dedekind cuts: defining addition and additive inverses

The note returns to Dedekind cuts. We denote by $\mathbb{R}$ the set of Dedekind cuts. For cuts X, Y, define an order by

$$X < Y \quad \Longleftrightarrow \quad X \subset Y, \quad X \neq Y.$$

This order is total because if $X \not\subseteq Y$, choose $x \in X \setminus Y$; the downward-closed property of cuts forces $Y \subset X$.

Define

$$X + Y = \{x + y : x \in X,\ y \in Y\}.$$

The zero cut is

$$\overline{0} = \{x \in \mathbb{Q} : x < 0\}.$$

One must check that $X + Y$ is again a Dedekind cut. It is nonempty; it is proper because rational upper bounds u_X, u_Y give an upper bound $u_X + u_Y$; it is downward closed because $q < x + y$ can be written as $(q - y) + y$ with $q - y < x \in X$; and it has no greatest element because X has none. The cut axioms below are all used:

(i) it is nonempty and not all of $\mathbb{Q}$;

(ii) it is downward closed;

(iii) it has no greatest element.

The page gives the standard proof. For downward closure, if $z < x+y$, then $z = x+(z-x)$ and $z - x < y$, so $z - x \in Y$. For no greatest element, choose larger elements $x' > x$ in X and $y' > y$ in Y; then $x' + y' > x + y$.

Associativity and commutativity follow directly:

$$(X + Y) + Z = X + (Y + Z), \qquad X + Y = Y + X.$$

Order compatibility is subtler. One needs the lemma: if X is a cut and $n \in \mathbb{N}$, then there exist $x \in X$ and $x' \in X'$ such that

$$0 < x' - x < \frac{1}{n}.$$

This approximation lemma lets one prove

$$X < Y \quad \Rightarrow \quad X + Z < Y + Z.$$

For a cut X, define

$$-X = \{y - x' : y \in \overline{0},\ x' \in X'\}.$$

Then $-X$ is the unique cut Z such that

$$X + Z = \overline{0}.$$

The sign construction uses rational approximation at the cut boundary: for a rational cut $X_a = \{q : q < a\}$, the set $\{y - x' : y < 0,\ x' \geq a\}$ is exactly $\{q : q < -a\}$, since one may choose y arbitrarily close to 0 and x' arbitrarily close to a from above. This reproduces the usual negative of a real number. In particular, if $p/q \in \mathbb{Q}$, then

$$-X_{p/q} = X_{-p/q}.$$

If A and B are lower cuts, their sum is

$$A + B = \{a + b : a \in A,\ b \in B\}.$$

The definition is not arbitrary. It is forced by wanting the embedding of rationals into cuts to preserve addition and order.

Conic Exercises: Tangency, Intersections at Infinity, and Polarity

N-20220716

Substituting a line into a conic produces a quadratic equation, but a repeated root and a single *finite* intersection are different statements. The distinction becomes clear in the projective description of conics, as developed in the Oxford projective geometry lectures.[1]

§124.1 A secant of an ellipse

Consider the line $y = 2x - 2$ and the ellipse

$$\frac{x^2}{5} + \frac{y^2}{4} = 1.$$

After substitution,

$$\frac{x^2}{5} + \frac{(2x-2)^2}{4} = 1 \quad \Longleftrightarrow \quad \frac{6}{5}x^2 - 2x = 0.$$

The roots $x = 0$ and $x = 5/3$ give the intersection points

$$(0, -2), \qquad (5/3, 4/3).$$

Because the equation is genuinely quadratic, the discriminant distinguishes two real intersections, a repeated contact point, and no real intersection.

§124.2 Tangents to an ellipse from a specified point

Let $E : x^2/a^2 + y^2/b^2 = 1$ with $a, b > 0$. For the point $P = (0, 2)$, write a possible tangent as $y = kx + 2$. Substitution gives

$$\left(\frac{1}{a^2} + \frac{k^2}{b^2}\right) x^2 + \frac{4k}{b^2} x + \frac{4}{b^2} - 1 = 0.$$

The discriminant vanishes exactly when

$$\boxed{k^2 = \frac{4 - b^2}{a^2}}.$$

Consequently there are two real tangents when $b < 2$, one horizontal tangent when $b = 2$, and no real tangents when $b > 2$. The particular numerical slopes depend on *both* semiaxes. A claimed pair of slopes without specifying a, b is insufficiently determined.

The same result follows by using the tangent at $(x_0, y_0) \in E$,

$$\frac{x x_0}{a^2} + \frac{y y_0}{b^2} = 1.$$

It passes through $(0, 2)$ precisely when $y_0 = b^2/2$. Substituting into the ellipse gives the same existence condition. This is a useful independent check of the discriminant calculation.

[1] https://courses.maths.ox.ac.uk/course/section.php?id=10755

§124.3 One Intersection without Tangency

For $x^2 - y^2 = 2$ and $y = x - 1$, substitution gives

$$x^2 - (x-1)^2 = 2, \qquad x = \frac{3}{2}, \quad y = \frac{1}{2}.$$

The resulting equation is linear, so its unique affine solution is not a double root. The projective hyperbola is $X^2 - Y^2 = 2Z^2$, and the line is $Y = X - Z$. At infinity $Z = 0$, these equations share the point $[1 : 1 : 0]$. The missing second projective intersection has moved to infinity, along an asymptotic direction. This explains why merely counting finite points is an unsafe test for tangency.

§124.4 Correcting a parabola tangent calculation

Take $y = 2x^2$ and $P = (2, 6)$. A line through P of slope k has equation $y = kx + 6 - 2k$. Contact requires a repeated root of

$$2x^2 - kx - 6 + 2k = 0.$$

The discriminant equation is

$$k^2 - 8(-6 + 2k) = k^2 - 16k + 48 = (k-4)(k-12) = 0.$$

Thus the actual tangents are

$$\boxed{y = 4x - 2, \qquad y = 12x - 18.}$$

The previously written negative slope does not satisfy the discriminant and is not a tangent through the stated point. We can verify the new answers by substitution: the first yields $2(x-1)^2 = 0$; the second yields $2(x-3)^2 = 0$.

§124.5 A Focus-and-Circle Maximum

Consider $x^2 = 2py$ with $p > 0$, focus $F = (0, p/2)$, and the circle

$$x^2 + (y+4)^2 = 1.$$

Its center is $(0, -4)$; because F lies above the circle, the shortest focus-to-circle distance is $p/2 + 3$. If this distance is 4, then $p = 2$, giving the parabola $x^2 = 4y$.

Parametrize tangent contact points by $T(t) = (t, t^2/4)$. Its tangent equation is

$$y = \frac{t}{2}x - \frac{t^2}{4}.$$

For an intersection point $P = (u, v)$, the two contact parameters t_1, t_2 solve

$$t^2 - 2ut + 4v = 0, \qquad t_1 + t_2 = 2u, \quad t_1 t_2 = 4v.$$

They are real and distinct when $u^2 - 4v > 0$. Directly computing the determinant of the two vectors from P to $T(t_1)$ and $T(t_2)$ gives

$$[PT(t_1)T(t_2)] = \frac{|t_1 - t_2|^3}{16} = \frac{1}{2}(u^2 - 4v)^{3/2}.$$

On the circle, $u^2 + (v+4)^2 = 1$, where $-5 \le v \le -3$. Therefore

$$u^2 - 4v = 1 - (v+4)^2 - 4v = 21 - (v+6)^2 \le 20.$$

Equality is attained at $P = (0, -5)$. Hence

$$\boxed{[PAB]_{\max} = \tfrac{1}{2}20^{3/2} = 20\sqrt{5}.}$$

The value comes from a verifiable extremum and explicit tangent geometry, not from an unexplained final number.

§124.6 An ellipse area constraint requires specified vertices

Suppose the quadrilateral is formed by the four axis-intercepts

$$(\pm a, 0), \qquad (0, \pm b)$$

of $x^2/a^2 + y^2/b^2 = 1$. The quadrilateral is a rhombus with diagonals $2a$ and $2b$, so its area is $2ab$. If it passes through $(0, -2)$ and the rhombus area is $4\sqrt{5}$, then $b = 2$, $a = \sqrt{5}$, and the ellipse is

$$\frac{x^2}{5} + \frac{y^2}{4} = 1.$$

> **Caution**
>
> The condition that an unspecified quadrilateral inscribed in an ellipse has area $4\sqrt{5}$ does not determine a from $b = 2$. The deduction $2ab = 4\sqrt{5}$ requires the stated axial vertices or an equivalent geometric condition.

For the line $y = kx - 3$ through $(0, -3)$, substituting into this ellipse produces

$$(4 + 5k^2)x^2 - 30kx + 25 = 0.$$

Its discriminant is

$$(-30k)^2 - 100(4 + 5k^2) = 400(k^2 - 1).$$

Thus two distinct real intersections occur exactly when $|k| > 1$, tangency when $|k| = 1$, and no real intersections when $|k| < 1$. Additional length constraints require the missing geometric definitions of their endpoints; no range for k should be quoted without them.

§124.7 Polarity packages tangent equations

Over a field of characteristic different from two, let a nonsingular projective conic be $X^T AX = 0$, with A symmetric and invertible. At a point T of the conic, its tangent line is

$$T^T AX = 0.$$

This follows by differentiating the quadratic equation:

$$d(X^T AX)_T(H) = 2T^T AH.$$

For an arbitrary projective point P, the line $P^T AX = 0$ is its *polar*. If P lies outside a real conic and has two real tangent-contact points T_1, T_2, then those points both satisfy $P^T AT_i = 0$, so the polar is the chord joining them.

The projective argument clarifies why tangent-chord relations can be handled independently of Euclidean distances. The discriminant counts multiplicity on a line; the bilinear form describes tangency; and affine normalization can simplify ellipse calculations when an area factor is tracked by the determinant of the change of variables. None of these tools requires replacing an actual calculation with a slogan.

Spatial Incidence, Skew Lines, Normals, and Orthogonal Projection

N-20220719

In a plane, two different nonparallel lines intersect. In three dimensions they may instead be skew, so a picture alone is no longer a reliable test for intersection. The elementary incidence rules become clearer when lines and planes are described as affine subspaces, as in Strang's MIT 18.06 linear-algebra lectures.[1]

§125.1 Lines and planes are affine spans

Given distinct points $p, q \in \mathbb{R}^3$, their line is

$$L = p + \mathbb{R}(q - p).$$

Given p and two linearly independent directions u, v, the plane they span is

$$\Pi = p + \operatorname{span}\{u, v\}.$$

These formulas prove several incidence statements at once. Three noncollinear points determine a unique affine plane; a line and a point not on it determine a unique plane; two intersecting lines with different directions determine a unique plane.

When directions are independent, uniqueness follows because any plane containing the supplied points must contain their difference vectors and therefore their span. Dependence is the exceptional case: three collinear points lie on infinitely many planes.

§125.2 Skew lines are detected by a scalar triple product

Consider two lines

$$L_1 = p + tu, \qquad L_2 = q + sv, \qquad u, v \neq 0.$$

If u and v are linearly dependent, the lines are parallel or coincident. Suppose instead $u \times v \neq 0$. The two lines lie in a common plane exactly when the displacement $q - p$ is in $\operatorname{span}\{u, v\}$, equivalently

$$\boxed{(q - p) \cdot (u \times v) = 0.}$$

If that scalar triple product vanishes, the nonparallel lines are coplanar and therefore intersect. Otherwise they are skew.

[1] https://web.mit.edu/18.06/www/index.shtml

Example 125.2.1

Take $L_1 = (t, 0, 0)$ and $L_2 = (0, s, 1)$. Here $u = (1, 0, 0)$, $v = (0, 1, 0)$, and $q - p = (0, 0, 1)$. Thus

$$(q - p) \cdot (u \times v) = (0, 0, 1) \cdot (0, 0, 1) = 1,$$

so the lines are skew even though their horizontal projections intersect.

A unit normal to both direction vectors is $n = (u \times v)/\|u \times v\|$. The shortest distance between nonparallel lines is

$$d(L_1, L_2) = \frac{|(q - p) \cdot (u \times v)|}{\|u \times v\|}.$$

To derive it, decompose the displacement $q - p$ into a component in the plane spanned by u, v and a perpendicular component along n. Varying s, t cancels the in-plane component, but cannot affect the perpendicular component. Hence its absolute length is the minimum distance.

§125.3 A normal vector classifies the line–plane relation

A plane has equation

$$\Pi = \{x \in \mathbb{R}^3 : n \cdot (x - p_0) = 0\}, \qquad n \neq 0.$$

Substitute the line $x = p + tu$. Its intersection parameters satisfy

$$n \cdot (p - p_0) + t(n \cdot u) = 0.$$

Three cases are now explicit. If $n \cdot u \neq 0$, there is exactly one solution; the line crosses the plane. If $n \cdot u = 0$ but $n \cdot (p - p_0) \neq 0$, no solution exists, so the line is parallel to the plane. If both vanish, the entire line lies in the plane.

In particular, saying that the line's direction is perpendicular to the normal establishes only that it is *parallel to or contained in* the plane. It does not distinguish the two without a point test.

Two planes have parallel normal vectors if and only if their directions agree; the geometric planes are either coincident or disjoint parallel planes. If their normals are independent, the intersection is a line since the two linear equations have rank two.

§125.4 Perpendicularity to two directions is enough

Theorem 125.4.1

Let $\Pi = p + \operatorname{span}\{u, v\}$, where u, v are linearly independent. A direction vector w is perpendicular to the plane if and only if

$$w \cdot u = w \cdot v = 0.$$

Proof. Every direction of Π has the form $au + bv$. The assumptions imply

$$w \cdot (au + bv) = a(w \cdot u) + b(w \cdot v) = 0$$

for all scalars a, b. The converse follows by applying perpendicularity to the particular directions u, v. □

This is the algebraic content of the traditional test using two intersecting lines in a plane. The intersection requirement guarantees independent directions; two parallel lines would give redundant information.

§125.5 Projection provides the closest point

Let W be a linear subspace of a finite-dimensional inner-product space. Every x decomposes uniquely as

$$x = p + r, \qquad p \in W, \quad r \in W^{\perp}.$$

For any $w \in W$, the Pythagorean theorem gives

$$\|x - w\|^2 = \|r + (p - w)\|^2 = \|r\|^2 + \|p - w\|^2 \geq \|r\|^2.$$

Thus p is the unique closest point of W to x. For an affine plane $p_0 + W$, project $x - p_0$ onto W and then add p_0.

If Π has normal n, the projection formula becomes

$$\operatorname{proj}_{\Pi}(x) = x - \frac{n \cdot (x - p_0)}{\|n\|^2}\, n,$$

and the distance from x to the plane is

$$d(x, \Pi) = \frac{|n \cdot (x - p_0)|}{\|n\|}.$$

These are two views of the same orthogonal decomposition.

§125.6 The three-perpendicular theorem as a dot product

Let $x \notin \Pi$, let $p = \operatorname{proj}_{\Pi}(x)$, and let $q \in \Pi$. The segment $x - q$ decomposes as

$$x - q = (x - p) + (p - q).$$

For any vector v tangent to Π,

$$(x - q) \cdot v = (x - p) \cdot v + (p - q) \cdot v = (p - q) \cdot v.$$

Hence xq is perpendicular in direction to a line of Π with direction v if and only if its projected segment pq is perpendicular to that line. This proves the exact direction comparison underlying the three-perpendicular theorem, without relying on a potentially misleading sketch.

§125.7 Incidence has an affine and a projective layer

The line $p + tu$ can be completed to a projective line, in which the direction u determines a point at infinity. Two parallel affine lines share that projective point; two skew lines remain disjoint as projective lines in three-space. They do not automatically intersect merely because the space was projectively completed.

In $\mathbb{P}^3$, a projective plane is the zero set of a nonzero linear functional ℓ on $\mathbb{R}^4$, and incidence of the plane with a point $[X]$ is simply $\ell(X) = 0$. The dual language adds no metric: distance and perpendicularity still require an inner product, while intersection and parallel completion require only linear dependence.

The practical distinction is now precise. Rank tests decide incidence, normal vectors decide orientation relative to planes, dot products decide perpendicularity, and orthogonal decomposition computes shortest distances.

Trigonometric Functions, Radians, Identities, and Triangle Formulas

N-20220802

§126.1 An angle is a motion before it is a number

An angle begins with two rays, but the most useful picture is a rotation. Fix an initial ray and rotate it until it reaches a terminal ray. Counterclockwise rotations are positive and clockwise rotations are negative. A full turn changes the amount of rotation without changing the terminal direction, so

$$\theta \sim \theta + 2k\pi, \qquad k \in \mathbb{Z}.$$

This is why an angle can be treated either as a real number recording a motion or as a point of the quotient

$$\mathbb{R}/2\pi\mathbb{Z}$$

recording only the final direction.

Radian measure is not chosen merely to replace degrees by another table of numbers. On a circle of radius r, an arc of length s subtends the angle

$$\theta = \frac{s}{r}.$$

The quotient is independent of the size of the circle. On the unit circle, the numerical value of the angle is exactly the signed arc length travelled. One full revolution therefore has length and angle

$$2\pi.$$

This normalization makes small rotations look linear. For $0 < \theta < \pi/2$, the standard comparison of an inscribed triangle, a circular sector, and a tangent triangle gives

$$\sin\theta < \theta < \tan\theta.$$

Dividing by $\sin\theta$ yields

$$1 < \frac{\theta}{\sin\theta} < \frac{1}{\cos\theta}.$$

As $\theta \to 0$, the two outer quantities tend to 1, so

$$\lim_{\theta\to 0} \frac{\sin\theta}{\theta} = 1.$$

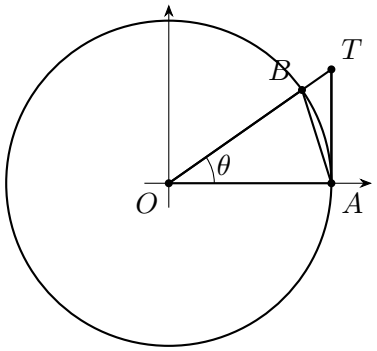


Figure 126.1: The areas of $\triangle OAB$, sector OAB, and $\triangle OAT$ give $\sin\theta < \theta < \tan\theta$.

The limit would contain a conversion constant if the variable were measured in degrees. Radians are therefore the coordinate in which the circle has unit speed at the identity. A later calculus chapter will use this limit to derive the trigonometric derivatives; here it already explains why the geometric and analytic normalizations coincide.

§126.2 Coordinates on the unit circle

A point on the unit circle has the form

$$p(\theta) = (\cos\theta, \sin\theta).$$

This is not merely notation for two unrelated functions. Sine and cosine are the coordinate functions of one moving point. The equation of the circle immediately gives

$$\cos^2\theta + \sin^2\theta = 1.$$

Several identities are visible before any algebra is done. Reversing the direction of travel reflects the point across the horizontal axis:

$$\cos(-\theta) = \cos\theta, \qquad \sin(-\theta) = -\sin\theta.$$

A half turn changes both coordinates:

$$\cos(\theta+\pi) = -\cos\theta, \qquad \sin(\theta+\pi) = -\sin\theta.$$

A quarter turn exchanges the coordinates up to sign:

$$\cos\left(\theta+\frac{\pi}{2}\right) = -\sin\theta, \qquad \sin\left(\theta+\frac{\pi}{2}\right) = \cos\theta.$$

The familiar quadrant signs are simply the signs of the two coordinates.

The tangent function records the slope of the terminal direction:

$$\tan\theta = \frac{\sin\theta}{\cos\theta}.$$

It is defined whenever the direction is not vertical. Its poles at

$$\theta = \frac{\pi}{2} + k\pi$$

are not defects of the circle; they are defects of using slope as a coordinate. In the real projective line, the vertical direction is an ordinary point at infinity, and tangent becomes a local coordinate on directions rather than a globally defined real function.

The graphs also come directly from the motion. Sine and cosine have period 2π, while a direction and its opposite have the same slope, so tangent has period π:

$$\sin(heta + 2\pi) = \sin\theta, \qquad \cos(heta + 2\pi) = \cos\theta,$$

$$\tan(heta + \pi) = \tan\theta.$$

Special-angle values should be reconstructed from symmetric points of the circle rather than memorized as isolated entries. For example,

$$p\left(\frac{2\pi}{3}\right) = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right),$$

so

$$\cos\frac{2\pi}{3} = -\frac{1}{2}, \qquad \sin\frac{2\pi}{3} = \frac{\sqrt{3}}{2}, \qquad \tan\frac{2\pi}{3} = -\sqrt{3}.$$

§126.3 Rotations contain the addition formulas

The rotation through angle θ is represented by

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

Its columns are the images of the coordinate basis vectors. Performing a rotation by β and then one by α is the same motion as rotating once by $\alpha + \beta$:

$$R(\alpha)R(\beta) = R(\alpha + \beta).$$

Multiplying the left-hand side gives

$$\begin{pmatrix} \cos\alpha\cos\beta - \sin\alpha\sin\beta & -\cos\alpha\sin\beta - \sin\alpha\cos\beta \\ \sin\alpha\cos\beta + \cos\alpha\sin\beta & -\sin\alpha\sin\beta + \cos\alpha\cos\beta \end{pmatrix}.$$

Comparing entries with $R(\alpha + \beta)$ yields

$$\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta,$$

$$\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta.$$

Replacing β by $-\beta$ and using parity gives the subtraction formulas.

There is also a direct geometric proof of the cosine difference identity. The dot product of two unit vectors is the cosine of the angle between them:

$$p(\alpha) \cdot p(\beta) = \cos(\alpha - \beta).$$

Expanding the coordinates gives

$$\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta.$$

The matrix proof emphasizes composition; the dot-product proof emphasizes relative angle. They prove the same formula for different reasons.

Dividing the sine and cosine formulas gives

$$\tan(\alpha + \beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta},$$

whenever both sides are defined. The denominator condition matters. If

$$1 - \tan\alpha \tan\beta = 0,$$

then the sum direction is vertical and the tangent is not a finite real number.

Setting $\beta = \alpha$ gives

$$\sin 2\alpha = 2\sin\alpha\cos\alpha,$$

$$\cos 2\alpha = \cos^2\alpha - \sin^2\alpha = 2\cos^2\alpha - 1 = 1 - 2\sin^2\alpha.$$

Solving the last two expressions for the squares gives

$$\cos^2\frac{\theta}{2} = \frac{1+\cos\theta}{2}, \qquad \sin^2\frac{\theta}{2} = \frac{1-\cos\theta}{2}.$$

The square-root versions require a sign chosen from the quadrant of $\theta/2$. Omitting that sign is one of the most common ways a correct identity becomes a false formula.

§126.4 Complex multiplication and multiple angles

Identify $(x, y) \in \mathbb{R}^2$ with $x + iy \in \mathbb{C}$. The unit-circle point becomes

$$z(\theta) = \cos\theta + i\sin\theta.$$

The addition formulas say exactly that

$$z(\alpha+\beta) = z(\alpha)z(\beta).$$

With the exponential notation

$$z(\theta) = e^{i\theta},$$

angle addition becomes ordinary multiplication.

Repeated multiplication gives De Moivre's formula:

$$(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta), \qquad n \in \mathbb{Z}.$$

For $n = 3$, expanding the left-hand side and comparing real and imaginary parts gives

$$\cos 3\theta = 4\cos^3\theta - 3\cos\theta,$$

$$\sin 3\theta = 3\sin\theta - 4\sin^3\theta.$$

Multiple-angle identities are therefore polynomial consequences of multiplication on the circle.

The roots of unity are equally transparent. Solving

$$z^n = 1$$

gives

$$z_k = e^{2\pi i k/n}, \qquad k = 0, 1, \ldots, n-1.$$

They are the vertices of a regular n-gon. Their sum is zero for $n > 1$, either by the geometric-series identity or because the centroid of a regular polygon is the origin.

This chapter uses roots of unity only as the natural endpoint of multiple-angle geometry. Their arithmetic and finite Fourier structure are developed in separate notes.

§126.5 Chebyshev polynomials turn angles into algebra

Define T_n by

$$T_n(\cos\theta) = \cos(n\theta).$$

The addition formula

$$\cos((n+1)\theta) + \cos((n-1)\theta) = 2\cos\theta\cos(n\theta)$$

shows that

$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x),$$

with

$$T_0(x) = 1, \qquad T_1(x) = x.$$

Thus

$$T_2(x) = 2x^2 - 1,$$
$$T_3(x) = 4x^3 - 3x,$$
$$T_4(x) = 8x^4 - 8x^2 + 1.$$

The roots are obtained without solving a general polynomial equation. Since

$$T_n(\cos\theta) = 0$$

when

$$n\theta = \frac{(2k-1)\pi}{2},$$

we have

$$x_k = \cos\frac{(2k-1)\pi}{2n}, \qquad k = 1, \dots, n.$$

Because the leading coefficient of T_n is 2^{n-1} for $n \geq 1$,

$$T_n(x) = 2^{n-1}\prod_{k=1}^{n}\left(x - \cos\frac{(2k-1)\pi}{2n}\right).$$

A trigonometric zero set has become an exact polynomial factorization.

Chebyshev polynomials also solve an extremal problem. On $[-1, 1]$,

$$|T_n(x)| \leq 1.$$

After normalizing the leading coefficient, no monic polynomial of degree n has smaller maximum absolute value on this interval. The alternating values

$$T_n\left(\cos\frac{k\pi}{n}\right) = (-1)^k$$

force this minimax behavior. This is why the same polynomials appear in interpolation nodes, approximation theory, and stable numerical algorithms. Multiple-angle formulas are not merely simplification tricks; they encode an optimal oscillation pattern.

§126.6 Triangle formulas are inner-product geometry

Let a Euclidean triangle have side lengths a, b, c opposite angles A, B, C. Place two side vectors u, v at the vertex with angle C, with

$$\|u\| = a, \qquad \|v\| = b.$$

The third side is $u - v$, so

$$c^2 = \|u - v\|^2 = \|u\|^2 + \|v\|^2 - 2u \cdot v.$$

Since

$$u \cdot v = ab \cos C,$$

we obtain the law of cosines:

$$\boxed{c^2 = a^2 + b^2 - 2ab \cos C.}$$

The Pythagorean theorem is the special case $C = \pi/2$.

The area K can be written using any included angle:

$$K = \frac{1}{2}bc \sin A = \frac{1}{2}ca \sin B = \frac{1}{2}ab \sin C.$$

Dividing these expressions gives

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

To identify the common value, place the triangle on its circumcircle. A chord of length a subtending inscribed angle A has length

$$a = 2R \sin A,$$

so

$$\boxed{\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R.}$$

Heron's formula is also an inner-product identity. Using the vectors u, v with lengths b, c and included angle A, the Gram determinant is

$$\det \begin{pmatrix} u \cdot u & u \cdot v \\ v \cdot u & v \cdot v \end{pmatrix} = b^2c^2 - (bc \cos A)^2 = 4K^2.$$

The law of cosines gives

$$bc \cos A = \frac{b^2 + c^2 - a^2}{2}.$$

Therefore

$$16K^2 = 4b^2c^2 - (b^2 + c^2 - a^2)^2.$$

Factoring the right-hand side yields

$$16K^2 = (a + b + c)(-a + b + c)(a - b + c)(a + b - c).$$

With

$$s = \frac{a + b + c}{2},$$

we recover

$$\boxed{K = \sqrt{s(s - a)(s - b)(s - c)}.}$$

The four factors are not mysterious. They are the triangle inequalities written symmetrically.

§126.7 The ambiguous triangle is a real geometric bifurcation

Three sides determine a Euclidean triangle up to congruence, and two sides with the included angle do the same. Two sides with a nonincluded angle can behave differently.

Take

$$a = 7, \qquad b = 10, \qquad A = 30^\circ.$$

The sine law gives

$$\sin B = \frac{b \sin A}{a} = \frac{5}{7}.$$

There are two angles in $(0, \pi)$ with this sine:

$$B_1 = \arcsin \frac{5}{7}, \qquad B_2 = \pi - B_1.$$

Both satisfy

$$A + B_j < \pi,$$

so both define triangles. Their third angles are

$$C_1 = \pi - A - B_1, \qquad C_2 = B_1 - A,$$

and their third sides are

$$c_j = \frac{a \sin C_j}{\sin A}.$$

Numerically,

$$B_1 \approx 45.58^\circ, \qquad B_2 \approx 134.42^\circ,$$

producing one obtuse triangle and one narrow acute triangle.

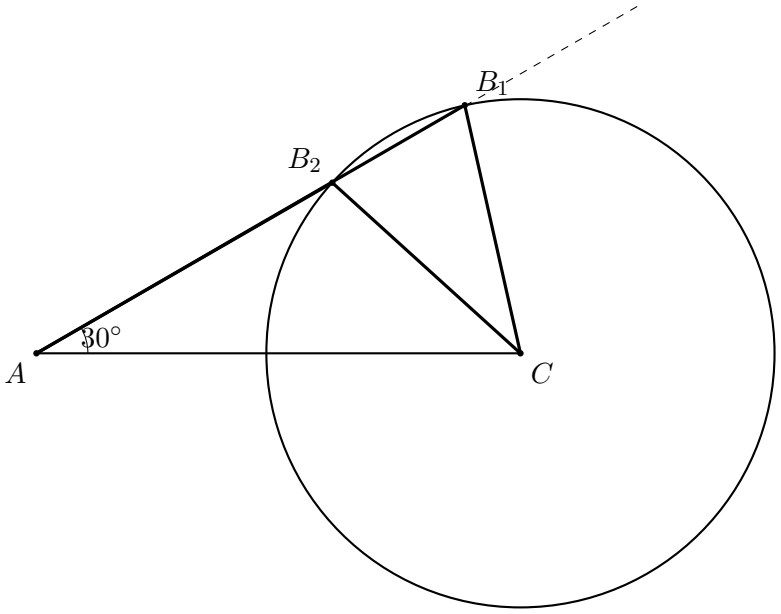


Figure 126.2: A ray from A can meet the circle centered at C twice. The two intersections are the two SSA triangles.

The ambiguity is not caused by a defective formula. It is the geometry of a circle intersecting a ray in two points. The cases of zero, one, or two triangles correspond to no intersection, tangency, or two intersections.

§126.8 The circle group explains why frequencies add

The rotation matrices form the group

$$SO(2) = \{Q \in \mathbb{R}^{2\times 2} : Q^T Q = I,\ \det Q = 1\}.$$

Let

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Since $J^2 = -I$, its exponential separates into even and odd powers:

$$e^{\theta J} = I\cos\theta + J\sin\theta = R(\theta).$$

Thus

$$SO(2) = \{e^{\theta J} : \theta \in \mathbb{R}\}.$$

The matrix J is the infinitesimal generator of rotations. Differentiating this identity gives

$$R'(\theta) = JR(\theta) = R(\theta)J.$$

This yields the coupled system

$$\cos'\theta = -\sin\theta, \qquad \sin'\theta = \cos\theta.$$

Logically, this is an alternative analytic construction once the matrix exponential has been defined; it is not a substitute for the elementary limit proof used in an introductory calculus development.

Under the identification $SO(2) \cong S^1$, the continuous one-dimensional complex characters are

$$\chi_n(e^{i\theta}) = e^{in\theta}, \qquad n \in \mathbb{Z}.$$

A short reason is that a character lifted to $\mathbb{R}$ must have the form $e^{ic\theta}$, and periodicity at 2π forces $c \in \mathbb{Z}$.

Multiplying characters adds their indices:

$$\chi_m\chi_n = \chi_{m+n}.$$

Taking real and imaginary parts gives the product-to-sum identities. For example,

$$\cos m\theta\cos n\theta = \frac{1}{2}\cos((m+n)\theta) + \frac{1}{2}\cos((m-n)\theta).$$

A product of oscillations becomes a sum of two frequencies because multiplication in the character group adds frequency labels.

The same characters are orthogonal on the circle:

$$\int_{-\pi}^{\pi} e^{im\theta}\overline{e^{in\theta}}\,d\theta = \begin{cases} 2\pi, & m = n, \\ 0, & m \neq n. \end{cases}$$

This is the minimal bridge from elementary trigonometry to Fourier analysis. The deeper theory belongs elsewhere, but the reason trigonometric modes are natural is already visible: they are the irreducible characters of the circle group.

§126.9 Changing curvature changes triangle trigonometry

Euclidean triangle formulas are not universal laws of all geometries. On the unit sphere, triangle sides are great-circle arc lengths. The spherical law of cosines is

$$\boxed{\cos c = \cos a \cos b + \sin a \sin b \cos C.}$$

For a right angle $C = \pi/2$, this becomes

$$\cos c = \cos a \cos b,$$

which is visibly different from the Euclidean Pythagorean theorem.

A particularly clean example is the spherical octant triangle with vertices at the north pole and two equatorial points separated by 90°. Its three side lengths are all $\pi/2$, and its three angles are also all $\pi/2$. Hence

$$A + B + C = \frac{3\pi}{2}.$$

The angular excess is

$$A + B + C - \pi = \frac{\pi}{2}.$$

The triangle occupies one eighth of the unit sphere, whose area is 4π, so its area is also $\pi/2$. This is a special case of the spherical excess formula

$$\text{area} = A + B + C - \pi.$$

It is the first visible shadow of Gauss–Bonnet: angle excess measures positive curvature.

The Euclidean law of cosines reappears at small scale. Replace a, b, c by $\varepsilon a, \varepsilon b, \varepsilon c$ in the spherical formula. Using

$$\cos(\varepsilon x) = 1 - \frac{\varepsilon^2 x^2}{2} + O(\varepsilon^4),$$

$$\sin(\varepsilon x) = \varepsilon x + O(\varepsilon^3),$$

we obtain

$$c^2 = a^2 + b^2 - 2ab\cos C + O(\varepsilon^2).$$

After zooming in, curvature disappears to first order and the geometry becomes Euclidean.

In hyperbolic geometry the corresponding formula is

$$\cosh c = \cosh a \cosh b - \sinh a \sinh b \cos C.$$

Its small-scale expansion gives the same Euclidean law, but hyperbolic triangles have angle sum less than π. Spherical, Euclidean, and hyperbolic trigonometry are therefore the positive-, zero-, and negative-curvature versions of one question: how do distances and directions interact in the ambient geometry?

§126.10 A product of sines that knows an integer

Place the n-th roots of unity at the vertices of a regular n-gon. Each vertex $e^{2\pi ik/n}$ has Euclidean distance

$$|1 - e^{2\pi ik/n}| = 2\sin\frac{\pi k}{n}$$

from the distinguished vertex 1. So the lengths of the $n-1$ chords from 1 are trigonometric numbers. Their *product*, surprisingly, is an integer.

Factor the geometric-series polynomial:

$$\frac{z^n - 1}{z - 1} = 1 + z + \cdots + z^{n-1} = \prod_{k=1}^{n-1}(z - e^{2\pi ik/n}).$$

Evaluate at $z = 1$ and take absolute values. Because all the angles $\pi k/n$ lie in $(0, \pi)$, their sines are positive, giving

$$\boxed{\prod_{k=1}^{n-1} 2\sin\frac{\pi k}{n} = n.}$$

For $n = 4$, the four vertices are the cardinal directions. The three chord lengths from 1 are $\sqrt{2}, 2, \sqrt{2}$, and their product is indeed 4.

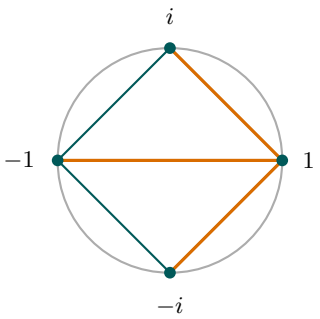


Figure 126.3: For $n = 4$, three chords from the point 1 have lengths $\sqrt{2}$, 2, and $\sqrt{2}$. The product is 4, as predicted by the factorization of $z^4 - 1$.

The identity is at once a calculation about circle geometry, a polynomial factorization, and a statement about roots of unity. There is nothing additional to memorize: the product formula is what the equation $z^n = 1$ says when viewed from the vertex $z = 1$.

Analytic Geometry through Normals, Line Pencils, and Quadratic Forms

N-20220911

§127.1 Lines as Affine Level Sets

A line in the plane can be written

$$H = \{x \in \mathbb{R}^2 : n \cdot x = c\}, \qquad n \neq 0.$$

The vector $n = (a, b)$ is normal to the line, and a direction vector is

$$d = (-b, a).$$

If $p \in H$, every point of the line has the parametric form

$$x = p + td, \qquad t \in \mathbb{R}.$$

Conversely, eliminating t gives

$$a(x_1 - p_1) + b(x_2 - p_2) = 0.$$

This single description includes vertical lines, where slope notation fails.

The familiar equations are coordinate choices for the same object. If $b \neq 0$,

$$ax + by = c \quad \Longleftrightarrow \quad y = -\frac{a}{b}x + \frac{c}{b}.$$

If a line passes through p_1, p_2, its direction is $p_2 - p_1$. If it meets the coordinate axes away from the origin, dividing by its intercepts produces

$$\frac{x}{a} + \frac{y}{b} = 1.$$

None of these forms is universally best; the normal form survives every orientation, while slope form is efficient only in the nonvertical chart.

In homogeneous coordinates, the line is represented by the coefficient triple

$$\ell = [a : b : -c],$$

and incidence with a projective point $X = [x : y : z]$ is the bilinear equation

$$ax + by - cz = 0.$$

The homogeneous form will later make points at infinity and polar lines ordinary parts of the calculation.

§127.2 Distances and Normal Vectors

Two lines with normals n_1, n_2 are parallel exactly when the normals are proportional. They are perpendicular exactly when

$$n_1 \cdot n_2 = 0,$$

because the direction of each line is a quarter-turn of its normal.

For

$$H = \{x : n \cdot x = c\}$$

and a point x_0, the signed displacement in the normal direction is

$$\frac{n \cdot x_0 - c}{\|n\|_2}.$$

Hence

$$\boxed{\operatorname{dist}_2(x_0, H) = \frac{|n \cdot x_0 - c|}{\|n\|_2}.}$$

To derive it, let

$$p = x_0 - \frac{n \cdot x_0 - c}{\|n\|_2^2} n.$$

Then $n \cdot p = c$, so $p \in H$, and $x_0 - p$ is normal to the line. Any other point of H differs from p by a tangent vector, so the Pythagorean theorem makes p the closest point.

For parallel lines

$$n \cdot x = c_1, \qquad n \cdot x = c_2,$$

the same formula gives

$$\operatorname{dist}_2(H_1, H_2) = \frac{|c_1 - c_2|}{\|n\|_2}.$$

The denominator is not a memorized normalization constant; it compensates for rescaling the defining equation.

§127.3 Hyperplane Distance and Dual Norms

The Euclidean formula has a normed-space version. Let $\|\cdot\|$ be any norm on $\mathbb{R}^n$, and let φ be a nonzero linear functional. Its dual norm is

$$\|\varphi\|_* = \sup_{\|u\| \le 1} |\varphi(u)|.$$

For the hyperplane

$$H = \{x : \varphi(x) = c\},$$

one has

$$\boxed{\operatorname{dist}(x_0, H) = \frac{|\varphi(x_0) - c|}{\|\varphi\|_*}.}$$

Indeed, for every $x \in H$,

$$|\varphi(x_0) - c| = |\varphi(x_0 - x)| \le \|\varphi\|_* \|x_0 - x\|,$$

which gives the lower bound. In finite dimensions, the supremum defining the dual norm is attained by a unit vector u; moving from x_0 in the signed u-direction reaches equality.

For $\varphi(x,y) = ax + by$, the dual of ℓ^1 is ℓ^∞, so

$$\operatorname{dist}_1((x_0,y_0),H) = \frac{|ax_0 + by_0 - c|}{\max(|a|,|b|)}.$$

The dual of ℓ^∞ is ℓ^1, giving

$$\operatorname{dist}_\infty((x_0,y_0),H) = \frac{|ax_0 + by_0 - c|}{|a| + |b|}.$$

The familiar square root in the Euclidean formula is therefore a special case of dual-norm geometry.

§127.4 Circles as Quadratic Level Sets

A circle with center $m = (a,b)$ and radius r is

$$\|x - m\|_2^2 = r^2.$$

Expanding gives

$$x^2 + y^2 + Dx + Ey + F = 0,$$

while completing squares recovers

$$m = \left(-\frac{D}{2}, -\frac{E}{2}\right),$$

$$r^2 = \frac{D^2 + E^2 - 4F}{4}.$$

The sign of this final quantity distinguishes a real circle, a point circle, and no real locus.

For a point p, define its power with respect to the circle by

$$\operatorname{Pow}_C(p) = \|p - m\|^2 - r^2.$$

If a line through p meets the circle at a, b, then

$$\operatorname{Pow}_C(p) = \overline{pa}\,\overline{pb}$$

using directed lengths. This follows by substituting the line parametrization $p + td$ into the circle equation: the product of the two parameter roots is the constant term divided by the leading coefficient.

For two circles, the locus of equal powers is their radical axis. Subtracting

$$C_1 : x^2 + y^2 - 4x = 0$$

and

$$C_2 : x^2 + y^2 - 2y - 3 = 0$$

cancels the quadratic terms and gives the line

$$-4x + 2y + 3 = 0.$$

The radical axis is linear because it compares two quadratic functions with the same leading form.

§127.5 Fractional Ranges and Pencils of Lines

Consider

$$\mu = \frac{x}{y-3}$$

under the constraint

$$|x| + |y| \le 1.$$

For fixed μ, rewrite the equation as

$$x = \mu(y-3).$$

This is a line through

$$P = (0,3).$$

The range problem asks which lines in the pencil through P meet the diamond

$$K = \{(x,y) : |x| + |y| \le 1\}.$$

Because K is convex and $P \notin K$, the extreme parameters occur at supporting lines from P to K.

The support function of the ℓ^1-unit ball is

$$h_K(n) = \max_{x \in K} n \cdot x = \|n\|_\infty.$$

The line can be written

$$(1, -\mu) \cdot (x,y) = -3\mu.$$

It meets K precisely when

$$|3\mu| \le h_K(1,-\mu) = \max(1, |\mu|).$$

If $|\mu| > 1$, this is impossible. If $|\mu| \le 1$, it becomes

$$3|\mu| \le 1.$$

Therefore

$$\boxed{\mu \in \left[-\frac{1}{3}, \frac{1}{3}\right].}$$

The endpoints correspond to the supporting lines through the vertices $(1,0)$ and $(-1,0)$. Convex duality has not replaced the elementary picture; it has explained why the endpoints are support data.

§127.6 Ratios on a graph are radial coordinates

For a point $(x, f(x))$ with $x \neq 0$, the ratio

$$\frac{f(x)}{x}$$

is the slope of the ray from the origin to the graph point. Counting solutions of

$$\frac{f(x)}{x} = m$$

is therefore the same as counting intersections between the graph and the line

$$y = mx.$$

A rational expression has become a radial-projection problem.

This viewpoint also identifies tangency thresholds. As m varies, the number of intersections can change only when the line becomes tangent to the graph or passes through a singular point. Algebraically, solving

$$f(x) = mx$$

together with

$$f'(x) = m$$

detects those critical slopes. The graphical and derivative descriptions are the same incidence event viewed globally and locally.

§127.7 The Moving Line and Its Circle

Let fixed points be

$$A = (-3, 0), \qquad B = (1, 6).$$

Suppose lines through A and B meet at P and are perpendicular. Then

$$\angle APB = 90^\circ,$$

so Thales' theorem places P on the circle with diameter AB.

The right triangle gives

$$PA^2 + PB^2 = AB^2.$$

Hence

$$PA\,PB \le \frac{PA^2 + PB^2}{2} = \frac{AB^2}{2}.$$

Since

$$AB^2 = (1+3)^2 + 6^2 = 52,$$

one obtains

$$\boxed{PA\,PB \le 26.}$$

Equality occurs when $PA = PB$, namely at the intersections of the diameter circle with the perpendicular bisector of AB.

The coordinate version would introduce two slopes and a quadratic constraint. The circle formulation identifies the locus first, after which the optimization is a one-line inequality.

§127.8 A general conic is an affine quadratic form

Write an affine conic as

$$F(x) = x^T Q x + 2\ell^T x + f = 0, \qquad Q = Q^T.$$

An orthogonal change of coordinates diagonalizes Q, removing the cross term without changing Euclidean angles. If Q is invertible, translating by

$$x = u - Q^{-1}\ell$$

removes the linear term. Rotation handles directions; translation handles the center.

Consider

$$xy - x - y = 0.$$

Set

$$u = \frac{x+y}{\sqrt{2}}, \qquad v = \frac{x-y}{\sqrt{2}}.$$

Then

$$xy = \frac{u^2 - v^2}{2}, \qquad x + y = \sqrt{2}\,u.$$

The equation becomes

$$u^2 - v^2 - 2\sqrt{2}\,u = 0,$$

or

$$\boxed{(u - \sqrt{2})^2 - v^2 = 2.}$$

The original rectangular-looking equation is a hyperbola whose principal axes are the eigenvectors of the quadratic matrix

$$Q = \begin{pmatrix} 0 & 1/2 \\ 1/2 & 0 \end{pmatrix}.$$

§127.9 Rank and the line at infinity distinguish conic types

Homogenizing a conic gives

$$X^T A X = 0, \qquad X = (x, y, z)^T,$$

for a symmetric 3×3 matrix A. The conic is degenerate when A loses rank. For example,

$$xy = 0$$

is the union of two lines and has a rank-deficient homogeneous matrix.

The intersection with the line at infinity $z = 0$ records affine directions.

- An ellipse
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = z^2$$
has no real point at infinity.
- A parabola
$$y^2 = 2pxz$$
meets the line at infinity at one double point $[1 : 0 : 0]$.
- A hyperbola
$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = z^2$$
has two real points at infinity, corresponding to its asymptotic directions.

For

$$xy - x - y = 0,$$

the projective closure is

$$XY - XZ - YZ = 0.$$

At infinity,

$$Z = 0 \quad \Longrightarrow \quad XY = 0,$$

so the two points are

$$[1 : 0 : 0], \qquad [0 : 1 : 0].$$

They are the projective endpoints of the two asymptote directions revealed by the standard hyperbola form.

§127.10 Conic Polarity and Tangents

Let a nondegenerate projective conic be

$$F(X) = X^T AX = 0, \qquad A = A^T.$$

For a point p, its polar line is

$$p^T AX = 0.$$

If p lies on the conic, this polar is the tangent line. Indeed, the differential is

$$dF_p(H) = 2p^T AH.$$

The tangent condition at p is

$$p^T A(X - p) = 0.$$

Since $p^T Ap = 0$, this reduces to

$$p^T AX = 0.$$

For the conic

$$XY - XZ - YZ = 0,$$

use the symmetric matrix

$$A = \begin{pmatrix} 0 & 1 & -1 \\ 1 & 0 & -1 \\ -1 & -1 & 0 \end{pmatrix},$$

which represents twice the equation. The affine point $(2, 2)$ corresponds to

$$p = [2 : 2 : 1]$$

and lies on the conic. Since

$$Ap = \begin{pmatrix} 1 \\ 1 \\ -4 \end{pmatrix},$$

its polar, hence its tangent, is

$$X + Y - 4Z = 0.$$

In the affine chart $Z = 1$,

$$\boxed{x + y = 4.}$$

The same answer follows from the gradient of $xy - x - y$ at $(2, 2)$, but the polar calculation remains valid in homogeneous coordinates and transforms naturally with the conic under projective changes of coordinates.

Tarski's Circle-Squaring Problem: From Polygon Dissection to Borel Equidecomposition

N-20221214

§128.1 The route of the note

Tarski's circle-squaring problem unfolds through a sequence of increasingly subtle notions:

$$\text{Tarski circle-squaring} \longrightarrow \text{Hilbert's third problem}$$
$$\longrightarrow \text{measure and paradoxical decomposition}$$
$$\longrightarrow \text{Laczkovich's theorem} \longrightarrow \text{Borel circle squaring.}$$

The visual configurations below support theorem statements whose algebraic and measure-theoretic content is developed directly in the prose.

Tarski asked in 1925 whether a disk and a square of equal area in the plane are equidecomposable: can the disk be partitioned into finitely many pieces, and can those pieces be moved by isometries to partition the square? The link between discrepancy, bounded flows and definability follows Laczkovich and the Borel theorem of Marks and Unger.

§128.2 Equidecomposability and dissection congruence

Let a group G act on a set X. Two subsets $A, B \subseteq X$ are G-equidecomposable if there are finite partitions

$$A = A_1 \sqcup \cdots \sqcup A_m, \qquad B = B_1 \sqcup \cdots \sqcup B_m,$$

and elements $g_i \in G$ such that

$$g_i(A_i) = B_i.$$

For plane polygons, one usually lets G be the group of Euclidean isometries. In that setting the classical language is dissection congruence: cut one polygon into finitely many polygonal pieces and rearrange them to obtain the other.

The first example is the familiar parallelogram-to-rectangle cut.

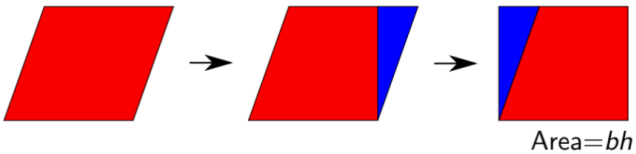


Figure 128.1: A parallelogram dissected and rearranged into a rectangle.

Area is obviously preserved by dissection congruence. The Wallace–Bolyai–Gerwien theorem says that, for plane polygons, area is also sufficient.

Theorem 128.2.1 (Wallace–Bolyai–Gerwien)
Any two plane polygons of the same area are dissection congruent.

One small point in the note is transitivity. If polygon P is dissection congruent to Q, and Q is dissection congruent to R, then P is dissection congruent to R. One refines the two dissections so that the intermediate polygon Q is cut compatibly, then composes the rearrangements.

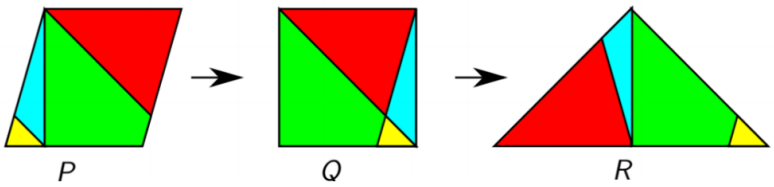


Figure 128.2: The transitivity intuition: a dissection from P to Q and one from Q to R compose.

The proof of the theorem is constructive. First chop a polygon into triangles.

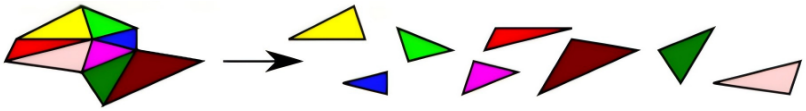


Figure 128.3: The first reduction: chop a polygon into triangles.

Then dissect each triangle into a parallelogram and then a rectangle; dissect each rectangle into a square; combine squares by the Pythagorean dissection.

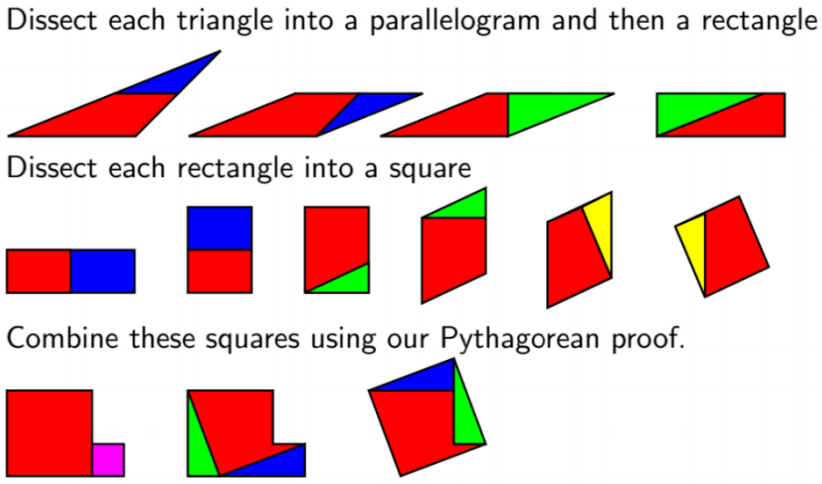


Figure 128.4: The visual chain: triangle to parallelogram/rectangle, rectangles to squares, then combine squares.

The theorem establishes the baseline: for tame plane polygons, equal area exactly characterizes finite dissection congruence.

§128.3 Hilbert's third problem and the Dehn invariant

Hilbert's third problem asks whether equal volume similarly characterizes dissection congruence of three-dimensional polytopes. Dehn's theorem says no: a cube and a regular tetrahedron of the same volume are not dissection congruent.

The extra obstruction is the Dehn invariant. If a polyhedron has edge lengths ℓ_i and corresponding dihedral angles θ_i, the invariant has the form

$$D(P) = \sum_i \ell_i \otimes \theta_i$$

with values in $\mathbb{R} \otimes_{\mathbb{Q}} (\mathbb{R}/\pi\mathbb{Q})$, where angles differing by rational multiples of π represent the same class. The invariant records length-angle data that volume does not see; the tensor product signals that this is no longer elementary area bookkeeping.

Sydler's theorem says that volume and Dehn invariant together are complete in dimension three. Thus the two-dimensional polygon theorem has no naive three-dimensional analogue.

§128.4 Measure-theoretic background

Measure introduces a fundamental obstruction. Vitali sets imply that for every $n \geq 1$, there is no extension of Lebesgue measure to all subsets of $\mathbb{R}^n$ that is both countably additive and invariant under isometries.

If countable additivity is weakened to finite additivity, the dimension matters. For $n \geq 3$, the Banach–Tarski paradox rules out a finitely additive isometry-invariant measure on all bounded sets compatible with volume, whereas Banach measures exist for $n \leq 2$. The reason is group-theoretic: in dimension at least three, the isometry group contains enough free-group behavior for paradoxical decompositions; in dimensions one and two it does not.

For Tarski's problem in the plane, this means that equal area is still a necessary condition. A disk and square that are equidecomposable by isometries must have the same Lebesgue measure, even if the pieces themselves are nonmeasurable.

§128.5 Tarski's circle-squaring problem

The central question is then:

> Are a disk and a square in $\mathbb{R}^2$ of the same area equidecomposable?

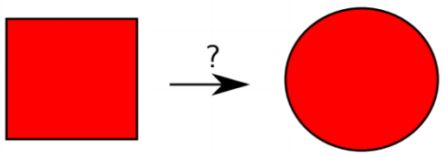


Figure 128.5: The central visual question: equal-area disk and square.

At first sight the boundary seems to obstruct the construction. A disk has circular boundary; a square has straight boundary. This intuition is correct for a stricter notion, but not for arbitrary equidecomposition.

§128.6 Scissors congruence and why nice pieces fail

The note distinguishes equidecomposability from scissors congruence. In the scissors-congruence version, the pieces must be tame: each piece is homeomorphic to a disk and has boundary given by curves of finite length. Under such restrictions, the disk and square are not scissors congruent.

The relevant signed boundary quantity is:

$$\text{convex circular perimeter} - \text{concave circular perimeter}.$$

When a curved circular arc is created inside a dissection, it is paired with an opposite concave arc, so the internal contributions cancel. The remaining outer boundary contribution is preserved. A disk has a nonzero circular-boundary contribution, while a square has none. Hence a disk and square cannot be scissors congruent in this tame sense.

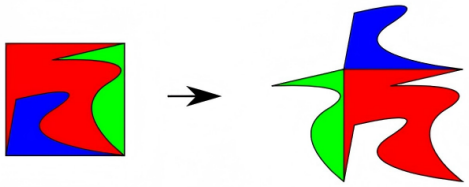


Figure 128.6: Tame scissors pieces preserve circular-boundary information.

This is the first major subtlety: the positive answer to Tarski's problem requires pieces much wilder than ordinary scissors pieces.

§128.7 Laczkovich's positive theorem

Laczkovich proved in 1990 that Tarski's circle-squaring problem has a positive answer. In 1992 he proved a broader theorem. A simplified statement is:

Theorem 128.7.1 (Laczkovich)
Let $A, B \subseteq \mathbb{R}^k$ be bounded sets with the same positive Lebesgue measure. If the boundaries of A and B have upper Minkowski dimension strictly less than k, then A and B are equidecomposable by translations.

For a disk and square in the plane, the boundaries are one-dimensional, so the hypothesis applies.

The proof does not draw a literal jigsaw. It converts the problem into a bounded matching problem in a graph generated by translations.

§128.8 First idea: work in the torus

Scale and translate A and B so that they lie inside $[0,1)^k$. Identify this cube with the k-torus

$$\mathbb{T}^k = (\mathbb{R}/\mathbb{Z})^k.$$

Then A and B are equidecomposable by translations as subsets of the torus iff, after possibly using more pieces, they are equidecomposable by translations in $\mathbb{R}^k$.

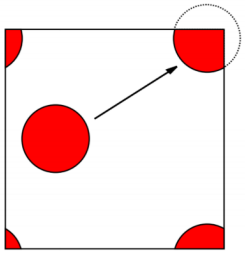


Figure 128.7: Working on the torus: translations wrap around the fundamental cube.

This torus reduction matters because translations become an action on a compact group. Compactness and periodic boundary conditions make the orbit structure manageable.

§128.9 Random translations and a free action

Choose a sufficiently large integer d and random translations

$$u_1, \dots, u_d \in \mathbb{T}^k.$$

They define an action of $\mathbb{Z}^d$ on $\mathbb{T}^k$ by

$$(n_1, \dots, n_d) \cdot x = n_1 u_1 + \cdots + n_d u_d + x.$$

For almost every choice of the u_i, this action is free: no nonzero element of $\mathbb{Z}^d$ fixes a point. Thus each orbit can be visualized as a copy of $\mathbb{Z}^d$.

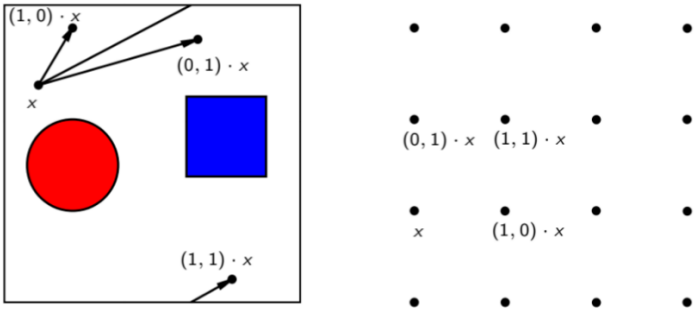


Figure 128.8: Random translations generate lattice-like orbits on the torus.

§128.10 The graph and bounded-distance bijection

Define a graph G with vertex set $\mathbb{T}^k$. Two points x, y are adjacent if there exists $g \in \mathbb{Z}^d$ with

$$g \cdot x = y, \qquad \|g\|_\infty = 1.$$

So edges correspond to single generator moves.

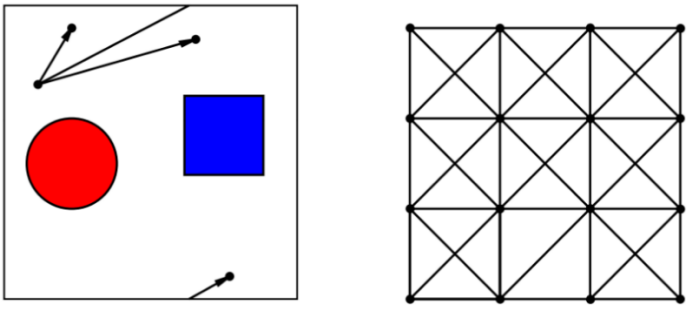


Figure 128.9: The generator graph: locally it looks like a lattice with bounded moves.

To show A and B are equidecomposable, it suffices to find a bijection

$$f : A \to B$$

of bounded distance in G: there is some fixed N such that

$$d_G(x, f(x)) \leq N \qquad \text{for all } x \in A.$$

If such an f exists, then only finitely many group elements g can occur. Put

$$A_g = \{x \in A : f(x) = g \cdot x\}.$$

The sets A_g partition A, and the translated sets $g \cdot A_g$ partition B. This gives the desired finite equidecomposition.

This is the conceptual heart of the proof:

$$\text{finite translation equidecomposition} \quad \Longleftarrow \quad \text{bounded-distance orbit matching.}$$

§128.11 Orbit matching

Inside a single orbit of the $\mathbb{Z}^d$ action, the problem becomes matching red A-points to blue B-points by bounded paths.

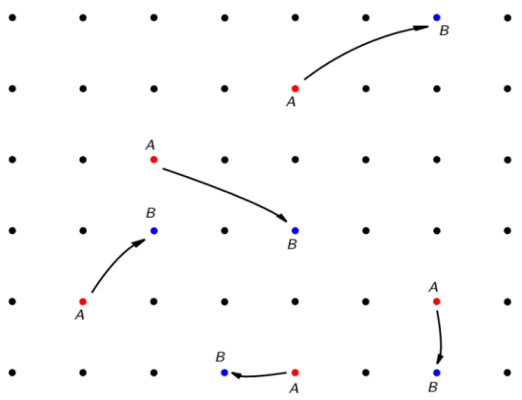


Figure 128.10: Within one orbit, equidecomposition becomes a matching problem.

A bounded matching can exist only if large finite regions of the orbit contain approximately the same number of A-points and B-points. This leads to discrepancy estimates.

§128.12 Discrepancy estimates

For an orbit point x, define a finite box

$$S_N(x) = \{(n_1, \dots, n_d) \cdot x : 0 \le n_i < N\}.$$

Ergodic intuition suggests

$$|S_N(x) \cap A| \approx \lambda(A) N^d.$$

This is only a heuristic: ordinary ergodic theorems alone are not strong enough. Laczkovich needs a uniform quantitative estimate from Diophantine approximation and discrepancy theory.

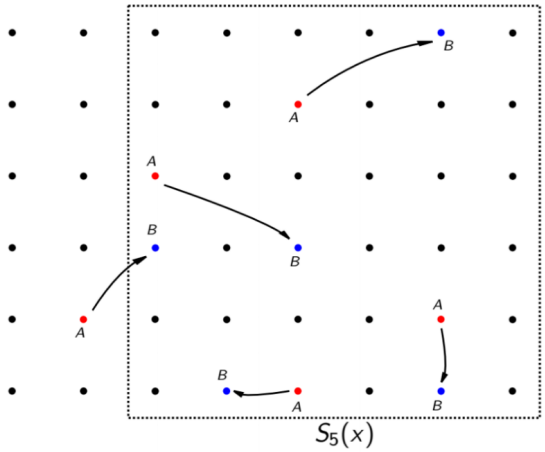


Figure 128.11: A finite orbit box $S_N(x)$; discrepancy compares observed counts with expected measure.

The key lemma states that there exist $\varepsilon > 0$ and M such that for all x and N,

$$\Big||S_N(x) \cap A| - \lambda(A) N^d\Big| \le M N^{d-1-\varepsilon},$$

and similarly for B:

$$\Big||S_N(x) \cap B| - \lambda(B) N^d\Big| \le M N^{d-1-\varepsilon}.$$

Since $\lambda(A) = \lambda(B)$, large boxes contain nearly equal numbers of A and B points. The error term grows more slowly than the volume term N^d, which is exactly what makes matching possible.

§128.13 From discrepancy to a bounded transport flow

The discrepancy estimate is useful because it places a quantitative restriction on how much mass can accumulate in one part of an orbit. Write

$$b(x) = \mathbf{1}_A(x) - \mathbf{1}_B(x)$$

for the surplus at a vertex of the orbit graph. A point of $A \setminus B$ contributes $+1$, a point of $B \setminus A$ contributes -1, and all other vertices have zero net surplus. A transport flow is an antisymmetric function on oriented edges,

$$\varphi(x, y) = -\varphi(y, x),$$

with discrete divergence

$$\boxed{\sum_{y\sim x}\varphi(x,y)=b(x).}$$

This is an exact conservation equation, not a statistical approximation.

Its meaning is already clear on four consecutive vertices v_0, v_1, v_2, v_3. Suppose $A = \{v_0, v_1\}$ and $B = \{v_2, v_3\}$. Sending one unit across v_0v_1, two across v_1v_2, and one across v_2v_3 gives

x	v_0	v_1	v_2	v_3
$b(x)$	1	1	-1	-1

because the outgoing minus incoming flows are exactly $1, 1, -1, -1$. The amount crossing each edge equals the cumulative imbalance on one side. The one-dimensional model explains the difficulty in higher dimensions: if a region has a large surplus, that surplus must be carried across the region's boundary.

Now use Laczkovich's discrepancy estimate for the same box $S_N(x)$, both for A and for B. Since $\lambda(A) = \lambda(B)$, subtraction gives

$$\left|\sum_{y\in S_N(x)} b(y)\right| \le 2MN^{d-1-\varepsilon}.$$

A d-dimensional box has $O(N^{d-1})$ boundary edges, so this surplus is smaller than the boundary scale by a factor $O(N^{-\varepsilon})$. At dyadic scales $N = 2^j$, the corresponding scale factors are summable:

$$\sum_{j\ge 0} 2^{-j\varepsilon} < \infty.$$

This is the critical quantitative feature. A mere ergodic estimate $o(N^d)$ would not control the amount of flow that must cross boundaries at successive scales. The stronger discrepancy estimate makes uniformly bounded multi-scale corrections plausible.

Constructing such a flow is a theorem-level step: the corrections must be routed consistently across overlapping boxes and across every orbit. The bound above explains the role of the exponent; it is not, by itself, a proof that the infinite network has a bounded solution. Laczkovich's method and its Borel refinement supply the additional combinatorial construction.

§128.14 Borel Pieces of a Real Flow

A uniformly bounded real-valued flow still does not immediately give a translation partition. A single point cannot be divided into a fraction of a point, whereas a real flow is permitted to carry noninteger quantities. An integer-valued flow is better adapted to matching discrete vertices.

In finite networks, one can see the difference by the integral max-flow theorem: when supplies and capacities are integers, a feasible flow can be chosen integer valued. For the infinite orbit graph there is an extra compatibility problem. Independently choosing an integral flow on each finite subgraph may fail to produce a globally bounded or definable choice.

There is also a second, logically separate issue. Even if every orbit admits an appropriate matching, making one arbitrary choice on each of uncountably many orbits need not yield a Borel function of the starting vertex. A valid equidecomposition by arbitrary pieces need not be a Borel equidecomposition.

The theorem of Marks and Unger solves both obstacles for bounded Borel sets $A, B \subseteq \mathbb{R}^k$ with

$$\lambda(A) = \lambda(B) > 0, \qquad \overline{\dim}_M(\partial A),\ \overline{\dim}_M(\partial B) < k.$$

Their proof first builds a bounded *Borel real flow* using discrepancy. It then uses special finite Borel partitions of free $\mathbb{Z}^d$-orbits, together with integral-flow arguments, to obtain a bounded *integer-valued Borel flow*. The final conversion produces a Borel bijection $f : A \to B$ that moves every point by one of finitely many translations. The underlying orbit partitions are a substantial input from Borel graph theory; bounded real flow alone is not the conclusion.

To see why this last property gives the desired pieces, let $F \subset \mathbb{Z}^d$ be the finite set of displacement vectors used by f. Define

$$A_g = \{x \in A : f(x) = g \cdot x\}, \qquad g \in F.$$

The sets A_g are Borel because f and the group action are Borel. They partition A, while the translates $g \cdot A_g$ partition B because f is a bijection. Only finitely many translations are required.

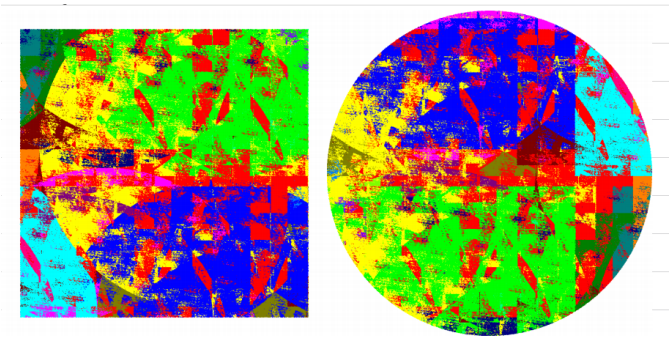


Figure 128.12: A schematic depiction of intricate Borel pieces: definability does not require polygonal or smooth boundaries.

For the circle and square, both boundaries have Minkowski dimension one, strictly below the ambient dimension two, so the theorem applies. The pieces are Borel, but they need not have finite-perimeter boundaries or resemble jigsaw pieces. A Borel decomposition is thus substantially more regular than a partition into arbitrary subsets, yet much less geometrically tame than ordinary scissors congruence.

One question has revealed three distinct invariance mechanisms. Euclidean polygon dissections are governed by area; polyhedral scissors congruence detects additional Dehn length-angle data; circle squaring is controlled by arithmetic discrepancy and graph transport once sufficiently complicated pieces are admitted. The regularity of those pieces determines which theorems can be used, and which obstructions survive.

Three Ways to Read a Triangle: Ratios, Circles, and Centers

N-20230119

A triangle can be treated as three points in an affine plane, as three points on a circle, or as three vectors measured from a metric center. Each description sees different phenomena. Ratios tell us when lines meet; complex multiplication tells us when four points share a circle; vector sums explain the otherwise mysterious alignment of triangle centers. The pleasure is in choosing the viewpoint in which a hard-looking theorem becomes a one-line computation.[1]

§129.1 Ceva and Menelaus in Coordinates

Let A, B, C be the vertices of a nondegenerate triangle. A point in the plane has *barycentric coordinates* $(\alpha : \beta : \gamma)$, meaning

$$P = \frac{\alpha A + \beta B + \gamma C}{\alpha + \beta + \gamma}$$

when the denominator is nonzero. In the projective completion, the triple remains meaningful even when its entries sum to zero. The coordinate α vanishes exactly on the line BC, and similarly for the other sidelines.

Put

$$D = (0 : 1 - t : t) \in BC,$$
$$E = (u : 0 : 1 - u) \in CA,$$
$$F = (1 - v : v : 0) \in AB.$$

For interior points, $0 < t, u, v < 1$; the formulas also describe extensions of the sides. The side ratios are

$$\frac{BD}{DC} = \frac{t}{1-t}, \qquad \frac{CE}{EA} = \frac{u}{1-u}, \qquad \frac{AF}{FB} = \frac{v}{1-v},$$

where ratios use directed coordinates along $B \to C$, $C \to A$, and $A \to B$.

When do the three cevians AD, BE, CF meet? If their intersection is $P = (\alpha : \beta : \gamma)$, the first cevian gives

$$\frac{\gamma}{\beta} = \frac{t}{1-t},$$

and the others give

$$\frac{\alpha}{\gamma} = \frac{u}{1-u}, \qquad \frac{\beta}{\alpha} = \frac{v}{1-v}.$$

Multiplying, every coordinate cancels:

$$\boxed{\frac{BD}{DC}\frac{CE}{EA}\frac{AF}{FB} = 1.}$$

Conversely, if this product equals one and the denominators are nonzero, choose nonzero coordinates α, β, γ satisfying the first two ratios. Their product condition forces the

[1] For the classical results and their elementary proofs see Coxeter and Greitzer, *Geometry Revisited*, and R. A. Johnson, *Advanced Euclidean Geometry*.

third, giving a common projective point. For interior side points this is the usual affine concurrence of Ceva's theorem.

Now ask the *dual* question: when do the three points D, E, F lie on one line? Collinearity means the determinant of their coordinate triples is zero:

$$\det \begin{pmatrix} 0 & 1-t & t \\ u & 0 & 1-u \\ 1-v & v & 0 \end{pmatrix} = (1-t)(1-u)(1-v) + tuv.$$

Thus

$$\boxed{\frac{BD}{DC}\frac{CE}{EA}\frac{AF}{FB} = -1}$$

when the three denominators are nonzero. This is Menelaus' theorem with the same ratio convention. What changes from $+1$ to -1 is the geometric question, not an arbitrary mnemonic about signs.

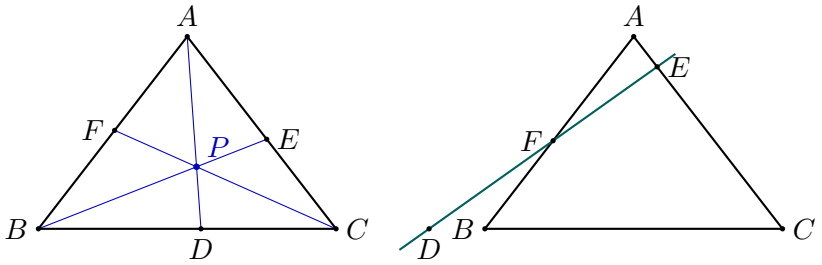


Figure 129.1: The same three side points can be studied by cevians (a common solution of line equations) or by a transversal (a vanishing determinant).

Barycentric coordinates have stripped the triangle of its Euclidean metric: no lengths, angles, or perpendiculars appeared. The resulting statements are projective incidence conditions. Their expression in products of ordinary side ratios, however, assumes an affine chart and directed coordinates; an arbitrary projective transformation need not preserve each individual affine ratio.

§129.2 Ptolemy from Triangle Inequality

Take four points A, B, C, D in this cyclic order on a convex quadrilateral. Ptolemy's theorem claims

$$AC \cdot BD = AB \cdot CD + BC \cdot AD$$

if the quadrilateral is cyclic. But why should a product of diagonals equal a *sum* of products of sides?

Represent the vertices by complex numbers a, b, c, d. The elementary identity

$$(a-b)(c-d) + (a-d)(b-c) = (a-c)(b-d)$$

holds for any four complex numbers. Taking absolute values gives

$$\boxed{|a-c||b-d| \le |a-b||c-d| + |a-d||b-c|.}$$

We have obtained the *Ptolemy inequality* for arbitrary four points in the Euclidean plane, without drawing any extra circle.

Equality in the triangle inequality occurs when the two summands $(a-b)(c-d)$ and $(a-d)(b-c)$ have the same argument and orientation. For distinct points in a convex configuration, this is the cyclicity condition. Indeed their ratio

$$\frac{(a-b)(c-d)}{(a-d)(b-c)}$$

is a positive real number precisely when the directed inscribed angles cut out the same circle with the stated cyclic order. Thus the cyclic case is the equality case of an elementary complex inequality.

There is an even more visual version: invert about A. A circle through A becomes a line, so B, C, D turn into three collinear points B', C', D'. Ptolemy's equality then becomes the addition of the two shorter segments on a line to obtain the longest one.

To see this algebraically, invert with radius ρ. For any $U, V \neq A$, inversion gives

$$U'V' = \frac{\rho^2 UV}{AU \cdot AV}.$$

If A, B, C, D occur on the circle in that order, the image points B', C', D' lie in order along a line, so

$$B'D' = B'C' + C'D'.$$

Multiply by $AB \cdot AC \cdot AD/\rho^2$ to recover

$$AC \cdot BD = AD \cdot BC + AB \cdot CD.$$

This is a memorable reason why Ptolemy has an addition sign: it is the segment-addition axiom after a circle is straightened.

§129.3 The Simson Line

Let P lie on the circumcircle of ABC. Project P perpendicularly onto the full lines AB, BC, CA, obtaining F, D, E. Simson's theorem says that D, E, F are collinear.

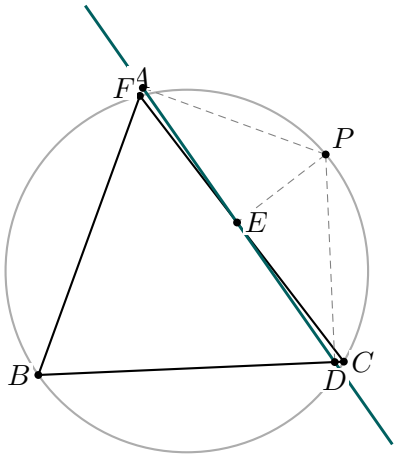


Figure 129.2: Project one point of the circumcircle onto all three sidelines; the perpendicular feet unexpectedly lie on a single line.

A clean proof uses the unit circle. A similarity lets us take its center as the origin and radius as one, so the complex coordinates of a, b, c, p have absolute value one. The perpendicular foot from p to the line through u, v has complex coordinate

$$\boxed{h_{uv} = \frac{1}{2}(p + u + v - uv\overline{p}).}$$

To verify this, note that the line through u, v is

$$z + uv\overline{z} = u + v.$$

The proposed h_{uv} satisfies the equation, and $p - h_{uv}$ is perpendicular to the line.

Subtract two foot coordinates:

$$h_{ab} - h_{ac} = \frac{1}{2}(b - c)(1 - a\overline{p}), \qquad h_{ab} - h_{bc} = \frac{1}{2}(a - c)(1 - b\overline{p}).$$

Since $|p| = 1$, we have $1 - a\overline{p} = (p - a)/p$. Therefore

$$\frac{h_{ab} - h_{ac}}{h_{ab} - h_{bc}} = \frac{(b - c)(p - a)}{(a - c)(p - b)}.$$

This is a cross ratio of four points on the unit circle, hence real. Two complex difference vectors have real quotient exactly when they point along the same undirected line. It follows that h_{ab}, h_{ac}, h_{bc} are collinear. The special cases in which two feet coincide follow by continuity.

This proof identifies the hidden bridge: perpendicular projection is a *metric* operation, but the final collinearity is detected by a *circle cross ratio*. The same reality criterion also underlies the equality case of Ptolemy's inequality.

§129.4 The Euler Line from Vectors

Let O be the circumcenter and use it as the vector origin. Write the position vectors of the vertices as a, b, c, with

$$|a| = |b| = |c| = R.$$

Consider the vector

$$h = a + b + c.$$

The line through A in direction $h - a = b + c$ is perpendicular to BC, because

$$(b + c) \cdot (b - c) = |b|^2 - |c|^2 = 0.$$

Thus the point represented by h lies on the altitude from A; the same calculation for B and C proves that h is the orthocenter H.

The centroid has vector

$$g = \frac{a + b + c}{3} = \frac{h}{3}.$$

Therefore O, G, H lie on one line and

$$\boxed{\overrightarrow{OG} = \frac{1}{3}\overrightarrow{OH}, \qquad OG : GH = 1 : 2,}$$

whenever $O \neq H$. In the equilateral case, all three centers coincide. Euler's line is simply the line along a vector sum taken from the circumcenter.

The same origin gives a striking proof of Euler's relation between the incenter I, inradius r, circumradius R, and their distance $OI = d$. Let the side lengths opposite A, B, C be a_0, b_0, c_0, and write $s = a_0 + b_0 + c_0$. The incenter is the weighted average

$$i = \frac{a_0 a + b_0 b + c_0 c}{s}.$$

The elementary weighted-variance identity, valid when the weights sum to one, reads

$$\sum_j w_j |v_j|^2 - \left|\sum_j w_j v_j\right|^2 = \sum_{j<k} w_j w_k |v_j - v_k|^2.$$

Apply it to $v_1 = a, v_2 = b, v_3 = c$, with weights $a_0/s, b_0/s, c_0/s$. All three squared vertex norms are R^2, and the three squared pairwise distances are c_0^2, a_0^2, b_0^2. Hence

$$\begin{aligned} R^2 - |i|^2 &= \frac{a_0 b_0 c_0^2 + b_0 c_0 a_0^2 + c_0 a_0 b_0^2}{s^2} \\ &= \frac{a_0 b_0 c_0}{s}. \end{aligned}$$

The area Δ satisfies

$$\Delta = \frac{a_0 b_0 c_0}{4R} = r\frac{s}{2}.$$

Consequently $a_0 b_0 c_0 / s = 2Rr$, giving

$$\boxed{OI^2 = R^2 - 2Rr.}$$

In particular $R \geq 2r$, with equality only if $O = I$, i.e. the triangle is equilateral.

An elementary triangle-center formula has emerged from the same operation that appears in statistics as a variance decomposition. It measures how far the weighted center I lies from the equal-radius vertex circle.

§129.5 What changes when the allowed transformations change?

The computations distinguish three useful geometries. Barycentric coordinates reveal concurrence and collinearity without requiring a metric. Inversion and complex coordinates reveal why cyclicity supports Ptolemy and why projected feet form a Simson line. Euclidean inner products are needed to define perpendiculars and circle centers, but they make Euler's center relations almost effortless.

The distinction matters in practice. A general affine transformation preserves the centroid and collinearity, but it need not preserve circumcenters or perpendicular feet. A projective transformation preserves incidences, but ordinary ratios on a line need not remain unchanged. A Möbius transformation preserves generalized circles and directed angles, but it does not preserve ordinary distances. Recognizing which transformations preserve a theorem lets us simplify the drawing without accidentally destroying the property we are trying to prove.

Special Proofs of Trigonometric Addition Formulas

N-20230220

§130.1 Why prove the same identity more than once?

The addition formulas are familiar:

$$\cos(\alpha+\beta)=\cos\alpha\cos\beta-\sin\alpha\sin\beta,$$

$$\sin(\alpha+\beta)=\sin\alpha\cos\beta+\cos\alpha\sin\beta.$$

A second proof is worthwhile only when it reveals something the first proof hides. A Euclidean proof shows how angle and signed area enter. A rotation-matrix proof identifies the formulas with a group law. A complex-exponential proof packages both identities into one multiplicative equation. A projective calculation explains why the tangent formula has a denominator and what happens when it vanishes.

These proofs also differ in logical cost. Some begin from synthetic Euclidean geometry. Some assume a classification of plane rotations. Some require power series or differential equations. The aim of this chapter is therefore not to accumulate tricks, but to compare proof architectures:

$$\text{assumptions}\longrightarrow\text{mechanism}\longrightarrow\text{conclusion}\longrightarrow\text{possible generalization}.$$

Throughout, put

$$u(\theta)=(\cos\theta,\sin\theta).$$

Sine and cosine are initially the coordinates of the unit-circle point at directed angle θ.

§130.2 A chord-length proof of the cosine difference formula

Take two points

$$P=u(\alpha),\qquad Q=u(\beta)$$

on the unit circle. The angle at the center between the two radii is $\alpha-\beta$, up to orientation. The law of cosines in the isosceles triangle OPQ gives

$$|P-Q|^2=1^2+1^2-2\cos(\alpha-\beta).$$

Thus

$$|P-Q|^2=2-2\cos(\alpha-\beta).$$

The same squared distance can be computed from coordinates:

$$|P-Q|^2=(\cos\alpha-\cos\beta)^2+(\sin\alpha-\sin\beta)^2.$$

Expanding and using

$$\cos^2\theta+\sin^2\theta=1$$

gives

$$|P-Q|^2=2-2(\cos\alpha\cos\beta+\sin\alpha\sin\beta).$$

Comparing the two expressions yields

$$\boxed{\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta.}$$

Replacing β by $-\beta$ and using the parity of sine and cosine gives the cosine addition formula.

This proof does not use rotation matrices or complex numbers. It does use the Euclidean distance formula, the Pythagorean identity, and the law of cosines. If the law of cosines was itself proved from the dot-product identity

$$u \cdot v = |u||v| \cos\angle(u, v),$$

then the proof is really an inner-product proof in geometric clothing. That is not a defect, but it should be stated honestly.

§130.3 Signed area proves the sine difference formula

The determinant of two plane vectors is their signed parallelogram area:

$$\det(P, Q) = \det \begin{pmatrix} \cos\alpha & \cos\beta \\ \sin\alpha & \sin\beta \end{pmatrix}.$$

Computing the determinant gives

$$\det(P, Q) = \cos\alpha\sin\beta - \sin\alpha\cos\beta.$$

Geometrically, P and Q are unit vectors whose directed angle from P to Q is $\beta - \alpha$. The signed parallelogram area is therefore

$$\det(P, Q) = \sin(\beta - \alpha).$$

Consequently,

$$\boxed{\sin(\beta - \alpha) = \sin\beta\cos\alpha - \cos\beta\sin\alpha.}$$

Changing signs or interchanging α and β produces all sine addition and subtraction formulas.

Orientation is essential here. Ordinary unsigned area would give only

$$|\sin(\beta - \alpha)|.$$

The determinant remembers whether the second vector lies counterclockwise or clockwise from the first. The sign in the addition formula is therefore an orientation statement, not an arbitrary convention.

The cosine and sine proofs can be combined in one identity. In an oriented Euclidean plane,

$$P \cdot Q + i\det(P, Q) = e^{i(\beta-\alpha)}.$$

The real part records metric information; the imaginary part records oriented area.

§130.4 The rotation proof begins from a group law

Let $R(\theta)$ be the unique orientation-preserving Euclidean isometry fixing the origin and sending the first basis vector to $u(\theta)$. Its second column must be the positive quarter-turn of the first, so

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

Rotating first by β and then by α adds directed angles. Hence

$$R(\alpha)R(\beta) = R(\alpha+\beta).$$

This statement is geometric: it says that composition in the rotation group corresponds to addition of angle parameters.

Matrix multiplication gives

$$R(\alpha)R(\beta) = \begin{pmatrix} \cos\alpha\cos\beta - \sin\alpha\sin\beta & -\cos\alpha\sin\beta - \sin\alpha\cos\beta \\ \sin\alpha\cos\beta + \cos\alpha\sin\beta & -\sin\alpha\sin\beta + \cos\alpha\cos\beta \end{pmatrix}.$$

Comparing with $R(\alpha+\beta)$ proves both addition formulas at once.

The proof is short, but it is not assumption-free. One must know that rotations compose by adding directed angles and that the displayed matrix represents the geometric rotation independently of the formulas being proved. If the matrix formula for $R(\theta)$ was itself derived by applying the addition formulas to a general vector, the reasoning would be circular.

The structural advantage is substantial. The formulas are now matrix coefficients of a representation

$$R : \mathbb{R}/2\pi\mathbb{Z} \longrightarrow SO(2).$$

The identity

$$R(\alpha+\beta) = R(\alpha)R(\beta)$$

is a homomorphism law, and the trigonometric identities are its coordinate expansion.

§130.5 An analytic Euler proof with no hidden circularity

Euler's formula is often used in one line:

$$e^{i\theta} = \cos\theta + i\sin\theta.$$

Then

$$e^{i(\alpha+\beta)} = e^{i\alpha}e^{i\beta}$$

immediately yields the addition formulas by comparing real and imaginary parts. This is valid only if Euler's formula and the exponential law were established without already using the addition formulas.

One noncircular route begins from power series. Define

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!},$$

$$\cos z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!}, \qquad \sin z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!}.$$

Absolute convergence justifies regrouping terms, and separating even and odd powers gives

$$e^{iz} = \cos z + i \sin z.$$

The Cauchy product of the exponential series proves

$$e^{z+w} = e^z e^w.$$

Therefore

$$\cos(\alpha + \beta) + i \sin(\alpha + \beta) = (\cos \alpha + i \sin \alpha)(\cos \beta + i \sin \beta).$$

Multiplying the right-hand side and comparing components proves the formulas.

A second noncircular analytic route uses differential equations. Let $z(t)$ solve

$$z'(t) = iz(t), \qquad z(0) = 1.$$

Uniqueness implies the flow law

$$z(s + t) = z(s)z(t).$$

Writing $z = x + iy$ gives

$$x' = -y, \qquad y' = x, \qquad x(0) = 1, \quad y(0) = 0.$$

If cosine and sine are defined as the solutions of this system, then

$$z(t) = \cos t + i \sin t,$$

and the flow law again yields the addition formulas.

These analytic proofs generalize well because exponentials and flows exist in many algebras and Lie groups. Their cost is that they depend on analysis rather than elementary circle geometry.

§130.6 Where the circular Euler proof enters

There is another common derivation of Euler's formula. One starts with the trigonometric addition formulas, proves that

$$\theta \longmapsto \cos \theta + i \sin \theta$$

is multiplicative, and then identifies this multiplicative map with $e^{i\theta}$. This is an excellent derivation of Euler's formula from trigonometry.

It cannot then be reversed and advertised as an independent proof of the addition formulas. The logical loop would be

$$\text{addition formulas} \Longrightarrow \text{Euler formula} \Longrightarrow \text{addition formulas}.$$

The final implication is correct, but the chain as a whole proves nothing new.

The issue is not that a theorem may never be used in both directions. Equivalent statements often illuminate each other. The issue is whether one is claiming an independent foundation. To break the loop, Euler's formula must enter from a separate construction such as power series, an ODE, or the complex exponential defined by functional analysis.

§130.7 Tangent lives on a projective line

Once the sine and cosine formulas are known, division gives

$$\tan(\alpha+\beta)=\frac{\tan\alpha+an\beta}{1-an\alpha an\beta},$$

provided the expressions are finite. A direct slope calculation explains the fractional-linear shape.

Represent a nonvertical direction by a vector $(1,m)^T$, where m is its slope. Rotating it through α gives

$$R(\alpha)\begin{pmatrix}1\\m\end{pmatrix}=\begin{pmatrix}\cos\alpha-m\sin\alpha\\ \sin\alpha+m\cos\alpha\end{pmatrix}.$$

The new slope is

$$m'=\frac{\sin\alpha+m\cos\alpha}{\cos\alpha-m\sin\alpha}=\frac{\tan\alpha+m}{1-man\alpha}.$$

Putting $m=\tan\beta$ gives the tangent addition formula.

If

$$1-man\alpha=0,$$

the rotated direction is vertical. The denominator does not signal a breakdown of rotation; it says that the affine slope chart has reached the point at infinity of the projective line

$$\mathbb{RP}^1.$$

The transformation

$$m\longmapsto\frac{\tan\alpha+m}{1-man\alpha}$$

is a Möbius transformation on that projective line. This is why proving tangent first is usually awkward: tangent is not a global coordinate, whereas sine and cosine remain finite everywhere.

§130.8 A dependency graph for the proofs

The four routes can be displayed as a directed acyclic graph only after the foundational choices are fixed.

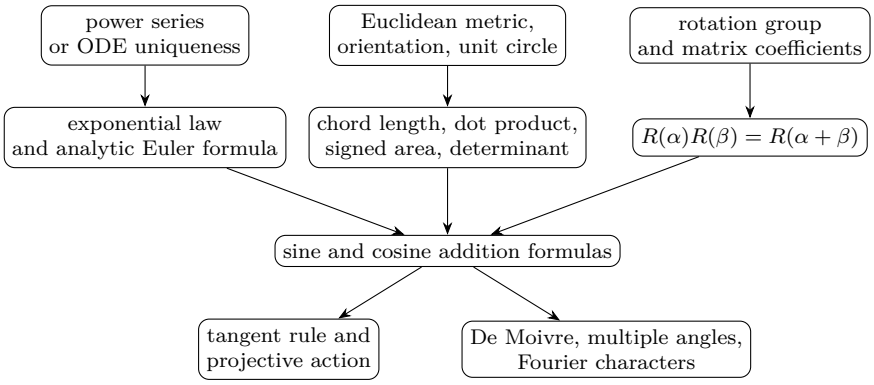


Figure 130.1: Three independent routes to the addition formulas. A geometrically derived Euler formula would lie downstream of the central box and cannot be moved upstream without circularity.

The graph also clarifies equivalence. Once the addition formulas are known, they can reconstruct the rotation homomorphism and the multiplicativity of $\cos\theta + i\sin\theta$. Those reverse arrows are mathematically useful, but they belong to an equivalence diagram, not to a foundational proof DAG.

§130.9 The same group template produces hyperbolic identities

The rotation matrix is built from an element whose square is $-I$. Hyperbolic functions arise from a generator whose square is I. Let

$$K = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad K^2 = I.$$

Then

$$e^{tK} = I\cosh t + K\sinh t = \begin{pmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{pmatrix}.$$

Write this matrix as $H(t)$. The exponential law gives

$$H(s+t) = H(s)H(t).$$

Multiplying matrices yields

$$\boxed{\cosh(s+t) = \cosh s\cosh t + \sinh s\sinh t,}$$

$$\boxed{\sinh(s+t) = \sinh s\cosh t + \cosh s\sinh t.}$$

The sign difference from circular cosine is explained by

$$K^2 = I \qquad \text{instead of} \qquad J^2 = -I.$$

Moreover,

$$H(t)^T \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} H(t) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Thus $H(t)$ preserves the Minkowski quadratic form $x^2 - y^2$, just as $R(t)$ preserves the Euclidean form $x^2 + y^2$. Circular and hyperbolic addition laws are two real two-dimensional representations of one-parameter groups.

Spherical trigonometry does not arise by replacing every symbol in this calculation. Rotations about different axes in three dimensions generally do not commute, and the sides of a spherical triangle are not one additive parameter along a fixed subgroup. Genuine generalization requires identifying the preserved structure, not merely changing sin into another named function.

§130.10 Which proof should be used?

The answer depends on what has already been developed and what one wants next.

Proof route	Main prerequisites	What becomes transparent	Best continuation
Chord and area	Euclidean congruence, distance, orientation	metric and sign	triangle geometry
Rotation matrices	classification and composition of rotations	group law and representation	linear algebra, Lie groups
Power series or ODE	convergence or uniqueness theory	exponential flow	complex analysis, differential equations
Projective slope	rotation action on directions	tangent denominator and infinity	projective geometry

For formal verification, the analytic series route has explicit algebraic subgoals but requires substantial infrastructure for convergence and rearrangement. The matrix route is short once a library already contains rotations as orientation-preserving isometries. The synthetic route may look elementary on paper but demands careful encoding of directed angles, congruence, and orientation. A proof assistant rewards explicit dependencies; it does not automatically make the visually shortest proof the formally shortest one.

The important lesson is not that one proof wins. The geometric proof identifies the quantities being measured. The matrix proof identifies the composition law. The analytic proof exposes a flow and generalizes to exponentials in other algebras. The projective calculation explains a singular denominator. Together they show how one identity can sit at the intersection of geometry, algebra, analysis, and representation theory without those subjects becoming logically indistinguishable.

Trigonometric Systems, Chebyshev Polynomials, and Tridiagonal Determinants

N-20230223

§131.1 Constraints in Trigonometric Systems

Consider the original problem

$$\sin\alpha+\sin\beta=\frac{1}{2},\qquad \cos\alpha-\cos\beta=\frac{1}{3}.$$

The desired quantity is $\cos\alpha\cos\beta$. Introducing four unknowns $a=\sin\alpha$, $b=\sin\beta$, $c=\cos\alpha$, $d=\cos\beta$ gives the two linear equations

$$a+b=\frac{1}{2},\qquad c-d=\frac{1}{3},$$

but they cannot be solved independently: one must retain the circle equations

$$a^2+c^2=1,\qquad b^2+d^2=1.$$

A determinant argument treating a,b,c,d as free coordinates would introduce spurious solutions. The trigonometric variables must instead be linked through the same pair of angles; their circle constraints are part of the algebraic system.

The right substitution uses the shared geometry of the angles:

$$S=\frac{\alpha+\beta}{2},\qquad D=\frac{\alpha-\beta}{2}.$$

Sum-to-product identities give

$$2\sin S\cos D=\frac{1}{2},\qquad -2\sin S\sin D=\frac{1}{3}.$$

Thus

$$\sin S\cos D=\frac{1}{4},\qquad \sin S\sin D=-\frac{1}{6}.$$

Without dividing by any potentially zero factor, square and add to obtain

$$\sin^2 S=\frac{1}{16}+\frac{1}{36}=\frac{13}{144}.$$

Consequently

$$\cos(\alpha+\beta)=\cos 2S=1-2\sin^2 S=\frac{59}{72}.$$

To find the difference angle, use

$$\cos 2D=\frac{(\sin S\cos D)^2-(\sin S\sin D)^2}{\sin^2 S}=\frac{1/16-1/36}{13/144}=\frac{5}{13}.$$

Now the product-to-sum identity completes the problem:

$$\boxed{\cos\alpha\cos\beta=\frac{\cos(\alpha+\beta)+\cos(\alpha-\beta)}{2}=\frac{1}{2}\left(\frac{59}{72}+\frac{5}{13}\right)=\frac{1127}{1872}.}$$

In contrast, dividing the original equations to get $-\tan D=2/3$ is legitimate here only after proving that the common factor $\sin S\cos D$ is nonzero; the first displayed equation supplies exactly that proof.

§131.2 Rational parametrization and multiple-angle formulas

An alternative solution method for some circle equations is

$$t = \tan\frac{x}{2}, \qquad \cos x = \frac{1-t^2}{1+t^2}, \qquad \sin x = \frac{2t}{1+t^2}.$$

It comes from the standard stereographic parametrization of the unit circle. The point $(-1, 0)$ is missing for finite real t, so a rationally transformed equation must either handle it separately or impose the appropriate domain. This matters when clearing denominators.

The same circle structure produces multiple-angle identities. De Moivre's formula is

$$(\cos x + i \sin x)^n = \cos nx + i \sin nx.$$

Taking real and imaginary parts of the binomial expansion yields

$$\cos nx = \sum_{j=0}^{\lfloor n/2 \rfloor} (-1)^j \binom{n}{2j} \cos^{n-2j} x \sin^{2j} x,$$

$$\sin nx = \sum_{j=0}^{\lfloor (n-1)/2 \rfloor} (-1)^j \binom{n}{2j+1} \cos^{n-2j-1} x \sin^{2j+1} x.$$

These formulas answer the original question about the alternating signs and binomial coefficients: both are forced by powers of i.

The addition identities also give second-order recurrences:

$$\cos nx = 2 \cos x \cos((n-1)x) - \cos((n-2)x),$$

$$\sin nx = 2 \cos x \sin((n-1)x) - \sin((n-2)x).$$

The tangent recurrence

$$\tan nx = \frac{\tan((n-1)x) + \tan x}{1 - \tan((n-1)x) \tan x}$$

holds only where all displayed tangent expressions and the quotient are defined. It is often better to work with sine and cosine when the denominator may vanish.

§131.3 Determinants and Trigonometric Recurrences

Let $A_n(\theta)$ be the $n \times n$ tridiagonal matrix with $2\cos\theta$ on its diagonal and 1 on the two adjacent diagonals:

$$A_n(\theta) = \begin{pmatrix} 2\cos\theta & 1 & & \\ 1 & 2\cos\theta & \ddots & \\ & \ddots & \ddots & 1 \\ & & 1 & 2\cos\theta \end{pmatrix}.$$

With $D_n = \det A_n$ and the empty determinant $D_0 = 1$, expansion along the last row gives

$$D_n = 2\cos\theta \, D_{n-1} - D_{n-2}, \qquad D_1 = 2\cos\theta.$$

Define $U_0(x) = 1$, $U_1(x) = 2x$ and

$$U_n(x) = 2xU_{n-1}(x) - U_{n-2}(x).$$

By matching both initial values and the recurrence,

$$\boxed{D_n(\theta) = U_n(\cos\theta) = \frac{\sin((n+1)\theta)}{\sin\theta}} \qquad (\sin\theta \neq 0).$$

The trigonometric quotient extends to $\sin\theta = 0$ by continuity, whereas U_n is a polynomial valid for all x. For example,

$$D_2 = 4\cos^2\theta - 1, \qquad D_3 = 8\cos^3\theta - 4\cos\theta.$$

The recurrence and the two initial values determine the determinant uniquely. It is also easy to incorporate a modified first diagonal entry. Let $A_n(\delta, t)$ have off-diagonal entries 1, diagonal entries $2t$ except for the first entry δ, and write $E_n = \det A_n(\delta, t)$. Cofactor expansion yields

$$E_0 = 1, \quad E_1 = \delta, \qquad E_n = 2tE_{n-1} - E_{n-2}.$$

With the convention $U_{-1}(t) = 0$, the unique solution is

$$\boxed{E_n = \delta U_{n-1}(t) - U_{n-2}(t)} \qquad (n \geq 1).$$

For $n = 1$ this gives δ, and for $n = 2$ it gives $2t\delta - 1$. Changing the boundary entry changes the initial conditions, while the same Chebyshev recurrence governs the interior.

The recurrence and special-function formulas above follow the NIST *Digital Library of Mathematical Functions*, Sections 18.5 and 18.9.[1]

§131.4 Why the algebra and trigonometry agree

The common ingredient is a linear recurrence with fixed coefficients. The sequences $\cos nx$, $\sin nx$, and the tridiagonal determinants D_n all satisfy

$$u_n = 2c\,u_{n-1} - u_{n-2}, \qquad c = \cos x.$$

A second-order recurrence is determined by two initial values, and differences between the sequences come from those values rather than a different governing law.

One can make this structural connection explicit with the companion matrix

$$\begin{pmatrix} u_n \\ u_{n-1} \end{pmatrix} = \begin{pmatrix} 2c & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} u_{n-1} \\ u_{n-2} \end{pmatrix}.$$

Its eigenvalues are $e^{\pm ix}$ when $c = \cos x$. Powers of these eigenvalues explain the appearance of $\sin nx$ and $\cos nx$, while the determinant recurrence arises from Laplace expansion. This is the precise bridge between the seemingly unrelated pages of trigonometry and linear algebra.

[1] NIST Digital Library of Mathematical Functions, Chapter 18, especially §§18.5 and 18.9.

Projective Geometry: Completing Lines and Preserving Incidence

N-20240518

Two parallel affine lines never meet in their familiar plane. The projective plane is built so that the apparent exception becomes an ordinary incidence statement. Its construction, and the geometry of cross-ratios and conics below, follow the standard sequence in the University of Oxford's projective geometry lectures.[1]

§132.1 Projective points are one-dimensional subspaces

Over a field k, define

$$\mathbb{P}^2(k) = (k^3 \setminus \{0\})/k^\times, \qquad [x : y : z] = [\lambda x : \lambda y : \lambda z] \quad (\lambda \neq 0).$$

A projective point represents an entire one-dimensional subspace of k^3; a projective line represents a two-dimensional subspace. Thus the projective line with equation

$$ax + by + cz = 0$$

is not merely an affine line extended very far away: it is the set of all one-dimensional subspaces contained in the indicated plane through the origin of k^3.

In the chart $z \neq 0$, rescale to $z = 1$ and obtain the affine coordinates $(x/z, y/z)$. The projective points with $z = 0$ constitute the line at infinity relative to *this chosen chart*.

§132.2 Where two parallel affine lines intersect

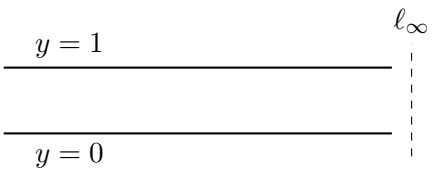
The affine equations $y = 0$ and $y = 1$ become the homogeneous projective equations

$$y = 0, \qquad y - z = 0.$$

Their intersection satisfies $y = z = 0$ and is therefore the single projective point

$$[1 : 0 : 0].$$

More generally, two distinct projective lines in $\mathbb{P}^2(k)$ arise from distinct two-dimensional subspaces of k^3; their intersection has dimension one. This linear-algebra argument proves that they meet in exactly one projective point.



The dashed line is a mnemonic for the missing directions, not a Euclidean line sitting at a large finite distance.

[1] https://courses.maths.ox.ac.uk/course/section.php?id=10755

§132.3 The dual plane and incidence

The coefficients $[a : b : c]$ of a line are themselves projective coordinates in the dual vector space $(k^3)^*$. The incidence relation

$$ax + by + cz = 0$$

pairs a point $[x : y : z]$ and a dual point $[a : b : c]$, and is independent of how either is scaled.

For distinct projective points u, v, a linear functional vanishing on their span describes their unique joining line. Dually, the two line functionals ℓ_1, ℓ_2 have a one-dimensional common kernel giving their unique intersection point. The dual formulations follow from the same rank-two linear algebra. No metric notions were needed.

§132.4 Projective Changes of Coordinates

An invertible matrix $A \in GL(3, k)$ acts by $[v] \mapsto [Av]$; scalar multiples of A give the same projective map, so the group is $PGL(3, k)$. As A takes subspaces to subspaces, it preserves incidence and collinearity.

It does not preserve Euclidean distances or angles. For example the affine shear

$$(x, y) \mapsto (x + y, y)$$

extends to a projective transformation but sends the perpendicular coordinate axes to nonperpendicular lines. Projective geometry studies properties surviving the larger group, not arbitrary measurements of a drawing.

§132.5 The cross-ratio really survives fractional linear maps

Write four distinct points on $\mathbb{P}^1(k)$ in an affine chart as a, b, c, d. Their cross-ratio is

$$[a, b; c, d] = \frac{(c - a)(d - b)}{(c - b)(d - a)}.$$

For $T(t) = (\alpha t + \beta)/(\gamma t + \delta)$ with $\alpha\delta - \beta\gamma \neq 0$, direct subtraction gives

$$T(u) - T(v) = \frac{(\alpha\delta - \beta\gamma)(u - v)}{(\gamma u + \delta)(\gamma v + \delta)}.$$

Substitute the four differences into the cross-ratio: every determinant factor and denominator cancels. Thus $[T(a), T(b); T(c), T(d)] = [a, b; c, d]$, wherever the affine expressions are defined. The homogeneous version extends this identity to points at infinity.

This calculation explains why cross-ratio, rather than length or angle, is a natural projective invariant.

§132.6 Conics and the choice of an affine window

A plane conic has a homogeneous quadratic equation $v^T Q v = 0$, with Q symmetric when the characteristic is not two. Under an invertible coordinate change $v = Au$, the same conic is represented by $u^T A^T Q A u = 0$. For nonsingular Q, this describes a nonsingular projective conic.

The familiar real distinctions among ellipse, parabola, and hyperbola arise from intersections with the chosen line at infinity. For instance, homogenizing

$$x^2 + y^2 = 1, \qquad y = x^2, \qquad x^2 - y^2 = 1$$

gives

$$x^2 + y^2 = z^2, \qquad x^2 = yz, \qquad x^2 - y^2 = z^2.$$

On $z = 0$, these have respectively no real points, one real point, and two real points (the point of tangency in the middle case has multiplicity two). A projective transformation can move the line at infinity, so these affine classifications need not survive it.

§132.7 A conic can be parametrized by a projective line

Consider the nonsingular conic $xz - y^2 = 0$. There is a map

$$\nu_2 : \mathbb{P}^1 \longrightarrow \mathbb{P}^2, \qquad [s : t] \longmapsto [s^2 : st : t^2].$$

Its image lies on the conic because $s^2t^2 - (st)^2 = 0$. Conversely, if $x \neq 0$ on the conic, rescale to $[1 : y/x : (y/x)^2]$; the remaining point $[0 : 0 : 1]$ is the image of $[0 : 1]$. Thus ν_2 is an isomorphism onto the conic (over any field, with the usual scheme interpretation).

> **Caution**
>
> The conic is isomorphic to $\mathbb{P}^1$ as a projective algebraic curve; it cannot be mapped onto a projective *line in* $\mathbb{P}^2$ by a projective linear automorphism of $\mathbb{P}^2$. Such an automorphism preserves polynomial degree and sends lines to lines, whereas this embedding is quadratic.

The completion of affine lines, duality of points and lines, invariance of cross-ratio, and quadratic parametrization of a conic are different consequences of a single choice: use homogeneous coordinates and linear subspaces to record incidence without making a particular affine chart fundamental.

Transformations of Conics and the Erlangen View of Geometry

N-20250211

§133.1 The organizing question

This chapter studies conic sections through transformations. The guiding question is not only how an ellipse, parabola, or hyperbola is described by an equation, but what kind of geometry regards two such objects as the same.

For transformations of the *same real affine plane*, the first part of a useful hierarchy really is

$$\text{Euclidean} \subset \text{similarity} \subset \text{affine} \subset \text{projective}.$$

Conformal and inversive geometry take another direction. A shear is affine and projective but does not preserve angles; inversion in a circle preserves angles (up to orientation) but generally takes straight lines to circles. The two families are therefore not successive levels of the same inclusion chain.

The interesting question is what each transformation actually preserves. This is the spirit of Klein's Erlangen program: a geometry concerns properties invariant under a specified group of transformations. We will test that principle with concrete conics, a cross-ratio calculation and an honest example of inversion.[1]

§133.2 Affine transformations

Definition 133.2.1. An affine transformation of $\mathbb{R}^n$ is a map

$$T(x) = Ax + b,$$

where A is a linear map and $b \in \mathbb{R}^n$. If $A \in GL(n, \mathbb{R})$, then T is an affine automorphism.

Affine transformations preserve straight lines, parallelism, ratios along a line, convexity, and barycentric combinations. They do not preserve lengths, angles, or circles.

Example 133.2.2

An ellipse can be sent to a circle by an affine transformation. If

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1,$$

then the affine map

$$(x, y) \mapsto (x/a, y/b)$$

sends it to the unit circle.

[1] For real projective conics, cross-ratios and polarity see Bobenko's *Geometry I* lecture notes, Sections 2.5 and 2.10–2.13.

§133.3 Conics in matrix form

A plane conic may be written as

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0.$$

In homogeneous coordinates $[x : y : z]$, this becomes

$$Ax^2 + Bxy + Cy^2 + Dxz + Eyz + Fz^2 = 0.$$

Equivalently,

$$X^T Q X = 0, \qquad X = (x, y, z)^T,$$

where Q is a symmetric 3×3 matrix.

Under a projective transformation $X \mapsto MX$, the conic transforms by

$$Q \mapsto M^{-T} Q M^{-1}.$$

Thus projective classification of conics becomes a question about symmetric bilinear forms up to change of coordinates.

A concrete matrix is the best antidote to the vague sentence that all nondegenerate conics are projectively equivalent. The parabola

$$C_1 : \quad y = x^2$$

has homogeneous equation

$$X^2 - YZ = 0,$$

so it is represented by

$$A_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -\frac{1}{2} \\ 0 & -\frac{1}{2} & 0 \end{pmatrix}.$$

The unit circle

$$C_2 : \quad X^2 + Y^2 = Z^2$$

is represented by

$$A_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Now take

$$M = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & 1 \end{pmatrix}.$$

A direct multiplication gives

$$M^T A_1 M = A_2.$$

Indeed, if $U = (u, v, w)^T$ and $X = MU = (u, v + w, -v + w)^T$, then

$$X^T A_1 X = u^2 - (v + w)(-v + w) = u^2 + v^2 - w^2 = U^T A_2 U.$$

Thus the projective change of coordinates $U = M^{-1} X$ sends the parabola equation to the circle equation.

This is not an affine transformation. The inverse matrix is

$$M^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix}.$$

The unique real point at infinity of the parabola,

$$[0 : 1 : 0],$$

is sent to

$$M^{-1}[0 : 1 : 0] = [0 : 1 : 1],$$

a finite real point on the circle. Meanwhile the usual circular points at infinity of the real circle are the complex points

$$[1 : i : 0], \qquad [1 : -i : 0].$$

So the projective transformation has moved the affine line at infinity itself; this is exactly why a parabola and a circle can be the same projective conic while remaining different in affine geometry. The difference between them is absolute for affine geometry, but only a coordinate choice for projective geometry.

§133.4 Projective geometry

Projective geometry adds points at infinity. In the real projective plane,

$$\mathbb{P}^2(\mathbb{R}) = \{[x : y : z] \mid (x, y, z) \neq 0\}/\sim,$$

where

$$[x : y : z] = [\lambda x : \lambda y : \lambda z], \qquad \lambda \neq 0.$$

The affine plane is the chart $z \neq 0$, and the line at infinity is $z = 0$.

Definition 133.4.1. A projective transformation of $\mathbb{P}^2$ is induced by an invertible matrix

$$M \in GL(3, \mathbb{R}),$$

acting by

$$[X] \mapsto [MX].$$

Scalar multiples of M induce the same projective transformation. Thus the transformation group is

$$PGL(3, \mathbb{R}).$$

Projective transformations preserve incidence and cross-ratio, but not parallelism, distance, or angle. In projective geometry, all nondegenerate conics over an algebraically closed field are projectively equivalent.

§133.5 Cross-ratio

For four collinear points A, B, C, D, the cross-ratio is denoted

$$(A, B; C, D).$$

In an affine coordinate it has the form

$$(a, b; c, d) = \frac{(c-a)(d-b)}{(c-b)(d-a)}.$$

Theorem 133.5.1

A projective transformation of a projective line preserves the cross-ratio of four ordered distinct points. The restriction of a projective transformation of $\mathbb{P}^2$ to any line has the same property.

Proof. In an affine chart, a projective change of line coordinate has the form

$$F(t) = \frac{\alpha t + \beta}{\gamma t + \delta}, \qquad \alpha\delta - \beta\gamma \neq 0.$$

For any two finite coordinates where the denominators do not vanish,

$$F(s) - F(t) = \frac{(\alpha\delta - \beta\gamma)(s-t)}{(\gamma s + \delta)(\gamma t + \delta)}.$$

The cross-ratio is a quotient of four such differences. Each factor $\alpha\delta - \beta\gamma$ and each denominator $\gamma t + \delta$ appears equally often in its numerator and denominator, and cancels. Hence

$$(F(a), F(b); F(c), F(d)) = (a, b; c, d).$$

Points at infinity are handled by taking limits in projective coordinates. For a transformation of a projective plane, restrict its defining linear map to the two-dimensional vector space representing the given line; the same calculation applies. □

For an actual numerical example, take $(a, b, c, d) = (0, 1, 2, 3)$. Then

$$(0, 1; 2, 3) = \frac{(2-0)(3-1)}{(2-1)(3-0)} = \frac{4}{3}.$$

The non-affine projective map $F(t) = t/(t+1)$ sends these four positions to $0, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}$, whose cross-ratio is

$$\frac{(\frac{2}{3} - 0)(\frac{3}{4} - \frac{1}{2})}{(\frac{2}{3} - \frac{1}{2})(\frac{3}{4} - 0)} = \frac{4}{3}.$$

The individual interval lengths have changed. The special combination has not. Since a projectivity of a line is determined by the images of three distinct points, the cross-ratio measures the position of a fourth point relative to those three in a way that survives perspective.

Remark 133.5.2 — The cross-ratio is the prototype of a projective invariant. Once one understands it, projective geometry stops looking like distorted Euclidean geometry and becomes an invariant theory of incidence.

§133.6 Inversion: when a straight line turns into a circle

Inversion about the unit circle is defined on $\mathbb{R}^2 \setminus \{0\}$ by

$$I(x) = \frac{x}{\|x\|^2}.$$

It is an involution: $I(I(x)) = x$. The unit circle remains pointwise fixed; points inside it exchange places with points outside. In complex notation $I(z) = 1/\overline{z}$.

Take the line $x = c$, $c \neq 0$, and write the coordinates of its inverted image as (u, v). Since inversion is its own inverse,

$$\frac{u}{u^2 + v^2} = c \quad \Longleftrightarrow \quad \boxed{\left(u - \frac{1}{2c}\right)^2 + v^2 = \frac{1}{4c^2}.}$$

The image is a circle of radius $1/(2|c|)$ centered at $(1/(2c), 0)$, with the origin omitted. The origin is its limiting point corresponding to points running infinitely far out along the original line. In the other direction, circles passing through the center of inversion become lines.

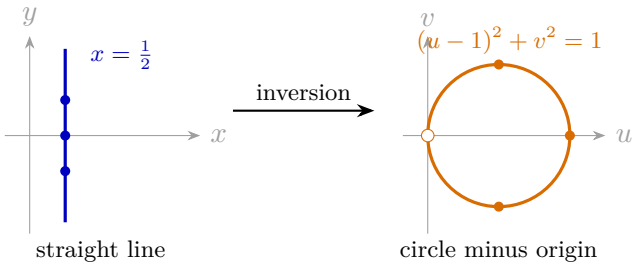


Figure 133.1: Under unit-circle inversion, $(\frac{1}{2}, 0)$ maps to $(2, 0)$ and $(\frac{1}{2}, \pm\frac{1}{2})$ map to $(1, \pm 1)$. The hollow origin is excluded, though the image circle passes through it as a limiting point.

But inversion has an entirely different invariant from projective maps. Put $u = x/\|x\|$. Differentiation gives

$$DI_x = \frac{1}{\|x\|^2}(I_2 - 2uu^T).$$

The matrix $I_2 - 2uu^T$ is a reflection: its square is the identity, it has eigenvalues $1, -1$, and it preserves inner products. Hence the derivative of inversion is a positive scale factor followed by an orthogonal reflection. It preserves *unoriented* angles locally, while reversing orientation. In complex geometry it is anti-conformal (anti-holomorphic), rather than an orientation-preserving holomorphic map.

This already refutes the proposed inclusion of all projective maps inside the conformal group, or vice versa. The affine shear $(x, y) \mapsto (x + y, y)$ is projective but changes a right angle. Inversion preserves angles but maps our straight line to a genuine circle, and so cannot be a projective plane transformation, which must send lines to lines.

§133.7 Möbius transformations

A Möbius transformation has the form

$$z \mapsto \frac{az + b}{cz + d}, \qquad ad - bc \neq 0.$$

It acts naturally on the Riemann sphere

$$\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}.$$

The group of Möbius transformations is

$$PGL(2, \mathbb{C}).$$

Möbius transformations send generalized circles to generalized circles and preserve the *complex* cross-ratio. They are projective transformations of the complex projective line $\mathbb{P}^1(\mathbb{C})$ and precisely the orientation-preserving conformal automorphisms of the Riemann sphere. This is a projective space of complex dimension one, not the real projective plane $\mathbb{P}^2(\mathbb{R})$ used for ordinary conics.

Circle inversion $z \mapsto 1/\overline{z}$, on the other hand, is anti-Möbius: conjugation reverses orientation and conjugates the complex cross-ratio. It preserves the underlying real-circle geometry without belonging to the holomorphic Möbius group. These distinctions explain how projective and conformal structures can meet in one complex dimension while remaining incomparable as transformation groups of the Euclidean plane.

§133.8 A hierarchy of invariants

Transformation group	Preserved	Forgotten
Euclidean	Distance, angle, incidence	Coordinates
Similarity	Angle, ratios of lengths	Absolute scale
Affine	Incidence, parallelism, ratios on a line	Angle and length
Projective	Incidence, cross-ratio	Parallelism and affine ratio
Möbius	Generalized circles, angles, cross-ratio on $\widehat{\mathbb{C}}$	Euclidean size

§133.9 Conics and the Line at Infinity

Fix the affine chart $Z = 1$, with line at infinity $Z = 0$. A real affine ellipse, parabola and hyperbola can have projective completions

$$\begin{aligned} X^2 + Y^2 - Z^2 &= 0 \quad \text{ellipse (unit circle),} \\ X^2 - YZ &= 0 \quad \text{parabola,} \\ X^2 - Y^2 - Z^2 &= 0 \quad \text{hyperbola.} \end{aligned}$$

Their intersection with the *fixed* line $Z = 0$ gives the three calculations

	equation at infinity	real intersections
ellipse	$X^2 + Y^2 = 0$	none
parabola	$X^2 = 0$	one double point
hyperbola	$X^2 - Y^2 = 0$	two distinct points

A projective parabola is tangent to the line at infinity, while a projective hyperbola meets it in the two directions of its asymptotes. The circle has two complex points at infinity $[1 : \pm i : 0]$, and no real ones. The intersection pattern is preserved under affine transformations because the line at infinity stays where it is; a general projective map may move this line.

Over the reals one must also be careful with the claim that all nonsingular conics are projectively equivalent. A nondegenerate symmetric quadratic form in three variables has a signature: the numbers of positive and negative squares after a real invertible change of coordinates. Sylvester's law of inertia says that signature is preserved. If the form is definite, the equation $X^TQX = 0$ has no nonzero real solution, and its projective conic is empty. If it is indefinite, multiplying its equation by -1 when necessary and changing coordinates brings it to

$$U^2 + V^2 - W^2 = 0.$$

Thus all *nonempty nonsingular real* projective conics are equivalent; an empty definite conic forms a separate case. Over an algebraically closed field of characteristic different from 2, nonsingular plane conics do form one projective equivalence class.

This explains precisely what was achieved in the parabola-to-circle matrix calculation: we changed the chosen affine chart while staying within the real indefinite projective class. We did not claim an arbitrary real quadratic form can be changed into an ellipse.

§133.10 Polar lines

For a nonsingular conic $X^TQX = 0$, a point P has a polar line

$$P^TQX = 0.$$

If P lies on the conic, the polar line is the tangent line at P. If P lies outside, the polar encodes the chord of contact of tangents from P.

The pole-polar relation is one of the best examples of why projective conic theory is richer than coordinate classification. It converts points to lines and lines to points in a way controlled by the conic.

The definition actually comes from differentiation. If $F(X) = X^TQX$ with symmetric Q, then

$$DF_P(V) = 2P^TQV.$$

At a point P on the conic, the projective tangent line is therefore the equation $P^TQX = 0$. Since Q is invertible, the assignment from a point P to the covector QP is itself an isomorphism (up to scale). Thus a conic establishes a precise duality between points and lines.

Here is a calculation that makes the polar line visible. For the unit circle $X^2 + Y^2 - Z^2 = 0$, choose the exterior point

$$P = [2 : 0 : 1], \qquad Q = \operatorname{diag}(1, 1, -1).$$

Its polar is $2X - Z = 0$, or $x = 1/2$ in the affine plane. The polar meets the circle at

$$T_{\pm} = \left(\frac{1}{2}, \pm\frac{\sqrt{3}}{2}\right).$$

The tangent at $T_{\pm}$ has equation

$$\frac{1}{2}x \pm \frac{\sqrt{3}}{2}y = 1.$$

Both tangent lines pass through $(2, 0)$: substituting $x = 2$, $y = 0$ makes either left-hand side equal to 1. Hence the polar of this exterior point is exactly the chord of contact

of its two tangents. This picture is an incidence statement controlled entirely by the bilinear form.

The reciprocity of polarity is just the symmetry of Q. If R belongs to the polar line of P, then $P^TQR = 0$. Symmetry gives $R^TQP = 0$, so P belongs to the polar line of R. What sounds like a rather mystical point–line duality is a one-line identity in linear algebra.

§133.11 Parametrizing a circle by lines through one point

The same circle connects projective geometry to rational functions. Fix the point $(0, -1)$ on $x^2 + y^2 = 1$, and pass the line $y = tx - 1$ through it. Substitution gives

$$x^2 + (tx - 1)^2 = 1 \quad \Longleftrightarrow \quad x((1 + t^2)x - 2t) = 0.$$

The root $x = 0$ is the fixed base point; the second intersection is

$$\boxed{x = \frac{2t}{1 + t^2}, \qquad y = \frac{t^2 - 1}{1 + t^2}.}$$

The missing top point $(0, 1)$ is reached at the projective parameter $t = \infty$. In homogeneous coordinates the parametrization is

$$[S : T] \longmapsto [2ST : T^2 - S^2 : S^2 + T^2].$$

We have exhibited a rational parametrization of a nonsingular conic by the projective line of slopes through one point. The half-angle formulas of trigonometry conceal the same idea: $\tan(\theta/2)$ turns points on a circle into rational coordinates.

§133.12 How to choose the right geometry

The Erlangen viewpoint becomes useful only when it tells us which tools are allowed. For conic problems, a practical guide is:

Question being asked	Natural geometry	Reason
length, angle, focus, eccentricity	Euclidean/similarity	metric data is essential
midpoints, parallelism, area ratios	affine	affine maps preserve barycentric structure
incidence, tangency, conic equivalence	projective	the line at infinity may move
cross-ratio or four points on a line	projective	cross-ratio is projectively invariant
angle with generalized circles	conformal/inversive	inversion preserves angles and generalized circles

Remark 133.12.1 (A common mistake) — Saying that two conics are projectively equivalent does not mean they are the same in Euclidean or affine geometry. A parabola and a circle may be projectively equivalent, but the projective map can move the line at infinity. Once that happens, affine notions such as parallelism and the distinction between ellipse/parabola/hyperbola need not be preserved.

XI

Topology and Homotopy Theory

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Group Actions, Homotopy, and the Hopf Fibration

N-20210806

§134.1 The orbit geometry behind a nontrivial bundle

The free circle action on a three-sphere leads to an unexpectedly nontrivial fibration over a two-sphere. Group actions describe the symmetry, homotopy controls deformation, and lifting properties reveal why the circle fibers fit together globally.[1]

§134.2 Group actions

A group action is one of the first ways in which abstract algebra touches geometry and topology. A group can be understood as acting by symmetries or permutations of a set.

Definition 134.2.1. Let G be a group and X a set. A left action of G on X is a map

$$G \times X \to X, \qquad (g, x) \mapsto g \cdot x,$$

such that

$$e \cdot x = x, \qquad (gh) \cdot x = g \cdot (h \cdot x).$$

Equivalently, it is a homomorphism

$$\varphi : G \to S_X,$$

where S_X is the permutation group of X.

Remark 134.2.2 — One may also use right actions, written $x \cdot g$, satisfying

$$x \cdot 1 = x, \qquad (x \cdot g) \cdot h = x \cdot (gh).$$

Unless stated otherwise, the left-action convention will be used throughout.

Example 134.2.3

The symmetric group S_n acts on $X = \{1, 2, \dots, n\}$. Cayley's theorem says that every group G can be realized as a subgroup of the permutation group of its underlying set, by the action of left multiplication.

Example 134.2.4

The symmetry group of a cube acts on the cube and may also be regarded as a group of linear transformations of $\mathbb{R}^3$ preserving the cube.

[1] For the homotopy exact sequence and the Hopf fibration, see Hatcher, *Algebraic Topology*, Chapter 4, especially the discussion of fiber bundles.

Example 134.2.5 (Conjugation)
A group G acts on itself by conjugation:

$$g \cdot x = gxg^{-1}.$$

For a fixed $a \in G$, its orbit is

$$\mathrm{Orb}_G(a) = \{gag^{-1} \mid g \in G\},$$

which is the conjugacy class of a. Its stabilizer is

$$\mathrm{Stab}_G(a) = \{g \in G \mid gag^{-1} = a\},$$

which is the centralizer of a.

Definition 134.2.6. For an action of G on X, the stabilizer and orbit of $x \in X$ are

$$\mathrm{Stab}_G(x) = \{g \in G \mid g \cdot x = x\},$$

$$\mathrm{Orb}_G(x) = \{g \cdot x \mid g \in G\}.$$

Theorem 134.2.7 (Orbit-stabilizer)
If G is finite, then

$$|G| = |\mathrm{Stab}_G(x)|\,|\mathrm{Orb}_G(x)|.$$

Proof. The map $G \to \mathrm{Orb}_G(x)$, $g \mapsto g \cdot x$, has fibres equal to left cosets of $\mathrm{Stab}_G(x)$. Hence the orbit has $[G : \mathrm{Stab}_G(x)]$ elements. □

§134.3 Homotopy

Algebraic topology tries to classify topological spaces and continuous maps using algebraic invariants. Before doing that, one needs a flexible notion of sameness. Homotopy is such a notion.

Definition 134.3.1. Let X, Y be topological spaces and $f, g : X \to Y$ continuous maps. A homotopy from f to g is a continuous map

$$F : X \times I \to Y, \qquad I = [0, 1],$$

such that

$$F(x, 0) = f(x), \qquad F(x, 1) = g(x).$$

When such an F exists, write $f \simeq g$.

Equivalently, a homotopy may be viewed as a map

$$F : X \to Y^I,$$

where $Y^I = \mathrm{Top}(I, Y)$ is the path space, and the endpoint evaluation maps recover f and g.

Proposition 134.3.2

Homotopy is an equivalence relation on maps $X \to Y$, and it is compatible with composition. If $f \simeq g$, then

$$k \circ f \circ h \simeq k \circ g \circ h$$

whenever the compositions are defined.

Definition 134.3.3. Two spaces X and Y have the same homotopy type if there are maps $f : X \to Y$ and $g : Y \to X$ such that

$$g \circ f \simeq \mathrm{id}_X, \qquad f \circ g \simeq \mathrm{id}_Y.$$

§134.4 Cofibrations, fibrations, and deformation retracts

Cofibrations and fibrations are most transparent through their extension and lifting diagrams. Each arrow is part of the data, and writing the diagrams explicitly prevents the simple geometric idea from disappearing behind the abstraction.

§134.4.i Homotopy extension property

A map $j : A \to X$ is a cofibration if it has the homotopy extension property. The situation is this. We know a map

$$g : X \to Z,$$

and we also know a homotopy on the smaller space

$$G : A \times I \to Z.$$

These two pieces of data must agree at time 0: for every $a \in A$,

$$G(a, 0) = g(j(a)).$$

If this compatibility condition holds, the homotopy extension property asks for a map

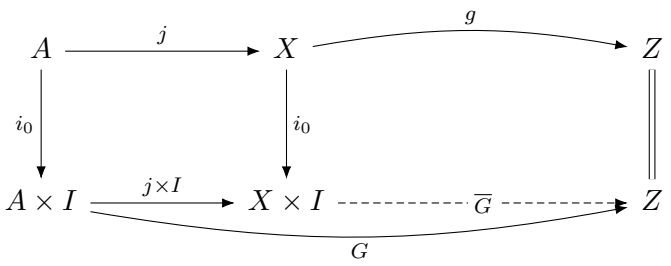
$$\overline{G} : X \times I \to Z$$

such that

$$\overline{G}(x, 0) = g(x), \qquad \overline{G}(j(a), t) = G(a, t).$$

In other words, a homotopy that has only been prescribed on the subspace A must extend to a homotopy on all of X, while keeping the original time-zero map on X.

The corresponding diagram is



Here i_0 denotes insertion at time zero: $a \mapsto (a, 0)$ or $x \mapsto (x, 0)$, according to the source. The bottom arrow $j \times I$ says that the cylinder over A sits inside the cylinder over X.

The two solid maps into Z are the information already given: g on X, and G on $A \times I$. The rightmost equality means that both pieces are maps into the same target Z, not two different copies of Z. The dashed arrow is not part of the data. It is exactly the thing whose existence is required.

This is why the word "extension" is literal. We first define a map on the union

$$X \times \{0\} \cup A \times I.$$

On $X \times \{0\}$ it is g. On $A \times I$ it is G. The compatibility condition $G(a, 0) = g(j(a))$ says these two formulas agree on the overlap $A \times \{0\}$. The H.E.P. says this partial map extends across the whole cylinder $X \times I$.

A useful mental picture is: $A \subset X$ is a part of the space whose motion has already been specified. A cofibration is an inclusion good enough that this partial motion can be filled in over the rest of X. Thus cofibrations are the maps along which homotopies extend outward.

§134.4.ii Homotopy lifting property

A fibration is the dual-looking story. Let

$$p : X \to A$$

be a map. The homotopy lifting property says that if a homotopy has been drawn downstairs in A, and if its starting point has already been lifted upstairs to X, then the whole homotopy can be lifted upstairs.

Using path spaces makes the diagram compact. The notation A^I means the space of paths $I \to A$, and X^I means the space of paths $I \to X$. Evaluation at time 0 gives

$$ev_0 : A^I \to A, \qquad ev_0 : X^I \to X.$$

The map $p : X \to A$ also induces a map on path spaces

$$p^I : X^I \to A^I, \qquad \gamma \mapsto p \circ \gamma.$$

Now take a parameter space Z. A map

$$F : Z \to A^I$$

is a Z-indexed family of paths downstairs in A. A map

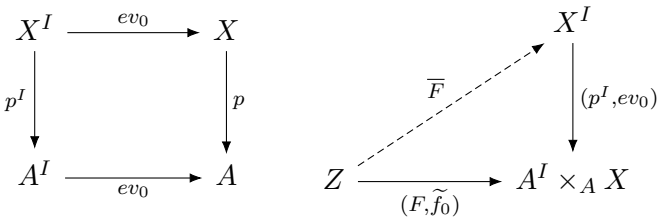
$$\widetilde{f_0} : Z \to X$$

chooses a starting lift upstairs. The compatibility condition is

$$p \circ \widetilde{f_0} = ev_0 \circ F.$$

This simply says: the chosen point upstairs really lies over the initial point of the path downstairs.

The lifting diagram is



The square on the left is the structural square. The horizontal map $p^I : X^I \to A^I$ sends an upstairs path to its downstairs projection, and the two ev_0 arrows take initial points. The square commutes because evaluating $p \circ \gamma$ at time 0 gives $p(\gamma(0))$. The diagram on the right packages the actual lifting problem. A pair $(F, \widetilde{f}_0)$ lands in the fiber product $A^I \times_A X$ exactly when $p \circ \widetilde{f}_0 = ev_0 \circ F$. The dashed arrow $\overline{F} : Z \to X^I$ is the desired lifted family of paths.

Unpacking the dashed arrow, for each $z \in Z$ it gives a path

$$\overline{F}(z) : I \to X.$$

The two required equations are

$$p \circ \overline{F}(z) = F(z), \qquad \overline{F}(z)(0) = \widetilde{f}_0(z).$$

So the lifted path projects to the given downstairs path and starts at the chosen upstairs point. This is the exact formal meaning of "lift upward from the base."

The relationship with the earlier definition is now transparent. A cofibration problem begins with a homotopy on a subspace and asks to extend it sideways across a larger cylinder. A fibration problem begins with a homotopy downstairs and asks to lift it vertically to a path or homotopy upstairs. Both are extension problems, but in different directions: H.E.P. fills missing points in the domain; H.L.P. fills missing lifts in the total space.

Definition 134.4.1. A map $j : A \to X$ is a cofibration if it has the homotopy extension property described above.

Definition 134.4.2. A map $p : X \to A$ is a fibration if it has the homotopy lifting property described above.

Remark 134.4.3 — Cofibration and fibration are dual-looking ideas. A cofibration lets homotopies extend outward from a subspace; a fibration lets homotopies lift upward from the base. In the Hopf fibration $S^3 \to S^2$, the word "fibration" means exactly that small motions of a point in the base sphere can be lifted to motions in S^3, once a starting point in the circle fiber has been chosen.

Definition 134.4.4. A deformation retract of X onto a subspace A is a homotopy

$$H : X \times I \to X$$

such that

$$H(x, 0) = x, \qquad H(x, 1) \in A, \qquad H(a, t) = a$$

for all $x \in X$, $a \in A$, and $t \in I$.

Proposition 134.4.5

The pair (X, A) has the homotopy extension property if and only if

$$X \times \{0\} \cup A \times I$$

is a retract of $X \times I$.

§134.5 Spheres and homotopy groups

The n-sphere is

$$S^n = \{(x_1, \dots, x_{n+1}) \in \mathbb{R}^{n+1} \mid x_1^2 + \cdots + x_{n+1}^2 = 1\}.$$

The n-torus is

$$T^n = (S^1)^n.$$

Thus a point of T^2 may be described by a pair of angles, or equivalently by a point of $S^1 \times S^1$.

Stereographic projection identifies a sphere minus one point with Euclidean space. For example,

$$S^2 \setminus \{\text{north pole}\} \cong \mathbb{R}^2.$$

The equator is the region where the projection is least distorted; away from it, shapes become increasingly distorted.

Definition 134.5.1. Let X be a based space with basepoint x_0. The n-th homotopy group or homotopy set is

$$\pi_n(X, x_0) = [(S^n, *), (X, x_0)],$$

the set of based homotopy classes of based maps $S^n \to X$. For $n \geq 1$, it has a natural group structure.

Using the model

$$S^n \simeq I^n / \partial I^n,$$

the group operation for $n \geq 1$ may be described by pinching the sphere along an equator:

$$S^n \to S^n \vee S^n.$$

Given two based maps $f, g : S^n \to X$, the product first pinches S^n into two copies, then applies f on one copy and g on the other.

Remark 134.5.2 — For $n = 0$, $\pi_0(X)$ records path components and is not usually a group. For $n \geq 1$, $\pi_n(X, x_0)$ is a group; for $n \geq 2$, it is abelian.

§134.6 The Hopf fibration

The Hopf fibration is a map

$$S^3 \to S^2.$$

It is one of the first examples showing that spheres are connected by unexpectedly rich maps.

Describe

$$S^3 = \{(z_1, z_2) \in \mathbb{C}^2 \mid |z_1|^2 + |z_2|^2 = 1\}.$$

A clean formula for the Hopf map is

$$h : S^3 \to \mathbb{C}P^1 \cong S^2, \qquad h(z_1, z_2) = [z_1 : z_2].$$

On the chart $z_1 \neq 0$, this looks like

$$z_2 / z_1.$$

The case $z_1 = 0$ is the point at infinity.

Caution

It is tempting to say $z_2/0 = \infty$, and that intuition is useful. The mathematically clean way is to use projective coordinates $[z_1 : z_2]$, which handle infinity automatically.

The fibers over three simple points can be written without any abstraction:

$$F_1 = h^{-1}([1:0]) = \{(e^{it}, 0) : 0 \le t < 2\pi\},$$

$$F_2 = h^{-1}([0:1]) = \{(0, e^{it}) : 0 \le t < 2\pi\},$$

and

$$F_3 = h^{-1}([1:1]) = \left\{ \left(\frac{e^{it}}{\sqrt{2}}, \frac{e^{it}}{\sqrt{2}} \right) : 0 \le t < 2\pi \right\}.$$

To see them inside ordinary three-space, use stereographic projection

$$\Pi(z_1, z_2) = \left(\frac{\mathrm{Re}(z_1)}{1 - \mathrm{Im}(z_2)}, \frac{\mathrm{Im}(z_1)}{1 - \mathrm{Im}(z_2)}, \frac{\mathrm{Re}(z_2)}{1 - \mathrm{Im}(z_2)} \right).$$

Writing $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$, this is the projection from the point $(0, i) \in S^3$.

For F_1, substitution gives

$$\Pi(e^{it}, 0) = (\cos t, \sin t, 0),$$

the standard unit circle in the plane $z = 0$. For F_2,

$$\Pi(0, e^{it}) = \left(0, 0, \frac{\cos t}{1 - \sin t} \right),$$

so this fiber becomes the vertical line together with the point at infinity; the missing value $t = \pi/2$ is precisely the projection point. For F_3,

$$\Pi\left(\frac{e^{it}}{\sqrt{2}}, \frac{e^{it}}{\sqrt{2}} \right) = \left(\frac{\cos t/\sqrt{2}}{1 - \sin t/\sqrt{2}}, \frac{\sin t/\sqrt{2}}{1 - \sin t/\sqrt{2}}, \frac{\cos t/\sqrt{2}}{1 - \sin t/\sqrt{2}} \right).$$

This curve lies in the plane $X = Z$. Eliminating t gives

$$2X^2 + (Y - 1)^2 = 2, \qquad Z = X,$$

so it is again a circle, not a random distorted loop.

The important topological fact is that $\Pi(F_1)$ and $\Pi(F_3)$ form a Hopf link. With the orientations induced by increasing t, their linking number is $+1$; reversing one orientation changes the sign. The computation matches the picture: one circle passes once through a spanning disk of the other. The core geometric intuition of the Hopf fibration is that any two distinct circle fibers in S^3 become linked once after stereographic projection to $\mathbb{R}^3$.

There is also a quaternionic picture. Regard S^3 as the unit quaternions and S^2 as the purely imaginary unit quaternions. Fix $p \in S^2$. For a unit quaternion $q \in S^3$, define

$$q \mapsto qpq^{-1}.$$

This gives another description of a map $S^3 \to S^2$, equivalent to the Hopf fibration.

Remark 134.6.1 — The main fact to remember is simple but powerful: the Hopf fibration has total space S^3, base S^2, and fibre S^1. A full understanding needs pictures, but the formula already shows why complex numbers, quaternions, and topology appear together.

§134.7 Why distinct Hopf fibers are linked

The Hopf map admits a useful explicit form:

$$H : S^3 \subset \mathbb{C}^2 \longrightarrow S^2 \subset \mathbb{C} \times \mathbb{R}, \qquad H(z_1, z_2) = (2z_1\overline{z_2},\ |z_1|^2 - |z_2|^2).$$

Since $|z_1|^2 + |z_2|^2 = 1$, its image lies on S^2:

$$|2z_1\overline{z_2}|^2 + (|z_1|^2 - |z_2|^2)^2 = 1.$$

Multiplication by a common unit complex number does not change the image,

$$H(e^{it}z_1, e^{it}z_2) = H(z_1, z_2).$$

Conversely, any two points of S^3 with the same image lie on the same common-phase orbit. Thus each fiber is a circle.

The fibers above the north and south poles are especially simple:

$$C_1 = \{(e^{it}, 0) : 0 \le t < 2\pi\}, \qquad C_2 = \{(0, e^{it}) : 0 \le t < 2\pi\}.$$

They are disjoint embedded circles in S^3. To compute their linking number, regard S^3 as the boundary of the unit ball $B^4 \subset \mathbb{C}^2$. The circles bound the two properly embedded complex disks

$$D_1 = \{(z, 0) : |z| \le 1\}, \qquad D_2 = \{(0, w) : |w| \le 1\}.$$

Their interiors meet transversely at the origin, with positive intersection sign in the complex orientation. The intersection-number formula for linking in the boundary of a four-ball therefore gives

$$\boxed{\mathrm{lk}(C_1, C_2) = 1}.$$

The ordered configuration space of two distinct points of S^2 is path connected. Moving the two base points continuously without letting them meet deforms their fibers through disjoint circles, so any two distinct Hopf fibers have the same linking number. What looks like a collection of ordinary circles is globally organized by a nontrivial linking invariant.

§134.8 The long exact sequence records the same twist

The Hopf map is a fiber bundle

$$S^1 \longrightarrow S^3 \xrightarrow{H} S^2.$$

Its long exact sequence in homotopy contains

$$\pi_2(S^3) \longrightarrow \pi_2(S^2) \xrightarrow{\partial} \pi_1(S^1) \longrightarrow \pi_1(S^3).$$

Since $\pi_2(S^3) = \pi_1(S^3) = 0$, exactness gives

$$\boxed{\partial : \pi_2(S^2) \xrightarrow{\sim} \pi_1(S^1)}.$$

Both groups are isomorphic to $\mathbb{Z}$. The connecting homomorphism measures how the fiber coordinate winds when local lifts of a sphere in the base are compared; it has degree 1 or -1, depending on the orientation convention. In bundle terminology, its absolute value is the first Chern number of the Hopf line bundle.

The fibration cannot be the trivial product $S^2 \times S^1 \to S^2$, since that product has fundamental group $\mathbb{Z}$, while S^3 is simply connected. The local picture is unremarkable: over a small disk of the base, the total space is a disk times a circle. The obstruction only appears when the local trivializations are glued globally.

Group actions supply the circle fibers, the lifting property gives the homotopy sequence, and the linking computation reveals the same nontriviality in a concrete geometric picture. The transition from group actions to homotopy and fiber bundles is thus driven by one object whose local and global descriptions disagree in a precise way.

Metric Spaces: Topology, Completion, and the Price of Infinity

N-20221218

Two spaces may have the same open sets and yet disagree about which sequences are Cauchy. This small surprise marks the boundary between topology and metric geometry. Starting with convergence and open balls, we can locate exactly where completeness and compactness use information that open sets alone cannot provide.[1]

§135.1 What a metric contributes

A metric $d : X \times X \to [0, \infty)$ is symmetric, vanishes precisely on the diagonal, and satisfies

$$d(x, z) \le d(x, y) + d(y, z).$$

Its open balls $B_r(x) = \{y : d(x, y) < r\}$ define a collection of open sets. A sequence converges to x exactly when its terms eventually lie in every ball around x. Continuity of $f : X \to Y$ means preimages of open sets are open, or equivalently in metric spaces, that $x_n \to x$ implies $f(x_n) \to f(x)$.

For a normed vector space, the metric is $d(x, y) = \|x - y\|$. The Euclidean, taxicab and maximum norms on $\mathbb{R}^n$ generate the same open sets because

$$\|v\|_\infty \le \|v\|_2 \le \sqrt{n}\,\|v\|_\infty, \qquad \|v\|_2 \le \|v\|_1 \le \sqrt{n}\,\|v\|_2.$$

These inequalities explicitly allow a ball for one metric to contain a suitably smaller ball for the other. Their exact distances differ, but they agree on all convergent sequences and continuous functions.

§135.2 One topology, two notions of being complete

Let $X = \mathbb{R}$. Besides the ordinary metric $d(x, y) = |x - y|$, define

$$\rho(x, y) = |\arctan x - \arctan y|.$$

Since $\arctan : \mathbb{R} \to (-\pi/2, \pi/2)$ is a strictly increasing homeomorphism, ρ is a metric inducing the usual open sets.

Nevertheless, $x_n = n$ is a ρ-Cauchy sequence: $\arctan n \to \pi/2$, so

$$\rho(n, m) = |\arctan n - \arctan m| \longrightarrow 0 \qquad (n, m \to \infty).$$

It does not converge in X for ρ, since no real number has arctangent $\pi/2$. Therefore $(\mathbb{R}, \rho)$ is incomplete, although $(\mathbb{R}, d)$ is complete. Even boundedness changes: $\rho(x, y) < \pi$ for all x, y.

A topological homeomorphism preserves neighborhood structure, but need not preserve the comparison of two points moving together. Uniformly continuous maps do preserve Cauchy sequences; ordinary continuous maps need not. That is the extra structure carried by the uniformity of a metric.

[1] See MIT 18.S190, *Introduction to Metric Spaces*, Lectures 2–5: https://ocw.mit.edu/courses/18-s190-introduction-to-metric-spaces-january-iap-2023/pages/lecture-notes/.

§135.3 Completion adds the missing limits in a canonical way

For any metric space (X, d), consider the set $\mathcal{C}(X)$ of Cauchy sequences and identify

$$(x_n) \sim (y_n) \quad \Longleftrightarrow \quad d(x_n, y_n) \to 0.$$

The distance

$$\widehat{d}([(x_n)], [(y_n)]) = \lim_{n\to\infty} d(x_n, y_n)$$

is well defined: the absolute difference between successive cross-distances is bounded by $d(x_n, x_m) + d(y_n, y_m)$, so those distances form a Cauchy sequence in $\mathbb{R}$. Equivalent representatives give the same limit by the triangle inequality.

Constant sequences embed X isometrically as a dense subset of its completion $\widehat{X}$. Every Cauchy sequence in $\widehat{X}$ converges; one proves this by approximating its successive terms with points of X to within 2^{-n}, and taking the corresponding diagonal sequence. Thus $\widehat{X}$ is complete.

The metric ρ above makes this construction almost visible: $(\mathbb{R}, \rho)$ is isometric to $(-\pi/2, \pi/2)$ with ordinary distance, so its completion is the closed interval $[-\pi/2, \pi/2]$. Two ideal endpoints are added at what had been $+\infty$ and $-\infty$.

§135.4 Compactness requires finite resolution at every scale

A metric space is totally bounded if for each $\varepsilon > 0$ it can be covered by finitely many balls of radius ε. Mere boundedness only provides one large ball. An infinite set with the discrete metric is bounded but not totally bounded: every ball of radius less than 1 contains a single point.

Theorem 135.4.1

A metric space is compact if and only if it is complete and totally bounded.

The constructive direction is particularly instructive. Suppose X is totally bounded. Cover X by finitely many balls of radius $1/2$; infinitely many terms of any sequence (x_n) lie in one of them. From those indices, choose an infinite subsequence lying in one ball of radius $1/4$. Repeat with radii 2^{-k} and take a diagonal subsequence. Its tail lies in a ball of radius 2^{-k} for every k, so it is Cauchy. If X is complete it converges.

Thus complete plus totally bounded implies sequential compactness, which is equivalent to open-cover compactness for metric spaces. In the other direction, compactness forces finite ball covers immediately, and compactness also implies completeness by extracting a convergent subsequence from each Cauchy sequence.

§135.5 Closed and Bounded in Infinite Dimensions

In $\mathbb{R}^n$, closed and bounded sets are compact because bounded sets are totally bounded and closed subsets of a complete space are complete. The missing implication in infinite dimensions is total boundedness.

In the Hilbert space ℓ^2, let e_k be the sequence with a single 1 in position k. Every e_k belongs to the closed unit ball, but

$$\|e_j - e_k\|_2 = \sqrt{2} \qquad (j \neq k).$$

No ball of radius $1/2$ can contain two of these points, so the unit ball cannot be covered by finitely many such balls. It is closed and bounded, yet not compact.

This obstruction motivates weak topologies in functional analysis. The closed unit ball of a Hilbert space need not be compact in the norm topology, while it is compact in the weak topology by a more sophisticated theorem (Banach–Alaoglu, together with reflexivity). Changing the topology deliberately changes which kinds of compactness are available.

§135.6 Completeness also makes dense properties coexist

The Baire category theorem states that a complete metric space is a Baire space: a countable intersection of dense open subsets is dense. To see the mechanism, take dense open sets $U_1, U_2, \dots$ and any nonempty open ball B_0. Since U_1 is dense and open, one can choose a closed ball B_1 of positive radius, contained in $B_0 \cap U_1$. Continue recursively with

$$B_n \subseteq B_{n-1} \cap U_n, \qquad \operatorname{radius}(B_n) < 2^{-n},$$

and nonempty interior for each ball. Their centers form a Cauchy sequence. Completeness gives a limit lying in every closed B_n, hence in $B_0 \cap \bigcap_n U_n$.

The proof shows exactly what completeness buys: a nested sequence of increasingly accurate approximations produces an actual point. For Baire category, this point simultaneously satisfies countably many dense conditions; for metric completion, it realizes a Cauchy sequence; for compactness, finite approximations at every scale provide a convergent subsequence. The three theorems share a concrete construction rather than just a similar-sounding notion of convergence.

Completion, Uniform Structure, and Quotient Topology

N-20221221

§136.1 Cauchy Sequences and Missing Points

A sequence (x_n) in a metric space (X, d) is Cauchy when for every $\varepsilon > 0$ there is N such that

$$d(x_m, x_n) < \varepsilon \qquad (m, n \geq N).$$

It records points that eventually become mutually indistinguishable at every metric scale. A complete space contains a limit for every such sequence.

The rational sequence

$$q_n = \frac{\lfloor 10^n \sqrt{2} \rfloor}{10^n}$$

is Cauchy in $\mathbb{Q}$, but it has no rational limit. Nothing is wrong with the sequence; the intended limit is absent from the space. Completion constructs a space in which every coherent approximation process has a point to name.

Two Cauchy sequences should represent the same new point when their mutual distance tends to zero:

$$(x_n) \sim (y_n) \quad \Longleftrightarrow \quad d(x_n, y_n) \longrightarrow 0.$$

Reflexivity and symmetry are immediate. If $(x_n) \sim (y_n)$ and $(y_n) \sim (z_n)$, then

$$d(x_n, z_n) \leq d(x_n, y_n) + d(y_n, z_n) \longrightarrow 0,$$

so the relation is transitive.

§136.2 The completion metric is a limit of mutual distances

Let $\widehat{X}$ be the set of equivalence classes of Cauchy sequences. Define

$$\widehat{d}([(x_n)], [(y_n)]) = \lim_{n \to \infty} d(x_n, y_n).$$

The scalar sequence on the right is Cauchy because

$$|d(x_n, y_n) - d(x_m, y_m)| \leq d(x_n, x_m) + d(y_n, y_m).$$

Hence the limit exists in $\mathbb{R}$.

The value does not depend on the chosen representatives. If

$$(x_n) \sim (x'_n), \qquad (y_n) \sim (y'_n),$$

then

$$|d(x_n, y_n) - d(x'_n, y'_n)| \leq d(x_n, x'_n) + d(y_n, y'_n) \longrightarrow 0.$$

Thus both representative pairs give the same limit.

Nonnegativity and symmetry pass directly to the limit. If $\widehat{d} = 0$, then $d(x_n, y_n) \to 0$, so the classes are equal. Finally,

$$d(x_n, z_n) \leq d(x_n, y_n) + d(y_n, z_n)$$

passes to the limit and gives the triangle inequality. Therefore $\widehat{d}$ is a genuine metric.

§136.3 Embedding into the Completion

Send $x \in X$ to the constant sequence:

$$\iota(x) = [(x, x, x, \ldots)].$$

Then

$$\widehat{d}(\iota(x), \iota(y)) = d(x, y),$$

so ι is an isometric embedding.

Its image is dense. Let

$$\xi = [(x_n)] \in \widehat{X}.$$

For fixed N,

$$\widehat{d}(\xi, \iota(x_N)) = \lim_{m \to \infty} d(x_m, x_N).$$

Because (x_n) is Cauchy, this quantity tends to zero as $N \to \infty$. Thus

$$\iota(x_N) \longrightarrow \xi.$$

Every completed point is a limit of original points.

For $X = \mathbb{Q}$, the decimal truncations (q_n) of $\sqrt{2}$ define a class in $\widehat{\mathbb{Q}}$. It cannot equal any constant rational class: if it equalled $r \in \mathbb{Q}$, then

$$|q_n - r| \longrightarrow 0,$$

forcing $r = \sqrt{2}$. The new class is exactly the missing irrational point described by rational approximations.

§136.4 Completeness by a Diagonal Argument

Let (ξ_r) be a Cauchy sequence in $\widehat{X}$. Passing to a subsequence if necessary, arrange that

$$\widehat{d}(\xi_r, \xi_{r+1}) < 2^{-r}.$$

Density of $\iota(X)$ lets us choose $y_r \in X$ with

$$\widehat{d}(\iota(y_r), \xi_r) < 2^{-r}.$$

Then

$$\begin{aligned} d(y_r, y_{r+1}) &\leq \widehat{d}(\iota(y_r), \xi_r) + \widehat{d}(\xi_r, \xi_{r+1}) \\ &\quad + \widehat{d}(\xi_{r+1}, \iota(y_{r+1})) \\ &< 2^{-r} + 2^{-r} + 2^{-(r+1)}. \end{aligned}$$

The geometric tail shows that (y_r) is Cauchy in X. Let

$$\xi = [(y_r)] \in \widehat{X}.$$

The construction gives

$$\xi_r \longrightarrow \xi$$

along the chosen subsequence, and since the original sequence was Cauchy, the whole sequence has the same limit. Hence $\widehat{X}$ is complete.

The argument exposes the mechanism: approximate each abstract completed point by an original point with rapidly shrinking error, then use one diagonal Cauchy sequence in X to encode the limit.

§136.5 Extending Uniformly Continuous Maps

Let Y be complete and let

$$f : X \to Y$$

be uniformly continuous. If (x_n) is Cauchy, then $(f(x_n))$ is Cauchy. Indeed, for every $\varepsilon > 0$, choose one $\delta > 0$ working simultaneously for all points:

$$d_X(x, x') < \delta \quad \Longrightarrow \quad d_Y(f(x), f(x')) < \varepsilon.$$

The Cauchy condition eventually places every pair x_m, x_n within δ.

Define

$$\widehat{f} : \widehat{X} \to Y, \qquad \widehat{f}([(x_n)]) = \lim_{n\to\infty} f(x_n).$$

If $(x_n) \sim (y_n)$, uniform continuity gives

$$d_Y(f(x_n), f(y_n)) \longrightarrow 0,$$

so the definition is independent of representatives. It extends f because constant sequences retain their values.

The extension is unique. If $g : \widehat{X} \to Y$ is continuous and agrees with f on X, then for

$$\xi = \lim_n \iota(x_n)$$

one has

$$g(\xi) = \lim_n g(\iota(x_n)) = \lim_n f(x_n) = \widehat{f}(\xi).$$

Thus completion is characterized by the extension property

uniformly continuous maps into complete spaces extend uniquely.

Uniform continuity is essential. The continuous map

$$f : (0, 1) \to \mathbb{R}, \qquad f(x) = \frac{1}{x}$$

has no continuous extension to the completion $[0, 1]$, because it diverges along $x_n = 1/n \to 0$.

§136.6 One topology, two different completions

A homeomorphism need not preserve Cauchy sequences. In fact, the same underlying set can carry two metrics with exactly the same open sets and different completions. On $\mathbb{R}$, compare

$$d_{\mathrm{E}}(x, y) = |x - y|, \qquad d_{\mathrm{A}}(x, y) = |\arctan x - \arctan y|.$$

The strictly increasing homeomorphism $\arctan : \mathbb{R} \to (-\pi/2, \pi/2)$ shows that d_{A} induces the usual topology on $\mathbb{R}$. Yet the sequence $x_n = n$ is not Cauchy for d_{E}, while it is Cauchy for d_{A}, because

$$|\arctan n - \arctan m| \longrightarrow 0 \quad (n, m \to \infty).$$

Similarly $-n$ is d_{A}-Cauchy. The isometry furnished by arctangent identifies $(\mathbb{R}, d_{\mathrm{A}})$ with the open interval $(-\pi/2, \pi/2)$ with its usual metric. Thus

$$\widehat{(\mathbb{R}, d_{\mathrm{A}})} \cong [-\pi/2, \pi/2],$$

whereas $(\mathbb{R}, d_{\mathrm{E}})$ is already complete. The first completion adds two endpoints and the second adds nothing.

The distinction does not come from pointwise convergence to existing points: the two metrics agree about that. It comes from whether *pairs of points far out in the space* become mutually close. A homeomorphism preserves local neighborhoods but not uniform closeness. This is exactly the extra information encoded by a uniform structure.

§136.7 The universal property makes completion unique

Suppose

$$\iota_X : X \hookrightarrow Y, \qquad j_X : X \hookrightarrow Z$$

are dense isometric embeddings into complete metric spaces. Apply the extension property to the uniformly continuous map

$$j_X \circ \iota_X^{-1} : \iota_X(X) \to Z.$$

It extends uniquely to an isometry

$$F : Y \to Z.$$

Reversing the roles gives

$$G : Z \to Y.$$

Both composites agree with the identity on the dense copy of X. By uniqueness of extension,

$$GF = \mathrm{id}_Y, \qquad FG = \mathrm{id}_Z .$$

Therefore any two completions are uniquely isometric in a way that fixes X.

The quotient of Cauchy sequences is one concrete construction; the universal property is what makes the result canonical.

§136.8 Homeomorphism forgets the uniform scale of closeness

The interval $(0, 1)$ and the real line are homeomorphic via

$$h(x) = \tan\bigl(\pi(x - 1/2)\bigr).$$

Yet $(0, 1)$ is bounded and incomplete in its Euclidean metric, while $\mathbb{R}$ is unbounded and complete. Boundedness and completeness are therefore not topological invariants.

The map is not uniformly continuous. Take

$$x_n = 1 - \frac{1}{n}, \qquad y_n = 1 - \frac{2}{n}.$$

Then

$$|x_n - y_n| = \frac{1}{n} \longrightarrow 0,$$

but

$$h(x_n) = \cot\frac{\pi}{n}, \qquad h(y_n) = \cot\frac{2\pi}{n},$$

and their difference grows on the order of n. No single δ controls the map near both endpoints.

Topology records which sets are neighborhoods. Uniform structure additionally records which pairs are uniformly close across the whole space. Completion depends on the latter, because Cauchy sequences are uniform objects.

§136.9 When a Bijection Is a Homeomorphism

The map

$$f : [0, 2\pi) \to S^1, \qquad f(t) = (\cos t, \sin t)$$

is continuous and bijective, but its inverse is not continuous at $(1, 0)$. Points approaching $(1, 0)$ from the lower semicircle have parameters tending to 2π, while the point itself has parameter 0.

A useful positive theorem is the compact–Hausdorff criterion.

Theorem 136.9.1

If X is compact, Y is Hausdorff, and $f : X \to Y$ is a continuous bijection, then f is a homeomorphism.

Proof. A closed subset $C \subseteq X$ is compact. Its image $f(C)$ is compact in Y, hence closed because Y is Hausdorff. Thus f is a closed map. For any closed $C \subseteq X$,

$$(f^{-1})^{-1}(C) = f(C)$$

is closed, so f^{-1} is continuous. □

The failed half-open-interval example lacks compactness. The theorem identifies exactly which global hypotheses make inverse continuity automatic.

§136.10 Quotient Topology and Gluing

Let $q : X \to Y$ be a surjection. The quotient topology on Y is defined by

$$U \subseteq Y \text{ open} \quad \Longleftrightarrow \quad q^{-1}(U) \text{ open in } X.$$

It is the strongest topology on Y making q continuous.

Its universal property is the factorization test:

$$g : Y \to Z \text{ is continuous} \quad \Longleftrightarrow \quad g \circ q : X \to Z \text{ is continuous.}$$

Thus a map out of a quotient is constructed upstairs, where the original points are still visible, provided the map is constant on equivalence classes.

A subset $A \subseteq X$ is saturated when it is a union of equivalence classes:

$$A = q^{-1}(q(A)).$$

For arbitrary A,

$$q(A) \text{ is open in } Y \quad \Longleftrightarrow \quad q^{-1}(q(A)) \text{ is open in } X.$$

A quotient map need not send every open set to an open set. On $[0, 1]$, collapse the entire interval $[0, 1/2]$ to one point. The open set $(0, 1/2)$ has saturation $[0, 1/2]$, which is not open, so its image is not open in the quotient.

The quotient topology controls precisely the saturated information that survives the identification.

§136.11 A circle can be made by completing rational angles

Completion and quotient topology meet very naturally in the simplest periodic space. Consider the additive quotient group

$$X = \mathbb{Q}/\mathbb{Z}.$$

Here $[q]$ means that q and $q+k$ represent the same point for every integer k. The usual metric $|q-r|$ cannot simply be written on quotient classes, because it depends on which representatives we pick. The right metric is the circular distance

$$\boxed{d_X([q],[r]) = \min_{k\in\mathbb{Z}} |q-r-k|.}$$

Replacing either representative by an integer translate merely reindexes the minimum. Also $d_X = 0$ exactly when $q - r \in \mathbb{Z}$, so it is a genuine metric. The triangle inequality follows by choosing integer translates for each of the two legs and adding their distances.

The map

$$\iota : \mathbb{Q}/\mathbb{Z} \longrightarrow S^1, \qquad [q] \longmapsto e^{2\pi i q}$$

is well defined and injective. Equip S^1 with the angular metric

$$d_{S^1}(e^{2\pi i x}, e^{2\pi i y}) = \min_{k\in\mathbb{Z}} |x-y-k|.$$

By construction, ι is an *isometry*. Rational angles are dense in all real angles, so its image is dense in S^1. The circle with its angular metric is compact and hence complete. By the uniqueness of metric completion,

$$\boxed{\widehat{\mathbb{Q}/\mathbb{Z}} \cong S^1.}$$

An actual missing point is easy to exhibit. Let

$$q_n = \frac{\lfloor 10^n \sqrt{2} \rfloor}{10^n} \in \mathbb{Q}.$$

The residue classes $[q_n]$ form a Cauchy sequence in $\mathbb{Q}/\mathbb{Z}$ whose image on the circle approaches $e^{2\pi i\sqrt{2}}$. That limit cannot arise from a single rational class, since $e^{2\pi i\sqrt{2}} = e^{2\pi i q}$ would imply $\sqrt{2} - q \in \mathbb{Z}$.

This construction differs from identifying the endpoints of $[0,1]$. The map $[0,1] \to S^1$, $t \mapsto e^{2\pi i t}$, realizes a *topological quotient* by gluing 0 and 1; the isometric embedding above realizes a *metric completion* by adding all the missing irrational angle classes. Both routes lead to the same familiar circle, but they answer different universal problems. That is a satisfying concrete explanation of why completion and quotient both appear in the same chapter.

§136.12 The Circle as an Endpoint Quotient

Let

$$q : [0,1] \to [0,1]/(0 \sim 1)$$

be the endpoint quotient, and define

$$F(t) = e^{2\pi i t}.$$

Since

$$F(0) = F(1),$$

the map is constant on equivalence classes. The quotient universal property produces a unique continuous map

$$\overline{F} : [0,1]/(0 \sim 1) \to S^1$$

with

$$\overline{F} \circ q = F.$$

It is bijective: every point of the circle has a parameter in $[0,1]$, and the only repeated parameter values are the identified endpoints.

The quotient of the compact interval is compact, while S^1 is Hausdorff. By the compact–Hausdorff theorem,

$$\boxed{[0,1]/(0 \sim 1) \cong S^1.}$$

A neighborhood of the glued point has preimage

$$[0, \varepsilon) \cup (1 - \varepsilon, 1],$$

so two one-sided endpoint neighborhoods become one two-sided circular neighborhood. The concrete local picture, the quotient factorization property, and the compactness theorem all describe the same gluing.

Open-Set Calculus: Continuity, Closure, Nets, and Connected Components

N-20221222

§137.1 Continuity is the stability of tests under preimage

Let $f : X \to Y$ be a map between topological spaces. The definition

$$V \subseteq Y \text{ open} \quad \Longrightarrow \quad f^{-1}(V) \subseteq X \text{ open}$$

uses preimages because preimages preserve all Boolean operations:

$$f^{-1}\left(\bigcup_{\alpha} V_\alpha\right) = \bigcup_{\alpha} f^{-1}(V_\alpha), \qquad f^{-1}(V \cap W) = f^{-1}(V) \cap f^{-1}(W).$$

Consequently continuous maps compose without an additional theorem:

$$(g \circ f)^{-1}(U) = f^{-1}(g^{-1}(U)).$$

The open-set definition has equivalent closed-set and neighborhood forms.

Proposition 137.1.1

For a map $f : X \to Y$, the following are equivalent.

(i) f is continuous.

(ii) The preimage of every closed set is closed.

(iii) For every $x \in X$ and every neighborhood V of $f(x)$, there is a neighborhood U of x such that $f(U) \subseteq V$.

Proof. The equivalence of (i) and (ii) follows from

$$f^{-1}(Y \setminus C) = X \setminus f^{-1}(C).$$

Assume (i). If V is a neighborhood of $f(x)$, choose an open set W with

$$f(x) \in W \subseteq V.$$

Then $U = f^{-1}(W)$ is a neighborhood of x and $f(U) \subseteq V$. Conversely, apply (iii) to an open set V and every point of $f^{-1}(V)$. The resulting neighborhoods show that $f^{-1}(V)$ is open. □

Images do not enjoy this stability. The continuous map $x \mapsto x^2$ sends $(-1, 1)$ to $[0, 1)$, which is not open in $\mathbb{R}$.

§137.2 The Sierpiński space classifies open subsets

Let

$$\Sigma = \{0, 1\}, \qquad \tau_\Sigma = \{\varnothing, \{1\}, \Sigma\}.$$

This is the Sierpiński space. For a subset $U \subseteq X$, define

$$\chi_U(x) = \begin{cases} 1, & x \in U, \\ 0, & x \notin U. \end{cases}$$

Then

$$\chi_U^{-1}(\{1\}) = U.$$

Since $\{1\}$ is the only nontrivial open test in Σ, one obtains

$$\boxed{U \text{ is open in } X \quad \Longleftrightarrow \quad \chi_U : X \to \Sigma \text{ is continuous.}}$$

Thus open subsets of X are in bijection with continuous maps $X \to \Sigma$. The preimage operation is literally composition:

$$\chi_{f^{-1}(U)} = \chi_U \circ f.$$

This gives a concrete meaning to the contravariance of topology: a map of spaces pulls open tests backward.

§137.3 Recovering Topology from Closure

For $A \subseteq X$, the closure is

$$\overline{A} = \bigcap \{C \subseteq X : C \text{ closed and } A \subseteq C\}.$$

Equivalently,

$$x \in \overline{A} \quad \Longleftrightarrow \quad U \cap A \neq \varnothing \text{ for every open neighborhood } U \ni x.$$

The closure operator satisfies the Kuratowski rules

$$\overline{\varnothing} = \varnothing, \qquad A \subseteq \overline{A}, \qquad \overline{\overline{A}} = \overline{A}, \qquad \overline{A \cup B} = \overline{A} \cup \overline{B}.$$

Monotonicity follows: $A \subseteq B$ implies $\overline{A} \subseteq \overline{B}$.

Conversely, suppose an operation $c : \mathcal{P}(X) \to \mathcal{P}(X)$ satisfies these four rules. Declare C closed exactly when $c(C) = C$. Arbitrary intersections of such fixed points are fixed, because if $C = \bigcap_i C_i$, then monotonicity gives

$$c(C) \subseteq \bigcap_i c(C_i) = C,$$

while extensivity gives the reverse inclusion. Finite unions are fixed by the fourth rule. Hence the fixed points define the closed sets of a topology, and c is its closure operation.

Closure is therefore not merely derived notation; it contains the same information as the topology.

§137.4 The Subspace Topology

For $Y \subseteq X$, the subspace topology is

$$\tau_Y = \{Y \cap U : U \in \tau_X\}.$$

It is the weakest topology on Y for which the inclusion

$$\iota : Y \hookrightarrow X$$

is continuous. More precisely, for any space Z and map $g : Z \to Y$,

$$g : Z \to Y \text{ is continuous} \iff \iota \circ g : Z \to X \text{ is continuous.}$$

Closure behaves exactly as expected.

Proposition 137.4.1
If $A \subseteq Y \subseteq X$, then

$$\boxed{\overline{A}^{Y} = Y \cap \overline{A}^{X}.}$$

Proof. A point $y \in Y$ lies in $\overline{A}^{Y}$ iff every subspace neighborhood $Y \cap U$ of y meets A. This is equivalent to saying that every neighborhood U of y in X meets A, hence $y \in \overline{A}^{X}$. □

For example, $[0,1)$ is open in $[0,2]$ because

$$[0,1) = [0,2] \cap (-1,1),$$

even though it is not open in $\mathbb{R}$.

§137.5 Nets recover closure in every topological space

A directed set is a preorder $(D, \leq)$ such that every two elements have a common upper bound. A net in X is a function

$$x_\bullet : D \to X.$$

It converges to x if for every neighborhood $U \ni x$, there exists $d_0 \in D$ such that

$$d \geq d_0 \implies x_d \in U.$$

Sequences are the special case $D = \mathbb{N}$.

Theorem 137.5.1 (Net characterization of closure)
For $A \subseteq X$,

$$x \in \overline{A} \iff \text{some net in } A \text{ converges to } x.$$

Proof. If a net (a_d) in A converges to x, every neighborhood of x eventually contains some a_d, so it meets A.

Conversely, suppose $x \in \overline{A}$. Direct the set $\mathcal{N}(x)$ of neighborhoods of x by reverse inclusion:

$$U \preceq V \iff V \subseteq U.$$

For each $U \in \mathcal{N}(x)$, choose $a_U \in A \cap U$. Given a neighborhood $W \ni x$, whenever $U \succeq W$ one has $U \subseteq W$, hence $a_U \in W$. Therefore (a_U) converges to x. □

Sequences may fail to detect closure. Let $[0,\omega_1]$ carry the order topology, where ω_1 is the first uncountable ordinal, and set

$$A = [0,\omega_1).$$

Every neighborhood of ω_1 contains a tail $(\alpha,\omega_1]$, so $\omega_1 \in \overline{A}$. But a sequence $(\alpha_n) \subset A$ has countable supremum

$$\alpha = \sup_n \alpha_n < \omega_1,$$

and therefore cannot converge to ω_1. The net indexed by ordinals $\alpha < \omega_1$, ordered in the usual way, does converge to ω_1.

§137.6 Connectedness and Discrete Separators

A separation of X is a decomposition

$$X = U \sqcup V$$

into nonempty open sets. A space is connected when no separation exists. Equivalently, the only clopen subsets are $\varnothing$ and X.

Let $D = \{0,1\}$ have the discrete topology.

Proposition 137.6.1

A space X is connected iff every continuous map

$$f : X \to D$$

is constant.

Proof. If f is nonconstant, then $f^{-1}(0)$ and $f^{-1}(1)$ are nonempty disjoint open sets covering X. Conversely, a separation $X = U \sqcup V$ defines a nonconstant continuous characteristic map taking U to 0 and V to 1. □

Continuous images preserve connectedness. If $f(X) = A \sqcup B$ were separated, then $f^{-1}(A)$ and $f^{-1}(B)$ would separate X.

This gives the intermediate value theorem in topological form. Since an interval $I \subseteq \mathbb{R}$ is connected, the image $f(I)$ of a continuous real-valued function is connected, hence an interval. Therefore every value between $f(a)$ and $f(b)$ is attained.

§137.7 Connected versus Path-Connected

A path from x to y is a continuous map

$$\gamma : [0,1] \to X, \qquad \gamma(0) = x, \quad \gamma(1) = y.$$

Since $[0,1]$ is connected, its image is connected. If a path-connected space had a separation, no path could join points in the two pieces. Hence

$$\text{path-connected} \quad \Longrightarrow \quad \text{connected}.$$

The converse fails for the topologist's sine curve

$$T = \{(x, \sin(1/x)) : x > 0\} \cup (\{0\} \times [-1,1]).$$

The graph

$$G = \{(x, \sin(1/x)) : x > 0\}$$

is the continuous image of $(0, \infty)$, so it is connected. Its closure is T, and the closure of a connected set is connected; thus T is connected.

It is not path-connected. Suppose a path $\gamma(t) = (x(t), y(t))$ joined a point of the vertical segment to a point of G. Let

$$t_0 = \sup\{t : x(t) = 0\}.$$

Immediately after t_0, the coordinate $x(t) > 0$ and tends to zero, while

$$y(t) = \sin(1/x(t)).$$

By the intermediate value property of x, the quantity $1/x(t)$ crosses arbitrarily large intervals, forcing $y(t)$ to oscillate between values near 1 and -1 arbitrarily close to t_0. This contradicts continuity of y at t_0.

§137.8 Local Path-Connectedness

The connected component of x is the union of all connected subsets containing x. The path component of x is the set of points joined to x by paths. Every path component lies inside a connected component.

A space is locally path-connected if every point has a neighborhood basis of path-connected open sets.

Theorem 137.8.1

In a locally path-connected space, path components are open and coincide with connected components.

Proof. Let P be a path component and let $x \in P$. Choose a path-connected open neighborhood $U \ni x$. Every point of U can be joined to x, and x can be joined to the chosen basepoint of P. Hence $U \subseteq P$. Thus P is open.

Path components partition X, so the complement of P is a union of other open path components. Therefore P is also closed. A connected component cannot meet two disjoint clopen path components, so it is contained in one path component. The reverse containment always holds, giving equality. □

Open subsets of $\mathbb{R}^n$, manifolds, and CW complexes are locally path-connected. The topologist's sine curve fails precisely at the limiting vertical segment, where arbitrarily small neighborhoods inherit infinitely many oscillating pieces. Local hypotheses explain why ordinary geometric spaces obey the simpler intuition while general topology does not.

Paths to Groups: Homotopy, Coverings, Fundamental Groups, and Presentations

N-20221223

§138.1 Composing Paths up to Homotopy

A path in a space X is a continuous map

$$\gamma : [0,1] \to X.$$

Its initial and terminal points are

$$s(\gamma) = \gamma(0), \qquad t(\gamma) = \gamma(1).$$

If $t(\gamma) = s(\eta)$, define the concatenation

$$(\gamma * \eta)(u) = \begin{cases} \gamma(2u), & 0 \le u \le \frac{1}{2}, \\ \eta(2u-1), & \frac{1}{2} \le u \le 1. \end{cases}$$

The reverse path is

$$\overline{\gamma}(u) = \gamma(1-u).$$

At the level of parametrized paths, concatenation is not strictly associative. The paths

$$(\alpha * \beta) * \gamma \qquad \text{and} \qquad \alpha * (\beta * \gamma)$$

traverse the same geometric pieces but allocate different subintervals of $[0,1]$ to them. Likewise, $\gamma * \overline{\gamma}$ is not literally the constant path. The desired algebra appears only after quotienting by a suitable deformation relation.

§138.2 Homotopy Relative to Endpoints

Two paths $\gamma_0, \gamma_1 : [0,1] \to X$ with the same endpoints are homotopic relative endpoints if there is a continuous map

$$H : [0,1] \times [0,1] \to X$$

such that

$$H(u,0) = \gamma_0(u), \qquad H(u,1) = \gamma_1(u),$$

and

$$H(0,v) = s(\gamma_0), \qquad H(1,v) = t(\gamma_0)$$

for every v.

This is an equivalence relation. It is also compatible with concatenation: if

$$\gamma_0 \simeq \gamma_1 \text{ rel } \partial I, \qquad \eta_0 \simeq \eta_1 \text{ rel } \partial I,$$

then

$$\gamma_0 * \eta_0 \simeq \gamma_1 * \eta_1 \text{ rel } \partial I.$$

Indeed, concatenate the two homotopies at each time v. Thus homotopy relative endpoints is a congruence for path composition.

Reparametrization through an increasing map $r : [0,1] \to [0,1]$ with $r(0) = 0$, $r(1) = 1$ does not change the homotopy class. A linear interpolation between $r(u)$ and u gives the homotopy. This removes the associativity defect:

$$[\alpha] * ([\beta] * [\gamma]) = ([\alpha] * [\beta]) * [\gamma].$$

Similarly,

$$[\gamma] * [\overline{\gamma}] = [c_{s(\gamma)}], \qquad [\overline{\gamma}] * [\gamma] = [c_{t(\gamma)}].$$

§138.3 The Fundamental Groupoid

The fundamental groupoid $\Pi_1(X)$ has

- points of X as objects;
- homotopy classes relative endpoints of paths as morphisms;
- concatenation as composition.

Every morphism is invertible, with inverse represented by the reversed path. This is why a groupoid, rather than a monoid, is the natural first algebraic object attached to a space.

Fixing a basepoint x_0 and taking endomorphisms gives the fundamental group

$$\pi_1(X, x_0) = \mathrm{Aut}_{\Pi_1(X)}(x_0).$$

Its elements are based loops modulo based homotopy.

If ρ is a path from x_0 to x_1, basepoint change is

$$\rho_{\#} : \pi_1(X, x_0) \to \pi_1(X, x_1), \qquad [\gamma] \longmapsto [\overline{\rho} * \gamma * \rho].$$

A different choice of ρ changes this isomorphism by an inner automorphism. The groupoid keeps this dependence visible instead of hiding it behind a noncanonical identification.

§138.4 Loops Lift to Integers

Consider

$$p : \mathbb{R} \to S^1, \qquad p(t) = e^{2\pi i t}.$$

For each $t_0 \in \mathbb{R}$, restriction to

$$\left(t_0 - \frac{1}{2}, t_0 + \frac{1}{2}\right)$$

is a homeomorphism onto $S^1 \setminus \{-p(t_0)\}$. Thus p is a covering map.

Let $\gamma : [0,1] \to S^1$ and choose $a \in \mathbb{R}$ with $p(a) = \gamma(0)$. Subdivide $[0,1]$ so that the image of each subinterval lies in one evenly covered arc. On the first subinterval use the inverse branch whose value at 0 is a; on each later subinterval choose the unique branch agreeing at the common endpoint. This constructs a unique lift

$$\widetilde{\gamma} : [0,1] \to \mathbb{R}, \qquad p \circ \widetilde{\gamma} = \gamma, \qquad \widetilde{\gamma}(0) = a.$$

If γ is a loop based at 1 and $\widetilde{\gamma}(0) = 0$, then

$$e^{2\pi i\widetilde{\gamma}(1)} = 1,$$

so

$$\widetilde{\gamma}(1) \in \mathbb{Z}.$$

Define the winding number by

$$\operatorname{wind}(\gamma) = \widetilde{\gamma}(1).$$

Concatenation adds endpoints of lifts:

$$\operatorname{wind}(\gamma * \eta) = \operatorname{wind}(\gamma) + \operatorname{wind}(\eta).$$

A based homotopy of loops lifts after fixing the initial lift. The terminal lift is a continuous integer-valued function of the homotopy parameter, hence constant. Therefore winding number depends only on the based homotopy class.

§138.5 The covering computation gives $\pi_1(S^1) \cong \mathbb{Z}$

The map

$$W : \pi_1(S^1, 1) \to \mathbb{Z}, \qquad W([\gamma]) = \operatorname{wind}(\gamma)$$

is a homomorphism.

It is surjective because

$$\gamma_n(u) = e^{2\pi i n u}$$

has lift $\widetilde{\gamma}_n(u) = nu$ and winding number n.

It is injective. Suppose $W([\gamma]) = 0$. The lift $\widetilde{\gamma}$ is then a loop in $\mathbb{R}$ based at 0. The linear homotopy

$$\widetilde{H}(u, v) = (1 - v)\widetilde{\gamma}(u)$$

contracts it to the constant loop. Projecting by p gives a based null-homotopy of γ. Hence

$$\ker W = 0.$$

Therefore

$$\boxed{\pi_1(S^1, 1) \cong \mathbb{Z}.}$$

The group operation is addition of winding numbers, and reversal changes the sign.

§138.6 The punctured plane retracts onto the circle

Let

$$\mathbb{C}^* = \mathbb{C} \setminus \{0\}.$$

The map

$$r : \mathbb{C}^* \to S^1, \qquad r(z) = \frac{z}{|z|}$$

is a deformation retraction. Define

$$H(z, v) = \left((1 - v) + \frac{v}{|z|}\right) z.$$

The scalar coefficient is positive, so $H(z,v) \neq 0$. Moreover,

$$H(z,0) = z, \qquad H(z,1) = \frac{z}{|z|},$$

and if $|z| = 1$, then $H(z,v) = z$ for all v. Thus the inclusion $S^1 \hookrightarrow \mathbb{C}^*$ and r are homotopy inverses.

Consequently,

$$\boxed{\pi_1(\mathbb{C}^*, 1) \cong \mathbb{Z}.}$$

The missing point is measured exactly by winding number.

§138.7 Continuous maps induce homomorphisms

A based continuous map

$$f : (X, x_0) \to (Y, y_0)$$

sends a loop γ to $f \circ \gamma$. It preserves homotopies and concatenation, hence induces

$$f_* : \pi_1(X, x_0) \to \pi_1(Y, y_0), \qquad f_*([\gamma]) = [f \circ \gamma].$$

Functoriality is immediate:

$$(g \circ f)_* = g_* \circ f_*, \qquad (\mathrm{id}_X)_* = \mathrm{id}_{\pi_1(X)}.$$

For

$$f_m : S^1 \to S^1, \qquad f_m(z) = z^m,$$

one has

$$f_m \circ \gamma_n(u) = e^{2\pi i m n u}.$$

Under $\pi_1(S^1) \cong \mathbb{Z}$, the induced homomorphism is therefore

$$(f_m)_* : \mathbb{Z} \to \mathbb{Z}, \qquad n \longmapsto mn.$$

This computation distinguishes the maps $z \mapsto z^m$ up to based homotopy.

§138.8 Van Kampen assembles groups from overlapping pieces

Let $X = U \cup V$, where $U, V, U \cap V$ are path-connected open sets containing a basepoint x_0. The Seifert–van Kampen theorem states that the diagram

$$\begin{array}{ccc} \pi_1(U \cap V, x_0) & \longrightarrow & \pi_1(U, x_0) \\ \downarrow & & \downarrow \\ \pi_1(V, x_0) & \longrightarrow & \pi_1(X, x_0) \end{array}$$

is a pushout in the category of groups. Concretely,

$$\pi_1(X, x_0) \cong \frac{\pi_1(U, x_0) * \pi_1(V, x_0)}{\langle\langle i_{U*}(\omega) i_{V*}(\omega)^{-1} : \omega \in \pi_1(U \cap V, x_0) \rangle\rangle}.$$

The free product records loops from the two pieces; the normal closure imposes the identifications already visible in the overlap.

For the figure-eight

$$X = S^1 \vee S^1,$$

choose neighborhoods retracting onto the two circles whose intersection is contractible. The overlap contributes no relation, so

$$\boxed{\pi_1(S^1 \vee S^1) \cong \mathbb{Z} * \mathbb{Z} = F(a, b).}$$

The words

$$ab, \quad ba, \quad aba^{-1}b^{-1}$$

represent genuinely different loop classes; no commutativity relation is present.

§138.9 Attaching a two-cell adds one normal relation

Let X be path-connected and attach a disk along a loop

$$\varphi : S^1 \to X.$$

The resulting space

$$X \cup_\varphi D^2$$

contains a new disk across which the attaching loop can contract. Van Kampen gives

$$\pi_1(X \cup_\varphi D^2) \cong \pi_1(X)/\langle\langle [\varphi] \rangle\rangle.$$

Thus cell attachment translates directly into a group presentation.

A torus has a CW structure with one vertex, two one-cells a, b, and one two-cell attached along

$$aba^{-1}b^{-1}.$$

Starting from the figure-eight gives

$$\pi_1(T^2) \cong \langle a, b \mid aba^{-1}b^{-1} = 1 \rangle \cong \mathbb{Z}^2.$$

The two basic loops commute because the square filling of the commutator becomes the two-cell of the torus.

§138.10 Coverings of the Circle

The classification of coverings becomes transparent once we use the additive description of the circle,

$$S^1 \cong \mathbb{R}/\mathbb{Z}, \qquad [t] \longmapsto e^{2\pi i t}.$$

Every subgroup of its fundamental group $\mathbb{Z}$ is $n\mathbb{Z}$ for a unique *nonnegative* integer n. The value $n = 0$ is an important case: it denotes the zero subgroup, not a covering with zero sheets.

All the corresponding coverings fit in one construction:

$$\boxed{\begin{aligned} X_n &= \mathbb{R}/n\mathbb{Z}, \\ p_n : X_n &\longrightarrow \mathbb{R}/\mathbb{Z}, \qquad [t]_{n\mathbb{Z}} \longmapsto [t]_{\mathbb{Z}}. \end{aligned}}$$

For $n \geq 1$, the space X_n is a circle whose coordinate has period n. The covering has n sheets: the fiber over $[s]_{\mathbb{Z}}$ consists of the n classes represented by $s, s+1, \ldots, s+n-1$. After identifying X_n with the unit circle by

$$[t]_{n\mathbb{Z}} \longmapsto e^{2\pi i t/n},$$

the map p_n is exactly $z \mapsto z^n$. The induced subgroup is $n\mathbb{Z}$, because a closed lifted loop traversing the domain circle once maps to a loop of winding number n.

What about $n = 0$? By definition

$$X_0 = \mathbb{R}/0\mathbb{Z} = \mathbb{R}.$$

The corresponding map is

$$p_0 : \mathbb{R} \longrightarrow S^1, \qquad p_0(t) = e^{2\pi i t}.$$

It has infinitely many sheets, is simply connected upstairs, and is therefore the *universal covering*. Its fundamental group has trivial image, exactly the subgroup $0\mathbb{Z}$ missing from the finite-degree list.

The same construction also computes its deck transformations. A homeomorphism of X_n commuting with p_n must preserve each fiber, and translations by an integer do just that:

$$\tau_k([t]_{n\mathbb{Z}}) = [t+k]_{n\mathbb{Z}}.$$

For $n \geq 1$, $\tau_k = \tau_\ell$ precisely when $k \equiv \ell \pmod{n}$. These are all the deck transformations, so

$$\boxed{\mathrm{Deck}(p_n) \cong \mathbb{Z}/n\mathbb{Z} \quad (n \geq 1), \qquad \mathrm{Deck}(p_0) \cong \mathbb{Z}.}$$

For the universal cover this says that integer translations of the real line are its entire symmetry group over the circle.

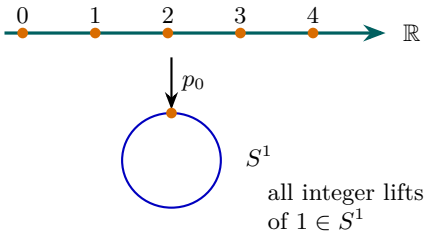


Figure 138.1: The universal circle cover. All integers in $\mathbb{R}$ lie above the same point 1 of S^1. Translation by one unit permutes the fiber, generating the deck group $\mathbb{Z}$.

The classification theorem for connected based coverings of sufficiently nice spaces now has no missing case: every subgroup $n\mathbb{Z}$, including $n = 0$, has an explicitly described connected covering. For unbased coverings one usually takes subgroups up to conjugacy; $\mathbb{Z}$ is abelian, so this makes no difference here. Hatcher's *Algebraic Topology*, Section 1.3, explains why the correspondence holds for general locally path-connected, semilocally simply connected spaces.

The progression is now concrete:

$$\text{paths} \longrightarrow \text{homotopy classes} \longrightarrow \text{groups} \longrightarrow \text{coverings and presentations}.$$

A single integer n has acquired three compatible meanings: a subgroup $n\mathbb{Z}$ of $\pi_1(S^1)$, the number of sheets (for $n > 0$), and the cyclic deck group of the covering. The zero subgroup explains why a simply connected real line must appear beside all the finite circle covers.

Quotient and Product Topologies: Universal Properties and Gluing

N-20221228

§139.1 Gluing Endpoints by Quotients

Start with $I = [0, 1]$ and identify $0 \sim 1$. The map

$$q : I \longrightarrow S^1, \qquad q(t) = e^{2\pi i t}$$

is continuous, surjective, and identifies exactly the two endpoints. It therefore induces a continuous bijection $I/(0 \sim 1) \to S^1$. The quotient is compact and the circle is Hausdorff, so the induced bijection is a homeomorphism. This last compact-to-Hausdorff argument is the rigorous step behind the familiar picture of bending an interval into a circle.

The general definition for a surjection $q : X \to Y$ is

$$U \subseteq Y \text{ is open} \quad \Longleftrightarrow \quad q^{-1}(U) \subseteq X \text{ is open.}$$

This is the finest topology on Y for which q is continuous. A map $f : Y \to Z$ is continuous precisely when $f \circ q : X \to Z$ is continuous. Both parts follow immediately by testing preimages of open sets in Z.

A subset $A \subseteq X$ is called *saturated* when

$$A = q^{-1}(q(A)).$$

Thus the open subsets of the quotient correspond to the saturated open subsets upstairs. Near the point obtained from identifying 0 and 1, a neighborhood pulls back to the union of small half-intervals near *both* ends. Keeping only one end is not saturated and therefore does not describe a quotient neighborhood.

The presentation and notation follow the quotient-space and product-space chapters in Munkres' *Topology*, with the universal-property formulation also given in Hatcher's *Algebraic Topology*, Section 0.[1]

§139.2 Collapsing a boundary and reading a glued surface

The quotient D^n/S^{n-1} collapses the boundary of the closed n-disk to a single point. The resulting space is homeomorphic to S^n: the disk interior maps onto a sphere minus one point, while its boundary maps to the missing point. This can be described by stereographic projection, or by regarding the disk as the one-point compactification of its interior $\mathbb{R}^n$. The map is a quotient map because the compact disk maps onto the Hausdorff sphere.

A square provides another family of examples. Identifying one pair of opposite edges with matching directions gives a cylinder; identifying the other pair in the same direction then gives a torus. Reversing one gluing leads to a Klein bottle, whereas a single twisted

[1] See A. Hatcher, *Algebraic Topology*, Section 0, freely available at https://pi.math.cornell.edu/~hatcher/AT/AT.pdf.

identification on a strip gives a Möbius band. It is essential to specify *which* edges and the direction of their identifications; a sketch with unlabeled arrows is not enough.

The same gluing can be encoded by an attaching word. For a torus it is

$$aba^{-1}b^{-1},$$

while a Klein bottle can be presented by

$$aba^{-1}b.$$

For the torus, a square is attached as a two-cell to a wedge of two circles. Its boundary loop is a commutator, yielding

$$\pi_1(T^2) \cong \langle a, b \mid aba^{-1}b^{-1} = 1 \rangle \cong \mathbb{Z}^2.$$

Thus the arrow labels in the square are the start of a genuine algebraic calculation of a fundamental group.

§139.3 Products use the opposite continuity test

For topological spaces X, Y, the product topology on $X \times Y$ is generated by the rectangles

$$U \times V, \qquad U \subseteq X,\ V \subseteq Y \text{ open}.$$

Indeed,

$$U \times V = \pi_X^{-1}(U) \cap \pi_Y^{-1}(V),$$

so it is the coarsest topology making both projections π_X, π_Y continuous. Its mapping property reads

$$h : Z \to X \times Y \text{ continuous} \quad \Longleftrightarrow \quad \pi_X \circ h,\ \pi_Y \circ h \text{ continuous}.$$

Compare this with the quotient criterion, where the universal test concerns *maps out of* the quotient. The two constructions are both specified by functions, but by different directions of functions:

construction	canonical maps	continuity checked by
product	$X \times Y \to X, Y$	component maps into the product
quotient	$X \to X/\sim$	maps defined on the quotient

A continuous map into a product requires coordinatewise continuity; a continuous map out of a quotient can be checked before identifying the points.

§139.4 When quotients behave badly

Let $q : \mathbb{R} \to Y$ identify all rational numbers as one point and all irrational numbers as another. Neither $\mathbb{Q}$ nor its complement is open in $\mathbb{R}$, and neither has open nonempty saturated proper subsets. Therefore Y carries the *indiscrete* two-point topology. It cannot be Hausdorff. This is a concrete warning that quotient spaces of Hausdorff spaces need not remain Hausdorff.

There is a second pitfall in mixing the two constructions. Given quotient maps

$$q : X \to X/\sim, \qquad r : Y \to Y/\approx,$$

the product $q \times r : X \times Y \to (X/\!\sim) \times (Y/\!\approx)$ is a continuous surjection, but it need not be a quotient map for arbitrary spaces. The induced set-theoretic bijection

$$(X \times Y)/(\sim \times \approx) \longrightarrow (X/\!\sim) \times (Y/\!\approx)$$

may therefore fail to be a homeomorphism. If both q and r are *open quotient maps*, however, their product is open and surjective, hence a quotient map; the displayed bijection is then a homeomorphism. The endpoint-identification map above is closed rather than necessarily open, but compactness/Hausdorffness supplies an alternative direct proof for the torus.

The point of the universal-property approach is that it forces these hypotheses into view. A drawing can correctly predict a circle or torus while hiding the conditions under which its proposed quotient carries the intended topology.

Algebraic Topology and Algebra: Products, Wedge Sums, CW Complexes, Cosets, and Surface Gluings

N-20221229

§140.1 From quotient pictures to algebraic invariants

Several threads meet in one progression: the torus as $(\mathbb{R}/\mathbb{Z})^2$, the product $S^1 \times S^1$, disjoint unions and wedge sums, CW complexes, cosets, and polygonal gluing descriptions of surfaces. The important experiment is to attach a single two-cell to the same figure-eight one-skeleton in two different ways. The resulting torus and Klein bottle have identical cell counts but different fundamental groups and homology.[1]

A useful geometric realization of the product is

$$S^1 \times S^1 \cong T^2.$$

Think of a torus as a vertical circle carried around a horizontal circle. The path traced by that moving circle is the torus. This is not merely a picture: it is a parametrization.

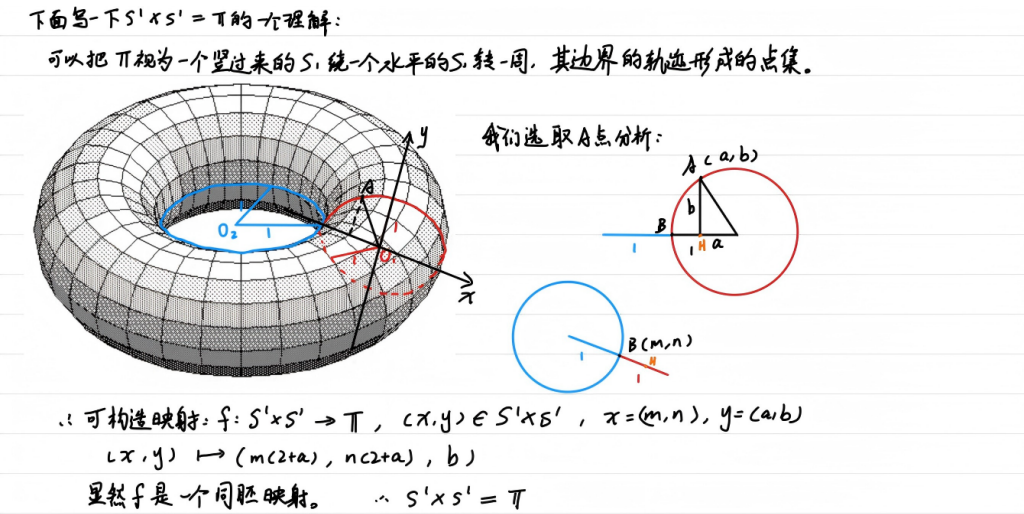


Figure 140.1: The two circle coordinates of the torus: one travels around its major circle, the other around its cross-section.

§140.2 $S^1 \times S^1$ as a parametrized torus

Write a point of the first circle as an angle θ, and a point of the second circle as an angle ϕ. For a torus with major radius R and minor radius r, a standard parametrization is

$$F(\theta, \phi) = ((R + r\cos\phi)\cos\theta, (R + r\cos\phi)\sin\theta, r\sin\phi).$$

[1]The CW construction, attaching words, cellular boundaries and projective-space examples follow A. Hatcher, *Algebraic Topology*, Chapters 0–2.

This realizes $S^1 \times S^1 \to T^2$. The first angle decides where the circular cross-section is placed around the central hole; the second decides where we are on that cross-section.

A point A may be decomposed into two circle parameters, but the invariant statement is cleaner: the torus is the space of ordered pairs of circle-points, realized geometrically by the map above.

§140.3 Disjoint union

For topological spaces X and Y, the disjoint union $X \amalg Y$ keeps the two spaces formally separate. A subset $U \subseteq X \amalg Y$ is open iff

$$U \cap X \text{ is open in } X, \qquad U \cap Y \text{ is open in } Y.$$

If X and Y are nonempty, then $X \amalg Y$ is disconnected. Indeed, X and Y are both open in the disjoint union topology, disjoint, nonempty, and their union is the whole space. This proof is more precise than the vague statement that the spaces are simply "side by side". The topology, not only the picture, gives the separation.

§140.4 Wedge sum

The wedge sum is obtained from a disjoint union by identifying chosen basepoints:

$$X \vee Y = (X \amalg Y)/(x_0 \sim y_0).$$

Thus disjoint union places spaces side by side, while wedge sum fuses them at one point. The prototype is $S^1 \vee S^1$, the figure-eight space.

Example 57.4.4 ($S^1 \vee S^1$ is a figure eight)
Let $X = S^1$ and $Y = S^1$, and let $x_0 \in X$ and $y_0 \in Y$ be any points. Then $X \vee Y$ is a "figure eight": it is two circles fused together at one point.

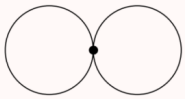


Figure 140.2: The wedge sum $S^1 \vee S^1$ as two circles fused at one point.

This construction matters because loops in the two circles remain visibly independent. Algebraically, the fundamental group of the figure eight is the free group on two generators. So the picture is not a toy: it is one of the first spaces where topology produces noncommutative algebra.

§140.5 CW complexes: building spaces by cells

A CW complex is built inductively. Start with a set of 0-cells. Attach 1-cells by gluing endpoints to the 0-skeleton. Attach 2-cells by gluing the boundary circle of a disk to the 1-skeleton. Continue dimension by dimension:

$$X^0 \subseteq X^1 \subseteq X^2 \subseteq \cdots.$$

A k-cell is a copy of D^k attached along S^{k-1} by an attaching map

$$\varphi_\alpha : S^{k-1}_\alpha \to X^{k-1}.$$

Informally: attach disks along their boundaries.

The disk D^2 illustrates that the same space may have different CW structures. One uses two 0-cells, two 1-cells, and one 2-cell. The other uses one 0-cell, one 1-cell whose two endpoints are glued to that point, and one 2-cell. Both build a disk, but the second is more economical.

Example 57.5.2 (D^2 with $2+2+1$ and $1+1+1$ cells)

(a) First, we start with X^0 having two points e_a^0 and e_b^0. Then, we join them with two 1-cells D^1 (green), call them e_c^1 and e_d^1. The endpoints of each 1-cell (the copy of S^0) get identified with distinct points of X^0; hence $X^1 \cong S^1$. Finally, we take a single 2-cell e^2 (yellow) and weld it in, with its boundary fitting into the copy of S^1 that we just drew. This gives the figure on the left.

(b) In fact, one can do this using just $1+1+1=3$ cells. Start with X^0 having a single point e^0. Then, use a single 1-cell e^1, fusing its two endpoints into the single point of X^0. Then, one can fit in a copy of S^1 as before, giving D^2 as on the right.

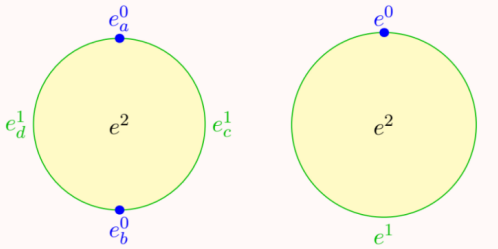


Figure 140.3: Two CW descriptions of D^2: one more literal, one more economical.

The lesson is that a CW decomposition is not unique. It is chosen because it makes later topology and algebra computable.

§140.6 The torus as a quotient of a square

The torus can be formed from a square by identifying opposite edges in the same direction. Walking off the right edge returns one at the corresponding point on the left edge; walking off the top returns one at the bottom. Algebraically,

$$T^2 \cong (\mathbb{R}/\mathbb{Z})^2 \cong S^1 \times S^1.$$

The quotient square viewpoint treats $[0,1] \times [0,1]$ as a fundamental domain for $(\mathbb{R}/\mathbb{Z})^2$.

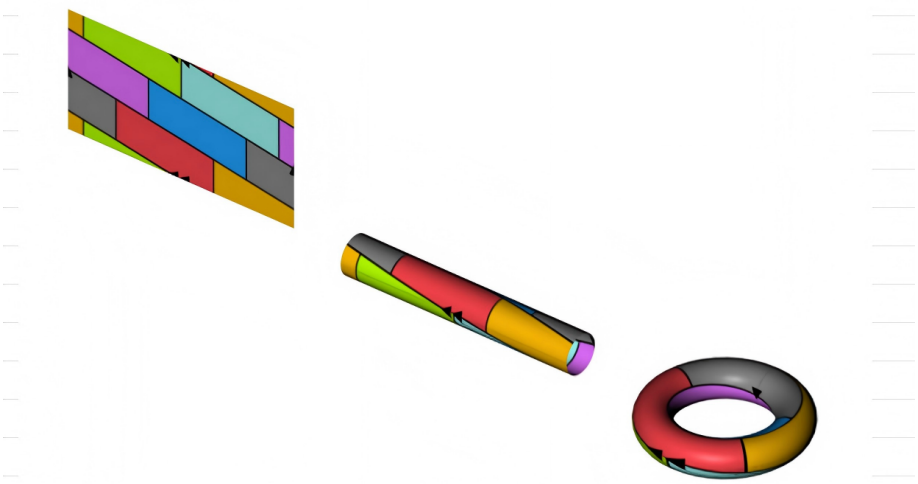


Figure 140.4: The quotient square/cylinder/torus picture: $(\mathbb{R}/\mathbb{Z})^2$ as a glued square.

As a CW complex, the torus has one 0-cell, two 1-cells a, b, and one 2-cell. The boundary of the 2-cell is attached along the word

$$aba^{-1}b^{-1}.$$

This gives the familiar presentation

$$\pi_1(T^2) = \langle a, b \mid aba^{-1}b^{-1} = 1 \rangle \cong \mathbb{Z}^2.$$

The commutativity of $\mathbb{Z}^2$ is encoded by the boundary word of the square.

§140.7 Cosets as algebraic quotients

Cosets are the algebraic analogue of quotienting. If H is a subgroup of G, right congruence modulo H is

$$a \equiv_r b \pmod{H} \iff ab^{-1} \in H.$$

It is an equivalence relation. Reflexivity uses $aa^{-1} = e \in H$; symmetry uses $(ab^{-1})^{-1} = ba^{-1}$; transitivity uses

$$ac^{-1} = (ab^{-1})(bc^{-1}).$$

The equivalence class of a is the right coset

$$Ha = \{ha : h \in H\}.$$

The map $H \to Ha$, $h \mapsto ha$, is a bijection, so $|Ha| = |H|$. The group G is partitioned into cosets, and two cosets are either equal or disjoint. This is exactly the algebraic version of cutting a space into equivalence classes.

The quotient $(\mathbb{R}/\mathbb{Z})$ illustrates the same principle in a geometric setting. Both are quotient constructions: identify things differing by a subgroup.

§140.8 The Klein bottle

The Klein bottle is built like the torus, except one pair of edges is identified in the opposite direction. If one first glues a pair of edges, one gets a cylinder. But gluing

the resulting boundary circles in opposite directions cannot be embedded in ordinary three-dimensional space without self-intersection. The usual picture in $\mathbb{R}^3$ is therefore an immersion, not a faithful embedding.

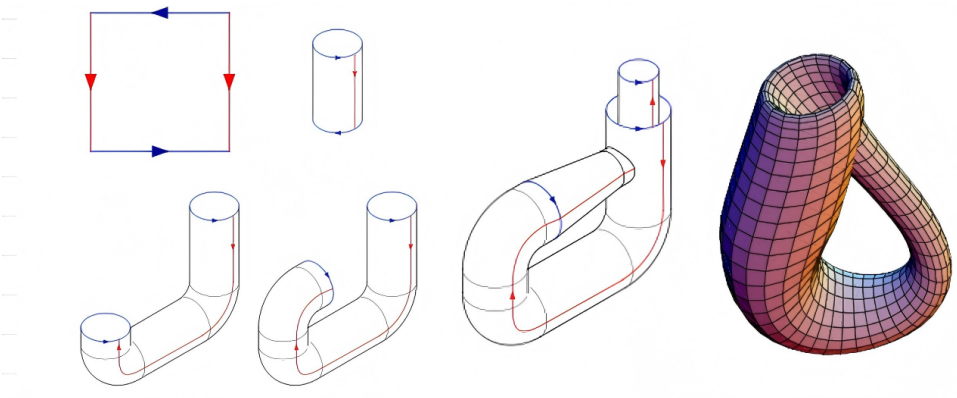


Figure 140.5: The Klein bottle: one edge identification is reversed, so the usual 3D picture self-intersects.

As a CW complex, the Klein bottle again has one 0-cell, two 1-cells, and one 2-cell, but the attaching word changes. A common presentation is

$$\pi_1(K) = \langle a, b \mid aba^{-1} = b^{-1} \rangle.$$

The relation says that going around one direction reverses the other. Thus the torus and Klein bottle have the same cell counts but different attaching words, and this difference controls orientability and fundamental group.

§140.9 Polygon schemata for surfaces

Labelled polygons give concrete gluing schemata for surfaces. A polygon with labelled, oriented boundary edges can be glued according to those labels. The resulting quotient is a surface. For example, the torus corresponds to

$$aba^{-1}b^{-1},$$

while an orientable surface of genus g has a boundary word of the form

$$a_1 b_1 a_1^{-1} b_1^{-1} \cdots a_g b_g a_g^{-1} b_g^{-1}.$$

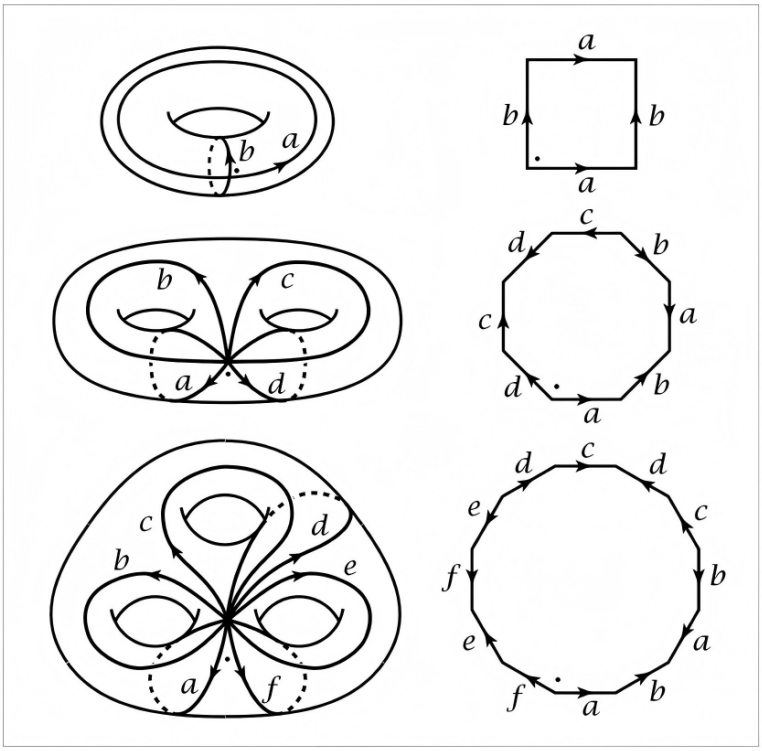


Figure 140.6: Polygon schemata and surface gluings: topology encoded by labelled boundary words.

The picture compresses several ideas at once: cut a surface into a polygon, label the boundary, glue equal labels, and read those labels as algebraic generators and relations.

§140.10 Projective spaces as quotient spaces

Real and complex projective spaces give two further quotients with strikingly different cellular structures. Real projective space is

$$\mathbb{RP}^n = S^n/(x \sim -x),$$

the space of lines through the origin in $\mathbb{R}^{n+1}$. Complex projective space is the space of complex lines through the origin in $\mathbb{C}^{n+1}$, equivalently

$$\mathbb{CP}^n = S^{2n+1}/S^1.$$

A useful CW fact is that $\mathbb{RP}^n$ has one cell in each dimension $0, 1, \dots, n$, while $\mathbb{CP}^n$ has one cell in each even real dimension $0, 2, 4, \dots, 2n$.

§140.11 An attaching word has two algebraic interpretations

Begin with the figure eight

$$X^1 = S^1 \vee S^1.$$

Its fundamental group is the free group $F(a, b)$ on the loops a, b. A cellular map $S^1 \to X^1$ can be described, up to homotopy, by a word $w(a, b)$: as one goes around the boundary circle, the map traverses the two loops with the orientations specified by the letters.

Attach a two-cell along this map, forming $X = X^1 \cup_w D^2$. Van Kampen's theorem says that the attaching disk makes the loop w null-homotopic:

$$\boxed{\pi_1(X) \cong F(a,b)/\langle\langle w \rangle\rangle,}$$

where the denominator is the normal closure of the word. The normal closure matters: if a loop becomes null-homotopic, so do all its conjugates.

Now compute the same attaching map in cellular homology. Since X has one cell in dimension zero, two in dimension one, and one in dimension two, its cellular chain complex is

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\partial_2} \mathbb{Z}^2 \xrightarrow{\partial_1 = 0} \mathbb{Z} \longrightarrow 0.$$

The boundary of the two-cell is obtained by *abelianizing the word*: if w has exponent sums r in a and s in b, then

$$\boxed{\partial_2(1) = (r, s).}$$

The group computation remembers the order of the letters; the homology computation remembers only their total signed counts. This difference is exactly why the same attaching diagram contains more information than the resulting chain complex.

§140.12 Torus versus Klein Bottle

For the torus, the boundary word is

$$w_T = aba^{-1}b^{-1}.$$

Its exponent sums are both zero, hence $\partial_2 = 0$. It follows that

$$\begin{aligned} H_0(T^2;\mathbb{Z}) &\cong \mathbb{Z}, \\ H_1(T^2;\mathbb{Z}) &\cong \mathbb{Z}^2, \\ H_2(T^2;\mathbb{Z}) &\cong \mathbb{Z}. \end{aligned}$$

In the fundamental group, however, the attaching relation is $aba^{-1}b^{-1} = 1$, or $ab = ba$, so

$$\pi_1(T^2) = \langle a, b \mid [a,b] = 1 \rangle \cong \mathbb{Z}^2.$$

For the Klein bottle, take the standard word

$$w_K = aba^{-1}b.$$

It has exponent sum zero in a and two in b. Therefore

$$\partial_2(1) = (0,2), \qquad H_1(K;\mathbb{Z}) = \frac{\mathbb{Z}^2}{\langle (0,2) \rangle} \cong \mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z},$$

whereas

$$H_2(K;\mathbb{Z}) = \ker \partial_2 = 0.$$

The corresponding fundamental-group presentation is

$$\pi_1(K) = \langle a, b \mid aba^{-1} = b^{-1} \rangle.$$

It records a conjugation that reverses one generator. That reversal is the algebraic shadow of the twist in the edge identification and of the failure of orientability.

Both spaces have Euler characteristic

$$\chi = 1 - 2 + 1 = 0.$$

Thus χ is incapable of distinguishing the two gluing patterns. The chain complex can distinguish them because it remembers ∂_2, and the fundamental group distinguishes them even more sharply because it remembers the noncommutative attaching word.

There is a second instructive comparison. The figure eight itself has

$$H_1(S^1 \vee S^1; \mathbb{Z}) = \mathbb{Z}^2$$

just like the torus. The difference is the extra two-cell: its attaching commutator is invisible after abelianization, so H_1 does not change, but $H_2 = \mathbb{Z}$ appears and π_1 becomes abelian. This shows exactly which information each invariant forgets.

§140.13 Projective Spaces in All Dimensions

Real and complex projective spaces are quotient spaces:

$$\mathbb{RP}^n = S^n/(x \sim -x), \qquad \mathbb{CP}^n = S^{2n+1}/S^1.$$

Their natural CW decompositions are rather different.

The space $\mathbb{RP}^n$ has one cell e^k in every dimension $0 \le k \le n$. Its k-skeleton is $\mathbb{RP}^k$. The attaching map of e^k is the double covering $S^{k-1} \to \mathbb{RP}^{k-1}$. When the cellular attaching map is projected onto the quotient of the $(k-1)$-skeleton by its $(k-2)$-skeleton, the resulting degree is

$$\boxed{\partial_k = 1 + (-1)^k.}$$

One way to see the sign is to compare the two hemispheres of S^{k-1}: the second contributes the degree of the antipodal map, which is $(-1)^k$. Thus the cellular boundaries alternate between multiplication by 2 and the zero map:

$$\partial_{2j} = 2, \qquad \partial_{2j+1} = 0.$$

For $\mathbb{RP}^2$, the chain complex is therefore

$$0 \longrightarrow \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \longrightarrow 0,$$

and

$$\boxed{H_1(\mathbb{RP}^2; \mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}, \qquad H_2(\mathbb{RP}^2; \mathbb{Z}) = 0.}$$

The 2-torsion is the homological residue of the antipodal identification.

Complex projective space has instead one cell in every even real dimension $0, 2, \dots, 2n$, and no odd-dimensional cells. Every cellular differential therefore vanishes for dimension reasons. Hence

$$\boxed{H_{2j}(\mathbb{CP}^n; \mathbb{Z}) \cong \mathbb{Z} \quad (0 \le j \le n),} \qquad H_{2j+1}(\mathbb{CP}^n; \mathbb{Z}) = 0.$$

The difference in cell dimensions explains the difference in homology with almost no additional computation.

§140.14 Orientability is encoded by the top boundary

The chain computations give a uniform way to detect orientability for connected closed surfaces. A coherent orientation on all top-dimensional cells produces a nonzero fundamental cycle, which generates $H_2(M;\mathbb{Z}) \cong \mathbb{Z}$. If going around a loop reverses the local orientation, such a global cycle cannot exist, and $H_2(M;\mathbb{Z}) = 0$.

The torus has a top cycle because the opposite oriented edge contributions cancel. The Klein bottle does not: its top two-cell has boundary $2b$, so it cannot be a cycle over $\mathbb{Z}$. Nevertheless, with coefficients in $\mathbb{F}_2$, that boundary becomes zero:

$$2b = 0 \quad \text{in } \mathbb{F}_2.$$

Thus

$$H_2(K;\mathbb{F}_2) \cong \mathbb{F}_2,$$

as for every connected closed surface. Reduction modulo two forgets the distinction between positive and negative orientation.

The labelled polygon has led to a chain of increasingly sensitive invariants. The cell counts give Euler characteristic, exponent sums give cellular boundary maps, and the full boundary word gives the fundamental-group presentation. The arithmetic of 2-torsion and the sign of an attaching map retain information that an unlabelled drawing or an Euler-characteristic calculation cannot.

From Extremal Triangles to Euler Characteristic and Degree

N-20240330

§141.1 Maximum-area triangles live on the convex hull

Let S be a set of n points in the plane, not all collinear. A triangle determined by S is maximal when its area is the largest area attained by any triangle with vertices in S.

Fix two vertices a, b. The area of abp is

$$\operatorname{area}(abp) = \frac{1}{2}|a-b|\,\operatorname{dist}(p, \overleftrightarrow{ab}).$$

As a function of p, the signed height above the line ab is affine. Its maximum and minimum over the convex hull of S occur at hull vertices. Therefore, if p lies strictly inside the convex hull, replacing it by a suitable hull vertex does not decrease the area and, unless the height was already extremal along an entire hull edge, increases it.

Under the standing assumption that no three points are collinear, every vertex of a maximum-area triangle is therefore a vertex of the convex hull. Interior points may be deleted without changing the family of maximal triangles. The counting problem reduces to the vertices

$$v_0, v_1, \dots, v_{m-1}$$

of a strictly convex m-gon, where $m \le n$. The proof below establishes the upper bound required by the original problem in this general-position setting.

§141.2 Alternating Maximal Triangles

The key fact is an alternation lemma.

Proposition 141.2.1 (Alternation of maximal triangles)

Let Δ and Δ' be two maximum-area triangles whose vertices lie on a convex polygon. Every side of Δ' meets some side of Δ.

Proof. Write

$$\Delta = a_1a_2a_3.$$

Through each a_i, draw the line parallel to the opposite side $a_{i+1}a_{i+2}$, with indices modulo three. These three lines bound a larger triangle T.

Every point of the original set lies in T. If a point lay beyond, say, the line through a_1 parallel to a_2a_3, then its distance from the line a_2a_3 would exceed the height of a_1. Replacing a_1 by that point would produce a triangle of larger area.

The central triangle Δ divides $T \setminus \Delta$ into three corner triangles T_1, T_2, T_3, each a translate of $-\Delta$ and therefore of the same area as Δ. Let the vertices of Δ' lie in these four regions.

If all three lie in one T_i, equality of areas forces $\Delta' = T_i$; its sides are parallel to and meet the corresponding boundary sides of Δ. If one vertex lies in each T_i, the three connecting sides cross the three sides of Δ.

The remaining possibility places two vertices, say p, q, in one corner triangle and the third vertex r in another. A line through p parallel to qr separates one vertex of Δ farther from the line qr than p. Replacing p by that vertex produces a triangle with base qr and strictly greater height, contradicting maximality of pqr.

Thus the nonalternating configuration is impossible, proving the claim. □

The proposition is stronger than "all vertices lie on the hull." It says that two extremal triples cannot occupy unrelated arcs of the cyclic order; their vertices must interlace strongly enough that their sides encounter one another.

§141.3 Cyclic lifts turn alternation into the bound n

Index the vertices of the convex m-gon periodically by all integers:

$$v_{i+m} = v_i.$$

Call $(i, j, k) \in \mathbb{Z}^3$ a proper triple when

$$i < j < k < i + m$$

and $v_i v_j v_k$ is a maximal triangle. Order triples coordinatewise:

$$(i, j, k) \le (i', j', k') \quad \Longleftrightarrow \quad i \le i',\ j \le j',\ k \le k'.$$

The alternation proposition implies that any two proper triples are comparable. To see the cyclic-order step, place the six lifted indices on the real line. If neither

$$(i, j, k) \le (i', j', k')$$

nor the reverse inequality holds, their coordinate order changes sign at least twice: one lifted vertex of the second triangle lies after the corresponding vertex of the first, while another lies before it. After a cyclic relabelling, two consecutive vertices of one triangle then lie strictly in the same open boundary arc between two consecutive vertices of the other. The side joining those two vertices is contained in the corresponding sector of the convex polygon and cannot meet any side of the other triangle. This contradicts the alternation proposition. Therefore all proper triples form a chain $C \subset \mathbb{Z}^3$.

Fix one proper triple (a, b, c), and consider the half-open subchain

$$\begin{aligned} C_0 = \{(i, j, k) \in C : {} & (a, b, c) \le (i, j, k), \\ & (i, j, k) < (a + m, b + m, c + m)\}. \end{aligned}$$

Here the final strict inequality means coordinatewise $\le$, but not equality of all three coordinates.

Every geometric maximal triangle has exactly three proper representatives in C_0, corresponding to choosing each of its three vertices in turn as the first cyclic vertex. On the other hand, distinct integer points in a coordinatewise chain have strictly increasing coordinate sum. Inside the displayed half-open interval, that sum ranges over fewer than $3m$ integer increments. Hence

$$|C_0| \le 3m.$$

If N is the number of maximal triangles, then

$$3N = |C_0| \le 3m,$$

so

$$\boxed{N \le m \le n.}$$

The bound is sharp. For a regular m-gon with m not divisible by three, the m cyclically shifted triangles whose vertices are as evenly spaced as possible all have maximum area.

§141.4 Euler's Formula via Spanning Trees

Let a connected graph be embedded in the sphere without edge crossings. Choose a spanning tree T. The tree has

$$E_T = V - 1$$

edges and only one face on the sphere, so

$$V - E_T + F = V - (V - 1) + 1 = 2.$$

Now add the remaining edges of the graph one at a time. Each new edge joins two vertices already connected by the tree-and-previous-edges subgraph. Drawn without crossings, it lies inside one existing face and splits that face into two. Thus

$$E \longmapsto E + 1, \qquad F \longmapsto F + 1,$$

while V is unchanged. The quantity $V - E + F$ remains constant.

Therefore every connected planar embedding satisfies

$$\boxed{V - E + F = 2.}$$

For a graph with c connected components in the sphere,

$$V - E + F = 1 + c.$$

The proof identifies the invariant move: a non-tree edge creates exactly one independent cycle and exactly one new face.

Subdivision also preserves the alternating count. Inserting a new vertex into an edge changes

$$(V, E, F) \longmapsto (V + 1, E + 1, F),$$

while drawing a diagonal across a face changes

$$(V, E, F) \longmapsto (V, E + 1, F + 1).$$

Both leave $V-E+F$ unchanged. Euler characteristic therefore belongs to the underlying cell decomposition, not to the accidental fineness of the drawing.

§141.5 Incidence counting leaves only five regular polyhedra

Suppose every face of a regular convex polyhedron is a p-gon and exactly q faces meet at every vertex. Counting edge–face incidences and vertex–edge incidences gives

$$pF = 2E, \qquad qV = 2E.$$

Substitute

$$F = \frac{2E}{p}, \qquad V = \frac{2E}{q}$$

into Euler's formula:

$$\frac{2E}{q} - E + \frac{2E}{p} = 2.$$

Since $E > 0$,

$$\frac{1}{p} + \frac{1}{q} > \frac{1}{2}.$$

With $p, q \geq 3$, the only possibilities are

$$(p, q) = (3, 3), (3, 4), (4, 3), (3, 5), (5, 3).$$

They correspond to

(p, q)	V	E	F
$(3, 3)$	4	6	4
$(3, 4)$	6	12	8
$(4, 3)$	8	12	6
$(3, 5)$	12	30	20
$(5, 3)$	20	30	12.

Duality interchanges vertices and faces while preserving edges, so it swaps (p, q) with (q, p). The cube and octahedron are dual, as are the dodecahedron and icosahedron; the tetrahedron is self-dual. The classification is therefore an interaction between Euler characteristic, local incidence, and duality.

§141.6 Euler Characteristics from Polygon Gluings

A topological quotient identifies points rather than cosets. Begin with a polygon as one two-cell and glue its boundary edges according to labels.

For the torus, use a square with boundary word

$$aba^{-1}b^{-1}.$$

All four vertices are identified, the four sides become two one-cells, and the interior is one two-cell. Hence

$$(V, E, F) = (1, 2, 1), \qquad \chi(T^2) = 0.$$

For the real projective plane, glue a two-gon with boundary word

$$aa.$$

There is one vertex, one edge, and one face:

$$\chi(\mathbb{R}P^2) = 1 - 1 + 1 = 1.$$

For the Klein bottle, use the square word

$$aba^{-1}b.$$

Again there is one vertex, two edges, and one face, so

$$\chi(K) = 0.$$

The torus and Klein bottle have the same Euler characteristic but are not homeomorphic; Euler characteristic is powerful but not complete.

Three quotient constructions should not be conflated:

$$G/N$$

is a quotient group whose well-defined multiplication requires $N \trianglelefteq G$;

$$X/\sim$$

is a quotient topological space with the quotient topology; and

$$H_k(X) = Z_k(X)/B_k(X)$$

is an algebraic quotient of cycles by boundaries. They share the act of identifying equivalent elements, but the structures being preserved are different.

§141.7 Euler Characteristic from Chain Complexes

For a finite cell complex, let C_k be the vector space spanned by its k-cells, with boundary maps

$$\cdots \longrightarrow C_{k+1} \xrightarrow{\partial_{k+1}} C_k \xrightarrow{\partial_k} C_{k-1} \longrightarrow \cdots$$

satisfying

$$\partial_k \partial_{k+1} = 0.$$

Set

$$Z_k = \ker \partial_k, \qquad B_k = \operatorname{im} \partial_{k+1}, \qquad H_k = Z_k / B_k.$$

Then

$$\dim C_k = \dim Z_k + \operatorname{rank} \partial_k,$$

and

$$\dim Z_k = \dim H_k + \operatorname{rank} \partial_{k+1}.$$

Therefore

$$\dim C_k = \dim H_k + \operatorname{rank} \partial_k + \operatorname{rank} \partial_{k+1}.$$

Taking the alternating sum over k, the two boundary-rank sums cancel after shifting the index:

$$\boxed{\sum_k (-1)^k \dim C_k = \sum_k (-1)^k \dim H_k.}$$

The left side is the alternating cell count; the right side is the alternating sum of Betti numbers

$$b_k = \dim H_k.$$

For the sphere,

$$(b_0, b_1, b_2) = (1, 0, 1), \qquad \chi(S^2) = 2.$$

For the torus,

$$(b_0, b_1, b_2) = (1, 2, 1), \qquad \chi(T^2) = 0.$$

Over $\mathbb{Q}$, the projective plane has Betti numbers

$$(1, 0, 0),$$

so $\chi(\mathbb{R}P^2) = 1$. Its integral homology contains torsion, but torsion does not affect the alternating ranks.

§141.8 Degree on the circle is winding made algebraic

For

$$f_n : S^1 \to S^1, \qquad f_n(e^{it}) = e^{int},$$

the point travels around the target n times as t runs from 0 to 2π. Its degree is

$$\deg f_n = n.$$

For a smooth map and a regular value y, degree can be computed by oriented preimages:

$$\deg f = \sum_{x \in f^{-1}(y)} \operatorname{sign}(df_x).$$

For f_n, a regular target point has $|n|$ preimages, each with the sign of n.

The homological definition generalizes to every sphere. Since

$$H_d(S^d;\mathbb{Z}) \cong \mathbb{Z},$$

a continuous map $f : S^d \to S^d$ induces multiplication by one integer:

$$f_*([S^d]) = (\deg f)[S^d].$$

Functoriality gives

$$\deg(g \circ f) = \deg g \, \deg f.$$

The multiplicativity is simply the multiplication of the two induced endomorphisms of $\mathbb{Z}$.

§141.9 Nonzero Degree Implies Surjectivity

Suppose $f : S^d \to S^d$ omits a point p. Then it factors through

$$S^d \setminus \{p\} \cong \mathbb{R}^d,$$

whose top homology vanishes:

$$H_d(\mathbb{R}^d) = 0.$$

Consequently the induced map on $H_d(S^d)$ is zero, so

$$\deg f = 0.$$

Taking the contrapositive,

$$\boxed{\deg f \neq 0 \quad \Longrightarrow \quad f \text{ is surjective.}}$$

Degree is therefore an existence invariant. It does not locate a preimage, but it prevents all preimages from disappearing under continuous deformation.

As a simple application, every map homotopic to the identity has degree one and is surjective. No such map can be continuously deformed into a constant map, whose degree is zero.

§141.10 Brouwer via No Retraction

There is no continuous retraction

$$r : B^{d+1} \to S^d$$

from the closed ball to its boundary. If $i : S^d \hookrightarrow B^{d+1}$ is the inclusion and $r \circ i = \mathrm{id}_{S^d}$, then on top homology

$$r_* \circ i_* = \mathrm{id}_{H_d(S^d)}.$$

But

$$H_d(B^{d+1}) = 0,$$

so i_* is the zero map, making the composition zero. This contradiction proves the no-retraction theorem.

Now let

$$F : B^{d+1} \to B^{d+1}$$

be continuous and suppose it has no fixed point. Put

$$y = F(x), \qquad d = x - y \neq 0.$$

The ray starting at y and passing through x is

$$y + td, \qquad t \geq 0.$$

Its intersection with the boundary satisfies

$$\|y + td\|^2 = 1,$$

or

$$\|d\|^2 t^2 + 2(y \cdot d)t + \|y\|^2 - 1 = 0.$$

The forward intersection is obtained from the positive root

$$t(x) = \frac{-y \cdot d + \sqrt{(y \cdot d)^2 + \|d\|^2(1 - \|y\|^2)}}{\|d\|^2}.$$

Define

$$r(x) = y + t(x)d.$$

Because $d \neq 0$, the formula is continuous and satisfies $\|r(x)\| = 1$. If $x \in S^d$, then $t = 1$ is the forward root, so

$$r(x) = x.$$

Thus a fixed-point-free F would construct a forbidden retraction. Therefore every continuous self-map of the closed ball has a fixed point.

The passage from cell counts to this existence theorem is now explicit. Euler characteristic survives subdivision because boundary ranks cancel in a chain complex. Degree records the action on the surviving top homology class. A nonzero class obstructs both omission of a target point and retraction of a ball onto its boundary.

Hopf Fibration, Homotopy, and the Infinite-Stability Reading

N-20240727

§142.1 From the meme to infinite stability

A lecture poster by Keyao Peng placed elementary arithmetic, the Hopf fibration, homotopy groups, configuration spaces, algebraic K-theory, and the line

$$0 = 1 - 1 = -1 + 1 = 0$$

on the same ladder. The Hopf fibration recalled the linked-circle picture in S^3, one of the first images that made topology feel deeper than point-set definitions. The equation itself, however, looked analytic and algebraic before it looked topological. The middle expressions suggested rearrangement, cancellation, and ambiguity of formal sums, so I read the ladder as

$$\begin{aligned} &\text{alternating sums} \longrightarrow \text{regularization} \\ &\longrightarrow \text{Eilenberg swindle} \\ &\longrightarrow \text{vanishing under infinite stability.} \end{aligned}$$

The rest of the picture then began to line up: the Hopf map suggested stable homotopy, K-theory suggested addition and group completion, and the final equality looked like the endpoint of a stabilization trick.

§142.2 The analytic reflex

The most immediate association is the infinite alternating series

$$1 - 1 + 1 - 1 + \cdots .$$

Its partial sums are

$$1,\ 0,\ 1,\ 0,\ \ldots,$$

so the ordinary limit does not exist.

Definition 142.2.1 (Cesàro summation)**.** Let $s_n = a_0 + \cdots + a_n$ be the partial sums of a series. If the averages

$$\sigma_N = \frac{s_0 + s_1 + \cdots + s_N}{N + 1}$$

converge to L, then the series is Cesàro summable to L.

For the alternating series, the Cesàro averages of the partial sums tend to

$$\frac{1}{2}.$$

Thus the expression is not convergent in the usual sense, but it has a regularized value in the Cesàro sense.

Definition 142.2.2 (Abel summation)**.** Given a formal series $\sum_{n\geq 0} a_n$, form its power series

$$F(r) = \sum_{n\geq 0} a_n r^n$$

for $0 < r < 1$. If $F(r)$ has a limit L as $r \to 1^-$, then the original series is Abel summable to L.

For

$$1 - 1 + 1 - 1 + \cdots,$$

one has

$$1 - r + r^2 - r^3 + \cdots = \frac{1}{1+r},$$

so Abel summation again gives $1/2$.

Remark 142.2.3 — The same cultural memory also points toward the analytic continuation of the Dirichlet eta function. The important point is not the exact analytic formalism, but the habit of reading a formally divergent expression through a more flexible rule.

I therefore read the line

$$0 = 1 - 1 = -1 + 1 = 0$$

as a statement about how formal algebra changes when one passes from finite arithmetic to limiting procedures. In ordinary arithmetic, the line is trivial. In the world of infinite expressions, however, the order of grouping and the rule of interpretation may change what the expression seems to mean.

Claim 142.2.4 (Analytic reading)**.** I read the equation not as a numerical equality, but as evidence that cancellation becomes unstable once infinite formal processes are allowed.

§142.3 Why Hopf and K-theory require more than summation

Regularization alone does not explain why the meme includes the Hopf fibration and K-theory. Those are not the usual tools for assigning values to divergent series. The Hopf fibration suggests homotopy and stable phenomena. K-theory suggests addition, cancellation, and group completion of mathematical objects.

The analytic reflex therefore becomes a structural question. Instead of asking only whether

$$1 - 1 + 1 - 1 + \cdots$$

has a regularized value, I ask whether a categorical or K-theoretic setting can make cancellation legitimate through infinite size or stability.

That is exactly the kind of situation where the Eilenberg swindle appears.

§142.4 The Eilenberg swindle

The Eilenberg swindle is the mechanism that made the analytic reflex look relevant to K-theory.

Theorem 142.4.1 (Eilenberg swindle)

Suppose an additive category has countable direct sums and an object

$$S = A \oplus A \oplus A \oplus \cdots .$$

Then

$$S \cong A \oplus S.$$

Consequently, any Grothendieck-type invariant K_0 compatible with direct sums satisfies

$$[S] = [A] + [S],$$

and therefore forces

$$[A] = 0.$$

Idea. The isomorphism is the shift of the infinite direct sum:

$$A \oplus A \oplus A \oplus \cdots \cong A \oplus (A \oplus A \oplus A \oplus \cdots).$$

The finite summand A is absorbed by the infinite tail. Passing to an additive invariant converts this absorption into a vanishing statement. □

This is the algebraic counterpart of the alternating-series picture. One can group an infinite expression in different ways:

$$(1 - 1) + (1 - 1) + \cdots = 0,$$

but also

$$1 + (-1 + 1) + (-1 + 1) + \cdots = 1.$$

The paradox is not that arithmetic is broken. The point is that infinite formal manipulation can erase the distinction between adding one copy and adding no copy, because the tail of the expression absorbs the finite change.

Remark 142.4.2 — This makes the line $0 = 1-1 = -1+1 = 0$ look like a compact symbolic form of infinite absorption. The two middle expressions are different finite presentations, but in an infinitely stable setting they may collapse to the same zero class.

§142.5 How the swindle resembles a K-theory statement

In algebraic K-theory, one often begins with a category of objects equipped with a direct-sum operation. The first invariant records formal additive differences.

Definition 142.5.1 (Grothendieck group)**.** Let M be a commutative monoid. Its Grothendieck group $K_0(M)$ is the universal abelian group receiving a monoid homomorphism from M. Concretely, elements may be represented by formal differences

$$[A] - [B],$$

subject to the relation

$$[A \oplus C] - [B \oplus C] = [A] - [B].$$

The expression $1-1$ then resembles a formal difference between one positive copy and one negative copy. A sufficiently infinite environment can force such formal differences to vanish.

Proposition 142.5.2 (Vanishing by absorption, heuristic form)
If a category contains an object S with

$$S \cong S \oplus A,$$

then any additive invariant that sees direct sum as addition tends to identify the class of A with zero.

This is why swindle arguments prove vanishing theorems in categories with infinite coproducts or infinite-dimensional stabilization. The chain I see is: Hopf fibration points toward stable homotopy; stable homotopy points toward group completion in K-theory; group completion suggests infinite stability and the Eilenberg swindle; the final line becomes symbolic infinite cancellation. After enough stabilization, the complicated machinery collapses into the fact that infinity can absorb a finite summand.

§142.6 Where the Hopf fibration enters

The Hopf fibration serves as the geometric entrance into stable homotopy. The Hopf map

$$S^3 \to S^2$$

is a concrete example of a map whose nontriviality is not visible from ordinary dimension-counting. It teaches that topology can remember a hidden process.

From there, stable homotopy suggests that some phenomena persist after suspending or stabilizing. I connect this with the word “stable” in algebra: after passing to a sufficiently stabilized category, hidden structure becomes algebraic cancellation and ultimately a vanishing statement.

The ladder of associations is

$$\text{Hopf} \quad \rightsquigarrow \quad \text{stable homotopy} \quad \rightsquigarrow \quad \text{stable algebra} \quad \rightsquigarrow \quad \text{swindle}.$$

The Hopf fibration stands for the beginning of stable topology; the equation stands for the end of a stabilization trick.

§142.7 Reading the ladder

The diagram compresses elementary cancellation and progressively more structural forms of stability into one ladder.



Figure 142.1: The meme that compressed Hopf fibration, stable homotopy, K-theory, and the equation $0 = 1 - 1 = -1 + 1 = 0$ into one ladder.

The Hopf fibration represents hidden topology; homotopy groups record deformation; stable homotopy records persistence under stabilization; K-theory organizes addition and cancellation; and the Eilenberg swindle produces vanishing through infinite absorption. In this way the visible clues fit a genuine mechanism in K-theory rather than a stray analogy with divergent series.

§142.8 The infinite-stability reading

The reading can be stated cleanly.

Theorem 142.8.1 (Infinite-stability interpretation)

In elementary arithmetic, $0 = 1 - 1 = -1 + 1 = 0$ is trivial. In an infinitely stable categorical setting, however, finite additions and cancellations can be absorbed by an infinite tail. The Eilenberg swindle expresses this absorption. Therefore the equation may be read as a symbolic way to say that certain K-theoretic invariants vanish once infinite direct-sum operations are allowed.

This reading emphasizes three words:

infinity, stability, vanishing.

The line $0 = 1-1 = -1+1 = 0$ then becomes an emblem of the idea that a sophisticated invariant may collapse because the ambient category is too large.

§142.9 The finite clues

The chain now explains why alternating sums make $1 - 1$ and $-1 + 1$ feel like more than ordinary arithmetic, why K-theory brings in direct sum and group completion, why infinite stabilization can force an additive invariant to vanish, and why the Hopf fibration points toward stable homotopy. The central claim is not that 0 is secretly nonzero. It is that in an infinitely stabilized algebraic world, the distinction between adding and not adding a finite piece can disappear.

What I still cannot place is the repeated appearance of finite sets and configuration spaces. The mechanism above needs an infinite tail, while both of those clues seem stubbornly finite. Perhaps the equation is not only recording what cancels, but also remembering how the cancellation happens. I do not yet see where such a memory would live.

Nikonov's Seminar Prelude: Knots, Diagrams, Probes, and the Space Between the Lines

N-20250705

§143.1 First orientation

Shortly after the Gaokao, I listened to an online seminar by I. M. Nikonov on partial tribrackets. Quasigroups, biquandloids, thickened surfaces, crossings, arcs, regions, and probes arrived almost at once. The first point that began to hold together was an old tension in knot theory:

a knot is a topological object, but calculations use diagrams.

Nikonov's paper does more than introduce another coloring invariant. It asks whether the pieces of a diagram—arcs, semiarcs, regions, and crossings—can themselves be described topologically. Once that question is asked, the move from diagrammatic pieces to probes in a thickened surface becomes the natural place to begin.

§143.2 The topological object and its planar shadow

Topologically, a knot is not first of all a picture on paper. It is an embedded circle in three-dimensional space:

$$K : S^1 \hookrightarrow \mathbb{R}^3,$$

or, more generally, a tangle in the thickening $F \times I$ of a surface F. The equivalence relation is ambient isotopy: the whole surrounding space is allowed to deform, carrying one knot to the other.

The story begins with the familiar planar shadow of a knot. A diagram is what we can draw, but the actual knot lives one dimension higher.

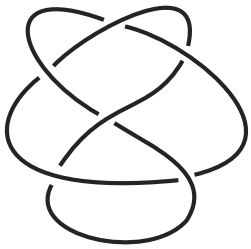


Figure 143.1: From Nikonov's paper: a knot diagram. It is a planar record of a three-dimensional object.

This distinction matters. The diagram has crossings, but the knot in space has no literal self-intersection. At a crossing, one strand passes over the other. The diagram remembers this by marking over/under information.

§143.3 Some basic knot-theory language

A few standard definitions make the paper much less mysterious.

Definition 143.3.1 (Knot and ambient isotopy)**.** A knot is a tame embedding $K : S^1 \hookrightarrow S^3$, usually drawn in $\mathbb{R}^3 \subset S^3$. Two knots K_0, K_1 are equivalent if there is an ambient isotopy $H_t : S^3 \to S^3$ with $H_0 = \mathrm{id}$ and $H_1(K_0) = K_1$.

The word "ambient" is important. We do not merely slide points of the curve along the curve. We deform the whole surrounding three-dimensional space. This is why a knot is a topological object and not just a parametrized curve.

Definition 143.3.2 (Tangle in a thickened surface)**.** Let F be an oriented compact surface and $I = [0, 1]$. A tangle in $F \times I$ is a properly embedded compact one-dimensional manifold, possibly with boundary on $\partial F \times I$. A knot is the closed one-component special case.

The thickened-surface language is the natural setting of Nikonov's paper. Instead of drawing everything in the plane, one works over a surface F, and the knot or tangle floats in the slab $F \times I$.

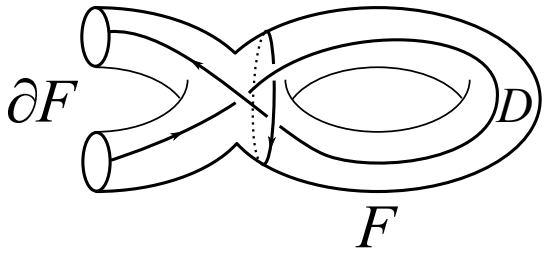


Figure 143.2: From Nikonov's paper: a tangle diagram on a surface.

Definition 143.3.3 (Diagram)**.** A diagram of a tangle is the projection of the tangle to F, with crossing data recording which strand passes over the other. Thus a diagram is a two-dimensional shadow plus over/under information.

The basic theorem justifying all diagrammatic knot theory is Reidemeister's theorem.

Theorem 143.3.4 (Reidemeister theorem)

Two tame knot diagrams represent ambient isotopic knots if and only if they are related by a finite sequence of planar isotopies and Reidemeister moves.

This theorem is the bridge between topology and combinatorics. It says that if a construction on diagrams is unchanged under the three Reidemeister moves, then it really defines a knot invariant.

The combinatorial definition says that a knot can be represented by a diagram modulo Reidemeister moves. These are the local moves that change the drawing without changing the underlying knot.

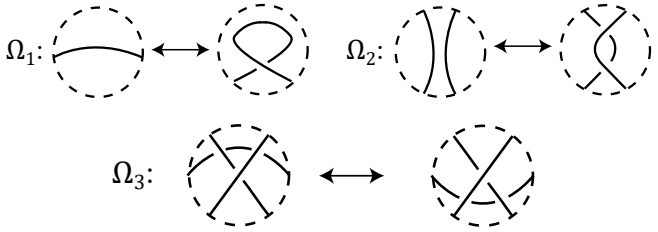


Figure 143.3: From Nikonov's paper: Reidemeister moves. These are the local equivalences that make diagrams represent knots.

So the first dictionary is:

$$\text{knot in space} \quad \longleftrightarrow \quad \text{diagram modulo Reidemeister moves.}$$

This is powerful because diagrams are finite and computable. But it also creates the problem that motivates the paper: if a calculation assigns labels to arcs or regions of a diagram, what exactly are those arcs and regions as topological objects?

A few famous theorems live on the topological side of this dictionary. They are not the main subject of the note, but they explain the background against which the diagrammatic approach became so attractive.

Definition 143.3.5 (Knot complement and knot group)**.** The complement of a knot $K \subset S^3$ is $S^3 \setminus N(K)$, where $N(K)$ is a small tubular neighbourhood of the knot. The knot group is

$$\pi_1(S^3 \setminus K),$$

or equivalently the fundamental group of the complement after removing a tubular neighbourhood.

The complement is often more informative than the curve itself. A loop in the complement asks: can this loop be contracted to a point without crossing the knot? The knot group packages the obstruction algebraically.

Theorem 143.3.6 (Schubert prime decomposition, informal form)

Every nontrivial knot decomposes as a connected sum of prime knots, and this decomposition is unique up to order.

Here the connected sum operation is the knot-theoretic analogue of multiplication: cut out a small arc from each knot and splice the loose ends together. Prime knots are the knots that cannot be decomposed further.

Another landmark is the geometric decomposition picture. Very roughly, many knot complements split into pieces that are torus-like, satellite-like, or hyperbolic. The figure-eight knot is the standard first example of a hyperbolic knot. This is the global topological world behind knot diagrams: elegant, deep, and often difficult to compute with directly.

§143.4 The pieces of a diagram

A diagram is not just a curve with crossings. It has several kinds of local pieces. Four basic types are especially important here:

$$\text{arcs}, \qquad \text{semiarcs}, \qquad \text{crossings}, \qquad \text{regions}.$$

These are the pieces that appear in coloring rules, parity theories, skein relations, and many diagrammatic constructions.

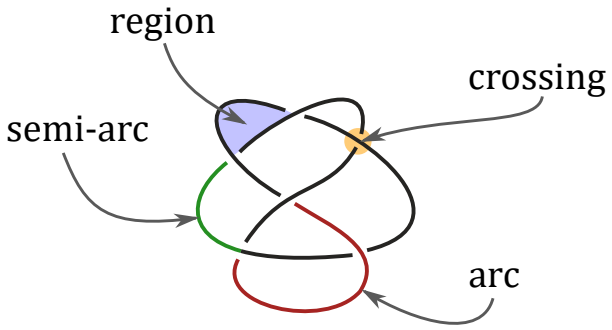


Figure 143.4: From Nikonov's paper: the elementary pieces of a knot diagram.

The naive view is that these are pieces of ink on the page. An arc is a piece of curve between undercrossings. A semiarc is a smaller piece between consecutive crossings. A region is a component of the complement of the diagram. A crossing is a vertex with over/under data.

This is good enough for drawing, but it is not yet good enough for topology. A Reidemeister move can create or destroy a crossing, split an arc, merge two regions, or change how pieces correspond. If one wants to build invariants by labeling these pieces, one needs a way to say when a piece before the move is the same as a piece after the move.

This is exactly where many elementary knot invariants already give hints. For example, the linking number of a two-component link is computed from signs of mixed crossings. A quandle coloring assigns algebraic labels to arcs. A shadow or region coloring assigns labels to complementary regions. These methods work only because the labels transform correctly under Reidemeister moves.

So there are two levels:

$$\begin{array}{c}\text{first choose pieces of a diagram}\\ \Downarrow\\ \text{then assign algebraic labels to those pieces}\\ \Downarrow\\ \text{prove the labels survive Reidemeister moves.}\end{array}$$

Nikonov's paper attacks the first level. Before asking which labels to assign, ask what the pieces are.

This is the hidden question behind the paper:

Can a diagram element be defined without choosing a particular diagram?

§143.5 Nikonov's bridge: diagram elements as probes

Nikonov's answer is to replace a two-dimensional piece of a diagram by a three-dimensional path in a thickened surface. Work in

$$F \times I, \qquad I = [0,1],$$

where the knot or tangle lies between the bottom surface $F \times \{0\}$ and the top surface $F \times \{1\}$. A diagram is obtained by projecting down to F.

The key idea is that a diagram element can be detected by a probe: a path travelling through the thickened surface and interacting with the knot in a controlled way.

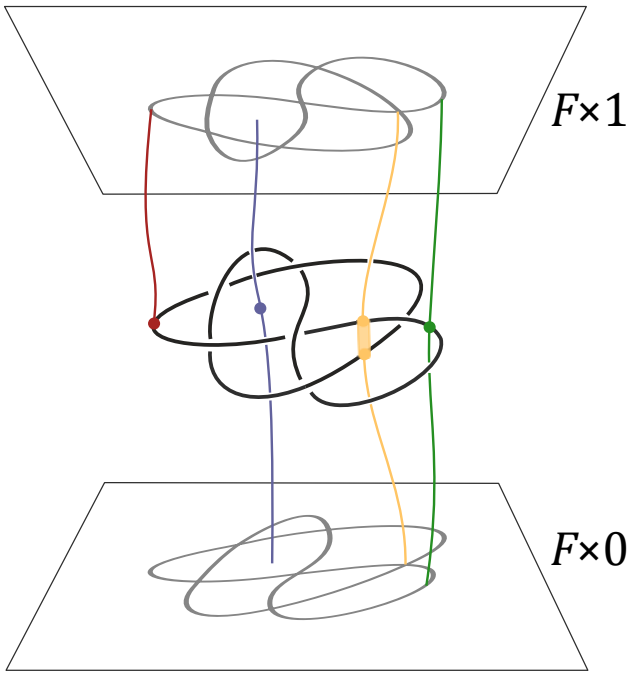


Figure 143.5: From Nikonov's paper: topological interpretation of diagram elements by probes in $F \times I$.

The slogan is:

diagram element	topological probe
arc	a path from the knot to the top surface
semiarc	a bottom-to-top path meeting the knot once
crossing	a bottom-to-top path meeting the knot twice
region	a bottom-to-top path avoiding the knot

This was the point at which the terminology from the seminar became geometric. The word "arc" is no longer merely a segment in a planar picture: it is represented by a way of reaching the knot from the ambient three-dimensional space. A "region" is no longer just an empty component on paper, but a path through the complement of the knot.

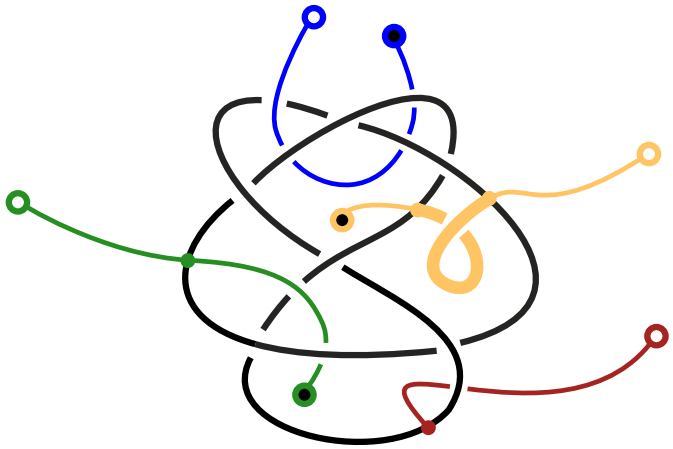


Figure 143.6: From Nikonov's paper: probe diagrams for arc, semiarc, region, and crossing elements.

The great advantage of this formulation is that isotopy becomes natural. If the tangle moves in $F \times I$, the probes can move with it. Thus the correspondence of diagram elements under Reidemeister moves is no longer a purely combinatorial bookkeeping rule; it is the shadow of a topological motion.

§143.6 Invariants, coinvariants, and labels

The seminar also made the distinction between invariants and coinvariants feel less formal: it clarifies what kind of object survives a change of diagram.

An invariant is something that gives the same value for every diagram of the same knot. For example, a polynomial invariant or a numerical invariant should not depend on which diagram we draw.

A coinvariant is subtler. It may give different-looking objects for different diagrams, but these objects are canonically isomorphic once we account for the moves between diagrams. A vector space is a good mental model: its dimension is an invariant, while the vector space itself, together with the way it changes under coordinate changes, is closer to a coinvariant.

For diagram elements, the relevant object is often not one number but a whole set of possible classes. The set of arc probes, region probes, or crossing probes should be thought of as a universal object from which more concrete labels can be obtained.

In this language, Nikonov's first main conceptual claim is:

isotopy classes of probes give universal homotopical coinvariants of diagram elements.

This sentence is dense, but the intuition is simple. Instead of assigning a label to an arc immediately, first ask what the arc is intrinsically. The answer is a class of probes. Any later coloring rule is a way of reading algebraic information from these probe classes.

> **Remark 143.6.1 —** The word "universal" should be read in the usual algebraic way. A universal object is not one more label among many labels. It is the object from which all compatible labels should factor. Thus the universal arc object is like the raw space of possible arc labels before choosing a particular quandle, polynomial, or coefficient set.

This is also why the paper talks about coinvariants rather than only invariants. A set of probes may change its concrete presentation when the diagram changes, but the induced object is naturally identified through the isotopy or through the homotopy class of Reidemeister moves. The object is stable, even when its drawing is not literally identical.

§143.7 From probes to colorings

Once diagram pieces have become probes, the later algebraic structures—quandles, partial ternary quasigroups, biquandloids, crossoids, and the multicrossing complex—no longer look like independent tricks. They are different algebraic shadows of the same topological idea:

diagram pieces become probes, and probes have isotopy or homotopy classes.

The resulting chain is

$$\begin{array}{c}\text{topological knot or tangle in } F \times I \\ \Downarrow \\ \text{planar diagram with arcs, regions, crossings} \\ \Downarrow \\ \text{topological reinterpretation of those pieces as probes} \\ \Downarrow \\ \text{algebraic colorings extracted from homotopy classes of probes.}\end{array}$$

A ternary operation on regions can only be understood after the underlying notion of a region has been made topological. Nikonov's paper answers: a region is not merely a blank patch in a diagram; it is a topological probe through the complement of the knot.

Nikonov's Paper at One Crossing: Region Probes, Partial Tribrackets, and What the Seminar Was About

N-20250707

§144.1 Following the region probes

Returning to Nikonov's paper after the seminar, the sequence that finally made the terminology cohere was the passage from regions of diagrams to topological probes and then to an operation at one crossing:

$$\begin{array}{c}\text{regions of a diagram}\\ \Downarrow\\ \text{region probes}\\ \Downarrow\\ \text{partial ternary operation at one crossing}\end{array}$$

The word "partial" and the appearance of a ternary operation both become visible along this path.

§144.2 The local dictionary

The paper has two large movements. The first movement is topological: arcs, regions, semiarcs, and crossings of diagrams are reinterpreted as isotopy classes of probes. The second movement is algebraic: homotopy classes of these probes naturally carry coloring structures.

The local dictionary is:

diagram element	algebraic shadow
arc	quandle
region	partial ternary quasigroup / partial tribracket
semiarc	biquandloid
crossing	crossoid

The second row is where the title of the seminar enters the picture: region probes should carry a partial ternary operation.

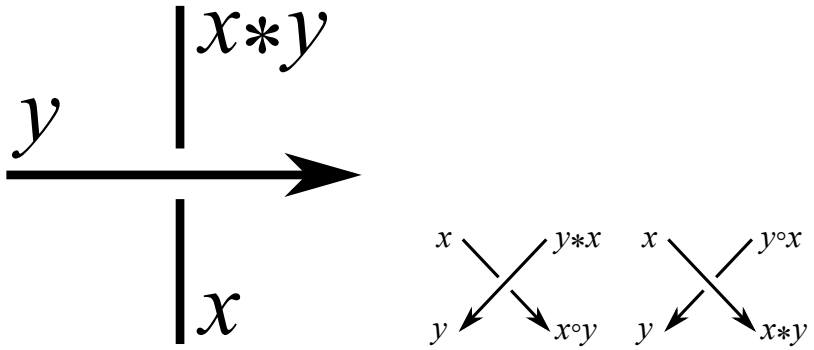


Figure 144.1: From Nikonov's paper: arc and semiarc colorings are the neighbouring examples around the region story.

The figures above place the region construction beside the more familiar arc and semiarc colorings. Quandles and biquandles are standard algebraic gadgets for diagram colorings; Nikonov's point is that the less familiar partial tribracket and crossoid fit into the same topological pattern.

§144.3 A region is not just a blank patch

In an ordinary knot diagram, a region is a connected component of the plane or surface after removing the diagram. Informally, it is a blank patch between strands. But this definition depends on the drawing.

Nikonov replaces the patch by a topological path. In a thickened surface $F \times I$, a region probe is a path from the bottom $F \times \{0\}$ to the top $F \times \{1\}$ that does not intersect the knot. In other words, it is a way to pass through the empty space around the knot.

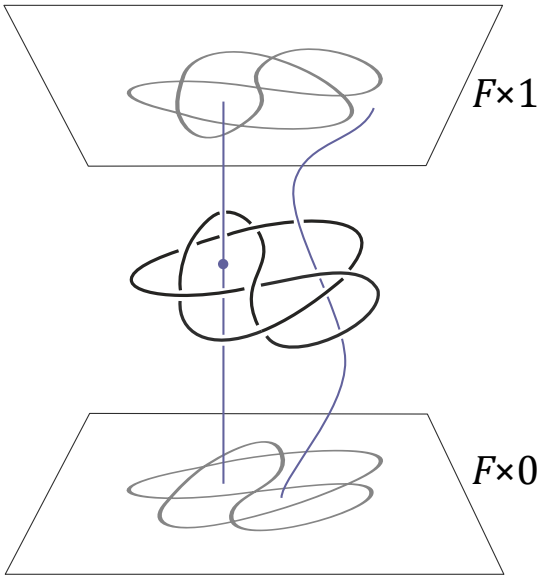


Figure 144.2: From Nikonov's paper: a region probe passes through the complement of the knot.

This is the most important mental shift. In the diagram, a region looks passive. In the thickened surface, a region is represented by an actual route through space. If the knot moves by isotopy, the route can move too. Thus a region can be followed through Reidemeister moves without pretending that it is only a two-dimensional patch of paper.

For a quick analogy, imagine the knot as a collection of wires suspended in a slab of transparent jelly. A region probe is a thin needle pushed from the bottom of the jelly to the top without touching any wire. Moving the knot deforms the possible needle paths. The algebra of regions is really an algebra of these possible needle paths.

§144.4 Four Regions around a Crossing

A tribracket is a ternary operation. Instead of combining two inputs, it combines three:

$$[a, b, c] = d.$$

In knot diagrams, the reason for a ternary operation is visual. Around one crossing there are four adjacent regions. If three of their colors are known, the coloring rule determines the fourth.

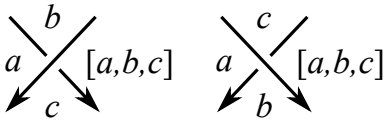


Figure 144.3: From Nikonov's paper: the tribracket coloring rule near a crossing.

This is the cleanest way to remember the algebra. A quandle is suited to arc colorings because arcs meet at a crossing in a binary-looking way. A tribracket is suited to region colorings because four regions meet around a crossing, so knowing three should determine the fourth.

The operation is designed so that the coloring survives Reidemeister moves. That is the usual diagrammatic requirement: if two diagrams represent the same knot, their coloring data should correspond.

Definition 144.4.1 (Horizontal tribracket, informal version)**.** A tribracket is a set X with a ternary operation

$$[\ ,\ ,\ ] : X^3 \to X$$

such that in an equation

$$[a, b, c] = d$$

any three of a, b, c, d determine the fourth, and such that the operation satisfies the compatibility identity forced by the third Reidemeister move.

The first condition is the analogue of invertibility. If a crossing tells us the fourth region from three known regions, then we must also be able to solve backwards when the diagram is read in another local direction.

The second condition is the real knot-theoretic condition. Around a third Reidemeister move, there are two ways of propagating region colors across three nearby crossings. These two ways must give the same final color. In Nikonov's source, this is expressed by the identity

$$[b, [a, b, c], [a, b, d]] = [c, [a, b, c], [a, c, d]] = [d, [a, b, d], [a, c, d]].$$

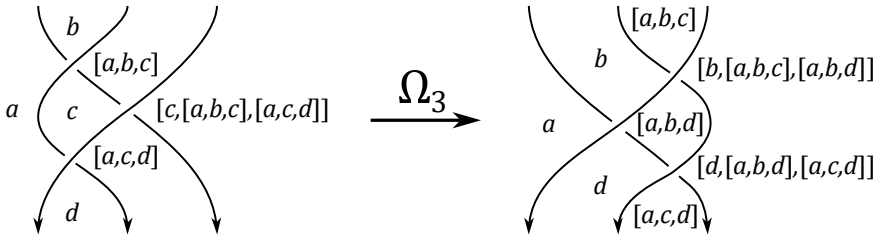


Figure 144.4: From Nikonov's paper: the third Reidemeister move is the source of the tribracket identity.

Theorem 144.4.2 (Coloring invariance for tribrackets)

If X is a tribracket, then tribracket colorings of a diagram correspond bijectively under Reidemeister moves.

This theorem is the usual coloring-invariant mechanism. The algebraic axioms are not arbitrary: they are exactly the local rules needed so that the diagram move does not change the set of colorings.

Example 144.4.3 (Two useful models)

An Alexander-type tribracket has the form

$$[a, b, c] = tb + sc - tsa$$

on a module over $\mathbb{Z}[t^{\pm 1}, s^{\pm 1}]$. A group-based Dehn-type model, in the convention used later for the partial construction, has the form

$$[a, b, c] = ba^{-1}c.$$

The first example is linear and computational; the second example is closer to the topology of paths in a complement.

§144.5 Why the operation is partial

In the plane, it is tempting to think that any three region colors can be inserted into the operation. Nikonov's point is that, on a fixed surface and from the topological viewpoint, not every formal triple of regions is genuinely realizable around a crossing.

A region is represented by a probe. Three regions can be composed at a crossing only when the corresponding probes can be arranged in the correct local configuration. If topology prevents that configuration, then the operation should simply not be defined. This is why the algebra is partial.

The paper includes a useful warning example: a formal composition may look legal if one only stares at symbols, but may fail to be realizable on the actual diagram.

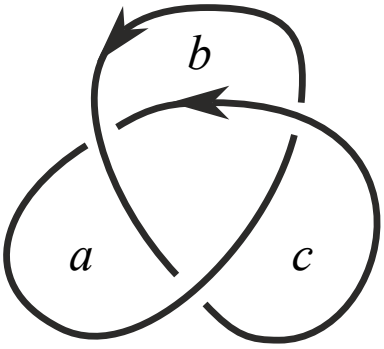


Figure 144.5: From Nikonov's paper: an indicated composition of regions that is not realizable on a classical diagram.

This figure is the whole story in miniature. A purely algebraic operation would like to accept any triple. The topology says: no, only triples that can actually occur as neighbouring regions around a crossing should be allowed. The adjective "partial" is not a technical nuisance; it records genuine geometric obstruction.

Definition 144.5.1 (Partial ternary quasigroup, notebook version)**.** A partial ternary quasigroup is a set X, a compatibility relation $\uparrow$, and a ternary operation

$$[a, b, c]$$

defined only when $b \uparrow a$ and $c \uparrow a$. The axioms say that the output is compatible with the neighbouring inputs, that the operation can be solved uniquely for any missing entry whenever the compatibility conditions make sense, and that the third-Reidemeister identity still holds for compatible quadruples.

The compatibility relation is the essential new ingredient. One should read

$$b \uparrow a$$

as "the region b is allowed to sit next to the region a in the required local way." In a flat beginner picture this sounds unnecessary, but in a fixed thickened surface it records real topology.

Example 144.5.2 (Alexander numbering)
The paper gives a very small partial example:

$$X = \mathbb{Z}, \qquad a \uparrow b \iff b = a - 1, \qquad [a, b, c] = a + 2.$$

This is not meant to be the most exciting invariant. It is a memory aid: partiality can be as simple as allowing only neighbouring integer levels to interact.

§144.6 Probe clearing and color rules

The paper also explains how a region probe can be cleared through the diagrammatic situation. The figure below should be read as a topological mechanism behind the coloring rule. We are not just writing $[a, b, c] = d$; we are moving a probe through the thickened surface and watching how its class changes.

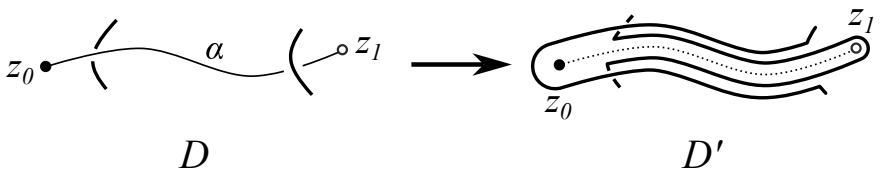


Figure 144.6: From Nikonov's paper: clearing a region probe in the construction of the partial tribracket.

This is the connection with the first note. Diagrammatic elements become probes; homotopy classes of probes become the raw material for algebra; algebraic coloring rules become shadows of probe motion.

The theorem proved in this part of the paper can be read as follows.

Theorem 144.6.1 (Nikonov's fundamental partial tribracket, informal form)

For a tangle $T \subset F \times I$ with diagram D, colorings of D by a partial ternary quasigroup correspond to homomorphisms out of the topological partial tribracket built from homotopy classes of region probes, with the natural $\pi_1(F)$-invariance condition.

The exact statement has more notation, but the meaning is very clean. A diagram coloring is not merely a coloring of a planar picture. It is the same thing as a compatible algebraic map from the topological object made of region probes.

In the proof, "clearing a probe" is the operational step. One pulls the diagram away from the probe so that the probe's value can be read as the color of a region after the transformation. Reidemeister invariance then says that this value does not depend on the particular clearing process.

A small local rule can also be expressed as propagation of colors between neighbouring regions.

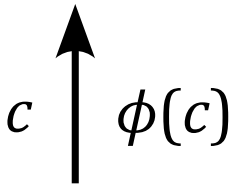


Figure 144.7: From Nikonov's paper: a local propagation rule for neighbouring regions.

This is the sentence I had been trying to extract from the seminar:

> a partial tribracket is the algebra of how region probes interact near crossings.

§144.7 What about crossings themselves?

The paper does not stop at regions. Crossings also become topological objects: a crossing probe is a bottom-to-top path intersecting the tangle twice, with extra framing data on the segment between the two intersection points.

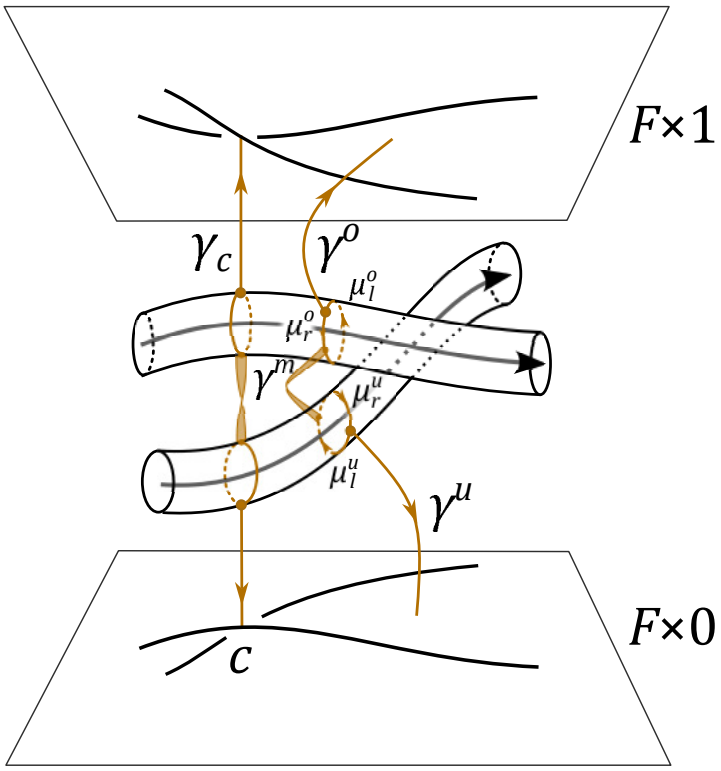


Figure 144.8: From Nikonov's paper: a crossing probe. Crossings are treated as probe classes rather than merely as vertices of a drawing.

This leads toward crossoids, which generalize parity-type structures on knots. Without developing their full definition, the common pattern is already visible:

choose a diagram element
⇓
replace it by a probe
⇓
take homotopy classes
⇓
read off an algebraic coloring structure.

The partial tribracket is simply the region case of this general machine.

§144.8 One crossing, one allowed computation

The small Alexander-numbering model from above is enough to put the word *partial* to work. Recall the exact rule

$$X = \mathbb{Z}, \qquad a \uparrow b \iff b = a - 1, \qquad [a, b, c] = a + 2,$$

where $[a, b, c]$ is defined when $b \uparrow a$ and $c \uparrow a$. In other words, given a first input a, the other two inputs must each equal $a + 1$.

For instance, the triple $(a, b, c) = (0, 1, 1)$ is admissible because

$$1 \uparrow 0, \qquad 1 \uparrow 0, \qquad \text{and then} \qquad \boxed{[0, 1, 1] = 2.}$$

The superficially similar expression $[0, 2, 1]$ has no value: $2 \not\uparrow 0$. It is important *not* to write an arbitrary number on its right-hand side.

three inputs	compatibility	fourth color
$(0,1,1)$	$1 \uparrow 0,\ 1 \uparrow 0$	2
$(0,2,1)$	$2 \not\uparrow 0$	undefined

The local rule is deliberately simple. In the topological construction, the compatibility condition records whether three region probes can really be arranged around a crossing in the required way. When they can, the fourth class is determined; when they cannot, the operation has no business pretending otherwise.

Nikonov's *The Crossing and the Arc from the Topological Viewpoint* gives the larger context: homotopy classes of probes also lead to quandles for arcs, biquandloids for semiarcs, and crossoids for crossings. The concrete distinction between $[0,1,1]$ and $[0,2,1]$ is already enough to remember what a *partial* algebra is doing here.

A Nontrivial Loop at Zero: FinSet K-Theory and Stable Homotopy

N-20251217

§145.1 From the meme to the lecture's explanation

In July 2024, after seeing a meme by Keyao Peng, I read the line

$$0 = 1 - 1 = -1 + 1 = 0$$

through alternating-series regularization and the Eilenberg swindle. A few days before writing this note, while I was reorganizing my homepage, Su Yuyang saw the same line and immediately thought of

$$\sum_{k\geq 0}(-1)^k = 1 - 1 + 1 - 1 + \cdots .$$

That was also my first association. The series has Cesàro and Abel value $1/2$, and it is related to the analytic continuation of the Dirichlet eta function, so the middle expressions naturally look like rearrangement and regularization before they look like topology.

The Hopf-fibration and K-theory clues in the meme led me from analysis to a second mechanism. If an object admits an infinite decomposition

$$S = A \oplus A \oplus A \oplus \cdots ,$$

then

$$S \cong A \oplus S,$$

and a Grothendieck-type invariant may force

$$[A] = 0.$$

This is the Eilenberg swindle: an infinite tail absorbs a finite summand. It gave a coherent explanation of why addition, cancellation, group completion, stability, and vanishing might all sit behind the same arithmetic-looking line.

Keyao Peng's lecture gives the official explanation, and it corrects the object being studied. The two middle expressions are not competing values of an infinite sum, and the equality is not a symbol for absorption at infinity. They are successive frames in a finite motion: create a positive and a negative point, exchange them, and annihilate them again. The comparison is

meme-based infinite-stability reading	lecture's finite-loop explanation
regularization or absorption	creation–swap–annihilation
an infinite object erases a difference	a finite path preserves a distinction
triviality and vanishing	nontriviality of a loop

The more accurate object is therefore not the endpoint 0, but a loop based at 0:

$$\varnothing \sim \{+,-\} \sim \{-,+\} \sim \varnothing.$$

Under the K-theory of finite sets, this loop becomes a nontrivial element. Under the Barratt–Priddy–Quillen theorem, it corresponds to the stable Hopf class.

§145.2 The lecture starts from ordinary equations

The slide deck begins with familiar equations: Pythagoras, Euler's identity, Stokes' theorem, the Navier–Stokes equation, the Langlands slogan, and then the strange meme equation. This is pedagogically important. It suggests that an equation is not merely a string of symbols with a truth value. An equation may also encode a method, a structure, or a path.

In elementary arithmetic,

$$0 = 1 - 1 = -1 + 1 = 0$$

is boring. Both sides are zero. But the lecture asks us to stop looking only at the value. In a higher structure, there can be more than one way for an object to equal itself. The identity loop at 0 and the creation–swap–annihilation loop at 0 need not be the same.

Remark 145.2.1 — This is the conceptual bridge to homotopy type theory and higher category theory. A statement $a = b$ can be represented by a path from a to b. Then two proofs of equality may themselves be compared by a homotopy. The lecture uses this philosophy in a concrete topological model rather than as formal type theory.

§145.3 The Hopf fibration as the first nontrivial map

The first mathematical anchor is the Hopf fibration. Define

$$S^3 = \{(z_1, z_2) \in \mathbb{C}^2 : |z_1|^2 + |z_2|^2 = 1\}.$$

The Hopf map is

$$H : S^3 \to \mathbb{CP}^1 \cong S^2, \qquad (z_1, z_2) \mapsto [z_1 : z_2].$$

For every point of S^2, its preimage is a circle. These circles are linked in a highly organized way.

The Hopf fibration matters because it is not null-homotopic. It represents a nontrivial element of

$$\pi_3(S^2).$$

In beginner language: even though S^3 and S^2 are both spheres, the map $H : S^3 \to S^2$ cannot be continuously shrunk to a constant map.

Definition 145.3.1 (Null-homotopic map)**.** A map $f : X \to Y$ is null-homotopic if it is homotopic to a constant map. Equivalently, there is a continuous family of maps $f_t : X \to Y$ with $f_0 = f$ and f_1 constant.

The Hopf map is the first warning: a process can be topologically nontrivial even if it looks, from far away, like a map between simple spaces.

§145.4 Homotopy groups and the first stable class

For a based space X, the group $\pi_n(X)$ is made from based maps $S^n \to X$, modulo based homotopy. The lecture reviews the low-dimensional intuition:

$$\pi_0(X) = \text{connected components}, \qquad \pi_1(X) = \text{loops modulo homotopy}.$$

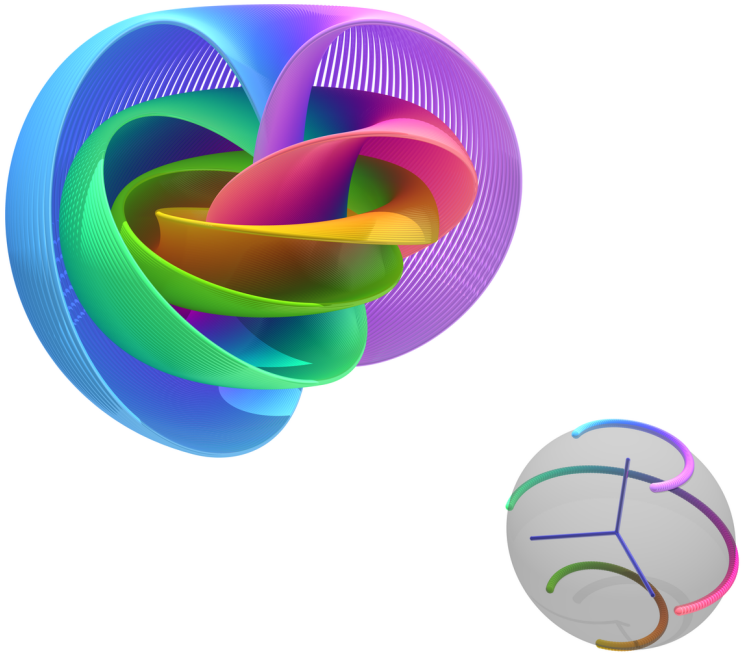


Figure 145.1: the Hopf fibration visualized by linked fibres.

Higher homotopy groups generalize this to maps from higher-dimensional spheres.

For spheres, the groups are easy below the diagonal:

$$\pi_i(S^n) = 0 \qquad (i < n).$$

On the diagonal, one has the degree:

$$\pi_n(S^n) \cong \mathbb{Z}.$$

But above the diagonal the world becomes subtle. The Hopf map gives the famous example

$$\pi_3(S^2) \neq 0.$$

The stable homotopy groups of spheres are obtained by repeatedly suspending:

$$\pi_i^s(\mathbb{S}) = \lim_{n\to\infty} \pi_{i+n}(S^n).$$

The Hopf map contributes the first stable nontrivial class

$$\eta \in \pi_1^s(\mathbb{S}) \cong \mathbb{Z}/2.$$

This class has order two: doing it twice becomes trivial, but doing it once is not trivial.

§145.5 From natural numbers to finite sets

The lecture then changes viewpoint. Instead of defining the natural numbers as primitive symbols, define them as isomorphism classes of finite sets:

$$\mathbb{N} \cong \pi_0(\mathrm{FinSet}).$$

Here a finite set of size n represents the number n. Disjoint union represents addition:

$$A + B := A \sqcup B.$$

To get the integers, one takes a group completion. Informally, one allows formal differences

$$A - B.$$

Thus the number $1 - 1$ is represented by a pair consisting of one positive point and one negative point.

Remark 145.5.1 — The corresponding slide phrases this as a move from numbers to finite sets: the natural number n is represented by an n-element set, and addition is represented by disjoint union. This is the first categorification step in the lecture: instead of remembering only the cardinality, one keeps the finite set itself and later also its symmetries.

Definition 145.5.2 (Grothendieck group)**.** Given a commutative monoid M, its Grothendieck group $K_0(M)$ is the universal abelian group receiving a monoid homomorphism from M. For finite sets under disjoint union this gives

$$K_0(\mathrm{FinSet}) \cong \mathbb{Z}.$$

So far, this still only explains the value of $1 - 1$. It does not yet explain why the path

$$0 \to 1 - 1 \to -1 + 1 \to 0$$

can be interesting. For that, we must remember not just the set of isomorphism classes, but the space of finite sets and isomorphisms.

§145.6 The space of finite sets remembers its symmetries

A set and its cardinality are different amounts of information. Let $\mathcal{F}$ be the *groupoid* of finite sets and bijections. All finite sets of size n are isomorphic, but each such set has an automorphism group Σ_n. Taking the classifying space (the realization of the nerve) gives

$$\boxed{B\mathcal{F} \simeq \coprod_{n \geq 0} B\Sigma_n.}$$

Here $B\Sigma_n$ has one connected component and fundamental group Σ_n. A loop in it remembers an automorphism—an actual permutation of n points—rather than only the number n. In particular,

$$\pi_0(B\mathcal{F}) \cong \mathbb{N}, \qquad \pi_1(B\Sigma_2) \cong \Sigma_2 \cong \mathbb{Z}/2.$$

The crucial distinction is that the empty set belongs to the $n = 0$ component, whose automorphism group is trivial. The transposition loop initially lives in the component of a *two-point* set, not at the empty set. Disjoint union gives $B\mathcal{F}$ the compatible, homotopy-commutative multiplication needed to pass from $\mathbb{N}$ to $\mathbb{Z}$ *without forgetting those loops.* To obtain a loop based in the zero component, we must perform group completion *at the level of spaces*, not merely take the Grothendieck group of the set of cardinalities.

§145.7 The K-space of finite signed sets

Let $K(\mathcal{F})$ denote the E_∞ group completion of $B\mathcal{F} = \coprod_{n\geq 0} B\Sigma_n$ under disjoint union. Ordinary group completion of cardinalities gives only

$$\pi_0 K(\mathcal{F}) \cong \mathbb{Z}.$$

The space-level construction retains homotopies and automorphisms as well. Its correct mathematical input is the groupoid of finite sets and bijections; a model made from signed point configurations visualizes the inverse objects but requires coherent rules for moving and cancelling points. Merely writing $\{+,-\} = \varnothing$ as a formal relation would not determine a homotopy type.

In the signed-configuration picture used by the lecture, the empty configuration is a base point. A positive and a negative point can be created as a cancelling pair, exchanged, and annihilated. This provides a concrete representative of a loop in the zero component of the group-completed space.

The Barratt–Priddy–Quillen theorem, in a precise infinite-loop-space form, identifies this group completion with the underlying infinite-loop space of the sphere spectrum:

$$\boxed{K(\mathcal{F}) \simeq Q(S^0) = \operatorname*{colim}_n \Omega^n S^n.}$$

One useful equivalent description is

$$Q(S^0) \simeq \mathbb{Z} \times (B\Sigma_\infty)^+,$$

where the superscript $+$ denotes the plus construction. In particular,

$$\boxed{\pi_1 K(\mathcal{F}) \cong \pi_1^s(\mathbb{S}) \cong \mathbb{Z}/2.}$$

The group completion has gained negative *components* and kept a nontrivial *loop*. These are two different operations on different layers of information.[1]

§145.8 The loop h

Now the lecture constructs the loop

$$h : \varnothing \sim \{+,-\} \sim \{-,+\} \sim \varnothing.$$

It has three stages.

First, create a cancelling pair:

$$\varnothing \longrightarrow \{+,-\}.$$

Second, swap the two points:

$$\{+,-\} \longrightarrow \{-,+\}.$$

Third, annihilate the cancelling pair:

$$\{-,+\} \longrightarrow \varnothing.$$

[1] For the group-completion and plus-construction descriptions, see Stefan Schwede, *Global Algebraic K-Theory*, Section 8.

K-groups of finite sets

Let $K_i(\mathrm{FinSet}) = \pi_i(K(F), \emptyset)$ to be the higher K-groups, we have the following observations:

- $\pi_0 K(F) = \mathbb{Z}$, by $x \mapsto \|x\|$. This coincides with $K_0(\mathrm{FinSet})$.
- We can construct a loop $h \in \pi_1(K(F), \emptyset)$ by using identifications

$$h : \emptyset \sim \{+, -\} \sim \{-, +\} \sim \emptyset$$

- In fact $\pi_1(K(F), \emptyset) = \mathrm{S}_n^{ab} = \mathbb{Z}/2$ with h as the generator. That is to say, h is a non-trivial identification.

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Figure 145.2: From the lecture slides: the loop h represented by creation, swap, and annihilation.

This is the corrected meaning of

$$0 = 1 - 1 = -1 + 1 = 0.$$

It is not the claim that zero equals zero. It is the claim that there is a particular loop at zero. The middle step, the swap, is the nontrivial part.

Remark 145.8.1 — If the two points were not moved, the movie would look like a boring creation followed by immediate annihilation. The swap forces a braid-like motion. In configuration space, this motion wraps around the collision locus where the two points would meet.

The slides record the key computation:

$$K_1(\mathrm{FinSet}) \cong \mathbb{Z}/2,$$

and the loop h gives the nontrivial element. Doing the swap twice is null-homotopic, but doing it once is not.

§145.9 The parity of a swap is what survives

Why should the exchange of two points represent the nontrivial class? Before group completion there is already an elementary invariant that detects it. Every permutation can be expressed as a product of transpositions. Define

$$\mathrm{sgn} : \Sigma_n \to \{\pm 1\}, \qquad \mathrm{sgn}(\sigma) = (-1)^{\#\text{ transpositions}}.$$

Parity is independent of the chosen expression: for example, it equals the sign of the corresponding permutation matrix's determinant. Thus

$$\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}(\sigma)\operatorname{sgn}(\tau).$$

A single swap has sign -1; two successive swaps have sign $+1$.

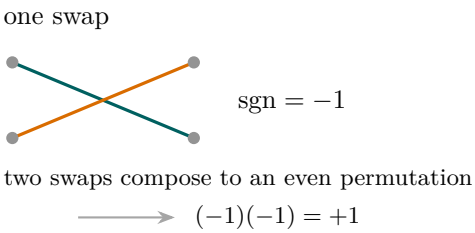


Figure 145.3: A transposition has odd parity, while its square is even. In the stabilized finite-set construction this parity detects the first nontrivial homotopy class.

There is a clean group-theoretic check. For $n \geq 2$, all transpositions in Σ_n are conjugate. Consequently they have the same image in any *abelian* quotient; that common image has order at most two because a transposition squares to the identity. Since transpositions generate Σ_n, its abelianization has at most two elements. The sign map is a surjection onto $\mathbb{Z}/2$, so

$$\boxed{\Sigma_n^{\mathrm{ab}} \cong \mathbb{Z}/2 \qquad (n \geq 2).}$$

Adding an untouched extra point sends Σ_n into Σ_{n+1}, and this stabilization preserves the sign.

The sign calculation alone does not prove the Barratt–Priddy–Quillen theorem or identify *every* higher homotopy group. It does make the first loop intelligible: the nontrivial permutation class survives stabilization and appears, via the theorem, as the nonzero element of

$$\pi_1 K(\mathcal{F}) \cong \mathbb{Z}/2.$$

The lecture's creation–swap–annihilation movie represents this class in a signed-configuration model. The identity movie gives zero; the one-swap movie does not; concatenating two copies gives zero again.

There is a key difference from the Eilenberg swindle. An infinite absorber can make an additive Grothendieck class vanish, but group completion of a homotopy-coherent space remembers the automorphisms of its finite objects. The cancellation of cardinalities has not cancelled the parity of their permutations.

§145.10 Finite sets and stable spheres

The group-completion model above describes an *infinite loop space*. The underlying theorem concerns the corresponding *spectrum*, and this distinction is worth making explicit.

Theorem 145.10.1 (Barratt–Priddy–Quillen)

The connective K-theory spectrum of the symmetric monoidal groupoid of finite sets and bijections (with disjoint union) is equivalent to the sphere spectrum $\mathbb{S}$. In particular,

$$K_i(\mathrm{FinSet}) \cong \pi_i^s(\mathbb{S}) \qquad (i \geq 0).$$

At the bottom of the spectrum we have

$$K_0(\mathrm{FinSet}) \cong \mathbb{Z}, \qquad K_1(\mathrm{FinSet}) \cong \mathbb{Z}/2.$$

The first group keeps track of signed cardinality, while the second remembers the parity detected above. Under the equivalence, the nontrivial one-swap class becomes

$$\eta \in \pi_1^s(\mathbb{S}) \cong \mathbb{Z}/2,$$

the stable Hopf element.

There is a further geometric interpretation, but one should state it at the correct level. By the Pontryagin–Thom theorem, stable homotopy groups of spheres are framed cobordism groups,

$$\pi_i^s(\mathbb{S}) \cong \Omega_i^{\mathrm{fr}}.$$

For $i = 1$, a nonbounding stable framing on a circle represents the nontrivial class. The creation–swap–annihilation picture is a convenient representative of the corresponding finite-set loop; its passage to framed cobordism is justified by the stable homotopy equivalence, not by declaring the naive space of all signed configurations to be literally equal to a framed configuration space.

§145.11 Relation with the classical Hopf map

The classical Hopf fibration represents a nontrivial element of $\pi_3(S^2)$. Stabilizing the Hopf map gives the stable element

$$\eta \in \pi_1^s(\mathbb{S}).$$

The finite-set loop h gives the same class under the BPQ theorem.

Thus the meme ladder is not merely free association. The chain is:

$$\begin{gathered}\text{Hopf fibration } S^3 \to S^2 \\ \Downarrow \\ \text{stable Hopf element } \eta \in \pi_1^s(\mathbb{S}) \\ \Downarrow \\ K_1(\mathrm{FinSet}) \cong \pi_1^s(\mathbb{S}) \\ \Downarrow \\ \text{the loop } h : \varnothing \sim \{+,-\} \sim \{-,+\} \sim \varnothing.\end{gathered}$$

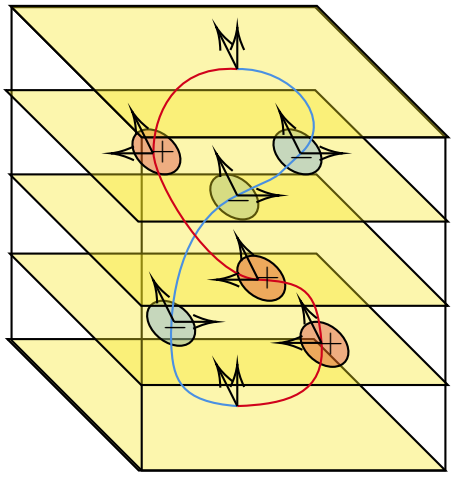
This is the clean mathematical sense in which the arithmetic-looking line is the Hopf map in disguise.

§145.12 Cobordism: the movie becomes a circle

The slides then place the motion in three-dimensional space. As time evolves, the moving point configuration sweeps out a one-dimensional manifold. The creation of the pair, their swap, and their annihilation form a closed loop. Because the points carry signs or framings, the resulting circle is a framed one-manifold.

What about cobordism?

If we place this motion in 3-dimensional space, we obtain a circle:



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Figure 145.4: From the lecture slides: the creation–swap–annihilation movie becomes a framed circle.

Framed cobordism asks whether this framed circle is the boundary of a framed disk. The answer is no for the loop h. This is another expression of the same nontrivial $\mathbb{Z}/2$-class.

Definition 145.12.1 (Framed cobordism, informal)**.** A framed n-manifold is an n-manifold equipped with a trivialization of its normal bundle. Two framed manifolds are framed cobordant if they form the boundary of a framed manifold one dimension higher.

This is the geometric face of the equation. The path starts and ends at the empty configuration, but its trace is not the boundary of a trivial framed disk. The loop is geometrically nonzero.

§145.13 Algebraic K-theory of a field

After finite sets, the lecture repeats the construction with finite-dimensional vector spaces over a field k. Let

$$\mathcal{V}_k = \mathrm{FinVect}_k/\sim$$

be the moduli space of finite-dimensional vector spaces. Under direct sum, it behaves like a categorified version of addition.

The low K-groups are familiar:

$$K_0(k) \cong \mathbb{Z}, \qquad K_1(k) \cong k^\times.$$

The first says that a vector space is measured by its dimension after group completion. The second says that the determinant detects the abelianized automorphism information:

$$\mathrm{GL}_\infty(k)^{\mathrm{ab}} \cong k^\times.$$

This parallels the finite-set case:

object	K_0	K_1
FinSet	$\mathbb{Z}$	$\mathbb{Z}/2$
$\mathrm{FinVect}_k$	$\mathbb{Z}$	$k^\times$

The lecture now asks for a geometric model of these algebraic K-groups beyond degree one.

§145.14 Intrinsic skein-lasagna

The advanced second half of the slide deck introduces intrinsic skein-lasagna. This is not merely decorative. It is a proposed geometric language for K-groups analogous to how framed cobordism gives a geometric language for stable homotopy.

The first definition is an n-treefold: a space locally homeomorphic to

$$T \times \mathbb{R}^{n-1},$$

where T is a tree. The branching models a controlled kind of singularity.

Given a symmetric monoidal category $(\mathcal{C}, \otimes)$, an n-$(\mathcal{C}, \otimes)$-lasagna is, roughly, a framed n-treefold with labels from $\mathcal{C}$, satisfying compatibility at the branching loci.

Definition 145.14.1 (Intrinsic skein-lasagna, informal)**.** An intrinsic lasagna is a geometric object built from treefold-like pieces, with labels coming from a symmetric monoidal category. The word "intrinsic" means that the object is studied on its own, rather than as something embedded in a previously chosen ambient manifold.

The slide illustrates n-treefolds as spaces locally modelled on $T \times \mathbb{R}^{n-1}$, where T is a tree. For the notebook, the useful point is that branching is allowed but controlled.

The word "intrinsic" is important. Classical skein-lasagna modules often involve embedded objects inside an ambient manifold. Here the lecture emphasizes that one can study the geometry of the lasagna itself, without first choosing an ambient space. If desired, one may imagine the ambient space as $\mathbb{R}^\infty$, but it is no longer part of the primary data.

§145.15 The $(\mathcal{V}_k, \oplus)$-cobordism group

The lecture specializes the lasagna story to

$$(\mathcal{V}_k, \oplus),$$

where $\mathcal{V}_k$ is the category or moduli object of finite-dimensional vector spaces under direct sum. One obtains groups denoted

$$\Omega_n^{\mathcal{V}_k, \oplus}.$$

The examples in low degree match algebraic K-theory:

$$\Omega_0^{\mathcal{V}_k, \oplus} \cong K_0(k) \cong \mathbb{Z},$$

$$\Omega_1^{\mathcal{V}_k, \oplus} \cong K_1(k) \cong k^\times.$$

Remark 145.15.1 — The slide summarizes the first two $(\mathcal{V}_k, \oplus)$-cobordism groups: degree 0 recovers dimension, and degree 1 recovers determinant. In formulas,

$$\Omega_0^{\mathcal{V}_k,\oplus} \cong K_0(k) \cong \mathbb{Z}, \qquad \Omega_1^{\mathcal{V}_k,\oplus} \cong K_1(k) \cong k^\times.$$

This is why vector-space-labelled circles are the K_1 analogue of the finite-set swap loop.

The degree-one case is especially helpful. A circle labelled by an automorphism $f \in \mathrm{GL}(V)$ represents a class, and the determinant gives its value in $k^\times$. A commutator becomes null-cobordant, matching the abelianization of $\mathrm{GL}_\infty(k)$.

This is the lasagna analogue of the finite-set story. In finite sets, a swap gives the nontrivial element of $\mathbb{Z}/2$. In vector spaces, an automorphism gives an element of $k^\times$.

§145.16 Generalized Pontryagin–Thom

The slide deck then states the central claim:

$$K_n(k) \xrightarrow{\cong} \Omega_n^{\mathcal{V}_k,\oplus} \qquad (n \geq 0).$$

This is presented as a generalized Pontryagin–Thom isomorphism.

Generalized Pontryagin–Thom Isomorphism

Recall that $K_0(k) = \mathbb{Z}$ and $K_1(k) = k^\times$. This is not just a coincidence:

Generalized Pontryagin–Thom Isomorphism (Claim)

For all $n \geq 0$, we have the isomorphism

$$K_n(k) \xrightarrow{\cong} \Omega_n^{\mathcal{V}_k,\oplus}$$

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Figure 145.5: From the lecture slides: the generalized Pontryagin–Thom isomorphism identifies $K_n(k)$ with a lasagna-cobordism group.

The analogy is:

classical/stable homotopy	algebraic K-theory side
$\pi_n^s(\mathbb{S})$	$K_n(k)$
Ω_n^{fr}	$\Omega_n^{\mathcal{V}_k,\oplus}$
framed manifolds	vector-space-labelled lasagnas

This is why the lecture moves from elementary arithmetic to Hopf fibration and then to intrinsic skein-lasagna. It is building a geometric model of algebraic K-theory.

§145.17 The K_2 slide and Steinberg relations

In degree two, the lecture recalls the classical presentation

$$K_2(k) \cong k^\times \otimes k^\times / \langle a \otimes (1-a) \rangle.$$

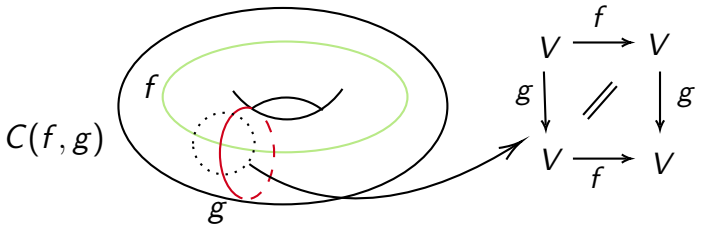
It then considers matrices $f, g \in \mathrm{GL}(k)$ with commutator

$$[f, g] = 1.$$

Such a commuting pair defines a two-dimensional lasagna $C(f, g)$.

$\Omega_2^{\mathcal{V}_k,\oplus}$ **and** $K_2(k)$

For matrices $f, g \in \mathrm{GL}(k)$ such that $[f, g] = 1$, one can define the following 2-lasagna $C(f, g)$:



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Figure 145.6: From the lecture slides: a two-lasagna $C(f, g)$ associated to a commuting pair.

The next slide gives a more concrete example using elementary matrices. For $i \neq j$, let

$$e_{ij}(a)$$

be the elementary matrix with a in the (i, j)-entry and ones on the diagonal. These matrices satisfy Steinberg relations, such as

$$[e_{ij}(a), e_{kl}(b)] = \begin{cases} e_{il}(ab), & j = k, \\ \mathbf{1}, & \text{otherwise.} \end{cases}$$

The slide then uses Hall–Witt-type commutator identities to produce a nontrivial proof that a certain commutator is the identity. This proof is not just algebraic decoration; it is a blueprint for constructing a bounding three-lasagna.

3-**Lasagna bounds** $C(e_{12}(a), e_{12}(b))$

For $i \neq j$, let the elementary matrices $e_{ij}(a)$ be

$$(e_{ij}(a))_{kl} = \begin{cases} 1, & \text{if } k = l, \\ a, & \text{if } k = i, l = j, \\ 0, & \text{otherwise.} \end{cases}$$

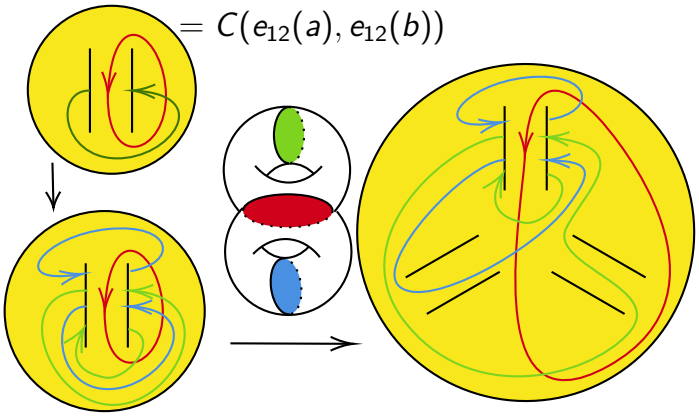
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Figure 145.7: From the lecture slides: elementary matrices, Steinberg relations, and the algebraic pattern behind the three-lasagna.

§145.18 The final lasagna diagrams

The final nontrivial slides display a Heegaard splitting and a boundary relation. They are not necessary for understanding the original meme, but they are essential for understanding where the lecture is going. The creation–swap–annihilation loop was the baby example. The lasagna diagrams are the research-level continuation of the same philosophy: algebraic K-theory classes are represented by geometric/cobordism-like objects.

Following this blueprint, we obtain the following Heegaard splitting to construct a lasagna:



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Figure 145.8: From the lecture slides: a Heegaard-splitting picture used to construct a three-lasagna.

At this point the lecture has travelled very far from the original equation. But the path is coherent:

equation as path
⇓
finite signed configurations
⇓
K-theory of finite sets
⇓
stable homotopy and framed cobordism
⇓
algebraic K-theory of fields
⇓
intrinsic skein-lasagna as a geometric model.

§145.19 What the equation finally says

The equation

$$0 = 1 - 1 = -1 + 1 = 0$$

should be read as the name of a loop:

$$h : \varnothing \to \{+, -\} \to \{-, +\} \to \varnothing.$$

It is a loop at zero in the K-space of finite sets. It is not the identity loop. It represents the generator of

$$K_1(\mathrm{FinSet}) \cong \mathbb{Z}/2.$$

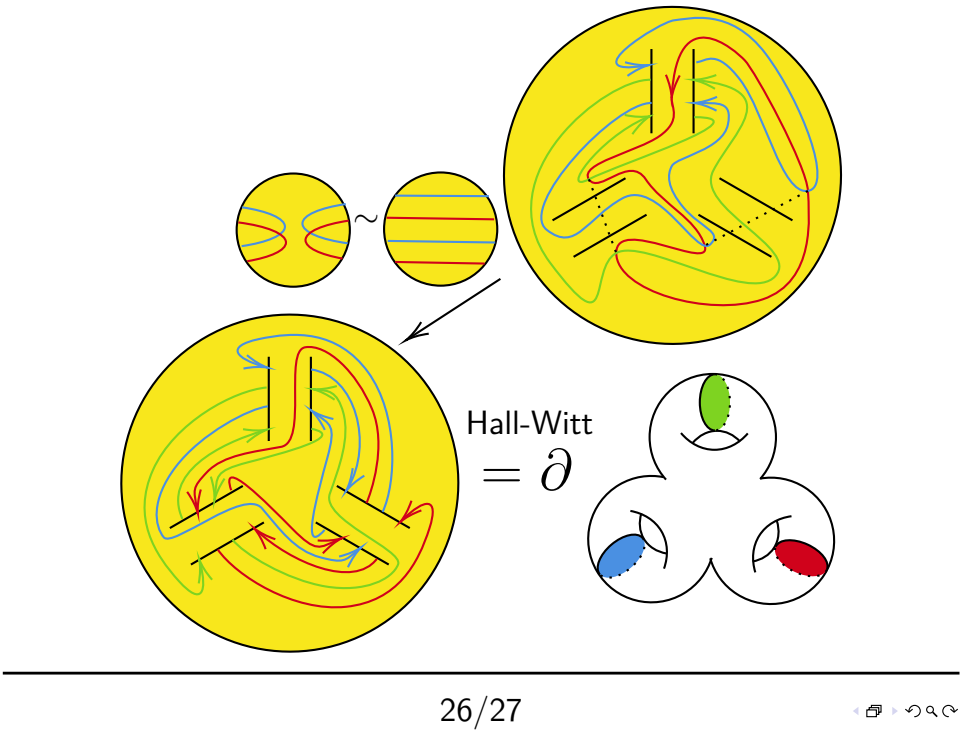


Figure 145.9: From the lecture slides: the boundary relation for the three-lasagna construction.

By Barratt–Priddy–Quillen, this group is

$$\pi_1^s(\mathbb{S}) \cong \mathbb{Z}/2,$$

and h corresponds to the stable Hopf class η.

This is why the Hopf fibration belongs in the same story as a signed finite-set calculation. The shared object is not an ordinary number. It is a stable homotopy class.

Almost Nowhere, Yet Everywhere

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Can a subset of the real line have measure zero, intersect every open interval, and still deserve to be called *typical*? We will construct one that does. Then we will construct a closed set with positive measure and no interior, and discover that these are only the beginning of the disagreement between two notions of size.

One way of describing a small set is to ask how much length it occupies. Another is to ask whether it leaves an open region in which it is absent, and whether it can be built by combining sets with that property. The first leads to Lebesgue measure; the second leads to Baire category. They support remarkably similar theorems and sometimes pronounce opposite verdicts on the very same subset of $\mathbb{R}$. The classical source for this subject is Oxtoby's *Measure and Category*, especially its opening chapters on the real line and Liouville numbers.[1]

I have a rather silly history with this little book. When I was much younger, I was so taken with Springer's Graduate Texts in Mathematics that I wanted the small personal triumph of finishing *one whole GTM*. I chose GTM 2, partly because Oxtoby's volume looked like one of the thinnest on the shelf. The title made the bargain even sweeter: *Measure and Category* sounded to me like a remarkably fancy pocket-sized tour of measure theory *and* category theory. Well, it turns out that this category is Baire's category—meager sets, nowhere-dense sets, and all that—rather than the world of objects, arrows, and functors. Seven years later, opening the same slender volume again feels a little like meeting an old accomplice in a very innocent mathematical misunderstanding. At least this time I know what the title means.

Just to settle the misunderstanding properly, here are the two meanings side by side.

Definition 146.0.1 (Baire category)**.** Let X be a topological space. A subset $A \subseteq X$ is *nowhere dense* if

$$\operatorname{int}_X(\overline{A}) = \varnothing.$$

It is of *first category*, or *meager*, if it is a countable union of nowhere-dense subsets of X; otherwise it is of *second category*. A subset is *comeager* (or *residual*) if its complement is meager.

Definition 146.0.2 (Category in category theory)**.** A *category* $\mathcal{C}$ consists of a collection of objects, sets of morphisms $\operatorname{Hom}_{\mathcal{C}}(A, B)$ for each pair of objects, identity morphisms $\mathrm{id}_A : A \to A$, and associative composition

$$\operatorname{Hom}_{\mathcal{C}}(B, C) \times \operatorname{Hom}_{\mathcal{C}}(A, B) \longrightarrow \operatorname{Hom}_{\mathcal{C}}(A, C).$$

Composition satisfies $h \circ (g \circ f) = (h \circ g) \circ f$ whenever defined, and $\mathrm{id}_B \circ f = f = f \circ \mathrm{id}_A$ for every $f : A \to B$.

§146.1 The Cantor set is already quite strange

Start with the interval $C_0 = [0, 1]$. Remove its open middle third to obtain

$$C_1 = [0, \tfrac{1}{3}] \cup [\tfrac{2}{3}, 1].$$

[1] John C. Oxtoby, *Measure and Category: A Survey of the Analogies between Topological and Measure Spaces*, 2nd ed., Graduate Texts in Mathematics 2 (1980).

From each of these two intervals remove its open middle third. Continue forever, and write C_n for the nth surviving set:

$$C = \bigcap_{n=0}^{\infty} C_n.$$

At stage n, C_n consists of 2^n closed intervals, each of length 3^{-n}. Thus

$$\lambda(C_n) = 2^n 3^{-n} = \left(\frac{2}{3}\right)^n, \qquad \boxed{\lambda(C) = 0}.$$

The last equality follows from continuity of measure from above, since C_0 has finite measure. The set C is closed as an intersection of closed sets.

Yet C is uncountable. Every binary sequence $b = (b_1, b_2, \ldots)$ determines a point

$$x_b = \sum_{n=1}^{\infty} \frac{2b_n}{3^n} \in C.$$

Different binary sequences give different points: at the first place where two sequences differ, the discrepancy $2 \cdot 3^{-n}$ exceeds the greatest possible contribution 3^{-n} from all later terms. This gives an injection $\{0,1\}^{\mathbb{N}} \hookrightarrow C$. A zero-measure set may contain uncountably many points.

There is a different sense in which C is small. It contains no nonempty open interval. If an interval $I \subseteq C$ existed, then for large n it would have to fit inside a component of C_n, each of which has length $3^{-n} < |I|$, an impossibility. Since C is closed, this says that C is *nowhere dense*:

$$\operatorname{int} \overline{C} = \varnothing.$$

A nowhere-dense set is one whose closure fails to contain even one nonempty open interval. Notice that it can still have many accumulation points; indeed the Cantor set is perfect, with no isolated points.



Figure 146.1: The first two deletions in the middle-thirds Cantor construction. After n stages, 2^n short intervals remain, while their total length decays as $(2/3)^n$.

§146.2 A fat Cantor set

Could the deletion scheme leave a set with *positive* measure, yet still no interior? Yes. Keep the left-right splitting procedure but remove less length.

Begin again with $F_0 = [0,1]$. At stage $n \geq 1$, remove from the middle of each of the 2^{n-1} surviving closed intervals an open interval of length 4^{-n}. The first stage removes an interval of length $1/4$; the second removes two intervals of length $1/16$; the third removes four intervals of length $1/64$, and so forth. Define $F = \bigcap_{n=0}^{\infty} F_n$.

There is room for all these deletions. At stage $n-1$, each of the 2^{n-1} remaining intervals has the same length

$$\ell_{n-1} = \frac{1 - \sum_{k=1}^{n-1} 2^{k-1} 4^{-k}}{2^{n-1}} = \frac{1/2 + 2^{-n}}{2^{n-1}},$$

which is strictly greater than the next deletion length 4^{-n}.

The total removed length is

$$\sum_{n=1}^{\infty} 2^{n-1} 4^{-n} = \sum_{n=1}^{\infty} 2^{-n-1} = \frac{1}{2}.$$

Because the deleted intervals are disjoint,

$$\boxed{\lambda(F) = \frac{1}{2}}.$$

Each component of F_n has length at most 2^{-n}. For every interval I of positive length, sufficiently late components are shorter than I, so $I \not\subseteq F$. The set F is closed with empty interior: it is nowhere dense and occupies exactly half the unit interval.



Figure 146.2: The first two stages of a Smith–Volterra–Cantor construction. It has the same endlessly branching structure as the previous set, but its total remaining length tends to 1/2.

We could have made the retained measure as close to 1 as desired. For $0 < \varepsilon < 1$, remove length $2\varepsilon\, 4^{-n}$ from the middle of each surviving interval at stage n. The total removed length is

$$\sum_{n \geq 1} 2^{n-1} (2\varepsilon\, 4^{-n}) = \varepsilon.$$

The resulting closed nowhere-dense set has measure $1 - \varepsilon$. Thus nowhere density places no useful upper bound on Lebesgue measure below the measure of the whole interval.

The two Cantor-type sets have brought us to a question. A nowhere-dense set is small in the topology of open intervals, whether its measure is 0, 1/2, or 0.999. What happens if we allow *countably many* nowhere-dense pieces?

§146.3 Completeness and the Baire theorem

A set A is *meager* (of first category) if it is a countable union of nowhere-dense sets, or is contained in such a union. A set is *comeager* or *residual* if its complement is meager. Here category is Baire's classical terminology; it has no connection with the notion of a category in homological algebra or category theory.

Finite unions of nowhere-dense sets are nowhere dense. A countable union is a different matter. Every singleton $\{q\} \subseteq \mathbb{R}$ is nowhere dense, so the rational numbers

$$\mathbb{Q} = \bigcup_{q \in \mathbb{Q}} \{q\}$$

are meager. But $\mathbb{Q}$ is dense in $\mathbb{R}$. Being nowhere dense is not a property preserved under a countable union.

Could all of $\mathbb{R}$ be obtained as a countable union of nowhere-dense sets? The Baire category theorem rules that out.

Theorem 146.3.1 (Baire category theorem)

If (X, d) is a complete metric space and $U_1, U_2, \ldots$ are open dense subsets of X, then

$$\bigcap_{n=1}^{\infty} U_n$$

is dense in X. In particular, every nonempty open subset of X is nonmeager in itself.

Proof. Let $W \subseteq X$ be an arbitrary nonempty open set. We must find a point of W belonging to every U_n.

Because U_1 is open and dense, $W \cap U_1$ is nonempty and open. Choose a closed ball

$$\overline{B_1} = \overline{B}(x_1, r_1) \subseteq W \cap U_1, \qquad 0 < r_1 < \frac{1}{2}.$$

The open set $B_1 \cap U_2$ is also nonempty. Choose a second closed ball

$$\overline{B_2} \subseteq B_1 \cap U_2, \qquad 0 < r_2 < \frac{1}{4}.$$

Continue, choosing

$$\overline{B_{n+1}} \subseteq B_n \cap U_{n+1}, \qquad 0 < r_{n+1} < 2^{-n-1}.$$

The closed balls are nested, and their diameters tend to zero. Their centers form a Cauchy sequence: if $m > n$, then $x_m \in B_n$, so $d(x_m, x_n) \leq r_n$. Completeness gives a limit $x \in X$.

For each fixed n, all later centers belong to $\overline{B_n}$, a closed set, so $x \in \overline{B_n}$. Hence $x \in U_n$ for every n and $x \in W$. Since W was arbitrary, the intersection is dense. □

The key geometric action is to keep choosing a smaller closed ball inside a new open dense set. The radii go to zero, and completeness guarantees that this infinite sequence of compatible choices still leaves an actual point. Without completeness, the procedure may converge to a point missing from the space: $\mathbb{Q}$, with its usual metric, is itself a countable union of nowhere-dense singletons.

A set of the form $\bigcap_n U_n$, where the U_n are open, is called a G_δ set. A countable intersection of *open dense* sets is a particularly useful kind of G_δ: Baire's theorem says it remains dense. Moreover any countable intersection of dense G_δ sets is dense G_δ, since we may collect all the open dense factors into one countable family.

§146.4 A set that is negligible and generic

We are ready for the most conspicuous disagreement between measure and category. First enumerate the rationals as

$$\mathbb{Q} = \{q_1, q_2, \ldots\}.$$

For every $n \geq 1$, place a tiny interval around each rational and take their union:

$$V_n = \bigcup_{j=1}^{\infty} \left(q_j - 2^{-n-j-2},\, q_j + 2^{-n-j-2}\right). \tag{146.1}$$

Each V_n is open and contains all rationals, so it is dense. Yet a countable subadditivity calculation gives

$$\lambda(V_n) \le \sum_{j=1}^{\infty} 2^{-n-j-1} = 2^{-n-1}.$$

Thus V_n is open and dense while occupying a very small total length. As n grows, this length shrinks to zero.

Consider the set

$$G = \bigcap_{n=1}^{\infty} V_n.$$

By Baire's theorem, G is dense G_δ; in particular it is comeager. But $G \subseteq V_n$ for every n, so

$$0 \le \lambda(G) \le \inf_n 2^{-n-1} = 0.$$

We have constructed

$$\boxed{G \text{ is comeager and dense, yet } \lambda(G) = 0.} \tag{146.2}$$

Its complement $H = \mathbb{R} \setminus G$ is meager, since it is the union of the nowhere-dense closed sets $\mathbb{R} \setminus V_n$. Meanwhile

$$\lambda(\mathbb{R} \setminus H) = \lambda(G) = 0.$$

Thus H has *full Lebesgue measure*, in the sense that its complement is null, even though H is meager. Both G and H are dense: every nonempty open interval has positive length and cannot be contained in the null set G.

Here is a precise answer to the opening question. The statement that a property holds *almost everywhere* means that its failure set has measure zero. The statement that it holds *generically* in the Baire sense means that its failure set is meager. On the real line these two notions can make opposite claims about the same property.

A concrete property having this behavior is waiting for us in number theory.

§146.5 Numbers that are exceptionally easy to approximate

A real number x is a *Liouville number* if it can be approximated by rational numbers to an arbitrarily high algebraic order: for every integer $k \ge 1$ and every $N \ge 2$, there exist integers p and $q \ge N$ such that

$$0 < \left| x - \frac{p}{q} \right| < \frac{1}{q^k}. \tag{146.3}$$

The requirement that the inequality hold for arbitrarily large denominators rules out merely repeating a single rational approximation. It is equivalent to the usual Liouville condition for irrational x, with the additional insistence that x itself is not rational.

There are explicit examples. Take

$$L = \sum_{j=1}^{\infty} 10^{-j!} = 0.110001000000000000000001\ldots$$

and its rational truncations $L_m = \sum_{j=1}^{m} 10^{-j!}$. The denominator of L_m is $q_m = 10^{m!}$; the omitted tail is bounded by

$$0 < L - L_m < 2 \cdot 10^{-(m+1)!} = 2q_m^{-(m+1)}.$$

For any prescribed k, choose m so large that $m+1 > k$ and $2q_m^{-(m+1)} < q_m^{-k}$. This exhibits rational approximations satisfying (146.3) with denominators tending to infinity.

Why are these numbers called especially well approximated? Suppose instead that α is algebraic of degree d, with a nonzero polynomial $P \in \mathbb{Z}[x]$ of degree d satisfying $P(\alpha) = 0$. For any rational $p/q \neq \alpha$, where $q \geq 1$,

$$q^d P(p/q) \in \mathbb{Z} \setminus \{0\}, \qquad |P(p/q)| \geq q^{-d}.$$

If p/q is near α, the derivative P' is bounded by some constant $M > 0$ on the interval between them. The mean value theorem gives

$$|P(p/q)| = |P(p/q) - P(\alpha)| \leq M|p/q - \alpha|.$$

Therefore

$$\left|\alpha - \frac{p}{q}\right| \geq \frac{1}{Mq^d}. \tag{146.4}$$

For sufficiently large q, this contradicts a requirement such as $|\alpha - p/q| < q^{-(d+1)}$. Consequently every Liouville number is transcendental.

This establishes an unexpected kind of rarity, but the measure and category of the *whole set* of Liouville numbers are even more interesting.

§146.6 The Liouville numbers form a dense null G_δ

For integers $k \geq 1$ and $N \geq 2$, define

$$U_{k,N} = \bigcup_{\substack{q \geq N \\ p \in \mathbb{Z}}} \left(\left(\frac{p}{q} - q^{-k}, \frac{p}{q} \right) \cup \left(\frac{p}{q}, \frac{p}{q} + q^{-k} \right) \right). \tag{146.5}$$

This is open. It is also dense: for every open interval I, choose a sufficiently large denominator q and a rational p/q strictly inside I; points immediately to either side of p/q also lie in $I \cap U_{k,N}$. The set $\mathcal{L}$ of Liouville numbers is exactly

$$\mathcal{L} = \bigcap_{k=1}^{\infty} \bigcap_{N=2}^{\infty} U_{k,N}. \tag{146.6}$$

It is a countable intersection of open dense subsets of $\mathbb{R}$. Therefore Baire's theorem says that $\mathcal{L}$ is a dense G_δ set, hence comeager.

To prove it has measure zero, fix $R > 0$, and consider $U_{4,N} \cap [-R, R]$. If one of the intervals centered at p/q meets $[-R, R]$, then for $q \geq 2$ we have $|p/q| \leq R + 1$. For each q, at most $2(R+1)q + 3$ integers p have this property, and each punctured interval has total length $2q^{-4}$. Hence, for a constant K_R independent of N,

$$\lambda(U_{4,N} \cap [-R, R]) \leq K_R \sum_{q=N}^{\infty} q^{-3} \xrightarrow[N \to \infty]{} 0.$$

But $\mathcal{L} \subseteq U_{4,N}$ for every N; thus

$$\lambda(\mathcal{L} \cap [-R, R]) = 0 \qquad \text{for every } R > 0.$$

Taking a countable union of such bounded intervals yields

$$\boxed{\lambda(\mathcal{L}) = 0, \qquad \mathcal{L} \text{ is dense and comeager in } \mathbb{R}.} \tag{146.7}$$

In fact the complement $\mathbb{R}\setminus\mathcal{L}$ has full Lebesgue measure while being meager. Almost every real number is *not* a Liouville number, and a Baire-generic real number *is* a Liouville number.

The result goes further than a curious classification of sets. It gives a very short proof of a remarkable statement about addition.

Proposition 146.6.1

Every real number is the sum of two Liouville numbers. Equivalently,

$$\mathcal{L} + \mathcal{L} = \mathbb{R}.$$

Proof. Fix $t \in \mathbb{R}$. Since $\mathcal{L}$ is dense G_δ, so is the translate and reflection $t - \mathcal{L} = \{t - y : y \in \mathcal{L}\}$. To see this directly, apply the homeomorphism $x \mapsto t - x$ to each open dense set in (146.6). By Baire's theorem,

$$\mathcal{L} \cap (t - \mathcal{L}) \neq \varnothing.$$

Choose x in this intersection. Then $x \in \mathcal{L}$ and $t - x \in \mathcal{L}$, giving $t = x + (t - x)$. □

Every Liouville number is transcendental, so even *every algebraic number* can be written as the sum of two particular transcendental numbers. A measure-zero subset of the line has sumset equal to the entire line. Cardinality and length cannot predict this additive behavior; category provides an elegant existence proof.

§146.7 What does a homeomorphism remember?

There is a second reason to separate topology from measure. The two Cantor-type sets we constructed at the beginning are actually *homeomorphic*. To see this without a classification theorem, label the left and right surviving intervals at each stage by 0 and 1. An infinite binary sequence $b = (b_1, b_2, \ldots)$ singles out one nested sequence of intervals in C, hence one point c_b. The same sequence singles out one point f_b of the fat Cantor set F.

Both interval trees shrink in diameter to zero, so the correspondence $c_b \mapsto f_b$ is continuous, bijective and has a continuous inverse. It preserves order: if two binary sequences first differ at position n, the point with digit 0 lies to the left of the point with digit 1 in both constructions.

In fact it extends to an increasing homeomorphism

$$h : [0,1] \longrightarrow [0,1], \qquad h(C) = F.$$

Each complementary interval of C corresponds, under the left-right address system, to one complementary interval of F. Map their endpoints accordingly, and extend the map linearly across each gap. The resulting strictly increasing function is continuous, fixes $0, 1$, and therefore has a continuous inverse.

We have produced a homeomorphism with

$$\lambda(C) = 0, \qquad \lambda(h(C)) = \lambda(F) = \frac{1}{2}.$$

Thus Lebesgue measure zero is *not* invariant under arbitrary homeomorphisms. In contrast, homeomorphisms preserve the property of being nowhere dense, since they preserve closures and interiors, and therefore preserve meager and comeager sets.

There is a useful boundary on the example. If a map $h : [0,1] \to \mathbb{R}$ is K-Lipschitz, then it maps Lebesgue-null sets to Lebesgue-null sets. Indeed, cover a null set A by intervals I_j with $\sum_j |I_j| < \varepsilon$. Every image $h(I_j)$ is contained in an interval of length at most $K|I_j|$, and so

$$\lambda^*(h(A)) \le K \sum_j |I_j| < K\varepsilon.$$

Letting $\varepsilon \to 0$ gives $\lambda(h(A)) = 0$. Consequently the homeomorphism taking C to our fat Cantor set cannot be Lipschitz. Topological continuity allows deformations that quantitative regularity would forbid.

§146.8 Two different ways to ignore small sets

Consider an open set $U \subseteq \mathbb{R}$ and an open family $U_i \subseteq U$. One way to call the family a cover is to require

$$\overline{\bigcup_i U_i} \supseteq U.$$

This says that the union of the pieces is *dense* in U: there is no nonempty open part of U that the family completely misses.

Another way, given a measure μ, is to require

$$\mu\left(U \setminus \bigcup_i U_i\right) = 0.$$

This says that the family covers U *almost everywhere*. Lafforgue's lecture on the measure-theoretic topology of language-model probability spaces uses this latter kind of measure-induced covering relation, and it is instructive to compare the two requirements in a familiar setting.

For Lebesgue measure on $(0,1)$, let F be the fat Cantor set of measure $1/2$. The relative complement

$$V = (0,1) \setminus F$$

is open and dense in $(0,1)$. But

$$\lambda(V) = \frac{1}{2} < 1 = \lambda((0,1)).$$

Thus the singleton family $\{V\}$ covers $(0,1)$ in the dense-cover sense and fails to cover it in the almost-everywhere sense.

Conversely, if $\lambda(U \setminus V) = 0$ with V open in U, then V must be dense in U. Otherwise some nonempty open interval $I \subseteq U \setminus \overline{V}$ would have positive measure and lie outside V, a contradiction. For Lebesgue measure, almost-everywhere open coverage is therefore stronger than dense open coverage.

This implication relies on the fact that every nonempty open interval has positive Lebesgue measure. For a probability measure concentrated at one point, even that can fail. On $X = [0,1]$, take $\mu = \delta_0$ and the relatively open set $V = [0, 1/4)$. Then $\mu(V) = \mu(X) = 1$, but V is far from dense in X.

The two covering rules discard different kinds of exceptional sets. In one, the missing pieces are invisible to the measure; in the other, the missing pieces have no open interior. The constructions in this chapter show why these are distinct mathematical judgements, even on the real line.

XII

Algebraic and Arithmetic Geometry

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From Diophantine Inequalities to Arithmetic Deformation

N-20211007

§147.1 The question is an inequality, not a slogan

Fesenko's survey presents inter-universal Teichmueller theory under another name: arithmetic deformation theory. This name is useful because it puts the reader's attention on the mathematical target before the new language appears. The target is not first of all a mystical "inter-universal" philosophy. It is a family of Diophantine inequalities in which one wants to control the arithmetic size of a curve, an elliptic curve, or a rational point.

The basic shape is always the same. One side measures size: height, discriminant, or a quantity built out of bad reduction. The other side measures the places where the object is allowed to be complicated: conductor, ramification, different, or a divisor of excluded points. The problem is to prove that the first kind of size cannot grow too fast compared with the second kind of size.

Definition 147.1.1 (Number field and absolute Galois group)**.** A number field is a finite extension $K/\mathbb{Q}$. If K^{alg} is an algebraic closure of K, its absolute Galois group is

$$G_K = \mathrm{Gal}(K^{\mathrm{alg}}/K),$$

regarded as a profinite topological group.

The appearance of G_K is not decoration. A large part of the story leading to IUT asks how much arithmetic geometry can be recovered from groups of symmetries. In ordinary algebraic geometry one starts with equations and then studies automorphisms or fundamental groups. Anabelian geometry tries to reverse this direction: from a suitable fundamental group, recover the geometric or arithmetic object.

§147.2 A first chain of rigidity results

The first rigidity theorem in this note is about number fields alone.

Theorem 147.2.1 (Neukirch–Ikeda–Uchida, schematic form)

Let K_1 and K_2 be number fields. An isomorphism of topological groups

$$G_{K_1} \cong G_{K_2}$$

comes from a unique field-theoretic identification of the corresponding algebraic closures, in a way compatible with the two number fields. Thus, for number fields, the absolute Galois group remembers the field.

This theorem should be read as a warning against the naive view that a Galois group is only an auxiliary invariant. In the number-field case, the group is so rigid that the field can be reconstructed from it.

The next theorem connects curves with a surprisingly small amount of ramification.

Theorem 147.2.2 (Belyi)

Let C be an irreducible smooth projective algebraic curve over a field of characteristic zero. Then C can be defined over $\overline{\mathbb{Q}}$ if and only if there exists a finite morphism

$$C \longrightarrow \mathbb{P}^1$$

which is ramified over at most three points of $\mathbb{P}^1$.

Such maps are called Belyi maps. For this note, the important point is not the proof but the direction of thought. Curves over number fields, coverings of $\mathbb{P}^1 \setminus \{0, 1, \infty\}$, and Galois/fundamental-group data are not separate topics. They are different faces of the same arithmetic geometry.

Remark 147.2.3 (Why this belongs in the IUT entrance) — Fesenko emphasizes that versions of Belyi maps are used later in arithmetic deformation theory and its application to Diophantine geometry. This is why the story does not begin directly with theta-links. Before one reaches theta-links, one has already entered a world where curves, number fields, and fundamental groups are designed to remember one another.

§147.3 From Mordell finiteness to effective inequalities

Faltings' theorem gives a finiteness result.

Theorem 147.3.1 (Faltings–Mordell)

If C is a smooth projective curve of genus > 1 over a number field K, then $C(K)$ is finite.

Finiteness is a qualitative statement. The effective Mordell problem asks for more: not only that rational points are finite, but that their heights can be bounded in terms of computable arithmetic data. In the background of Fesenko's survey, this effective direction sits beside several closely related conjectures:

effective Mordell, Szpiro, *abc*, Frey, Vojta for hyperbolic curves.

These are not identical statements, but Fesenko presents them as a closely connected cluster. The important organizing idea is that they all seek an inequality strong enough to control Diophantine complexity.

§147.4 The Szpiro inequality as a concrete target

The Szpiro conjecture is a good first model because it is phrased directly in terms of elliptic curves and bad reduction. Let E_K be an elliptic curve over a number field K. In the split multiplicative situation, the bad fibers of the minimal regular proper model have type I_{n_v}. Here v runs over nonarchimedean valuations of K, $k(v)$ is the residue field, and n_v is the number of components in the bad fiber.

Definition 147.4.1 (Conductor-size and discriminant-size, split multiplicative model)**.** In the split multiplicative case, the logarithmic discriminant-size and conductor-size are, schematically,

$$\sum_{v \text{ bad}} n_v \log |k(v)|, \qquad \sum_{v \text{ bad}} \log |k(v)|.$$

The first expression counts bad fibers with multiplicity n_v. The second remembers only the bad places themselves.

The conjectural Szpiro inequality then has the form

$$\sum_{v \text{ bad}} n_v \log |k(v)| \leq c + (6+\epsilon) \sum_{v \text{ bad}} \log |k(v)|,$$

for every $\epsilon > 0$, with a constant c depending on K and ϵ. Equivalently, using the absolute norms of the minimal discriminant and the conductor, one writes

$$N(\operatorname{Disc} E_K) \leq c' N(\operatorname{Cond} E_K)^{6+\epsilon}.$$

Remark 147.4.2 (The meaning of the exponent) — The number 6 is not a random constant inserted into the conjecture. In Fesenko's discussion it is connected with the geometric Szpiro-type inequality for elliptic surfaces and with the comparison between the arithmetic and geometric sides of the problem. The discriminant measures arithmetic degeneration, while the conductor records where and how severely bad reduction occurs. Szpiro-type inequalities constrain the former in terms of the latter.

A very simple reading is this: the conductor records where the elliptic curve behaves badly; the discriminant records how badly it behaves there. Szpiro predicts that "how badly" cannot be much larger than a fixed power of "where badly".

§147.5 Vojta's inequality and the hyperbolic curve viewpoint

The Vojta form is closer to the geometry of punctured curves.

Definition 147.5.1 (Hyperbolic curve)**.** Over a field of characteristic zero, a hyperbolic curve is a smooth projective geometrically connected curve of genus g with r points removed, such that

$$2 - 2g - r < 0.$$

Examples include $\mathbb{P}^1 \setminus \{0, 1, \infty\}$ and an elliptic curve with the origin removed.

The condition is not merely topological decoration. Hyperbolic curves have nonabelian algebraic fundamental groups, and these groups are the kind of objects anabelian geometry can hope to use for reconstruction.

Let C be a smooth proper geometrically connected curve over a number field K, and let D be a reduced effective divisor on C such that $C \setminus D$ is hyperbolic. Vojta-type inequalities compare a height attached to $\omega_C(D)$ with terms measuring ramification and contact with D:

$$h_{\omega_C(D)}(x) \leq c + (1+\epsilon)\big(\log\text{-diff}_C(x) + \log\text{-cond}_D(x)\big).$$

Here log-diff measures the arithmetic complexity of the field of definition of x, while log-cond measures intersection with the divisor D.

Remark 147.5.2 (Why Belyi maps reappear) — Fesenko notes that Belyi maps reduce the Vojta conjecture for such (C, D) to the special case

$$C = \mathbb{P}^1, \qquad D = [0] + [1] + [\infty].$$

Thus the same three-punctured projective line that appears in Belyi's theorem also acts as a universal testing ground for the Diophantine inequality.

§147.6 Arithmetic fundamental groups

For a geometrically integral scheme X over a perfect field K, there is a fundamental exact sequence

$$1 \longrightarrow \pi_1^{\mathrm{geom}}(X) \longrightarrow \pi_1(X) \longrightarrow G_K \longrightarrow 1,$$

where

$$\pi_1^{\mathrm{geom}}(X) = \pi_1(X \times_K K^{\mathrm{alg}}).$$

For schemes over number fields or related fields, $\pi_1(X)$ is often called an arithmetic fundamental group.

Definition 147.6.1 (Anabelian reconstruction, guiding form)**.** Anabelian reconstruction is the idea that, for suitable schemes X, one can recover the arithmetic or geometric object from its algebraic fundamental group together with the projection

$$\pi_1(X) \longrightarrow G_K.$$

The slogan is not that every space is determined by its fundamental group, but that certain highly nonabelian arithmetic schemes are rigid enough for this to happen.

In the IUT entrance, this matters because the later theory needs structures that survive when ordinary ring-theoretic comparison is no longer available. Fesenko summarizes the transition as follows: arithmetic deformation theory is related to anabelian geometry, but uses a different package of tools, including mono-anabelian geometry, nonarchimedean theta-functions, monoid-theoretic categories, and deconstruction/reconstruction of ring structures.

§147.7 Tate Curves and Local Parameters

The first truly arithmetic-deformation object in the survey is not the *abc* equation itself. It is an elliptic curve with split multiplicative reduction and its local q-parameters.

Let E_F be an elliptic curve over a number field F. Suppose that, at a bad nonarchimedean valuation v, the curve has split multiplicative reduction. Let F_v be the completion of F at v. Then locally one has a Tate uniformization

$$E_F(F_v) \cong F_v^\times / \langle q_v \rangle,$$

where $q_v \in F_v^\times$ is the local q-parameter and $\langle q_v \rangle$ is the cyclic subgroup generated by q_v.

Definition 147.7.1 (The idele q_{E_F})**.** Define an idele $q_{E_F} \in \mathbb{A}_F^\times$ by setting its component to be

$$(q_{E_F})_v = \begin{cases} q_v, & v \text{ a bad split multiplicative valuation,} \\ 1, & v \text{ archimedean or good reduction.} \end{cases}$$

after choosing the local q_v at each bad valuation.

The valuation of q_v records the number n_v of components of the corresponding bad fiber. Therefore the degree of this idele is not an artificial invariant: it packages the left-hand side of the Szpiro inequality.

§147.8 Normalised degree as log-volume

Fesenko uses ideles to define a normalised degree. Let k be a number field and $\mathbb{A}_k^\times$ its idele group. For $\alpha \in \mathbb{A}_k^\times$, the non-normalised degree is

$$\deg_k(\alpha) = -\log|\alpha|,$$

where $|\alpha|$ is the module by which multiplication by α scales Haar measure on the adele ring. The normalised degree is

$$\deg(\alpha) = \frac{1}{[k:\mathbb{Q}]} \deg_k(\alpha).$$

This normalisation is stable under finite field extension, which is why the same symbol deg can be used after passing to larger number fields.

Remark 147.8.1 (Why log-volume enters) — Because of the minus sign, $-\deg(\alpha)$ is naturally interpreted as a log-volume. In the later IUT inequality, the left side is written in terms of $-\deg q_E$. This is why the problem of bounding Szpiro-type data becomes a problem about bounding log-volumes.

For the elliptic curve above, Fesenko identifies the goal as giving a suitable upper bound on

$$\deg q_{E_F}.$$

This is the cleanest first target of arithmetic deformation theory in the survey.

§147.9 Why ordinary scheme-theoretic geometry is not enough

The next move is the point where the theory stops looking like ordinary algebraic geometry. Fix a prime $l > 3$, relatively prime to the bad valuations and to the relevant local valuation numbers. For a bad valuation v, Fesenko records a monoid-theoretic transformation on the submonoid of $F_v^\times$ generated by units and q_v:

$$q_v \longmapsto q_v^{m^2}, \qquad u \longmapsto u \quad (u \in \mathcal{O}_{F_v}^\times),$$

where

$$1 \le m \le \frac{l-1}{2}.$$

The element $q_v^{m^2}$ is later viewed as a special value of a nonarchimedean theta-function.

Proposition 147.9.1 (The first deformation warning)

The transformation

$$q_v \mapsto q_v^{m^2}, \qquad u \mapsto u$$

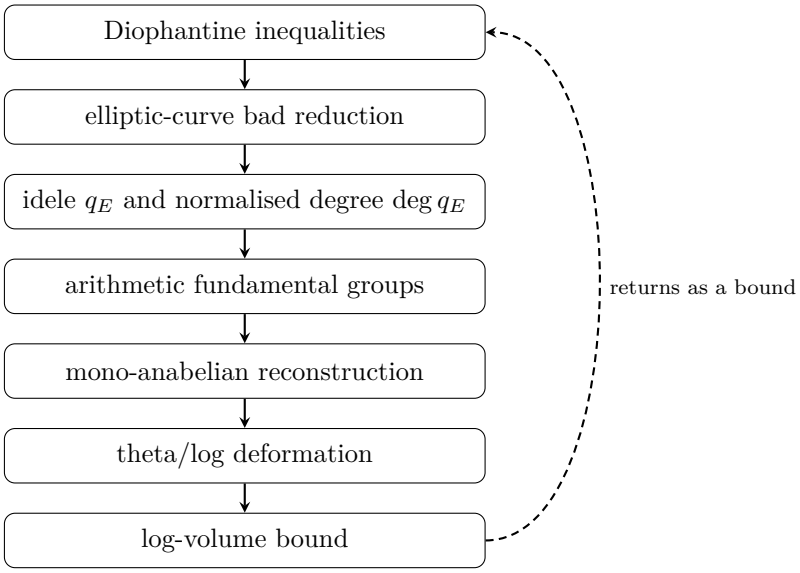
should be read as monoid-theoretic rather than ring-theoretic. It is not a morphism of schemes and does not preserve the full ring structure.

Proof. The point is conceptual rather than computational. A ring morphism must respect both multiplication and addition. The displayed transformation is defined on multiplicative data: the local units and the generator q_v of the Tate period subgroup. It is designed to manipulate multiplicative/theta data, not to preserve addition in F_v. Thus it belongs to the monoid-theoretic side of the theory. □

This is the first reason the word "deformation" is appropriate. The ring structure is not transported intact. It is partly dismantled so that certain multiplicative and theta-theoretic information can be moved, after which one needs reconstruction tools.

§147.10 The route from deformation to inequality

The full theta-link belongs to later notes. But the route can already be drawn.



The diagram is not a proof. It is a navigation map. Starting from Diophantine inequalities, one passes to elliptic curves and their bad reduction. The bad reduction is encoded by q_E. IUT then introduces a deformation machine involving arithmetic fundamental groups, mono-anabelian reconstruction, theta-links, log-links, and log-theta-lattices. The intended output is a log-volume inequality which returns to the original Diophantine problem.

Theorem 147.10.1 (Main IUT inequality, schematic form)

In Fesenko's account, the main theorem of IUT gives a bound on log-volumes of the form

$$-\deg q_E \leq -\deg \Theta_E,$$

where the right-hand side is the log-volume of a union of possible images of theta-data after applying the theta-link, subject to specified indeterminacies.

The statement above is deliberately schematic. It records the mathematical role of the theorem, not the construction of the objects. The important point for this first note is that the theorem does not directly say "abc is true". It gives a bound on a deformation volume. A further computation of the right-hand side is then used to obtain a bound of the desired Diophantine kind.

Remark 147.10.2 (The bridge to later notes) — The words theta-link, log-link, log-theta-lattice, indeterminacy, and multiradiality all belong to the mechanism behind Θ_E. They should not be treated as decorative terminology. They are the devices by which the theory tries to compare arithmetic data after the ordinary ring structure has been deconstructed.

§147.11 What the schematic comparison establishes

The first entrance to IUT should therefore be remembered through four questions.

1. What is being bounded? The initial concrete target is $\deg q_E$, which packages the bad-reduction side of a Szpiro-type inequality.

2. What is rigid? Absolute Galois groups and arithmetic fundamental groups can, in suitable settings, remember number fields or curves.

3. What is allowed to vary? The ordinary ring structure is no longer transported as a single indivisible object; multiplicative and additive aspects are separated.

4. Why is advanced machinery needed? Once ordinary scheme morphisms are not enough, one needs mono-anabelian reconstruction, Kummer-type methods, nonarchimedean theta-functions, and log-volume control.

This is already a mathematical picture, but not yet the full theory. The next note should begin where this one stops: with the moment when the map involving $q_v^{m^2}$ becomes part of the theta-link, and when the lost additive structure has to be handled through log-links and log-shells.

Arithmetic Symmetry: From Prime Ideals to Modular Forms

N-20220123

The equation

$$6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$$

already hints at a problem with applying ordinary prime factorization outside the integers. In a number field the right objects for factorization are ideals; they allow one to examine arithmetic one prime at a time. Much further along that road, elliptic curves provide local numerical data that can be read both from Galois representations and from modular forms. This gives a concrete glimpse of why two such different theories meet.[1]

§148.1 Why elements stop having unique prime coordinates

In $R = \mathbb{Z}[\sqrt{-5}]$, the norm is

$$N(a + b\sqrt{-5}) = a^2 + 5b^2.$$

It is multiplicative and takes positive integer values on nonzero elements. In particular, there is no element of norm 2 or 3. This makes 2, 3, and $1 \pm \sqrt{-5}$ irreducible: any nontrivial factorization of one of these elements would force a factor of norm 2 or 3. The two factorizations of 6 above are genuinely distinct, since the only units in R are ± 1.

The ring R is the full ring of integers of $\mathbb{Q}(\sqrt{-5})$. Although elements do not factor uniquely, nonzero ideals do:

$$\mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{v_{\mathfrak{p}}(\mathfrak{a})}$$

with finitely many nonzero exponents. This is a theorem for rings of integers of number fields, which are Dedekind domains.

The ideal $\mathfrak{p} = (2, 1 + \sqrt{-5})$ is a tangible new arithmetic object. The quotient $R/\mathfrak{p}$ is $\mathbb{F}_2$, so $\mathfrak{p}$ is prime. Yet it is not principal: a generator would have absolute norm 2, and $a^2 + 5b^2 = 2$ has no integer solutions. The ideal class group records exactly this failure of every ideal to have a single generator. It is a finite abelian group for every number field.

§148.2 Local arithmetic records what happens at one prime

For a rational prime p, the p-adic absolute value measures divisibility:

$$|x|_p = p^{-v_p(x)} \qquad (x \neq 0).$$

Completing $\mathbb{Q}$ for the metric $|x - y|_p$ gives the field $\mathbb{Q}_p$. A sequence such as

$$1, \quad 1 + p, \quad 1 + p + p^2, \quad \dots$$

[1] The arithmetic context follows Ireland and Rosen, *A Classical Introduction to Modern Number Theory*, and Silverman, *The Arithmetic of Elliptic Curves*. The numerical modular-form data are recorded in the LMFDB level-32 isogeny class.

converges p-adically because the successive differences become divisible by increasingly high powers of p. Its limit is $1/(1-p)$ in $\mathbb{Q}_p$, even though the partial sums grow in the usual real metric.

For a number field K, completions K_v at its places include complex or real fields as well as finite extensions of $\mathbb{Q}_p$. Passing to K_v lets one study arithmetic near a chosen prime or infinite place. The local data do not automatically solve every global problem, but they are the natural inputs for Euler products and adelic descriptions.

Example 148.2.1 (Building a 7-adic square root, one digit at a time)
Although $\sqrt{2}$ is irrational over $\mathbb{Q}$, the equation $f(x) = x^2 - 2 = 0$ has a solution in $\mathbb{Q}_7$. The first digit is $x_1 = 3$, since $f(3) = 7 \equiv 0 \pmod 7$. To keep the first digit fixed while determining the next one, write $x_2 = 3 + 7t$. Then

$$f(3+7t) = 7 + 42t + 49t^2 \equiv 7(1+6t) \pmod{49}.$$

The correction must satisfy $1 + 6t \equiv 0 \pmod 7$, so $t \equiv 1 \pmod 7$. Hence

$$x_2 = 10, \qquad x_2^2 - 2 = 98 = 2 \cdot 49.$$

For a third digit let $x_3 = 10 + 49t$. Dividing $f(x_3)$ by 49 and reducing modulo 7 now gives

$$2 + 20t \equiv 2 + 6t \equiv 0 \pmod 7, \quad t \equiv 2 \pmod 7.$$

Thus

$$\boxed{3 \pmod 7, \quad 10 \pmod{49}, \quad 108 \pmod{343}}, \qquad 108^2 - 2 = 34 \cdot 343.$$

The general step is equally concrete. Given $f(x_k) \equiv 0 \pmod{p^k}$, put $x_{k+1} = x_k + p^k t$ and solve

$$\frac{f(x_k)}{p^k} + t f'(x_k) \equiv 0 \pmod p.$$

If $f'(x_k)$ is a unit modulo p, exactly one correction $t \bmod p$ works. The successive residue classes define a Cauchy sequence in $\mathbb{Z}_p$ and converge to a root: this is the simple-root case of Hensel's lemma.
There is no contradiction with irrationality over $\mathbb{Q}$: solvability in one completion is local information, not a rational solution of the original equation.

§148.3 An elliptic curve gives a computable local fingerprint

Consider the elliptic curve over $\mathbb{Q}$

$$E: \quad y^2 = x^3 - x.$$

Its discriminant is 64, so the reduction is smooth at every odd prime. For such a prime p, define

$$a_p = p + 1 - \#E(\mathbb{F}_p),$$

where the point at infinity is included in the count.

At $p = 5$, the values of $x^3 - x$ for $x = 0, 1, 2, 3, 4$ are

$$0, \quad 0, \quad 1, \quad 4, \quad 0 \pmod 5.$$

The nonzero squares $1, 4$ each admit two square roots in $\mathbb{F}_5$, while 0 admits one. Thus there are $1 + 1 + 2 + 2 + 1 = 7$ affine points and one point at infinity:

$$\#E(\mathbb{F}_5) = 8, \qquad \boxed{a_5 = 5 + 1 - 8 = -2.}$$

Hasse's theorem ensures $|a_p| \le 2\sqrt{p}$ for every good prime. The point count is therefore a small integer deviation from the rough expectation $p + 1$.

§148.4 Finite Fields and Characteristic Polynomials

For a good prime p, the finite group $E(\mathbb{F}_p)$ is controlled by an arithmetic symmetry: the Frobenius map

$$(x, y) \longmapsto (x^p, y^p)$$

on the points over an algebraic closure of $\mathbb{F}_p$. Its fixed points are exactly the points defined over $\mathbb{F}_p$. This is why counting solutions of a cubic equation can reveal the trace of a linear operator.

To make the operator literal, choose a second prime $\ell \neq p$. The ℓ-adic Tate module $T_\ell(E)$, built by taking compatible ℓ^n-torsion points, is free of rank two over $\mathbb{Z}_\ell$. At good reduction the Frobenius action has characteristic polynomial

$$\boxed{P_p(T) = T^2 - a_p T + p}, \qquad a_p = p + 1 - \#E(\mathbb{F}_p).$$

This theorem includes two nontrivial assertions: the trace a_p is integral and independent of the auxiliary ℓ, and the determinant is p. Our count at $p = 5$ gives

$$P_5(T) = T^2 + 2T + 5.$$

Let its complex roots be α, β. Then

$$\alpha + \beta = -2, \qquad \alpha\beta = 5.$$

Over $\mathbb{F}_{p^r}$, the relevant Frobenius is the r-fold iterate, with eigenvalues α^r, β^r. Hence

$$\#E(\mathbb{F}_{p^r}) = p^r + 1 - (\alpha^r + \beta^r).$$

The point counts over *all* finite extensions are thereby encoded by one quadratic polynomial.

For example, with $p = 5$,

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = (-2)^2 - 10 = -6,$$

so

$$\boxed{\#E(\mathbb{F}_{25}) = 25 + 1 - (-6) = 32.}$$

The extension-field count required no enumeration of the 25^2 possible affine pairs. This also gives a useful sanity check on terminology: $\alpha^2 + \beta^2 = -6$ is the *Frobenius trace over* $\mathbb{F}_{25}$; it is distinct from the coefficient a_{25} of the modular form below.

Hasse's bound $|a_p| \le 2\sqrt{p}$ says the Frobenius roots have complex absolute value $\sqrt{p}$. Thus even the approximate count $p + 1 + O(\sqrt{p})$ reflects a spectral statement.

§148.5 A modular form with the same Euler factors

The Hasse–Weil L-function packages the point-counting data into local factors. At good primes,

$$L_p(E,s) = \frac{1}{1 - a_p p^{-s} + p^{1-2s}}.$$

Writing $X = p^{-s}$, the denominator is $1 - a_p X + pX^2$. Its reciprocal gives a recurrence:

$$\frac{1}{1 - a_p X + pX^2} = 1 + a_p X + (a_p^2 - p)X^2 + \cdots.$$

For our curve at $p = 5$, this predicts the coefficients

$$a_5 = -2, \qquad a_{25} = a_5^2 - 5 = -1.$$

Compare the latter with the trace -6 over $\mathbb{F}_{25}$. The two quantities are related by $a_{25} = (\alpha^2 + \beta^2) + p$, but they answer different questions: one is a Dirichlet-series coefficient, the other a trace of Frobenius squared.

The curve $E : y^2 = x^3 - x$ has conductor 32. The modularity theorem associates it with a normalized weight-two newform for $\Gamma_0(32)$. In this particular case the newform has the remarkably concrete product expression

$$\boxed{f(z) = \eta(4z)^2 \eta(8z)^2 = q \prod_{n=1}^{\infty} (1 - q^{4n})^2 (1 - q^{8n})^2}, \qquad q = e^{2\pi i z},$$

where $\eta(z) = q^{1/24} \prod_{n \geq 1} (1 - q^n)$ is the Dedekind eta function. Expanding the first few factors yields

$$f(z) = q - 2q^5 - 3q^9 + 6q^{13} + 2q^{17} - q^{25} + \cdots.$$

The coefficients $a_5 = -2$ and $a_{25} = -1$ agree exactly with the predictions from the elliptic curve.

What makes this an arithmetic theorem rather than a numerical coincidence is the equality of L-functions:

$$\boxed{L(E,s) = L(f,s).}$$

Equivalently, the Hecke eigenvalue $a_p(f)$ of the modular form matches $p + 1 - \#E(\mathbb{F}_p)$ at each good prime. The existence of a modular form with this property for every elliptic curve over $\mathbb{Q}$ is the deep content of modularity; the eta product allows us to inspect one instance term by term.

§148.6 Complex Multiplication and Symmetry

The curve has a particularly useful extra symmetry over $\mathbb{Q}(i)$:

$$(x, y) \longmapsto (-x, iy).$$

It respects the curve equation because

$$(iy)^2 = -y^2 = (-x)^3 - (-x).$$

Together with the usual integer multiplication maps, this gives complex multiplication by $\mathbb{Z}[i]$. At primes $p \equiv 3 \pmod 4$, a shadow of this symmetry is already visible in elementary arithmetic.

Let χ be the quadratic character on $\mathbb{F}_p$, with $\chi(0) = 0$. For each $x \in \mathbb{F}_p$, the equation $y^2 = x^3 - x$ has $1 + \chi(x^3 - x)$ solutions in y. Including the point at infinity, we get

$$\#E(\mathbb{F}_p) = p + 1 + \sum_{x \in \mathbb{F}_p} \chi(x^3 - x), \qquad a_p = -\sum_{x \in \mathbb{F}_p} \chi(x^3 - x).$$

Now $x^3 - x$ is odd in x. If $p \equiv 3 \pmod 4$, then $\chi(-1) = -1$, so the terms paired at x and $-x$ cancel:

$$\chi((-x)^3 - (-x)) = \chi(-1)\chi(x^3 - x) = -\chi(x^3 - x).$$

Consequently

$$\boxed{p \equiv 3 \pmod 4 \quad \Longrightarrow \quad a_p = 0.}$$

This predicts $a_3 = a_7 = a_{11} = 0$ without counting individual points. The modular form displays the same pattern: its eta product has only exponents $n \equiv 1 \pmod 4$, so its Fourier coefficients at indices $3, 7, 11, \ldots$ vanish.

There is a natural progression of ideas here. The factorization of ideals in a number field retains arithmetic data that elements can lose. Finite-field point counts extract local invariants of an elliptic curve. The Frobenius polynomial organizes these invariants over extension fields; the Euler factors collect them prime by prime. Modularity then places the same coefficients inside a holomorphic form, while complex multiplication explains a conspicuous pattern of vanishing coefficients on both sides. The general Langlands philosophy extends such matching of Galois and automorphic data, but the curve above already exhibits an explicit and verifiable instance.

Riemann Surfaces as One-Dimensional Complex Manifolds

N-20220212

§149.1 Structure beyond topology

The sphere and the torus are both smooth real surfaces, yet their meromorphic maps obey different global constraints. To see the source of the difference, begin with what local coordinates record. A topology determines which subsets are open; a smooth atlas makes differentiation possible; a complex atlas requires that every change of coordinates be holomorphic.[1]

Example 149.1.1

The interval $(0,2) \subset \mathbb{R}$ is a topological 1-manifold. One chart may be the identity map

$$\varphi_1(x) = x.$$

The alternative coordinate $\psi(x) = 1 + \sqrt[3]{x-1}$, using the real cube root, is also a homeomorphism from $(0,2)$ onto itself. Yet its derivative is unbounded at $x = 1$. The transition map between these two charts fails to be C^1, so the two charts are not compatible in a smooth atlas. Topology alone cannot decide smooth differentiability.

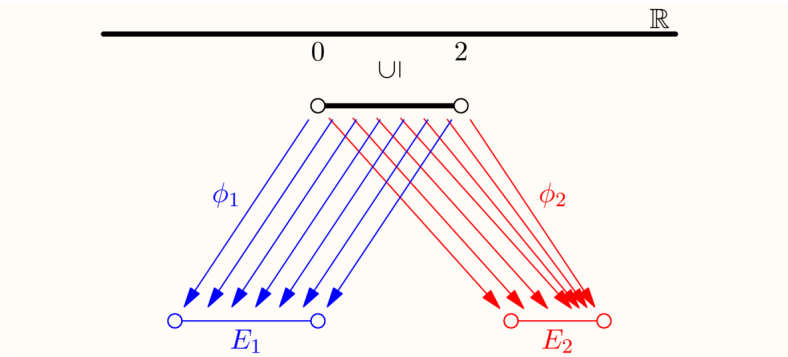


Figure 149.1: Two coordinate descriptions of the interval used to separate topology from smooth structure.

[1]The local normal form and branched-cover constructions follow O. Forster, *Lectures on Riemann Surfaces*. The topological proof of Riemann–Hurwitz is developed in E. Looijenga's Riemann-surface lectures.

Remark 149.1.2 — The lesson is that the same underlying topological space may admit different additional structures. Smoothness is not merely a property of the set; it is encoded by charts and transition maps.

§149.2 From smooth structure to complex structure

A smooth surface embedded in $\mathbb{R}^3$ can be described locally by diffeomorphisms to open subsets of $\mathbb{R}^2$. But this embedded definition is not sufficiently general: for example, the Klein bottle cannot be embedded in $\mathbb{R}^3$.

The abstract definition of a smooth manifold therefore uses charts

$$\varphi_i : U_i \to E_i \subseteq \mathbb{R}^n$$

whose transition maps are smooth. The same philosophy gives complex manifolds by replacing $\mathbb{R}^n$ with $\mathbb{C}^n$ and smooth transition maps with holomorphic transition maps.

Remark 149.2.1 — Because holomorphic functions are automatically analytic, a complex structure is very rigid. This is why a Riemann surface is much more than a real two-dimensional smooth manifold.

§149.3 Definition of a Riemann surface

Definition 149.3.1. A Riemann surface is a connected, Hausdorff, second-countable topological space X equipped with an atlas $\{(U_i, \varphi_i)\}$, where

$$\varphi_i : U_i \xrightarrow{\sim} E_i \subseteq \mathbb{C}$$

is a homeomorphism onto an open subset of $\mathbb{C}$, and all transition maps

$$\varphi_j \circ \varphi_i^{-1}$$

are holomorphic on their domains.

Each φ_i is a complex chart. A local coordinate near $p \in X$ is the complex number

$$z = \varphi_i(x).$$

If $\varphi_i(p) = 0$, the coordinate is centered at p.

Definition 149.3.2. More generally, a complex n-manifold is a Hausdorff space locally modeled on open subsets of $\mathbb{C}^n$, with holomorphic transition maps.

Every complex n-manifold is naturally a smooth real $2n$-manifold.

§149.4 Examples

Example 149.4.1 (Open subsets of the plane)
Every connected open subset $U \subseteq \mathbb{C}$ is a Riemann surface, with the identity chart.

Example 149.4.2 (The Riemann sphere)

The Riemann sphere $\mathbb{C}_\infty$ is the complex plane together with a point at infinity. One way to construct its complex structure is to cover it by two charts: one chart with coordinate z, and another chart near infinity with coordinate

$$t = 1/z.$$

On the overlap, the transition function is

$$z = 1/t,$$

which is holomorphic away from 0. Thus the two charts glue into a Riemann surface.

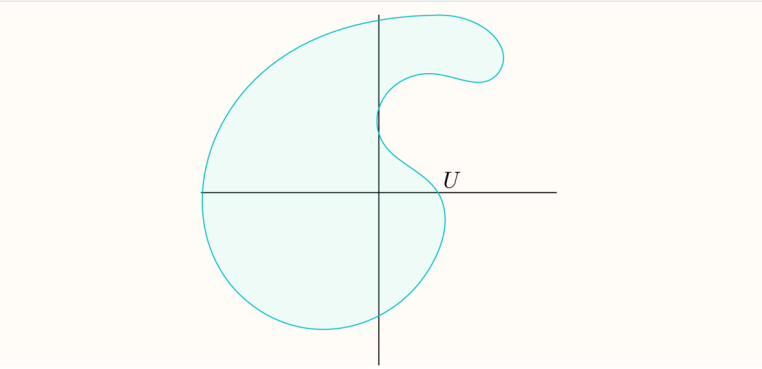


Figure 149.2: Stereographic charts used to define the complex structure on the Riemann sphere.

Caution

The sign choice in the stereographic chart matters. If the transition map became $z = 1/\bar{t}$, it would be antiholomorphic rather than holomorphic. The complex structure depends on getting this orientation right.

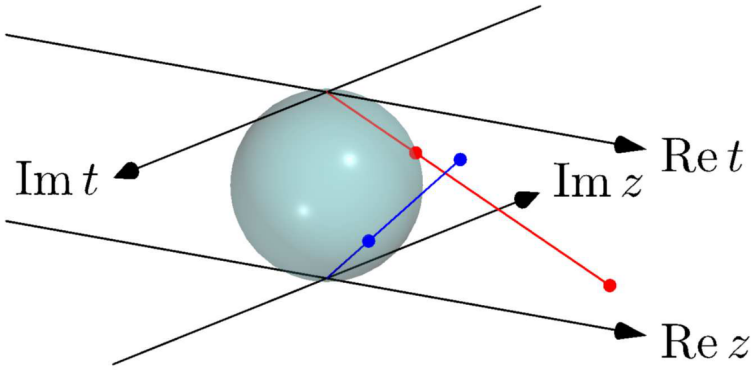


Figure 149.3: The second stereographic chart and the transition coordinate near infinity.

Example 149.4.3 (A complex torus)

Let $L = \mathbb{Z}[i] \subset \mathbb{C}$, a lattice under addition. The quotient

$$\mathbb{C}/L$$

is a complex torus. Locally, the quotient map from $\mathbb{C}$ gives complex coordinates, and different choices of lifts differ by translations

$$z \mapsto z + a, \qquad a \in L,$$

which are holomorphic.

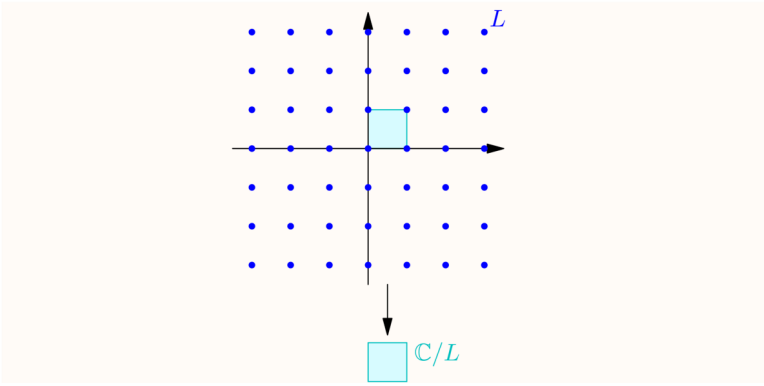


Figure 149.4: The quotient picture of a complex torus, with opposite sides identified.

Remark 149.4.4 — The complex torus is compact. Any holomorphic function on it is constant, but meromorphic functions on it are much more interesting.

Example 149.4.5

A disjoint union of two Riemann spheres is not a Riemann surface under the convention used here, because a Riemann surface is required to be connected. If one drops connectedness, each connected component is a Riemann surface.

§149.5 Holomorphic maps between Riemann surfaces

Definition 149.5.1. Let X, Y be Riemann surfaces. A map $f : X \to Y$ is holomorphic at $p \in X$ if, for charts

$$\varphi_1 : U_1 \to E_1, \qquad \varphi_2 : U_2 \to E_2$$

with $p \in U_1$ and $f(p) \in U_2$, the local expression

$$\varphi_2 \circ f \circ \varphi_1^{-1}$$

is holomorphic near $\varphi_1(p)$. The map is holomorphic if this holds at every point.

Example 149.5.2 (The cubic map)

The map

$$f : \mathbb{C} \to \mathbb{C}, \qquad f(z) = z^3$$

is holomorphic. In the usual global coordinate on $\mathbb{C}$, this is just a polynomial, hence holomorphic everywhere. At a point $a \neq 0$, the derivative

$$f'(a) = 3a^2$$

is nonzero, so the inverse function theorem gives a local holomorphic inverse near a. At 0, however, the derivative vanishes. The local model is exactly

$$z \mapsto z^3,$$

so the map has multiplicity 3 at the origin. This is the first concrete place where "holomorphic map" and "branched covering" begin to touch.

Example 149.5.3 (Inclusion into the Riemann sphere)

The inclusion

$$\mathbb{C} \hookrightarrow \mathbb{C}_\infty$$

is a holomorphic map of Riemann surfaces. On the finite chart of $\mathbb{C}_\infty$, this map is simply

$$z \mapsto z,$$

so it is holomorphic in the most literal possible sense. The point of the example is not the formula itself, but the target: the Riemann sphere adds one extra point ∞, and the ordinary plane sits inside it as the finite chart. Thus a map into $\mathbb{C}_\infty$ may be checked by using the coordinate z away from infinity and the coordinate $w = 1/z$ near infinity.

§149.6 Meromorphic functions as maps to the sphere

The Riemann sphere lets us treat poles as honest values. A meromorphic function

$$f : X \to \mathbb{C}$$

can be regarded as a holomorphic map

$$g : X \to \mathbb{C}_\infty$$

by sending poles to ∞.

Proposition 149.6.1

Nonconstant meromorphic functions $X \to \mathbb{C}$ correspond to holomorphic maps $X \to \mathbb{C}_\infty$ not identically equal to ∞.

Proof. Near a pole p, take a local coordinate z with $z(p) = 0$. A meromorphic function has the form $f(z) = z^{-m}h(z)$, where $m \geq 1$ and h is holomorphic with $h(0) \neq 0$. In the

target coordinate $t = 1/w$ near infinity, the composite is

$$t \circ f(z) = \frac{z^m}{h(z)},$$

which is holomorphic and sends p to $t = 0$. Thus f extends as a holomorphic map into the sphere. Conversely, for a nonconstant holomorphic map to the sphere, use $t = 1/w$ near a preimage of infinity. The function $t \circ f$ is holomorphic with an isolated zero of finite order m, so f has a pole of order m. At all other points it is an ordinary holomorphic function. □

§149.7 Degree and multiplicity

For nonconstant holomorphic maps between compact Riemann surfaces, fibres behave surprisingly well when counted with multiplicity.

Proposition 149.7.1

Let X, Y be compact Riemann surfaces and let $f : X \to Y$ be nonconstant holomorphic. For each $y \in Y$, count the points in $f^{-1}(y)$ with multiplicity. The resulting number is finite and independent of y. This number is the degree of f, denoted $\deg(f)$.

Remark 149.7.2 — This is much stronger than what one expects from arbitrary smooth maps. The rigidity of holomorphic maps makes the fibre count locally constant, and connectedness makes it globally constant.

Example 149.7.3 (The power map has degree k)

The map

$$z \mapsto z^k$$

extends to a holomorphic map $\mathbb{C}_\infty \to \mathbb{C}_\infty$ of degree k. On the finite chart this is a polynomial. Near infinity, use the coordinate $w = 1/z$. Then

$$z^k = \frac{1}{w^k}, \qquad \frac{1}{z^k} = w^k,$$

so the extended map is also holomorphic at ∞. For a generic value $a \in \mathbb{C}^*$, the equation

$$z^k = a$$

has k distinct solutions. Counting multiplicity, the same number remains true over the exceptional values 0 and ∞. This is why the degree is k, not merely because the formula contains the exponent k.

Definition 149.7.4. Let $f : X \to Y$ be a nonconstant holomorphic map and $p \in X$. The multiplicity of f at p is the unique integer $m \geq 1$ such that, after choosing suitable local coordinates centered at p and $f(p)$, the map is locally of the form

$$z \mapsto z^m.$$

Remark 149.7.5 — The guiding slogan is: every nonconstant holomorphic map locally looks like a power map. This is one of the reasons Riemann surfaces are so much cleaner than arbitrary smooth surfaces.

Idea of the local normal form. Choose local coordinates centered at p and $f(p)$, so the local expression becomes a nonconstant holomorphic function F with $F(0) = 0$. Its Taylor expansion has the form

$$F(z) = a_m z^m + a_{m+1} z^{m+1} + \cdots, \qquad a_m \neq 0.$$

Thus

$$F(z) = z^m u(z),$$

where u is holomorphic and $u(0) \neq 0$. After shrinking the coordinate neighborhood, u has a holomorphic m-th root v. Replacing the source coordinate z by $zv(z)$, the map becomes exactly

$$z \mapsto z^m.$$

The integer m is uniquely determined by the order of vanishing of F at 0, hence does not depend on the chosen coordinates. □

§149.8 Local power form is not global

The local normal form $z \mapsto z^m$ must be read with the word *local* emphasized. A useful test case is

$$U = \mathbb{C}^*, \qquad F : U \to \mathbb{C}^*, \qquad F(z) = z^2.$$

At $p = 1$, the derivative is nonzero, so in the local-normal-form sense the multiplicity is $m = 1$. Nearby, one can choose a local inverse

$$z = \sqrt{w}$$

with $w = F(z)$. But this square-root coordinate cannot be chosen globally on $\mathbb{C}^*$. If the target point travels once around the loop

$$w(t) = e^{2\pi i t}, \qquad 0 \leq t \leq 1,$$

then analytic continuation of the local branch starting at $\sqrt{1} = 1$ ends at -1, not 1. This sign change is monodromy.

There is a small trap here. If one instead moves the source point as

$$p(t) = e^{2\pi i t},$$

then $F(p(t)) = e^{4\pi i t}$ winds twice around the target origin, so the square root changes sign twice and returns to the original branch. The monodromy is seen by following a local inverse over a loop in the target, not by confusing the covering coordinate with a single global coordinate downstairs.

Thus the map $z \mapsto z^2$ is an unbranched two-fold covering of $\mathbb{C}^*$, while the missing point 0 is where the corresponding map on $\mathbb{C}$ would branch. To make $\sqrt{w}$ single-valued one must either cut the base, for example by removing a ray, or pass to a covering space. This is the global meaning of the failure: local coordinate charts glue with a monodromy representation, and on a non-simply-connected region those gluings need not collapse to one global coordinate.

§149.9 Why a compact surface forces meromorphic functions

The earlier examples distinguish the two basic compact Riemann surfaces: the sphere $\mathbb{P}^1(\mathbb{C})$, assembled from the z- and $1/z$-charts, and a complex torus $\mathbb{C}/\Lambda$, assembled from translations by a lattice. On either compact surface a global holomorphic function $f : X \to \mathbb{C}$ must be constant. Indeed, $|f|$ attains its maximum, and the maximum-modulus principle forces f to be constant on a connected surface.

Meromorphic functions avoid this obstruction by allowing isolated poles. A nonconstant meromorphic f determines a holomorphic map

$$f : X \longrightarrow \mathbb{P}^1(\mathbb{C}),$$

where each pole maps to ∞. Thus the poles are not merely defects of a function: they are precisely the points where the image enters the second coordinate chart on the sphere. For compact X, the resulting map has a finite degree d, and the local power map $z \mapsto z^{e_p}$ records its ramification index at p.

We can now ask a concrete topological question. If an analytic map has degree d and branches at some points, how can the number of handles of its source and target differ? The answer follows from counting cells over a covering.

§149.10 Riemann–Hurwitz by lifting a triangulation

Let $f : X \to Y$ be a nonconstant holomorphic map between connected compact Riemann surfaces, of degree d. Put

$$R(f) = \sum_{p \in X} (e_p - 1).$$

Only finitely many terms are nonzero: a nonconstant holomorphic map is locally a power map, and its critical points are isolated; compactness makes the total critical set finite.

Choose a triangulation of Y with every branch value among its vertices. Over an open face or an open edge, the map is an ordinary d-sheeted covering. Each open face and edge therefore has exactly d lifts. Over a vertex y, the number of distinct preimages is

$$\#f^{-1}(y) = d - \sum_{p \in f^{-1}(y)} (e_p - 1),$$

because the multiplicities of the points in the fiber sum to d.

Count Euler characteristics in the lifted triangulation:

$$\begin{aligned} \chi(X) &= V_X - E_X + F_X \\ &= dV_Y - R(f) - dE_Y + dF_Y \\ &= d\chi(Y) - R(f). \end{aligned}$$

Since an orientable compact surface of genus g has $\chi = 2 - 2g$, this yields

$$\boxed{2g_X - 2 = d(2g_Y - 2) + R(f).}$$

This is the Riemann–Hurwitz formula. The correction $R(f)$ counts how many distinct preimages of vertices are lost when several sheets merge at a ramification point. The proof explains why the local analytic multiplicities have a global topological cost.

§149.11 The power map checks every term

Take

$$f : \mathbb{P}^1 \to \mathbb{P}^1, \qquad z \longmapsto z^n \quad (n \geq 2).$$

A generic nonzero finite value has n distinct preimages, so $d = n$. At $z = 0$, the map has ramification index n. At infinity, use $w = 1/z$ in the source and $v = 1/f(z)$ in the target. Then $v = w^n$, so the index at infinity is also n. All other points have $e_p = 1$.

Both surfaces have genus zero, and the formula reads

$$-2 = n(-2) + (n-1) + (n-1) = -2.$$

The ramification at infinity is essential. Forgetting it would produce an apparent contradiction, precisely because an affine calculation sees only one of the two exceptional values.

§149.12 A square root of a polynomial produces a torus

Consider the compactification of the smooth affine curve

$$X : \quad y^2 = (x - a_1)(x - a_2)(x - a_3)(x - a_4),$$

where the complex numbers $a_1, \dots, a_4$ are distinct. The projection

$$\pi : X \longrightarrow \mathbb{P}^1, \qquad (x, y) \longmapsto x$$

has degree two: generically there are two possible square roots.

At $x = a_i$, write $x = a_i + t^2$ in a local parameter t. Since the other three factors are nonzero there, the equation determines y as t times a nonvanishing holomorphic function. The projection is locally $t \mapsto t^2$, so each of the four points above a_i has $e = 2$.

We must still inspect infinity. Put

$$u = \frac{1}{x}, \qquad v = \frac{y}{x^2}.$$

Then the curve equation becomes

$$v^2 = \prod_{i=1}^{4} (1 - a_i u).$$

At $u = 0$, the equation has two distinct solutions $v = \pm 1$. They are regular points, and u is a local coordinate near either of them; hence the double cover is unramified over infinity.

The total ramification is therefore $R(\pi) = 4$, and Riemann–Hurwitz gives

$$2g_X - 2 = 2(-2) + 4 = 0, \qquad \boxed{g_X = 1.}$$

The smooth compactification is topologically a torus. More generally, if the defining polynomial has $2m$ distinct roots, the same computation gives $R = 2m$ and

$$2g_X - 2 = -4 + 2m, \qquad g_X = m - 1.$$

An algebraic square-root equation has acquired a precise topological genus.

§149.13 Why a Torus Differs from a Sphere

The two principal examples now behave differently even before one studies their geometry explicitly. A degree-one nonconstant holomorphic map between compact Riemann surfaces is a biholomorphism: the sum of multiplicities in each fiber is one, so every fiber has one point with local index one, and the map is a bijective local holomorphic isomorphism. In particular, a genus-one surface cannot map biholomorphically to the sphere.

The genus-one double cover above must have degree at least two; its four simple branch points make that possible. Conversely, a degree-two meromorphic function on a torus records the torus as two copies of the sphere joined along branch cuts. The analytic data of sheets and critical points and the topological data of handles are linked by a single cell-counting identity.

This is the utility of treating complex structure as something encoded by coordinate changes. Locally a nonconstant holomorphic map is only a power map. Globally those local powers may connect sheets in a way that produces a new surface, and the amount of branching measures the resulting change in Euler characteristic.

Elliptic Curves from Smooth Cubics to Finite Groups

N-20221120

§150.1 Completing the Elliptic Curve

Over a field k of characteristic different from 2 and 3, a short Weierstrass equation has the form

$$E: \quad y^2 = x^3 + ax + b.$$

Two extra requirements turn this equation into an elliptic curve. First, the cubic must be nonsingular. Second, one chooses a distinguished point that will serve as the identity of a group law.

The affine equation does not visibly contain such an identity point. Homogenizing gives the projective cubic

$$Y^2Z = X^3 + aXZ^2 + bZ^3.$$

On the line at infinity $Z = 0$, the equation forces $X = 0$, leaving the single point

$$\mathcal{O} = [0 : 1 : 0].$$

This point is not attached by hand after the geometry is finished. It is already present in the projective closure, and every vertical affine line meets the cubic there.

§150.2 The Discriminant and Smoothness

Let

$$f(x) = x^3 + ax + b.$$

An affine point (x, y) on the curve is singular when both partial derivatives of

$$F(x, y) = y^2 - f(x)$$

vanish:

$$2y = 0, \qquad 3x^2 + a = 0.$$

Thus a singular point must have $y = 0$, while x is a common root of f and f'. Equivalently, the cubic polynomial has a repeated root.

For a depressed cubic $x^3 + ax + b$, the resultant of f and f' is, up to sign,

$$4a^3 + 27b^2.$$

Hence the curve is nonsingular precisely when

$$\boxed{4a^3 + 27b^2 \neq 0.}$$

The conventional discriminant is

$$\Delta = -16(4a^3 + 27b^2).$$

At the point at infinity, the projective partial derivative with respect to Z is nonzero, so $\mathcal{O}$ is smooth as well.

The condition is geometric, not decorative. At a node or cusp there is no single tangent direction with the behavior required by the chord-and-tangent construction. Smoothness is what allows the cubic to carry a regular group law.

§150.3 The Third Point and Vieta's Formulas

Take two affine points

$$P = (x_1, y_1), \qquad Q = (x_2, y_2)$$

with $x_1 \neq x_2$. The line through them is

$$y = m(x - x_1) + y_1, \qquad m = \frac{y_2 - y_1}{x_2 - x_1}.$$

Substituting into the cubic gives a polynomial equation of degree three in x. Two roots are already known: x_1, x_2. If the third intersection has coordinate x_R, comparison of the quadratic coefficient gives

$$x_1 + x_2 + x_R = m^2.$$

Therefore

$$x_R = m^2 - x_1 - x_2.$$

The third point itself is

$$R = (x_R, m(x_R - x_1) + y_1).$$

The group sum is defined by reflecting this point across the x-axis:

$$P + Q = -R.$$

Thus

$$\boxed{x_{P+Q} = m^2 - x_1 - x_2,}$$

$$\boxed{y_{P+Q} = m(x_1 - x_{P+Q}) - y_1.}$$

The formulas are Vieta's relations applied to the line–cubic intersection, not independent identities to memorize.

§150.4 Tangency and the Point at Infinity

When $P = Q$, the secant slope is replaced by the tangent slope. Implicit differentiation of

$$y^2 = x^3 + ax + b$$

gives

$$2y\frac{dy}{dx} = 3x^2 + a.$$

Hence, when $y_1 \neq 0$,

$$m = \frac{3x_1^2 + a}{2y_1}.$$

The same coordinate formulas then compute $2P$.

The reflection

$$-(x, y) = (x, -y)$$

is the group inverse. The line through P and $-P$ is vertical and meets the projective cubic at $\mathcal{O}$, so

$$P + (-P) = \mathcal{O}.$$

If $y = 0$, the tangent is vertical; consequently

$$2P = \mathcal{O},$$

and P is a point of order two.

Finally,

$$P + \mathcal{O} = P.$$

Projective geometry has removed the exceptional case: a vertical line still has its third intersection, but that intersection lies at infinity.

§150.5 A rational example shows both addition and doubling

Consider

$$E: \quad y^2 = x^3 - x + 1$$

over $\mathbb{Q}$. The points

$$P = (0, 1), \qquad Q = (1, 1)$$

lie on the curve. Their joining line is horizontal, so $m = 0$. Hence

$$x_{P+Q} = 0^2 - 0 - 1 = -1,$$

$$y_{P+Q} = 0(0 - (-1)) - 1 = -1.$$

Therefore

$$P + Q = (-1, -1).$$

A direct substitution verifies

$$(-1)^2 = (-1)^3 - (-1) + 1 = 1.$$

For doubling P,

$$m = \frac{3 \cdot 0^2 - 1}{2 \cdot 1} = -\frac{1}{2}.$$

Thus

$$x_{2P} = \frac{1}{4},$$

$$y_{2P} = -\frac{1}{2}\left(0 - \frac{1}{4}\right) - 1 = -\frac{7}{8}.$$

Indeed,

$$\left(-\frac{7}{8}\right)^2 = \left(\frac{1}{4}\right)^3 - \frac{1}{4} + 1 = \frac{49}{64}.$$

The geometry and the rational arithmetic agree exactly.

§150.6 Divisors explain why the chord rule is associative

The coordinate formulas make closure and inverses visible, but a direct verification of

$$(P + Q) + R = P + (Q + R)$$

would be unpleasant. The reason for associativity is better seen through divisors.

On the projective cubic, let L be a line meeting E at P, Q, R, counted with multiplicity. The line at infinity meets the Weierstrass cubic at $\mathcal{O}$ with multiplicity three. The ratio of the two linear forms therefore defines a rational function on E whose divisor is

$$P + Q + R - 3\mathcal{O}.$$

A principal divisor represents zero in the degree-zero Picard group, so

$$[P-\mathcal{O}]+[Q-\mathcal{O}]+[R-\mathcal{O}]=0 \qquad \text{in } \operatorname{Pic}^0(E).$$

Since the geometric rule defines $P+Q=-R$, it is exactly the rule required for

$$P \longmapsto [P-\mathcal{O}]$$

to respect addition.

For a smooth genus-one curve with chosen point $\mathcal{O}$, this map is an isomorphism

$$E \cong \operatorname{Pic}^0(E).$$

The Picard group is an abelian group because divisor classes add associatively. The chord-and-tangent operation inherits that associativity. Thus divisor theory does not merely restate the picture in advanced language; it supplies the group axiom that the picture alone does not prove.

§150.7 The same formulas work over finite fields

Let $p>3$ be prime and suppose $\Delta \not\equiv 0 \pmod p$. The set

$$E(\mathbb{F}_p)=\{(x,y)\in\mathbb{F}_p^2 : y^2=x^3+ax+b\}\cup\{\mathcal{O}\}$$

forms a finite abelian group. The addition formulas are unchanged, but division means multiplication by a modular inverse.

For example, on

$$E: \quad y^2=x^3-x+1 \qquad \text{over } \mathbb{F}_5,$$

the quadratic residues are $0,1,4$. Checking each x gives

x	0	1	2	3	4
x^3-x+1	1	1	2	0	1
$\#\{y : y^2=x^3-x+1\}$	2	2	0	1	2.

Including $\mathcal{O}$,

$$\#E(\mathbb{F}_5)=2+2+0+1+2+1=8.$$

Hasse's theorem predicts

$$\left|\#E(\mathbb{F}_5)-(5+1)\right|\le 2\sqrt{5},$$

and the count 8 satisfies this bound.

§150.8 One point generates the entire eight-element example

Take

$$P=(0,1)\in E(\mathbb{F}_5).$$

For doubling,

$$m=\frac{-1}{2}\equiv 4\cdot 3\equiv 2 \pmod 5,$$

so

$$2P=(4,1).$$

Doubling again,

$$m = \frac{3 \cdot 4^2 - 1}{2} \equiv \frac{2}{2} \equiv 1 \pmod 5,$$

and therefore

$$4P = (3, 0).$$

Because a point with $y = 0$ has order two,

$$8P = \mathcal{O}.$$

Since $4P \neq \mathcal{O}$, the order of P is exactly eight. The group has eight elements and contains an element of order eight, hence

$$E(\mathbb{F}_5) \cong \mathbb{Z}/8\mathbb{Z}.$$

This small example makes scalar multiplication concrete rather than treating the finite group as a black box.

§150.9 Scalar Multiplication and Asymmetry

For a positive integer n, scalar multiplication is

$$[n]P = P + \cdots + P.$$

It is computed efficiently by binary expansion. To evaluate $[13]P$, for example, one forms

$$P,\ 2P,\ 4P,\ 8P$$

and combines the terms corresponding to

$$13 = 8 + 4 + 1.$$

This double-and-add method requires only logarithmically many group operations.

The inverse problem asks for n given P and $Q = [n]P$. On suitably chosen large finite-field subgroups, no comparably efficient general method is known; this is the elliptic-curve discrete logarithm problem. The cryptographic use therefore rests on a precise asymmetry: projective algebra makes the forward group operation explicit and fast, while recovering the scalar remains difficult.

The path from the cubic equation to that asymmetry is now continuous. The discriminant guarantees a smooth projective curve, the point at infinity supplies an identity, line intersections give the addition formulas, divisor classes prove associativity, and reduction modulo a prime turns the same geometry into a finite abelian group.

From Self-Duality to Spectral Curves: A Computable Route through Hitchin's Equations

N-20230801

§151.1 Why the story begins in four dimensions

Hitchin's equations live on a Riemann surface, but their form is much easier to remember once one sees the four-dimensional equation from which they descend. Let $P \to M$ be a principal bundle over an oriented Riemannian four-manifold, and let $\mathcal{A}$ be a connection. Its curvature is

$$F_{\mathcal{A}} = d\mathcal{A} + \mathcal{A} \wedge \mathcal{A}.$$

The Hodge star acts on two-forms and satisfies $*^2 = 1$, so every curvature two-form splits as

$$F_{\mathcal{A}} = F_{\mathcal{A}}^{+} + F_{\mathcal{A}}^{-}, \qquad *F_{\mathcal{A}}^{\pm} = \pm F_{\mathcal{A}}^{\pm}.$$

An anti-self-dual connection satisfies

$$F_{\mathcal{A}}^{+} = 0, \qquad \text{equivalently} \qquad F_{\mathcal{A}} + *F_{\mathcal{A}} = 0.$$

This first-order equation is stronger than the Yang–Mills equation

$$d_{\mathcal{A}}^{*} F_{\mathcal{A}} = 0.$$

Indeed, the Bianchi identity $d_{\mathcal{A}} F_{\mathcal{A}} = 0$ and anti-self-duality give

$$d_{\mathcal{A}}^{*} F_{\mathcal{A}} = - * d_{\mathcal{A}} * F_{\mathcal{A}} = * d_{\mathcal{A}} F_{\mathcal{A}} = 0.$$

Thus the four-dimensional equation already combines geometry, gauge symmetry, and a variational problem. The two-dimensional Hitchin system appears when two directions are suppressed but their connection components are retained as fields.

§151.2 Dimensional reduction produces three coupled equations

Work locally on

$$M = \Sigma \times \mathbb{R}^2$$

with oriented coordinates (x, y, u, v), and assume that all fields are independent of u, v. Write the four-dimensional connection as

$$\mathcal{A} = A_x \, dx + A_y \, dy + \phi_1 \, du + \phi_2 \, dv.$$

The first two components form a connection

$$A = A_x \, dx + A_y \, dy$$

on Σ; the suppressed components ϕ_1, ϕ_2 become adjoint-valued functions on Σ.

The curvature components are now explicit:

$$F_{xy} = \partial_x A_y - \partial_y A_x + [A_x, A_y],$$

$$F_{xu} = D_x\phi_1, \qquad F_{xv} = D_x\phi_2,$$

$$F_{yu} = D_y\phi_1, \qquad F_{yv} = D_y\phi_2, \qquad F_{uv} = [\phi_1, \phi_2].$$

With the chosen orientation, the anti-self-duality equations are equivalent, up to the harmless global sign convention between self-dual and anti-self-dual, to

$$F_{xy} + [\phi_1, \phi_2] = 0,$$

$$D_x\phi_1 - D_y\phi_2 = 0, \qquad D_x\phi_2 + D_y\phi_1 = 0.$$

The first equation balances curvature against the commutator of the two reduced fields. The last two are a covariant Cauchy–Riemann system.

Introduce the complex coordinate

$$z = x + iy$$

and package the reduced fields into a Higgs field

$$\Phi = \frac{1}{2}(\phi_1 - i\phi_2)\, dz.$$

After choosing a Hermitian metric and writing Φ^{*h} for the adjoint, the three real equations become the two Hitchin equations

$$\boxed{F_A + [\Phi, \Phi^{*h}] = 0, \qquad \bar\partial_A \Phi = 0.}$$

Different normalizations of z, the metric, or the factor $1/2$ move constants between the two terms, but the structure is invariant: a real curvature equation and a complex holomorphicity equation.

§151.3 Gauge Symmetry and Quotients

A unitary gauge transformation g changes the pair by

$$A \longmapsto gAg^{-1} - dg\, g^{-1}, \qquad \Phi \longmapsto g\Phi g^{-1}.$$

Consequently,

$$F_A \longmapsto gF_Ag^{-1}, \qquad [\Phi, \Phi^{*h}] \longmapsto g[\Phi, \Phi^{*h}]g^{-1},$$

and

$$\bar\partial_A\Phi \longmapsto g(\bar\partial_A\Phi)g^{-1}.$$

The equations are therefore gauge invariant. Two solutions related by a gauge transformation describe the same geometric object, so the moduli space is not merely a zero set; it is a quotient

$$\mathcal{M}_H = \{(A, \Phi) : \text{Hitchin equations}\}/\mathcal{G}.$$

The quotient matters. A raw solution space contains whole gauge orbits, while stabilizers can create singular points. This is the first reason that deformation theory and stability cannot be postponed to an ornamental final section.

There is also a symplectic reading. The affine space of pairs (A, Φ) carries a natural hyperkähler structure, and the unitary gauge group acts by hyperkähler isometries. In one complex structure, the real moment map is

$$\mu_{\mathbb{R}}(A, \Phi) = F_A + [\Phi, \Phi^{*h}],$$

while the complex moment map is

$$\mu_{\mathbb{C}}(A, \Phi) = \bar{\partial}_A \Phi.$$

Thus Hitchin's moduli space is an infinite-dimensional hyperkähler quotient. The moment-map language does not replace the PDE; it explains why a metric equation, a holomorphic equation, and a quotient appear together.

§151.4 From Equations to Higgs Bundles

The operator $\bar{\partial}_A$ defines a holomorphic structure on the underlying complex bundle E. The equation

$$\bar{\partial}_A \Phi = 0$$

then says that

$$\Phi \in H^0(\Sigma, \operatorname{End}(E) \otimes K),$$

where K is the canonical bundle. A pair (E, Φ) of this kind is a Higgs bundle.

Not every holomorphic Higgs bundle corresponds to a solution of the real equation. The missing condition is stability. For a bundle E, define its slope by

$$\mu(E) = \frac{\deg E}{\operatorname{rk} E}.$$

A Higgs bundle is stable when every proper nonzero holomorphic subbundle $F \subset E$ satisfying

$$\Phi(F) \subset F \otimes K$$

obeys

$$\mu(F) < \mu(E).$$

Only Φ-invariant subbundles are tested, because a subbundle ignored by Φ is not a subsystem of the Higgs pair.

The Hitchin–Kobayashi correspondence says, in the appropriate degree-zero setting, that a polystable Higgs bundle admits a Hermitian metric solving

$$F_A + [\Phi, \Phi^{*h}] = 0.$$

The analytic quotient by unitary gauge and the algebraic quotient by complex gauge are therefore two descriptions of the same well-behaved moduli problem. Stability is not an extra adjective attached after the equations; it is the algebraic condition that replaces the missing compactness of complex gauge orbits.

§151.5 Rank-Two Reduction to a PDE

The real equation becomes concrete in rank two. Choose a local holomorphic frame in which

$$\Phi = \begin{pmatrix} 0 & Q \\ 1 & 0 \end{pmatrix} dz$$

for a local quadratic-differential coefficient Q, and choose a diagonal Hermitian metric

$$h = \begin{pmatrix} e^{-u} & 0 \\ 0 & e^{u} \end{pmatrix}.$$

The adjoint of the matrix part of Φ is

$$\Phi^{*h} = \begin{pmatrix} 0 & e^{2u} \\ \overline{Q}e^{-2u} & 0 \end{pmatrix} d\bar{z}.$$

Hence

$$[\Phi, \Phi^{*h}] = \begin{pmatrix} |Q|^2 e^{-2u} - e^{2u} & 0 \\ 0 & e^{2u} - |Q|^2 e^{-2u} \end{pmatrix} dz \wedge d\bar{z}.$$

The signs in the curvature calculation deserve care. In a holomorphic frame, the Chern connection has $(1,0)$-part $h^{-1}\partial h$, so

$$h^{-1}\partial h = \begin{pmatrix} -\partial u & 0 \\ 0 & \partial u \end{pmatrix}, \qquad F_h = \bar{\partial}(h^{-1}\partial h).$$

Because $\bar{\partial}\partial u = -\partial\bar{\partial}u$ for a scalar function, the curvature is

$$F_h = \begin{pmatrix} \partial\bar{\partial}u & 0 \\ 0 & -\partial\bar{\partial}u \end{pmatrix} = \begin{pmatrix} u_{z\bar{z}} & 0 \\ 0 & -u_{z\bar{z}} \end{pmatrix} dz \wedge d\bar{z}.$$

The diagonal entries of $F_h + [\Phi, \Phi^{*h}] = 0$ are negatives of one another, and either one now yields

$$\boxed{u_{z\bar{z}} + |Q|^2 e^{-2u} - e^{2u} = 0.}$$

Equivalently, $u_{z\bar{z}} = e^{2u} - |Q|^2 e^{-2u}$. The two exponential terms arise from the two off-diagonal entries of the Higgs field. Their competition produces a sinh–Gordon-type equation when Q has no zeros and the holomorphic coordinate is adjusted appropriately. Different curvature, wedge-order, and area-form conventions may change the overall factors and signs, so the derivation above fixes them explicitly.[1] Away from zeros of Q, a shift of u converts the equation into a sinh–Gordon type equation. The abstract matrix PDE has become a nonlinear scalar equation whose singular behavior is tied to the zeros of the spectral data.

§151.6 A solution also produces a family of flat connections

The Hitchin equations connect Higgs bundles to representations because they imply flatness of a one-parameter family. For $\lambda \in \mathbb{C}^*$, define

$$\nabla_\lambda = D_A + \lambda^{-1}\Phi + \lambda\Phi^{*h}.$$

[1] Compare the local rank-two reduction in R. S. Ward, *Geometry of Solutions of Hitchin Equations on* $\mathbb{R}^2$, Section 2.

Its curvature contains three different powers of λ:

$$F_{\nabla_\lambda} = \lambda^{-1} D_A^{0,1}\Phi + \left(F_A + [\Phi, \Phi^{*h}]\right) + \lambda D_A^{1,0}\Phi^{*h}.$$

The complex equation kills the λ^{-1} term, its adjoint kills the λ term, and the real equation kills the middle term. Thus

$$F_{\nabla_\lambda} = 0$$

for every $\lambda \neq 0$.

A flat connection has monodromy, giving a representation of the fundamental group. This is the operational core of nonabelian Hodge theory:

$$\begin{aligned} \text{harmonic bundle} &\longleftrightarrow \text{polystable Higgs bundle}, \\ \text{flat connection} &\longleftrightarrow \text{representation of } \pi_1(\Sigma). \end{aligned}$$

The arrows are not merely analogies. The harmonic metric is what combines a holomorphic Higgs pair with its adjoint to manufacture the flat family.

§151.7 The Hitchin map records global eigenvalue data

For a rank-n Higgs field, the coefficients of

$$\det(\eta I - \Phi) = \eta^n + a_1\eta^{n-1} + \cdots + a_n$$

are invariant under conjugation. Since Φ is a K-valued endomorphism,

$$a_i \in H^0(\Sigma, K^i).$$

They define the Hitchin map

$$h : \mathcal{M}_H \longrightarrow \bigoplus_{i=1}^{n} H^0(\Sigma, K^i).$$

For trace-free rank two, the characteristic polynomial is

$$\eta^2 - q, \qquad q \in H^0(\Sigma, K^2),$$

so the Hitchin base is simply $H^0(K^2)$.

The equation

$$\eta^2 = q$$

defines a curve $\widetilde{\Sigma}$ inside the total space of K. Here η is the tautological one-form: at a point $(x, \xi) \in K$, its value is the cotangent vector ξ. Over a point where $q(x) \neq 0$, the two square roots of $q(x)$ are the two eigenvalues of $\Phi(x)$. At a simple zero of q, the sheets meet and the projection

$$\pi : \widetilde{\Sigma} \to \Sigma$$

branches. The spectral curve is therefore the global replacement for diagonalizing a matrix point by point.

§151.8 Theta Characteristics in Rank Two

The local companion matrix contains entries of different bundle types, so it should not be placed globally on $\mathcal{O} \oplus \mathcal{O}$. Choose instead a theta characteristic

$$S^2 \cong K$$

and set

$$E = S \oplus S^{-1}.$$

Then the Higgs field

$$\Phi_q = \begin{pmatrix} 0 & q \\ 1 & 0 \end{pmatrix} : E \longrightarrow E \otimes K$$

is globally well defined. Indeed, the lower-left entry belongs to

$$\mathrm{Hom}(S, S^{-1} \otimes K) \cong S^{-2} \otimes K \cong \mathcal{O},$$

while the upper-right entry belongs to

$$\mathrm{Hom}(S^{-1}, S \otimes K) \cong S^2 \otimes K \cong K^2.$$

The determinant is trivial, the trace is zero, and

$$\det(\eta I - \Phi_q) = \eta^2 - q.$$

This is the rank-two Hitchin section. The bundle bookkeeping is not cosmetic: it is what turns a local matrix mnemonic into a global Higgs bundle.

§151.9 A Genus-Two Spectral Curve

Take the genus-two hyperelliptic curve

$$\Sigma : \quad y^2 = (x^2 - 1)(x^2 - 4)(x^2 - 9).$$

A basis of holomorphic one-forms is

$$\omega_0 = \frac{dx}{y}, \qquad \omega_1 = x\frac{dx}{y}.$$

Consider the quadratic differential

$$q = (x^2 - 2)\left(\frac{dx}{y}\right)^2.$$

It is holomorphic. Near either point at infinity, with $t = 1/x$, one has

$$\frac{dx}{y} \sim -t\,dt, \qquad x^2 - 2 \sim t^{-2},$$

so q is asymptotic to a nonzero multiple of dt^2.

The zeros occur at $x = \pm\sqrt{2}$. These are not branch values of the hyperelliptic map, so each value of x gives two points of Σ. Thus q has four simple zeros, matching

$$\deg K^2 = 4g - 4 = 4.$$

The spectral curve is

$$\widetilde{\Sigma}: \quad \eta^2 = q.$$

It is a double cover of Σ branched at those four zeros. Riemann–Hurwitz gives

$$2g(\widetilde{\Sigma}) - 2 = 2\big(2g(\Sigma) - 2\big) + 4 = 2 \cdot 2 + 4 = 8,$$

so

$$\boxed{g(\widetilde{\Sigma}) = 5.}$$

The slogan "the spectral curve is a branched cover" has now produced a concrete genus.

§151.10 Higgs Bundles from Spectral Curves

Let $\mathcal{L}$ be a line bundle on $\widetilde{\Sigma}$. Pushing forward gives a rank-two bundle

$$E = \pi_* \mathcal{L}.$$

Multiplication by the tautological eigenvalue defines

$$\eta : \mathcal{L} \longrightarrow \mathcal{L} \otimes \pi^* K.$$

After applying π_* and the projection formula,

$$\pi_*(\mathcal{L} \otimes \pi^* K) \cong (\pi_* \mathcal{L}) \otimes K,$$

this becomes a Higgs field

$$\Phi : E \longrightarrow E \otimes K.$$

Thus the nonabelian rank-two object is reconstructed from abelian line-bundle data on the eigenvalue cover.

For fixed determinant, the line bundle is constrained by a norm condition. The generic Hitchin fiber is the Prym variety

$$\operatorname{Prym}(\widetilde{\Sigma}/\Sigma) = \ker\Big(\operatorname{Nm} : \operatorname{Jac}(\widetilde{\Sigma}) \to \operatorname{Jac}(\Sigma)\Big)^0.$$

Its dimension is

$$g(\widetilde{\Sigma}) - g(\Sigma) = 5 - 2 = 3.$$

On the other hand,

$$\dim H^0(\Sigma, K^2) = 3g - 3 = 3.$$

The base and generic fiber therefore have equal complex dimension. This is the dimension count behind algebraic complete integrability: the generic Prym fibers are half-dimensional Lagrangian abelian varieties.

§151.11 The deformation complex computes the tangent space

To understand a moduli space locally, vary both the holomorphic structure and the Higgs field. An infinitesimal deformation is a pair

$$(a, \varphi) \in \Omega^{0,1}(\operatorname{End}_0 E) \oplus \Omega^{1,0}(\operatorname{End}_0 E).$$

An infinitesimal complex gauge transformation $s \in \Omega^0(\operatorname{End}_0 E)$ changes the pair by

$$D_0 s = (\bar{\partial}_E s, [\Phi, s]).$$

Linearizing the holomorphicity equation gives

$$D_1(a,\varphi) = \bar\partial_E \varphi + [a,\Phi].$$

Because $\bar\partial_E \Phi = 0$, one has $D_1 D_0 = 0$. The infinitesimal moduli problem is therefore controlled by the complex

$$\Omega^0(\mathrm{End}_0\, E) \xrightarrow{D_0} \Omega^{0,1}(\mathrm{End}_0\, E) \oplus \Omega^{1,0}(\mathrm{End}_0\, E) \xrightarrow{D_1} \Omega^{1,1}(\mathrm{End}_0\, E).$$

Equivalently, in holomorphic language one uses the two-term complex

$$\mathcal{C}^\bullet : \mathrm{End}_0\, E \xrightarrow{[\Phi,\cdot]} \mathrm{End}_0\, E \otimes K.$$

Its hypercohomology has a direct interpretation:

$$\mathbb{H}^0(\mathcal{C}^\bullet) = \text{infinitesimal automorphisms},$$

$$\mathbb{H}^1(\mathcal{C}^\bullet) = \text{tangent space}, \qquad \mathbb{H}^2(\mathcal{C}^\bullet) = \text{obstructions}.$$

For a stable trace-free Higgs bundle, the automorphism space is zero and, by duality, the obstruction space vanishes at a smooth point. Hence the tangent dimension is minus the Euler characteristic of the complex.

Since $\mathrm{End}_0\, E$ has rank 3 and degree 0, Riemann–Roch gives

$$\chi(\mathrm{End}_0\, E) = 3(1-g).$$

Also,

$$\chi(\mathrm{End}_0\, E \otimes K) = 3(2g-2) + 3(1-g) = 3(g-1).$$

Therefore

$$\chi(\mathcal{C}^\bullet) = 3(1-g) - 3(g-1) = -6(g-1),$$

and

$$\boxed{\dim_{\mathbb{C}} T_{(E,\Phi)}\mathcal{M}_H = 6g-6.}$$

For genus two this is 6, exactly the sum of the three-dimensional Hitchin base and the three-dimensional Prym fiber. The local deformation calculation and the global spectral description meet at the same dimension.

The agreement is more than a dimension check. It shows how the different descriptions solve different parts of the same problem. Four-dimensional anti-self-duality explains the origin of the equations. Dimensional reduction identifies the Higgs field as connection data in suppressed directions. Gauge theory supplies the real moment-map equation and the analytic quotient. Stability produces a workable algebraic quotient. The flat family records monodromy and representations. The Hitchin map extracts invariant eigenvalue data, while the spectral curve converts a nonabelian bundle into a line bundle on a branched cover. Finally, the deformation complex determines whether the quotient is smooth and computes its tangent dimension.

None of these viewpoints is an optional advanced appendix. The local scalar equation explains what the PDE asks a metric to do; the genus-two cover makes the spectral construction calculable; and the equality

$$6g-6 = (3g-3) + (3g-3)$$

shows why the Hitchin fibration has exactly the right size to be an integrable system.

Plane Curves: Singularities, Intersection Multiplicities, and Normalization

N-20240608

A polynomial equation in two variables draws a curve, but the equation retains information that may disappear in the picture. A tangent line may touch at two algebraic orders, a crossing point may contain two distinct local branches, and two curves whose affine drawings meet only once can have another intersection at infinity. The coordinate ring, local intersection multiplicity and normalization recover precisely this lost information.[1]

§152.1 Curved Images of an Affine Line

Work over the complex numbers. Consider the parabola

$$C = V(y - x^2) \subset \mathbb{A}^2_{\mathbb{C}}.$$

Its coordinate ring is

$$\mathbb{C}[C] = \mathbb{C}[x, y]/(y - x^2) \cong \mathbb{C}[x].$$

The isomorphism sends the class of y to x^2: every polynomial function on the parabola is uniquely represented by a polynomial in the x-coordinate alone.

Geometrically this comes from the parametrization

$$t \longmapsto (t, t^2)$$

with inverse given by projection onto the first coordinate. Although the parabola bends in the surrounding plane, it is isomorphic, as an affine algebraic variety, to the affine line.

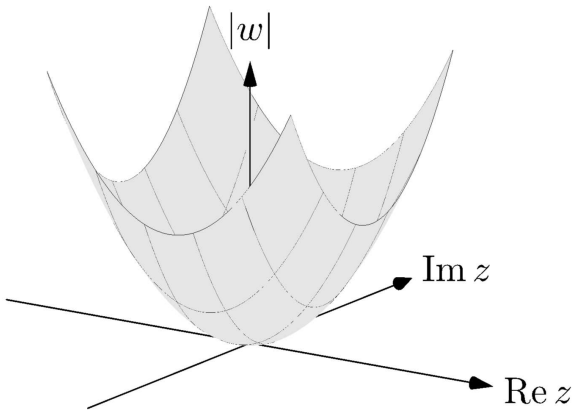


Figure 152.1: The parabola is an affine algebraic copy of the line; its apparent curvature depends on its embedding into the plane.

This distinction matters whenever a geometric claim is meant to survive a change of coordinates: being a parabola in a chosen plane is different from having the intrinsic function theory of $\mathbb{A}^1$.

[1] M. Artin, MIT 18.721, *Introduction to Algebraic Geometry*, Chapters 1 and 4; Stacks Project, §43.16.

§152.2 Tangent Cones and Singularities

For a plane curve $F(x,y) = 0$, a point $p = (a,b)$ is smooth when at least one of $F_x(p), F_y(p)$ is nonzero. The linear part

$$F_x(p)(x-a) + F_y(p)(y-b)$$

then defines its unique tangent line.

At a singular point both first derivatives vanish. Translate the point to the origin and write

$$F(x,y) = F_m(x,y) + F_{m+1}(x,y) + \cdots,$$

where each F_j is homogeneous of degree j and $F_m \neq 0$. The equation

$$F_m(x,y) = 0$$

is the tangent cone; the integer m is the multiplicity of the plane curve at the point.

For the cuspidal cubic

$$C_{\text{cusp}}: \quad y^2 = x^3,$$

the lowest term is y^2. Its tangent cone is the line $y = 0$ counted twice. For the nodal cubic

$$C_{\text{node}}: \quad y^2 = x^2(x+1),$$

the lowest terms are $y^2 - x^2 = (y-x)(y+x)$. Its tangent cone has two distinct lines, $y = x$ and $y = -x$.

Both points have multiplicity two, yet their local branches behave differently. The tangent cone makes the distinction before we attempt to draw either curve.

§152.3 Normalization separates a cusp from a node

The cusp has a polynomial parametrization

$$\nu_{\text{cusp}} : \mathbb{A}^1 \longrightarrow C_{\text{cusp}}, \qquad t \longmapsto (t^2, t^3).$$

On coordinate rings, this identifies

$$A = \mathbb{C}[x,y]/(y^2 - x^3) \cong \mathbb{C}[t^2, t^3] \subset \mathbb{C}[t] = B.$$

The element t belongs to the fraction field of A, since $t = y/x$ away from $x = 0$. It is integral over A, because it satisfies the monic equation

$$T^2 - x = 0.$$

Thus B is a finite integral extension of A. Since $\mathbb{C}[t]$ is integrally closed and has the same fraction field, it is the normalization of A. The map is one-to-one on complex points; the cusp point corresponds to $t = 0$, but the local ring becomes regular after normalization.

The node has a different parametrization:

$$\nu_{\text{node}}(t) = (t^2 - 1,\ t(t^2 - 1)).$$

Indeed,

$$\big(t(t^2-1)\big)^2 = (t^2-1)^2\big((t^2-1)+1\big).$$

Again $t = y/x$ away from $x = 0$, and $t^2 = x + 1$ shows that the normalization is $\mathbb{C}[t]$. This time, however,

$$\nu_{\text{node}}(1) = \nu_{\text{node}}(-1) = (0,0).$$

Two distinct points of the normalized line are identified at the node. Normalization therefore resolves the singularities in different ways: it restores a regular local parameter at the cusp and separates two branches at the node.

§152.4 Intersection Multiplicity

Two algebraic curves can meet at a single point and still contribute more than one to their intersection number. For plane curves without a common component, a local algebraic measure is

$$I_p(F,G) = \dim_{\mathbb{C}} \frac{\mathbb{C}[[x-a, y-b]]}{(F,G)}$$

at an isolated intersection $p = (a,b)$, when this quotient is finite-dimensional.

Take the cusp $F = y^2 - x^3$. Intersect it with the horizontal tangent $G = y$ at the origin:

$$\frac{\mathbb{C}[[x,y]]}{(y^2 - x^3, y)} \cong \frac{\mathbb{C}[[x]]}{(x^3)}.$$

The quotient has basis $1, x, x^2$, so

$$\boxed{I_{(0,0)}(C_{\text{cusp}}, V(y)) = 3.}$$

By contrast, intersecting with the vertical line $x = 0$ gives

$$\frac{\mathbb{C}[[x,y]]}{(y^2 - x^3, x)} \cong \frac{\mathbb{C}[[y]]}{(y^2)},$$

whose dimension is two:

$$\boxed{I_{(0,0)}(C_{\text{cusp}}, V(x)) = 2.}$$

The two lines meet the same visible point, but the tangent line has order-three contact. Under the parametrization $(x,y) = (t^2, t^3)$, this is especially transparent: the functions x and y vanish to orders two and three in t.

§152.5 Projective closure remembers a missing intersection

Homogenizing $y = x^2$ produces the projective conic

$$\overline{C}: \quad YZ = X^2 \quad \subset \mathbb{P}^2_{\mathbb{C}}.$$

The affine chart $Z = 1$ recovers $y = x^2$. Consider the vertical line $X = 0$. In affine coordinates, it meets the parabola at the origin $(0,0)$. In projective coordinates, the simultaneous equations

$$X = 0, \qquad YZ = 0$$

give two points:

$$[0:0:1] \qquad \text{and} \qquad [0:1:0].$$

The second lies on the line at infinity $Z = 0$.

Both intersections are transverse. At the affine point, use the chart $Z = 1$ with equation $y = x^2$; the line $x = 0$ cuts it simply. At the point at infinity, use the chart $Y = 1$, where $z = x^2$, and again the line $x = 0$ cuts simply. Their multiplicities are both one.

Bézout's theorem predicts the total intersection number

$$\deg(\overline{C}) \deg(V(X)) = 2 \cdot 1 = 2.$$

The projective calculation finds exactly those two intersections. The apparently missing contribution in the affine plane was at infinity, not an arithmetic error.

A different line gives another distribution of the same total count. The line $Y = 0$ is tangent to the conic at $[0:0:1]$. In the chart $Z = 1$, the quotient by $y = 0$ becomes $\mathbb{C}[[x]]/(x^2)$, so that one point contributes multiplicity two. The total remains two, but two simple intersections have merged into one tangent intersection.

§152.6 Elimination sees the same multiplicity in one variable

If $F(x,y)=0$ and $G(x,y)=0$, the resultant

$$\operatorname{Res}_y(F,G)$$

eliminates y and produces a polynomial condition on x. It can reveal possible intersection coordinates, with multiplicities and extraneous factors that must be interpreted according to the leading coefficients and the chosen projection.

For example, take the parabola $F=y-x^2$ and the horizontal line $G=y$. Eliminating y gives, up to a nonzero scalar,

$$\operatorname{Res}_y(y-x^2,y)=x^2.$$

The double root at $x=0$ is exactly the local tangency multiplicity found from the quotient ring. For the cusp y^2-x^3 intersecting $y=0$, elimination similarly gives x^3, reflecting the triple contact.

The resultant is a computational shadow of intersection theory: it remembers how many roots are merged by projection, while local rings explain the precise geometric meaning of those multiplicities.

A plane curve is therefore best studied through several cooperating descriptions. Its coordinate ring specifies polynomial functions intrinsic to the curve. The tangent cone detects the first nonzero local geometry at a singularity. Normalization recovers smooth local parameters and may separate branches. Intersection multiplicities measure tangencies algebraically, and projective completion accounts for contributions invisible in an affine drawing. Each additional tool answers a concrete question left unresolved by the preceding one.

Divisors on Curves: Zeros, Poles, and Riemann–Roch

N-20240914

On a compact connected Riemann surface X, a nonzero meromorphic function f has an isolated integer order at every point. In a local coordinate $z(p) = 0$,

$$f(z) = z^m u(z), \qquad u(0) \neq 0, \qquad \operatorname{ord}_p(f) = m.$$

A change of coordinate multiplies z by a holomorphic unit, so m is intrinsic. The basic treatment here follows Forster, *Lectures on Riemann Surfaces*, and the definitions and degree results of the Stacks Project.[1]

§153.1 A divisor remembers every local condition

A divisor is a finite integral sum $D = \sum_p n_p[p]$. Its degree is $\deg D = \sum_p n_p$. The principal divisor of a nonzero meromorphic function is

$$(f) = \sum_p \operatorname{ord}_p(f)[p], \qquad (fg) = (f) + (g).$$

On the Riemann sphere $\mathbb{P}^1$, the coordinate $w = 1/z$ near infinity gives

$$z - a = w^{-1}(1 - aw), \qquad (z - a) = [a] - [\infty].$$

In particular, the affine zero at a must be balanced by a pole at infinity.

Theorem 153.1.1 (Degree zero)
For a nonzero meromorphic function on a compact connected Riemann surface, $\deg(f) = 0$.

For a nonconstant f, the induced holomorphic map $X \to \mathbb{P}^1$ has the same degree when counted above 0 or above ∞. The respective local multiplicities are exactly the orders of its zeros and poles. For constant $f \neq 0$, the divisor is zero.

§153.2 Allowed poles form a vector space

For divisors write $D \geq 0$ when each coefficient is nonnegative. Set

$$L(D) = \{0\} \cup \{f \neq 0 \text{ meromorphic} : (f) + D \geq 0\}, \qquad \ell(D) = \dim_{\mathbb{C}} L(D).$$

With $D = n[\infty]$ on $\mathbb{P}^1$, the condition forbids finite poles and permits a pole of order at most n at infinity. Such rational functions are exactly polynomials of degree at most n:

$$L(n[\infty]) = \operatorname{span}\{1, z, \dots, z^n\}, \qquad \ell(n[\infty]) = n + 1 \quad (n \geq 0).$$

Prescribing a zero at 0 changes the divisor to $2[\infty] - [0]$, and gives

$$L(2[\infty] - [0]) = \operatorname{span}\{z, z^2\}.$$

[1] Stacks Project, Tag 02AQ, with related material in Tag 02RS.

Adding a further zero at 1 leaves just $\mathbb{C}z(z-1)$. This makes the bookkeeping precise: a pole allowance raises the possible degree, while a vanishing condition imposes a linear constraint.

Proposition 153.2.1

If $\deg D < 0$, then $L(D) = \{0\}$.

Proof. If $f \neq 0$ belonged to $L(D)$, the effective divisor $(f) + D$ would have nonnegative degree, whereas $\deg((f) + D) = \deg(f) + \deg D = \deg D < 0$, a contradiction. □

§153.3 Line bundles explain the local-to-global problem

On a small open set U, choose a meromorphic function g_U with orders prescribed by $D|_U$. The allowable local functions are $g_U^{-1}\mathcal{O}_X(U)$. On overlapping charts U, V, the ratio g_U/g_V is holomorphic and nowhere zero. Those ratios are precisely transition functions for a holomorphic line bundle $\mathcal{O}(D)$. Thus

$$L(D) = H^0(X, \mathcal{O}(D)).$$

The vector space is finite-dimensional for compact X. Divisors turn local restrictions on poles into globally compatible sections; the obstruction is that independently chosen local sections need not glue.

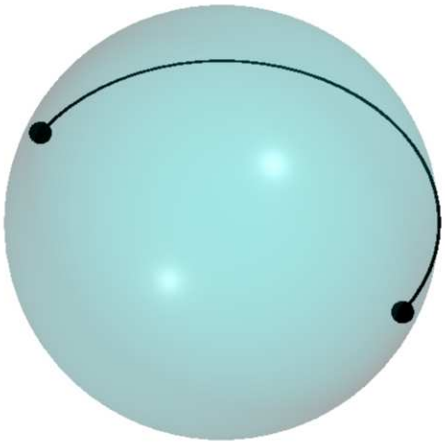


Figure 153.1: Local orders of functions on a compact curve are constrained by global analytic and topological data.

§153.4 Canonical divisors and the genus

A meromorphic differential is locally $h(z)\,dz$. Changing to another local coordinate multiplies the coefficient by a nonzero holomorphic factor, so $\operatorname{ord}_p(h(z)\,dz)$ is well defined. The divisor $K = (\omega)$ of a nonzero meromorphic differential is a canonical divisor. Different choices of ω differ by a meromorphic function, hence determine linearly equivalent divisors.

On $\mathbb{P}^1$ the calculation

$$dz = -w^{-2}\,dw, \qquad w = 1/z,$$

shows $K = -2[\infty]$ and $\deg K = -2$. A complex torus $\mathbb{C}/\Lambda$, by contrast, carries the nowhere-vanishing holomorphic differential dz, so $K = 0$. The general degree formula is

$$\deg K = 2g - 2$$

for a compact Riemann surface of genus g. This is a global theorem; the sphere and torus are tests of its normalization, not its proof.

§153.5 Riemann–Roch counts sections and obstructions

Theorem 153.5.1 (Riemann–Roch)
If D is a divisor on a compact connected Riemann surface of genus g, then

$$\ell(D) - \ell(K - D) = \deg D + 1 - g.$$

We quote the general theorem rather than disguise its cohomological proof as a calculation. The correction $\ell(K - D)$ counts meromorphic differentials whose divisor is at least D; these are the global constraints missing from a naive dimension count.

Take $X = \mathbb{P}^1$, $D = n[\infty]$ with $n \geq 0$, and $K = -2[\infty]$. Since $\deg(K - D) = -2 - n < 0$, the preceding proposition gives $\ell(K - D) = 0$. Hence

$$\ell(D) = n + 1,$$

recovering the explicit polynomial basis. The theorem has passed a nontrivial consistency check.

On an elliptic curve $g = 1$, $K \sim 0$. For a divisor of positive degree d, $\deg(-D) < 0$ gives $\ell(-D) = 0$, and so $\ell(D) = d$. In particular

$$\ell([p]) = 1, \qquad \ell(2[p]) = 2.$$

The first equality says that a nonconstant meromorphic function cannot have just one simple pole. The second says a function with at most one double pole does exist. This is the first dimension-count explanation for why elliptic functions naturally develop double poles.

§153.6 Ramification is another kind of local multiplicity

For a nonconstant holomorphic map $F : X \to Y$ of degree d, a local coordinate expression at p has the form $z \mapsto z^{e_p}$ after suitable changes of coordinates. The Riemann–Hurwitz theorem states

$$2g_X - 2 = d(2g_Y - 2) + \sum_{p \in X} (e_p - 1).$$

For the rational map $F(z) = z^d$ on $\mathbb{P}^1$, only 0 and ∞ ramify, each with index d. The right-hand side is

$$-2d + (d - 1) + (d - 1) = -2,$$

as required for a sphere.

Riemann–Roch and Riemann–Hurwitz answer different questions. One counts sections subject to allowed zeros and poles; the other compares the canonical divisors of curves connected by a map. Their common arithmetic is the addition of local integer orders to obtain a global constraint.

Zariski Geometry: When Functions Become the Space

N-20241026

§154.1 Which geometry can a polynomial equation remember?

Consider the two equations $x = 0$ and $x^2 = 0$ over a field k. They have the same solutions in k, but the quotient algebras

$$k[x]/(x) \cong k, \qquad k[x]/(x^2)$$

are different: the class ϵ of x in the second ring is nonzero with $\epsilon^2 = 0$. Before defining any topology, we have encountered a choice. Do we study *ordinary point sets*, where these equations agree, or an algebraic space retaining infinitesimal equations, where they differ? This note follows the standard construction of affine geometry from rings, as organized in the *Stacks Project*, especially the chapters on spectra and affine schemes.[1]

First let k be algebraically closed, and put $A = k[x_1, \dots, x_n]$. For an ideal $I \subseteq A$, define

$$V(I) = \{a \in k^n : f(a) = 0 \text{ for all } f \in I\}.$$

Conversely, a subset $X \subseteq k^n$ determines

$$I(X) = \{f \in A : f(a) = 0 \text{ for every } a \in X\}.$$

The operations reverse containment: if $I \subseteq J$ then $V(J) \subseteq V(I)$. They also exchange natural operations:

$$V(I + J) = V(I) \cap V(J), \qquad V(IJ) = V(I) \cup V(J).$$

For the two axes, $V(x) \cup V(y) = V(xy)$, whereas $V(x) \cap V(y) = V(x, y) = \{(0, 0)\}$.

§154.2 What Points Forget in Coordinate Rings

The coordinate ring of an algebraic set X is

$$k[X] = A/I(X).$$

Its elements are polynomial functions restricted to X; two polynomials define the same function on X precisely when their difference belongs to $I(X)$.

Hilbert's Nullstellensatz is the crucial bridge. For an ideal I over an algebraically closed field,

$$\boxed{I(V(I)) = \sqrt{I} = \{f \in A : \text{some } f^m \in I\}.}$$

Thus point sets detect radical ideals. For instance, $I = (x^2)$ and $\sqrt{I} = (x)$ yield identical classical zero sets, and the associated coordinate ring of the *set* is $k[x]/(x)$. The thicker quotient $k[x]/(x^2)$ retains information that classical points have forgotten.

[1] Stacks Project, Tag 00DY, and Tag 01HR on affine schemes.

Example 154.2.1 (The curved hyperbola is an affine punctured line)
The parabola $Y = V(y - x^2)$ has

$$k[Y] = k[x, y]/(y - x^2) \cong k[x]$$

by eliminating y. The hyperbola $H = V(xy - 1)$ has

$$k[H] = k[x, y]/(xy - 1) \cong k[x, x^{-1}].$$

The projection $(x, y) \mapsto x$ identifies H with $\mathbb{A}^1 \setminus \{0\}$; its inverse is $t \mapsto (t, t^{-1})$. The identity can be checked in either direction and illustrates why the algebra of functions can reveal structure obscured by a drawing.

§154.3 Closed sets and the deliberately coarse topology

The closed subsets of affine space are the sets $V(I)$. They form a topology because finite unions and arbitrary intersections of algebraic sets are again algebraic. Over an infinite field, a nonzero polynomial in one variable has only finitely many roots, so a proper Zariski-closed subset of $\mathbb{A}^1$ is finite. Consequently nonempty Zariski-open subsets of the affine line have finite complements. This topology is useful for algebraic dependence; it does not try to measure Euclidean distance.

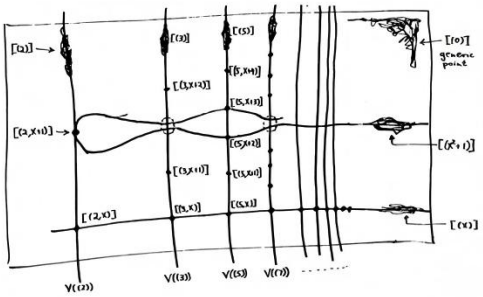


Figure 154.1: Geometric sketches can suggest shapes, while polynomial equations and their function rings determine the algebraic structure.

For $f \in k[X]$, the distinguished open set

$$D(f) = \{p \in X : f(p) \neq 0\}$$

is the region where division by f makes sense. In particular,

$$D(x) \subseteq \mathbb{A}^1, \qquad \mathcal{O}(D(x)) = k[x, x^{-1}].$$

This can be derived directly: localizing $k[x]$ at the multiplicative set $\{1, x, x^2, \ldots\}$ introduces fractions $g(x)/x^m$. Algebraic geometry uses these localizations in place of functions defined using arbitrary analytic neighborhoods.

§154.4 Sheaves, Stalks, and Gluing

To see how local data are organized, it is convenient to pass to the full spectrum. For any commutative ring R with 1,

$$\operatorname{Spec} R = \{\mathfrak{p} : \mathfrak{p} \text{ a prime ideal of } R\},$$

with closed sets $V(I) = \{\mathfrak{p} : I \subseteq \mathfrak{p}\}$ and basic opens $D(f) = \{\mathfrak{p} : f \notin \mathfrak{p}\}$. The *structure sheaf* on this space has the decisive local formulas

$$\boxed{\mathcal{O}_{\operatorname{Spec} R}(D(f)) = R_f, \qquad \mathcal{O}_{\operatorname{Spec} R,\mathfrak{p}} = R_{\mathfrak{p}}.}$$

The second is the stalk at $\mathfrak{p}$: denominators are allowed exactly when they lie outside $\mathfrak{p}$.

Why is a sheaf necessary? If $D(f) = \bigcup_i D(f_i)$, a function described by sections $s_i \in R_{f_i}$ must satisfy

$$s_i|_{D(f_i f_j)} = s_j|_{D(f_i f_j)}$$

on every overlap. The sheaf condition asserts that compatible descriptions glue uniquely. For a finite standard-open cover of $\operatorname{Spec} R$, the algebraic content is the exact sequence

$$0 \longrightarrow R \longrightarrow \bigoplus_i R_{f_i} \longrightarrow \bigoplus_{i,j} R_{f_i f_j},$$

where the last arrow sends (s_i) to the differences of its restrictions. This is not merely a slogan about gluing: it says exactly what agreement on overlaps means and why a global section is determined by its local pieces.

As a small check, $D(t) \cup D(1-t) = \operatorname{Spec} k[t]$, since the ideal $(t, 1-t)$ contains 1. The two representations $t/t \in k[t, t^{-1}]$ and $(1-t)/(1-t) \in k[t, (1-t)^{-1}]$ agree on the overlap and glue to the global function 1.

§154.5 Branches and Infinitesimal Directions

Consider again

$$X = V(xy) \subseteq \mathbb{A}^2, \qquad R = k[x, y]/(xy).$$

Its two irreducible components correspond to the minimal primes (x) and (y). The open $D(x)$ has ring

$$R_x \cong k[x, x^{-1}],$$

because $xy = 0$ and x has become invertible, forcing $y = 0$. Similarly $R_y \cong k[y, y^{-1}]$. On these opens, only one branch remains. Their intersection $D(xy)$ is empty.

The origin, however, lies in neither $D(x)$ nor $D(y)$. Its local ring is

$$R_{(x,y)} = \big(k[x, y]/(xy)\big)_{(x,y)}.$$

The maximal ideal $\mathfrak{m} = (x, y)R_{(x,y)}$ satisfies

$$\dim_k \mathfrak{m}/\mathfrak{m}^2 = 2,$$

since the linear classes of x and y survive independently; the relation $xy = 0$ only appears in degree two. A smooth curve has one-dimensional tangent space at a smooth closed point. Thus the local ring detects the crossing as singular, even though each punctured branch individually looks like a line.

The nilpotent example $k[\epsilon]/(\epsilon^2)$ makes a different point. Its spectrum has a single prime (ϵ), hence one underlying topological point, but its global sections have the form

$$a + b\epsilon, \qquad \epsilon^2 = 0.$$

Topology alone cannot distinguish it from $\operatorname{Spec} k$. The structure sheaf does. This is why the geometry of schemes retains both prime ideals as points and rings of local functions.

§154.6 Morphisms reverse arrows on rings

For affine schemes, a ring homomorphism $\varphi : B \to A$ induces a map of spectra

$$\operatorname{Spec} A \longrightarrow \operatorname{Spec} B, \qquad \mathfrak{p} \longmapsto \varphi^{-1}(\mathfrak{p}).$$

The inverse image of a prime ideal is prime, and the preimage of $D(b)$ is $D(\varphi(b))$; hence this map is continuous and compatible with local sections. Conversely, an affine-scheme morphism induces such a homomorphism on global functions. Schematically,

$$X \longrightarrow Y \quad \rightsquigarrow \quad \Gamma(Y, \mathcal{O}_Y) \longrightarrow \Gamma(X, \mathcal{O}_X).$$

One can see this without any categorical machinery. The map

$$\mathbb{A}^1 \longrightarrow \mathbb{A}^2, \qquad t \longmapsto (t, t^2)$$

pulls the coordinate functions x, y back to t, t^2; the induced ring map is

$$k[x, y] \longrightarrow k[t], \qquad x \mapsto t, \quad y \mapsto t^2,$$

whose kernel is $(y - x^2)$. This single calculation connects the geometry of the parabola, its coordinate ring, and the reversal of arrows. Localizations and stalks make the same correspondence meaningful near every point.

Schemes Gently: A Guided Tour of Spectrum, Nilpotents, and Local Rings

N-20250118

§155.1 Prime ideals, generic points, and specialization

The set of solutions of a polynomial equation does not, by itself, remember generic subvarieties or infinitesimal structure. Following the construction of affine schemes in the *Stacks Project*, we examine how a ring determines a topological space together with local rings.[1]

For a ring R, define

$$\operatorname{Spec} R = \{\mathfrak{p} \subset R : \mathfrak{p} \text{ is prime}\}.$$

This is the spectrum of R. Its points are prime ideals.

At first this feels like a category mistake: ideals are algebraic objects, not points. But this change is exactly what allows geometry to remember generic behavior, arithmetic information, and infinitesimal thickening.

For $R = k[x]$ over an algebraically closed field k, the maximal ideals

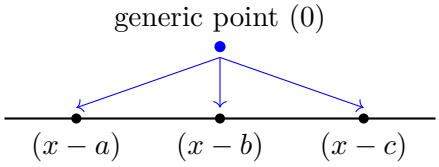
$$(x - a), \qquad a \in k,$$

correspond to ordinary closed points. This requires the algebraically closed hypothesis: over $\mathbb{R}$, the ideal $(x^2 + 1)$ is also maximal, and its residue field is $\mathbb{C}$. There is also the prime (0), the generic point.

The closure of the generic point is the whole space:

$$\overline{\{(0)\}} = \operatorname{Spec} k[x].$$

This is not Hausdorff behavior, but it is meaningful. The generic point represents the whole irreducible line, while $(x - a)$ represents a closed point.



The arrows are suggestive: the generic point specializes to closed points.

It is tempting to keep only maximal ideals, because over an algebraically closed field they correspond to ordinary closed points. But prime ideals contain information about irreducible subspaces.

In $k[x, y]$, the prime ideal (x) is not a closed point. It corresponds to the whole y-axis. The maximal ideal $(x, y - a)$ is a closed point on that axis.

Thus prime ideals let a scheme contain both points and subvarieties as points of different levels of generality. This is strange at first, but it is exactly what makes generic points possible.

[1] Stacks Project, *The spectrum of a ring* (Tag 00DY), and *Affine schemes* (Tag 01HR).

In $\operatorname{Spec} R$, one says that $\mathfrak{q}$ specializes to $\mathfrak{p}$ when

$$\mathfrak{q} \subseteq \mathfrak{p}.$$

The smaller prime is more generic; the larger prime is more special.

For $k[x]$,

$$(0) \subset (x-a),$$

so the generic point specializes to the closed point $(x-a)$. This order relation is the topological shadow of ideal containment.

For an ideal $I \subset R$, define

$$V(I) = \{\mathfrak{p} \in \operatorname{Spec} R : I \subseteq \mathfrak{p}\}.$$

These are closed sets.

For $f \in R$, define

$$D(f) = \{\mathfrak{p} : f \notin \mathfrak{p}\}.$$

This is the region where f is allowed to be inverted. Distinguished opens are the basic open sets of affine schemes.

§155.2 Functions near a point

At a prime ideal $\mathfrak{p}$, the local ring is

$$R_{\mathfrak{p}}.$$

It is obtained by inverting every element not in $\mathfrak{p}$. The interpretation is simple: near $\mathfrak{p}$, functions not vanishing there should be units.

For $R = k[x]$ and $\mathfrak{p} = (x-a)$, the ring

$$k[x]_{(x-a)}$$

contains rational functions whose denominators do not vanish at a. This is the algebraic version of functions defined near a.

A spectrum is not only a topological space. It carries functions. On a distinguished open $D(f)$, the structure sheaf satisfies

$$\mathcal{O}_{\operatorname{Spec} R}(D(f)) = R_f.$$

The stalk at $\mathfrak{p}$ is

$$\mathcal{O}_{\operatorname{Spec} R,\mathfrak{p}} = R_{\mathfrak{p}}.$$

These two formulas make local calculations possible; the sheaf condition supplies the missing compatibility. If $D(f) = \bigcup_i D(f_i)$, sections from the rings R_{f_i} glue to a unique section of R_f precisely when they agree after restricting to the overlaps $R_{f_i f_j}$. This gives the algebraic exactness pattern $0 \to R_f \to \bigoplus_i R_{f_i} \to \bigoplus_{i,j} R_{f_i f_j}$, where the final map records pairwise differences.

At a prime $\mathfrak{p}$, the local ring $R_{\mathfrak{p}}$ has maximal ideal $\mathfrak{p}R_{\mathfrak{p}}$. The residue field is

$$\kappa(\mathfrak{p}) = R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}}.$$

For a closed point $(x-a) \subset k[x]$, the residue field is k. For the generic point (0), the residue field is $k(x)$, the field of rational functions.

This is another way to see the meaning of generic points: at the generic point of the affine line, the coordinate x is not specialized to a number; it remains a variable.

§155.3 Arithmetic and local prime information

The spectrum of the integers has points

$$(p)$$

for prime numbers p, together with the generic point (0). An integer n vanishes at exactly the primes dividing it:

$$V(n) = \{(p) : p \mid n\}.$$

Thus divisibility becomes geometry.

This converts divisibility into a concrete local calculation. On $D(6)$, the primes (2) and (3) have been removed, and the ring of regular sections is $\mathbb{Z}[1/6]$. At (p), the local ring is $\mathbb{Z}_{(p)}$, whose denominators are exactly the integers not divisible by p. Its residue field is $\mathbb{F}_p$, while the generic point (0) has residue field $\mathbb{Q}$.

§155.4 Invisible Infinitesimal Structure

Consider

$$R = k[x]/(x^2).$$

The element x is nonzero but satisfies $x^2 = 0$. Topologically, $\operatorname{Spec} R$ has only one point. Algebraically, it remembers an infinitesimal thickening of that point.

A useful picture is a point with a first-order tangent fuzz. The fuzz cannot be seen as a second classical point, but it can be detected by functions.

This is one of the reasons schemes are powerful: they keep information that ordinary point sets discard.

Let

$$R = k[\epsilon]/(\epsilon^2).$$

A function on $\operatorname{Spec} R$ can have the form

$$a + b\epsilon.$$

The term $b\epsilon$ is invisible if one only looks at ordinary points, because ϵ is nilpotent. But algebraically it records first-order infinitesimal information.

The dual-number model detects a derivative directly. For a polynomial $f(x)$ and an element $a + b\epsilon$, the binomial theorem gives

$$f(a + b\epsilon) = f(a) + bf'(a)\epsilon$$

because all terms with ϵ^2 vanish. The coefficient b is a first-order tangent direction; ordinary points see only the value $f(a)$. This is a calculation, rather than a drawing of a point with vaguely described “fuzz”.

Projective Schemes and Gluing: Building Projective Space from Affine Windows

N-20250301

§156.1 The projective line is assembled from affine pieces

Earlier projective geometry used homogeneous coordinates to add points at infinity and remove affine exceptions. Scheme theory adds another layer: every affine window comes with a ring of functions, and the windows must be glued compatibly.

The construction that does this is Proj. It is to graded rings what Spec is to ordinary rings, but the grading forces an essential degree-zero condition on every chart. We follow the construction of Proj and twisting sheaves in the *Stacks Project*.[1]

The best first example is $\mathbb{P}^1$. Take two affine lines:

$$U_0 = \operatorname{Spec} R[t], \qquad U_1 = \operatorname{Spec} R[s].$$

On the region where both coordinates are nonzero, glue them by

$$s = \frac{1}{t}.$$

The two copies overlap on $\operatorname{Spec} R[t, t^{-1}] \cong \operatorname{Spec} R[s, s^{-1}]$. The transition homomorphism sends $s \mapsto t^{-1}$. In the homogeneous coordinates $[x_0 : x_1]$, the first chart has $t = x_1/x_0$ and the second has $s = x_0/x_1$, which verifies the relation $ts = 1$. The result is the projective line. The point is not the formula alone, but the construction style:

projective space is made by gluing affine charts.

§156.2 Homogeneous primes and the degree-zero chart algebra

Let

$$S = S_0 \oplus S_1 \oplus S_2 \oplus \cdots$$

be a graded ring. Projective geometry only respects homogeneous equations, because

$$[x_0 : \cdots : x_n] = [\lambda x_0 : \cdots : \lambda x_n].$$

The construction $\operatorname{Proj} S$ uses homogeneous prime ideals not containing the irrelevant ideal.

This definition sounds heavier than the picture, but it encodes a familiar rule: equations in projective space must be homogeneous.

The scheme-theoretic projective space is

$$\mathbb{P}^n_R = \operatorname{Proj} R[x_0, \ldots, x_n],$$

where the variables have degree 1.

[1] See https://stacks.math.columbia.edu/tag/01M3, especially the standard-open construction and the subsequent sections on invertible sheaves.

The standard affine open where $x_i \neq 0$ is written

$$D_+(x_i).$$

On it, the ratios

$$\frac{x_j}{x_i}$$

serve as affine coordinates.

Thus $\mathbb{P}^n$ is covered by $n+1$ affine schemes. Projective geometry is global, but it is computed through affine windows.

For $S = R[x_0, \ldots, x_n]$, the ideal

$$S_+ = (x_0, \ldots, x_n)$$

is called the irrelevant ideal. It corresponds to the forbidden point where all homogeneous coordinates vanish at once:

$$[0 : \cdots : 0],$$

which is not a projective point.

Thus $\operatorname{Proj} S$ uses homogeneous prime ideals that do not contain all positive-degree elements. If we included S_+ for a polynomial ring over a field, it would lie outside every standard open $D_+(x_i)$ and would represent the prohibited simultaneous zero of the homogeneous coordinates. Excluding every homogeneous prime containing S_+ ensures the standard opens cover $\operatorname{Proj} S$. This is the algebraic justification for the nonzero-coordinate rule.

For a homogeneous element $f \in S$, the standard open $D_+(f)$ is affine. Its coordinate ring is the degree-zero part of the localization:

$$\mathcal{O}(D_+(f)) = (S_f)_0.$$

This formula explains why ratios appear in projective coordinates. Localizing at x_i allows division by powers of x_i, and taking degree zero keeps only scale-invariant expressions such as

$$\frac{x_j}{x_i}.$$

Indeed, for $S = k[x_0, \ldots, x_n]$, the degree-zero localization is

$$\begin{aligned}(S_{x_i})_0 &\cong k\left[\frac{x_0}{x_i}, \ldots, \widehat{\frac{x_i}{x_i}}, \ldots, \frac{x_n}{x_i}\right] \\ &\cong k[t_1, \ldots, t_n].\end{aligned}$$

The hat omits $x_i/x_i = 1$. Its spectrum is the affine space $\mathbb{A}^n_k$, so the standard projective chart and its regular functions follow from the degree-zero localization.

§156.3 Three Charts of a Conic

For $\mathbb{P}^2$, the standard opens are

$$D_+(x), \qquad D_+(y), \qquad D_+(z).$$

On $D_+(z)$, use

$$X = \frac{x}{z}, \qquad Y = \frac{y}{z}.$$

On $D_+(x)$, use

$$U = \frac{y}{x}, \qquad V = \frac{z}{x}.$$

Each chart is affine, but the overlap maps must agree on triple intersections. On $D_+(z) \cap D_+(x)$, the coordinates satisfy

$$U = \frac{Y}{X}, \qquad V = \frac{1}{X}, \qquad X = \frac{1}{V}, \qquad Y = \frac{U}{V}.$$

Substitution in either direction gives the identity transition map upon composition; similar ratio identities handle the third chart. These are the actual gluing data defining $\mathbb{P}^2$.

This is the practical meaning of projective schemes: use affine algebra locally, then glue.

Consider the projective conic

$$xz - y^2 = 0.$$

On the chart $z \neq 0$, set $X = x/z$ and $Y = y/z$. The equation becomes

$$X = Y^2.$$

On the chart $x \neq 0$, set $U = y/x$ and $V = z/x$. The equation becomes

$$V = U^2.$$

The same projective curve has multiple affine faces. A single affine chart may hide what happens at infinity, but the projective object keeps all charts at once.

For the conic

$$xz - y^2 = 0,$$

the chart $y \neq 0$ uses coordinates

$$A = \frac{x}{y}, \qquad B = \frac{z}{y}.$$

The equation becomes

$$AB = 1.$$

So in this chart the conic looks like a hyperbola. In another chart it looked like a parabola. This is a concrete reminder that affine appearances depend on the chosen window.

The projective curve is the object obtained by holding all these windows together.

For a single affine hypersurface equation $f(x, y) = 0$, homogenizing to its total degree gives a projective equation

$$F(x, y, z) = 0.$$

For a general affine *ideal*, however, homogenizing a few chosen generators is not by itself a safe definition of scheme-theoretic projective closure. One must homogenize the ideal appropriately (or saturate the resulting homogeneous ideal with respect to the homogenizing coordinate). This distinction matters if embedded or nilpotent structure is to survive correctly.

So Proj is not merely a notation for old projective varieties. It is the projective version of the local-function viewpoint developed in scheme theory.

§156.4 From Functions to Sections

Projective geometry naturally leads to line bundles. The coordinates of $\mathbb{P}^n_k$ scale simultaneously; an ordinary polynomial of degree $d > 0$ is multiplied by λ^d under $[x_0 : \cdots : x_n] \mapsto [\lambda x_0 : \cdots : \lambda x_n]$, so it cannot be a globally defined scalar function on projective space. Instead, the transition factors $(x_i/x_j)^d$ between affine charts define the twisting sheaf $\mathcal{O}(d)$. For $d \geq 0$, its global sections are exactly homogeneous polynomials of degree d in $x_0, \ldots, x_n$. A homogeneous equation of degree d is better thought of as a section of $\mathcal{O}(d)$ than as an ordinary global function.

This explains why projective varieties have fewer global regular functions than affine varieties. Projective geometry shifts attention from functions to sections of line bundles.

This remark points back to divisors and Riemann–Roch: projective geometry, line bundles, and spaces of sections are parts of the same story.

The twisting sheaves therefore complete the passage from homogeneous coordinates to projective schemes. The same graded algebra that builds $\operatorname{Proj} S$ also supplies its affine windows and its line bundles: projective geometry has been reconstructed from homogeneous algebra and gluing while retaining the local functions invisible in a purely incidence-based picture.

When Addition Starts Floating: Theta-Links, Log-Links, and Arithmetic Deformation

N-20251113

§157.1 A multiplicative period and its arithmetic scale

Imagine a curve whose local points can be described by multiplying nonzero numbers, with one special number q acting as a period. Over a nonarchimedean local field L, an elliptic curve with split multiplicative reduction admits a Tate uniformization

$$E_q(\overline{L}) \simeq \overline{L}^{\times}/q^{\mathbb{Z}}, \qquad 0 < |q|_v < 1.$$

The quotient identifies u, qu, q^2u and so on, just as $\mathbb{R}/\mathbb{Z}$ identifies numbers differing by an integer. The operation here is multiplication, though, and the valuation $v(q)$ measures the size of the period in the nonarchimedean direction. It also records the valuation of the minimal discriminant of the curve.

A genuine example gives the notation some weight. Dupuy and Hilado construct initial theta data beginning with the elliptic curve $11a1$,

$$E: \quad y^2 + y = x^3 - x^2 - 10x - 20, \qquad v_{11}(q_{11}) = 5.$$

The valuations of q_{11}, q_{11}^4 and q_{11}^9 are consequently

$$\boxed{5, \quad 20, \quad 45.}$$

Here the squares $1, 4, 9$ are particularly suggestive. Special values of nonarchimedean theta-functions naturally produce q^{m^2}-type information; a theta-link is designed to compare this selected multiplicative information across arithmetic settings.

There is a catch with far-reaching consequences. A comparison that respects products and special theta-values has no automatic reason to respect sums. Once the additive structures can no longer be identified directly, it becomes necessary to control what survives: roots and Galois actions, logarithms of units, and the ambiguities in reconstructing numbers from those records. This is the motivation for theta-links, Kummer theory, log-links and the network of Hodge theaters. Their common goal is a numerical inequality extracted from arithmetic information that can still be compared.[1]

§157.2 The nonarchimedean theta-function

Let L be a local field of characteristic zero with finite residue field, and let C_L be the completion of an algebraic closure of L. Let $q \in L$ be a nonzero element of the maximal ideal of $\mathcal{O}_L$. In the application to elliptic curves, q is the Tate parameter q_v.

[1] See Ivan Fesenko, *Arithmetic Deformation Theory via Arithmetic Fundamental Groups and Nonarchimedean Theta-Functions*, and Taylor Dupuy and Anton Hilado, *The Statement of Mochizuki's Corollary 3.12: Initial Theta Data*.

Definition 157.2.1 (Theta-function of type au^m)**.** A holomorphic function $f : C_L^\times \to C_L$, defined over L, is a theta-function of type au^m, where $a \in C_L^\times$ and $m \in \mathbb{Z}$, if

$$f(u) = au^m f(qu).$$

A meromorphic function invariant under $u \mapsto qu$ descends to a function on the Tate quotient

$$C_L^\times / \langle q \rangle.$$

The basic nonarchimedean theta-function used in the survey is

$$\theta(u) = \sum_{n \in \mathbb{Z}} (-1)^n q^{n(n-1)/2} u^n.$$

It also has the product expansion

$$\theta(u) = (1-u) \prod_{n \geq 1} (1-q^n)(1-q^n u)(1-q^n u^{-1}),$$

which is the nonarchimedean analogue of the Jacobi triple product expression. The function is chosen because its zeros and poles interact well with the Tate curve and with the theta-values used in IUT.

Remark 157.2.2 (Why a theta-function appears) — The theta-function is not added to the story for analogy with complex analysis alone. On a Tate curve, elliptic functions with period q can be expressed in terms of theta-functions. Thus theta-functions are a natural language for functions on the local analytic uniformization

$$E(L) \cong L^\times / \langle q \rangle.$$

Fesenko also records a modified theta-function, written here schematically as $\ddot{\Theta}(u)$, whose special values at points separated by periods from $\pm i$ satisfy a relation of the form

$$q^{m^2/2} \ddot{\Theta}(i\sqrt{q^m}) = \ddot{\Theta}(i).$$

After choosing a $2l$-th root of q, this leads to theta-values of the shape

$$\Theta(\sqrt{-q^m}) = q^{m^2}, \qquad 1 \leq m \leq \frac{l-1}{2},$$

up to multiplication by roots of unity of order dividing $2l$.

Remark 157.2.3 (A controlled simplification) — The formula above is the correct structural feature needed here: special theta-values encode powers q^{m^2}. The complete IUT construction keeps track of coverings, labels, roots of unity, Galois actions, and compatible local-global data. Those details are not being suppressed because they are unimportant; they are being postponed because the first conceptual step is to see why theta-values can serve as deformation data.

§157.3 The local theta-link as a non-ring-theoretic map

At a bad reduction valuation of odd residue characteristic, the main version of the theta-link described by Fesenko revolves around a local monoid-theoretic transformation. In simplified notation it sends

$$q \longmapsto \{\Theta(\sqrt{-q^m}) = q^{m^2}\}_{1 \leq m \leq (l-1)/2},$$

and it sends units to units by the identity map, up to the appropriate torsion ambiguity.

Definition 157.3.1 (Local theta-link, structural form)**.** A local theta-link is a monoid-theoretic comparison which, at a bad split multiplicative valuation, keeps the unit part essentially fixed while replacing the Tate-period generator q by a family of theta-values behaving like q^{m^2}. It is a comparison of multiplicative and theta-theoretic data, not a homomorphism of local rings.

Thus the local theta-link acts on two kinds of local data:

$$\mathcal{O}_L^\times / \operatorname{Tor}(\mathcal{O}_L^\times), \qquad \{q^{m^2} : 1 \le m \le (l-1)/2\},$$

and it is glued to global monoid-theoretic data satisfying the product formula.

We can feel the difference between a multiplicative comparison and a ring map without knowing IUT. Work just in $\mathbb{Q}_5^\times$. Every nonzero element has a unique expression

$$x = 5^n u, \qquad n \in \mathbb{Z}, \quad u \in \mathbb{Z}_5^\times.$$

Define the following *toy* transformation, with exponent $4 = 2^2$:

$$T(5^n u) = 5^{4n} u.$$

It behaves impeccably with respect to multiplication:

$$T(5^n u \cdot 5^m v) = 5^{4(n+m)} uv = T(5^n u) T(5^m v).$$

It even fixes every unit, so $T(1) = 1$. But consider the entirely ordinary addition $1+5 = 6$. Since 6 is a 5-adic unit,

$$\boxed{T(1+5) = T(6) = 6 \qquad \text{while} \qquad T(1) + T(5) = 1 + 5^4 = 626.}$$

The map remembers products and valuations while failing to preserve sums. Every ring identity still holds *inside* $\mathbb{Q}_5$; it is the attempt to *transport* addition through T that fails.

This calculation is not a construction of a theta-link. It isolates the elementary feature behind Fesenko's explanation: a theta-link compares selected multiplicative/theta-theoretic structures across theaters and provides no ordinary ring isomorphism carrying every sum along with them. The real theory has considerably more data than this toy, but that basic distinction is already unavoidable.

This is the first honest meaning of the phrase "addition starts floating". It does not mean that the equation $1 + 1 = 2$ becomes false inside a fixed field. It means that after theta-deformation one is no longer allowed to identify the additive coordinates on the two sides as if a ring homomorphism had carried them across.

§157.4 The bridge supplied by Kummer theory

If the theta-link only destroyed additive compatibility, it would not be useful. The next ingredient explains how monoid-theoretic data can still communicate with Galois and fundamental-group data.

For an elementary glimpse, suppose a field L contains the ℓ-th roots of unity and we adjoin α with

$$\alpha^\ell = a \in L^\times.$$

Any Galois automorphism σ fixing L must send α to another root of the same equation. Therefore

$$\sigma(\alpha) = \zeta_\sigma \alpha, \qquad \zeta_\sigma^\ell = 1, \quad \text{or} \quad \boxed{\frac{\sigma(\alpha)}{\alpha} \in \mu_\ell.}$$

One piece of *multiplicative* information, the possibility of extracting an ℓ-th root, has produced a record of *Galois* action. For example, when $\alpha = q^{1/\ell}$, the possible ratios $\sigma(\alpha)/\alpha$ explain why extracting roots of a Tate parameter naturally creates torsion labels. If the roots of unity are fixed by the Galois group, the ratios even form a character; in the general situation they form a cocycle.

This is the small calculation behind the much more elaborate generalized Kummer correspondence. The point is to keep enough information about roots and Galois symmetries to compare structures even when an ordinary additive identification is unavailable.

Definition 157.4.1 (Kummer map, schematic form)**.** Let H be an open subgroup of a Galois or arithmetic fundamental group acting on an abelian group M. Kummer theory studies maps of the form

$$M^H \longrightarrow H^1(H, \mathrm{Hom}(\mathbb{Q}/\mathbb{Z}, M)),$$

obtained by considering divisibility of elements of M.

For ordinary intuition, Kummer theory turns the question of extracting roots into a cohomological Galois datum. In IUT, the relevant versions are generalized and truncated. They are applied not merely to elements of fields, but also to monoid-theoretic structures associated with theta-functions, theta-values, number fields, and their completions.

Remark 157.4.2 (Kummer theory as a translator) — The theta-link is monoid-theoretic. Mono-anabelian reconstruction works on the side of arithmetic fundamental groups. Kummer theory is one of the translators between these languages. It relates theta-values and monoid data to Galois or fundamental-group structures that can be reconstructed and compared.

A key issue here is cyclotomic rigidity. Various copies of $\widehat{\mathbb{Z}}$, or quotients of such copies, arise from roots of unity, from monoid-theoretic data, and from subquotients of fundamental groups. IUT needs algorithms that identify the relevant cyclotomic structures in a canonical way. Without such rigidity, the roots of unity introduced by extracting l-th or $2l$-th roots would create uncontrolled ambiguity.

§157.5 Two symmetries and the bookkeeping of labels

The theta-link is built after choosing a prime $l > 3$. The l-torsion points of the elliptic curve provide labels modeled on $\mathbb{F}_l$. Fesenko emphasizes two different symmetries.

Definition 157.5.1 (The two label symmetries)**.** The geometric-additive symmetry is

$$\mathbb{F}_l^{\times\pm} = \mathbb{F}_l \rtimes \{\pm 1\},$$

acting on labels by

$$z \longmapsto \pm z + a.$$

The arithmetic-multiplicative symmetry is

$$\mathbb{F}_l^{*\pm} = \mathbb{F}_l^{\times}/\{\pm 1\}.$$

The first symmetry is essentially geometric. It is related to the geometric part of the arithmetic fundamental groups and to Kummer theory around theta-values. It is also

used to synchronize conjugacy ambiguities of local Galois groups attached to different labels.

The second symmetry is essentially arithmetic. It is related to the Galois action of number fields and is multiplicative by definition. It helps separate the zero label from the nonzero labels and is more naturally multiradial.

Symmetry	Nature	Role
$\mathbb{F}_l^{\times\pm}$	geometric/additive	conjugate synchronization
$\mathbb{F}_l^{*\pm}$	arithmetic/multiplicative	label bookkeeping and descent

Remark 157.5.2 (Why labels are not cosmetic) — The labels are not merely indices in a long construction. They keep track of different copies of local Galois data and theta-values. If these labels are identified too naively, one loses the distinction between the geometric-additive and arithmetic-multiplicative parts of the construction.

§157.6 Conjugate synchronization and the basepoint problem

Fundamental groups depend on basepoints. In ordinary expositions this dependence is often hidden because different basepoints lead to groups identified up to inner automorphism. In IUT this ambiguity cannot simply be ignored, because the theta-link and log-link compare structures that live in different theaters.

Definition 157.6.1 (Conjugate synchronization, informal)**.** Conjugate synchronization is a system of identifications between local absolute Galois groups attached to different labels, arranged so that one does not introduce uncontrolled conjugacy indeterminacy.

The need for such synchronization is one reason the theory becomes much more rigid than a naive "copy-and-paste" of local fields. The comparison is not allowed to choose arbitrary basepoint identifications. It must respect the label symmetries and the Kummer-theoretic data surrounding the theta-values.

§157.7 The Log-Link and Additive Structure

The theta-link uses multiplicative theta-values. To regain access to additive structure, Fesenko turns to the nonarchimedean logarithm.

Definition 157.7.1 (Nonarchimedean logarithm)**.** For a local field L, the nonarchimedean logarithm

$$\log : \mathcal{O}_L^{\times} \longrightarrow L$$

is defined on elements $1-\alpha$, with α in the maximal ideal of $\mathcal{O}_L$, by

$$\log(1-\alpha) = -\sum_{n\geq 1} \frac{\alpha^n}{n},$$

and sends multiplicative representatives of the finite residue field to 0.

The key point is that log transforms multiplication into addition where it is defined. Thus the logarithm lets multiplicative unit data speak an additive language.

One computation is enough to show why a logarithm is the natural partner of multiplication. In $\mathbb{Q}_5$, take units close to 1, say $u, v \in 1 + 5\mathbb{Z}_5$. The convergent power series gives

$$\log_5(1+t) = t - \frac{t^2}{2} + \frac{t^3}{3} - \cdots, \qquad t \in 5\mathbb{Z}_5,$$

and consequently

$$\boxed{\log_5(uv) = \log_5 u + \log_5 v.}$$

For instance, $6 = 1 + 5$ and $36 = 6^2$ belong to this group, so

$$\log_5(36) = 2\log_5(6).$$

No numerical approximation is needed: this is an exact identity of 5-adic numbers.

A useful mental picture is that the logarithm changes the *language* of the data. What used to be multiplication of nearby units becomes addition of their logarithms. It does not magically transport all additions across a theta-link. In particular, the toy transformation T above still sends $1 + 5$ and $T(1) + T(5)$ to different numbers. The log-link helps compare a selected, logarithmically encoded piece of the additive story, while the theory has to control the pieces it cannot identify directly.

Definition 157.7.2 (Log-link, structural form)**.** A log-link is the comparison which shifts multiplicative unit data through the nonarchimedean logarithm, so that additive structures can be reconstructed from logarithmic images of multiplicative structures.

Fesenko explains the relation between the two links as follows. The multiplicative structures on the two sides of the theta-link are related through value-group data. The additive structures are related through unit-group data after applying the log-link.

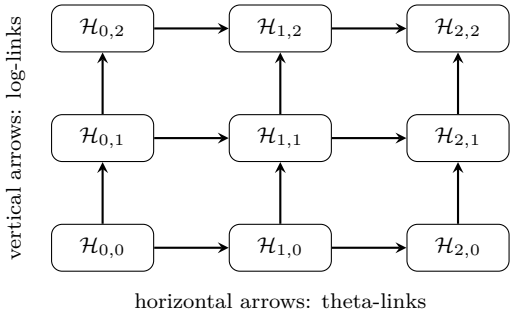
Remark 157.7.3 (Why the logarithm is not optional) — Theta-values naturally act on multiplicative data, not directly on the additive group of the local field. The logarithm creates a setting where multiplicative unit data has an additive image. This is why the log-link is not an auxiliary simplification; it is part of the reconstruction mechanism.

§157.8 The log-theta-lattice

The theta-link and log-link do not appear as isolated arrows. They are organized into a two-dimensional pattern called the log-theta-lattice.

Definition 157.8.1 (Log-theta-lattice, schematic)**.** The log-theta-lattice is a two-dimensional array of theaters. Vertical arrows are log-links, and horizontal arrows are theta-links. A horizontal theta-link at one level depends on the log-link structure immediately below it.

The picture is:



This diagram is only a guide. Fesenko notes that the main results use just two neighboring infinite vertical lines and the horizontal arrow between the lattice points $(0,0)$ and $(1,0)$. The important idea is that the theory does not compare a single source and target once. It studies a controlled pattern of comparisons.

§157.9 The Log-Shell

A technical difficulty now appears. The log-link does not commute with the theta-link. If one expected a clean commutative square, one would be disappointed. IUT instead introduces a common structure that can hold the relevant images even when elementwise compatibility fails.

Definition 157.9.1 (Log-shell)**.** For a nonarchimedean local field L, the nonarchimedean part of the log-shell is a compact subgroup of the form

$$(p^*)^{-1}\log(\mathcal{O}_L^\times),$$

where $p^* = p$ if p is odd and $2^* = 4$. At a complex archimedean place, the corresponding log-shell is the closed ball of radius π.

The log-shell should not be imagined as an error bar in a numerical approximation. It is a structural container. The relevant Kummer isomorphisms are not compatible with the log-link at the level of individual elements, but their images are contained in the log-shell. This is exactly the kind of weakened compatibility that arithmetic deformation needs.

Remark 157.9.2 (Compatibility without equality) — In ordinary algebra one often tries to prove that a diagram commutes element by element. The log-shell records a weaker but controlled form of compatibility: one compares inclusions of certain subsets or regions rather than exact equality of individual transported elements.

§157.10 Mono-anabelian reconstruction and multiradiality

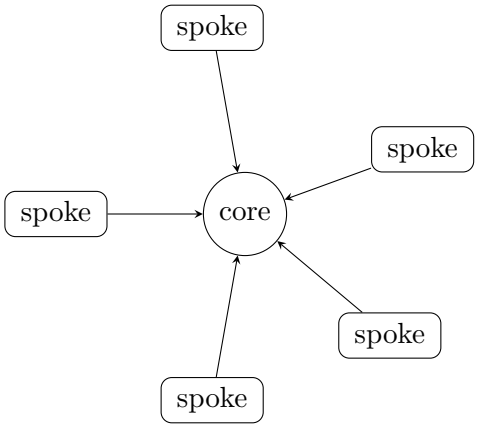
The words “reconstruction” and “multiradiality” are easy to mystify, but their roles are concrete.

Definition 157.10.1 (Mono-anabelian reconstruction)**.** Mono-anabelian reconstruction is the construction of ring-theoretic data from a single topological group that is known to be isomorphic to an arithmetic fundamental group of a suitable hyperbolic curve or orbicurve. The reconstruction is algorithmic and functorial in the sense needed for localization and change of base field.

The difference from ordinary anabelian comparison is important. Classical anabelian geometry often compares two geometric objects through an isomorphism of their fundamental groups. Mono-anabelian geometry asks for algorithms that start from one suitable group and recover the ring structure associated to it.

Definition 157.10.2 (Multiradial algorithm)**.** A reconstruction algorithm is multiradial if an object constructed from one spoke of a wheel-like system is described in terms that still make sense from the viewpoint of other spokes. Equivalently, it is compatible with simultaneous execution at multiple spokes.

Here the "wheel and spokes" language is only an analogy, but a useful one. A spoke corresponds to one theater or one radial perspective. The center corresponds to a core object reconstructed from data that can be described from more than one spoke.



multiradial descriptions must remain meaningful from several spokes at once

Remark 157.10.3 (Why indeterminacy appears) — A multiradial description is stronger than a one-spoke description. To make such descriptions possible, IUT allows certain controlled indeterminacies. These are not defects added after the fact; they are part of how the theory avoids making illegal identifications between different theaters.

§157.11 The three indeterminacies

Fesenko singles out three indeterminacies that enter the computation of volume deformation.

Definition 157.11.1 (The three indeterminacies, structural form)**.** The three main indeterminacies are:

1. Ind1: an indeterminacy related to permutation symmetries of Galois and arithmetic fundamental groups along vertical lines of the log-theta-lattice;
2. Ind2: an indeterminacy related to the horizontal theta-link and to the action of a compact group on $\log(\mathcal{O}_L^\times)$;
3. Ind3: an indeterminacy arising from upper semi-compatibility of Kummer isomorphisms with log-links in a single vertical line.

The third phrase, "upper semi-compatibility", is deliberately weaker than commutativity of a diagram. Instead of asking for equality of transported elements, one asks for controlled inclusion relations among the relevant subsets.

Why keep track of these ambiguities instead of just choosing convenient identifications? Think of trying to compare two measurements when their exact matching is unavailable. One can still compare the *sets of possible results*. For example, if different choices produce

values in $[1,2]$ and $[3/2,5/2]$, both sets fit inside the common interval $[1,5/2]$, even though no particular element of the first interval has been identified with a particular element of the second. The interval is deliberately larger than either set: it is a *container* in which a bound can be stated without pretending to have a pointwise correspondence.

This is only an analogy for the role of a log-shell and the allowed indeterminacies: the actual objects are logarithmic images, Galois-theoretic correspondences and adelic data. Yet it explains the logical advantage of controlling an entire region of possible images. An estimate about its size may survive changes of labels or choices, even when an equality between two chosen elements would make no sense.

The three indeterminacies track different sources of that freedom: the choice of synchronization along the log-links, the symmetries seen across a theta-link, and the weaker compatibility of Kummer-theoretic images with logarithmic data. One should not confuse the *existence of a container* with a proof of the difficult global inequality bounding its size; the latter is the substantive assertion of IUT.

This is where the intuitive phrase "floating addition" becomes mathematically safer. The additive structure is not discarded into chaos. It is reconstructed through logarithmic images and fundamental-group algorithms, while the unavoidable ambiguity is measured by the three indeterminacies.

§157.12 Where the comparison is supposed to lead

We can now follow the story without keeping nine new definitions in our heads. At a split multiplicative place the Tate parameter q records a local scale; theta-values supply the q^{m^2}-type data for the theta-link. The obstacle is already visible in the little map $T(5^n u) = 5^{4n}u$: knowing how to transport products gives no permission to transport sums.

The surrounding mechanisms address different parts of this obstacle. Kummer theory turns selected roots into Galois information. The two symmetries control geometric and arithmetic labels. The log-link expresses multiplication of units in an additive language, while log-shells and indeterminacies hold possible images in a common container. Mono-anabelian reconstruction and multiradial descriptions are meant to keep useful information accessible from several theaters. The log-theta-lattice makes these comparisons interact.

Fesenko's account also gives a strikingly simple piece of algebra behind the *goal* of this machinery. Write $D > 0$ for a normalized degree associated to Tate parameters and V for a logarithmic volume term constructed from the theta-data. The intended inequalities have the shape

$$-D \le V \le a(\ell) - b(\ell)D, \qquad b(\ell) > 1.$$

If these estimates can be established in the relevant arithmetic setting, a school-level rearrangement gives

$$\boxed{(b(\ell)-1)D \le a(\ell).}$$

The challenge lies in the first comparison: it relates data from different theaters while respecting only the allowed information. Dupuy and Hilado make a useful distinction here. Their computable initial theta data explain how the input to the claimed inequality is assembled; the intricate inter-theater estimate is treated as an assumption, not proved by the input computation.

The phrase *addition starts floating* now has a definite intuition: inside each field addition remains entirely ordinary, but transporting it through a multiplicative comparison

is delicate. The aim is to recover enough bounded, choice-independent numerical information for a Diophantine inequality while keeping the differences between the theaters visible.

Teichmueller Theory Without One Universe: Classical Deformation, Hodge Theaters, and Identification

N-20251114

§158.1 The same torus can have more than one shape

Draw a square and identify its opposite edges. It becomes a torus. Now shear the square into a parallelogram and identify the opposite edges again. The two tori have the same underlying topology: a loop traveling once horizontally and a loop traveling once vertically still generate the first homology. Their complex shapes, however, have changed.

How should we compare them? The answer is to remember which horizontal and vertical loops on the first torus correspond to those on the second. This additional information is a *marking*. Once it is fixed, quasiconformal maps can measure precisely how much a complex structure has been stretched. Classical Teichmueller theory is built around this simple but powerful idea: change the complex structure while retaining enough common information to compare the results.

Arithmetic deformation leads to a more unsettling version of the same question. Suppose two constructions retain chosen multiplicative, Galois-theoretic and theta-theoretic information, but there is no prescribed isomorphism of the ambient rings. What is the analogue of a marking, and what sort of numerical comparison can still be meaningful? The Hodge theaters, theta-links and log-links of inter-universal Teichmueller theory give a language for attempting such comparisons. The classical example will help us see exactly where the analogy is useful and where its limits lie.

§158.2 Teichmueller Theory and Markings

Let S be an oriented topological surface of finite type. To remember which complex surface is supposed to be a deformation of S, one uses a marking.

Definition 158.2.1 (Marked Riemann surface)**.** A marked Riemann surface modeled on S is a pair

$$(X, f),$$

where X is a Riemann surface and

$$f : S \longrightarrow X$$

is an orientation-preserving homeomorphism, or more generally a quasiconformal map, regarded up to homotopy.

The marking is the key device. Without it, one has only a collection of complex curves. With it, different complex structures can still be compared as structures on the same underlying topological surface.

Definition 158.2.2 (Teichmueller space)**.** The Teichmueller space $T(S)$ is the set of equivalence classes of marked Riemann surfaces (X, f), where

$$(X, f) \sim (Y, g)$$

if there is a biholomorphism $h : X \to Y$ such that $h \circ f$ is homotopic to g.

Thus classical Teichmueller theory separates two layers:

$$\text{topological surface } S \qquad \text{and} \qquad \text{complex structure on } S.$$

The topology is the reference frame. The complex structure is what moves.

Remark 158.2.3 (The role of marking) — A marking is an authorized identification. It tells us when two points of Teichmueller space represent two complex structures on the same underlying surface. This will be the main contrast with IUT: IUT cannot simply use a single marking to identify all ring structures across its theaters.

§158.3 Distortion is measured in one universe

Teichmueller theory does not only parametrize complex structures. It measures deformation by quasiconformal distortion.

Definition 158.3.1 (Quasiconformal distortion)**.** Let $f : X \to Y$ be a quasiconformal map between Riemann surfaces. In a local coordinate it has distributional derivatives f_z and $f_{\bar{z}}$. Its Beltrami coefficient is

$$\mu_f = \frac{f_{\bar{z}}}{f_z},$$

and the maximal dilatation is

$$K(f) = \frac{1 + \|\mu_f\|_\infty}{1 - \|\mu_f\|_\infty}.$$

The Teichmueller distance is then defined by minimizing distortion among maps in the correct homotopy class:

$$d_T((X, f), (Y, g)) = \frac{1}{2} \inf_h \log K(h),$$

where $h : X \to Y$ ranges over quasiconformal maps such that $h \circ f$ is homotopic to g.

The important point is structural: this distance makes sense because the comparison is made inside one clear framework. Both surfaces are marked by the same S. The phrase "same underlying topological surface" has a precise meaning.

Proposition 158.3.2 (Classical deformation has a common reference object)
In classical Teichmueller theory, the comparison of two complex structures is mediated by the fixed surface S. The marking supplies a legitimate way to say that the two complex structures are structures on the same object.

Proof. A point of $T(S)$ is not merely a Riemann surface X; it is a pair (X, f). Given two points (X, f) and (Y, g), the maps f and g identify both surfaces with the same topological source S, up to homotopy. Therefore quasiconformal maps $h : X \to Y$ can be tested against the homotopy condition $h \circ f \simeq g$. The fixed S is the common reference object. □

For example, take the same topological torus $S = \mathbb{R}^2/\mathbb{Z}^2$ and give it two complex structures:

$$X_i = \mathbb{C}/(\mathbb{Z} + i\mathbb{Z}), \qquad X_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z}), \quad \tau = \frac{1}{3} + i.$$

Both have a perfectly concrete marking from S:

$$f_\tau([a, b]) = [a + b\tau].$$

One is drawn with a square fundamental domain, the other with a sheared parallelogram. The horizontal and vertical loops of S are still remembered as the two marked fundamental cycles.

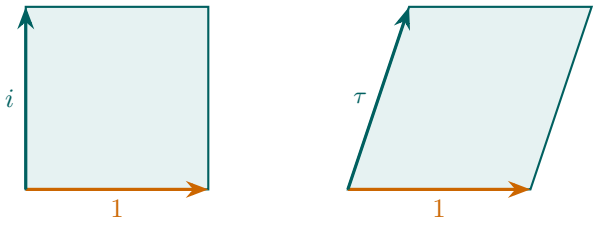


Figure 158.1: Two shapes of a marked torus. The colored arrows still identify the same two topological cycles, so a comparison is possible without pretending the two complex structures coincide.

The obvious real-linear map between the two fundamental parallelograms is

$$F(x + iy) = x + \frac{y}{3} + iy = \left(1 - \frac{i}{6}\right) z + \frac{i}{6}\bar{z}.$$

It sends the marked generators $1, i$ to $1, \tau$. Its Beltrami coefficient has size

$$|\mu_F| = \left|\frac{i/6}{1 - i/6}\right| = \frac{1}{\sqrt{37}}, \qquad K(F) = \frac{\sqrt{37} + 1}{\sqrt{37} - 1}.$$

This is the distortion of one explicit comparison map, not an assertion that it minimizes Teichmueller distance. It is enough to see how a marking makes a quantitative question possible: the two structures have changed, but we know exactly which cycles are being compared.

§158.4 Why the arithmetic analogy cannot be literal

In the arithmetic setting, the objects are not Riemann surfaces equipped with varying complex structures. The basic geometric objects are hyperbolic curves, elliptic curves with punctures or orbifold quotients, number fields, local fields, arithmetic fundamental groups, and Galois groups. The rigid structure is no longer a topological surface. It is closer to a package of fundamental-group-like data from which ring structures can be reconstructed.

Definition 158.4.1 (Arithmetic fundamental-group reference data, schematic)**.** For a hyperbolic curve X over a number field K, the arithmetic fundamental group fits into an exact sequence

$$1 \longrightarrow \pi_1^{\mathrm{geom}}(X) \longrightarrow \pi_1(X) \longrightarrow G_K \longrightarrow 1.$$

In the IUT entrance, such group-theoretic data acts as a rigid reference structure from which arithmetic and ring-theoretic data may be reconstructed by mono-anabelian methods.

The analogy with classical Teichmueller theory is therefore shifted:

classical theory	:	fixed topological type, variable complex structure,
IUT direction	:	rigid group-theoretic data, deformed ring-theoretic presentations.

This is only an analogy, but it is a mathematically meaningful one.

Remark 158.4.2 (What should not be inferred) — The word "Teichmueller" does not mean that IUT constructs a Teichmueller space of number fields in the classical sense. It means that the theory is organized around deformation, rigidity, and comparison of structures, but the structures being compared are arithmetic and group-theoretic rather than complex-analytic.

§158.5 Hodge theaters as arithmetic stages

The language of theaters is one of the points where IUT becomes unfamiliar. A theater should not be imagined as a parallel universe in a science-fiction sense. It is a package of data within which certain arithmetic, geometric, and group-theoretic structures can be discussed internally.

Definition 158.5.1 (Hodge theater, working schematic)**.** In this note, a Hodge theater means a package of arithmetic data containing several mutually related layers: local fields, number-field data, arithmetic fundamental groups, monoid-theoretic data, and theta-data. The full definition in IUT is much more elaborate. The working point is that a theater provides an internal stage on which ring structures, Galois data, theta-values, and reconstruction algorithms coexist.

The word "theater" is useful because IUT does not simply compare two rings by a ring isomorphism. It compares data internal to one theater with data internal to another theater through special links.

Definition 158.5.2 (Inter-universal comparison, schematic)**.** An inter-universal comparison is a comparison between data belonging to distinct Hodge theaters, where one does not assume a global ring-theoretic identification between the theaters. The comparison is mediated by structures such as theta-links, log-links, Kummer correspondences, and mono-anabelian reconstruction.

In ordinary algebraic geometry, the most comfortable comparison has the form

$$R \cong R'$$

or at least a morphism of schemes. In IUT, the central horizontal comparison is not of that kind. The theta-link is precisely a link that deforms ring-theoretic structure rather than preserving it.

§158.6 Etale-like and Frobenius-like directions

Fesenko's survey uses the distinction between structures that behave like rigid etale or fundamental-group data and structures that behave more like monoid-theoretic or Frobenius-type data. This is one of the cleanest ways to understand why ordinary scheme morphisms are not enough.

Definition 158.6.1 (Etale-like and Frobenius-like structures, schematic)**.** An etale-like structure is a structure on the fundamental-group side: it is functorial, rigid, and invariant enough to be used for reconstruction. A Frobenius-like structure is a monoid-theoretic structure used to build deformation links, especially when no ordinary Frobenius morphism of number fields exists.

The terminology is suggestive. Over a field of positive characteristic, the Frobenius map $x \mapsto x^p$ is a genuine morphism. For number fields, there is no such global Frobenius endomorphism. IUT instead builds categorical and monoid-theoretic analogues that can play a deformation role.

Remark 158.6.2 (The arithmetic reason this is radical) — The theory tries to get a Frobenius-like deformation effect in a setting where the usual Frobenius morphism does not exist. This is part of the reason the comparison cannot remain inside ordinary scheme-theoretic geometry.

§158.7 The theta-link is the point where one universe breaks

The local model already appeared in the previous two notes:

$$q_v \longmapsto q_v^{m^2}, \qquad u \longmapsto u.$$

This operation is meaningful on the multiplicative/monoid side, but it is not a ring homomorphism. It therefore cannot serve as an identification of local fields as rings.

The toy transformation from the preceding note captures the difficulty in a form we can check:

$$T(5^n u) = 5^{4n} u, \qquad T(1+5) = 6, \quad T(1) + T(5) = 626.$$

The first formula respects multiplication, but the last two numbers show that it does not transport ordinary addition. A theta-link is a far more sophisticated comparison built from theta-values and multiplicative data; it is useful to remember this tiny counterexample whenever an arrow between theaters starts to look like an ordinary map of rings.

This also explains the difference between two pictures we might otherwise confuse. In the marked-torus example, the common topological marking tells us how to compare points and cycles before measuring distortion. Across a theta-link, one has *selected* structures in common, not an all-purpose identification of the underlying arithmetic rings. A comparison that silently supplies a ring isomorphism would be using more information than the construction is meant to carry.

This is the exact mathematical sense in which the "one universe" picture breaks. If one tries to force the two sides of the theta-link into the same ring-theoretic universe, one loses the deformation that the theta-link was built to express.

§158.8 The log-link does not restore naive equality

The log-link exists because the theta-link moves multiplicative data while the final Diophantine inequality needs additive and volume information. The nonarchimedean logarithm

$$\log : \mathcal{O}_L^{\times} \longrightarrow L$$

translates unit data into additive data. This does not mean that the original ring structure has been transported unchanged.

Definition 158.8.1 (Recovered additive data, schematic)**.** Recovered additive data means additive structure obtained after applying logarithmic and mono-anabelian reconstruction procedures to multiplicative or group-theoretic data. It is not the same thing as an additive structure transported by a ring isomorphism.

The distinction is small in words but large in mathematics. Classical Teichmueller theory deforms complex structures while markings keep track of the same underlying topological surface. IUT deforms arithmetic structures in a way that prevents one from identifying the two sides as the same ring. Additive information must therefore be reconstructed rather than simply carried over.

Remark 158.8.2 (Why the log-shell is needed) — The log-link and the theta-link do not give a clean commutative square of elementwise identifications. The log-shell provides a controlled container for the resulting ambiguity. It is a way of replacing exact equality by a bounded or contained region in which the relevant images live.

§158.9 Classical marking versus multiradial reconstruction

The most useful comparison between classical Teichmueller theory and IUT is not a slogan about deformation. It is a comparison between two ways of controlling identity.

The comparison can be kept in five pairs.

1. Classical theory chooses a fixed topological surface S; IUT treats arithmetic fundamental-group-like data as rigid and reconstructive.
2. Classical theory uses a marking $f : S \to X$; IUT uses mono-anabelian algorithms to recover ring-theoretic data from group-theoretic input.
3. Classical theory moves the complex structure on S; IUT moves a ring-theoretic or monoid-theoretic presentation of arithmetic data.
4. Classical theory measures distortion by quasiconformal maps; IUT creates and measures arithmetic deformation by theta-links and log-links.
5. Classical comparison remains inside one marked topological setting; IUT comparison moves across distinct Hodge theaters.

This list shows why IUT is not merely difficult notation. The role played by marking in classical Teichmueller theory is not replaced by one simple arithmetic marking. Instead, it is replaced by a much more elaborate system: reconstruction algorithms, label symmetries, Kummer theory, log-shells, and indeterminacies.

Definition 158.9.1 (Multiradial identification, schematic)**.** A multiradial identification is not a direct equality of objects in one common ring universe. It is an identification of reconstructed structures that remains meaningful when viewed from several theaters or spokes at once.

This is why multiradiality is conceptually central. If a construction only makes sense from one spoke, it may depend on a hidden choice of coordinates. A multiradial construction is designed to survive the change of theater.

§158.10 The mathematical source of the controversy

This note should not become a news summary. The mathematical tension can be stated without naming any episode. It is a tension about legitimate identification.

A small thought experiment shows where the burden lies. Suppose each of two theaters produces a numerical set of possible logarithmic images, A and B. If we had a canonical identification of all arithmetic elements, we could attempt to pair their individual values. Without it, we can still ask a weaker question: do the images fit inside a common region C, and how large does C have to be?

$$A \subseteq C, \qquad B \subseteq C, \qquad \text{estimate the size of } C.$$

This is the intuitive role of the log-shell and the associated indeterminacies. It preserves a possibility for *inequalities* even if element-by-element equalities are unavailable. The construction of a suitable container and, especially, the final bound on its size require the actual IUT machinery. The little set diagram cannot establish that estimate; it explains what kind of statement the machinery is trying to make possible.

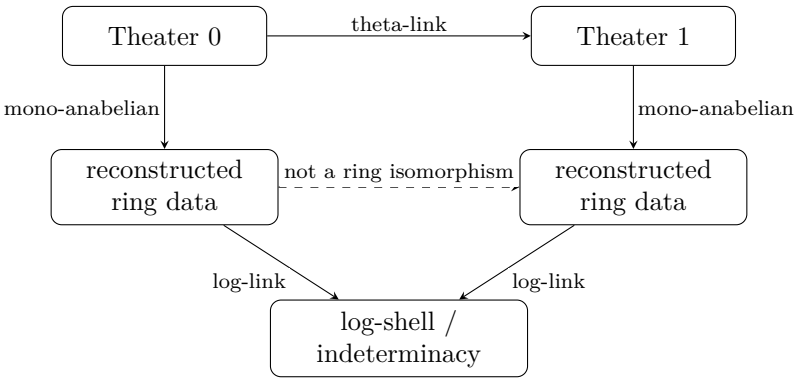
Dupuy and Hilado draw a helpful boundary in their account of initial theta data. For example, starting from the curve $11a1$ they show how one can choose the auxiliary elliptic-curve and torsion data needed to *state* Mochizuki's inequality. That explicit input is valuable ordinary arithmetic geometry. It leaves the core inter-theater comparison, and its proposed global bound, as a separate mathematical question. The distinction prevents an understandable example from being mistaken for a proof of the final Diophantine claim.

The controversial foundational move is therefore not merely "there are many universes". A vague multiverse would have no mathematical force. The real move is this: do not identify rings directly; instead recover and compare selected structures by controlled algorithms. The difficulty is that the replacement mechanisms must be strong enough to support the final log-volume inequality. If they are too weak, the deformation cannot be measured. If they are interpreted as ordinary equalities, the deformation disappears.

Remark 158.10.1 (Why a one-universe reading is dangerous) — A one-universe reading tries to place all theaters into a single background in which ordinary ring-theoretic equality is available. That may make the theory look more familiar, but it changes the object under study. The theta-link is designed precisely to avoid being a ring-theoretic identification.

§158.11 A compact model of the logic

The logical structure of the comparison can be summarized as follows:



The dashed arrow is the temptation that the theory resists. The comparison does not proceed by declaring the two reconstructed rings equal. It proceeds by controlling how their relevant images sit inside the logarithmic and theta-theoretic structures.

§158.12 What remains comparable between different theaters

For the two marked tori X_i and X_τ, we could follow the same two homology cycles through the shear and calculate its dilatation. The marking told us exactly which objects corresponded. Our formula for the shear even made that comparison numerical:

$$F(z) = \left(1 - \frac{i}{6}\right) z + \frac{i}{6}\bar{z}, \qquad K(F) = \frac{\sqrt{37}+1}{\sqrt{37}-1}.$$

All the coordinates involved belonged to a common topological surface.

The arithmetic comparison has a different kind of anchor. A theta-link retains carefully specified multiplicative and theta-theoretic information; Galois groups and Kummer theory record how roots can be compared; log-links and log-shells help organize the ambiguity in the remaining information. These structures make certain comparisons possible, even though a ring isomorphism carrying every addition from one theater into another has not been supplied.

A numerical inequality is meaningful only after the allowable correspondences and the effect of their ambiguities have been controlled. This is where the principal mathematical difficulty lies. The shared idea with classical Teichmueller theory is a measured deformation; the striking new constraint is that one must first construct a language in which the measurement can legitimately be compared across theaters.

When Roots Walk Around Holes

N-20261030

Choose a square root of 1, say $+1$, and let the number under the square root move once around the origin in the complex plane. Keep following the same square root continuously. By the time the input returns to 1, the output has become -1. The input is back where it started, but the root remembers the journey.

There is a very concrete way to make sense of this. We can replace the troublesome multivalued expression by an ordinary function on a space with more than one sheet. Closed paths on the base space then act on the possible values of the function. It is a short route from elementary complex numbers to covering spaces, fundamental groups and permutations of polynomial roots, and eventually to the Galois group of an equation. Kuga's *Galois' Dream* develops this geometric viewpoint through coverings, monodromy and differential equations.[1]

§159.1 One turn changes a square root

For a path that goes once around the origin, put

$$z(\theta) = e^{i\theta}, \qquad 0 \le \theta \le 2\pi.$$

At every θ there are two solutions of $w^2 = z(\theta)$. Starting from $w(0) = 1$, the continuous choice is

$$w(\theta) = e^{i\theta/2}. \tag{159.1}$$

It satisfies $w(\theta)^2 = z(\theta)$ at every instant. But

$$w(2\pi) = e^{i\pi} = -1 \neq w(0).$$

A second circuit returns the root to its original value. The two local square-root branches are exchanged by going once around the origin.

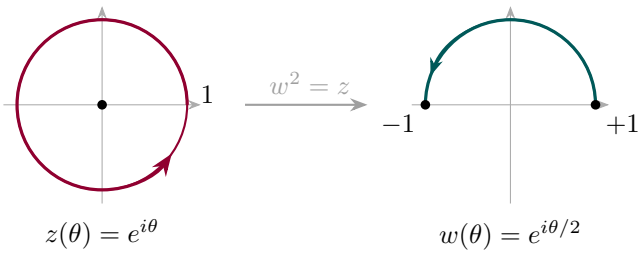


Figure 159.1: The input completes a circle while the continuously followed root completes only a semicircle. The two endpoints correspond to different square roots of the same input.

There is a simple obstruction to choosing a continuous square root on all of $\mathbb{C}^\times = \mathbb{C} \setminus \{0\}$. If such a function s existed with $s(z)^2 = z$, then applying it around the unit circle would give a continuous choice of w satisfying (159.1) and returning to its starting

[1] Michio Kuga, *Galois' Dream: Group Theory and Differential Equations* (Birkhäuser, 1993), especially the chapters on covering surfaces, their fundamental groups, and function theory on covering spaces. The explicit polynomial computations in the present note are carried out below.

value. The only continuous choices along that circle are $e^{i\theta/2}$ and $-e^{i\theta/2}$, and each ends at the opposite value. So no globally continuous branch exists.

The issue is the missing point at 0. On a simply connected region avoiding 0, we can choose a continuous logarithm and define

$$\sqrt{z} = \exp\left(\frac{1}{2}\log z\right).$$

The branch changes when the chosen logarithm gains $2\pi i$ after a circuit.

§159.2 The sheets form an ordinary space

Rather than treating the square root as a defective function, construct the space

$$E = \{(z, w) \in \mathbb{C}^\times \times \mathbb{C}^\times : w^2 = z\}$$

and the projection

$$p : E \longrightarrow \mathbb{C}^\times, \qquad (z, w) \longmapsto z.$$

Every z has exactly two preimages, $(z, \sqrt{z})$ and $(z, -\sqrt{z})$. The space E is naturally identified with $\mathbb{C}^\times$ through

$$w \longmapsto (w^2, w),$$

so p becomes the map $w \mapsto w^2$.

If $z_0 \neq 0$, choose a small disk D around z_0 avoiding the origin. The disk admits two continuous, indeed holomorphic, branches of the square root. Its inverse image under p splits into two disjoint open sets, each mapped homeomorphically to D. That is exactly the local property of a *two-sheeted covering map*.

A path in the base has a unique lift through each chosen starting point of the fiber. The function $w(\theta)$ in (159.1) is the lift of $z(\theta)$ that starts at $w = 1$.

An element of the fiber over $z = 1$ is simply a choice from $\{+1, -1\}$. Lifting a closed path permutes these two choices. A loop making k positive turns around the origin induces

$$w \longmapsto (-1)^k w. \tag{159.2}$$

Why does only the number of turns matter? Two loops in $\mathbb{C}^\times$ that can be continuously deformed into one another through loops avoiding 0 have endpoint-compatible lifts. Their winding number,

$$\operatorname{wind}(\gamma, 0) = \frac{1}{2\pi i}\oint_\gamma \frac{dz}{z} \in \mathbb{Z},$$

classifies their homotopy classes. Thus

$$\pi_1(\mathbb{C}^\times, 1) \cong \mathbb{Z}, \qquad k \longmapsto (w \mapsto (-1)^k w).$$

For $w^n = z$, the same argument gives

$$w(\theta) = e^{i\theta/n}, \qquad w(2\pi) = e^{2\pi i/n} w(0).$$

The n roots of unity are permuted cyclically. We obtain an action of $\mathbb{Z}$ on the n sheets through the cyclic group of order n.

The map $w \mapsto -w$ is an automorphism of the square-root covering: it exchanges the sheets and commutes with p. It is often called a *deck transformation*. In the square-root example, the loop permutation agrees with a deck transformation. That agreement requires care in general: a covering may have far more monodromy permutations than deck transformations.

§159.3 Following the roots of a polynomial

Let $P(z)$ be a polynomial of degree d, and consider its solutions as a parameter t varies:

$$P(z) = t.$$

A root z_0 of $P(z) - t_0$ is *simple* when $P'(z_0) \neq 0$. The holomorphic implicit-function theorem then supplies a unique holomorphic root $z(t)$ near t_0, satisfying

$$z(t_0) = z_0, \qquad P(z(t)) = t.$$

Differentiating the identity gives

$$z'(t) = \frac{1}{P'(z(t))}. \tag{159.3}$$

So the roots can move holomorphically as long as they stay simple.

The possible failures occur at critical values

$$B = \{P(c) : P'(c) = 0\}.$$

Remove these finitely many values from the parameter plane and write

$$X = \mathbb{C} \setminus B, \qquad E_P = \{(t, z) \in X \times \mathbb{C} : P(z) = t\}.$$

The projection $E_P \to X$ is a d-sheeted covering map. Every parameter in X has exactly d distinct roots, and every sufficiently small disk in X allows all of them to be labeled by holomorphic functions.

A path $t(s)$ in X can therefore be lifted to a continuously varying root $z(s)$. If the path is closed, its endpoint may be another root of the same equation. Fix $t_0 \in X$ and label the roots above it $r_1, \dots, r_d$. Every loop at t_0 produces a permutation of these labels, and homotopic loops produce the same permutation. This yields a *monodromy representation*

$$\rho : \pi_1(X, t_0) \longrightarrow S_d. \tag{159.4}$$

We choose a consistent convention for composing paths and permutations; a different convention merely reverses the order of products. The image of ρ is the monodromy group of the polynomial family.

§159.4 A cubic with two collisions

Take the explicit family

$$z^3 - 3z = t. \tag{159.5}$$

The critical points of $P(z) = z^3 - 3z$ are the solutions of

$$P'(z) = 3(z^2 - 1) = 0, \qquad z = \pm 1.$$

Their critical values are

$$P(-1) = 2, \qquad P(1) = -2.$$

Consequently the three roots are distinct for every

$$t \in X := \mathbb{C} \setminus \{-2, 2\}.$$

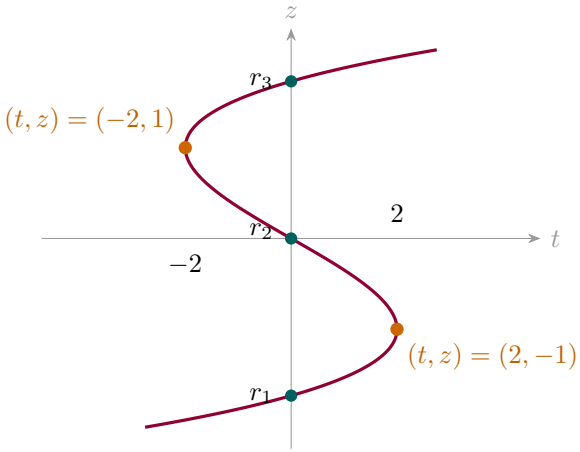


Figure 159.2: The real curve $t = z^3 - 3z$, with t drawn horizontally. Between the two critical parameters, a vertical line meets three real branches. The first two collide at $t = 2$, and the last two at $t = -2$. Following a complex loop around either critical parameter exchanges the corresponding pair.

Start at $t_0 = 0$. The three roots are particularly simple:

$$r_1 = -\sqrt{3}, \qquad r_2 = 0, \qquad r_3 = \sqrt{3}.$$

For real $-2 < t < 2$, all three roots remain real and distinct. We can follow them by increasing or decreasing t without crossing a collision.

At $t = 2$, the polynomial factors as

$$z^3 - 3z - 2 = (z + 1)^2(z - 2).$$

Thus r_1 and r_2 collide at $z = -1$, while r_3 tends to 2. To see what happens around that collision, write $z = -1 + u$. An exact expansion gives

$$t = 2 - 3u^2 + u^3. \tag{159.6}$$

For small u, the leading relation is

$$u^2 \sim \frac{2 - t}{3}.$$

In fact this resemblance to a square root can be made exact. Choose a holomorphic branch of $g(u) = \sqrt{u - 3}$ near $u = 0$, which is possible because $u - 3 \neq 0$ there. The change of local coordinate $v = u\, g(u)$ has nonzero derivative at the origin and satisfies $v^2 = t - 2$. Thus $v = \pm\sqrt{t - 2}$ are the two local branches, and a circuit of t around 2 exchanges them. Since $u \mapsto v$ is locally invertible, it also exchanges the corresponding two roots:

$$\gamma_2 : \quad (r_1\, r_2),$$

where γ_2 is a small positive loop around $t = 2$, connected to the base point 0 along a path inside X.

At $t = -2$, we have

$$z^3 - 3z + 2 = (z - 1)^2(z + 2).$$

Here r_2 and r_3 collide at $z = 1$. Writing $z = 1 + u$ gives

$$t = -2 + 3u^2 + u^3. \tag{159.7}$$

Here the exact local coordinate is $v = u\sqrt{3+u}$, for which $v^2 = t+2$. A small circuit about -2 exchanges the two square-root branches, and now the induced permutation is

$$\gamma_{-2} : \quad (r_2\, r_3).$$

These two transpositions generate the entire symmetric group S_3:

$$(12),\ (23),\ (12)(23) = (123), \qquad \langle (12), (23) \rangle = S_3.$$

Since the fundamental group of the plane punctured at ± 2 is generated by loops around these punctures, the monodromy group of (159.5) is exactly

$$\boxed{\mathrm{Mon}(z^3 - 3z - t) = S_3.} \tag{159.8}$$

With the roots labeled at $t = 0$, the group consists of all six permutations. We did not assume an abstract symmetry in advance: the two visible collisions in the parameter plane produced the group.

§159.5 The discriminant remembers the parity

There is an algebraic way of detecting the collisions, without plotting the roots. For a monic cubic

$$z^3 + pz + q,$$

its discriminant is

$$\Delta = -4p^3 - 27q^2.$$

With $p = -3$ and $q = -t$, we find

$$\Delta(t) = 27(4 - t^2). \tag{159.9}$$

It vanishes precisely at $t = \pm 2$, as the derivative calculation predicted.

Why is this a natural expression? If $r_1(t), r_2(t), r_3(t)$ are the three local root branches, put

$$D(t) = (r_1 - r_2)(r_1 - r_3)(r_2 - r_3).$$

Then

$$D(t)^2 = \Delta(t).$$

Exchanging any two roots changes the sign of D; an even permutation preserves it. Thus the two possible values of $\sqrt{\Delta(t)}$ keep track of the parity of the permutation obtained by continuation. A circuit around either simple zero of Δ changes the sign of its square root, just as our local root analysis predicted a transposition.

It is worth distinguishing two groups at this point. A loop in the parameter plane gives a permutation by *analytic continuation*. Galois theory begins with an algebraic extension of the rational-function field $\mathbb{C}(t)$ and asks for its field automorphisms. For algebraic functions of one complex variable, these descriptions agree after one passes to the splitting field.

Theorem 159.5.1 (Algebraic monodromy principle)

Let $F(z,t) \in \mathbb{C}(t)[z]$ be a separable polynomial, and remove from the complex parameter line the finitely many values where its roots are singular, collide, or the degree drops. Analytic continuation of the local root branches along loops produces a permutation group naturally isomorphic, as a permutation group of the roots, to the Galois group of the splitting field of F over $\mathbb{C}(t)$.

The comparison rests on the algebraic-function covering: continuations of algebraic branches act on their field of functions, while globally single-valued meromorphic algebraic functions of t are rational. Proving the general theorem uses the correspondence between finite branched coverings of Riemann surfaces and finite algebraic extensions of their function fields. In this example we can verify the conclusion independently, using only ordinary polynomial algebra.

The polynomial $z^3 - 3z - t$ is irreducible over $\mathbb{C}(t)$. Indeed, it is irreducible in $\mathbb{C}[z, t]$: because it is linear in t with coefficient -1, any nontrivial factorization would have a factor independent of t, and that factor would divide the unit -1. Gauss's lemma then gives irreducibility over $\mathbb{C}(t)$. Its Galois group is therefore a transitive subgroup of S_3.

The discriminant $\Delta(t) = 27(4 - t^2)$ is not a square in $\mathbb{C}(t)$: it has a zero of odd order at each of $t = \pm 2$, whereas a square rational function has zeros and poles of even order. Thus the Galois group contains an odd permutation and is not contained in the alternating group A_3. The only transitive subgroups of S_3 are A_3 and S_3. Therefore

$$\mathrm{Gal}\big(\mathrm{Spl}_{\mathbb{C}(t)}(z^3 - 3z - t)/\mathbb{C}(t)\big) \cong S_3, \tag{159.10}$$

in agreement with our monodromy calculation.

§159.6 A free group that produces six permutations

The parameter domain of the cubic,

$$X = \mathbb{C} \setminus \{-2, 2\},$$

has two punctures. It deformation retracts onto a figure-eight graph, so

$$\pi_1(X, 0) \cong F_2,$$

the free group on two generators. One generator can be taken to wind once around 2, the other once around -2. There is no relation forcing them to commute.

Our explicit continuation calculation defines a surjective homomorphism

$$F_2 \longrightarrow S_3, \qquad a \longmapsto (12), \qquad b \longmapsto (23).$$

The source contains infinitely many distinct homotopy classes of loops. The target contains only six permutations, because most of those different journeys have exactly the same effect on the three roots. For example, going twice around either individual branch point returns each root to itself.

The earlier distinction between monodromy and deck transformations can now be made concrete. A deck transformation $h : E_P \to E_P$ satisfies $p \circ h = p$. It sends every fiber to itself and commutes with lifting loops, so its permutation of the fiber over 0 must commute with the entire monodromy group S_3. The centralizer of S_3 in its natural action on three letters is trivial. Therefore this cubic covering has no nonidentity deck transformation, even though its monodromy group is all of S_3.

The square-root covering was quite different: its two sheets could be exchanged by the globally defined map $w \mapsto -w$, giving a nontrivial deck transformation. This is why it is useful to regard a covering as more than a collection of sheets with arbitrary labels.

§159.7 From two branch points to every permutation

The cubic suggests a more ambitious experiment. For an integer $n \geq 2$, set

$$P_n(z) = z^n - nz, \qquad P_n(z) = t. \tag{159.11}$$

The derivative is

$$P_n'(z) = n(z^{n-1} - 1).$$

Its critical points are the $(n-1)$st roots of unity ζ, and at each one

$$P_n(\zeta) = \zeta^n - n\zeta = -(n-1)\zeta,$$

because $\zeta^{n-1} = 1$. These critical values are distinct. Moreover,

$$P_n''(\zeta) = n(n-1)\zeta^{n-2} \neq 0,$$

so every critical point is simple. The local expansion near ζ has a nonzero quadratic leading term,

$$t - P_n(\zeta) = \frac{1}{2}P_n''(\zeta)(z-\zeta)^2 + O((z-\zeta)^3).$$

Therefore a loop around each critical value interchanges exactly two roots and fixes the others. Its monodromy is a transposition.

We can determine the full monodromy group without explicitly following all n roots.

Theorem 159.7.1

For every integer $n \geq 2$, the polynomial $z^n - nz - t$ has monodromy group S_n over the complex parameter plane with its critical values removed. Its Galois group over $\mathbb{C}(t)$ is also S_n.

Proof. Let B be the finite set of critical values above and let $X = \mathbb{C} \setminus B$. The total space

$$E_n = \{(t, z) \in X \times \mathbb{C} : P_n(z) = t\}$$

is isomorphic to the punctured plane $\mathbb{C} \setminus P_n^{-1}(B)$, by the map $z \mapsto (P_n(z), z)$. This space is path connected: removing finitely many points from $\mathbb{C}$ does not disconnect it. Hence the monodromy action on the n points of a fiber is *transitive*. Indeed, a path in the total space between two fiber points projects to a loop in the base whose lift carries one point to the other.

The fundamental group of the plane minus finitely many punctures is generated by loops around those punctures. We have seen that each such generator acts as a transposition. Therefore the monodromy group is a transitive subgroup of S_n generated by transpositions.

Here is a useful elementary group-theoretic lemma. Given a collection of transpositions on n letters, draw a graph whose vertices are the letters and whose edges join the two letters exchanged by each transposition. The generated group preserves each connected component of that graph. Conversely, transpositions along the edges of a connected graph generate the entire symmetric group on its vertices: follow paths to obtain any exchange, or choose a spanning tree and proceed inductively. Thus a transitive group generated by transpositions is necessarily S_n. Our monodromy group is S_n.

The polynomial $z^n - nz - t$ is irreducible in $\mathbb{C}(t)[z]$ by the same linear-in-t argument used for the cubic. The algebraic monodromy principle identifies its Galois group, as a permutation group on the roots, with S_n. □

At $n = 5$, we obtain a concrete one-parameter quintic

$$z^5 - 5z = t$$

with Galois group S_5 over $\mathbb{C}(t)$. The group S_5 is not solvable: its alternating subgroup A_5 is nonabelian and simple. By the classical Galois criterion, the roots of this polynomial cannot be expressed *as functions of the indeterminate* t using finitely many field operations and radical extractions over $\mathbb{C}(t)$.

That last qualification matters. Special values of t may produce equations with very simple roots; for instance $t = 0$ gives $z(z^4 - 5) = 0$. The obstruction concerns a formula valid for the generic parameter. We have watched the obstruction emerge from entirely explicit, ordinary square-root exchanges around the critical values.

§159.8 Differential equations remember loops too

There is a related viewpoint that makes Kuga's inclusion of differential equations in the story feel less mysterious. If an algebraic root branch satisfies

$$P(z(t)) = t,$$

then away from the critical values it satisfies the differential equation

$$\frac{dz}{dt} = \frac{1}{P'(z)}.$$

Different initial values corresponding to different roots give different local solutions. Continuing those solutions around a branch point permutes the initial values. For the cubic, this is the equation

$$\frac{dz}{dt} = \frac{1}{3z^2 - 3}.$$

Its singular behavior as $z \to \pm 1$ is exactly the local collision that produced the transpositions.

Linear differential equations display a close cousin of this phenomenon. On $\mathbb{C}^\times$, consider

$$\frac{dy}{dz} = \frac{\alpha}{z}\, y, \tag{159.12}$$

where $\alpha \in \mathbb{C}$ is fixed. On a simply connected region avoiding 0, its solutions are

$$y(z) = c \exp(\alpha \log z) = c z^\alpha.$$

After one positive circuit about 0, the logarithm gains $2\pi i$, so a continued solution changes by

$$y \longmapsto e^{2\pi i \alpha} y.$$

For $\alpha = \frac{1}{2}$, this factor is -1, and we recover the original square-root example as a first-order differential equation.

For a linear system $Y'(z) = A(z)Y(z)$ of rank r, local solutions form an r-dimensional vector space. Analytic continuation along closed paths acts by invertible linear maps on this space, giving a representation

$$\pi_1(X, z_0) \longrightarrow \mathrm{GL}_r(\mathbb{C}).$$

In the polynomial-root examples, the action was by permutations of a finite set; for linear differential equations the natural action is by matrices. Both record the same essential geometric event: a locally defined solution is transported around a hole and comes back transformed.

Twenty-Seven Lines on a Cubic Surface

N-20261109

A general surface of degree 3 in projective three-space contains exactly twenty-seven complex projective lines, provided the surface is smooth. The number has appeared in classical geometry for almost two centuries, and it looks unlikely at first sight. A plane contains far too many lines to count; a smooth quadric surface carries two entire one-parameter families of lines. Then, at degree three, the answer suddenly becomes a finite integer:

$$\boxed{27}.$$

The best way I know to make that number less mysterious is to leave three-dimensional space for a while and work in the projective plane. Six points, a small geometric surgery at each of them, and some elementary intersection arithmetic will produce every one of the twenty-seven lines. The geometry is classical; the modern language is that of blow-ups and del Pezzo surfaces.[1]

§160.1 Why a line on a cubic is special

Work over $\mathbb{C}$. A projective line in $\mathbb{P}^3$ is the set of one-dimensional linear subspaces of a fixed two-dimensional subspace of $\mathbb{C}^4$. A smooth cubic surface is the zero set

$$S = \{[x_0 : x_1 : x_2 : x_3] \in \mathbb{P}^3 : F(x_0, x_1, x_2, x_3) = 0\},$$

where F is homogeneous of degree three and its four partial derivatives never vanish simultaneously on S.

If we choose an arbitrary projective line $\ell \subset \mathbb{P}^3$, the restriction $F|_\ell$ is a homogeneous cubic in the line's two coordinates. For ℓ to lie *entirely* in S, all four coefficients of this restricted cubic must vanish.

The space of lines in $\mathbb{P}^3$, the Grassmannian $\mathrm{Gr}(2,4)$, has dimension four. Requiring four coefficients to vanish suggests a zero-dimensional set of solutions. This is a useful dimension count, though it does not prove that the set is finite or that its cardinality is 27.

An instructive comparison is a smooth quadric:

$$x_0 x_3 - x_1 x_2 = 0.$$

It is the image of the Segre map

$$([s_0 : s_1], [t_0 : t_1]) \longmapsto [s_0 t_0 : s_0 t_1 : s_1 t_0 : s_1 t_1],$$

so the surface is isomorphic to $\mathbb{P}^1 \times \mathbb{P}^1$. Holding either factor fixed gives a whole family of lines. The cubic surface is a different creature.

[1] For the classical cubic-surface geometry, see Igor Dolgachev, *Classical Algebraic Geometry: A Modern View* (2012), Chapters 8–9. The blow-up method and intersection-theoretic counting argument are also presented in Ravi Vakil's *Complex Algebraic Surfaces*, Class 14.

§160.2 Replacing a point by its directions

Take the affine plane with coordinates (x, y) and blow up the origin. The construction is surprisingly explicit:

$$\mathrm{Bl}_0 \mathbb{A}^2 = \{((x, y), [u : v]) \in \mathbb{A}^2 \times \mathbb{P}^1 : xv = yu\}. \tag{160.1}$$

There is a natural projection

$$\pi : \mathrm{Bl}_0 \mathbb{A}^2 \longrightarrow \mathbb{A}^2, \qquad ((x, y), [u : v]) \longmapsto (x, y).$$

Away from the origin, $[u : v] = [x : y]$ is uniquely determined. Over the origin, the relation $0 = 0$ imposes no condition, so

$$\pi^{-1}(0, 0) = \{(0, 0)\} \times \mathbb{P}^1.$$

The single point has been replaced by an entire projective line of directions. This new curve is the *exceptional divisor* E.

In the chart $u \neq 0$, write $s = v/u$. The blow-up equation becomes $y = xs$; the exceptional divisor is $x = 0$, and s records the slope of a line approaching the origin. Thus the point where several curves met is opened into a family of directions. Curves with distinct tangent directions now meet E at distinct points.

We can carry out this operation at six distinct points $p_1, \dots, p_6 \in \mathbb{P}^2$. Denote the resulting smooth surface by

$$\pi : S \longrightarrow \mathbb{P}^2.$$

The six exceptional curves are $E_1, \dots, E_6$.

Choose the points in *general position*: no three are collinear, and no conic passes through all six. These are the conditions that will make the anticanonical map an embedding into a smooth cubic surface.

§160.3 Counting using divisors

A divisor class on S can be expressed in terms of

$$H = \pi^*[\text{a line in } \mathbb{P}^2], \qquad E_1, \dots, E_6.$$

The intersection pairing is determined by

$$H^2 = 1, \qquad H \cdot E_i = 0, \qquad E_i \cdot E_j = -\delta_{ij}. \tag{160.2}$$

The sign in $E_i^2 = -1$ may look strange. Geometrically the exceptional curve is a projective line with a negative normal bundle; intersection theory keeps track of the fact that it came from contracting a point.

The canonical divisor satisfies the blow-up formula

$$K_S = \pi^* K_{\mathbb{P}^2} + \sum_{i=1}^{6} E_i = -3H + \sum_{i=1}^{6} E_i.$$

Put

$$A = -K_S = 3H - \sum_{i=1}^{6} E_i. \tag{160.3}$$

Then

$$A^2 = 9 - 6 = 3.$$

The linear system $|A|$ consists of plane cubic curves passing through our six points, with their proper transforms on S. Plane cubics form a ten-dimensional vector space of homogeneous forms; six general point conditions reduce the dimension to four. The resulting projective linear system maps S to $\mathbb{P}^3$.

A classical del Pezzo surface theorem says that, for six points in general position, the divisor A is very ample. Thus this map is an embedding. Its image has degree

$$\deg \varphi_A(S) = A^2 = 3,$$

so it is a smooth cubic surface in $\mathbb{P}^3$. We have found a way to *construct* a smooth cubic surface out of six points in a plane.

Moreover, the same classical classification theorem identifies every smooth cubic surface over $\mathbb{C}$ with a blow-up of six points in general position. The usual proof uses the existence and contraction of six mutually skew lines on the cubic. We take this geometric classification as an input; the numerical count below will be proved directly from the intersection pairing.

§160.4 Three kinds of curves

Once the six points have been chosen, three families of curves are immediately visible.

The six exceptional curves. Each blown-up point yields E_i, with

$$E_i^2 = -1, \qquad A \cdot E_i = 1.$$

Since its degree under the anticanonical embedding is one, E_i maps to a projective line in $\mathbb{P}^3$. We have six lines already.

The fifteen lines through pairs of points. For $1 \le i < j \le 6$, the unique plane line through p_i and p_j has proper transform

$$L_{ij} = H - E_i - E_j.$$

It has

$$L_{ij}^2 = 1 - 1 - 1 = -1, \qquad A \cdot L_{ij} = 3 - 1 - 1 = 1.$$

Thus each L_{ij} becomes a line on the cubic surface. There are

$$\binom{6}{2} = 15$$

of them.

The six conics through five points. For each i, the other five points determine a conic. Its proper transform has class

$$Q_i = 2H - \sum_{j \neq i} E_j.$$

General position ensures that it is irreducible and avoids the sixth point. Again,

$$Q_i^2 = 4 - 5 = -1, \qquad A \cdot Q_i = 6 - 5 = 1.$$

So these six conics also become projective lines under the anticanonical embedding.

We have produced

$$\underbrace{6}_{E_i} + \underbrace{15}_{L_{ij}} + \underbrace{6}_{Q_i} = 27 \tag{160.4}$$

distinct lines. The surprising number now has a very simple combinatorial origin. Yet a count of 27 *known* lines does not exclude the existence of a twenty-eighth. The exclusion is where intersection theory earns its keep.

§160.5 Why there cannot be a twenty-eighth

Let $C \subseteq S$ be any line in the cubic embedding, viewed as a curve on our blown-up plane. Because the embedding is given by $A = -K_S$, its degree is

$$A \cdot C = 1.$$

A line is a smooth rational curve, hence has genus 0. The adjunction formula reads

$$2g(C) - 2 = C^2 + K_S \cdot C.$$

Substituting $g(C) = 0$ and $K_S \cdot C = -1$ gives

$$C^2 = -1. \tag{160.5}$$

If C equals one of the exceptional curves E_i, it is already on our list. Otherwise C is an irreducible curve distinct from every E_i, so the intersection number $C \cdot E_i$ is nonnegative. Write its divisor class as

$$C = dH - \sum_{i=1}^{6} m_i E_i, \qquad d \in \mathbb{Z}, \quad m_i \in \mathbb{Z}_{\geq 0}.$$

The coefficients have a geometric interpretation: d is the degree of the curve's image in the plane, and m_i is its multiplicity at p_i. Because C is not exceptional, $d \geq 1$.

The two numerical conditions just obtained become

$$3d - \sum_{i=1}^{6} m_i = 1, \qquad d^2 - \sum_{i=1}^{6} m_i^2 = -1. \tag{160.6}$$

Thus

$$\sum_i m_i = 3d - 1, \qquad \sum_i m_i^2 = d^2 + 1.$$

Apply Cauchy–Schwarz to the six integers m_i:

$$\left(\sum_i m_i\right)^2 \leq 6 \sum_i m_i^2.$$

Substitution yields

$$(3d-1)^2 \leq 6(d^2+1), \qquad 3d^2 - 6d - 5 \leq 0.$$

For an integer $d \geq 1$, this forces

$$\boxed{d = 1 \quad \text{or} \quad d = 2.}$$

That bound is the decisive step. Every possible nonexceptional line on the cubic must arise from a plane curve of degree one or two.

If $d = 1$, the equations say

$$\sum_i m_i = 2, \qquad \sum_i m_i^2 = 2.$$

Since the m_i are nonnegative integers, $\sum_i m_i(m_i - 1) = 0$, so every m_i is 0 or 1. Exactly two of them equal 1. This recovers the fifteen classes $H - E_i - E_j$.

If $d = 2$, similarly,

$$\sum_i m_i = 5, \qquad \sum_i m_i^2 = 5.$$

Exactly five m_i equal 1, recovering the six conic classes $2H - \sum_{j\neq i} E_j$.

No further cases exist. The possibilities are exhausted by six exceptional curves, fifteen pair-lines and six five-point conics.

Theorem 160.5.1 (The twenty-seven lines)

Every smooth cubic surface in $\mathbb{P}^3_{\mathbb{C}}$ contains exactly 27 projective lines. Under its presentation as the blow-up of $\mathbb{P}^2$ at six points in general position, their classes are

$$\begin{aligned} E_i &\quad (1 \le i \le 6), \\ H - E_i - E_j &\quad (1 \le i < j \le 6), \\ 2H - \sum_{j\neq i} E_j &\quad (1 \le i \le 6). \end{aligned}$$

The proof has two logically distinct ingredients. The classical classification identifies an arbitrary smooth cubic with a six-point blow-up. Once that identification is made, the complete list and the absence of extra lines follow from the two intersection equations and one Cauchy–Schwarz inequality.

§160.6 The intersections are just arithmetic

The divisor classes do more than count the lines. They tell us precisely which pairs of lines meet. Distinct irreducible curves on a smooth surface have nonnegative intersection number, and in this configuration the number is either 0 or 1. Zero means the curves are disjoint; one means they meet at a single point, counted with multiplicity.

Let the symbols E_i, L_{ij}, Q_i denote the 27 lines as above. Using (160.2), we find

$$\begin{aligned} E_i \cdot E_j &= 0 && (i \neq j), \\ Q_i \cdot Q_j &= 0 && (i \neq j), \\ E_i \cdot Q_j &= \begin{cases} 0, & i = j, \\ 1, & i \neq j, \end{cases} \\ E_k \cdot L_{ij} &= \begin{cases} 1, & k \in \{i, j\}, \\ 0, & k \notin \{i, j\}, \end{cases} \\ L_{ij} \cdot Q_k &= \begin{cases} 1, & k \in \{i, j\}, \\ 0, & k \notin \{i, j\}, \end{cases} \\ L_{ij} \cdot L_{kl} &= \begin{cases} 1, & \{i, j\} \cap \{k, l\} = \varnothing, \\ 0, & |\{i, j\} \cap \{k, l\}| = 1, \end{cases} && (ij \neq kl). \end{aligned} \tag{160.7}$$

For instance, why do the two lines L_{12} and L_{34} meet? Their plane ancestors are lines through disjoint pairs of points. Two projective lines in the plane meet in one point, and that point is different from all six blow-up centers. The intersection survives the blow-up. On the other hand, L_{12} and L_{13} meet in the original plane at p_1. Blowing up p_1 separates their distinct tangent directions, so their proper transforms are disjoint. The algebra records this:

$$(H - E_1 - E_2) \cdot (H - E_3 - E_4) = 1, \qquad (H - E_1 - E_2) \cdot (H - E_1 - E_3) = 0.$$

Choose any exceptional line E_i. It meets five of the L_{ij}, one for each $j \neq i$, and five of the Q_j, again for $j \neq i$. So

$$\boxed{\text{every line meets exactly ten of the other 26 lines.}}$$

The same count follows for any L_{ij} or Q_i by the formulas, or from the symmetry discussed later. The graph whose vertices are the 27 lines and whose edges mean intersection is a highly regular finite graph: it has

$$\frac{27 \cdot 10}{2} = 135$$

edges.

§160.7 Forty-five tritangent planes

The intersection pattern contains another lovely number. A *tritangent plane* is a plane whose intersection with the cubic surface is the union of three lines:

$$S \cap \Pi = L_1 + L_2 + L_3.$$

Because a hyperplane section of S has class $A = -K_S$, three line classes form a tritangent section when their sum equals A.

For any two distinct indices i, j,

$$E_i + L_{ij} + Q_j = E_i + (H - E_i - E_j) + \left(2H - \sum_{k \neq j} E_k\right) = 3H - \sum_{k=1}^{6} E_k = A.$$

These three curves meet pairwise, so they indeed form a plane section. There are $6 \cdot 5 = 30$ choices of ordered pair (i, j), giving 30 tritangent planes.

There is one more family. Partition $\{1, 2, 3, 4, 5, 6\}$ into three unordered pairs, say $\{i, j\}, \{k, l\}, \{m, n\}$. Then

$$L_{ij} + L_{kl} + L_{mn} = A,$$

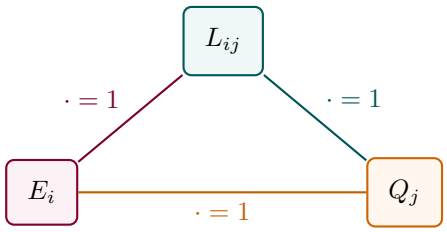
and the three lines meet pairwise because their index pairs are disjoint. The number of partitions into three unordered pairs is

$$\frac{6!}{(2!)^3\, 3!} = 15.$$

Together these give

$$\boxed{30 + 15 = 45 \text{ tritangent planes.}} \tag{160.8}$$

These are all the possibilities: any tritangent plane consists of three of our 27 line classes. If one is an E_i, the degree and intersection equations force the other two to have degrees 1 and 2 in the plane and to be L_{ij} and Q_j. If none is exceptional, the three possible degrees are 1 or 2, and their sum must equal the coefficient 3 of H in A; hence all three are L-type curves, giving a partition into three pairs.



$E_i + L_{ij} + Q_j = A$, for $i \neq j$

Figure 160.1: One of the 30 tritangent triples of mixed types. Its three lines meet pairwise and their divisor classes sum to a hyperplane section of the cubic surface.

§160.8 A double-six

Arrange the six exceptional lines and the six conic lines in two rows:

$$\begin{pmatrix} E_1 & E_2 & E_3 & E_4 & E_5 & E_6 \\ Q_1 & Q_2 & Q_3 & Q_4 & Q_5 & Q_6 \end{pmatrix}.$$

The lines within either row are mutually disjoint. Across the rows, E_i meets Q_j precisely when $i \neq j$; corresponding entries E_i, Q_i are disjoint. This arrangement is known as a *Schläfli double-six*.

It is quite a rigid object. Given one row of six mutually skew lines on a cubic surface, the complementary row in a double-six is determined. One can contract the first six lines to obtain a projective plane, and the second row then appears as the six conics through five of the contracted points.

There are 36 different double-sixes among the 27 lines. That number is a classical fact about the configuration, and it can be computed from its full symmetry group. One striking consequence is that the six-point blow-up presentation is far from unique: a single cubic surface admits many different choices of six skew lines to contract to a plane.

§160.9 Why an E_6 root system appears

The intersection form on divisor classes has signature $(1, 6)$:

$$\operatorname{Pic}(S) \cong \mathbb{Z}H \oplus \bigoplus_{i=1}^{6} \mathbb{Z}E_i, \qquad (H^2, E_1^2, \dots, E_6^2) = (1, -1, \dots, -1).$$

The canonical divisor $K_S = -3H + \sum_i E_i$ satisfies $K_S^2 = 3 > 0$. Its orthogonal complement

$$K_S^{\perp} = \{D \in \operatorname{Pic}(S) : D \cdot K_S = 0\}$$

has rank six and a negative-definite intersection pairing.

Consider six particularly simple classes:

$$\alpha_i = E_i - E_{i+1} \quad (1 \leq i \leq 5), \qquad \alpha_6 = H - E_1 - E_2 - E_3. \tag{160.9}$$

Direct calculation using (160.2) gives

$$K_S \cdot \alpha_i = 0, \qquad \alpha_i^2 = -2,$$

and

$$\alpha_i \cdot \alpha_j = \begin{cases} 1, & i,j \text{ adjacent in the diagram below,} \\ 0, & i \neq j \text{ and not adjacent.} \end{cases}$$

The diagram is the Dynkin diagram of the exceptional root system E_6.

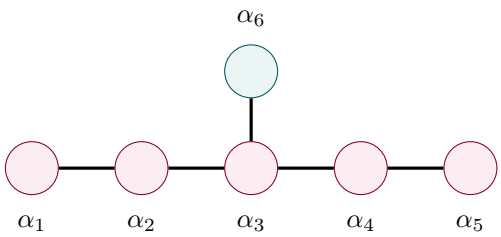


Figure 160.2: The six roots extracted from the intersection pairing form an E_6 Dynkin diagram. Adjacency means intersection number 1, and every root has self-intersection -2.

To turn these classes into symmetries, use the reflection associated to a root α with $\alpha^2 = -2$:

$$s_\alpha(D) = D + (D \cdot \alpha)\alpha. \tag{160.10}$$

It fixes every class orthogonal to α, sends α to $-\alpha$, and preserves the intersection pairing. Since $K_S \cdot \alpha = 0$, it also fixes the canonical class. Thus the reflection preserves the two numerical conditions

$$D^2 = -1, \qquad (-K_S) \cdot D = 1.$$

Our classification says that these conditions, together with the appropriate effective line classes, produce exactly the 27 possibilities. Indeed these root reflections permute that finite set.

One calculation makes the symmetry surprisingly concrete. For $\alpha_6 = H - E_1 - E_2 - E_3$, we have $E_1 \cdot \alpha_6 = 1$, and hence

$$\begin{aligned} s_{\alpha_6}(E_1) &= E_1 + (E_1 \cdot \alpha_6)\alpha_6 \\ &= H - E_2 - E_3 \\ &= L_{23}. \end{aligned}$$

A reflection in the root lattice exchanges the class of an exceptional curve with the class of a line through two of the blown-up points! It can mix the three families that gave the count $6 + 15 + 6$.

The group generated by the reflections in the E_6 roots is the Weyl group $W(E_6)$. A classical theorem identifies it with the full automorphism group of the intersection graph of the 27 lines. Its order is

$$|W(E_6)| = 2 \cdot 5 \cdot 6 \cdot 8 \cdot 9 \cdot 12 = 51840. \tag{160.11}$$

The factors are the degrees of the basic polynomial invariants of the E_6 reflection representation. Establishing the invariant-theoretic result is beyond the counting proof; here the visible geometric content is the explicit root lattice, its reflection formula and its action on the line classes.

It is important to identify which symmetry we are speaking about. The Weyl group describes the *incidence configuration* of the 27 lines, not necessarily the group of projective automorphisms of one fixed cubic surface. A general smooth cubic has no nontrivial projective automorphisms, even though its 27 lines have the entire abstract $W(E_6)$ incidence symmetry.

§160.10 Twenty-seven lines in one explicit equation

After all this intersection theory, it is pleasant to return to an equation one can check by substitution. Consider the Fermat cubic

$$S_F: \quad x_0^3 + x_1^3 + x_2^3 + x_3^3 = 0 \qquad \text{in } \mathbb{P}^3_{\mathbb{C}}. \tag{160.12}$$

It is smooth: the four partial derivatives are $3x_i^2$, and they cannot all vanish at a projective point.

Divide the four indices into two unordered pairs, for example $\{0,1\} \sqcup \{2,3\}$. For any two cube roots of unity $\zeta^3 = \eta^3 = 1$, impose the linear equations

$$x_0 + \zeta x_1 = 0, \qquad x_2 + \eta x_3 = 0. \tag{160.13}$$

They define a projective line. On that line,

$$x_0^3 + x_1^3 = (-\zeta)^3 x_1^3 + x_1^3 = 0,$$

and likewise $x_2^3 + x_3^3 = 0$, so the line lies entirely on S_F.

There are three ways to partition four indices into two pairs:

$$\{0,1\} \sqcup \{2,3\},$$
$$\{0,2\} \sqcup \{1,3\},$$
$$\{0,3\} \sqcup \{1,2\}.$$

For each partition there are three choices of ζ and three of η, yielding nine lines. The choices give distinct projective lines, hence

$$\underbrace{3}_{\text{pairings}} \times \underbrace{3}_{\zeta} \times \underbrace{3}_{\eta} = 27 \tag{160.14}$$

explicit lines on the smooth Fermat cubic. Our classification theorem says these are *all* of them.

This short calculation provides a complementary view of the famous number. In the blow-up picture, 27 arose as $6+15+6$; in one particularly symmetric equation it appears as $3 \cdot 3 \cdot 3$. The two enumerations concern the same geometric objects, although their indexing systems look completely different.

There is a small extra twist if we insist on *real* lines. The equation (160.12) has real coefficients, but most of its 27 complex lines do not lie in real projective space. Complex conjugation sends the line specified by (ζ, η) to the line specified by $(\overline{\zeta}, \overline{\eta})$. For a fixed pairing, the three cube roots of unity are

$$1, \qquad \omega = e^{2\pi i/3}, \qquad \omega^2 = \overline{\omega}.$$

Only 1 is real. Since all the lines in our explicit list are distinct, a line is fixed by conjugation precisely when both parameters equal 1. There is exactly one such line for each of the three pairings, so

$$\boxed{\#\{\text{real lines on } S_F\} = 3.}$$

Thus the same smooth cubic surface that has 27 lines over $\mathbb{C}$ has just three lines defined over $\mathbb{R}$.

The parity is no accident. For *any* smooth cubic surface defined by a real equation, complex conjugation permutes its 27 complex lines. Every orbit has one or two elements; a line fixed by conjugation is a real line, while a nonreal line belongs to a two-element conjugate pair. Because 27 is odd, the number of real lines must also be odd. This easy argument does not determine the number, but it extracts a genuine arithmetic constraint from a purely geometric count.

§160.11 Following twenty-seven lines around a loop

There is one last connection to the geometry of polynomial roots. In our earlier note on monodromy, the roots of $z^3 - 3z = t$ were continued along paths in the parameter plane. Two simple branch points produced transpositions and, together, the group S_3.

For cubic surfaces the *coefficients of the cubic equation* can also move. Why should its 27 lines then admit locally continuous labels, as the roots of a polynomial do? The self-intersection calculation provides the local rigidity needed for an answer.

A line $L \subset S$ is a smooth rational curve, so $L \cong \mathbb{P}^1$. Its normal bundle inside the surface is a line bundle whose degree is $L^2 = -1$:

$$N_{L/S} \cong \mathcal{O}_{\mathbb{P}^1}(-1).$$

Infinitesimal deformations of L *within a fixed surface* are parametrized by $H^0(L, N_{L/S})$. Since

$$H^0(\mathbb{P}^1, \mathcal{O}(-1)) = 0,$$

there is no nontrivial first-order deformation of a line while the surface remains fixed. This is the infinitesimal reason the lines form a discrete set.

The corresponding obstruction space is $H^1(L, N_{L/S})$, which also vanishes for $\mathcal{O}(-1)$. Standard deformation theory therefore says that when we vary the smooth cubic surface slightly, each of its lines moves uniquely and smoothly with that variation. Equivalently, the universal family of pairs

$$\{(S, L) : S \text{ a smooth cubic surface}, \quad L \subset S \text{ a line}\}$$

is locally a 27-sheeted covering over the space of smooth cubic equations (with scalar multiples identified). This is a geometric version of the implicit-function theorem for the roots of a polynomial.

A closed path through this parameter space may consequently return to the original cubic while permuting its 27 lines. We obtain a monodromy action on a set of 27 objects, just as we previously did for three polynomial roots.

For the universal family of smooth cubic surfaces, the monodromy group of the 27 lines is the full Weyl group $W(E_6)$. This classical result can be understood from the blow-up presentations and their Cremona transformations. The relevant paths are paths *in the space of cubic surfaces*; a closed path on a single fixed cubic surface is a different object.

The numerical count and the symmetry group thus come from the same elementary-looking starting data: six points in a plane. The six points give 27 line classes, the intersection pairing gives their incidences, and the classes orthogonal to the canonical divisor produce the E_6 root system. A question about twenty-seven very concrete straight lines has led, through a precise chain of calculations, to one of the exceptional Weyl groups.

XIII

Machine Learning and Inference

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Backpropagation in One MLP Layer: From a Three-Dimensional Tensor to an Outer Product

N-20250926

§161.1 The weight derivative and its three indices

Consider one middle layer of a multilayer perceptron:

$$z = Wx + b, \qquad y = \sigma(z).$$

Here

$$x \in \mathbb{R}^n, \qquad W \in \mathbb{R}^{m\times n}, \qquad b \in \mathbb{R}^m, \qquad z, y \in \mathbb{R}^m.$$

The activation function σ acts elementwise, for example sigmoid or ReLU. The goal of backpropagation is to compute

$$\frac{\partial L}{\partial W}$$

for a loss function L.

The organization of derivatives as linear maps and reverse-mode pullbacks follows MIT OCW, *Matrix Calculus for Machine Learning and Beyond*.[1]

The softmax–cross-entropy setting provides a standard example. If the final layer produces logits z_i, then

$$\hat{y}_i = \frac{e^{z_i}}{\sum_j e^{z_j}}, \qquad L = -\sum_i t_i \ln \hat{y}_i,$$

where t_i is a one-hot label. Backpropagation asks how this scalar loss changes when each weight $W_{j,k}$ changes.

Strictly speaking, the derivative of the vector $y \in \mathbb{R}^m$ with respect to the matrix $W \in \mathbb{R}^{m\times n}$ has three indices:

$$\frac{\partial y_i}{\partial W_{j,k}}.$$

The index i chooses the output component, j chooses the row of W, and k chooses the column of W. So the object has shape

$$m \times m \times n.$$

This is the source of the confusion recorded in the note. The derivative is theoretically a third-order tensor, but in neural-network code one usually sees only a matrix gradient. The matrix appears after chain-rule contraction with the upstream gradient.

Write the linear preactivation component as

$$z_r = \sum_{t=1}^{n} W_{r,t} x_t + b_r.$$

[1] MIT 18.S096 lecture notes.

Then

$$\frac{\partial z_r}{\partial W_{j,k}} = \begin{cases} x_k, & r = j, \\ 0, & r \neq j, \end{cases} = \delta_{rj} x_k.$$

This is sparse because z_r depends only on row r of W.

For $y_i = \sigma(z_i)$ with elementwise activation,

$$\frac{\partial y_i}{\partial z_r} = 0 \quad (i \neq r), \qquad \frac{\partial y_i}{\partial z_i} = \sigma'(z_i).$$

Thus

$$\frac{\partial y_i}{\partial W_{j,k}} = \sum_{r=1}^{m} \frac{\partial y_i}{\partial z_r} \frac{\partial z_r}{\partial W_{j,k}} = \frac{\partial y_i}{\partial z_j} x_k.$$

For elementwise activation this is nonzero only when $i = j$.

§161.2 Scalar losses contract the tensor

The loss L is scalar. Therefore

$$\frac{\partial L}{\partial W_{j,k}} = \sum_{i=1}^{m} \frac{\partial L}{\partial y_i} \frac{\partial y_i}{\partial W_{j,k}}.$$

Substituting the previous formula gives

$$\frac{\partial L}{\partial W_{j,k}} = \sum_{i=1}^{m} \frac{\partial L}{\partial y_i} \frac{\partial y_i}{\partial z_j} x_k.$$

The sum over i is exactly the j-th component of the upstream gradient with respect to z:

$$\delta_j = \frac{\partial L}{\partial z_j}.$$

Hence

$$\frac{\partial L}{\partial W_{j,k}} = \delta_j x_k.$$

Stacking all j, k gives the outer product

$$\boxed{\frac{\partial L}{\partial W} = \delta x^T}$$

where

$$\delta = \frac{\partial L}{\partial z}.$$

The original derivative $\partial y / \partial W$ is a tensor with shape (m, m, n). But backpropagation does not need to store it. The chain rule contracts it with the vector $\partial L / \partial y$, leaving a matrix of shape (m, n).

Equivalently:

$$\text{three-dimensional tensor} \xrightarrow{\text{chain-rule summation}} \text{matrix gradient}.$$

This is not a loss of information for gradient descent, because gradient descent only needs the derivative of the scalar loss with respect to each parameter.

For the final layer with softmax and cross entropy, the derivative simplifies to

$$\frac{\partial L}{\partial z} = \hat{y} - t.$$

Therefore the final-layer weight gradient is

$$\frac{\partial L}{\partial W} = (\hat{y} - t)x^T.$$

This is why implementation code often looks much simpler than the theoretical tensor derivation.

§161.3 Reverse-mode derivatives as cotangent pullbacks

The same computation can be described as a pullback of covectors. A forward layer is a map

$$f : \text{parameters and activations} \longrightarrow \text{outputs}.$$

Its differential sends small forward perturbations to small output perturbations:

$$df : T_x X \to T_{f(x)} Y.$$

But the loss supplies a covector at the output,

$$dL \in T^*_{f(x)} Y,$$

and backpropagation sends it backward by precomposition:

$$f^*(dL) = dL \circ df \in T^*_x X.$$

This is why reverse-mode differentiation is naturally contravariant.

For a batched linear layer in PyTorch convention,

$$X \in \mathbb{R}^{b\times n}, \qquad W \in \mathbb{R}^{m\times n}, \qquad Y = XW^T \in \mathbb{R}^{b\times m},$$

the component formula is

$$Y_{\alpha i} = \sum_{k=1}^{n} X_{\alpha k} W_{ik}.$$

Hence

$$\frac{\partial Y_{\alpha i}}{\partial W_{jk}} = \delta_{ij} X_{\alpha k},$$

a Jacobian tensor of shape (b, m, m, n). Contracting with the upstream gradient $G = \partial L/\partial Y$, of shape (b, m), gives

$$\begin{aligned}\frac{\partial L}{\partial W_{jk}} &= \sum_{\alpha,i} G_{\alpha i} \frac{\partial Y_{\alpha i}}{\partial W_{jk}} \\ &= \sum_{\alpha} G_{\alpha j} X_{\alpha k},\end{aligned}$$

so

$$\frac{\partial L}{\partial W} = G^T X.$$

If the weights are stored instead as $W_{\text{col}} \in \mathbb{R}^{n\times m}$ and $Y = XW_{\text{col}}$, the same formula appears as

$$\frac{\partial L}{\partial W_{\text{col}}} = X^T G.$$

The two expressions differ only by the convention for storing the weight matrix.

§161.4 A numerical gradient and a useful consistency check

Take $m = 2$, $n = 3$, input $x = (1, 2, 3)^T$, and preactivations $z = (0.1, -1)^T$. For sigmoid activation σ,

$$\sigma'(0.1) \approx 0.2494, \qquad \sigma'(-1) \approx 0.1966.$$

The matrix obtained by stacking the derivatives of each output y_i with respect to the entries in *its own row* $W_{i,:}$ is

$$\begin{pmatrix} 0.2494 & 0.4988 & 0.7482 \\ 0.1966 & 0.3932 & 0.5898 \end{pmatrix}.$$

This is a collection of two nonzero slices of the full (m, m, n) Jacobian, *not* the complete Jacobian itself. If the upstream loss gradient happens to be $\partial L/\partial y = (1, 1)^T$, the contracted *weight gradient* is numerically the same matrix. For a general upstream gradient u, however, it becomes

$$\frac{\partial L}{\partial W} = (u \odot \sigma'(z))x^T.$$

This distinction explains when the numerical outer product is a legitimate gradient and when it merely displays the local activation sensitivities.

In vectorized notation one may write $\mathrm{vec}(\delta x^T) = x \otimes \delta$ when columns are stacked, but constructing this Kronecker product gains nothing computationally. For batches the correct matrix formula in the convention $X \in \mathbb{R}^{b\times n}$, $G \in \mathbb{R}^{b\times m}$, $Y = XW^T$, is $\nabla_W L = G^T X$; its transpose $X^T G$ belongs to the alternative column-stored weight convention.

Latent Variable Models: Why Inference Is Harder Than Generation

N-20251008

A generative model describes how unobserved causes produce observations. The forward direction can be simple even when the reverse problem of inferring causes from observations is difficult. This asymmetry is what motivates variational inference. The encoder–decoder shape is shared with ordinary autoencoders, but probability distributions play a different role in the two architectures. A two-component Gaussian mixture already reveals Bayes' rule, marginalization, an exact variational objective and the meaning of its approximation gap.[1]

To picture the problem, imagine a drawing program trained on handwritten digits. A conventional autoencoder compresses an existing image and tries to reproduce it. A generative latent-variable model goes the other way: it first samples a hidden description z, then uses a decoder to produce a new image. The decoder must represent *uncertainty*; otherwise it merely outputs one deterministic reconstruction.

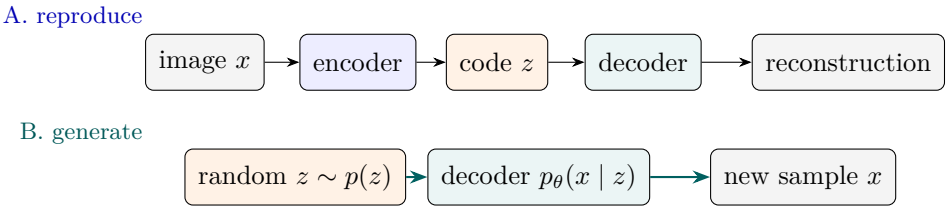


Figure 162.1: Compression and generation are different tasks. For a latent generative model, the lower row is the probability model; the upper encoder-like direction will become an approximation to inference.

In a real image model, z might encode a digit's identity, stroke thickness, slant, and other hidden factors. These coordinates need not correspond to named human concepts, but they offer a compact set of variables from which the model can generate high-dimensional observations. The toy models below use one-dimensional x, z, so every probability calculation remains visible.

§162.1 A hidden bit generates an observed real number

Let Z take values $-1, +1$ with equal probability, and fix $\mu > 0, \sigma > 0$. Generate a real observation according to

$$\Pr(Z = +1) = \Pr(Z = -1) = \frac{1}{2}, \qquad X \mid Z = z \sim \mathcal{N}(z\mu, \sigma^2).$$

Equivalently,

$$X = \mu Z + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2),$$

[1] The latent-variable perspective follows Bishop, *Pattern Recognition and Machine Learning*, Chapter 9, and Kingma and Welling, *Auto-Encoding Variational Bayes*.

with Z and ε independent. Sampling is trivial: flip a fair coin and add Gaussian noise to one of the two possible means.

Write

$$\varphi_\sigma(t) = \frac{1}{\sqrt{2\pi}\sigma} e^{-t^2/(2\sigma^2)}.$$

The joint density with respect to counting measure in z and Lebesgue measure in x is

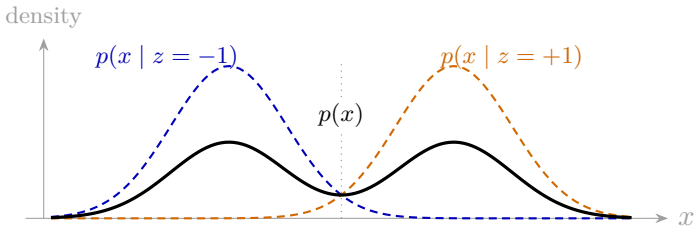
$$p(x, z) = \frac{1}{2}\varphi_\sigma(x - z\mu).$$

To obtain the marginal density of the observation we must sum over the hidden possibilities:

$$\boxed{p(x) = \frac{1}{2}\varphi_\sigma(x - \mu) + \frac{1}{2}\varphi_\sigma(x + \mu).}$$

This sum is a genuine probability density: integrating either Gaussian contributes $1/2$. The value $p(x)$ itself need not be at most one; the probabilities of events are integrals of $p(x)$.

A plot is more revealing than the symbols alone. Take $\mu = 1.55$ and $\sigma = 0.8$. The two colored curves represent the possible observations given each hidden state. The black curve is the actual distribution after the hidden state has been forgotten.



At $x = 0$, both explanations are equally plausible.

Figure 162.2: The mixture distribution (solid black) is the average of two conditional Gaussian densities (dashed). Two distinct hidden causes can generate overlapping observations. Curves are rescaled by a common factor for clarity.

The overlap is important. If we observe a value near the middle, a decoder tells us it *could* have come from either component. To choose between those explanations requires an additional computation, the posterior.

The model can produce a distribution with two peaks. Symmetry gives $p'(0) = 0$. Using $\varphi''_\sigma(t) = (t^2 - \sigma^2)\varphi_\sigma(t)/\sigma^4$, we obtain

$$p''(0) = \frac{\mu^2 - \sigma^2}{\sigma^4}\, p(0).$$

If $\mu > \sigma$, the origin is a strict local minimum. Since $p(x) \to 0$ as $|x| \to \infty$, there must be a peak on each side of the origin. A mixture of individually unimodal Gaussians can therefore be bimodal.

§162.2 Bayes' rule is a reverse generative computation

After observing $X = x$, the two possible latent explanations have posterior weights

$$\Pr(Z = z \mid X = x) = \frac{p(x, z)}{p(x)}.$$

The likelihood ratio cancels the common Gaussian normalizer:

$$\log \frac{\Pr(Z=+1 \mid x)}{\Pr(Z=-1 \mid x)} = \log \frac{\varphi_\sigma(x-\mu)}{\varphi_\sigma(x+\mu)} = \frac{(x+\mu)^2-(x-\mu)^2}{2\sigma^2} = \frac{2\mu x}{\sigma^2}.$$

Therefore, writing $\operatorname{sigmoid}(t) = 1/(1+e^{-t})$,

$$\boxed{\Pr(Z=+1 \mid x) = \operatorname{sigmoid}\left(\frac{2\mu x}{\sigma^2}\right).}$$

The posterior is most uncertain at $x = 0$; for large positive x, the explanation $Z = +1$ becomes strongly favored. At the same time, a positive observation is never *proof* that the positive latent state occurred: both Gaussian likelihoods have nonzero tails.

For example, at $\mu = \sigma = 1$,

observation x	-1	0	$+1$
$\Pr(Z=+1 \mid x)$	0.119	0.500	0.881

to three decimal places. This is a *soft assignment*. It reports how much of the posterior probability belongs to each possible cause, rather than declaring one cause with certainty.

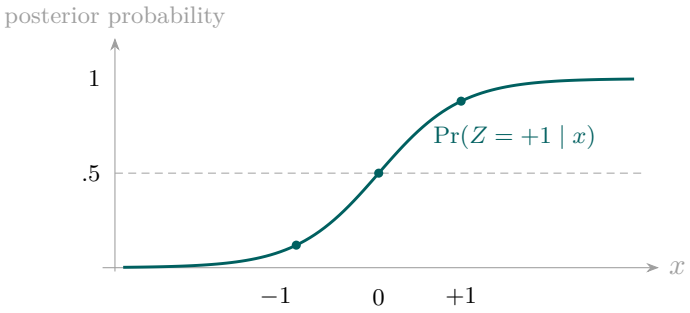


Figure 162.3: Bayesian inference reverses the model: after seeing x, it assigns probabilities to the possible latent explanations. Near the overlap, that assignment is uncertain.

The general latent-variable model replaces this two-term sum by

$$p_\theta(x) = \int p_\theta(x \mid z)p(z)\,dz, \qquad p_\theta(z \mid x) = \frac{p_\theta(x \mid z)p(z)}{p_\theta(x)}.$$

Generation uses $p(z)$ followed by $p_\theta(x \mid z)$. Inference needs the normalizing integral $p_\theta(x)$, which can become intractable when z is high-dimensional or the decoder is nonlinear.

§162.3 A soluble continuous model shows what is lost

There is one continuous example in which the inverse computation is still exact. Let

$$Z \sim \mathcal{N}(0,1), \qquad X \mid Z = z \sim \mathcal{N}(z, \sigma^2).$$

Completing the square in the joint density gives

$$-\frac{1}{2}z^2 - \frac{(x-z)^2}{2\sigma^2} = -\frac{1+\sigma^{-2}}{2}\left(z - \frac{x}{1+\sigma^2}\right)^2 - \frac{x^2}{2(1+\sigma^2)}.$$

Consequently,

$$\boxed{Z \mid X = x \sim \mathcal{N}\left(\frac{x}{1+\sigma^2}, \frac{\sigma^2}{1+\sigma^2}\right),}$$

and

$$X \sim \mathcal{N}(0, 1+\sigma^2).$$

The posterior mean is a shrinkage estimator: a noisy observation is pulled toward the prior mean zero. Its coefficient $1/(1+\sigma^2)$ decreases as the observation noise grows.

Now change only the decoder to

$$X = Z^2 + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2).$$

Here is the geometric reason the nonlinear case is harder. The decoder $z \mapsto z^2$ *folds* the latent line: two distant causes can produce the same observation.

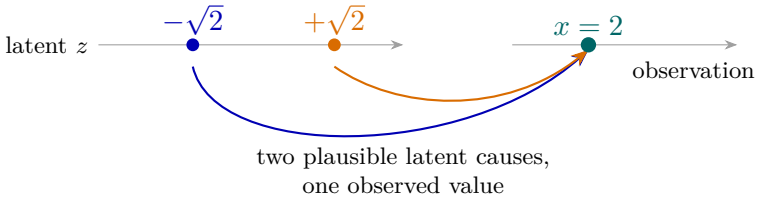


Figure 162.4: Nonlinear decoders can identify distant latent values. An inverse model must then represent multiple plausible explanations, not just one point.

For $x > 0$ and small noise, both $z \approx +\sqrt{x}$ and $z \approx -\sqrt{x}$ are plausible explanations, so the posterior is generally bimodal rather than Gaussian. Sampling from the generative model remains easy, but completing a single quadratic square no longer solves the inverse problem. A nonlinear neural decoder greatly increases this gap between forward sampling and posterior computation.

§162.4 An Exactly Solvable Variational Example

Return to the hidden-bit mixture. Suppose we approximate the true posterior with an arbitrary Bernoulli distribution

$$q_r(+1) = r, \qquad q_r(-1) = 1 - r, \qquad 0 < r < 1.$$

Define

$$\mathcal{L}(r; x) = \sum_{z \in \{-1,+1\}} q_r(z) \log p(x, z) - \sum_{z \in \{-1,+1\}} q_r(z) \log q_r(z).$$

Writing $p_\pm = p(x, \pm 1)$, we have

$$\mathcal{L}(r; x) = r \log p_+ + (1-r) \log p_- - r \log r - (1-r) \log(1-r).$$

Differentiation gives

$$\frac{\partial \mathcal{L}}{\partial r} = \log \frac{p_+}{p_-} - \log \frac{r}{1-r}, \qquad \frac{\partial^2 \mathcal{L}}{\partial r^2} = -\frac{1}{r(1-r)} < 0.$$

The objective is strictly concave. Its unique maximizer satisfies

$$\frac{r_*}{1-r_*} = \frac{p_+}{p_-} \quad \Longrightarrow \quad \boxed{r_* = \Pr(Z = +1 \mid X = x).}$$

Thus optimizing this objective recovers Bayes' posterior exactly in a problem small enough to solve on paper.

The value at the optimum is the marginal log-likelihood:

$$\mathcal{L}(r_*; x) = \log(p_+ + p_-) = \log p(x).$$

For an arbitrary r, direct subtraction gives the stronger identity

$$\boxed{\log p(x) - \mathcal{L}(r; x) = D_{\mathrm{KL}}(q_r \,\|\, p(\cdot \mid x)) \geq 0.}$$

Here $D_{\mathrm{KL}}(q\|p) = \sum_z q(z) \log(q(z)/p(z))$. The inequality follows from convexity of $t \log t$, or from the log-sum inequality; equality holds if and only if the approximate posterior equals the true one.

This finite example is the Gibbs variational principle in concrete form. It explains both the lower bound and its exact gap before we make either distribution an output of a neural network.

§162.5 Why a neural encoder is useful

For a large nonlinear generative model, solving the optimal posterior for every x separately would be expensive. A recognition model $q_\phi(z \mid x)$ instead learns a *function of the observation* that produces an approximate posterior. This is amortized inference: computational effort is shared across many data points.

A continuous VAE often chooses a diagonal Gaussian family

$$q_\phi(z \mid x) = \mathcal{N}(z; \mu_\phi(x), \mathrm{diag}(\sigma_\phi^2(x))).$$

The parameters $\mu_\phi(x)$ and $\sigma_\phi(x)$ are functions computed by an encoder. A diagonal Gaussian is convenient to sample and evaluate, but it cannot faithfully represent the two separated modes of a strongly multimodal posterior. The restriction of the variational family introduces a real approximation error, distinct from optimization error.

The encoder is useful because it learns to perform the approximate inverse operation *quickly for each new observation*. It usually does not output a single latent coordinate. For a diagonal-Gaussian VAE it outputs a mean and a vector of standard deviations, describing a region of plausible latent codes.

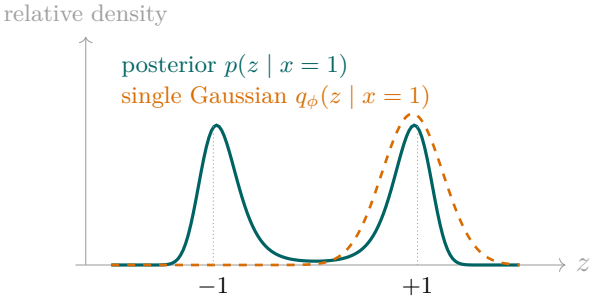


Figure 162.5: An inverse problem with two explanations: $Z \sim \mathcal{N}(0, 1)$, $X \mid Z = z \sim \mathcal{N}(z^2, 0.35^2)$, and $x = 1$. The posterior has two peaks near $z = \pm 1$, while one Gaussian approximation can cover only one peak narrowly or both peaks very broadly. Curves are rescaled vertically to compare their shapes; peak heights are not on a shared probability-density scale.

At generation time, the input image and inference network are not needed. Draw $z \sim p(z)$ from the prior and then $x \sim p_\theta(x \mid z)$ from the decoder. At inference time, start with an observed x and use $q_\phi(z \mid x)$ to approximate where it may have come from. During training, both paths cooperate through a likelihood lower bound.

The exact mixture and Gaussian calculations establish the core issue. A generative story determines a joint law, marginalization measures how likely an observation is, and Bayes' rule inverts the story. When that inversion is hard, a tractable family $q_\phi(z \mid x)$ replaces it. The next mathematical task is to optimize the resulting lower bound and differentiate it through random sampling.

From a Latent-Variable Model to a Trainable Generator

N-20251009

A latent-variable model has a simple forward recipe: sample a hidden code z, and then sample an observation x given that code. Even a mixture of simple conditional Gaussian laws can produce several peaks after the hidden cause is forgotten; determining which cause explains an observation then requires Bayes' rule. With images, the remaining questions are practical: what exactly does a neural decoder output, how do we measure an entire dataset under the model, and why do we need a recognition network when we can sample from the model already?[1]

§163.1 A decoder produces probabilities, not just pixels

For a latent code $z \in \mathbb{R}^d$, the standard VAE starts with a simple prior such as

$$p(z) = \mathcal{N}(z; 0, I).$$

A neural decoder maps z to the parameters of a conditional distribution $p_\theta(x \mid z)$. The model is

$$\boxed{p_\theta(x, z) = p(z)\, p_\theta(x \mid z).}$$

The direction $z \to x$ is the generative direction; unlike a reconstruction-only autoencoder, it can begin from a random latent draw without any observed image.

The observation distribution must match the data. If an image consists of binary pixels $x_j \in \{0, 1\}$, the decoder might output a number $\pi_j(z) \in (0, 1)$ for each pixel and define

$$p_\theta(x \mid z) = \prod_{j=1}^{D} \pi_j(z)^{x_j} (1 - \pi_j(z))^{1 - x_j}.$$

Each output is the probability that a pixel is on, not the pixel itself. The negative log-likelihood becomes

$$-\log p_\theta(x \mid z) = -\sum_j [x_j \log \pi_j(z) + (1 - x_j) \log(1 - \pi_j(z))],$$

which is binary cross-entropy.

For a real-valued feature vector, a Gaussian decoder might be preferable:

$$p_\theta(x \mid z) = \mathcal{N}(x; \mu_\theta(z), \sigma_x^2 I).$$

Up to terms independent of θ, the negative log-likelihood is

$$-\log p_\theta(x \mid z) = \frac{\|x - \mu_\theta(z)\|^2}{2\sigma_x^2} + \text{constant}.$$

So the familiar squared reconstruction error is a probabilistic modeling choice, with the noise variance determining its scale. The likelihood specification matters as much as the neural-network architecture.

[1] The model architecture and approximate-inference viewpoint follow Kingma and Welling, *An Introduction to Variational Autoencoders* (2019), and C. Doersch, *Tutorial on Variational Autoencoders*.

§163.2 Conditional Independence and Correlation

The product over pixels assumes that the pixels are independent *given* z. That is a computational simplification. It does not imply that pixels are independent in the generated dataset, because the same latent variable affects every pixel.

A tiny example makes the distinction sharp. Let Z be a fair bit. Given $Z = z$, set $X_1 = z$ and $X_2 = z$ deterministically. The two observations are trivially independent conditional on Z, but marginally

$$\Pr(X_1 = 1) = \Pr(X_2 = 1) = \frac{1}{2}, \qquad \Pr(X_1 = 1, X_2 = 1) = \frac{1}{2} \neq \frac{1}{4}.$$

Thus the shared latent cause has created dependence between the observed variables. This is why even a decoder using a factorized pixel likelihood can model correlated images after latent codes are marginalized out.

The price of a simple factorized decoder is also visible. Conditioned on a *fixed* code z, it cannot represent unexplained correlations between pixels beyond those already encoded by z. A better latent representation or a richer likelihood may be needed when the data exhibit residual structure.

§163.3 A dataset means many hidden causes

For observations $x^{(1)}, \ldots, x^{(N)}$, assume independent examples and give each example its own latent variable $z^{(i)}$. The joint law is

$$p_\theta(x^{(1:N)}, z^{(1:N)}) = \prod_{i=1}^{N} p(z^{(i)}) p_\theta(x^{(i)} \mid z^{(i)}).$$

The same prior and decoder are used for every observation, while each observation has its own latent code. A small continuous example also reveals the geometry of the marginal distribution produced by a shared latent cause.

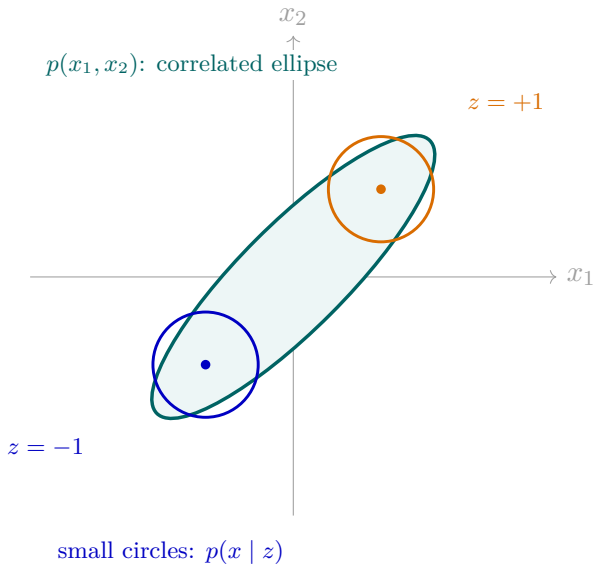


Figure 163.1: A one-dimensional latent $Z \sim \mathcal{N}(0,1)$ generates $X_1 = Z + \varepsilon_1$ and $X_2 = Z + \varepsilon_2$, where the independent noises have variance 0.4^2. Each conditional distribution $p(x \mid z)$ is a circular Gaussian (two examples are outlined). After forgetting z, $p(x)$ has covariance $\left(\begin{smallmatrix} 1.16 & 1 \\ 1 & 1.16 \end{smallmatrix}\right)$ and elongated elliptical contours. Conditional independence has produced strong marginal correlation. Ellipses mark equal Mahalanobis radii.

Learning by marginal maximum likelihood would maximize

$$\log p_\theta(x^{(1:N)}) = \sum_{i=1}^{N} \log p_\theta(x^{(i)}),$$
$$p_\theta(x) = \int p(z) p_\theta(x \mid z)\, dz.$$

The integral is the obstacle. Even if the decoder evaluates quickly for one z, integrating its output over all plausible latent codes can be very expensive.

§163.4 When Prior Sampling Fails

The identity

$$p_\theta(x) = \mathbb{E}_{z \sim p(z)}[p_\theta(x \mid z)]$$

suggests a Monte Carlo approximation by averaging $p_\theta(x \mid z)$ over prior samples. This works in principle as an unbiased estimator of the likelihood, but it can be practically useless when the observation is explained by only a tiny part of latent space.

Here is an exact discrete example. Take K independent fair latent bits, $Z = (Z_1, \dots, Z_K)$. Let the observed bit X equal one exactly when every hidden bit equals one:

$$\Pr(X = 1 \mid Z = z) = \begin{cases} 1, & z = (1, \dots, 1), \\ 0, & \text{otherwise.} \end{cases}$$

Generating a sample is trivial. Yet

$$\boxed{\Pr(X = 1) = 2^{-K}.}$$

If we draw M latent samples from the prior and average their likelihoods for the observed event $X = 1$, the estimate is exactly zero unless at least one sample hits the all-ones configuration. The probability of never hitting it is

$$\boxed{\left(1 - 2^{-K}\right)^{M}}.$$

For $K = 20$, even ten thousand prior samples miss the relevant latent state with probability about 0.9905. To obtain a reasonable probability of observing even one relevant sample, M must be of order 2^K.

Once $X = 1$ is known, however, the posterior is concentrated on the all-ones code. The difficulty was not evaluating the decoder; it was sampling latent states without regard to the observed data. This is the basic reason approximate posterior inference is so valuable.

§163.5 Recognition is a fast approximate inverse

The recognition network supplies

$$q_\phi(z \mid x) \approx p_\theta(z \mid x).$$

For a continuous latent code, a common choice is a diagonal Gaussian,

$$q_\phi(z \mid x) = \mathcal{N}\big(z; \mu_\phi(x), \operatorname{diag}(\sigma_\phi^2(x))\big).$$

The neural encoder computes $\mu_\phi(x)$ and $\sigma_\phi(x)$ from the input image, describing a whole distribution of plausible codes. The decoder and encoder answer different conditional questions:

$$\underbrace{p_\theta(x \mid z)}_{\text{what could this cause generate?}} \quad \text{and} \quad \underbrace{q_\phi(z \mid x)}_{\text{what might have caused this observation?}}.$$

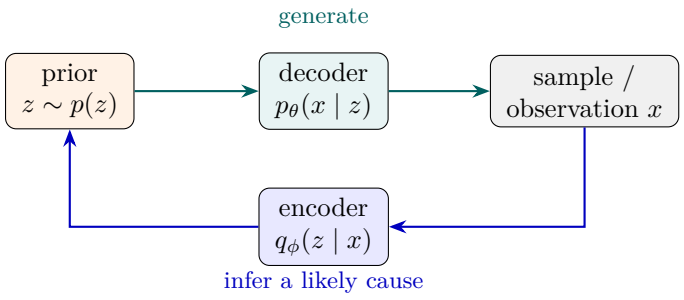


Figure 163.2: The probabilistic decoder defines the forward model. The recognition network approximates the reverse conditional distribution without computing its exact normalizing integral.

An ordinary variational method could optimize a separate approximate posterior for every training example. A VAE amortizes that computation: the same encoder parameters ϕ map every new input to posterior parameters in a single forward pass. This makes large-dataset inference fast, but it constrains all approximations to come from one shared learned function.

There are now two distinct approximation errors to keep in mind. A diagonal Gaussian family may be unable to represent a multimodal posterior, and even within that family a learned encoder may not output the best variational parameters for every example. The evidence lower bound addresses the training objective and quantifies the discrepancy with the true posterior; the next mathematical step is to optimize that bound with gradients.

Variational Inference: KL Divergence, Jensen, and the ELBO

N-20251010

§164.1 The approximate posterior as an optimization problem

The true posterior in a latent variable model is

$$p_\theta(z \mid x) = \frac{p_\theta(x, z)}{p_\theta(x)}.$$

It is usually intractable because the marginal likelihood

$$p_\theta(x) = \int p_\theta(x, z)\, dz$$

is hard to compute.

Variational inference introduces a tractable family

$$q_\phi(z \mid x)$$

and tries to make it close to the true posterior. A natural measure of closeness is

$$D_{\mathrm{KL}}(q_\phi(z \mid x) \| p_\theta(z \mid x)).$$

This is the reverse direction from what one might first guess: we place expectation under q_ϕ, because q_ϕ is the distribution from which we can sample and whose density we can evaluate.

The obstacle is that this KL divergence contains $p_\theta(z \mid x)$, which itself contains the intractable $p_\theta(x)$. The central algebraic trick is to rewrite it so that $\log p_\theta(x)$ appears as a constant plus a lower bound.

§164.2 The ELBO identity

Start from the KL divergence:

$$D_{\mathrm{KL}}(q_\phi(z \mid x) \| p_\theta(z \mid x)) = \mathbb{E}_{q_\phi(z|x)} \left[\log \frac{q_\phi(z \mid x)}{p_\theta(z \mid x)} \right].$$

Using

$$p_\theta(z \mid x) = \frac{p_\theta(x, z)}{p_\theta(x)},$$

we get

$$\begin{aligned} D_{\mathrm{KL}}(q_\phi(z \mid x) \| p_\theta(z \mid x)) &= \mathbb{E}_{q_\phi} \left[\log q_\phi(z \mid x) - \log p_\theta(x, z) + \log p_\theta(x) \right] \\ &= \log p_\theta(x) - \mathbb{E}_{q_\phi} \left[\log p_\theta(x, z) - \log q_\phi(z \mid x) \right]. \end{aligned}$$

Define

$$\mathcal{L}(\theta, \phi; x) = \mathbb{E}_{q_\phi(z|x)} \left[\log p_\theta(x, z) - \log q_\phi(z \mid x) \right].$$

Then the identity becomes

$$\log p_\theta(x) = \mathcal{L}(\theta, \phi; x) + D_{\mathrm{KL}}(q_\phi(z \mid x) \| p_\theta(z \mid x)).$$

Since KL divergence is nonnegative,

$$\mathcal{L}(\theta, \phi; x) \leq \log p_\theta(x).$$

This is why $\mathcal{L}$ is called the evidence lower bound, or ELBO.

Remark 164.2.1 — The ELBO is not introduced by magic. It is exactly what remains when the intractable posterior KL is separated from the log marginal likelihood.

§164.3 Jensen's inequality derivation

The same lower bound can be derived directly from Jensen's inequality. For the equality in the next display we assume $p_\theta(x, z) = 0$ almost everywhere that $q_\phi(z \mid x) = 0$: the trial density covers the support of the unnormalized target. (A Gaussian trial density with strictly positive variance has full support and satisfies this in ordinary continuous latent models.) If q misses some posterior mass, the intervening equality is replaced by an inequality, and Jensen still supplies a lower bound.

$$\begin{aligned}
\log p_\theta(x) &= \log \int p_\theta(x, z)\, dz \\
&= \log \int q_\phi(z \mid x) \frac{p_\theta(x, z)}{q_\phi(z \mid x)}\, dz \\
&= \log \mathbb{E}_{q_\phi(z|x)} \left[\frac{p_\theta(x, z)}{q_\phi(z \mid x)} \right].
\end{aligned}$$

Because log is concave,

$$\begin{aligned}
\log p_\theta(x) &\geq \mathbb{E}_{q_\phi(z|x)} \left[\log \frac{p_\theta(x, z)}{q_\phi(z \mid x)} \right] \\
&= \mathbb{E}_{q_\phi(z|x)} \left[\log p_\theta(x, z) - \log q_\phi(z \mid x) \right] \\
&= \mathcal{L}(\theta, \phi; x).
\end{aligned}$$

The two derivations emphasize different things. The KL derivation shows the gap between the bound and the true log-likelihood. The Jensen derivation shows why a lower bound exists at all.

§164.4 Reconstruction term and regularization term

Using the latent variable factorization

$$p_\theta(x, z) = p(z) p_\theta(x \mid z),$$

the ELBO becomes

$$\begin{aligned}
\mathcal{L}(\theta, \phi; x) &= \mathbb{E}_{q_\phi(z|x)} \left[\log p_\theta(x \mid z) + \log p(z) - \log q_\phi(z \mid x) \right] \\
&= \mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x \mid z)] - D_{\mathrm{KL}}(q_\phi(z \mid x) \| p(z)).
\end{aligned}$$

This is the familiar VAE decomposition:

$$\boxed{\mathcal{L} = \text{expected log-likelihood} - \text{KL to the prior.}}$$

The first term asks whether latent samples from $q_\phi(z \mid x)$ explain x well under the decoder. The second term asks whether the approximate posterior stays close to the prior.

It is common to call the first term the reconstruction term. But mathematically it is an expected log-likelihood. It becomes a squared-error reconstruction loss only under a Gaussian decoder with fixed variance. It becomes a cross-entropy-like loss under a Bernoulli decoder.

§164.5 The ELBO gap

The identity

$$\log p_\theta(x) = \mathcal{L}(\theta, \phi; x) + D_{\mathrm{KL}}(q_\phi(z \mid x) \| p_\theta(z \mid x))$$

shows that there are two goals mixed together.

Increasing $\mathcal{L}$ can improve the model likelihood $\log p_\theta(x)$. But it also tries to reduce the posterior approximation gap

$$D_{\mathrm{KL}}(q_\phi(z \mid x) \| p_\theta(z \mid x)).$$

If q_ϕ were flexible enough to equal the true posterior, then the bound would be tight:

$$\mathcal{L}(\theta, \phi; x) = \log p_\theta(x).$$

In practice, q_ϕ is restricted, often to a diagonal Gaussian, so the bound is generally not exact.

§164.6 A Gaussian ELBO Gap

We can measure how loose an ELBO is without training a neural network. Consider the continuous latent model

$$Z \sim \mathcal{N}(0, 1), \qquad X \mid Z = z \sim \mathcal{N}(z, 1),$$

and observe $x = 2$. Completing the square gives

$$\begin{aligned} -\frac{z^2}{2} - \frac{(2-z)^2}{2} &= -(z-1)^2 - 1, \\ Z \mid X = 2 &\sim \mathcal{N}(1, \tfrac{1}{2}), \\ X &\sim \mathcal{N}(0, 2). \end{aligned}$$

The exact log evidence is therefore

$$\log p(2) = -1 - \frac{1}{2}\log(4\pi).$$

Choose a variational family of Gaussians with the correct posterior variance but a movable mean:

$$q_\mu(z \mid 2) = \mathcal{N}(\mu, \tfrac{1}{2}).$$

The KL divergence between two normals of equal variance v is $(m_1 - m_2)^2/(2v)$. Here $v = 1/2$, so

$$\boxed{D_{\mathrm{KL}}(q_\mu \| p(z \mid 2)) = (\mu - 1)^2.}$$

The ELBO identity now describes a complete curve, not just an inequality:

$$\boxed{\mathcal{L}(\mu; 2) = \log p(2) - (\mu - 1)^2.}$$

At $\mu = 0$ the bound is one nat below the evidence. At $\mu = 1$ it touches the evidence exactly. In this example we can tighten the ELBO by improving the *inference* distribution while holding the *generative* model fixed.

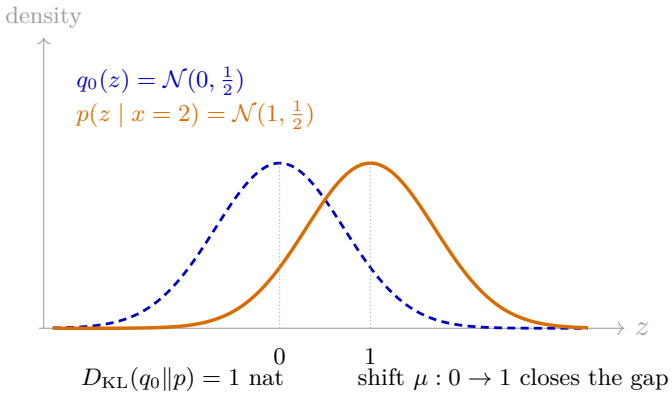


Figure 164.1: The variational gap is a mismatch of probability mass. The true posterior (solid orange) and the trial law $q_\mu = \mathcal{N}(\mu, \frac{1}{2})$ have equal variance. Moving the dashed mean from 0 to 1 makes the curves coincide and closes the one-nat ELBO gap.

This also clarifies a common inference mistake. Maximizing the ELBO over both θ and ϕ does *not* guarantee that every upward optimization step increases the true $\log p_\theta(x)$: the model evidence and approximation gap can change simultaneously. In a flexible posterior family, good inference makes the bound a better proxy for likelihood. In a restrictive family, model fitting and approximate inference remain entangled.[1]

§164.7 Closed-form KL for diagonal Gaussians

The standard VAE uses

$$q_\phi(z \mid x) = \mathcal{N}(z; \mu, \mathrm{diag}(\sigma^2)), \qquad p(z) = \mathcal{N}(0, I).$$

For this pair, the KL divergence has a closed form:

$$D_{\mathrm{KL}}(q \| p) = \frac{1}{2} \sum_{j=1}^{d} \left(\mu_j^2 + \sigma_j^2 - 1 - \log \sigma_j^2 \right).$$

Here is the one-dimensional calculation. Let

$$q(z) = \mathcal{N}(\mu, \sigma^2), \qquad p(z) = \mathcal{N}(0, 1).$$

[1] For the general variational derivation and its statistical interpretation, see Blei, Kucukelbir and McAuliffe, *Variational Inference: A Review for Statisticians*.

Then

$$\log q(z) = -\frac{1}{2}\log(2\pi\sigma^2) - \frac{(z-\mu)^2}{2\sigma^2},$$

and

$$\log p(z) = -\frac{1}{2}\log(2\pi) - \frac{z^2}{2}.$$

Therefore

$$\begin{aligned} D_{\mathrm{KL}}(q\|p) &= \mathbb{E}_q[\log q(z) - \log p(z)] \\ &= -\frac{1}{2}\log\sigma^2 - \frac{1}{2}\mathbb{E}_q\left[\frac{(z-\mu)^2}{\sigma^2}\right] + \frac{1}{2}\mathbb{E}_q[z^2] \\ &= -\frac{1}{2}\log\sigma^2 - \frac{1}{2} + \frac{1}{2}(\mu^2+\sigma^2) \\ &= \frac{1}{2}(\mu^2 + \sigma^2 - 1 - \log\sigma^2). \end{aligned}$$

The multivariate diagonal formula follows by summing over independent coordinates.

This is one reason the diagonal Gaussian encoder is practical: the regularization term does not need Monte Carlo estimation.

§164.8 Negative ELBO as a loss

Training usually minimizes the negative ELBO:

$$-\mathcal{L}(\theta,\phi;x) = -\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] + D_{\mathrm{KL}}(q_\phi(z \mid x)\|p(z)).$$

For a Bernoulli decoder with probabilities $\pi_\theta(z)$,

$$p_\theta(x \mid z) = \prod_i \pi_i(z)^{x_i}(1-\pi_i(z))^{1-x_i},$$

so

$$-\log p_\theta(x \mid z) = -\sum_i \left[x_i \log \pi_i(z) + (1-x_i)\log(1-\pi_i(z))\right].$$

For a Gaussian decoder

$$p_\theta(x \mid z) = \mathcal{N}(x; \mu_\theta(z), \sigma^2 I),$$

we have

$$-\log p_\theta(x \mid z) = \frac{1}{2\sigma^2}\|x - \mu_\theta(z)\|^2 + \text{constant}.$$

Thus the common reconstruction losses are not arbitrary engineering choices. They are negative log-likelihoods under different decoder distributions.

§164.9 Empirical lower bounds

The original AEVB experiments plotted the variational lower bound during training. The figure below is useful as a historical reminder: what is being optimized is not raw reconstruction error, but a lower bound $\mathcal{L}$ on marginal likelihood.

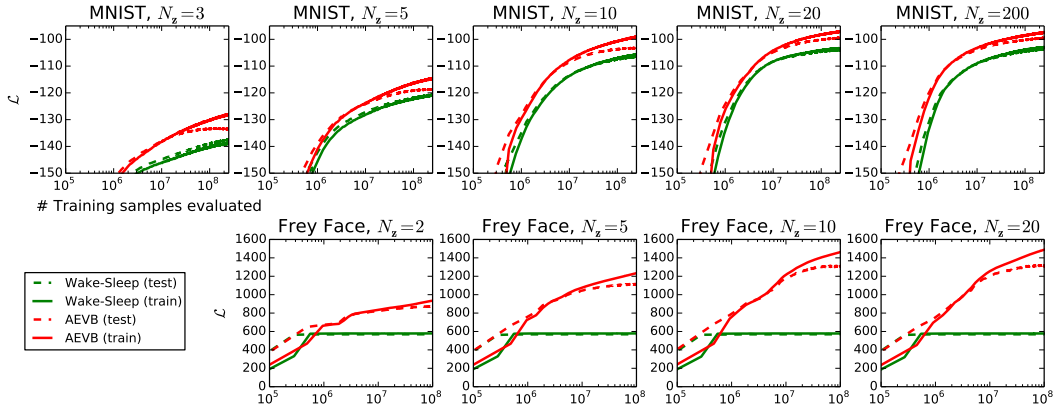


Figure 164.2: Variational lower bound curves in the AEVB experiments.

The mathematical significance of the plot comes from the identity

$$\log p_\theta(x) = \mathcal{L}(\theta, \phi; x) + D_{\mathrm{KL}}(q_\phi(z \mid x) \| p_\theta(z \mid x)).$$

§164.10 Information-theoretic reading

The ELBO also resembles a rate–distortion tradeoff. The expected log-likelihood asks the latent variable to preserve enough information to explain x. The KL term penalizes deviations of $q_\phi(z \mid x)$ from the prior $p(z)$.

Loosely:

term	role
$-\mathbb{E}_q[\log p_\theta(x \mid z)]$	distortion / reconstruction cost
$D_{\mathrm{KL}}(q_\phi(z \mid x) \| p(z))$	rate / information cost

This interpretation is helpful, but it should not replace the probabilistic derivation. The ELBO is first of all a lower bound on marginal likelihood.

There is an exact identity behind the rate interpretation, and it makes the limitations of that interpretation precise. Suppose inputs are drawn from $p_{\mathrm{data}}(x)$, and write the joint law induced by the encoder as

$$q(x, z) = p_{\mathrm{data}}(x) q_\phi(z \mid x), \qquad q(z) = \int p_{\mathrm{data}}(x) q_\phi(z \mid x)\, dx.$$

The latter $q(z)$ is the *aggregated posterior*: sample a training input and then a code from its encoder distribution. Insert $\log q(z) - \log q(z)$ into the KL term averaged over inputs:

$$\begin{aligned} &\mathbb{E}_{p_{\mathrm{data}}(x)} D_{\mathrm{KL}}(q_\phi(z \mid x) \| p(z)) \\ &= \mathbb{E}_{q(x,z)} \left[\log \frac{q_\phi(z \mid x)}{q(z)} + \log \frac{q(z)}{p(z)} \right] \\ &= \boxed{I_q(X; Z) + D_{\mathrm{KL}}(q(z) \| p(z)).} \end{aligned}$$

The first term is mutual information: how much the code tells us about the observed input. The second term measures how far the *overall* distribution of inferred codes is from the sampling prior. Thus the per-example KL penalty is an *upper bound* on encoded mutual information, and it also penalizes a mismatch between aggregate code usage and the prior.

A two-bit calculation distinguishes the terms. Let X be a fair bit, encode it perfectly by $Z = X$, and choose the prior $p(Z = 0) = p(Z = 1) = 1/2$. Then the aggregated posterior already equals the prior, so $D_{\mathrm{KL}}(q(z)\|p(z)) = 0$. Yet the encoder carries one bit about X, and

$$I_q(X; Z) = \log 2, \qquad \mathbb{E}_X D_{\mathrm{KL}}(q(z \mid X)\|p(z)) = \log 2.$$

Matching the aggregated posterior to the prior does not force the representation to forget the input. Conversely, if the encoder ignores x and always returns the prior, the KL is zero and $I_q(X; Z) = 0$. Such a representation can suffer posterior collapse if the decoder can explain the data unaided.

The rate–distortion analogy is therefore more than a slogan, but the word *rate* requires care: the KL to the prior contains mutual information *plus* aggregated-posterior mismatch. Neither term automatically measures the semantic usefulness of a latent coordinate.

§164.11 Limits of the ELBO

The ELBO is powerful, but it is not magic. Several limitations remain.

First, the variational family may be too small. A diagonal Gaussian $q_\phi(z \mid x)$ cannot represent a complicated multimodal posterior.

Second, the prior may be too simple. The standard normal prior is convenient, but the true aggregated latent structure of data may not be standard Gaussian.

Third, the decoder may be too powerful. If the decoder can model the data without using z, the approximate posterior may collapse toward the prior. This is posterior collapse.

Fourth, maximizing a lower bound is not the same as exactly maximizing $\log p_\theta(x)$. The gap is the posterior KL.

§164.12 What survives later

The VAE objective is not merely a heuristic sum of two losses. It is a variational identity:

$$\log p_\theta(x) = \mathcal{L}(\theta, \phi; x) + D_{\mathrm{KL}}(q_\phi(z \mid x)\|p_\theta(z \mid x)).$$

The ELBO can be written as

$$\mathcal{L}(\theta, \phi; x) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] - D_{\mathrm{KL}}(q_\phi(z \mid x)\|p(z)).$$

The first form explains why it is a lower bound. The second form explains why it looks like reconstruction plus regularization.

The remaining question is computational: how can we differentiate an expectation with respect to parameters that also control the sampling distribution? That is the role of the reparameterization trick.

The Reparameterization Trick and Auto-Encoding Variational Bayes

N-20251011

§165.1 The gradient problem

The ELBO for one data point is

$$\mathcal{L}(\theta, \phi; x) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] - D_{\mathrm{KL}}(q_\phi(z \mid x) \| p(z)).$$

For the diagonal Gaussian encoder and standard Gaussian prior, the KL term has a closed form. The difficult part is the expectation

$$\mathbb{E}_{z \sim q_\phi(z|x)}[\log p_\theta(x \mid z)].$$

We need gradients with respect to both θ and ϕ:

$$\nabla_{\theta,\phi} \mathbb{E}_{z \sim q_\phi(z|x)}[\log p_\theta(x \mid z)].$$

The subtlety is that ϕ controls the distribution from which z is sampled. The sample z is random, but its randomness depends on the encoder parameters.

The reparameterization trick is the mathematical device that turns this stochastic node into a differentiable transformation of parameter-free noise.

§165.2 Score-function estimator and pathwise estimator

There is a general identity for differentiating expectations:

$$\nabla_\phi \mathbb{E}_{q_\phi(z)}[f(z)] = \mathbb{E}_{q_\phi(z)}[f(z) \nabla_\phi \log q_\phi(z)].$$

This is often called a score-function estimator. It is very general because it does not require z to be a differentiable function of ϕ. But it often has high variance.

The reparameterization trick uses a more geometric idea. Instead of sampling

$$z \sim q_\phi(z \mid x),$$

write

$$z = g_\phi(\epsilon, x), \qquad \epsilon \sim p(\epsilon),$$

where the noise distribution $p(\epsilon)$ does not depend on ϕ.

Then

$$\mathbb{E}_{z \sim q_\phi(z|x)}[f(z)] = \mathbb{E}_{\epsilon \sim p(\epsilon)}[f(g_\phi(\epsilon, x))].$$

Now the parameters appear inside an ordinary differentiable function. Gradients can pass through g_ϕ.

§165.3 Two Unbiased Gradient Estimators

Consider the one-dimensional problem

$$Z \sim \mathcal{N}(\mu, 1), \qquad f(Z) = Z^2, \qquad F(\mu) = \mathbb{E}[Z^2] = \mu^2 + 1.$$

The correct derivative is $F'(\mu) = 2\mu$. This tiny example exposes the two common ways of obtaining stochastic derivatives when the *distribution*, rather than just the function, contains the parameter.

The score identity comes from differentiating the density:

$$\frac{d}{d\mu}\mathbb{E}_{q_\mu}[f(Z)] = \int f(z)\frac{\partial q_\mu(z)}{\partial \mu}\,dz = \mathbb{E}_{q_\mu}\left[f(Z)\frac{\partial \log q_\mu(Z)}{\partial \mu}\right].$$

For the unit-variance Gaussian, $\partial_\mu \log q_\mu(z) = z - \mu$. The score-function estimate from one sample is therefore

$$G_{\text{score}} = Z^2(Z - \mu).$$

It is unbiased: write $Z = \mu + \epsilon$, $\epsilon \sim \mathcal{N}(0, 1)$ and use the odd moments of ϵ:

$$\mathbb{E}[Z^2(Z - \mu)] = \mathbb{E}[(\mu + \epsilon)^2\epsilon] = 2\mu.$$

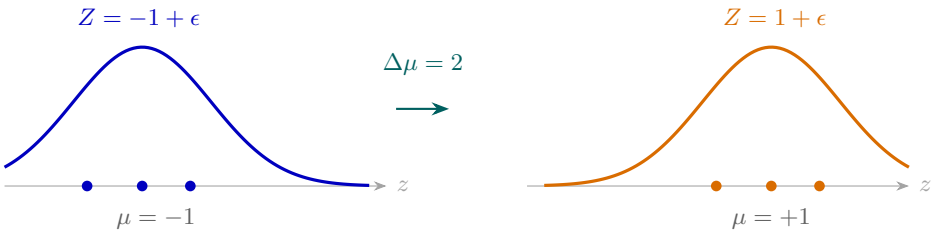
The pathwise estimator uses $Z = \mu + \epsilon$ *from the beginning*. The noise has a fixed distribution, so we differentiate the ordinary function $(\mu + \epsilon)^2$:

$$G_{\text{path}} = \frac{\partial}{\partial \mu}(\mu + \epsilon)^2 = 2(\mu + \epsilon) = 2Z, \qquad \mathbb{E}[G_{\text{path}}] = 2\mu.$$

At the transparent setting $\mu = 0$, both estimators have mean zero, but their variances are

$$\operatorname{Var}(G_{\text{score}}) = \mathbb{E}[Z^6] = 15,$$
$$\operatorname{Var}(G_{\text{path}}) = 4\mathbb{E}[Z^2] = 4.$$

Thus the pathwise estimator has lower variance *in this example*. This is not a universal variance-ordering theorem; it explains why moving randomness into a parameter-independent noise source can be so effective for smooth continuous models. The score identity also works for discrete random variables, where a differentiable sample path may not exist.



The dots use the same three noise draws $\epsilon = -0.8, 0, 0.7$.

Figure 165.1: Pathwise differentiation moves individual samples along with their density. The same noise value ϵ produces $Z_\mu = \mu + \epsilon$; increasing μ by 2 moves every sample by 2, so $\partial Z_\mu/\partial\mu = 1$. The fixed random seeds make the correspondence between the two densities visible.

Interchanging derivative and expectation requires suitable regularity, for instance differentiability together with an integrable dominating bound for the derivative. In a

neural decoder this condition is often satisfied locally under standard assumptions, but it is still a mathematical assumption. The original AEVB construction uses the lower path to make stochastic optimization of continuous latent-variable models practical.[1]

§165.4 Gaussian reparameterization

For the standard VAE encoder,

$$q_\phi(z \mid x) = \mathcal{N}(z; \mu_\phi(x), \operatorname{diag}(\sigma_\phi^2(x))).$$

A sample from this distribution can be written as

$$z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I).$$

Here $\odot$ denotes coordinatewise multiplication.

Thus

$$\begin{aligned}&\mathbb{E}_{z\sim q_\phi(z|x)}[\log p_\theta(x \mid z)] \\ &\quad = \mathbb{E}_{\epsilon\sim\mathcal{N}(0,I)}\left[\log p_\theta(x \mid \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon)\right].\end{aligned}$$

The distribution of ϵ is fixed. The dependence on ϕ is now inside μ_ϕ and σ_ϕ, so ordinary backpropagation can be used.

The trick is not a heuristic. It is a change of variables in the sampling procedure.

§165.5 The stochastic ELBO estimator

With L samples $\epsilon^{(1)}, \dots, \epsilon^{(L)} \sim \mathcal{N}(0, I)$, define

$$z^{(\ell)} = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon^{(\ell)}.$$

Then the Monte Carlo estimator of the ELBO is

$$\widehat{\mathcal{L}}(\theta, \phi; x) = \frac{1}{L} \sum_{\ell=1}^{L} \log p_\theta(x \mid z^{(\ell)}) - D_{\mathrm{KL}}(q_\phi(z \mid x) \| p(z)).$$

For minibatches, the dataset objective is approximated by averaging this quantity over the examples in the minibatch.

The original AEVB method is therefore a stochastic-gradient method for maximizing a variational lower bound. The randomness is not removed; it is organized so that differentiation is possible.

§165.6 Why implementations output log variance

Mathematically, the encoder needs $\sigma_\phi(x) > 0$. In practice, one often outputs

$$\log \sigma_\phi^2(x)$$

instead of $\sigma_\phi(x)$ directly. If

$$\ell_\phi(x) = \log \sigma_\phi^2(x),$$

then

$$\sigma_\phi(x) = \exp\left(\frac{1}{2}\ell_\phi(x)\right).$$

[1] See Kingma and Welling, *Auto-Encoding Variational Bayes*, especially the reparameterized lower-bound estimator and amortized inference construction.

This automatically keeps the standard deviation positive.

The Gaussian KL then becomes

$$D_{\mathrm{KL}}(q\|p) = \frac{1}{2}\sum_{j=1}^{d}\left(\mu_j^2 + \exp(\ell_j) - 1 - \ell_j\right).$$

This is the same formula as before, written in terms of log variance.

A single numerical forward pass makes the symbols operational. Suppose a one-dimensional encoder returns

$$\mu_\phi(x) = \frac{1}{2}, \qquad \ell_\phi(x) = \log\sigma_\phi^2(x) = \log\frac{1}{4}.$$

Then $\sigma_\phi(x) = 1/2$. If the fixed noise draw happens to be $\epsilon = -1$, the reparameterized latent sample is

$$z = \frac{1}{2} + \frac{1}{2}(-1) = 0.$$

The analytic KL penalty against $\mathcal{N}(0,1)$ is

$$\begin{aligned} D_{\mathrm{KL}}(q_\phi(z \mid x)\|p(z)) &= \frac{1}{2}\left(\frac{1}{4} + \frac{1}{4} - 1 - \log\frac{1}{4}\right) \\ &= \boxed{\log 2 - \frac{1}{4} \approx 0.4431}. \end{aligned}$$

For a simple Gaussian decoder $p_\theta(x \mid z) = \mathcal{N}(x; z, 1)$ and an observed $x = 1$, this particular noise draw produces a stochastic negative ELBO estimate

$$-\widehat{\mathcal{L}} = \frac{1}{2}\log(2\pi) + \frac{1}{2}(1-0)^2 + \left(\log 2 - \frac{1}{4}\right).$$

The sample-dependent reconstruction term changes from draw to draw, while the analytic Gaussian KL does not. A Monte Carlo estimate of the expectation is unbiased under independent sampling; a single random draw is not itself a deterministic lower bound on the evidence.

§165.7 Training as ELBO maximization

The mathematical training loop is:

encoder:	$x \mapsto \mu_\phi(x),\ \log\sigma_\phi^2(x),$
noise:	$\epsilon \sim \mathcal{N}(0, I),$
latent sample:	$z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon,$
decoder:	$z \mapsto$ parameters of $p_\theta(x \mid z),$
objective:	maximize $\widehat{\mathcal{L}}(\theta, \phi; x).$

If one minimizes a loss instead, the loss is the negative ELBO:

$$-\widehat{\mathcal{L}} = -\frac{1}{L}\sum_{\ell=1}^{L}\log p_\theta(x \mid z^{(\ell)}) + D_{\mathrm{KL}}(q_\phi(z \mid x)\|p(z)).$$

The first term depends on the decoder likelihood. The second term is analytic for the standard Gaussian encoder.

§165.8 Decoder likelihoods and reconstruction losses

For binary data, a Bernoulli decoder gives

$$p_\theta(x \mid z) = \prod_i \pi_i(z)^{x_i}(1-\pi_i(z))^{1-x_i}.$$

The negative log-likelihood is

$$-\log p_\theta(x \mid z) = -\sum_i \left[x_i \log \pi_i(z) + (1-x_i)\log(1-\pi_i(z))\right].$$

This is binary cross entropy.

For real-valued data, a Gaussian decoder with fixed variance gives

$$p_\theta(x \mid z) = \mathcal{N}(x; \mu_\theta(z), \sigma^2 I).$$

Then

$$-\log p_\theta(x \mid z) = \frac{1}{2\sigma^2}\|x - \mu_\theta(z)\|^2 + \text{constant}.$$

This is squared error up to scaling and constants.

Thus the word "reconstruction loss" hides a probabilistic choice. Different likelihood models produce different reconstruction terms.

§165.9 Generation after training

After learning θ, generation follows the model direction, not the inference direction:

$$z \sim p(z), \qquad x \sim p_\theta(x \mid z).$$

If the decoder likelihood is Gaussian, one may sample x from that Gaussian or use its mean as a representative output. If the decoder likelihood is Bernoulli, one samples binary coordinates or uses probabilities for visualization.

Latent interpolation uses points such as

$$z_t = (1-t)z_0 + tz_1, \qquad 0 \le t \le 1.$$

This often produces visually smooth transitions, but it is not a theorem that every linear path in latent space is semantically meaningful. It works when the learned latent geometry is reasonably aligned with the prior and decoder.

§165.10 Empirical marginal likelihood curves

The original experiments compared AEVB with older methods by plotting marginal log-likelihood estimates. The figure is included not as the main point of the theory, but as a reminder that the ELBO machinery was designed to make likelihood-based generative modeling trainable.

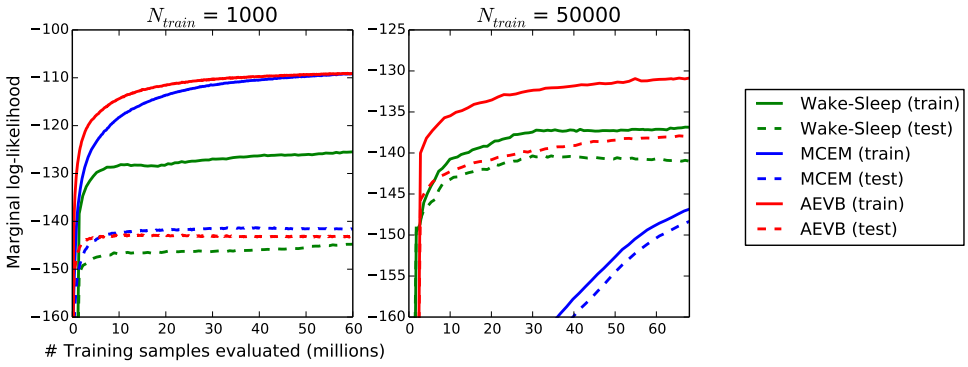


Figure 165.2: Marginal log-likelihood curves in the AEVB experiments.

The mathematical object behind these curves is still the same:

$$\log p_\theta(x) = \mathcal{L}(\theta, \phi; x) + D_{\mathrm{KL}}(q_\phi(z \mid x) \| p_\theta(z \mid x)).$$

Training improves a tractable stochastic estimate of $\mathcal{L}$, and the hope is that this also improves the marginal likelihood.

§165.11 Limitations and extensions

Several limitations are already visible from the mathematics.

First, the posterior family may be too simple. A diagonal Gaussian cannot capture all posterior shapes. Normalizing flows enrich $q_\phi(z \mid x)$ by transforming a simple base distribution through invertible maps.

Second, the prior may be too simple. The standard normal prior is convenient, but the aggregated posterior may have more complicated geometry.

Third, posterior collapse may occur. If the decoder is too expressive, the model may learn

$$q_\phi(z \mid x) \approx p(z),$$

so that z carries little information about x. Then the KL term is small, but the latent representation is weak.

Fourth, the usual ELBO uses one particular weighting between reconstruction and KL. Variants such as β-VAE modify the objective:

$$\mathbb{E}_q[\log p_\theta(x \mid z)] - \beta D_{\mathrm{KL}}(q_\phi(z \mid x) \| p(z)).$$

This changes the rate–distortion tradeoff and can encourage more disentangled latent coordinates, though not automatically.

§165.12 How the pieces fit

The VAE becomes trainable because the stochastic latent sample is rewritten as

$$z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I).$$

This makes the expectation differentiable with respect to the encoder parameters:

$$\mathbb{E}_{z \sim q_\phi(z|x)}[f(z)] = \mathbb{E}_{\epsilon \sim \mathcal{N}(0,I)}[f(\mu_\phi(x) + \sigma_\phi(x) \odot \epsilon)].$$

The full mathematical chain is now visible:

$$\begin{gathered}\text{latent variable model } p(z)p_\theta(x \mid z) \\ \Downarrow \\ \text{intractable posterior } p_\theta(z \mid x) \\ \Downarrow \\ \text{variational approximation } q_\phi(z \mid x) \\ \Downarrow \\ \text{ELBO lower bound} \\ \Downarrow \\ \text{reparameterized stochastic gradients.}\end{gathered}$$

This is why the VAE is more than an autoencoder with noise. It is a method for turning approximate Bayesian inference in a latent variable model into differentiable optimization.

Pure Mathematics as a Language for Machine Learning and Data Analysis

N-20260302

Suppose a neural network turns a million parameters into one loss. Computing its full derivative sounds like a million separate experiments, yet one backward pass gives all million partial derivatives. Suppose four data points lie at the corners of a square. A loop in their neighborhood graph exists at one spatial scale and vanishes at another. Or suppose two probability distributions have the same variance but different centers: there is a way to transport one into the other that is visibly optimal, without solving a huge optimization problem.

These puzzles look unrelated until we ask what is being *carried across a map*. A perturbation can be pushed forward, a covector pulled backward, a topological cycle followed through an inclusion, a unit of probability mass coupled to its destination, and a symmetry carried through a predictor. Each operation has a precise compatibility law. The pleasure comes from discovering that the compatibility law often does the hard calculation for us.[1]

§166.1 Backpropagation as cotangent pullback

A gradient is usually displayed as a column of numbers. The more useful geometric object is the differential: a linear functional that evaluates the first-order effect of any perturbation. A forward neural network is a composite of maps between parameter spaces, activation spaces, and finally the one-dimensional loss space. If

$$L : Y \to \mathbb{R}$$

is the loss, then its differential at the output is a cotangent vector

$$dL_y \in T_y^* Y.$$

Backpropagation pulls this covector backward along the forward computational graph. In this sense the derivative operation used in training is contravariant:

$$f : X \to Y \quad \Longrightarrow \quad f_x^* : T_{f(x)}^* Y \to T_x^* X.$$

Forward maps push tangent perturbations forward; reverse-mode automatic differentiation pulls covectors backward.

For a concrete PyTorch-shaped layer, take

$$X \in \mathbb{R}^{b \times d}, \qquad W \in \mathbb{R}^{n \times d}, \qquad Y = XW^T \in \mathbb{R}^{b \times n}.$$

In indices,

$$Y_{\alpha i} = \sum_{k=1}^{d} X_{\alpha k} W_{ik}.$$

[1] The differential viewpoint follows MIT OCW's *Matrix Calculus for Machine Learning and Beyond*. For persistence see Edelsbrunner and Harer, *Computational Topology*; for transport see Peyré and Cuturi, *Computational Optimal Transport*.

The full Jacobian with respect to the weight matrix has entries

$$\frac{\partial Y_{\alpha i}}{\partial W_{jk}} = \delta_{ij} X_{\alpha k},$$

so its tensor shape is

$$(b, n, n, d).$$

This is the honest tensor object. Neural-network code does not materialize it because the upstream gradient

$$G = \frac{\partial L}{\partial Y} \in \mathbb{R}^{b \times n}$$

immediately contracts it:

$$\begin{aligned}\frac{\partial L}{\partial W_{jk}} &= \sum_{\alpha=1}^{b} \sum_{i=1}^{n} \frac{\partial L}{\partial Y_{\alpha i}} \frac{\partial Y_{\alpha i}}{\partial W_{jk}} \\ &= \sum_{\alpha=1}^{b} G_{\alpha j} X_{\alpha k}.\end{aligned}$$

Therefore

$$\boxed{\frac{\partial L}{\partial W} = G^T X.}$$

If one stores weights in the column convention $W_{\text{col}} \in \mathbb{R}^{d \times n}$ and writes $Y = X W_{\text{col}}$, the same contraction is

$$\frac{\partial L}{\partial W_{\text{col}}} = X^T G.$$

The categorical phrase "pull back the covector" is exactly this matrix multiplication: the high-order Jacobian collapses because it is being paired with a covector from the scalar loss.

Why reverse differentiation reverses arrows

Let $f : X \to Y$ and $g : Y \to Z$ be smooth maps. The differential moves a tiny displacement forward:

$$D(g \circ f)_x = Dg_{f(x)}\, Df_x.$$

A covector at $g(f(x))$ is a linear rule for evaluating such displacements. Applying it after the composite is the same as first pulling it through Dg and then through Df:

$$\boxed{(g \circ f)^*_x = f^*_x \circ g^*_{f(x)}.}$$

The order reverses automatically. This identity is the precise meaning of reverse-mode differentiation; it is a property of dual maps.

For a small calculation, define

$$u_1 = x_1 + x_2, \qquad u_2 = x_1 x_2, \qquad L = u_1^2 + \tfrac{1}{2} u_2^2.$$

At $(x_1, x_2) = (1, 2)$, the forward pass gives $(u_1, u_2) = (3, 2)$ and $L = 11$. The loss first supplies the covector $(2u_1, u_2) = (6, 2)$. Pull it through

$$Du = \begin{pmatrix} 1 & 1 \\ x_2 & x_1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix}$$

to obtain

$$\boxed{\nabla_x L = (Du)^T \begin{pmatrix} 6 \\ 2 \end{pmatrix} = \begin{pmatrix} 10 \\ 8 \end{pmatrix}.}$$

We could expand $L = (x_1 + x_2)^2 + \frac{1}{2}x_1^2 x_2^2$ and differentiate directly; it gives the same answer. The two computations differ in what they remember: the backward calculation reuses the intermediate values of the forward computation.

There is also a useful pairing identity behind every implementation. If a forward perturbation is δx, then

$$\langle dL_y, Df_x(\delta x)\rangle = \langle (Df_x)^* dL_y, \delta x\rangle.$$

Both sides are the same first-order change in the loss. When Df is too large to write as a matrix, this identity defines the operation that the backward pass must compute. It also makes a finite-difference sanity check possible: replacing x by $x + h\delta x$ should change L by $h\,\langle (Df_x)^* dL, \delta x\rangle + O(h^2)$.

The Adjoint-State Method

Reverse-mode differentiation is even more striking when the layer is the solution of an equation. Suppose a parameter vector θ determines an invertible matrix and a right-hand side, and define $u(\theta)$ implicitly by

$$A(\theta)u(\theta) = b(\theta).$$

We care about one scalar objective $J(\theta) = \ell(u(\theta), \theta)$. Differentiating the linear system in a direction $\delta\theta$ yields

$$A\,\delta u = \delta b - (\delta A)u.$$

It looks as though we must solve another system for δu for *every* parameter. But the differential of the objective is only

$$\delta J = \nabla_u \ell^T \delta u + D_\theta \ell[\delta\theta].$$

Solve instead the single *adjoint system*

$$\boxed{A^T \lambda = \nabla_u \ell.}$$

Then the unwanted δu disappears:

$$\begin{aligned} \delta J &= \lambda^T A\,\delta u + D_\theta \ell[\delta\theta] \\ &= \boxed{\lambda^T(\delta b - (\delta A)u) + D_\theta \ell[\delta\theta].} \end{aligned}$$

One forward solve for u and one backward solve for λ give directional derivatives with respect to arbitrarily many parameters. This is the adjoint method used in differentiable physics and PDE-constrained optimization; backpropagation through a feedforward network is its finite computational-graph counterpart.

Here is a two-by-two check:

$$A(t) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad J(t) = \frac{1}{2}\|u(t)\|^2.$$

Solving directly gives $u(t) = (1 - t, 1)^T$, so

$$J'(t) = -(1 - t).$$

For the adjoint method, $A(t)^T\lambda = u(t)$ implies $\lambda_1 = 1 - t$. Since

$$A'(t)u(t) = \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

the boxed formula gives $J'(t) = -\lambda_1 = -(1-t)$ again. The little example demonstrates the point of a large theory: it is the *adjoint variable*, not a gigantic derivative array, that answers a scalar sensitivity question.

§166.2 Geometric data analysis

The same question reappears in data analysis: which mathematical language becomes necessary when coordinates alone no longer describe the structure?

A point cloud is often treated as a finite sample from a hidden space. Topological data analysis makes this idea precise by replacing the cloud with a filtered simplicial complex. For a scale parameter ε, one forms for instance the Vietoris–Rips complex

$$VR_\varepsilon(X) = \{\sigma \subseteq X : \ d(x,y) \le \varepsilon \text{ for all } x, y \in \sigma\}.$$

As ε grows, connected components merge, loops appear and disappear, and higher-dimensional holes may be born and die. Persistent homology records this evolution. The chain complex viewpoint is governed by the boundary identity

$$\partial_{k-1} \circ \partial_k = 0,$$

and homology is

$$H_k = \ker \partial_k / \operatorname{im} \partial_{k+1}.$$

This quotient is the mathematical form of the phrase "holes that are cycles but not boundaries."

Example 166.2.1 (A square-shaped hole with a finite lifetime)

Take the four vertices of the unit square with Euclidean distance. When $1 \le \varepsilon < \sqrt{2}$, their Vietoris–Rips complex is a four-edge cycle without any filled triangles. Its first homology over a field has dimension one. When $\varepsilon \ge \sqrt{2}$, both diagonals appear, so the complex is the complete 3-simplex and H_1 vanishes. The one-dimensional homology class therefore persists over $[1, \sqrt{2})$ under this filtration convention.

Four points, one hole, and one diagonal

Why can a loop be visible at one resolution and disappear at another? Work with coefficients in $\mathbb{F}_2$, so that reversing the orientation of an edge makes no difference. Label the corners of the unit square $0, 1, 2, 3$ in cyclic order. At scale $1 \le \varepsilon < \sqrt{2}$ the Rips complex contains the four edges but neither diagonal. Consider the 1-chain

$$c = [01] + [12] + [23] + [30].$$

The boundary of an edge is the sum of its two endpoints, hence

$$\partial_1 c = (0+1) + (1+2) + (2+3) + (3+0) = 0 \quad \text{in } \mathbb{F}_2.$$

It is a cycle. But the complex has *no triangles*, so $\operatorname{im}\partial_2 = 0$: there is no two-dimensional chain with boundary c. Consequently

$$\boxed{H_1(VR_\varepsilon(X);\mathbb{F}_2) \cong \mathbb{F}_2, \qquad 1 \le \varepsilon < \sqrt{2}.}$$

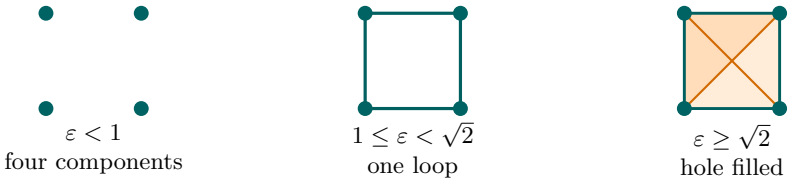


Figure 166.1: The Vietoris–Rips filtration of four square corners. Edges appear at distance 1; diagonals and all triangles appear at distance $\sqrt{2}$. The loop therefore survives for precisely the interval $[1, \sqrt{2})$. The rightmost complex also includes the tetrahedral 3-simplex, whose two-dimensional faces are represented by the filled regions.

When the diagonal [02] appears, the two triangles [012] and [023] are present. Their boundaries add to

$$\partial_2([012] + [023]) = [01] + [12] + [20] + [02] + [23] + [30] = c,$$

because the diagonal occurs twice and cancels in $\mathbb{F}_2$. The same chain that was a nontrivial hole has become a boundary. Nothing about the four points changed; only the scale of the question did. A persistent barcode draws the resulting H_1 class as one horizontal interval with birth 1 and death $\sqrt{2}$.

Why this picture survives small measurement errors

Suppose the point x_i moves to y_i with $d(x_i, y_i) \le \delta$. For every pair,

$$\bigl|d(x_i, x_j) - d(y_i, y_j)\bigr| \le 2\delta$$

by the triangle inequality. Thus every edge, triangle, or higher simplex present at scale ε in the first Rips complex is present at scale $\varepsilon + 2\delta$ in the second:

$$VR_\varepsilon(X) \longrightarrow VR_{\varepsilon+2\delta}(Y).$$

The reverse comparison has the same form. Such inclusions give a 2δ-interleaving of the two filtrations, so their persistence diagrams can be matched with bottleneck error at most 2δ (under the standard finite-filtration conventions). In particular, a loop with a lifetime much longer than 4δ cannot be matched to the diagonal by a perturbation of this size: its lifetime exceeds twice the bottleneck error.

This is the practical reason to measure *persistence* rather than count holes at one chosen resolution. One scale may be capricious; a long interval survives a quantified amount of noise. The general stability theorem of Cohen-Steiner, Edelsbrunner and Harer develops this idea for sublevel-set persistence. For finite point clouds, the simple edge-distance argument above already explains the stability mechanism.

§166.3 Manifold hypothesis and embeddings

The manifold hypothesis says that high-dimensional data often concentrate near a lower-dimensional manifold $M \subseteq \mathbb{R}^N$. Whitney's embedding theorem guarantees a smooth embedding of a smooth d-manifold into $\mathbb{R}^{2d}$. This is an existence theorem for smooth coordinates; it does not imply that a finite noisy dataset lies on such a manifold, or that a learned representation will recover an embedding. These conclusions require separate assumptions on sampling, geometry, and the optimization objective.

The practical warning is that an embedding preserves the manifold as a smooth object, but not necessarily distances, probability densities, or task-relevant semantics. PCA preserves a linear subspace model; manifold learning tries to preserve local neighborhoods; representation learning tries to learn coordinates useful for a loss function.

A Circle That Defeats PCA

Take points uniformly distributed on the unit circle,

$$X = (\cos\Theta, \sin\Theta), \qquad \Theta \sim \mathrm{Unif}[0, 2\pi).$$

The support is a smooth one-dimensional manifold. Yet its mean is zero and its covariance is

$$\mathrm{Cov}(X) = \begin{pmatrix} \mathbb{E}\cos^2\Theta & \mathbb{E}\sin\Theta\cos\Theta \\ \mathbb{E}\sin\Theta\cos\Theta & \mathbb{E}\sin^2\Theta \end{pmatrix} = \tfrac{1}{2} I_2.$$

Every line through the origin explains exactly the same variance. Therefore PCA has *no preferred first principal direction*, even though the data are intrinsically one-dimensional. The dimension of a linear span and the dimension of a curved manifold answer different questions.

There is a second difficulty. If we try to use the first coordinate $x = \cos\theta$ as a global parameter, it assigns the same value to θ and $-\theta$. At the leftmost and rightmost points, projection onto the horizontal axis has vanishing tangential derivative and fails to be a local coordinate. A single linear coordinate cannot traverse a circle once without identifying points.

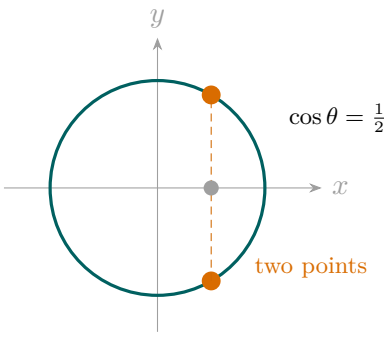


Figure 166.2: The circle has intrinsic dimension one but covariance $\frac{1}{2} I_2$, so its PCA directions are tied. The two orange points are distinct, yet their projection onto the x-axis is identical.

A genuinely local coordinate can be obtained by the half-angle substitution

$$t = \tan\frac{\theta}{2}, \qquad (\cos\theta, \sin\theta) = \left(\frac{1 - t^2}{1 + t^2}, \frac{2t}{1 + t^2}\right).$$

As t runs through $\mathbb{R}$, this parametrizes the circle except the point $(-1, 0)$, which is reached only as $|t| \to \infty$. A second chart handles that missing point. This illustrates exactly why differential geometry uses *atlases* of charts, and why a globally continuous one-to-one latent coordinate can be topologically impossible for a circular data manifold.

§166.4 Optimal transport and stochastic dynamics

For probability measures μ, ν on a metric space, the p-Wasserstein distance is

$$W_p(\mu,\nu) = \left(\inf_{\pi \in \Pi(\mu,\nu)} \int d(x,y)^p \, d\pi(x,y) \right)^{1/p}.$$

Here $\Pi(\mu,\nu)$ is the set of couplings with marginals μ and ν. The dual viewpoint is equally important. For $p = 1$,

$$W_1(\mu,\nu) = \sup_{\|f\|_{\mathrm{Lip}} \le 1} \left(\int f \, d\mu - \int f \, d\nu \right).$$

The uncrossing trick

Let two equally weighted particles sit at -1 and 1, with destinations 0 and 2:

$$\mu = \tfrac{1}{2}\delta_{-1} + \tfrac{1}{2}\delta_{1}, \qquad \nu = \tfrac{1}{2}\delta_{0} + \tfrac{1}{2}\delta_{2}.$$

Sending -1 to 0 and 1 to 2 costs $\frac{1}{2}(1^2 + 1^2) = 1$; crossing the assignments costs $\frac{1}{2}(3^2 + 1^2) = 5$. The reason extends beyond this example. If $x_1 < x_2$ and $y_1 < y_2$,

$$(x_1 - y_1)^2 + (x_2 - y_2)^2 - (x_1 - y_2)^2 - (x_2 - y_1)^2 = -2(x_2 - x_1)(y_2 - y_1) \le 0.$$

Thus uncrossing any two matched pairs cannot raise quadratic transport cost.

Write $F^{-1}(u) = \inf\{x : F(x) \ge u\}$ for a distribution's quantile function. Pairing the same uniform random number $U \in (0,1)$ across two distributions gives $X = F_\mu^{-1}(U)$ and $Y = F_\nu^{-1}(U)$. The no-crossing argument first proves optimality for finite ordered point clouds; approximating general one-dimensional distributions gives

$$\boxed{W_p^p(\mu,\nu) = \int_0^1 |F_\mu^{-1}(u) - F_\nu^{-1}(u)|^p \, du, \quad p \ge 1,}$$

when the p-th moments are finite. A minimization over every possible joint distribution has collapsed to one integral. Empirically, this says to sort both samples and pair their order statistics.

For $p = 2$ there is a useful one-dimensional closed form. Let

$$\mu = \mathcal{N}(a, \sigma_1^2), \qquad \nu = \mathcal{N}(b, \sigma_2^2), \qquad \sigma_1, \sigma_2 > 0.$$

In one dimension the optimal map between these two Gaussians is affine and monotone, so write

$$T(x) = \alpha x + \beta.$$

The condition $T_{\#}\mu = \nu$ forces

$$\alpha = \frac{\sigma_2}{\sigma_1}, \qquad \beta = b - \alpha a.$$

If $X = a + \sigma_1 Z$ with $Z \sim \mathcal{N}(0,1)$, then

$$T(X) = b + \sigma_2 Z.$$

Thus

$$\begin{aligned} W_2^2(\mu,\nu) &= \mathbb{E}[(X - T(X))^2] \\ &= \mathbb{E}[((a-b) + (\sigma_1 - \sigma_2)Z)^2] \\ &= (a-b)^2 + (\sigma_1 - \sigma_2)^2. \end{aligned}$$

This formula is small, but it explains a lot of geometric language around generative models: matching Gaussians means matching both their centers and their spreads.

A curved-looking space of measures hides a flat plane

The Gaussian formula does something more surprising than calculate a distance. Fix two one-dimensional Gaussian laws with means a, b and positive standard deviations σ_0, σ_1. Their optimal coupling uses the *same* standard Gaussian Z:

$$X_0 = a + \sigma_0 Z, \qquad X_1 = b + \sigma_1 Z.$$

Move each paired particle at constant speed:

$$X_s = (1-s)X_0 + sX_1 = \big((1-s)a + sb\big) + \big((1-s)\sigma_0 + s\sigma_1\big)Z.$$

The law μ_s is Gaussian for every $0 \le s \le 1$. Writing $a_s = (1-s)a + sb$ and $\sigma_s = (1-s)\sigma_0 + s\sigma_1$, the one-dimensional transport formula gives

$$\begin{aligned} W_2^2(\mu_s,\mu_t) &= (a_s - a_t)^2 + (\sigma_s - \sigma_t)^2 \\ &= |s-t|^2((a-b)^2 + (\sigma_0 - \sigma_1)^2). \end{aligned}$$

Thus

$$\boxed{W_2(\mu_s,\mu_t) = |s-t|\, W_2(\mu_0,\mu_1).}$$

We have constructed a constant-speed Wasserstein geodesic without invoking geodesic equations. In fact, the space of one-dimensional Gaussian distributions is isometric, under W_2, to the Euclidean half-plane with coordinates (mean, standard deviation) and $\sigma > 0$.

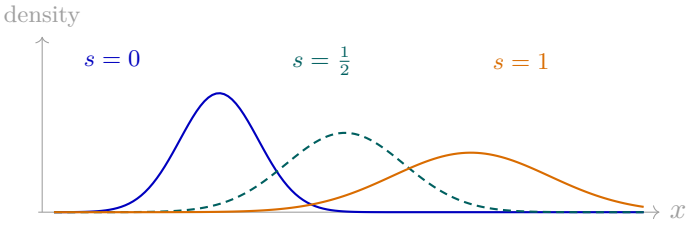


Figure 166.3: Three points of a Wasserstein geodesic between $\mathcal{N}(-1.6, 0.5^2)$ and $\mathcal{N}(1.6, 1^2)$. The intermediate law is $\mathcal{N}(0, 0.75^2)$: both the mean and the standard deviation interpolate linearly. This is the geometry of optimal coupling, distinct from merely adding Gaussian noise.

Diffusion models can be read through stochastic differential equations

$$dX_t = b(t, X_t)\, dt + \sigma(t)\, dW_t.$$

The density p_t evolves according to a Fokker–Planck equation, and the reverse-time dynamics involve the score $\nabla_x \log p_t(x)$. Thus score-based generative modeling is an attempt to learn the vector field that reverses a noisy transport of probability mass.

For DDPMs the training objective is usually written as a variational lower bound, or in its simplified form as a denoising mean-square loss. It is not literally a Wasserstein objective. Still, each reverse step compares Gaussian conditional distributions, and for Gaussians the geometry above says that a squared displacement of means and a mismatch of variances are exactly the ingredients of W_2^2. The ELBO uses KL divergences between Gaussian conditionals; when the variance is fixed or separately prescribed, minimizing that KL has the same practical center-matching behavior as minimizing the corresponding Gaussian W_2 term. The distinction matters: an optimal-transport cost minimizes over couplings, whereas the DDPM training objective comes from a likelihood-based variational construction. Similar quadratic penalties for fixed-variance Gaussian comparisons do not make the two optimization problems equivalent.

Score as a Denoising Direction

Let X_0 be any random vector in $\mathbb{R}^d$ with finite first moment. Corrupt it with Gaussian noise:

$$Y = X_0 + \sqrt{t}\,\epsilon, \qquad \epsilon \sim \mathcal{N}(0, I), \quad t > 0.$$

Its density is the Gaussian convolution

$$p_t(y) = \int (2\pi t)^{-d/2} e^{-\|y-x\|^2/(2t)}\, p_0(dx).$$

Differentiate *this* expression, using $\nabla_y e^{-\|y-x\|^2/(2t)} = -(y-x)e^{-\|y-x\|^2/(2t)}/t$:

$$t\nabla_y p_t(y) = \int (x-y)\,\varphi_t(y-x)\,p_0(dx),$$

$$t\nabla_y \log p_t(y) = \frac{\int x\varphi_t(y-x)\,p_0(dx)}{\int \varphi_t(y-x)\,p_0(dx)} - y.$$

The quotient is just the conditional expectation of the clean signal given its noisy observation. Therefore

$$\boxed{\mathbb{E}[X_0 \mid Y = y] = y + t\,\nabla_y \log p_t(y).}$$

This is Tweedie's formula. It explains the score as a geometrically meaningful vector field: t times the score is the correction from a noisy point to its *posterior mean* clean point. It is a theorem about conditional expectation, independent of any neural network.

Here is a complete discrete example. Let X_0 be equally likely to be $-a$ or $+a$. After observing $Y = y$, the posterior odds are

$$\frac{\Pr(X_0 = a \mid y)}{\Pr(X_0 = -a \mid y)} = \exp\left(\frac{2ay}{t}\right).$$

Hence

$$\mathbb{E}[X_0 \mid y] = a\tanh\frac{ay}{t}, \qquad \boxed{\frac{d}{dy}\log p_t(y) = \frac{a\tanh(ay/t) - y}{t}.}$$

For small noise, positive observations are corrected toward $+a$ and negative ones toward $-a$; near $y = 0$, the two explanations compete. As noise increases, the two peaks merge.

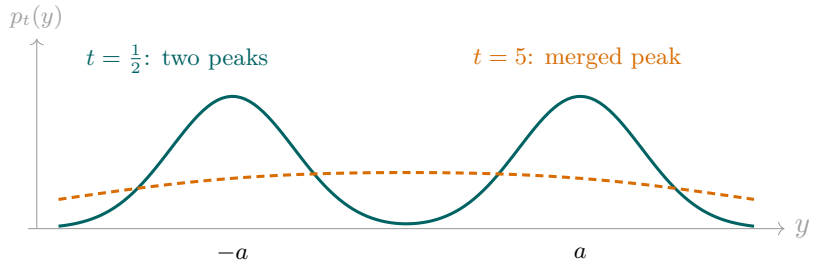


Figure 166.4: Two point masses at $\pm a$ with $a = 2$, blurred by Gaussian noise. The colored curves are the exact Gaussian-mixture densities at $t = \frac{1}{2}$ and $t = 5$, plotted with the same vertical multiplier for legibility. The score field is the logarithmic derivative of these changing distributions.

Optimal transport and diffusion can both describe movement of probability mass, but their mechanisms differ: the former chooses a coupling to minimize displacement, while heat diffusion continually mixes in new noise. The denoising formula makes precise how the latter noisy evolution can still carry information about the original distribution.

§166.5 Symmetry, equivariance, and category-like thinking

If a group G acts on an input space and an output space, a map F is equivariant if

$$F(gx) = gF(x).$$

Convolutional neural networks exploit translation equivariance; graph neural networks exploit permutation equivariance; gauge-equivariant networks try to respect local choices of frame on a bundle.

Here is the matrix calculation behind the slogan. Let the graph be the cycle C_4, with adjacency matrix

$$A = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix},$$

and let P be the permutation matrix rotating the four vertices by $90°$. Since P is a graph automorphism,

$$PA = AP.$$

The same is true for the usual normalized adjacency $\widehat{A}$, because normalization is built from A and the degree matrix, both invariant under the same rotation:

$$P\widehat{A} = \widehat{A}P.$$

A standard graph neural-network layer is

$$H^{(l+1)} = \sigma(\widehat{A}H^{(l)}W),$$

where σ acts entrywise. If the input features are rotated, then

$$\begin{aligned} F(PH^{(l)}) &= \sigma(\widehat{A}PH^{(l)}W) \\ &= \sigma(P\widehat{A}H^{(l)}W) \\ &= P\sigma(\widehat{A}H^{(l)}W) \\ &= PH^{(l+1)}. \end{aligned}$$

Equivariance is therefore not only a philosophical commitment to symmetry. In this example it is the algebraic fact that the permutation matrix commutes with the graph operator.

Functoriality has a testable meaning. A continuous map $f : X \to Y$ induces a homomorphism $H_k(f) : H_k(X) \to H_k(Y)$, and $H_k(g \circ f) = H_k(g) \circ H_k(f)$. This is a compatibility law between topological maps and homology, not an explanation of intelligence. Equivariance of a graph layer is likewise an explicit compatibility statement, expressed here by the commuting identity $F(PH) = PF(H)$.

An equivariant map can be made by averaging

There is a pleasing construction when the symmetry group is finite. Suppose G acts linearly on input and output spaces by representations ρ_{in} and ρ_{out}. Take *any* function h, possibly a function with no symmetry at all, and define

$$\boxed{(\mathcal{R}h)(x) = \frac{1}{|G|} \sum_{g \in G} \rho_{\text{out}}(g^{-1}) h\big(\rho_{\text{in}}(g)x\big).}$$

Why should this be equivariant? Apply the definition to a transformed input $\rho_{\text{in}}(k)x$, and rename the summation index $u = gk$:

$$\begin{aligned}(\mathcal{R}h)(\rho_{\text{in}}(k)x) &= \frac{1}{|G|} \sum_{g} \rho_{\text{out}}(g^{-1}) h(\rho_{\text{in}}(gk)x) \\ &= \rho_{\text{out}}(k) \frac{1}{|G|} \sum_{u} \rho_{\text{out}}(u^{-1}) h(\rho_{\text{in}}(u)x) \\ &= \rho_{\text{out}}(k)(\mathcal{R}h)(x).\end{aligned}$$

The proof consists of one change of summation variables. The group operation guarantees that the average sees every transformed input exactly once.

For example, let $G = \{e, s\}$ swap the coordinates of $\mathbb{R}^2$ and take

$$h(x_1, x_2) = (x_1^2 + x_2,\, 0).$$

The raw function does not commute with the swap. Its symmetrization is

$$(\mathcal{R}h)(x_1, x_2) = \frac{1}{2}\big(x_1^2 + x_2,\; x_2^2 + x_1\big),$$

which clearly swaps its two outputs whenever the inputs are swapped. This construction is often called *group averaging* or a Reynolds-type operator. It is an explicit way to turn a desired symmetry law into an exact property of a model.

There is a distinction worth keeping in the graph example. A *graph automorphism* satisfies $PAP^T = A$, so it may act on the node features while the adjacency matrix stays fixed. An arbitrary relabeling P usually changes the displayed adjacency matrix too. Under a simultaneous relabeling of graph and features, however,

$$\widehat{PAP^T} = P\widehat{A}P^T, \qquad \boxed{F(PAP^T, PH) = PF(A, H).}$$

This is permutation equivariance as a statement about the *representation of the same graph*, even when the permutation is not an automorphism of that graph.

A common law for maps of different kinds

The contravariance of backpropagation and the covariance of homology are different laws. A continuous map of spaces $f : X \to Y$ induces

$$H_k(f) : H_k(X) \to H_k(Y), \qquad H_k(g \circ f) = H_k(g) \circ H_k(f).$$

The directions here are *forward*: a chain is sent to the image chain. For cotangent vectors, dualizing reverses composition. The distinction is concrete. If an inclusion of simplicial complexes fills a hole, a nonzero cycle class may be carried *forward* to zero, whereas a sensitivity to an output is carried *backward* to an input.

The common principle is functoriality: what matters about a transformation is what it does to the structure under study, and a composite transformation must act compatibly with its two pieces. In machine learning this is useful only when the acting map and the preserved structure are named explicitly; otherwise the language has no computational content.

§166.6 Maps that remember the right information

The group-averaging construction hides one last satisfying fact. If h is already equivariant, every summand in $\mathcal{R}h$ equals $h(x)$, and therefore $\mathcal{R}h = h$. Because the first application of $\mathcal{R}$ makes *any* function equivariant, applying it again changes nothing:

$$\boxed{\mathcal{R}^2 = \mathcal{R}.}$$

Thus averaging is a *projection* in the space of functions, even when we have not chosen an inner product there. It takes an arbitrary predictor and projects it onto a class of predictors obeying an exact symmetry law. With a suitable invariant inner product, it becomes the familiar orthogonal projection onto the fixed subspace. A construction from group theory has become a geometric operation on models.

The other calculations have a similarly concrete payoff. The differential does not ask us to store a huge Jacobian: the cotangent pullback tells us which contraction to perform. A loop around four points becomes a boundary at a precise scale, and stability turns that observation into a noise-tolerant feature. Pairing quantiles solves the one-dimensional transport problem by forbidding crossings. Differentiating a Gaussian convolution turns an intractable-looking posterior expectation into a derivative of an observable density.

There is no need to treat these methods as interchangeable. Each has a different domain and a different invariant. What makes them powerful is that the difficult calculation is replaced by a structural identity whose assumptions can be checked and whose consequences can actually be computed.

Compression, Information, and Hidden Features

N-20261009

While working on BiSem, a ViT-based pipeline for compressing laboratory video and running its sender on a Raspberry Pi 5, I became interested in a peculiar discrepancy. The reconstructed images could lose quite visible details of a person or chair, while internal patch representations still carried information from which a small probe could predict person locations. Improving a local reconstruction score did not necessarily improve the probe. These are different tests of what a representation retains, and they have different mathematical answers.

I will begin with a tiny distribution that makes the distinction exact, then work through sufficient statistics, rate–distortion theory and the information bottleneck. After revisiting actual BiSem frames and latent measurements, we shall ask a second question: even if information survives, who decided which coordinates or sparse components deserve to be called *features*?[1]

§167.1 A perfect classifier and a poor reconstruction

Let Y and N be independent fair bits and put $X = (Y, N)$. Pretend that Y records the identity of an object and N records a nuisance detail. All four source outcomes have probability $1/4$:

$$X \in \{(0,0), (0,1), (1,0), (1,1)\}.$$

An encoder sends only

$$Z = f(X) = Y.$$

A classifier using Z predicts Y without error. A decoder trying to recover the *whole* X cannot guess N better than chance, because N is independent of Z.

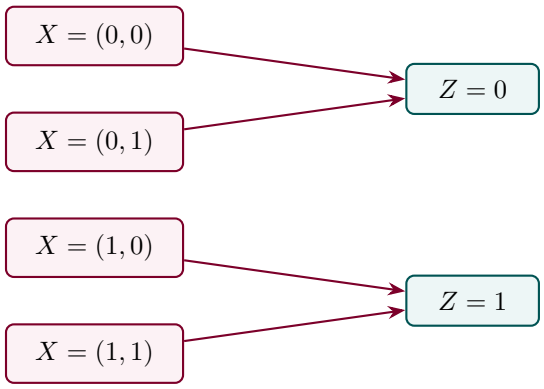


Figure 167.1: A task-perfect encoding that loses half of the source's two independent bits. Each codeword corresponds to two distinct possible inputs.

[1] For the information-bottleneck principle, see Tishby, Pereira and Bialek, *The Information Bottleneck Method* (1999/2000). For sparse representation uniqueness we follow Donoho and Elad, *Optimally Sparse Representation in General (Nonorthogonal) Dictionaries via ℓ^1 Minimization* (2003).

Measure reconstruction error by the number of incorrect bits:

$$d_{\text{rec}}(x, \widehat{x}) = \mathbf{1}_{\{y \neq \widehat{y}\}} + \mathbf{1}_{\{n \neq \widehat{n}\}}.$$

The optimal decoder using $Z = Y$ can reconstruct Y perfectly, but has

$$\Pr(\widehat{N} \neq N) = \frac{1}{2}, \qquad \mathbb{E}[d_{\text{rec}}(X, \widehat{X})] = \frac{1}{2}.$$

The task loss $d_{\text{task}}(y, \widehat{y}) = \mathbf{1}_{\{y \neq \widehat{y}\}}$, by contrast, is exactly zero.

This reconstruction error is the best possible for *any* deterministic one-bit code, not merely $Z = Y$. A deterministic decoder has only two possible outputs, one for each codeword. At most two of the four equally likely binary pairs can be reconstructed exactly. Each of the other two differs from its reconstruction in at least one bit, so the average number of erroneous bits is at least $2/4 = 1/2$. Our task-based encoder attains this bound while also solving the classification problem perfectly.

A representation can be excellent for one loss function and poor for another. The word *information* becomes useful only when we specify what information is being requested.

§167.2 Sufficiency as an information identity

For finite random variables, write $H(X)$ for Shannon entropy and

$$I(X; Y) = H(Y) - H(Y \mid X)$$

for mutual information. In the example,

$$H(X) = H(Y, N) = 2, \qquad H(Z) = 1,$$

with logarithms to base 2. Moreover

$$I(Z; Y) = 1 = I(X; Y), \qquad I(Z; X) = 1.$$

The one-bit representation retains *all* information about Y, although it retains only one of the two bits in X.

For a general encoder specified by a conditional distribution $p(z \mid x)$, assume the Markov relation

$$Y \longrightarrow X \longrightarrow Z : \qquad Y \perp Z \mid X.$$

The chain rule for mutual information gives

$$\begin{aligned} I(Y; X, Z) &= I(Y; X) + I(Y; Z \mid X) = I(Y; X), \\ I(Y; X, Z) &= I(Y; Z) + I(Y; X \mid Z). \end{aligned}$$

Therefore

$$\boxed{I(Y; X) - I(Y; Z) = I(Y; X \mid Z) \geq 0.} \tag{167.1}$$

This is the data processing inequality for the Markov chain. No downstream computation of Z can increase the information that X provided about Y.

Equality has a particularly useful interpretation:

$$I(Y; Z) = I(Y; X) \quad \Longleftrightarrow \quad I(Y; X \mid Z) = 0.$$

For finite alphabets, the latter is equivalent to $Y \perp X \mid Z$. Thus knowing Z makes the original input X irrelevant for predicting Y. This is the statistical sense in which Z is *sufficient* for Y.

In our example $Z = Y$ is sufficient. So is $Z = (Y, N)$, but the latter remembers unnecessary information. One can ask for a sufficient representation that carries as little information about X as possible. Since

$$I(X; Z) \geq I(Y; Z) = I(Y; X) = 1$$

for every sufficient Z, the simple choice $Z = Y$ is optimal for this particular distribution.

There is an important qualification in applications. A representation that is sufficient for a given label distribution need not be sufficient for every future task. If tomorrow's task asks for N, our one-bit representation loses it entirely. Sufficiency is always relative to the target being predicted.

§167.3 How expensive is it to remember more?

Rate–distortion theory gives a general language for the tension between compression and reconstruction. For a source X, a reconstruction $\widehat{X}$, and distortion $d(X, \widehat{X})$, define

$$R(D) = \inf_{\substack{p(\widehat{x}|x) \\ \mathbb{E}d(X,\widehat{X}) \leq D}} I(X; \widehat{X}). \tag{167.2}$$

The infimum runs over all (possibly randomized) reconstruction channels satisfying a distortion budget. The single-letter expression is the information-theoretic rate–distortion function of a memoryless source with additive distortion; operational coding statements about long blocks require a separate theorem.

A complete calculation is possible for one fair bit. Let $X \sim \mathrm{Bernoulli}(1/2)$ and use Hamming distortion $d(x, \widehat{x}) = \mathbf{1}_{\{x \neq \widehat{x}\}}$. If $\Pr(X \neq \widehat{X}) = d \leq D \leq 1/2$, define the error bit $E = X \oplus \widehat{X}$. Since X is determined by the pair $(\widehat{X}, E)$,

$$H(X \mid \widehat{X}) = H(E \mid \widehat{X}) \leq H(E) = h_2(d) \leq h_2(D),$$

where $h_2(u) = -u \log_2 u - (1-u)\log_2(1-u)$ is binary entropy. Hence

$$I(X; \widehat{X}) = 1 - H(X \mid \widehat{X}) \geq 1 - h_2(D).$$

To attain the bound, take an independent noise bit $E \sim \mathrm{Bernoulli}(D)$ and set $\widehat{X} = X \oplus E$. Then the reconstruction error is D, while $\widehat{X}$ is fair and E is independent of $\widehat{X}$. Thus $H(X \mid \widehat{X}) = h_2(D)$, giving

$$\boxed{R(D) = 1 - h_2(D) \quad (0 \leq D \leq \tfrac{1}{2}),} \tag{167.3}$$

and $R(D) = 0$ for $D \geq 1/2$. The answer is a convex decreasing curve: allowing progressively more reconstruction mistakes decreases the amount of information that must be transmitted.

For $X = (Y, N)$, the no-distortion requirements of our two tasks are already enough to exhibit different rate requirements:

$$R_{\mathrm{rec}}(0) = H(Y, N) = 2, \qquad R_{\mathrm{task}}(0) = H(Y) = 1.$$

An exact reconstruction of both independent bits requires two bits of information; an exact prediction of the first requires only one. The distortion function determines which bits are expensive to forget.

§167.4 The information bottleneck

The information bottleneck makes the target Y explicit in the compression objective. For an encoder $p(z \mid x)$ obeying $Y - X - Z$, consider

$$\mathcal{L}_\beta(Z) = I(X;Z) - \beta I(Y;Z), \qquad \beta \geq 0. \tag{167.4}$$

The first term penalizes retaining information about X. The second rewards retaining information useful for predicting Y.

For the two-bit example, we can solve this optimization *globally*, even allowing randomized encoders. The Markov-chain inequality gives $I(X;Z) \geq I(Y;Z)$, so

$$\mathcal{L}_\beta(Z) \geq (1-\beta) I(Y;Z).$$

When $0 \leq \beta \leq 1$, this is nonnegative, and a constant Z attains zero. When $\beta > 1$, use $I(Y;Z) \leq H(Y) = 1$ and note that multiplying by $1 - \beta < 0$ reverses the inequality. Hence

$$\mathcal{L}_\beta(Z) \geq 1 - \beta,$$

and $Z = Y$ attains it. Therefore

$$\boxed{\min_Z \mathcal{L}_\beta(Z) = \begin{cases} 0, & 0 \leq \beta \leq 1, \\ 1-\beta, & \beta > 1. \end{cases}} \tag{167.5}$$

In this tiny world the solution makes an abrupt transition: with a weak reward for relevance, the best code says nothing; with a sufficiently strong reward, it communicates precisely the task bit. More complicated sources yield richer trade-off curves. This example is useful because the entire optimization can be checked without invoking a neural architecture or an approximation to mutual information.

§167.5 Three measurements from a video encoder

The distinction between retained information and visible reconstruction quality is not confined to toy distributions. The following are later measurements from my BiSem/NineLab experiments, collected on October 30–31. They are included retrospectively in this dated note because they make the mathematical questions unusually concrete.[2]

Averages hide a badly reconstructed person. One high-motion frame from the laboratory videos is shown below. We used a person region detected in the *original* high-resolution input, then mapped the same coordinates to the different reconstructions. The person was not rediscovered independently in every reconstructed frame: doing so would systematically discard regions in which the detector failed after compression.

A peak signal-to-noise ratio is a monotone transform of mean squared pixel error:

$$\mathrm{PSNR}(x, \hat{x}) = 10 \log_{10} \frac{L^2}{|R|^{-1} \sum_{i \in R} (x_i - \hat{x}_i)^2}, \tag{167.6}$$

where R is the compared pixel region and L is the maximum intensity range. Changing R from the full frame to a person box changes the distortion measure. A localized failure may have little effect on the average error over a large, mostly static background.

[2] The panels come directly from the BiSem/NineLab experimental figures in the weekly report of October 26–November 1. Numerical results are from *BISEM_THREE_AXES_PERSON_ROI_LATENT_SIZE_20261030* and the audited E9 closeout of October 31. The figures are measurements on specific videos and checkpoints, not general benchmark claims.

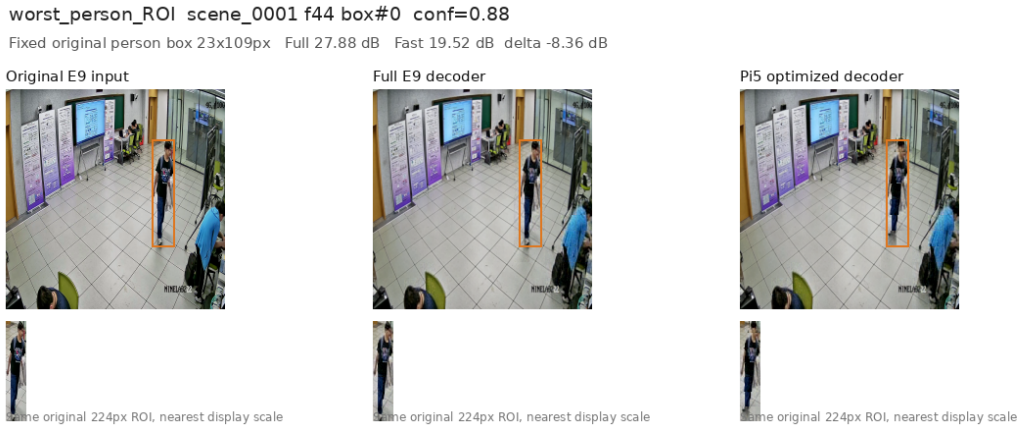


Figure 167.2: A real BiSem/NineLab high-motion case. The same person region is followed from the source frame to the original E9 reconstruction and a faster Pi 5 variant. In this particular frame the region PSNR drops from 27.88 to 19.52 dB, much more sharply than a full-frame average suggests. Image from the project's October 30 visual audit.

Information can become easier to read out. A separate experiment tested person-location information in hidden states of the frozen E9 ViT encoder. Labels were derived from YOLO detections on the original 1080*p* videos, projected to the 14×14 patch grid. A linear probe was trained on one set of recordings and evaluated on a held-out recording of 41 frames, giving 7,697 evaluated patches, 732 of them positive under the chosen pseudo-label rule.

Features supplied to probe	Person-patch AP	AUROC
Fixed spatial coordinates only	0.430	0.733
ViT patch embedding	0.338	0.707
ViT block 3	**0.789**	0.943
ViT block 6	0.777	**0.949**
ViT block 9	0.772	0.940
ViT block 12	0.743	0.933

Table 167.1: Measured E9 linear-probe results for original-video YOLO-derived person-patch labels. AP is average precision and AUROC is area under the receiver operating characteristic curve. The coordinate-only baseline is particularly relevant because the camera view is fixed.

The intermediate hidden states are much more *linearly decodable* under this particular evaluation. One might worry that this contradicts the data processing inequality: how could a later layer have more information than an earlier layer?

The answer is that a restricted linear probe measures *accessibility of information to a particular model class*, not mutual information. A nonlinear map can make a fact linearly accessible while creating no new information at all. The XOR example makes this precise. Take independent fair bits B_1, B_2, let

$$Y = B_1 \oplus B_2, \qquad X = (B_1, B_2).$$

The pair X determines Y exactly, so $I(X;Y) = 1$ bit. Yet no affine threshold can

separate the two XOR classes in the plane: labeling $(0,0),(1,1)$ negative and $(1,0),(0,1)$ positive would require

$$b < 0, \quad w_1 + w_2 + b < 0, \quad w_1 + b > 0, \quad w_2 + b > 0.$$

The last two inequalities give $w_1 + w_2 + 2b > 0$; the first two imply $w_1 + w_2 + 2b < b < 0$, a contradiction. After the nonlinear transformation $Z = B_1 \oplus B_2$, a one-dimensional linear threshold predicts Y perfectly, while

$$I(Y;Z) = I(Y;X) = 1.$$

No information was created. It merely became easier for a linear reader to extract.

The BiSem probe is not a proof that the encoder captures an invariant notion of personhood. The labels are detector-generated, the scene comes from a restricted laboratory camera, and a coordinate-only classifier already scores 0.430 AP. The observation is more modest and more useful: probe performance and reconstruction fidelity answer different questions.

A local repair is not a universal feature improvement. In the later VAL115 comparison, a residual correction with coefficient $\gamma = 0.5$ was tested against a sparse K12 baseline. Over 144 frames, the full-frame PSNR increased from 31.800 to 31.869 dB, a difference of only 0.069 dB. Across 515 repeated person-region observations, the mean ROI PSNR gain was 0.174 dB and the median gain was 0.038 dB. The improvement was concentrated in the highly dynamic first scene.

The corresponding final-layer person-patch probes, trained and assessed in the project's VAL115 comparison, told a different story:

$$\begin{aligned} \mathrm{AP}(\text{original E9}) &= 0.7416, \\ \mathrm{AP}(\text{sparse K12}) &= 0.7340, \\ \mathrm{AP}(\text{sparse K12} + \text{residual}) &= 0.7329. \end{aligned} \tag{167.7}$$

In this experiment, a small gain in person-region pixel fidelity did not come with an improvement in the probe. These VAL115 scores belong to a separate final-layer evaluation and should not be treated as a repeated measurement of the layer-by-layer probe in Table 167.1.

We also compared a framewise error in the person region with drift in the ViT hidden states under sparse temporal updates. A 41-frame exploratory probe showed a negative Spearman association (approximately -0.65) between latent drift and the change in local PSNR. Its frames are temporally correlated, so this is diagnostic evidence, not a causal claim.

These experiments suggest several possible distortion measures: full-image pixel error, error restricted to objects of interest, predictive error of a learned probe, and discrepancy between internal states. Optimizing one may improve or worsen the others. The distinction leads naturally to our next question: even when a useful signal is present in h, which hidden coordinates represent it?

§167.6 Coordinates are negotiable

Now consider a hidden representation $h(x) \in \mathbb{R}^d$ inside a neural computation. If its next layer is linear, with logits $s(x) = Wh(x)$, then any invertible matrix $A \in \mathrm{GL}_d(\mathbb{R})$ gives an equivalent factorization:

$$h'(x) = Ah(x), \qquad W' = WA^{-1}, \qquad W'h'(x) = Wh(x). \tag{167.8}$$

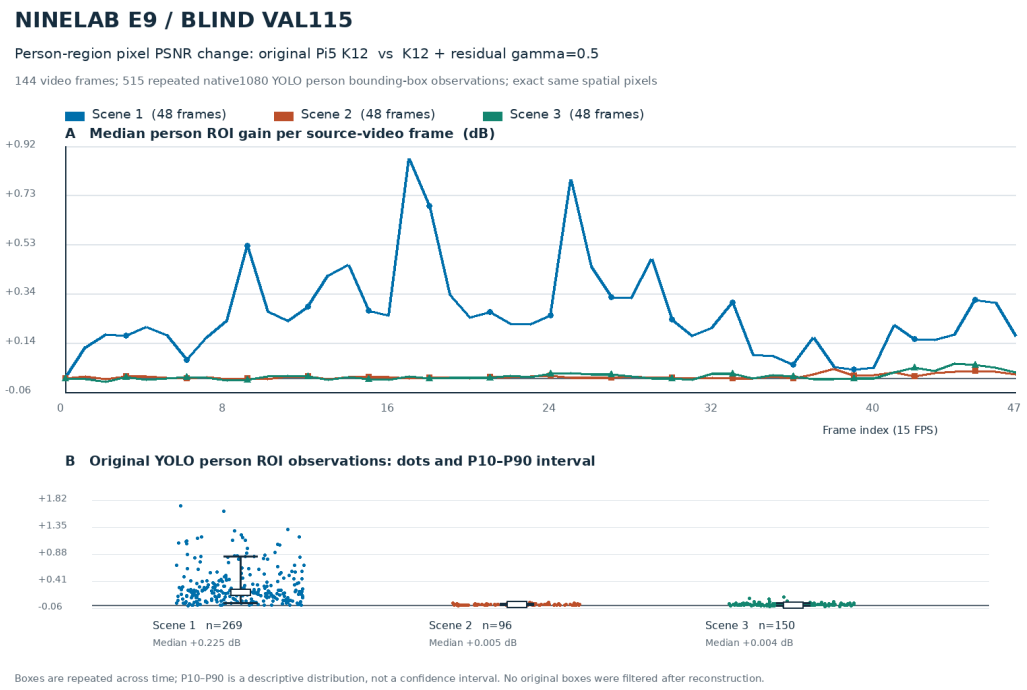


Figure 167.3: Measured person-region PSNR changes across the 144-frame VAL115 clip, grouped by scene. The local gains vary sharply; the 515 person observations are repeated detections across frames, not 515 independent videos. BiSem/NineLab E9 closeout, October 31.

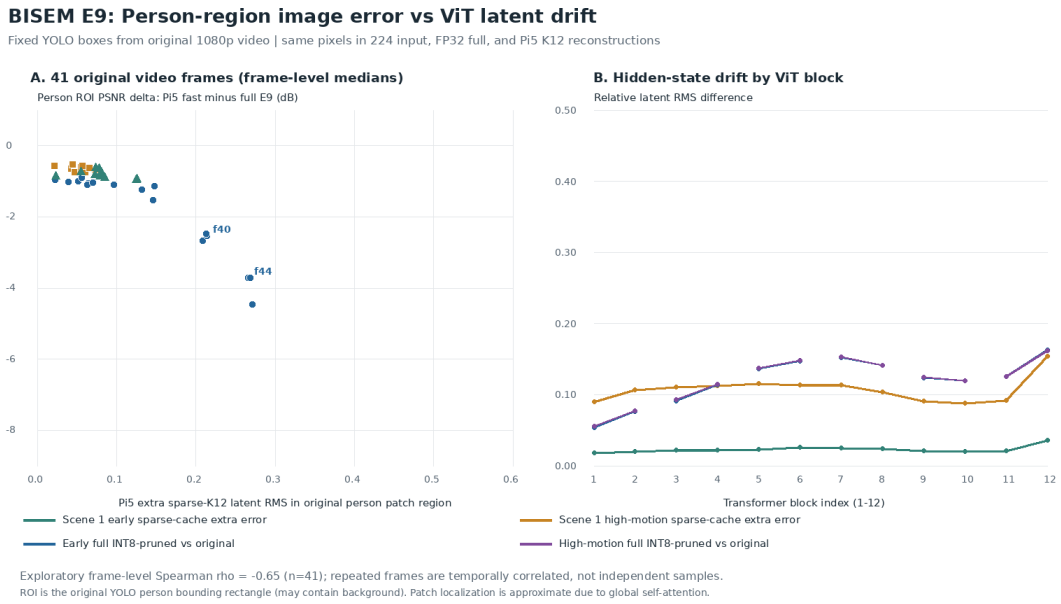


Figure 167.4: An actual BiSem E9 study of person-region image error and ViT hidden-state drift in a laboratory video. The plotted association motivates a task-sensitive distortion analysis; it does not show that a particular hidden feature caused the reconstruction failure.

A change of hidden coordinates need not change the output function at all. It may, however, completely change which individual coordinate appears to correspond to a concept.

For instance, suppose

$$h(x) = \begin{pmatrix} a(x) \\ b(x) \end{pmatrix}, \qquad A = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Then

$$h'_1(x) = \frac{a(x) + b(x)}{\sqrt{2}}, \qquad h'_2(x) = \frac{a(x) - b(x)}{\sqrt{2}}.$$

Neither of the original two coordinates survives unchanged, despite identical linear predictions after the corresponding change of W.

This is a genuine mathematical ambiguity, though *not* a license to insert arbitrary basis changes into every layer of a network. A nonlinear activation or a fixed architectural restriction may prevent a general A from being absorbed by neighboring weights. The observation is that the behavior of a model alone does not always select a preferred coordinate system for its internal states.

A geometric quantity such as the span of all hidden states, or the dimension of that span, behaves more predictably under invertible changes of coordinates than the interpretation of a single neuron. If we want individual *features*, some further principle must make a decomposition special.

§167.7 Looking for a sparse description

Suppose the hidden vectors are approximately constructed from reusable directions:

$$h \simeq Dz, \qquad D = [d_1 \ \cdots \ d_p] \in \mathbb{R}^{d \times p}, \qquad z \in \mathbb{R}^p. \tag{167.9}$$

The columns d_j form a *dictionary*; the components of z are coefficients. We might have many more dictionary atoms than hidden-state dimensions, $p > d$, and still hope that only a few z_j are nonzero for any particular input.

For an exact observation h, the idealized sparse recovery problem is

$$\min_{z:Dz=h} \|z\|_0, \qquad \|z\|_0 = \#\{j : z_j \neq 0\}.$$

The count $\|z\|_0$ is not a norm; minimizing it generally involves combinatorial search. Two popular alternatives are

$$\min_{z:Dz=h} \|z\|_1 \quad \text{and} \quad \min_z (\|h - Dz\|_2^2 + \lambda \|z\|_1),$$

where $\|z\|_1 = \sum_j |z_j|$. The first seeks a sparse exact representation; the second permits reconstruction error. They need not recover the same coefficients unless additional assumptions hold.

A sparse autoencoder (SAE) learns an approximate version of this decomposition over a distribution of hidden states. Its encoder assigns coefficients $z = f_\phi(h)$, while a decoder reconstructs

$$\widehat{h} = D f_\phi(h) + b.$$

A typical training objective balances reconstruction and sparsity:

$$\mathbb{E}_h \left[\|h - D f_\phi(h) - b\|_2^2 + \lambda \|f_\phi(h)\|_1 \right].$$

The encoder may use ReLU or another nonlinearity, and the actual sparsity penalty varies with the implementation. A constraint or normalization on the decoder columns is important: without it, scaling D upward and the coefficients downward can reduce an ℓ^1 penalty without changing reconstruction.

The decoder's reconstructed state has a practical purpose. It must retain enough information to substitute for the original hidden state in the downstream computation. Reconstructing h accurately checks one condition; checking whether the downstream prediction survives this substitution checks a different, more behaviorally relevant condition.

§167.8 Can two sparse explanations agree exactly?

Let us leave the training process aside and study an exact algebraic question. Fix a dictionary D. Can the same vector h have two different k-sparse representations?

Define the *spark* of D as the smallest number of its columns that are linearly dependent:

$$\operatorname{spark}(D) = \min\{|S| : \{d_j : j \in S\} \text{ is linearly dependent}\}.$$

Assume the dictionary has more columns than its rank, so this minimum exists. For a full-column-rank dictionary one may adopt the convention $\operatorname{spark}(D) = p + 1$.

Theorem 167.8.1 (Uniqueness of sparse coding)

Every vector h has at most one representation $h = Dz$ with $\|z\|_0 \le k$ if and only if

$$\operatorname{spark}(D) > 2k. \tag{167.10}$$

Proof. Suppose $Dz = Dz'$, where both z, z' have at most k nonzero entries. Then

$$D(z - z') = 0, \qquad \|z - z'\|_0 \le 2k.$$

If $\operatorname{spark}(D) > 2k$, no nonzero vector with support of size at most $2k$ lies in the kernel, so $z = z'$.

Conversely, if $\operatorname{spark}(D) \le 2k$, there is a nonzero kernel vector v supported on some set S with $|S| \le 2k$. Partition S into S_1, S_2, each with at most k elements. Define $z = v$ on S_1 and zero elsewhere, and $z' = -v$ on S_2 and zero elsewhere. Then $z - z' = v$, so $Dz = Dz'$, while $z \neq z'$. Uniqueness fails. □

The condition is sharp as a uniform guarantee. It does not say that every vector is k-sparse, and it does not say that solving an ℓ^1 optimization problem necessarily finds the sparse code.

Example 167.8.2 (Three directions in the plane)

Let

$$D = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \qquad h = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

The same vector has the representations

$$h = 1 \cdot d_3 = 1 \cdot d_1 + 1 \cdot d_2.$$

Any two dictionary columns are independent, but all three are linearly dependent,

so $\operatorname{spark}(D) = 3$. The theorem guarantees uniqueness among one-sparse representations, and here the one-sparse code $z = (0, 0, 1)$ is indeed unique. At sparsity level 2, uniqueness is already lost.

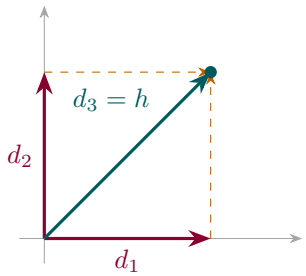


Figure 167.5: A vector assembled either from one diagonal atom or from two coordinate atoms. Sparse coding chooses the shorter representation only under assumptions that make such a choice meaningful and recoverable.

It is useful to see a more checkable sufficient condition for (167.10). Suppose all columns of D have unit Euclidean norm, and define their *mutual coherence*

$$\mu(D) = \max_{i \neq j} |\langle d_i, d_j \rangle|.$$

If s columns were dependent, let G_S be their Gram matrix. Its diagonal is 1 and each off-diagonal entry has absolute value at most $\mu(D)$. Whenever

$$(s-1)\mu(D) < 1,$$

the matrix is strictly diagonally dominant and therefore invertible, a contradiction. Hence

$$\operatorname{spark}(D) \geq 1 + \frac{1}{\mu(D)}$$

in the usual real-valued inequality sense. In particular,

$$2k < 1 + \frac{1}{\mu(D)} \quad \Longrightarrow \quad \text{all } k\text{-sparse representations are unique.} \tag{167.11}$$

The smaller the correlations between distinct atoms, the larger the guaranteed range of identifiable sparse codes. The condition is sufficient rather than necessary.

§167.9 What should a reconstructed state preserve?

The reconstructed state in an SAE is often evaluated using Euclidean error, $\|h - \widehat{h}\|_2^2$. But the rest of a network does not read a hidden state by measuring its Euclidean distance from some reference point. It performs a computation on it.

For a linear downstream layer $g(h) = Wh$, the output error is

$$\|g(h) - g(\widehat{h})\|_2 = \|W(h - \widehat{h})\|_2 \leq \|W\|_{\mathrm{op}} \|h - \widehat{h}\|_2.$$

The inequality is useful: small Euclidean reconstruction error guarantees small output error when the operator norm is controlled. Its converse generally fails.

To see why, let $h = (Y, N)^\top$ in our opening example and let $W = (1 \;\; 0)$, so the next layer only reads the task bit. The reconstructed state

$$\widehat{h} = (Y, 0)^\top$$

loses the nuisance bit and has positive mean squared reconstruction error:

$$\mathbb{E}\|h - \widehat{h}\|_2^2 = \mathbb{E}[N^2] = \frac{1}{2}.$$

Nevertheless $W\widehat{h} = Wh = Y$ for every input. The error vector lies in $\ker W$, which the downstream layer cannot detect. Indeed the downstream computation depends only on the equivalence class

$$[h] \in \mathbb{R}^d / \ker W.$$

This quotient is the exact linear algebraic space of information relevant to W.

At the opposite extreme, a small perturbation in a direction heavily amplified by W can be consequential. If $W = (M \;\; 0)$ and $\widehat{h} - h = (\varepsilon, 0)^\top$, the output changes by $M\varepsilon$. A task-sensitive distortion such as

$$d_W(h, \widehat{h}) = \|W(h - \widehat{h})\|_2^2$$

therefore measures a different kind of fidelity from ordinary hidden-state reconstruction. For a nonlinear downstream map, a local Jacobian plays a similar role, although the linear approximation may fail under large interventions.

This is why an SAE can have an excellent reconstruction score while the interpretation of its features remains unconvincing, and why a task-preserving compression can have a poor reconstruction score. The two scores answer different mathematical questions. They become especially informative when reported side by side.

§167.10 Which feature have we identified?

The previous theorem concerns coefficients when the dictionary is *fixed*. Learning the dictionary itself poses a different identifiability question. If P is a permutation matrix and S is an invertible diagonal matrix, then

$$Dz = (DPS)(S^{-1}P^{-1}z).$$

The same reconstruction can be described by reordered and rescaled atoms. Sparsity is unchanged by permutations and nonzero rescaling of coordinates. These ambiguities are usually regarded as harmless after normalizing atoms and forgetting their order.

There can be more serious ambiguities. For a general invertible A,

$$Dz = (DA^{-1})(Az),$$

and no observation of h alone distinguishes the two factorizations. Sparsity restricts such transformations and, under appropriate assumptions on the distribution of z, can make the generating dictionary identifiable up to permutation and scaling. A uniqueness theorem for a particular sparse code under a known dictionary does not by itself establish identifiability of an entire trained SAE.

Nor does mathematical uniqueness automatically give a chemical or linguistic interpretation to an atom. A feature may correlate with an experimental property merely

because both are driven by a third variable. To test a mechanistic claim, one can intervene on a coefficient, reconstruct the hidden state with that intervention, and measure a downstream change. For instance, with $z = f_\phi(h)$ and chosen coordinate j, compare

$$\widehat{h} = Dz + b, \qquad \widehat{h}^{(j \leftarrow 0)} = D(z - z_j e_j) + b,$$

then pass each state through the *same* downstream network. This defines a controlled computational intervention, though its effects must still be interpreted cautiously when the reconstructed states are off the model's normal hidden-state distribution.

The elementary $X = (Y, N)$ example and the sparse dictionary example illuminate different obstacles. A bottleneck may discard information needed for another task. A hidden vector may retain information while admitting competing decompositions. A unique sparse code may still lack a validated interpretation. These are separate mathematical questions, and none is answered merely by naming an axis a feature.

How Should a Neural Network Take a Step? Muon and Matrix Geometry

N-20261017

A neural network has millions or billions of numbers to adjust. It is tempting to think that an optimizer merely decides how far to move each one. But those numbers do not live in isolation. Many form matrices, and a matrix acts on an entire space of activations. The question of how large an update is already has several quite different answers.

This becomes particularly vivid when we compare AdamW and Muon. Muon keeps only a momentum state for the hidden weight matrices it handles, often makes rapid progress early in language-model pretraining, and yet does not always produce a substantially better model at the end. The practical gain may be enough to prefer it. The mathematical puzzle is more interesting: *what aspect of the geometry of a neural network does Muon manage to respect?*

We will begin with two numbers, pass through matrix duality and the polar decomposition, and then return to the whole Transformer. Throughout, a good update is good *relative to a specified norm, local objective, and budget.* Choosing those three ingredients is precisely the difficulty.[1]

§168.1 Two directions hidden in a tiny gradient

Suppose the gradient of a loss with respect to a 2×2 weight matrix is

$$G = \begin{pmatrix} 8 & 0 \\ 0 & 1 \end{pmatrix}.$$

The first diagonal entry asks for an update eight times as large as the second one. For now, forget all momentum and training noise and consider two possible unit-scale steps:

$$D_{\mathrm{grad}} = -\frac{1}{8}\begin{pmatrix} 8 & 0 \\ 0 & 1 \end{pmatrix} = -\begin{pmatrix} 1 & 0 \\ 0 & 1/8 \end{pmatrix}, \qquad D_{\mathrm{polar}} = -\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Both have largest singular value 1. The first respects the relative size of the two gradient entries. The second gives the two singular directions equal strength.

To first order, the predicted loss change is $\langle G, D \rangle = \operatorname{tr}(G^\top D)$. Hence

$$\langle G, D_{\mathrm{grad}} \rangle = -8 - \frac{1}{8} = -\frac{65}{8}, \qquad \langle G, D_{\mathrm{polar}} \rangle = -9.$$

With the *same upper bound on the spectral norm*, the second step achieves a larger first-order decrease. But its Frobenius length is $\sqrt{2}$, whereas that of the first step is only $\sqrt{65}/8$. We must not quietly switch from one notion of size to another halfway through a comparison.

[1] For the algorithms and scaling observations, see Keller Jordan and collaborators, *Muon: An optimizer for hidden layers in neural networks* (2024), and Jingyuan Liu and collaborators, *Muon is Scalable for LLM Training* (2025). The sources describe related but distinct practical recipes.

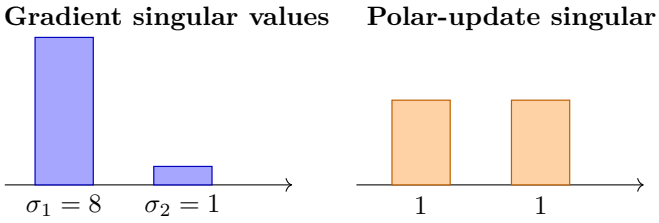


Figure 168.1: The polar factor preserves the singular directions while flattening the nonzero singular values. Bar heights are schematic; labels give the exact values.

The example poses a useful question. Is it better to keep the information in the gradient magnitudes, or to prevent a few large singular modes from swallowing nearly the entire update? Neither preference makes sense until we decide what constitutes a unit step.

§168.2 A norm tells us how to spend a step

For a differentiable loss $L(W)$, a first-order approximation near W gives

$$L(W + D) = L(W) + \langle G, D\rangle + o(\|D\|), \qquad G = \nabla_W L(W).$$

Fix a radius $\eta > 0$ and a matrix norm $\|\cdot\|_{\mathcal{X}}$. The associated first-order update solves

$$\min_{\|D\|_{\mathcal{X}} \le \eta} \langle G, D\rangle. \tag{168.1}$$

Geometrically, we slide a hyperplane perpendicular to G across a norm ball until it first touches the ball.

In two dimensions, a Euclidean ball is round, while an ℓ^∞ ball is a square. Their supporting points generally lie in different directions. The corners of the square make it possible to move at full speed in several coordinates at once.

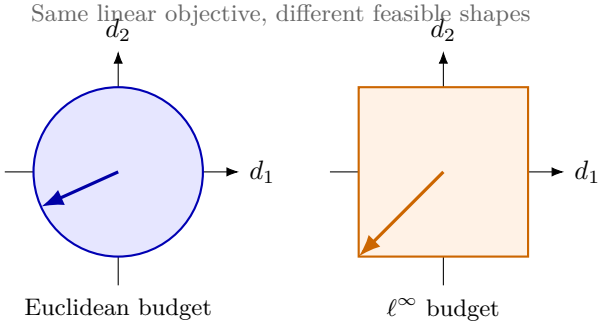


Figure 168.2: A unit ball is part of the definition of an optimizer's geometry. The arrows illustrate minimizing directions for a gradient with positive, unequal coordinates; the drawing is schematic.

To state the calculation precisely, the *dual norm* is

$$\|G\|_{\mathcal{X},*} = \max_{\|D\|_{\mathcal{X}} \le 1} \langle G, D\rangle.$$

Symmetry of a norm ball immediately gives

$$\min_{\|D\|_{\mathcal{X}} \le \eta} \langle G, D \rangle = -\eta \|G\|_{\mathcal{X},*}. \tag{168.2}$$

The value comes from the dual norm; a maximizing vector determines the direction. This distinction matters when a practical optimizer discards the dual-norm magnitude and prescribes a learning rate instead.

For a vector gradient g, the standard cases are easy to check:

Budget on d	Optimal direction	First-order reduction per unit radius
$\|d\|_2 \le 1$	$-g/\|g\|_2$	$\|g\|_2$
$\|d\|_\infty \le 1$	$-\operatorname{sign}(g)$	$\|g\|_1$
$\|d\|_1 \le 1$	$-e_j \operatorname{sign}(g_j),\ j \in \arg\max_i \lvert g_i \rvert$	$\|g\|_\infty$

Thus the familiar $\ell^1, \ell^2, \ell^\infty$ geometries make three different decisions: concentrate the budget, preserve the gradient direction, or update all coordinates at full permitted magnitude. The formula for ℓ^2 assumes $g \neq 0$; at $g = 0$ every feasible update minimizes the linear term.

One can also replace the hard constraint by

$$\min_D \left\{ \langle G, D \rangle + \frac{\lambda}{2} \|D\|_{\mathcal{X}}^2 \right\}, \qquad \lambda > 0.$$

If D_1 is a unit-ball minimizer and $G \neq 0$, the optimum has the form

$$D_\lambda = \frac{\|G\|_{\mathcal{X},*}}{\lambda} D_1. \tag{168.3}$$

Indeed, any $D = rT$ with $\|T\|_{\mathcal{X}} = 1$ reduces the problem to minimizing $-r\|G\|_{\mathcal{X},*} + \lambda r^2/2$. Choosing a *direction* by a norm and choosing its *scale* by a learning-rate schedule are separable modeling decisions.[2]

§168.3 Which two-norm? A matrix has several

We should settle a terminological trap before going further. For a vector, $\|x\|_2$ has an unambiguous meaning. For a matrix $A \in \mathbb{R}^{m \times n}$, both of these are common:

$$\|A\|_{\mathrm{F}} = \left(\sum_{i,j} A_{ij}^2 \right)^{1/2}, \qquad \|A\|_{\mathrm{op}} = \max_{x \neq 0} \frac{\|Ax\|_2}{\|x\|_2}.$$

The first treats a matrix as a long vector of mn entries. The second treats it as an operator on input activations. Some authors write $\|A\|_2$ for the *second* quantity, even though the first is the Schatten-2 norm.

If $A = U \operatorname{diag}(\sigma_1, \dots, \sigma_r) V^\top$ is a compact singular value decomposition, then

$$\|A\|_{\mathrm{op}} = \max_i \sigma_i, \qquad \|A\|_{\mathrm{F}} = \Big(\sum_i \sigma_i^2 \Big)^{1/2}, \qquad \|A\|_* = \sum_i \sigma_i.$$

Here $\|\cdot\|_*$ is the nuclear or trace norm. For the matrix of all ones in dimension n,

$$\|A\|_{\mathrm{op}} = n, \qquad \|A\|_{\mathrm{F}} = n, \qquad \|A\|_* = n;$$

[2] The geometric viewpoint follows Jeremy Bernstein and Laker Newhouse, *Old Optimizer, New Norm: An Anthology* (2024). The steepest-descent identities used here also follow directly from finite-dimensional norm duality.

but for the identity I_n,

$$\|I_n\|_{\mathrm{op}} = 1, \qquad \|I_n\|_{\mathrm{F}} = \sqrt{n}, \qquad \|I_n\|_* = n.$$

The identity has no particularly large effect on any *single* input direction. Nevertheless, it acts nontrivially on *all* n directions, a distinction detected by the Frobenius and nuclear norms.

These quantities are special cases of the Schatten norms

$$\|A\|_{S_p} = \left(\sum_i \sigma_i^p\right)^{1/p}, \quad 1 \le p < \infty, \qquad \|A\|_{S_\infty} = \max_i \sigma_i.$$

In particular, S_2 is the Frobenius norm and S_∞ is the spectral norm. The phrase about learning in a Euclidean two-norm rather than a spectral norm only becomes meaningful once this distinction is fixed.

§168.4 Why spectral steepest descent is a matrix sign

The dual norm of the spectral norm is the nuclear norm. An especially transparent proof also gives the update direction.

Proposition 168.4.1

Let $G \in \mathbb{R}^{m\times n}$ have compact SVD $G = U_r \operatorname{diag}(\sigma_1, \dots, \sigma_r) V_r^\top$, where $\sigma_i > 0$. Then

$$\min_{\|D\|_{\mathrm{op}} \le \eta} \langle G, D\rangle = -\eta \sum_{i=1}^{r} \sigma_i,$$

and a minimizer is

$$D_{\mathrm{sp}} = -\eta U_r V_r^\top. \tag{168.4}$$

Proof. Write u_i, v_i for the singular vectors. For every admissible D,

$$\langle G, D\rangle = \sum_i \sigma_i\, u_i^\top D v_i \ge -\sum_i \sigma_i \|D v_i\|_2 \ge -\eta \sum_i \sigma_i.$$

For $D = -\eta U_r V_r^\top$, the identity $U_r^\top U_r = V_r^\top V_r = I_r$ shows $D v_i = -\eta u_i$, so both inequalities are equalities. □

The matrix $U_r V_r^\top$ is the *partial polar factor*, also called the matrix sign in this context. All its nonzero singular values are 1. For an invertible square matrix, it is orthogonal; for a rectangular matrix, it has orthonormal columns or rows on the relevant subspace. There is no requirement that the weight matrix itself become orthogonal.

Contrast this with the Frobenius-constrained solution:

$$D_{\mathrm{F}} = -\eta \frac{G}{\|G\|_{\mathrm{F}}}.$$

The spectral optimizer keeps the singular vectors and flattens the singular values; the Frobenius optimizer keeps the ratios between singular values. Both are exact optimizers of their respective *linearized* problems.

We also have a precise answer to the opening calculation. For $G = \operatorname{diag}(8, 1)$, spectral descent gives $-\eta I_2$, while Frobenius descent gives $-\eta \operatorname{diag}(8, 1)/\sqrt{65}$. The gradient-scaled step of our opening example used a *spectral*, rather than Frobenius, radius. Comparing the three objects prevents two meanings of a normalized gradient from being conflated.

§168.5 Between preserving and erasing the spectrum

The difference between those two updates suggests that the choice need not be all-or-nothing. For $1 < p < \infty$, write $q = p/(p-1)$. Schatten-p and Schatten-q norms are dual, just as vector ℓ^p and ℓ^q norms are dual.

Proposition 168.5.1

Under the constraint $\|D\|_{S_p} \le \eta$, one optimal update is

$$D_p = -\eta U_r \operatorname{diag}\left(\frac{\sigma_i^{q-1}}{(\sum_j \sigma_j^q)^{1/p}}\right)_{i=1}^{r} V_r^\top. \tag{168.5}$$

Its first-order decrease is $-\eta\|G\|_{S_q}$.

Proof. The singular-value inequality gives $\langle G, D\rangle \ge -\sum_i \sigma_i s_i(D)$. By Hölder's inequality,

$$\sum_i \sigma_i s_i(D) \le \left(\sum_i \sigma_i^q\right)^{1/q} \left(\sum_i s_i(D)^p\right)^{1/p}.$$

Equality holds when the singular vectors are aligned in opposite directions and $s_i(D)$ is proportional to σ_i^{q-1}. Setting the p-norm of that singular-value vector to η gives the stated normalization. □

It is illuminating to suppress the overall scalar and look only at

$$\Psi_a(G) = U_r \operatorname{diag}(\sigma_i^a) V_r^\top, \qquad 0 \le a \le 1.$$

When $a = 1$, the spectral magnitudes are preserved. When $a = 0$, every nonzero singular value becomes 1. Intermediate exponents gently compress the dynamic range. More precisely, the *direction* $\Psi_a(G)$ is that of Schatten-p steepest descent for $p = 1 + 1/a$, when $a > 0$; the limit $a \to 0$ corresponds to S_∞.

For our two-singular-value example:

a	relative singular magnitudes	ratio of large to small
1	$(8, 1)$	8
1/2	$(\sqrt{8}, 1)$	$\sqrt{8}$
1/4	$(8^{1/4}, 1)$	$8^{1/4}$
0	$(1, 1)$	1

This is a family of *directions*. A practical implementation must still say how the direction is normalized and how the step size is selected.

There is an important caveat. Applying Ψ_a to a momentum estimate, or to an Adam-normalized update, is not generally the same as solving the original loss's steepest-descent problem in a single fixed Schatten geometry. The transformed input already depends on optimization history. The spectral family remains useful precisely because it lets us study *how much* of that input's spectrum should be kept.

§168.6 AdamW lives in moving coordinates

Let g_t be the stochastic gradient at iteration t; write all products and divisions below entrywise. Adam maintains

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1-\beta_1) g_t, \\ v_t &= \beta_2 v_{t-1} + (1-\beta_2) g_t \odot g_t, \\ \widehat{m}_t &= \frac{m_t}{1-\beta_1^t}, \qquad \widehat{v}_t = \frac{v_t}{1-\beta_2^t}. \end{aligned}$$

With decoupled weight decay, the update is

$$W_{t+1} = (1-\eta_t\lambda) W_t - \eta_t \frac{\widehat{m}_t}{\sqrt{\widehat{v}_t} + \varepsilon}. \tag{168.6}$$

The first state tracks a preferred direction; the second tracks the recent size of gradient coordinates. Coordinates with large historical squared gradients are damped.

It would be inaccurate to say that full AdamW is simply Euclidean gradient descent. Freeze v_t, set $D_t = \operatorname{diag}((\sqrt{\widehat{v}_{t,i}} + \varepsilon)^{-1})$ after vectorizing the matrix, and ignore weight decay. The update is then

$$d_t = -\eta_t D_t \widehat{m}_t.$$

This is the minimizer of

$$\min_d \left\{ \widehat{m}_t^\top d + \frac{1}{2\eta_t} d^\top D_t^{-1} d \right\}. \tag{168.7}$$

Thus, at a frozen instant, Adam behaves like steepest descent in a *weighted quadratic geometry*. The geometry moves with v_t; the momentum $\widehat{m}_t$ is also generally different from the instantaneous gradient. Equation (168.7) is an interpretation of the direction conditional on these states, not an exact static-norm description of the entire Adam trajectory.

There is a second, simpler connection. If $\beta_1 = \beta_2 = 0$ and $\varepsilon = 0$ on nonzero gradient coordinates, Adam's update direction is

$$\frac{g_t}{\sqrt{g_t \odot g_t}} = \operatorname{sign}(g_t).$$

This is the ℓ^∞-budget direction from our norm-ball picture. With realistic momentum and second-moment estimates, however, Adam's direction need not be a sign vector.

The distinction is both mathematically and experimentally important. Muon and AdamW do not form a clean opposition between spectral norm and ordinary L^2. A cleaner comparison is between *spectral normalization of a matrix-valued momentum* and *history-dependent coordinatewise preconditioning*. Their norm geometries are related but their actual optimization procedures contain additional ingredients.

§168.7 Muon orthogonalizes a momentum matrix

For a hidden matrix W_t, a typical Muon step maintains one momentum buffer

$$M_t = \beta M_{t-1} + G_t$$

and forms a Nesterov-style combination such as $\widetilde{M}_t = \beta M_t + G_t$. It then computes an approximation to the polar factor

$$\operatorname{polar}(\widetilde{M}_t) = U_r V_r^\top$$

and uses a shape-dependent scale $s_{m,n}$:

$$W_{t+1} = (1 - \eta_t \lambda) W_t - \eta_t s_{m,n} \operatorname{polar}(\widetilde{M}_t). \tag{168.8}$$

Momentum conventions, the exact normalization of the gradient, weight decay and scaling factors vary across implementations. It is the *polar operation on a matrix update* that identifies the family.

Why not compute a full SVD at every step? Muon uses a small number of Newton–Schulz-type matrix iterations. Starting with a normalized matrix X_0, a popular form applies a polynomial

$$X_{k+1} = aX_k + b(X_k X_k^\top) X_k + c(X_k X_k^\top)^2 X_k. \tag{168.9}$$

If $X_k = U \operatorname{diag}(s_i) V^\top$, the next iterate has the same singular vectors and transformed singular values

$$s_i \longmapsto a s_i + b s_i^3 + c s_i^5.$$

We can think of the iteration as repeatedly redrawing the *heights* of the singular-value bars, ideally pushing them toward a common level. The real computation uses matrix multiplications and avoids explicitly extracting those bars.

A finite number of polynomial iterations only *approximates* the polar factor. The approximation can also matter for small singular values; the different published coefficient choices and scaling rules should not be treated as mathematically identical. In particular, the efficient GPU implementation is an algorithmic achievement in its own right, distinct from the abstract solution of a spectral-norm trust-region problem.

The standard practice is hybrid: Muon handles selected two-dimensional hidden weights, while embeddings, normalization scales, biases and frequently the output head remain on AdamW. A Transformer therefore uses more than one update geometry at the same time.

The polar step also does not make second-moment information intrinsically useless. For example, the 2026 preprint *Muon*[2] applies an Adam-style second-moment preconditioner *before* orthogonalization, partly to improve the conditioning of Newton–Schulz inputs. Its factorized variant trades additional state against cheaper approximations. This is a different algorithm from the one-state Muon discussed here, and is another reason to separate geometry from implementation.

§168.8 The memory advantage has an exact accounting

Suppose a hidden weight matrix contains $N = mn$ parameters and its optimizer states are stored in 32-bit floating-point numbers. Ignoring distributed sharding, the persistent state count for that matrix is

Optimizer on the matrix	Persistent state tensors	State bytes
AdamW	$m_t,\ v_t$	$8N$
Muon	M_t	$4N$

That is a genuine reduction of *one full state tensor* for eligible matrix parameters. It does not require any claim about validation loss to be valuable.

For a concrete scale, a matrix with $N = 10^8$ entries costs about 800 MB of such AdamW optimizer state and 400 MB of Muon momentum state (decimal megabytes). But this is deliberately a *state-only* calculation.

Real training also stores parameter values, gradients, possibly FP32 master weights, activations, communication buffers and allocator overhead. Muon's matrix iterations

require temporary workspace, and the nonmatrix parameters still use AdamW. ZeRO or tensor parallelism can change which rank holds the persistent state and how frequently it is communicated. Therefore the per-matrix state saving is certain under the stated storage assumptions; a particular reduction in *peak GPU memory* must be measured rather than inferred by multiplying the number of model parameters by four bytes.

A similar separation is needed for speed. Muon can use fewer optimization steps to reach a loss threshold and still take more time per step, because its orthogonalization involves extra matrix multiplications. The two comparisons to report are

$$\text{loss versus tokens seen}, \qquad \text{loss versus actual wall-clock time}.$$

The first measures sample efficiency; the second captures the computer on which one actually trains.

§168.9 Why rectangular matrices need different scales

Even a perfect polar factor has a surprisingly shape-dependent *entrywise* size. Let $P = U_r V_r^\top$ be the polar factor of a rank-r matrix of shape $m \times n$. Since the singular values of P are 1 on its active rank,

$$\|P\|_{\mathrm{F}}^2 = r.$$

Define the root-mean-square entry magnitude by $\mathrm{RMS}(P) = \|P\|_{\mathrm{F}}/\sqrt{mn}$. We obtain the exact identity

$$\mathrm{RMS}(P) = \sqrt{\frac{r}{mn}}. \tag{168.10}$$

For a full-rank matrix, $r = \min(m, n)$, so

$$\mathrm{RMS}(P) = \frac{1}{\sqrt{\max(m, n)}}. \tag{168.11}$$

A square $d \times d$ hidden matrix has polar-update RMS $1/\sqrt{d}$. A $d \times 4d$ expansion matrix has RMS $1/(2\sqrt{d})$. If we used the same raw multiplier for both, the rectangular layer would receive an update with half the RMS, even though the matrices are participating in the same network.

Multiplying by $\sqrt{\max(m, n)}$ aligns the full-rank RMS at 1. Liu and collaborators use a multiplier of the form $0.2\sqrt{\max(m, n)}$ to aim for an AdamW-like update RMS in their training recipe. The factor 0.2 is an *empirical scale choice*; the square-root dimension factor follows from the rank calculation.

Two cautions follow. The formula is exact for the ideal rank-r polar factor, while practical Newton–Schulz outputs have approximately flattened singular values. And equal RMS does not mean equal effect on the network. The latter issue takes us from the optimizer to the actual role of a matrix.

§168.10 A layer is a map, not a table of parameters

A hidden linear layer transforms an activation $x \in \mathbb{R}^{d_{\mathrm{in}}}$ into

$$y = Wx, \qquad W \in \mathbb{R}^{d_{\mathrm{out}} \times d_{\mathrm{in}}}.$$

If the parameters change by D, the layer output changes by

$$\delta y = Dx.$$

The spectral norm has a direct functional meaning:

$$\sup_{x\neq 0}\frac{\|Dx\|_2}{\|x\|_2}=\|D\|_{\mathrm{op}}. \tag{168.12}$$

Constraining $\|D\|_{\mathrm{op}}$ limits the worst amplification of an input through the update map. It is a plausible geometric prior for a linear layer whose inputs and outputs are measured by Euclidean norms.

This argument is suggestive, not conclusive. Embedding tables are typically used by *selecting rows*, not by multiplying arbitrary dense input vectors. Attention weights are embedded in bilinear interactions, a softmax, and residual connections. Normalization parameters act by coordinatewise multiplication. Treating every parameter block as an arbitrary linear operator throws away those different roles.

Even among linear maps, the notion of a unit step can depend on width. A vector RMS is

$$\|x\|_{\mathrm{rms}}=\frac{\|x\|_2}{\sqrt{d_{\mathrm{in}}}},\qquad \|y\|_{\mathrm{rms}}=\frac{\|y\|_2}{\sqrt{d_{\mathrm{out}}}}.$$

The induced RMS-to-RMS operator norm is

$$\|D\|_{\mathrm{rms}\to\mathrm{rms}}=\sqrt{\frac{d_{\mathrm{in}}}{d_{\mathrm{out}}}}\,\|D\|_{\mathrm{op}}. \tag{168.13}$$

Thus a stable worst-case *RMS* effect across rectangular layers suggests a width correction. The appropriate factor depends on which spaces and normalization conventions we regard as physically meaningful. This is one reason that matrix shapes and parameterization rules must be part of a serious optimizer comparison.

§168.11 The same network can have very different coordinates

There is a simple obstruction to choosing an optimizer geometry from flattened parameters alone. Consider a two-layer linear network,

$$f(x)=W_2W_1x.$$

For any nonzero scalar c,

$$W_1'=cW_1,\qquad W_2'=c^{-1}W_2$$

produces exactly the same function. Yet an entrywise or Frobenius update with the same raw numerical scale can have a different effect in these new coordinates.

To see this without matrix notation, set $f(x)=abx$ with scalar parameters a,b. Let the loss depend on $z=ab$ through some differentiable function $\ell(z)$. Its parameter gradients are

$$\frac{\partial L}{\partial a}=b\ell'(ab),\qquad \frac{\partial L}{\partial b}=a\ell'(ab).$$

Under the equivalent parameterization $a'=ca,\ b'=b/c$, these become

$$\frac{\partial L}{\partial a'}=\frac{b}{c}\ell'(ab),\qquad \frac{\partial L}{\partial b'}=ca\ell'(ab).$$

Euclidean gradient descent therefore takes different *functional* steps in two descriptions of the same current model. Indeed, to first order,

$$\delta z=b\,\delta a+a\,\delta b,$$

and the two gradient steps induce

$$\delta z = -\eta(a^2 + b^2)\ell'(z), \qquad \delta z' = -\eta\left(c^2a^2 + \frac{b^2}{c^2}\right)\ell'(z).$$

For large c, the second expression may be arbitrarily larger. The ambiguity is already present in the simplest possible neural network.

This calculation does not prove that there exists one perfectly parameterization-invariant practical optimizer for modern Transformers. It does show why geometry tied to the *function performed* by the model is a more principled target than geometry assigned solely to a list of entries.

§168.12 Typical activations and worst-case activations disagree

The spectral norm considers the worst input direction. But the training data do not necessarily explore every direction equally often.

Let x have finite second moment $C_x = \mathbb{E}[xx^\top]$, which is positive semidefinite. The expected squared output change caused by D is

$$\begin{aligned}\mathbb{E}\|Dx\|_2^2 &= \mathbb{E}[x^\top D^\top Dx] \\ &= \operatorname{tr}(D^\top D\,\mathbb{E}[xx^\top]) \\ &= \operatorname{tr}(DC_xD^\top) = \|DC_x^{1/2}\|_{\mathrm{F}}^2. \end{aligned} \tag{168.14}$$

This is an exact identity, with no Gaussian assumption. If the mean of x is nonzero, C_x is the *second-moment* matrix; it is not the centered covariance unless the mean is zero.

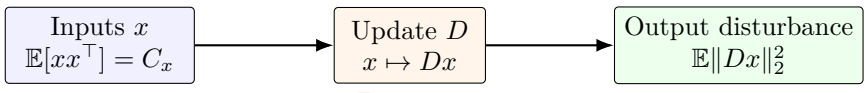
For a numerical example, suppose

$$C_x = \begin{pmatrix} 100 & 0 \\ 0 & 1 \end{pmatrix}, \qquad D_1 = \varepsilon\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad D_2 = \varepsilon\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Both updates have operator norm $|\varepsilon|$ and Frobenius norm $|\varepsilon|$. Yet

$$\mathbb{E}\|D_1x\|_2^2 = 100\varepsilon^2, \qquad \mathbb{E}\|D_2x\|_2^2 = \varepsilon^2.$$

A unit update in a frequently excited input direction can be a hundred times as consequential, in expected squared output change, as the same-sized update in a quiet direction.



$\operatorname{tr}(DC_xD^\top)$ weighs an update
by the directions the inputs actually visit.

Figure 168.3: An input-dependent geometry counts how the update acts on the observed activation distribution. Operator norm instead bounds the most amplified direction, whether or not that direction is common.

If $C_x = I$, the data-weighted norm reduces to Frobenius norm. If C_x is strongly anisotropic, an unweighted Frobenius ball can be a poor model of functional change. The spectral norm remains valuable as a worst-case bound, but the example explains why it should not automatically be called the unique natural geometry of training.

§168.13 From change in activations to change in loss

Even equal output disturbance need not mean equal *loss* disturbance. Some output directions are much more consequential than others.

Let a layer's output be $y = Wx$, and suppose that, for a particular input x, a smooth local loss as a function of y has Hessian $H_y(x)$. Its second-order response to D contains the term

$$\frac{1}{2}\mathbb{E}_x\left[(Dx)^\top H_y(x)(Dx)\right]. \tag{168.15}$$

An arbitrary Hessian need not be positive semidefinite; a positive metric is obtained only when one uses an appropriate positive-semidefinite curvature surrogate, such as a Gauss–Newton or Fisher-type matrix.

In the instructive case where a fixed positive-definite output metric A is appropriate and the input second moment is $B = C_x$, the associated quadratic size becomes

$$\|D\|_{A,B}^2 = \operatorname{tr}(ADBD^\top) = \|A^{1/2}DB^{1/2}\|_{\mathrm{F}}^2. \tag{168.16}$$

The input side accounts for *where the network is used*; the output side accounts for *which changes matter*. This is a concrete reason to consider geometry on *both* sides of a weight matrix.

There is a familiar connection to natural gradient. A Fisher quadratic form measures the local change in a model's predictive distribution, often expressed through a second-order approximation to a Kullback–Leibler divergence. Kronecker-factored approximations in layered networks replace a large curvature matrix by smaller input and output factors. The product form (168.16) has the same algebraic shape.

But the factorization is an *approximation* in general: the exact expectation of $H_y(x) \otimes xx^\top$ need not equal the product of the expectations. A metric estimated from mini-batches may also be singular or noisy. None of these equations declares an optimizer optimal before its estimation cost and errors have been accounted for.

§168.14 The layers also compete for an update budget

Even after selecting a norm for each matrix, we must decide how to combine them. A network parameter space is a product:

$$\Theta = \mathbb{R}^{m_1\times n_1} \times \cdots \times \mathbb{R}^{m_L\times n_L} \times (\text{vector and scalar parameters}).$$

For simplicity, take just the matrix factors, with gradient G_ℓ at layer ℓ, and consider the weighted product norm

$$\|D\|_{\text{block}}^2 = \sum_{\ell=1}^{L} a_\ell \|D_\ell\|_{\text{op}}^2, \qquad a_\ell > 0.$$

The regularized first-order model is separable:

$$\min_{D_1,\dots,D_L} \sum_{\ell=1}^{L} \langle G_\ell, D_\ell\rangle + \frac{\lambda}{2}\sum_{\ell=1}^{L} a_\ell \|D_\ell\|_{\text{op}}^2. \tag{168.17}$$

Using the single-matrix duality result, a minimizer for every nonzero G_ℓ is

$$D_\ell = -\frac{\|G_\ell\|_*}{\lambda a_\ell}\,\operatorname{polar}(G_\ell). \tag{168.18}$$

In other words, a matrix's polar factor fixes its *direction*, while the weight a_ℓ and the layerwise nuclear norm decide *how much of the network budget* that matrix receives.

Compare this with a separate rule assigning each layer exactly the same scalar learning rate. Such a rule can use the same polar *directions*, yet it generally does not solve (168.17). With two layers whose gradient nuclear norms are 10 and 1, respectively, the first-order model with equal a_ℓ allots ten times the spectral step magnitude to the first layer. Equal fixed per-layer radii allot them equal magnitudes instead.

There is no theorem saying that proportional-to-gradient budgets will train an LLM best; it is an exact result for a particular local model. It exposes a choice that is easy to mistake for a mere implementation detail: *how are the norms of different modules aggregated?*

Recent studies make this issue concrete. Crawshaw and collaborators explore product norms and Muon variants with improved learning-rate robustness; He and collaborators demonstrate that learning rates adapted to layers' spectral statistics can help *both* Muon and AdamW. Better layerwise step allocation is therefore a serious alternative explanation for some improvements attributed entirely to the matrix orthogonalization operation.[3]

§168.15 A small singular value can carry loud noise

There is a second, less obvious difficulty with full spectral flattening: polar normalization may treat a tiny component as though it were as reliable as a large one.

Consider

$$G_\delta = \begin{pmatrix} 1 & 0 \\ 0 & \delta \end{pmatrix}, \qquad \delta \neq 0.$$

Its partial polar factor is

$$\operatorname{polar}(G_\delta) = \begin{pmatrix} 1 & 0 \\ 0 & \operatorname{sign}(\delta) \end{pmatrix}.$$

Changing δ from 10^{-6} to -10^{-6} changes the gradient matrix by only $2\cdot 10^{-6}$ in Frobenius norm. Yet it changes the polar factor by 2 in operator norm. As $\delta \to 0^{\pm}$, the two one-sided polar limits disagree.

This is an exact discontinuity in the ideal matrix sign at a rank change. It does *not* prove that a particular finite-step Muon implementation is unstable: the polynomial iteration, momentum, gradient noise distribution and finite precision all influence what happens. It does reveal an obvious potential cost of discarding the magnitude of every small singular component. A weak, unreliable singular mode may be raised to a full-sized update direction.

Partially compressing the spectrum, clipping a few large spectral spikes, or applying coordinatewise RMS normalization before spectral processing are different responses to this issue. They answer distinct questions: should we suppress dominant modes, preserve small ones, or estimate their reliability first?

[3] See Michael Crawshaw and collaborators, *An Exploration of Non-Euclidean Gradient Descent: Muon and its Many Variants* (ICML 2026), and Di He and collaborators, *One LR Doesn't Fit All* (ICML 2026).

§168.16 Why a larger first-order win can lose the actual step

A more dramatic example shows why the best *linearized* update need not be the best finite update. Consider an exact quadratic loss in two parameters,

$$\ell(z_1, z_2) = 8z_1 + z_2 + \frac{1}{2}(z_1^2 + 100z_2^2), \qquad z = (0, 0).$$

The gradient at the origin is $g = (8, 1)$. Put these parameters on the diagonal of a 2×2 matrix; the two singular directions are then precisely the coordinate axes. The curvature of the second direction is a hundred times that of the first.

At an operator-norm step budget of $\eta = 0.1$, consider

$$D_{\text{polar}} = -0.1\,\text{diag}(1, 1), \qquad D_{\text{scaled}} = -0.1\,\text{diag}(1, 1/8).$$

Both have operator norm 0.1. The polar step wins on the first-order term:

$$\langle G, D_{\text{polar}} \rangle = -0.9, \qquad \langle G, D_{\text{scaled}} \rangle = -0.8125.$$

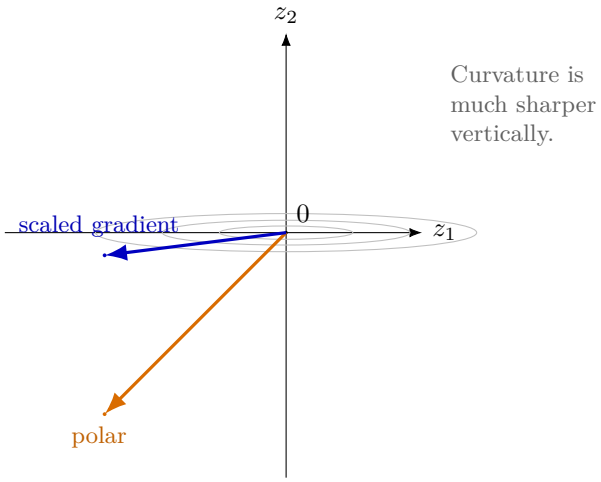


Figure 168.4: Two updates of equal spectral norm on the explicitly specified quadratic. Ellipses indicate the *curvature part* $\frac{1}{2}(z_1^2 + 100z_2^2)$, centered at the origin; the full loss has additional linear terms.

But the *actual* loss changes are

$$\ell(-0.1, -0.1) - \ell(0, 0) = -0.9 + \frac{1}{2}(0.01 + 100 \cdot 0.01) = -0.395,$$
$$\ell(-0.1, -0.0125) - \ell(0, 0) = -0.8125 + \frac{1}{2}(0.01 + 100 \cdot 0.0125^2) = -0.7996875.$$

The step with the *worse first-order prediction* lowers the true quadratic loss by roughly twice as much. Every number here comes from the stated quadratic; there is no neural-network experiment hidden inside the example.

What happened? The polar step spent much of its second-order curvature budget moving in the z_2 direction, where even a small displacement is expensive. In general,

$$L(W + D) - L(W) \approx \langle G, D \rangle + \frac{1}{2}\,\text{vec}(D)^\top H_W\,\text{vec}(D). \tag{168.19}$$

One can factor the second-order term as

$$\frac{1}{2}\|D\|_{\mathrm{F}}^2 \underbrace{\frac{\operatorname{vec}(D)^\top H_W \operatorname{vec}(D)}{\|D\|_{\mathrm{F}}^2}}_{\text{directional curvature, for } D\neq 0}.$$

Thus a method can make a favorable first-order step, yet incur a large curvature penalty. The Hessian need not be positive in general; here it is positive definite, so the mechanism is particularly clear.

Shuche Wang and collaborators report smaller normalized directional sharpness for Muon than for Adam in their studied language-model settings.[4] That finding makes a plausible bridge from spectral update geometry to faster training, but it is an empirical and model-dependent observation. The quadratic above explains why such a curvature comparison is worth making, and why one should not infer a universally superior optimizer solely from the matrix sign formula.

§168.17 Early speed, eventual loss, and model quality

Several different empirical statements tend to be rolled together when discussing new optimizers. We should pull them apart.

First, reaching a validation-loss threshold in fewer tokens is a claim about data or optimization-step efficiency. *Second*, reaching it in less wall-clock time also depends on kernel speed, precision, communication and sharding. *Third*, reaching a smaller final loss depends on the training horizon, learning-rate schedule, weight decay, regularization and tuning budget. *Fourth*, a model's useful capabilities may not be accurately ordered by a small difference in aggregate next-token loss.

The published record illustrates these distinctions:

- In the large-scale Moonlight study, Muon was made practical with weight decay and matrix-shape-aware scaling, and showed strong training-efficiency results. Those findings concern a particular distributed implementation and training recipe.
- Wen, Hall, Ma and Liang compared optimizers across several language-model scales and training horizons with more systematic hyperparameter searches. They found that rankings can change between intermediate and final checkpoints; the advantage over tuned AdamW narrows at larger model sizes in their evaluated regime.
- Qi and collaborators evaluated a family $U\Sigma^a V^\top$ on a controlled 124M-parameter GPT-2 setup. In that setting, the completely flattened momentum update was not consistently best. RMS-normalized inputs and partial spectral compression were competitive, and the amount of compression mattered.
- Layerwise learning-rate studies and product-norm studies show that a large part of the optimization problem lies *outside* the normalization of any one matrix: how the method distributes step sizes over heterogeneous layers changes the comparison.

These observations are compatible. A method may win on the time needed to reach an early loss threshold, show a narrow final loss advantage under a different budget, and yield models whose downstream accuracies are statistically indistinguishable on the tested tasks. A good comparison must say which outcome matters and hold the other resources approximately fixed.

[4] See *Why Muon Outperforms Adam: A Curvature Perspective* (2026 preprint).

In particular, tuning one AdamW learning rate globally and giving Muon separate matrix/vector learning rates is an unfair test of the full recipes. But assigning separate learning rates to *every* AdamW layer also costs tuning resources, and Muon may benefit from the same freedom. A credible comparison reports its search space, search cost and held-out selection rule.[5]

§168.18 A more meaningful matrix geometry

We can now make a constructive guess. If the input statistics B and output sensitivity A matter, perhaps a useful update norm should be placed on

$$Z = A^{1/2} D B^{1/2}$$

rather than on the raw entries of D.

This suggestion leads to two precise first-order models. Suppose for the moment that $A \in \mathbb{R}^{m\times m}$ and $B \in \mathbb{R}^{n\times n}$ are symmetric positive definite and held fixed during the step.

A data-weighted quadratic budget. Consider

$$\min_D \left\{ \langle G, D\rangle + \frac{\lambda}{2} \|A^{1/2} D B^{1/2}\|_{\mathrm{F}}^2 \right\}. \tag{168.20}$$

Differentiating with respect to D gives

$$G + \lambda A D B = 0.$$

Hence

$$D_{\mathrm{quad}} = -\frac{1}{\lambda} A^{-1} G B^{-1}. \tag{168.21}$$

This is a two-sided preconditioning rule, not a mysterious new optimizer. It follows directly from the geometry of the chosen quadratic form.

A data-weighted spectral budget. Alternatively, constrain the worst-case singular gain of the *whitened* update:

$$\|A^{1/2} D B^{1/2}\|_{\mathrm{op}} \le \eta. \tag{168.22}$$

Set $Z = A^{1/2} D B^{1/2}$. Cyclicity of the trace gives

$$\langle G, D\rangle = \langle A^{-1/2} G B^{-1/2}, Z\rangle.$$

The spectral-norm result proved earlier applies immediately:

$$D_{\mathrm{wsp}} = -\eta A^{-1/2} \operatorname{polar}(A^{-1/2} G B^{-1/2}) B^{-1/2}. \tag{168.23}$$

These two formulas connect an Adam-like idea—use statistics to change the coordinates in which an update is measured—with a Muon-like idea—control the singular gains of the update as a map. They also show why a universal choice between Frobenius and spectral is too crude: *which input and output metrics* are used is part of the question.

The derivations are exact for fixed positive-definite A, B. A real implementation would need to estimate and regularize these factors, handle changing activation distributions, exploit block structure, distribute the extra state, and approximate inverse square roots. If those costs outweigh the benefit, a simpler optimizer may still be preferable.

[5] For controlled optimizer comparisons, see Kaiyue Wen, David Hall, Tengyu Ma and Percy Liang, *Fantastic Pretraining Optimizers and Where to Find Them* (ICLR 2026), and Xianbiao Qi and collaborators, *Delving into Muon and Beyond* (ICML 2026). Their setups and conclusions should not be collapsed into a single universal speedup.

This view puts several existing research directions in perspective. Schatten interpolation trades off preservation and flattening of singular values; spectral clipping can impose operator-norm control on the updates of other optimizers; joint constraints on *weights and updates* can limit activation drift. These are different ways of injecting matrix geometry into a practical training procedure, with different computational costs and guarantees.[6]

§168.19 How would we tell which geometry fits an LLM?

There is an experimental way to make the mathematics answerable. Consider a single reproducible Transformer architecture and a fixed training corpus. Compare AdamW, matrix/vector-decoupled AdamW, a properly tuned layerwise AdamW, Muon with the same access to layerwise tuning, and at least one intermediate spectral-power variant. The parameter groups, weight decay rules, data order, precision and random seeds must be stated.

For each recipe, distinguish three budgets: *tokens*, *wall-clock time* and *tuning computation*. A fourth resource, *peak GPU memory*, is especially important when the optimizer is being considered because of its state size. Record entire learning curves and final checkpoints instead of selecting only an early interval where one optimizer leads.

The geometry itself admits more informative probes than loss alone. On representative layers, measure the singular values of the gradient momentum and the *actual applied updates*; the ratio between update RMS and weight RMS; the distribution of activation second moments; and the layerwise output changes Dx induced by each update. If curvature-vector products are affordable, estimate directional curvature along the actual optimizer steps. This helps separate three possible explanations:

1. spectral flattening changes which singular directions are explored;
2. shape and layerwise learning-rate rules change where the update budget is spent;
3. the resulting steps align better with the local curvature or with activation directions that matter.

These explanations are not mutually exclusive. A method can save optimizer-state memory independently of all three, and a lower final validation loss remains only one measurement of the trained model.

A particularly diagnostic control is to keep *the same polar direction* but vary only the layerwise scaling, then to keep *the same layerwise scaling* while varying spectral compression. The two comparisons isolate effects that a single Muon-versus-AdamW curve otherwise mixes together.

§168.20 Returning to the first two singular values

The matrix $\mathrm{diag}(8, 1)$ gave us a choice between following a strong gradient mode and distributing our effort over two modes. With a spectral-norm budget and a linearized loss, the polar step wins; that is a theorem. With a nontrivial curvature, the same step can lose at finite learning rates; we exhibited an exact quadratic. With anisotropic

[6] See Xiaowen Jiang, Andrei Semenov and Sebastian Stich, *Enhancing LLM Training via Spectral Clipping* (ICML 2026), and Tian Xie and collaborators, *Controlled LLM Training on Spectral Sphere* (2026 preprint). The second work proposes constraints on both parameters and their updates; its empirical scope should be kept distinct from a general theorem of optimality.

activations, equal-sized matrix steps can change the network's outputs by very different amounts; this too followed from a one-line trace identity.

The original practical observations now fit into a sharper picture. Muon genuinely reduces persistent state for the hidden matrices to which it applies. Its spectral update is often an effective way to spend a matrix-shaped step, and relatively little machinery is required to implement it efficiently on modern accelerators. Its early loss advantage, when observed, does not settle which geometry would be best for every module or throughout training. Similar end-of-run losses need not imply identical optimization dynamics, and small differences in aggregate loss need not imply a materially better downstream model.

If there is a particularly natural geometry for language-model training, it must respect at least *what the matrices do*, *how their inputs are distributed*, *what their outputs influence*, and *how step sizes are coordinated across the network*. Spectral norm is a strong and elegantly motivated starting point. The remaining mathematical task is to understand when its worst-case notion of size captures the perturbations a Transformer actually cares about.

Toposes, Probability, and Language Models: Notes on Lafforgue's Lecture

N-20261026

On October 20, I attended Laurent Lafforgue's Top Talk at Tsinghua University, *A Topological Point of View on Probability Spaces: The Example of Large Language Models*. I also had the chance to speak with Lafforgue in person and obtained his permission to translate the lecture slides. The lecture begins with coverings on a partially ordered set, constructs a topos of infinite token sequences, equips it with the probability law of a language model, passes from measure to Grothendieck topology and geometric logic, and finally goes in the opposite direction: a law of large numbers recovers a numerical measure from topology on an infinite space of samples.

We follow the lecture in its original order. I supply a few missing definitions and calculations, with more detailed proofs for the law of large numbers at the end. Questions about language models, representation and interpretability are reserved until the lecture has been fully developed.

§169.1 Coverings on an ordered set

Let $(C, \leq)$ be a partially ordered set. We think of $x' \leq x$ as a piece of x, and of a family $(x_i \leq x)_{i \in I}$ as a possible covering of x. A *Grothendieck topology* J specifies which families are coverings, subject to four conditions.

Definition 169.1.1 (Grothendieck topology on a poset)**.** For each $x \in C$, choose a collection $J(x)$ of families $(x_i \leq x)_{i \in I}$. The following rules hold.

1. *Identity.* The singleton family $x \leq x$ covers x.
2. *Enlargement.* If $(x_i \leq x)_{i \in I}$ covers x, and another family $(x'_j \leq x)_{j \in J}$ satisfies
$$\forall i \in I \ \exists j \in J : \qquad x_i \leq x'_j \leq x,$$
then the second family also covers x.
3. *Restriction.* If $(x_i \leq x)_{i \in I}$ covers x, then for every $y \leq x$ there is a covering $(y_j \leq y)_{j \in J}$ such that each y_j lies under some x_i:
$$\forall j \in J \ \exists i \in I : \qquad y_j \leq x_i \leq x.$$
4. *Composition.* If $(x_i \leq x)_{i \in I}$ covers x, and for every i the family $(x_{ij} \leq x_i)_{j \in J_i}$ covers x_i, then
$$(x_{ij} \leq x)_{i \in I,\ j \in J_i}$$
covers x.

For ordinary open sets ordered by inclusion, these axioms say that a cover remains a cover when we enlarge its pieces, restrict to a smaller open set, or cover each member of a cover in turn. The categorical generalization replaces inequalities $x' \leq x$ by morphisms $x' \to x$ and chains of inequalities by composition. In the categorical formulation, covering sieves give a particularly convenient language.

§169.2 From a topology to a topos

The covering prescription tells us what it means for local data to glue together.

Definition 169.2.1. Let (C, J) be a small site. A presheaf $F : C^{\mathrm{op}} \to \mathbf{Set}$ is a *J-sheaf* when compatible local sections over a covering family glue uniquely over the covered object. The category

$$\mathcal{E} = \mathrm{Sh}(C, J)$$

is the *Grothendieck topos* presented by the site.

The lecture also denotes the presheaf category by $\widehat{C}$, and the sheaf topos by $\widehat{C}_J$. Different sites can present equivalent toposes.

For example, let X be a topological space and C a basis of open sets ordered by inclusion. With the usual covers

$$U = \bigcup_{i \in I} U_i,$$

we obtain

$$\mathcal{E}_X = \mathrm{Sh}(X),$$

independent, up to equivalence, of the chosen basis.

There is also an intrinsic notion of a *point* of a topos: a geometric morphism $\mathbf{Set} \to \mathcal{E}$. The points form a topological space denoted $|\mathcal{E}|$. Geometric morphisms $f : \mathcal{E}' \to \mathcal{E}$ induce continuous maps

$$|f| : |\mathcal{E}'| \longrightarrow |\mathcal{E}|.$$

The lecture records a canonical continuous map $X \to |\mathcal{E}_X|$ and an equivalence

$$\mathcal{E}_X \simeq \mathcal{E}_{|\mathcal{E}_X|}.$$

Here $|\mathcal{E}_X|$ is the sobrification of X. In particular, if Y is sober, geometric morphisms $\mathcal{E}_X \to \mathcal{E}_Y$ correspond to continuous maps $X \to Y$. These statements explain in what sense a topos extends the idea of a topological space.

§169.3 The topological space of texts

Fix a finite vocabulary M of tokens. Write C_M for the set of nonempty finite words

$$m = (m_1, \ldots, m_\ell), \qquad m_i \in M, \qquad |m| = \ell.$$

The crucial choice of order is *reverse prefix order*:

$$m \leq m' \quad \Longleftrightarrow \quad |m| \geq |m'| \text{ and } m_i = m'_i \quad (1 \leq i \leq |m'|).$$

Thus a longer word is smaller: a word such as *a dog barks* lies under *a dog*, since it specifies a more restrictive condition on a continuation.

The lecture endows this ordered set with a Grothendieck topology J. A family $(m_i \leq m)_{i \in I}$ covers m precisely when there is some length $L \geq |m|$ such that every word w of length L extending m extends at least one m_i. Equivalently, for every word w of length L,

$$w \leq m \quad \Longleftrightarrow \quad \exists i \in I : \ w \leq m_i.$$

The cover eventually accounts for every continuation of the given prefix.

Now form the set of infinite sequences

$$M^{\mathbb{N}} = \{(a_1, a_2, \ldots) : a_i \in M\}.$$

To a finite word m attach its *cylinder*

$$U_m = \{a \in M^{\mathbb{N}} : a_i = m_i,\ 1 \le i \le |m|\}.$$

The cylinders form a basis of the product topology, where M is discrete. A longer prefix determines a smaller cylinder:

$$m \le m' \implies U_m \subseteq U_{m'}.$$

Proposition 169.3.1 (The text-space identification)
The points of the topos

$$\mathcal{E}_M = \mathrm{Sh}(C_M, J)$$

may be identified with $M^{\mathbb{N}}$, with basic open sets U_m. Moreover,

$$\mathcal{E}_M \simeq \mathrm{Sh}(M^{\mathbb{N}}).$$

Each cylinder splits into finitely many disjoint cylinders one token longer:

$$U_m = \coprod_{a \in M} U_{(m,a)}.$$

Repeated refinement gives every length L cylinder below m. Since M is finite, $M^{\mathbb{N}}$ is compact, and each cylinder is compact as well. An open cover of a cylinder by cylinders therefore has a finite subcover; a sufficiently long common prefix length then detects the cover. This explains the choice of J.

The space $M^{\mathbb{N}}$ is compact, Hausdorff and totally disconnected: it is *profinite*. This property will matter when extending measures to subtoposes.

§169.4 Next-token probabilities and their measure

A language model is now reduced to a precise mathematical input. For each finite prefix m, it supplies

$$p(a \mid m) \in [0,1] \qquad (a \in M), \qquad \sum_{a \in M} p(a \mid m) = 1.$$

The lecture writes these as conditional probabilities $\mu(U_{(m,a)} \mid U_m)$ associated with the decomposition

$$U_m = \coprod_{a \in M} U_{(m,a)}.$$

For the first token we also allow the empty prefix $\varnothing$, with $U_\varnothing = M^{\mathbb{N}}$.

Set

$$\mu(U_{(m_1,\ldots,m_\ell)}) = \prod_{i=1}^{\ell} p(m_i \mid m_1, \ldots, m_{i-1}).$$

The normalization of next-token probabilities gives

$$\sum_{a \in M} \mu(U_{(m,a)}) = \mu(U_m).$$

These consistent cylinder probabilities determine a unique Borel probability measure on $M^{\mathbb{N}}$, by the extension theorem for finite-dimensional distributions. Thus

$$\underbrace{p(a \mid m)}_{\text{one-step rule}} \rightsquigarrow \underbrace{\mu(U_m)}_{\text{prefix probability}} \rightsquigarrow \underbrace{\mu}_{\text{measure on infinite texts}}.$$

Only this family of conditional probabilities is used here. The construction requires no information about embeddings, hidden states or the network architecture.

§169.5 Subtoposes and finer coverings

If $\mathcal{E} = \mathrm{Sh}(C, J)$, a *subtopos* $\mathcal{E}' \hookrightarrow \mathcal{E}$ is the topos analogue of a subspace. Every Grothendieck topology J' on C finer than J gives one:

$$J' \supseteq J \quad \longmapsto \quad \mathrm{Sh}(C, J') \hookrightarrow \mathrm{Sh}(C, J).$$

Conversely, all subtoposes arise this way. A stronger covering rule imposes more gluing conditions, so fewer presheaves qualify as sheaves.

For a subset $Z \subseteq X$, the familiar subspace gives

$$\mathcal{E}_Z \hookrightarrow \mathcal{E}_X.$$

On the site of open sets of X, its covering condition is

$$(U_i \subseteq U)_{i \in I} \text{ covers for } J_Z \quad \Longleftrightarrow \quad U \cap Z = \bigcup_{i \in I} (U_i \cap Z).$$

Only what happens on Z matters.

But subtoposes have more possibilities than ordinary point-set subspaces. For example, declare a family $U_i \subseteq U$ to cover when $\bigcup_i U_i$ is *dense* in U. This dense topology can yield a subtopos not presented by a subset $Z \subseteq X$ in the elementary way. It is the first indication that subtoposes contain information invisible from point sets alone.

§169.6 Operations on subtoposes

The subtoposes of a fixed topos $\mathcal{E}$ form an ordered set under inclusion. This ordered set admits arbitrary unions and intersections. The lecture emphasizes two distributivity properties:

$$\mathcal{E}' \cup \bigcap_{\alpha} \mathcal{F}_\alpha = \bigcap_{\alpha} (\mathcal{E}' \cup \mathcal{F}_\alpha),$$

and, for finite families,

$$\mathcal{E}' \cap (\mathcal{F}_1 \cup \mathcal{F}_2) = (\mathcal{E}' \cap \mathcal{F}_1) \cup (\mathcal{E}' \cap \mathcal{F}_2).$$

For two subtoposes $\mathcal{E}_1, \mathcal{E}_2 \hookrightarrow \mathcal{E}$, their difference is characterized without pointwise set subtraction:

$$\mathcal{E}_1 \setminus \mathcal{E}_2 \subseteq \mathcal{F} \quad \Longleftrightarrow \quad \mathcal{E}_1 \subseteq \mathcal{E}_2 \cup \mathcal{F} \qquad \text{for every subtopos } \mathcal{F}. \tag{169.1}$$

It is the smallest remainder which, when joined to $\mathcal{E}_2$, contains $\mathcal{E}_1$.

There is a warning in the slides. Although

$$Z \subseteq X \longmapsto \mathcal{E}_Z \hookrightarrow \mathcal{E}_X$$

preserves unions, it does not generally preserve intersections, even finite ones. One may have $Z_1 \cap Z_2 = \varnothing$ but

$$\mathcal{E}_{Z_1} \cap \mathcal{E}_{Z_2} \neq 0;$$

the intersection subtopos may have no points and nevertheless be nontrivial. Geometric intuition based only on sets of points must therefore be used carefully.

A geometric morphism $f : \mathcal{E}' \to \mathcal{E}$ also induces direct- and inverse-image operations on subtoposes:

$$f_* : \mathrm{Subtop}(\mathcal{E}') \to \mathrm{Subtop}(\mathcal{E}), \qquad f^{-1} : \mathrm{Subtop}(\mathcal{E}) \to \mathrm{Subtop}(\mathcal{E}'),$$

with the adjunction

$$f_*(\mathcal{F}') \subseteq \mathcal{F} \quad \Longleftrightarrow \quad \mathcal{F}' \subseteq f^{-1}(\mathcal{F}).$$

§169.7 Extending a measure from open sets

We next need a notion of measure suited to the lattice of open sets. In the lecture, a bounded measure on the opens of X is a map

$$\mu : \mathcal{O}(X) \longrightarrow \mathbb{R}_{\geq 0}$$

satisfying the valuation identity

$$\mu(U) + \mu(V) = \mu(U \cap V) + \mu(U \cup V), \tag{169.2}$$

as well as continuity for directed increasing families:

$$\mu\left(\bigcup_{i \in I} U_i\right) = \sup_{i \in I} \mu(U_i). \tag{169.3}$$

We understand $\mu(\varnothing) = 0$. For an ordinary finite Borel measure these properties follow from finite additivity and continuity from below. In locale theory this is often called a *continuous valuation*.

For a subtopos $\mathcal{E}' \hookrightarrow \mathcal{E}_X$, define

$$\mu(\mathcal{E}') = \inf\{\mu(U) : U \subseteq X \text{ open and } \mathcal{E}' \subseteq \mathcal{E}_U\}. \tag{169.4}$$

A generalized subspace is evaluated by the measures of its open neighborhoods.

The slide cites a theorem of Olivier Leroy (1996 or earlier). First, for any increasing sequence of subtoposes $(\mathcal{E}_n)$,

$$\mu\left(\bigcup_{n \geq 1} \mathcal{E}_n\right) = \sup_{n \geq 1} \mu(\mathcal{E}_n). \tag{169.5}$$

Second, suppose X is regular: every open U is covered by open V with $\overline{V} \subseteq U$. Then the extension satisfies

$$\mu(\mathcal{E}_1) + \mu(\mathcal{E}_2) = \mu(\mathcal{E}_1 \cap \mathcal{E}_2) + \mu(\mathcal{E}_1 \cup \mathcal{E}_2), \tag{169.6}$$

and for every subtopos $\mathcal{E}'$ there is a smallest subtopos

$$\mathcal{E}'' \subseteq \mathcal{E}' \quad \text{with} \quad \mu(\mathcal{E}'') = \mu(\mathcal{E}').$$

The lecture remarks that this theorem addresses the Banach–Tarski paradox. This concerns a change of setting from arbitrary point-sets to subtoposes and valuations. It does not construct a countably additive, Euclidean-isometry-invariant extension of volume to every subset of $\mathbb{R}^3$; the classical obstruction remains under those requirements.

For a finite vocabulary M, $M^{\mathbb{N}}$ is profinite and hence regular. It therefore satisfies the hypothesis used in the second part of Leroy's theorem.

§169.8 The Grothendieck topology induced by a measure

Suppose X is a regular topological space with a bounded measure μ. On the ordered set of its open subsets—or on the associated lattice of subtoposes—the lecture defines J_μ by declaring

$$(U_i \subseteq U)_{i\in I} \text{ to be a } J_\mu\text{-cover} \quad \Longleftrightarrow \quad \mu\left(\bigcup_{i\in I} U_i\right) = \mu(U). \tag{169.7}$$

For ordinary measures the uncovered part is thus null:

$$\mu\left(U \setminus \bigcup_i U_i\right) = 0.$$

The covering members need account for U only *almost everywhere*, a weaker requirement than ordinary topological coverage. This gives a subtopos

$$\mathcal{E}_\mu \hookrightarrow \mathcal{E}_X.$$

The lecture observes that knowing J_μ is equivalent to knowing which locally closed subsets are negligible for μ. It also identifies the points of $\mathcal{E}_\mu$ with the positive-mass atoms represented by points:

$$\{x \in X : \mu(\{x\}) > 0\}.$$

Pointless portions of a subtopos may still be nontrivial, so its point set alone does not capture the complete structure.

For $X = M^{\mathbb{N}}$, the language-model interpretation is that this topology reveals which finite continuations $m' \leq m$ can carry positive probability, together with the more general notion of almost-sure coverage. Whether that information captures the model's judgements is a separate question, to be considered after the mathematical construction is complete.

§169.9 Geometric logic and classifying toposes

The next step is logical. A *geometric first-order theory* T is formulated using finite conjunctions, arbitrary disjunctions and existential quantifiers, together with the ordinary structural operations of first-order logic. These operations are special because their interpretation is preserved by inverse images of geometric morphisms.

For any topos $\mathcal{E}$, one can speak of models of T *inside* $\mathcal{E}$; they form a category

$$\mathrm{Mod}_T(\mathcal{E}).$$

If $f : \mathcal{E}' \to \mathcal{E}$ is a geometric morphism, its inverse-image functor induces

$$f^* : \mathrm{Mod}_T(\mathcal{E}) \longrightarrow \mathrm{Mod}_T(\mathcal{E}').$$

In this sense, a geometric theory can be interpreted in many different mathematical universes.

Theorem 169.9.1 (Classifying topos)
For every geometric theory T there exists a topos $\mathcal{E}_T$ and a universal model

$$\mathcal{U}_T \in \mathrm{Mod}_T(\mathcal{E}_T)$$

such that, naturally in $\mathcal{E}$,

$$\mathrm{Geom}(\mathcal{E}, \mathcal{E}_T) \simeq \mathrm{Mod}_T(\mathcal{E}), \qquad f \longmapsto f^*\mathcal{U}_T.$$

In particular, points of $\mathcal{E}_T$, namely geometric morphisms $\mathbf{Set} \to \mathcal{E}_T$, correspond to set-valued models of T.

One can regard the classifying topos as a space whose points are models of a theory. The universal model is the single object whose pullbacks give all its interpretations. The lecture attributes the systematic development of these ideas to categorical logicians including Lawvere, Joyal, Makkai and Reyes in the 1970s, with early precedents in Monique Hakim's thesis.

There is a striking converse: a given Grothendieck topos can be represented as the classifying topos of infinitely many geometric theories. The same topos may therefore be reached by different logical presentations.

§169.10 Subtoposes as additional axioms

A theorem of Olivia Caramello gives a precise correspondence between strengthening a theory and passing to a subtopos.

Theorem 169.10.1 (Subtopos–quotient theory correspondence)
Let T be a geometric theory in a language Σ.

1. Adjoining geometric axioms in the same language produces a quotient theory T', and its classifying topos is a subtopos
$$\mathcal{E}_{T'} \hookrightarrow \mathcal{E}_T.$$
2. Two such families of added axioms determine the same subtopos exactly when the resulting theories are provably equivalent over T.
3. Every subtopos of $\mathcal{E}_T$ arises from some such quotient theory, unique up to provable equivalence.

The direction of the arrows is worth remembering: more axioms mean fewer models and a *smaller* classifying topos. If $\mathcal{E}_T \simeq \mathrm{Sh}(C, J)$, the earlier description of subtoposes turns this into the correspondence

$$\{\text{additional geometric axioms}\}/\text{provable equivalence}$$
$$\longleftrightarrow$$
$$\{J' \supseteq J : \ J' \text{ a Grothendieck topology on } C\}.$$

This is the mathematical mechanism behind the lecture's translation of logical provability questions into questions about coverings.

§169.11 The logic of a measure

The abstract correspondence becomes particularly transparent for an ordered set $(C, \leq)$. Consider a propositional language containing one nullary relation symbol R_x for each $x \in C$. The basic theory has the sequents

$$R_{x'} \vdash R_x \qquad \text{whenever } x' \leq x.$$

Here R_x can be read as saying that the condition represented by x holds; if the finer condition x' holds, so does x. The lecture presents this as the propositional description of the presheaf topos $\widehat{C}$.

A small qualification makes the presentation mathematically precise: the order implications alone do not axiomatize $\widehat{C}$ for an arbitrary poset. The background flatness conditions for its points must also be encoded. For example, when C consists of two incomparable objects, $\widehat{C} \simeq \mathbf{Set} \times \mathbf{Set}$ has two points up to isomorphism, whereas a theory of two independent nullary predicates has four set-valued models. The displayed sequents record the order relations stressed in the lecture; the covering sequents below are imposed on the appropriate underlying classifying presentation.

For every J-cover $(x_i \leq x)_{i \in I}$, add the axiom

$$R_x \vdash \bigvee_{i \in I} R_{x_i}. \tag{169.8}$$

The new theory classifies $\mathrm{Sh}(C, J)$. A cover is thus read logically: whenever x holds, at least one of the pieces x_i must hold.

If C is a basis of open sets of a space X, use a symbol R_U for each $U \in C$. The order gives

$$U' \subseteq U \quad \Longrightarrow \quad R_{U'} \vdash R_U,$$

while an ordinary open cover $U = \bigcup_i U_i$ gives

$$R_U \vdash \bigvee_i R_{U_i}.$$

For the measure-induced topology J_μ, the additional axioms are exactly

$$R_U \vdash \bigvee_i R_{U_i} \qquad \text{when} \qquad \mu\left(U \setminus \bigcup_i U_i\right) = 0. \tag{169.9}$$

The interpretation is precise: a measure-zero omission does not prevent the family from being declared a cover. The resulting geometric theory records *almost-everywhere* covering relations as axioms. Nothing in this construction has yet referred to a training procedure or the internal computation of a neural network; its logical content is derived from the chosen measure and its null sets.

§169.12 Sequence spaces and product measures

We now begin the final part of the lecture. Let X be an arbitrary topological space and set

$$X^{\mathbb{N}} = \prod_{n \in \mathbb{N}} X.$$

Its points are sequences $x_\bullet = (x_0, x_1, \ldots)$. The product topology has a basis consisting of sets

$$U = \prod_{n \in \mathbb{N}} U_n, \qquad U_n \subseteq X \text{ open}, \qquad U_n = X \text{ for all sufficiently large } n. \tag{169.10}$$

When $X = M^{\mathbb{N}}$ is the space of infinite texts, a basic open subset of $X^{\mathbb{N}}$ specifies finitely many prefixes:

$$U = \left(\prod_{0 \le i < N} U_{m_i} \right) \times \left(\prod_{i \ge N} X \right).$$

One may picture a point of $X^{\mathbb{N}}$ as an infinite library of infinite texts. A basic open set puts restrictions on only finitely many entries in the library.

If μ is a probability measure on X, there is a product measure $\mu^{\mathbb{N}}$ on $X^{\mathbb{N}}$. On product cylinders it is given by

$$\mu^{\mathbb{N}} \left(\prod_{n \in \mathbb{N}} U_n \right) = \prod_{n \in \mathbb{N}} \mu(U_n), \tag{169.11}$$

with all but finitely many factors equal to 1. Under this measure, the coordinates $x_0, x_1, \ldots$ are independent, identically distributed samples with law μ.

§169.13 Occurrence frequencies

Fix an open set $U \subseteq X$. For each sample sequence $x_\bullet \in X^{\mathbb{N}}$, the proportion of its first n entries that belong to U is

$$p_n^U(x_\bullet) = \frac{\#\{0 \le i < n : x_i \in U\}}{n} = \frac{1}{n} \sum_{i=0}^{n-1} \mathbf{1}_U(x_i). \tag{169.12}$$

The last expression will be the bridge to the law of large numbers.

A sequence of frequencies need not converge for a particular $x_\bullet$. Define its upper and lower limiting frequencies by

$$p_+^U(x_\bullet) = \lim_{n \to \infty} \sup_{k \ge n} p_k^U(x_\bullet), \qquad p_-^U(x_\bullet) = \lim_{n \to \infty} \inf_{k \ge n} p_k^U(x_\bullet).$$

Always

$$0 \le p_-^U(x_\bullet) \le p_+^U(x_\bullet) \le 1.$$

They are equal precisely when the sequence $p_n^U(x_\bullet)$ converges. At this stage these are *pointwise* definitions for arbitrary sequences; randomness has not yet supplied a limit.

§169.14 Subspaces defined by limiting frequencies

Choose a dense subset $Q \subseteq [0,1]$; to make countability explicit we may take $Q = \mathbb{Q} \cap [0,1]$. For every open $U \subseteq X$ and $q \in Q$, define

$$P_{\ge q}^U(X^{\mathbb{N}}) = \{x_\bullet : p_-^U(x_\bullet) \ge q\}, \tag{169.13}$$

$$P_{\le q}^U(X^{\mathbb{N}}) = \{x_\bullet : p_+^U(x_\bullet) \le q\}. \tag{169.14}$$

These are the two families of subspaces used on slide 15.

The quantifiers hidden in these definitions are important. For $q' < q$,

$$x_\bullet \in P_{\ge q}^U \quad \Longleftrightarrow \quad \#\{n \ge 1 : p_n^U(x_\bullet) \le q'\} < \infty \quad \text{for every } q' < q.$$

In words: eventually the observed frequency stays above every threshold strictly smaller than q. Dually,

$$x_\bullet \in P_{\le q}^U \quad \Longleftrightarrow \quad \#\{n \ge 1 : p_n^U(x_\bullet) \ge q'\} < \infty \quad \text{for every } q' > q.$$

Lemma 169.14.1 (Measurability of the frequency subspaces)
For each open U and $q \in [0,1]$, both $P^U_{\geq q}$ and $P^U_{\leq q}$ belong to the Borel σ-algebra of $X^{\mathbb{N}}$.

Proof. The coordinate maps $x_\bullet \mapsto x_i$ are continuous, so $\mathbf{1}_U(x_i)$ is Borel measurable. Each finite average p^U_n is therefore Borel. Moreover

$$P^U_{\geq q} = \bigcap_{r=1}^{\infty} \bigcup_{N=1}^{\infty} \bigcap_{n \geq N} \{p^U_n > q - 1/r\},$$

and

$$P^U_{\leq q} = \bigcap_{r=1}^{\infty} \bigcup_{N=1}^{\infty} \bigcap_{n \geq N} \{p^U_n < q + 1/r\}.$$

These are countable combinations of Borel sets. □

This lemma is necessary before assigning a product-measure probability to the new subspaces. They are defined by infinite limiting conditions, yet remain measurable.

§169.15 The law of large numbers and measure reconstruction

The last slide contains the central result. The first part states that for $q > q'$,

$$\mu^{\mathbb{N}}(P^U_{\geq q}(X^{\mathbb{N}}) \cap P^U_{\leq q'}(X^{\mathbb{N}})) = 0. \tag{169.15}$$

In fact the intersection is empty: no sequence can have $p^U_- \geq q > q' \geq p^U_+$, since the lower limit cannot exceed the upper limit.

The substantive part states that these subspaces have *zero-or-one probabilities* determined exactly by $\mu(U)$:

Theorem 169.15.1 (Law of large numbers in the form used by the lecture)
Let X be a regular topological space carrying a Borel probability measure μ, and give $X^{\mathbb{N}}$ the independent product measure $\mu^{\mathbb{N}}$. For every open $U \subseteq X$, writing $p = \mu(U)$, we have

$$\mu^{\mathbb{N}}(P^U_{\geq q}) = \begin{cases} 1, & q \leq p, \\ 0, & q > p, \end{cases} \qquad (q \in Q), \tag{169.16}$$

and

$$\mu^{\mathbb{N}}(P^U_{\leq q}) = \begin{cases} 1, & q \geq p, \\ 0, & q < p. \end{cases} \qquad (q \in Q). \tag{169.17}$$

Let us unfold the probabilistic argument rather than treating the theorem as a black box. For a fixed open U, define random variables on $X^{\mathbb{N}}$

$$Y_i(x_\bullet) = \mathbf{1}_U(x_i).$$

Because the coordinates are independent with law μ, the Y_i are i.i.d. Bernoulli random variables with parameter $p = \mu(U)$. Therefore

$$\mathbb{E}Y_i = p, \qquad \operatorname{Var}(Y_i) = p(1-p), \qquad p^U_n = \frac{1}{n} \sum_{i=0}^{n-1} Y_i.$$

By independence,

$$\mathbb{E}p_n^U = p, \qquad \mathrm{Var}(p_n^U) = \frac{p(1-p)}{n}. \tag{169.18}$$

Chebyshev's inequality immediately gives

$$\mu^{\mathbb{N}}(|p_n^U - p| \geq \varepsilon) \leq \frac{p(1-p)}{n\varepsilon^2}.$$

This proves convergence *in probability*. To identify the sets $P_{\geq q}^U$ and $P_{\leq q}^U$, however, we need convergence *almost surely*.

One route to the stronger statement uses the exponential tail bound for bounded independent variables:

$$\mu^{\mathbb{N}}(|p_n^U - p| \geq \varepsilon) \leq 2e^{-2n\varepsilon^2}. \tag{169.19}$$

For completeness, its mechanism is a one-line Chernoff argument after the bounded-variable moment estimate $\mathbb{E}e^{t(Y_i-p)} \leq e^{t^2/8}$. Independence and Markov's inequality give

$$\Pr(p_n^U - p \geq \varepsilon) \leq \inf_{t>0} \exp\left(-nt\varepsilon + \frac{nt^2}{8}\right) = e^{-2n\varepsilon^2},$$

and the lower tail is identical. For each fixed $\varepsilon > 0$,

$$\sum_{n=1}^{\infty} \mu^{\mathbb{N}}(|p_n^U - p| \geq \varepsilon) \leq \sum_{n=1}^{\infty} 2e^{-2n\varepsilon^2} < \infty.$$

The first Borel–Cantelli lemma now says that the deviation event occurs only finitely often, almost surely. Applying this for $\varepsilon = 1, 1/2, 1/3, \ldots$ and intersecting those countably many probability-one events yields

$$\mu^{\mathbb{N}}\left(\left\{x_\bullet : \lim_{n\to\infty} p_n^U(x_\bullet) = p\right\}\right) = 1. \tag{169.20}$$

On this full-measure set, both p_-^U and p_+^U equal p. The two zero-or-one formulas in the theorem follow at once. Notice that the proof requires *independent sampled elements of* X. When X is itself a space of infinite texts, the x_i are independently sampled *texts*, not successive tokens in a single generated text.

Now comes the converse direction which motivates the entire lecture.

Proposition 169.15.2 (Recovering a measure from full-measure frequency events)

For every open $U \subseteq X$, the number $\mu(U)$ is uniquely determined by which of the measurable sets $P_{\geq q}^U$ and $P_{\leq q}^U$ have full $\mu^{\mathbb{N}}$-measure. In fact,

$$\mu(U) = \sup\{q \in Q : \mu^{\mathbb{N}}(P_{\geq q}^U) = 1\}, \tag{169.21}$$

$$= \inf\{q \in Q : \mu^{\mathbb{N}}(P_{\leq q}^U) = 1\}. \tag{169.22}$$

Proof. The first probability equals 1 exactly for $q \leq \mu(U)$, and the second exactly for $q \geq \mu(U)$. Density of Q in $[0,1]$ identifies the common boundary. We include $0, 1 \in Q$, so the formulas also cover the endpoint cases. □

This is where the measure-induced topology reappears. Apply the construction of J_μ to the measurable subsets of $X^{\mathbb{N}}$, equipped with $\mu^{\mathbb{N}}$. Its covering relation tells us

which measurable subsets are null; equivalently, by taking complements, which have full measure. Thus it tells us the truth values of

$$\mu^{\mathbb{N}}(P^U_{\geq q}) = 1, \qquad \mu^{\mathbb{N}}(P^U_{\leq q}) = 1.$$

The reconstruction formulas then recover $\mu(U)$ for every open U. This is the final assertion of the lecture: the numerical probability measure on X is completely determined by a Grothendieck topology $J_{\mu^{\mathbb{N}}}$ on an appropriate ordered collection of measurable subsets of the *sequence space* $X^{\mathbb{N}}$.

Example 169.15.3 (Two coins with the same finite support)

Let $X = \{0, 1\}$, take $U = \{1\}$, and let $\mu_p(\{1\}) = p$ for $0 < p < 1$. For both $p = 1/3$ and $p = 2/3$, every finite binary word has positive probability. Merely asking which finite cylinders are possible cannot distinguish the two measures. But consider the subset

$$A_{1/3} = \left\{ x_\bullet : \lim_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} x_i = \frac{1}{3} \right\}.$$

Then

$$\mu^{\mathbb{N}}_{1/3}(A_{1/3}) = 1, \qquad \mu^{\mathbb{N}}_{2/3}(A_{1/3}) = 0.$$

The two product measures have different null sets, even though their finite-prefix supports coincide. Indeed, the same argument shows that if two probability measures $\mu \neq \nu$ differ on some open U, their infinite independent product measures are mutually singular: the set where the empirical frequency of U converges to $\mu(U)$ is full under one and null under the other.

For the language-model space $X = M^{\mathbb{N}}$, there is a useful further observation. The family of prefix cylinders

$$\mathcal{B} = \{U_m : m \in C_M\}$$

is countable. Hence the intersection over all $U \in \mathcal{B}$ of the probability-one events from (169.20) still has probability one. With probability one, a single infinite independent library of texts has the correct limiting occurrence frequency for *every finite prefix simultaneously*. Those cylinder probabilities determine the Borel measure μ uniquely.

The result is mathematically exact, but its infinite-sample formulation should be kept distinct from finite statistical estimation. For any one fixed open U, the concentration inequality (169.19) also yields an operational bound:

$$n \geq \frac{\log(2/\delta)}{2\varepsilon^2} \quad \Longrightarrow \quad \Pr(|p^U_n - \mu(U)| \geq \varepsilon) \leq \delta.$$

Estimating many events simultaneously requires an additional bound on the class of events being considered. This finite-sample result is a supplementary consequence of the ordinary concentration argument; the topological reconstruction theorem itself is an assertion about exact limiting events and their nullity.

§169.16 What does the construction reveal about a model?

The mathematical result is quite beautiful: an exact real-valued probability law can be recovered from a yes-or-no notion of negligible subsets, provided we are allowed to pass

to an infinite independent sample space. The interesting question for AI begins when we ask what information about a *trained model* survives this construction.

Let $\mathfrak{M}$ denote a class of language-model implementations. The lecture defines a chain of maps of the form

$$\begin{aligned} &\mathfrak{M} \longrightarrow \mathrm{Prob}(M^{\mathbb{N}}) \longrightarrow \{\text{Grothendieck topologies}\}, \\ &F \longmapsto \mu_F \longmapsto J_{\mu_F}. \end{aligned} \tag{169.23}$$

There is also the refined map $\mu_F \mapsto J_{\mu_F^{\mathbb{N}}}$, from which the preceding section reconstructs μ_F. Every construction in this chain factors through the model's next-token conditional probabilities. It makes sense for an autoregressive Markov model, a probabilistic grammar, or a completely different implementation with the same conditionals. The mathematics does not use any special property of the Transformer.

That immediately raises an *identifiability* problem. Suppose two implementations F, G satisfy

$$p_F(a \mid m) = p_G(a \mid m) \qquad \text{for every finite history } m \text{ and token } a.$$

Then $\mu_F = \mu_G$, hence both $J_{\mu_F} = J_{\mu_G}$ and $J_{\mu_F^{\mathbb{N}}} = J_{\mu_G^{\mathbb{N}}}$. The latter construction can recover their complete common output law, but it cannot tell the two implementations apart.

There are elementary examples of this ambiguity even within a simple neural layer. If logits depend on a hidden vector through

$$z = Wh, \qquad h = Vx,$$

then for any invertible matrix A we can replace these parameters by

$$h' = Ah, \qquad W' = WA^{-1}$$

and obtain exactly the same z, and therefore the same probabilities after softmax. The coordinates of the hidden representation have changed. More radically, different circuits or architectures may implement exactly the same input-output function. Knowing that function does not recover the way it was computed.

The resulting limitation is mathematical, quite apart from whether toposes are useful in other contexts: every property extracted *solely* from μ_F is constant on the observational equivalence class

$$[F]_{\mathrm{out}} = \{G : \mu_G = \mu_F\}.$$

A theory intending to recover specific internal features, causal circuits or training-induced representations must use additional information, assumptions, or interventions that distinguish members of this class.

§169.17 What is possible after a prefix?

Slide 9 says that the measure topology tells us which continuations of a text are possible. For an ordinary softmax output

$$p_\theta(a \mid m) = \frac{e^{z_a(m)}}{\sum_{b \in M} e^{z_b(m)}},$$

every token has strictly positive probability, provided the logits are finite. Therefore every finite word $m = (m_1, \ldots, m_\ell)$ has

$$\mu_\theta(U_m) = \prod_{i=1}^{\ell} p_\theta(m_i \mid m_{<i}) > 0.$$

All finite continuations are possible in this support-theoretic sense, including incoherent and factually mistaken ones. For a fixed ordinary softmax model, the question of *which finite continuation has positive probability* tells us essentially nothing about its graded judgements.

There is an important distinction here. Full support on finite cylinders does *not* imply that the measure-induced topology J_μ is trivial. A particular infinite sequence can have probability zero:

$$\mu(\{a\}) = \lim_{\ell\to\infty} \mu(U_{a_1,\dots,a_\ell}),$$

and null sets need not be open. The infinite-sample construction detects more than finite-word support, as the two-coin example showed. My concern is instead the inference from these mathematical relations to the model's linguistic or conceptual competence. The extra information supplied by the infinite-sample topology is the model's *output probability law*; it does not disclose the internal computations that produced the law.

A further choice is usually left implicit: which distribution are we analyzing? The unmodified model defines $p_\theta(\cdot \mid m)$, but an actual decoder may transform it. For top-k sampling, for example,

$$p_{\theta,k}(a \mid m) = \frac{\mathbf{1}\{a \in K_k(m)\}\, p_\theta(a \mid m)}{\sum_{b\in K_k(m)} p_\theta(b \mid m)},$$

where $K_k(m)$ is the selected set of k tokens. Tokens outside $K_k(m)$ now have exactly zero probability. Top-p sampling changes the support as well. The same trained parameters θ can consequently produce different measures and different associated topologies under different decoding rules.

One may decide to study the raw model distribution and avoid that ambiguity. The object under study is then unambiguously the pre-decoding output law. A different tokenizer introduces another representation choice: text strings are encoded into different token spaces, and a comparison between their measures requires specifying a map between those spaces and, where appropriate, pushing measures forward along it. A topological construction does not by itself guarantee invariance under such choices.

§169.18 Logical semantics and learned semantics

The use of geometric logic in the lecture is mathematically exact. The axiom

$$R_U \vdash \bigvee_i R_{U_i}$$

means that the family covers U; for the measure topology, it expresses a relation holding up to a null set. A classifying topos supplies the appropriate semantic universe for a formal theory, and quotient theories correspond to subtoposes. These are substantive connections between logic, geometry and measure.

The word *semantics*, however, refers to another problem in machine learning. Formal semantics begins with a language and an interpretation: it asks whether a formula is true in a model. Representation learning asks how a system acquires useful concepts from experience, whether those concepts transfer across tasks, and whether they causally participate in computation.

If a formal language already contains a predicate $\mathrm{Cat}(x)$, the theory may encode relations involving that predicate. It has not thereby explained how a visual model learns to recognize cats from pixels, or why one learned feature is stable under changes of viewpoint and another is not. Likewise, naming a predicate after a scientific mechanism

does not discover that mechanism in data. The interpretation connecting the symbol to observations is an essential part of the problem.

This matters especially when the concepts themselves are unknown. In scientific learning we may not know in advance which properties or interactions deserve a predicate. We want a method that can discover useful distinctions, discover that a proposed distinction fails, or form a more appropriate representation. Describing already-interpreted propositions inside a rich logic is valuable, but the correspondence between those propositions and empirical phenomena still needs to be established.

There is a sharp mathematical version of this concern. Given finitely many observations D, consider the version space

$$\mathcal{V}(D) = \{T : \ T \text{ is consistent with all observations in } D\}.$$

Many incompatible theories can lie in $\mathcal{V}(D)$, some generalizing to new observations and others fitting accidents of the sample. The topos–quotient correspondence can represent families of added axioms, but representation of the candidates does not select the right one. Generalization requires an inductive bias, objective, prior, additional evidence or some other selection principle.

Lafforgue's 2024 Barcelona talk, *Some Sketches for a Topos-Theoretic AI*, approaches this problem through a chosen descriptive vocabulary, relations among its predicates, quotient theories and information-extraction morphisms.[1] It is a more explicit proposal for symbolic induction than the probability-space construction discussed in Tsinghua. The specification of a language and a family of morphisms is still only part of a learning procedure: one would also need a way to discover or revise vocabulary, search for useful morphisms, validate hypotheses and compare their predictions.

§169.19 Numerical representations and the origin of meaning

In an earlier presentation, Lafforgue expressed concern that reducing descriptions of the world to collections of numbers makes their meaning disappear, and that enormous numerical ambient spaces contain many points with no meaningful interpretation.[2] This is an interesting motivation for seeking a different mathematical object on which learning could take place.

I find it hard to derive an inefficiency or loss of meaning from the mere use of coordinates. A numerical representation can preserve information perfectly, and a learned representation can organize that information in ways that support recognition, generalization and intervention. The relevant mathematical questions concern the map from data to representations, the structure of the image of that map, the inductive bias of the architecture and the behavior of the optimization algorithm.

For example, a hidden state may live in $\mathbb{R}^d$ while the set of states actually reached on data occupies a highly constrained subset

$$\mathcal{H}_\theta = \{h_\theta(x) : x \in \operatorname{supp}(P_{\text{data}})\} \subseteq \mathbb{R}^d.$$

Counting arbitrary points of the ambient space that have no semantic interpretation says little about the computational cost of evaluating h_θ. A numerical optimizer does not enumerate all of $\mathbb{R}^d$; and modern numerical hardware exploits the parallelism of matrix

[1] See Laurent Lafforgue, *Some sketches for a topos-theoretic AI* (Barcelona Mathematics and Machine Learning Colloquium, 2024).

[2] Laurent Lafforgue, *Some possible roles for AI of Grothendieck topos theory* (ETH Zürich, 2022).

operations. Efficiency is a property of the algorithm, representation, data distribution and machine together.

Interpretability research can at least formulate testable claims about hidden computation. Suppose a feature $f(h)$ is proposed to represent a concept. High correlation between $f(h)$ and the concept may motivate the hypothesis, but a causal claim requires an intervention. Schematically, for an output event A, one can compare

$$\Delta_f(x) = P_\theta(A \mid x, \mathrm{do}(f(h) := c)) - P_\theta(A \mid x),$$

with the intervention defined by the actual computational graph rather than by a merely symbolic manipulation. Activation patching, representation analysis and causal interventions try to determine whether a feature plays an operative role in a model's behavior. Work such as emphLocating and Editing Factual Associations in GPT (ROME) and Anthropic's emphMapping the Mind of a Large Language Model illustrates why a learned feature and a possible causal intervention are experimentally meaningful objects. Their empirical conclusions are limited, but they ask questions whose answers may be supported or refuted by experiments.

For a theory of topological or logical representations to address the same phenomenon, it would be helpful to identify a corresponding invariant, causal mechanism, predicted intervention effect, or learning rule. A formal interpretation of a predicate and an observed learned feature may eventually be connected; that connection requires an actual construction.

§169.20 Learning, generalization, and computational claims

Other remarks in earlier talks suggested that language models might be approaching a limit because available written text had already been used in training, and that topological methods might be closer to natural human thought. Those suggestions raise questions quite distinct from the topology of measures.

A limit on additional pretraining text does not imply a limit on all procedures that improve a model. Optimization, architecture, synthetic tasks, verification and reinforcement learning introduce other sources of learning signal. In a schematic policy-gradient calculation, for example,

$$\nabla_\theta J(\theta) = \mathbb{E}_{x,y\sim\pi_\theta}\left[(R(x,y) - b(x))\nabla_\theta \log \pi_\theta(y \mid x)\right].$$

Multiple sampled solutions to the same problem, followed by task-based reward or verification, can produce parameter updates without supplying a new human-written answer to every sample. This does not establish unlimited improvement: useful rewards and meaningful tasks remain necessary. It does show why corpus size alone cannot settle the question.

Likewise, the fact that topological language can describe a cognitive phenomenon does not imply that a human mind implements the corresponding mathematical operations. Such a claim would require independent evidence from cognitive modeling, neuroscience or a predictive account of behavior. A useful mathematical description and an account of the mechanism are different levels of explanation.

My broader interest in the proposed Topos-for-AI program can be expressed by separating several possible goals. One might try to replace vector spaces by new learnable objects while retaining data-driven representation discovery; this would require explicit learning algorithms and a theory of what is preserved by abstraction. One might keep neural networks and use toposes to organize the symbolic component of a neuro-symbolic

system; this would require an interface between continuous representations and interpreted predicates. Another possibility is to learn additional axioms inside an already chosen formal language; its successes and limits would depend on the search method and inductive principles. One may also study categorical structures attached to an existing computation or to its output law, as in the Tsinghua lecture. Each project has distinct mathematical obligations, and the existence of a topos presentation alone does not discharge them.

The lecture establishes a precise relation among probability laws, almost-sure covers, geometric theories and long-run frequencies. Its final reconstruction theorem is a satisfying demonstration of how much quantitative information can be encoded in apparently qualitative structure. The question that remains with me is what further structure, beyond the output law, would let us study the concepts and computations actually learned by a neural network.